UNITED STATES Coast Pilot®

1

Atlantic Coast: Eastport, Maine to Cape Cod, Massachusetts

2025 (55th) Edition

This edition cancels the 54th Edition and includes all previously published corrections.

Weekly updates to this edition are available at:
nauticalcharts.noaa.gov/publications/coast-pilot/index.html

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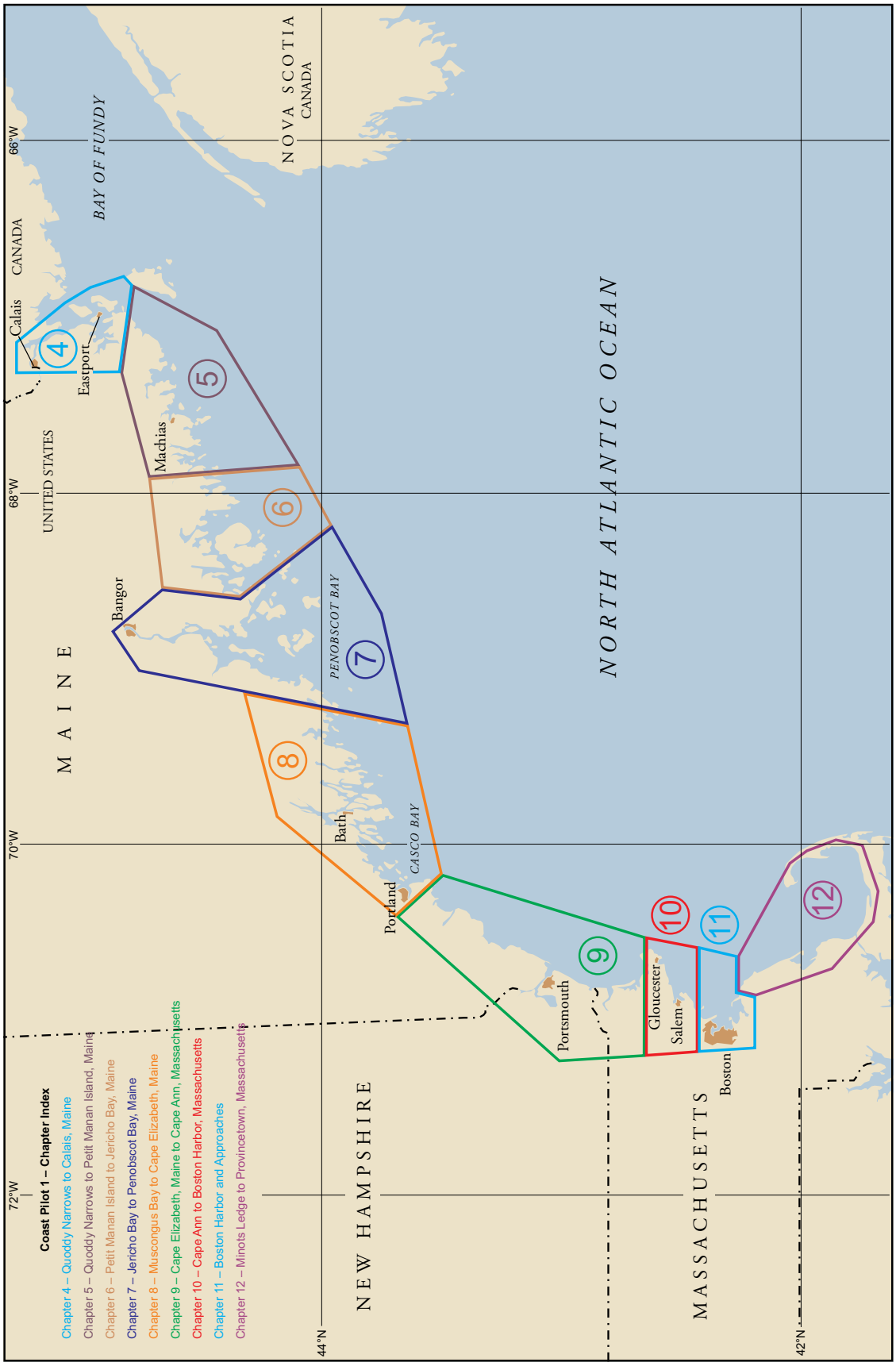

Coast Pilot 1 – Chapter Index
Chapter 4 – Quoddy Narrows to Calais, Maine
Chapter 5 – Quoddy Narrows to Petit Manan Island, Maine
Chapter 6 – Petit Manan Island to Jericho Bay, Maine
Chapter 7 – Jericho Bay to Penobscot Bay, Maine
Chapter 8 – Muscongus Bay to Cape Elizabeth, Maine
Chapter 9 – Cape Elizabeth, Maine to Cape Ann, Massachusetts
Chapter 10 – Cape Ann to Boston Harbor, Massachusetts
Chapter 11 – Boston Harbor and Approaches
Chapter 12 – Minots Ledge to Provincetown, Massachusetts
72°W
70°W
68°W
66°W
44°N
42°N
CANADA
UNITED STATES
MAINE
NEW HAMPSHIRE
MASSACHUSETTS
NOVA SCOTIA
CANADA
BAY OF FUNDY
PENOBSCOT BAY
CASCO BAY
NORTH ATLANTIC OCEAN
Calais
Eastport
Machias
Bangor
Bath
Portland
Portsmouth
Gloucester
Salem
Boston
4
5
6
7
8
9
10
11
12

Preface

The United States Coast Pilot is published by the National Ocean Service (NOS), National Oceanic and Atmospheric Administration (NOAA), pursuant to the Act of 6 August 1947 (33 U.S.C. 883a and b), and the Act of 22 October 1968 (44 U.S.C. 1310).

The Coast Pilot supplements the navigational information shown on NOAA nautical charts. The publication is continually updated and maintained from inspections conducted by NOAA survey vessels and field parties, corrections published in Notices to Mariners, information from other Federal agencies, State and local governments, maritime and pilots' associations, port authorities, and concerned mariners.

NOAA's Office of Coast Survey encourages public feedback regarding its suite of nautical charting products and services through **ASSIST**, Coast Survey's stakeholder engagement and feedback tool. This allows customers to submit questions or comments or to report an error with NOAA's nautical charts and products.
Customers can access **ASSIST** at *www.nauticalcharts.noaa.gov/customer-service/assist/*
Those who prefer to communicate by telephone can contact Coast Survey at 1–888–990–6622.

Coast Pilot corrections are no longer published in the NGA Notice to Mariners effective 01 January 2021. Additional information regarding the NGA policy change can be referenced at *msi.nga.mil/NTM* in the Notice to Mariners 52/20 Hydrogram and Marine Information sections.

Coast Pilot Updates

Check for weekly critical updates for this edition at *nauticalcharts.noaa.gov/publications/coast-pilot/index.html*
(See **33 CFR 164.33 Charts and Publications**, chapter 2, for regulations.)

Customers may print the specifically affected paragraphs to revise this book, or download an updated file (PDF) of the entire volume.

A *Weekly Record of Updates* is provided directly preceding the index.

Contents

Preface III

Chapter 1: General Information 1

Chapter 2: Navigation Regulations 35

Chapter 3: Eastport to Cape Cod 133

Chapter 4: Quoddy Narrows to Calais, Maine 161

Chapter 5: Quoddy Narrows to Petit Manan Island, Maine 171

Chapter 6: Petit Manan Island to Jericho Bay, Maine 187

Chapter 7: Jericho Bay to Penobscot Bay, Maine 213

Chapter 8: Muscongus Bay to Cape Elizabeth, Maine 251

Chapter 9: Cape Elizabeth, Maine to Cape Ann, Massachusetts. 295

Chapter 10: Cape Ann to Boston Harbor, Massachusetts. 331

Chapter 11: Boston Harbor and Approaches 345

Chapter 12: Minots Ledge to Provincetown, Massachusetts 363

Navigation Rules 381

Appendix A 407

Weekly Record of Updates. 413

Index 417

General Information

(1)

UNITED STATES COAST PILOT®

(2) The United States Coast Pilot, published by the National Oceanic and Atmospheric Administration (NOAA), is a series of ten nautical books (volumes) that encompasses a wide variety of information important to navigators of U.S. coastal/intracoastal waters and the waters of the Great Lakes. The Coast Pilot is intended to be used as a supplement to NOAA nautical charts. Much of the content cannot be shown graphically on the charts and is not readily available elsewhere. Topics which are covered include environmental factors of weather, climate, ice conditions, tides, water levels, currents, prominent coastal features and landmarks. Specific information on vertical clearances, wharf descriptions, small-craft facilities, hazards, dredged channels and depths are also provided. Navigation services and regulations are also identified including pilotage, towing, anchorages, routes and traffic separation schemes, environmental protection, and other Federal laws.

(3) New editions of each volume are issued annually. Fully updated files are posted weekly on the Internet, and are also available through NOAA Certified Chart Agents at *www.nauticalcharts.noaa.gov*.

(4) **Amendments** to this publication are available at *nauticalcharts.noaa.gov/publications/coast-pilot/index.html.*

(5)

Using the Coast Pilot

(6) **Chapter 1** contains definitions of general and standard terms used throughout the volume, discussions of NOAA charting products and services, descriptions of maritime services by various U.S. Government agencies, Notices to Mariners and other information pertinent to safe navigation.

(7) **Chapter 2** contains selected extracts from the Code of Federal Regulations (CFR) that affect mariners.

(8) **Chapter 3** contains general information that is peculiar to the region covered by a particular Coast Pilot volume. For example, practical information regarding offshore currents and dangers, coastal aids to navigation, prominent landmarks and the general character of the coast and depths helpful in approaching the region.

(9) In **Chapter 4 and the remaining numbered chapters**, the detailed description of the region begins. A map precedes each chapter and outlines the nautical charts used in the area to be discussed. In these chapters, as much as possible, the coastal description is in geographic sequence, north to south on the east coast, east to west on the gulf coast, clockwise around each of the Great Lakes and south to north on the west coast and Alaskan coast. Features are described as they appear on the largest scale chart, with that chart number prominently shown in blue.

(10) **Appendix A** contains contact information regarding the various products, services and agencies detailed throughout the volume.

(11) **Navigation Rules**— preceding Appendix A, contains the International (72 COLREGS) and Inland Navigation Rules, technical Annexes, and associated Federal rules and regulations.

(12) The **Weekly Record of Updates** is intended as a log for critical updates applied to this volume.

(13) The **Index** contains geographic names mentioned throughout a Coast Pilot volume. These names are boldfaced and indexed along with the number of the largest scale chart on which the entire feature appears. Asterisks preceding a chart number in the index of Coast Pilot 5 indicate charts published by the National Geospatial-Intelligence Agency, and in the index of Coast Pilot 6, charts published by the Canadian Hydrographic Service.

(14)

Bearings

(15) Bearings and courses are in degrees true and are measured clockwise from **000°** (north) to **359°**. The bearings of an aid to navigation (e.g., directional light, light sector, range) are given as viewed from the bridge of a vessel toward the light.

(16)

Bridges and Cables

(17) Vertical clearances of bridges and overhead cables are in feet above mean high water unless otherwise stated; clearances in Coast Pilot 6 are in feet above Low Water Datum unless otherwise stated. When the water level is above Low Water Datum, the bridge and overhead cable clearances given in the Coast Pilot and shown on the charts should be reduced accordingly. Clearances of drawbridges are for the closed position, although the open clearances are also given for vertical-lift bridges. Whenever a bridge span over a channel does not open fully to an unlimited clearance position, a minimum clearance for the sections over the channel is given; the same applies to swing and pontoon bridges with openings less than 50 feet horizontally. Clearances given in the Coast Pilot are those approved for nautical charting and are supplied by the U.S. Coast Guard (bridges) and U.S. Army Corps of Engineers (cables). See charts for horizontal clearances

of bridges, as these are generally given in the Coast Pilot only when they are less than 50 feet (15 meters). Tables listing structures across waterways, found in some Coast Pilots, show both horizontal and vertical clearances. Submarine cables are rarely mentioned.

(18)

Cable ferries

(19) Cable ferries are guided by cables fastened to shore and sometimes propelled by a cable rig attached to the shore. Generally, the cables are suspended during crossings and dropped to the bottom when the ferries dock. Where specific operating procedures are known they are mentioned in the text. Since operating procedures vary, mariners are advised to exercise extreme caution and seek local knowledge. **DO NOT ATTEMPT TO PASS A MOVING CABLE FERRY.**

(20)

Courses

(21) These are true and are given in degrees clockwise from **000°** (north) to **359°**. The courses given are the courses to be made good.

(22)

Currents

(23) Stated current velocities are the averages at strength. Velocities are in knots, which are nautical miles per hour. Directions are the true directions to which the currents set (see chapter 3, this book).

(24)

Depths

(25) Depth is the vertical distance from the chart datum to the bottom and is expressed in the same units (feet, meters or fathoms) as those soundings found on the chart. (See Chart Datum, this chapter, for further detail.) The **controlling depth** is the least known depth of a channel. This depth is determined by periodic hydrographic surveys and restricts use of the channel to drafts less than that depth. The **centerline controlling depth** applies only to the channel centerline or close proximity; lesser depths may exist in the remainder of the channel. The **midchannel controlling depth** is the controlling depth of only the middle half of the channel. **Federal project depth** is the original design dredging depth of a channel planned by the U.S. Army Corps of Engineers (USACE) and may be deeper than current conditions. For this reason, project depth must not be confused with controlling depth. **Depths alongside** wharves usually have been reported by owners and/or operators of the waterfront facilities and have not been verified by Government surveys. Since these depths may be subject to change, local authorities should be consulted for the latest controlling depths.

(26) For all maintained channels with controlling depths detailed on charts in tabular form, the Coast Pilot usually states only the project depths. For all other channels which may be depicted on charts with depth legends, notes or soundings, the Coast Pilot will list where to find the most recent information on the latest known surveys. Depths may vary considerably between maintenance dredging.

(27)

Under-keel clearances

(28) It is becoming increasingly evident that economic pressures are causing mariners to navigate through waters of barely adequate depth, with under-keel clearances being finely assessed from the charted depths, predicted tide levels and depths recorded by echo sounders.

(29) It cannot be too strongly emphasized that even charts based on modern surveys may not show all sea-bed obstructions or the shoalest depths, and actual tide levels may be appreciably lower than those predicted.

(30) In many ships an appreciable correction must be applied to shoal soundings recorded by echo sounders due to the horizontal distance between the transducers. This separation correction, which is the amount by which recorded depths therefore exceed true depths, increases with decreasing depths to a maximum equal to half the distance apart of the transducers; at this maximum the transducers are aground. Ships whose transducers are more than 6 feet (1.8 meters) apart should construct a table of true and recorded depths using the Traverse Tables. (Refer to the topic on echo soundings elsewhere in chapter 1.)

(31) Other appreciable corrections, which must be applied to many ships, are for settlement and squat. These corrections depend on the depth of water below the keel, the hull form and the speed of the ship.

(32) Settlement causes the water level around the ship to be lower than would otherwise be the case. It will always cause echo soundings to be less than they would otherwise be. Settlement is appreciable when the depth is less than seven times the draft of the ship and increases as the depth decreases and the speed increases.

(33) Squat denotes a change in trim of a ship underway, relative to her trim when stopped. It usually causes the stern of a vessel to sit deeper in the water. However, it is reported that in the case of mammoth ships, squat causes the bow to sit deeper. Depending on the location of the echo sounding transducers, this may cause the recorded depth to be greater or less than it ought to be. **Caution and common sense are continuing requirements for safe navigation.**

(34)

Distances

(35) These are in nautical miles unless otherwise stated. A nautical mile is one minute of latitude, or approximately 2,000 yards, and is about 1.15 statute miles.

(36) Coast Pilot 6 is in statute miles unless otherwise stated. A statute mile is 5,280 feet or about 0.87 nautical mile.

(37)

Geographic Coordinates

(38) Geographic coordinates listed in the Coast Pilot are referred to North American Datum of 1983 (NAD 83) unless otherwise noted for certain CFR extracts in chapter 2.

(39)

Heights

(40) These are in feet (meters) above the tidal datum used for that purpose on the charts, usually mean high water. However, the heights of the decks of piers and wharves are given in feet (meters) above the chart datum for depths.

(41) Coast Pilot 6 is in feet (meters) above the chart datum used for that purpose on the charts, usually Low Water Datum.

(42)

Light and Sound Signal Characteristics

(43) These are not described in the Coast Pilot. Also, light sectors and visible ranges are generally not fully described. This information can be found in U.S. Coast Guard Light Lists.

(44)

Obstructions

(45) Wrecks and other obstructions are mentioned only if they are relatively permanent and in or near normal traffic routes.

(46)

Radio Navigational Aids

(47) For detailed information on Radio Navigation Aids see the **United States Coast Guard Light Lists** and the National Geospatial-Intelligence Agency's **Radio Navigational Aids, Publication 117**.

(48)

Ranges

(49) These are not fully described. "**A 339°** Range" means that the rear structure bears **339°** from the front structure. (See United States Coast Guard Light Lists.)

(50)

Reported information

(51) Information received by NOAA from various sources concerning depths, dangers, currents, facilities, and other topics, which has not been verified by Government surveys or inspections, is often included in the Coast Pilot; such **unverified information** is qualified as "reported" and should be regarded with caution.

(52)

Tides

(53) Tidal information, including real-time water levels, tide predictions and tidal current predictions are available at *tidesandcurrents.noaa.gov*.

(54)

Time

(55) Unless otherwise stated, all times are given in local standard time in the 24-hour system. (Noon is 1200, 2:00 p.m. is 1400 and midnight is 0000.)

(56)

Winds

(57) Directions are the true directions from which the winds blow; however, sometimes (rarely) compass points are used. Unless otherwise indicated, speeds are given in knots, which are nautical miles per hour.

(58)

NAUTICAL CHARTS

(59) NOAA produces and maintains a suite of over 1,000 nautical charts that cover the U.S. coastal waters, the Great Lakes and U.S. territories. These charts provide a graphic representation of water depths, the shoreline, prominent topographic and man-made features, aids to navigation and other navigational information useful to the mariner. NOAA's charts are available in a variety of digital formats designed to meet the specific requirements of all mariners. Paper copies may also be obtained through one of NOAA's Print-on-Demand partners.

(60)

Paper Print on Demand Nautical Charts

(61) The content of Print-On-Demand (POD) charts is updated weekly by NOAA with the most current U.S. Coast Guard Local Notice to Mariners and other critical safety information. POD charts are printed under the authority of NOAA and shipped through partnerships between NOAA and commercial providers. POD information and a list of participating POD chart agents can be found at *nauticalcharts.noaa.gov/publications/print-agents.html#paper-charts-mobile*.

(62)

Portable Document Format (PDF) Nautical Charts

(63) Almost all of NOAA's nautical charts may be downloaded for free as Portable Document Format (PDF) files at *nauticalcharts.noaa.gov/charts/noaa-raster-charts.html#full-size-nautical-charts*. The PDF nautical charts are exact replicas of the images used to produce POD and Raster Navigational Charts (RNC). As such, they also have all the latest updates based on U.S. Coast Guard Local Notices to Mariners, National Geospatial-Intelligence Agency Notices to Mariners and other critical safety information.

(64) Most PDF charts can be printed at the proper scale from any plotter accommodating a 36-inch paper width. When printed properly, PDF charts and POD charts are very similar, but PDF charts have not yet been approved to meet Federal regulations for paper chart carriage requirements as POD charts have.

(65)

BookletCharts

(66) The NOAA BookletChart™ is a product that can be printed by the users for free. They are made to help recreational boaters locate themselves on the water. BookletCharts are reduced in scale and divided into pages for convenience but otherwise contain all the information of the full-scale nautical charts and are updated weekly. For more information visit *nauticalcharts.noaa.gov/charts/noaa-raster-charts.html#booklet-charts*.

(67)

Raster Navigational Charts (NOAA RNC®)

(68) NOAA Raster Navigational Charts (NOAA RNC®) are geo-referenced digital images of NOAA's entire suite of paper charts. NOAA RNCs are official data that can be used in many types of electronic charting systems (ECS), including Raster Chart Display Systems (RCDS) and some Electronic Chart Display and Information Systems (ECDIS). Current regulations support the use of RNCs as a primary means of navigation when ENCs are not available, but they require an accompanying minimal set of up-to-date paper charts. They can integrate position information from the Global Positioning System (GPS) and other navigational sensors, such as radar and automatic identification systems (AIS) to show a vessel's track, waypoints, and planned routes. NOAA RNCs and their weekly updates are available free of charge at *nauticalcharts.noaa.gov/charts/noaa-raster-charts.html*.

(69)

Electronic Navigational Charts (NOAA ENC®)

(70) NOAA Electronic Navigational Charts (NOAA ENC®) are databases of charted objects and their attributes with standardized content, structure and format. They comply with International Hydrographic Organization (IHO) specifications stated in IHO Publication S-57. They may be used as an alternative to paper charts required on SOLAS class vessels.

(71) ENCs are intended for use in electronic charting systems (ECS) as well as Electronic Chart Display and Information Systems (ECDIS). ECDIS are programmable to show as much or as little data as the user requires. They can integrate position information from the Global Positioning System (GPS) and other navigational sensors, such as radar and automatic identification systems (AIS) to show a vessel's track, waypoints and planned routes. Using this information ECDIS can use ENCs to give warning of impending danger in relation to the vessel's position and movement. NOAA ENCs and their updates are available free of charge at *nauticalcharts.noaa.gov/charts/noaa-enc.html*.

(72)

Nautical Chart—New Editions and Corrections

(73) New editions of paper Print-on-Demand (POD) charts are available on the Monday after NOAA clears a new edition for release. Once the authorized POD chart is available, it meets federal chart carriage requirements, and should be put into service immediately. It should be updated from the *last correction and cleared through* dates shown in the lower left corner of the chart.

(74) The chart date is of vital importance to the navigator. When charted information becomes obsolete, further use of the chart for navigation is dangerous. Natural and artificial changes, many of them critical, are occurring constantly; therefore it is important that navigators use up-to-date charts. Nautical charts and publications are available for purchase from authorized POD agents and their sales outlets.

(75) NOAA's "Nautical Chart Update" website allows mariners to update their nautical charts from one database that includes information from NOAA, NGA U.S. Notice to Mariners, U.S. Coast Guard Local Notices to Mariners and the Canadian Coast Guard Notices to Mariners at: *nauticalcharts.noaa.gov/charts/chart-updates.html*.

(76)

Nautical Chart Numbering System

(77) This chart numbering system, adopted by NOAA and National Geospatial-Intelligence Agency (NGA), provides for a uniform method of identifying charts published by both agencies. Nautical charts published by NGA and by the Canadian Hydrographic Service are identified in the Coast Pilot by an asterisk preceding the chart number.

(78)

Chart Scale

(79) The scale of a chart is the ratio of a given distance on the chart to the actual distance that it represents on the earth. For example, one unit of measurement on a 1:10,000 scale chart is equal to 10,000 of the same unit on the earth's surface. Large scale charts show greater detail of a relatively small area. Small scale charts show less detail but cover a larger area. Certain hydrographic information may be omitted on smaller scale charts. **Mariners should always obtain the largest scale coverage for near shore navigation**.

(80) The scales of nautical charts range from 1:2,500 to about 1:5,000,000. Graphic scales are generally shown on charts with scales of 1:80,000 or larger, and numerical scales are given on smaller scale charts. NOAA charts are classified according to scale as follows:

(81) **Sailing charts**, scales 1:600,000 and smaller, are for use in fixing the mariner's position approaching the coast from the open ocean or for sailing between distant coastwise ports. On such charts the shoreline and topography are generalized and only offshore soundings, principal lights, outer buoys and landmarks visible at considerable distances are shown.

(82) **General charts**, scales 1:150,000 to 1:600,000, are for coastwise navigation outside of outlying reefs and shoals.

(83) **Coast charts**, scales 1:50,000 to 1:150,000, are for inshore navigation leading to bays and harbors of considerable width and for navigating large inland waterways.

(84) **Harbor charts**, scales larger than 1:50,000, are for harbors, anchorage areas and the smaller waterways.

(85) **Special charts**, at various scales, cover the Intracoastal waterway and miscellaneous small-craft areas.

(86)

Chart Projections

(87) The **Mercator projection** used on most nautical charts has straight-line meridians and parallels that

intersect at right angles. On any particular chart the distances between meridians are equal throughout, but distances between parallels increase progressively from the equator toward the poles so that a straight line between any two points is a rhumb line. This unique property of the Mercator projection is one of the main reasons why it is preferred by the mariner.

(88) The **Polyconic projection** is used on most U.S. nautical charts of the Great Lakes. On this projection, parallels of latitude appear as non-concentric circles, and meridians appear as curved lines converging toward the pole and concave to the central meridian. The scale is correct along any parallel and along the central meridian of the projection. Along other meridians the scale increases with increased difference of longitude from the central meridian.

(89)

Chart Datum, Tidal Waters

(90) Chart Datum is the particular tidal level to which soundings and depth curves on a nautical chart or bathymetric map are referred. The tidal datum of **Mean Lower Low Water** is used on all NOAA charts, except for charts in the Great Lakes and non-tidal inland waterways. For information on **Chart Datum, Great Lakes System**, see Coast Pilot 6, chapter 3.

(91)

Horizontal Datum

(92) Nautical charts are constructed based on one of a number of horizontal datums which are adopted to best represent individual regions around the world. Note that the terms horizontal datum, horizontal geodetic datum, and horizontal control datum are synonymous.

(93) The exact placement of lines of latitude and longitude on a nautical chart is dependent on the referenced horizontal datum. Charts of the United States are currently referenced primarily to the North American Datum of 1983 (NAD 83), and the World Geodetic System 1984 (WGS 84). WGS 84 is equivalent to the NAD 83 for charting purposes.

(94) NAD 83 and WGS 84 have replaced the North American Datum of 1927 and other regional datums as the primary horizontal datum to which NOAA charts are referenced. Since some geographic positions may still be referenced to the older datums, NOAA has included notes on charts which show the amount to shift those positions in latitude and longitude to fit the chart's NAD 83 or WGS 84 projection.

(95) It should be noted that the physical shift between positions on older datums and NAD 83/WGS 84 was significant. Mariners should always be certain the positions they are plotting on a nautical chart are on the same datum as the chart.

(96)

Chart Accuracy

(97) The value of a nautical chart depends upon the accuracy of the surveys on which it is based. The chart reflects what was found by field surveys and what has been reported to NOAA. It also represents general conditions at the time of surveys or reports and does not necessarily portray present conditions. Significant changes may have taken place since the date of the last survey or report.

(98) Each sounding represents an actual measure of depth and location at the time the survey was made, and each bottom characteristic represents a sampling of the surface layer of the sea bottom at the time of the sampling. Areas where sand and mud prevail, especially the entrances and approaches to bays and rivers exposed to strong tidal current and heavy seas, are subject to continual change.

(99) In coral regions and where rocks and boulders abound, it is always possible that surveys may have failed to find every obstruction. Thus, when navigating such waters, customary routes and channels should be followed, and areas where irregular and sudden changes in depth indicate conditions associated with pinnacle rocks, coral heads, or boulders should be avoided.

(100) Information charted as "reported" should be treated with caution when navigating the area, because the actual conditions have not been verified by government surveys.

(101)

Source Diagrams and Zone of Confidence Diagrams

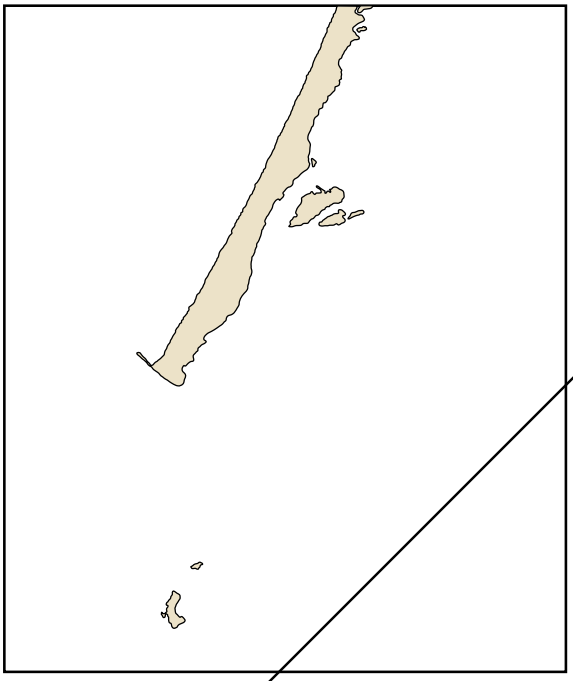
(102) The age and accuracy of hydrographic survey data that support nautical charts can vary. Depth information on nautical charts, paper or digital, is based on data from the latest available hydrographic survey, which in many cases may be quite old. Diagrams are provided on nautical charts to assist mariners in assessing hydrographic survey data and the associated level of risk to navigate in a particular area. There are currently two types of diagrams shown on NOAA paper and raster navigational charts (RNCs) of 1:500,000 scale and larger—**Zone of Confidence (ZOC) Diagrams** and **Source Diagrams**. ZOC information (designated CATZOC) is also found on electronic navigational charts (ENCs). This provides consistency in the display of source data between ENCs and newer paper charts.

(103) Both types of diagrams consist of a graphic representation of the extents of hydrographic surveys within the chart and accompanying table of related survey quality categories. CATZOC information on an ENC, unlike the diagrams on a paper chart or RNC, is displayed over the ENC data using symbols rather than letters. These symbols are displayed on a separate layer, which can be viewed when planning a route, then switched off until needed again at another time.

(104) On **ZOC Diagrams**, the quality of the hydrographic data is assessed according to six categories; five quality categories for assessed data (A1, A2, B, C and D) and a sixth category (U) for data that has not yet been assessed. On the ENC, the categories are shown using a rating system of stars—the higher the quality, the greater the number of stars. Assessment of hydrographic data quality and classification into zones of confidence is based on a combination of: survey date, position accuracy, depth

Zone of Confidence Diagrams

ZOC CATEGORIES				
ZOC	DATE	POSITION ACCURACY	DEPTH ACCURACY	SEAFLOOR COVERAGE
A1	2008–2016	± 16.4 ft	= 1.6 feet + 1% depth	All significant seafloor features detected
A2	—	± 65.6 ft	= 3.3 feet + 2% depth	All significant seafloor features detected
B	2005	± 164.0 ft	= 3.3 feet + 2% depth	Uncharted features hazardous to surface navigation are not expected but may exist
C	—	± 1640.4 ft	= 6.6 feet + 2% depth	Depth anomalies may be expected
D	—	Worse than ZOC C	Worse than ZOC C	Large depth anomolies may be expected
U	Unassessed – The quality of the bathymetric data has yet to be assessed.			



accuracy and sea floor coverage (the survey's ability to detect objects on the seafloor.)

(105) **Source Diagrams** provide the mariner with additional information about the density and adequacy of the sounding data depicted on the chart. The adequacy with which sounding data reflects the configuration of the bottom depends on the following factors: survey technology employed (sounding and navigation equipment), survey specifications in effect (prescribed survey line spacing and sounding interval) and type of bottom (e.g., rocky with existence of submerged pinnacles, flat sandy, coastal deposits subject to frequent episodes of deposition and erosion). Source diagrams will be replaced with ZOC diagrams as new editions are created.

(108)

Chart Symbols, Abbreviations and Terms

(109) The standard symbols and abbreviations approved for use on nautical charts produced by the U.S. Government are described in **U.S. Chart No. 1: Symbols, Abbreviations and Terms used on Paper and Electronic Navigational Charts.** This reference, jointly maintained by the National Geospatial-Intelligence Agency (NGA) and NOAA, is available at *nauticalcharts.noaa.gov/publications/us-chart-1.html.*

(110) The publication **Chart 1: Symbols, Abbreviations and Terms** published by the Canadian Hydrographic Service, is available online at *charts.gc.ca/publications/chart1-carte1/index-eng.asp.*

(111) Some symbols and abbreviations used on foreign charts, including reproductions of foreign charts made by NGA, are different than those used on U.S. charts. It is recommended that mariners who use foreign charts also obtain the symbol sheet or Chart No. 1 produced by the appropriate foreign agency.

(112) Mariners are warned that the buoyage systems, shapes and colors used by other countries often have a different significance than the U.S. system.

(113)

Areas with Blue Tint

(114) A blue tint is shown in water areas on many charts to accentuate shoals and other areas considered dangerous for navigation when using that particular chart. Since the danger curve varies with the intended purpose of a chart a careful inspection should be made to determine the contour depth of the blue tint areas.

(115)

Bridge and Cable Clearances

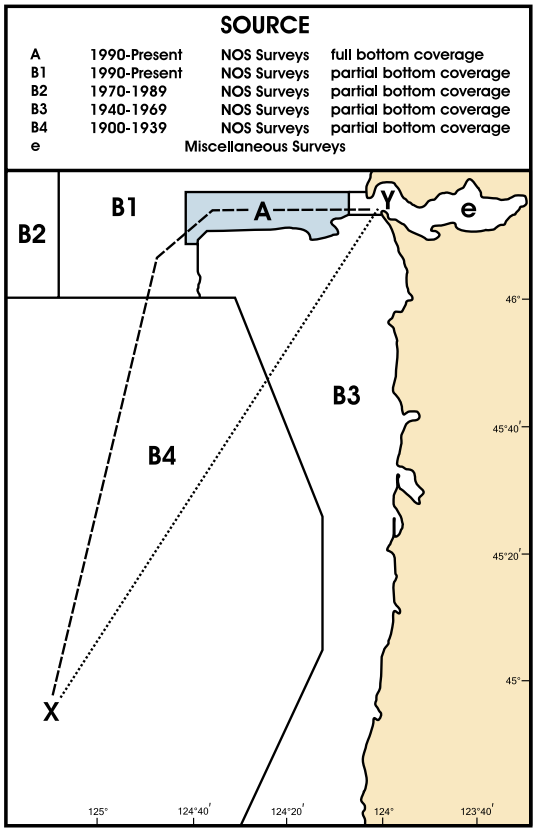
(116) For bascule bridges whose spans do not open to a full vertical position, unlimited overhead clearance is not available for the entire charted horizontal clearance when the bridge is open, due to the inclination of the drawspans over the channel.

(117) Charted in black text, vertical clearances of overhead cables are for the lowest wires at mean high water as authorized and permitted by the U.S. Army Corps of Engineers (USACE). Reported clearances received from sources other than the USACE are labeled as such. When

(107)

Source Diagrams

Referring to the accompanying sample Source Diagram to the right and the previous discussion of survey methods over time, transiting from Point X to Point Y, along the track indicated by the dotted line, would have the following information available about the relative quality of the depth information shown on the chart.



Point X lies in an area surveyed by NOAA between 1900-1939. The sounding data in this area would have been collected by leadline. Depths between sounding points can only be inferred, and undetected features might exist between the sounding points in areas of irregular relief — caution should be exercised.

The transit then crosses an area surveyed by NOAA between 1940-1969. The sounding data in this area would have been collected by continuous recording single beam echo sounder. It is possible that features could have been missed between sounding lines, although echo sounders record all depths along a sounding line with varying beam widths.

The transit ends in an area charted from miscellaneous surveys. These surveys may be too numerous to depict or may vary in age, reliability, origin or technology used. No inferences about the fitness of the data can be made in this area from the diagram.

Referring again to the accompanying sample Source Diagram, and the previous discussion of survey methods over time, a mariner could choose to transit from Point X to Point Y, along the track shown with a dashed line.

The transit starts again in an area surveyed by NOAA between 1900-1939. The sounding data in this area would have been collected by leadline. Depths between sounding points can only be inferred, and undetected features might still exist between the sounding points in areas of irregular relief — caution should be exercised.

The transit then crosses an area surveyed by NOAA between 1990–present, with partial bottom coverage. The data is collected in metric units and acquired by continuous recording single beam echo sounder. It is possible that features could have been missed between the sounding lines, although echo sounders record all depths along a sounding line with varying beam widths.

The transit then crosses into an area surveyed by NOAA etween 1990–present, having full bottom coverage. This area of the charted diagram is shaded with a blue screen to draw attention to the fact that full bottom coverage has been achieved. The data in this area would have been collected in metric units and acquired by side scan sonar or multibeam sonar technology. Undetected features in this area, at the time of the survey, would be unlikely.

The transit ends in an area charted from miscellaneous surveys. These surveys may be too numerous to depict or may vary in age, reliability, origin or technology used. No inferences about the fitness of the data can be made in this area from the diagram. By choosing to transit along the track shown by the dashed line, the mariner would elect to take advantage of survey information that is more recent and collected with modern technology.

provided, safe vertical clearances are shown in magenta text and indicate the highest points of a ship that can pass under an overhead power cable without risk of electrical discharge from the cable to the ship or without making contact with a bridge. **Vessels with masts, stacks, booms or antennas should allow sufficient clearance under power cables to avoid arcing.**

(118)

Submarine Cables and Submerged Pipelines

(119) **Submarine cables** and **submerged pipelines** cross many waterways used by both large and small vessels, but all of them may not be charted. For inshore areas, they usually are buried beneath the seabed, but for offshore areas they may lie on the ocean floor. Warning signs are often posted to warn mariners of their existence.

(120) The installation of submarine cables or pipelines in U.S. waters or the Continental Shelf of the United States is under the jurisdiction of one or more Federal agencies, depending on the nature of the installation. They are shown on the charts when the necessary information is reported to NOAA and they have been recommended for charting by the responsible agency. The chart symbols for submarine cable and pipeline areas are usually shown for inshore areas, whereas chart symbols for submarine cable and pipeline routes may be shown for offshore areas. Submarine cables and pipelines are not described in the Coast Pilots.

(121) In view of the serious consequences resulting from damage to submarine cables and pipelines, vessel operators should take special care when anchoring, fishing or engaging in underwater operations near areas where these cables or pipelines may exist or have been reported to exist. Mariners are also warned that the areas where cables and pipelines were originally buried may have changed and they may be exposed; extreme caution should be used when operating vessels in depths of water comparable to the vessel's draft.

(122) Certain cables carry high voltage, while many pipelines carry natural gas under high pressure or petroleum products. Electrocution, fire or explosion with injury, loss of life or a serious pollution incident could occur if they are broached.

(123) Vessels fouling a submarine cable or pipeline should attempt to clear without undue strain. Anchors or gear that cannot be cleared should be slipped, but no attempt should be made to cut a cable or a pipeline.

(124)

Artificial Obstructions to Navigation

(125) **Disposal areas** are designated by the U.S. Army Corps of Engineers for depositing dredged material where there is sufficient depth not to cause shoaling or create a danger to surface navigation. The areas are charted without blue tint, and soundings and depth curves are retained.

(126) **Disposal sites** are areas established by Federal regulation (**40 CFR 220** through **228**) in which dumping of dredged and fill material and other nonbuoyant objects is allowed with the issuance of a permit. Dumping of dredged and fill material is supervised by the U.S. Army Corps of Engineers and all other dumping by the Environmental Protection Agency (EPA). (See U.S. Army Corps of Engineers and Environmental Protection Agency, this chapter, and Appendix A for office addresses.)

(127) **Dumping grounds** are also areas that were established by Federal regulation (**33 CFR 205**). However, these regulations have been revoked and the use of the areas discontinued. These areas will continue to be shown on nautical charts until such time as they are no longer considered to be a danger to navigation.

(128) Disposal Sites and Dumping Grounds are rarely mentioned in the Coast Pilot, but are shown on nautical charts. **Mariners are advised to exercise caution in the vicinity of all dumping areas.**

(129) **Spoil areas** are for the purpose of depositing dredged material, usually near and parallel to dredged channels. Spoil areas are usually charted from survey drawings from U.S. Army Corps of Engineers after-dredging surveys, though they may originate from private or other Government agency surveys. On nautical charts, spoil areas are tinted blue, labeled and have all soundings and depth curves omitted from within their boundaries. Spoil areas present a hazard to navigation and even the smallest craft should avoid crossing them.

(130) **Fish havens** are artificial shelters constructed of various materials including rocks, rubble, derelict barges/oil rigs and specially designed precast structures. This material is placed on the sea floor to simulate natural reefs and attract fish. Fish havens are often located near fishing ports or major coastal inlets and are usually considered hazards to shipping. Before such a reef may be built, the U.S Army Corps of Engineers must issue a permit specifying the location and depth over the reef. Constructed of rigid material and projecting above the bottom, they can impede surface navigation and therefore represent an important feature for charting. Fish havens may be periodically altered by the addition of new material, thereby possibly increasing the hazard. They are outlined and labeled on charts and show the minimum authorized depth when known. Fish havens are tinted blue if they have a minimum authorized depth of 11 fathoms or less. If the minimum authorized depth is unknown and they are in depths greater than 11 fathoms, they are considered a danger to navigation. Navigators should be cautious about passing over fish havens or anchoring in their vicinity.

(131) **Fishtrap areas** are areas established by the U.S. Army Corps of Engineers, or State or local authority, in which traps may be built and maintained according to established regulations. The fish stakes that may exist in these areas are obstructions to navigation and may be dangerous. The limits of fishtrap areas and a cautionary note are usually charted. Navigators should avoid these areas.

(132)

Local Magnetic Disturbances

(133) If measured values of magnetic variation differ from the expected (charted) values by several degrees, a magnetic disturbance note will be printed on the chart. The note will indicate the location and magnitude of the disturbance, but the indicated magnitude should not be considered as the largest possible value that may be encountered. Large disturbances are more frequently detected in the shallow waters near land masses than on the deep sea. Generally, the effect of a local magnetic disturbance diminishes rapidly with distance, but in some locations there are multiple sources of disturbances and the effects may be distributed for many miles.

(134)

Compass Roses

(135) Each compass rose shows the date, magnetic variation and the annual change in variation. Prior to the new edition of a nautical chart, the compass roses are reviewed. Corrections for annual change and other revisions may be made as a result of newer and more accurate information. On some general and sailing charts, the magnetic variation is shown by isogonic lines in addition to the compass roses.

(136)

Echo Soundings

(137) The echo sounder on a ship may indicate small variations from charted soundings; this may be due to the fact that various corrections (instrument corrections, settlement and squat, draft and velocity corrections) are made to echo soundings in surveying which are not normally made in ordinary navigation, or to observational errors in reading the echo sounder. Instrument errors vary between different equipment and must be determined by calibration aboard ship. Most types of echo sounders are factory calibrated for a velocity of sound in water of 800 fathoms per second, but the actual velocity may differ from the calibrated velocity by as much as 5 percent, depending upon the temperature and salinity of the waters in which the vessel is operating; the highest velocities are found in warm, highly saline water and the lowest in icy freshwater. Velocity corrections for these variations are determined and applied to echo soundings during hydrographic surveys. All echo soundings must be corrected for the vessel's draft, unless the draft observation has been set on the echo sounder.

(138) Observational errors include misinterpreting false echoes from schools of fish, seaweed, etc., but the most serious error that commonly occurs is where the depth is greater than the scale range of the instrument; a 400–fathom scale indicates 15 fathoms when the depth is 415 fathoms. Caution in navigation should be exercised when wide variations from charted depths are observed.

(139)

NOTICES TO MARINERS

(140) **Notices to Mariners** are published to advise operators of marine information affecting the safety of navigation. The notices include changes in aids to navigation, depths in channels, bridge and overhead cable clearances, reported dangers and other useful marine information. They should be used routinely for updating the latest editions of nautical charts and related publications.

(141) **Local Notices to Mariners** are issued by each Coast Guard District Commander for the waters under their jurisdiction. (See Appendix A for Coast Guard district(s) covered by this volume.) These notices are usually published weekly and are available at *navcen.uscg.gov*.

(142) **U.S. Notice to Mariners**, published weekly by the National Geospatial-Intelligence Agency, are prepared jointly with NOAA and the Coast Guard. These notices contain selected items from the Local Notices to Mariners and other reported marine information required by oceangoing vessels operating in both foreign and domestic waters. Special items covering a variety of subjects and generally not discussed in the Coast Pilot or shown on nautical charts are published annually in Notice to Mariners No. 1. These items are important to the mariner and should be read for future reference. These notices are available at *msi.nga.mil/NGAPortal/MSI.portal*.

(143) **Broadcast Notices to Mariners** are made by the Coast Guard to report deficiencies and important changes in aids to navigation. (See Navigational Warnings, Information and Weather, this chapter.)

(144) The **Special Notice to Mariners** is an annual publication containing important information for mariners on a variety of subjects which supplements information not usually found on charts and in navigational publications. It includes excerpts from various Federal laws and regulations regarding marine pollution reporting, aids to navigation and Vessel Traffic Service (VTS) procedures. There are tips for trip planning, updates to the Rules of the Road and information on local hazards. Also included are points of contact, phone numbers and email addresses for various subject matter experts to assist the mariner in locating further information.

(145) Vessels operating within the limits of the Coast Guard districts can obtain information affecting NOAA charts and related publications from the Local Notices to Mariners. Small craft using the Intracoastal Waterway and other waterways and small harbors within the United States that are not normally used by oceangoing vessels will require the Local Notices to Mariners to keep charts and related publications up to date.

(146)

AIDS TO NAVIGATION

(147)

U.S. Aids to Navigation System

(148) The navigable waters of the United States are marked to assist navigation using the U.S. Aids to Navigation System, a system consistent with the International Association of Marine Aids to Navigation and Lighthouse Authorities (IALA) Maritime Buoyage System. The **IALA Maritime Buoyage System** is followed by most of the world's maritime nations and will improve maritime safety by encouraging conformity in buoyage systems worldwide. IALA buoyage is divided into two regions made up of Region A and Region B. All navigable waters of the United States follow IALA Region B, except U.S. possessions west of the International Date Line and south of 10° north latitude, which follow IALA Region A. Lateral aids to navigation in Region A vary from those located within Region B. Nonlateral aids to navigation are the same as those used in Region B. Appropriate nautical charts and publications should be consulted to determine whether the Region A or Region B marking schemes are in effect for a given area.

(149) As standard protocol, the U.S. Coast Guard reported assigned positions of aids to navigation uses the North American Datum of 1983 (NAD 83). Due to the development of new navigational systems and the retirement of old systems, the World Geodetic System 1984 (WGS 84) has become the preferred standard. In 2020, the U.S. Coast Guard Chief of the Office of Navigation Systems (CG-NAV) announced that all geographic coordinates for aids to navigation assigned positions will be reported using WGS 84.

(150)

Reporting Defects in Aids to Navigation

(151) Promptly notify the nearest Coast Guard District Commander if an aid to navigation is observed to be missing, sunk, capsized, out of position, damaged, extinguished or showing improper characteristics.

(152) **Aids to navigation** in United States waters of the Great Lakes and their connecting waters, except for the St. Lawrence River, are maintained by the U.S. Coast Guard. Local jurisdiction for the region is assigned to the Commander, Ninth Coast Guard District. The Lake Champlain region and the Hudson River are under the jurisdiction of the Commander, First Coast Guard District. (See Appendix A for the addresses.)

(153) It is unlawful to establish or maintain any aid similar to those maintained by the U.S. Coast Guard without first obtaining permission from the Coast Guard District Commander. The licensed officer in command of a vessel which collides with any aid must report the fact promptly to the nearest U.S. Coast Guard Sector.

(154)

Lights

(155) **Nominal range** is the term for the range of visibility of lights as defined in the U.S. Coast Guard Light List and shown on nautical charts. It is the maximum distance a light can be seen in clear weather (meteorological visibility of 10 nautical miles). Nominal range is listed for all lighted aids to navigation except range lights, directional lights, and private aids to navigation.

(156) **Luminous range** is the greatest distance a light may be seen given its nominal range and the prevailing meteorological visibility. The Luminous Range Diagram, found in the U.S. Coast Guard Light List, enables the mariner to determine the approximate luminous range of a light when the nominal range and the prevailing meteorological visibility are known. The nominal range and the luminous range do not take into account elevation, observer's height of eye, or the curvature of the earth.

(157) **Geographic range** is the greatest distance the curvature of the earth permits an object, of a given height, to be seen from a particular height of eye without regard to luminous intensity or visibility conditions. To determine the actual geographic range for height of eye, the geographic range must be corrected by a distance corresponding to the height difference. The Geographic Range Table, found in the U.S. Coast Guard Light List, gives the approximate geographic range of visibility for an object which may be seen by an observer at sea level.

(158) The maximum distances at which lights can be seen may at times be increased by abnormal atmospheric refraction and may be greatly decreased by unfavorable weather conditions such as fog, rain, haze or smoke. All except the most powerful lights are easily obscured by such conditions. In some conditions of the atmosphere white lights may have a reddish hue. During weather conditions which tend to reduce visibility, colored lights are more quickly lost to sight than white lights. Navigational lights should be used with caution because of the following conditions that may exist.

(159) A light may be extinguished and the fact not reported to the Coast Guard for correction, or a light may be located in an isolated area where it will take time to correct.

(160) In regions where ice conditions prevail the lantern panes of unattended lights may become covered with ice or snow, which will greatly reduce the visibility and may also cause colored lights to appear white.

(161) Brilliant shore lights used for advertising and other purposes, particularly those in densely populated areas, make it difficult to identify a navigational light.

(162) At short distances flashing lights may show a faint continuous light between flashes.

(163) The distance of an observer from a light cannot be estimated by its apparent intensity. The characteristics of lights in an area should always be checked in order that powerful lights visible in the distance not be mistaken for nearby lights showing similar characteristics at low intensity such as those on lighted buoys.

(164) The apparent characteristic of a complex light may change with the distance of the observer, due to color and intensity variations among the different lights of the group. The characteristic as charted and shown in the Light List may not be recognized until nearer the light.

(165) Motion of a vessel in a heavy sea may cause a light to alternately appear and disappear, and thus give a false characteristic.

(166) Where lights have different colored sectors, be guided by the correct bearing of the light; do not rely on being able to accurately observe the point at which the color changes. On either side of the line of demarcation of colored sectors there is always a small arc of uncertain color.

(167) On some bearings from the light, the range of visibility of the light may be reduced by obstructions. In such cases, the obstructed arc might differ with height of eye and distance. When a light is cut off by adjoining land and the arc of visibility is given, the bearing on which the light disappears may vary with the distance of the vessel from which observed and with the height of eye. When the light is cut off by a sloping hill or point of land, the light may be seen over a wider arc by a ship far off than by one closer.

(168) Arcs of circles drawn on charts around a light are not intended to give information as to the distance at which it can be seen, but solely to indicate, in the case of lights which do not show equally in all directions, the bearings between which the variation of visibility or obscuration of the light occurs.

(169) Lights of equal candlepower but of different colors may be seen at different distances. This fact should be considered not only in predicting the distance at which a light can be seen, but also in identifying it.

(170) Lights should not be passed close aboard, because in many cases riprap mounds are maintained to protect the structure against ice damage and scouring action.

(171) Many prominent towers, tanks, smokestacks, buildings and other similar structures, charted as landmarks, display flashing and/or fixed red aircraft obstruction lights. Lights shown from landmarks are charted only when they have distinctive characteristics to enable the mariner to positively identify the location of the charted structure.

(172)

Articulated Lights

(173) An articulated light is a vertical pipe structure supported by a submerged buoyancy chamber and attached by a universal coupling to a weighted sinker on the seafloor. The light, allowed to move about by the universal coupling, is not as precise as a fixed aid. However, it has a much smaller watch circle than a conventional buoy, because the buoyancy chamber tends to force the pipe back to a vertical position when it heels over under the effects of wind, wave or current.

(174) Articulated lights are primarily designed to mark narrow channels with greater precision than conventional buoys.

(175)

Daybeacons

(176) Daybeacons are unlighted aids affixed to stationary structures. They are marked with dayboards for daytime identification. The dayboards aid navigation by presenting one of several standard shapes and colors which have navigational significance. Dayboards are sometimes referred to as daymarks.

(177) Daybeacons are found on-shore and in shallow water. They are frequently used to mark channel edges.

(178)

Articulated Daybeacons

(179) Articulated daybeacons are similar to articulated lights, described above, except they are unlighted.

(180)

Buoys

(181) The aids to navigation depicted on charts comprise a system consisting of fixed and floating aids with varying degrees of reliability. Therefore, prudent mariners will not rely solely on any single aid to navigation, particularly a floating aid.

(182) The approximate position of a buoy is represented by the dot or circle associated with the buoy symbol. The approximate position is used because of practical limitations in positioning and maintaining buoys and their sinkers in precise geographical locations. These limitations include, but are not limited to, inherent imprecisions in position fixing methods, prevailing atmospheric and sea conditions, the slope of and the material making up the seabed, the fact that buoys are moored to sinkers by varying lengths of chain and the fact that buoy body and/or sinker positions are not under continuous surveillance, but are normally checked only during periodic maintenance visits which often occur more than a year apart. The position of the buoy body can be expected to shift inside and outside of the charting symbol due to the forces of nature. The mariner is also cautioned that buoys are liable to be carried away, shifted, capsized, sunk, etc. Lighted buoys may be extinguished or sound signals may not function as a result of ice, running ice or other natural causes, collisions or other accidents.

(183) For the foregoing reasons, a prudent mariner must not rely completely upon the charted position or operation of floating aids to navigation but will also utilize bearings from fixed objects and aids to navigation on shore. Further, a vessel attempting to pass close aboard always risks collision with a yawing buoy or with the obstruction the buoy marks.

(184) Buoys may not always properly mark shoals or other obstructions due to shifting of the shoals or of the buoys. Buoys marking wrecks or other obstructions are usually placed on the seaward or channelward side and not directly over a wreck. Since buoys may be located some distance

from a wreck they are intended to mark, and since sunken wrecks are not always static, extreme caution should be exercised when operating in the vicinity of such buoys.

(185)

Automatic Identification System (AIS) Aids to Navigation

(186) AIS is an automatic communication and identification system intended to improve the safety of navigation by assisting the efficient operation of a Vessel Traffic Services (VTS), ship reporting, ship-to-ship and ship-to-shore operations. AIS is increasingly being used as an aid to navigation. An AIS-equipped aid to navigation may provide a positive identification of the aid. It may also have the capability to transmit an accurate position and provide additional information such as actual tide height and/or weather information.

(187) The AIS message may represent an aid to navigation that physically exists (physical AIS Aid to Navigation) or the message, transmitted from a remote location, may represent an aid to navigation that does not physically exist (virtual AIS Aid to Navigation). A virtual aid to navigation is a digital information object promulgated by an authorized service provider that can be presented on navigational systems.

(188) Physical AIS aids to navigation are charted with the symbol for the physical aid (such as a buoy or light) with a magenta circle surrounding the symbol and labeled AIS. Virtual aids to navigation are charted with a small central dot with a topmark symbol indicating the purpose of the aid, surrounded by a magenta circle and labeled V-AIS. Temporary AIS aids to navigation and stations remotely transmitting an AIS signal are not charted. See U.S. Chart No. 1, Section S, for additional information and examples.

(189)

Examples of Charted AIS Aids to Navigation

(190)

Bridge Lights and Clearance Gages

(191) The Coast Guard regulates marine obstruction lights and clearance gages on bridges across navigable waters. Where installed, clearance gages are generally vertical numerical scales, reading from top to bottom, and show the actual vertical clearance between the existing water level and the lowest point of the bridge over the channel; the gages are normally on the right-hand pier or abutment of the bridge, on both the upstream and downstream sides.

(192) Bridge lights are fixed red or green and are privately maintained; they are generally not charted or described in the text of the Coast Pilot. All bridge piers (and their protective fenders) and abutments that are in or adjacent to a navigation channel are marked on all channel sides by red lights. On each channel span of a fixed bridge, there is a range of two green lights marking the center of the channel and a red light marking both edges of the channel, except that when the margins of the channel are confined by bridge piers, the red lights on the span are omitted, since the pier lights then mark the channel edges. For multiplespan fixed bridges, the main-channel span may also be marked by three white lights in a vertical line above the green range lights.

(193) On all types of drawbridges, one or more red lights are shown from the drawspan (higher than the pier lights) when the span is closed; when the span is open, the higher red lights are obscured and one or two green lights are shown from the drawspan, higher than the pier lights. The number and location of the red and green lights depend upon the type of drawbridge.

(194) Bridges and their lighting, construction and maintenance are set forth in **33 CFR 114**, **115**, **116**, and **118** (not carried in this Coast Pilot). Aircraft obstruction lights prescribed by the Federal Aviation Administration may operate at certain bridges.

(195)

Sound Signals

(196) Caution should be exercised in the use of sound signals for navigation purposes. They should be considered solely as warning devices.

(197) Sound travels through the air in a variable manner, even without the effects of wind, and, therefore the hearing of sound signals cannot be implicitly relied upon.

(198) Experience indicates that distances must not be judged only by the intensity of the sound; that occasionally there may be areas close to a sound signal in which it is not heard; and that fog may exist not far from a station, yet not be seen from it, so the signal may not be operating. It is not always possible to start a sound signal immediately when fog is observed.

(199)

Channel Markers

(200) Lights, daybeacons, and buoys along dredged channels do not always mark the bottom edges. Due to local conditions, aids may be located inside or outside the channel limits shown by dashed lines on a chart. The Light List tabulates the offset distances for these aids in many instances.

(201) Aids may be moved, discontinued or replaced by other types to facilitate dredging operations. Mariners should exercise caution when navigating areas where dredges with auxiliary equipment are working.

(202) Temporary changes in aids are not included on the charts.

(203)

Light Lists

(204) The Coast Guard Light Lists are a means for communicating aids to navigation information to the maritime public. They are updated weekly and

available for download on the United States Coast Guard Navigation Center's website at *www.navcen.uscg.gov*. Mariners should refer to these lists for detailed information regarding the characteristics and visibility of lights, and the description of light structures, buoys, sound signals and electronic aids.

(205)

ELECTRONIC POSITIONING SYSTEMS

(206) **Global Positioning System (GPS)**permits land, sea, and airborne users to determine their three-dimensional position, velocity and time 24 hours a day, in all weather, anywhere in the world. The basic system is defined as a constellation of satellites, the navigation payloads which produce the GPS signals, ground stations, data links and associated command and control facilities, that are operated and maintained by the Department of Defense. Please report GPS problems or anomalies at *navcen.uscg.gov* or contact the USCG Navigation Information Service at 703–313–5900.

(207)

LORAN-C

(208) LORAN, an acronym for LOng RAnge Navigation, was an electronic aid to navigation consisting of shore-based radio transmitters. In accordance with the Department of Homeland Security Appropriations Act, the U.S. Coast Guard terminated the transmission of all LORAN-C signals as of August 2010, rendering them unusable and permanently discontinued. For more details, visit *navcen.uscg.gov*. The Coast Guard strongly urges mariners accustomed to using LORAN-C for navigation to shift to a GPS navigation system and become familiar with its operation. NOAA is removing LORAN-C lines of position from all of its charts as new editions are published.

(209)

SEARCH AND RESCUE

(210)

Coast Guard Search and Rescue

(211) The Coast Guard conducts and/or coordinates search and rescue operations for surface vessels or aircraft that are in distress or overdue. Search and rescue vessels and aircraft have special markings, including a wide slash of red-orange and a small slash of blue on the forward portion of the hull or fuselage. Other parts of aircraft, normally painted white, may have other areas painted red to facilitate observation. The cooperation of vessel operators with Coast Guard helicopters, fixed-wing aircraft, and vessels may mean the difference between life and death for some seaman or aviator; such cooperation is greatly facilitated by the prior knowledge on the part of vessel operators of the operational requirements of Coast Guard equipment and personnel, of the international distress signals and procedures and of good seamanship.

(212)

Search and Rescue Great Lakes

(213) The United States Coast Guard has established a toll-free search and rescue telephone number for the Great Lakes. The number is intended for use when the telephone number of the nearest Coast Guard station is unknown or when that station cannot be contacted. The toll-free number should not be used without first attempting to contact the nearest Coast Guard station. In all Great Lakes States the telephone number is 800-321-4400. This number is to be used for public reports of distress incidents, suspicious sightings, pollution or other maritime concerns.

(214)

Radiotelephone Distress Message

(215) Distress calls indicate a vessel or aircraft is threatened by grave and imminent danger and requests immediate assistance. They have absolute priority over all other transmissions. All stations which hear a distress call must immediately cease any transmission capable of interfering with the distress traffic and continue to listen on the frequency used for the emission of the distress call. This call should not be addressed to a particular station, and acknowledgment of receipt should not be given before the distress message which follows it is sent.

(216) Distress calls are made on VHF-FM channel 16 (MAYDAY). For less serious situations than warrant the distress procedure, the radiotelephone urgency signal consisting of three repetitions of the word PAN-PAN (pronounced PAWN-PAWN), or the safety signal SECURITE (pronounced SECURITAY) spoken three times, are used as appropriate. For complete information on emergency radio procedures, see **47 CFR 80** or **Radio Navigational Aids, Pub. 117**.

(217)

Global Maritime Distress and Safety System (GMDSS)

(218) This international system, developed by the International Maritime Organization (IMO), is based on a combination of satellite and terrestrial radio services and has changed international distress communications from being primarily ship-to-ship based to primarily ship-to-shore (Rescue Coordination Center) based. Prior to the GMDSS, the number and types of radio safety equipment required to be carried by vessels depended upon the tonnage. Under GMDSS, the number and type of radio safety equipment vessels are required to carry depend on the areas in which they travel; GMDSS sea areas are defined by governments. All GMDSS-regulated ships must carry a satellite Emergency Position Indicating Radio Beacon (EPIRB), a NAVTEX receiver (if they travel in any areas served by NAVTEX), an Inmarsat-C SafetyNET receiver (if they travel in any areas not served by NAVTEX), a DSC-equipped VHF radiotelephone, two or more VHF handhelds and a search and rescue radar transponder (SART).

(219)

Automated Mutual Assistance Vessel Rescue System (AMVER)

(220) AMVER is a worldwide voluntary ship reporting system operated by the United States Coast Guard to promote safety of life and property at sea. AMVER's mission is to quickly provide search and rescue (SAR) authorities, on demand, accurate information on the positions and characteristics of vessels near a reported distress. Any merchant vessel anywhere on the globe, on a voyage of greater than 24 hours duration, is welcome in the AMVER system and family. International participation is voluntary regardless of the vessel's flag of registry, the nationality of the owner or company or ports of call.

(221) According to U.S. Maritime Administration (MARAD) regulations, U.S. flag merchant vessels of 1,000 gross tons or more operating in foreign commerce and foreign flag vessels of 1,000 gross tons or more for which an Interim War Risk Insurance Binder has been issued under the provisions of Title XII, Merchant Marine Act, 1936, must report and regularly update their voyages and positions to AMVER in accordance with instructions set forth in the AMVER Ship Reporting System Manual. For more information contact AMVER Maritime Relations U.S. Coast Guard, 1 South Street Battery Park Building, New York, NY 10004; Phone: 212–668–7764, Fax: 212-668-7684, Telex: 127594-AMVER NYK, or go to *amver.com*.

(222)

COSPAS-SARSAT

(223) COSPAS: Space System for Search of Distress Vessels - SARSAT: Search and Rescue Satellite-Aided Tracking. COSPAS-SARSAT is an international satellite system designed to provide distress alert and location data to assist search and rescue operations using satellites and ground facilities to detect and locate the signals of distress beacons operating on 406 MHz. For more information on the Cospas-Sarsat System go to *cospas-sarsat.int*.

(224)

Digital Selective Calling (DSC)

(225) The U.S. Coast Guard offers VHF and MF/HF radiotelephone service to mariners as part of the Global Maritime Distress and Safety System. This service, called digital selective calling (DSC), allows mariners to instantly send an automatically formatted distress alert to the Coast Guard or other rescue authority anywhere in the world. Digital selective calling also allows mariners to initiate or receive distress, urgency, safety and routine radiotelephone calls to or from any similarly equipped vessel or shore station, without requiring either party to be near a radio loudspeaker. Each ship or shore station equipped with a DSC terminal has a unique Maritime Mobile Station Identity (MMSI). This is a nine-digit number that specifically identifies a ship, coast station, or group of stations. The DSC system alerts an operator when a distress call is received. It will provide the operator with a pre-formatted message that can include the distressed vessel's nine-digit MMSI, location, nature of distress, desired mode of communication and preferred working frequency.

(226)

Emergency Position Indicating Radiobeacons (EPIRB)

(227) EPIRBs emit a radio signal that can be used to locate mariners in distress. SARSAT satellites can locate the position of a 406 MHz EPIRB which greatly increases a mariner's chances of survival. While orbiting the earth, the satellites continuously monitor EPIRB frequencies. When SARSAT receives an EPIRB signal, it determines the beacon's position that is ultimately relayed to the nearest Coast Guard Rescue Coordination Center where rescue units are dispatched to the scene.

(228) Mariners should ensure that their EPIRB is in working condition and stowed properly at all times to avoid non-distress emissions. Mariners are required to register their 406 MHz EPIRBs for improved search and rescue response and keep the registration current at all times. Registration can be accomplished online at *beaconregistration.noaa.gov*.

(229)

EPIRB Types

Type	Frequency	Description
Cat I	406 MHz	Float-free, automatically activated EPIRB. Detectable by satellite anywhere in the world. Recognized by the Global Maritime and Distress Safety System (GMDSS).
Cat II	406 MHz	Similar to Category I, except is manually activated. Some models are also water activated.

(230)

Medical Advice

(231) Ships at sea with no medical personnel embarked and experiencing a medical emergency onboard can receive medical advice via radiotelex, radiotelephony or Inmarsat. Messages are generally addressed RADIOMEDICAL followed by the name of the coast station to which the message is sent. The priority of the message should depend on the severity of the ailment. In extreme emergency, the urgency signal (PAN-PAN) should precede the address. Messages are sent using distress and safety frequencies.

(232)

Vessel Identification

(233) Coast Guard search and rescue aircraft and surface craft use radar to assist in locating disabled vessels. Wooden and fiberglass vessels are often poor radar targets. Operators of disabled craft that are the object of a search are requested to hoist, as high above the waterline as possible, a radar-reflecting device. If no special radar-reflecting device is aboard, an improvised device can be used. This should consist of metallic objects of irregular shape. The more irregular the shape, the better will be the radar-reflective quality. For quick identification at night, shine spotlights straight up. If aircraft are involved, once

(241)

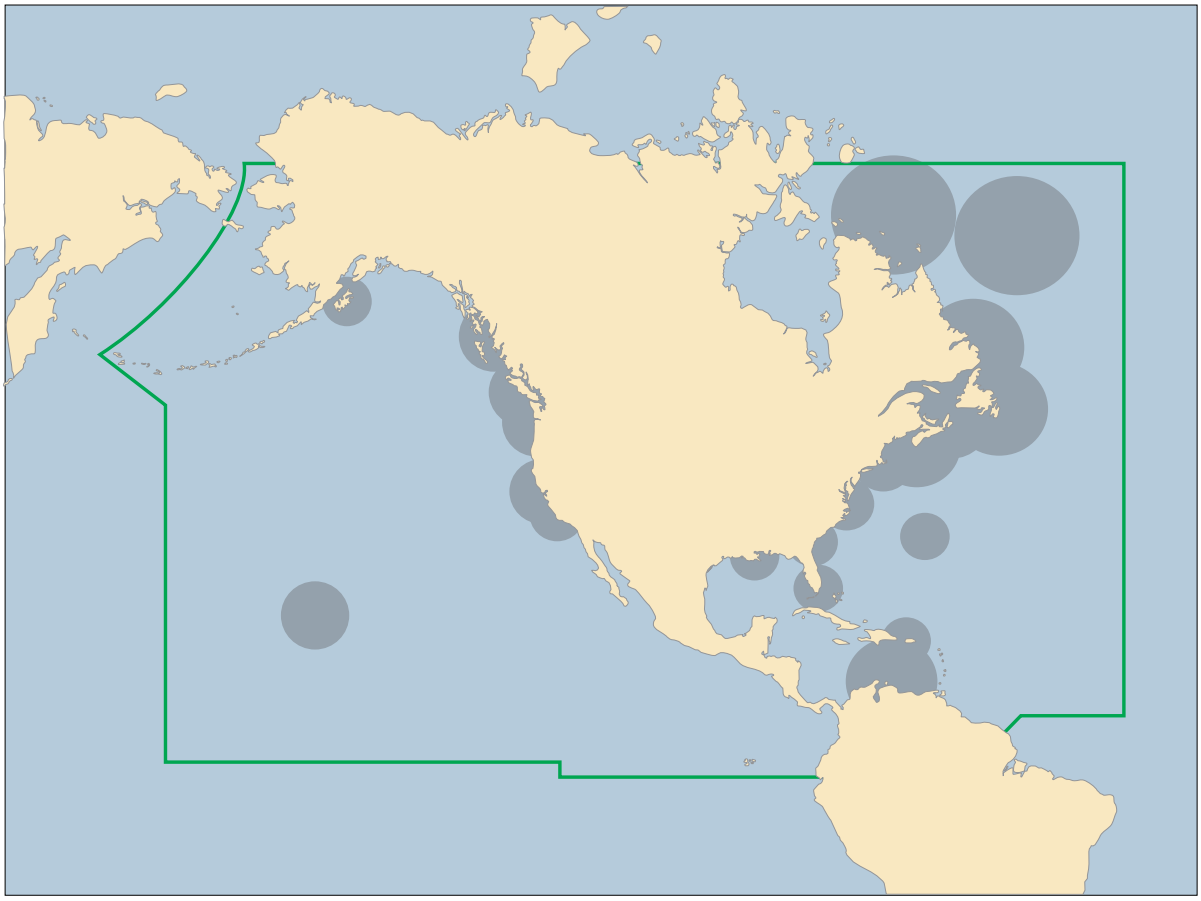
U.S. VHF Channel Information

New Channel	Old Channel	Ship Frequency (MHz)		Channel Usage
		Transmit	Receive	
1001	01A	156.050	156.050	Port Operations and Commercial, VTS. Available only in New Orleans / Lower Mississippi area.
1005	05A	156.250	156.250	Port Operations or VTS in the Houston, New Orleans and Seattle areas
06	06	156.300	156.300	Intership Safety
1007	07A	156.350	156.350	Commercial. VDSMS
08	08	156.400	156.400	Commercial (Intership only) VDSMS
09	09	156.450	156.450	Boater Calling; Commercial and Non-commercial. VDSMS
10	10	156.500	156.500	Commercial. VDSMS
11	11	156.550	156.550	Commercial; VTS in selected areas. VDSMS
12	12	156.600	156.600	Port Operations; VTS in selected areas
13	13	156.650	156.650	Intership Navigation Safety (Bridge-to-Bridge). Ships greater than 20m maintain a listening watch on this channel in U.S. waters.
14	14	156.700	156.700	Port Operations; VTS in selected areas
15	15	- -	156.750	Environmental (Receive only) Used by Class C EPIRBs
16	16	156.800	156.800	International Distress, Safety and Calling. Ships required to carry radio, USCG, and most coast stations maintain a listening watch on this channel. *(Refer to: Radio Watchkeeping Regulations).*
17	17	156.850	156.850	State and local government maritime control
1018	18A	156.900	156.900	Commercial. VDSMS
1019	19A	156.950	156.950	Commercial. VDSMS
20	20	157.000	161.600	Port Operations (duplex)
1020	20A	157.000	157.000	Port Operations
1021	21A	157.050	157.050	U.S. Coast Guard only
1022	22A	157.100	157.100	Coast Guard Liaison and Maritime Safety Information Broadcasts. *(Broadcasts announced on Channel 16)*
1023	23A	157.150	157.150	U.S. Coast Guard only
24	24	157.200	161.800	Public Correspondence (Marine Operator). VDSMS
25	25	157.250	161.850	Public Correspondence (Marine Operator). VDSMS
26	26	157.300	161.900	Public Correspondence (Marine Operator). VDSMS
27	27	157.350	161.950	Public Correspondence (Marine Operator). VDSMS
28	28	157.400	162.000	Public Correspondence (Marine Operator). VDSMS
1063	63A	156.175	156.175	Port Operations and Commercial, VTS. Available only in New Orleans / Lower Mississippi area.
1065	65A	156.275	156.275	Port Operations
1066	66A	156.325	156.325	Port Operations
67	67	156.375	156.375	Commercial. Used for Bridge-to-Bridge communications in lower Mississippi River. (Intership only.)
68	68	156.425	156.425	Non-Commercial. VDSMS
69	69	156.475	156.475	Non-Commercial. VDSMS
70	70	156.525	156.525	Digital Selective Calling (voice communications not allowed)
71	71	156.575	156.575	Non-Commercial. VDSMS
72	72	156.625	156.625	Non-Commercial (Intership only). VDSMS
73	73	156.675	156.675	Port Operations
74	74	156.725	156.725	Port Operations
77	77	156.875	156.875	Port Operations (Intership only)
1078	78A	156.925	156.925	Non-Commercial. VDSMS
1079	79A	156.975	156.975	Commercial. Non-commercial in Great Lakes only. VDSMS
1080	80A	157.025	157.025	Commercial. Non-commercial in Great Lakes only. VDSMS
1081	81A	157.075	157.075	U.S. Government only - Environmental protection operations.
1082	82A	157.125	157.125	U.S. Government only
1083	83A	157.175	157.175	U.S. Coast Guard only
84	84	157.225	161.825	Public Correspondence (Marine Operator). VDSMS
85	85	157.275	161.875	Public Correspondence (Marine Operator). VDSMS
86	86	157.325	161.925	Public Correspondence (Marine Operator). VDSMS
87	87	157.375	157.375	Public Correspondence (Marine Operator). VDSMS
88	88A	157.425	157.425	Commercial (Intership only). VDSMS
AIS 1	AIS 1	161.975	161.975	Automatic Identification System (AIS)
AIS 2	AIS 2	162.025	162.025	Automatic Identification System (AIS)

Boaters should normally use channels listed as Non-Commercial. Channel 16 is used for calling other stations or for distress alerting. Channel 13 should be used to contact a ship when there is danger of collision. All ships of length 20m or greater are required to guard VHF channel 13, in addition to VHF channel 16, when operating within U.S. territorial waters. Users may be fined by the FCC for improper use of these channels. Frequencies are in megahertz (MHz). Modulation is 16KF3E or 16KG3E. VDSMS (VHF Digital Small Message Services). Transmissions of short digital messages in accordance with RTCM Standard 12301.1 is allowed. Four-digit VHF maritime channel numbers: US channel 05A is now designated and recognized internationally by the channel 1005, and the U.S. Coast Guard channel 22A is now designated and recognized internationally by the channel "1022". These new channel number designations are being displayed on new models of VHF marine radios.

Further information can be obtained by visiting the following:

U.S. VHF Channel Information - *https://www.navcen.uscg.gov/us-vhf-channel-information*
Radio Watchkeeping Regulations - *https://www.navcen.uscg.gov/radio-watchkeeping-regulations*
International Telecommunications Union (ITU) Radio Regulations Appendix 18 - *https://navcen.uscg.gov/international-vhf-marine-radio-channels-freq*



you are identified, turn lights away so as not to blind aircraft crew.

(234)

Float Plan

(235) Small craft operators should prepare a float plan before starting a trip and leave it ashore with a yacht club, marina, friend or relative. It is advisable to regularly use a checking-in procedure by radio or telephone for each point specified in the float plan. A float plan is vital for determining if a boat is overdue and will assist in locating a missing vessel in the event search and rescue operations become necessary.

(236)

NAVIGATIONAL WARNINGS, INFORMATION AND WEATHER

(237) Marine radio warnings and weather are disseminated by many sources and through several types of transmissions. For complete information on radio warnings and weather, see **Radio Navigational Aids, Pub. 117** and the National Weather Service (NWS) publication **Worldwide Marine Radiofacsimile Broadcast Schedules**.

(238) Radio navigational warning broadcasts are designed to provide the mariner with up-to-date marine information vital to safe navigation. There are three types of broadcasts: coastal and local, long range and worldwide.

(239) Coastal and local warnings are generally restricted to ports, harbors and coastal waters and involve items of local interest. Usually, local or short-range warnings are broadcast from a single coastal station, frequently by voice and also radiotelegraph, to assist small craft operators in the area. The information is often quite detailed. Foreign area broadcasts are frequently in English as well as the native language. In the United States, short-range radio navigational warnings are broadcast by the U.S. Coast Guard Districts via NAVTEX and subordinate coastal radio stations.

(240) Long range warnings are intended primarily to assist mariners on the high seas by promulgating navigational safety information concerning port and harbor approaches, coastlines and major ocean areas. Long-range radio navigational warnings are usually broadcast by means of radiotelegraphy and in many instances by radio-teletypewriter. A NAVAREA system of navigational warning areas has been developed providing worldwide coverage using standard format and procedures. The U.S. participates as Area Coordinator for both NAVAREA IV

(Western North Atlantic) and NAVAREA XII (Eastern North Pacific).

(242) The United States also maintains worldwide coverage using the HYDROLANT/HYDROPAC Navigational Warning System outside of NAVAREAs IV and XII.

(243)

NAVTEX

(244) NAVTEX is a standard international method of broadcasting notices to mariners and marine weather forecasts using small, low cost receivers designed to be installed in the pilothouse of a vessel. NAVTEX receivers screen incoming messages, inhibiting those which had been previously received or are of a category not of interest to the user, and print the rest on adding machine-size paper. NAVTEX not only provides marine information previously available only to those knowledgeable in Morse code but also allows any mariner who cannot man a radio full time to receive safety information at any hour. All NAVTEX transmissions are made on 518 kHz. Mariners who do not have NAVTEX receivers but have Simplex Teletype Over Radio (SITOR) radio equipment can also receive these broadcasts by operating it in the Forward Error Correction (FEC) mode and tuning to 518 kHz.

(245) Information broadcast over NAVTEX includes offshore weather forecasts, offshore marine advisory warnings, search and rescue information and navigational information that applies to waters from the line of demarcation (separating Inland Rules from COLREG Rule waters) to 200 miles offshore. Navigational information that affects the safety of navigation of deep draft (15 feet or more) vessels within the U.S. Inland Rules waters will also be included. Gulf Stream location is also included from Miami and Portsmouth. Coastal and high seas weather forecasts are not being broadcast over NAVTEX. The Safety of Life at Sea Convention, as amended in 1988, requires vessels regulated by that convention to carry NAVTEX receivers.

(246) See Appendix A, U.S. NAVTEX Transmitting Stations, for a list of NAVTEX broadcast stations and message content covered by this Coast Pilot.

(248)

Broadcast Notice to Mariners

(249) The U.S. Coast Guard transmits **Urgent Marine Information Broadcast** (UMIB) safety warnings and communicates with mariners on VHF channel 1022 (previously channel 22A). These safety broadcasts contain information such as notices to mariners, storm warnings, distress warnings and other pertinent information that is vital for safe navigation. Following a preliminary call on VHF-FM channel 16 (156.800 MHz), mariners are instructed to shift to VHF-FM channel 1022 (157.100 MHz). A shipboard radio tuned to channel 1022 (U.S. mode) both transmits and listens on 157.100 MHz, and can receive UMIBs transmitted by the U.S Coast Guard. A shipboard radio tuned to channel 22 (international mode) transmits on 157.100 MHz and listens on 161.700 MHz, and cannot receive these safety warnings. In accordance with **33 CFR 26.03(d)**, "The radiotelephone required by (this Bridge-to-bridge radiotelephone regulation) must be capable of transmitting and receiving on VHF FM channel 22A (157.100 MHz)." This regulation applies to foreign ships in U.S. waters as well as to U.S. ships.

(250)

NOAA Weather Radio Broadcasts

(251) NOAA Weather Radio provides continuous broadcasts of the latest weather information directly from (NWS) offices. In addition to general weather information, marine weather is provided by stations along the sea coasts and the Great Lakes. During severe weather, NWS forecasters can interrupt the regular broadcasts and substitute special warning messages. The stations operate 24 hours daily, and messages are repeated every 4 to 6 minutes and are routinely revised every 1 to 3 hours or more frequently if necessary. The broadcasts are made on seven VHF-FM frequencies, 162.40 to 162.55 MHz. The 162.475 MHz frequency is only used in special cases where needed to avoid channel interference. They can usually be heard as far as 40 miles from the antenna site, sometimes more. The effective range depends on many factors, including the height of the broadcast antenna, terrain, quality of the receiver and the type of receiving antenna. As a general rule, listeners close to or perhaps beyond the 40 mile range should have a good quality receiver system to get reliable reception. (See Appendix A for a list of these stations in the area covered by this Coast Pilot.)

(252)

Commercial Maritime Coast Stations and Weather Nets

(253) Commercial maritime coast stations, which provide communications services, broadcast weather information to ships at sea as a public service, or make forecast information available on demand, either free or for a nominal fee. These transmissions are most commonly performed using HF SITOR and Pactor/E-Mail; however, several of these stations also offer services via Inmarsat satellite and other means.

(254) There are also a number of maritime weather *nets* operating on commercial marine VHF, MF and HF, where weather information is exchanged. These *nets* are extremely popular in areas of the world that have a large yachting population and where weather is dynamic, such as in the Caribbean, and typically incorporate volunteers ashore.

(255) Information on commercial maritime coast stations, including schedules and frequencies, is available in the **Radio Navigational Aids, Pub. 117**.

(256)

Standard Abbreviations for Broadcasts

(257) A listing of Standard Abbreviations for Textual Maritime Safety Broadcasts can be found in this chapter. These abbreviations were jointly approved by the U.S.

(258)

Standard Abbreviations Used in Broadcasts

Aids to Navigation

AERO RBN — Aeronautical Radiobeacon
ART DBN — Articulated Daybeacon
ART LT — Articulated Light
DESTR — Destroyed
DISCONTD — Discontinued
ESTAB — Established
ELB — Exposed Location Buoy
FOG SIG — Fog Signal Station
LNB — Large Navigation Buoy
LT — Light
LLNR — Light List Number
LBB — Lighted Bell Buoy
LB — Lighted Buoy
LGB — Lighted Gong Buoy
LHB — Lighted Horn Buoy
LWB — Lighted Whistle Buoy
ODAS — Ocean Data Acquisition System
PRIV MAINTD — Privately Maintained
RACON — Radar Beacon
RA REF — Radar Refector
TRUB — Temporarily Replaced by Unlighted Buoy
TRLB — Temporarily Replaced by Lighted Buoy
WHIS — Whistle

Light Characteristics

AL — Alternating
CHAR — Characteristic
FL(2+1) — Composite Group-Flashing
OC(2+1) — Composite Group-Occulting
Q — Continuous Quick-Flashing
FFL — Fixed and Flashing
F — Fixed
FL(3) — Group-Flashing
OC(2) — Group-Occulting
IQ — Interrupted Quick-Flashing
ISO — Isophase
MO(A) — Morse Code
OC — Occulting
FL — Single-Flashing

Colors (Color refers to light characteristics of Aids to Navigation only)

B — Black
BU — Blue
G — Green
OR — Orange
R — Red
W — White
Y — Yellow

Organizations

CCGD(#) — Commander, Coast Guard District (#)
CG — Coast Guard
COE — Corps of Engineers
NGA — National Geospatial-Intelligence Agency
NOS — National Ocean Service
NWS — National Weather Service

Vessels

A/C — Aircraft
F/V — Fishing Vessel
LNG — Liquified Natural Gas Carrier
M/V — Motor Vessel*
P/C — Pleasure Craft
R/V — Research Vessel
S/V — Sailing Vessel

* M/V includes: Steam Ship, Container Vessel, Cargo Vessel, etc.

Compass Directions

N — North
S — South
E — East
W — West
NE — Northeast
NW — Northwest
SE — Southeast
SW — Southwest

Various

ANCH — Anchorage
ANCH PROHIB — Anchorage Prohibited
APPROX — Approximate
ATLC — Atlantic
AUTH — Authorized
AVG — Average
BRG — Bearing
BKW — Breakwater
BNM — Broadcast Notice to Mariners
CHAN — Channel
CFR — Code of Federal Regulations
CONT — Continue
DEG — Degrees (temp, geo-position)
DIA — Diameter
ED — Edition
EFF — Effect/Effective
ENTR — Entrance
EXPLOS ANCH — Explosive Anchorage
FM(S) — Fathoms
FT — Foot/Feet
HBR — Harbor
HT — Height
HZ — Hertz
HOR CL — Horizontal Clearance
HR — Hour
COLREGS — International Regulations for Preventing Collisions at Sea
KHZ — Kilohertz
KM — Kilometer
KT(S) — Knot(s)
LAT — Latitude
LNM — Local Notice to Mariners
LONG — Longitude
MAINTD — Maintained
MAX — Maximum
MHZ — Megahertz
MB — Millibar
MM — Millimeter
MIN — Minute (time, geo position)
MOD — Moderate
MT —Mountain, Mount
NM — Nautical Mile(s)
NTM — Notice to Mariners
OBSTR — Obstruction
OCCASION — Occasion/Occasionally
OPAREA — Operating Area
PAC — Pacific
PT(S) — Point(s)
POS — Position
PA —Position Approximate
PRES — Pressure
PRIV — Private/Privately
PROHIB — Prohibited
PUB — Publication
RGE — Range
REP — Reported
RESTR — Restricted
RK — Rock
ST — Saint
SEC — Second (time, geo position)
SIG STA — Signal Station
STA — Station
SM — Statute Mile(s)
S SIG STA — Storm Signal Station
TEMP — Temporary
TSTORM — Thunderstorm
THRU — Through
T — True
UNCOV — Uncovers
UTC — Universal Coordinate Time
UMIB — Urgent Marine Information Broadcast
VEL — Velocity
VERT CL — Vertical Clearance
VIS — Visibility
YD — Yard(s)
WARN — Warning
WX — Weather
WK — Wreck

Coast Guard, National Weather Service, National Geospatial-Intelligence Agency and the Radio Technical Commission for Maritime Services. In addition to appearing in radio broadcasts of the U.S. Coast Guard and National Weather Service, they appear in Notices to Mariners of the U.S. Coast Guard and National Geospatial-Intelligence Agency and in NAVTEX.

(259)

Voluntary Observing Ship Program (VOS)

(260) The Voluntary Observing Ship program is organized for the purpose of obtaining weather and oceanographic observations from moving ships. An international program under World Meteorological Organization auspices, the VOS has over 5000 vessels participating from 23 countries. Any vessel willing to take and transmit observations in marine areas can join the program. Weather observations are essential to meteorologists preparing weather forecasts for coastal, offshore and high seas areas. For more information on the VOS, including a comprehensive observing handbook, visit *vos.noaa.gov*.

(261)

National Institute of Standards and Technology (NIST)

(262) The National Institute of Standards and Technology maintains the standards for time and frequency for most users in the United States. NIST provides a variety of services designed to deliver time and frequency signals to the people who need them. The signals are broadcast via several mediums, including high and low frequency radio, the Internet and telephone lines. Broadcasts of time and frequency signals are made by stations operating in the part of the radio spectrum that is properly known as high frequency (HF) but is commonly called shortwave. Station WWV is located just north of Fort Collins, Colorado, and station WWVH is located on the island of Kaua'i, Hawaii. Both stations broadcast continuous time and frequency signals on 2.5, 5, 10 and 15 MHz; WWV also broadcasts on 20 MHz.

(263) **NIST Time and Frequency Services, Special Publication 432** gives a detailed description of the signals and services offered by NIST, how they work and how you can use them. The publication is available for download at *nist.gov/pml/div688/generalpubs.cfm*.

(264)

CAUTIONARY INFORMATION

(265)

Hurricanes and Tropical Storms

(266) Hurricanes, tropical storms and other major storms may cause considerable damage to marine structures, aids to navigation and moored vessels, resulting in submerged debris in unknown locations. Fixed aids to navigation may have been damaged or destroyed. Buoys may have been moved from charted positions, damaged, sunk, extinguished or otherwise made inoperative. Mariners should not rely upon the position or operation of an aid to navigation. Charted soundings, channel depths and shoreline may not reflect actual conditions following these storms. Wrecks and submerged obstructions may have been displaced from charted locations. Pipelines may have become uncovered or moved. Mariners are urged to exercise extreme caution and are requested to report aids to navigation discrepancies and hazards to navigation to the U.S. Coast Guard.

(267)

Destructive Waves

(268) Unusual sudden changes in water level can be caused by tsunamis or violent storms. These two types of destructive waves have become commonly known as **tidal waves,** a name which is technically incorrect as they are not the result of tide-producing forces.

(269) **Tsunamis** (seismic sea waves) are ocean waves generated by any rapid large-scale disturbance of the sea water. Most tsunamis are generated by earthquakes, but they may also be caused by volcanic eruptions, landslides, undersea slumps or meteor impacts.

(270) The waves radiate outward in all directions from the disturbance and can propagate across entire ocean basins. Tsunami waves are distinguished from ordinary ocean waves by their great length between peaks, often exceeding 100 miles in the deep ocean, and by the long interval of time between these peaks, ranging from five minutes to an hour. The speed at which tsunamis travel depends on the ocean depth. A tsunami can exceed 500 knots in the deep ocean but slows to 20 or 30 knots in the shallow water near land. In less than 24 hours, a tsunami can cross the entire Pacific Ocean.

(271) In the deep ocean, a tsunami is barely noticeable and will only cause a small and slow rising and falling of the sea surface as it passes. Only as it approaches land does a tsunami become a hazard. As the tsunami approaches land and shallow water, the waves slow down and become compressed, causing them to grow in height. In the best of cases, the tsunami comes onshore like a quickly rising tide and causes a gentle flooding of low-lying coastal areas. In the worst of cases, a bore will form.

(272) A bore is a wall of turbulent water that can exceed several yards in height and can rush onshore with great destructive power. Behind the bore is a deep and fast-moving flood that can pick up and sweep away almost anything in its path. Minutes later, the water will drain away as the trough of the tsunami wave arrives, sometimes exposing great patches of the sea floor, then the water will rush in again as before, causing additional damage. This destructive cycle may repeat many times before the hazard finally passes. Sometimes the first noticeable part of the wave is the trough, which causes a recession of the water from shore, and people who have gone out to investigate this unusual exposure of the beach have been engulfed by the oncoming crest. Such an unexplained withdrawal of the sea should be considered as nature's warning of an approaching wave.

(273) Tsunamis do not have a season and do not occur regularly or frequently. Yet they pose a major threat to the coastal populations of the Pacific and other world oceans and seas. Nothing can be done to prevent them, but their adverse impact can be reduced with proper planning. The loss of life and property can be lessened if shipmasters and others acquaint themselves with the behavior of these waves so that intelligent action can be taken when they become imminent.

(274) NOAA oversees the U.S. Tsunami Program with its mission to provide a 24-hour detection and warning system and increase public awareness about the threat of tsunamis. The NOAA National Weather Service operates two tsunami warning centers The West Coast/Alaska Tsunami Warning Center in Palmer, Alaska, and the Richard H. Hagemeyer Pacific Tsunami Warning Center in ʻEwa Beach, Hawaii: *www.tsunami.gov*. These centers continuously monitor data from seismological and tidal stations, evaluate earthquakes that have the potential to generate tsunamis and disseminate tsunami information and warning bulletins to government authorities and the public.

(275) A tsunami warning is issued when a potential tsunami with significant inundation is imminent or expected. Warnings alert the public that widespread, dangerous coastal flooding accompanied by powerful currents is possible and may continue for several hours after arrival of the initial wave. Warnings also alert emergency management officials to take action for the entire tsunami hazard zone. When a tsunami warning has been issued, use a NOAA Weather Radio or stay tuned to a Coast Guard emergency frequency station or a local radio or television station for updated emergency information.

(276)

Storm Surge

(277) A considerable rise or fall in the level of the sea along a particular coast may result from strong winds and sharp change in barometric pressure. In cases where the water level is raised, higher waves can form with greater depth, and the combination can be destructive to low regions, particularly at high stages of tide. Extreme low levels can result in depths which are considerably less than those shown on nautical charts. This type of wave occurs especially in coastal regions bordering on shallow waters which are subject to tropical storms.

(278) **Seiche** is a stationary vertical wave oscillation with a period varying from a few minutes to an hour or more but somewhat less than the tidal periods. It is usually attributed to external forces such as strong winds, changes in barometric pressure, swells or tsunamis disturbing the equilibrium of the water surface. Seiche is found both in enclosed bodies of water and superimposed upon the tides of the open ocean. When the external forces cause a short-period horizontal oscillation on the water, it is called **surge**.

(279) The combined effect of seiche and surge sometimes makes it difficult to maintain a ship in its position alongside a pier even though the water may appear to be completely undisturbed, and heavy mooring lines have been parted repeatedly under such conditions. Pilots advise taut lines to reduce the effect of the surge.

(280)

Immersion Hypothermia

(281) Immersion hypothermia is the loss of heat when a body is immersed in water. With few exceptions, humans die if their core temperature of approximately 99.7° F drops below 78.6° F. Cardiac arrest is the most common direct cause of death. During prolonged immersion, the main threat to life is cold or cold and drowning combined.

(282)

SURVIVAL TIME VERSUS WATER TEMPERATURE

Water Temperature (°F)	Exhaustion or Unconsciousness	Expected Time of Survival
32	15 minutes	15 to 45 minutes
32 to 41	15-30 minutes	30 to 90 minutes
41 to 50	30-60 minutes	1 to 3 hours
50 to 59	1-2 hours	1 to 6 hours
59 to 68	2-7 hours	2 to 40 hours
68 to 77	3-12 hours	3 hours to indefinite
77 and above	indefinite	indefinite

(283) The length of time that a human survives in water depends on the water temperature and to a lesser extent on the person's behavior and body type. The table shows approximate human survival time in the sea. Body type can cause deviations, as small people become hypothermic more rapidly than large people. The cooling rate can be slowed by the person's behavior and insulated gear. The Heat Escape Lessening Posture (HELP) was developed for those in the water alone and the huddle for small groups. Both require a PFD (personal flotation device), or life preserver. HELP involves holding the arms close to the body, keeping the thighs together, and raising the knees to protect the groin area. In the huddle, people face each other and keep their bodies as close together as possible. These positions improve survival time to approximately two times that of a swimmer and one and a half times that of a person in the passive position.

(284) Near-drowning victims in cold water (less than 70° F) are revivable for much longer periods than usual. Keys to a successful revival are immediate cardiopulmonary resuscitation (CPR) and administration of pure oxygen. Total re-warming is not necessary at first. The whole revival process may take hours and require medical help.

(285)

Wind Chill and Frostbite

(286) When the body is warmer than its surroundings, it begins to lose heat. The rate of loss depends on barriers such as clothing and insulation, the speed of air movement and air temperature. Heat loss increases dramatically in moving air that is colder than skin temperature (91.4° F). Even a light wind increases heat loss, and a strong

wind can lower the body temperature if the rate of loss is greater than the body's heat replacement rate.

(287) When skin temperature drops below 50° F, there is a marked constriction of blood vessels, leading to vascular stagnation, oxygen want and cellular damage. The first indication that something is wrong is a painful tingling. Swelling of varying extent follows, provided freezing has not occurred. Excruciating pain may be felt if the skin temperature is lowered rapidly, but freezing of localized portions of the skin may be painless when the rate of change is slow. Possible effects of cold include cold allergy (welts), chilblains, which appear as reddened, warm, itching, swollen patches on the fingers and toes, and trench foot and immersion foot, which present essentially the same picture. Both result from exposure to cold and lack of circulation. Wetness can add to the problem as water and wind soften the tissues and accelerate heat loss.

(288) Frostbite usually begins when the skin temperature falls within the range of 14° to 4° F. Ice crystals form in the tissues and small blood vessels. The rate of heat loss determines the rate of freezing, which is accelerated by wind, wetness, extreme cold and poor blood circulation. Parts of the body susceptible to freezing are those with surfaces large in relation to their volume, such as toes, fingers, ears, nose, chin and cheeks.

(289) Injuries from the cold may, to a large extent, be prevented by maintaining natural warmth through the use of proper footgear and adequate, dry clothing, by avoiding cramped positions and constricting clothing and by active exercise of the hands, legs and feet.

(290)

MARINE POLLUTION

(291)

The Federal Water Pollution Control Act (Clean Water Act)

(292) The Federal Water Pollution Control Act (FWPCA) or Clean Water Act (CWA) was passed to restore and maintain the chemical, physical and biological integrity of the waters within the United States.

(293)

No-Discharge Zones

(294) Section 312 of the FWPCA gives the Environmental Protection Agency (EPA) and States the authority to designate certain areas as No-Discharge Zones (NDZ) for vessel sewage. Freshwater lakes, freshwater reservoirs or other freshwater impoundments whose entrances and exits prohibit traffic by regulated vessels (vessels with installed toilets) are, by regulation, NDZs. Rivers that do not support interstate navigation vessel traffic are also NDZs by regulation. Water bodies that can be designated as NDZs by States and EPA include the Great Lakes and their connecting waterways, freshwater lakes and impoundments accessible through locks and other flowing waters that support interstate navigation by vessels subject to regulation.

(295) Inside NDZ waters, discharge of any sewage, whether treated or untreated, is completely prohibited.

(296) Discharge of sewage in waters not designated as NDZs is regulated by the Marine Sanitation Device Standard (see **40 CFR 140** in chapter 2.)

(297) Additional information concerning the regulations may be obtained from *water.epa.gov*.

(298)

Oil Spill Reporting

(299) Reporting requirements for any oil discharge, noxious liquid substance or harmful substance occurring in waters under U.S. jurisdiction are found in **33 CFR 153**, subpart B (not in this Coast Pilot.) Any person in charge of a vessel or an onshore/offshore facility must, as soon as they have knowledge of any discharge of oil or a hazardous substance, immediately notify the National Response Center (NRC) at 800-424-8802 or NRC@uscg.mil.

(300)

Ocean Dumping

(301) The Marine Protection Research and Sanctuaries Act of 1972, as amended (33 USC 1401 et seq.), regulates the dumping of all material, except fish waste, into ocean waters. Radiological, chemical and biological warfare agents and other high level radioactive wastes are expressly banned from ocean disposal. The USACE issues permits for the disposal of dredged spoils; the EPA is authorized to issue permits for all other dumping activities. Surveillance and enforcement to prevent unlawful transportation of material for dumping or unlawful dumping under the Act has been assigned to the U.S. Coast Guard. The Act provides civil penalties of up to $50,000 and criminal penalties of up to $50,000 and/or one year imprisonment.

(302)

SELECT NAVIGATION RULES

(303)

Improper use of searchlights

(304) No person shall flash or cause to be flashed the rays of a searchlight or other blinding light onto the bridge or into the pilothouse of any vessel underway. The International Code Signal "PG2" may be made by a vessel inconvenienced by the glare of a searchlight in order to apprise the offending vessel of the fact.

(305)

Use of Radar

(306) Navigation Rules, International-Inland, Rule 7, states, in part, that every vessel shall use all available means appropriate to the prevailing circumstances and conditions to determine if risk of collision exists. If there is any doubt such risk shall be deemed to exist. Proper use shall be made of radar equipment if fitted and operational, including long-range scanning to obtain early warning of risk of collision and radar plotting or equivalent systematic observation of detected objects.

(306) <Deleted Paragraph>

(307) This rule places an additional responsibility on vessels that are equipped and manned to use radar to do so while underway during periods of reduced visibility without in any way relieving commanding officers of the responsibility of carrying out normal precautionary measures.

(308) Navigation Rules, International-Inland, Rules 6, 7, 8, and 19 apply to the use of radar.

(309)

Danger signal

(310) Navigation Rules, International-Inland, Rule 34(d), states that when vessels in sight of one another are approaching each other and from any cause either vessel fails to understand the intentions or actions of the other or is in doubt whether sufficient action is being taken by the other to avoid collision, the vessel in doubt shall immediately indicate such doubt by giving at least five short and rapid blasts on the whistle. Such signal may be supplemented by a light signal of at least five short and rapid flashes.

(311)

Narrow channels

(312) Navigation Rules, International-Inland, Rule 9(b) states that a vessel of less than 20 meters in length or a sailing vessel shall not impede the passage of a vessel that can safely navigate only within a narrow channel or fairway.

(313)

REGULATED WATERS

(314)

Traffic Separation Schemes (Traffic Lanes)

(315) To increase the safety of navigation, particularly in converging areas of high traffic density, routes incorporating traffic separation have been adopted by the IMO in certain areas of the world. In the interest of safe navigation, it is recommended that through traffic use these schemes, as far as circumstances permit, by day and by night and in all weather conditions. When approved or established, traffic separation scheme details are announced in Notice to Mariners and later depicted on appropriate charts and included in the U.S. Coast Pilot. See **33 CFR 167**, chapter 2, for regulations.

(316) The IMO is recognized as the only international body responsible for establishing and recommending measures on an international level concerning ships' routing. In deciding whether or not to adopt or amend a traffic separation scheme, IMO will consider whether the scheme complies with the design criteria for traffic separation schemes and with the established methods of routing. IMO also considers whether the aids to navigation proposed will enable mariners to determine their position with sufficient accuracy to navigate the scheme in accordance with Rule 10 of the International Regulations for Preventing Collisions at Sea (72 COLREGS).

(317) The IMO approved routing measures which affect shipping in or near U.S. waters are:

(318)

IMO-Approved Traffic Separation Routes
Portland, Maine (approaches to)
Boston, Massachusetts (approaches to)
Narragansett Bay, Rhode Island (approaches to)
Buzzards Bay, Massachusetts (approaches to)
New York, New York
Delaware Bay
Chesapeake Bay (approaches to)
Cape Fear River (approaches to)
Galveston Bay (approaches to)
Of San Francisco, California
Los Angeles/Long Beach, California (approaches to)
Strait of Juan de Fuca (approaches to and in)
Puget Sound (approaches to and in)
Haro Strait, Boundary Pass and the Strait of Georgia
Prince William Sound, Alaska

(318.001)

MARITIME ZONES AND BOUNDARIES

(318.002) The maritime zones recognized under international law include internal waters, the territorial sea, the contiguous zone, the exclusive economic zone (EEZ), the continental shelf, the high seas, and the Area (see Figure 1). The breadth of the territorial sea, contiguous zone, and EEZ (and in some cases the continental shelf) is measured from the baseline determined in accordance with customary international law as reflected in the 1982 Law of the Sea Convention.

(318.003) The limits of these zones are officially depicted on NOAA nautical charts. The limits shown on the most recent chart edition takes precedence. The boundaries of maritime zones between coastal States are established through international agreements entered into by those nations. For the official description of the U.S. maritime boundaries with other nations, contact the U.S. Department of State. For more information on NOAA's U.S. Maritime Limits & Boundaries visit: *https://www.noaa.gov/maritime-zones-and-boundaries*

(319)

Baseline

(320) Generally speaking, the normal baseline is the low-water line along the coast as marked on large-scale charts officially recognized by the coastal State. (LOSC art. 5). Special rules for determining the baseline apply in a variety of circumstances, such as with bays, ports, mouths of rivers, deeply indented coastlines, fringing reefs, and roadsteads. (LOSC arts. 6-15). Consistent with these rules, the U.S. baselines are the mean of the lower low tides as depicted on the largest-scale NOAA nautical charts. The U.S. normal baselines are ambulatory and subject to changes as the coastline accretes and erodes. NOAA's nautical charts depict the baselines from which

the seaward limits of the U.S. territorial sea, contiguous zone, and exclusive economic zone are measured as well as the seaward boundary of the Three Nautical Mile Line and the Natural Resources Boundary, as described below.

(322)

Internal Waters

(323) Internal waters are the waters on the landward side of the baselines from which the breadth of the territorial sea is measured. (LOSC art. 8). The United States has full sovereignty over its internal waters as if they were part of its land territory and may exclude foreign flag vessels from its internal waters subject to the right of entry of vessels in distress. The right of innocent passage does not apply in internal waters. Ships and aircraft may not enter or overfly internal waters without permission of the coastal State. Examples of internal waters include rivers, harbors, lagoons, some bays and canals, and lakes, including the Great Lakes.

(324)

Territorial Sea

(325) Each coastal State may claim a territorial sea that extends seaward up to 12 nautical miles (nm) from its baselines. (LOSC arts. 3, 4). The coastal State exercises sovereignty over its territorial sea, the airspace above it, and the seabed and subsoil beneath it. (LOSC art. 2). Foreign flagged ships enjoy the right of innocent passage while transiting the territorial sea subject to laws and regulations adopted by the coastal State that are in conformity with the **Law of the Sea Convention** (LOSC arts. 17-26) and other rules of international law relating to such passage.

(325.001) In 1988, the United States claimed a 12 nm territorial sea. The extension of the territorial sea of the United States includes the belt of seas around the Commonwealth of Puerto Rico, Guam, American Samoa, the U.S. Virgin Islands, the Commonwealth of the Northern Mariana Islands and any other territory or possession over which the United States exercises sovereignty. (Presidential Proclamation No. 5928 of December 27, 1988, 54 Fed. Reg. 777 (Dec. 27, 1988)). The territorial sea of the United States extends seaward to 12 nm from the baselines, which is determined in accordance with the Law of the Sea Convention except as otherwise established in a maritime boundary treaty of the United States. Vessels of all States navigating through the territorial sea enjoy the right of innocent passage. (LOSC art. 17). However, as a coastal State, the United States may adopt certain laws and regulations relating to innocent passage so long as they are in conformity with the provisions of LOSC and other rules of international law. (LOSC art. 21(1)).

(326)

Contiguous Zone

(327) Each coastal State may claim a contiguous zone adjacent to and beyond its territorial sea that extends seaward up to 24 nm from its baselines. (LOSC art. 33(1) & (2)). In its contiguous zone, a coastal State may exercise the control necessary to prevent the infringement of its customs, fiscal, immigration, or sanitary laws and regulations within its territory or territorial sea, and punish infringement of those laws and regulations committed within its territory or territorial sea. (LOSC art. 33(1) (a) & (b)). In addition, in order to control traffic in archeological and historical objects, a coastal State may presume that the removal of archeological and historical objects (e.g., underwater cultural heritage) found at sea within its contiguous zone without its approval would result in an infringement of its laws and regulations. (LOSC art. 303).

(321)

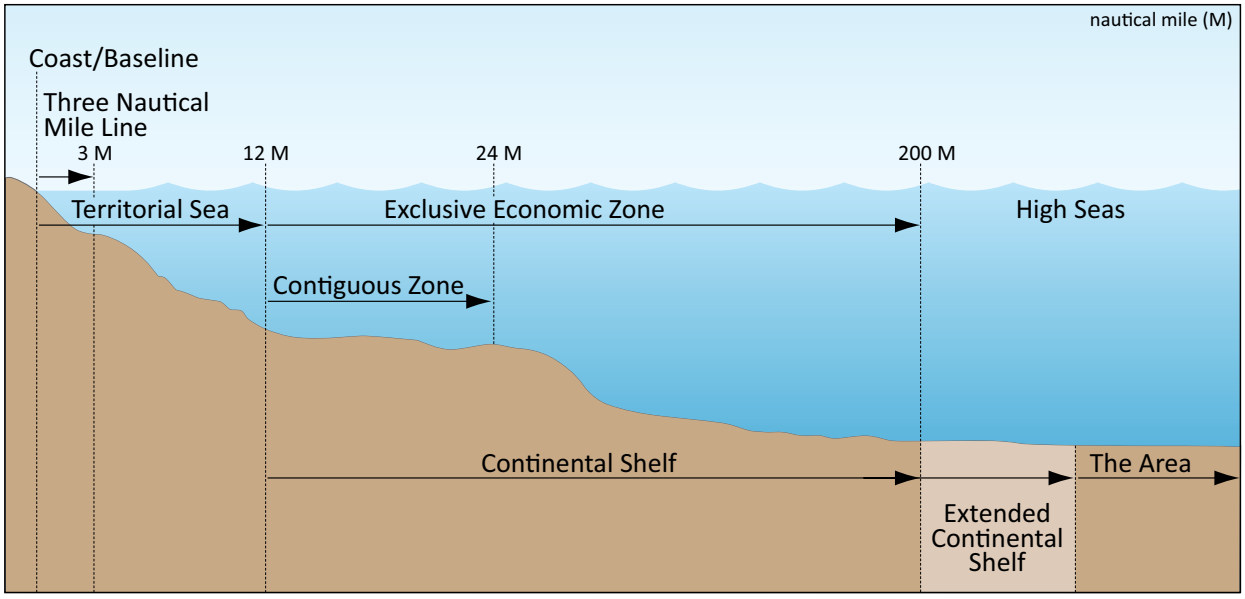


Figure 1: Offshore extent of maritime zones beyond internal waters
https://www.noaa.gov/maritime-zones-and-boundaries

(327.001) The contiguous zone of the United States includes the waters off of all U.S. coastal states, the Commonwealth of Puerto Rico, Guam, American Samoa, the U.S. Virgin Islands, the Commonwealth of the Northern Mariana Islands and any other territory or possession over which the United States exercises sovereignty. In 1999, the United States claimed a contiguous zone extending from 12 to 24 nm offshore. (Presidential Proclamation No. 7219 of August 2, 1999, Contiguous Zone of the United States, 64 Fed. Reg. 48,701 (Aug. 8, 1999)).

(328)

Exclusive Economic Zone

(329) Each coastal State may claim an exclusive economic zone (EEZ) beyond and adjacent to its territorial sea that extends seaward up to 200 nm from its baselines (or out to a maritime boundary with another coastal State). (LOSC art. 55). Within its EEZ, a coastal State has: (a) sovereign rights for the purpose of exploring, exploiting, conserving, and managing natural resources, whether living or nonliving, of the seabed and subsoil and the superjacent waters and with regard to other activities for the economic exploitation and exploration of the zone, such as the production of energy from the water, currents, and winds; (b) jurisdiction as provided for in the relevant provisions of the LOSC with regard to the establishment and use of artificial islands, installations, and structures, marine scientific research, and the protection and preservation of the marine environment; and (c) other rights and duties provided for in the LOSC. (LOSC art. 56).

(330) The United States claimed a 200 nm EEZ in 1983. The U.S. EEZ extends no more than 200 nm from the territorial sea baselines and is adjacent to the 12 nm territorial sea of the United States, including the Commonwealth of Puerto Rico, Guam, American Samoa, the U.S. Virgin Islands, the Commonwealth of the Northern Mariana Islands and any other territory or possession over which the United States exercises sovereignty. (Presidential Proclamation No. 5030 of March 10, 1983; Exclusive Economic Zone and Maritime Boundaries; Notice of Limits; 60 Fed. Reg. 43,825 (Aug. 23, 1995)). As such, the exclusive economic zone overlaps the 12 nm-24 nm contiguous zone. In December 2023, the U.S. Department of State released the geographic coordinates defining the outer limits of the U.S. extended continental shelf. (U.S. Dep't of State, Announcement of U.S. Extended Continental Shelf Outer Limits (Dec. 19, 2023); see also Continental Shelf and Maritime Boundaries; Notice of Limits; 88 Fed. Reg. 88,470 (Dec. 21, 2023)). In addition, the U.S. Department of State also updated information pertaining to the geographic coordinates defining the outer limits of the U.S. EEZ. (Exclusive Economic Zone and Maritime Boundaries; Notice of Limits; 88 Fed. Reg. 88,477 (Dec. 21, 2023)).

(331) Note: Under certain U.S. fisheries laws, such as the Magnuson-Stevens Fishery Conservation and Management Act, the term EEZ is defined as having an inner boundary that is coterminous with the seaward (or outer) boundary of each of the individual coastal states of the U.S. See 16 U.S.C. § 1802(11). Under the Submerged Lands Act, the seaward boundary of each of the individual coastal states is generally three nautical (or geographic) miles from the coast line. The seaward boundaries of Florida (Gulf of America coast only), Texas, and Puerto Rico extend nine nautical miles from the coast line. In the Great Lakes, each U.S. state's seaward boundary may extend to the international maritime boundary with Canada. See 43 U.S.C. § 1312. Under the Submerged Lands Act, a coastal state's seaward boundary may be fixed by Supreme Court decree. (See below for further information on the Three Nautical Mile Line and the Natural Resources Boundary).

(331.001)

Continental Shelf

(331.002) The continental shelf of a coastal State is comprised of the seabed and subsoil of the submarine areas that extend beyond its territorial sea throughout the natural prolongation of its land territory to the outer edge of the continental margin, or to a distance of 200 nm from its baselines where the outer edge of the continental margin does not extend up to that distance. (LOSC art. 76(1)). The extent of the continental shelf can also be limited by a maritime boundary with another coastal State. (LOSC art. 76(10)).

(331.003) Where the outer edge of a coastal State's continental margin extends beyond 200 nm from its baselines, the outer limits of its continental shelf are determined in accordance with Article 76 paragraphs 2-7 of the **Law of the Sea Convention.** (LOSC art. 76 (2-7)). The portion of a coastal State's continental shelf that lies beyond the 200 nm limit is often called the extended continental shelf (ECS).

(331.004) A coastal State exercises sovereign rights and exclusive jurisdiction over its continental shelf for the purpose of exploring it and exploiting its natural resources, as well as for other purposes specified in the Law of the Sea Convention. The natural resources of the continental shelf consist of the mineral and other non-living resources of the seabed and subsoil together with living organisms belonging to sedentary species, that is to say, organisms which, at the harvestable stage, either are immobile on or under the seabed or are unable to move except in constant physical contact with the seabed or subsoil. (LOSC art. 77).

(331.005) In 1945, the United States proclaimed jurisdiction and control over its continental shelf (Presidential Proclamation No. 2667 of Sept. 28, 1945; 10 Fed. Reg. 12,303 (Oct. 2, 1945)). Consistent with international law, the United States exercises its continental shelf rights out to a distance of at least 200 nautical miles from the baselines through several domestic laws. The U.S. Extended Continental Shelf Project, led by the U.S. Department of State, NOAA, and the U.S. Geological Survey, determines the outer limits of the U.S. continental

shelf beyond 200 nautical miles (i.e., extended continental shelf). In December 2023, the U.S. Department of State released the geographic coordinates defining the outer limits of the U.S. extended continental shelf. (U.S. Dep't of State, Announcement of U.S.Extended Continental Shelf Outer Limits (Dec. 19, 2023); see also Continental Shelf and Maritime Boundaries; Notice of Limits; 88 Fed. Reg. 88,470 (Dec. 21, 2023)).

(331.006)

High Seas

(331.007) The high seas comprise all parts of the sea that are not included in the exclusive economic zone, the territorial sea or the internal waters of a State, or in the archipelagic waters of an archipelagic State. (LOSC art. 86).

(331.008)

Area

(331.009) The Area is comprised of the seabed and ocean floor and subsoil thereof beyond the limits of national jurisdiction. (LOSC art. 1(1)). It does not include superjacent waters (i.e., the water column) or the air space above those waters. (LOSC arts. 1(1), 135). No State may claim or exercise sovereignty or sovereign rights over any part of the Area or its resources. (LOSC art. 137(1)). The term "resources" means all solid, liquid or gaseous mineral resources in situ in the Area at or beneath the seabed, including polymetallic nodules. (LOSC art. 133(a)).

(331.010)

Straits Used for International Navigation

(331.011) "Straits used for international navigation" are those that are used or are capable of use for international navigation between one area of the high seas or exclusive economic zone ("EEZ") and another area of the high seas or EEZ. (LOSC art. 37). Part III of the Law of the Sea Convention (LOSC arts. 34-45) describes the regime of transit passage through such straits and the rights, jurisdiction, and duties of the States bordering such straits. Transit passage means the exercise in accordance with Part III of the Law of the Sea Convention of the freedom of navigation and overflight solely for the purpose of continuous and expeditious transit of the strait. (LOSC arts. 37, 38). The right of transit passage applies throughout straits used or capable of use for international navigation, including to all normally used approaches to and from such straits. Ships and aircraft in transit passage must comply with the duties outlined in LOSC article 39, which include proceeding without delay and refraining from any activities other than those incident to their normal modes of continuous and expeditious transit. Ships in transit passage may not carry out any research or survey activities without the prior authorization of the States bordering the strait. (LOSC art. 40). States bordering straits used for international navigation may designate sea lanes and prescribe traffic separation schemes for navigation in accordance with Part III where necessary to promote safe passage of ships. (LOSC art. 41). They may also adopt laws and regulations relating to transit passage in respect of certain activities, such as fishing. (LOSC art. 42). States bordering straits used for international navigation may not hamper transit passage. (LOSC art. 44). The transit passage regime does not otherwise affect the legal status of the waters forming an international strait or the exercise of sovereignty or jurisdiction by the bordering States over the waters, air space, seabed, and subsoil of the strait. (LOSC art. 34).

(332)

Three Nautical Mile Line

(333) The Three Nautical Mile Line, as measured from the territorial sea baselines and previously identified as the outer limit of the U.S. territorial sea, is retained on NOAA charts because it continues to be used in certain federal laws.

(334) Note: Since the "coast line," a term used in the Submerged Lands Act (43 USC Section 1301 et seq.), and the baselines are determined using the same criteria under international law, the Three Nautical Mile Line is generally the same as the seaward boundaries of individual U.S. coastal states under the Submerged Lands Act. There are exceptions; therefore, the Three Nautical Mile Line does not necessarily depict the seaward boundaries of all U.S. coastal states in all circumstances under the Submerged Lands Act.

(335)

Natural Resources Boundary

(336) The nine (9) nm Natural Resources Boundary is the seaward boundary of the submerged lands of Puerto Rico, Texas and the Gulf coast of Florida. It coincides with the inner limit of the U.S. outer continental shelf under the Outer Continental Shelf Lands Act (43 U.S.C. Section 1331 et seq.).

(336.001)

SUPPLEMENTAL INFORMATION

(337)

Notification of Arrival and Vessel Response Plans

(338) A **Notification of Arrival (NOA)** must be submitted by all U.S. and foreign vessels bound for or departing from ports or places in the United States. (See **33 CFR 160 – Subpart C**, chapter 2). Additionally, tank vessels and non-tank vessels are required to submit an oil spill response plan. (See **33 CFR 155**– Subparts D and J, not contained in this Coast Pilot.)

(339)

Marine Protected Area (MPA)

(340) Marine Protected Areas (MPAs) are particular places in ocean, coastal and estuarine ecosystems where vital natural and cultural resources are given greater protection than in surrounding waters. MPAs have been established in the U.S. for more than a century. Currently, there are over 1,700 MPAs in U.S. marine waters and the Great Lakes, with levels of protection ranging from a few "no-take" areas that prohibit all extractive uses to the more common multiple use areas that allow vessel access,

anchoring, fishing and non-consumptive activities. MPAs are managed by dozens of Federal, state, tribal and local authorities. For detailed information on MPA locations, types, interactive map, purposes and legal restrictions, visit *marineprotectedareas.noaa.gov.*

(341)

Archaeological Resource Preservation

(342) Under Federal and state laws it is illegal to destroy, deface, collect, transport, sell or trade archaeological, cultural, submerged and historic resources without authorization. Applicable laws include, but are not limited to, the Historic Sites Act, the Archaeological Resource Protection Act, the National Historic Preservation Act the Abandoned Shipwreck Act, and the Sunken Military Craft Act. These laws protect archaeological resources on lands administered by the National Park Service, U.S. Fish and Wildlife Service, Bureau of Land Management, and National Marine Sanctuaries as well as state, private and Native lands.

(343)

DEPARTMENT OF AGRICULTURE

(344)

Animal and Plant Health Inspection Service

(345) The Animal and Plant Health Inspection Service is responsible for protecting the Nation's animal population, food and fiber crops and forests from invasion by foreign pests. They administer agricultural quarantine and restrictive orders issued under authority provided in various acts of Congress. The regulations prohibit or restrict the importation or interstate movement of live animals, meats, animal products, plants, plant products, soil, injurious insects, and associated items that may introduce or spread plant pests and animal diseases which may be new to or not widely distributed within the United States or its territories. Inspectors examine imports at ports of entry as well as the vessel, its stores and crew or passenger baggage.

(346) The Service also provides an inspection and certification service for exporters to assist them in meeting the quarantine requirements of foreign countries. (See **Appendix A** for a list of ports where agricultural inspectors are located and inspections conducted.)

(347)

DEPARTMENT OF COMMERCE

(348)

National Oceanic and Atmospheric Administration (NOAA)

(349) The National Oceanic and Atmospheric Administration (NOAA) conducts research and gathers data about the global oceans, atmosphere, space and sun, and applies this knowledge to improve our understanding and stewardship of the environment.

(350) NOAA provides services to the nation and the public through five major organizations: the National Ocean Service; the National Weather Service; the National Marine Fisheries Service; the National Environmental Satellite, Data and Information Service (NESDIS); and NOAA Research; and numerous special program units. In addition, NOAA research and operational activities are supported by the Nation's seventh uniformed service, the NOAA Corps, a commissioned officer corps of men and women who operate NOAA ships and aircraft and serve in scientific and administrative positions.

(351)

National Ocean Service (NOS)

(352) The National Ocean Service's primary concern is the health and safety of our Nation's coastal and oceanic environment. Within NOS, the **Office of Coast Survey** is responsible for producing and maintaining the suite of over 1000 nautical charts and the Coast Pilots that cover the coastal waters of the U.S. and its territories. Nautical charts are published primarily for the use of the mariner but serve the public interest in many other ways. Cartographers in Coast Survey receive and compile information from a variety of government and non-governmental sources for portrayal on nautical charts and the Coast Pilots. In addition, Coast Survey hydrographers, as well as private contractors, conduct new surveys that are used to update these products. The principal facilities of Coast Survey are located at NOAA headquarters in Silver Spring, MD; Norfolk, VA (Marine Operations Center Atlantic); and Seattle, WA (Western Regional Center).

(353) The **Center for Operational Oceanographic Products and Services (CO-OPS)** collects and distributes observations and predictions of water levels and currents to ensure safe, efficient and environmentally sound maritime commerce. Users can find a variety of information, including observed water level and currents data, tide and current predictions, sea level trends and coastal inundation information. Tides and Currents information is available at *tidesandcurrents.noaa.gov.*

(354) **PORTS® (Physical Oceanographic Real-Time System)** is a decision support tool that improves the safety and efficiency of maritime commerce and coastal resource management. Data from PORTS® supports navigation safety, improves the efficiency of U.S. ports and harbors, and ensures the protection of coastal marine resources. PORTS® collects and disseminates observations of water levels, currents, salinity, bridge air gap and meteorological parameters (e.g., winds, atmospheric pressure, air and water temperatures) that mariners need to navigate safely and allows seaport and terminal facilities to make good decisions. PORTS® data and information is provided via the internet at *tidesandcurrents.noaa.gov/ports_info.html* and, in some areas, via telephone voice response.

(355)

NOAA Tide Predictions and Tidal Current Predictions

(356) NOAA discontinued the annual printed Tide Tables and Tidal Current Tables in 2020, and has transitioned to

providing this information digitally online. The online prediction service equals or exceeds the accuracy of the historically printed publications. Tide and tidal current predictions needed for navigation can be generated in real-time for any time period required by the mariner, for as short as one day, or as long as an entire year. All predictions for U.S. waters are available at *tidesandcurrents.noaa.gov*.

(357) U.S. Coast Guard regulations do not consider access to NOAA's online prediction service "while navigating" as meeting carriage requirements. In order to use predictions from these services, the predictions must be generated in advance and either be stored on the user's device as an electronic file (PDF, screen image, data table, etc.), or as a printed page.

(358) **NOAA Tide Predictions** - *tidesandcurrents.noaa.gov/tide_predictions.html* - allows the generation of predicted tides for more than 3000 locations along the U.S. coastline. Tide predictions may be generated as times and heights of high and low tides for all locations, or as interval predictions (hourly, 15-minute, 6-minute, etc.) for many locations. Tide predictions may be generated for past, present, or future dates; with lengths of 1 day to 1 month, or the full calendar year. The Users Guide - *tidesandcurrents.noaa.gov/PageHelp.html* - describes the displays, formats, additional capabilities, and uses of this online service.

(359) **Caution** –When using Tide Predictions, slack water should not be confused with high or low water. For ocean stations there is often little difference between the time of high or low water and the start of flood/ebb currents; but for places in narrow channels, landlocked harbors or on tidal rivers, the time of slack current may differ by several hours from the time of high or low water. The relationship of the times of high or low water to the flood and ebb of the current depends upon a number of factors unique to each location; no simple general rule can be given which applies to every location. For navigation or other activities which depend on slack water, tidal current predictions should be used to provide times of slack water.

(360) **NOAA Tidal Current Predictions** –*https://tidesandcurrents.noaa.gov/* - allows the generation of predicted currents for more than 2500 locations along the U.S. coastline. Tidal current predictions may be generated as times and speeds of flood/ebb currents and times of slack water for all locations; or as interval predictions of speed (hourly, 30-minute, 6-minute) for many locations. Tidal current predictions may be generated for past, present or future dates; with length of 1 day to 2 weeks, or the full calendar year. The Users Guide - *https://tidesandcurrents.noaa.gov/PageHelp.html* - describes the displays, formats, additional capabilities, and uses of this online service.

(361)

National Weather Service (NWS)

(362)

National Data Buoy Center Meteorological Buoys

(363) The National Data Buoy Center (NDBC) deploys moored meteorological buoys that provide weather data directly to the mariner as well as to marine forecasters.

(364) These buoys have a watch circle radius (WCR) of 2,000 to 4,000 yards from assigned position (AP). In addition, any mooring in waters deeper than 1,000 feet will have a floating "loop" or catenary that may be as little as 500 feet below the surface. This catenary could be anywhere within the buoy's WCR. Any underwater activity within this radius may contact the mooring, causing a failure.

(365) To avoid cutting or damaging a mooring, mariners are urged to exercise extreme caution when navigating in the vicinity of meteorological buoys and to remain well clear of the watch circle. If a mooring is accidentally contacted or cut, please notify NDBC at 228-688-2835 or 228-688-2436.

(366) For further information relating to these buoys visit *ndbc.noaa.gov*.

(367)

Marine Weather Forecasts

(368) The NWS provides marine weather forecasts and warnings for the U.S. coastal waters, the Great Lakes, offshore waters and high seas areas. Scheduled marine forecasts are issued four times daily from **National Weather Service Offices** with local areas of responsibility around the United States, Guam, American Samoa and Puerto Rico. (See Appendix A for NWS Offices located in the area covered by this Coast Pilot.)

(369) Typically, marine forecasts contain information on wind speed and direction, wave heights, visibility, weather and a general synopsis of weather patterns affecting the region. The forecasts are supplemented with special marine warnings and statements, radar summaries, marine observations, small-craft advisories, gale warnings, storm warnings and various categories of tropical cyclone warnings, e.g., tropical depression, tropical storm and hurricane warnings. Specialized products such as coastal flood, seiche, and tsunami warnings, heavy surf advisories, low water statements, ice forecasts and outlooks and lake shore warnings and statements are issued as necessary. For further information, visit: *https://www.weather.gov/marine/* .

(369) <Deleted Paragraph>

(370) The principal means of disseminating marine weather services and products in coastal areas is **NOAA Weather Radio**. This network of more than 900 transmitters, covering all 50 states, adjacent coastal waters, Puerto Rico, the U.S. Virgin Islands and the U.S. Pacific Territories, is operated by the NWS and provides continuous broadcasts of weather information for the general public. These broadcasts repeat recorded messages every 4 to 6 minutes. Messages are updated

periodically, usually every 2-3 hours and amended as required to include the latest information. When severe weather threatens, routine transmissions are interrupted and the broadcast is devoted to emergency warnings. (See Appendix A for NOAA Weather Radio Stations covered by this Coast Pilot.)

(371) In coastal areas, the programming is tailored to the needs of the marine community. Each coastal marine forecast covers a specific area. For example, "Cape Henlopen to Virginia Beach, out 20 miles." The broadcast range is about 40 miles from the transmitting antenna site, depending on terrain and quality of the receiver used. When transmitting antennas are on high ground, the range is somewhat greater, reaching 60 miles or more. Some receivers are equipped with a warning alert device that can be turned on by means of a tone signal controlled by the NWS office concerned. This signal is transmitted for 13 seconds preceding an announcement of a severe weather warning.

(372) Marine weather warnings are displayed to small-craft operators and others within sight of the shore by the flags, pennants and lights of the **Coastal Warning Display** program. These displays are meant to warn the public of approaching storm conditions and visually communicate that citizens should take personal responsibility for individual safety in the face of an approaching storm. Anyone observing the signals displayed by the program is urged to tune to the NWS radio broadcasts for the latest information. (See **National Weather Service Coastal Warning Displays** illustration for additional information.)

(374) NWS marine weather products are also disseminated to marine users through the broadcast facilities of the Coast Guard, Navy and commercial marine radio stations. Details on these broadcasts including times, frequencies and broadcast content are listed on the NWS internet site, **Marine Product Dissemination Information**, *https://www.weather.gov/marine/nws_dissemination* .

(375) Ships of all nations share equally in the effort to report weather observations. These reports enable meteorologists to create a detailed picture of wind, wave and weather patterns over the open waters that no other data source can provide and upon which marine forecasts are based. The effectiveness and reliability of these forecasts and warnings plus other services to the marine community are strongly linked to the observations received from mariners. There is an especially urgent need for ship observations in the coastal waters, and the NWS asks that these be made and transmitted whenever possible. Many storms originate and intensify in coastal areas. There may be a great difference in both wind direction and speed between the open sea, the offshore waters and on the coast itself.

(376) Information on how ships, commercial fishermen, offshore industries and others in the coastal zone may participate in the marine observation program is available from **National Weather Service Port Meteorological Officers (PMOs)**. PMOs are located in major U.S. port cities where they visit ships in port to assist masters and mates with the weather observation program, provide instruction on the interpretation of weather charts, calibrate barometers and other meteorological instruments and discuss marine weather communications and marine weather requirements affecting the ships' operations. (For further information on the Voluntary Observing Ship Program and PMOs, go to *vos.noaa.gov*.)

(377)

Space Weather Prediction Center (SWPC)

(378) The Space Weather Prediction Center provides real-time monitoring and forecasting of solar and geophysical events that impact satellites, power grids, communications, navigation and many other technological systems. (See Space Weather Prediction Center in Appendix A.)

(379)

National Environmental Satellite, Data, and Information Service (NESDIS)

(380) Among its functions, NESDIS archives, processes and disseminates the non-real-time meteorological and oceanographic data collected by government agencies and private institutions. Marine weather observations are collected from ships at sea on a voluntary basis. About one million observations are received annually at NESDIS's National Climatic Center. They come from vessels representing every maritime nation. These observations, along with land data, are returned to the mariners in the form of climatological summaries and atlases for coastal and ocean areas. They are available in such NOAA publications as the **U.S. Coast Pilot, Mariners Weather Log** and **Local Climatological Data, Annual Summary**. They also appear in the National Geospatial-Intelligence Agency's **Pilot Chart Atlases** and **Sailing Directions Planning Guides**.

(381)

DEPARTMENT OF DEFENSE

(382)

National Geospatial-Intelligence Agency (NGA)

(383) The National Geospatial-Intelligence Agency provides hydrographic, navigational, topographic, and geodetic data, charts, maps and related products and services to the Armed Forces, other Federal Agencies, the Merchant Marine and mariners in general. Publications include Sailing Directions, List of Lights, Distances Between Ports, Radio Navigational Aids, International Code of Signals, American Practical Navigator (Bowditch) and Notice to Mariners. (See NGA Procurement Information in Appendix A.)

(384)

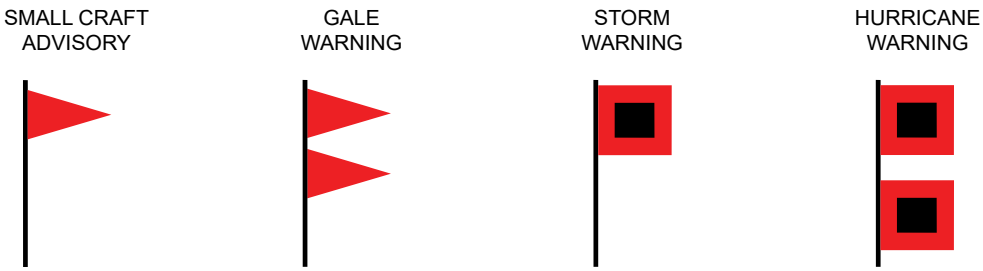
Army Corps of Engineers

(385) The U.S. Army Corps of Engineers has charge of the improvement of the rivers and harbors of the United States and of miscellaneous other civil works, which include the administration of certain Federal laws enacted for the protection and preservation of navigable waters

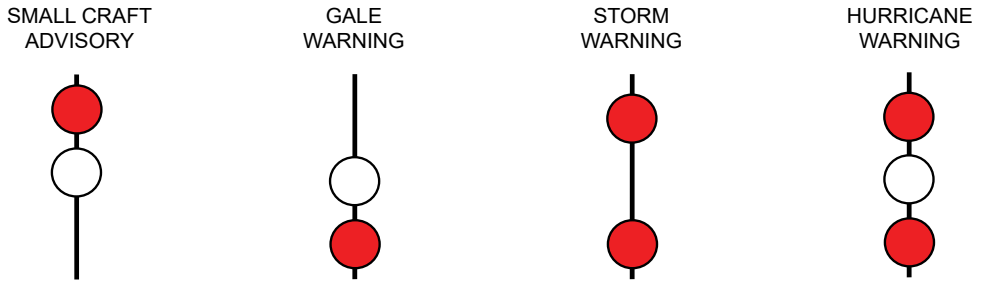
(373)

NATIONAL WEATHER SERVICE COASTAL WARNING DISPLAYS

DAYTIME SIGNALS



NIGHT (LIGHT) SIGNALS



SMALL CRAFT ADVISORY: An advisory issued by coastal and Great Lakes Weather Forecast O f ces (WFO) for areas included in the Coastal Waters Forecast or Nearshore Marine Forecast (NSH) products. Thresholds governing the issuance of small craft advisories are speci f c to geographic areas. A Small Craft Advisory may also be issued when sea or lake ice exists that could be hazardous to small boats. There is no precise de f nition of a small craft. Any vessel that may be adversely a f ected by Small Craft Advisory criteria should be considered a small craft. Other considerations include the experience of the vessel operator, and the type, overall size, and sea worthiness of the vessel. There is no legal de f nition of "small craft". The Small Craft Advisory is an advisory in Coastal Waters and Nearshore forecasts for sustained winds, frequent gusts, or sea/wave conditions, exceeding de f ned thresholds speci f c to geographic areas. A Small Craft Advisory may also be issued when sea or lake ice exists that could be hazardous to small boats.

Eastern (ME to SC, Lake Erie, Lake Ontario) – Sustained winds or frequent gusts ranging between 25 and 33 knots (except 20 to 25 knots, lower threshold area dependent, to 33 knots for harbors, bays, etc.) and/or seas or waves 5 to 7 feet and greater, area dependent.

Central (MN to OH) – Sustained winds or frequent gusts (on the Great Lakes) between 22 and 33 knots inclusive, and/or seas or waves greater than 4 feet.

Southern (GA to TX and Caribbean) – Sustained winds of 20 to 33 knots, and/or forecast seas 7 feet or greater that are expected for more than 2 hours.

Western (WA..CA) - Sustained winds of 21 to 33 knots, potentially in combination with wave heights exceeding 10 feet (or wave steepness values exceeding local thresholds).

Alaska (AK) – Sustained winds or frequent gusts of 23 to 33 knots. A small craft advisory for rough seas may be issued for sea/wave conditions deemed locally signi f cant, based on user needs, and should be no lower than 8 feet.

Hawaii (HI), Samoa – Sustained winds 25 knots or greater and seas 10 feet or greater.

Guam and the Northern Mariana Islands – Sustained winds 22 to 33 knots and/or combined seas of 10 feet or more. "Frequent gusts"are typically long duration conditions (greater than 2 hours).

For a list of NWS Weather O f ces by Region, refer to the following website: http://www.nws.noaa.gov/organization.php

GALE WARNING: To indicate winds within the range 34 to 47 knots are forecast for the area.

STORM WARNING: To indicate winds 48 knots and above, no matter how high the speed, are forecast for the area. However, if the winds are associated with a tropical cyclone (hurricane), the STORM WARNING indicates that winds within the range 48-63 knots are forecast.

HURRICANE WARNING: Issued only in connection with a tropical cyclone (hurricane) to indicate that winds 64 knots and above are forecast for the area.

NOTE: A "HURRICANE WATCH" is an announcement issued by the National Weather Service via press and television broadcasts whenever a tropical storm or hurricane becomes a threat to a coastal area. The "Hurricane Watch" announcement is not a warning, rather it indicates that the hurricane is near enough that everyone in the area covered by the "Watch" should listen to their radios for subsequent advisories and be ready to take precautionary action in case hurricane warnings are issued.

NOTE: A SPECIAL MARINE WARNING is issued whenever a severe local storm or strong wind of brief duration is imminent and is not covered by existing warnings or advisories. No visual displays will be used in connection with the Special Marine Warning Bulletin; boaters will be able to receive thesespecial warnings by keeping tuned to a NOAA Weather Radio station or to Coast Guard and commercial radio stations that transmit marine weather information.

of the United States; the establishment of regulations for the use, administration, and navigation of navigable waters; the establishment of harbor lines; the removal of sunken vessels obstructing or endangering navigation; and the granting of permits for structures or operations in navigable waters and for discharges and deposits of dredged and fill materials in these waters.

(386) **Restricted areas** in most places are defined and regulations governing them are established by the USACE. The regulations are enforced by the authority designated in the regulations, and the areas are shown on the large-scale charts of the National Ocean Service. Copies of the regulations may be obtained at the District offices of the USACE. The regulations also are included in the appropriate Coast Pilot.

(387) Information concerning the various ports, improvements, channel depths, navigable waters and the condition of the Intracoastal Waterways in the areas under their jurisdiction may be obtained direct from the District Engineer Offices. (See Appendix A for addresses.)

(388) The USACE has general supervision of location, construction and manner of maintenance of all **fishtraps,** weirs, pounds or other fishing structures in the navigable waters of the United States. Where state and/or local controls are sufficient to regulate these structures, including that they do not interfere with navigation, the USACE leaves such regulation to the state or local authority. (See **33 CFR 330** (not carried in this Pilot) for applicable Federal regulations.) Construction permits issued by the Engineers specify the lights and signals required for the safety of navigation.

(389) **Fish havens**, artificial reefs constructed to attract fish, can be established in U.S. coastal waters only as authorized by a USACE permit; the permit specifies the location, extent and depth over these mounds of rubble.

(390)

Naval Observatory

(391) The United States Naval Observatory (USNO) provides a wide range of astronomical data and products and serves as the official source of time for the U.S. Department of Defense and a standard of time for the entire United States. The USNO provides earth orientation products such as the latest 24-hour and 48-hour sets of GPS satellite orbits, the latest determinations and predictions for polar motion and information for GPS users. The USNO also maintains a reference for precise time (USNO Master Clock) and monitors the GPS constellation. For extensive information on the USNO products available, visit: *https://www.cnmoc.usff.navy.mil/usno/* or contact by telephone at 202-762-1467.

(392)

DEPARTMENT OF HEALTH AND HUMAN SERVICES

(393)

Food and Drug Administration (FDA)

(394) Under the provisions of the Control of Communicable Diseases Regulations (**21 CFR 1240**) and Interstate Conveyance Sanitation Regulations (**21 CFR 1250**), vessel companies operating in interstate traffic must obtain potable water for drinking and culinary purposes only at watering points found acceptable to the FDA. Water supplies used in watering point operations must also be inspected to determine compliance with applicable Interstate Quarantine Regulations (**42 CFR 72**). These regulations are based on authority contained in the Public Health Service Act (PL 78–410). Penalties for violation of any regulation prescribed under authority of the Act are provided for under Section 368 (42 USC 271) of the Act.

(395)

Public Health Service

(396) The Public Health Service administers foreign quarantine procedures at U.S. ports of entry.

(397) All vessels arriving in the United States are subject to public health inspection. Vessels subject to routine boarding for quarantine inspection are only those which have had on board during the 15 days preceding the date of expected arrival or during the period since departure (whichever period of time is shorter) the occurrence of any death or ill person among passengers or crew (including those who have disembarked or have been removed). The master of a vessel must report such occurrences immediately by radio to the quarantine station at or nearest the port at which the vessel will arrive.

(398) In addition, the master of a vessel carrying 13 or more passengers must report by radio 24 hours before arrival the number of cases (including zero) of diarrhea in passengers and crew recorded in the ship's medical log during the current cruise. All cases that occur after the 24 hour report must also be reported not less than 4 hours before arrival.

(399) *Ill person* means a person who:

(400) 1. Has a temperature of 100°F (or 38°C) or greater, accompanied by a rash, glandular swelling or jaundice, or which has persisted for more than 48 hours; or

(401) 2. Has diarrhea, defined as the occurrence in a 24 hour period of three or more loose stools or of a greater than normal (for the person) amount of loose stools.

(402) Vessels arriving at ports under control of the United States are subject to sanitary inspection to determine whether measures should be applied to prevent the introduction, transmission or spread of communicable disease.

(403) Specific public health laws, regulations, policies and procedures may be obtained by contacting U.S. Quarantine Stations, U.S. Consulates or the Chief

Program Operations, Division of Quarantine, Centers for Disease Control, Atlanta, GA 30333. (See Appendix A for addresses of U.S. Public Health Service Quarantine Stations.)

(404)

DEPARTMENT OF HOMELAND SECURITY

(405)

Citizenship and Immigration Services

(406) The **U.S. Citizenship and Immigration Service** (USCIS) is the federal agency that oversees lawful immigration to the United States. the Service enhances security and improves the efficiency of national immigration services by exclusively focusing on the administration of benefit applications. No person may enter the United States until they have been inspected by an immigration officer. A list of the offices covered by this Coast Pilot is given in Appendix A.

(407)

U.S. Coast Guard

(408) The U.S. Coast Guard has among its duties the enforcement of the laws of the United States on the high seas and in coastal and inland waters of the U.S. and its possessions; enforcement of navigation and neutrality laws and regulations; establishment and enforcement of navigational regulations upon the Inland Waters of the United States, including the establishment of a demarcation line separating the high seas from waters upon which U.S. navigational rules apply; administration of the Oil Pollution Act of 1990, as amended; establishment and administration of vessel anchorages; approval of bridge locations and clearances over navigable waters; administration of the alteration of obstructive bridges; regulation of drawbridge operations; inspection of vessels of the Merchant Marine; admeasurement of vessels; documentation of vessels; preparation and publication of merchant vessel registers; registration of stack insignia; port security; issuance of Merchant Marine licenses and documents; search and rescue operations; investigation of marine casualties and accidents and suspension and revocation proceedings; destruction of derelicts; operation of aids to navigation; maintenance and issuance of Light Lists and Local Notices to Mariners; and operation of ice-breaking facilities.

(409) Issuance of certificates of registry (more commonly referred to as Certificates of Documentation) with endorsements indicating eligibility of vessels that measure at least 5 net tons to engage in various trades for commercial vessels and certain recreational vessels that are numbered either by the Coast Guard or by a state having an approved numbering system (the latter is the most common) and the administration of the various laws pertaining thereto are functions of the Coast Guard and specifically the National Vessel Documentation Center. Owners of vessels may obtain the necessary information from the National Vessel Documentation Center either by mail to the National Vessel Documentation Center, 792 T.J. Jackson Drive, Falling Waters, WV 25419-9502; via toll free number: 800-799-8362; or via online at: *dco.uscg.mil/Our-Organization/Deputy-for-Operations-Policy-and-Capabilities-DCO-D/National-Vessel-Documentation-Center/.*

(410)

U.S. Customs and Border Protection

(411) The U.S. Customs and Border Protection administers certain laws relating to:

(412) – entry and clearance of vessels and permits for certain vessel movements between points in the United States

(413) – prohibitions against coastwise transportation of passengers and merchandise

(414) – salvage

(415) – dredging and towing by foreign vessels

(416) – certain activities of vessels in the fishing trade

(417) – regular and special tonnage taxes on vessels

(418) – landing and delivery of foreign merchandise (including unlading, appraisement, lighterage, drayage, warehousing and shipment in bond)

(419) – collection of customs duties, including duty on imported pleasure boats and yachts and 50% duty on foreign repairs to American vessels engaged in trade

(420) – customs treatment of sea and ship's stores while in port and the baggage of crewmen and passengers

(421) – illegally imported merchandise

(422) – remission of penalties or forfeiture if customs or navigation laws have been violated.

(423) Customs and Border Protection also cooperates with many other Federal agencies in the enforcement of statutes for which they are responsible for. Customs districts and ports of entry are listed in Appendix A.

(424) The Customs and Border Protection office may issue, without charge, a **cruising license**, normally valid for one year, to a yacht of a foreign country that has a reciprocal agreement with the United States. A foreign yacht holding a cruising license is exempt from having to undergo formal entry and clearance procedures such as filing manifests and obtaining permits to proceed as well as from payment of tonnage tax and entry and clearance fees at all but the first port of entry. These vessels must not engage in trade, violate the laws of the United States or visit a vessel not yet inspected by a Customs Agent and does, within 24 hours of arrival at each port or place in the United States, report the fact of arrival to the nearest customhouse. Countries that have reciprocal agreements granting these privileges to U.S. yachts are:

(425)

Countries with U.S. Cruising License Reciprocity	
Argentina	Honduras
Australia	Ireland
Austria	Italy
Bahama Islands	Jamaica
Belguim	Liberia
Bermuda	Marshall Islands

Canada	Netherlands
Denmark	New Zealand
Finland	Norway
France	Sweden
Germany	Switzerland
Great Britain	Turkey
Greece	

(426) Further information concerning cruising licenses may be obtained from the headquarters port for the customs district in which the license is desired or at *cbp.gov*. U.S. yacht owners planning cruises to foreign ports may contact the nearest customs district headquarters as to customs requirements.

(427)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

(428) The U.S. EPA provides coordinated governmental action to ensure the protection of the environment by abating and controlling pollution on a systematic basis. The ocean dumping permit program of the EPA provides that except when authorized by permit, the dumping of any material into the ocean is prohibited by the "Marine Protection, Research, and Sanctuaries Act of 1972, Public Law 92–532," as amended (33 USC 1401 et seq.).

(429) Permits for the **dumping of dredged material** into waters of the United States, including the territorial sea, and into ocean waters are issued by the U.S. Army Corps of Engineers. Permits for the dumping of fill material into waters of the United States, including the territorial sea, are also issued by the U.S. Army Corps of Engineers. Permits for the dumping of other material in the territorial sea and ocean waters are issued by the EPA.

(430) U.S. Army Corps of Engineers regulations relating to the above are contained in **33 CFR 323** and **324**; EPA regulations are in **40 CFR 220** though **228**. (See Disposal Sites, this chapter.)

(431) Persons or organizations who want to file for an application for an ocean dumping permit should write the EPA Regional Office for the region in which the port of departure is located. (See Appendix A for addresses of regional offices and States in the EPA coastal regions.)

(432) The letter should contain the name and address of the applicant, name and address of person or firm, the name and usual location of the conveyance to be used in the transportation and dumping of the material involved, a physical description where appropriate, and the quantity to be dumped and proposed dumping site.

(433) Everyone who writes EPA will be sent information about a final application for a permit as soon as possible. This final application is expected to include questions about the description of the process or activity giving rise to the production of the dumping material, information on past activities of applicant or others with respect to the disposal of the type of material involved, and a description about available alternative means of disposal of the material with explanations about why an alternative is thought by the applicant to be inappropriate.

(433.001)

FEDERAL AVIATION ADMINISTRATION (FAA)

(433.002)

Obstruction Evaluation/Airport Airspace Analysis (OE/AAA)

(433.003) The **Federal Aviation Administration (FAA)** has established comprehensive policies and procedures to evaluate potential hazards to air navigation posed by construction or alteration projects. These assessments are critical to maintaining the safety and efficiency of the National Airspace System (NAS). To support this effort, the FAA's Air Traffic Organization (ATO) developed the Obstruction Evaluation and Airport Airspace Analysis (OE/AAA) System, a web-based platform that enables public submission and FAA processing of proposed construction or alteration notices. The collection and processing of this information are authorized under **49 U.S.C. § 44718** and **14 CFR Part 77**, which pertains to the notice and study of proposed construction or alteration near navigable airspace.

(433.004) Maritime vessels are required to comply with the provisions of **14 CFR Part 77 Subpart B**. In consideration of procedures and for potential conflict between aircraft and maritime vessels, waterways of the United States with vessel height greater than 200 feet (Air Draft) above the waterline, must be assessed by the FAA. Due to airport, airspace, and waterway proximity, lesser heights may also require assessment. Owners, operators, agents, voyage planners, and masters may consult the **Obstruction Evaluation/Airport Airspace Analysis (OE/AAA) Pre-Screening Tool** at *https://oeaaa.faa.gov/oeaaa/oe3a/main/#/home* to assess applicability for lesser (Air Draft) heights.

(434)

FEDERAL COMMUNICATIONS COMMISSION (FCC)

(435) The Federal Communications Commission controls non-government radio communications in the United States, Guam, Puerto Rico and the Virgin Islands. Commission inspectors have authority to board ships to determine whether their radio stations comply with international treaties, Federal laws and Commission regulations. The commission has field offices in the principal U.S. ports. (See Appendix A for addresses.) Information concerning ship radio regulations and service documents may be obtained from the Federal Communications Commission, Washington, DC 20554, or from any of the field offices.

(436)

Measurements and Equivalencies

nautical mile — 1,852 meters / 6,076.12 feet
statute mile — 5,280 feet / 1,609.3 meters / 1.6093 kilometers
cable — 0.1 nautical mile (CN) / 720 feet (US)
fathom — 6 feet / 1.8288 meters
foot — 0.3048 meter
inch — 2.54 centimeters
pound (avoirdupois) — 453.59 gram
kilometer — 1,000 meters
knot — 1.6877 feet per second / 0.5144 meters per second
miles/hour (statute) — 1.466 feet per second / 0.44704 meters per second
acre — 43,560 square feet / 4,046.82 square meters
gram — 0.0022046 pound (avoirdupois) / 0.035274 ounce
meter — 39.37 inches / 3.281 feet / 1.0936 yards
short ton — 2,000 pounds
long ton — 2,240 pounds
metric ton — 2,204.6 pounds
kilogram — 2.2 pounds
liter — 1.0567 quarts
barrel (petroleum) — 42 gallons (US)

Conversion Factors

Linear

inches — muiltiply by 25.40 — millimeters
inches — multiply by 2.540 — centimeters
centimeters — multiply by 0.032808 — feet
feet — multiply by 30.48 — centimeters
feet — multiply by 0.3048 — meters
feet — multiply by 0.00016458 — nautical miles
yard — multiply by 0.9144 — meters
meters — multiply by 3.2808 — feet
meters — multiply by 1.094 — yards
meters — multiply by 0.0005399 — nautical miles
statute miles — multiply by 0.86897 — nautical miles
statute miles — multiply by 1.6093 — kilometers
statute miles — multiply by 1,609.3 — meters
nautical miles — multiply by 1.151 — statute miles

Area

acres — multiply by 4,046.9 — square meters
acres — multiply by 43,560 — square feet
acres — multiply by 0.404685 — hectare
hectare — multiply by 2.471054 — acres
hectare — multiply by 10,000 — square meters
hectare — multiply by 1.07639×10^5 — square feet
square feet — multiply by 0.0929 — square meters
square feet — multiply by 0.00002296 — acres
square meters — multiply by 10.764 — square feet
square meters — multiply by 0.0002471 — acres

Depths

fathoms — multiply by 1.8288 — meters
feet — multiply by 0.3048 — meters
meters — multiply by 0.54681 — fathoms
meters — multiply by 3.2808 — feet

Rate

feet/second — multiply by 0.5925 — knots
feet/second — multiply by 0.6818 — miles/hour
feet/second — multiply by 30.48 — centimeters/second
statute miles/hour — multiply by 0.8689 — knots
statute miles/hour — multiply by 1.467 — feet/second
statute miles/hour — multiply by 0.447 — meters/second
knots — multiply by 1.151 — miles/hour
knots — multiply by 0.5144 — meters/second
knots — multiply by 1.6878 — feet/second
centimeters/second — multiply by 0.01944 — miles/hour
centimeters/second — multiply by 0.02237 — miles/hour
centimeters/second — multiply by 0.032808 — feet/second

Mass

grams — multiply by 0.035275 — ounces
grams — multiply by 0.002205 — pounds
ounces — multiply by 28.349 — grams
pounds — multiply by 0.45359 — kilograms
short tons — multiply by 2,000 — pounds
short tons — multiply by 0.89286 — long tons
short tons — multiply by 0.9072 — metric tons
long tons — multiply by 2,240 — pounds
long tons — multiply by 1.12 — short tons
long tons — multiply by 1.016 — metric tons
metric tons — multiply by 1,000 — kilograms
metric tons — multiply by 0.9842 — long tons
metric tons — multiply by 1.1023 — short tons
metric tons — multiply by 2,204.6 — pounds

Volume

barrels (petroleum) — multiply by 42 — gallons (US)
barrels (petroleum) — multiply by 158.99 — liters
barrels (liquid, US) — multiply by 31.5 — gallons (US)
barrels (liquid, US) — multiply by 26.229 — gallons (British)
barrels (liquid, US) — multiply by 119.24 — liters
gallons (US) — multiply by 0.02381 — barrels (petroleum)
gallons (US) — multiply by 3.7854 — liters
liters — multiply by 0.26417 — gallons (US)

(437)

Navigation Regulations

(1) This chapter contains extracts from **Code of Federal Regulations (CFR)** that are of importance to mariners in the area covered by this Coast Pilot. Sections of little value to the mariner are sometimes omitted. Omitted sections are signified by the following [...]

(2) Extracts from the following titles are contained in this chapter.

(3)

Title 15: Commerce and Foreign Trade
- Part 922—National Marine Sanctuary Program Regulations

Title 33: Navigation and Navigable Waters
- Part 26—Vessel Bridge-to-Bridge Radiotelephone Regulations
- Part 80—COLREGS Demarcation Lines
- Part 81—72 COLREGS: Implementing Rules
- Part 82—72 COLREGS: Interpretive Rules
- Part 88—Annex V: Pilot Rules
- Part 89—Inland Navigation Rules: Implementing Rules
- Part 90—Inland Rules: Interpretive Rules
- Part 110—Anchorage Regulations
- Part 117—Drawbridge Operation Regulations
- Part 150—Deepwater Ports: Operations
- Part 157—Rules for the Protection of the Marine Environment Relating to Tank Vessels Carrying Oil in Bulk
- Part 160—Ports and Waterways Safety-General
- Part 164—Navigation Safety Regulations (in part)
- Part 165—Regulated Navigation Areas and Limited Access Areas
- Part 167—Offshore Traffic Separation Schemes
- Part 169—Ship Reporting Systems
- Part 207—Navigation Regulations
- Part 334—Danger Zones and Restricted Area Regulations

Title 40: Protection of Environment
- Part 140—Marine Sanitation Device Standard

Title 46: Shipping
- Part 15—Manning Requirements

Title 50: Wildlife and Fisheries
- Part 222—General Endangered and Threatened Marine Species
- Part 224—Endangered Marine and Anadromous Species
- Part 226—Designated Critical Habitat

(4) These regulations can only be amended by the enforcing agency or other authority cited in the regulations. Accordingly, requests for changes to these regulations should be directed to the appropriate agency for action. In those regulations where the enforcing agency is not cited or is unclear, recommendations for changes should be directed to the following Federal agencies for action:

(5) **National Oceanic and Atmospheric Administration**—15 CFR 922; 50 CFR 222, 224 and 226

(6) **United States Coast Guard**—33 CFR 26, 80, 81, 82, 88, 89, 110, 117, 150, 157, 160, 164, 165, 167 and 169; 46 CFR 15

(7) **United States Army Corps of Engineers**—33 CFR 207 and 334

(8) **Environmental Protection Agency**—40 CFR 140

(9)

TITLE 15–COMMERCE AND FOREIGN TRADE

(10)

Part 922–NationalMarine Sanctuary Program Regulations

(11)

Subpart A-General

(12)

§ 922.1 Applicability of regulations.

(13) Unless noted otherwise, the regulations in subparts A, D, and E of this part apply to all National Marine Sanctuaries and related site-specific regulations set forth in this part. Subparts B and C of this part apply to the sanctuary nomination process and to the designation of future Sanctuaries.

(14)

§922.2 Mission, goals, and special policies.

(15) (a) In accordance with the standards set forth in title III of the Marine Protection, Research, and Sanctuaries Act of 1972, as amended, also known as the National Marine Sanctuaries Act (Act) the mission of the National Marine Sanctuary program (Program) is to identify, designate and manage areas of the marine environment of special national, and in some cases international, significance due to their conservation, recreational, ecological, historical, research, educational, or aesthetic qualities.

(16) (b) The goals of the Program are to carry out the mission to:

(17) (1) Identify and designate as National Marine Sanctuaries areas of the marine environment which are of special national significance;

(18) (2) Provide authority for comprehensive and coordinated conservation and management of these marine areas, and activities affecting them, in a manner which complements existing regulatory authorities;

(19) (3) Support, promote, and coordinate scientific research on, and monitoring of, the resources of these

marine areas, especially long-term monitoring and research of these areas:

(20) (4) Enhance public awareness, understanding, appreciation, and wise use of the marine environment;

(21) (5) Facilitate to the extent compatible with the primary objective of resource protection, all public and private uses of the resources of these marine areas not prohibited pursuant to other authorities;

(22) (6) Develop and implement coordinated plans for the protection and management of these areas with appropriate Federal agencies, State and local governments, Native American tribes and organizations, international organizations, and other public and private interests concerned with the continuing health and resilience of these marine areas;

(23) (7) Create models of, and incentives for, ways to conserve and manage these areas;

(24) (8) Cooperate with global programs encouraging conservation of marine resources; and

(25) (9) Maintain, restore, and enhance living resources by providing places for species that depend upon these marine areas to survive and propagate.

(26) (c) To the extent consistent with the policies set forth in the Act, in carrying out the Program's mission and goals:

(27) (1) Particular attention will be given to the establishment and management of marine areas as National Marine Sanctuaries for the protection of the area's natural resource and ecosystem values; particularly for ecologically or economically important or threatened species or species assemblages, and for offshore areas where there are no existing special area protection mechanisms;

(28) (2) The size of a National Marine Sanctuary, while highly dependent on the nature of the site's resources, will be no larger than necessary to ensure effective management;

(29) (d) Management efforts will be coordinated to the extent practicable with other countries managing marine protected areas;

(30) (e) Program regulations, policies, standards, guidelines, and procedures under the Act concerning the identification, evaluation, registration, and treatment of historical resources shall be consistent, to the extent practicable, with the declared national policy for the protection and preservation of these resources as stated in the National Historic Preservation Act of 1966, 16 U.S.C. 470 et seq., the Archeological and Historical Preservation Act of 1974, 16 U.S.C. 469 et seq., and the Archeological Resources Protection Act of 1979 (ARPA), 16 U.S.C. 470aa et seq. The same degree of regulatory protection and preservation planning policy extended to historical resources on land shall be extended, to the extent practicable, to historical resources in the marine environment within the boundaries of designated National Marine Sanctuaries. The management of historical resources under the authority of the Act shall be consistent, to the extent practicable, with the Federal archeological program by consulting the Uniform Regulations, ARPA (43 CFR part 7) and other relevant Federal regulations. The Secretary of the Interior's Standards and Guidelines for Archeology may also be consulted for guidance. These guidelines are available from the Office of Ocean and Coastal Management at (301) 713-3125.

(31)

§922.3 Definitions.

(32) *Act* means title III of the Marine Protection, Research, and Sanctuaries Act of 1972, as amended, 16 U.S.C. 1431 et seq., also known as the National Marine Sanctuaries Act.

(33) *Assistant Administrator* means the Assistant Administrator for Ocean Services and Coastal Zone Management, National Oceanic and Atmospheric Administration (NOAA), or designee.

(34) *Benthic community* means the assemblage of organisms, substrate, and structural formations found at or near the bottom that is periodically or permanently covered by water.

(35) *Commercial fishing* means any activity that results in the sale or trade for intended profit of fish, shellfish, algae, or corals.

(36) *Conventional hook and line gear* means any fishing apparatus operated aboard a vessel and composed of a single line terminated by a combination of sinkers and hooks or lures and spooled upon a reel that may be hand or electrically operated, hand-held or mounted. This term does not include bottom longlines.

(37) *Cultural resources* means any historical or cultural feature, including archaeological sites, historic structures, shipwrecks, and artifacts.

(38) *Director* means, except where otherwise specified, the Director of the Office of Ocean and Coastal Resource Management, NOAA, or designee.

(39) *Exclusive economic zone* means the exclusive economic zone as defined in the Magnuson Fishery Conservation and Management Act, 16 U.S. 1801 et seq.

(40) *Fish wastes* means waste materials resulting from commercial fish processing operations.

(41) *Historical resource* means a resource possessing historical, cultural, archaeological or paleontological significance, including sites, structures, districts, and objects significantly associated with or representative of earlier people, cultures, maritime heritage, and human activities and events. Historical resources include "submerged cultural resources," and also include "historical properties," as defined in the National Historic Preservation Act, as amended, and its implementing regulations, as amended.

(42) *Indian tribe* means any American Indian tribe, band, group, or community recognized as such by the Secretary of the Interior.

(43) *Injure* means to change adversely, either in the long or short term, a chemical, biological or physical attribute of, or the viability of. This includes, but is not limited to, to cause the loss of or destroy.

(44) *Inventory* means a list of nominated areas selected by the Director as qualifying for future consideration of designation as a national marine sanctuary.

(45) *Lightering* means at-sea transfer of petroleum-based products, materials or other matter from vessel to vessel.

(46) *Marine* means those areas of coastal and ocean waters, the Great Lakes and their connecting waters, and submerged lands over which the United States exercises jurisdiction, including the exclusive economic zone, consistent with international law.

(47) *Mineral* means clay, stone, sand, gravel, metalliferous ore, non-metalliferous ore, or any other solid material or other matter of commercial value.

(48) *National historic landmark* means a district, site, building, structure or object designated as such by the Secretary of the Interior under the National Historic Landmarks Program (36 CFR part 65).

(49) *National Marine Sanctuary* means an area of the marine environment of special national significance due to its resource or human-use values, which is designated as such to ensure its conservation and management.

(50) *Person* means any private individual, partnership, corporation or other entity; or any officer, employee, agent, department, agency or instrumentality of the Federal Government, of any State or local unit of government, or of any foreign government.

(51) *Regional Fishery Management Council* means any fishery council established under section 302 of the Magnuson Fishery Conservation and Management Act, 16 U.S.C. 1801 et seq.

(52) *Sanctuary quality* means any of those ambient conditions, physical-chemical characteristics and natural processes, the maintenance of which is essential to the ecological health of the Sanctuary, including, but not limited to, water quality, sediment quality and air quality.

(53) *Sanctuary resource* means any living or non-living resource of a National Marine Sanctuary that contributes to the conservation, recreational, ecological, historical, research, educational, or aesthetic value of the Sanctuary, including, but not limited to, the substratum of the area of the Sanctuary, other submerged features and the surrounding seabed, carbonate rock, corals and other bottom formations, coralline algae and other marine plants and algae, marine invertebrates, brineseep biota, phytoplankton, zooplankton, fish, seabirds, sea turtles and other marine reptiles, marine mammals and historical resources. For Thunder Bay National Marine Sanctuary and Underwater Preserve, Sanctuary resource means an underwater cultural resource as defined at **§922.191**. For Mallows Bay-Potomac River National Marine Sanctuary, Sanctuary resource is defined at **§922.201(a)**.

(54) *Secretary* means the Secretary of the United States Department of Commerce, or designee.

(55) *Shunt* means to discharge expended drilling cuttings and fluids near the ocean seafloor.

(56) *State* means each of the several States, the District of Columbia, the Commonwealth of Puerto Rico, the Commonwealth of the Northern Mariana Islands, American Samoa, the United States Virgin Islands, Guam, and any other commonwealth, territory, or possession of the United States.

(57) *Subsistence use* means the customary and traditional use by rural residents of areas near or in the marine environment for direct personal or family consumption as food, shelter, fuel, clothing, tools, or transportation; for the making and selling of handicraft articles; and for barter, if for food or non-edible items other than money, if the exchange is of a limited and non-commercial nature.

(58) *Take or taking* means:

(59) (1) For any marine mammal, sea turtle, or seabird listed as either endangered or threatened pursuant to the Endangered Species Act, to harass, harm, pursue, hunt, shoot, wound, kill, trap, capture, collect or injure, or to attempt to engage in any such conduct;

(60) (2) For any other marine mammal, sea turtle, or seabird, to harass, hunt, capture, kill, collect or injure, or to attempt to engage in any such conduct.

(61) For the purpose of both (1) and (2) of this definition, this includes, but is not limited to, to collect any dead or injured marine mammal, sea turtle or seabird, or any part thereof; to restrain or detain any marine mammal, sea turtle or seabird, or any part thereof, no matter how temporarily; to tag any sea turtle, marine mammal or seabird; to operate a vessel or aircraft or to do any other act that results in the disturbance or molestation of any marine mammal, sea turtle or seabird.

(62) *Tropical fish* means fish or minimal sport and food value, usually brightly colored, often used for aquaria purposes and which lives in a direct relationship with live bottom communities.

(63) *Vessel* means a watercraft of any description capable of being used as a means of transportation in/on the waters of the Sanctuary.

(64)

§922.4 Effect of National Marine Sanctuary designation.

(65) The designation of a National Marine Sanctuary, and the regulations implementing it, are binding on any person subject to the jurisdiction of the United States. Designation does not constitute any claim to territorial jurisdiction on the part of the United States for designated sites beyond the U.S. territorial sea, and the regulations implementing the designation shall be applied in accordance with generally recognized principles of international law, and in accordance with treaties, conventions, and other agreements to which the United States is a party. No regulation shall apply to a person who is not a citizen, national, or resident alien of the United States, unless in accordance with:

(66) (a) Generally recognized principles of international law;

(67) (b) An agreement between the United States and the foreign state of which the person is a citizen; or

(68) (c) An agreement between the United States and the flag state of the foreign vessel, if the person is a crew member of the vessel.

(69)

Subpart D–Management Plan Development and Implementation

(70)

§922.30 General.

(71) (a) The Secretary shall implement each management plan, and applicable regulations, including carrying out surveillance and enforcement activities and conducting such research, monitoring, evaluation, and education programs as are necessary and reasonable to carry out the purposes and policies of the Act.

(72) (b) Consistent with Sanctuary management plans, the Secretary shall develop and implement site-specific contingency and emergency-response plans designed to protect Sanctuary resources. The plans shall contain alert procedures and actions to be taken in the event of an emergency such as a shipwreck or an oil spill.

(73)

§922.31 Promotion and coordination of Sanctuary use.

(74) The Secretary shall take such action as is necessary and reasonable to promote and coordinate the use of National Marine Sanctuaries for research, monitoring, and education purposes. Such action may include consulting with Federal agencies, or other persons to promote use of one or more Sanctuaries for research, monitoring and education, including coordination with the National Estuarine Research Reserve System.

(75)

Subpart E–Regulations of General Applicability

(76)

§922.40 Purpose.

(77) The purpose of the regulations in this subpart and in the site-specific subparts in this part is to implement the designations of the National Marine Sanctuaries by regulating activities affecting them, consistent with their respective terms of designation in order to protect, preserve and manage and thereby ensure the health, integrity and continued availability of the conservation, ecological, recreational, research, educational, historical and aesthetic resources and qualities of these areas. Additional purposes of the regulations implementing the designation of the Florida Keys and Hawaiian Islands Humpback Whale National Marine Sanctuaries are found at **§§922.160** and **922.180**, respectively.

(78)

§922.41 Boundaries.

(79) The boundary for each of the National Marine Sanctuaries is set forth in the site-specific regulations covered by this part.

(80)

§922.42 Allowed activities.

(81) All activities (e.g., fishing, boating, diving, research, education) may be conducted unless prohibited or otherwise regulated in the site-specific regulations covered by this part, subject to any emergency regulations promulgated under this part, subject to all prohibitions, regulations, restrictions, and conditions validly imposed by any Federal, State, or local authority of competent jurisdiction, including but not limited to, Federal, Tribal, and State fishery management authorities, and subject to the provisions of section 312 of the National Marine Sanctuaries Act (NMSA) (16 U.S.C. 1431 et seq.). The Assistant Administrator may only directly regulate fishing activities pursuant to the procedure set forth in section 304(a)(5) of the NMSA.

(82)

§922.43 Prohibited or otherwise regulated activities.

(83) The site-specific regulations applicable to the activities specified therein are set forth in the subparts covered by this part.

(84)

§922.44 Emergency regulations.

(85) (a) Where necessary to prevent or minimize the destruction of, loss of, or injury to a Sanctuary resource or quality, or minimize the imminent risk of such destruction, loss, or injury, any and all such activities are subject to immediate temporary regulation, including prohibition.

(86) (b) The provisions of this section do not apply to the following national marine sanctuaries with site-specific regulations that establish procedures for issuing emergency regulations:

(87) (1) Cordell Bank National Marine Sanctuary, §922.112(e).

(88) (2) Florida Keys National Marine Sanctuary, §922.165.

(89) (3) Hawaiian Islands Humpback Whale National Marine Sanctuary, §922.185.

(90) (4) Thunder Bay National Marine Sanctuary, §922.196.

(91) (5) Mallows Bay-Potomac River National Marine Sanctuary, §922.204.

(92) (6) [Reserved]

(93)

§922.45 Penalties.

(94) (a) Each violation of the NMSA or FKNMSPA, any regulation in this part, or any permit issued pursuant thereto, is subject to a civil penalty of not more than $100,000. Each day of a continuing violation constitutes a separate violation.

(95) (b) Regulations setting forth the procedures governing administrative proceedings for assessment of civil penalties, permit sanctions, and denials for enforcement reasons, issuance and use of written warnings, and release or forfeiture of seized property appear at 15 CFR part 904.

(96)

§922.46 Response costs and damages.

(97) Under section 312 of the Act, any person who destroys, causes the loss of, or injures any Sanctuary resource is liable to the United States for response costs and damages resulting from such destruction, loss or injury, and any vessel used to destroy, cause the loss of, or injure any Sanctuary resource is liable in rem to the United States for response costs and damages resulting from such destruction, loss or injury.

(98)

§922.47 Pre-existing authorizations or rights and certifications of pre-existing authorizations or rights.

(99) (a) Leases, permits, licenses, or rights of subsistence use or access in existence on the date of designation of any National Marine Sanctuary shall not be terminated by the Director. The Director may, however, regulate the exercise of such leases, permits, licenses, or rights consistent with the purposes for which the Sanctuary was designated.

(100) (b) The prohibitions listed in subparts F through P and R through T of this part do not apply to any activity authorized by a valid lease, permit, license, approval or other authorization in existence on the effective date of Sanctuary designation, or in the case of the Florida Keys National Marine Sanctuary the effective date of the regulations in this subpart P, and issued by any Federal, State or local authority of competent jurisdiction, or by any valid right of subsistence use or access in existence on the effective date of Sanctuary designation, or in the case of the Florida Keys National Marine Sanctuary the effective date of the regulations in this part P, provided that the holder of such authorization or right complies with certification procedures and criteria promulgated at the time of Sanctuary designation, or in the case of the Florida Keys National Marine Sanctuary the effective date of the regulations in subpart P, and with any terms and conditions on the exercise of such authorization or right imposed by the Director as a condition of certification as the Director deems necessary to achieve the purpose for which the Sanctuary was designated.

(101)

§922.48 National Marine Sanctuary permits-application procedures and issuance criteria.

(102) (a) A person may conduct an activity prohibited by subparts F through O and S and T of this part, if conducted in accordance with the scope, purpose, terms and conditions of a permit issued under this section and subparts F through O and S and T, as appropriate. For the Florida Keys National Marine Sanctuary, a person may conduct an activity prohibited by subpart P of this part if conducted in accordance with the scope, purpose, terms and conditions of a permit issued under §922.166. For the Thunder Bay National Marine Sanctuary and Underwater Preserve, a person may conduct an activity prohibited by subpart R of this part in accordance with the scope, purpose, terms and conditions of a permit issued under §922.195.

(103) (b) Applications for permits to conduct activities otherwise prohibited by subparts F through O and S and T of this part, should be addressed to the Director and sent to the address specified in subparts F through O of this part, or subparts R through T of this part, as appropriate. An application must include:

(104) (1) A detailed description of the proposed activity including a timetable for completion;

(105) (2) The equipment, personnel and methodology to be employed;

(106) (3) The qualifications and experience of all personnel;

(107) (4) The potential effects of the activity, if any, on Sanctuary resources and qualities; and

(108) (5) Copies of all other required licenses, permits, approvals or other authorizations.

(109) (c) Upon receipt of an application, the Director may request such additional information from the applicant as he or she deems necessary to act on the application and may seek the views of any persons or entity, within or outside the Federal government, and may hold a public hearing, as deemed appropriate.

(110) (d) The Director, at his or her discretion, may issue a permit, subject to such terms and conditions as he or she deems appropriate, to conduct a prohibited activity, in accordance with the criteria found in subparts F through O of this part, or subparts R through T of this part, as appropriate. The Director shall further impose, at a minimum, the conditions set forth in the relevant subpart.

(111) (e) A permit granted pursuant to this section is nontransferable.

(112) (f) The Director may amend, suspend, or revoke a permit issued pursuant to this section for good cause. The Director may deny a permit application pursuant to this section, in whole or in part, if it is determined that the permittee or applicant has acted in violation of the terms and conditions of a permit or of the regulations set forth in this section or subparts F through O of this part, or subparts R through T of this part or for other good cause. Any such action shall be communicated in writing to the permittee or applicant by certified mail and shall set forth the reason(s) for the action taken. Procedures governing permit sanctions and denials for enforcement reasons are set forth in subpart D of 15 CFR part 904.

(113)

§922.49 Notification and review of applications for leases, licenses, permits, approvals or other authorizations to conduct a prohibited activity.

(114) (a) A person may conduct an activity prohibited by subparts L through P of this part, or subparts R through T of this part, if such activity is specifically authorized by any valid Federal, State, or local lease, permit, license, approval, or other authorization issued after the effective date of Sanctuary designation, or in the case of the Florida Keys National Marine Sanctuary after the effective date of the regulations in subpart P, provided that:

(115) (1) The applicant notifies the Director, in writing, of the application for such authorization (and of any application for an amendment, renewal, or extension of such authorization) within fifteen (15) days of the date of filing of the application or of the effective date of Sanctuary designation, or in the case of the Florida Keys National Marine Sanctuary the effective date of the regulations in subpart P, whichever is later;

(116) (2) The applicant complies with the other provisions of this section;

(117) (3) The Director notifies the applicant and authorizing agency that he or she does not object to issuance of the authorization (or amendment, renewal, or extension); and

(118) (4) The applicant complies with any terms and conditions the Director deems reasonably necessary to protect Sanctuary resources and qualities.

(119) (b) Any potential applicant for an authorization described in paragraph (a) of this section may request the Director to issue a finding as to whether the activity for which an application is intended to be made is prohibited by subparts L through P of this part, or subparts R through T of this part, as appropriate.

(120) (c) Notification of filings of applications should be sent to the Director, Office of National Marine Sanctuaries at the address specified in subparts L through P of this part, or subparts R through T of this part, as appropriate. A copy of the application must accompany the notification.

(121) (d) The Director may request additional information from the applicant as he or she deems reasonably necessary to determine whether to object to issuance of an authorization described in paragraph (a) of this section, or what terms and conditions are reasonably necessary to protect Sanctuary resources and qualities. The information requested must be received by the Director within 45 days of the postmark date of the request. The Director may seek the views of any persons on the application.

(122) (e) The Director shall notify, in writing, the agency to which application has been made of his or her pending review of the application and possible objection to issuance. Upon completion of review of the application and information received with respect thereto, the Director shall notify both the agency and applicant, in writing, whether he or she has an objection to issuance and what terms and conditions he or she deems reasonably necessary to protect Sanctuary resources and qualities, and reasons therefor.

(123) (f) The Director may amend the terms and conditions deemed reasonably necessary to protect Sanctuary resources and qualities whenever additional information becomes available justifying such an amendment.

(124) (g) Any time limit prescribed in or established under this section may be extended by the Director for good cause.

(125) (h) The applicant may appeal any objection by, or terms or conditions imposed by, the Director to the Assistant Administrator or designee in accordance with the provisions of §922.50.

(126)

§922.50 Appeals of administrative action.

(127) (a)(1) Except for permit actions taken for enforcement reasons (see subpart D of 15 CFR part 904 for applicable procedures), an applicant for, or a holder of, a National Marine Sanctuary permit; an applicant for, or a holder of, a Special Use permit issued pursuant to section 310 of the Act; a person requesting certification of an existing lease, permit, license or right of subsistence use or access under § 922.47; or, for those Sanctuaries described in subparts L through P and R through T of this part, an applicant for a lease, permit, license or other authorization issued by any Federal, State, or local authority of competent jurisdiction (hereinafter appellant) may appeal to the Assistant Administrator:

(128) (i) The granting, denial, conditioning, amendment, suspension or revocation by the Director of a National Marine Sanctuary or Special Use permit;

(129) (ii) The conditioning, amendment, suspension or revocation of a certification under §922.47; or

(130) (iii) For those Sanctuaries described in subparts L through P and R through T of this part, the objection to issuance of or the imposition of terms and conditions on a lease, permit, license or other authorization issued by any Federal, State, or local authority of competent jurisdiction.

(131) (2) For those National Marine Sanctuaries described in subparts F through K and S and T of this part, any interested person may also appeal the same actions described in paragraphs (a)(1)(i) and (ii) of this section. For appeals arising from actions taken with respect to these National Marine Sanctuaries, the term ''appellant'' includes any such interested persons.

(132) (b) An appeal under paragraph (a) of this section must be in writing, state the action(s) by the Director appealed and the reason(s) for the appeal, and be received within 30 days of receipt of notice of the action by the Director. Appeals should be addressed to the Assistant Administrator for Ocean Services and Coastal Zone Management, NOAA 1305 East-West Highway, 13th Floor, Silver Spring, MD 20910.

(133) (c)(1) The Assistant Administrator may request the appellant to submit such information as the Assistant Administrator deems necessary in order for him or her to decide the appeal. The information requested must be received by the Assistant Administrator within 45 days of the postmark date of the request. The Assistant Administrator may seek the views of any other persons. For the Monitor National Marine Sanctuary, if the appellant has requested a hearing, the Assistant Administrator shall grant an informal hearing. For all other National Marine Sanctuaries, the Assistant Administrator may determine whether to hold an informal hearing on the appeal. If the Assistant Administrator determines that an informal hearing should be held, the Assistant Administrator may designate an officer before whom the hearing shall be held.

(134) (2) The hearing officer shall give notice in the **Federal Register** of the time, place and subject matter of the hearing. The appellant and the Director may appear personally or by counsel at the hearing and submit such material and present such arguments as deemed appropriate by the hearing officer. Within 60 days after the record for the hearing closes, the hearing officer shall recommend a decision in writing to the Assistant Administrator.

(135) (d) The Assistant Administrator shall decide the appeal using the same regulatory criteria as for the initial decision and shall base the appeal decision on the record before the Director and any information submitted regarding the appeal, and, if a hearing has been held, on the record before the hearing officer and the hearing officer's recommended decision. The Assistant Administrator shall notify the appellant of the final decision and the reason(s) therefore in writing. The Assistant Administrator's decision shall constitute final agency action for the purpose of the Administrative Procedure Act.

(136) (e) Any time limit prescribed in or established under this section other than the 30-day limit for filing an appeal may be extended by the Assistant Administrator or hearing office for good cause.

(137)

Subpart N–Stellwagen Bank National Marine Sanctuary

(138)

§922.140 Boundary

(139) (a) The Stellwagen Bank National Marine Sanctuary (Sanctuary) consists of an area of approximately 638 square nautical miles (NM) of Federal marine waters and the submerged lands thereunder, over and around Stellwagen Bank and other submerged features off the coast of Massachusetts. The boundary encompasses the entirety of Stellwagen Bank; Tillies Bank, to the northeast of Stellwagen Bank; and portions of Jefferys Ledge, to the north of Stellwagen Bank.

(140) (b) The Sanctuary boundary is identified by the following coordinates, indicating the most northeast, southeast, southwest, west-northwest, and north-northwest points:

(141) 42°45'59.83"N., 70°13'01.77"W. (NE);

(142) 42°05'35.51"N., 70°02'08.14"W. (SE);

(143) 42°07'44.89"N., 70°28'15.44"W. (SW);

(144) 42°32'53.52"N., 70°35'52.38"W. (WNW); and

(145) 42°39'04.08"N., 70°30'11.29"W. (NNW).

(146) The western border is formed by a straight line connecting the most southwest and the west-northwest points of the Sanctuary. At the most west-northwest point, the Sanctuary border follows a line contiguous with the three-mile jurisdictional boundary of Massachusetts to the most north-northwest point. From this point, the northern border is formed by a straight line connecting the most north-northwest point and the most northeast point. The eastern border is formed by a straight line connecting the most northeast and the most southeast points of the Sanctuary. The southern border follows a straight line between the most southwest point and a point located at 42°06'54.57"N., 70°16'42.7"W. From that point, the southern border then continues in a west-to-east direction along a line contiguous with the three-mile jurisdictional boundary of Massachusetts until reaching the most southeast point of the Sanctuary. The boundary coordinates are listed in appendix A to this subpart.

(147)

§922.141 Definitions.

(148) In addition to those definitions found at §922.3, the following definitions apply to this subpart:

(149) *Industrial* material means mineral, as defined in §922.3.

(150) *Traditional fishing* means those commercial or recreational fishing methods which have been conducted in the past within the Sanctuary.

(151)

§922.142 Prohibited or otherwise regulated activities.

(152) (a) Except as specified in paragraphs (b) through (f) of this section, the following activities are prohibited and thus are unlawful for any person to conduct or cause to be conducted:

(153) (1)(i) Discharging or depositing, from within the boundary of the Sanctuary, any material or other matter except:

(154) (A) Fish, fish parts, chumming materials or bait used in or resulting from traditional fishing operations in the Sanctuary;

(155) (B) Biodegradable effluent incidental to vessel use and generated by marine sanitation devices approved in accordance with Section 312 of the Federal Water Pollution Control Act, as amended, (FWPCA), 33 U.S.C. 1322 *et seq.;*

(156) (C) Water generated by routine vessel operations (e.g. cooling water, deck wash down and graywater as defined by Section 312 of the FWPCA) excluding oily wastes from bilge pumping; or

(157) (D) Engine exhaust;

(158) (ii) Discharging or depositing, from beyond the boundary of the Sanctuary, any material or other matter, except those listed in paragraphs (a)(1)(i)(A) through (D) of this section, that subsequently enters the Sanctuary and injures as Sanctuary resource or quality.

(159) (2) Exploring for, developing or producing industrial materials within the Sanctuary.

(160) (3) Drilling into, dredging or otherwise altering the seabed of the Sanctuary; or constructing, placing or abandoning any structure, material or other matter on the seabed of the Sanctuary, except as incidental result of:

(161) (i) Anchoring vessels;

(162) (ii) Traditional fishing operations; or

(163) (iii) Installation of navigation aids.

(164) (4) Moving, removing or injuring, or attempting to move, remove or injure, a Sanctuary historical resource. This prohibition does not apply to moving, removing or injury resulting incidentally from traditional fishing operations.

(165) (5) Taking any marine reptile, marine mammal or seabird in or above the Sanctuary, except as permitted by the Marine Mammal Protection Act, as amended, (MMPA), 16 U.S.C. 1361 *et seq.,* the Endangered Species Act, as amended, (ESA), 16 U.S.C. 1531 *et seq.,* and the Migratory Bird Treaty Act, as amended, (MBTA), 16 U.S.C. 703 *et seq.*

(166) (6) Lightering in the Sanctuary.

(167) (7) Possessing within the Sanctuary (regardless of where taken, moved or removed from), except as necessary for valid law enforcement purposes, any historical resource, or any marine mammal, marine reptile or seabird taken in violation of the MMPA, ESA or MBTA.

(168) (8) Interfering with, obstructing, delaying or preventing an investigation, search, seizure or disposition of seized property in connection with enforcement of the Act or any regulation or permit issued under the Act.

(169) (b) The prohibitions in paragraphs (a)(1), (3) through (8) of this section do not apply to any activity necessary to respond to an emergency threatening life, property or the environment.

(170) (c)(1)(i) All Department of Defense military activities shall be carried out in a manner that avoids to the maximum extent practicable any adverse impacts on Sanctuary resources and qualities.

(171) (ii) Department of Defense military activities may be exempted from the prohibitions in paragraphs (a)(1) and (3) through (7) of this section by the Director after consultation between the Director and the Department of Defense.

(172) (iii) If it is determined that an activity may be carried out, such activity shall be carried out in a manner that avoids to the maximum extent practicable any adverse impact on Sanctuary resources and qualities. Civil engineering and other civil works projects conducted by the U.S. Army Corps of Engineers are excluded from the scope of this paragraph (c).

(173) (2) In the event of threatened or actual destruction of, loss of, or injury to a Sanctuary resource or quality resulting from an untoward incident, including but not limited to spills and groundings caused by the Department of Defense, the Department of Defense shall promptly coordinate with the Director for the purpose of taking appropriate actions to respond to and mitigate the harm and, if possible, restore or replace the Sanctuary resource or quality.

(174) (d) The prohibitions in paragraphs (a)(1) and (3) through (7) of this section do not apply to any activity executed in accordance with the scope, purpose, terms and conditions of a National Marine Sanctuary permit issued pursuant to §922.48 and §922.143 or a Special Use permit issued pursuant to Section 310 of the Act.

(175) (e) The prohibitions in paragraphs (a)(1) and (3) through (7) of this section do not apply to any activity authorized by any lease, permit, license, approval or other authorization issued after the effective date of Sanctuary designation (November 4, 1992) and issued by any Federal, State or local authority of competent jurisdiction, provided that the applicant complies with §922.49, the Director notifies the applicant and authorizing agency that he or she does not object to issuance of the authorization, and the applicant complies with any terms and conditions the Director deems necessary to protect Sanctuary resources and qualifies. Amendments, renewals and extensions of authorizations in existence on the effective date designation constitute authorizations issued after the effective date.

(176) (f) Notwithstanding paragraphs (d) and (e) of this section, in no event may the Director issue a permit under §922.48 and §922.143, or under section 310 of the act, authorizing, otherwise approving, the exploration for, development or production of industrial materials within the Sanctuary, or the disposal of dredged materials within the Sanctuary (except by certification, pursuant to §922.47, of valid authorizations in existence on November 4, 1992) and any leases, licenses, permits, approvals or other authorizations authorizing the exploration, for development or production of industrial materials in the Sanctuary issued by other authorities after November 4, 1992, shall be invalid.

(177)

§922.143 Permit procedures and criteria.

(178) (a) A person may conduct an activity prohibited by §922.142 (a)(1) and (3) through (7) if conducted in accordance with scope, purpose, manner, terms and conditions of a permit issued under this section and §922.48.

(179) (b) Applications for such permits should be addressed to the Director, Office of Ocean and Coastal Resource Management, ATTN: Manager, Stellwagen Bank National Marine Sanctuary, 14 Union Street, Plymouth, MA 02360.

(180) (c) The Director, at his or her discretion may issue a permit, subject to such terms and conditions as he or she deems appropriate, to conduct an activity prohibited by §922.142(a)(1) and (3) through (7), if the Director finds that the activity will have only negligible short-term adverse effects on Sanctuary resources and qualities and will: further research related to Sanctuary resources and qualities; further the educational, natural or historical resource value of the Sanctuary; further salvage or recovery operations in or near the Sanctuary in connection with a recent air or marine casualty; or assist in managing the Sanctuary. In deciding whether to issue a permit, the Director may consider such factors as: the professional qualifications and financial ability of the applicant as related to the proposed activity; the duration of the activity and the duration of its effects; the appropriateness of the methods and procedures

proposed by the applicant for the conduct of the activity; the extent to which the conduct of the activity may diminish or enhance Sanctuary resources and qualities;

port entry to report beyond this area to facilitate traffic management within the VTS area.

(204)

§26.03 Radiotelephone required.

(205) (a) Unless an exemption is granted under §26.09 and except as provided in paragraph (a)(4) of this section, this part applies to:

(206) (1) Every power-driven vessel of 20 meters or over in length while navigating;

(207) (2) Every vessel of 100 gross tons and upward carrying one or more passengers for hire while navigating;

(208) (3) Every towing vessel of 26 feet or over in length while navigating; and

(209) (4) Every dredge and floating plant engaged in or near a channel or fairway in operations likely to restrict or affect navigation of other vessels except for an unmanned or intermittently manned floating plant under the control of a dredge.

(210) (b) Every vessel, dredge, or floating plant described in paragraph (a) of this section must have a radiotelephone on board capable of operation from its navigational bridge, or in the case of a dredge, from its main control station, and capable of transmitting and receiving on the frequency or frequencies within the 156-162 Mega-Hertz band using the classes of emissions designated by the Federal Communications Commission for the exchange of navigational information.

(211) (c) The radiotelephone required by paragraph (b) of this section must be carried on board the described vessels, dredges, and floating plants upon the navigable waters of the United States.

(212) (d) The radiotelephone required by paragraph (b) of this section must be capable of transmitting and receiving on VHF-FM channel 1022 (157.1 MHz).

(213) (e) While transiting any of the following waters, each vessel described in paragraph (a) of this section also must have on board a radiotelephone capable of transmitting and receiving on VHF-FM channel 67 (156.375 MHz):

(214) (1) The lower Mississippi River from the territorial sea boundary, and within either the Southwest Pass safety fairway or the South Pass safety fairway specified in 33 CFR 166.200, to mile 242.4 AHP (Above Head of Passes) near Baton Rouge;

(215) (2) The Mississippi River-Gulf Outlet from the territorial sea boundary, and within the Mississippi River-Gulf outlet Safety Fairway specified in 33 CFR 166.200, to that channel's junction with the Inner Harbor Navigation Canal; and

(216) (3) The full length of the Inner Harbor Navigation Canal from its junction with the Mississippi River to that canal's entry to Lake Pontchartrain at the New Seabrook vehicular bridge.

(217) (f) In addition to the radiotelephone required by paragraph (b) of this section, each vessel described in paragraph (a) of this section while transiting any waters within a Vessel Traffic Service Area, must have on board a radiotelephone capable of transmitting and receiving on the VTS designated frequency in Table 161.12(c) (VTS and VMRS Centers, Call Signs/MMSI, Designated Frequencies, and Monitoring Areas).

(218) **Note:** A single VHF-FM radio capable of scanning or sequential monitoring (often referred to as "dual watch" capability) will not meet the requirements for two radios.

(219)

§26.04 Use of the designated frequency.

(220) (a) No person may use the frequency designated by the Federal Communications Commission under section 8 of the Act, 33 U.S.C. section 1207(a), to transmit any information other than information necessary for the safe navigation of vessels or necessary tests.

(221) (b) Each person who is required to maintain a listening watch under section 5 of the Act shall, when necessary, transmit and confirm, on the designated frequency, the intentions of his vessel and any other information necessary for the safe navigation of vessels.

(222) (c) Nothing in these regulations may be construed as prohibiting the use of the designated frequency to communicate with shore stations to obtain or furnish information necessary for the safe navigation of vessels.

(223) (d) On the navigable waters of the United States, channel 13 (156.65 MHz) is the designated frequency required to be monitored in accordance with §26.05(a) except that in the area prescribed in §26.03(e), channel 67 (156.375 MHz) is the designated frequency.

(224) (e) On those navigable waters of the United States within a VTS area, the designated VTS frequency is an additional designated frequency required to be monitored in accordance with §26.05.

(225)

§26.05 Use of radiotelephone.

(226) Section 5 of the Act states that the radiotelephone required by this Act is for the exclusive use of the master or person in charge of the vessel, or the person designated by the master or person in charge to pilot or direct the movement of the vessel, who shall maintain a listening watch on the designated frequency. Nothing herein shall be interpreted as precluding the use of portable radiotelephone equipment to satisfy the requirements of this act.

(227)

§26.06 Maintenance of radiotelephone; failure of radiotelephone.

(228) Section 6 of the Act states–(a) Whenever radiotelephone capability is required by this Act, a vessel's radiotelephone equipment shall be maintained in effective operating condition. If the radiotelephone equipment carried aboard a vessel ceases to operate, the master shall exercise due diligence to restore it or cause it to be restored to effective operating condition at the earliest practicable time. The failure of a vessel's radiotelephone equipment shall not, in itself, constitute a violation of this Act, nor shall it obligate the master of any vessel to moor or anchor his vessel; however, the loss

of radiotelephone capability shall be given consideration in the navigation of the vessel.

(229)

§26.07 Communications.

(230) No person may use the services of, and no person may serve as, a person required to maintain a listening watch under section 5 of the Act, 33 U.S.C. 1204, unless the person can communicate in the English language.

(231)

§26.08 Exemption procedures.

(232) (a) The Commandant has redelegated to the Assistant Commandant for Prevention Policy, U.S. Coast Guard Headquarters, with the reservation that this authority shall not be further redelegated, the authority to grant exemptions from provisions of the Vessel Bridge-to-Bridge Radiotelephone Act and this part.

(233) (b) Any person may petition for an exemption from any provision of the Act or this part;

(234) (c) Each petition must be submitted in writing to Commandant (CG–DCO–D), Attn: Deputy for Operations Policy and Capabilities, U.S. Coast Guard Stop 7318, 2703 Martin Luther King Jr. Avenue SE., Washington, DC 20593–7318, and must state:

(235) (1) The provisions of the Act or this part from which an exemption is requested; and

(236) (2) The reasons why marine navigation will not be adversely affected if the exemption is granted and if the exemption relates to a local communication system how that system would fully comply with the intent of the concept of the Act but would not conform in detail if the exemption is granted.

(237)

§26.09 List of exemptions.

(238) (a) All vessels navigating on those waters governed by the navigation rules for Great Lakes and their connecting and tributary waters (33 U.S.C. 241 et seq.) are exempt from the requirements of the Vessel Bridge-to-Bridge Radiotelephone Act and this part until May 6, 1975.

(239) (b) Each vessel navigating on the Great Lakes as defined in the Inland Navigational Rules Act of 1980 (33 U.S.C. 2001 et seq.) and to which the Vessel Bridge-to-Bridge Radiotelephone Act (33 U.S.C. 1201–1208) applies is exempt from the requirements in 33 U.S.C. 1203, 1204, and 1205 and the regulations under §§26.03, 26.04, 26.05, 26.06, and 26.07. Each of these vessels and each person to whom 33 U.S.C. 1208(a) applies must comply with Articles VII, X, XI, XII, XIII, XV, and XVI and Technical Regulations 1-9 of "The Agreement Between the United States of America and Canada for Promotion of Safety on the Great Lakes by Means of Radio, 1973."

(240)

Part 80–COLREGSDemarcation Lines

(241)

§80.01 General basis and purpose of demarcation lines.

(242) (a) The regulations in this part establish the lines of demarcation delineating those waters upon which mariners shall comply with the International Regulations for Preventing Collisions at Sea, 1972 (72 COLREGS) and those waters upon which mariners shall comply with the Inland Navigation Rules.

(243) (b) The waters inside of the lines are Inland Rules Waters. The waters outside the lines are COLREGS Waters.

(244) (c) Geographic coordinates expressed in terms of latitude or longitude, or both, are not intended for plotting on maps or charts whose referenced horizontal datum is the North American Datum of 1983 (NAD 83), unless such geographic coordinates are expressly labeled NAD 83. Geographic coordinates without the NAD 83 reference may be plotted on maps or charts referenced to NAD 83 only after application of the appropriate corrections that are published on the particular map or chart being used.

(245)

§80.105 Calais, Maine to Cape Small, Maine.

(246) The 72 COLREGS shall apply on the harbors, bays, and inlets on the east coast of Maine from International Bridge at Calais, ME to the southwesternmost extremity of Bald Head at Cape Small.

(247)

§80.110 Casco Bay, Maine.

(248) (a) A line drawn from the southwesternmost extremity of Bald Head at Cape Small to the southeasternmost extremity of Ragged Island; thence to the southern tangent of Jaquish Island thence to Little Mark Island Monument Light; thence to the northernmost extremity of Jewell Island.

(249) (b) A line drawn from the tower on Jewell Island charted in approximate position latitude 43°40.6'N., longitude 70°05.9'W. to the northeasternmost extremity of Outer Green Island.

(250) (c) A line drawn from the southwesternmost extremity of Outer Green Island to Ram Island Ledge Light; thence to Portland Head Light.

(251)

§80.115 Portland Head, Maine to Cape Ann, Mass

(252) (a) Except inside lines specifically described in this section, the 72 COLREGS shall apply on the harbors, bays, and inlets on the east coast of Maine, New Hampshire, and Massachusetts from Portland Head to Halibut Point at Cape Ann.

(253) (b) A line drawn from the southernmost tower on Gerrish Island charted in approximate position latitude 43°04.0'N., longitude 70°41.2'W. to Whaleback

Light; thence to Jaffrey Point Light 2A; thence to the northeasternmost extremity of Frost Point.

(254) (c) A line drawn from the northernmost extremity of Farm Point to Annisquam Harbor Light.

(255)

§80.120 Cape Ann, MA to Marblehead Neck, MA

(256) (a) Except inside lines specifically described in this section, the 72 COLREGS shall apply on the harbors, bays and inlets on the east coast of Massachusetts from Halibut Point at Cape Ann to Marblehead Neck.

(257) (b) A line drawn from Gloucester Breakwater Light to the twin towers charted in approximate position latitude 42°35.1'N., longitude 70°41.6'W.

(258) (c) A line drawn from the westernmost extremity of Gales Point to the easternmost extremity of House Island; thence to Bakers Island Light; thence to Marblehead Light.

(259)

§80.125 Marblehead Neck, MA to Nahant, MA

(260) The 72 COLREGS apply on the harbors, bays, and inlets on the east coast of Massachusetts from Marblehead Neck to the easternmost tower at Nahant, charted in approximate position latitude 42°25.4'N., longitude 70°54.6'W.

(261)

§80.130 Boston Harbor entrance.

(262) A line drawn from the easternmost tower at Nahant, charted in approximate position latitude 42°25.4'N., longitude 70°54.6'W., to Boston Lighted Horn Buoy "B"; thence to the easternmost radio tower at Hull, charted in approximate position latitude 42°16.7'N., longitude 70°52.6'W.

(263)

§80.135 Hull, MA to Race Point, MA

(264) (a) Except inside lines described in this section, the 72 COLREGS apply on the harbors, bays, and inlets on the east coast of Massachusetts from the easternmost radio tower at Hull, charted in approximate position latitude 42°16.7'N., longitude 70°52.6'W., to Race Point on Cape Cod.

(265) (b) A line drawn from Canal Breakwater Light 6 south to the shoreline.

(266)

Part 81—72 COLREGS: IMPLEMENTING RULES

(267)

§81.1 Definitions.

(268) As used in this part:

(269) *72 COLREGS* refers to the International Regulations for Preventing Collisions at Sea, 1972, done at London, October 20, 1972, as rectified by the Proces-Verbal of December 1, 1973, as amended.

(270) *A vessel of special construction or purpose* means a vessel designed or modified to perform a special function and whose arrangement is thereby made relatively inflexible.

(271) *Interference with the special function of the vessel* occurs when installation or use of lights, shapes, or sound-signaling appliances under 72 COLREGS prevents or significantly hinders the operation in which the vessel is usually engaged.

(272)

§81.3 General.

(273) Vessels of special construction or purpose which cannot fully comply with the light, shape, and sound signal provisions of 72 COLREGS without interfering with their special function may instead meet alternative requirements. The Chief of the Prevention Division in each Coast Guard District Office makes this determination and requires that alternative compliance be as close as possible with the 72 COLREGS. These regulations set out the procedure by which a vessel may be certified for alternative compliance. The information collection and recordkeeping requirements in §§81.5 and 81.18 have been approved by the Office of Management and Budget under OMB control No. 1625-0019.

(274)

Alternative Compliance

(275)

§81.5 Application for a Certificate of Alternative Compliance.

(276) (a) The owner, builder, operator, or agent of a vessel of special construction or purpose who believes the vessel cannot fully comply with the 72 COLREGS light, shape, or sound signal provisions without interference with its special function may apply for a determination that alternative compliance is justified. The application must be in writing, submitted to the Chief of the Prevention Division of the Coast Guard District in which the vessel is being built or operated, and include the following information:

(277) (1) The name, address, and telephone number of the applicant.

(278) (2) The identification of the vessel by its:

(279) (i) Official number;

(280) (ii) Shipyard hull number;

(281) (iii) Hull identification number; or

(282) (iv) State number, if the vessel does not have an official number or hull identification number.

(283) (3) Vessel name and home port, if known.

(284) (4) A description of the vessel's area of operation.

(285) (5) A description of the provision for which the Certificate of Alternative Compliance is sought, including:

(286) (i) The 72 COLREGS Rule or Annex section number for which the Certificate of Alternative Compliance is sought;

(287) (ii) A description of the special function of the vessel that would be interfered with by full compliance with the provision of that Rule or Annex section; and

(288) (iii) A statement of how full compliance would interfere with the special function of the vessel.

(289) (6) A description of the alternative installation that is in closest possible compliance with the applicable 72 COLREGS Rule or Annex section.

(290) (7) A copy of the vessel's plans or an accurate scale drawing that clearly shows:

(291) (i) The required installation of the equipment under the 72 COLREGS,

(292) (ii) The proposed installation of the equipment for which certification is being sought, and

(293) (iii) Any obstructions that may interfere with the equipment when installed in:

(294) (A) The required location; and

(295) (B) The proposed location.

(296) (b) The Coast Guard may request from the applicant additional information concerning the application.

(297)

§81.9 Certificate of Alternative Compliance: Contents.

(298) The Chief of the Prevention Division issues the Certificate of Alternative Compliance to the vessel based on a determination that it cannot comply fully with 72 COLREGS light, shape, and sound signal provisions without interference with its special function. This Certificate includes—

(299) (a) Identification of the vessel as supplied in the application under §81.5(a)(2);

(300) (b) The provision of the 72 COLREGS for which the Certificate authorizes alternative compliance;

(301) (c) A certification that the vessel is unable to comply fully with the 72 COLREGS lights, shape, and sound signal requirements without interference with its special function;

(302) (d) A statement of why full compliance would interfere with the special function of the vessel;

(303) (e) The required alternative installation;

(304) (f) A statement that the required alternative installation is in the closest possible compliance with the 72 COLREGS without interfering with the special function of the vessel;

(305) (g) The date of issuance;

(306) (h) A statement that the Certificate of Alternative Compliance terminates when the vessel ceases to be usually engaged in the operation for which the certificate is issued.

(307)

§81.17 Certificate of Alternative Compliance: Termination.

(308) The Certificate of Alternative Compliance terminates if the information supplied under §81.5(a) or the Certificate issued under §81.9 is no longer applicable to the vessel.

(309)

§81.18 Notice and record of certification of vessels of special construction or purpose.

(310) (a) In accordance with 33 U.S.C. 1605(c), a notice is published in the Federal Register of the following:

(311) (1) Each Certificate of Alternative Compliance issued under §81.9; and

(312) (2) Each Coast Guard vessel determined by the Commandant to be a vessel of special construction or purpose.

(313) (b) Copies of Certificate of Alternative Compliance and documentation concerning Coast Guard vessels are available for inspection at Marine Transportation Systems Directorate, U.S. Coast Guard Headquarters, (CG-5PW), Stop 7509, 2703 Martin Luther King Avenue SE., Washington, DC 20593-7509.

(314) (c) The owner or operator of a vessel issued a Certificate shall ensure that the vessel does not operate unless the Certificate of Alternative Compliance or a certified copy of that Certificate is on board the vessel and available for inspection by Coast Guard personnel.

(315)

Exemptions

(316)

§81.20 Lights and sound signal appliances.

(317) Each vessel under the 72 COLREGS, except the vessels of the Navy, is exempt from the requirements of the 72 COLREGS to the limitation for the period of time stated in Rule 38 (a), (b), (c), (d), (e), (f), and (g) if:

(318) (a) Her keel is laid or is at a corresponding stage of construction before July 15, 1977; and

(319) (b) She meets the International Regulations for Preventing Collisions at Sea, 1960 (77 Stat. 194, 33 U.S.C. 1051-1094).

(320)

Part 82—72 COLREGS: INTERPRETATIVE RULES

(321)

§82.1 Purpose.

(322) This part contains the interpretative rules concerning the 72 COLREGS that are adopted by the Coast Guard for the guidance of the public.

(323)

§82.3 Pushing vessel and vessel being pushed: Composite unit.

(324) Rule 24(b) of the 72 COLREGS states that when a pushing vessel and a vessel being pushed ahead are rigidly connected in a composite unit, they are regarded as a power-driven vessel and must exhibit the lights under Rule 23. A "composite unit" is interpreted to be a pushing vessel that is rigidly connected by mechanical means to a vessel being pushed so they react to sea and swell as one vessel. "Mechanical means" does not include the following:

(325) (a) Lines.

(326) (b) Hawsers.

(327) (c) Wires.

(328) (d) Chains.

(329)

§82.5 Lights for moored vessels.

(330) For the purposes of Rule 30 of the 72 COLREGS, a *vessel at anchor* includes a barge made fast to one or more mooring buoys or other similar device attached to the sea or river floor. Such a barge may be lighted as a vessel at anchor in accordance with Rule 30, or may be lighted on the corners in accordance with 33 CFR 83.30(h) through (l).

(331)

§82.7 Sidelights for unmanned barges.

(332) An unmanned barge being towed may use the exception of COLREGS Rule 24(h). However, this exception only applies to the vertical sector requirements.

(333)

Part 88—ANNEX V: PILOT RULES

(334)

§88.01 Purpose and applicability.

(335) This part applies to all vessels operating on United States inland waters and to United States vessels operating on the Canadian waters of the Great Lakes to the extent there is no conflict with Canadian law.

(336)

§88.03 Definitions.

(337) The terms used in this part have the same meaning as the terms defined in part 83 of this subchapter.

(338)

§88.05 Law enforcement vessels.

(339) (a) Law enforcement vessels may display a flashing blue light when engaged in direct law enforcement or public safety activities. This light must be located so that it does not interfere with the visibility of the vessel's navigation lights.

(340) (b) The blue light described in this section may be displayed by law enforcement vessels of the United States and the States and their political subdivisions.

(341)

§88.07 Public safety activities.

(342) (a) Vessels engaged in government sanctioned public safety activities, and commercial vessels performing similar functions, may display an alternately flashing red and yellow light signal. This identification light signal must be located so that it does not interfere with the visibility of the vessel's navigation lights. The identification light signal may be used only as an identification signal and conveys no special privilege. Vessels using the identification light signal during public safety activities must abide by the Inland Navigation Rules, and must not presume that the light or the exigency gives them precedence or right of way.

(343) (b) Public safety activities include but are not limited to patrolling marine parades, regattas, or special water celebrations; traffic control; salvage; firefighting; medical assistance; assisting disabled vessels; and search and rescue.

(344)

Part 89—INLAND NAVIGATION RULES: IMPLEMENTING RULES

(345)

Subpart A—Certificate of Alternative Compliance

(346)

§89.1 Definitions.

(347) As used in this subpart:

(348) *Inland Rules* refers to the Inland Navigation Rules contained in the Inland Navigational Rules Act of 1980 (Pub. L. 96-591) and the technical annexes established under that act.

(349) *A vessel of special construction or purpose* means a vessel designed or modified to perform a special function and whose arrangement is thereby made relatively inflexible.

(350) *Interference with the special function of the vessel* occurs when installation or use of lights, shapes, or sound-signaling appliances under the Inland Rules prevents or significantly hinders the operation in which the vessel is usually engaged.

(351)

§89.3 General.

(352) Vessels of special construction or purpose which cannot fully comply with the light, shape, and sound signal provisions of the Inland Rules without interfering with their special function may instead meet alternative requirements. The Chief of the Prevention Division in each Coast Guard District Office makes this determination and requires that alternative compliance be as close as possible with the Inland Rules. These regulations set out the procedure by which a vessel may be certified for alternative compliance. The information collection and recordkeeping requirements in §§89.5 and 89.18 have been approved by the Office of Management and Budget under OMB control No. 1625-0019.

(353)

§89.5 Application for a Certificate of Alternative Compliance.

(354) (a) The owner, builder, operator, or agent of a vessel of special construction or purpose who believes the vessel cannot fully comply with the Inland Rules light, shape, or sound signal provisions without interference with its special function may apply for a determination that alternative compliance is justified. The application must be in writing, submitted to the Chief of the Prevention Division of the Coast Guard District in which the vessel is being built or operated, and include the following information:

(355) (1) The name, address, and telephone number of the applicant.

(356) (2) The identification of the vessel by its:

(357) (i) Official number;

(358) (ii) Shipyard hull number;

(359) (iii) Hull identification number; or

(360) (iv) State number, if the vessel does not have an official number or hull identification number.

(361) (3) Vessel name and home port, if known.

(362) (4) A description of the vessel's area of operation.

(363) (5) A description of the provision for which the Certificate of Alternative Compliance is sought, including:

(364) (i) The Inland Rules Rule or Annex section number for which the Certificate of Alternative Compliance is sought;

(365) (ii) A description of the special function of the vessel that would be interfered with by full compliance with the provision of that Rule or Annex section; and

(366) (iii) A statement of how full compliance would interfere with the special function of the vessel.

(367) (6) A description of the alternative installation that is in closest possible compliance with the applicable Inland Navigation Rules Rule or Annex section.

(368) (7) A copy of the vessel's plans or an accurate scale drawing that clearly shows:

(369) (i) The required installation of the equipment under the Inland Rules,

(370) (ii) The proposed installation of the equipment for which certification is being sought, and

(371) (iii) Any obstructions that may interfere with the equipment when installed in:

(372) (A) The required location; and

(373) (B) The proposed location.

(374) (b) The Coast Guard may request from the applicant additional information concerning the application.

(375)

§89.9 Certificate of Alternative Compliance: Contents.

(376) The Chief of the Prevention Division issues the Certificate of Alternative Compliance to the vessel based on a determination that it cannot comply fully with Inland Rules light, shape, and sound signal provisions without interference with its special function. This Certificate includes:

(377) (a) Identification of the vessel as supplied in the application under §89.5(a)(2);

(378) (b) The provision of the Inland Rules for which the Certificate authorizes alternative compliance;

(379) (c) A certification that the vessel is unable to comply fully with the Inland Rules light, shape, and sound signal requirements without interference with its special function;

(380) (d) A statement of why full compliance would interfere with the special function of the vessel;

(381) (e) The required alternative installation;

(382) (f) A statement that the required alternative installation is in the closest possible compliance with the Inland Rules without interfering with the special function of the vessel;

(383) (g) The date of issuance;

(384) (h) A statement that the Certificate of Alternative Compliance terminates when the vessel ceases to be usually engaged in the operation for which the certificate is issued.

(385)

§89.17 Certificate of Alternative Compliance: Termination.

(386) The Certificate of Alternative Compliance terminates if the information supplied under §89.5(a) or the Certificate issued under §89.9 is no longer applicable to the vessel.

(387)

§89.18 Record of certification of vessels of special construction or purpose.

(388) (a) Copies of Certificates of Alternative Compliance and documentation concerning Coast Guard vessels are available for inspection at the offices of the Marine Transportation Systems Directorate, U.S. Coast Guard Headquarters (CG-5PW), Stop 7509, 2703 Martin Luther King Avenue SE., Washington, DC 20593-7509.

(389) (b) The owner or operator of a vessel issued a Certificate shall ensure that the vessel does not operate unless the Certificate of Alternative Compliance or a certified copy of that Certificate is on board the vessel and available for inspection by Coast Guard personnel.

(390)

Subpart B—Waters Upon Which Certain Inland Navigation Rules Apply

(391)

§89.21 Purpose.

(392) Inland Navigation Rules 9(a)(ii), 14(d), and 15(b) apply to the Great Lakes, and along with 24(i), apply on the "Western Rivers" as defined in Rule 3(1), and to additional specifically designated waters. The purpose of this Subpart is to specify those additional waters upon which Inland Navigation Rules 9(a)(ii), 14(d), 15(b), and 24(i) apply.

(393)

§89.23 Definitions.

(394) As used in this subpart:

(395) *Inland Rules* refers to the Inland Navigation Rules contained in the Inland Navigational Rules Act of 1980 (Pub. L. 96-591, 33 U.S.C. 2001 et. seq.) and the technical annexes established under that Act.

(396)

§89.25 Waters upon which Inland Rules 9(a)(ii), 14(d), and 15(b) apply.

(397) Inland Rules 9(a)(ii), 14(d), and 15(b) apply on the Great Lakes, the Western Rivers, and the following specified waters:

(398) (a) Tennessee-Tombigbee Waterway.

(399) (b) Tombigbee River.

(400) (c) Black Warrior River.

(401) (d) Alabama River.

(402) (e) Coosa River.

(403) (f) Mobile River above the Cochrane Bridge at St. Louis Point.

(404) (g) Flint River.

(405) (h) Chattahoochee River.

(406) (i) The Apalachicola River above its confluence with the Jackson River.

(407)

§89.27 Waters upon which Inland Rule 24(j) applies.

(408) (a) Inland Rule 24(j) applies on the Western Rivers and the specified waters listed in §89.25 (a) through (i).

(409) (b) Inland Rule 24(j) applies on the Gulf Intracoastal Waterway from St. Marks, Florida, to the Rio Grande, Texas, including the Morgan City-Port Allen Alternate Route and the Galveston-Freeport Cutoff, except that a power-driven vessel pushing ahead or towing alongside shall exhibit the lights required by Inland Rule 24(c), while transiting within the following areas:

(410) (1) St. Andrews Bay from the Hathaway Fixed Bridge at Mile 284.6 East of Harvey Locks (EHL) to the DuPont Fixed Bridge at Mile 295.4 EHL.

(411) (2) Pensacola Bay, Santa Rosa Sound and Big Lagoon from the Light "10" off of Trout Point at Mile 176.9 EHL to the Pensacola Fixed Bridge at Mile 189.1 EHL.

(412) (3) Mobile Bay and Bon Secour Bay from the Dauphin Island Causeway Fixed Bridge at Mile 127.7 EHL to Little Point Clear at Mile 140 EHL.

(413) (4) Mississippi Sound from Grand Island Waterway Light "1" at Mile 53.8 EHL to Light "40" off the West Point of Dauphin Island at Mile 118.7 EHL.

(414) (5) The Mississippi River at New Orleans, Mississippi River-Gulf Outlet Canal and the Inner Harbor Navigation Canal from the junction of the Harvey Canal and the Algiers Alternate Route at Mile 6.5 West of Harvey Locks (WHL) to the Michoud Canal at Mile 18 EHL.

(415) (6) The Calcasieu River from the Calcasieu Lock at Mile 238.6 WHL to the Ellender Lift Bridge at Mile 243.6 WHL.

(416) (7) The Sabine Neches Canal from mile 262.5 WHL to mile 291.5 WHL.

(417) (8) Bolivar Roads from the Bolivar Assembling Basin at Mile 346 WHL to the Galveston Causeway Bridge at Mile 357.3 WHL.

(418) (9) Freeport Harbor from Surfside Beach Fixed Bridge at Mile 393.8 WHL to the Bryan Beach Pontoon Bridge at Mile 397.6 WHL.

(419) (10) Matagorda Ship Channel area of Matagorda Bay from Range "K" Front Light at Mile 468.7 WHL to the Port O'Connor Jetty at Mile 472.2 WHL.

(420) (11) Corpus Christi Bay from Redfish Bay Day Beacon "55" at Mile 537.4 WHL when in the Gulf Intracoastal Waterway main route or from the north end of Lydia Ann Island Mile 531.1A when in the Gulf Intracoastal Waterway Alternate Route to Corpus Christi Bay LT 76 at Mile 543.7 WHL.

(421) (12) Port Isabel and Brownsville Ship Channel south of the Padre Island Causeway Fixed Bridge at Mile 665.1 WHL.

(422)

Part 90—INLAND RULES: INTERPRETATIVE RULES

(423)

§90.1 Purpose.

(424) This part contains the interpretative rules for the Inland Rules. These interpretative rules are intended as a guide to assist the public and promote compliance with the Inland Rules.

(425)

§90.3 Pushing vessel and vessel being pushed: Composite unit.

(426) Rule 24(b) of the Inland Rules states that when a pushing vessel and a vessel being pushed ahead are rigidly connected in a composite unit, they are regarded as a power-driven vessel and must exhibit the lights prescribed in Rule 23. A "composite unit" is interpreted to be the combination of a pushing vessel and a vessel being push ahead that are rigidly connected by mechanical means so they react to sea and swell as one vessel. Mechanical means does not include lines, wires, hawsers, or chains.

(427)

§90.5 Lights for moored vessels.

(428) A *vessel at anchor* includes a vessel made fast to one or more mooring buoys or other similar device attached to the ocean floor. Such vessels may be lighted as a vessel at anchor in accordance with Rule 30, or may be lighted on the corners in accordance with 33 CFR 88.30(h) through (l).

(429)

§90.7 Sidelights for unmanned barges.

(430) An unmanned barge being towed may use the exception of COLREGS Rule 24(h). However, this exception only applies to the vertical sector requirements for sidelights.

(431)

Part 110–AnchorageRegulations

(432)

§110.1 General.

(433) (a) The areas described in subpart A of this part are designated as special anchorage areas for the purposes of rule 30 (33 CFR 83.30) and rule 35 (33 CFR 83.35) of the Inland Navigation Rules, 33 CFR Chapter I, Subchapter E. Vessels of less than 20 meters in length; and barges, canal boats, scows, or other nondescript craft, are not required to sound signals required by rule 35 of the Inland Navigation Rules. Vessels of less than 20 meters are not required to exhibit anchor lights or shapes required by rule 30 of the Inland Navigation Rules.

(434) (b) The anchorage grounds for vessels described in Subpart B of this part are established, and the rules and regulations in relation thereto adopted, pursuant to the

authority contained in section 7 of the act of March 4, 1915, as amended (38 Stat. 1053; 33 U.S.C. 471).

(435) (c) All bearings in the part are referred to true meridian.

(436) (d) Geographic coordinates expressed in terms of latitude or longitude, or both, are not intended for plotting on maps or charts whose reference horizontal datum is the North American Datum of 1983 (NAD 83), unless such geographic coordinates are expressly labeled NAD 83. Geographic coordinates without the NAD 83 reference may be plotted on maps or charts referenced to NAD 83 only after application of the appropriate corrections that are published on the particular map or chart being used.

(437)

Subpart A–Special Anchorage Areas

(438)

§110.4 Penobscot Bay, Maine.

(439) (a) *Rockland Harbor.* Beginning at a point bearing 244°, 1,715 yards, from Rockland Breakwater Light; thence 260°, 490 yards, to a point bearing 248° from Rockland Breakwater Light; thence 350°, 580 yards, to a point bearing 263° from Rockland Breakwater Light; thence 83°, 480 yards, to a point bearing 263° from Rockland Breakwater Light; and thence 169°, 550 yards, to the point of beginning. This area is limited to vessels no greater than 20 meters in length.

(440) (b) *Camden Harbor, Sherman Cove and adjacent waters.*

(441) (1) *Anchorage A.* All of the waters enclosed by a line beginning at Eaton Point at

(442) 44°12'31"N., 069°03'34"W.; thence to

(443) 44°12'28"N., 069°03'33"W.; thence to

(444) 44°12'32"N., 069°02'49"W.; thence along the shoreline to the point of beginning. DATUM: NAD83

(445) (2) *Anchorage B.* All of the waters enclosed by a line beginning at Dillingham Point at

(446) 44°12'12"N., 069°03'20"W.; thence to

(447) 44°12'14"N., 069°02'58"W.; thence to

(448) 44°12'19"N., 069°03'08"W.; thence to

(449) 44°12'28"N., 069°03'13"W.; thence to

(450) 44°12'26"N., 069°03'39"W.; thence along the shoreline to the point of beginning. DATUM: NAD83

(451) (c) *Stonington Harbor, Deer Island Thorofare.*

(452) (1) *Crotch Island*. All of the waters bound by the following points beginning at the northeast shore of Crotch Island located at

(453) 44°08'51.0"N., 068°40'06.0"W.; thence southerly along the shoreline to

(454) 44°08'36.0"N., 068°40'07.02"W.; thence to

(455) 44°08'36.0"N., 068°40'04.02"W.; thence to

(456) 44°08'46.98"N., 068°40'00.0"W.; thence to

(457) 44°08'55.02"N., 068°39'49.02"W.; thence to

(458) 44°08'54.0"N., 068°40'06.0"W.; thence back to origin. DATUM: NAD 83

(459) (2) [Reserved]

(460) (d) *Passagassawakeag River, Belfast Bay, Belfast, Maine*—(1) Special anchorage area A. All of the waters enclosed by a line beginning at 44°25′47.2458”N., 069°00′7.5943”W.; thence to 44°25′48”N., 068°59′57”"W.; thence to 44°25′39’N., 068°59′17”W.; thence to 44°25′33”N., 068°59′15”W.; thence to 44°25′30”N., 068°58′48”W.; thence to 44°25′23.9162”N., 068°58′54.0838”W.; thence to 44°25′42.7050”N., 068°59′55.2686”W., thence to the point of beginning.

(461) (2) *Special anchorage area B*. All of the waters enclosed by a line beginning at 44°25′45.3309″N., 069°00′09.0265″W.; thence to 44°25′41.1720″N., 068°59′58.2017″W.; thence to 44°25′27.7645″N., 068°59′23.3130″W.; thence to 44°25′18.2707″N., 068° 58′58.6083″W.; thence to 44°24′56″N., 068°59′23″W.; thence to 44°25′ 21.0416″N., 068°59′37.5019″W.; thence to 44°25′35.5413″N., 068°59′58.1933″W.; thence to the point of beginning.

(462)

§110.5 Casco Bay, Maine

(463) (a) *Beals Cove, west side of Orrs Island, Harpswell*. The entire cove as defined by the shoreline and a line across the entrance bearing 215° and tangent to the shore on the north side.

(464) (a–1) *Merriconeag Sound, Harpswell*. The area comprises that portion of the Sound beginning at a point on the shoreline about 1,000 feet northeasterly from the southwesterly extremity of Orr’s Island at

(465) 43°45'09"N., 69°59'14"W., thence extending 290° to a point at

(466) 43°45'10"N., 69°59'20"W., thence extending 20° to a point at

(467) 43°45'34"N., 69°59'05"W., thence extending 110° to a point on the shoreline at

(468) 43°45'33"N., 69°58'58"W., thence along the shoreline to the point of beginning.

(469) (a–2) *Mackerel Cove, Bailey Island, Harpswell*. The water area of Mackerel Cove lying northeasterly of a line from a point on Abner Point at

(470) 43°43'28"N., 70°00'19"W., to a point on Bailey Island at

(471) 43°43'18.2"N., 70°00'12.2"W.

(472) (b) *Harpswell Harbor, east side of Harpswell Neck, Harpswell*. The entire area lying westerly of a line bearing 8° from the eastern extremity of Stovers Point to the point of land at the northerly end of the harbor, said point of land bearing approximately 275° from the observatory on Orrs Island.

(473) (c) *Basin Cove, west side of Harpswell Neck, Harpswell*. All of the area lying northeasterly of a line bearing 350° from the northwest corner of the entrance to the cove.

(474) (c–1) *Basin Point, Potts Harbor, east side of Basin Point*. The water area east of Basin Point enclosed by a line beginning at the southernmost extremity of Basin Point at

(475) 43°44'17"N., 70°02'36"W.; thence easterly to

(476) 43°44'17"N., 70°02'19"W.; thence north northeasterly to a point on the shoreline at
(477) 43°44'43"N., 70°02'05"W.; thence following the shoreline to the point of beginning.
(478) (d) *Mussel Cove and adjacent waters at Falmouth Foreside, Falmouth*. All of the waters enclosed by a line beginning at the Dock House (F.S.) located at
(479) 43°44'22"N., 70°11'41"W.; thence to
(480) 43°44'19"N., 70°11'33"W.; thence to
(481) 43°44'00"N., 70°11'44"W.; thence to
(482) 43°43'37"N., 70°11'37"W.; thence to
(483) 43°43'04"N., 70°12'13"W.; thence to
(484) 43°41'56"N., 70°12'53"W.; thence to
(485) 43°41'49"N., 70°13'05"W.; thence to
(486) 43°42'11"N., 70°13'30"W.; thence along the shoreline to the point of beginning. DATUM: NAD 83.
(487) (e) *Harraseeket River*. That portion of the Harraseeket River within the mean low water lines, between Stockbridge Point and Weston Point, excluding therefrom a thoroughfare, 100 feet wide, the center line of which follows the natural channel.
(488) (f) Yarmouth Harbor and adjacent waters. (1) Littlejohn Island/Doyle Point Cousins Island Special Anchorage. All of the waters enclosed by a line connecting the following points: Starting from the northernmost point of Littlejohn Island at
(489) 43°45'51.6"N., 70°06'57.0"W.; thence to
(490) 43°45'46.8"N., 70°06'53.4"W.; thence to
(491) 43°45'25.8"N., 70°07'22.8"W.; thence to
(492) 43°45'16.8"N., 70°07'40.8"W.; thence to
(493) 43°44'57.0"N., 70°08'27.0"W.; thence to
(494) 43°44'59.9"N., 70°08'30.0"W. DATUM: NAD 83.
(495) (2) Madeleine and Sandy Point Special Anchorage. All of the waters enclosed by a line connecting the following points: Starting from a point northeast of Birch Point on Cousins Island at
(496) 43°45'15.1"N., 70°09'16.8"W.; thence to
(497) 43°45'21.0"N., 70°09'30.0"W.; thence to
(498) 43°45'37.8"N., 70°09'10.9"W.; thence to
(499) 43°45'57.0"N., 70°08'58.8"W.; thence to
(500) 43°46'01.3"N., 70°08'45.0"W. DATUM: NAD 83.
(501) (3) Drinkwater Point and Princes Point Special Anchorage. All of the waters enclosed by a line connecting the following points: Starting south of Drinkwater Point in Yarmouth, Maine at
(502) 43°46'26.8"N., 70°09'17.0"W.; thence to
(503) 43°46'21.0"N., 70°09'09.6"W.; thence to
(504) 43°46'04.2"N., 70°09'46.2"W.; thence to
(505) 43°45'28.8"N., 70°10'24.0"W.; thence to
(506) 43°45'43.2"N., 70°10'24.0"W. DATUM: NAD 83.
(507)

§110.6 Portland Harbor, Portland, Maine (between Little Diamond Island and Great Diamond Island).

(508) Beginning at the southeasterly corner of the wharf, at the most southerly point of Great Diamond Island at
(509) 43°40'13"N., 70°12'00"W.; thence extending southwesterly to the northeasterly corner of the wharf on the easterly side of Little Diamond Island at
(510) 43°40'03"N., 70°12'15"W.; thence extending along the northerly side of the wharf to its shoreward end at
(511) 43°40'03"N., 70°12'17"W.; thence extending along the shoreline of Little Diamond Island to
(512) 43°40'11"N., 70°12'20"W.; thence extending northeasterly to the shoreline of the southerly side of Great Diamond Island at
(513) 43°40'21"N., 70°12'06"W.; thence extending along the shoreline of Great Diamond Island to the shoreward end of a wharf at
(514) 43°40'15"N., 70°12'02"W.; thence extending along the southwesterly side of the wharf to the point of beginning.
(515)

§110.6a Fore River, Portland Harbor, Portland, Maine.

(516) The water area beginning at a point on the shoreline near the Coast Guard Base in position
(517) 43°38'43"N., 70°14'49"W., thence 319° to
(518) 43°38'55"N., 70°15'03"W., thence 050° to
(519) 43°39'06"N., 70°14'43"W., thence 161° to mainland; and thence southwesterly along the shore to the point of beginning.
(520)

§110.9 Wells Harbor, Maine.

(521) (a) *Anchorage "A".* All of the waters enclosed by a line beginning at
(522) 43°19'15.7"N., 70°33'42.1"W.; thence to
(523) 43°19'15.7"N., 70°33'40.3"W.; thence to
(524) 43°19'03.7"N., 070°33'42.6"W.; thence to
(525) 43°19'02.6"N., 70°33'45.7"W.; thence to the point of beginning. This area is approximately 5,800 sq. yards, encompassing the central portion of Wells Harbor.
(526) (b) *Anchorage "B".* All of the waters enclosed by a line beginning at
(527) 43°19'11.1"N., 70°33'49.8"W.; thence to
(528) 43°19'10.5"N., 70°33'47.3"W.; thence to
(529) 43°19'08.3"N., 070°33'47.3"W.; thence to
(530) 43°19'08.7"N., 070°33'50.6"W.; thence to the point of beginning. This area is approximately 25,000 sq. yards, encompassing the western portion of Wells Harbor.
(531) (c) *Anchorage "C".* All of the waters enclosed by a line beginning at
(532) 43°19'17.7"N., 70°33'34.0"W.; thence to
(533) 43°19'18.4"N., 70°33'32.9"W.; thence to
(534) 43°19'13.8"N., 070°33'25.5"W.; thence to
(535) 43°19'13.0"N., 070°33'26.2"W.; thence to the point of beginning. This area is approximately 8,200 sq. yards, encompassing the eastern portion of Wells Harbor.
(536)

§110.10 Portsmouth Harbor, N.H., north of Newcastle Island.

(537) From the northernmost point of Goat Island to 43°04'25"N., 070°43'37"W.; thence 089°30' for 1025

yards; thence 120° for 285 yards, thence 213° to the shoreline of Newcastle Island, thence along the shoreline of Newcastle Island and across the breakwater to Goat Island and to the point of beginning.

(538)

§110.25 Salem Sound, MA

(539) (a) *Beverly Harbor, north of Salem Neck, Salem, MA*. A line extending from the northerly end of the Salem Willows Yacht Club House 360 yards bearing 281° true to

(540) 42°32'14.3"N., 70°52'24.17"W.; thence north 275 yards to Monument Bar Beacon thence 540 yards bearing 080° to

(541) 42°32'25.3"N., 70°52'2.1"W., thence 365 yards bearing 175° to

(542) 42°32'14.3"N., 70°52'1.1"W.; thence 237° to the shore. [NAD83]

(543) (b) *Bass River*. All of the area upstream of the highway bridge (Popes Bridge) outside of the dredged channel.

(544) (c) *South Channel*. Bounded by a line commencing at the northern most point of Peach's Point at

(545) 42°31'08.6"N., 70°50'32.8"W.; thence westerly to a point, at

(546) 42°31'21.9"N., 70°51'15.1"W. off Fluen Point; thence westerly to a point at

(547) 42°31'19.3"N., 70°51'47.4"W. off Naugus Head; thence southwesterly to a point at

(548) 42°31'00.3" N., 70°51'16.6" W. east of Folger Point; thence to a point at

(549) 42°30'38.3"N.,70°52'34.6"W.; thence easterly to a point on Long Point at

(550) 42°30'52.6"N., 70°53'05"W.

(551) (d) *Beverly and Mackerel Coves, north side of Beverly Harbor*. The water area enclosed by a line commencing at the southernmost point of Curtis Point in Beverly; thence bearing 238°, 1,400 yards to

(552) 42°32'29.7"N., 70°51'32.1"W.; thence 284°, 1,475 yards to the western shoreline of Mackerel Cove; thence north northeasterly to the point of beginning.

(553) (e) *Collins Cove, Salem, MA*. The water area enclosed by a line beginning at Monument Bar Beacon; thence 242°, 580 yards to

(554) 42°32'14.5"N., 70°52'46.3"W.; thence 284°, 220 yards to

(555) 42°32'16"N., 70°52'55"W.; thence 231°, 525 yards to a point on the shoreline; thence following the shoreline and the western boundary of the special anchorage area as described in 33 CFR 110.25(a) to the point of beginning.

(556)

§110.26 Marblehead Harbor, Marblehead, Mass.

(557) The area comprises that portion of the harbor lying between the extreme low water line and southwestward of a line bearing 336° from Marblehead Neck Light to a point on Peachs Point at latitude 42°31'03", longitude 70°50'30".

(558)

§110.27 Lynn Harbor in Broad Sound, MA.

(559) North of a line bearing 244° from the tower of the Metropolitan District Building, extending from the shore to a point 100 feet from the east limit of the channel; east of a line bearing 358°, extending thence to a point 100 feet east of the northeast corner of the turning basin; south of a line bearing 88°, extending thence to the shore; and south and west of the shoreline to its intersection with the south boundary.

(560)

§110.29 Boston Inner Harbor, MA.

(561) (a) *Vicinity of Pleasant Park Yacht Club, Winthrop*. Southerly of a line bearing 276° from a point on the west side of Pleasant Street, Winthrop, 360 feet from the southwest corner of its intersection with Main Street; westerly of a line bearing 186° from a point on the south side of Main Street 140 feet from the southwest corner of its intersection with Pleasant Street; northerly of a line bearing 256° from a point on the west side of Pleasant Street 550 feet from the southwest corner of its intersection with Main Street and easterly of a line bearing 182° from a point on the south side of Main Street 640 feet from the southwest corner of its intersection with Pleasant Street.

(562) (b) *Mystic River, east side of Tobin Bridge*. Beginning at a line running from a point on the Tobin Bridge at

(563) 42°23'08.5"N., 071°02'48.2"W. to a point at

(564) 42°23'06.4"N., 071°02'43.7"W.; thence northwest to a point at

(565) 42°23'09.1"N., 071°02'43.2"W. along the shoreline to the western side of Tobin Bridge, thence to the point of origin.

(566) (c) *Mystic River, west side of Tobin Bridge*. Beginning at a line running from a point on the Tobin Bridge at

(567) 42°23'08.8"N., 071°02'48.6"W. to a point at

(568) 42°23'10.5"N., 071°05'52"W.; thence northwest to the southeasterly corner of the pier at

(569) 42°23'13.4"N. 071°02'57.1"W. along the pier to the shoreline to the eastern side of Tobin Bridge, thence to the point of origin.

(570)

§110.30 Boston Harbor, MA.

(571) (a) *Vicinity of South Boston Yacht Club, South Boston*. Northerly of a line bearing 96° from the stack of the heating plant of the Boston Housing Authority in South Boston; easterly of a line bearing 5° from the west shaft of the tunnel of the Boston Main Drainage Pumping Station; southerly of the shoreline; and westerly of a line bearing 158° from the northeast corner of the iron fence marking the east boundary of the South Boston Yacht Club property.

(572) (b) *Dorchester Bay, in vicinity of Savin Hill Yacht Club*. Northerly of a line bearing 64° from the stack of the old power plant of the Boston Elevated Railway on Freeport Street in Dorchester; westerly of a line bearing 163° from the stack of the Boston Main Drainage Pumping

Station on the Cow Pasture in Dorchester; and southerly and easterly of the shoreline.

(573) (c) *Dorchester Bay, in vicinity of Dorchester Yacht Club*. Eastward of a line bearing 21° from the stack located a short distance northwestward of the Dorchester Yacht Club; southward of a line bearing 294° from the southerly channel pier of the highway bridge; westward of the highway bridge and the shoreline; and northward of the shoreline.

(574) (d) *Quincy Bay, in vicinity of Wollaston and Squantum Yacht Clubs*. Northwesterly of a line bearing 36°30' from a point on the shore 2,600 feet easterly of the east side of the Wollaston Yacht Club landing; southwesterly of a line bearing 129°15' from the water tank in Squantum; and southeasterly and northeasterly of the shoreline.

(575) (e) *Quincy Bay, in vicinity of Merrymount Yacht Club*. South of a line starting from a point bearing 246°, 3,510 yards, from the stack of the pumping station on Nut Island, and extending thence 306° to the shore; west of a line bearing 190° from the aforesaid point to the shore; and north and east of the shoreline.

(576) (f) *Weymouth Fore River, in vicinity of Quincy Yacht Club*. A line from

(577) 42°16'46.9"N., 70°57'12.5"W. to

(578) 42°16'48.8"N., 70°57'5.5"W.; thence to

(579) 42°16'31"N., 70°56'23.1"W. to the northerly end of Raccoon Island to

(580) 42°15'48"N., 70°56'43.4"W.; thence along the western shoreline of Raccoon Island to

(581) 42°15'46.4"N., 70°56'55.4"W.; thence to

(582) 42°15'43"N. 70°57'5.8"W.; thence along the shoreline to the point of origin. [NAD83]

(583) (g) *Weymouth Fore River, in vicinity of Wessagussett Yacht Club*. Southwesterly of a line bearing 117° from channel light "4"; southeasterly of a line 150 feet from and parallel to the meandering easterly limit of the dredged channel; easterly of a line bearing 188° from the eastern extremity of Rock Island Head; and northwesterly of the shoreline.

(584) (h) *Weymouth Fore River, in the vicinity of Gull Point (PT).* All of the waters bound by the following points beginning at

(585) 42°15'05"N., 70°57'26"W.; thence to

(586) 42°15'00"N., 70°57'26"W.; thence to

(587) 42°15'15"N., 70°56'50"W.; thence to

(588) 42°15'18"N., 70°56'50"W.; thence to the point of the beginning. [NAD83]

(589) (i) *Weymouth Back River, in vicinity of Eastern Neck*. The cove on the north side of the river lying northerly of a line bearing 264°30' from the southwesterly corner of the American Agricultural Chemical Company's wharf (Bradley's Wharf) to the shore of Eastern Neck, about 2,200 feet distant.

(590) (j) *Area No. 1 in Allerton Harbor*. That area north of Spinnaker Island beginning at

(591) 42°18'15.3"N., 70°53'44.1"W.; thence due east to

(592) 42°18'15.3"N., 70°53'27.6"W.; thence due south to

(593) 42°18'07.8"N., 70°53'27.6"W.; thence due west to

(594) 42°18'07.8"N., 70°53'44.1"W.; thence due north to the point of beginning. [NAD83]

(595) (k) *Area No. 2 in Hull Bay*. That area south of Hog Island beginning at

(596) 42°17'50.8"N., 70°54'05.1"W.; thence due east to

(597) 42°17'50.8"N., 70°53'27.6"W.; thence due south to

(598) 42°17'30.3"N., 70°53'27.6"W.; thence due west to

(599) 42°17'30.3"N., 70°54'5.1"W.; thence due north to the point of beginning. [NAD83]

(600) (l) *Area No. 3 in Hull Bay*. That area north of Bumkin Island beginning at

(601) 42°17'22.3"N., 70°54'5.1"W.; thence due east to

(602) 42°17'22.3"N., 70°53'15.6"W.; thence due south to

(603) 42°17'01.3"N., 70°53'15.6"W.; thence due west to

(604) 42°17'01.3"N., 70°54'5.17"W.; thence due north to the point of beginning. [NAD83].

(605) (m) *Hingham Harbor Area 1*. Beginning at

(606) 42°15'39.3"N., 70°53'22.1"W.; thence to

(607) 42°15'53.8"N., 70°53'30.1"W.; thence to

(608) 42°15'56.3"N., 70°53'21.1"W.; thence to

(609) 42°15'42.3"N., 70°53'13.1"W.; thence to point of beginning. [NAD83]

(610) (n) *Hingham Harbor Area 2*. Beginning at

(611) 42°15'30.6"N., 70°53'0.5"W.; thence to

(612) 42°15'30.3"N., 70°53'11.6"W.; thence to

(613) 42°15'27.8"N., 70°53'16.1"W.; thence to

(614) 42°15'28.8"N., 70°53'29.1"W.; thence to

(615) 42°15'35.3"N., 70°53'32.1"W.; thence to

(616) 42°15'36.3"N., 70°53'34.6"W.; thence to

(617) 42°15'41.3"N., 70°53'32.6.5"W.; thence to

(618) 42°15'31.3"N., 70°53'26.1"W.; thence to

(619) 42°15'31.8"N., 70°53'01.1"W.; thence to point of beginning. [NAD83]

(620) (o) *Hingham Harbor Area 3*. Beginning at

(621) 42°15'33.3"N., 70°52'59.6"W.; thence to

(622) 42°15'33.8"N., 70°53'17.1"W.; thence to

(623) 42°15'35.8"N., 70°53'00.1"W.; thence to point of beginning. [NAD83]

(624) (p) *Hingham Harbor Area 4*. Beginning at

(625) 42°14'47.3"N., 70°53'07.6"W.; thence

(626) 42°14'48.8"N., 70°53'9.6"W.; thence to

(627) 42°14'54.3"N., 70°53'6.1"W.; thence to

(628) 42°14'56.9"N., 70°52'56.6"W.; thence to point of beginning. [NAD83]

(629) (q) *Hingham Harbor Area 5*. Beginning at

(630) 42°14'48.3"N., 70°52'55.1"W.; thence to

(631) 42°14'48.8"N., 70°53'0.1" W.; thence to

(632) 42°14'58.3"N., 70°52'49.1"W.; thence to

(633) 42°14'53.8"N., 70°52'48.1"W.; thence to point of beginning. [NAD83]

(634)

§110.31 Hull Bay and Allerton Harbor at Hull, MA

(635) (a) *Area No. 1 in Allerton Harbor*. That area north of Hog Island beginning at

(636) 42°18'15"N., 70°53'46"W.; thence due east to

(637) 42°18'15"N., 70°53'29.5"W.; thence due south to

(638) 42°18'07.5"N., 70°53'29.5"W.; thence due west to

(639) 42°18'07.5"N., 70°53'46"W.; thence due north to the point of beginning.

(640) (b) *Area No. 2 in Hull Bay*. That area south of Hog Island beginning at

(641) 42°17'50.5"N., 70°54'07"W.; thence due east to

(642) 42°17'50.5"N., 70°53'29.5"W.; thence due south to

(643) 42°17'30"N., 70°53'29.5"W.; thence due west to

(644) 42°17'30"N., 70°54'07"W.; thence due north to the point of beginning.

(645) (c) *Area No. 3 in Hull Bay*. That area north of Bumkin Island beginning at

(646) 42°17'22"N., 70°54'07"W.; thence due east to

(647) 42°17'22"N., 70°53'17.5"W.; thence due south to

(648) 42°17'01"N., 70°53'17.5"W.; thence due west to

(649) 42°17'01"N., 70°54'07"W.; thence due north to the point of beginning.

(650) **Note**: The areas will be principally for use by yachts and other recreational craft. Temporary floats or buoys for marking anchors will be allowed. Fixed mooring piles or stakes are prohibited. The anchoring of vessels and the placing of temporary moorings is under the jurisdiction, and at the discretion, of the local Harbor Master, Hull, MA.

(651)

§110.32 Hingham Harbor, Hingham, MA

(652) (a) *Area 1*. Beginning at

(653) 42°15'39"N., 70°53'24"W.; thence to

(654) 42°15'53.5"N., 70°53'32.0"W.; thence to

(655) 42°15'56.0"N., 70°53'23.0"W.; thence to

(656) 42°15'42.0"N., 70°53'15.0"W., thence to point of beginning.

(657) (b) *Area 2*. Beginning at

(658) 42°15'30.0"N., 70°53'02.5"W.; thence to

(659) 42°15'30.0"N., 70°53'13.5"W.; thence to

(660) 42°15'27.5"N., 70°53'18.0"W.; thence to

(661) 42°15'28.5"N., 70°53'31.0"W.; thence to

(662) 42°15'35.0"N., 70°53'34.0"W.; thence to

(663) 42°15'36.0"N., 70°53'36.5"W.; thence to

(664) 42°15'41.0"N., 70°53'34.5"W.; thence to

(665) 42°15'31.0"N., 70°53'28.0"W.; thence to

(666) 42°15'31.5"N., 70°53'03.0"W.; thence to point of beginning.

(667) (c) *Area 3*. Beginning at

(668) 42°15'33.0"N., 70°53'01.5"W.; thence to

(669) 42°15'33.5"N., 70°53'19.0"W.; thence to

(670) 42°15'35.5"N., 70°53'02.0"W.; thence to point of beginning.

(671) (d) *Area 4*. Beginning at

(672) 42°14'47.0"N., 70°53'09.5"W.; thence to

(673) 42°14'48.5"N., 70°53'11.5"W.; thence to

(674) 42°14'54.0"N., 70°53'08.0"W.; thence to

(675) 42°14'56.5"N., 70°52'58.5"W.; thence to point of beginning.

(676) (e) *Area 5*. Beginning at

(677) 42°14'48.0"N., 70°52'57.0"W.; thence to

(678) 42°14'48.5"N., 70°53'02.0"W.; thence to

(679) 42°14'58.0"N., 70°52'51.0"W.; thence to

(680) 42°14'53.5"N., 70°52'50.0"W.; thence to point of beginning.

(681) **Note:** The areas will be principally for use by yachts and other recreational craft. Temporary floats or buoys for marking anchors will be allowed in the areas but fixed piles or stakes may not be placed. The anchoring of vessels and the placing of moorings will be under the jurisdiction of the local Harbor Master.

(682)

§110.37 Sesuit Harbor, Dennis, MA

(683) All the waters of Sesuit Harbor southerly of a line extending between the outer end of the jetties on each side of the entrance to the Harbor.

(684)

Subpart B–Anchorage Grounds

(685)

§110.130 Bar Harbor, Maine.

(686) (a) *Anchorage grounds*. (1) Anchorage "A" is that portion of Frenchman Bay, Bar Harbor, ME enclosed by a rhumb line connecting the following points:

(687) 44°23'43"N., 68°12'00"W.; thence to

(688) 44°23'52"N., 68°11'22"W.; thence to

(689) 44°23'23"N., 68°10'59"W.; thence to

(690) 44°23'05"N., 68°11'32"W.; returning to start.

(691) (2) Anchorage "B" is that portion of Frenchman Bay, Bar Harbor, ME enclosed by a rhumb line connecting the following points:

(692) 44°24'33"N., 068°13'09"W.; thence to

(693) 44°24'42"N., 068°11'47"W.; thence to copied

(694) 44°24'11"N., 068°11'41"W.; thence to

(695) 44°24'02"N., 68°13'03"W.; returning to start.

(696) (b) *Regulations*. (1) Anchorage A is a general anchorage ground reserved for passenger vessels, small commercial vessels and pleasure craft. Anchorage B is a general anchorage ground reserved primarily for passenger vessels 200 feet and greater.

(697) (2) These anchorage grounds are authorized for use year round.

(698) (3) Temporary floats or buoys for marking anchors will be allowed in all anchorage areas.

(699) (4) Fixed mooring, piles or stakes are prohibited.

(700) (5) Any vessels anchored in this area shall be capable of moving and when ordered to move by the Captain of the Port shall do so with reasonable promptness.

(701) (6) The anchoring of vessels is under the coordination of the local Harbormaster.

(702)

§110.131 Sheepscot River in the vicinity of Edgecomb, Maine.

(703) (a) *Anchorage grounds*. All of the waters enclosed by a line starting from a point located at the southwestern end of Davis Island at

(704) 43°59.655'N., 69°39.617'W., thence to

(705) 43°59.687'N., 69°39.691'W.; thence to

(706) 43°59.847'N., 69°39.743'W.; thence to

(707) 43°59.879'N., 69°39.559'W.; thence to
(708) 43°59.856'N., 69°39.488'W.; thence to
(709) 43°59.771'N., 69°39.585'W.; thence to the point of beginning. DATUM: NAD 83

(710) (b) *Regulations*. (1) This anchorage is reserved for vessels of all types, with drafts of 3 to 12 feet.

(711) (2) These anchorage grounds are authorized for use from May through October.

(712) (3) Vessels are limited to a maximum stay of 1 week.

(713) (4) Fixed moorings, piles of stakes are prohibited.

(714) (5) Vessels must not anchor so as to obstruct the passage of other vessels proceeding to or from other anchorage spaces.

(715) (6) Anchors must not be placed in the channel and no portion of the hull or rigging of any anchored vessel shall extend outside the limits of the anchorage area.

(716) (7) The anchorage of vessels is under the coordination of the local Harbormaster.

(717)

§110.132 Rockland Harbor, Maine.

(718) (a) *The anchorage grounds*–(1) *Anchorage A*. Beginning at a point bearing 158°, 1,075 yards, from Rockland Breakwater Light; thence 252°, 2,020 yards, to a point bearing 224° from Rockland Breakwater Light; thence 345°, 740 yards, to a point bearing 242° from Rockland Breakwater Light; thence 72°, 1,300 yards , to a point bearing 222° from Rockland Breakwater Light; and thence 120°, 1,000 yards, to the point of beginning.

(719) (2) *Anchorage B*. Beginning at a point bearing 273°, 400 yards, from Rockland Breakwater Light; thence 273°, 700 yards, to a point bearing 273° from Rockland Breakwater Light; thence 349°, 850 yards, to a point bearing 305° from Rockland Breakwater Light; thence 89°, 700 yards, to a point bearing 328° from Rockland Breakwater Light; and thence 169°, 900 yards, to the point of beginning.

(720) (b) *The regulations*. (1) Anchorages A and B are general anchorage grounds reserved for merchant vessels, commercial vessels or passenger vessels over 65 feet in length. Fixed moorings, piles or stakes are prohibited.

(721) (2) A distance of approximately 500 yards shall be left between Anchorages A and B for vessels entering or departing from the Port of Rockland. A distance of approximately 100 yards shall be left between Anchorage A and the Special Anchorage Area for vessels entering or departing facilities in the vicinity of Atlantic Point. Any vessel anchored in these anchorages shall be capable of moving and when ordered to move by the Captain of the Port shall do so with reasonable promptness.

(722) (3) All other vessels within the Rockland Harbor area are prohibited from anchoring within 300 yards or operating within 100 feet of any navy yard, shipbuilding plant, power plant, oil terminal, marine terminal, munitions plant, military or naval arsenal or depot, warehouse, or freight pier without permission from the Captain of the Port, Rockland, Maine, or his authorized representative.

(723)

§110.133 Kennebec River in vicinity of Bath, Maine.

(724) (a) *The anchorage grounds*. Vessels may anchor only within the following limits:

(725) (1) Northward of a line bearing 54° true and extending from a point on Passmore's wharf in prolongation with the north side of Commerce Street, Bath, Maine, to a point on the shore in Woolwich, approximately 1,200 feet north of the Maine Central Railroad wharf.

(726) (2) Southward of a line drawn from the derrick on the Bath Iron Works wharf to Sassanoa Point in Woolwich.

(727) (b) *The regulations*. (1) Vessels in the north anchorage shall be so anchored as to leave a clear fairway of 150 feet channelward of the established harbor lines at Bath, and a clear fairway 200 feet from the east or Woolwich shore, for the passage of steamers, tows, rafts, and other watercraft.

(728) (2) The launching of vessels into the waters between the anchorages or the bringing up of such vessels by their anchors will be permitted: *Provided,* that the vessels so launched shall be removed there from within 12 hours from the time of anchorage.

(729)

§110.134 Portland Harbor, Maine.

(730) (a) *The anchorage grounds*–(1) *Anchorage A (general).* Beginning at 43°39'37"N., 070°14'35"W.; thence approximately 090° for 1550 yards to Fort Gorges Island Ledge Buoy 4; thence 350° for 300 yards; thence 025° for 780 yards; thence 303° for 750 yards; thence 254° for 560 yards; thence 186° for 750 yards; and thence to the point of beginning.

(731) (2) *Anchorage B (general-primarily intended for deep draft vessels).* Beginning at Fort Gorges Island Ledge Buoy 4; thence 062° to Little Diamond Island; thence along the southwestern shore to the pier on the southern end of Little Diamond Island; 133° for 1200 yards; 270° to House Island Light; thence along the western shore of House Island to Fort Scammel Point Light; thence 325° for 1700 yards to the point of beginning.

(732) (3) *Anchorage C*. Bounded on the northwest by House Island; on the north by a line running 90° from House Island Light to Peak Island; on the east by the western shore of Peak Island, by a line running 198° from the westernmost point on Peak Island to Cushing Island, and by the shore of Cushing Island to its westernmost point; and on the southwest by a line running from the westernmost point on Cushing Island to Fort Scammel Point Light.

(733) (b) *The regulations*. (1) Anchorage B is intended for general purposes, but especially for use by oil tankers and other large deep-draft ships entering harbor at night and intending to proceed to the dock allotted at daylight the following morning or as soon as practicable. This area is also to be used for quarantine anchorage. Vessels must be so anchored in this area as to leave at all times an open usable channel at least 100 feet wide for passage of ferry and other boats between Portland, Peak Island, and Bay

Points. Any vessels anchored in this area shall be ready to move on short notice when ordered to do so by the Captain of the Port.

(734) (2) Anchorage C is intended for use only by small vessels and for temporary anchorage.

(735)

§110.138 Boston Harbor, MA

(736) (a) *The anchorage grounds*–(1) *Bird Island Anchorage*. Beginning at a point bearing 93°, 1,400 yards, from the aerial beacon on top of the Boston Custom House tower; thence to a point bearing 81°, 1,600 yards, from the aerial beacon on top of the Boston Custom House tower; thence to a point bearing 102°, 3,100 yards, from the aerial beacon on top of the Boston Custom House tower; thence to a point bearing 109°, 3,050 yards, from the aerial beacon on top of the Boston Custom House tower; and thence to the point of beginning.

(737) (2) *President Roads Anchorage*–(i) *40-foot anchorage*. Beginning at a point bearing 237°, 522 yards from Deer Island Light; thence to a point bearing 254°, 2,280 yards from Deer Island Light; thence to a point bearing 261°, 2,290 yards from Deer Island Light; thence to a point bearing 278°, 2,438 yards from Deer Island Light; thence to a point bearing 319°, 933 yards from Deer Island Light; thence to a point bearing 319°, 666 yards from Deer Island Light; and thence to point of beginning.

(738) (ii) *35-foot anchorage*. Beginning at a point bearing 256°, 2,603 yards from Deer Island Light; thence to a point bearing 258°30', 3,315 yards from Deer Island Light; thence to a point bearing 264°, 3,967 yards from Deer Island Light; thence to a point bearing 261°, 2,290 yards from Deer Island Light; and thence to point of beginning.

(739) (3) *Long Island Anchorage*. East of Long Island, bounded as follows: Beginning at the southwesternmost point of Gallups Island; thence 270° to Long Island; thence southerly along the eastern shore line of Long Island to Bass Point; thence to the northernmost point of Rainsford Island; thence to Georges Island Gong Buoy 6; and thence to the point of beginning.

(740) (4) *Castle Island Anchorage*. Bounded on the north by Castle Island and adjacent land; on the east by a line between Castle Rocks Fog Signal Light and Old Harbor Shoal Buoy 2; on the southeast by a line between Old Harbor Shoal Buoy 2 and Old Harbor Buoy 4; and on the west by a line running due north from Old Harbor Buoy 4 to the shore line at City Point.

(741) (5) *Explosives anchorage*. In the lower harbor, bounded on the northeast by a line between the northeast end of Peddocks Island and the northeast end of Rainsford Island; on the northwest by Rainsford Island; on the southwest by a line between the western extremity of Rainsford Island and the westernmost point of Peddocks Island; and on the southeast by Peddocks Island.

(742) (b) *The regulations*. (1) The Captain of the Port may authorize the use of the President Roads Anchorage as an explosives anchorage when he finds that the interests of commerce will be promoted and that safety will not be prejudiced thereby. Vessels anchored in this area shall move promptly upon notification by the Captain of the Port.

(743) (2) In the Long Island Anchorage vessels shall anchor in the position designated by the Captain of the Port.

(744) (3) Floats or buoys for marking anchors or moorings in place will be allowed in all areas. Fixed mooring piles or stakes are prohibited.

(745)

Part 117–DrawbridgeOperation Regulations

(746)

Subpart A–General Requirements

(747)

§117.1 Purpose.

(748) (a) This part prescribes the general and special drawbridge operating regulations that apply to the drawbridges across the navigable waters of the United States and its territories. The authority to regulate drawbridges across the navigable waters of the United States is vested in the Secretary of Homeland Security.

(749) (b) Subpart A contains the general operation requirements that apply to all drawbridges.

(750) (c) Subpart B contains specific requirements for operation of individual drawbridges. These requirements are in addition to or vary from the general requirements in Subpart A. Specific sections in subpart B that vary from a general requirement in Subpart A supersede the general requirement. All other general requirements in Subpart A, that are not at variance, apply to the drawbridges and removable span bridges listed in Subpart B.

(751)

§117.4 Definitions.

(752) The following definitions apply to this part:

(753) *Appurtenance* means an attachment or accessory extending beyond the hull or superstructure that is not an integral part of the vessel and is not needed for a vessel's piloting, propelling, controlling, or collision avoidance capabilities.

(754) *Automated drawbridge* means a drawbridge that is operated by an automated mechanism, not a drawtender. An automated drawbridge is normally kept in the open to navigation position and closes when the mechanism is activated.

(755) *Deviation* means a District Commander's action authorizing a drawbridge owner to temporarily not comply with the drawbridge opening requirements in this part.

(756) *Drawbridge* means a bridge with an operational span that is intended to be opened for the passage of waterway traffic.

(757) *Drawspan* means the operational span of a drawbridge.

(758) *Lowerable* means a non-structural vessel appurtenance that is or can be made flexible, hinged, collapsible, or telescopic so that it can be mechanically or manually lowered.

(759) *Nonstructural* means that the item is not rigidly fixed to the vessel and can be relocated or altered.

(760) *Not essential to navigation* means that a nonstructural vessel appurtenance, when in the lowered position, would not adversely affect the vessel's piloting, propulsion, control, or collision-avoidance capabilities.

(761) *Public vessel* means a vessel that is owned and operated by the United States Government and is not engaged in commercial service, as defined in 46 U.S.C. 2101.

(762) *Remotely operated drawbridge* means a drawbridge that is operated by remote control from a location away from the drawbridge.

(763) *Removable span bridge* means a bridge that requires the complete removal of a span by means other than machinery installed on the bridge to open the bridge to navigation.

(764) *Untended* means that there is no drawtender at the drawbridge.

(765)

§117.5 When the drawbridge must open.

(766) Except as otherwise authorized or required by this part, drawbridges must open promptly and fully for the passage of vessels when a request or signal to open is given in accordance with this subpart.

(767)

§117.7 General requirements of drawbridge owners.

(768) Except for drawbridges that have been authorized, before January 3, 2007, to remain closed to navigation or as otherwise specified in subpart B, drawbridge owners must:

(769) (a) Provide the necessary drawtender(s) for the safe and prompt opening of the drawbridge.

(770) (b) Maintain the working machinery of the drawbridge in good operating condition.

(771) (c) Cycle the drawspan(s) periodically to ensure operation of the drawbridge.

(772) (d) Ensure that the drawbridge operates in accordance with the requirements of this part.

(773) (e) Any drawbridge allowed to remain closed to navigation prior to January 3, 2007, when necessary, must be returned to operable condition within the designated time set forth by the District Commander and will become subject to the requirements of this part.

(774)

§117.8 Permanent changes to drawbridge operation.

(775) (a) Anyone may submit a written request to the District Commander for a permanent change to a drawbridge operating requirement. The request must include documentation supporting or justifying the requested change.

(776) (b) If after evaluating the request, the District Commander determines that the requested change is not needed, he or she will respond to the request in writing and provide the reasons for denial of the requested change.

(777) (c) If the District Commander decides that a change may be needed, he or she will begin a rulemaking to implement the change.

(778)

§117.9 Delaying opening of a draw.

(779) No person shall unreasonably delay the opening of a draw after the signals required by §117.15 have been given.

(780) **Note:** Trains are usually controlled by the block method. That is, the track is divided into blocks or segments of a mile or more in length. When a train is in a block with a drawbridge, the draw may not be able to open until the train has passed out of the block and the yardmaster or other manager has "unlocked" the drawbridge controls. The maximum time permitted for delay is defined in Subpart B for each affected bridge. Land and water traffic should pass over or through the draw as soon as possible in order to prevent unnecessary delays in the opening and closure of the draw.

(781)

§117.11 Unnecessary opening of the draw.

(782) No vessel owner or operator shall –

(783) (a) Signal a drawbridge to open if the vertical clearance is sufficient to allow the vessel, after all lowerable nonstructural vessel appurtenances that are not essential to navigation have been lowered, to safely pass under the drawbridge in the closed position; or

(784) (b) Signal a drawbridge to open for any purpose other than to pass through the drawbridge opening.

(785)

§117.15 Signals.

(786) (a) *General.* (1) The operator of each vessel requesting a drawbridge to open shall signal the drawtender and the drawtender shall acknowledge that signal. The signal shall be repeated until acknowledged in some manner by the drawtender before proceeding.

(787) (2) The signals used to request the opening of the draw and to acknowledge that request shall be sound signals, visual signals, or radiotelephone communications described in this subpart.

(788) (3) Any of the means of signaling described in this subpart sufficient to alert the party being signaled may be used.

(789) (b) *Sound signals*. (1) Sound signals shall be made by whistle, horn, megaphone, hailer, or other device capable of producing the described signals loud enough to be heard by the drawtender.

(790) (2) As used in this section, prolonged blast means a blast of four to six seconds duration and short blast means a blast of approximately one second duration.

(791) (3) The sound signal to request the opening of a draw is one prolonged blast followed by one short blast sounded not more than three seconds after the prolonged blast. For vessels required to be passed through a draw during a scheduled closure period, the sound signal to request the opening of the draw during that period is five short blasts sounded in rapid succession.

(792) (4) When the draw can be opened immediately, the sound signal to acknowledge a request to open the draw is one prolonged blast followed by one short blast sounded not more than 30 seconds after the requesting signal.

(793) (5) When the draw cannot be opened immediately, or is open and shall be closed promptly, the sound signal to acknowledge a request to open the draw is five short blasts sounded in rapid succession not more than 30 seconds after the vessel's opening signal. The signal shall be repeated until acknowledged in some manner by the requesting vessel.

(794) (c) *Visual signals*. (1) The visual signal to request the opening of a draw is–

(795) (i) A white flag raised and lowered vertically; or

(796) (ii) A white, amber, or green light raised and lowered vertically.

(797) (2) When the draw can be opened immediately, the visual signal to acknowledge a request to open the draw, given not more than 30 seconds after the vessel's opening signal, is–

(798) (i) A white flag raised and lowered vertically;

(799) (ii) A white, amber, or green light raised and lowered vertically, or

(800) (iii) A fixed or flashing white, amber, or green light or lights.

(801) (3) When the draw cannot be opened immediately, or is open and must be closed promptly, the visual signal to acknowledge a request to open the draw is–

(802) (i) A red flag or red light swung back and forth horizontally in full sight of the vessel given not more than 30 seconds after the vessel's opening signal; or

(803) (ii) A fixed or flashing red light or lights given not more than 30 seconds after the vessel's opening signal.

(804) (4) The acknowledging signal when the draw cannot open immediately or is open and must be closed promptly shall be repeated until acknowledged in some manner by the requesting vessel.

(805) (d) *Radio telephone communications*. (1) Radiotelephones may be used to communicate the same information provided by sound and visual signals.

(806) (2) The vessel and the drawtender shall monitor the frequency used until the vessel has cleared the draw.

(807) (3) When radiotelephone contact cannot be initiated or maintained, sound or visual signals under this section shall be used.

(808)

§117.17 Signalling for contiguous drawbridges.

(809) When a vessel must pass two or more drawbridges close together, the opening signal is given for the first bridge. After acknowledgment from the first bridge that it will promptly open, the opening signal is given for the second bridge, and so on until all bridges that the vessel must pass have been given the opening signal and have acknowledged that they will open promptly.

(810)

§117.19 Signalling when two or more vessels are approaching a drawbridge.

(811) When two or more vessels are approaching the same drawbridge at the same time, or nearly the same time, whether from the same or opposite directions, each vessel shall signal independently for the opening of the draw and the drawtender shall reply in turn to the signal of each vessel. The drawtender need not reply to signals by vessels accumulated at the bridge for passage during a scheduled open period.

(812)

§117.21 Signalling for an opened drawbridge.

(813) When a vessel approaches a drawbridge with the draw in the open position, the vessel shall give the opening signal. If no acknowledgment is received within 30 seconds, the vessel may proceed, with caution, through the open draw.

(814)

§117.23 Installation of radiotelephones.

(815) (a) When the District Commander deems it necessary for reasons of safety of navigation, the District Commander may require the installation and operation of a radiotelephone on or near a drawbridge.

(816) (b) The District Commander gives written notice of the proposed requirement to the bridge owner.

(817) (c) All comments the owner wishes to submit shall be submitted to the District Commander within 30 days of receipt of the notice under paragraph (b) of this section.

(818) (d) If, upon consideration of the comments received, the District Commander determines that a radiotelephone is necessary, the District Commander notifies the bridge owner that a radiotelephone shall be installed and gives a reasonable time, not to exceed six months, to install the radiotelephone and commence operation.

(819)

§117.24 Radiotelephone installation identification.

(820) (a) The Coast Guard authorizes, and the District Commander may require the installation of a sign on drawbridges, on the upstream and downstream sides, indicating that the bridge is equipped with and operates a VHF radiotelephone in accordance with §117.23.

(821) (b) The sign shall give notice of the radiotelephone and its calling and working channels–

(822) (1) In plain language; or

(823) (2) By a sign consisting of the outline of a telephone handset with the long axis placed horizontally and a vertical three-legged lightning slash superimposed over the handset. The slash shall be as long vertically as the handset is wide horizontally and normally not less than 27 inches and no more than 36 inches long. The preferred calling channel should be shown in the lower left quadrant

and the preferred working channel should be shown in the lower right quadrant.

(824)

§117.31 Drawbridge operations for emergency vehicles and emergency vessels.

(825) (a) Upon receiving notification that an emergency vehicle is responding to an emergency situation, a drawtender must make all reasonable efforts to have the drawspan closed at the time the emergency vehicle arrives.

(826) (b) When a drawtender receives notice, or a proper signal as provided in §117.15 of this part, the drawtender shall take all reasonable measures to have the draw opened, regardless of the operating schedule of the draw, for passage of the following, provided this opening does not conflict with local emergency management procedures which have been approved by the cognizant Coast Guard Captain of the Port:

(827) (1) Federal, State, and local government vessels used for public safety;

(828) (2) vessels in distress where a delay would endanger life or property;

(829) (3) commercial vessels engaged in rescue or emergency salvage operations; and

(830) (4) vessels seeking shelter from severe weather.

(831)

§117.33 Closure of draw for natural disasters or civil disorders.

(832) Drawbridges need not open for the passage of vessels during periods of natural disasters or civil disorders declared by the appropriate authorities unless otherwise provided for in Subpart B or directed to do so by the District Commander.

(833)

§117.35 Temporary change to a drawbridge operating schedule.

(834) (a) For any temporary change to the operating schedule of a drawbridge, lasting less than or equal to 180 days, the District Commander may issue a deviation approval letter to the bridge owner and publish a "Notice of temporary deviation from regulations" in the Federal Register.

(835) (b) If the time period for a temporary change to the operating schedule of a drawbridge will be greater then 180 days, the District Commander will follow appropriate rulemaking procedures and publish a temporary rule in the Federal Register prior to the start of the action.

(836) (c) *Request for change*. (1) To temporarily change the drawbridge-operating requirements the bridge owner must submit a written request to the District Commander for approval of the change.

(837) (2) The request must describe the reason for the deviation and the dates and times scheduled for the start and end of the change.

(838) (3) Requests should be submitted as early as possible, preferably 90 days before the start of the action. District Commanders have discretion to accept requests submitted less than 90 days before a needed change if those requests can be processed before the date of the needed change.

(839) (d) *Determination*. The District Commander's determination to allow the schedule change is normally forwarded to the bridge owner within ten working days after receipt of the request. If the request is denied, the reasons for the denial will be set out in the District Commander's decision letter.

(840) (e) The drawbridge must return to its regular operating schedule immediately at the end of the designated time period.

(841) (f) If the authorized deviation period for an event is broken into separate time periods on the same day or on consecutive days, the drawbridge must provide openings for navigation between authorized schedule changes.

(842) (g) The District Commander will also announce the change to the operating schedule in the Local Notice to Mariners and other appropriate local media.

(843)

§117.36 Closure of drawbridge for emergency repair.

(844) (a) When a drawbridge unexpectedly becomes inoperable, or should be immediately rendered inoperable because of mechanical failure or structural defect, the drawbridge owner must notify the District Commander of the closure without delay and give the reason for the emergency closure of the drawbridge and an estimated time when the drawbridge will be returned to operating condition.

(845) (b) The District Commander will notify mariners about the drawbridge status through Broadcast Notices to Mariners, Local Notice to Mariners and any other appropriate local media.

(846) (c) Repair work under this section must be performed with all due speed in order to return the drawbridge to operation as soon as possible.

(847)

§117.39 Authorized closure of drawbridge due to infrequent requests for openings.

(848) (a) When there have been no requests for drawbridge openings for at least two years, a bridge owner may request in writing that the District Commander authorize the drawbridge to remain closed to navigation and to be untended.

(849) (b) The District Commander may:

(850) (1) Authorize the closure of the drawbridge;

(851) (2) Set out any conditions in addition to the requirement in paragraph (d): and

(852) (3) Revoke an authorization and order the drawbridge returned to operation when necessary.

(853) (c) All drawbridges authorized to remain closed to navigation, under this section, must be maintained in operable condition.

(854) (d) Authorization under this section does not:

(855) (1) Authorize physical changes to the drawbridge structure, or

(856) (2) Authorize removal of the operating machinery.

(857) (e) Drawbridges authorized under this section to remain closed to navigation and to be untended are identified in subpart B of this part.

(858)

§117.40 Advance notice for drawbridge opening.

(859) (a) Upon written request by the owner of a drawbridge, the District Commander may authorize a drawbridge to operate under an advance notice for opening. The drawbridge tender, after receiving the advance notice, must open the drawbridge at the requested time and allow for a reasonable delay in arrival of the vessel giving the advance notice.

(860) (b) If the request is approved, a description of the advanced notice for the drawbridge will be added to subpart B of this part.

(861)

§117.41 Maintaining drawbridges in the fully open position.

(862) (a) Drawbridges permanently maintained in the fully open to navigation position may discontinue drawtender service as long as the drawbridge remains fully open to navigation. The drawbridge must remain in the fully open position until drawtender service is restored.

(863) (b) If a drawbridge is normally maintained in the fully open to navigation position, but closes to navigation for the passage of pedestrian, vehicular, rail, or other traffic, the drawbridge must be tended unless:

(864) (1) Special operating requirements are established in subpart B of this part for that drawbridge; or

(865) (2) The drawbridge is remotely operated or automated.

(866)

§117.42 Remotely operated and automated drawbridges.

(867) (a) Upon written request by the owner of a drawbridge, the District Commander may authorize a drawbridge to operate under an automated system or from a remote location.

(868) (b) If the request is approved, a description of the full operation of the remotely operated or automated drawbridge will be added to subpart B of this part.

(869)

§117.47 Clearance gauges.

(870) (a) Clearance gauges are required for drawbridges across navigable waters of the United States discharging into the Atlantic Ocean south of Delaware Bay (including the Lewes and Rehoboth Canal, DE) or into the Gulf of America (including coastal waterways contiguous thereto and tributaries to such waterways and the Lower Atchafalaya River, LA), except the Mississippi River and its tributaries and outlets.

(871) (b) Except for provisions in this part which specify otherwise for particular drawbridges, clearance gauges shall be designed, installed, and maintained according to the provisions of 33 CFR 118.160 (not carried in this Coast Pilot).

(872) **Note**: Clearance gauge requirements, if any, for drawbridges other than those referred to in this section are listed in Subpart B under the appropriate bridge.

(873)

§117.49 Process of violations.

(874) (a) Complaints of alleged violations under this part are submitted to the District Commander of the Coast Guard District in which the drawbridge is located.

(875) (b) Penalties for violations under this part are assessed and collected under Subpart 1.07 of Part 1 of this chapter (not published in this Coast Pilot; see 33 CFR 1.07).

(876)

Subpart B–Specific Requirements

(877)

§117.51 General.

(878) The drawbridges in this subpart are listed by the state in which they are located and by the waterway they cross. Waterways are arranged alphabetically by state. The drawbridges listed under a waterway are generally arranged in order from the mouth of the waterway moving upstream. The drawbridges on the Atlantic Intracoastal Waterway are listed from north to south and on the Gulf Intracoastal Waterway from east to west.

(879)

§117.55 Posting of requirements.

(880) (a) The owner of each drawbridge under this subpart, other than removable span bridges, must ensure that a sign summarizing the requirements in this subpart applicable to the drawbridge is posted both upstream and downstream of the drawbridge. The requirements to be posted need not include those in Subpart A or §§117.51 through 117.59 of this part.

(881) (b) The signs shall be of sufficient size and so located as to be easily read at any time from an approaching vessel.

(882) (c) If advance notice is required to open the draw, the signs shall also state the name, address, and telephone number of the person to be notified.

(883)

§117.59 Special requirements due to hazards.

(884) For the duration of occurrences hazardous to safety or navigation, such as floods, freshets, and damage to the bridge or fender system, the District Commander may require the owner of an operational drawbridge listed in this subpart to have the bridge attended full time and open on signal.

(885)

MAINE

(886)

§117.521 [Removed].

(887)

§117.523 Back River.

(888) The draw of the Barter's Island Bridge, mile 2.0, between Hodgdon and Barters Island at Boothbay, shall open on signal from June 1 through October 31; except that, from 5 p.m. to 8 a.m., the draw shall be opened on signal if notice was given to the drawtender from 8 a.m. to 5 p.m. From November 1 through May 31 the draw shall open on signal if at least 24 hours notice is given to the drawtender or to the Maine Department of Transportation at Augusta.

(889)

§117.525 Kennebec River.

(890) The draw of the Carlton Bridge, mile 14.0, between Bath and Woolwich shall operate as follows:

(891) (a) From May 15 through September 30 the draw shall open on signal; except that, from 5 p.m. to 8 a.m., the draw shall open on signal if a two-hour notice is given by calling the number posted at the bridge.

(892) (b) From October 1 through May14 the draw shall open on signal; except that, from 5 p.m. to 8 a.m., the draw shall open on signal after a twenty-four hours notice is given from 8 a.m. to 5 p.m., on Saturday and Sunday, after an eight-hour notice is given by calling the number posted at the bridge.

(893)

§117.527 [Removed].

(894)

§117.529 [Removed].

(895)

§117.531 Piscataqua River.

(896) (a) The following requirements apply to all bridges across the Piscataqua River.

(897) (1) Public vessels of the United States, commercial vessels over 100 gross tons, inbound ferry service vessels and inbound commercial fishing vessels must be passed through the drawspan of each drawbridge as soon as possible. The opening signal from these vessels is four or more short blasts of a whistle, horn or a radio request.

(898) (2) The owners of these bridges shall provide and keep in good legible condition clearance gauges for each draw with figures not less than 18 inches high designed, installed and maintained according to the provisions of §118.160 (not carried in this Coast Pilot).

(899) (3) Trains and locomotives shall be controlled so that any delay in opening the draw shall not exceed five minutes. However, if a train moving toward the bridge has crossed the home signal for the bridge before the signal requesting opening of the bridge is given, that train may continue across the bridge and must clear the bridge interlocks before stopping.

(900) (4) Except as provided in paragraphs (b) through (c) of this section the draws shall open on signal.

(901) (b) The draw of the Memorial (US 1) bridge, mile 1.9, shall open on signal; except that from 15 May through 31 October, from 7 a.m. to 7 p.m., the draw need be opened only on the hour and half hour for recreational vessels and commercial vessels less than 100 gross tons except as provided in (a)(1).

(902) (c) The draw of the Sarah M. Long (Route 1 Bypass) bridge, mile 2.5, shall open as follows:

(903) (1) The main ship channel draw shall open on signal; except that from 15 May through 31 October, from 7 a.m. to 7 p.m., the draw need be opened only at quarter of and quarter after the hour for recreational vessels and commercial vessels less than 100 gross tons except as provided in (a)(1).

(904) (2) [Reserved]

(905)

§117.533 Sheepscot River.

(906) The draw of the Maine Central Railroad Bridge, mile 15.0, between Wiscasset and North Edgecombe, Maine, need not be opened for the passage of vessels. The draw of the Maine Central Railroad Bridge shall be returned to operable condition within six months after notification by the District Commander to do so.

(907)

§117.537 Townsend Gut

(908) The draw of the Southport (SR27) Bridge, at mile 0.7, across Townsend Gut between Boothbay Harbor and Southport, Maine shall open on signal; except that, from April 29 through September 30, between 6 a.m. and 6 p.m, the draw shall open on signal on the hour and half hour only, after an opening request is given.

(909)

MASSACHUSETTS

(910)

§117.586 Annisquam River and Blynman Canal.

(911) The draw of the Blynman (SR127) Bridge shall open on signal, except that, from noon to 6 p.m. on Thanksgiving Day, 6 p.m. on December 24 to midnight on December 25, and from 6 p.m. on December 31 to midnight on January 1, the draw shall open on signal if at least a two-hour advance notice is given by calling the number posted at the bridge.

(912)

§117.588 Base River.

(913) The Hall Whitaker Bridge, mile 0.6 at Beverly, shall operate as follows:

(914) (a) Public vessels of the United States must be passed as soon as possible.

(915) (b) The owners of this bridge shall provide and keep in good legible condition clearance gauges for each draw with figures not less than 12 inches high designed, installed and maintained according to the provisions of §118.160 of this chapter.

(916) (c) That the drawspan for the Hall Whitaker Drawbridge must open on signal if at least 24 hours notice is given.

(917)

§117.591 Charles River and its tributaries.

(918) (a) The following requirements apply to all bridges across the Charles River and its tributaries:

(919) (1) Public vessels of the United States, state or local vessels used for public safety; and vessels in distress shall be passed through the draw of each bridge as soon as possible without delay at any time. The opening signal from these vessels is four or more short blasts of a whistle or horn, or a radio request.

(920) (2) The owners of these bridges shall provide and keep in good legible condition clearance gauges for each draw with figures not less than 12 inches high designed, installed and maintained according to the provisions of §118.160 of this chapter.

(921) (3) Trains and locomotives shall be controlled so that any delay in opening the draw span shall not exceed ten minutes. However, if a train moving toward the bridge has crossed the home signal for the bridge before the signal requesting opening of the bridge is given, that train may continue across the bridge and must clear the bridge interlocks before stopping.

(922) (4) Except as provided in paragraph (b) through (f) of this section, the draws shall open on signal.

(923) (b) The draw of the Massachusetts Bay Transportation Authority (MBTA/Amtrak Bridge, mile 0.8, at Boston, shall open on signal; except that from 6:15 a.m. to 9:10 a.m. and 4:15 p.m. to 6:30 p.m., Monday through Friday, except holidays, the draw need not be opened for the passage of vessels, except as stated in paragraph (a)(1) of this section.

(924) (c) The draw of the Massachusetts Bay Transportation Authority (East Cambridge Viaduct) Railroad Bridge, mile 1.0 at Boston, need not be opened for the passage of vessels. However, the operating machinery of the draw shall be maintained in an operable condition.

(925) (d) The draw of the Massachusetts Department of Transportation (Craigie) Bridge, mile 1.0 at Boston, shall operate as follows:

(926) (1) Open on signal; except that from 6:15 a.m. to 9:10 a.m. and 3:15 p.m. to 6:30 p.m., Monday through Friday, except holidays, the draw need not open for the passage of vessels, except as stated in paragraph (a)(1) of this section.

(927) (2) From December 1 to March 31, the draw shall open on signal after a 24 hour advance notice is given.

(928) (3) From midnight to 8 a.m., April, May, October, and November, the draw shall open on signal after at least an eight-hour advance notice is given.

(929) (e) The draws of the bridges across Broad Canal, mile 0.0, need not open for the passage of vessels. However, the draws shall be returned to operable condition within one year after notification by the District Commander to do so.

(930)

§117.593 Chelsea River.

(931) (a) All drawbridges across Chelsea River shall open on signal. The opening signal for each drawbridge is two prolonged blasts followed by two short blasts and one prolonged blast. The acknowledging signal is three prolonged blasts when the draw can be opened immediately and is two prolonged blasts when the draw cannot be open or is open and must be closed.

(932) (b) The draw of the Chelsea Street Bridge, mile 1.3, at Chelsea, shall open as follows:

(933) (1) The draw shall open on signal to 139 feet above mean high water for all vessel traffic unless a full bridge opening to 175 feet above mean high water is requested.

(934) (2) The 139 foot opening will be signified by a range light display with one solid green light and one flashing green light and the full 175 foot opening will be signified with two solid green range lights.

(935)

§117.595 Danvers River.

(936) (a) The requirements in this paragraph apply to all bridges across the Danvers River:

(937) (1) The owners of these bridges shall provide and keep in good legible condition clearance gauges for each draw with figures not less than 12 inches high, designed, installed, and maintained according to the provisions of §118.160 of this chapter.

(938) (2) Trains and locomotives shall be controlled so that any delay in opening the draw span shall not exceed ten minutes. However, if a train moving toward the bridge has crossed the home signal for the bridge before the signal requesting opening of the bridge is given, that train may continue across the bridge and must clear the bridge interlocks before stopping.

(939) (b) The draw of the Massachusetts Bay Transportation Authority (MBTA)/AMTRAK Bridge, at mile 0.05, between Salem and Beverly, shall open on signal; except that, from midnight to 5 a.m., daily, and on December 25 and January 1, the draw shall open as soon as possible, but not more than one hour after notice is given to the drawtenders either at the bridge during the time the drawtenders are on duty or by calling the number posted at the bridge.

(940) (c) The Kernwood Bridge, at mile 1.0, shall operate as follows:

(941) (1) From May 1 through September 30, midnight to 5 a.m., and from October 1 through April 30, 7 p.m. to 5 a.m., draw shall open on signal after at least a one-hour advance notice is given by calling the number posted at the bridge.

(942) (2) From noon to 6 p.m. on Thanksgiving Day and all day on Christmas and New Years Day, the draw shall open on signal after at least a one-hour advance notice is given by calling the number posted at the bridge.

(943)

§117.597 Dorchester Bay.

(944) The draw of the William T. Morrisey Boulevard Bridge, mile 0.0, at Boston, shall operate as follows:

(945) (a) From 7:30 a.m. to 9 a.m. and 4:30 p.m. to 6 p.m., Monday through Friday, except holidays, the draw need not open for the passage of vessel traffic.

(946) (b) The draw shall open on signal from April 16 through May 31, from 8 a.m. through midnight, except as provided in paragraph (a) of this section. From midnight through 8 a.m. at least an eight-hour advance notice is required for bridge openings.

(947) (c) The draw shall open on signal at all times from June 1 through September 30, except as provided in paragraph (a) of this section.

(948) (d) The draw shall open on signal from October 1 through October 14, 8 a.m. through midnight, except as provided in paragraph (a) of this section. From midnight through 8 a.m. at least an eight-hour advance notice is required for bridge openings.

(949) (e) The draw shall open on signal from October 15 through April 15, after at least a 24 hours notice is given, except as provided in paragraph (a) of this section.

(950)

§117.599 [Removed].

(951)

§117.601 [Removed].

(952)

§117.603 Manchester Harbor.

(953) The Massachusetts Bay Transportation Authority Bridge at mile 1.0 in Manchester, shall operate as follows:

(954) (a) The draw shall open on signal–

(955) (1) From Memorial Day through September 30 from 7 a.m. to 11 p.m.;

(956) (2) From April 1 to Memorial Day and from October 1 to November 1 from 9 a.m. to 1 p.m. and 2 p.m. to 6 p.m.

(957) (b) At all other times, the draw shall open on signal with at least four hours notice.

(958) (c) The owner of this bridge shall provide and keep in good legible condition, clearance gauges for each draw with figures not less than twelve (12) inches high designed, installed and maintained in accordance with the provisions of section 118.160 of this chapter.

(959)

§117.605 Merrimack River.

(960) (a) The draw of the Newburyport US1 Bridge, mile 3.4, shall operate as follows:

(961) (1) From May 1 through November 15, from 6 a.m. to 10 p.m., the draw shall open on signal; except that, from Memorial Day through Labor Day, from 6 a.m. to 10 p.m., the draw shall open on signal only on the hour and half hour.

(962) (2) At all other times the draw shall open on signal after at least a one-hour advance notice is given by calling the number posted at the bridge.

(963) (b) The drawspans for the Massachusetts Department of Transportation drawbridges, mile 5.8 at Newburyport and mile 12.6 at Rock Village, and Groveland Drawbridge, mile 16.5 at Groveland, must open on signal if at least two hours notice is given. Public vessels of the United States must be passed through the drawspans as soon as possible.

(964)

§117.609 Mystic River.

(965) The draw of the S99 Alford Street Bridge, mile 1.4, shall open on signal; except that, Monday through Saturday, excluding holidays, the draw need not open for the passage of vessel traffic from 7:45 a.m. to 9 a.m., 9:10 a.m. to 10 a.m., and 5 p.m. to 6 p.m., daily. From November 1 through March 31, between 3 p.m. and 7 a.m., at least an eight-hour advance notice is required for bridge openings by calling the number posted at the bridge.

(966)

§117.611 Neponset River.

(967) The Granite Avenue Bridge, mile 2.5, between Boston and Milton, Massachusetts, shall operate as follows:

(968) (a) The draw of the Granite Avenue Bridge shall open on signal from May 1 through October 31; 6 a.m. to 12 midnight. At all other times the draw shall open on signal if at least one hour advance notice is given by calling the number posted at the bridge.

(969) (b) The owners of this bridge shall provide and keep in good legible condition, clearance gauges for each draw with figures not less than twelve (12) inches high designed, installed and maintained according to the provisions of §118.160 of this chapter.

(970)

§117.613 [Removed].

(971)

§117.615 Plum Island River.

(972) The draw of the Plum Island Turnpike Bridge, mile 3.3, between Newburyport and Plum Island, shall operate as follows:

(973) (a) From April 1 through November 30, 5 a.m. to 9 p.m., the draw shall open on signal if at least one hour advance notice is given by calling the number posted at the bridge. At all other times the draw shall open on signal if at least three hours advance notice is given.

(974) (b) The owners of this bridge shall provide and keep in good legible condition, clearance gauges for each draw with figures not less than twelve (12) inches high, designed, installed and maintained according to the provisions of §118.160 of this chapter

(975)

§117.618 Saugus River

(976) (a) The following requirements apply to all bridges across the Saugus River:

(977) (1) Public vessels of the United States, state or local vessels used for public safety, and vessels in distress

shall be passed through the draw of each bridge as soon as possible at any time. The opening signal from these vessels is four or more short blasts of a whistle or horn or a radio request.

(978) (2) The owners of these bridges shall provide and keep in good legible condition clearance gauges with figures not less than 12 inches high designed, installed and maintained according to provisions of §118.160 of this chapter.

(979) (3) Trains and locomotives shall be controlled so that any delay in opening the draw span shall not exceed seven minutes. However, if a train moving toward the bridge has crossed the home signal for the bridge before the signal requesting opening of the bridge is given, the train may continue across the bridge and must clear the bridge interlocks before stopping.

(980) (b) The draw of the General Edwards SR1A Bridge, mile 1.7, between Revere and Lynn, shall open on signal at all times if at least a two-hour advance notice is given by calling the number posted at the bridge.

(981) (c) The Fox Hill (SR107) Bridge, at mile 2.5, shall operate as follows:

(982) (1) The draw shall open on signal, except that, from October 1 through May 31, from 7 p.m. to 5 a.m., the draw shall open after at least a one-hour advance notice is given by calling the number posted at the bridge.

(983) (2) From noon to 6 p.m. on Thanksgiving Day, and all day on Christmas, and New Years Day, the draw shall open on signal after at least a one-hour advance notice is given by calling the number posted at the bridge.

(984)

§117.621 West Bay.

(985) The draw of the West Bay Bridge, mile 1.2, at Osterville, shall operate as follows:

(986) (a) From November 1 through April 30, the draw shall open on signal if at least a twenty-four hours advance notice is given.

(987) (b) From May 1 through June 15, the draw shall open on signal from 8 a.m. to 6 p.m.

(988) (c) From June 16 through September 30, the draw shall open on signal from 7 a.m. to 9 p.m.

(989) (d) From October 1 through October 31, the draw shall open on signal from 8 a.m. to 6 p.m.

(990) (e) At all other times from May 1 through October 31, the draw shall open on signal if at least a four-hours advance notice is given by calling the number posted at the bridge.

(991)

§117.622 Weymouth Fore River.

(992) The draw of the Quincy Weymouth SR3A bridge, mile 3.5 between Quincy Point and North Weymouth, Massachusetts, shall open on signal, except that:

(993) (a) From 6:30 a.m. to 9 a.m. and from 4:30 p.m. to 6:30 p.m., Monday through Friday, except holidays observed in the locality, the draw need not be opened.

(994) (b) The draw shall open on signal at all times for self-propelled vessels greater than 10,000 gross tons.

(995) (c) From noon to 6 p.m. on Thanksgiving Day, from 6 p.m. on December 24 to midnight on December 25, and from 6 p.m. on December 31 to midnight on January 1, the draw shall open on signal after at least a two-hour advance notice is given by calling the number posted at the bridge.

(996)

NEW HAMPSHIRE

(997)

§117.697 Hampton River.

(998) The SR1A bridge, mile 0.0 at Hampton, operates as follows:

(999) (a) The draw shall open on signal from April 1 through October 31 for the passage of vessels during daylight hours from three hours before to three hours after each high tide. Daylight hours means one-half hour before sunrise to one-half hour after sunset. High tide occurs one-half hour later than the time of high tide for Portland, Maine, as published in the tide tables published by private entities using data provided by the National Ocean Service. At all other times, the draw shall open on signal if at least three hours notice is given.

(1000) (b) The owners of the bridge shall provide and keep in good legible condition two board gages painted white with black figures not less than six inches high to indicate the vertical clearance under the closed draw at all stages of the tide. The gages shall be so placed on the bridge that they are plainly visible to operators of vessels approaching the bridge either up or downstream.

(1001) (c) Vessels which can pass under the closed draw with a clearance of one foot or more shall not signal for the opening of the draw. In case a vessel gives the prescribed signal and the drawtender is uncertain as to whether the vessel can safely pass, the drawtender shall open the draw. If the drawtender finds that there would have been a clearance of one foot or more had the draw remained closed, the matter shall be reported immediately to the District Commander, giving the name of the vessel, the time of opening the draw, the clearance under the bridge as indicated by the gage at the time of opening the draw, and the approximate vertical clearance required by the vessel.

(1002)

§117.699 Little Harbor.

(1003) The draw of the SR1B bridge, mile 1.0 between New Castle and Rye, shall open on signal from April 1 through October 31 from 6 a.m. to 10 p.m. if at least four hours notice is given. At all other times, the draw shall open as soon as possible only for emergencies.

(1004)

§117.700 Piscataqua River.

(1005) See §117.531, Piscataqua River, listed under Maine.

(1006)

Note

(1007) Call signs and radio channels for drawbridges equipped with radiotelephones are included with the bridge descriptions in chapters 4 through 12.

(1008)

Part 150—Deepwater Ports: Operations

(1009)

Subpart D—Vessel Navigation

(1010)

§150.300 What does this subpart do?

(1011) This subpart supplements the International Regulations for Prevention of Collisions at Sea, 1972 (72 COLREGS) described in subchapter D of this chapter, and prescribes requirements that:

(1012) (a) Apply to the navigation of all vessels at or near a deepwater port; and

(1013) (b) Apply to all vessels while in a safety zone, area to be avoided, or no anchoring area.

(1014)

§150.305 How does this subpart apply to unmanned deepwater ports?

(1015) The master of any tanker calling at an unmanned deepwater port is responsible for the safe navigation of the vessel to and from the deepwater port, and for the required notifications in §150.325 of this part. Once the tanker is connected to the unmanned deepwater port, the master must maintain radar surveillance in compliance with the requirements of §150.310 of this part.

(1016)

§150.310 When is radar surveillance required?

(1017) A manned deepwater port's person in charge of vessel operations must maintain radar surveillance of the safety zone or area to be avoided when:

(1018) (a) A tanker is proceeding to the safety zone after submitting the report required in §150.325 in this part;

(1019) (b) A tanker or support vessel is underway in the safety zone or area to be avoided;

(1020) (c) A vessel other than a tanker or support vessel is about to enter or is underway in the safety zone or area to be avoided; or

(1021) (d) As described in the deepwater port security plan.

(1022)

§150.320 What advisories are given to tankers?

(1023) A manned deepwater port's person in charge of vessel operations must advise the master of each tanker underway in the safety zone or area to be avoided of the following:

(1024) (a) At intervals not exceeding 10 minutes, the vessel's position by range and bearing from the pumping platform complex; and

(1025) (b) The position and the estimated course and speed, if moving, of all other vessels that may interfere with the movement of the tanker within the safety zone or area to be avoided.

(1026)

§150.325 What is the first notice required before a tanker enters the safety zone or area to be avoided?

(1027) (a) The owner, master, agent, or person in charge of a tanker bound for a manned deepwater port must comply with the notice of arrival requirements in subpart C of part 160 of this chapter.

(1028) (b) The owner, master, agent, or person in charge of a tanker bound for a manned deepwater port must report the pertinent information required in §150.15(i)(4)(vi) of this part for the vessel, including:

(1029) (1) The name, gross tonnage, and draft of the tanker;

(1030) (2) The type and amount of cargo in the tanker;

(1031) (3) The location of the tanker at the time of the report;

(1032) (4) Any conditions on the tanker that may impair its navigation, such as fire, or malfunctioning propulsion, steering, navigational, or radiotelephone equipment. The testing requirements in §164.25 of this chapter are applicable to vessels arriving at a deepwater port;

(1033) (5) Any leaks, structural damage, or machinery malfunctions that may impair cargo transfer operations or cause a product discharge; and

(1034) (6) The operational condition of the equipment listed under §164.35 of this chapter on the tanker.

(1035) (c) If the estimated time of arrival changes by more than 6 hours from the last reported time, the National Vessel Movement Center (NVMC) and the port's person in charge of vessel operations must be notified of the correction as soon as the change is known.

(1036) (d) If the information reported in paragraphs (b)(4) or (b)(5) of this section changes at any time before the tanker enters the safety zone or area to be avoided at the deepwater port, or while the tanker is in the safety zone or area to be avoided, the master of the tanker must report the changes to the NVMC and port's person in charge of vessel operations as soon as possible.

(1037)

§150.330 What is the second notice required before a tanker enters the safety zone or area to be avoided?

(1038) When a tanker bound for a manned deepwater port is 20 miles from entering the port's safety zone or area to be avoided, the master of the tanker must notify the port's person in charge of vessel operations of the tanker's name and location.

(1039)

§150.340 What are the rules of navigation for tankers in the safety zone or area to be avoided?

(1040) (a) A tanker must enter or depart the port's safety zone or area to be avoided in accordance with the navigation procedures in the port's approved operations manual as described in §150.15(i) of this part.

(1041) (b) A tanker must not anchor in the safety zone or area to be avoided, except in a designated anchorage area.

(1042) (c) A tanker may not enter a safety zone or area to be avoided in which another tanker is present, unless it has been cleared by the person in charge of the port and no other tankers are underway.

(1043) (d) A tanker must not operate, anchor, or moor in any area of the safety zone or area to be avoided in which the net under keel clearance would be less than 5 feet.

(1044)

§150.345 How are support vessels cleared to move within the safety zone or area to be avoided?

(1045) All movements of support vessels within a manned deepwater port's safety zone or area to be avoided must be cleared in advance by the port's person in charge of vessel operations.

(1046)

§150.350 What are the rules of navigation for support vessels in the safety zone or area to be avoided?

(1047) A support vessel must not anchor in the safety zone or area to be avoided, except:

(1048) (a) In an anchorage area; or

(1049) (b) For vessel maintenance, which, in the case of a manned deepwater port, must be cleared by the port's person in charge of vessel operations.

(1050)

§150.355 How are other vessels cleared to move within the safety zone?

(1051) (a) Clearance by a manned deepwater port's person in charge of vessel operations is required before a vessel, other than a tanker or support vessel, enters the safety zone.

(1052) (b) The port's person in charge of vessel operations may clear a vessel under paragraph (a) of this section only if its entry into the safety zone would not:

(1053) (1) Interfere with the purpose of the deepwater port;

(1054) (2) Endanger the safety of life, property, or environment; or

(1055) (3) Be prohibited by regulation.

(1056) (c) At an unmanned deepwater port, such as a submerged turret landing (STL) system, paragraphs (a) and (b) of this section apply once a tanker connects to the STL buoy.

(1057)

§ 150.380 Under what circumstances may vessels operate within the safety zone or area to be avoided?

(1058) (a) Table 150.380(a) of this section lists both the areas within a safety zone where a vessel may operate and the clearance needed for that location.

(1059)

Table 150.380(a)—Regulated Activities of Vessels at Deepwater Ports

Regulated activities	Safety zone	Areas to be avoided around each deepwater port component[1]	Anchorage areas	Other areas within and adjacent to the safety zone (e.g., no anchoring area)
Tankers calling at port	C	C	C	C
Support vessel movements	C	C	C	C
Transit by vessels other than tankers or support vessels	F	D	P	P
Mooring to surface components (for example an SPM) by vessels other than tankers or support vessels	N	N	N	N
Anchoring by vessels other than tankers or support vessels	N	F	C	F
Fishing, including bottom trawl (shrimping)	N	D	P	N
Mobile drilling operations or erection of structures[2]	N	R	N	N
Lightering/trans-shipment	N	N	N	N

[1]Areas to be avoided are in subpart J of this part.
[2]Not part of Port Installation.

Key to regulated activities for Table 150.380(a):
C–Movement of the vessel is permitted when cleared by the person in charge of vessel operations.
D–Movement is not restricted, but recommended transit speed not to exceed 10 knots. Communication with the person in charge of vessel operations.
F–Only in an emergency. Anchoring will be avoided in a no anchoring area except in the case of immediate danger to the ship or persons on board.
N–Not permitted.
P–Transit is permitted when the vessel is not in the immediate area of a tanker, and when cleared by the vessel tra f c supervisor.
R–Permitted only if determined that operation does not create unacceptable risk to personnel safety and security and operation. For transiting foreign-f ag vessels, the requirement for clearance to enter the area to be avoided and no anchoring area is advisory in nature, but mandatory for an anchorage area established within 12 nautical miles.

(1060) (b) If the activity is not listed in table 150.380(a) of this section, or otherwise provided for in this subpart, the Sector Commander's, or MSU Commander's, with COTP and OCMI authority permission is required before operating in the safety zone or regulated navigation area.

(1061)

§150.385 What is required in an emergency?

(1062) In an emergency, for the protection of life or property, a vessel may deviate from a vessel movement requirement in this subpart without clearance from a manned deepwater port's person in charge of vessel operations if the master advises the port person in charge of the reasons for the deviation at the earliest possible moment.

(1063)

Subpart J—Safety Zones, No Anchoring Areas, and Areas to be Avoided

(1064)

§150.900 What does this subpart do?

(1065) (a) This subpart provides requirements for the establishment, restrictions, and location of safety zones,

no anchoring areas (NAAs), and areas to be avoided (ATBAs) around deepwater ports.

(1066) (b) Subpart D of this part, concerning vessel navigation and activities permitted and prohibited at deepwater ports, applies to safety zones, NAAs, ATBAs, and their adjacent waters; and supplements the International Regulations for Preventing Collisions at Sea.

(1067) (c) Recommended shipping safety fairways associated with deepwater ports are described in part 166 of this chapter.

(1068)

§150.905 Why are safety zones, no anchoring areas, and areas to be avoided established?

(1069) (a) Safety zones, no anchoring areas (NAAs) and areas to be avoided (ATBAs) under this subchapter are established to promote safety of life and property, marine environmental protection, and navigational safety at deepwater ports and adjacent waters.

(1070) (b) Safety zones are the only federally regulated navigation areas. They accomplish these objectives by preventing or controlling specific activities, limiting access by vessels or persons, and by protecting the living resources of the sea from harmful agents.

(1071) (c) The NAAs and ATBAs are established via the International Maritime Organization (IMO). An NAA, specifically established to protect vessels in transit and sub-surface deepwater port components, will be mandatory. An ATBA will be a recommendatory routing measure.

(1072) (d) The sizes of restricted areas will be the minimum size needed to ensure safety, while at the same time considering potential impacts on other activities, including recreational boating, fishing, and Outer Continental Shelf activity.

(1073)

§150.910 What installations, structures, or activities are prohibited in a safety zone?

(1074) No installations, structures, or activities that are incompatible with or that present an unacceptable risk to safety of the deepwater port's operations or activity are allowed in the safety zone of a deepwater port.

(1075)

§150.915 How are safety zones, no anchoring areas, and areas to be avoided established and modified?

(1076) (a) Safety zones are developed and designated during the application process for a deepwater port license, and may be established or modified through rulemaking. Rulemakings will afford prior public notice and comment, except when there is good cause not to do so, for example due to an imminent threat to the safety of life and property.

(1077) (b) Before a safety zone, no anchoring area (NAA), or area to be avoided (ATBA) is established, all factors detrimental to safety are considered, including but not limited to:

(1078) (1) The scope and degree of the risk or hazard involved;

(1079) (2) Vessel traffic characteristics and trends, including traffic volume, the sizes and types of vessels involved, potential interference with the flow of commercial traffic, the presence of any unusual cargoes, and other similar factors;

(1080) (3) Port and waterway configurations and variations in local conditions of geography, climate and other similar factors;

(1081) (4) The need for granting exemptions for the installation and use of equipment or devices for use with vessel traffic services for certain classes of small vessels, such as self-propelled fishing vessels and recreational vessels;

(1082) (5) The proximity of fishing grounds, oil and gas drilling and production operations, or other potential or actual conflicting activity;

(1083) (6) Environmental factors;

(1084) (7) Economic impact and effects;

(1085) (8) Existing vessel traffic services; and

(1086) (9) Local practices and customs, including voluntary arrangements and agreements within the maritime community.

(1087) (c) The Executive Branch, acting through the Secretary of State and Commandant (CG–5P) proposes NAAs and ATBAs for deepwater ports to the International Maritime Organization (IMO) for approval. The ATBAs will be implemented after IMO approval is granted and announced in an IMO Circular, and after publication of a notice in the Federal Register.

(1088)

§150.920 How can I find notice of new or proposed safety zones?

(1089) In addition to documents published in the Federal Register under §150.915 of this part, the District Commander may provide public notice of new or proposed safety zones by Broadcast Notices to Mariners, Notices to Mariners, Local Notices to Mariners, newspapers, broadcast stations, or other means.

(1090)

§150.925 How long may a safety zone, no anchoring area, or area to be avoided remain in place?

(1091) A safety zone, no anchoring area, or area to be avoided may go into effect as early as initial delivery of construction equipment and materials to the deepwater port site, and may remain in place until the deepwater port is removed.

(1092)

§150.930 What datum is used for the geographic coordinates in this subpart?

(1093) The geographic coordinates used in this subpart have been revised to enable plotting using the North American Datum of 1983 (NAD 83) and no longer require the use of any further conversion factors for correction.

(1094)

§150.940 Safety zones for specific deepwater ports (in part).

(1095) (b) *Northeast Gateway Deepwater Port (NEGDWP)*—(1) *Location.* The safety zones for the NEGDWP consist of circular zones, each with a 500-meter radius and centered on each of the deepwater port's two submerged turret loading (STL) buoys. STL Buoy "A" is centered at the following coordinates: 42°23'38" N, 070°35'31" W. STL Buoy "B" is centered at the following coordinates: 42°23'56" N, 070°37'00" W. Each safety zone is located approximately 13 miles south-southeast of the City of Gloucester, Massachusetts, in Federal waters.

(1096) (2) *No anchoring areas.* Two mandatory no anchoring areas for NEGDWP are established for all waters within circles of 1,000-meter radii centered on the submerged turret loading buoy positions set forth in paragraph (b)(1) of this section.

(1097) (3) *Area to be avoided.* An area to be avoided (ATBA) for NEGDWP is as described in Table 150.940(B):

(1098)

Table 150.940(B)—ATBA for NEGDWP

Plotting guidance	Latitude (N)	Longitude (W)
(i) Starting at	42°24'17"	70°35'16"
(ii) A rhumb line to	42°24'35"	70°36'46"
(iii) Then an arc with a 1250 meter	42°23'56"	70°37'00"
(iv) To a point	42°23'17"	70°37'15"
(v) Then a rhumb	42°22'59"	70°35'45"
(vi) Then an arc with a 1250 meter	42°23'38"	70°35'31"
(vii) To the point of	42°24'17"	70°35'16"

(1099) (4) *Regulations*. (i) In accordance with the general regulations set forth in 33 CFR 165.23 and elsewhere in this part, no person or vessel may enter the waters within the boundaries of the safety zones described in paragraph (b)(1) of this section unless previously authorized by the Captain of the Port (COTP) Boston, or his/her authorized representative.

(1100) (ii) Notwithstanding paragraph (b)(4)(i) of this section, tankers and support vessels, as defined in 33 CFR 148.5, operating in the vicinity of NEGDWP are authorized to enter and move within such zones in the normal course of their operations following the requirements set forth in 33 CFR 150.340 and 150.345, respectively.

(1101) (iii) All other vessel operators desiring to enter, operate or conduct diving operations within a safety zone described in paragraph (b)(1) of this section must contact the COTP, or the COTP's authorized representative, to obtain permission by contacting the Sector Boston Command Center at 617–223–5761 or via VHF–FM Channel 16 (156.8 MHZ). Vessel operators given permission to enter, operate, or conduct diving operations in a safety zone must comply with all directions given to them by the COTP or the COTP's authorized representative.

(1102) (iv) No vessel, other than a support vessel or tanker calling on NEGDWP may anchor in the area described in paragraph (b)(2) of this section.

(1103) (c) *Neptune Deepwater Port (Neptune)*—(1) *Location*. The safety zones for Neptune consist of circular zones, each with a 500-meter radius and centered on each of Neptune's two submerged turret loading (STL) buoys. STL Buoy "A" is centered at the following coordinates: Latitude 42°29'12.3" N, Longitude 070°36'29.7" W; and STL Buoy "B": Latitude 42°27'20.5" N, Longitude 070°36'07.3" W. Each safety zone encompasses, within the respective 500-meter circles, the primary components of Neptune, including a submerged turret loading buoy and a pipeline end manifold. Each safety zone is located approximately eight nautical miles south-southeast of Gloucester, Massachusetts, in Federal waters.

(1104) (2) *No anchoring areas.* Two mandatory no anchoring areas for Neptune are established for all waters within circles of 1,000-meter radii centered on the submerged turret loading buoy positions set forth in paragraph (c)(1) of this section.

(1105) (3) *Area to be avoided.* An area to be avoided (ATBA) for Neptune is as described in Table 150.940(C):

(1106)

Table 150.940(C)—ATBA for NEGDWP

Plotting guidance	Latitude (N)	Longitude (W)
(i) Starting at	42°27'26.6"	70°35'13.1"
(ii) A rhumb line to	42°29'18.3"	70°35'35.4"
(iii) Then an arc with a 1250 meter radius centered at point	42°29'12.3"	70°36'29.7"
(iv) To a point	42°29'06.3"	70°37'24.0"
(v) Then a rhumb line to	42°27'14.7"	70°37'01.6"
(vi) Then an arc with a 1250 meter radius centered at point	42°27'20.5"	70°36'07.3"
(vii) To the point of starting	42°27'26.6"	70°35'13.1"

(1107) (4) *Regulations*. (i) In accordance with the general regulations set forth in 33 CFR 165.23 and elsewhere in this part, no person or vessel may enter the waters within the boundaries of the safety zones described in paragraph (c)(1) of this section unless previously authorized by the Captain of the Port (COTP) Boston, or the COTP's authorized representative.

(1108) (ii) Notwithstanding paragraph (c)(4)(i) of this section, liquefied natural gas carriers (LNGCs) and support vessels, as defined in 33 CFR 148.5, calling on Neptune, are authorized to enter and move within such zones in the normal course of their operations following the requirements set forth in 33 CFR 150.340 and 150.345, respectively.

(1109) (iii) All other vessel operators desiring to enter, operate or conduct diving operations within a safety zone described in paragraph (c)(1) of this section must contact the COTP or the COTP's authorized representative to obtain permission by contacting the Sector Boston

Command Center at 617–223–5761 or via VHF–FM Channel 16 (156.8 MHZ). Vessel operators given permission to enter or operate in a safety zone must comply with all directions given to them by the COTP or the COTP's authorized representative.

(1110) (iv) No vessel, other than an LNGC or support vessel calling on Neptune, may anchor in the area described in paragraph (c)(2) of this section.

(1111)

Part 157–Rules for the Protection of the Marine Environment relating to Tank Vessels Carrying Oil in Bulk.

(1112)

Subpart A–General

(1113)

§157.01 Applicability.

(1114) (a) Unless otherwise indicated, this part applies to each vessel that carries oil in bulk as cargo and that is:

(1115) (1) Documented under the laws of the United States (a U.S. vessel); or

(1116) (2) Any other vessel that enters or operates in the navigable waters of the United States, or that operates, conducts lightering under 46 U.S.C. 3715, or receives cargo from or transfers cargo to a deepwater port under 33 U.S.C. 1501 et seq., in the United States Exclusive Economic Zone, as defined in 33 U.S.C. 2701(8).

(1117) (b) This part does not apply to a vessel exempted under 46 U.S.C. 2109 or 46 U.S.C. 3702.

(1118)

§157.02 Incorporation by reference: Where can I get a copy of the publications mentioned in this part?

(1119) (a) Certain material is incorporated by reference into this part with the approval of the Director of the Federal Register under 5 U.S.C. 552(a) and 1 CFR part 51. To enforce any edition other than that specified in this section, the Coast Guard must publish notice of change in the Federal Register and the material must be available to the public. All approved material is available for inspection at the National Archives and Records Administration (NARA). For information on the availability of this material at NARA, call 202–741–6030 or go to *archives.gov/federal-register/cfr/ibr-locations.html*. Also, it is available for inspection at the Coast Guard Headquarters. Contact Commandant (CG–ENG), Attn: Office of Design and Engineering Standards, U.S. Coast Guard Stop 7509, 2703 Martin Luther King Jr. Avenue SE., Washington, DC 20593–7509; telephone 202–372–1375. The material is also available from the sources indicated in this section.

(1120) (b) *International Maritime Organization (IMO)*—4 Albert Embankment, London SE1 7SR, United Kingdom.

(1121) (1) IMCO Assembly Resolution A.393(X), adopted on 14 November 1977, Recommendation on International Performance and Test Specifications For Oily Water Separating Equipment and Oil Content Meters ("A.393(x)"), incorporation by reference approved for §157.12.

(1122) (2) IMO Assembly Resolution A.496(XII), Adopted on 19 November 1981, Agenda Item 11, Guidelines and Specifications for Oil Discharge Monitoring and Control Systems for Oil Tankers ("A.496(XII)"), incorporation by reference approved for §157.12.

(1123) (3) IMO Assembly Resolution A.586(14), Adopted on 20 November 1985, Agenda item 12, Revised Guidelines and Specifications for Oil Discharge Monitoring and Control Systems for Oil Tankers ("A.586(14)"), incorporation by reference approved for §157.12.

(1124) (4) IMO Marine Environment Protection Committee Resolution MEPC.13 (19), adopted on 9 December 1983, Guidelines for Plan Approval and Installation Survey of Oil Discharge Monitoring and Control Systems for Oil Tankers and Environmental Testing of Control Sections Thereof ("MEPC.13(19)"), incorporation by reference approved for §157.12.

(1125) (5) IMO Marine Environment Protection Committee Resolution MEPC.108(49), Adopted on 18 July 2003, Revised Guidelines and Specifications for Oil Discharge Monitoring and Control Systems for Oil Tankers ("MEPC.108(49)"), incorporation by reference approved for §157.12.

(1126) (6) IMO Assembly Resolution A.601(15), Provision and Display of Manoeuvring Information on Board Ships, Annex sections 1.1, 2.3, 3.1, and 3.2 with appendices adopted on 19 November 1987 ("A.601(15)"), incorporation by reference approved for §157.450.

(1127) (7) IMO Assembly Resolution A.744(18), Guidelines on the Enhanced Programme of Inspections During Surveys of Bulk Carriers and Oil Tankers, Annex B sections 1.1.3-1.1.4, 1.2-1.3, 2.1, 2.3-2.6, 3-8 and Annexes 1-10 with appendices, adopted 4 November 1993 ("A.744(18)"), incorporation by reference approved for §157.430.

(1128) (8) IMO Assembly Resolution A.751(18), Interim Standards for Ship Manoeuvrability, Annex sections 1.2, 2.3-2.4, 3-4.2, and 5, adopted 4 November 1993 with Explanatory Notes in MSC/Circ. 644 dated 6 June 1994 ("A.751(18)"), incorporation by reference approved for §157.445.

(1129) (9) MARPOL Consolidated Edition 2011, Annex I, Regulations for the prevention of pollution by oil, Chapter 4—Requirements for the cargo area of oil tankers, Part A—Construction, Regulation 22, "Pump-room bottom protection," (Annex I, Regulation 22) incorporation by reference approved for §157.14.

(1130) (10) MARPOL Consolidated Edition 2011, Annex I, Regulations for the prevention of pollution by oil, Chapter 4—Requirements for the cargo area of oil tankers, Part A—Construction, Regulation 23, "Accidental oil outflow performance," (Annex I, Regulation 23) incorporation by reference approved for §157.20.

(1131) (c) *Oil Companies International Marine Forum (OCIMF)* 27 Queen Anne's Gate, London, SW1H 9BU, England.

(1132) (1) International Safety Guide for Oil Tankers and Terminals, Fourth Edition, Chapters 6, 7 and 10, 1996 incorporation by reference approved for §157.435.

(1133) (2) [Reserved]

(1134)

§157.03 Definitions.

(1135) Except as otherwise stated in a subpart:

(1136) *Amidships* means the middle of the length.

(1137) *Animal fat* means a non-petroleum oil, fat, or grease derived from animals and not specifically identified elsewhere in this part.

(1138) *Ballast voyage* means the voyage that a tank vessel engages in after it leaves the port of final cargo discharge.

(1139) *Breadth or B* means the maximum molded breadth of a vessel in meters.

(1140) *Cargo tank length* means the length from the forward bulkhead of the forwardmost cargo tanks, to the after bulkhead of the aftermost cargo tanks.

(1141) *Center tank* means any tank inboard of a longitudinal bulkhead.

(1142) *Clean ballast* means ballast which:

(1143) (1) If discharged from a vessel that is stationary into clean, calm water on a clear day, would not–

(1144) (i) Produce visible traces of oil on the surface of the water or on adjoining shore lines; or

(1145) (ii) Cause a sludge or emulsion to be deposited beneath the surface of the water or upon adjoining shore lines; or

(1146) (2) If verified by an approved oil discharge monitoring and control system, has an oil content that does not exceed 15 p.m.

(1147) *Combination carrier* means a vessel designed to carry oil or solid cargoes in bulk.

(1148) *Crude oil* means any liquid hydrocarbon mixture occurring naturally in the earth, whether or not treated to render it suitable for transportation, and includes crude oil from which certain distillate fractions may have been removed, and crude oil to which certain distillate fractions may have been added.

(1149) *Deadweight or DWT* means the difference in metric tons between the lightweight displacement and the total displacement of a vessel measured in water of specific gravity 1.025 at the load waterline corresponding to the assigned summer freeboard.

(1150) *Dedicated clean ballast tank* means a cargo tank that is allocated solely for the carriage of clean ballast.

(1151) *Domestic trade* means trade between ports or places within the United States, its territories and possessions, either directly or via a foreign port including trade on the navigable rivers, lakes, and inland waters.

(1152) *Double bottom* means watertight protective spaces that do not carry any oil and which separate the bottom of tanks that hold any oil within the cargo tank length from the outer skin of the vessel.

(1153) *Double hull* means watertight protective spaces that do not carry any oil and which separate the sides, bottom, forward end, and aft end of tanks that hold any oil within the cargo tank length from the outer skin of the vessel as prescribed in §157.10d.

(1154) *Doubles sides* means watertight protective spaces that do not carry any oil and which separate the sides of tanks that hold any oil within the cargo tank length from the outer skin of the vessel.

(1155) *Existing vessel* means any vessel that is not a new vessel.

(1156) *Fleeting or assist towing vessel* means any commercial vessel engaged in towing astern, alongside, or pushing ahead, used solely within a limited geographic area, such as a particular barge fleeting area or commercial facility, and used solely for restricted service, such as making up or breaking up larger tows.

(1157) *Foreign trade* means any trade that is not domestic trade.

(1158) *From the nearest land* means from the baseline from which the territorial sea of the United States is established in accordance with international law.

(1159) *Fuel oil* means any oil used as fuel for machinery in the vessel in which it is carried.

(1160) *Inland vessel* means a vessel that is not oceangoing and that does not operate on the Great Lakes.

(1161) *Instantaneous rate of discharge of oil content* means the rate of discharge of oil in liters per hour at any instant, divided by the speed of the vessel in knots at the same instant.

(1162) *Integrated tug barge* means a tug and a tank barge with a mechanical system that allows the connection of the propulsion unit (the tug) to the stern of the cargo carrying unit (the tank barge) so that the two vessels function as a single self-propelled vessel.

(1163) Large primary structural member includes any of the following:

(1164) (1) Web frames.

(1165) (2) Girders.

(1166) (3) Webs.

(1167) (4) Main brackets.

(1168) (5) Transverses.

(1169) (6) Stringers.

(1170) (7) Struts in transverse web frames when there are 3 or more struts and the depth of each is more than 1/15 of the total depth of the tank.

(1171) *Length or L* means the distance in meters from the fore side of the stem to the axis of the rudder stock on a waterline at 85 percent of the least molded depth measured from the molded baseline, or 96 percent of the total length on that waterline, whichever is greater. In vessels designed with drag, the waterline is measured parallel to the designed waterline.

(1172) *Lightweight* means the displacement of a vessel in metric tons without cargo, fuel oil, lubricating oil, ballast water, fresh water, and feedwater in tanks, consumable stores, and any persons and their effects.

(1173) *Major conversion* means a conversion of an existing vessel that:

(1174) (1) Substantially alters the dimensions or carrying capacity of the vessel, except a conversion that includes only the installation of segregated ballast tanks, dedicated clean ballast tanks, a crude oil washing system, double sides, a double bottom, or a double hull;

(1175) (2) Changes the type of vessel;

(1176) (3) Substantially prolongs the vessel's service life; or

(1177) (4) Otherwise so changes the vessel that it is essentially a new vessel, as determined by the Commandant (CG–CVC).

(1178) *MARPOL 73/78* means the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating to that Convention. A copy of MARPOL 73/78 is available from the International Maritime Organization, 4 Albert Embankment, London, SE1

(1179) *New vessel* means:

(1180) (1) A U.S. vessel in domestic trade that:

(1181) (i) Is constructed under a contract awarded after December 31, 1974;

(1182) (ii) In the absence of a building contract, has the keel laid or is at a similar stage of construction after June 30, 1975;

(1183) (iii) Is delivered after December 31, 1977; or

(1184) (iv) Has undergone a major conversion for which:

(1185) (A) The contract is awarded after December 31, 1974;

(1186) (B) In the absence of a contract, conversion is begun after June 30, 1975; or

(1187) (C) Conversion is completed after December 31, 1977; and

(1188) (2) A foreign vessel or a U.S. vessel in foreign trade that;

(1189) (i) Is constructed under a contract awarded after December 31, 1975;

(1190) (ii) In the absence of a building contract, has the keel laid or is at a similar stage of construction after June 30, 1976;

(1191) (iii) Is delivered after December 31, 1979; or

(1192) (iv) Has undergone a major conversion for which:

(1193) (A) The contract is awarded after December 31, 1975;

(1194) (B) In the absence of a contract, conversion is begun after June 30, 1976; or

(1195) (C) Conversion is completed after December 31, 1979.

(1196) *Non-petroleum oil* means oil of any kind that is not petroleum-based. It includes, but is not limited to, animal fat and vegetable oil.

(1197) *Oceangoing* has the same meaning as defined in §151.05 of this chapter.

(1198) *Officer in charge of a navigational watch* means any officer employed or engaged to be responsible for navigating or maneuvering the vessel and for maintaining a continuous vigilant watch during his or her periods of duty and following guidance set out by the master, international or national regulations, and company policies.

(1199) *Oil* means oil of any kind or in any form including, but not limited to, petroleum, fuel oil, sludge, oil refuse, and oil mixed with wastes other than dredged spoil. This includes liquid hydrocarbons as well as animal and vegetable oils.

(1200) *Oil cargo residue* means any residue of oil cargo whether in solid, semi-solid, emulsified, or liquid form from cargo tanks and cargo pump room bilges, including but not limited to, drainages, leakages, exhausted oil, muck, clingage, sludge, bottoms, paraffin (wax), and any constituent component of oil. The term "oil cargo residue" is also known as "cargo oil residue."

(1201) *Oil residue* means–

(1202) (1) Oil cargo residue; and

(1203) (2) Other residue of oil whether in solid, semi-solid, emulsified, or liquid form resulting from drainages, leakages, exhausted oil and other similar occurrences from machinery spaces.

(1204) *Oil spill response vessel* means a vessel that is exclusively dedicated to operations to prevent or mitigate environmental damage due to an actual or impending accidental oil spill. This includes a vessel that performs routine service as an escort for a tank vessel, but excludes a vessel that engages in any other commercial activity, such as the carriage of any type of cargo.

(1205) *Oil tanker* means a vessel that is constructed or adapted primarily to carry crude oil or products in bulk as cargo. This includes a tank barge, a tankship, and a combination carrier, as well as a vessel that is constructed or adapted primarily to carry noxious liquid substances in bulk as cargo and which also carries crude oil or products in bulk as cargo.

(1206) Oily mixture means a mixture, in any form, with any oil content. "Oily mixture" includes, but is not limited to—

(1207) (1) Slops from bilges;

(1208) (2) Slops from oil cargoes (such as cargo tank washings, oily waste, and oily refuse);

(1209) (3) Oil residue; and

(1210) (4) Oily ballast water from cargo or fuel oil tanks, including any oil cargo residue.

(1211) Oily mixture means a mixture with any oil content.

(1212) *Other non-petroleum oil* means an oil of any kind that is not petroleum oil, an animal fat, or a vegetable oil.

(1213) *Permeability of a space* means the ratio of volume within a space that is assumed to be occupied by water to the total volume of that space.

(1214) *Petroleum oil* means petroleum in any form, including but not limited to, crude oil, fuel oil, sludge, oil residue, and refined products.

(1215) *Primary towing vessel* means any vessel engaged in towing astern, alongside, or pushing ahead and includes the tug in an integrated tug barge. It does not include fleeting or assist towing vessels.

(1216) *Product* means any liquid hydrocarbon mixture in any form, except crude oil, petrochemicals, and liquefied gases.

(1217) *Segregated ballast* means the ballast water introduced into a tank that is completely separated from the cargo oil and fuel oil system and that is permanently allocated to the carriage of ballast.

(1218) *Slop tank* means a tank specifically designated for the collection of cargo drainings, washings, and other oily mixtures.

(1219) *Tank* means an enclosed space that is formed by the permanent structure of a vessel, and designed for the carriage of liquid in bulk.

(1220) *Tank barge* means a tank vessel not equipped with a means of self-propulsion.

(1221) *Tank vessel* means a vessel that is constructed or adapted primarily to carry, or that carries, oil or hazardous material in bulk as cargo or cargo residue, and that–

(1222) (1) Is a vessel of the United States;

(1223) (2) Operates on the navigable waters of the United States; or

(1224) (3) Transfers oil or hazardous material in a port or place subject to the jurisdiction of the United States. This does not include an offshore supply vessel, or a fishing vessel or fish tender vessel of not more than 750 gross tons when engaged only in the fishing industry.

(1225) *Tankship* means a tank vessel propelled by mechanical power or sail.

(1226) *Vegetable oil* means a non-petroleum oil or fat not specifically identified elsewhere in this part that is derived from plant seeds, nuts, kernels, or fruits.

(1227) *Wing tank* means a tank that is located adjacent to the side shell plating.

(1228)

§157.04 Authorization of classification societies.

(1229) (a) The Coast Guard may authorize any classification society (CS) to perform certain plan reviews, certifications, and inspections required by this part on vessels classed by that CS except that only U.S. classification societies may be authorized to perform those plan reviews, inspections, and certifications for U.S. vessels.

(1230) (b) If a CS desires authorization to perform the plan reviews, certifications, and inspections required under this part, it must submit to the Commandant (CG–CVC), Attn: Office of Commercial Vessel Compliance, U.S. Coast Guard Stop 7501, 2703 Martin Luther King Jr. Avenue SE., Washington, DC 20593–7501, evidence from the governments concerned showing that they have authorized the CS to inspect and certify vessels on their behalf under the MARPOL 73/78.

(1231) (c) The Coast Guard notifies the CS in writing whether or not it is accepted as an authorized CS. If authorization is refused, reasons for the refusal are included.

(1232) (d) Acceptance as an authorized CS terminates unless the following are met:

(1233) (1) The authorized CS must have each Coast Guard regulation that is applicable to foreign vessels on the navigable waters of the United States.

(1234) (2) Each issue concerning equivalents to the regulations in this part must be referred to the Coast Guard for determination.

(1235) (3) Copies of any plans, calculations, records of inspections, or other documents relating to any plan review, inspection, or certification performed to meet this part must be made available to the Coast Guard.

(1236) (4) Each document certified under §§157.116(a)(2), 157.118(b)(1)(ii), and 157.216(b)(1)(ii) must be marked with the name or seal of the authorized CS.

(1237) (5) A copy of the final documentation that is issued to each vessel that is certified under this part must be referred to the Commandant (CG–CVC), Attn: Office of Commercial Vessel Compliance, U.S. Coast Guard Stop 7501, 2703 Martin Luther King Jr. Avenue SE., Washington, DC 20593–7501.

(1238)

Subpart B–Design, Equipment, and Installation

(1239)

§157.08 Applicability of Subpart B.

(1240) NOTE: An "oil tanker" as defined in §157.03 includes barges as well as self-propelled vessels.

(1241) (a) Sections 157.10d and 157.11(g) apply to each vessel to which this part applies.

(1242) (b) Sections 157.11 (a) through (f), 157.12, 157.15, 157.19(b)(3), 157.33, and 157.37 apply to each vessel to which this part applies that carries 200 cubic meters or more of crude oil or products in bulk as cargo, as well as to each oceangoing oil tanker to which this part applies of 150 gross tons or more. These sections do not apply to a foreign vessel which remains beyond the navigable waters of the United States and does not transfer oil cargo at a port or place subject to the jurisdiction of the United States.

(1243) (c) Section 157.21 applies to each oil tanker to which this part applies of 150 gross tons or more that is oceangoing or that operates on the Great Lakes. This section does not apply to a foreign vessel which remains beyond the navigable waters of the United States and does not transfer oil cargo at a port or place subject to the jurisdiction of the United States.

(1244) (d) Sections in subpart B of 33 CFR part 157 that are not specified in paragraphs (a) through (c) of this section apply to each oceangoing oil tanker to which this part applies of 150 gross tons or more, unless otherwise indicated in paragraphs (e) through (m) of this section. These sections do not apply to a foreign vessel which remains beyond the navigable waters of the United States and does not transfer oil cargo at a port or place subject to the jurisdiction of the United States.

(1245) (e) Sections 157.11 (a) through (f), 157.12, and 157.15 do not apply to a vessel, except an oil tanker, that carries less than 1,000 cubic meters of crude oil or

products in bulk as cargo and which retains oily mixtures on board and discharges them to a reception facility.

(1246) (f) Sections 157.11 (a) through (f), 157.12, 157.13, and 157.15 do not apply to a tank vessel that carries only asphalt, carbon black feedstock, or other products with similar physical properties, such as specific gravity and cohesive and adhesive characteristics, that inhibit effective product/water separation and monitoring.

(1247) (g) Sections 157.11 (a) through (f), 157.12, 157.13, 157.15, and 157.23 do not apply to a tank barge that cannot ballast cargo tanks or wash cargo tanks while underway.

(1248) (h) Sections 157.19 and 157.21 do not apply to a tank barge that is certificated by the Coast Guard for limited short protected coastwise routes if the barge is otherwise constructed and certificated for service exclusively on inland routes.

(1249) (i) Section 157.09(d) does not apply to any:

(1250) (1) U.S. vessel in domestic trade that is constructed under a contract awarded before January 8, 1976;

(1251) (2) U.S. vessel in foreign trade that is constructed under a contract awarded before April 1, 1977; or

(1252) (3) Foreign vessel that is constructed under a contract awarded before April 1, 1977.

(1253) (j) Sections 157.09 and 157.10a do not apply to a new vessel that:

(1254) (1) Is constructed under a building contract awarded after June 1, 1979;

(1255) (2) In the absence of a building contract, has the keel laid or is at a similar stage of construction after January 1, 1980;

(1256) (3) Is delivered after June 1, 1982; or

(1257) (4) Has undergone a major conversion for which:

(1258) (i) The contract is awarded after June 1, 1979;

(1259) (ii) In the absence of a contract, conversion is begun after January 1, 1980; or

(1260) (iii) Conversion is completed after June 1, 1982.

(1261) (k) Sections 157.09(b)(3), 157.10(c)(3), 157.10a(d)(3), and 157.10b(b)(3) do not apply to tank barges.

(1262) (1) Section 157.10b does not apply to tank barges if they do not carry ballast while they are engaged in trade involving the transfer of crude oil from an offshore oil exploitation or production facility on the Outer Continental Shelf of the United States.

(1263) (m) Section 157.12 does not apply to a U.S. vessel that:

(1264) (1) Is granted an exemption under Subpart F of this part; or

(1265) (2) Is engaged solely in voyages that are:

(1266) (i) Between ports or places within the United States, its territories or possessions;

(1267) (ii) Of less than 72 hours in length; and

(1268) (iii) At all times within 50 nautical miles of the nearest land.

(1269) (n) Section 157.10d does not apply to:

(1270) (1) A vessel that operates exclusively beyond the navigable waters of the United States and the United States Exclusive Economic Zone, as defined in 33 U.S.C. 2701(8);

(1271) (2) An oil spill response vessel;

(1272) (3) Before January 1, 2015–

(1273) (i) A vessel unloading oil in bulk as cargo at a deepwater port licensed under the Deepwater Port Act of 1974 (33 U.S.C. 1501 et seq.); or

(1274) (ii) A delivering vessel that is offloading oil in bulk as cargo in lightering activities–

(1275) (A) Within a lightering zone established under 46 U.S.C. 3715(b)(5); and

(1276) (B) More than 60 miles from the territorial sea base line, as defined in 33 CFR 2.20.

(1277) (4) A vessel documented under 46 U.S.C., Chapter 121, that was equipped with a double hull before August 12, 1992;

(1278) (5) A barge of less than 1,500 gross tons as measured under 46 U.S.C., Chapter 145, carrying refined petroleum in bulk as cargo in or adjacent to waters of the Bering Sea, Chukchi Sea, and Arctic Ocean and waters tributary thereto and in the waters of the Aleutian Islands and the Alaskan Peninsula west of 155 degrees west longitude; or

(1279) (6) A vessel in the National Defense Reserve Fleet pursuant to 50 App. U.S.C. 1744.

(1280) (o) Section 157.11(h) applies to every oil tanker delivered on or after January 1, 2010, meaning an oil tanker—

(1281) (1) For which the building contract is placed on or after January 1, 2007;

(1282) (2) In the absence of a building contract, the keel of which is laid or which is at a similar stage of construction on or after July 1,2007;

(1283) (3) The delivery of which is on or after January 1, 2010; or

(1284) (4) That has undergone a major conversion—

(1285) (i) For which the contract is placed on or after January 1, 2007;

(1286) (ii) In the absence of a contract, the construction work of which is begun on or after July 1, 2007; or

(1287) (iii) That is completed on or after January 1, 2010.

(1288)

§157.10d Double hulls on tank vessels.

(1289) (a) With the exceptions stated in §157.08(n), this section applies to a tank vessel–

(1290) (1) For which the building contract is awarded after June 30, 1990; or

(1291) (2) That is delivered after December 31, 1993;

(1292) (3) That undergoes a major conversion for which;

(1293) (i) The contract is awarded after June 30, 1990; or

(1294) (ii) Conversion is completed after December 31, 1993; or

(1295) (4) That is otherwise required to have a double hull by 46 U.S.C. 3703a(c).

(1296) NOTE: 46 U.S.C. 3703a(c) is shown in appendix G to this part.

(1297) (b) Each vessel to which this section applies must be fitted with:

(1298) (1) A double hull in accordance with this section; and

(1299) (2) If §157.10 applies, segregated ballast tanks and a crude oil washing system in accordance with that section.

(1300) (c) Except on a vessel to which §157.10d(d) applies, tanks within the cargo tank length that carry any oil must be protected by double sides and a double bottom as follows:

(1301) (1) Double sides must extend for the full depth of the vessel's side or from the uppermost deck, disregarding a rounded gunwale where fitted, to the top of the double bottom. At any cross section, the molded width of the double side, measured at right angles to the side shell plating, from the side of tanks containing oil to the side shell plating, must not be less than the distance w, as shown in Figure 157.10d(c) and specified as follows:

(1302) (i) For a vessel of 5,000 DWT and above: w=[0.5+(DWT/20,000)] meters; or, w=2.0 meters (79 in.), whichever is less, but in no case less than 1.0 meter (39 in.).

(1303) (ii) For a vessel of less than 5,000 DWT: w=[0.4+(2.4)(DWT/20,000)] meters, but in no case less than 0.76 meter (30 in.).

(1304) (iii) For a vessel to which Paragraph (a)(4) of this section applies: w=0.76 meter (30 in.), provided that the double side was fitted under a construction or conversion contract awarded prior to June 30, 1990.

(1306) (2) At any cross section, the molded depth of the double bottom, measured at right angles to the bottom shell plating, from the bottom of tanks containing oil to the bottom shell plating, must not be less than the distance h, as shown in Figure 157.10d(c) and specified as follows:

(1307) (i) For a vessel of 5,000 DWT and above: h=B/15; or, h=2.0 meters (79 in.), whichever is less, but in no case less than 1.0 meter (39 in.).

(1305)

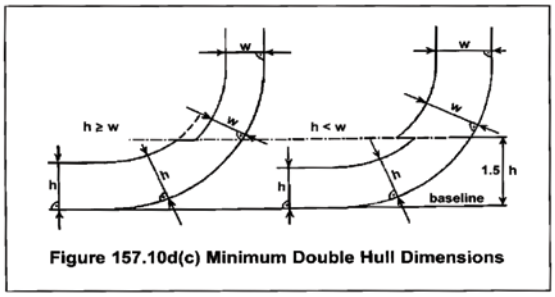


Figure 157.10d(c) Minimum Double Hull Dimensions

(1308) (ii) For a vessel of less than 5,000 DWT: h=B/15, but in no case less than 0.76 meter (30 in.).

(1309) (iii) For a vessel to which paragraph (a)(4) of this section applies: h=B/15; or, h=2.0 meters (79 in.), whichever is the lesser, but in no case less than 0.76 meter (30 in.), provided that the double bottom was fitted under a construction or conversion contract awarded prior to June 30, 1990.

(1310) (3) For a vessel built under a contract awarded after September 11, 1992, within the turn of the bilge or at cross sections where the turn of the bilge is not clearly defined, tanks containing oil must be located inboard of the outer shell–

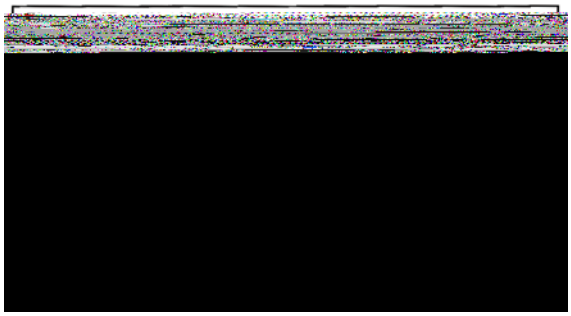
(1311) (i) For a vessel of 5,000 DWT and above: At levels up to 1.5h above the base line, not less than the distance h, as shown in Figure 157.10d(c) and specified in Paragraph (c)(2) of this section. At levels greater than 1.5h above the base line, not less than the distance w, as shown in Figure 157.10d(c) and specified in Paragraph (c)(1) of this section.

(1312) (ii) For a vessel of less than 5,000 DWT: Not less than the distance h above the line of the mid-ship flat bottom, as shown in Figure 157.10d(c)(3)(ii) and specified in Paragraph (c)(2) of this section. At levels greater than h above the line of the mid-ship flat bottom, not less than the distance w, as shown in Figure 157.10d(c)(3)(ii) and specified in Paragraph (c)(1) of this section.

(1314) (4) For a vessel to which §157.10(b) applies that is built under a contract awarded after September 11, 1992.

(1315) (i) The aggregate volume of the double sides, double bottom, forepeak tanks, and afterpeak tanks must not be less than the capacity of segregated ballast tanks required under §157.10(b). Segregated ballast tanks that may be provided in addition to those required under §157.10(b) may be located anywhere within the vessel.

(1313)



(1316) (ii) Double side and double bottom tanks used to meet the requirements of §157.10(b) must be located as uniformly as practicable along the cargo tank length. Large inboard extensions of individual double side and double bottom tanks, which result in a reduction of overall side or bottom protection, must be avoided.

(1317) (d) A vessel of less than 10,000 DWT that is constructed and certificated for service exclusively on inland or limited short protected coastwise routes must be fitted with double sides and a double bottom as follows:

(1318) (1) A minimum of 61 cm. (2 ft.) from the inboard side of the side shell plate, extending the full depth of the side or from the main deck to the top of the double bottom, measured at right angles to the side shell; and

(1319) (2) A minimum of 61 cm. (2 ft.) from the top of the bottom shell plating, along the full breadth of the vessel's bottom, measured at right angles to the bottom shell.

(1320) (3) For a vessel to which paragraph (a)(4) of this section applies, the width of the double sides and the depth of the double bottom may be 38 cm. (15 in.), in lieu of the dimensions specified in paragraphs (d)(1) and

(d)(2) of this section, provided that the double side and double bottom tanks were fitted under a construction or conversion contract awarded prior to June 30, 1990.

(1321) (4) For a vessel built under a contract awarded after September 11, 1992, a minimum 46 cm. (18 in.) clearance for passage between framing must be maintained throughout the double sides and double bottom.

(1322) (e) Except as provided in paragraph (e)(3) of this section, a vessel must not carry any oil in any tank extending forward of:

(1323) (1) The collision bulkhead; or

(1324) (2) In the absence of a collision bulk-head, the transverse plane perpendicular to the centerline through a point located:

(1325) (i) The lesser of 10 meters (32.8 ft.) or 5 percent of the vessel length, but in no case less than 1 meter (39 in.), aft of the forward perpendicular;

(1326) (ii) On a vessel of less than 10,000 DWT tons that is constructed and certificated for service exclusively on inland or limited short protected coastwise routes, the lesser of 7.62 meters (25 ft.) or 5 percent of the vessel length, but in no case less than 61 cm. (2 ft.), aft of the headlog or stem at the freeboard deck; or

(1327) (iii) On each vessel which operates exclusively as a box or trail barge, 61 cm. (2 ft.) aft of the headlog.

(1328) (3) This paragraph does not apply to independent fuel oil tanks that must be located on or above the main deck within the areas described in paragraphs (e)(1) and (e)(2) of this section to serve adjacent deck equipment that cannot be located further aft. Such tanks must be as small and as far aft as is practicable.

(1329) (f) On each vessel, the cargo tank length must not extend aft to any point closer to the stern than the distance equal to the required width of the double side, as prescribed in §157.10d(c)(1) or §157.10d(d)(1).

(1330)

Subpart G–Interim Measures for Certain Tank Vessels Without Double Hulls Carrying Petroleum Oils

(1331)

§157.400 Purpose and applicability.

(1332) (a) The purpose of this subpart is to establish mandatory safety and operational requirements to reduce environmental damage resulting from petroleum oil spills.

(1333) (b) This subpart applies to each tank vessel specified in §157.01 of this part that–

(1334) (1) Is 5,000 gross tons or more;

(1335) (2) Carries petroleum oil in bulk as cargo or oil cargo residue; and

(1336) (3) Is not equipped with a double hull meeting §157.10d of this part, or an equivalent to the requirements of §157.10d, but required to be equipped with a double hull at a date set forth in 46 U.S.C. 3703a (b)(3) and (c)(3).

(1337)

§157.445 Maneuvering performance capability.

(1338) (a) A tankship owner or operator shall ensure that maneuvering tests in accordance with IMO Resolution A.751(18), sections 1.2, 2.3-2.4, 3-4.2, and 5 (with Explanatory Notes in MSC/Circ. 644) have been conducted by July 29, 1997. Completion of maneuvering performance tests must be shown by–

(1339) (1) For a foreign flag tankship, a letter from the flag administration or an authorized classification society, as described in §157.04 of this part, stating the requirements in paragraph (a) of this section have been met; or

(1340) (2) For a U.S. flag tankship, results from the vessel owner confirming the completion of the tests or a letter from an authorized classification society, as described in §157.04 of this part, stating the requirements in paragraph (a) of this section have been met.

(1341) (b) If a tankship undergoes a major conversion or alteration affecting the control systems, control surfaces, propulsion system, or other areas which may be expected to alter maneuvering performance, the tankship owner or operator shall ensure that new maneuvering tests are conducted as required by paragraph (a) of this section.

(1342) (c) If a tankship is one of a class of vessels with identical propulsion, steering, hydrodynamic, and other relevant design characteristics, maneuvering performance test results for any tankship in the class may be used to satisfy the requirements of paragraph (a) of this section.

(1343) (d) The tankship owner or operator shall ensure that the performance test results, recorded in the format of Appendix 6 of the Explanatory Notes in MSC/Circ. 644., are prominently displayed in the wheelhouse.

(1344) (e) Prior to entering the port or place of destination and prior to getting underway, the tankship master shall discuss the results of the performance tests with the pilot while reviewing the anticipated transit and the possible impact of the tankship's maneuvering capability on the transit.

(1345)

Part 160–Portsand Waterways Safety–General

(1346)

Subpart A–General

(1347)

§160.1 Purpose.

(1348) This subchapter contains regulations implementing 46 U.S.C. chapter 700 "Ports and Waterways Safety" and related statutes.

(1349)

§160.3 Definitions.

(1350) For the purposes of this subchapter:

(1351) *Bulk* means material in any quantity that is shipped, stored, or handled without the benefit of package, label, mark or count and carried in integral or fixed independent tanks.

(1352) *Captain of the Port* means the Coast Guard officer designated by the Commandant to command a Captain of the Port Zone as described in part 3 of this chapter.

(1353) *Commandant* means the Commandant of the United States Coast Guard.

(1354) *Deviation* means any departure from any rule in this subchapter.

(1355) *Director, Vessel Traffic Services* means the Coast Guard officer designated by the Commandant to command a Vessel Traffic Service (VTS) as described in part 161 of this chapter.

(1356) *District Commander* means the Coast Guard officer designated by the Commandant to command a Coast Guard District as described in part 3 of this chapter.

(1357) *ETA* means estimated time of arrival.

(1358) *Length of Tow* means, when towing with a hawser, the length in feet from the stern of the towing vessel to the stern of the last vessel in tow. When pushing ahead or towing alongside, length of tow means the tandem length in feet of the vessels in tow excluding the length of the towing vessel.

(1359) *Person* means an individual, firm, corporation, association, partnership, or governmental entity.

(1360) *State* means each of the several States of the United States, the District of Columbia, the Commonwealth of Puerto Rico, Guam, American Samoa, the United States Virgin Islands, the Trust Territories of the Pacific Islands, the Commonwealth of the Northern Marianas Islands, and any other commonwealth, territory, or possession of the United States.

(1361) *Tanker* means a self-propelled tank vessel constructed or adapted primarily to carry oil or hazardous materials in bulk in the cargo spaces.

(1362) *Tank Vessel* means a vessel that is constructed or adapted to carry, or that carries, oil or hazardous material in bulk as cargo or cargo residue.

(1363) *Vehicle* means every type of conveyance capable of being used as a means of transportation on land.

(1364) *Vessel* means every description of watercraft or other artificial contrivance used, or capable of being used, as a means of transportation on water.

(1365) *Vessel Traffic Services (VTS)* means a service implemented under part 161 of this chapter by the United States Coast Guard designed to improve the safety and efficiency of vessel traffic and to protect the environment. The VTS has the capability to interact with marine traffic and respond to traffic situations developing in the VTS area.

(1366) *Vessel Traffic ServiceArea* or *VTS Area* means the geographical area encompassing a specific VTS area of service as described in part 161 of this chapter. This area of service may be subdivided into sectors for the purpose of allocating responsibility to individual Vessel Traffic Centers or to identify different operating requirements.

(1367) **Note:** Although regulatory jurisdiction is limited to the navigable waters of the United States, certain vessels will be encouraged or may be required, as a condition of port entry, to report beyond this area to facilitate traffic management within the VTS area.

(1368) *VTS Special Area* means a waterway within a VTS area in which special operating requirements apply.

(1369)

§160.5 Delegations.

(1370) (a) District Commanders and Captains of the Ports are delegated the authority to establish safety zones.

(1371) (b) Under the provisions of 33 CFR 6.04-1 and 6.04-6, District Commanders and Captains of the Ports have been delegated authority to establish security zones.

(1372) (c) Under the provisions 33 CFR §1.05-1, District Commanders have been delegated authority to establish regulated navigation areas.

(1373) (d) Subject to the supervision of the cognizant Captain of the Port and District Commander, Directors, Vessel Traffic Services are delegated authority under 33 CFR 1.01-30 to discharge the duties of the Captain of the Port that involve directing the operation, movement and anchorage of vessels within a Vessel Traffic Service area including management of vessel traffic within anchorages, regulated navigation areas and safety zones, and to enforce Vessel Traffic Service and ports and waterways safety regulations. This authority may be exercised by Vessel Traffic Center personnel. The Vessel Traffic Center may, within the Vessel Traffic Service area, provide information, make recommendations, or, to a vessel required under Part 161 of this chapter to participate in a Vessel Traffic Service, issue an order, including an order to operate or anchor as directed; require the vessel to comply with orders issued; specify times of entry, movement or departure; restrict operations as necessary for safe operation under the circumstances; or take other action necessary for control of the vessel and the safety of the port or of the marine environment.

(1374)

§160.7 Appeals.

(1375) (a) Any person directly affected by a safety zone or an order or direction issued under this subchapter may request reconsideration by the official who issued it or in whose name it was issued. This request may be made orally or in writing, and the decision of the official receiving the request may be rendered orally or in writing.

(1376) (b) Any person directly affected by the establishment of a safety zone or by an order or direction issued by, or on behalf of, a Captain of the Port may appeal to the District Commander through the Captain of the Port. The appeal must be in writing, except as allowed under paragraph (e) of this section, and shall contain complete supporting documentation and evidence which the appellant wishes to have considered. Upon receipt of the appeal, the District Commander may direct a representative to gather and submit documentation or other evidence which would be necessary or helpful to a resolution of the appeal. A copy of this documentation and evidence is made available to the appellant. The appellant is afforded five working days from the date of receipt to submit rebuttal materials.

Following submission of all materials, the District Commander issues a ruling, in writing, on the appeal. Prior to issuing the ruling, the District Commander may, as a matter of discretion, allow oral presentation on the issues.

(1377) (c) Any person directly affected by the establishment of a safety zone or by an order or direction issued by, or on behalf of, a District Commander, or who receives an unfavorable ruling on an appeal taken under paragraph (b) of this section may appeal to the Area Commander through the District Commander. The appeal must be in writing, except as allowed under paragraph (e) of this section, and shall contain complete supporting documentation and evidence which the appellant wishes to have considered. Upon receipt of the appeal, the Area Commander may direct a representative to gather and submit documentation or other evidence which would be necessary or helpful to a resolution of the appeal. A copy of this documentation and evidence is made available to the appellant. The appellant is afforded five working days from the date of receipt to submit rebuttal materials. Following submission of all materials, the Area Commander issues a ruling, in writing, on the appeal. Prior to issuing the ruling, the Area Commander may, as a matter of discretion, allow oral presentation on the issues.

(1378) (d) Any person who receives an unfavorable ruling on an appeal taken under paragraph (c) of this section, may appeal to the Commandant (CG–5P), Attn: Assistant Commandant for Prevention, U.S. Coast Guard Stop 7501, 2703 Martin Luther King Jr. Avenue SE., Washington, DC 20593–7501. The appeal must be in writing, except as allowed under paragraph (e) of this section. The Area Commander forwards the appeal, all the documents and evidence which formed the record upon which the order or direction was issued or the ruling under paragraph (c) of this section was made, and any comments which might be relevant, to the Assistant Commandant for Prevention. A copy of this documentation and evidence is made available to the appellant. The appellant is afforded 5 working days from the date of receipt to submit rebuttal materials to the Assistant Commandant for Prevention. The decision of the Assistant Commandant for Prevention is based upon the materials submitted, without oral argument or presentation. The decision of the Assistant Commandant for Prevention is issued in writing and constitutes final agency action.

(1379) (e) If the delay in presenting a written appeal would have significant adverse impact on the appellant, the appeal under paragraphs (b) and (c) of this section may initially be presented orally. If an initial presentation of the appeal is made orally, the appellant must submit the appeal in writing within five days of the oral presentation to the Coast Guard official to whom the presentation was made. The written appeal must contain, at a minimum, the basis for the appeal and a summary of the material presented orally. If requested, the official to whom the appeal is directed may stay the effect of the action while the ruling is being appealed.

(1380)

Subpart B–Control of Vessel and Facility Operations

(1381)

§160.101 Purpose.

(1382) This subpart describes the authority exercised by District Commanders and Captains of the Ports to insure the safety of vessels and waterfront facilities, and the protection of the navigable waters and the resources therein. The controls described in this subpart are directed to specific situations and hazards.

(1383)

§160.103 Applicability.

(1384) (a) This subpart applies to any–

(1385) (1) Vessel on the navigable waters of the United States, except as provided in paragraphs (b) and (c) of this section;

(1386) (2) Bridge or other structure on or in the navigable waters of the United States; and

(1387) (3) Land structure or shore area immediately adjacent to the navigable waters of the United States.

(1388) (b) This subpart does not apply to any vessel on the Saint Lawrence Seaway.

(1389) (c) Except pursuant to international treaty, convention, or agreement, to which the United States is a party, this subpart does not apply to any foreign vessel that is not destined for, or departing from, a port or place subject to the jurisdiction of the United States and that is in:

(1390) (1) Innocent passage through the territorial sea of the United States;

(1391) (2) Transit through the navigable waters of the United States which form a part of an international strait.

(1392)

§160.105 Compliance with orders.

(1393) Each person who has notice of the terms of an order issued under this subpart must comply with that order.

(1394)

§160.107 Denial of entry.

(1395) Each District Commander or Captain of the Port, subject to recognized principles of international law, may deny entry into the navigable waters of the United States or to any port or place under the jurisdiction of the United States, and within the district or zone of that District Commander or Captain of the Port, to any vessel not in compliance with the provisions of the Port and Tanker Safety Act (46 U.S.C. chapter 700) or the regulations issued thereunder.

(1396)

§160.109 Waterfront facility safety.

(1397) (a) To prevent damage to, or destruction of, any bridge or other structure on or in the navigable waters of the United States, or any land structure or shore area immediately adjacent to those waters, and to protect the navigable waters and the resources therein from harm

resulting from vessel or structure damage, destruction, or loss, each District Commander or Captain of the Port may:

(1398) (1) Direct the handling, loading, unloading, storage, stowage, and movement (including the emergency removal, control, and disposition) of explosives or other dangerous articles and substances, including oil or hazardous material as those terms are defined in 46 U.S.C. 2101 on any structure on or in the navigable waters of the United States, or any land structure or shore area immediately adjacent to those waters; and

(1399) (2) Conduct examinations to assure compliance with the safety equipment requirements for structures.

(1400)

§160.111 Special orders applying to vessel operations.

(1401) Each District Commander or Captain of the Port may order a vessel to operate or anchor in the manner directed when:

(1402) (a) The District Commander or Captain of the Port has reasonable cause to believe that the vessel is not in compliance with any regulation, law or treaty;

(1403) (b) The District Commander or Captain of the Port determines that the vessel does not satisfy the conditions for vessel operation and cargo transfers specified in §160.113; or

(1404) (c) The District Commander or Captain of the Port has determined that such order is justified in the interest of safety by reason of weather, visibility, sea conditions, temporary port congestion, other temporary hazardous circumstances, or the condition of the vessel.

(1405)

§160.113 Prohibition of vessel operation and cargo transfers.

(1406) (a) Each District Commander or Captain of the Port may prohibit any vessel, subject to the provisions of chapter 37 of Title 46, U.S. Code, from operating in the navigable waters of the United States, or from transferring cargo or residue in any port or place under the jurisdiction of the United States, and within the district or zone of that District Commander or Captain of the Port, if the District Commander or the Captain of the Port determines that the vessel's history of accidents, pollution incidents, or serious repair problems creates reason to believe that the vessel may be unsafe or pose a threat to the marine environment.

(1407) (b) The authority to issue orders prohibiting operation of the vessels or transfer of cargo or residue under paragraph (a) of this section also applies if the vessel:

(1408) (1) Fails to comply with any applicable regulation;

(1409) (2) Discharges oil or hazardous material in violation of any law or treaty of the United States;

(1410) (3) Does not comply with applicable vessel traffic service requirements;

(1411) (4) While underway, does not have at least one deck officer on the navigation bridge who is capable of communicating in the English language.

(1412) (c) When a vessel has been prohibited from operating in the navigable waters of the United States under paragraphs (a) or (b) of this section, the District Commander or Captain of the Port may allow provisional entry into the navigable waters of the United States, or into any port or place under the jurisdiction of the United States and within the district or zone of that District Commander or Captain of the Port, if the owner or operator of such vessel proves to the satisfaction of the District Commander or Captain of the Port, that the vessel is not unsafe or does not pose a threat to the marine environment, and that such entry is necessary for the safety of the vessel or the persons on board.

(1413) (d) A vessel which has been prohibited from operating in the navigable waters of the United States, or from transferring cargo or residue in a port or place under the jurisdiction of the United States under the provisions of paragraph (a) or (b)(1), (2) or (3) of this section, may be allowed provisional entry if the owner or operator proves, to the satisfaction of the District Commander or Captain of the Port that has jurisdiction, that the vessel is no longer unsafe or a threat to the environment, and that the condition which gave rise to the prohibition no longer exists.

(1414)

§160.115 Withholding of clearance.

(1415) (a) Each District Commander or Captain of the Port may request the Secretary of the Treasury, or the authorized representative thereof, to withhold or revoke the clearance required by 46 U.S.C. App. 91 of any vessel, the owner or operator of which is subject to any penalties under 46 U.S.C. 70036.

(1416)

Subpart C–Notification of Arrival, Hazardous Conditions, and Certain Dangerous Cargoes

(1417)

§160.201 General.

(1418) This subpart contains requirements and procedures for submitting a notice of arrival (NOA), and a notice of hazardous condition. The sections in this subpart describe:

(1419) (a) Applicability and exemptions from requirements in this subpart;

(1420) (b) Required information in an NOA;

(1421) (c) Required updates to an NOA;

(1422) (d) Methods and times for submission of an NOA, and updates to an NOA;

(1423) (e) How to obtain a waiver; and

(1424) (f) Requirements for submission of the notice of hazardous condition.

(1425) **Note to §160.201**. For notice-of-arrival requirements for the U.S. Outer Continental Shelf, see 33 CFR part 146.

(1426)

§160.202 Definitions.

(1427) Terms in this subpart that are not defined in this section or in §160.3 have the same meaning as those terms in 46 U.S.C. 2101. As used in this subpart—

(1428) *Agent* means any person, partnership, firm, company or corporation engaged by the owner or charterer of a vessel to act in their behalf in matters concerning the vessel.

(1429) *Barge* means a non-self propelled vessel engaged in commerce.

(1430) *Boundary waters* mean the waters from main shore to main shore of the lakes and rivers and connecting waterways, or the portions thereof, along which the international boundary between the United States and the Dominion of Canada passes, including all bays, arms, and inlets thereof, but not including tributary waters which in their natural channels would flow into such lakes, rivers, and waterways, or waters flowing from such lakes, rivers, and waterways, or the waters of rivers flowing across the boundary.

(1431) *Carried in bulk* means a commodity that is loaded or carried on board a vessel without containers or labels and received and handled without mark or count.

(1432) *Certain dangerous cargo (CDC)* includes any of the following:

(1433) (1) Division 1.1 or 1.2 explosives as defined in 49 CFR 173.50.

(1434) (2) Division 1.5D blasting agents for which a permit is required under 49 CFR 176.415 or, for which a permit is required as a condition of a Research and Special Programs Administration exemption.

(1435) (3) Division 2.3 "poisonous gas", as listed in 49 CFR 172.101 that is also a "material poisonous by inhalation" as defined in 49 CFR 171.8, and that is in a quantity in excess of 1 metric ton per vessel.

(1436) (4) Division 5.1 oxidizing materials for which a permit is required under 49 CFR 176.415 or for which a permit is required as a condition of a Research and Special Programs Administration exemption.

(1437) (5) A liquid material that has a primary or subsidiary classification of Division 6.1 "poisonous material" as listed in 49 CFR 172.101 that is also a "material poisonous by inhalation," as defined in 49 CFR 171.8 and that is in a bulk packaging, or that is in a quantity in excess of 20 metric tons per vessel when not in a bulk packaging.

(1438) (6) Class 7, "highway route controlled quantity" radioactive material or "fissile material, controlled shipment," as defined in 49 CFR 173.403.

(1439) (7) All bulk liquefied gas cargo carried under46 CFR 151.50-31 or listed in 46 CFR 154.7 that is flammable and/or toxic and that is not carried as certain dangerous cargo residue (CDC residue).

(1440) (8) The following bulk liquids except when carried as CDC residue:

(1441) (i) Acetone cyanohydrin;

(1442) (ii) Allyl alcohol;

(1443) (iii) Chlorosulfonic acid;

(1444) (iv) Crotonaldehyde;

(1445) (v) Ethylene chlorohydrin;

(1446) (vi) Ethylene dibromide;

(1447) (vii) Methacrylonitrile;

(1448) (viii) Oleum (fuming sulfuric acid); and

(1449) (ix) Propylene oxide, alone or mixed with ethylene oxide.

(1450) (9) The following bulk solids:

(1451) (i) Ammonium nitrate listed as Division 5.1 (oxidizing) material in 49 CFR 172.101 except when carried as CDC residue; and

(1452) (ii) Ammonium nitrate based fertilizer listed as a Division 5.1 (oxidizing) material in 49 CFR 172.101 except when carried as CDC residue.

(1453) *Certain dangerous cargo residue (CDC residue)* includes any of the following:

(1454) (1) Ammonium nitrate in bulk or ammonium nitrate based fertilizer in bulk remaining after all saleable cargo is discharged, not exceeding 1,000 pounds in total and not individually accumulated in quantities exceeding two cubic feet.

(1455) (2) For bulk liquids and liquefied gases, the cargo that remains onboard in a cargo system after discharge that is not accessible through normal transfer procedures, with the exception of the following bulk liquefied gas cargoes carried under 46 CFR 151.50-31 or listed in 46 CFR 154.7:

(1456) (i) Ammonia, anhydrous;

(1457) (ii) Chlorine;

(1458) (iii) Ethane;

(1459) (iv) Ethylene oxide;

(1460) (v) Methane (LNG);

(1461) (vi) Methyl bromide;

(1462) (vii) Sulfur dioxide; and

(1463) (viii) Vinyl chloride.

(1464) *Charterer* means the person or organization that contracts for the majority of the carrying capacity of a ship for the transportation of cargo to a stated port for a specified period. This includes "time charterers" and "voyage charterers."

(1465) *Crewmember* means all persons carried on board the vessel to provide navigation and maintenance of the vessel, its machinery, systems, and arrangements essential for propulsion and safe navigation or to provide services for other persons on board.

(1466) *Embark* means when a crewmember or a person in addition to the crew joins the vessel.

(1467) *Ferry schedule* means a published document that:

(1468) (1) Identifies locations a ferry travels to and from;

(1469) (2) Lists the times of departures and arrivals; and

(1470) (3) Identifies the portion of the year in which the ferry maintains this schedule.

(1471) *Foreign vessel* means a vessel of foreign registry or operated under the authority of a country except the United States.

(1472) *Great Lakes* means Lakes Superior, Michigan, Huron, Erie, and Ontario, their connecting and tributary

waters, the Saint Lawrence River as far as Saint Regis, and adjacent port areas.

(1473) *Gross tons* means the tonnage determined by the tonnage authorities of a vessel's flag state in accordance with the national tonnage rules in force before the entry into force of the International Convention on Tonnage Measurement of Ships, 1969 ("Convention"). For a vessel measured only under Annex I of the Convention, gross tons means that tonnage. For a vessel measured under both systems, the higher gross tonnage is the tonnage used for the purposes of the 300-gross-ton threshold.

(1474) *Hazardous condition* means any condition that may adversely affect the safety of any vessel, bridge, structure, or shore area or the environmental quality of any port, harbor, or navigable waterway of the United States. It may, but need not, involve collision, allision, fire, explosion, grounding, leaking, damage, injury or illness of a person aboard, or manning-shortage.

(1475) *Nationality* means the state (nation) in which a person is a citizen or to which a person owes permanent allegiance.

(1476) *Operating exclusively within a single Captain of the Port zone* refers to vessel movements within the boundaries of a single COTP zone, *e.g.*, from one dock to another, one berth to another, one anchorage to another, or any combination of such transits. Once a vessel has arrived in a port in a COPT zone, it would not be considered as departing from a port or place simply because of its movements within that specific port.

(1477) *Operator* means any person including, but not limited to, an owner, a charterer, or another contractor who conducts, or is responsible for, the operation of a vessel.

(1478) *Persons in addition to crewmembers* mean any person onboard the vessel, including passengers, who are not included on the list of crewmembers.

(1479) *Port or place of departure* means any port or place in which a vessel is anchored or moored.

(1480) *Port or place of destination* means any port or place in which a vessel is bound to anchor or moor.

(1481) *Public vessel* means a vessel that is owned or demise-(bareboat) chartered by the government of the United States, by a State or local government, or by the government of a foreign country and that is not engaged in commercial service.

(1482) *Time charterer* means the party who hires a vessel for a specific amount of time. The owner and his crew manage the vessel, but the charterer selects the ports of destination.

(1483) *Voyage charterer* means the party who hires a vessel for a single voyage. The owner and his crew manage the vessel, but the charterer selects the ports of destination.

(1484)

§160.203 Applicability.

(1485) (a) This subpart applies to the following vessels that are bound for or departing from ports or places within the navigable waters of the United States, as defined in 33 CFR 2.36(a), which includes internal waters and the territorial seas of the United States, and any deepwater port as defined in 33 CFR 148.5:

(1486) (1) U.S. vessels in commercial service, and

(1487) (2) All foreign vessels.

(1488) (b) Unless otherwise specified in this subpart, the owner, agent, master, operator, or person in charge of a vessel regulated by this subpart is responsible for compliance with the requirements in this subpart.

(1489) (c) Towing vessels controlling a barge or barges required to submit an NOA under this subpart must submit only one NOA containing the information required for the towing vessel and each barge under its control.

(1490)

§160.204 Exemptions and exceptions.

(1491) (a) Except for reporting notice of hazardous conditions, the following vessels are exempt from requirements in this subpart:

(1492) (1) A passenger or offshore supply vessel when employed in the exploration for or in the removal of oil, gas, or mineral resources on the continental shelf.

(1493) (2) An oil spill response vessel (OSRV) when engaged in actual spill response operations or during spill response exercises.

(1494) (3) After December 31, 2015, a vessel required by 33 CFR 165.830 or 165.921 to report its movements, its cargo, or the cargo in barges it is towing.

(1495) (4) A United States or Canadian vessel engaged in the salving operations of any property wrecked, or rendering aid and assistance to any vessels wrecked, disabled, or in distress, in waters specified in Article II of the 1908 Treaty of Extradition, Wrecking and Salvage (35 Stat. 2035; Treaty Series 502).

(1496) (5) The following vessels neither carrying certain dangerous cargo nor controlling another vessel carrying certain dangerous cargo:

(1497) (i) A foreign vessel 300 gross tons or less not engaged in commercial service.

(1498) (ii) A vessel operating exclusively within a single Captain of the Port zone. Captain of the Port zones are defined in 33 CFR part 3.

(1499) (iii) A U.S. towing vessel and a U.S. barge operating solely between ports or places of the contiguous 48 states, Alaska, and the District of Columbia.

(1500) (iv) A public vessel.

(1501) (v) Except for a tank vessel, a U.S. vessel operating solely between ports or places of the United States on the Great Lakes.

(1502) (vi) A U.S. vessel 300 gross tons or less, engaged in commercial service not coming from a foreign port or place.

(1503) (vii) Each ferry on a fixed route that is described in an accurate schedule that is submitted by the ferry operator, along with information in paragraphs (a)(5)(vii)(A) through (J) of this section, to the Captain of the Port for each port or place of destination listed in the schedule at least 24 hours in advance of the first date

(1520)

Table 160.206 – NOA Information Items		
Required Information	**Vessels neither carrying CDC nor controlling another vessel carrying CDC**	**Vessels carrying CDC or controlling another vessel carrying CDC**
(1) Vessel Information		
(i) Name	X	X
(ii) Name of the registered owner	X	X
(iii) Country of registry	X	X
(iv) Call sign	X	X
(v) International Maritime Organization (IMO) international number or, if vessel does not have an assigned IMO international number, substitute with o f cial number	X	X
(vi) Name of the operator	X	X
(vii) Name of the charterer	X	X
(viii) Name of classifcation society or recognized organization	X	X
(ix) Maritime Mobile Service Identity (MMSI) number, if applicable	X	X
(x) Whether the vessel is 300 gross tons or less (yes or no)	X	X
(xi) USCG Vessel Response Plan Control Number, if applicable	X	X
(2) Voyage Information		
(i) Names of last fve foreign ports or places visited	X	X
(ii) Dates of arrival and departure for last fve foreign ports or places visited	X	X
(iii) For the port or place of the United States to be visited, list the names of the receiving facility, the port or place, the city, and the state	X	X
(iv) For the port or place in the United States to be visited, the estimated date and time of arrival	X	X
(v) For the port or place in the United States to be visited, the estimated date and time of departure	X	X
(vi) The location (port or place and country) or position (latitude and longitude or waterway and mile marker) of the vessel at the time of reporting	X	X
(vii) The name and telephone number of a 24-hour point of contact	X	X
(viii) Whether the vessel's voyage time is less than 24 hours (yes or no)	X	X
(ix) Last port or place of departure	X	X
(x) Dates of arrival and departure for last port or place of departure	X	X
(3) Cargo Information		
(i) A general description of cargo, other than CDC, on board the vessel (e.g. grain, container, oil, etc.)	X	X
(ii) Name of each CDC carried, including cargo UN number, if applicable	–	X
(iii) Amount of each CDC carried	–	X
(4) Information for each Crewmember On Board		
(i) Full name	X	X
(ii) Date of birth	X	X
(iii) Nationality	X	X
(iv) Passport* or mariners document number (type of identifcation and number)	X	X
(v) Position or duties on the vessel	X	X
(vi) Where the crewmembers embarked (list port or place and country)	X	X
(5) Information for each Person On Board in Addition to Crew		
(i) Full name	X	X
(ii) Date of birth	X	X
(iii) Nationality	X	X
(iv) Passport number*	X	X
(v) Where the person embarked (list port or place and country)	X	X
(6) Operational condition of equipment required by 33 CFR part 164 of this chapter (see note to table)	X	X
(7) International Safety Management (ISM) Code Notice		
(i) The date of expiration for the company's Document of Compliance certifcate that covers the vessel	X	X
(ii) The date of expiration for the vessel's Safety Management Certifcate	X	X
(iii) The name of the Flag Administration, or the recognized organization(s) representing the vessel Flag Administration, that issued those certifcates	X	X
(8) International Ship and Port Facility Code (ISPS) Notice		
(i) The date of issuance for the vessel's International Ship Security Certifcate (ISSC), if any	X	X
(ii) Whether the ISSC, if any, is an initial Interim ISSC, subsequent and consecutive Interim ISSC, or fnal ISSC	X	X
(iii) Declaration that the approved ship security plan, if any, is being implemented	X	X
(iv) If a subsequent and consecutive Interim ISSC, the reasons therefore	X	X
(v) The name and 24-hour contact information for the Company Security O f cer	X	X
(vi) The name of the Flag Administration, or the recognized security organization(s) representing the vessel Flag Administration that issued the ISSC	X	X

Note to Table 160.206. For items with an asterisk (*), see paragraph (b) of this section. Submitting a response for item 6 indicating that navigation equipment is not operating properly does not serve as notice to the District Commander, Captain of the Port, or Vessel Tra f c Center, under 33 CFR 164.53.

and time of arrival listed on the schedule. At least 24 hours before the first date and time of arrival listed on the ferry schedule, each ferry operator who submits a schedule under paragraph (a)(5)(vii) of this section must also provide the following information to the Captain of the Port for each port or place of destination listed in the schedule for the ferry, and if the schedule or the following submitted information changes, the ferry operator must submit an updated schedule at least 24 hours in advance of the first date and time of arrival listed on the new schedule and updates on the following items whenever the submitted information is no longer accurate:

(1504) (A) Name of the vessel;

(1505) (B) Country of registry of the vessel;

(1506) (C) Call sign of the vessel;

(1507) (D) International Maritime Organization (IMO) international number or, if the vessel does not have an assigned IMO international number, the official number of the vessel;

(1508) (E) Name of the registered owner of the vessel;

(1509) (F) Name of the operator of the vessel;

(1510) (G) Name of the vessel's classification society or recognized organization, if applicable;

(1511) (H) Each port or place of destination;

(1512) (I) Estimated dates and times of arrivals at and departures from these ports or places; and

(1513) (J) Name and telephone number of a 24-hour point of contact.

(1514) (b) A vessel less than 500 gross tons is not required to submit the International Safety Management (ISM) Code Notice (Entry 7 in Table 160.206 of §160.206).

(1515) (c) A U.S. vessel is not required to submit the International Ship and Port Facility Security (ISPS) Code Notice information (Entry 8 in Table 160.206 of §160.206).

(1516)

§160.205 Notices of arrival.

(1517) The owner, agent, Master, operator, or person in charge of a vessel must submit notices of arrival consistent with the requirements in this subpart.

(1518)

§160.206 Information required in an NOA.

(1519) (a) Information required. With the exceptions noted in paragraph (b) of this section, each NOA must contain all of the information items specified in Table 160.206. Vessel owners and operators should protect any personal information they gather in preparing notices for transmittal to the National Vessel Movement Center (NVMC) to prevent unauthorized disclosure of that information.

(1521) (b) Exceptions. If a crewmember or person on board other than a crewmember is not required to carry a passport for travel, then passport information required in Table 160.206 by items (4)(iv) and (5)(iv) need not be provided for that person.

(1522)

§ 160.208 Updates to a submitted NOA.

(1523) (a) Unless otherwise specified in this section, whenever events cause NOA information submitted for a vessel to become inaccurate, or the submitter to realize that data submitted was inaccurate, the owner, agent, Master, operator, or person in charge of that vessel must submit an update within the times required in §160.212.

(1524) (b) Changes in the following information need not be reported:

(1525) (1) Changes in arrival or departure times that are less than six (6) hours;

(1526) (2) Changes in vessel location or position of the vessel at the time of reporting (entry (2)(vi) to Table 160.206); and

(1527) (3) Changes to crewmembers' position or duties on the vessel (entry (4)(vii) to Table 160.206).

(1528) (c) When reporting updates, revise and resubmit the NOA.

(1529)

§160.210 Methods for submitting an NOA.

(1530) (a) *National Vessel Movement Center (NVMC)*. Except as otherwise provided in this paragraph or paragraph (b) of this section, vessels must submit NOA information required by §160.206 to the NVMC using methods currently specified at *www.nvmc.uscg.gov*, which includes submission through the NVMC electronic Notice of Arrival and Departure (eNOAD) World Wide Web site, and XML, which includes the Excel Workbook format. These data may also be submitted using other methods that may be added as future options on *www.nvmc.uscg.gov*. XML spreadsheets may be submitted via email to *enoad@nvmc.uscg.gov*. If a vessel operator must submit an NOA or an update, for a vessel in an area without internet access or when experiencing technical difficulties with an onboard computer, and he or she has no shore-side support available, the vessel operator may fax or phone the submission to the NVMC. Fax at 1–800–547–8724 or 304–264–2684. Workbook available at *www.nvmc.uscg.gov*; or, telephone at 1–800–708–9823 or 304–264–2502.

(1531) (b) *Saint Lawrence Seaway*. Those vessels transiting the Saint Lawrence Seaway inbound, bound for a port or place in the United States, may meet the submission requirements of paragraph (a) of this section by submitting the required information to the Saint Lawrence Seaway Development Corporation and the Saint Lawrence Seaway Management Corporation of Canada using methods specified at *www.nvmc.uscg.gov*.

(1532)

§160.212 When to submit an NOA.

(1533) (a) *Submission of an NOA*. (1) Except as set out in paragraphs (a)(2) and (a)(3) of this section, all vessels must submit NOAs within the times required in paragraph (a)(4) of this section.

(1534) (2) Towing vessels, when in control of a vessel carrying CDC and operating solely between ports or

places of the contiguous 48 states, Alaska, and the District of Columbia, must submit an NOA before departure but at least 12 hours before arriving at the port or place of destination.

(1535) (3) U.S. vessels 300 gross tons or less, arriving from a foreign port or place, and whose voyage time is less than 24 hours must submit an NOA at least 60 minutes before departure from the foreign port or place. Also, Canadian vessels 300 gross tons or less, arriving directly from Canada, via boundary waters, to a United States port or place on the Great Lakes, whose voyage time is less than 24 hours must submit an NOA at least 60 minutes before departure from the Canadian port or place.

(1536) (4) Times for submitting NOAs are as follows:

(1537)

If your voyage time is –	Then you must submit an NOA –
(i) 96 hours or more; or	At least 96 hours before arriving at the port or place of destination; or
(ii) Less than 96 hours	Before departure but at least 24 hours before arriving at the port or place of destination.

(1538) (b) *Submission of updates to an NOA*. (1) Except as set out in paragraphs (b)(2) and (b)(3) of this section, vessels must submit updates in NOA information within the times required in paragraph (b)(4) of this section.

(1539) (2) Towing vessels, when in control of a vessel carrying CDC and operating solely between ports or places in the contiguous 48 states, Alaska, and the District of Columbia, must submit updates to an NOA as soon as practicable but at least 6 hours before entering the port or place of destination.

(1540) (3) U.S. vessels 300 gross tons or less, arriving from a foreign port or place, whose voyage time is—

(1541) (i) Less than 24 hours but greater than 6 hours, must submit updates to an NOA as soon as practicable, but at least 6 hours before entering the port or place of destination.

(1542) (ii) Less than or equal to 6 hours, must submit updates to an NOA as soon as practicable, but at least 60 minutes before departure from the foreign port or place.

(1543) (4) Times for submitting updates to NOAs are as follows:

(1544)

If your remaining voyage time is –	Then you must submit updates to an NOA –
(i) 96 hours or more;	As soon as practicable, but at least 24 hours before arriving at the port or place of destination;
(ii) Less than 96 hours but not less than 24 hours; or	As soon as practicable, but at least 24 hours before arriving at the port or place of destination; or
(iii) Less than 24 hours	As soon as practicable, but at least 12 hours before arriving at the port or place of destination.

(1545)

§160.214 Waivers.

(1546) The Captain of the Port may waive, within that Captain of the Port's designated zone, any of the requirements of this subpart for any vessel or class of vessels upon finding that the vessel, route, area of operations, conditions of the voyage, or other circumstances are such that application of this subpart is unnecessary or impractical for purposes of safety, environmental protection, or national security.

(1547)

§160.215 Force majeure.

(1548) When a vessel is bound for a port or place of the United States under force majeure, it must comply with the requirements in this section, but not other sections of this subpart. The vessel must report the following information to the nearest Captain of the Port as soon as practicable:

(1549) (a) The vessel Master's intentions;

(1550) (b) Any hazardous conditions as defined in §160.202; and

(1551) (c) If the vessel is carrying certain dangerous cargo or controlling a vessel carrying certain dangerous cargo, the amount and name of each CDC carried, including cargo UN number if applicable.

(1552)

§160.216 Notice of hazardous conditions.

(1553) (a) Whenever there is a hazardous condition either on board a vessel or caused by a vessel or its operation, the owner, agent, master, operator, or person in charge must immediately notify the nearest Coast Guard Sector Office or Group Office, and in addition submit any report required by 46 CFR 4.05-10.

(1554) (b) When the hazardous condition involves cargo loss or jettisoning as described in 33 CFR 97.115, the notification required by paragraph (a) of this section must include—

(1555) (1) What was lost, including a description of cargo, substances involved, and types of packages;

(1556) (2) How many were lost, including the number of packages and quantity of substances they represent;

(1557) (3) When the incident occurred, including the time of the incident or period of time over which the incident occurred;

(1558) (4) Where the incident occurred, including the exact or estimated location of the incident, the route the ship was taking, and the weather (wind and sea) conditions at the time or approximate time of the incident; and

(1559) (5) How the incident occurred, including the circumstances of the incident, the type of securing equipment that was used, and any other material failures that may have contributed to the incident.

(1560)

Part 164–Navigation Safety Regulations (in part). For a complete description of this part see 33 CFR 164.

(1561)

§164.01 Applicability.

(1562) (a) This part (except as specifically limited by this section) applies to each self-propelled vessel of 1600 or

more gross tons (except as provided in paragraphs (c) and (d) of this section, or for foreign vessels described in §164.02) when it is operating in the navigable waters of the United States except the St. Lawrence Seaway.

(1563) (b) Sections 164.70 through 164.82 of this part apply to each towing vessel of 12 meters (39.4 feet) or more in length operating in the navigable waters of the United States other than the St. Lawrence Seaway; except that a towing vessel is exempt from the requirements of §164.72 if it is–

(1564) (1) Used solely within a limited geographic area, such as a fleeting-area for barges or a commercial facility, and used solely for restricted service, such as making up or breaking up larger tows;

(1565) (2) Used solely for assistance towing as defined by 46 CFR 10.103;

(1566) (3) Used solely for pollution response; or

(1567) (4) Any other vessel exempted by the Captain of the Port (COTP). The COTP, upon written request, may, in writing, exempt a vessel from §164.72 for a specified route if he or she decides that exempting it would not allow its unsafe navigation under anticipated conditions.

(1568) (c) Provisions of §§164.11(a)(2) and (c), 164.30, 164.33, and 164.46 do not apply to warships or other vessels owned, leased, or operated by the United States Government and used only in government noncommercial service when these vessels are equipped with electronic navigation systems that have met the applicable agency regulations regarding navigation safety.

(1569) (d) Provisions of §164.46 apply to some self-propelled vessels of less 1600 gross tonnage.

(1570)

§164.02 Applicability exception for foreign vessels.

(1571) (a) Except for §164.46(c), none of the requirements of this part apply to foreign vessels that:

(1572) (1) Are not destined for, or departing from, a port or place subject to the jurisdiction of the United States; and

(1573) (2) Are in:

(1574) (i) Innocent passage through the territorial sea of the United States; or

(1575) (ii) Transit through navigable waters of the United States which form a part of an international strait.

(1576)

§164.03 Incorporation by reference.

(1577) (a) Certain material is incorporated by reference into this part with the approval of the Director of the Federal Register under 5 U.S.C. 552(a) and 1 CFR part 51. To enforce any edition other than that specified in this section, the Coast Guard must publish notice of the change in the Federal Register and the material must be available to the public. All approved material is available for inspection at the National Archives and Records Administration (NARA). For more information on the availability of this material at NARA, call 202–741–6030, or go to: *www.archives.gov/federal-register/cfr/ibr-locations.html*. Also, it is available for inspection at the Commandant (CG-NAV), U.S. Coast Guard Stop 7418, Attn: Office of Navigation Systems, 2703 Martin Luther King Jr. Ave. SE., Washington, DC 20593-7418, telephone 202–372–1565, and is available from the sources listed below.

(1578) (b) American Petroleum Institute (API), 1220 L Street NW., Washington, DC 20005-4070, 202–682–8000, *www.api.org*:

(1579) (1) API Specification 9A, Specification for Wire Rope, Section 3, Properties and Tests for Wire and Wire Rope, May 28, 1984, IBR approved for §164.74.

(1580) (2) [Reserved]

(1581) (c) ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959, 610–832–9585, *www.astm.org*:

(1582) (1) ASTM D4268-93, Standard Test Method for Testing Fiber Rope, IBR approved for §164.74.

(1583) (2) [Reserved]

(1584) (d) Cordage Institute, 350 Lincoln Street, Hingham, MA 02043.

(1585) (1) CIA-3, Standard Test Methods for Fiber Rope Including Standard Terminations, Revised, June 1980, IBR approved for §164.74.

(1586) (2) [Reserved]

(1587) (e) International Maritime Organization (IMO), 4 Albert Embankment, London SE1 7SR, United Kingdom, *www.imo.org*:

(1588) (1) IMO Resolution A342(IX), Recommendation on Performance Standards for Automatic Pilots, November 12, 1975, IBR approved for §164.13.

(1589) (2) IMO Resolution A.917(22), Guidelines for the Onboard Operational Use of Shipborne Automatic Identification System (AIS), January 25, 2002, IBR approved for §164.46.

(1590) (3) SN/Circ.227, Guidelines for the Installation of a Shipborne Automatic Identification System (AIS), January 6, 2003, IBR approved for §164.46.

(1591) (4) SN/Circ.244, Guidance on the Use of the UN/LOCODE in the Destination Field in AIS Messages, December 15, 2004, IBR approved for §164.46.

(1592) (5) SN/Circ.245, Amendments to the Guidelines for the Installation of a Shipborne Automatic Identification System (AIS)(SN/Circ.227), December 15, 2004, IBR approved for §164.46.

(1593) (6) SOLAS, International Convention for the Safety of Life at Sea, 1974, and 1988 Protocol relating thereto, 2000 Amendments, effective January and July 2002, (SOLAS 2000 Amendments), IBR approved for §164.46.

(1594) (7) Conference resolution 1, Adoption of amendments to the Annex to the International Convention for the Safety of Life at Sea, 1974, and amendments to Chapter V of SOLAS 1974, adopted on December 12, 2002, IBR approved for §164.46.

(1595) (8) SN.1/Circ.289, Guidance on the Use of AIS Application-Specific Messages, June 2, 2010, IBR approved for §164.46.

(1596) (f) National Marine Electronics Association (NMEA), 7 Riggs Avenue, Severna Park, MD 21146, 800–808–6632, *www.nmea.org*:

(1597) (1) NMEA 0400, Installation Standard for Marine Electronic Equipment used on Moderate-Sized Vessels, Version 3.10, February 2012, IBR approved for §164.46.

(1598) (2) [Reserved]

(1599) (g) Radio Technical Commission for Maritime Services (RTCM), 1611 N. Kent St., Suite 605, Arlington, VA 22209, 703–527–2000, *www.rtcm.org*:

(1600) (1) RTCM Paper 12-78/DO-100, Minimum Performance Standards, Loran C Receiving Equipment, 1977, IBR approved for §164.41.

(1601) (2) RTCM Paper 71-95/SC112-STD, RTCM Recommended Standards for Marine Radar Equipment Installed on Ships of Less Than 300 Tons Gross Tonnage, Version 1.1, October 10, 1995, IBR approved for §164.72.

(1602) (3) RTCM Paper 191-93/SC112-X, RTCM Recommended Standards for Maritime Radar Equipment Installed on Ships of 300 Tons Gross Tonnage and Upwards, Version 1.2, December 20, 1993, IBR approved for §164.72.

(1603) (h) International Electrotechnical Commission (IEC), 3, rue de Varembe, Geneva, Switzerland, +41 22 919 02 11, *http://www.iec.ch/.* Email: info@iec.ch.

(1604) (1) IEC 62065 (IEC 62065 2002–03), Maritime navigation and radiocommunication equipment and systems—Track control systems— Operational and performance requirements, methods of testing and required test results, First Edition, dated 2002, IBR approved for **§ 164.13(d)**.

(1605) (2) IEC 62065 (IEC 62065 2014–02), Maritime navigation and radiocommunication equipment and systems—Track control systems— Operational and performance requirements, methods of testing and required test results, Edition 2.0, dated 2014, IBR approved for **§ 164.13(d)**.

(1606)

§164.11 Navigation under way; General.

(1607) The owner, master, or person in charge of each vessel underway shall ensure that:

(1608) (a) The wheelhouse is constantly manned by persons who:

(1609) (1) Direct and control the movement of the vessel; and

(1610) (2) Fix the vessel's position;

(1611) (b) Each person performing a duty described in paragraph (a) of this section is competent to perform that duty;

(1612) (c) The position of the vessel at each fix is plotted on a chart of the area and the person directing the movement of the vessel is informed of the vessel's position;

(1613) (d) Electronic and other navigational equipment, external fixed aids to navigation, geographic reference points, and hydrographic contours are used when fixing the vessel's position;

(1614) (e) Buoys alone are not used to fix the vessel's position;

(1615) **Note:** Buoys are aids to navigation placed in approximate positions to alert the mariner to hazards to navigation or to indicate the orientation of a channel. Buoys may not maintain an exact position because strong or varying currents, heavy seas, ice, and collisions with vessels can move or sink them or set them adrift. Although buoys may corroborate a position fixed by other means, buoys cannot be used to fix a position: however, if no other aids are available, buoys alone may be used to establish an estimated position.

(1616) (f) The danger of each closing visual or each closing radar contact is evaluated and the person directing the movement of the vessel knows the evaluation;

(1617) (g) Rudder orders are executed as given;

(1618) (h) Engine speed and direction orders are executed as given;

(1619) (i) Magnetic variation and deviation and gyrocompass errors are known and correctly applied by the person directing the movement of the vessel;

(1620) (j) A person whom he has determined is competent to steer the vessel is in the wheelhouse at all times (See also 46 U.S.C. 8702(d), which requires an able seaman at the wheel on U.S. vessels of 100 gross tons or more in narrow or crowded waters during low visibility.);

(1621) (k) If a pilot other than a member of the vessel's crew is employed, the pilot is informed of the draft, maneuvering characteristics, and peculiarities of the vessel and of any abnormal circumstances on the vessel that may affect its safe navigation.

(1622) (l) Current velocity and direction for the area to be transited are known by the person directing the movement of the vessel;

(1623) (m) Predicted set and drift are known by the person directing movement of the vessel;

(1624) (n) Tidal state for the area to be transited is known by the person directing movement of the vessel;

(1625) (o) The vessel's anchors are ready for letting go;

(1626) (p) The person directing the movement of the vessel sets the vessel's speed with consideration for:

(1627) (1) The prevailing visibility and weather conditions;

(1628) (2) The proximity of the vessel to fixed shore and marine structures;

(1629) (3) The tendency of the vessel underway to squat and suffer impairment of maneuverability when there is small underkeel clearance;

(1630) (4) The comparative proportions of the vessel and the channel;

(1631) (5) The density of marine traffic;

(1632) (6) The damage that might be caused by the vessel's wake;

(1633) (7) The strength and direction of the current; and

(1634) (8) Any local vessel speed limit;

(1635) (q) The tests required by §164.25 are made and recorded in the vessel's log; and

(1636) (r) The equipment required by this part is maintained in operable condition.

(1637) (s) Upon entering U.S. waters, the steering wheel or lever on the navigating bridge is operated to determine if the steering equipment is operating properly under manual control, unless the vessel has been steered under

manual control from the navigating bridge within the preceding 2 hours, except when operating on the Great Lakes and their connecting and tributary waters.

(1638) (t) At least two of the steering gear power units on the vessel are in operation when such units are capable of simultaneous operation, except when the vessel is sailing on the Great Lakes and their connecting and tributary waters, and except as required by paragraph (u) of this section.

(1639) (u) On each passenger vessel meeting the requirements of the International Convention for the Safety of Life at Sea, 1960 (SILAS 60) and on each cargo vessel meeting the requirements of SILAS 74 as amended in 1981, the number of steering-gear power units necessary to move the rudder from 35° on either side to 30° on the other in not more than 28 seconds must be in simultaneous operation.

(1640)

§164.13 Navigation underway: tankers.

(1641) (a) As used in this section, "tanker" means a self-propelled tank vessel, including integrated tug barge combinations, constructed or adapted primarily to carry oil or hazardous material in bulk in the cargo spaces and inspected and certificated as a tanker.

(1642) (b) Each tanker must have an engineering watch capable of monitoring the propulsion system, communicating with the bridge, and implementing manual control measures immediately when necessary. The watch must be physically present in the machinery spaces or in the main control space and must consist of at least an engineer with an appropriately endorsed license or merchant mariner credential.

(1643) (c) Each tanker must navigate with at least two deck officers with an appropriately endorsed license or merchant mariner credential on watch on the bridge, one of whom may be a pilot. In waters where a pilot is required, the second officer, must be an individual holding an appropriately endorsed license or merchant mariner credential and assigned to the vessel as master, mate, or officer in charge of a navigational watch, who is separate and distinct from the pilot.

(1644) (d) This paragraph (d) has preemptive effect over State or local regulation within the same field. A tanker may navigate using a heading or track control system only if:

(1645) (1) The tanker is at least one-half nautical mile (1,012 yards) beyond the territorial sea baseline, as defined in **33 CFR 2.20**;

(1646) (i) Not within waters specified in **33 CFR part 110** (anchorages), or; (ii) Not within waters specified as precautionary areas in **33 CFR part 167**, and;

(1647) (2) There is a person, competent to steer the vessel, present to assume manual control of the steering station at all times including, but not limited to, the conditions listed in **46 CFR 35.20–45(a)** through **(c)**; and

(1648) (3) The system meets the heading or track control specifications of either IEC 62065 (2002–03) or IEC 62065 (2014–02) (incorporated by reference, see **§ 164.03**).

(1649)

§164.15 Navigation bridge visibility.

(1650) (a) The arrangement of cargo, cargo gear, and trim of all vessels entering or departing from U.S. ports must be such that the field of vision from the navigation bridge conforms as closely as possible to the following requirements:

(1651) (1) From the conning position, the view of the sea surface must not be obscured by more than the lesser of two ship lengths or 500 meters (1640 feet) from dead ahead to 10 degrees on either side of the vessel. Within this arc of visibility any blind sector caused by cargo, cargo gear, or other permanent obstruction must not exceed 5 degrees.

(1652) (2) From the conning position, the horizontal field of vision must extend over an arc from at least 22.5 degrees abaft the beam on one side of the vessel, through dead ahead, to at least 22.5 degrees abaft the beam on the other side of the vessel. Blind sectors forward of the beam caused by cargo, cargo gear, or other permanent obstruction must not exceed 10 degrees each, nor total more than 20 degrees, including any blind sector within the arc of visibility described in paragraph (a)(1) of this section.

(1653) (3) From each bridge wing, the field of vision must extend over an arc from at least 45 degrees on the opposite bow, through dead ahead, to at least dead astern.

(1654) (4) From the main steering position, the field of vision must extend over an arc from dead ahead to at least 60 degrees on either side of the vessel.

(1655) (b) A clear view must be provided through at least two front windows at all times regardless of weather conditions.

(1656)

§164.19 Requirements for vessels at anchor.

(1657) The master or person in charge of each vessel that is anchored shall ensure that:

(1658) (a) A proper anchor watch is maintained;

(1659) (b) Procedures are followed to detect a dragging anchor; and

(1660) (c) Whenever weather, tide, or current conditions are likely to cause the vessel's anchor to drag, action is taken to ensure the safety of the vessel, structures, and other vessels, such as being ready to veer chain, let go a second anchor, or get underway using the vessel's own propulsion or tug assistance.

(1661)

§164.25 Tests before entering or getting underway.

(1662) (a) Except as provided in paragraphs (b) and (c) of this section no person may cause a vessel to enter into or get underway on the navigable waters of the United States unless no more than 12 hours before entering or getting underway, the following equipment has been tested:

(1663) (1) Primary and secondary steering gear. The test procedure includes a visual inspection of the steering

gear and its connecting linkage, and where applicable, the operation of the following:

(1664) (i) Each remote steering gear control system.

(1665) (ii) Each steering position located on the navigating bridge.

(1666) (iii) The main steering gear from the alternative power supply, if installed.

(1667) (iv) Each rudder angle indicator in relation to the actual position of the rudder.

(1668) (v) Each remote steering gear control system power failure alarm.

(1669) (vi) Each remote steering gear power unit failure alarm.

(1670) (vii) The full movement of the rudder to the required capabilities of the steering gear.

(1671) (2) All internal vessel control communications and vessel control alarms.

(1672) (3) Standby or emergency generator, for as long as necessary to show proper functioning, including steady state temperature and pressure readings.

(1673) (4) Storage batteries for emergency lighting and power systems in vessel control and propulsion machinery spaces.

(1674) (5) Main propulsion machinery, ahead and astern.

(1675) (b) Vessels navigating on the Great Lakes and their connecting and tributary waters, having once completed the test requirements of this subpart, are considered to remain in compliance until arriving at the next port of call on the Great Lakes.

(1676) (c) Vessels entering the Great Lakes from the St. Lawrence Seaway are considered to be in compliance with this subpart if the required tests are conducted preparatory to or during the passage of the St. Lawrence Seaway or within one hour of passing Wolfe Island.

(1677) (d) No vessel may enter, or be operated on the navigable waters of the United States unless the emergency steering drill described below has been conducted within 48 hours prior to entry and logged in the vessel logbook, unless the drill is conducted and logged on a regular basis at least once every three months. This drill must include at a minimum the following:

(1678) (1) Operation of the main steering gear from within the steering gear compartment.

(1679) (2) Operation of the means of communications between the navigating bridge and the steering compartment.

(1680) (3) Operation of the alternative power supply for the steering gear if the vessel is so equipped.

(1681)

§164.30 Charts, publications, and equipment: General.

(1682) No person may operate or cause the operation of a vessel unless the vessel has the marine charts, publications, and equipment as required by §§164.33 through 164.41 of this part.

(1683)

§164.33 Charts and publications.

(1684) (a) Each vessel must have the following:

(1685) (1) Marine charts of the area to be transited, published by the National Ocean Service, U.S. Army Corps of Engineers, or a river authority that–

(1686) (i) Are of a large enough scale and have enough detail to make safe navigation of the area possible; and

(1687) (ii) Are currently corrected.

(1688) (2) For the area to be transited, a currently corrected copy of, or applicable currently corrected extract from, each of the following publications:

(1689) (i) U.S. Coast Pilot.

(1690) (ii) Coast Guard Light List.

(1691) (3) For the area to be transited, the current edition of, or applicable current extract from:

(1692) (i) Tide tables published by private entities using data provided by the National Ocean Service.

(1693) (ii) Tidal current tables published by private entities using data provided by the National Ocean Service, or river current publication issued by a river authority.

(1694) (b) As an alternative to the requirements for paragraph (a) of this section, a marine chart or publication, or applicable extract, published by a foreign government may be substituted for a U.S. chart and publication required by this section. The chart must be of large enough scale and have enough detail to make safe navigation of the area possible, and must be currently corrected. The publication, or applicable extract, must singly or in combination contain similar information to the U.S. Government publication to make safe navigation of the area possible. The publication, or applicable extract must be currently corrected, with the exceptions of tide and tidal current tables, which must be the current editions.

(1695) (c) As used in this section, "currently corrected" means corrected with changes contained in all Notices to Mariners published by National Geospatial-Intelligence Agency, or an equivalent foreign government publication, reasonably available to the vessel, and that is applicable to the vessel's transit.

(1696)

§164.35 Equipment: All vessels.

(1697) Each vessel must have the following:

(1698) (a) A marine radar system for surface navigation.

(1699) (b) An illuminated magnetic steering compass, mounted in a binnacle, that can be read at the vessel's main steering stand.

(1700) (c) A current magnetic compass deviation table or graph or compass comparison record for the steering compass, in the wheelhouse.

(1701) (d) A gyrocompass.

(1702) (e) An illuminated repeater for the gyrocompass required by paragraph (d) of this section that is at the main steering stand, unless that gyrocompass is illuminated and is at the main steering stand.

(1703) (f) An illuminated rudder angle indicator in the wheelhouse.

(1704) (g) The following maneuvering information prominently displayed on a fact sheet in the wheelhouse:

(1705) (1) A turning circle diagram to port and starboard that shows the time and distance and advance and transfer required to alter course 90 degrees with maximum rudder angle and constant power settings, for either full and half speeds, or for full and slow speeds. For vessels whose turning circles are essentially the same for both directions, a diagram showing a turning circle in one direction, with a note on the diagram stating that turns to port and starboard are essentially the same, may be substituted.

(1706) (2) The time and distance to stop the vessel from either full and half speeds, or from full and slow speeds, while maintaining approximately the initial heading with minimum application of rudder.

(1707) (3) For each vessel with a fixed propeller, a table of shaft revolutions per minute for a representative range of speeds.

(1708) (4) For each vessel with a controllable pitch propeller, a table of control settings for a representative range of speeds.

(1709) (5) For each vessel that is fitted with an auxiliary device to assist in maneuvering, such as a bow thruster, a table of vessel speeds at which the auxiliary device is effective in maneuvering the vessel.

(1710) (6) The maneuvering information for the normal load and normal ballast condition for:

(1711) (i) Calm weather-wind 10 knots or less, calm sea;

(1712) (ii) No current;

(1713) (iii) Deep water conditions-water depth twice the vessel's draft or greater; and

(1714) (iv) Clean hull.

(1715) (7) At the bottom of the fact sheet, the following statement:

(1716)

WARNING

The response of the (name of the vessel) may be different from that listed above if any of the following conditions, upon which the maneuvering information is based, are varied:
(1) Calm weather—wind 10 knots or less, calm sea;
(2) No current;
(3) Water depth twice the vessel's draft or greater;
(4) Clean hull; and
(5) Intermediate drafts or unusual trim.

(1717)

(1718) (h) An echo depth sounding device.

(1719) (i) A device that can continuously record the depth readings of the vessel's echo depth sounding device, except when operating on the Great Lakes and their connecting and tributary waters.

(1720) (j) Equipment on the bridge for plotting relative motion.

(1721) (k) Simple operating instructions with a block diagram, showing the changeover procedures for remote steering gear control systems and steering gear power units. permanently displayed on the navigating bridge and in the steering gear compartment.

(1722) (l) An indicator readable from the centerline conning position showing the rate of revolution of each propeller, except when operating on the Great Lakes and their connecting and tributary waters.

(1723) (m) If fitted with controllable pitch propellers, an indicator readable from the centerline conning position showing the pitch and operational mode of such propellers, except when operating on the Great Lakes and their connecting and tributary waters.

(1724) (n) If fitted with lateral thrust propellers, an indicator readable from the centerline conning position showing the direction and amount of thrust of such propellers, except when operating on the Great Lakes and their connecting and tributary waters.

(1725) (o) A telephone or other means of communication for relaying headings to the emergency steering station. Also, each vessel of 500 gross tons and over and constructed on or after June 9, 1995 must be provided with arrangements for supplying visual compass-readings to the emergency steering station.

(1726)

§164.37 Equipment: Vessels of 10,000 gross tons or more.

(1727) (a) Each vessel of 10,000 gross tons or more must have, in addition to the radar system under §164.35(a), a second marine radar system that operates independently of the first.

(1728) **Note:** Independent operation means two completely separate systems, from separate branch power supply circuits or distribution panels to antennas, so that failure of any component of one system will not render the other system inoperative.

(1729) (b) On each tanker of 10,000 gross tons or more that is subject to 46 U.S.C. 3708, the dual radar system required by this part must have a short range capability and a long range capability and each radar must have true north features consisting of a display that is stabilized in azimuth.

(1730)

§164.38 Automatic radar plotting aids (ARPA). (See 33 CFR 164.)

(1731)

§164.39 Steering gear: Foreign tankers.

(1732) (a) This section applies to each foreign tanker of 10,000 gross tons or more, except a public vessel, that–

(1733) (1) Transfers oil at a port or place subject to the jurisdiction of the United States; or

(1734) (2) Otherwise enters or operates in the navigable waters of the United States, except a vessel described by §164.02 of this part.

(1735) (b) *Definitions*. The terms used in this section are as follows:

(1736) *Constructed* means the same as in Chapter II-1, Regulations 1.1.2 and 1.1.3.1, of SILAS 74.

(1737) *Existing tanker* means a tanker–

(1738) (1) For which the building contract is placed on or after June 1, 1979;

(1739) (2) In the absence of a building contract, the keel of which is laid or which is at a similar stage of construction on or after January 1, 1980;

(1740) (3) The delivery of which occurs on or after June 1, 1982; or

(1741) (4) That has undergone a major conversion contracted for on or after June 1, 1979; or construction of which was begun on or after January 1, 1980, or completed on or after June 1, 1982.

(1742) *Public vessel, oil, hazardous materials, and foreign vessel* mean the same as in 46 U.S.C. 2101.

(1743) *SOLAS 74* means the International Convention for the Safety of Life at Sea, 1974, as amended.

(1744) *Tanker* means a self-propelled vessel defined as a tanker by 46 U.S.C. 2101(38) or as a tank vessel by 46 U.S.C. 2101(39).

(1745) (c) Each tanker constructed on or after September 1, 1984, must meet the applicable requirements of Chapter II-1, Regulations 29 and 30, of SOLAS 74.

(1746) (d) Each tanker constructed before September 1, 1984, must meet the requirements of Chapter II-1, Regulation 29.19, of SOLAS 74.

(1747) (e) Each tanker of 40,000 gross tons or more, constructed before September 1, 1984, that does not meet the single-failure criterion of Chapter II-1, Regulation 29.16, of SOLAS 74, must meet the requirements of Chapter II-1, Regulation 29.20, of SOLAS 74.

(1748) (f) Each tanker constructed before September 1, 1984, must meet the applicable requirements of Chapter II-1, Regulations 29.14 and 29.15, of SOLAS 74.

(1749)

§164.40 Devices to indicate speed and distance.

(1750) (a) Each vessel required to be fitted with an Automatic Radar Plotting Aid (ARPA) under §164.38 of this part must be fitted with a device to indicate speed and distance of the vessel either through the water or over the ground.

(1751) (b) The device must meet the following specifications:

(1752) (1) The display must be easily readable on the bridge by day or night.

(1753) (2) Errors in the indicated speed, when the vessel is operating free from shallow water effect, and from the effects of wind, current, and tide, should not exceed 5 percent of the speed of the vessel, or 0.5 knot, whichever is greater.

(1754) (3) Errors in the indicated distance run, when the vessel is operating free from shallow water effect, and from the effects of wind, current, and tide, should not exceed 5 percent of the distance run of the vessel in one hour or 0.5 nautical mile in each hour, whichever is greater.

(1755)

§164.41 Electronic position fixing devices.

(1756) (a) Each vessel calling at a port in the continental United States, including Alaska south of Cape Prince of Wales, except each vessel owned or barefoot chartered and operated by the United States, or by a state or its political subdivision, or by a foreign nation, and not engaged in commerce, must have a satellite navigation receiver with—

(1757) (1) Automatic acquisition of satellite signals after initial operator settings have been entered; and

(1758) (2) Position updates derived from satellite information during each usable satellite pass.

(1759) (b) A system that is found by the Commandant to meet the intent of the statements of availability, coverage, and accuracy for the U.S. Coastal Confluence Zone (CCZ) contained in the U.S. "Federal Radionavigation Plan" (Report No. DOD–NO 4650.4–P, I or No. DOT–TSC–RSPA–80–16, I). A person desiring a finding by the Commandant under this subparagraph must submit a written application describing the device to the Commandant (CG-DCO-D), Attn: Deputy for Operations Policy and Capabilities, U.S. Coast Guard Stop 7318, 2703 Martin Luther King Jr. Avenue SE., Washington, DC 20593-7318. After reviewing the application, the Commandant may request additional information to establish whether or not the device meets the intent of the Federal Radionavigation Plan.

(1760) **Note**: The Federal Radionavigation Plan is available from the National Technical Information Service, Springfield, VA 22161, with the following Government Accession Numbers:

(1761) Vol 1, ADA 116468

(1762) Vol 2, ADA 116469

(1763) Vol 3, ADA 116470

(1764) Vol 4, ADA 116471

(1765)

§164.42 Rate of turn indicator.

(1766) Each vessel of 100,000 gross tons or more constructed on or after September 1, 1984, shall be fitted with a rate of turn indicator.

(1767)

§164.43 [Removed]

(1768)

§164.46 Automatic Identification System (AIS).

(1769) (a) *Definitions*. As used in this section—Automatic Identification Systems or AIS means a maritime navigation safety communications system standardized by the International Telecommunication Union (ITU), adopted by the International Maritime Organization (IMO), that—

(1770) (1) Provides vessel information, including the vessel's identity, type, position, course, speed, navigational status and other safety-related information automatically to appropriately equipped shore stations, other ships, and aircraft;

(1771) (2) Receives automatically such information from similarly fitted ships, monitors and tracks ships; and

(1772) (3) Exchanges data with shore-based facilities.

(1773) *Gross tonnage* means tonnage as defined under the International Convention on Tonnage Measurement of Ships, 1969.

(1774) *International voyage* means a voyage from a country to which the present International Convention

for the Safety of Life at Sea applies to a port outside such country, or conversely.

(1775) *Properly installed, operational* means an Automatic Identification System (AIS) that is installed and operated using the guidelines set forth by the International Maritime Organization (IMO) Resolution A.917(22) and Safety of Navigation Circulars (SN/Circ.) 227, 244, 245, and SN.1/Circ.289; or National Marine Electronics Association (NMEA) Installation Standard 0400-3.10 in lieu of SN/Circ.227 and 245 (incorporated by reference, see §164.03).

(1776) (b) *AIS carriage*—(1) *AIS Class A device*. The following vessels must have on board a properly installed, operational Coast Guard type-approved AIS Class A device:

(1777) (i) A self-propelled vessel of 65 feet or more in length, engaged in commercial service.

(1778) (ii) A towing vessel of 26 feet or more in length and more than 600 horsepower, engaged in commercial service.

(1779) (iii) A self-propelled vessel that is certificated to carry more than 150 passengers.

(1780) (iv) A self-propelled vessel engaged in dredging operations in or near a commercial channel or shipping fairway in a manner likely to restrict or affect navigation of other vessels.

(1781) (v) A self-propelled vessel engaged in the movement of—

(1782) (A) Certain dangerous cargo as defined in subpart C of part 160 of this chapter, or

(1783) (B) Flammable or combustible liquid cargo in bulk that is listed in 46 CFR 30.25-1, Table 30.25-1.

(1784) (2) *AIS Class B device*. Use of a Coast Guard type-approved AIS Class B device in lieu of an AIS Class A device is permissible on the following vessels if they are not subject to pilotage by other than the vessel Master or crew:

(1785) (i) Fishing industry vessels;

(1786) (ii) Vessels identified in paragraph (b)(1)(i) of this section that are certificated to carry less than 150 passengers and that—

(1787) (A) Do not operate in a Vessel Traffic Service (VTS) or Vessel Movement Reporting System (VMRS) area defined in Table 161.12(c) of §161.12 of this chapter, and

(1788) (B) Do not operate at speeds in excess of 14 knots; and

(1789) (iii) Vessels identified in paragraph (b)(1)(iv) of this section engaged in dredging operations.

(1790) **Note to paragraph (b):** Under 46 U.S.C. 70002 and 33 CFR 160.111, a Coast Guard Captain of the Port (COTP) may restrict the operation of a vessel if he or she determines that by reason of weather, visibility, sea conditions, port congestion, other hazardous circumstances, or the condition of such vessel, the restriction is justified in the interest of safety. In certain circumstances, if a COTP is concerned that the operation of a vessel not subject to §164.46 would be unsafe, the COTP may determine that voluntary installation of AIS by the operator would mitigate that concern. Fishing industry vessels include fishing vessels, fish processing vessels, and fish tender vessels as defined in 46 U.S.C. 2101.

(1791) (c) *SOLAS provisions*. The following self-propelled vessels must comply with International Convention for Safety of Life at Sea (SOLAS), as amended, Chapter V, regulation 19.2.1.6 (Positioning System), 19.2.4 (AIS Class A), and 19.2.3.5 (Transmitting Heading Device) or 19.2.5.1 (Gyro Compass) as applicable (Incorporated by reference, see §164.03):

(1792) (1) A vessel of 300 gross tonnage or more, on an international voyage.

(1793) (2) A vessel of 150 gross tonnage or more, when carrying more than 12 passengers on an international voyage.

(1794) (d) *Operations*. The requirements in this paragraph are applicable to any vessel equiped with AIS.

(1795) (1) Use of AIS does not relieve the vessel of the requirements to sound whistle signals or display lights or shapes in accordance with the International Regulations for Preventing Collisions at Sea, 1972 (72 COLREGS), 28 U.S.T. 3459, T.I.A.S. 8587, or Inland Navigation Rules, 33 CFR part 83; nor of the radio requirements of the Vessel Bridge-to-Bridge Radiotelephone Act, 33 U.S.C. 1201-1208, part 26 of this chapter, and 47 CFR part 80.

(1796) (2) AIS must be maintained in effective operating condition, which includes—

(1797) (i) The ability to reinitialize the AIS, which requires access to and knowledge of the AIS power source and password;

(1798) (ii) The ability to access AIS information from the primary conning position of the vessel;

(1799) (iii) The accurate broadcast of a properly assigned Maritime Mobile Service Identity (MMSI) number;

(1800) (iv) The accurate input and upkeep of all AIS data fields and system updates; and

(1801) (v) For those vessels denoted in paragraph (b) of this section, the continual operation of AIS and its associated devices (*e.g.*, positioning system, gyro, converters, displays) at all times while the vessel is underway or at anchor, and, if moored, at least 15 minutes prior to getting underway; except when its operation would compromise the safety or security of the vessel or a security incident is imminent. The AIS should be returned to continuous operation as soon as the compromise has been mitigated or the security incident has passed. The time and reason for the silent period should be recorded in the ship's official log and reported to the nearest Captain of the Port or Vessel Traffic Center (VTC).

(1802) (3) AIS safety-related text messaging must be conducted in English and solely to exchange or communicate pertinent navigation safety information (analogous to a SECURITE broadcast). Although not prohibited, AIS text messaging should not be relied upon as the primary means for broadcasting distress (MAYDAY)

or urgent (PAN PAN) communications. (47 CFR 80.1109, Distress, urgency, and safety communications).

(1803) (4) AIS application-specific messaging (ASM) is permissible, but is limited to applications adopted by the International Maritime Organization (such as IMO SN.1/Circ.289) or those denoted in the International Association of Marine Aids to Navigation and Lighthouse Authorities' (IALA) ASM Collection for use in the United States or Canada, and to no more than one ASM per minute.

(1804) **Note 1 to §164.46(d):** The Coast Guard has developed the "USCG AIS Encoding Guidance" to help ensure consistent and accurate data encoding (input) by AIS users. This Guide is available at our "AIS Frequently Asked Questions" (FAQ #2) World Wide Web page at *www.navcen.uscg.gov/ais-frequently-asked-questions#2*. Although of great benefit, the interfacing or installation of other external devices or displays (e.g., transmitting heading device, gyro, rate of turn indicator, electronic charting systems, and radar), is not currently required except as denoted in §164.46(c). Most application-specific messages require interfacing to an external system that is capable of their portrayal, such as equipment certified to meet Radio Technical Commission for Maritime Services (RTCM) electronic chart system (ECS) standard 10900 series.

(1805) (e) *Watchkeeping*. AIS is primarily intended for use by the Master or person in charge of the vessel, or by the person designated by the Master or person in charge to pilot or direct the movement of the vessel, who must maintain a periodic watch for AIS information.

(1806) (f) *Portable AIS*. The use of a portable AIS is permissible only to the extent that electromagnetic interference does not affect the proper function of existing navigation and communication equipment on board and such that only one AIS device may be transmitting on board a vessel at any one time.

(1807) (g) *AIS Pilot Plug*. The AIS Pilot Plug on any vessel subject to pilotage by other than the vessel Master or crew must be readily available and easily accessible from the primary conning position of the vessel and permanently affixed (not an extension cord) and adjacent (within 3 feet) to a 120-volt 50/60 Hz AC power receptacle (NEMA 5-15).

(1808) (h) *Exceptions*. The following vessels may seek up to a 5-year deviation from the AIS requirements of this section by requesting a deviation under §164.55.

(1809) (1) Vessels that operate solely within a very confined area (e.g., less than a 1 nautical-mile radius, shipyard, or barge fleeting facility);

(1810) (2) Vessels that conduct only short voyages (less than 1 nautical mile) on a fixed schedule (e.g., a bank-to-bank river ferry service or a tender vessel);

(1811) (3) Vessels that are not likely to encounter other AIS-equipped vessels;

(1812) (4) Vessels whose design or construction makes it impracticable to operate an AIS device (e.g., those that lack electrical power, have an exposed or open cabin, or are submersible); or

(1813) (5) Vessels denoted in paragraph (b)(2) that seek a deviation from requirements in paragraphs (d)(2)(ii) and (e) of this section because their AIS Class B device lacks a display.

(1814) (i) *Prohibition*. Except for maritime support stations (see 47 CFR 80.5) licensed by the Federal Communications Commission (FCC), broadcasts from AIS Class A or B devices on aircraft, non-self propelled vessels or from land are prohibited.

(1815) (j) *Implementation date*. Those vessels identified in paragraphs (b) and (c) of this section that were not previously subject to AIS carriage must install AIS no later than March 1, 2016.

(1816)

§164.51 Deviations from rules: Emergency.

(1817) Except for the requirements of §164.53(b), in an emergency, any person may deviate from any rule in this part to the extent necessary to avoid endangering persons, property, or the environment.

(1818)

§164.53 Deviations from rules and reporting: Non-operating equipment.

(1819) (a) If during a voyage any equipment required by this part stops operating properly, the person directing the movement of the vessel may continue to the next port of call, subject to the directions of the District Commander or the Captain of the Port, as provided by 33 CFR 160.

(1820) (b) If the vessel's automatic identification system (AIS), radar, radio navigation receivers, gyrocompass, echo depth sounding device, or primary steering gear stops operating properly, the person directing the movement of the vessel must report or cause to be reported that it is not operating properly to the nearest Captain of the Port, District Commander, or, if participating in a Vessel Traffic Service, to the Vessel Traffic Center, as soon as possible.

(1821)

§164.55 Deviations from rules: Continuing operation or period of time.

(1822) The Captain of the Port, upon written application, may authorize a deviation from any rule in this part if he determines that the deviation does not impair the safe navigation of the vessel under anticipated conditions and will not result in a violation of the rules for preventing collisions at sea. The authorization may be issued for vessels operating in the waters under the jurisdiction of the Captain of the Port for any continuing operation or period of time the Captain of the Port specifies.

(1823)

§164.61 Marine casualty reporting and record retention.

(1824) When a vessel is involved in a marine casualty as defined in 46 CFR 4.03-1, the master or person in charge of the vessel shall:

(1825) (a) Ensure compliance with 46 CFR 4.05, "Notice of Marine Casualty and Voyage Records," and

(1826) (b) Ensure that the voyage records required by 46 CFR 4.05-15 are retained for:

(1827) (1) 30 days after the casualty if the vessel remains in the navigable waters of the United States; or

(1828) (2) 30 days after the return of the vessel to a United States port if the vessel departs the navigable waters of the United States within 30 days after the marine casualty.

(1829)

§164.70 Definitions.

(1830) For purposes of §§164.72 through 164.82, the term–

(1831) *Current edition* means the most recent published version of a publication, chart, or map required by §164.72.

(1832) *Currently corrected edition* means a current or previous edition of a publication required by §164.72, corrected with changes that come from Notice to Mariners (NTMs) or Notices to Navigation reasonably available and that apply to the vessel's transit. Hand-annotated river maps from U.S. Army Corps of Engineers (USACE) are currently corrected editions if issued within the previous 5 years.

(1833) *Great Lakes* means the Great Lakes and their connecting and tributary waters including the Calumet River as far as the Thomas J. O'Brien Lock and Controlling Works (between miles 326 and 327), the Chicago River as far as the east side of the Ashland Avenue Bridge (between miles 321 and 322), and the Saint Lawrence River as far east as the lower exit of Saint Lambert Lock.

(1834) *Merchant mariner credential* or *MMC* means the credential issued by the Coast Guard under 46 CFR part 10. It combines the individual merchant mariner's document, license, and certificate of registry enumerated in 46 U.S.C. subtitle II part E as well as the STCW endorsement into a single credential that serves as the mariner's qualification document, certificate of identification, and certificate of service.

(1835) *Swing-meter* means an electronic or electric device that indicates that rate of turn of the vessel on board which it is installed.

(1836) *Towing vessel* means a commercial vessel engaged in or intending to engage in pulling, pushing or hauling alongside, or any combination of pulling, pushing, or hauling alongside.

(1837) *Western Rivers* means the Mississippi River, its tributaries, South Pass, and Southwest Pass, to the navigational-demarcation lines dividing the high seas from harbors, rivers, and other inland waters of the United States, and the Port Allen-Morgan City Alternative Route, and that part of the Atchafalaya River above its junction with the Port Allen-Morgan City Alternative Route including the Old River and the Red River and those waters specified by §§89.25 and 89.27 of this chapter, and such other, similar waters as are designated by the COTP.

(1838)

§164.72 Navigational-safety equipment, charts or maps, and publications required on towing vessels.

(1839) (a) Except as provided by §164.01(b), each towing vessel must be equipped with the following navigational-safety equipment:

(1840) (1) *Marine Radar*. By August 2, 1997, a marine radar that meets the following applicable requirements:

(1841) (i) For a vessel of less than 300 tons gross tonnage that engages in towing on navigable waters of the U.S., including Western Rivers, the radar must meet–

(1842) (A) The requirements of the Federal Communications Commission (FCC) specified by 47 CFR part 80; and

(1843) (B) RTCM Standard for Marine Radar Equipment Installed on Ships of Less Than 300 Tons Gross Tonnage, RTCM Paper-71-95/SC112-STD, Version 1.1, display Category II and stabilization Category Bravo.

(1844) (ii) For a vessel of less than 300 tons gross tonnage that engages in towing seaward of navigable waters of the U.S. or more than three nautical miles from shore on the Great Lakes, the radar must meet–

(1845) (A) The requirements of the FCC specified by 47 CFR part 80; and

(1846) (B) RTCM Standard for Marine Radar Equipment Installed on Ships of Less Than 300 Tons Gross Tonnage, RTCM Paper 71-95/SC112-STD, Version 1.1, display Category I and stabilization Category Alpha.

(1847) (iii) For a vessel of 300 tons gross tonnage or more that engages in towing on navigable waters of the U.S. including Western rivers, the radar must meet–

(1848) (A) The requirements of the Federal Communications Commission (FCC) specified by a 47 CFR part 80; and

(1849) (B) RTCM Recommended Standards for Marine Radar Equipment Installed on Ships of 300 Tons Gross Tonnage and Upwards, RTCM Paper 191-93/SC112-X, Version 1.2 except the requirements for azimuth stabilization in paragraph 3.10.

(1850) (iv) For a vessel of 300 tons gross tonnage or more that engages in towing seaward of navigable waters of the U.S. or more than three nautical miles from shore on the Great Lakes, the radar must meet–

(1851) (A) The requirements of the FCC specified by 47 CFR Part 80; and

(1852) (B) RTCM Recommended Standards for Marine Radar Equipment Installed on Ships of 300 Tons Gross Tonnage and Upwards, RTCM Paper 191-93/SC112-X, Version 1.2.

(1853) (v) A towing vessel with an existing radar must meet the applicable requirements of paragraphs (a)(1)(i) through (iv) of this section by August 2, 1998; except that a towing vessel with an existing radar must meet the display and stabilization requirements of paragraph (a)(1)(ii)(B) of this section by August 2, 2001.

(1854) (2) *Searchlight*. A searchlight, directable from the vessel's main steering station and capable of illuminating objects at a distance of at least two times the length of the tow.

(1880)

TABLE 164.72 – Equipment, Charts or Maps, and Publications of Towing Vessels for 12 Meters or More in Length

	Western Rivers	U.S. Navigable Waters (other than Western Rivers)	Waters seaward of Navigable Waters and 3 NM or more from shore on the Great Lakes
Marine Radar: Towing Vessels of less than 300 GT	RTCM Paper 71-95/SC112-STD Version 1.1 Display Category II[1] Stabilization Category BRAVO	RTCM Paper 71-95/SC112-STD Version 1.1 Display Category II[1] Stabilization Category BRAVO	RTCM Paper 71-95/SC112-STD Version 1.1 Display Category I[2] Stabilization Category ALPHA
Towing Vessels of 300 GT or more	RTCM Paper 191-93/SC112-X Version 1.2 (except the Azimuth stabilization requirement in paragraph 3.10)[1]	RTCM Paper 191-93/SC112-X Version 1.2 (except the Azimuth stabilization requirement in paragraph 3.10)[1]	RTCM Paper 191-93/SC112-X Version 1.2[1]
Searchlight	X	X	X
VHF-FM Radio	X	X	X
Magnetic Compass	X[3]	X	X
Swing Meter	X[3]		
Echo Depth-sounding Device		X	X
Electronic Position Fixing Device			X
Charts or Maps	(1) Large enough scale (2) Current edition or currently corrected edition	(1) Large enough scale (2) Current edition or currently corrected edition	(1) Large enough scale (2) Currently corrected edition
General Publications	(1) U.S. Coast Guard Light List (2) Notices to Navigation or Local Notices to Mariners (3) River-current Tables	(1) U.S. Coast Guard Light List (2) Local Notices to Mariners (3) Tidal-current Tables (4) Tide Tables (5) U.S. Coast Pilot	(1) U.S. Coast Guard Light List (2) Local Notices to Mariners (3) Tidal-current Tables (4) Tide Tables (5) U.S. Coast Pilot

Notes:
[1] Towing vessels with existing radar must meet this requirement by August 2, 1998.
[2] Towing vessels with existing radar must meet this requirement by August 2, 1998 but do not need to meet the display and stabilization requirements until August 2, 2001.
[3] A towing vessel may carry either a swing-meter or a magnetic compass.

(1855) (3) *VHF-FM Radio*. An installation or multiple installations of VHF-FM radios as prescribed by part 26 of this chapter and 47 CFR part 80, to maintain a continuous listening watch on the designated calling channel, VHF-FM Channel 13 (except on portions of the Lower Mississippi River, where VHF-FM Channel 67 is the designated calling channel), and to separately monitor the International Distress and Calling Channel, VHF-FM Channel 16, except when transmitting or receiving traffic on other VHF-FM channels or when participating in a Vessel Traffic Service (VTS) or monitoring a channel of a VTS. (Each U.S. towing vessel of 26 feet (about 8 meters) or more in length, except a public vessel, must hold a ship-radio-station license for radio transmitters (including radar and EPIRBs), and each operator must hold a restricted operator's license or higher. To get an application for either license, call (800) 418-FORM or (202) 418-FORM, or write to the FCC; Wireless Bureau, Licensing Division; 1270 Fairfield Road; Gettysburg, PA 17325-7245.)

(1856) (4) *Magnetic Compass*. Either–

(1857) (i) An illuminated swing-meter or an illuminated card-type magnetic steering compass readable from the vessel's main steering station, if the vessel engages in towing exclusively on Western Rivers; or

(1858) (ii) An illuminated card-type magnetic steering compass readable from the vessel's main steering station.

(1859) (5) *Echo Depth-Sounding Device*. By August 2, 2001, an echo depth-sounding device readable from the vessel's main steering station, unless the vessel engages in towing exclusively on Western Rivers.

(1860) (6) *Electronic Position-Fixing Device*. An electronic position-fixing device, a satellite navigational system such as the Global Positioning System (GPS) as required by §164.41, if the vessel engages in towing seaward of navigable waters of the U.S. or more than three nautical miles from shore on the Great Lakes.

(1861) (b) Each towing vessel must carry on board and maintain the following:

(1862) (1) *Charts or maps*. Marine charts or maps of the areas to be transited, published by the National Ocean Service (NOS), the ACOE, or a river authority that satisfy the following requirements.

(1863) (i) The charts or maps must be of a large enough scale and have enough detail to make safe navigation of the areas possible.

(1864) (ii) The charts or maps must be either–

(1865) (A) Current editions or currently corrected editions, if the vessel engages in towing exclusively on navigable waters of the U.S., including Western Rivers; or

(1866) (B) Currently corrected editions, if the vessel engages in towing seaward of navigable waters of the U.S. or more than three nautical miles from shore on the Great Lakes.

(1867) (iii) The charts or maps may be, instead of charts or maps required by paragraphs (b)(1) (i) and (ii) of this section, currently corrected marine charts or maps, or applicable extracts, published by a foreign government. These charts or maps, or applicable extracts, must contain information similar to that on the charts or maps required by paragraphs (b)(1) (i) and (ii) of the section, be of large enough scale, and have enough detail to make safe navigation of the areas possible, and must be currently corrected.

(1868) (2) *General publications*. A currently corrected edition of, or an applicable currently corrected extract from, each of the following publications for the area to be transited:

(1869) (i) If the vessel is engaged in towing exclusively on Western Rivers–

(1870) (A) U.S. Coast Guard Light List;

(1871) (B) Applicable Notices to Navigation published by the ACOE, or Local Notices to Marines (LNMs) published by the Coast Guard, for the area to be transited, when available; and

(1872) (C) River-current tables published by a river authority, if available.

(1873) (ii) if the vessel is engaged other than in towing exclusively on Western Rivers–

(1874) (A) Coast Guard Light List;

(1875) (B) Notices to Mariners published by the National Geospatial-Intelligence Agency, or LNMs published by the Coast Guard;

(1876) (C) Tidal-Current tables published by private entities using data provided by the NOS, or river-current tables published by a river authority;

(1877) (D) Tide tables published by private entities using data provided by the NOS; and

(1878) (E) U.S. Coast Pilot.

(1879) (c) Table 164.72, following, summarizes the navigational-safety equipment, charts or maps, and publications required for towing vessels of 12 meters or more in length engaged in towing:

(1881)

§164.74 Towline and terminal gear for towing astern.

(1882) (a) *Towline*. The owner, master, or operator of each vessel towing astern shall ensure that the strength of each towline is adequate for its intended service, considering at least the following factors:

(1883) (1) The size and material of each towline must be–

(1884) (i) Appropriate for the horsepower or bollard pull of the vessel;

(1885) (ii) Appropriate for the static loads and dynamic loads expected during the intended service;

(1886) (iii) Appropriate for the sea conditions expected during the intended service;

(1887) (iv) Appropriate for exposure to the marine environment and to any chemicals used or carried on board the vessel;

(1888) (v) Appropriate for the temperatures of normal stowage and service on board the vessel;

(1889) (vi) Compatible with associated navigational-safety equipment; and

(1890) (vii) Appropriate for the likelihood of mechanical damage.

(1891) (2) Each towline as rigged must be–

(1892) (i) Free of knots;

(1893) (ii) Spliced with a thimble, or have a poured socket at its end; and

(1894) (iii) Free of wire clips except for temporary repair, for which the towline must have a thimble and either five wire clips or as many wire clips as the manufacturer specifies for the nominal diameter and construction of the towline, whichever is more.

(1895) (3) The condition of each towline must be monitored through the–

(1896) (i) Keeping on board the towing vessel or in company files of a record of the towline's initial minimum breaking strength as determined by the manufacturer, by a classification ("class") society authorized in §157.04 of this chapter, or by a tensile test that meets API Specifications 9A, Specification for Wire Rope, Section 3; ASTM D 4268 (incorporated by reference, see §164.03), Standard Test Method for Testing Fiber Ropes; or Cordage Institute CIA 3, Standard Test Methods for Fiber Rope Including Standard Terminations;

(1897) (ii) If the towline is purchased from another owner, master, or operator of a vessel with the intent to use it as a towline or if it is retested for any reason, keeping on board the towing vessel or in company files of a record of each retest of the towline's minimum breaking strength as determined by a class society authorized in §157.04 of this chapter or by a tensile test that meets API Specification 9A, Section 3; ASTM D 4268 (incorporated by reference, see §164.03); or Cordage Institute CIA 3, Standard Test Methods;

(1898) (iii) Conducting visual inspections of the towline in accordance with the manufacturer's recommendations, or at least monthly, and whenever the serviceability of the towline is in doubt (the inspections being conducted by the owner, master, or operator, or by a person on whom the owner, master, or operator confers the responsibility to take corrective measures appropriate for the use of the towline);

(1899) (iv) Evaluating the serviceability of the whole towline or any part of the towline, and removing the whole or part from service either as recommended by the manufacturer or a class society authorized in §157.04 of this chapter or in accordance with a replacement schedule developed by the owner, master, or operator that accounts for at least the–

(1900) (A) Nautical miles on, or time in service of, the towline;

(1901) (B) Operating conditions experienced by the towline;

(1902) (C) History of loading of the towline;

(1903) (D) Surface condition, including corrosion and discoloration, of the towline;

(1904) (E) Amount of visible damage to the towline;

(1905) (F) Amount of material deterioration indicated by measurements of diameter and, if applicable, measurements of lay extension of the towline; and

(1906) (G) Point at which a tensile test proves the minimum breaking strength of the towline inadequate by the standards of paragraph (a)(1) of this section, if necessary; and

(1907) (v) Keeping on board the towing vessel or in company files of a record of the material condition of the towline when inspected under paragraphs (a)(3)(iii) and (iv) of this section. Once this record lapses for three months or more, except when a vessel is laid up or out of service or has not deployed its towline, the owner, master, or operator shall retest the towline or remove it from service.

(1908) (b) *Terminal gear*. The owner, master, or operator of each vessel towing astern shall ensure that the gear used to control, protect, and connect each towline meets the following criteria:

(1909) (1) The material and size of the terminal gear are appropriate for the strength and anticipated loading of the towline and for the environment;

(1910) (2) Each connection is secured by at least one nut with at least one cotter pin or other means of preventing its failure;

(1911) (3) The lead of the towline is appropriate to prevent sharp bends in the towline from fairlead blocks, chocks, or tackle;

(1912) (4) There is provided a method, whether mechanical or non-mechanical, that does not endanger operating personnel but that easily releases the towline;

(1913) (5) The towline is protected from abrasion or chafing by chafing gear, lagging, or other means;

(1914) (6) Except on board a vessel towing in ice on Western Rivers or one using a towline of synthetic or natural fiber, there is fitted a winch that evenly spools and tightly winds the towline; and

(1915) (7) If a winch is fitted, there is attached to the main drum a brake that has holding power appropriate for the horsepower or bollard pull of the vessel and can be operated without power to the winch.

(1916)

§164.76 Towline and terminal gear for towing alongside and pushing ahead.

(1917) The owner, master, or operator of each vessel towing alongside or pushing ahead shall ensure the face wires, spring lines, and push gear used–

(1918) (a) Are appropriate for the vessel's horsepower;

(1919) (b) Are appropriate for the arrangement of the tow;

(1920) (c) Are frequently inspected; and

(1921) (d) Remain serviceable.

(1922)

§164.78 Navigation under way: Towing vessels.

(1923) (a) The owner, master, or operator of each vessel towing shall ensure that each person directing and controlling the movement of the vessel–

(1924) (1) Understands the arrangement of the tow and the effects of maneuvering on the vessel towing and on the vessel, barge, or object being towed;

(1925) (2) Can fix the position of the vessel using installed navigational equipment, aids to navigation, geographic reference-points, and hydrographic contours;

(1926) (3) Does not fix the position of the vessel using buoys alone (Buoys are aids to navigation placed in approximate positions either to alert mariners to hazards to navigation or to indicate the orientation of a channel. They may not maintain exact charted positions, because strong or varying currents, heavy seas, ice and collisions with vessels can move or sink them or set them adrift. Although they may corroborate a position fixed by other means, they cannot fix a position; however, if no other aids are available, buoys alone may establish an estimated position.);

(1927) (4) Evaluates the danger of each closing visual or radar contact;

(1928) (5) Knows and applies the variation and deviation, where a magnetic compass is fitted and where charts or maps have enough detail to enable this type of correction;

(1929) (6) Knows the speed and direction of the current, and the set, drift, and tidal state for the area to be transited;

(1930) (7) Proceeds at a safe speed taking into account the weather, visibility, density of traffic, draft of tow, possibility of wake damage, speed and direction of the current, and local speed-limits; and

(1931) (8) Monitors the voyage plan required by §164.80.

(1932) (b) The owner, master, or operator of each vessel towing shall ensure that the tests and inspections required by §164.80 are conducted and that the results are entered in the log or other record carried on board.

(1933)

§164.80 Tests, inspections, and voyage planning.

(1934) (a) The owner, master, or operator of each towing vessel of less than 1,600 GT shall ensure that the following tests and inspections of gear occur before the vessel embarks on a voyage of more than 24 hours or when each new master or operator assumes command:

(1935) (1) *Steering-systems*. A test of the steering-gear-control system; a test of the main steering gear from the alternative power supply, if installed; a verification of the rudder-angle indicator relative to the actual position of the rudder; and a visual inspection of the steering gear and its linkage.

(1936) (2) *Navigational equipment*. A test of all installed navigational equipment.

(1937) (3) *Communications*. Operation of all internal vessel control communications and vessel-control alarms, if installed.

(1938) (4) *Lights*. Operation of all navigational lights and all searchlights.

(1939) (5) *Terminal gear*. Visual inspection of tackle; of connections of bridle and towing pendant, if applicable; of chafing gear; and the winch brake, if installed.

(1940) (6) *Propulsion systems*. Visual inspection of the spaces for main propulsion machinery, of machinery, and of devices for monitoring machinery.

(1941) (b) The owner, master, or operator of each towing vessel of 1,600 GT or more shall ensure that the following tests of equipment occur at the frequency required by §164.25 and that the following inspections of gear occur before the vessel embarks on a voyage of more than 24 hours or when each new master or operator assumes command:

(1942) (1) *Navigational equipment*. Tests of onboard equipment as required by §164.25.

(1943) (2) *Terminal gear*. Visual inspection of tackle; of connections of bridle and towing pendant, if applicable; of chafing gear; and of the winch brake, if installed.

(1944) (c)(1) The voyage-planning requirements outlined in this section do not apply to you if your towing vessel is–

(1945) (i) Used solely for any of the following services or any combination of these services–

(1946) (A) Within a limited geographic area, such as fleeting-area for barges or a commercial facility, and used for restricted service, such as making up or breaking up larger tows:

(1947) (B) For harbor assist;

(1948) (C) For assistance towing as defined by 46 CFR 10.103;

(1949) (D) For response to emergency or pollution;

(1950) (ii) A public vessel that is both owned, or demise chartered, and operated by the United States Government or by a government of a foreign country; and that is not engaged in commercial service;

(1951) (iii) A foreign vessel engaged in innocent passage; or

(1952) (iv) Exempted by the Captain of the Port (COTP).

(1953) (2) If you think your towing vessel should be exempt from these voyage planning requirements for a specified route, you should submit a written request to the appropriate COTP. The COTP will provide you with a written response granting or denying your request.

(1954) (3) If any part of a towing vessel's intended voyage is seaward of the baseline (i.e. the shoreward boundary) of the territorial sea of the U.S.,then the owner, master, or operator of the vessel, employed to tow a barge or barges, must ensure that the voyage with the barge or barges is planned, taking into account all pertinent information before the vessel embarks on the voyage. The master must check the planned route for proximity to hazards before the voyage begins. During a voyage, if a decision is made to deviate substantially from the planned route, then the master or mate must plan the new route before deviating from the planned route. The voyage plan must follow company policy and consider the following (related requirements noted in parentheses):

(1955) (i) Applicable information from nautical charts and publication (also see paragraph (b) of section 164.72), including Coast Pilot, Coast Guard Light List, and Coast Guard Local Notice to Mariners for the port of departures, all ports of call, and the destination;

(1956) (ii) Current and forecast weather, including visibility, wind, and sea state for the port of departure, all ports of call, and the destination (also see paragraphs (a)(7) of section 164.78 and (b) of section 164.82);

(1957) (iii) Data on tides and currents for the port of departure, all ports of call, and the destination, and the river staged and forecast, if appropriate;

(1958) (iv) Forward and after drafts of the barge or barges and under-keel and vertical clearances (air-gaps) for all bridges, ports, and berthing areas;

(1959) (v) Pre-departure checklists;

(1960) (vi) Calculated speed and estimated time of arrival at proposed waypoints;

(1961) (vii) Communication contacts at any Vessel Traffic Services, bridges, and facilities, and any port specific requirements for VHF radio;

(1962) (viii) Any master's or operator's standing orders detailing closest points of approach, special conditions, and critical maneuvers; and

(1963) (ix) Whether the towing vessel has sufficient power to control the tow under all foreseeable circumstances.

(1964)

§164.82 Maintenance, failure, and reporting.

(1965) (a) *Maintenance*. The owner, master, or operator or each towing vessel shall maintain operative the navigational-safety equipment required by §164.72.

(1966) (b) *Failure*. If any of the navigational-safety equipment required by §164.72 fails during a voyage, the owner, master, or operator of the towing vessel shall exercise due diligence to repair it at the earliest practicable time. He or she shall enter its failure in the log or other record carried on board. The failure of equipment, in itself, does not constitute a violation of this rule; nor does it constitute unseaworthiness; nor does it obligate an owner, master, or operator to moor or anchor the vessel. However, the owner, master, or operator shall consider the state of the equipment-along with such factors as weather, visibility, traffic, and the dictates of good seamanship-in deciding whether it is safe for the vessel to proceed.

(1967) (c) *Reporting*. The owner, master, or operator of each towing vessel whose equipment is inoperative or otherwise impaired while the vessel is operating within a Vessel Traffic Service (VTS) Area shall report the fact as required by 33 CFR Table 161.18(a) row Q.

(1968) (d) *Deviation and authorization*. The owner, master, or operator of each towing vessel unable to repair within 96 hours an inoperative marine radar required by §164.72(a) shall so notify the Captain of the Port (COTP) and shall seek from the COTP both a deviation from the requirements of this section and an authorization for continued operation in the area to be transited. Failure of redundant navigational-safety equipment, including but not limited to failure of one of two installed radars, where

each satisfies §164.72(a), does not necessitate either a deviation or an authorization.

(1969) (1) The initial notice and request for a deviation and an authorization may be spoken, but the request must also be written. The written request must explain why immediate repair is impracticable, and state when and by whom the repair will be made.

(1970) (2) The COTP, upon receiving even a spoken request, may grant a deviation and an authorization from any of the provisions of §§164.70 through 164.82 for a specified time if he or she decides that they would not impair the safe navigation of the vessel under anticipated conditions.

(1971)

Part 165–Regulated Navigation Areas and Limited Access Areas

(1972)

Subpart A–General

(1973)

§165.1 Purpose of part.

(1974) The purpose of this part is to:

(1975) (a) Prescribe procedures for establishing different types of limited or controlled access areas and regulated navigation areas;

(1976) (b) Prescribe general regulations for different types of limited or controlled access areas and regulated navigation areas;

(1977) (c) Prescribe specific requirements for established areas; and

(1978) (d) List specific areas and their boundaries.

(1979)

§165.3 Definitions.

(1980) The following definitions apply to this part:

(1981) *Credential* means any or all of the following:

(1982) (1)Merchant mariner's document.

(1983) (2)Merchant mariner's license.

(1984) (3)STCW endorsement.

(1985) (4)Certificate of registry.

(1986) (5)Merchant mariner credential.

(1987) *Merchant mariner credential* or *MMC* means the credential issued by the Coast Guard under 46 CFR part 10. It combines the individual merchant mariner's document, license, and certificate of registry enumerated in 46 U.S.C. subtitle II part E as well as the STCW endorsement into a single credential that serves as the mariner's qualification document, certificate of identification, and certificate of service.

(1988)

§165.5 Establishment procedures.

(1989) (a) A safety zone, security zone, or regulated navigation area may be established on the initiative of any Coast Guard official authorized to issue such an order in accordance with 33 CFR 1.05-1.

(1990) (b) Any person may request that a safety zone, security zone, or regulated navigation area be established. Except as provided in paragraph (c) of this section, each request must be submitted in writing to either the Captain of the Port or District Commander having jurisdiction over the location as described in 33 CFR 3, and include the following:

(1991) (1) The name of the person submitting the request;

(1992) (2) The location and boundaries of the safety zone, security zone, or regulated navigation area;

(1993) (3) The date, time, and duration that the safety zone, security zone, or regulated navigation area should be established;

(1994) (4) A description of the activities planned for the safety zone, security zone, or regulated navigation area;

(1995) (5) The nature of the restrictions or conditions desired; and

(1996) (6) The reason why the safety zone, security zone, or regulated navigation area is necessary.

(1997) (c) Safety Zones and Security Zones. If, for good cause, the request for a safety zone or security zone is made less than 5 working days before the zone is to be established, the request may be made orally, but it must be followed by a written request within 24 hours.

(1998)

§165.7 Notification.

(1999) (a) The establishment of these limited access areas and regulated navigation areas is considered rulemaking. The procedures used to notify persons of the establishment of these areas vary depending upon the circumstances and emergency conditions. Notification may be made by marine broadcasts, local notice to mariners, local news media, distribution in leaflet form, and on-scene oral notice, as well as publication in the Federal Register.

(2000) (b) Notification normally contains the physical boundaries of the area, the reasons for the rule, its estimated duration, and the method of obtaining authorization to enter the area, if applicable, and special navigational rules, if applicable.

(2001)

§165.8 Geographic coordinates.

(2002) Geographic coordinates expressed in terms of latitude or longitude, or both, are not intended for plotting on maps or charts whose referenced horizontal datum is the North American Datum of 1983 (NAD 83), unless such geographic coordinates are expressly labeled NAD 83. Geographic coordinates without the NAD 83 reference may be plotted on maps or charts referenced to NAD 83 only after application of the appropriate corrections that are published on the particular map or chart being used.

(2003)

§165.9 Geographic application of limited and controlled access areas and regulated navigation areas.

(2004) (a) *General*. The geographic application of the limited and controlled access areas and regulated navigation areas in this part are determined based on the statutory authority under which each is created.

(2005) (b) *Safety zones and regulated navigation areas*. These zones and areas are created under the authority of 46 U.S.C. 70001–70041. Safety zones established under 46 U.S.C. 70116 and regulated navigation areas may be established in waters subject to the jurisdiction of the United States as defined in §2.38 of this chapter, including the territorial sea to a seaward limit of 12 nautical miles from the baseline.

(2006) (c) *Security zones*. These zones have two sources of authority–46 U.S.C. Chapter 700, and the Act of June 15, 1917, as emended by both the Magnuson Act of August 9, 1950 ("Magnuson Act"), 46 U.S.C. 70051–54, and sec. 104 the Maritime Transportation Security Act of 2002 (Pub. L. 107-295, 116 Stat. 2064). Security zones established under either 46 U.S.C. 70116 or 46 U.S.C. 70051 may be established in waters subject to the jurisdiction of the United States as defined in §2.38 of this chapter, including the territorial sea to a seaward limit of 12 nautical miles from the baseline.

(2007) (d) *Naval vessel protection zones*. These zones are issued under the authority of 14 U.S.C. 503 and 527 and may be established in waters subject to the jurisdiction of the United States as defined in §2.38 of this chapter, including the territorial sea to a seaward limit of 12 nautical miles from the baseline.

(2008)

Subpart B–Regulated Navigation Areas

(2009)

§165.10 Regulated navigation area.

(2010) A regulated navigation area is a water area within a defined boundary for which regulations for vessels navigating within the area have been established under this part.

(2011)

§165.11 Vessel operating requirements (regulations).

(2012) Each District Commander may control vessel traffic in an area which is determined to have hazardous conditions, by issuing regulations:

(2013) (a) Specifying times of vessel entry, movement, or departure to, from, within, or through ports, harbors, or other waters;

(2014) (b) Establishing vessel size, speed, draft limitations, and operating conditions; and

(2015) (c) Restricting vessel operation, in a hazardous area or under hazardous conditions, to vessels which have particular operating characteristics or capabilities which are considered necessary for safe operation under the circumstances.

(2016)

§165.13 General regulations.

(2017) (a) The master of a vessel in a regulated navigation area shall operate the vessel in accordance with the regulations contained in Subpart F.

(2018) (b) No person may cause or authorize the operation of a vessel in a regulated navigation area contrary to the regulations in this Part.

(2019)

Subpart C–Safety Zones

(2020)

§165.20 Safety zones.

(2021) A Safety Zone is a water area, shore area, or water and shore area to which, for safety or environmental purposes, access is limited to authorized persons, vehicles, or vessels. It may be stationary and described by fixed limits or it may be described as a zone around a vessel in motion.

(2022)

§165.23 General regulations.

(2023) Unless otherwise provided in this part:

(2024) (a) No person may enter a safety zone unless authorized by the COTP or the District Commander;

(2025) (b) No person may bring or cause to be brought into a safety zone any vehicle, vessel, or object unless authorized by the COTP or the District Commander;

(2026) (c) No person may remain in a safety zone or allow any vehicle, vessel, or object to remain in a safety zone unless authorized by the COTP or the District Commander; and

(2027) (d) Each person in a safety zone who has notice of a lawful order or direction shall obey the order or direction of the COTP or District Commander issued to carry out the purposes of this subpart.

(2028)

Subpart D–Security Zones

(2029)

§165.30 Security zones.

(2030) (a) A security zone is an area of land, water, or land and water which is so designated by the Captain of the Port or District Commander for such time as is necessary to prevent damage or injury to any vessel or waterfront facility, to safeguard ports, harbors, territories, or waters of the United States or to secure the observance of the rights and obligations of the United States.

(2031) (b) The purpose of a security zone is to safeguard from destruction, loss, or injury from sabotage or other subversive acts, accidents, or other causes of a similar nature:

(2032) (1) Vessels,

(2033) (2) Harbors,

(2034) (3) Ports and

(2035) (4) Waterfront facilities:

(2036) In the United States and all territory and water, continental or insular, that is subject to the jurisdiction of the United States.

(2037)

§165.33 General regulations.

(2038) Unless otherwise provided in the special regulations in Subpart F of this part:

(2039) (a) No person or vessel may enter or remain in a security zone without the permission of the Captain of the Port;

(2040) (b) Each person and vessel in a security zone shall obey any direction or order of the Captain of the Port;

(2041) (c) The Captain of the Port may take possession and control of any vessel in the security zone;

(2042) (d) The Captain of the Port may remove any person, vessel, article, or thing from a security zone;

(2043) (e) No person may board, or take or place any article or thing on board, any vessel in a security zone without the permission of the Captain of the Port; and

(2044) (f) No person may take or place any article or thing upon any waterfront facility in a security zone without the permission of the Captain of the Port.

(2045)

Subpart E–Restricted Waterfront Areas

(2046)

§165.40 Restricted Waterfront Areas.

(2047) The Commandant, may direct the COTP to prevent access to waterfront facilities, and port and harbor areas, including vessels and harbor craft therein. This section may apply to persons who do not possess the credentials outlined in 33 CFR 125.09 when certain shipping activities are conducted that are outlined in 33 CFR 125.15.

(2048)

Subpart F–Specific Regulated Navigation Areas and Limited Access Areas

(2049)

§165.100 Regulated Navigation Area: Navigable waters within the First Coast Guard District.

(2050) (a) *Regulated navigation area*. All navigable waters of the United States, as that term is used in 33 CFR 2.36, within the geographic boundaries of the First Coast Guard District, as defined in 33 CFR 3.05-1(b).

(2051) (b) *Definitions*. Terms used in this section have the same meaning as those found in 33 CFR 157.03. Single-hull identifies any tank barge that is not a double-hull tank barge.

(2052) (c) *Applicability*. This section applies to primary towing vessels engaged in towing tank barges carrying petroleum oil in bulk as cargo in the regulated navigation area, or as authorized by the District commander.

(2053) (d) *Regulations*–(1) *Positive control for barges*. (i) Except as provided in paragraph (d)(1)(iii) and paragraph 5 of this section, each single hull tank barge, unless being towed by a primary towing vessel with twin-screw propulsion and with a separate system for power to each screw, must be accompanied by an escort tug of sufficient capability to promptly push or tow the tank barge away from danger of grounding or collision in the event of–

(2054) (A) A propulsion failure;

(2055) (B) A parted towing line;

(2056) (C) A loss of tow;

(2057) (D) A fire;

(2058) (E) Grounding;

(2059) (F) A loss of steering; or

(2060) (G) Any other time a vessel may be operating in a Hazardous Vessel Operating Condition as defined in §161.2 of this Chapter.

(2061) (ii) Double–hull tank barges are exempt from paragraph (d)(1)(i) of this section

(2062) (iii) The cognizant Captain of the Port (COTP) upon written application, may authorize an exemption from the requirements of paragraph (d)(1)(i) of this section for–

(2063) (A) Any tank barge with a capacity of less than 25,000 barrels, operating in an area with limited depth or width such as a creek or small river; or

(2064) (B) Any tank barge operating on any waters within the COTP Zone, if the operator demonstrates to the satisfaction of the COTP that the barge employs an equivalent level of safety to that provided by the positive control provisions of this section. Each request for an exemption under this paragraph must be submitted in writing to the cognizant COTP no later than 7 days before the intended transit.

(2065) (iv) The operator of a towing vessel engaged in towing any tank barge must immediately call for an escort or assist tug to render assistance in the event of any of the occurrences identified in paragraph (d)(1)(i) of this section.

(2066) (2) *Enhanced communications*. Each vessel engaged in towing a tank barge must communicate by radio on marine band or Very High Frequency (VHF) channel 13 or 16, and issue security calls on marine band or VHF channel 13 or 16, upon approach to the following places:

(2067) (i) Execution Rocks Light (USCG Light List No. [LLNR] 21440).

(2068) (ii) Matinecock Point Shoal Lighted Gong Buoy 21 (LLNR 21420).

(2069) (iii) 32A Buoy (LLNR 21380).

(2070) (iv) Cable and Anchor Reef Lighted Bell Buoy 28C (LLNR 21330).

(2071) (v) Stratford Shoal (Middle Ground) Light (LLNR 21260).

(2072) (vi) Old Field Point Light (LLNR 21275).

(2073) (vii) Approach to Stratford Point from the south (NOAA Chart 12370).

(2074) (viii) Falkner Island Light (LLNR 21170).

(2075) (ix) TE Buoy (LLNR 21160).

(2076) (x) PI Buoy (LLNR 21080).

(2077) (xi) Race Rock Light (LLNR 19815).

(2078) (xii) Valiant Rock Lighted Whistle Buoy 11 (LLNR 19825).

(2079) (xiii) Approach to Point Judith in vicinity of Block Island ferry route.

(2080) (xiv) Buzzards Bay Entrance Light (LLNR 630).

(2081) (xv) Buzzards Bay Midchannel Lighted Buoy BB (LLNR 16055).

(2082) (xvi) Cleveland East Ledge Light (LLNR 16080).

(2083) (xvii) Hog Island Channel Lighted Buoys 1 (LLNR 16130) and 2 (LLNR 16135).

(2084) (xviii) Approach to the Bourne Bridge.

(2085) (xix) Approach to the Sagamore Bridge.

(2086) (xx) Approach to the eastern entrance of Cape Cod Canal.

(2087) (3) *Voyage planning*. (i) Each owner or operator of a towing vessel employed to tow a tank barge shall prepare a written voyage plan for each transit of the tank barge.

(2088) (ii) The watch officer is authorized to make modifications to the plan and validate it as necessary.

(2089) (iii) Except as provided in paragraph (d)(3)(iv) of this section, each voyage plan must contain:

(2090) (A) A description of the type, volume, and grade of cargo.

(2091) (B) Applicable information from nautical charts and publications, including Coast Pilot, Coast Guard Light List, and Coast Guard Local Notice of Mariners, for the destination(s).

(2092) (C) Current and forecasted weather, including visibility, wind, and sea state for the destination(s).

(2093) (D) Data on tides and tidal currents for the destination(s).

(2094) (E) Forward and after drafts of the tank barge, and under-keel and vertical clearances for each port and berthing area.

(2095) (F) Pre-departure checklists.

(2096) (G) Calculated speed and estimated times of arrival at proposed waypoints.

(2097) (H) Communication contacts at Vessel Traffic Service (VTS) (If applicable), bridges, and facilities, and port-specific requirements for VHF radio.

(2098) (I) The master's standing orders detailing closest points of approach, special conditions, and critical maneuvers.

(2099) (iv) Each owner or operator of a tank barge on an intra-port transit of not more than four hours may prepare a voyage plan that contains:

(2100) (A) The information described in paragraphs (d)(3)(iii)(D) and (E) of this section.

(2101) (B) Current weather conditions including visibility, wind, and sea state. This information may be entered in either the voyage plan or towing vessel's log book.

(2102) (C) The channels of VHF radio to monitor.

(2103) (D) Other considerations such as availability of pilot, assist tug, berth and line-handlers, depth of berth at mean low water, danger areas, and securité calls

(2104) (4) *Navigation restriction areas*. Unless authorized by the cognizant COTP, no tank barge may operate in–

(2105) (i) The waters of Cape Cod Bay south of latitude 42°05' North and east of longitude 70°25' West; or

(2106) (ii) The waters of Fishers Island Sound east of longitude 72°02' West, and west of longitude 71°55' West.

(2107) (5) *Special Buzzards Bay regulations*. (i) For the purposes of this section, "Buzzards Bay" is the body of water east and north of a line drawn from the southern tangent of Sakonnet Point, Rhode Island, in approximate position 41°27.2'N., 71°11.7'W., to the Buzzards Bay Entrance Light in approximate position 41°23.48'N., 71°02.5'W., and then to the southwestern tangent of Cuttyhunk Island, Massachusetts, at approximate position 41°24.6'N., 70°57.0'W., and including all of the Cape Cod Canal to its eastern entrance, except that the area of New Bedford harbor within the confines (north) of the hurricane barrier, and the passages through the Elizabeth Islands, is not considered to be "Buzzards Bay".

(2108) (ii) *Additional Positive Control for Barges*. Except as provided in paragraph (d)(1)(iii) of this section, each single hull tank barge transiting Buzzards Bay and carrying 5,000 or more barrels of oil or other hazardous material must, in addition to its primary tug, be accompanied by an escort tug of sufficient capability to promptly push or tow the tank barge away from danger of grounding or collision in the event of–

(2109) (A) A propulsion failure;

(2110) (B) A parted tow line;

(2111) (C) A loss of tow;

(2112) (D) A fire;

(2113) (E) Grounding;

(2114) (F) A loss of steering; or

(2115) (G) Any other time a vessel may be operating in a Hazardous Vessel Operating Condition as defined in §161.2 of this subchapter.

(2116) (iii) *Federal Pilotage*. Each single hull tank barge transiting Buzzards Bay and carrying 5,000 or more barrels of oil or other hazardous material must be under the direction and control of a pilot, who is not a member of the crew, operating under a valid, appropriately endorsed, Federal first class pilot's license issued by the Coast Guard ("federally licensed pilot"). Pilots are required to embark, direct, and control from the primary tug during transits of Buzzards Bay.

(2117) (iv) In addition to the vessels denoted in §161.16 of this chapter, requirements set forth in subpart B of Part 161 also apply to any vessel transiting VMRS Buzzards Bay required to carry a bridge-to-bridge radiotelephone by part 26 of this chapter.

(2118) (A) A VMRS Buzzards Bay user must:

(2119) (1) Not enter or get underway in the area without first notifying the VMRS Center;

(2120) (2) Not enter VMRS Buzzards Bay if a Hazardous Vessel Operating Condition or circumstance per §161.2 of this Subchapter exists:

(2121) (3) If towing astern, do so with as short a hawser as safety and good seamanship permits;

(2122) (4) Not meet, cross, or overtake any other VMRS user in the area without first notifying the VMRS center;

(2123) (5) Before meeting, crossing, or overtaking any other VMRS user in the area, communicate on the designated vessel bridge-to-bridge radiotelephone frequency, intended navigation movements, and any

other information necessary in order to make safe passing arrangements. This requirement does not relieve a vessel of any duty prescribed by the Navigation Rules (COLREGS and their associated Annexes and Inland Navigation Rules (33 CFR subchapter E)).

(2124) (B) [Reserved]

(2125) (e) In addition to the authority for this part 165, this section is also authorized under authority of section 311, Pub. L. 105-383.

(2126)

§165.101 Kittery, Maine–regulated navigation area.

(2127) (a) The following is a regulated navigation area–Waters within the boundaries of a line beginning at

(2128) 43°04'50"N., 70°44'52"W.; then to

(2129) 43°04'52"N., 70°44'53"W.; then to

(2130) 43°04'59"N., 70°44'46"W.; then to

(2131) 43°05'05"N., 70°44'32"W.; then to

(2132) 43°05'03"N., 70°44'30"W.; then to the beginning point.

(2133) (b) *Regulations*–No vessel may operate in this area at a speed in excess of five miles per hour.

(2134)

§165.102 Security Zone; Walkers Point, Kennebunkport ME.

(2135) (a) *Location*. The following area is a security zone: From point of land located on Cape Arundel at 43°20.4'N., 70°28.0'W.; thence to a point approximately 500 yards southwest of Walkers Point located at 43°20.2'N., 70°27.9'W.; thence to a point located approximately 500 yards south of Walkers Point at 43°20.1'N., 70°27.6'W.; thence to a point located approximately southeast of Walkers Point at 43°20.4'N., 70°27.2'W.; thence to an unnamed point of land located at 43°20.9'N., 70°27.1'W.; thence along the shoreline of Walkers Point to the beginning point. The aforementioned offshore positions are approximated by white buoys marked in orange indicating an exclusionary area.

(2136) (b) *Regulations*. (1) In accordance with the general regulations in §165.33 of this part, entry into this zone is prohibited unless authorized by the Captain of the Port, Portland, Maine. §165.33 also contained other general requirements.

(2137) (2) No person may swim upon or below the surface of the water within the boundaries of this security zone.

(2138)

§165.103 Safety and Security Zones; LPG Vessel Transits in Portland, Maine, Captain of the Port Zone, Portsmouth Harbor, Portsmouth, New Hampshire.

(2139) (a) *Location*. The following areas are safety and security zones: (1) Except as provided in paragraph (a) (2) of this section, all navigable waters of the Portland, Maine, Captain of the Port zone, as defined in 33 CFR 3.05–15, one mile ahead, one half mile astern, and 1000-yards on either side of any Liquefied Petroleum Gas vessel.

(2140) (2) All waters of the Piscataqua River within a 500-yard radius of any Liquefied Petroleum Gas (LPG) vessel while it is moored at the LPG receiving facility on the Piscataqua River, Newington, New Hampshire.

(2141) (b) *Definitions*. For purposes of this section, navigable waters of the United States includes all waters of the territorial sea as described in Presidential Proclamation No. 5928 of December 27, 1988. Presidential Proclamation No. 5928 of December 27, 1988 declared that the territorial sea of the United States extends to 12 nautical miles from the baseline of the United States.

(2142) (c) *Regulations*. (1) In accordance with the general regulations in §§165.23 and 165.33 of this part, entry into or movement within these zones is prohibited unless previously authorized by the Captain of the Port (COTP), Portland, Maine.

(2143) (2) All persons and vessels shall comply with the instructions of the COTP or the designated on-scene U.S. Coast Guard patrol personnel. On-scene Coast Guard patrol personnel include commissioned, warrant, and petty officers of the Coast Guard on board Coast Guard, Coast Guard Auxiliary, and local, state, and federal law enforcement vessels. Emergency response vessels are authorized to move within the zone, but must abide by restrictions imposed by the Captain of the Port.

(2144) (3) No person may swim upon or below the surface of the water within the boundaries of the safety and security zones unless previously authorized by the Captain of the Port, Portland, Maine or his authorized patrol representative.

(2145) (d) The Captain of the Port will notify the maritime community and local agencies of periods during which these safety and security zones will be in effect by providing notice of arrivals and departures of LPG vessels via the telephone and/or Marine Safety Information Radio Broadcasts.

(2146)

§165.105 Security Zones; Passenger Vessels, Portland, Maine, Captain of the Port Zone.

(2147) (a) *Definition*. "Passenger vessel" as used in this section means a passenger vessel over 100 gross tons authorized to carry more than 500 passengers for hire making voyages, any part of which is on the high seas, and for which passengers are embarked, disembarked or pay a port call, in the Portland, Maine, Captain of the Port zone as delineated in 33 CFR 3.05-15.

(2148) (b) *Location*. The following areas are security zones:

(2149) (1) All navigable waters within the Portland, Maine, Captain of the Port Zone, extending from the surface to the sea floor, within a 100-yard radius of any passenger vessel that is anchored, moored, or in the process of mooring.

(2150) (2) All navigable waters, within the Portland, Maine, Captain of the Port Zone, extending from the surface to the sea floor, extending 200 yards ahead, and 100 yards aside and astern of any passenger vessel that is underway.

(2151) (c) *Regulations*. (1) In accordance with the general regulations in §165.33 of this part, entry into or movement within these zones is prohibited unless previously authorized by the Coast Guard Captain of the Port, Portland, Maine (COTP) or his designated representative.

(2152) (2) All persons and vessels must comply with the instructions of the COTP or the designated on-scene Coast Guard patrol personnel. On-scene Coast Guard patrol personnel include commissioned, warrant and petty officers of the Coast Guard on board Coast Guard, Coast Guard Auxiliary, and local, state and federal law enforcement vessels. Emergency response vessels are authorized to move within the zone, but must abide by restrictions imposed by the COTP or his designated representative.

(2153) (3) No person may swim upon or below the surface of the water within the boundaries of these security zones unless previously authorized by the COTP or his designated representative.

(2154) (d) *Enforcement*. The Captain of the Port will enforce these zones and may enlist the aid and cooperation of any Federal, state, county, municipal, or private agency to assist in the enforcement of the regulation.

(2155)

§165.110 Safety and Security Zone; Liquefied Natural Gas Carrier Transits and Anchorage Operations, Boston, Massachusetts.

(2156) (a) *Definitions*. As used in this section-

(2157) *Authorized representative* means a Coast Guard commissioned, warrant, or petty officer or a Federal, State, or local law enforcement officer designated by or assisting the Captain of the Port (COTP) Boston.

(2158) *Deepwater port* means any facility or structure meeting the definition of deepwater port in 33 CFR 148.5.

(2159) *Support vessel* means any vessel meeting the definition of support vessel in 33 CFR 148.5.

(2160) (b) *Location*. The following areas are safety and security zones:

(2161) (1) *Vessels underway*. All navigable waters of the United States within the Captain of the Port (COTP) Boston zone, as defined in 33 CFR 3.05–10, two miles ahead and one mile astern, and 500 yards on each side of any liquefied natural gas carrier (LNGC) vessel while underway.

(2162) (2) *Vessels anchored in Broad Sound*. All waters within a 500-yard radius of any anchored LNGC vessel located in the waters of Broad Sound bounded by a line starting at position 42°25'N., 070°58'W.; then running southeast to 42°22'N., 070°56'W.; then running east to 42°22'N., 070°50'W.; then running north to 42°25'N., 070°50'W.; then running west back to the starting point (NAD 83).

(2163) (3) *Vessels moored at the Distrigas LNG facility*. All waters within a 400-yard radius of any LNGC vessel moored at the Distrigas LNG facility in Everett, MA.

(2164) (4) *Vessels calling on a deepwater port*. All waters within a 500-meter radius of any LNGC engaged in regasification or transfer, or otherwise moored, anchored, or affixed to a deepwater port listed in 33 CFR 150.490 and falling within the waters of the Boston COTP Zone, as defined in 33 CFR 3.05–10.

(2165) (c) *Regulations*.

(2166) (1) In accordance with the general regulations in 165.23 and 165.33 of this chapter, entry into or movement within these zones is prohibited unless authorized by the Captain of the Port Boston, or his authorized representative.

(2167) (2) No person or vessel may enter the waters within the boundaries of the safety and security zones described in paragraph (b) of this section unless previously authorized by the COTP Boston, or his/her authorized representative. However, LNGCs and support vessels, as defined in 33 CFR 148.5, operating in the vicinity of NEGDWP are authorized to enter and move within such zones in the normal course of their operations following the requirements set forth in 33 CFR 150.340 and 150.345, respectively.

(2168) (3) All vessels operating within the safety and security zones described in paragraph (b) of this section must comply with the instructions of the COTP or his/her authorized representative.

(2169)

§165.111 Safety Zone: Boston Harbor; Boston, Massachusetts.

(2170) (a) The following areas are established as safety zones during the conditions specified:

(2171) (1) Around the U.S.S. *Constitution* or any accompanying parade vessels when *Constitution* is underway-300 yards in all directions in the waters around the U.S.S. *Constitution* and each parade vessel accompanying *Constitution* whenever the U.S.S. *Constitution* is underway in Boston Harbor from the time such vessels depart their respective berths until the time they complete their transit and are safely moored.

(2172) (2) Whenever *Constitution* is moored at Pier 1, Charlestown Navy Yard-the waters between Hoosac Pier and Pier 1, Charlestown Navy Yard, from the imaginary line connecting the outer easternmost point protruding into Boston Harbor from Hoosac Pier to the outer westernmost point protruding into Boston Harbor from Pier 1, Charlestown Navy Yard, extending inbound along the face of both piers to the landside points where both piers end.

(2173) (3) Around the U.S.S. *Constitution*-fifty yards in all directions in the waters around Constitution when the vessel is moored at any Boston berthing location other than Pier 1, Charlestown Navy Yard.

(2174) (b) The general regulations governing safety zones as contained in **33 CFR 165.23** apply.

(2175)

§165.112 Safety Zone: USS CASSIN YOUNG, Boston, Massachusetts.

(2176) (a) *Location*. The following area is a safety zone:

(2177) Around the USS CASSIN YOUNG (DD-793) and any accompanying parade vessels when the USS CASSIN YOUNG is underway. The zone extends 100 yards in all directions in the waters around the USS CASSIN YOUNG and accompanying parade vessels whenever the USS CASSIN YOUNG is underway in Boston Harbor from the time the USS CASSIN YOUNG departs its berth until it is safely moored.

(2178) (b) *Regulations*. The general regulations governing safety zones as contained in 33 CFR 165.23 apply.

(2179)

§165.114 Safety and Security Zones: Escorted Vessels–Boston Harbor, Massachusetts.

(2180) (a) *Location*. The following waters within the Boston Captain of the Port Zone, 1000 yards ahead and astern, and 100 yards on each side of any designated escorted vessel, are established as safety and security zones: All waters of Boston Inner Harbor, including the waters of the Mystic River, Chelsea River, and Reserved Channel west of a line running from Deer Island Light, at position 42°20'25"N., 070°57'15"W., and west of the Long Island Bridge, running from Long Island to Moon Head.

(2181) (b) *Escorted vessel defined*. For the purposes of this section, escorted vessels operating in Boston Harbor include the following: Any vessels deemed to be in need of escort protection by the Captain of the Port, Boston for security reasons or under other circumstances.

(2182) (c) *Regulations*.

(2183) (1) In accordance with the general regulations in §§**165.23** and **165.33** of this part, entry into or movement within this zone is prohibited unless authorized by the Captain of the Port, Boston.

(2184) (2) All vessel operators shall comply with the instructions of the COTP or the designated on-scene U.S. Coast Guard patrol personnel. On-scene Coast Guard patrol personnel include commissioned, warrant, and petty officers of the Coast Guard on board Coast Guard, Coast Guard Auxiliary, local, state, and federal law enforcement vessels.

(2185)

§165.115 Safety and Security Zones; Pilgrim Nuclear Power Plant, Plymouth, Massachusetts.

(2186) (a) *Location*. All waters of Cape Cod Bay and land adjacent to those waters enclosed by a line beginning at position

(2187) 41°56'59.3"N., 070°34'58.5"W.; thence to

(2188) 41°57'12.2"N., 070°34'41.9"W.; thence to

(2189) 41°56'42.3"N., 070°34'00.1"W.; thence to

(2190) 41°56'29.5"N., 070°34'14.5"W.

(2191) (b) *Regulations*.

(2192) (1) In accordance with the general regulations in §§165.23 and 165.33 of this part, entry into or movement within these zones is prohibited unless authorized by the Captain of the Port, Boston.

(2193) (2) All vessel operators shall comply with the instructions of the COTP or the designated on-scene U.S. Coast Guard patrol personnel. On-scene Coast Guard patrol personnel include commissioned, warrant, and petty officers of the Coast Guard on board Coast Guard, Coast Guard Auxiliary, local, state, and federal law enforcement vessels.

(2194) (3) No person may enter the waters or land area within the boundaries of the safety and security zones unless previously authorized by the Captain of the Port, Boston or his authorized patrol representative.

(2195)

§165.116 Safety and Security Zones; Salem and Boston Harbors, Massachusetts.

(2196) (a) *Location*. The following areas are permanent safety and security zones:

(2197) (1) *Reserved Channel, Boston Harbor*. All waters of Boston Harbor within one hundred fifty (150) yards off the bow and stern and one hundred (100) yards abeam of any vessel moored at the Massachusetts Port Authority Black Falcon Terminal;

(2198) (2) *Boston Inner Harbor*. All waters of Boston Harbor within one hundred (100) feet of the Coast Guard Integrated Support Command (ISC) Boston piers and;

(2199) (3) *Salem Harbor*. All waters of Salem Harbor within a two-hundred and fifty (250) yard radius of the center point of the PG & E Power Plant Terminal Wharf, Salem, MA, located at 42°31.33'N., 070°52.67'W., when a vessel is moored at this pier. All coordinates are North American Datum 1983.

(2200) (b) *Regulations*.

(2201) (1) In accordance with the general regulations in §165.23 and §165.33 of this part, entry into or movement within these zones is prohibited unless authorized by the Captain of the Port, Boston.

(2202) (2) All vessel operators shall comply with the instructions of the Captain of the Port or the designated on-scene U.S. Coast Guard patrol personnel. On-scene Coast Guard patrol personnel include commissioned, warrant, and petty officers of the Coast Guard on board Coast Guard, Coast Guard Auxiliary, local, state, and federal law enforcement vessels.

(2203) (3) No person may enter the waters or land area within the boundaries of the safety and security zones unless previously authorized by the Captain of the Port, Boston or his authorized patrol representative.

(2204)

§165.117 Regulated Navigation Areas, Safety and Security Zones: Deepwater Ports, First Coast Guard District.

(2205) (a) *Location*. (1) *Regulated navigation areas*. All waters within a 1,000 meter radius of the geographical positions set forth in paragraph (a)(3) of this section are designated as regulated navigation areas.

(2206) (2) *Safety and security zones*. All waters within a 500-meter radius of the geographic positions set forth in paragraph (a)(3) of this section are designated as safety and security zones.

(2207) (3) *Coordinates*. (i) The geographic coordinates forming the loci for the regulated navigation areas, safety and security zones for the Northeast Gateway Deepwater Port are: 42°23′38″N., 070°35′31″W.; and 42°23′56″N., 070°37′00″W. (NAD 83).

(2208) (ii) The geographic coordinates forming the loci for the regulated navigation areas, safety, and security zones for Neptune Deepwater Port are: 42°29′12.3″N., 70°36′29.7″W.; and 42°27′20.5″N., 70°36′07.3″W. (NAD 83).

(2209) (iii) [Reserved]

(2210) (b) *Definitions*. As used in this section–

(2211) *Authorized representative* means a Coast Guard commissioned, warrant, or petty officer or a Federal, State, or local law enforcement officer designated by or assisting the Captain of the Port (COTP) Boston.

(2212) *Deepwater port* means any facility or structure meeting the definition of deepwater port 33 CFR 148.5.

(2213) *Dredge* means fishing gear consisting of a mouth frame attached to a holding bag constructed of metal rings or mesh.

(2214) *Support vessel* means any vessel meeting the definition of support vessel in 33 CFR 148.5.

(2215) *Trap* means a portable, enclosed device with one or more gates or entrances and one or more lines attached to surface floats used for fishing. Also called a pot.

(2216) (c) *Applicability*. This section applies to all vessels operating in the regulated navigation areas set forth in paragraph (a) of this section, except–

(2217) (1) Those vessels conducting cargo transfer operations with the deepwater ports whose coordinates are provided in paragraph (a)(3) of this section.

(2218) (2) Support vessels operating in conjunction therewith, and

(2219) (3) Coast Guard vessels or other law enforcement vessels operated by or under the direction of an authorized representative of the COTP Boston.

(2220) (d) *Regulations*. (1) No vessel may anchor, engage in diving operations, or commercial fishing using nets, dredges, traps (pots), or use of remotely operated vehicles (ROVs) in the regulated navigation areas set forth in paragraph (a)(1) of this section.

(2221) (2) In accordance with the general regulations in §§165.23 and 165.33 of this part, entry into or movement within the safety and security zones designated in paragraph (a)(2) of this section is prohibited unless authorized by the COTP Boston, or his/her authorized representative.

(2222) (3) Notwithstanding paragraph (d)(2) of this section, tankers and support vessels, as defined in 33 CFR 148.5, operating in the vicinity of NEGDWP are authorized to enter and move within such zones in the normal course of their operations following the requirements set forth in 33 CFR 150.340 and 150.345, respectively.

(2223) (4) All vessels operating within the safety and security zones described in paragraph (a)(2) of this section must comply with the instructions of the COTP or his/her authorized representative.

(2224)

§ 165.120 Security Zone, John Joseph Moakley United States Courthouse, Boston, MA.

(2225) (a) *Location*. This security zone encompasses all U.S. navigable waters, from surface to bottom, within five hundred (500) yards of the John Joseph Moakley United States Courthouse (Moakley Courthouse) in Boston, MA, and following any natural waterside seawall configuration.

(2226) (b) *Regulations*. While this security zone is being enforced, the following regulations, along with those contained in 33 CFR 165.33, apply:

(2227) (1) No person or vessel may enter or remain in this security zone without the permission of the Captain of the Port (COTP), Sector Boston. However, the COTP hereby grants vessels permission to enter this security zone as long as such vessels proceed through the area with caution and operate at a speed no faster than that speed necessary to maintain a safe course, unless otherwise required by the Navigation Rules as published in 33 CFR part 83 and remain beyond one hundred (100) yards of the Moakley Courthouse in Boston, MA, following any natural waterside seawall configuration enclosed by a line connecting the following points:

(2228)

Latitude	Longitude
42°21'15"N.	71°02'54"W.; Bounded by the curvature of the seawall, thence to
42°21'18"N.	71°02'43"W.; thence to
42°21'20"N.	71°02'40"W.; Bounded by 100 yards of the curvature of the seawall, thence to
42°21'16"N.	71°02'57"W.; thence to point of origin.

(2229) (2) Although vessels have permission to enter the five hundred (500) yards security zone under the conditions mentioned in the preceding paragraph, no person or vessel may come within one hundred (100) yards of the Moakley Courthouse under any conditions unless given express permission from the COTP or the COTP's designated representatives.

(2230) (3) Any person or vessel permitted to enter the security zone shall comply with the directions and orders of the COTP or the COTP's representatives. Upon being hailed by siren, radio, flashing lights, or other means, the operator of a vessel within the zone shall proceed as directed. Any person or vessel within the security zone shall exit the zone when directed by the COTP or the COTP's representatives.

(2231) (4) To obtain permissions required by this regulation, individuals may reach the COTP or a COTP representative

via VHF channel 16 or 617–223–5757 (Sector Boston Command Center) to obtain permission.

(2232) (5) *Penalties*. Those who violate this section are subject to the penalties set forth in 46 U.S.C. 70036 and 46 U.S.C. 70052.

(2233) (c) *Effective and enforcement period*. This security zone is in effect permanently but will only be enforced when deemed necessary by the COTP. Anyone, including members of federal, state or local law enforcement agencies, may request that this security zone be enforced.

(2234) (d) *Notification*. The COTP will notify the public of the enforcement of this security zone by publishing a Notice of Enforcement (NOE) in the **Federal Register** and via the other means listed in 33 CFR 165.7. Such notifications will include the date and times of enforcement, along with any predetermined conditions of entry.

(2235) (e) *COTP representative*. The COTP's representative may be any Coast Guard commissioned, warrant, or petty officer or any Federal, state, or local law enforcement officer who has been designated by the COTP to act on the COTP's behalf. The COTP's representative may be on a Coast Guard vessel, a Coast Guard Auxiliary vessel, federal, state or local law enforcement or safety vessel, or a location on shore.

(2236)

§165.123 Cruise Ships, Sector Southeastern New England Captain of the Port (COTP) Zone.

(2237) (a) *Location*. The following areas are security zones: All navigable waters within the Southeastern New England Captain of the Port (COTP) Zone, extending from the surface to the sea floor:

(2238) (1) Within a 200-yard radius of any cruise ship that is underway and is under escort of U.S. Coast Guard law enforcement personnel or designated representative, or

(2239) (2) Within a 100-yard radius of any cruise ship that is anchored, at any berth or moored.

(2240) (b) *Definitions*. For the purposes of this section—

(2241) *Cruise ship* means a passenger vessel as defined in 46 U.S.C. 2101(22), that is authorized to carry more than 400 passengers and is 200 or more feet in length. A cruise ship under this section will also include ferries as defined in 46 CFR 2.10–25 that are authorized to carry more than 400 passengers and are 200 feet or more in length.

(2242) *Designated representative* means any Coast Guard commissioned, warrant, or petty officer who has been designated by the COTP to act on the COTP's behalf. The designated representative may be on a Coast Guard vessel, or onboard Federal, state, or a local agency vessel that is authorized to act in support of the Coast Guard.

(2243) *Southeastern New England COTP Zone* is as defined in 33 CFR 3.05–20.

(2244) (c) *Enforcement*. The security zones described in this section will be activated and enforced upon entry of any cruise ship into the navigable waters of the United States (see 33 CFR 2.36(a) to include the 12 NM territorial sea) in the Southeastern New England COTP zone. This zone will remain activated at all times while a cruise ship is within the navigable waters of the United States in the Sector Southeastern New England COTP Zone. In addition, the Coast Guard may broadcast the area designated as a security zone for the duration of the enforcement period via Broadcast Notice to Mariners.

(2245) (d) *Regulations*. (1) In accordance with the general regulations in 33 CFR part 165, subpart D, no person

Coast Guard. Upon being hailed by a U.S. Coast Guard vessel via siren, radio, flashing light, or other means, the operator of a vessel shall proceed as directed.

(2257)

Subpart G–Protection of Naval Vessels

(2258)

§165.2010 Purpose.

(2259) This subpart establishes the geographic parameters of naval vessel protection zones surrounding U.S. naval vessels in the navigable waters of the United States. This subpart also establishes when the U.S. Navy will take enforcement action in accordance with the statutory guideline of 14 U.S.C. 91. Nothing in the rules and regulations contained in this subpart shall relieve any vessel, including U.S. naval vessels, from the observance of the Navigation Rules. The rules and regulations contained in this subpart supplement, but do not replace or supercede, any other regulation pertaining to the safety or security of U.S. naval vessels.

(2260)

§165.2015 Definitions.

(2261) The following definitions apply to this subpart:

(2262) *Atlantic Area* means that area described in 33 CFR 3.04–1 Atlantic Area.

(2263) *Large U.S. naval vessel* means any U.S. naval vessel greater than 100 feet in length overall.

(2264) *Naval defensive sea area* means those areas described in 32 CFR part 761.

(2265) *Naval vessel protection zone* is a 500-yard regulated area of water surrounding large U.S. naval vessels that is necessary to provide for the safety or security of these U.S. naval vessels.

(2266) *Navigable waters of the United States* means those waters defined as such in 33 CFR part 2.

(2267) *Navigation rules* means the Navigation Rules, International-Inland.

(2268) *Official patrol* means those personnel designated and supervised by a senior naval officer present in command and tasked to monitor a naval vessel protection zone, permit entry into the zone, give legally enforceable orders to persons or vessels within the zone, and take other actions authorized by the U.S. Navy.

(2269) *Pacific Area* means that area described in 33 CFR 3.04–3 Pacific Area.

(2270) *Restricted area* means those areas established by the Army Corps of Engineers and set out in 33 CFR part 334.

(2271) *Senior naval officer present in command* is, unless otherwise designated by competent authority, the senior line officer of the U.S. Navy on active duty, eligible for command at sea, who is present and in command of any part of the Department of Navy in the area.

(2272) *U.S. naval vessel* means any vessel owned, operated, chartered, or leased by the U.S. Navy; any pre-commissioned vessel under construction for the U.S. Navy, once launched into the water; and any vessel under the operational control of the U.S. Navy or a Combatant Command.

(2273) *Vessel* means every description of watercraft or other artificial contrivance used, or capable of being used, as a means of transportation on water, except U.S. Coast Guard or U.S. naval vessels.

(2274)

§165.2020 Enforcement authority.

(2275) (a) *Coast Guard*. Any Coast Guard commissioned, warrant or petty officer may enforce the rules and regulations contained in this subpart.

(2276) (b) *Senior naval officer present in command*. In the navigable waters of the United States, when immediate action is required and representatives of the Coast Guard are not present or not present in sufficient force to exercise effective control in the vicinity of large U.S. naval vessels, the senior naval officer present in command is responsible for the enforcement of the rules and regulations contained in this subpart to ensure the safety and security of all large naval vessels present. In meeting this responsibility, the senior naval officer present in command may directly assist any Coast Guard enforcement personnel who are present.

(2277)

§165.2025 Atlantic Area.

(2278) (a) This section applies to any vessel or person in the navigable waters of the United States within the boundaries of the U.S. Coast Guard Atlantic Area, which includes the First, Fifth, Seventh, Eighth and Ninth U.S. Coast Guard Districts.

(2279) **Note to §165.2025 paragraph (a):** The boundaries of the U.S. Coast Guard Atlantic Area and the First, Fifth, Seventh, Eighth and Ninth U.S. Coast Guard Districts are set out in 33 CFR part 3.

(2280) (b) A naval vessel protection zone exists around U.S. naval vessels greater than 100 feet in length overall at all times in the navigable waters of the United States, whether the large U.S. naval vessel is underway, anchored, moored, or within a floating dry dock, except when the large naval vessel is moored or anchored within a restricted area or within a naval defensive sea area.

(2281) (c) The Navigation Rules shall apply at all times within a naval vessel protection zone.

(2282) (d) When within a naval vessel protection zone, all vessels shall operate at the minimum speed necessary to maintain a safe course, unless required to maintain speed by the Navigation Rules, and shall proceed as directed by the Coast Guard, the senior naval officer present in command, or the official patrol. When within a naval vessel protection zone, no vessel or person is allowed within 100 yards of a large U.S. naval vessel unless authorized by the Coast Guard, the senior naval officer present in command, or official patrol.

(2283) (e) To request authorization to operate within 100 yards of a large U.S. naval vessel, contact the Coast Guard, the senior naval officer present in command, or the official patrol on VHF-FM channel 16.

(2284) (f) When conditions permit, the Coast Guard, senior naval officer present in command, or the official patrol should:

(2285) (1) Give advance notice on VHF-FM channel 16 of all large U.S. naval vessel movements; and

(2286) (2) Permit vessels constrained by their navigational draft or restricted in their ability to maneuver to pass within 100 yards of a large U.S. naval vessel in order to ensure a safe passage in accordance with the Navigation Rules; and

(2287) (3) Permit commercial vessels anchored in a designated anchorage area to remain at anchor when within 100 yards of passing large U.S. naval vessels; and

(2288) (4) Permit vessels that must transit via a navigable channel or waterway to pass within 100 yards of a moored or anchored large U.S. naval vessel with minimal delay consistent with security.

(2289) **Note to §165.2025 paragraph (f):** The listed actions are discretionary and do not create any additional right to appeal or otherwise dispute a decision of the Coast Guard, the senior naval officer present in command, or the official patrol.

(2290)

Part 167–Offshore Traffic Separation Schemes

(2291)

Subpart A-General

(2292)

§167.1 Purpose.

(2293) The purpose of the regulations in this part is to establish and designate traffic separation schemes and precautionary areas to provide access routes for vessels proceeding to and from U.S. ports.

(2294)

§167.3 Geographic coordinates.

(2295) Geographic coordinates are defined using North American 1927 Datum (NAD 27) unless indicated otherwise.

(2296)

§167.5 Definitions.

(2297) (a) *Area to be avoided* means a routing measure comprising an area within defined limits in which either navigation is particularly hazardous or it is exceptionally important to avoid casualties and which should be avoided by all ships or certain classes of ships.

(2298) (b) *Traffic separation scheme (TSS)* means a designated routing measure which is aimed at the separation of opposing streams of traffic by appropriate means and by the establishment of traffic lanes.

(2299) (c) *Traffic lane* means an area within defined limits in which one-way traffic is established. Natural obstacles, including those forming separation zones, may constitute a boundary.

(2300) (d) *Separation zone or line* means a zone or line separating the traffic lanes in which ships are proceeding in opposite or nearly opposite directions; or separating a traffic lane from the adjacent sea area; or separating traffic lanes designated for particular classes of ships proceeding in the same direction.

(2301) (e) *Precautionary area* means a routing measure comprising an area within defined limits where ships must navigate with particular caution and within which the direction of traffic flow may be recommended.

(2302) (f) *Deep-water route* means an internationally recognized routing measure primarily intended for use by ships that, because of their draft in relation to the available depth of water in the area concerned, require the use of such a route.

(2303) (g) *Two-way route* means a route within defined limits inside which two-way traffic is established, aimed at providing safe passage of ships through waters where navigation is difficult or dangerous.

(2304)

§167.10 Operating rules.

(2305) The operator of a vessel in a TSS shall comply with Rule 10 of the International Regulations for Preventing Collisions at Sea, 1972, as amended.

(2306)

§167.15 Modification of schemes.

(2307) (a) A traffic separation scheme or precautionary area described in this part may be permanently amended in accordance with 46 U.S.C. 70003 (92 Stat. 1473), and with international agreements.

(2308) (b) A traffic separation scheme or precautionary area in this part may be temporarily adjusted by the Commandant of the Coast Guard in an emergency, or to accommodate operations which would create an undue hazard for vessels using the scheme or which would contravene Rule 10 of the International Regulations for Preventing Collisions at Sea, 1972. Adjustment may be in the form of a temporary traffic lane shift, a temporary suspension of a section of the scheme, a temporary precautionary area overlaying a lane, or other appropriate measure. Adjustments will only be made where, in the judgment of the Coast Guard, there is no reasonable alternative means of conducting an operation and navigation safety will not be jeopardized by the adjustment. Notice of adjustments will be made in the appropriate Notice to Mariners and in the Federal Register Requests by members of the public for temporary adjustments to traffic separation schemes must be submitted 150 days prior to the time the adjustment is desired. Such Requests, describing the interference that would otherwise occur to a TSS, should be submitted to the District Commander of the Coast Guard District in which the TSS is located.

(2309)

Subpart B-Description of Traffic Separation Schemes and Precautionary Areas

(2310)

§167.50 In the approaches to Portland, ME: General.

(2311) The traffic separation scheme in the approaches to Portland, ME, consists of three parts: A precautionary area, an Eastern approach and a Southern approach. The specific areas in the traffic separation scheme in the approaches to Portland, ME, are described in §§167.51 through 167.53.

(2312)

§ 167.51 In the approaches to Portland, ME: Precautionary area.

(2313) A precautionary area is established with a radius of 5.45 miles centered upon geographical position 43°31.60'N., 70°05.53'W., the areas within separation zones and traffic lanes excluded.

(2314)

§167.52 In the approaches to Portland, ME: Eastern approach.

(2315) (a) A separation zone 1 mile wide is established and is centered upon the following geographical positions:

(2316) 43°30.18'N., 69°59.17'W.

(2317) 43°24.28'N., 69°32.70'W.

(2318) (b) A traffic lane ½ miles wide is established on each side of the separation zone.

(2319)

§167.53 In the approaches to Portland, ME: Southern approach.

(2320) (a) A separation zone 1 mile wide is established and is centered upon the following geographical positions:

(2321) 43°27.00'N., 70°03.48'W.

(2322) 43°07.82'N., 69°54.95'W.

(2323) (b) A traffic lane 1½ mile wide is established on each side of the separation zone.

(2324)

§167.75 In the approach to Boston, MA: General.

(2325) The traffic separation scheme (TSS) in the approach to Boston, MA, consists of three parts: Two precautionary areas and a TSS. The specific areas in the TSS in the approach to Boston, MA, are described in §§167.76 and 167.77. The geographic coordinates in §§167.76 and 167.77 are defined using North American Datum 1983 (NAD 83), which is equivalent to WGS 1984 datum.

(2326)

§167.76 In the approach to Boston, MA: Precautionary areas.

(2327) (a) A precautionary area is established with a radius of 6.17 nautical miles centered upon geographical position 42°22.71'N., 70°46.97'W.

(2328) (b) (1) A precautionary area is established and is bounded to the east by a circle of radius 15.5 miles, centered upon geographical position 40°35.01'N., 68°59.96'W., intersected by the traffic separation schemes in the approach to Boston, MA, and Eastern approach, off Nantucket (part II of the TSS off New York) at the following geographical positions:

(2329) 40°50.47'N., 68°58.67'W.

(2330) 40°23.75'N., 69°13.95'W.

(2331) (2) The precautionary area is bounded to the west by a line connecting the two TSSs between the following geographical positions:

(2332) 40°48.03'N., 69°02.95'W.

(2333) 40°36.76'N., 69°15.13'W.

(2334)

§167.77 In the approach to Boston, MA: Traffic Separation Scheme.

(2335) (a) A separation zone 1 mile wide is established and is centered upon the following geographic positions:

(2336) 42°20.73'N., 70°39.06'W.

(2337) 42°18.28'N., 70°01.14'W.

(2338) 40°49.25'N., 69°00.81'W.

(2339) (b) A traffic lane for northbound traffic is established between the separation zone and a line connecting the following geographical positions:

(2340) 40°50.47'N., 68°58.67'W.

(2341) 42°20.17'N., 69°59.40'W.

(2342) 42°22.71'N., 70°38.62'W.

(2343) (c) A traffic lane for southbound traffic is established between the separation zone and a line connecting the following geographical positions:

(2344) 42°18.82'N., 70°40.49'W.

(2345) 42°16.39'N., 70°02.88'W.

(2346) 40°48.03N., 69°02.95'W.

(2347)

Part 169–ShipReporting Systems

(2348)

Subpart A–General

(2349)

§169.1 What is the purpose of this subpart?

(2350) This subpart prescribes the requirements for mandatory ship reporting systems. Ship reporting systems are used to provide, gather, or exchange information through radio reports. The information is used to provide data for many purposes including, but not limited to: navigation safety, maritime security and domain awareness, environmental protection, vessel traffic services, search and rescue, weather forecasting and prevention of marine pollution.

(2351)

§169.5 How are terms used in this part defined?

(2352) As used in this part-

(2353) *Administration* means the Government of the State whose flag the ship is entitled to fly.

(2354) *Cargo ship* means any ship which is not a passenger ship.

(2355) *Flag Administration* means the Government of a State whose flag the ship is entitled to fly.

(2356) *Gross tonnage* means tonnage as defined under the International Convention on Tonnage Measurement of Ships, 1969 (Incorporated by reference, see §169.15).

(2357) *Gross tons* means vessel tonnage measured in accordance with the method utilized by the flag state administration of that vessel.

(2358) *High speed craft* means a craft that is operable on or above the water and is capable of a maximum speed equal to or exceeding V=3.7xdispl.1667, where "V" is the maximum speed and "displ" is the vessel displacement corresponding to the design waterline in cubic meters.

(2359) *High speed passenger craft* means a high speed craft carrying more than 12 passengers.

(2360) *International voyage* means a voyage from a country to which the present International Convention for the Safety of Life at Sea (SOLAS), 1974 applies to a port outside such country, or conversely. For U.S. ships, such voyages will be considered to originate at a port in the United States, regardless of when the voyage actually began. Such voyages for U.S. ships will continue until the ship returns to the United States from its last foreign port.

(2361) *Long range identification and tracking (LRIT) information or position report* means report containing the following information:

(2362) (1) The identity of the ship;

(2363) (2) The position of the ship (latitude and longitude); and

(2364) (3) The date and time of the position provided.

(2365) *LRIT Data Center* means a center established by a SOLAS Contracting Government or a group of Contracting Governments, or in the case of International Data Center, by IMO, to request, receive, process, and archive LRIT information. An LRIT Data Center may be National, Regional, Co-operative or International.

(2366) *Mandatory ship reporting system* means a ship reporting system that requires the participation of specified vessels or classes of vessels, and that is established by a government or governments after adoption of a proposed system by the International Maritime Organization (IMO) as complying with all requirements of regulation V/8-1 of the International Convention for the Safety of Life at Sea, 1974, as amended (SOLAS), except paragraph (e) thereof.

(2367) *Mobile offshore drilling unit* means a self-propelled vessel capable of engaging in drilling operations for the exploration or exploitation of subsea resources.

(2368) *Passenger ship* means a ship that carries more than 12 passengers.

(2369) *Self-propelled ships* means ships propelled by mechanical means.

(2370) *Shore-based authority* means the government appointed office or offices that will receive the reports made by ships entering each of the mandatary ship reporting systems. The office or offices will be responsible for the management and coordination of the system, interaction with participating ships, and the safe and effective operation of the system. Such an authority may or may not be an authority in charge of a vessel traffic service.

(2371) *United States* means the States of the United States, the District of Columbia, Guam, Puerto Rico, the Virgin Islands, American Samoa, the Northern Mariana Islands, and any other territory or possession of the United States.

(2372)

§169.10 What geographic coordinates are used?

(2373) Geographic coordinates expressed in terms of latitude or longitude, or both, are not intended for plotting on maps or charts where the referenced horizontal datum is the North American Datum of 1983 (NAD 83), unless such geographic coordinates are expressly labeled NAD 83. Geographic coordinates without the NAD 83 reference may be plotted on maps or charts referenced to NAD 83 only after application of the appropriate corrections that are published on the particular map or chart being used.

(2374)

§169.15 Incorporation by reference: Where can I get a copy of the publications mentioned in this part?

(2375) (a) Certain material is incorporated by reference into this part with the approval of the Director of the Federal Register under 5 U.S.C. 552(a) and 1 CFR part 51. To enforce any edition other than that specified in this section, the Coast Guard must publish notice of change in the **Federal Register** and the material must be available to the public. All approved material is available for inspection at the National Archives and Records Administration (NARA). For information on the availability of this material at NARA, call 202-741-6030 or go to *http://archives.gov/federal_register/code_of_federal_regulations/ibr_locations.html*. Also, it is available for inspection at the Coast Guard, Headquarters. Contact Commandant (CG-NAV), Attn: Office of Navigation Systems, 2703 Martin Luther King Jr. Avenue SE., Stop 7418, Washington, DC 20593-7418, and is available from the sources indicated in this section.

(2376) (b) *International Electrotechnical Commission (IEC) Bureau Central de la Commission Electrotechnique Internationale,* 3 rue de Varembé, P.O. Box 131, 1211 Geneva 20, Switzerland.

(2377) (1) IEC 60945, Fourth edition 2002-08, Maritime navigation and radiocommunication equipment and systems–General requirements–Methods of testing and required test results, incorporation by reference approved for §169.215.

(2378) (2) [Reserved]

(2379) (c) *International Maritime Organization (IMO),* 4 Albert Embankment, London SE1 7SR, U.K.

(2380) (1) IMO Resolution MSC.202(81), adopted on May 19, 2006, Adoption of Amendments to the International Convention for the Safety of Life at Sea, 1974, as Amended, incorporation by reference approved for §169.240.

(2381) (2) IMO Resolution MSC. 210(81), adopted on May 19, 2006, Performance Standards and Functional Requirements for the Long-Range Identification and

Tracking of Ships, incorporation by reference approved for §§169.215 and 169.240.

(2382) (3) IMO Resolution MSC.254(83), adopted on October 12, 2007, Adoption of Amendments to the Performance Standards and Functional Requirements for the Long-Range Identification and Tracking of Ships, incorporation by reference approved for §§169.215 and 169.240.

(2383) (4) IMO Resolution A.694(17), adopted on November 6, 1991, General Requirements for Shipborne Radio Equipment Forming Part of the Global Maritime Distress and Safety System (GMDSS) and for Electronic Navigational Aids, incorporation by reference approved for §165.215.

(2384) (5) International Convention on Tonnage Measurement of Ships, 1969, incorporation by reference approved for §169.5.

(2385)

Subpart B–Establishment of Two Mandatory Ship Reporting Systems for the Protection of Northern Right Whales

(2386)

§169.100 What mandatory ship reporting systems are established by this subpart?

(2387) This subpart prescribes requirements for the establishment and maintenance of two mandatory ship reporting systems for the protection of the endangered northern right whale (also known as the North Atlantic right whale). These two systems are designated for certain areas of the East Coast of the United States. One system is located in the northeast and is identified as WHALESNORTH. The other system is located in the southeast and is identified as WHALESSOUTH.

(2388) **Note:** 50 CFR 224.103(c) contains requirements and procedures concerning northern right whale approach limitations and avoidance procedures.

(2389)

§169.102 Who is the shore-based authority?

(2390) The U.S. Coast Guard is the shore-based authority for these mandatory ship reporting systems.

(2391)

§169.105 Where is the northeastern reporting system located?

(2392) Geographical boundaries of the northeastern area include the waters of Cape Cod Bay, Massachusetts Bay, and the Great South Channel east and southeast of Massachusetts. The coordinates (NAD 83) of the area are as follows: from a point on Cape Ann, Massachusetts at

(2393) 42°39'N, 70°37'W; then northeast to

(2394) 42°45'N, 70°13'W; then southeast to

(2395) 42°10'N, 68°31'W; then south to

(2396) 41°00'N, 68°31'W; then west to

(2397) 41°00'N, 69°17'W; then northwest to

(2398) 42°05'N, 70°02'W, then west to

(2399) 42°04'N, 70°10'W; and then along the Massachusetts shoreline of Cape Cod Bay and Massachusetts Bay back to the point on Cape Ann at

(2400) 42°39'N, 70°37'W.

(2401)

§169.110 When is the northeastern reporting system in effect?

(2402) The mandatory ship reporting system in the northeastern United States operates year-round.

(2403)

§169.115 Where is the southeastern reporting system located?

(2404) Geographical boundaries of the southeastern area include coastal waters within about 25 nautical miles (45 kilometers) along a 90-nautical mile (170-kilometer) stretch of the Atlantic seaboard in Florida and Georgia. The area coordinates (NAD 83) extends from the shoreline east to longitude 80°51.6'W with the southern and northern boundaries at latitude 30°00'N and 31°27'N., respectively.

(2405)

§169.120 When is the southeastern reporting system in effect?

(2406) The mandatory ship reporting system in the southeastern United States operates during the period beginning on November 15 each year through April 16 of the following year.

(2407)

§169.125 What classes of ships are required to make reports?

(2408) Each self-propelled ship of 300 gross tons or greater must participate in the reporting systems, except government ships exempted from reporting by regulation V/8-1(c) of SOLAS. However, exempt ships are encouraged to participate in the reporting systems.

(2409)

§169.130 When are ships required to make reports?

(2410) Participating ships must report to the shore-based authority upon entering the area covered by a reporting system. Additional reports are not necessary for movements made within a system or for ships exiting a system.

(2411)

§169.135 How must the reports be made?

(2412) (a) A ship equipped with INMARSAT C must report in IMO standard format as provided in §169.140 in table 169.140.

(2413) (b) A ship not equipped with INMARSAT C must report to the Coast Guard using other means, listed below in order of precedence–

(2414) (1) By email to rightwhale.msr@noaa.gov;

(2415) (2) HF voice communication; or

(2416) (3) VHF voice communications.

(2417) (c) HF reports made directly to the Coast Guard's Communications Command (COMMCOM) in Chesapeake, VA, or VHF reports made to Coast Guard

shore units, should only be made by ships not equipped with INMARSAT C. Ships in this category must provide all the required information to the Coast Guard watchstander.

(2418)

§169.140 What information must be included in the report?

(2419) Each ship report made to the shore-based authority must follow the standard reporting and format requirements listed this section in table 169.140. Current email address and telex numbers are published annually in the U.S. Coast Pilot.

(2420)

TABLE 169.140 – Requirements for Ship Reports

Telegraphy	Function	Information required
Name of system	System identifer	Ship reporting system WHALESNORTH or WHALESSOUTH
M	INMARSAT Number	Vessel INMARSAT number
A	Ship	The name, call sign or ship station identity, IMO number, and fag of the vessel.
B	Date and time of event	A 6-digit group giving day of month (frst two digits), hours and minutes (last four digits)
E	True course	A 3-digit group indicating true course.
F	Speed in knots and tenths of knots	A 3-digit group.
H	Date, time and point of entry into system	Entry time expressed as in (B) and entry position expressed as-(1) a 4-digit group giving latitude in degrees and minutes su f xed with N (north) or S (south) and a 5-digit group giving longitude in degrees and minutes su f xed with E (east) or W (west); or (2) True bearing (frst 3 digits) and distance (state distance) in nautical miles from a clearly identifed landmark (state landmark)
I	Destination and expected time of arrival	Name of port and date time group expressed as in (B)
L	Route information	Intended track.

(2421)

Subpart C-Transmission of Long Range Identification and Tracking Information

(2422)

§169.200 What is the purpose of this subpart?

(2423) This subpart implements Regulation 19-1 of SOLAS Chapter V (SOLAS V/19-1) and requires certain ships engaged on an international voyage to transmit vessel identification and position information electronically. This requirement enables the Coast Guard to obtain long range identification and tracking (LRIT) information and thus heightens our overall maritime domain awareness, enhances our search and rescue operations, and increases our ability to detect anomalies and deter transportation security incidents.

(2424)

§169.205 What types of ships are required to transmit LRIT information (position reports)?

(2425) The following ships, while engaged on an international voyage, are required to transmit position reports:

(2426) (a) A passenger ship, including high speed passenger craft.

(2427) (b) A cargo ship, including high speed craft, of 300 gross tonnage or more.

(2428) (c) A mobile offshore drilling unit while underway and not engaged in drilling operations.

(2429)

§169.210 Where during its international voyage must a ship transmit position reports?

(2430) The requirements for the transmission of position reports, imposed by the United States, vary depending on the relationship of the United States to a ship identified in §169.205.

(2431) (a) *Flag State relationship*. A U.S. flag ship engaged on an international voyage must transmit position reports wherever they are located.

(2432) (b) *Port State relationship*. A foreign flag ship engaged on an international voyage must transmit position reports after the ship has announced its intention to enter a U.S. port or place under requirements in 33 CFR part 160, subpart C.

(2433) (c) *Coastal State relationship*. A foreign flag ship engaged on an international voyage must transmit position reports when the ship is within 1,000 nautical miles of the baseline of the United States, unless their Flag Administration, under authority of SOLAS V/19-1.9.1, has directed them not to do so.

(2434)

§169.215 How must a ship transmit position reports?

(2435) A ship must transmit position reports using Long Range Identification and Tracking (LRIT) equipment that has been type-approved by their Administration. To be type-approved by the Coast Guard, LRIT equipment must meet the requirements of IMO Resolutions A.694(17), MSC.210(81), and MSC.254(83), and IEC standard IEC 60945 (Incorporated by reference, see §169.15).

(2436)

§169.220 When must a ship be fitted with LRIT equipment?

(2437) A ship identified in §169.205 must be equipped with LRIT equipment–

(2438) (a) Before getting underway, if the ship is constructed on or after December 31, 2008.

(2439) (b) By the first survey of the radio installation after December 31, 2008, if the ship is–

(2440) (1) Constructed before December 31, 2008, and

(2441) (2) Operates within–

(2442) (i) One hundred (100) nautical miles of the United States baseline, or

(2443) (ii) Range of an Inmarsat geostationary satellite, or other Application Service Provider recognized by the Administration, with continuous alerting is available.

(2444) (c) By the first survey of the radio installation after July 1, 2009, if the ship is–

(2445) (1) Constructed before December 31, 2008, and

(2446) (2) Operates within the area or range specified in paragraph (b)(2) of this section as well as outside the range of an Inmarsat geostationary satellite with which continuous alerting is available. While operating in the area or range specified in paragraph (b)(2) of this section, however, a ship must install LRIT equipment by the first survey of the radio installation after December 31, 2008.

(2447)

§169.225 Which Application Service Providers may a ship use?

(2448) A ship may use an application Service Provider (ASP) recognized by its administration. Some Communication Service Providers may also serve as an ASP.

(2449)

§169.230 How often must a ship transmit position reports?

(2450) A ship's LRIT equipment must transmit position reports at 6-hour intervals unless a more frequent interval is requested remotely by an LRIT Data Center.

(2451)

§169.235 What exemptions are there from reporting?

(2452) A ship is exempt from this subpart if it is–

(2453) (a) Fitted with an operating automatic identification system (AIS), under 33 CFR 164.46, and operates only within 20 nautical miles of the United States baseline,

(2454) (b) A warship, naval auxiliaries or other ship owned or operated by a SOLAS Contracting Government and used only on Government non-commercial service, or

(2455) (c) A ship solely navigating the Great Lakes of North America and their connecting and tributary waters as far east as the lower exit of the St. Lambert Lock at Montreal in the Province of Quebec, Canada.

(2456)

§169.240 When may LRIT equipment be switched off?

(2457) A ship engaged on an international voyage may switch off its LRIT equipment only when it is permitted by its Flag Administration, in circumstances detailed in SOLAS V/19–1.7, or in paragraph 4.4.1, of resolution MSC.210(81), as amended by resolution MSC.254(83) (Incorporated by reference, see §169.15).

(2458)

§169.245 What must a ship master do if LRIT equipment is switched off or fails to operate?

(2459) (a) If a ship's LRIT equipment is switched off or fails to operate, the ship's master must inform his or her Flag Administration without undue delay.

(2460) (b) The master must also make an entry in the ship's logbook that states–

(2461) (1) His or her reason for switching the LRIT equipment off, or an entry that the equipment has failed to operate, and

(2462) (2) The period during which the LRIT equipment was switched off or non-operational.

(2463) Note to §169.245: for U.S. vessels, the U.S. Coast Guard serves as the Flag Administration for purposes of this section. All LRIT notifications for the U.S. Flag administration, in addition to requests or questions about LRIT, should be communicated to the U.S. Coast Guard by e-mail addressed to *LRIT@uscg.mil*.

(2464)

Part 207–NavigationRegulations

(2465)

§207.9 Mystic River, MA; dam of Commonwealth of Massachusetts, Metropolitan District Commission.

(2466) (a) *Definition and authority of superintendent*. The term superintendent as used in the regulations in this section shall mean himself and/or his personnel then on duty at the dam. The positioning and movements of all watercraft of every description while in the locks or within 100 yards of the locks or dam shall be subject to the direction of the superintendent whose orders must be obeyed. This order does not relieve the master of the responsibility for the safety of his vessel.

(2467) (b) *Description of Locks*. There are three (3) locks to be used for the passage of vessels; one large lock 325 feet long, 45 feet wide, shall be used for vessels with draft up to seventeen (17) feet; two small locks each 120 feet long and 22 feet wide shall be used for boats up to six (6) feet draft.

(2468) (c) *Maximum draft*. Vessels drawing within six (6) inches of depth over the sills shall not be permitted lockage except under special permission of the superintendent. Every vessel using the locks and drawing more than ten (10) feet shall be accurately and distinctly marked at bow and stern showing the exact draft of water at such portions of the vessel. Gages set into the walls or the locks, both upstream and downstream of each gate, indicate the depth in feet of water over the sill of the gate.

(2469) (d) *Vessels denied lockage*. The superintendent may deny passage through the locks to any craft with sharp, rough projecting corners, overhanging equipment or cargo, or any craft or tow that is in sinking condition or in any way unseaworthy or insufficiently manned and equipped, or any craft failing to comply with the regulations in this section or with any orders given in pursuance thereof.

(2470) (e) *Protection of lock gates*. (1) In no case shall boats be permitted to enter or leave any of the locks until directed to do so by the superintendent. Boats shall not be permitted to enter or start to leave until the lock gates are at rest within the gate recesses. All persons, whether in charge of vessels or not, are prohibited from willfully or carelessly damaging the locks or any of the appurtenances or the grounds adjacent thereto, and from throwing or allowing any material of any kind to fall from the barge, scow or other vessel into the locks.

(2471) (2) No person shall permit or suffer any vessel, scow, raft, or float to come in contact with any gate or any of the locks of the Amelia Earhart Dam.

(2472) (f) *Damage to walls*. The sides of all craft passing through the locks must be free from projection of any kind which might injure the lock walls. All craft must be provided with suitable fenders. One or more men as the superintendent may direct shall be kept at the head of every tow until it has cleared the lock and guide walls, and shall protect the walls by use of the fenders.

(2473) (g) *Unnecessary delay at locks*. No person shall cause or permit any craft of which he is in charge to remain in the locks or their approaches for a longer period of time than is necessary for the passage of the locks unless he is especially permitted to do so by the superintendent, and if such craft is, in the opinion of such superintendent, in a position to obstruct navigation, it shall be removed at once as requested or directed by the superintendent.

(2474) (h) *Procedure at locks*. The locks shall be operated promptly for the passage of all craft upon signal, excepting only in such cases as are specifically provided for in the regulations in this section. All registered merchant vessels shall pass through the locks in the order directed by the superintendent. Other craft shall be allowed to pass through the locks at the discretion of the superintendent.

(2475) (i) *Navigation of the locks*. (1) All barges navigating the locks whether approaching or leaving the locks are required to be assisted by one or more tugs of sufficient power to insure full control at all times. All craft approaching the locks while any other vessel going in the opposite direction is in or about to enter shall be stopped where they will not obstruct the free passage of such other vessel.

(2476) (2) All vessels over 100 gross tons including those which are accompanied by towboats must attach not less than two good and sufficient lines, cables, or hawsers to the bollards or other fixtures provided for the purpose to check the speed of the vessel and to stop it as soon as it has gone far enough to permit the lock gate behind it to be closed. Each line, cable, or hawser shall be attended on board while passing into the lock by one or more of the vessel's crew. Where vessels are so long that in order to get them wholly within the locks it is necessary to go within 100 feet of the lock gate ahead, the speed of the vessel must be slow and the vessel must be fully under control at all times by the lines, cables or hawsers. All towboats and vessels less than 100 gross tons may enter the locks without having lines out subject to the discretion of the superintendent. The master or person in charge of a vessel shall arrange to have any line, cable, or hawser handed or thrown from the lock walls by the superintendent, or his assistants, made fast on the vessel as requested or directed, so that in cases of emergency such line, cable, or hawser may also be used to check the speed of and stop the vessel.

(2477) (3) Operators of vessels less than 200 gross tons may use the floating moorings in the large lock to fasten lines or hawsers, but they shall not be used to check the way on any vessel greater than 30 gross tons.

(2478) (4) Vessels less than 30 gross tons may fasten lines to the floating moorings in the large or small locks. All persons shall keep off the floating moorings at all times.

(2479) (5) No line shall be attached to anything on or a part of the dam except the fixtures provided for this purpose.

(2480) (6) Equipment of each craft shall include a sufficient bow line and stern line.

(2481) (j) *Mooring*. When a craft is in position in the lock, it shall be securely fastened in a manner satisfactory to the superintendent to prevent the craft moving about while the lock is being filled or emptied, and the lines, cables, or hawsers used for this purpose shall be attended as far as is necessary or required while the filling or emptying is in progress.

(2482) (k) [Reserved]

(2483) (l) *Signals*. (1) All craft desiring lockage shall, on approaching the locks, signal by two long and two short blasts of a whistle or other sound device. Two long blasts from the lock in reply will indicate a delayed opening and direct the craft not to enter the lock.

(2484) (2) Lights are located at each end of each lock and will normally show red. No vessel shall come within 100 feet of the outside of any gate when the signal is red except when so directed by the superintendent.

(2485) (3) Fireboats and craft owned by the U.S. Government shall be given prompt and preferential lockage when they sound four long blasts.

(2486) (4) No vessel shall move into or out of any lock until the controlling signal is green. A green light in addition to audio loud speakers, operated by the superintendent or his assistants, will direct craft through the locks.

(2487) (5) It shall be the duty of every master or person in charge of any vessel to ascertain by personal observation that the lock gate is fully open before proceeding.

(2488) (m) *Operating machinery*. Lock employees only shall be permitted to operate the lock gates, signals or other appliances. No person shall deface or injure any part of the Amelia Earhart Dam, or any pier, wall or other structure or any mechanism connected therewith; nor shall any person, without the consent of the superintendent, make fast to the dam, guard, guide wall, pier, or any appurtenance thereof any vessel, scow, raft, or float.

(2489) (n) *Vessel to carry regulations*. A copy of the regulations in this section shall be kept at all times on board each vessel regularly engaged in navigating the locks. Copies may be obtained without charge from the superintendent; the Commonwealth of Massachusetts,

M.D.C. Parks Division, Boston, MA; New England Division, Corps of Engineers, Division Engineer, Waltham, MA.

(2490)

§207.10 Charles River, MA; dam of Charles River Basin Commission.

(2491) (a) The movements of all vessels or boats in and near the lock shall be under the direction of the superintendent in charge of these structures and his assistants, whose orders and signals shall be obeyed.

(2492) (b) Every vessel using the lock and drawing more than 10 feet shall be accurately and distinctly marked at the bow and stern, showing the exact draft of water at such portions of the vessel.

(2493) (c) All steam vessels desiring to pass through the lock shall signal for the same by two long and two short blasts of the whistle.

(2494) (d) (1) All vessels passing through the lock shall have their outboard spars, if any, rigged in, and booms amidships, and secured. All standing and running rigging must be triced in to keep it from blowing out and fouling the drawbridge. Every vessel of 200 tons and under shall be provided with at least two, and every vessel of more than 200 tons shall be provided with at least four good and sufficient lines, cables, or hawsers. Anchors shall either be stowed or shall hang from hawse pipes, hauled up close, clear of the water if possible. Vessels with anchors under foot or hanging from catheads will not be permitted to enter the lock.

(2495) (2) All vessels must be sufficiently manned and must have a sufficient number of round and fore-and-aft fenders to protect the lock from injury. All heavy rope fenders must be securely lashed to prevent their falling into the lock and interfering with the gates.

(2496) (e) All vessels approaching the lock while any other vessel going in the opposite direction is in or about to enter it shall be stopped where they will not obstruct the free passage of such other vessel.

(2497) (f) It shall be the duty of every master or person in charge of any vessel upon approaching the lock from the upstream end to ascertain by personal observation whether or not the upper lock gate is open, and a vessel shall not be permitted to come within 100 feet of the upper lock gate until the gate has been wholly withdrawn into its recess.

(2498) (g) All towboats, whether towing or not, and other steam vessels of less than 100 tons gross may enter the lock under their own power and without having lines out, but all other vessels, including those which are accompanied by towboats, must attach not less than two good and sufficient lines, cables, or hawsers to the bollards or other fixtures provided for the purpose to check the speed of the vessel and to stop it as soon as it has gone far enough to permit the lock gate behind it to be closed, and each line, cable, or hawser shall be attended on board while passing into the lock by one or more of the vessel's crew. Where vessels are so long that in order to get them wholly within the lock it is necessary to approach within 150 feet of the lock gate ahead, the speed of the vessel must be slow and fully under control by the lines, cables, or hawsers. Steam vessels of more than 100 tons gross, not including towboats, will not be permitted to turn their propellers on entering the lock after the bow of the vessel has entered, but will be drawn in by means of capstans on the lock walls or otherwise, and their speed must be checked and the vessel stopped by lines, cables, or hawsers as in other cases. All steam vessels may leave the lock under their own power. The master or person in charge of a vessel shall arrange to have any line, cable, or hawser handed or thrown from the lock walls by the superintendent or his assistants, made fast on the vessel as requested or directed, so that in cases of emergency such line, cable, or hawser may also be used to check the speed of and stop the vessel.

(2499) (h) When a vessel is in position in the lock it shall be securely fastened in a manner satisfactory to the superintendent, or his assistant in charge of the lock at the time, to prevent the vessel from moving about while the lock is being filled or emptied, and the lines, cables, and hawsers used for this purpose shall be attended as far as is necessary or required while the filling or emptying is in progress.

(2500) (i) No vessel which has iron or irons projecting from it or lumber or other cargo projecting over its sides shall enter the lock, except at such time and with such precautions to prevent damage to the lock or its appurtenances as the superintendent, or the assistant in charge of the lock at the time, may require.

(2501) (j) All persons, whether in charge of vessels or not, are prohibited from willfully or carelessly damaging the lock, any of its appurtenances or the grounds adjacent thereto, and from throwing any material of any kind into the lock. No line shall be attached to anything except the bollards and other fixtures provided for the purpose.

(2502) (k) Upon each passage through the lock, the master or clerk of a vessel shall make a statement of the kind and tonnage of the freight carried.

(2503) (l) No person shall cause or permit any vessel or boat of which he is in charge to remain in the lock or its approaches for a longer time than is necessary for the passage of the lock, unless he is especially permitted to do so by the superintendent or the assistant in charge of the lock at the time, and if such vessel or boat is, in the opinion of such superintendent or assistant, in a position to obstruct navigation it shall be removed at once as requested or directed by such superintendent or assistant.

(2504) (m) All registered merchant vessels shall pass through the lock in the order directed by the superintendent or the assistant in charge of the lock at the time. Unregistered craft will not be allowed to pass through the lock separately unless especially permitted by such superintendent or assistant.

(2505) (n) The lock shall be operated promptly for the passage of all vessels upon signal, excepting only in such cases as are specifically provided for in this section.

(2506)

§207.20 Cape Cod Canal, MA; use, administration, and navigation.

(2507) (See United States Coast Pilot 2, Atlantic Coast, Cape Cod to Sandy Hook.)

(2508)

§207.800 Collection of navigation statistics.

(2509) (a) *Definitions*. For the purpose of this regulation the following terms are defined:

(2510) (1) *Navigable waters of the United States* means those waters of the United States that are subject to the ebb and flow of the tide shoreward to the mean high water mark, and/or are presently used, or have been used in the past, or may be susceptible to use to transport interstate or foreign commerce. (See 33 CFR part 329 for a more complete definition of this term.)

(2511) (2) *Offenses and Violations* mean:

(2512) (i) Failure to submit a required report.

(2513) (ii) Failure to provide a timely, accurate, and complete report.

(2514) (iii) Failure to submit monthly listings of idle vessels or vessels in transit.

(2515) (iv) Failure to submit a report required by the lockmaster or canal operator.

(2516) (3) *Leased or chartered vessel* means a vessel that is leased or chartered when the owner relinquishes control of the vessel through a contractual agreement with a second party for a specified period of time and/or for a specified remuneration from the lessee. Commercial movements on an affreightment basis are not considered a lease or charter of a particular vessel.

(2517) (4) *Person or entity* means an individual, corporation, partnership, or company.

(2518) (5) *Timely* means vessel and commodity movement data must be received by the Waterborne Commerce Statistics Center within 30 days after the close of the month in which the vessel movement or nonmovement takes place.

(2519) (6) *Commercial vessel* means a vessel used in transporting by water, either merchandise or passengers for compensation or hire, or in the course of business of the owner, lessee, or operator of the vessel.

(2520) (7) *Reporting situation* means a vessel movement by an operator that is required to be reported. Typical examples are listed in the instructions on the various ENG Forms. Five typical movements that are required to be reported by vessel operating companies include the following examples: Company A is the barge owner, and the barge transports corn from Minneapolis, MN to New Orleans, LA, with fleeting at Cairo, IL.

(2521) (i) *Lease/Charter*: If Company A leases or charters the barge to Company B, then Company B is responsible for reporting the movements of the barge until the lease/charter expires.

(2522) (ii) *Interline Movement*: A barge is towed from Minneapolis to Cairo by Company A, and from Cairo to New Orleans by Company B. Since Company A is the barge owner, and the barge is not leased. Company A reports the entire movement of the barge with an origin of Minneapolis and a destination of New Orleans.

(2523) (iii) *Vessel Swap/Trade*: Company A swaps barge with Company B to allow Company B to meet a delivery commitment to New Orleans. Since Company A has not leased/chartered the barge, Company A is responsible for filing the report. Company B is responsible for filing the report on the barge which is traded to Company A. The swap or trade will not affect the primary responsibility for reporting the individual vessel movements.

(2524) (iv) *Re-Consignment*: Barge is reconsigned to Mobile, AL. Company A reports the movements as originating in Minneapolis and terminating in Mobile. The point from which barge is reconsigned is not reported, only points of loading and unloading.

(2525) (v) *Fleeting*: Barge is deposited at a New Orleans fleeting area by Company A and towed by Company B from fleeting area to New Orleans area dock for unloading. Company A, as barge owner, reports entire movements from Minneapolis to the unloading dock in New Orleans. Company B does not report any barge movement.

(2526) (b) Implementation of the waterborne commerce statistics provisions of the River and Harbor Act of 1922, as amended by the Water Resources Development Act of 1986 (Pub. L. 99-662), mandates the following.

(2527) (1) *Filing requirements*. Except as provided in paragraph (b)(2) of this section, the person or entity receiving remuneration for the movement of vessels or for the transportation of goods or passengers on the navigable waters is responsible for assuring that the activity report of commercial vessels is timely filed.

(2528) (i) For vessels under lease/charter agreements, the lessee or charterer of any commercial vessel engaged in commercial transportation will be responsible for the filing of said reports until the lease/charter expires.

(2529) (ii) The vessel owner, or his designated agent, is always the responsible party for ensuring that all commercial activity of the vessel is timely reported.

(2530) (2) The following Vessel Information Reports are to be filed with the Army Corps of Engineers, at the address specified on the ENG Form, and are to include:

(2531) (i) *Monthly reports*. These reports shall be made on ENG Forms furnished upon written request of the vessel operating companies to the Army Corps of Engineers. The forms are available at the following address: U.S. Army Corps of Engineers, Waterborne Commerce Statistics Center, Post Office Box 62180, New Orleans, LA 70161-1280.

(2532) (A) All movements of domestic waterborne commercial vessels shall be reported, including but not limited to: Dry cargo ship and tanker moves, loaded and empty barge moves, towboat moves, with or without barges in tow, fishing vessels, movements of crew boats and supply boats to offshore locations, tugboat moves and movements of newly constructed vessels from the shipyard to the point of delivery.

(2533) (B) Vessels idle during the month must also be reported.

(2534) (C) Notwithstanding the above requirements, the following waterborne vessel movements need not be reported:

(2535) *(1)* Movements of recreational vessels.

(2536) *(2)* Movements of fire, police, and patrol vessels.

(2537) *(3)* Movements of vessels exclusively engaged in construction (e.g., piledrivers and crane barges). **Note:** however, that movements of supplies, materials, and crews to or from the construction site must be timely reported.

(2538) *(4)* Movements of dredges to or from the dredging site. However, vessel movements of dredged material from the dredging site to the disposal site must be reported.

(2539) *(5)* Specific movements granted exemption in writing by the Waterborne Commerce Statistics Center.

(2540) (D) ENG Forms 3925 and 3925b shall be completed and filed by vessel operating companies each month for all voyages or vessel movements completed during the month. Vessels that did not complete a move during the month shall be reported as idle or in transit.

(2541) (E) The vessel operating company may request a waiver from the Army Corps of Engineers, and upon written approval by the Waterborne Commerce Center, the company may be allowed to provide the requisite information of the above paragraph (D), on computer printouts, magnetic tape, diskettes, or alternate medium approved by the Center.

(2542) (F) Harbor Maintenance Tax information is required on ENG Form 3925 for cargo movements into or out of ports that are subject to the provisions of section 1402 of the Water Resources Development Act of 1986 (Pub. L. 99-662).

(2543) *(1)* The name of the shipper of the commodity, and the shipper's Internal Revenue Service number or Social Security number, must be reported on the form.

(2544) *(2)* If a specific exemption applies to the shipper, the shipper should list the appropriate exemption code. The specific exemption codes are listed in the directions for ENG Form 3925.

(2545) *(3)* Refer to 19 CFR part 24 for detailed information on exemptions and ports subject to the Harbor Maintenance Tax.

(2546) (ii) *Annual reports*. Annually an inventory of vessels available for commercial carriage of domestic commerce and vessel characteristics must be filed on ENG Forms 3931 and 3932.

(2547) (iii) *Transaction reports*. The sale, charter, or lease of vessels to other companies must also be reported to assure that proper decisions are made regarding each company's duty for reporting vessel movements during the year. In the absence of notification of the transaction, the former company of record remains responsible until proper notice is received by the Corps.

(2548) (iv) *Reports to lockmasters and canal operators*. Masters of self-propelled non-recreational vessels which pass through locks and canals operated by the Army Corps of Engineers will provide the data specified on ENG Forms 3102b, 3102c, and/or 3102d to the lockmaster, canal operator, or his designated representative in the manner and detail dictated.

(2549) (c) *Penalties for noncompliance*. The following penalties for noncompliance can be assessed for offenses and violations.

(2550) (1) *Criminal penalties*. Every person or persons violating the provisions of this regulation shall, for each and every offenses, be liable to a fine of not more than $5,000, or imprisonment not exceeding two months, to be enforced in any district court in the United States within whose territorial jurisdiction such offense may have been committed.

(2551) (2) In addition, any person or entity that fails to provide timely, accurate, and complete statements or reports required to be submitted by the regulation in this section may also be assessed a civil penalty of up to $6,270 per violation under 33 U.S.C. 555, as amended.

(2552) (3) *Denial of passage*. In addition to these fines, penalties, and imprisonments, the lockmaster or canal operator can refuse to allow vessel passage.

(2553) (d) *Enforcement policy*. Every means at the disposal of the Army Corps of Engineers will be utilized to monitor and enforce these regulations.

(2554) (1) To identify vessel operating companies that should be reporting waterborne commerce data, The Corps will make use of, but is not limited to, the following sources.

(2555) (i) Data on purchase and sale of vessels.

(2556) (ii) U.S. Coast Guard vessel documentation and reports.

(2557) (iii) Data collected at Locks, Canals, and other facilities operated by the Corps.

(2558) (iv) Data provided by terminals on ENG Form 3926.

(2559) (v) Data provided by the other Federal agencies including the Internal Revenue Service, Customs Service, Maritime Administration, Department of Transportation, and Department of Commerce.

(2560) (vi) Data provided by ports, local facilities, and State or local governments.

(2561) (vii) Data from trade journals and publications.

(2562) (viii) Site visits and inspections.

(2563) (2) *Notice of violation*. Once a reporting violation is determined to have occurred, the Chief of the Waterborne Commerce Statistics Center will notify the responsible party and allow 30 days for the reports to be filed after the fact. If the reports are not filed within this 30-day notice period, then appropriate civil or criminal actions will be undertaken by the Army Corps of Engineers, including the proposal of civil or criminal penalties for noncompliance. Typical cases for criminal or civil action include, but are not limited to, those violations which are willful, repeated, or have a substantial impact in the opinion of the Chief of the Waterborne Commerce Statistics Center.

(2564) (3) *Administrative assessment of civil penalties*. Civil penalties may be assessed in the following manner.

(2565) (i) *Authorization*. If the Chief of the Waterborne Commerce Statistics Center finds that a person or entity has failed to comply with any of the provisions specified herein, he is authorized to assess a civil penalty in accordance with the Class I penalty provisions of 33 CFR part 326. Provided, however, that the procedures in 33 CFR part 326 specifically implementing the Clean Water Act (33 U.S.C. 1319(g)(4)), public notice, comment period, and state coordination, shall not apply.

(2566) (ii) *Initiation*. The Chief of the Waterborne Commerce Statistics Center will prepare and process a proposed civil penalty order which shall state the amount of the penalty to be assessed, describe by reasonable specificity the nature of the violation, and indicate the applicable provisions of 33 CFR part 326.

(2567) (iii) *Hearing requests*. Recipients of a proposed civil penalty order may file a written request for a hearing or other proceeding. This request shall be as specified in 33 CFR part 326 and shall be addressed to the Director of the Water Resources Support Center, Casey Building, Fort Belvoir, VA 22060-5586, who will provide the requesting person or entity with a reasonable opportunity to present evidence regarding the issuance, modification, or revocation of the proposed order. Thereafter, the Director of the Water Resources Center shall issue a final order.

(2568) (4) *Additional remedies*. Appropriate cases may also be referred to the local U.S. Attorney for prosecution, penalty collection, injunctive, and other relief by the Chief of the Waterborne Commerce Statistics Center.

(2569)

Part 334–Danger Zones and Restricted Area Regulations

(2570)

§334.1 Purpose.

(2571) The purpose of this part is to:

(2572) (a) Prescribe procedures for establishing, amending and disestablishing danger zones and restricted areas;

(2573) (b) List the specific danger zones and restricted areas and their boundaries; and

(2574) (c) Prescribe specific requirements, access limitations and controlled activities within the danger zones and restricted areas.

(2575)

§334.2 Definitions.

(2576) (a) *Danger zone*. A defined water area (or areas) used for target practice, bombing, rocket firing or other especially hazardous operations, normally for the armed forces. The danger zones may be closed to the public on a full-time or intermittent basis, as stated in the regulations.

(2577) (b) *Restricted area*. A defined water area for the purpose of prohibiting or limiting public access to the area. Restricted areas generally provide security for Government property and/or protection to the public from the risks of damage or injury arising from the Government's use of that area.

(2578)

§334.3 Special policies.

(2579) (a) *General*. The general regulatory policies stated in 33 CFR part 320 will be followed as appropriate. In addition, danger zone and restricted area regulations shall provide for public access to the area to the maximum extent practicable.

(2580) (b) *Food fishing industry*. The authority to prescribe danger zone and restricted area regulations must be exercised so as not to unreasonably interfere with or restrict the food fishing industry. Whenever the proposed establishment of a danger zone or restricted area may affect fishing operations, the District Engineer will consult with the Regional Director, U.S. Fish and Wildlife Service, Department of the Interior and the Regional Director, National Marine Fisheries Service, National Oceanic & Atmospheric Administration (NOAA).

(2581) (c) *Temporary, occasional or intermittent use*. If the use of the water area is desired for a short period of time, not to exceed thirty days in duration, and that planned operations can be conducted safely without imposing unreasonable restrictions on navigation, and without promulgating restricted area regulations in accordance with the regulations in this section, applicants may be informed that formal regulations are not required. Activities of this type shall not reoccur more often than biennially (every other year), unless danger zone/restricted area rules are promulgated under this Part. Proper notices for mariners requesting that vessels avoid the area will be issued by the Agency requesting such use of the water area, or if appropriate, by the District Engineer, to all known interested persons. Copies will also be sent to appropriate State agencies, the Commandant, U.S. Coast Guard, Washington, DC 20590, and Director, National Geospatial-Intelligence Agency, Hydrographic Center, Washington, DC 20390, ATTN: Code NS 12. Notification to all parties and Agencies shall be made at least two weeks prior to the planned event, or earlier, if required for distribution of Local Notice to Mariners by the Coast Guard.

(2582)

§334.4 Establishment and amendment procedures.

(2583) (a) *Application*. Any request for the establishment, amendment or revocation of a danger zone or restricted area must contain sufficient information for the District Engineer to issue a public notice, and as a minimum must contain the following:

(2584) (1) Name, address and telephone number of requestor including the identity of the command and DoD facility and the identity of a point of contact with phone number.

(2585) (2) Name of waterway and if a small tributary, the name of a larger connecting waterbody.

(2586) (3) Name of closest city or town, county/parish and state.

(2587) (4) Location of proposed or existing danger zone or restricted area with a map showing the location, if possible.

(2588) (5) A brief statement of the need for the area, its intended use and detailed description of the times, dates and extent of restriction.

(2589) (b) *Public notice*. (1) The Corps will normally publish public notices and **Federal Register** documents concurrently. Upon receipt of a request for the establishment, amendment or revocation of a danger zone or restricted area, the District Engineer should forward a copy of the request with his/her recommendation, a copy of the draft public notice and a draft **Federal Register** document to the Office of the Chief of Engineers, ATTN: CECW-OR. The Chief of Engineers will publish the proposal in the **Federal Register** concurrent with the public notice issued by the District Engineer.

(2590) (2) *Content*. The public notice and **Federal Register** documents must include sufficient information to give a clear understanding of the proposed action and should include the following items of information:

(2591) (i) Applicable statutory authority or authorities; (40 Stat. 266; 33 U.S.C. 1) and (40 Stat. 892; 33 U.S.C. 3)

(2592) (ii) A reasonable comment period. The public notice should fix a limiting date within which comments will be received, normally a period not less than 30 days after publication of the notice.

(2593) (iii) The address of the District Engineer as the recipient of any comments received.

(2594) (iv) The identity of the applicant/proponent;

(2595) (v) The name or title, address and telephone number of the Corps employee from whom additional information concerning the proposal may be obtained;

(2596) (vi) The location of the proposed activity accompanied by a map of sufficient detail to show the boundaries of the area(s) and its relationship to the surrounding area.

(2597) (3) *Distribution*. Public notice will be distributed in accordance with 33 CFR 325.3(d)(1). In addition to this general distribution, public notices will be sent to the following Agencies:

(2598) (i) The Federal Aviation Administration (FAA) where the use of airspace is involved.

(2599) (ii) The Commander, Service Force, U.S. Atlantic Fleet, if a proposed action involves a danger zone off the U.S. Atlantic coast.

(2600) (iii) Proposed danger zones on the U.S. Pacific coast must be coordinated with the applicable commands as follows:

(2601) Alaska, Oregon and Washington:

(2602) Commander, Naval Base, Seattle

(2603) California:

(2604) Commander, Naval Base, San Diego

(2605) Hawaii and Trust Territories:

(2606) Commander, Naval Base, Pearl Harbor

(2607) (c) *Public hearing*. The District Engineer may conduct a public hearing in accordance with 33 CFR part 327.

(2608) (d) *Environmental documentation*. The District Engineer shall prepare environmental documentation in accordance with appendix B to 33 CFR part 325.

(2609) (e) *District Engineer's recommendation*. After closure of the comment period, and upon completion of the District Engineer's review he/she shall forward the case through channels to the Office of the Chief of Engineers, ATTN: CECW-OR with a recommendation of whether or not the danger zone or restricted area regulation should be promulgated. The District Engineer shall include a copy of environmental documentation prepared in accordance with appendix B to 33 CFR part 325, the record of any public hearings, if held, a summary of any comments received and a response thereto, and a draft of the regulation as it is to appear in the **Federal Register**.

(2610) (f) *Final decision*. The Chief of Engineers will notify the District Engineer of the final decision to either approve or disapprove the regulations. The District Engineer will notify the applicant/proponent and publish a public notice of the final decision. Concurrent with issuance of the public notice the Office of the Chief of Engineers will publish the final decision in the **Federal Register** and either withdraw the proposed regulation or issue the final regulation as appropriate. The final rule shall become effective no sooner than 30 days after publication in the **Federal Register** unless the Chief of Engineers finds that sufficient cause exists and publishes that rationale with the regulations.

(2611)

§334.5 Disestablishment of a danger zone.

(2612) (a) Upon receipt of a request from any agency for the disestablishment of a danger zone, the District Engineer shall notify that agency of its responsibility for returning the area to a condition suitable for use by the public. The agency must either certify that it has not used the area for a purpose that requires cleanup or that it has removed all hazardous materials and munitions, before the Corps will disestablish the area. The agency will remain responsible for the enforcement of the danger zone regulations to prevent unauthorized entry into the area until the area is deemed safe for use by the public and the area is disestablished by the Corps.

(2613) (b) Upon receipt of the certification required in paragraph (a) of this section, the District shall forward the request for disestablishment of the danger zone through channels to CECW-OR, with its recommendations. Notice of proposed rulemaking and public procedures as outlined in §334.4 are not normally required before publication of the final rule revoking a restricted area or danger zone regulation. The disestablishment/revocation of the danger zone or restricted area regulation removes a restriction on a waterway.

(2614)

§334.6 Datum.

(2615) (a) Geographic coordinates expressed in terms of latitude or longitude, or both, are not intended for plotting

on maps or charts whose reference horizontal datum is the North American Datum of 1983 (NAD 83), unless such geographic coordinates are expressly labeled NAD 83. Geographic coordinates without the NAD 83 reference may be plotted on maps or charts referenced to NAD 83 only after application of the appropriate corrections that are published on the particular map or chart being used.

(2616) (b) For further information on NAD 83 and National Service nautical charts please contact:

(2617) Director, Coast Survey (N/CG2)

(2618) National Ocean Service, NOAA

(2619) 1315 East-West Highway, Station 6147

(2620) Silver Spring, MD 20910-3282.

(2621)

§334.10 Gulf of Maine off Seal Island, Maine; naval aircraft bombing target area.

(2622) (a) *The danger zone*. A circular area with a radius of 1.5 nautical miles, having its center just easterly of Seal Island at latitude 43°53'00" and longitude 68°44'00".

(2623) (b) *The regulations*. (1) No aerial bombing practice will take place in the danger zone after 5:00 p.m. Mondays through Saturdays, at any time on Sundays, or during foggy or inclement weather.

(2624) (2) Vessels or other watercraft will be allowed to enter the danger zone any time there are no aerial bombing exercises being conducted.

(2625) (3) No live ammunition or explosives will be dropped in the area.

(2626) (4) Suitable Notice to Mariners, by appropriate methods, will be issued by the Commander, First Coast Guard District, Boston, Massachusetts; upon request of the Commandant, First Naval District, Boston, Massachusetts, or his designated agent.

(2627) (5) Prior to the conducting of each bombing practice, the area will be patrolled by a naval aircraft or surface vessel to ensure that no persons or watercraft are within the danger zone.

(2628) Vessels may be requested to veer off when drops are to be made, however, drops will be made only when the area is clear. The patrol aircraft will employ the method of warning known as "buzzing" which consists of low flight by the airplane and repeated opening and closing of the throttle.

(2629) (6) Any such watercraft shall, upon being so warned, immediately leave the designated area and, until the conclusion of the practice, shall remain at such distance that it will be safe from falling projectiles.

(2630) (7) The regulations of this section shall be enforced by the Commandant, First Naval District, Boston, Massachusetts, or such agencies as he may designate.

(2631)

§334.20 Gulf of Maine off Cape Small, Maine; naval aircraft practice mining range area.

(2632) (a) *The danger zone*. Within an area bounded as follows: Beginning at

(2633) 43°43'00"N., 69°46'00"W.; thence to

(2634) 43°38'30"N., 69°46'00"W.; thence to

(2635) 43°38'30"N., 69°49'30"W.; thence to

(2636) 43°42'10"N., 69°49'30"W.; thence to the point of beginning.

(2637) (b) *The regulations*. (1) Test drops from aircraft will be made within the area at intermittent periods from noon until sunset local time and only during periods of good visibility.

(2638) (2) Testing will not restrict any fishing, recreational, or commercial activities in the testing area.

(2639) (3) Aircraft will patrol the area prior to and during test periods to insure that no surface vessels are within the area. No test drops will be made while surface vessels are transitting the area.

(2640) (4) No live ammunition or explosives will be dropped in the area.

(2641) (5) The regulations of this section shall be enforced by the Commandant, First Naval District, Boston, MA, or such agencies as he may designate.

(2642)

§334.30 Gulf of Maine off Pemaquid Point, Maine; naval Sonobuoy Test Area.

(2643) (a) *The area*. The test area or "Foul Area" encompasses a circular area one nautical mile in radius, the center of which is located 7.9 nautical miles, bearing 187° magnetic from Pemaquid Light.

(2644) (b) *The regulations*. (1) Sonobuoy drops will be made only in the designated area and when visibility is at least three miles.

(2645) (2) Sonobuoy drop tests will normally be conducted at intermittent periods on a 5-day week basis, Monday through Friday. However, on occasion tests may be conducted intermittently on a seven-day week basis.

(2646) (3) Prior to and during the period when sonobuoys are being dropped, an escort vessel or naval aircraft will be in the vicinity to ensure that no persons or vessels are in the testing area. Vessels may be requested to veer off when sonobuoys are about to be dropped, however, drops will be made only when the area is clear.

(2647) (4) The sonobuoys drops will be made in connection with the production and experimentation of sonobuoys.

(2648) (5) No live ammunition or explosives will be involved.

(2649) (6) The regulations in this section shall be enforced by the Commanding Officer, U.S. Naval Air Station, Brunswick, Maine, or such agencies as he may designate.

(2650)

§334.40 Atlantic Ocean in vicinity of Duck Island, Maine, Isles of Shoals; naval aircraft bombing target area.

(2651) (a) *The danger zone*. A circular area with a radius of 500 yards having its center on Shag Rock in the vicinity of Duck Island at latitude 43°00'12", longitude 70°36'12".

(2652) (b) *The regulations*. (1) No person or vessel shall enter or remain in the danger zone from 8:00 a.m. to 5:00 p.m. (local time) daily, except as authorized by the enforcing agency.

(2653) (2) This section shall be enforced by the Commandant, First Naval District, and such agencies as he may designate.

(2654)

§334.45 Kennebec River, Bath Iron Works Shipyard, Naval Restricted Area, Bath, Maine.

(2655) (a) *The area*. The waters within a coffin shaped area on the west side of the river south of the Carlton (Route 1) highway bridge beginning on the western shore at

(2656) 43°54'40.7"N., 069°48'44.8"W.; thence easterly to

(2657) 43°54'40.7"N.,069°48'36.8"W.;thencesoutheasterly to

(2658) 43°54'10.4"N., 069°48'34.7"W.; thence southwesterly to

(2659) 43°53'55.1"N., 069°48'39.1"W.; thence westerly to

(2660) 43°53'55.1"N., 69°48'51.8"W.; thence northerly along the westerly shoreline to the point of origin.

(2661) (b) *The regulation*. All persons, swimmers, vessels and other craft, except those vessels under the supervision or contract to local military or Naval authority, vessels of the United States Coast Guard, and local or state law enforcement vessels, are prohibited from entering the restricted areas without permission from the Supervisor of Shipbuilding, USN Bath, Maine or his authorized representative.

(2662) (c) *Enforcement*. The regulation in this section, promulgated by the United States Army Corps of Engineers, shall be enforced by the, Supervisor of Shipbuilding, Conversion and Repair, Bath, United States Navy and/or such agencies or persons as he/she may designate.

(2663)

§334.50 Piscataqua River at Portsmouth Naval Shipyard, Kittery, Maine; restricted areas.

(2664) (a) *The areas*. *Area No. 1:* The area bounded by a line beginning at a point on the easterly side of Seavey Island at

(2665) 43°04'37"N., 70°43'44"W.; thence to

(2666) 43°04'36"N., 70°43'40"W.; thence to the pier on the westerly side of Clark Island at

(2667) 43°04'36.5"N., 70°43'34"W.; thence along the northerly side of Clark Island to a point on the easterly side at

(2668) 43°04'37"N., 70°43'25"W.; thence northeasterly to the easterly side of Jamaica Island at

(2669) 43°04'49"N., 70°43'24"W.; thence along the southerly and westerly sides of Jamaica Island and thence generally along the easterly side of Seavey Island to the point of beginning.

(2670) *Area No. 2:* The area bounded by a line beginning at a point on the southerly side of Seavey Island at Henderson Point at

(2671) 43°04'29"N., 70°44'14"W.; thence to

(2672) 43°04'29.5"N., 70°44'17.4"W.; thence to

(2673) 43°04'36.6"N., 70°44'22.6"W.; thence to

(2674) 43°04'44.8"N., 70°44'33.2"W.; thence to

(2675) 43°04'47.4"N., 70°44'42.1"W.; thence to

(2676) 43°04'48"N., 70°44'52"W.; thence to

(2677) 43°04'49"N., 70°44'54"W.; thence to

(2678) 43°04'51"N., 70°44'55"W.; thence to

(2679) 43°04'53"N., 70°44'53"W.; thence to

(2680) 43°04'57"N., 70°44'47"W.; thence to

(2681) 43°04'58"N., 70°44'46"W.; thence to

(2682) 43°05'02"N., 70°44'36"W.; thence to

(2683) 43°05'04"N., 70°44'31"W.; thence along the westerly side of Seavey Island to the beginning point.

(2684) (b) *The regulations*. All persons, vessels and other craft, except those vessels under the supervision of or contract to local military or naval authority, are prohibited from entering the restricted areas without permission from the Commander, Portsmouth Naval Shipyard or his/her authorized representative.

(2685)

§334.60 Cape Cod Bay south of Wellfleet Harbor, MA; naval aircraft bombing target area.

(2686) (a) *The danger zone*. A circular area with a radius of 1,000 yards having its center on the aircraft bombing target hulk James Longstreet in Cape Cod Bay at 41°49'46"N., 70°02'54"W.

(2687) (b) *The regulations*. (1) No person or vessel shall enter or remain in the danger zone at any time, except as authorized by the enforcing agency.

(2688) (2) This section shall be enforced by the Commandant, First Naval District, and such agencies as he may designate.

(2689)

TITLE 40–PROTECTION OF ENVIRONMENT

(2690)

Part 140–MarineSanitation Device Standard

(2691)

§140.1 Definitions.

(2692) For the purpose of these standards the following definitions shall apply:

(2693) (a) *Sewage* means human body wastes and the wastes from toilets and other receptacles intended to receive or retain body wastes;

(2694) (b) *Discharge* includes, but is not limited to, any spilling, leaking, pumping, pouring, emitting, emptying, or dumping;

(2695) (c) *Marine sanitation device* includes any equipment for installation onboard a vessel and which is designed to receive, retain, treat, or discharge sewage and any process to treat such sewage;

(2696) (d) *Vessel* includes every description of watercraft or other artificial contrivance used, or capable of being used, as a means of transportation on waters of the United States;

(2697) (e) *New vessel* refers to any vessel on which construction was initiated on or after January 30, 1975;

(2698) (f) *Existing vessel* refers to any vessel on which construction was initiated before January 30, 1975;

(2699) (g) *Fecal coliform bacteria* are those organisms associated with the intestines of warm-blooded animals that are commonly used to indicate the presence of fecal material and the potential presence of organisms capable of causing human disease.

(2700)

§140.2 Scope of standard.

(2701) The standard adopted herein applies only to vessels on which a marine sanitation device has been installed. The standard does not require the installation of a marine sanitation device on any vessel that is not so equipped. The standard applies to vessels owned and operated by the United States unless the Secretary of Defense finds that compliance would not be in the interest of national security.

(2702)

§140.3 Standard.

(2703) (a) (1) In freshwater lakes, freshwater reservoirs or other freshwater impoundments whose inlets or outlets are such as to prevent the ingress or egress by vessel traffic subject to this regulation, or in rivers not capable of navigation by interstate vessel traffic subject to this regulation, marine sanitation devices certified by the U.S. Coast Guard (see 33 CFR part 159, published in 40 FR 4622, January 30, 1975), installed on all vessels shall be designed and operated to prevent the overboard discharge of sewage, treated or untreated, or of any waste derived from sewage. This shall not be construed to prohibit the carriage of Coast Guard-certified flow-through treatment devices which have been secured so as to prevent such discharges.

(2704) (2) In all other waters, Coast Guard-certified marine sanitation devices installed on all vessels shall be designed and operated to either retain, dispose of, or discharge sewage. If the device has a discharge, subject to paragraph (d) of this section, the effluent shall not have a fecal coliform bacterial count of greater than 1,000 per 100 milliliters nor visible floating solids. Waters where a Coast Guard-certified marine sanitation device permitting discharge is allowed include coastal waters and estuaries, the Great Lakes and inter-connected waterways, fresh-water lakes and impoundments accessible through locks, and other flowing waters that are navigable interstate by vessels subject to this regulation.

(2705) (b) This standard shall become effective on January 30, 1977 for new vessels and on January 30, 1980 for existing vessels (or, in the case of vessels owned and operated by the Department of Defense, two years and five years, for new and existing vessels, respectively, after promulgation of implementing regulations by the Secretary of Defense under section 312(d) of the Act).

(2706) (c) Any vessel which is equipped as of the date of promulgation of this regulation with a Coast Guard-certified flow-through marine sanitation device meeting the requirements of paragraph (a)(2) of this section, shall not be required to comply with the provisions designed to prevent the overboard discharge of sewage, treated or untreated, in paragraph (a)(1) of this section, for the operable life of that device.

(2707) (d) After January 30, 1980, subject to paragraphs (e) and (f) of this section, marine sanitation devices on all vessels on waters that are not subject to a prohibition of the overboard discharge of sewage, treated or untreated, as specified in paragraph (a)(1) of this section, shall be designed and operated to either retain, dispose of, or discharge sewage, and shall be certified by the U.S. Coast Guard. If the device has a discharge, the effluent shall not have a fecal coliform bacterial count of greater than 200 per 100 milliliters, nor suspended solids greater than 150 mg/1.

(2708) (e) Any existing vessel on waters not subject to a prohibition of the overboard discharge of sewage in paragraph (a)(1) of this section, and which is equipped with a certified device on or before January 30, 1978, shall not be required to comply with paragraph (d) of this section, for the operable life of that device.

(2709) (f) Any new vessel on waters not subject to the prohibition of the overboard discharge of sewage in paragraph (a)(1) of this section, and on which construction is initiated before January 31, 1980, which is equipped with a marine sanitation device before January 31, 1980, certified under paragraph (a)(2) of this section, shall not be required to comply with paragraph (d) of this section, for the operable life of that device.

(2710) (g) The degrees of treatment described in paragraphs (a) and (d) of this section are "appropriate standards" for purposes of Coast Guard and Department of Defense certification pursuant to section 312(g)(2) of the Act.

(2711)

§140.4 Complete prohibition.

(2712) (a) Prohibition pursuant to CWA section 312(f)(3): a State may completely prohibit the discharge from all vessels of any sewage, whether treated or not, into some or all of the waters within such State by making a written application to the Administrator, Environmental Protection Agency, and by receiving the Administrator's affirmative determination pursuant to section 312(f)(3) of the Act. Upon receipt of an application under section 312(f)(3) of the Act, the Administrator will determine within 90 days whether adequate facilities for the safe and sanitary removal and treatment of sewage from all vessels using such waters are reasonably available. Applications made by States pursuant to section 312(f)(3) of the Act shall include:

(2713) (1) A certification that the protection and enhancement of the waters described in the petition require greater environmental protection than the applicable Federal standard;

(2714) (2) A map showing the location of commercial and recreational pump-out facilities;

(2715) (3) A description of the location of pump-out facilities within waters designated for no discharge;

(2716) (4) The general schedule of operating hours of the pump-out facilities;

(2717) (5) The draught requirements on vessels that may be excluded because of insufficient water depth adjacent to the facility;

(2718) (6) Information indicating that treatment of wastes from such pump-out facilities is in conformance with Federal law; and

(2719) (7) Information on vessel population and vessel usage of the subject waters.

(2720) (b) Prohibition pursuant to CWA section 312(f)(4)(A): a State may make a written application to the Administrator, Environmental Protection Agency, under section 312(f)(4)(A) of the Act, for the issuance of a regulation completely prohibiting discharge from a vessel of any sewage, whether treated or not, into particular waters of the United States or specified portions thereof, which waters are located within the boundaries of such State. Such application shall specify with particularly the waters, or portions thereof, for which a complete prohibition is desired. The application shall include identification of water recreational areas, drinking water intakes, aquatic sanctuaries, identifiable fish-spawning and nursery areas, and areas of intensive boating activities. If, on the basis of the State's application and any other information available to him, the Administrator is unable to make a finding that the waters listed in the application require a complete prohibition of any discharge in the waters or portions thereof covered by the application, he shall state the reasons why he cannot make such a finding, and shall deny the application. If the Administrator makes a finding that the waters listed in the application require a complete prohibition of any discharge in all or any part of the waters or portions thereof covered by the State's application, he shall publish notice of such findings together with a notice of proposed rule making, and then shall proceed in accordance with 5 U.S.C. 553. If the Administrator's finding is that applicable water quality standards require a complete prohibition covering a more restricted or more expanded area than that applied for by the State, he shall state the reasons why his finding differs in scope from that requested in the State's application.

(2721) (1) For the following waters the discharge from a vessel of any sewage (whether treated or not) is completely prohibited pursuant to CWA section 312(f)(4)(A):

(2722) (i) Boundary Waters Canoe Area, formerly designated as the Superior, Little Indian Sioux, and Caribou Roadless Areas, in the Superior National Forest, Minnesota, as described in 16 U.S.C. 577-577d1.

(2723) (ii) Waters of the State of Florida within the boundaries of the Florida Keys National Marine Sanctuary as delineated on a map of the Sanctuary at http://www.fknms.nos.noaa.gov /.

(2724) (2)(i) For the marine waters of the State of California, the following vessels are completely prohibited from discharging any sewage (whether treated or not):

(2725) (A) A large passenger vessel;

(2726) (B) A large oceangoing vessel equipped with a holding tank which has not fully used the holding tank's capacity, or which contains more than de minimis amounts of sewage generated while the vessel was outside of the marine waters of the State of California.

(2727) (ii) For purposes of paragraph (b)(2) of this section:

(2728) (A) "Marine waters of the State of California" means the territorial sea measured from the baseline as determined in accordance with the Convention on the Territorial Sea and the Contiguous Zone and extending seaward a distance of three miles, and all enclosed bays and estuaries subject to tidal influences from the Oregon border (41.999325 North Latitude, 124.212110 West Longitude, decimal degrees, NAD 1983) to the Mexican border (32.471231 North Latitude, 117.137814 West Longitude, decimal degrees, NAD 1983). A map illustrating these waters can be obtained from EPA or viewed at *http://www.epa.gov/region9/water/no-discharge/overview.html*.

(2729) (B) A "large passenger vessel" means a passenger vessel, as defined in section 2101(22) of title 46, United States Code, of 300 gross tons or more, as measured under the International Convention on Tonnage Measurement of Ships, 1969, measurement system in 46 U.S.C. 14302, or the regulatory measurement system of 46 U.S.C. 14502 for vessels not measured under 46 U.S.C. 14302, that has berths or overnight accommodations for passengers.

(2730) (C) A "large oceangoing vessel" means a private, commercial, government, or military vessel of 300 gross tons or more, as measured under the International Convention on Tonnage Measurement of Ships, 1969, measurement system in 46 U.S.C. 14302, or the regulatory measurement system of 46 U.S.C. 14502 for vessels not measured under 46 U.S.C.14302, that is not a large passenger vessel.

(2731) (D) A "holding tank" means a tank specifically designed, constructed, and fitted for the retention of treated or untreated sewage, that has been designated and approved by the ship's flag Administration on the ship's stability plan; a designated ballast tank is not a holding tank for this purpose.

(2732) (c)(1) *Prohibition pursuant to CWA* section 312(f)(4)(B): A State may make written application to the Administrator of the Environmental Protection Agency under section 312(f)(4)(B) of the Act for the issuance of a regulation establishing a drinking water intake no discharge zone which completely prohibits discharge from a vessel of any sewage, whether treated or untreated, into that zone in particular waters, or portions thereof, within such State. Such application shall:

(2733) (i) Identify and describe exactly and in detail the location of the drinking water supply intake(s) and the community served by the intake(s), including average and maximum expected amounts of inflow;

(2734) (ii) Specify and describe exactly and in detail, the waters, or portions thereof, for which a complete prohibition is desired, and where appropriate, average, maximum and low flows in million gallons per day (MGD) or the metric equivalent;

(2735) (iii) Include a map, either a USGS topographic quadrant map or a NOAA nautical chart, as applicable,

clearly marking by latitude and longitude the waters or portions thereof to be designated a drinking water intake zone; and

(2736) (iv) Include a statement of basis justifying the size of the requested drinking water intake zone, for example, identifying areas of intensive boating activities.

(2737) (2) If the Administrator finds that a complete prohibition is appropriate under this paragraph, he or she shall publish notice of such finding together with a notice of proposed rulemaking, and then shall proceed in accordance with 5 U.S.C. 553. If the Administrator's finding is that a complete prohibition covering a more restricted or more expanded area than that applied for by the State is appropriate, he or she shall also include a statement of the reasons why the finding differs in scope from that requested in the State's application.

(2738) (3) If the Administrator finds that a complete prohibition is inappropriate under this paragraph, he or she shall deny the application and state the reasons for such denial.

(2739) (4) For the following waters the discharge from a vessel of any sewage, whether treated or not, is completely prohibited pursuant to CWA section 312(f)(4)(B):

(2740) (i) Two portions of the Hudson River in New York State, the first is bounded by an east-west line through the most northern confluence of the Mohawk River which will be designated by the Troy-Waterford Bridge (126th Street Bridge) on the south and Lock 2 on the north, and the second of which is bounded on the north by the southern end of Houghtaling Island and on the south by a line between the Village of Roseton on the western shore and Low Point on the eastern shore in the vicinity of Chelsea, as described in Items 2 and 3 of 6 NYCRR Part 858.4.

(2741) (ii) [Reserved]

(2742)

§140.5 Analytical procedures.

(2743) In determining the composition and quality of effluent discharge from marine sanitation devices, the procedures contained in 40 CFR part 136, "Guidelines Establishing Test Procedures for the Analysis of Pollutants," or subsequent revisions or amendments thereto, shall be employed.

(2744)

TITLE 46–SHIPPING

(2745)

Part 15–Manning Requirements (in part)

(2746)

Subpart 1–Vessels in Foreign Trade

(2747)

§15.1001 General.

(2748) Self-propelled vessels engaged in foreign commerce are required to use a pilot holding a valid MMC or license with appropriate endorsement as a first-class pilot when operating in the navigable waters of the United States specified in this subpart.

(2749)

§15.1040 Massachusetts.

(2750) The following U.S. navigable waters located within the State of Massachusetts when the vessel is in transit, but not bound to or departing from a port within the following listed operating areas:

(2751) (a) Cape Cod Bay south of latitude 41°48'54"N;

(2752) (b) The Cape Cod Canal; and

(2753) (c) Buzzards Bay east of a line extending from the southernmost point of Wilbur Point (41°34'55"N., 70°51'15"W.) to the easternmost point of Pasque Island (41°26'55"N., 70°50'30"W.)

(2754)

TITLE 50–WILDLIFEAND FISHERIES

(2755)

Part 222–GeneralEndangered and Threatened Marine Species

(2756)

Subpart A–Introduction and General Provisions

(2757)

§222.101 Purpose and scope of regulations.

(2758) (a) The regulations of parts 222, 223, and 224 of this chapter implement the Endangered Species Act (Act), and govern the taking, possession, transportation, sale, purchase, barter, exportation, importation of, and other requirements pertaining to wildlife and plants under the jurisdiction of the Secretary of Commerce and determined to be threatened or endangered pursuant to section 4(a) of the Act. These regulations are implemented by the National Marine Fisheries Service, National Oceanic and Atmospheric Administration, U.S. Department of Commerce. This part pertains to general provisions and definitions. Specifically, parts 223 and 224 pertain to provisions to threatened species and endangered species, respectively. Part 226 enumerates designated critical habitat for endangered and threatened species. Certain of the endangered and threatened marine species enumerated in §§224.102 and 223.102 are included in Appendix I or II to the Convention on International Trade of Endangered Species of Wild Fauna and Flora. The importation, exportation, and re-exportation of such species are subject to additional regulations set forth at 50 CFR part 23, chapter I.

(2759) (b) For rules and procedures relating to species determined to be threatened or endangered under the jurisdiction of the Secretary of the Interior, see 50 CFR parts 10 through 17. For rules and procedures relating to the general implementation of the Act jointly by the Departments of the Interior and Commerce and for certain species under the joint jurisdiction of both the

Secretaries of the Interior and Commerce, see 50 CFR Chapter IV. Marine mammals listed as endangered or threatened and subject to these regulations may also be subject to additional requirements pursuant to the Marine Mammal Protection Act (for regulations implementing that act, see 50 CFR part 216).

(2760) (c) No statue or regulation of any state shall be construed to relieve a person from the restrictions, conditions, and requirements contained in parts 222, 223, and 224 of this chapter. In addition, nothing in parts 222, 223, and 224 of this chapter, including any permit issued pursuant thereto, shall be construed to relieve a person from any other requirements imposed by a statute or regulation of any state or of the United States, including any applicable health, quarantine, agricultural, or customs laws or regulations, or any other National Marine Fisheries Service enforced statutes or regulations.

(2761)

Part 224–EndangeredMarine and Anadromous Species

(2762)

§224.103 Special prohibitions for endangered marine mammals.

(2763) (c) *Approaching right whales.*

(2764) (1) *Prohibitions*. Except as provided under paragraph (c)(3) of this section, it is unlawful for any person subject to the jurisdiction of the United States to commit, attempt to commit, to solicit another to commit, or cause to be committed any of the following acts:

(2765) (i) Approach (including by interception) within 500 yards (460 m) of a right whale by vessel, aircraft, or any other means;

(2766) (ii) Fail to undertake required right whale avoidance measures specified under paragraph (c)(2) of this section.

(2767) (2) *Right whale avoidance measures*. Except as provided under paragraph (c)(3) of this section, the following avoidance measures must be taken if within 500 yards (460 m) of a right whale:

(2768) (i) If underway, a vessel must steer a course away from the right whale and immediately leave the area at a slow safe speed.

(2769) (ii) An aircraft must take a course away from the right whale and immediately leave the area at a constant airspeed.

(2770) (3) *Exceptions*. The following exceptions apply to this section, but any person who claims the applicability of an exception has the burden of proving that the exception applies:

(2771) (i) Paragraphs (c)(1) and (c)(2) of this section do not apply if a right whale approach is authorized by the National Marine Fisheries Service through a permit issued under part 222, subpart C, of this chapter (General Permit Procedures) or through a similar authorization.

(2772) (ii) Paragraphs (c)(1) and (c)(2) of this section do not apply where compliance would create an imminent and serious threat to a person, vessel, or aircraft.

(2773) (iii) Paragraphs (c)(1) and (c)(2) of this section do not apply when approaching to investigate a right whale entanglement or injury, or to assist in the disentanglement or rescue of a right whale, provided that permission is received from the National Marine Fisheries Service or designee prior to the approach.

(2774) (iv) Paragraphs (c)(1) and (c)(2) of this section do not apply to an aircraft unless the aircraft is conducting whale watch activities.

(2775) (v) Paragraph (c)(2) of this section does not apply to the extent that a vessel is restricted in her ability to maneuver and, because of the restriction, cannot comply with paragraph (c)(2) of this section.

(2776)

§224.105 Speed restrictions to protect North Atlantic Right Whales.

(2777) (a) The following restrictions apply to: All vessels greater than or equal to 65 ft (19.8 m) in overall length and subject to the jurisdiction of the United States, and all other vessels greater than or equal to 65 ft (19.8 m) in overall length entering or departing a port or place subject to the jurisdiction of the United States. These restrictions shall not apply to U.S. vessels owned or operated by, or under contract to, the Federal Government. This exemption extends to foreign sovereign vessels when they are engaging in joint exercises with the U.S. Department of the Navy. In addition, these restrictions do not apply to law enforcement vessels of a State, or political subdivision thereof, when engaged in law enforcement or search and rescue duties.

(2778) (1) *Southeast U.S.* (south of St. Augustine, FL to north of Brunswick, GA): Vessels shall travel at a speed of 10 knots or less over ground during the period of November 15 to April 15 each year in the area bounded by the following: Beginning at 31°27'00.0"N., 80°51'36.0"W.; thence west to charted mean high water line then south along charted mean high water line and inshore limits of COLREGS limit to a latitude of 29°45'00.0"N., thence east to 29°45'00.0"N., 80°51'36.0"W.; thence back to starting point. (Fig. 1).

(2779) (2) *Mid-Atlantic U.S.* (from north of Brunswick, Georgia to Rhode Island): Vessels shall travel 10 knots or less over ground in the period November 1 to April 30 each year:

(2780) (i) In the area bounded by the following: 33°56'42.0"N., 77°31'30.0"W.; thence along a NW bearing of **313.26°** True to charted mean high water line then south along mean high water line and inshore limits of COLREGS limit to a latitude of 31°27'00.0"N.; thence east to

(2781) 31°27'00.0"N., 80°51'36.0"W.; thence to

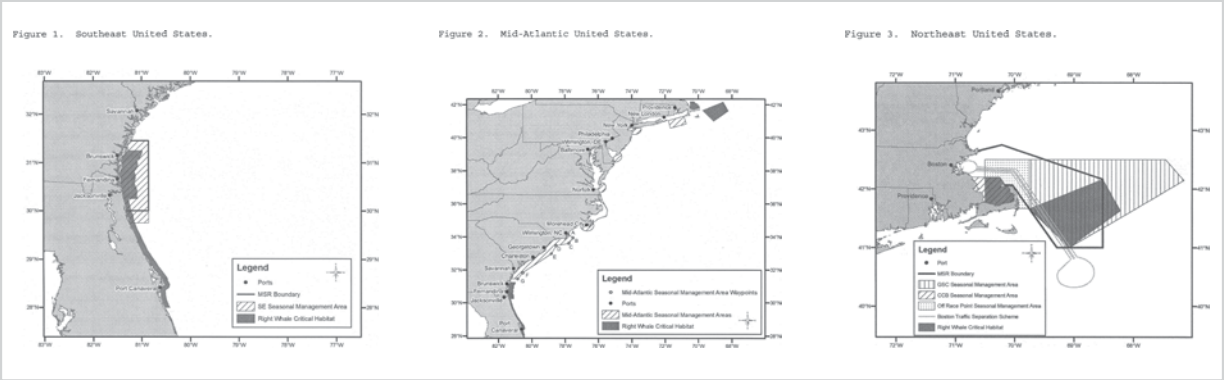
(2782) 31°50'00.0"N., 80°33'12.0"W.; thence to

(2783) 32°59'06.0"N., 78°50'18.0"W.; thence to

(2784) 33°28'24.0"N., 78°32'30.0"W.; thence to

(2785) 33°36'30.0"N., 77°47'06.0"W.; thence back to starting point.;

(2817)



(2786) (ii) Within a 20-nm (37 km) radius (as measured seaward from COLRGES delineated coast lines and the center point of the port entrance) (Fig. 2) at the

(2787) (A) Ports of New York/New Jersey:

(2788) 40°29'42.2"N., 73°55'57.6"W.;

(2789) (B) Delaware Bay (Ports of Philadelphia and Wilmington): 38°52'27.4"N., 75°01'32.1"W.;

(2790) (C) Entrance to the Chesapeake Bay (Ports of Hampton Roads and Baltimore): 37°00'36.9"N., 75°57'50.5"W.; and

(2791) (D) Ports of Morehead City and Beaufort, NC: 34°41'32.0"N., 76°40'08.3"W.; and

(2792) (iii) In Block Island Sound, in the area bounded by the following coordinates: Beginning at

(2793) 40°51'53.7"N., 70°36'44.9"W.; thence to

(2794) 41°20'14.1"N., 70°49'44.1"W.; thence to

(2795) 41°04'16.7"N., 71°51'21.0"W.; thence to

(2796) 40°35'56.5"N., 71°38'25.1"W.; thence back to starting point. (Fig. 2).

(2797) (3) *Northeast U.S. (north of Rhode Island):*

(2798) (i) *In Cape Cod Bay, MA:* Vessels shall travel at a speed of 10 knots or less over ground during the period of January 1 to May 15 in Cape Cod Bay, in an area beginning at 42°04'56.5"N., 70°12'00.0"W.; thence north to 42°12'00.0"N., 70°12'00.0"W.; thence due west to charted mean high water line; thence along charted mean high water within Cape Cod Bay back to beginning point. (Fig. 3).

(2799) (ii) *Off Race Point:* Vessels shall travel at a speed of 10 knots or less over ground during the period of March 1 to April 30 each year in waters bounded by straight lines connecting the following points in the order stated (Fig. 3):

(2800) 42°30'00.0"N., 69°45'00.0"W.; thence to

(2801) 42°30'00.0"N., 70°30'00.0"W.; thence to

(2802) 42°12'00.0"N., 70°30'00.0"W.; thence to

(2803) 42°12'00.0"N., 70°12'00.0"W.; thence to

(2804) 42°04'56.5"N., 70°12'00.0"W.; thence along charted mean high water line and inshore limits of COLREGS limit to a latitude of 41°40'00.0"N., thence due east to 41°41'00.0"N., 69°45'00.0"W.; thence back to starting point.

(2805) (iii) *Great South Channel:* Vessels shall travel at a speed of 10 knots or less over ground during the period of April 1 to July 31 each year in all waters bounded by straight lines connecting the following points in the order stated (Fig. 3):

(2806) 42°30'00.0"N., 69°45'00.0"W.

(2807) 41°40'00.0"N., 69°45'00.0"W.

(2808) 41°00'00.0"N., 69°05'00.0"W.

(2809) 42°09'00.0"N., 67°08'24.0"W.

(2810) 42°30'00.0"N., 67°27'00.0"W.

(2811) 42°30'00.0"N., 69°45'00.0"W.

(2812) (b) Except as noted in paragraph (c) of this section, it is unlawful under this section:

(2813) (1) For any vessel subject to the jurisdiction of the United States to violate any speed restriction established in paragraph (a) of this section; or

(2814) (2) For any vessel entering or departing a port or place under the jurisdiction of the United States to violate any speed restriction established in paragraph (a) of this section.

(2815) (c) A vessel may operate at a speed necessary to maintain safe maneuvering speed instead of the required ten knots only if justified because the vessel is in an area where oceanographic, hydrographic and/or meteorological conditions severely restrict the maneuverability of the vessel and the need to operate at such speed is confirmed by the pilot on board or, when a vessel is not carrying a pilot, the master of the vessel. If a deviation from the ten-knot speed limit is necessary, the reasons for the deviation, the speed at which the vessel is operated, the latitude and longitude of the area, and the time and duration of such deviation shall be entered into the logbook of the vessel. The master of the vessel shall attest to the accuracy of the logbook entry by signing and dating it.

(2816) (d) No later than January 1, 2019, the National Marine Fisheries Service will publish and seek comment on a report evaluating the conservation value and economic and navigational safety impacts of this section, including any recommendations to minimize burden of such impacts.

(2818)

Part 226–DesignatedCritical Habitat

(2819)

§226.101 Purpose and scope.

(2820) The regulations contained in this part identify those habitats designated by the Secretary of Commerce as critical, under section 4 of the Act, for endangered and threatened species under the jurisdiction of the Secretary of Commerce. Those species are enumerated at §223.102 of this chapter if threatened and at §224.101 of this chapter if endangered. For regulations pertaining to the designation of critical habitat, see part 424 of this title; for regulations pertaining to prohibitions against the adverse modification or destruction of critical habitat, see part 402 of this title. Additional information regarding designated critical habitats that is not provided in this section may be obtained upon request to the Office of Protected Resources (*see* §222.102, definition of "Office of Protected Resources").

(2821)

§ 226.203 Critical habitat for North Atlantic right whales (*Eubalaena glacialis*).

(2822) Critical habitat is designated for North Atlantic right whales as described in this section. The textual descriptions in paragraph (b) of this section are the definitive source for determining the critical habitat boundaries. The maps of the critical habitat units provided in paragraph (c) of this section are for illustrative purposes only.

(2823) (a) Physical and biological features essential to the conservation of endangered North Atlantic right whales.

(2824) (1) *Unit 1*. The physical and biological features essential to the conservation of the North Atlantic right whale, which provide foraging area functions in Unit 1 are: The physical oceanographic conditions and structures of the Gulf of Maine and Georges Bank region that combine to distribute and aggregate *C.finmarchicus* for right whale foraging, namely prevailing currents and circulation patterns, bathymetric features (basins, banks, and channels), oceanic fronts, density gradients, and temperature regimes; low flow velocities in Jordan, Wilkinson, and Georges Basins that allow diapausing *C.finmarchicus* to aggregate passively below the convective layer so that the copepods are retained in the basins; late stage *C.finmarchicus* in dense aggregations in the Gulf of Maine and Georges Bank region; and diapausing *C.finmarchicus* in aggregations in the Gulf of Maine and Georges Bank region.

(2825) (2) *Unit 2*. The physical features essential to the conservation of the North Atlantic right whale, which provide calving area functions in Unit 2, are:

(2826) (i) Sea surface conditions associated with Force 4 or less on the Beaufort Scale,

(2827) (ii) Sea surface temperatures of 7°C to 17°C, and

(2828) (iii) Water depths of 6 to 28 meters, where these features simultaneously co-occur over contiguous areas of at least 231 nmi2 of ocean waters during the months of November through April. When these features are available, they are selected by right whale cows and calves in dynamic combinations that are suitable for calving, nursing, and rearing, and which vary, within the ranges specified, depending on factors such as weather and age of the calves.

(2829) (b) *Critical habitat boundaries*. Critical habitat includes two areas (Units) located in the Gulf of Maine and Georges Bank Region (Unit 1) and off the coast of North Carolina, South Carolina, Georgia and Florida (Unit 2).

(2830) (1) *Unit 1*. The specific area on which are found the physical and biological features essential to the conservation of the North Atlantic right whale include all waters, seaward of the boundary delineated by the line connecting the geographic coordinates and landmarks identified herein:

(2831) (i) The southern tip of Nauset Beach (Cape Cod) (41°38.39′N., 69°57.32′W.).

(2832) (ii) From this point, southwesterly to 41°37.19′N., 69°59.11′W.

(2833) (iii) From this point, southward along the eastern shore of South Monomoy Island to 41°32.76′N., 69°59.73′W.

(2834) (iv) From this point, southeasterly to 40°50′N., 69°12′W.

(2835) (v) From this point, east to 40°50′N., 68°50′W.

(2836) (vi) From this point, northeasterly to 42°00′N., 67°55′W.

(2837) (vii) From this point, east to 42°00′N., 67°30′W.

(2838) (viii) From this point, northeast to the intersection of the U.S.-Canada maritime boundary and 42°10′N.

(2839) (ix) From this point, following the U.S.-Canada maritime boundary north to the intersection of 44°49.727′N., 66°57.952′W.; From this point, moving southwest along the coast of Maine, the specific area is located seaward of the line connecting the following points:

(2840)

Latitude	Longitude
44°49.727'N.	66°57.952'W.
44°49.67'N.	66°57.77'W.
44°48.64'N.	66°56.43'W.
44°47.36'N.	66°59.25'W.
44°45.51'N.	67°02.87'W.
44°37.07'N.	67°09.75'W.
44°27.77'N.	67°32.86'W.
44°25.74'N.	67°38.39'W.
44°21.66'N.	67°51.78'W.
44°19.08'N.	68°02.05'W.
44°13.55'N.	68°10.71'W.
44°08.36'N.	68°14.75'W.
43°59.36'N.	68°37.95'W.
43°59.83'N.	68°50.06'W.
43°56.72'N.	69°04.89'W.

Latitude	Longitude
43°50.28'N.	69°18.86'W.
43°48.96'N.	69°31.15'W.
43°43.64'N.	69°37.58'W.
43°41.44'N.	69°45.27'W.
43°36.04'N.	70°03.98'W.
43°31.94'N.	70°08.68'W.
43°27.63'N.	70°17.48'W.
43°20.23'N.	70°23.64'W.
43°04.06'N.	70°36.70'W.
43°02.93'N.	70°41.47'W.

(2841) (x) From this point (43°2.93′ N/70°41.47′ W.) on the coast of New Hampshire south of Portsmouth, the boundary of the specific area follows the coastline southward along the coasts of New Hampshire and Massachusetts along Cape Cod to Provincetown southward along the eastern edge of Cape Cod to the southern tip of Nauset Beach (Cape Cod) (41°38.39′N., 69°57.32′W.) with the exception of the area landward of the lines drawn by connecting the following points:

(2842)

Latitude	Longitude		
42°59.986'N.	70°44.654'W.	to	Rye Harbor
42°59.956'N.	70°44.737'W.		Rye Harbor
42°53.691'N.	70°48.516'W.	to	Hampton Harbor
42°53.519'N.	70°48.748'W.		Hampton Harbor
42°49.136'N.	70°48.242'W.	to	Newburyport Harbor
42°48.964'N.	70°48.282'W.		Newburyport Harbor
42°42.145'N.	70°46.995'W.	to	Plum Island Sound
42°41.523'N.	70°47.356'W.		Plum Island Sound
42°40.266'N.	70°43.838'W.	to	Essex Bay
42°39.778'N.	70°43.142'W.		Essex Bay
42°39.645'N.	70°36.715'W.	to	Rockport Harbor
42°39.613'N.	70°36.60'W.		Rockport Harbor
42°20.665'N.	70°57.205'W.	to	Boston Harbor
42°20.009'N.	70°55.803'W.		Boston Harbor
42°19.548'N.	70°55.436'W.	to	Boston Harbor
42°18.599'N.	70°52.961'W.		Boston Harbor
42°15.203'N.	70°46.324'W.	to	Cohasset Harbor
42°15.214'N.	70°47.352'W.		Cohasset Harbor
42°12.09'N.	70°42.98'W.	to	Scituate Harbor
42°12.211'N.	70°43.002'W.		Scituate Harbor
42°09.724'N.	70°42.378'W.	to	New Inlet
42°10.085'N.	70°42.875'W.		New Inlet
42°04.64'N.	70°38.587'W.	to	Green Harbor
42°04.583'N.	70°38.631'W.		Green Harbor
41°59.686'N.	70°37.948'W.	to	Duxbury Bay/ Plymouth Harbor
41°58.75'N.	70°39.052'W.		Duxbury Bay/ Plymouth Harbor
41°50.395'N.	70°31.943'W.	to	Ellisville Harbor
41°50.369'N.	70°32.145'W.		Ellisville Harbor
41°45.87'N.	70°28.62'W.	to	Sandwich Harbor
41°45.75'N.	70°28.40'W.		Sandwich Harbor
41°44.93'N.	70°25.74'W.	to	Scorton Harbor
41°44.90'N.	70°25.60'W.		Scorton Harbor
41°44.00'N.	70°17.50'W.	to	Barnstable Harbor
41°44.00'N.	70°13.90'W.		Barnstable Harbor
41°45.53'N.	70°09.387'W.	to	Sesuit Harbor
41°45.523'N.	70°09.307'W.		Sesuit Harbor
41°45.546'N.	70°07.39'W.	to	Quivett Creek
41°45.551'N.	70°07.32'W.		Quivett Creek
41°47.269'N.	70°01.411'W.	to	Namskaket Creek
41°47.418'N.	70°01.306'W.		Namskaket Creek
41°47.961'N.	70°0.561'W.	to	Rock Harbor Creek
41°48.07'N.	70°0.514'W.		Rock Harbor Creek
41°48.432'N.	70°0.286'W.	to	Boat Meadow River
41°48.483'N.	70°0.216'W.		Boat Meadow River
41°48.777'N.	70°0.317'W.	to	Herring River
41°48.983'N.	70°0.196'W.		Herring River
41°55.501'N.	70°03.51'W.	to	Herring River, inside Wellfeet Harbor
41°55.322'N.	70°03.191'W.		Herring River, inside Wellfeet Harbor
41°53.922'N.	70°01.333'W.	to	Blackfish Creek/ Loagy Bay
41°54.497'N.	70°01.182'W.		Blackfish Creek/ Loagy Bay
41°55.503'N.	70°02.07'W.	to	Duck Creek
41°55.753'N.	70°02.281'W.		Duck Creek
41°59.481'N.	70°04.779'W.	to	Pamet River
41°59.563'N.	70°04.718'W.		Pamet River
41°03.601'N.	70°14.269'W.	to	Hatches Harbor
41°03.601'N.	70°14.416'W.		Hatches Harbor
41°48.708'N.	69°56.319'W.	to	Nauset Harbor
41°48.554'N.	69°56.238'W.		Nauset Harbor
41°40.685'N.	69°56.781'W.	to	Chatham Harbor
41°40.884'N.	69°56.28'W.		Chatham Harbor

(2843) (xi) In addition, the specific area does not include waters landward of the 72 COLREGS lines (33 CFR part 80) described below.

(2844) (A) *Portland Head, ME to Cape Ann, MA*.

(2845) (1) A line drawn from the northernmost extremity of Farm Point to Annisquam Harbor Light.

(2846) (2) [Reserved]

(2847) (B) *Cape Ann MA to Marblehead Neck, MA*.

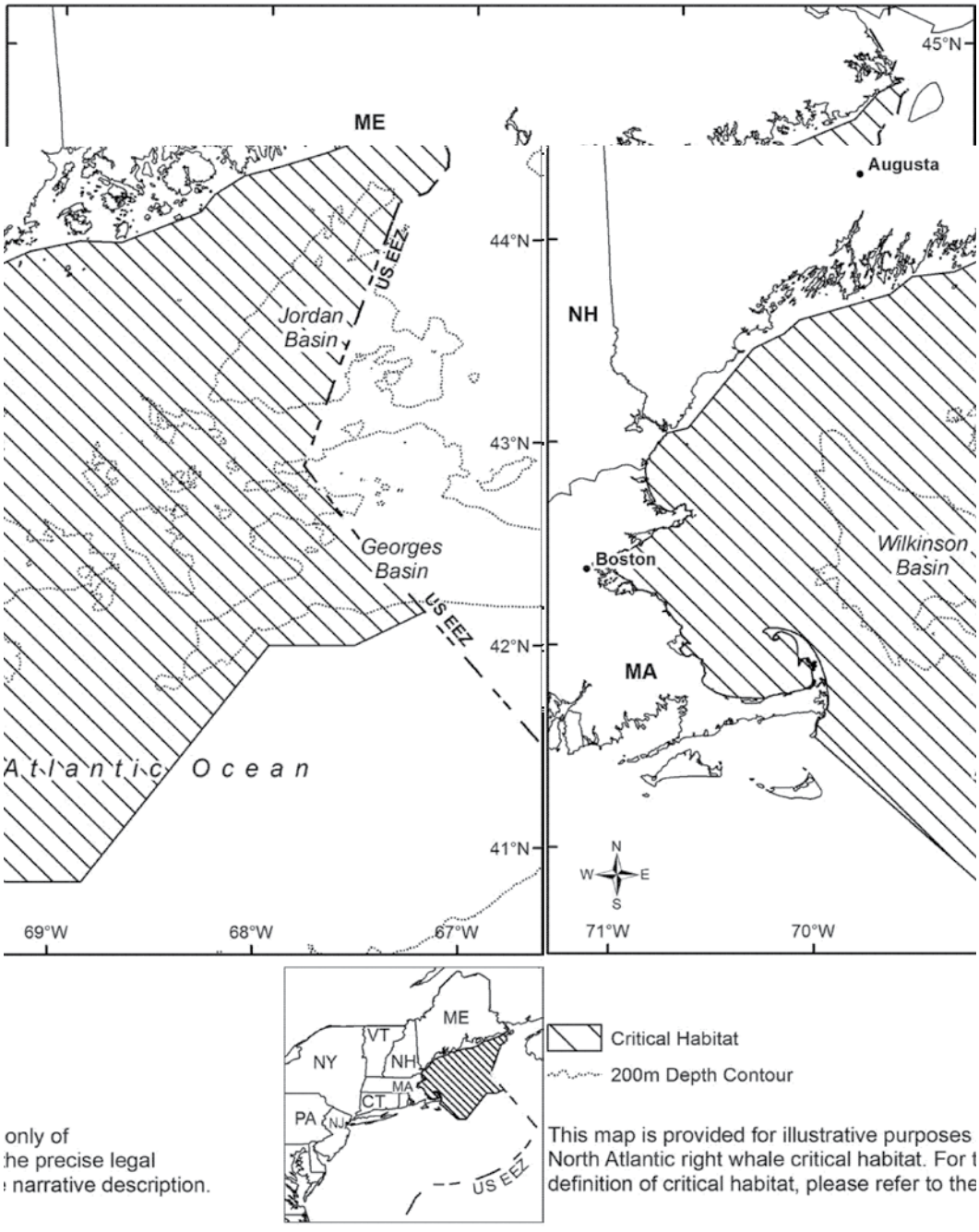
(2848) (1) A line drawn from Gloucester Harbor Breakwater Light to the twin towers charted at latitude 42°35′06.177"N., longitude 70°41′32.330"W.

(2849) (2) A line drawn from the westernmost extremity of Gales Point to the easternmost extremity of House Island; thence to Bakers Island Light; thence to Marblehead Light.

(2850) (C) *Hull, MA to Race Point, MA*.

(2856)

North Atlantic Right Whale Critical Habitat
Northeastern U.S. Foraging Area **Unit 1**



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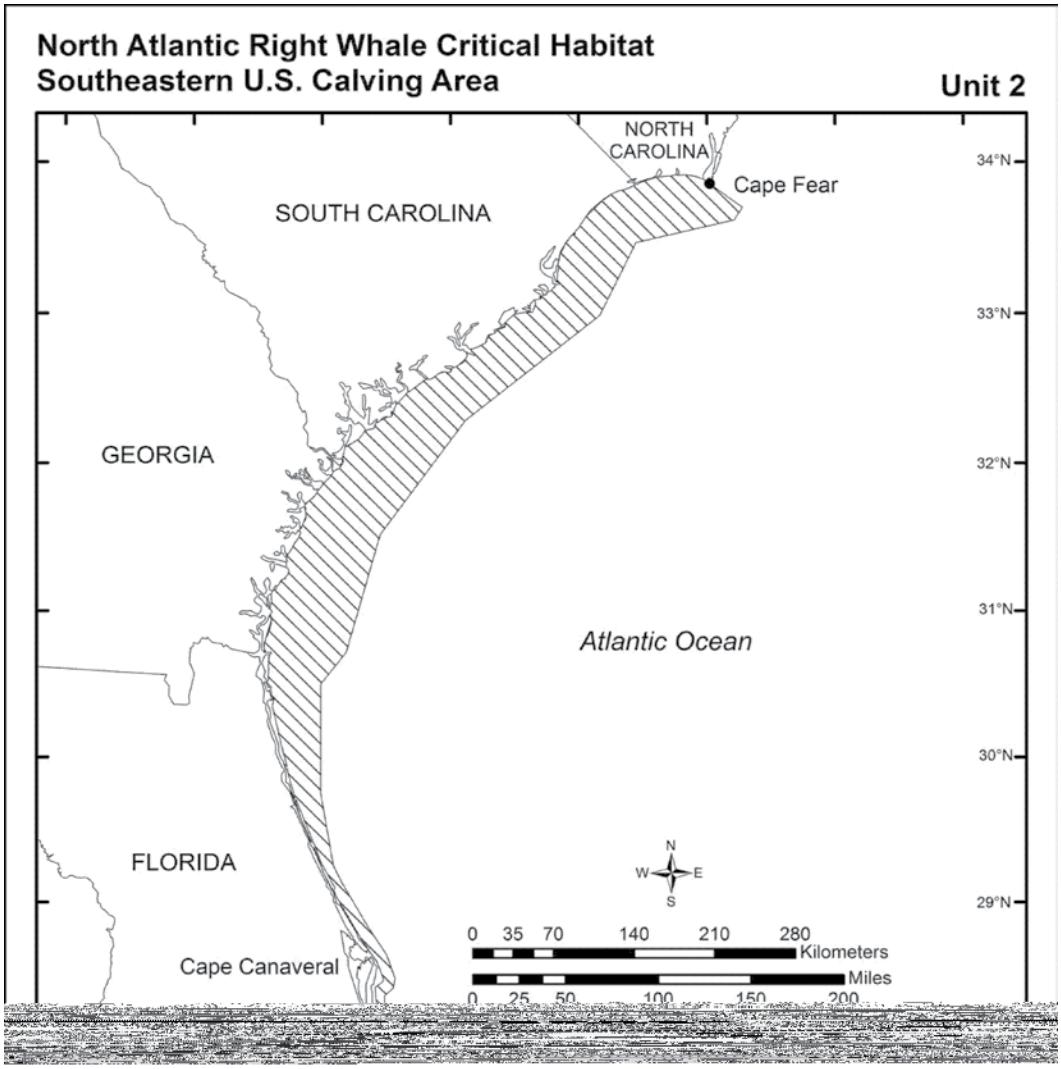
he precise legal
narrative description.

This map is provided for illustrative purposes
North Atlantic right whale critical habitat. For t
definition of critical habitat, please refer to the

(2857)


North Atlantic Right Whale Critical Habitat
Southeastern U.S. Calving Area
Unit 2
NORTH CAROLINA
Cape Fear
SOUTH CAROLINA
GEORGIA
FLORIDA
Atlantic Ocean
Cape Canaveral
34°N
33°N
32°N
31°N
30°N
29°N
N
W
E
S
0 35 70 140 210 280
Kilometers
Miles
0 25 50 100 150 200

(2851) (1) A line drawn from Canal Breakwater Light 4 south to the shoreline.

(2852) (2) [Reserved]

(2853) (2) *Unit 2*. Unit 2 includes marine waters from Cape Fear, North Carolina, southward to 28°N latitude (approximately 31 miles south of Cape Canaveral, Florida) within the area bounded on the west by the shoreline and the 72 COLREGS lines, and on the east by rhumb lines connecting the following points in the order stated from north to south.

(2854)

Latitude	Longitude
33°51'N.	at shoreline
33°42'N.	77°43'W.
33°37'N.	77°47'W.
33°28'N.	78°33'W.
32°59'N.	78°50'W.
32°17'N.	79°53'W.
31°31'N.	80°33'W.
30°43'N.	80°49'W.
30°30'N.	81°01'W.
29°45'N.	81°01'W.
29°15'N.	80°55'W.
29°08'N.	80°51'W.
28°50'N.	80°39'W.
28°38'N.	80°30'W.
28°28'N.	80°26'W.
28°24'N.	80°27'W.
28°21'N.	80°31'W.
28°16'N.	80°31'W.
28°11'N.	80°33'W.
28°00'N.	80°29'W.
28°00'N.	at shoreline

(2855) (c) Overview maps of the designated critical habitat for the North Atlantic right whale follow.

Eastport to Cape Cod

(1) The coasts of Maine, New Hampshire and part of Massachusetts lie between West Quoddy Head in Maine and Provincetown in Massachusetts. Most of the Maine coast is irregular, rocky and bold with numerous islands, bays, rivers and coves. There are numerous fishing villages and towns along the Maine coast that are frequented by tourists during the summer months. The primary deep-draft ports in Maine are at Eastport, Searsport and Portland. The more densely populated coasts of New Hampshire and Massachusetts have numerous sandy beaches and fewer of the islands, bays and coves that characterize the Maine coast. Major ports are at Portsmouth, NH, and Boston, MA.

(2) The **Gulf of Maine** is the great indentation of the coast between the Canadian province of Nova Scotia on the northeast and Massachusetts on the southwest. It includes the Bay of Fundy and Massachusetts Bay as subsidiary features. Because of its changeable weather, frequent fogs and strong tidal currents, this area has a bad reputation among mariners.

(3) From West Quoddy Head to Penobscot Bay the coast is mostly rocky and is indented by numerous large bays and excellent harbors. Among the many islands along this coast are passages that are much used, usually by vessels with less than 12 feet in draft, as they afford anchorage in headwinds or thick weather. The many boulders, rocks and ledges along and off this coast require the closest attention of the navigator, as in many cases they rise abruptly from deep water and soundings do not generally indicate their proximity until it is too late to avoid them. Prudent mariners are reminded that spring tides greatly effect both high and low tides outside of the normal tidal range.

(4) Between Penobscot Bay and Cape Elizabeth the coast is rocky and broken by numerous bays and rivers, many of which are excellent harbors. In Muscongus and Casco Bays good channels lead between the islands, affording inside passages that are used by the smaller class of vessels passing along the coast. Extreme caution should be exercised when approaching the bays, sounds and rivers in this area due to the inset of the flood tidal currents. Particular caution is necessary for small craft crossing Penobscot Bay and the mouths of the Kennebec, Sheepscot and New Meadows Rivers when the wind is contrary to the current because heavy tide rips are encountered. Great caution is also necessary when standing along this stretch of coast in thick weather due to the numerous dangers that in some places lie nearly 10 miles offshore.

(5) Between Cape Elizabeth and Portsmouth there are fewer harbors and marked indentations. The shore is more thickly settled than farther eastward, and several of the beaches are popular summer resorts. The outlying dangers are well marked and fewer in number.

(6) Southward of Portsmouth the coast is low and mostly sandy, with a few outcropping ledges and outlying dangers, but the northern shore of Cape Ann is high and rocky.

(7) Between Cape Ann and Plymouth the coast is rocky, mostly bold and has numerous islands, dry rocks, boulders and covered ledges near the shore, with deep channels between them. The shores of Cape Cod Bay are mostly sandy, with extensive sand shoals extending out well from the shore in many places. Boulders also occur in places in Cape Cod Bay.

(8)

Prominent features

(9) The 14-mile coast between West Quoddy Head and Little River presents no special features. From Little River westward to Portland the shore is broken by numerous bays and islands. Grand Manan Island has nearly perpendicular, dark, rocky faces about 200 feet high on its western side.

(10) The numerous radio towers on the peninsula north of Cross Island on the east side of Machias Bay are prominent. The radar domes on Howard Mountain west of Machias Bay can be seen many miles at sea.

(11) Pigeon Hill, on the western side of Pigeon Hill Bay near the head, is 317 feet high. Schoodic Head, near the south end of Schoodic Peninsula, on the eastern side of the entrance to Frenchman Bay, is 440 feet high. An elevated water tank on Big Moose Island, at the south end of the peninsula, is prominent and reported to be a good radar target.

(12) Cadillac Mountain, the highest on Mount Desert Island, is 1,530 feet high and the most prominent landmark on this part of the coast; near it are other mountains nearly as high. Isle au Haut is 543 feet high near its northern end and is on the eastern side of the entrance to East Penobscot Bay. The Camden Hills (Mount Megunticook, 1,385 feet) are on the western side of Penobscot Bay above the town of Camden. Monhegan Island, 9.3 miles from the mainland, is 165 feet high and is a mark for all vessels bound into Penobscot Bay from westward. Seguin Island, about 2.3 miles off the mouth of the Kennebec River, is about 145 feet high and is a mark for vessels bound into the river or standing along the coast. Observation towers

may be seen along the coast west of the Kennebec River to Boston.

(13) Cape Elizabeth, on the southern side of the entrance to Portland Harbor, is about 90 feet high and is marked by a light and an abandoned lighthouse. Two tall elevated water tanks, one near the mouth of Saco River and one at Cape Porpoise Harbor, are the most prominent landmarks between Portland and Portsmouth. Mount Agamenticus, 691 feet high and the most prominent land feature between Portland and Cape Ann, is about 4.5 miles inland and 9 miles northward of Portsmouth. A ski lodge on the mountain is reported to be prominent. The Isles of Shoals, lying about 6 miles from the coast and southeastward of Portsmouth Harbor entrance, can be seen a long distance, the large hotel on Star Island and an observation tower on Appledore Island being conspicuous marks. Boon Island Light is about 9 miles northeastward of the Isles of Shoals and about 6.5 miles offshore. Cape Ann is high at its northern end, but its eastern end is comparatively low. The two lighthouses on Thatcher Island, one of which is abandoned, are the most conspicuous marks seen when approaching the cape.

(14) The land southward of Cape Ann is comparatively low, is well settled and has numerous artificial marks. A strobe-lighted stack at Salem is the most prominent.

(15) In the approaches to Boston Harbor, the most prominent landmarks are a standpipe on Winthrop Head, the control tower of Logan International Airport, the Customhouse tower, several very high office buildings, a tower on Telegraph (Nantasket) Hill and two lighted radio towers at Nantasket Beach.

(16) In the approaches and on the shores of Cape Cod Bay, the most prominent landmarks are a pointed tower west of Scituate Harbor, the cliffs between Scituate and New Inlet, the Pilgrim Nuclear Power Plant at Rocky Point, a strobe-lighted stack at the entrance to Cape Cod Canal, a standpipe at Barnstable and Pilgrim Monument at Provincetown.

(17) Approaching Cape Cod from the east or south, the most outstanding marks are Highland Light, Nauset Beach Light and Chatham Light.

(18)

Disposal Sites and Dumping Grounds

(19) These areas are rarely mentioned in the Coast Pilot but are shown on the nautical charts. (See Disposal Sites and Dumping Grounds, Chapter 1, and charts for limits.)

(20)

Aids to navigation

(21) Lights are numerous, both on the mainland and offshore islands, along the section of coast covered by this Coast Pilot. Most of the principal light stations are equipped with sound signals. Many coastal and harbor buoys are equipped with radar reflectors, which greatly increase the range at which the buoys may be detected on the radarscope. Most of the critical dangers are marked.

(22)

Radar

(23) Radar is an important navigation aid in this area, since the shoreline of many of the offshore islands and much of the mainland coast is bold and presents good radar targets. Radar is of particular importance due to the extended periods of low visibility that is common in this area.

(24)

COLREGS Demarcation Lines

(25) Lines have been established to delineate those waters upon which mariners must comply with the International Regulations for Preventing Collisions at Sea, 1972 (72 COLREGS) and those waters upon which mariners must comply with the Inland Navigational Rules Act of 1980 (Inland Rules). The waters inside of the lines are **Inland Rules Waters,** and the waters outside of the lines are **COLREGS Waters.** (See **33 CFR Part 80**, chapter 2, for specific lines of demarcation.)

(26)

Ports and Waterways Safety

(27) (See **33 CFR Part 160**, chapter 2, for regulations governing vessel operations and requirements for notification of arrivals, departures, hazardous conditions, and certain dangerous cargoes to the Captain of the Port.)

(28)

Regulated Navigation Areas

(29) Regulated Navigation Areas have been established within the navigable waters of the First Coast Guard District to increase operational safety for towing vessels and tank barges. (See **33 CFR 165.100**, chapter 2, for limit and regulations.)

(30)

Harbor and river entrances

(31) The deepwater ports are approached through deep and stable natural channels. The approaches to the major ports are generally wide, but the channels inside the harbor are generally narrow and strong currents develop, making tugs necessary for large vessels. Those with deepest drafts usually enter these ports at or near high water slack.

(32) Most of the small craft harbors in Maine have entrance channels that are generally deep and stable with numerous submerged, partially submerged and bare rocks. Most of these dangers are marked, and the chart should be followed closely. Along the New Hampshire and Massachusetts coasts, comparatively shallow channels through shifting bars, common at many of the small-craft harbor and river entrances, make caution and current local knowledge advisable for safe entry. Waves break across many of these bars during certain conditions of wind and current; strangers should not attempt to enter under these conditions. On many of the bars the buoys are moved from time to time to mark the shifting channels. The most favorable time to enter most of these harbors is on a rising tide with a smooth sea.

(33)

Traffic Separation Schemes

(34) Traffic Separation Schemes (Traffic Lanes) have been established in the approaches to Frenchman Bay/ Bar Harbor; Penobscot Bay; Portland, ME; and Boston, MA. (See chapters 6, 7, 8 and 11, respectively, for details.)

(35)

Anchorages

(36) Between West Quoddy Head and Portland, anchorages are numerous, those most frequently used by coasting vessels being Little River, Starboard Cove, Englishman Bay, Narraguagus Bay, Prospect Harbor, Winter Harbor, Southwest Harbor, Rockland Harbor, Port Clyde, Boothbay Harbor and Portland Harbor. Southward of Portland the only anchorages available for large vessels are in the harbors of Portsmouth, Gloucester, Salem, Boston, Plymouth and Provincetown. Other harbors available for small vessels and motorboats are described in the text. Anchorage areas established by Federal Regulations are given in **33 CFR 110**, chapter 2.

(37)

Marine Protected Areas

(38) The chapters that follow may contain references to Marine Protected Areas (MPAs) occurring in navigable coastal waters. The critical environmental information is intended to inform readers about the location, purpose and legal restrictions of coastal MPAs, with an emphasis on activities of interest to the maritime community. For detailed information on MPAs, visit www.*marineprotectedareas.noaa.gov*.

(39)

Dangers

(40) The Gulf of Maine is a region of ledges and boulders. The ledges rise abruptly from deep water and the boulders ordinarily lie singly or in clusters on an otherwise flat bottom, so that the navigator cannot depend on soundings to avoid them. The depths are so variable that it is quite impossible to determine a vessel's position by soundings alone, but the navigator will find a frequent use of the sounding apparatus of the greatest assistance in approaching both Georges and Browns Bank from southward and eastward because the bottom slope in that area is well defined.

(41) As a measure of safety, vessels should avoid broken ground where abrupt changes are indicated by the chart to depths less than 10 to 12 fathoms. Dangers have been found in places where least depths of as much as 20 fathoms were the only indications found by the survey. It is always safest, therefore, to select from the chart a sailing line that leads in the deepest water and well clear of broken ground.

(42) The principal offshore dangers are Ammen Rock, a part of Cashes Ledge; Georges and Cultivator Shoals, both a part of Georges Bank; and Nantucket Shoals.

(43)

Pipelaying Barges

(44) With the increased number of pipeline laying operations, operators of all types of vessels should be aware of the dangers of passing close aboard, close ahead or close astern of a jetbarge or pipelaying barge. Pipelaying barges and jetbarges usually move at 0.5 knot or less and have anchors that extend out about 3,500 to 5,000 feet in all directions and that may be marked by lighted anchor buoys. The exposed pipeline behind the pipelaying barge and the area in the vicinity of anchors are hazardous to navigation and should be avoided. The pipeline and anchor cables also represent a submerged hazard to navigation. It is suggested, if safe navigation permits, for all types of vessels to pass well ahead of the pipelaying barge or well astern of the jetbarge. The pipelaying barge, jetbarge and attending vessels may be contacted on VHF-FM channel 16 for passage instructions.

(45)

North Atlantic Rightwhales

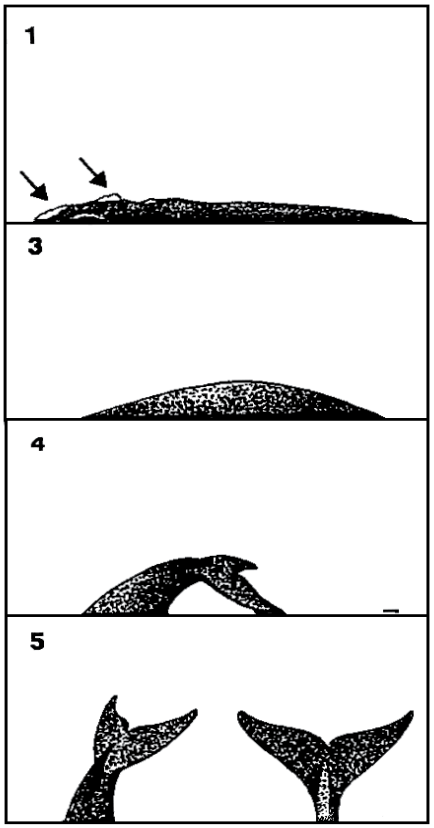
(46) The North Atlantic right whale is one of the world's most endangered large whale species. North Atlantic right whales are found primarily in continental shelf waters between Florida and Nova Scotia. They migrate annually along the east coast between the feeding grounds off New England and Canada and the calving grounds off Florida, Georgia and South Carolina. Because right whales mate, rest, feed and nurse their young at the surface and often do not move out of the way of oncoming ships, they are highly vulnerable to being struck. Pregnant females and females with nursing calves appear to be particularly vulnerable to collisions with ships. Ship strikes and fishing gear entanglements are the two known sources of human-related mortality. Intentionally approaching within 500 yards of right whales is prohibited and is a violation of federal law. (See **50 CFR 224.103**, chapter 2 for limits, regulations and exceptions.)

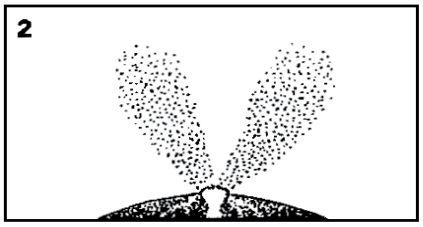
(47) **Description of North Atlantic right whale.** Right whales are large baleen whales. Adults are generally 45 to 55 feet in length and can weigh up to 70 tons. The body is mostly black, but irregularly shaped white patches may be present on the ventral surface. The best field identification marks are a broad back with no dorsal fin, irregular bumpy white patches (callosities) on the head, and a distinctive two-column V-shaped blow when viewed from directly behind or in front of the whale. The whales have broad, paddle-shaped flippers and a broad, deeply notched tail. (See diagrams and photographs.) Right whales are slow moving and seldom travel faster than 5 or 6 knots. They can stay submerged for 10 to 20 minutes and may appear suddenly when surfacing to breathe. They are often seen alone or in small groups. At times, right whales form large courtship groups of 20 to 30 animals.

(49) **Seasonal occurrence of North Atlantic right whales.** During seasons and in areas where right whales

(48)

North Atlantic right whale





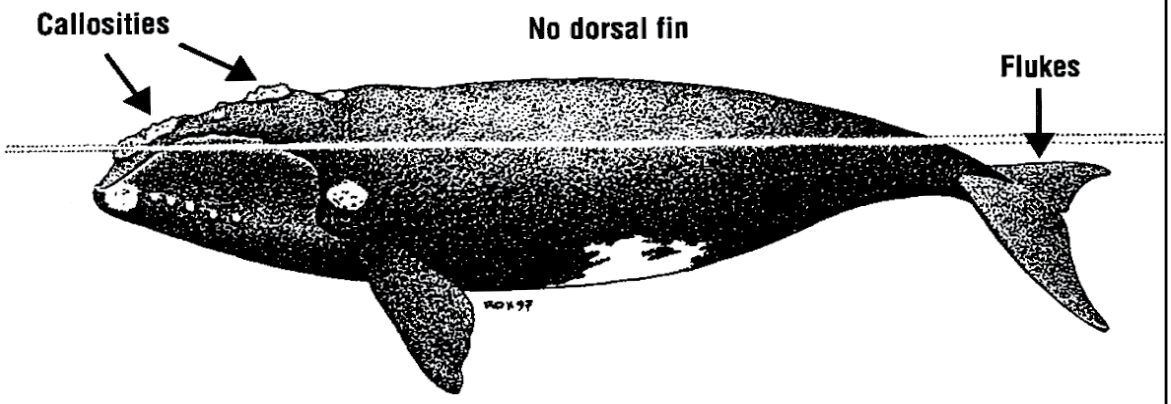
1) Whitish patches of raised and roughened skin (called callosities) on top of the head (see arrows)

2) V-shaped blow easily visible from in front or behind the whale

3) No dorsal fin on the back

4) Tail flukes often lifted vertically when the animal dives

5) All black tail on the top and underside



may occur, vessel operators should maintain a sharp lookout for whales and reduce speeds when consistent with safe navigation. In any given year oceanographic variability may affect the seasonal distribution of right whales. In 1986, right whales were frequently sighted within the Stellwagen Bank National Marine Sanctuary throughout the summer, and in the early spring of 1998 a large number of right whales were documented near the Narragansett/Buzzards Bay Traffic Separation Scheme. Two areas in U.S. waters have been designated as critical habitats for North Atlantic right whales; the northeastern foraging area and southeastern calving area. (See **50 CFR 226.203**, chapter 2 for limits, regulations and exceptions).

(50)

Seasonal occurrence of North Atlantic right whales

Location	Season	Comments
Central Gulf of Maine (Jordan Basin, Cashes Ledge)	April to June October to December	
Cape Cod Bay	December to May	
Great South Channel, northern edge of Georges Bank	March to July	
Bay of Fundy, Scotian Shelf (Browns Bank, Roseway Basin)	July to October	Most of the population can be found in this area during this time.
Jeffreys Ledge	October to December	Whales are frequently sighted in this area.
Stellwagen Bank National Marine Sanctuary	Year-round	Peak sightings occur in the early spring with infrequent sightings in the summer.
New York to North Carolina	November to April	The migration corridor between right whale habitats is within 30 miles of the Atlantic coast.
South Carolina, Georgia and Florida calving area	November to April	Calving right whale sightings have occurred as far north as Cape Fear, NC and as far south as Miami, FL. Sightings in Gulf waters are rare.

(51) **Mandatory Speed Restrictions:** Vessels 65 feet or greater in length overall (LOA) are subject to mandatory speed restrictions of 10 knots or less in seasonal management areas (SMA) along the U.S. East Coast during times when right whales are likely to be present (See map above for locations of SMAs). The Northeastern SMA speed restrictions are in place from January 1 through May 15 in Cape Cod Bay, from March 1 through April 30 off Race Point and from April 1 through July 31 in the Great South Channel. Speed restrictions in the Mid-Atlantic U.S. SMAs are in place from November 1 to April 30 and include Block Island Sound; entry into the Ports of New York/New Jersey; Delaware Bay, Entrance to Chesapeake Bay; and the Ports of Morehead City and Beaufort, NC; and within a continuous boundary approximately 20 nautical miles from shore around the major ports of Wilmington, NC; Charleston, SC; and Savannah, GA. Speed restrictions are in place in the southeastern U.S. SMA from November 15 to April 15; this area extends from shore approximately 30 nautical miles eastward and contains the major ports of Brunswick, GA, Fernandina Beach, FL and Jacksonville, FL. (See **50 CFR 224.105**, chapter 2 for regulations, limitations, and exceptions and complete description of the SMAs.) Boundaries of the SMAs are shown on NOAA Electronic Navigational Charts. NOAA Fisheries may also establish voluntary Dynamic Management Areas (DMAs) when right whales are present in areas and times not covered by the SMAs. Information about established DMAs will be announced over NOAA's customary maritime communication media. Mariners are encouraged to avoid or reduce speeds to 10 knots or less while transiting through DMAs.

(52) **Area to be avoided.** In order to significantly reduce the risk of ship strikes to the North Atlantic right whale, an area to be avoided was established in the Great South Channel, east of the Boston Harbor traffic lanes. Ships of 300 gross tons and above should avoid the area bounded by lines connecting the following geographical positions:

(53) 41°44'08"N., 69°34'50"W.;

(54) 42°10'00"N., 68°31'00"W.;

(55) 41°24'53"N., 68°31'00"W.; and

(56) 40°50'28"N., 68°58'40"W. between the period of April 1 through July 31.

(57) **Early Warning and Sighting Advisory Systems:** As weather and conditions permit, dedicated seasonal programs of aerial and vessel surveys are conducted in the northeast and southeast U.S. to provide whale sighting information to mariners. Surveys typically occur in the following locations at the specified times: a) Cape Cod Bay, the Gulf of Maine, the Great South Channel, and Rhode Island, Block Island and Long Island Sounds from January through July: b) South Carolina/North Carolina border south to Crescent Beach, FL, from December through March. Survey planes occasionally use VHF-FM channel 16 to contact ships directly if whales have been spotted in close proximity to that vessel. However, many right whales go undetected by surveys. Seasonal right whale advisories and sighting reports are broadcast periodically for these and surrounding areas by Coast Guard Broadcast Notice to Mariners, NAVTEX, NOAA Weather Radio, Cape Cod Canal Vessel Traffic Control and the Bay of Fundy Vessel Traffic Control and are included in the return message from the Right Whale Mandatory Ship Reporting (MSR) systems. General sighting information may be obtained by sending an e-mail to ne.rw.sightings@noaa.gov (Northeast) or se.rw.sightings@noaa.gov (Southeast).

(58) **Precautions when transiting right whale habitat and areas of recently reported right whale sightings.**

(59) NOAA recommends the following precautionary measures be taken to avoid adverse interactions with North Atlantic right whales:

(60) Before entering right whale habitat (See "Seasonal Occurrence" table), check Coast Guard Broadcast Notices to Mariners, NAVTEX, NOAA Weather Radio, Mandatory Ship Reporting (MSR) system, Cape Cod Canal Vessel Traffic Control and the Bay of Fundy Vessel Traffic Control as well as other sources for recent right whale sighting reports. Local ship pilots also have information on whale sightings and safe local operating procedures.

(61) Review right whale identification materials and maintain a sharp watch with lookouts familiar with spotting whales. Although right whales are large, their dark color and lack of a dorsal fin can make them difficult to spot.

(62) Avoid transiting through the right whale habitats and areas where right whales have recently been sighted. If transiting between ports within critical habitats, minimize transit distance. Route around observed or recently reported right whales and anticipate delays due to prudent seamanship in response to whale sightings. Avoid transits at night or during periods of low visibility.

(63) If a right whale is sighted from the ship or reported along the intended track of the ship, mariners should exercise caution, post a lookout and reduce speed to 10 knots when consistent with safe navigation. If a right whale is sighted, a vessel must steer a course away from the right whale and immediately leave the area at slow safe speed. Do not assume right whales will move out of the way of an approaching vessel. Mariners should keep in mind that it is illegal to approach a right whale closer than 500 yards. (See **50 CFR 224.103**, chapter 2 for limits, regulations and exceptions.)

(64) Any whale accidentally struck, dead whale carcasses or sighting of an injured or entangled whale should be reported immediately to the Coast Guard or NOAA National Marine Fisheries Service noting the precise location, date and time of the accident or sighting. Call 866–755–6622 for reports to NOAA for the area from Virginia to Maine or 877–942–5343 (877–WHALE–HELP) for the area from North Carolina to Florida. In the event of a strike or sighting of a dead, injured or entangled whale, the following information should be provided:

(65) location, date, and time of the accident or sighting of a carcass or an entangled whale,

(66) speed and course of the vessel,

(67) vessel specifications such as size and propulsion,

(68) water depth,

(69) environmental conditions such as visibility, wind speed and direction,

(70) description of the impact,

(71) fate of the animal, and

(72) species and size, if known.

(73) **Recommended Two-Way Routes to Avoid Whales.** To reduce the possibility of vessel strikes with right whales, Two-Way Routes were developed for vessels entering and transiting through Cape Cod Bay and arriving and departing the ports of Brunswick, GA; Fernandina Beach; FL and Jacksonville, FL. The routes were developed from an analysis of historical right whale sightings and are designed to reduce the likelihood of adverse interactions between large vessels and right whales. The routes are found on the latest NOAA Nautical Charts. In July 2007, the northern leg of the Boston Traffic Separation Scheme (TSS) was shifted to direct ship traffic away from an area of high whale density. Use of the modified TSS is expected to considerably reduce the risk of striking a whale.

(74)

Mandatory Ship Reporting (MSR) Systems WHALESNORTH and WHALESSOUTH

(75) Mandatory Ship Reporting (MSR) systems require all vessels, 300 gross tons or greater, to report to the U.S. Coast Guard upon entering two designated reporting areas off the east coast of the United States. (See **33 CFR 169**, chapter 2, for limits and regulations.) Sovereign immune vessels are exempt from the requirement to report but are encouraged to participate.

(76) The two reporting systems will operate independently of each other. The system in the northeastern United States will operate year round and the system in the southeastern United States will operate each year from November 15 through April 15. Reporting ships are only required to make reports when entering a reporting area during a single voyage (that is, a voyage in which a ship is in the area). Ships are not required to report when leaving a port in the reporting area nor when exiting the system.

(77) Mariners should check all MSR messages carefully before transmittal to ensure the message includes the correct address and format. Additional greeting or comments in the message will preclude message receipt by the MSR system. Failure to receive a timely return message from the MSR system that provides locations of recent right whale sightings and precautionary guidance should be reported to the local U.S. Coast Guard Sector Office.

(78) **Northeastern reporting system/Southeastern reporting system** (See **33 CFR 169.105** and**169.115**, chapter 2, for limits.)

(79) Vessels shall make reports in accordance with the format in IMO Resolution A.858 (20) in accordance with the International Convention for the Safety of Life at Sea 1974 (SOLAS 74). (See **33 CFR 169.135** and **169.140**, chapter 2, for additional information.) Vessels should report via INMARSAT C or via alternate satellite communications to one of the following addresses:

(80) Email: RightWhale.MSR@noaa.gov or Telex: 48156090.

(81) Vessels not equipped with INMARSAT C or Telex should submit reports to the U.S. Coast Guard's Communication Area Master Station Atlantic (CAMSLANT) via HF voice frequencies on 4125 kHz, 6215 kHz, 8291 kHz, 12290 kHz and 16420 kHz or by

calling 800-742-8519x0. Vessels equipped only with VHF-FM voice communications should submit reports to the nearest U.S. Coast Guard Sector Office.

(82)

Example Report
WHALESNORTH
WHALESNORTH//
M/487654321//
A/CALYPSO/NRUS//
B/031401Z APR//
E/345//
F/10.0//
H/031410Z APR/4104N/06918W//
I/BOSTON/032345Z APR//
L/WP/4104N/06918W/15.5//
L/WP/4210N/06952W/15.5//
L/WP/4230N/07006W/15.5//
WHALESSOUTH
WHALESSOUTH//
M/412345678//
A/BEAGLE/NVES//
B/270810Z MAR//
E/250//
F/10.0//
H/270810Z MAR/3030N/08052W//
I/MAYPORT/271215Z MAR//
L/RL/17.0//

(83) **Browns Bank** (42°38'N., 65°52'W.) as defined by the 50-fathom curve is 56 miles long east to west and has an average width of 15 miles. Near the western end of the bank is a sandy ridge with depths of 16 to 28 fathoms. Between the inner 50-fathom curve of Browns Bank and the coastal bank at the southwestern end of Nova Scotia are depths of 47 to 88 fathoms. Browns Bank is a feeding and mating habitat for endangered North Atlantic right whales in late summer and early fall (peak season: July through October).

(84) **Cape Sable** (43°24'N., 65°37'W.), the southern extremity of Nova Scotia, is marked with a light and a sound signal; a racon is at the light. The principal dangers off Cape Sable are Brazil Rock and Blonde Rock; Brazil Rock is marked by a lighted buoy. Seal Island, 17.5 miles west of Cape Sable is marked by a light and a sound signal near the southern end.

(85) **Lurcher Shoal** (43°50'N., 66°30'W.), 13 miles off the west coast of Nova Scotia, has a least depth of 1¼ fathoms. It is the most westerly danger off the coast of Nova Scotia in the approaches to the Bay of Fundy. It is marked by lighted buoys on its southwestern and northeastern ends; the southwestern buoy has a racon. Lurcher Shoal is a feeding habitat for endangered North Atlantic right whales in late summer and early fall (peak season: July through October).

(86) **Grand Manan Banks** (44°12'N., 67°05'W.), 19 miles southward of Grand Manan Island, have an extent of about 16 miles in a northeast-southwest direction and consist of two sections, **Northeast Bank** and **Southwest Bank**, with a channel 2.5 miles wide between them. The bottom is rocky and the least depth, 19 fathoms, is found on Northeast Bank. The tidal currents on the banks attain a velocity of 1.5 knots at strength, at which time there are extensive tide rips with both flood and ebb. The flood current sets to the north-northeast and the ebb to the south-southwest. A good check on the position of a vessel may be obtained by soundings on these banks.

(87) Nova Scotia and the Bay of Fundy are described in **Pub. No. 145, Sailing Directions (Enroute), Nova Scotia and the St. Lawrence,** published by the National Geospatial-Intelligence Agency, Washington, D.C. The Bay of Fundy (Grand Manan Basin) is a feeding and nursery area for endangered North Atlantic right whales in the summer and fall (peak season: July through October).

(88) **Jeffreys Bank** (43°22'N., 68°44'W.), with a least found depth of 35 fathoms, lies about 26 miles southward of Matinicus Rock Light.

(89) **Platts Bank** (43°09'N., 69°37'W.), has a least found depth of 27 fathoms and lies about 40 miles southeast of Portland between the two Portland Approach Traffic Lanes. North Atlantic right whales occur around Platts Bank (peak season: April through June and October through December). Banks with a least depth of 40 fathoms are about 6 miles northwestward and 15 miles eastward (**Three Dory Ridge**) of the shoalest part of Platts Bank.

(90) **Cashes Ledge** (42°54'N., 68°57'W.), with depths of 13 fathoms in places, is about 27 miles long. **Ammen Rock,** covered 4¼ fathoms, is near the middle of the ledge. The sea breaks over this rock in heavy weather. Cashes Ledge is a feeding area for the endangered North Atlantic right whale in the summer and fall (peak season: June through mid-December).

(91) **Fippennies Ledge** (42°47'N., 69°18'W.), with a least known depth of 37 fathoms, lies about 16 miles southwest of Ammen Rock. Fippennies Ledge is a feeding area for the endangered North Atlantic right whale in the summer and fall (peak season: June through mid-December).

(92) **Jeffreys Ledge** extends northeastward from Cape Ann and has general depths of 15 to 30 fathoms. The northeastern point of the ledge is 20 miles eastward of Boon Island Light. Jeffreys Ledge is a feeding area for endangered North Atlantic right whales in late fall (peak season: October through December, although right whales have been sighted in this area year round.

(93) **Stellwagen Bank** lies northward of Cape Cod and off the entrance to Massachusetts Bay; depths found over it are 10 to 20 fathoms. Stellwagen Bank is a feeding area for endangered North Atlantic right whales year round.

sometimes difficult to see and a vessel is liable to be taken into shoaler water than was intended.

(111) Calm, clear days are few; when the sea is calm it is usually foggy, and when clear, it is usually rough. Also, a considerable amount of hazy weather is to be expected, and this limits visibility.

(112) Should it become necessary to anchor in this area, open sea anchorage may be had anywhere that depths permit. Consideration should be given to the proximity of shoals and to the possibility of dragging due to winds and currents. Generally it has been found best to avoid the deeper channels and, when rougher water is experienced, to anchor in the lee of a shoal which would tend to knock down the heavier swells. A scope of five to one or greater should always be used.

(113) Endangered North Atlantic right whales may occur and have been reported off the south side of Nantucket Island.

(114) **Seasonal Management Areas** are located in the waters of Georges Bank.

(115)

Area to be avoided

(116) Because of the great danger of stranding and for reasons of environmental protection, the International Maritime Organization (IMO) has established an area to be avoided in the area of Nantucket Shoals. All vessels carrying cargoes of oil or hazardous materials and all other vessels of more than 1,000 gross tons should avoid the area bounded by the following points:

(117) 41°16.5'N., 70°12.5'W.;

(118) 40°43.2'N., 70°00.5'W.;

(119) 40°44.5'N., 69°19.0'W.;

(120) 41°04.5'N., 69°19.0'W.;

(121) 41°23.5'N., 69°31.5'W.; and

(122) 41°23.4'N., 70°02.8'W.

(123) **Phelps Bank**, the southeasternmost part of the Nantucket Shoals, is about 6.5 miles long and 2.5 miles wide. **Asia Rip**, the shoalest point of the bank, covered 5¾ fathoms, is at the southern end. A wreck, covered 3½ fathoms, is 3 miles south-southeastward of Asia Rip in 40°45'N., 69°19'W. The other shoals and rips of Nantucket Shoals are described in **United States Coast Pilot 2, Atlantic Coast, Cape Cod to Sandy Hook**.

(124) Deep-draft vessels should pass southward and eastward of the wreck off Asia Rip and eastward of the easterly edge of the shoals as defined above. For a distance of 15 miles eastward and southeastward and 17 miles southward from Nantucket Island, the shoals have depths less than 16 feet, and this area should be avoided by all vessels. The tidal currents are strong and variable in direction, forming extensive rips and broken water over the shoals.

(125) A large wreck area is near the southern part of **Fishing Rip**. A wreck and a submerged obstruction are also near the southern part of the rip in about 41°00.0'N., 69°27.0'W. and 41°01.0'N., 69°29.7'W., respectively.

(126)

Caution

(127) Eastbound and westbound vessels should not pass within a 1.5 mile radius from position 40°30'09"N., 69°14'48"W.; if they pass closer, they are inside the traffic separation zone and are in violation of Rule 10(e) of the International Regulations for Preventing Collisions at Sea 1972 (72 COLREGS).

(128) Vessels crossing the traffic separation scheme shall do so in compliance with Rule 10(c) (72 COLREGS) and should not pass closer than 1 mile from position 40°30'09"N., 69°14'48"W.

(129) **Great South Channel** is the passage between the easternmost of the Nantucket Shoals and the westernmost shoal spots of Georges Bank. The channel is about 27 miles wide and has depths of 19 fathoms and greater throughout, with lesser depths along the eastern and western edges. The Great South Channel is a feeding area for endangered North Atlantic right whales in spring (peak season: March through July).

(130) All vessels 65 feet or greater in length overall (LOA) and subject to the jurisdiction of the United States are restricted to speeds of 10 knots or less in the Great South Channel Seasonal Management Area between April 1 and July 31. The area is defined as the waters bounded by:

(131) 42°30'N., 69°45'W.;

(132) 42°30'N., 67°27'W.;

(133) 42°09'N., 67°08.4'W.;

(134) 41°00'N., 69°05'W.;

(135) 41°40'N., 69°45'W.; thence back to starting point. (See **50 CFR 224.105**, chapter 2, for regulations, limitations and exceptions.)

(136) Vessels transiting within the Great South Channel may transit into the WHALESNORTH Mandatory Ship Reporting Area. Each self-propelled ship of 300 gross tons or greater entering WHALESNORTH must participate in the Mandatory Ship Reporting System (See **33 CFR 169**, chapter 2, for limits and regulations, and chapter 3 for sample reports). Sovereign immune vessels are exempt from the requirement to report but are encouraged to participate.

(137)

Submarine Canyons

(138) Submarine canyons are indentations in the edge of the **Continental Shelf**, which is bounded on its seaward side by the 100-fathom curve. They may be traced from depths of 1,000 fathoms or more to the shoaler areas of the shelf. The navigator who has available some means of echo sounding should have in mind the various canyons in this locality. The soundings in crossing them are very characteristic in each case and such soundings may be used to determine the vessel's position with considerable accuracy.

(139) Some of the more important canyons in the region are listed here: **Corsair Canyon**, in approximate longitude 66°10'W., on the eastern side of Georges Bank, has a northwesterly trend. On the southern side

and toward the western end of Georges Bank, and with a northerly trend, are **Lydonia Canyon**, 67°40'W.; **Gilbert Canyon**, 67°50'W.; **Oceanographer Canyon**, 68°05'W.; and **Welker Canyon**, 68°30'W. Southeastward and southward of Nantucket Shoals, and with a northerly trend, are **Hydrographer Canyon**, 69°00'W.; **Veatch Canyon**, 69°35'W.; and **Atlantis Canyon**, 70°15'W.

(140)

Wrecks

(141) An examination of the record of wrecks along the coast of Maine eastward of Portland shows that wrecks have occurred on practically all of the off-lying islands and rocks between Portland and Machias Bay, most of them in thick weather, either fog or snow. Many of the wrecks could have been prevented if frequent soundings had been taken or due allowance had been made for the tidal currents setting into or out of bays or rivers.

(142) During thick weather great caution is necessary when approaching the coast, especially eastward of Petit Manan Island, where the tidal currents have considerable velocity. If one of the offshore lights has not been made and the position accurately determined before the fog shuts in, it is advisable to keep well outside until it clears. Between Machias Bay and Seguin Island a landfall will be made in clear weather before the outlying dangers are encountered.

(143) South of Portland the wrecks have occurred most frequently on the prominent headlands or the shoals off them, namely, Cape Elizabeth, Cape Ann and the north side of Cape Cod, with less frequent wrecks on the less prominent headlands. Numerous wrecks have also occurred on the dangers in the approaches to Boston Harbor, more frequently on the south side from Scituate to Point Allerton. Most of the wrecks have occurred during thick weather.

(144) Between Portland and Boston the most dangerous points for coasting vessels are the dangers off Cape Elizabeth, Boon Island, Isles of Shoals, Cape Ann and the dangers in the entrance to Boston Harbor. The soundings in the vicinity of Cape Ann are very irregular and cannot be depended upon to fix even approximately the vessel's position.

(145) The numerous strandings on the north end of Cape Cod between Highland Light and Race Point Light have usually occurred to vessels approaching Massachusetts Bay or Cape Cod Bay from southward or eastward in thick weather. Keeping in a depth greater than 20 fathoms will ensure giving the eastern side of Cape Cod a berth of 3 miles and, if this depth is followed, will lead to Peaked Hill Bar Lighted Whistle Buoy 2PH, northeastward of the end of the cape.

(146)

Lobster pots

(147) The extent of lobster traps, even at high densities, has expended beyond nearshore or inland waters, particularly those from St. Croix River to the vicinity of Portland. Small painted wooden, closed cell foam, and rigid plastic buoys of various designs and colors, secured by small lines, float on the surface; in some cases a second buoy, usually an unpainted bottle and hard to see, is attached to the lobster pot. These buoys extend from the shore out to, and in many cases across, the sailing routes. Small yachts and motorboats are cautioned against fouling, which is liable to result in a sprung shaft or propeller.

(148) Fishtraps and fish havens are discussed in chapter 1.

(149)

Danger zones

(150) Danger zones have been established within the area of this Coast Pilot. (See **33 CFR Part 334,** chapter 2, for limits and regulations.)

(151)

Drawbridges

(152) The general regulations that apply to all drawbridges are given in **33 CFR 117.1** through **117.49**, chapter 2, and the specific regulations that apply only to certain drawbridges are given in **33 CFR Part 117**, **Subpart B,** chapter 2. Where these regulations apply, references to them are made in the Coast Pilot under the name of the bridge or the waterway over which the bridge crosses.

(153) The drawbridge opening signals (see **33 CFR 117.15,** chapter 2) have been standardized for most drawbridges within the United States. The opening signals for those few bridges that are nonstandard are given in the specific drawbridge regulations. The specific regulations also address matters such as restricted operating hours and required advance notice for openings.

(154) The mariner should be acquainted with the general and specific regulations for drawbridges over waterways to be transited.

(155)

Routes

(156) **Approaching or standing along the coast of Maine eastward of Portland**. This section of the coast is dangerous on account of the strong tidal currents, frequent fog and numerous off-lying dangers. Soundings are of little assistance to locate the position of a vessel, but they should be taken at frequent intervals to prevent too close an approach to dangers.

(157) Endangered North Atlantic right whales may occur year round in the Gulf of Maine (peak season: December through March).

(158)

Coming from the vicinity of Cape Sable

(159) Vessels bound to Machias or ports eastward of it should make Machias Seal Island Light and pass westward of it. If bound to Eastport or Calais, the route through Grand Manan Channel is preferable to passing eastward of Grand Manan Island, because in bad weather an anchorage may be made either at Little River or in Machias Bay.

(160) It is not advisable for a stranger to pass eastward of Machias Seal Island or between it and Grand Manan Island, due to the number of ledges on which the sea

breaks in heavy weather, including Bull Rock, a buoyed danger awash at low water.

(161) If bound to ports in Penobscot Bay, vessels should steer so as to make either Mount Desert Light on Mount Desert Rock or Matinicus Rock Light. On a clear day Cadillac Mountain, the highest part of Mount Desert Island, may be sighted before Mount Desert Light, and Isle au Haut may be seen about the same time as Matinicus Rock.

(162)

Coming from the vicinity of Cape Cod or Cape Ann

(163) Vessels, both steamers and large tows, bound into Penobscot Bay, including those coming from Boston and Cape Cod Canal, and also those passing eastward of Cape Cod, usually make the lighted whistle buoy off Cape Ann and then shape course for Manana Island Lighted Whistle Buoy 14M and enter through Two Bush or Muscle Ridge Channels. In the winter and in bad weather the small class of vessels follow the coast, sighting the principal lights, and making an anchorage on approach of bad weather. Vessels bound from Cape Cod or Cape Ann to points eastward of Penobscot Bay usually shape the course from Cape Ann to either Monhegan Island or Matinicus Rock Light.

(164)

Standing along the coast

(165) In clear weather, vessels stand along the coast close enough to make the lights and to recognize the principal landmarks on shore. In thick weather they aim to make the sound signals or the whistle or bell buoys; these buoys are placed close enough to one another and to the sound signals to be readily followed up by vessels if not set too much off their course by the tidal currents. When running in thick weather a vessel should verify its position as often as possibly by the aids, and when approaching a sound signal or buoy should proceed slowly, taking soundings, and if necessary stop until the looked-for aid is found and recognized before it continues for the next aid. Three good harbors that a stranger, standing along the coast in their vicinity, can make in thick weather and enter with ordinary precautions are Machias Bay, Winter Harbor, and Boothbay Harbor.

(166)

Approaching or standing along the coast between Portland and Cape Cod – Approaching Massachusetts Bay from sea

(167) The approach to the coast of Massachusetts north of Cape Cod is through the Gulf of Maine. Nantucket Shoals and Georges Bank, because of their many shoal spots and the strong tidal currents setting over them, are a menace to navigators approaching the coast or standing from Canadian ports to New York. Browns Bank need not be avoided, for its soundings may assist in determining a vessel's approximate position.

(168) The part of Georges Bank lying between latitude 41°05'N., and 42°00'N., and longitude 67°17'W., and 68°35'W. should be avoided. In heavy weather the sea breaks on the spots with 10 fathoms or less, and strong tide rips are encountered. The tide rips do not always indicate shoal water.

(169) Vessels passing south of the dangerous part of Georges Bank should keep in 25 fathoms or more. Approaching this part of the bank from eastward or southward, the water shoals gradually. Approaching from westward, the depths are irregular and the water shoals abruptly in places of 20 fathoms or less. On the north side of Georges Bank, between longitudes 66°00'W., and 68°00'W., the 100-fathom and 50-fathom curves are only a few miles apart, and when approaching the dangerous part of the bank from northward 50 fathoms may be taken as a good depth to avoid the shoals.

(170) Vessels equipped with echo sounding and following the 100-fathom curve along the south side of Georges Bank can frequently verify their position when crossing the several submarine gorges.

(171) The only known outlying danger in the Gulf of Maine to be avoided by vessels bound to ports in Massachusetts is Ammen Rock, which is a part of Cashes Ledge and is covered 4¼ fathoms.

(172) **Vessels from ports in northern Europe or the British Provinces and bound to ports in the United States north of Cape Cod** approach the coast passing Cape Sable and Georges Bank, between latitudes 42°00'N., and 43°10'N. If bound to Boston, they cross Browns Bank and shape the course for Boston Lighted Whistle Buoy B.

(173) The **North Atlantic Lane Routes** are described in **Pub. No. 140, Sailing Directions, North Atlantic Ocean (Planning Guide)**, published by the National Geospatial-Intelligence Agency, Washington, D.C. They are shown on Pilot Charts of the North Atlantic Ocean.

(174) **Vessels approaching the Gulf of Maine from southwest** sometimes endeavor to make the 50-fathom curve on the southern edge of Georges Bank, in 40°20'N., 68°50'W., then stand 000° on soundings of over 30 and less than 50 fathoms for about 50 miles, and then shape a 323° course, taking care to keep in a greater depth than 20 fathoms until the course is laid to sight Highland Light. This light, Nauset Beach Light, and the Pilgrim Monument at Provincetown are the most prominent marks on Cape Cod.

(175) Deep-draft vessels coming from Cape Hatteras, Chesapeake Bay, Delaware Bay or New York make Nantucket Shoals Lighted Horn Buoy N, thence through Great South Channel to the Gulf of Maine.

(176) Vessels of medium draft coming from southward or alongshore may use the Cape Cod Canal or enter the Gulf of Maine through Vineyard and Nantucket Sounds. The controlling depth for these passages is 32 feet. These routes avoid Nantucket Shoals and are followed by vessels in the coasting trade.

(197)

Mean Surface Water Temperatures (°C) and Densities

	Years		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Mean
Eastport, ME	32	Temp	3.3	1.7	1.7	2.9	4.7	6.8	9.2	10.7	11.0	10.3	8.6	5.9	6.4
		Density	23.8	23.7	23.6	23.3	23.2	23.5	23.8	24.0	24.1	24.2	24.0	23.8	23.8
Bar Harbor, ME	56	Temp	1.4	0.4	1.7	4.6	7.8	11.3	13.7	14.2	13.1	11.1	8.3	5.9	7.6
		Density	23.7	23.7	23.5	23.2	23.2	23.3	23.5	23.6	23.8	24.0	23.9	23.7	23.6
Portland, ME	25	Temp	0.7	0.1	1.5	4.6	8.8	12.7	15.2	15.6	14.4	11.2	7.3	3.1	7.9
		Density	22.3	22.1	21.3	20.5	20.8	21.6	22.3	22.5	22.5	22.5	22.2	22.0	21.9
Portsmouth, NH	39	Temp	2.1	1.3	2.3	5.6	9.2	12.7	15.0	15.8	14.6	11.3	8.2	4.5	8.6
		Density	21.7	21.5	19.7	18.1	19.7	21.4	22.5	22.9	23.0	23.0	22.1	21.2	21.4
Boston, MA	19	Temp	1.8	1.2	3.4	7.4	11.9	16.0	18.3	18.8	17.5	13.5	9.1	4.3	10.3
		Density	21.0	20.9	19.1	18.8	19.8	20.9	21.8	22.0	21.8	22.0	21.5	21.3	20.9

Temperature (Celsius)
F (Fahrenheit) = 1.8C (Celsius) + 32

Density as used in this table is the specifc gravity of the sea water or the ratio between the weight of a sea-water sample and the weight of an equal volume of distilled water at 15°C (59°F).

(177)

Standing along the coast between Portland and Cape Cod

(178) The lights and other aids to navigation are sufficiently numerous to enable a stranger to run either at night or the daytime in clear weather. There are numerous anchorages where a vessel with good ground tackle can ride out any gale. Of these, Provincetown Harbor is the harbor of refuge most frequently used by vessels approaching Massachusetts Bay from seaward. The navigator, when crossing the banks and when approaching the coast, should not neglect to take soundings at frequent intervals.

(179)

Currents

(180) The Tidal Current prediction service at *tidesandcurrents.noaa.gov* should be consulted for specific information about times, directions, and velocities of the current at the numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(181) The current movement is very nearly simultaneous throughout the offshore area of the Gulf of Maine. It is generally rotary in character, the direction of flow changing continuously in a clockwise movement with no period of slack water.

(182) The velocity at strength over Georges Bank varies from about 1 knot to 2 knots. The velocity of the minimum current that occurs midway between the times of strength is usually about one-half the velocity at strength.

(183) Between Georges Bank and Browns Bank the velocity at strength is about 1.5 knots, and there is a like velocity between Browns Bank and Cape Sable Bank.

(184) Off Nova Scotia, outside the 50-fathom curve, the velocity at strength is about 1.5 knots; inside the 50-fathom curve the velocity is between 1.5 and 2.5 knots. The tidal currents offshore from Cape Sable are very uncertain, both in velocity and direction. It is reported that the tidal current on Browns Bank occasionally runs to the northeastward for 15 hours continuously with a velocity of 2 knots, while at other times the set is as strong to the southwestward.

(185) In Grand Manan Channel the average velocity at strength of the current is about 2.5 knots. The currents set approximately parallel to the channel, the flood setting northeastward and the ebb southwestward.

(186) At the entrance to the Bay of Fundy, 5 miles southeastward of Gannet Rock, the flood current has an average velocity at strength of about 2.5 knots and sets 040°. The ebb has an average velocity at strength of about 4 knots and sets 230°.

(187) Along the axis of the Bay of Fundy from Grand Manan Island to Cape Spencer the currents have an average velocity at strength of from 1.5 to 2 knots. The flood sets northeastward and the ebb southwestward.

(188) Eastward of Mount Desert Island the tidal currents along the coast are stronger and more regular than those farther west. Between Mount Desert Island and Portland there is a westward resultant drift along the coast.

(189) With easterly or southeasterly winds the currents have a tendency to set toward the shore.

(190) At Portland Lighted Whistle Buoy P the tidal current is weak, being on the average less than 0.3 knot at time of strength, setting 335° on the flood and 140° on the ebb. Since the tidal current is weak, currents of 1 knot or more occur only with strong winds. The largest velocity likely to occur is about 1.5 knots.

(191) At Boston Lighted Whistle Buoy B the tidal current averages about 0.8 knot. The velocity and direction of the current is therefore greatly influenced by the wind. The largest velocity likely to occur is about 1.4 knot.

(192) Over Stellwagen Bank, and in the channel between it and Cape Cod, the flood current sets westward and the ebb northeastward to eastward. The velocity at strength increases from about 0.2 knot at the northern end of the bank to over 1 knot at the southern end.

(193) Along the coast of Maine eastward of Portland the flood sets eastward and has greater velocity than the ebb, which sets westward. In passing from one headland to another it is always necessary to make allowance for the current setting into or out of the bays or rivers, according to the stage of the tide; such allowance frequently amounts to as much as 5°.

(194)

Weather, Atlantic Coast, Eastport to Cape Cod

(195) This section presents an overall, seasonal picture of the weather that can be expected in the near and offshore waters along the United States east coast from Eastport, Maine, to Cape Cod, Massachusetts, as well as coastal and near-coastal sites. Detailed information, particularly concerning navigational weather hazards, can be found in the weather articles in the following chapters.

(196) All weather articles in this volume are the product of the National Oceanographic Data Center (NODC) and the National Climatic Data Center (NCDC). The meteorological and climatological tables are the product of the NCDC. Both centers are entities of the National Environmental Satellite, Data, and Information Service (NESDIS) of the National Oceanic and Atmospheric Administration (NOAA). If further information is needed in relation to the content of the weather articles, meteorological tables or climatological tables, contact the National Climatic Data Center, Federal Building, 151 Patton Avenue, Asheville, NC 28801-5001. You may also contact the NCDC at 828-271-4800, or fax your request to 828-271-4876.

(198) Climatological tables and meteorological tables for coastal locations relevant to discussions within this volume are located in this chapter (chapter 3) and in following chapters within the appropriate port text description. The climatological tables are a special extraction from the International Station Meteorological Climate Summary (ISMCS). The ISMCS is a CD-ROM jointly produced by the National Climatic Data Center (NCDC), Fleet Numerical Meteorology and Oceanography Detachment-Asheville, and the U.S. Air Force Environmental Technical Applications Center, Operating Location–A. The meteorological tables for the ocean areas are compiled from observations made by ships in passage and extracted from the National Climatic Data Center's Tape Deck-1129, Surface Marine Observations. Listed in Appendix A are National Weather Service offices and radio stations that transmit weather information.

(199) From winter blizzards to summer thunderstorms, a variety of weather plagues the Gulf of Maine. Tricky currents, large tidal ranges and numerous shoals complicate matters. The following text describes the weather problems that face the mariner when navigating these waters. The individual chapters contain information on local weather hazards.

(201)

Extratropical Cyclones

(202) One of the biggest problems in these waters is the winter storm; the most powerful of these is the "Nor'easter." It generates rough seas, strong winds and high tides that threaten safety at sea and cause damage in port. These storms do not often come without warning. They are usually well forecasted, whether approaching from the U.S. mainland or from the seas to the south.

(203) Difficulty arises when they develop or deepen explosively off the mid-Atlantic coast. Sometimes called "Hatteras Storms," these lows can grow from small, weak frontal waves to full blown systems in less than 24 hours. Not only can their circulation expand to cover most of the western North Atlantic, but they often accelerate rapidly northeastward. Within the Gulf of Maine, these storms can generate 30-foot waves and hurricane-force winds. Each year more than 40 extratropical systems move across or close to the Gulf of Maine. They average about 2 to 4 per month, but as many as 10 can affect the region in a single month. Most systems are weak, but a few generate gales and rough seas for hundreds of miles, particularly from September through April.

(204) Signals from a distant "Hatteras Storm" include 5 to 10-foot (1.5 to 3 m) swells, with periods of 10 seconds or more, rolling in from the southeast. The most dependable early indicator is falling pressure. A definite weather change is likely if you observe pressure falls exceeding 2 mb every 3 hours. A drop of 5 mb in 3 hours indicates a strong change, while 10 mb in 3 hours warns of an impending extreme event.

(205) As a storm approaches, winds strengthen, clouds thicken and lower and precipitation begins. Early in the storm's life, wind waves can very quickly become steep. This can make it difficult to reach port, especially when you have to navigate an inlet with treacherous breaking waves. In deeper waters, waves can build to over 20 feet (over 6 m). During winter, the possibility of superstructure icing calls for early action based on the latest forecast and a knowledge of your vessel.

(206)

Cold Fronts

(207) These usually approach from west through north. Ahead of the front, winds are usually squally and often blow out of the south through southwest. Cirrus clouds give way to altocumulus or altostratus and nimbostratus, then cumulonimbus. Pressure falls moderately, seas become choppy and showers, perhaps thunderstorms, occur. With the frontal passage, winds shift rapidly to the west and northwest. Strong gusts and squalls continue. Clearing usually occurs a short distance behind the front as the cold air moves in. Cold fronts can move through the area quite rapidly; their speed varies from about 10 to 20 knots in summer up to 40 knots in winter. From spring through fall, these fronts are often preceded by dense fog.

(208) During the spring and summer when the air ahead of the cold front may be very unstable, a line of

(200)

METEOROLOGICAL TABLE – COASTAL AREA OFF JONESPORT, ME (Between 44°N to 45°N and 66°W to 69°W)

WEATHER ELEMENTS	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	ANNUAL
Wind > 33 knots [1]	8.7	7.9	7.3	3.8	1.4	0.5	0.2	0.3	0.9	2.5	5.8	9.3	3.7
Wave Height > 9 feet [1]	4.6	2.9	3.1	1.5	0.3	0.1	0.0	0.4	0.4	0.9	3.6	3.6	1.4
Visibility < 2 nautical miles [1]	13.6	11.1	11.8	12.9	18.4	26.7	35.7	30.9	14.9	9.0	7.2	11.4	17.9
Precipitation [1]	23.8	21.6	18.6	15.1	12.8	9.6	7.0	7.4	9.9	11.2	15.6	22.1	13.7
Temperature > 69° F	0.0	0.0	0.0	0.0	0.0	0.7	1.6	1.7	0.5	0.1	0.0	0.0	0.5
Mean Temperature (°F)	27.2	27.6	33.5	40.3	46.9	53.1	57.6	59.0	56.0	49.8	42.6	32.8	45.3
Temperature < 33° F [1]	65.4	67.0	37.8	5.4	0.1	0.0	0.0	0.0	0.0	0.2	9.1	45.3	16.6
Mean RH (%)	83	83	83	83	85	88	90	90	87	82	81	83	86
Overcast or Obscured [1]	43.1	37.5	38.6	39.9	40.3	41.3	41.2	38.5	32.0	34.1	41.8	44.0	39.3
Mean Cloud Cover (8ths)	5.8	5.4	5.2	5.3	5.5	5.6	5.5	5.2	4.6	5.1	5.7	5.9	5.4
Mean SLP (mbs)	1013	1014	1013	1013	1014	1013	1014	1015	1017	1017	1015	1015	1014
Ext. Max. SLP (mbs)	1047	1052	1049	1041	1043	1036	1037	1036	1040	1044	1047	1046	1052
Ext. Min. SLP (mbs)	967	966	959	974	974	981	992	987	980	977	966	969	959
Prevailing Wind Direction	NW	NW	NW	SW	SW	SW	SW	SW	SW	SW	NW	NW	SW
Thunder and Lightning [1]	0.2	0.2	0.3	0.3	0.2	0.6	0.6	0.5	0.3	0.3	0.2	0.2	0.4

METEOROLOGICAL TABLE – COASTAL AREA OFF PORTLAND, ME (Between 43°N to 44°N and 66° to 71°W)

WEATHER ELEMENTS	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	ANNUAL
Wind > 33 knots [1]	10.1	10.2	7.0	3.8	1.0	0.4	0.1	0.3	0.9	2.3	6.0	8.9	3.8
Wave Height > 9 feet [1]	8.8	8.2	4.3	2.0	1.1	0.8	0.3	0.3	0.8	2.0	6.1	6.1	3.2
Visibility < 2 nautical miles [1]	13.0	11.8	11.8	13.3	20.9	29.2	38.9	33.0	17.2	10.0	7.6	10.5	19.5
Precipitation [1]	24.8	23.1	16.2	12.5	9.3	8.3	5.4	5.5	6.9	9.0	13.8	21.5	12.3
Temperature > 69° F	0.0	0.0	0.0	0.0	0.1	0.8	3.0	3.9	0.9	0.1	0.0	0.1	0.9
Mean Temperature (°F)	31.4	30.8	35.5	40.8	46.9	53.4	59.0	60.3	57.4	51.7	44.2	36.1	47.1
Temperature < 33° F [1]	51.1	53.5	25.7	3.2	0.1	0.0	0.0	0.0	0.0	0.1	4.6	30.0	12.0
Mean RH (%)	82	81	82	83	87	89	91	90	87	83	80	81	85
Overcast or Obscured [1]	45.0	40.1	38.5	38.0	39.1	39.5	41.7	38.4	31.0	32.1	41.6	44.7	39.2
Mean Cloud Cover (8ths)	5.9	5.5	5.2	5.0	5.2	5.3	5.4	5.0	4.5	4.7	5.6	6.0	5.3
Mean SLP (mbs)	1014	1013	1014	1014	1015	1014	1015	1016	1018	1018	1016	1016	1015
Ext. Max. SLP (mbs)	1050	1048	1046	1041	1041	1036	1037	1041	1046	1045	1043	1046	1050
Ext. Min. SLP (mbs)	970	960	961	976	973	984	986	987	990	977	976	971	960
Prevailing Wind Direction	NW	NW	NW	W	SW	SW	SW	SW	SW	SW	NW	NW	SW
Thunder and Lightning [1]	0.2	0.2	0.0	0.1	0.2	0.3	0.3	0.3	0.2	0.1	0.2	0.1	0.2

METEOROLOGICAL TABLE – COASTAL AREA OFF BOSTON, MA (Between 42°N to 43°N and 66°W to 71°W)

WEATHER ELEMENTS	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	ANNUAL
Wind > 33 knots [1]	6.6	6.5	5.1	2.8	0.8	0.5	0.2	0.5	0.6	2.0	4.8	8.4	2.8
Wave Height > 9 feet [1]	8.9	9.4	7.5	4.9	2.9	1.6	0.2	1.1	2.2	5.1	8.1	10.1	4.7
Visibility < 2 nautical miles [1]	8.1	9.2	9.6	12.0	22.5	26.2	29.6	25.4	14.3	8.8	5.9	6.1	15.2
Precipitation [1]	18.8	19.9	12.7	9.8	7.3	7.2	4.2	4.7	5.5	6.7	11.3	18.0	9.4
Temperature > 69° F	0.1	0.1	0.0	0.0	0.3	2.3	8.9	11.0	2.4	0.4	0.1	0.1	2.6
Mean Temperature (°F)	34.2	33.9	37.6	42.8	48.7	56.9	63.8	64.9	60.8	54.5	46.9	38.7	50.6
Temperature < 33° F [1]	39.4	39.5	17.5	1.4	0.2	0.0	0.0	0.0	0.0	0.0	1.9	21.4	7.9
Mean RH (%)	83	83	83	84	88	88	88	87	85	82	80	80	84
Overcast or Obscured [1]	45.3	43.2	38.2	37.0	39.7	36.2	36.9	34.0	31.2	29.2	39.7	44.8	37.0
Mean Cloud Cover (8ths)	5.8	5.6	5.1	4.8	5.1	5.0	5.1	4.8	4.5	4.4	5.5	5.8	5.0
Mean SLP (mbs)	1015	1014	1014	1014	1015	1014	1015	1016	1018	1018	1017	1016	1016
Ext. Max. SLP (mbs)	1046	1060	1053	1042	1050	1044	1034	1036	1038	1048	1053	1055	1060
Ext. Min. SLP (mbs)	957	966	957	967	977	985	990	988	980	978	970	972	957
Prevailing Wind Direction	NW	NW	NW	SW	SW	SW	SW	SW	SW	SW	NW	NW	SW
Thunder and Lightning [1]	0.1	0.5	0.4	0.4	0.3	0.5	0.8	0.6	0.4	0.2	0.3	0.4	0.4

[1] Percentage Frequency

thunderstorms, known as a squall line, may develop. These instability lines can form 50 to 300 miles ahead of a fast-moving front. They may even contain tornados or waterspouts and can inflict considerable damage on fishing vessels and small craft.

(209)

Tropical Cyclones

(210) This section is condensed from the Hurricane Havens Handbook for the North Atlantic Ocean published by the Naval Environmental Prediction Research Facility at Monterey, CA. While this study concentrates on Boston, the climatology and principles of navigation can be applied to the entire region. The navigation information can be applied to winter storms as well. Data is also incorporated from the Global Tropical/Extratropical Cyclone Climatic Atlas CD-ROM jointly produced by the National Climatic Data Center and the Fleet Numerical Meteorology and Oceanography Detachment-Asheville.

(211) The Gulf of Maine is not in the primary hurricane belt, but some of the most destructive hurricanes have occurred along its shores. For the purposes of this study, any hurricane that approaches within 180 miles is considered a "threat." Of the 105 known hurricanes that threatened Boston from 1842 to 1995, 91 occurred from August through October, with the main threat in September. The hurricane (winds of 64 knots or more) threat has a peak in August and September; 89 hurricanes occurred in those months. Tropical cyclones usually move in from the south or southwest.

(212) Because of the natural protection offered by the shape of the coast from Cape Cod to Cape Hatteras, most recurving storms either make landfall south of Hatteras or pass New England well offshore to the southeast. The majority of storms pass well to the southeast of New England, following the Gulf Stream. Occasionally storms accelerate on a more northerly track, similar to the disastrous hurricane of 1938. That storm advanced rapidly up the east coast, offshore near Hatteras, across central Long Island, into Connecticut and finally through Vermont. This hurricane's speed of advance reached 52 knots, an advance that would be difficult to prepare for even with today's sophisticated warning methods. Exceptionally fast-moving storms pose the greatest threat. For example, based on climatology, a September storm located near 27°N., 74°W. would reach Boston in about 3 or 4 days. However, the 1938 hurricane is believed to have traveled this distance in about 30 hours.

(213) Tropical cyclones tend to accelerate as they move north of about 30°N. Speeds of advance range from 25 to 30 knots for those crossing the New England coast, compared to 20 to 25 knots for those passing offshore to the southeast.

(214) Between 1946 and 1995, 43 hurricanes passed within 180 miles of Boston. Sustained winds in the Boston area ranged from about 20 to 75 knots, with gusts up to 87 knots. Winds near the center of these hurricanes ranged from 74 to 117 knots. The main threats from these storms were high winds and seas, heavy rainfall, and rising sea level. In the open Gulf of Maine, seas of 15 to 20 feet (4.6 to 6 m) are likely, particularly with winds from an easterly direction. Inner Boston Harbor is somewhat protected by Deer Island to the north and east and by Long Island to the southeast. Most of the lower inner harbor is exposed to the east, and vessels in this area, unless berthed at well-protected piers, will experience high wind waves. Storm surge during hurricanes has not been a major problem for the Boston Harbor area. The highest hurricane-generated surge of 3.9 feet (1.2 m) was measured in the September 1938 hurricane. A 4.9-foot (1.5 m) surge occurred in an extra-tropical storm on November 30, 1945. With a normal tide range of 9 to 10 feet (2.7 to 3 m), surge heights of 4 feet (1.2 m) are not of great concern unless they occur at extreme high tide. Then flooding would be considerable. A slow-moving storm with a persistent onshore flow that causes a second high tide while keeping water levels high is the biggest flood threat.

(215) The inner harbor at Boston is considered a hurricane haven if suitable berthing is available. Anchorages in the harbor are unsuitable because of their limited size; surrounding shoals, ledges and rocks; and variable holding characteristics. The numerous shoals and rocks in the outer harbor make navigation during heavy weather particularly dangerous, so early arrival in the harbor is advantageous. The best anchorages are about 40 miles southeast of Boston Harbor in Cape Cod Bay, within the Hook of the Cape. Small craft should either be removed to positions above projected flood levels or bottom moored in protected areas. There are no recommended small craft mooring facilities in the main harbor. During hurricane Donna in 1960, hundreds of small boats were ripped from their moorings and smashed against rocks or seawalls. Small craft berthed along the western bank of the inner harbor, below the confluence of the Mystic and Chelsea Rivers, appear particularly vulnerable to damage from passing hurricanes.

(216) If evasion at sea is necessary, it is advisable to clear the shoal areas and reach the deep water beyond the continental shelf. The shoal areas should be avoided, because they combine the hazards of limited draft and shallow water wave action in a totally exposed ocean area.

(217) Tropical cyclones that pass near the Bahamas, then stay close to the east coast, pose the greatest threat for a dangerous, completely overwater approach to Boston. These storms are likely to accelerate rapidly northward. They should be evaded either within the confines of Massachusetts Bay or by heading southeastward to clear well to the east. Time is important. From the latitude of South Carolina, a storm usually reaches its closest point of approach to Boston in 15 to 27 hours. Heading southeastward requires crossing in front of a northeastward moving storm; extra caution is advised. Since a majority of these storms accelerate northeastward, vessels must take early action. If the storm becomes one of the unusual northward-accelerating hurricanes that are

particularly dangerous to Boston, less time is available to evade, but it is easier to clear well east of the storm. Hurricanes that move ashore along the Atlantic coast usually weaken considerably before reaching the region.

(218) Based on climatology, tropical cyclones north of about 27°N. and east of 70°W. have a low probability of being a destructive threat to Boston. If a major storm in this area does pose a threat, it will most probably move toward the north-northeast. The best evasion route would then lie to the southwest along the coast. If the forecast route is correct, this would keep you in the safe semicircle and allow you to evade the storm that curves to the northeast. Tropical cyclones approaching from the Gulf region and western Caribbean are not usually a threat to shipping in the Boston area. While passage of such storms within 180 miles of Boston occurs fairly often, their long track over land greatly reduces the wind threat and nearly assures their change to a an extratropical system. They may cause local flooding due to heavy precipitation.

(219)

Waves

(220) In March 1984, a 968-mb low off the New Jersey coast generated a 33-foot (10 m) wave at Buoy 44005 (42°53.9'N., 68°56.6'W.). Systems similar to this are partly responsible for the rough seas encountered in the Gulf of Maine from September through April.

(221) Water depth, fetch and wind duration influence wave heights. Most vulnerable are waters exposed to the flow through the Northeast and Great South Channels. Just east of Massachusetts Bay, seas usually remain below 10 feet (3 m), although a 26-foot (8 m) wave has been observed. A near shore buoy about 7 miles southeast of Cape Elizabeth recorded a 23-foot (7 m) wave in February 1983. The 50-year maximum significant wave heights in the Gulf of Maine range from 30 to 35 feet (9 to 11 m), except near the approaches to Boston Harbor and off southwestern Nova Scotia where they drop to 15 to 25 feet (5 to 8 m). Significant wave height (SWH) is defined as the average height of the highest one-third of the waves of a given wave group.

(222) The table below, from Marine Weather of Western Washington by Kenneth E. Lilly, Jr., Commander, NOAA, shows the relationship between significant wave height and the heights of other waves.

(223)

Wave Heights from Significant Wave Heights (SWH)	
Most frequent wave heights	0.5 x SWH
Average wave heights	0.6 x SWH
Significant wave height (average height of highest 33%)	1.0 x SWH
Height of highest 10% of the waves	1.3 x SWH
One wave in 1,175 waves	1.9 x SWH
One wave in 300,000 waves	2.5 x SWH

(224) This table can be used to project a range of wave heights that might be expected in deep water. If significant wave heights of 10 feet (3 m) are forecast, then the most frequently observed waves should be in the 5- to 6-foot (1.5 to 1.8 m) range while one wave in 100 should reach 17 feet (5.2 m). A giant or rogue wave might reach 25 feet (7.6 m) in these circumstances. These rogue or "killer" waves occur when the large number of different waves that make up a sea occasionally reinforce each other. This action creates a wave that is much steeper and higher than the surrounding waves. These rogue waves often occur in a stormy sea and are described by mariners who have experienced them as coming out of nowhere and disappearing just as quickly. If significant wave heights are observed at 20 feet (6.1 m), then a rogue wave could reach 50 feet (15.2 m) if the water depth could support it.

(225) Rough sea conditions are usually generated by gales out of the northwest through northeast. Waves greater than 10 feet (3 m) occur about 10 to 15 percent of the time in winter. From fall through spring, wave heights of more than 7 feet (2.1 m) frequently last one day or more; in midwinter they often last 2 days or more.

(226) In addition to coastal storms, cold fronts with rapidly shifting winds can also create dangerous seas.

(227) Steep waves are often more dangerous than high waves with a gentle slope. Waves appear menacing when the ratio of wave height to length reaches about 1/18. They begin to break when this ratio is about 1/10. Steepest waves develop when strong winds first begin to blow or early in a storm's life. The ship no longer rides easily but is slammed. Steep waves are particularly dangerous to small craft. When wave heights are greater than 5 feet (1.5 m), periods of less than 6 seconds can create problems for boats under 100 feet (under 31 m) long. Waves of 10 feet (3 m) or more with periods of 6 to 10 seconds can affect comfort in 100- to 200-foot (31 to 61 m) vessels. When wind waves reach 20 feet (6.1 m), they become hazardous to vessels under 200 feet (61 m) long and provide a rough ride for larger ships. Waves moving into shallow water become steeper and break when the depth is about 1.3 times the wave height. Areas such as Cultivator Shoal and Georges Shoal are dangerous in heavy weather. Wave steepness is also increased by tidal currents, particularly when they oppose the wind.

(228) Swells can create problems for larger vessels. In these waters, about one-half of the waves of 10 feet (3 m) or more are swells from distant storms. They are uncomfortable to ships that roll or pitch in sympathy. Swells with 500- to 1,000-foot (153 to 305 m) wave lengths affect ships of these lengths. When steaming into such swells a resonance is set up until the bow digs into the waves. The resulting pitch will cause more of a power loss than a roll caused by a sea. Swells with wave lengths that range from about three-fourths to twice the ship's length can have this effect. Pitching is heaviest when the ship's speed produces synchronism between the period of encounter and the ship's natural pitching period; this often occurs at or near normal ship speeds.

(229) When running before a following sea, the greatest danger arises when ship speed is equal to that of the waves or when the waves overtake the ship so slowly that an

almost static situation is created with the vessel lying on the wave crest. In this latter case, stability is so reduced that a small vessel could capsize. Waves on the quarter or stern can also result in very poor steering quality. As seas move along the vessel from aft to forward, the rudder is less effective and the boat may be slewed across the face of a sea, filling the decks with water as it broaches. It could lose its stability and capsize, particularly if the boat is trimmed by the head.

(230)

Winds

(231) Migratory weather systems cause winds that frequently change in strength and direction. Gulf of Maine winds are generally westerly but often take on a northerly component in winter and a southerly one in summer. Strongest winds are generated by lows and cold fronts in fall and winter and by fronts and thunderstorms during spring and summer. Extreme winds are usually associated with a hurricane or severe northeaster and can reach 125 knots. Sustained winds of 100 knots occur about every 50 years on average; gusts are usually about 30 percent higher.

(232) In the open seas, away from the influence of land, winds are stronger and less complex. From December through March, winds are mainly out of the west through north with gales occurring about 6 to 12 percent of the time. In general, windspeeds increase with distance from the coast. If winds persist for a long time over a long fetch they will generate rough seas. In the Gulf of Maine, winter windspeeds of 15 knots or more persist for more than 12 hours about 70 to 80 percent of the time. However these winds often shift and a new fetch is established. Summer winds are usually out of the south through southwest, and gales are infrequent. During the spring and fall, winds are more variable.

(233) Coastal winds are complex since they are influenced by the topography. Over land speeds are reduced. However, channels and headlands can redirect the wind and even increase the speed by funneling the wind. In general, winds have southerly components in summer and northerly ones in winter. In sheltered waters near Rockland, Portland and Brunswick, there are a large percentage of calms, particularly during the morning hours. When the existing circulation is weak and there is a difference between land and water temperatures, a land-sea breeze circulation may be set up. As the land heats faster than the water, a sea breeze is established during the day; this onshore flow may reach 15 knots or more. At night, the land cools more rapidly, often resulting in a weak breeze off the land. In many locations, the sea breeze serves to reinforce the prevailing summer wind.

(234)

Visibilities

(235) Fog, precipitation, smoke, and haze all reduce visibilities. Fog is the most restrictive and persistent. It forms when warm, moist air moves across colder water, when very cold air moves over warmer water, or when moist air is cooled to near its dew point by radiation or rainfall. These conditions can be triggered by a number of weather situations.

(236) Prior to the arrival of a cold front, southerly flow of warm air across cool Gulf waters often results in dense fog. Warm or stationary fronts can also bring fog. Rainfall from lows and fronts can create an evaporation fog. Along the coast, radiation fog is common on clear, calm nights, but it usually burns off during the morning. In the spring, coastal fog may occur near the mouths of rivers and streams that are fed by cold snow melt.

(237) As a result of mixing and circulation patterns, water temperatures decrease northeasterly and easterly across the Gulf of Maine. Temperature differences of 5 °F to 10 °F are common. Water temperatures in summer range from 50 °F to 60 °F. This is when fog is most frequent. Warm air from the south or southwest can create large patches which may persist for days at a time. Just south of Nova Scotia and in the entrance to the Bay of Fundy, visibilities drop to 0.5 mile or below on an average of 18 days in July. These frequencies fall off dramatically to the southwest.

(238) Areas along the coast, at the heads of bays and within rivers may be comparatively clear while fog is very thick outside. The frequency of fog over land and water is usually in opposition. Land fog is often most frequent in fall and winter; the maximum of sea fog is in spring and summer. Consequently, figures for fog or poor visibility at inland or sheltered harbors are no guide to conditions at sea or in the approaches.

(239)

Superstructure icing

(240) Heavy winter weather can cause ice to collect on ships sailing these waters. At worst, superstructure icing can sink a vessel. When air temperature drops below the freezing point of sea water (about 28.6 °F), strong winds and rough seas will cause large amounts of sea spray to freeze to the superstructure and parts of the hull that are not frequently washed by the sea. Ice amounts increase rapidly with falling air and sea temperatures and increasing wind speeds. The most dangerous conditions exist when gales last for several days in temperatures of 28 °F or lower. The ice buildup on a trawler can exceed 5 tons per hour.

(241) A moderate rate of ice accumulation usually occurs when air temperatures are less than 29 °F with winds of 13 knots or more. When air temperatures drop to 16 °F or below and winds reach 30 knots or greater, ice collects more rapidly. On a 300- to 500-ton vessel, it would accumulate at more than 4 tons per hour and is called severe. In the Gulf of Maine, the potential for superstructure icing is present from November through April, particularly north of Portsmouth. December, January and February are the worst months. The potential for moderate icing exists about 10 to 15 percent of the time.

(242) In addition to sea spray, ice is also caused by freezing rain or drizzle and fog in freezing conditions. While these two causes could create enough weight on the rigging to cause it to fall, this is minor in comparison with the freezing spray hazard. Icing on the superstructure elevates the center of gravity, decreasing the metacentric height. It increases the sail area and heeling moment due to wind action. Its non-uniform distribution changes the trim. It can hamper steerability and lower ship speed. Icing also creates hazardous deck conditions.

(243) Experience and research have helped develop some guidelines for use in weather conditions that cause icing. However, there are no hard and fast rules to guide a skipper through these conditions since no two ships or storms are the same. What follows should just be considered general observations. Common sense dictates that when encountering potential icing situations two prudent courses of action would be to seek shelter from the sea and steer toward warmer water, if feasible. In the Gulf of Maine, warmer water is usually found to the south in winter. Once icing has begun, it is prudent to slow down enough so that little or no spray is taken on board. It is also important to use whatever means are available to keep ice from building up. This includes crewmen using tools or baseball bats to remove ice from the deck and superstructure. In general, heaving to with the bow into the wind and sea as much as possible with the intent of minimizing buildup and varying the course slightly to ensure a symmetrical buildup is a good rule.

(244) The main threat to trawlers is loss of stability. As ice forms, the boat's center of gravity is raised, freeboard is decreased by the added weight and the vessel may eventually be in danger of capsizing. Model experiments indicate that the center of gravity tends to become highest with the wind 30 degrees off the bow and is lowest with the stern to the wind. This is due to spray reaching the superstructure. However good navigational practice dictates that it is usually not recommended to exchange a bow on sea for a following sea unless the safety of the ship is in jeopardy. When ice builds up significantly, it is important to remember that the removal of one ton of ice 50 feet (15.2 m) from the vessel's center of gravity is as effective as removing 10 tons of ice 5 feet (1.5 m) above the center of gravity.

(245)

Optical Phenomena

(246) Optical phenomena range from electromagnetic displays to intricate geometrical patterns. The aurora and Saint Elmo's fire are electromagnetic displays. Halos, coronas, parhelia, sun pillars and related effects are optical phenomena associated with the refraction and the diffraction of light through suspended cloud particles; mirages, looming and twilight phenomena such as the "green flash" are associated with refraction of light through air of varying density. Occasionally, sunlight is refracted simultaneously by cloud suspensions and by dense layers of air, producing complex symmetric patterns of light around the sun.

(247) A mirage is caused by refraction of light rays in a layer of air whose density increases or decreases rapidly near the surface. A marked decrease in air density with increasing altitude causes looming, towering and superior mirages. Looming occurs when objects appear to rise above their true elevation. Objects below the horizon may actually be brought into view. This effect often leads to a serious underestimation of horizontal distances. Unimpressive landmarks and distant ships may acquire startling characteristics through apparent vertical stretching; this phenomenon is known as towering. A superior mirage is so named because of the appearance of an image above the actual object. Ships have been seen with an inverted image above and an upright image floating above that. Another type, inferior mirages, result from the upward bending of light rays in an unstable air mass. This phenomenon is observed locally whenever a superheated land mass or a wide expanse of open water is overrun by cold air. Sinking below the horizon of relatively close objects may result in an overestimation of horizontal distances.

(248) Occasionally, a complicated vertical temperature distribution may transform hilly coastlines into impressive walls of lofty pinnacles. This phenomenon is known as Fata Morgana.

(249) On clear days, just as the upper rim of the sun disappears below the horizon, green light is sometimes refracted from the solar spectrum. This brief phenomenon is called the green flash.

(250) Floating ice crystals (cirriform clouds, light snow flakes, ice fog or drifting snow) may cause the refraction of light into a variety of faintly colored arcs and halos. This phenomenon, which may be recognized from the fact that the red band is closest to the light source, includes halos, arcs that open toward or away from the sun, mock images and various geometrical figures that may be located in various parts of the sky with reference to the sun.

(251) Fogbows, resulting from refraction through suspended water particles, are seen in the region of the sky directly opposite from the sun, or the antisolar point. These bows, although occasionally brilliantly colored, are normally seen as broad white bands with faintly colored borders. Rainbows are also observed.

(252) When atmospheric particles are about equal in size to the wavelength of light, diffraction is likely to occur. Diffractional phenomena frequently show properties similar to those of refraction except for reversal in the spectrum colors, violet being closest to the source of light. The Brocken bow, or glory, appears on clouds or fog banks as a colored ring around the projected shadow of the observers head. Solar and lunar coronas, which are observed only through high clouds, resemble halos except that they may assume increasingly larger diameters as the size of the particles decrease. When the light from the sun or the moon is diffracted by cirrus or cirrostratus,

iridescence may sharply delineate the outline of clouds in brilliant green, blue, pink, orange or purple.

(253) Refraction of sunlight takes place whenever the intervening particles are larger than the wavelength. Sunlight that is reflected from ice crystals is transformed into sun pillars and parhelic circles. When both phenomena occur in combination, they form the remarkable sun cross. Paricelenci circles are observed with moonlight.

(254) The **aurora borealis** (northern lights) and St. Elmo's fire are two types of electrical phenomena frequently observed in this region. The zone of maximum auroral frequency extends along the periphery of a 20- to 25-degree circle whose center is at the magnetic pole. Auroras are generally associated with moonless nights. An artificial maximum exists in winter because of the longer hours of darkness. No conclusive evidence is available to show that a seasonal variation in the frequency of auroras exists. However, periods of intense sunspot activity are reflected in a maximum occurrence of this electrical phenomenon. Generally auroras may be classified as having either a ray structure (rays, streams, draperies, corona) or a nebulous appearance (homogenous quiet arc, homogenous band, pulsating arcs, pulsating surfaces, diffuse luminous surfaces and feeble glow). Flaming auroras, which fall in neither category, may be added to this list. Auroras may remain uniformly red, green or purple or assume a rapid succession of these colors. Brilliant shifting auroras are invariably accompanied by magnetic storms and electrical interference with communications.

(255) **St. Elmo's fire** is occasionally observed in this area, but because of its faintness it is most commonly observed during the night hours and on dark overcast days. These eerie flickers of bluish light are usually caused by the unusual electrification of snow-filled air, which is most likely when the wind is strong. St. Elmo's fire is restricted to the tips of such objects as ship masts, wind vanes and airplane wings.

(256)

Dew Point

(257) The temperature at which condensation to water droplets occurs is called the dew point. If the dew point is above freezing, condensation will be in the form of water. When the dew point reaches freezing, ice crystals will be deposited on cold surfaces. Knowledge of the dew point, along with cargo temperature and moisture content, is vital for hold ventilation decisions. It is also a parameter used in forecasting fog formation.

(258)

Cargo Care

(259) When free air has a dew point temperature higher than the temperature of the surface with which it comes in contact, the air is often cooled sufficiently below its dew point to release moisture. When this happens aboard ship, condensation will take place on relatively cold cargo or on the ship's structure within the hold, where it later drips onto the cargo. Thus, if cargo is stowed in a cool climate and the vessel sails into warmer waters, ventilation of the hold with outside air will likely lead to sweat damage in any cargo sensitive to moisture. Under such conditions, external ventilation should, as a rule, be closed off entirely, unless the cargo generates internal heat, that hazard being greater than sweat damage. In the opposite case, when a vessel is loaded during a warm period and moves into cooler weather, vulnerable cargo should be ventilated.

(260) A safe rule for ventilation directed toward moisture control may be stated as follows: Whenever accurate measurements show the outside air has a dew point below the dew point of the air surrounding the cargo to be protected, such outside air is capable of removing moisture from the hold and the ventilation process can be safely started. Whenever the reverse is true, and the outside dew point is higher than the dew point temperature around the cargo, then ventilation will increase the moisture content of the hold and may readily result in sweating within the ship. The above does not take into account possible fumes or gases in the compartment. In such cases discretion must be used.

(261)

Precipitation

(262) Although precipitation amounts at sea are not measured, the ship observations reporting precipitation show a maximum in winter and spring, ranging from a high of near 20 percent in January to less than 10 percent in July and August. Some 5 to 10 percent of all observations report snow in January and February.

(263) In the winter when a cyclone passes to the south or southeast, precipitation over the coastal area generally falls as snow. Along the coastal area precipitation amounts are fairly uniformly distributed throughout the year, ranging from about 2.5 to 4.5 inches (64 to 115 mm) per month.

(264) Thunderstorms are not frequent, occurring on an average of less than 20 per year, mainly during June, July and August. Over the sea their frequency and severity decrease.

(265)

Cloudiness

(266) Low clouds covering 0.6 or more of the sky are reported in nearly one-half the vessel observations in the New England offshore area from November through March, while only 20 to 30 percent of the July-October observations report this condition. Overcast conditions of 0.8 to 1.0 sky cover at the coastal stations range from about 55 to 60 percent in winter to 30 to 40 percent in summer.

(267)

Tropical Cyclones

(268) Tropical cyclones, although much rarer than the extratropical variety, occasionally move northward in late summer and autumn. The storm centers generally move through the region in a northeastward direction toward and across Nova Scotia or over the adjacent ocean, but some have passed northward onto the southern New

England coast. As a rule, these tropical storms are much more violent than the extratropical storms of the same season. Many of them have taken on some characteristics of extratropical cyclones before reaching the area and are less intense than in more southerly latitudes.

(269) A tropical cyclone is a warm-core, low-pressure system that develops over the warm waters of the tropical oceans and exhibits a rotary, counterclockwise circulation in the northern hemisphere (clockwise in the southern hemisphere). Although relatively small in area coverage, this storm can attain awesome strength, with winds near its center reaching 175 knots or more. Tropical cyclones occur almost entirely in six rather distinct regions of the world; one of these, the **North Atlantic Region** (West Indies, Caribbean Sea, Gulf region and waters off the east coast of the United States), includes the area covered by this Coast Pilot. In this region, tropical cyclones with winds of 34–63 knots are called **tropical storms,** while tropical cyclones with winds greater than 63 knots are called hurricanes. **Hurricanes** are infrequent in comparison with middle- and high-latitude storms, but they have a record of destruction far exceeding that of any other type of storm. Because of their fury, and the fact that they are predominately oceanic, they merit the special attention of all mariners, whether professional or amateur.

(270) Rarely do mariners who have experienced a fully developed tropical cyclone (hurricane) at sea wish to encounter a second one. They have learned the wisdom of avoiding them if possible. The uninitiated may be misled by the deceptively small size of a tropical cyclone as it appears on a weather map, and by the fine weather experienced only a few hundred miles from the reported center of such a storm. The rapidity with which the weather can deteriorate with approach of the storm, and the violence of the hurricane, are difficult to visualize if they have not been experienced.

(271) As a tropical cyclone moves out of the tropics to higher latitudes, it normally loses energy slowly, expanding in area until it gradually dissipates or acquires the characteristics of extratropical cyclones. At any stage, a tropical cyclone normally loses energy at a much faster rate if it moves over land. As a general rule, tropical cyclones of the North Atlantic region move with the prevailing winds of the area. In small hurricanes the diameter of the area of destructive winds may not exceed 25 miles while in some of the greatest storms the diameter may be as much as 400 to 500 miles.

(272) At the center is a comparative calm known as the "eye of the storm." The diameter of this "eye" varies with individual storms and may be as little as 7 miles but is rarely more than 30 miles. The average is 15 to 20 miles. This center is the region of low atmospheric pressure around which winds blow in a more or less circular course, spiraling inward in a counterclockwise direction. Winds at the outer edge of the storm area are light to moderate and gusty and often increase toward the center to speeds too high for instrument recording. Although the air movement near the center of the hurricane is usually light and fitful, the seas in this area are in most cases very heavy and confused, rendered so by the violent shifting winds that surround it. Furthermore, after the center has passed a vessel, she may expect a sharp renewal of the gales, with winds from a more or less opposite direction. The hurricane may affect an area covering tens of thousands of square miles.

(273) In the North Atlantic, tropical cyclones form over a wide range of ocean between the Cape Verde Islands and the Windward Island, over the western part of the Caribbean Sea and Gulf waters. While some may initially move northward, especially those that form southeast of Bermuda, the majority take a westerly to northwesterly course. Of these, some curve gradually northward, either east of or above the larger islands of the West Indies, then turn northeastward or eastward for varying distances from the Atlantic coast of the United States. Some pass over or to the south of the larger islands and enter the Gulf region, then curve northward or northeastward and land somewhere on eastern Gulf coast. While others may continue westward and land on the western Gulf coast.

(274) The most common path is curved, the storms moving generally in a westward direction at first, turning later to the northwestward and finally to the northeastward. A considerable number, however, remain in low latitudes and do not turn appreciably to the northward. Freak movements are not uncommon, and there have been storms that described loops, hairpin-curved paths and other irregular patterns. Movement toward the southeast is rare and in any case of short duration. The expanse of the Caribbean and Gulf waters, the coastal regions bordering these bodies of water, and the Atlantic coast are subject to these storms during the hurricane season.

(275) Hurricanes develop over the southern portions of the North Atlantic, including the Caribbean Sea and Gulf waters, **mostly from June through October, infrequently in May and November and rarely in other months;** the hurricane season reaches its peak in September. An average of nine tropical cyclones form each year (reaching at least tropical storm intensity) and five of these reach hurricane strength. June and July storms tend to develop in the northwestern Caribbean or Gulf waters. During August there is an increase in number and intensity, the area of formation extends east of the Lesser Antilles. September storms develop between 50°W. and the Lesser Antilles, in southern Gulf waters, the western Caribbean, near the Bahamas, and around the Cape Verde Islands. Formation in October shifts primarily to the western Caribbean, and off-season storms are widespread with a slight concentration in the southwestern Caribbean.

(276) The average speed of movement of tropical cyclones in the tropics is about 10 to 15 knots. This speed, however, varies considerably according to the location of the storm, its development and attendant meteorological conditions. The highest rates of progression usually occur when

the storm is moving northward or northeastward in the middle or higher latitudes.

(277)

Locating and tracking tropical cyclones

(278) The National Hurricane Center/Tropical Prediction Center located near Miami, Florida collects weather observations hourly, depending on the source, from land stations, ships at sea, aircraft and satellites. When a tropical cyclone is located, usually in its early formative stage (a tropical "wave"), it is followed closely. In the North Atlantic, U.S. Navy, Air Force and NOAA aircraft make frequent flights to the vicinity of such storms to provide information needed for tracking the tropical cyclone and determining its intensity. With the implementation of the NEXt Generation Weather RADar (NEXRAD), coastal radar sites follow the movement of the storm's precipitation area when it is in range. The network provides total coastal coverage from Eastport, Maine, through Brownsville, Texas. Advisories from the Hurricane Center are made available on a 6-hour basis giving information on each storm's location, intensity and movement. These advisories become more frequent if landfall is imminent. As a further aid, the mariner may obtain weather reports by radio directly from other ships in the vicinity of a tropical cyclone.

(279)

Signs of approach

(280) Although radio reports and satellite data, if available, normally prove adequate for locating and avoiding a tropical cyclone, knowledge of the appearance of the sea and sky in the vicinity of such a storm is useful to the mariner. The passage of a hurricane at sea is an experience not soon to be forgotten.

(281) An early indication of the approach of such a storm is the presence of a long swell. In the absence of a tropical cyclone, the crests of swell in the deep waters of the Atlantic pass at the rate of perhaps eight per minute. Swell generated by a tropical cyclone is about twice as long, the crests passing at the rate of perhaps four per minute. Swell may be observed several days before arrival of the storm.

(282) When the storm center is 500 to 1,000 miles away, the barometer usually rises a little, and the skies are relatively clear. Cumulus clouds, if present at all, are few in number, and their vertical development appears suppressed. Nearly perfect tropical blue skies are usually present. The barometer usually appears restless, pumping up and down a few hundredths of an inch. You are in the subsidence sector of the storm, under the influence of the upper-level high pressure system that is acting as the exhaust system for the storm.

(283) As the tropical cyclone comes nearer, a cloud sequence begins which resembles that associated with the approach of a warm front in middle latitudes. Snow-white, fibrous "mare's tails" (cirrus at about 22,000 to 30,000 feet in altitude (6,700 to 9,100 m)) appear when the storm is about 300 to 600 miles away. Usually these seem to converge, more or less, in the direction from which the storm is approaching. This convergence is particularly apparent at about the time of sunrise and sunset.

(284) Shortly after the cirrus appears, but sometimes before, the barometer starts a long, slow fall. At first the fall is so gradual that it only appears to alter somewhat the normal daily cycle (two maximums and two minimums in the tropics). As the rate of fall increases, the daily pattern is completely lost in the more or less steady fall.

(285) The cirrus becomes more confused and tangled and then gradually gives way to a continuous veil of cirrostratus. Below this veil, altostratus forms, and then stratocumulus. These clouds gradually become more dense, and as they do so, the weather becomes unsettled. A fine, mistlike rain begins to fall, interrupted from time to time by showers. The barometer has fallen perhaps a tenth of an inch.

(286) As the fall becomes more rapid, the wind increases in gustiness, and its speed becomes greater, reaching a value of perhaps 22 to 40 knots (Beaufort 6-8). On the horizon appears a dark wall of heavy cumulonimbus, the **bar** of the storm. Portions of this heavy cloud become detached from time to time and drift across the sky, accompanied by rain squalls and wind of increasing speed. Between squalls, the cirrostratus can be seen through breaks in the stratocumulus.

(287) As the bar approaches, the barometer falls more rapidly and wind speed increases. The seas, which have been gradually mounting, become tempestuous, and squall lines, one after another, sweep past in ever increasing number and intensity.

(288) With the arrival of the bar, the day becomes very dark, squalls become virtually continuous and the barometer falls precipitously, with a rapid increase in the wind speed. The center may still be 100 to 200 miles away in a hurricane. As the center of the storm comes closer, the ever-stronger wind shrieks through the rigging and about the superstructure of the vessel. As the center approaches, rain falls in torrents. The wind fury increases. The seas become mountainous. The tops of huge waves are blown off to mingle with the rain and fill the air with water. Objects at a short distance are not visible. Even the largest and most seaworthy vessels become virtually unmanageable and may sustain heavy damage. Less sturdy vessels do not survive. Navigation virtually stops as safety of the vessel becomes the prime consideration. The awesome fury of this condition can only be experienced. Words are inadequate to describe it.

(289) If the eye of the storm passes over the vessel, the winds suddenly drop to a breeze as the wall of the eye passes. The rain stops, and skies clear sufficiently to permit the sun to shine through holes in the comparatively thin cloud cover. Visibility improves. Mountainous seas approach from all sides, apparently in complete confusion. The barometer reaches its lowest point, which may be 1½ or 2 inches below normal in hurricanes. As the wall on the opposite side of the eye arrives, the full

fury of the wind strikes as suddenly as it ceased, but from the opposite direction. The sequence of conditions that occurred during approach of the storm is reversed and passes more quickly, as the various parts of the storm are not as wide in the rear of a storm as on its forward side.

(290)

Locating the center of a tropical cyclone

(291) If intelligent action is to be taken to avoid the full fury of a tropical cyclone, early determination of its location and direction of travel relative to the vessel is essential. The bulletins and forecasts are an excellent general guide, but they are not infallible and may be sufficiently in error to induce a mariner in a critical position to alter course so as to unwittingly increase the danger of the vessel. Often it is possible, using only those observations made aboard ship, to obtain a sufficiently close approximation to enable the vessel to maneuver to the best advantage.

(292) As previously stated, the presence of an exceptionally long swell is usually the first visible indication of the existence of a tropical cyclone. In deep water it approaches from the general direction of origin (the position of the storm center when the swell was generated). However, in shoaling water this is a less reliable indication because the direction is changed by refraction, the crests being more nearly parallel to the bottom contours.

(293) When the cirrus clouds appear, their point of convergence provides an indication of the direction of the storm center. If the storm is to pass well to one side of the observer, the point of convergence shifts slowly in the direction of storm movement. If the storm center will pass near the observer, this point remains steady. When the bar becomes visible, it appears to rest upon the horizon for several hours. The darkest part of this cloud is in the direction of the storm center. If the storm is to pass to one side, the bar appears to drift slowly along the horizon. If the storm is heading directly toward the observer, the position of the bar remains fixed. Once within the area of the dense, low clouds, one should observe their direction of movement, which is almost exactly along the isobars, with the center of the storm being 90° from the direction of cloud movement (left of direction of movement in the Northern Hemisphere.)

(294) The winds are probably the best guide to the direction of the center of a tropical cyclone. The circulation is cyclonic, but because of the steep pressure gradient near the center, the winds there blow with greater violence and are more nearly circular than in extratropical cyclones.

(295) According to Buys Ballot's law, an observer who faces into the wind has the center of the low pressure on his right (northern hemisphere) and somewhat behind him. If the wind followed circular isobars exactly, the center would be exactly eight points, or 90°, from dead ahead when facing into the wind. However, the track of the wind is usually inclined somewhat toward the center, so that the angle dead ahead varies between perhaps 8 and 12 points (90° to 135°). The inclination varies in different parts of the same storm. It is least in front of the storm and greatest in the rear, since the actual wind is the vector sum of that due to the pressure gradient and the motion of the storm along the track. A good average is perhaps 10 points in front, and 11 or 12 points in the rear. These values apply when the storm center is still several hundred miles away. Closer to the center, the wind blows more nearly along the isobars, the inclination being reduced by one or two points at the wall of the eye. Since wind direction usually shifts temporarily during a squall, its direction at this time should not be used for determining the position of the center.

(296) When the center is within radar range, it might be located by this equipment. However, since the radar return is predominately from the rain, results can be deceptive, and other indications should not be neglected.

(297) Distance from the storm center is more difficult to determine than direction. Radar is perhaps the best guide. The rate of fall of the barometer is of some help; this is only a rough indication, however, for the rate of fall may be quite erratic and will vary somewhat with the depth of the low at the center, the speed of the storm center along its track and the stage in the life cycle of the storm.

(298)

Maneuvering to avoid the storm center

(299) The safest procedure with respect to tropical cyclones is to avoid them. If action is taken sufficiently early, this is simply a matter of setting a course that will take the vessel well to one side of the probable track of the storm and then continuing to plot the position of the storm center, as given in the weather bulletins, revising the course as needed. Detailed information on the vulnerability of North Atlantic ports to hurricanes may be found in the **Hurricane Havens Handbook for the North Atlantic Ocean** published by the Marine Meteorology Division, Naval Research Laboratory, Monterey, CA 93943 and at *nrlmry.navy.mil/pubs.htm*.

(300) However, such action is not always possible. Within the storm area, the proper action to take depends in part upon the vessel's position relative to the storm center and its direction of travel. It is customary to divide the circular area of the storm into two parts. In the northern hemisphere, that part to the **right** of the storm track (facing in the direction toward which the storm is moving) is called the **dangerous semicircle**. It is considered dangerous because (1) the actual wind **speed** is greater than that due to the pressure gradient alone, since it is augmented by the forward motion of the storm, and (2) the **direction** of the wind and sea is such as to carry a vessel into the path of the storm (in the forward part of the semicircle). The part to the **left** of the storm track is called the **navigable semicircle.** In this part, the wind is decreased by the forward motion of the storm, and the wind blows vessels away from the storm track (in the forward part). Because of the greater wind speed in the dangerous semicircle, the seas are higher here than in the navigable semicircle.

(301) A plot of successive positions of the storm center should indicate the semicircle in which a vessel is located. However, if this is based upon weather bulletins, it is not a reliable guide because of the lag between the observations upon which the bulletin is based and the time of reception of the bulletin, with the ever present possibility of a change in the direction of motion of the storm. The use of radar eliminates this lag, but the return is not always a true indication of the center. Perhaps the most reliable guide is the wind. Within the cyclonic circulation, a **veering wind** (one changing direction to the right in the northern hemisphere and to the left in the southern hemisphere) indicates a position in the dangerous semicircle, and a **backing wind** (one changing in a direction opposite to a veering wind) indicates a position in the navigable semicircle. However, if a vessel is underway, its motion should be considered. If it is outrunning the storm or pulling rapidly toward one side (which is not difficult during the early stages of a storm, when its speed is low), the opposite effect occurs. This should usually be accompanied by a rise in atmospheric pressure, but if motion of the vessel is nearly along an isobar, this may not be a reliable indication. If in doubt, the safest action is usually to stop long enough to determine definitely the semicircle. The loss in valuable time may be more than offset by the minimizing of the possibility of taking the wrong action and increasing the danger to the vessel. If the wind direction remains steady (for a vessel which has stopped), with increasing speed and falling barometer, the vessel is in or near the path of the storm. If it remains steady with decreasing speed and rising barometer, the vessel is on the storm track, behind the center.

(302) The first action to take if the vessel is within the cyclonic circulation is to determine the position of the vessel with respect to the storm center. **While the vessel can still make considerable way through the water, a course should be selected to take it as far as possible from the center**. If the vessel can move faster than the storm, it is a relatively simple matter to outrun the storm if sea room permits. But when the storm is faster, the solution is not as simple. In this case, the vessel, if ahead of the storm, will approach nearer to the center. The problem is to select a course that will produce the greatest possible minimum distance. This is best determined by means of a relative movement plot.

(303) As a very general rule, for a vessel in the Northern Hemisphere, safety lies in placing the wind on the starboard bow in the dangerous semicircle and on the starboard quarter in the navigable semicircle. If on the storm track ahead of the storm, the wind should be put about two points on the starboard quarter until the vessel is well within the navigable semicircle, and the rule for that semicircle then followed. With a faster than average vessel, the wind can be brought a little farther aft in each case. However, as the speed of the storm increases along its track, the wind should be brought farther forward. If land interferes with what would otherwise be the best maneuver, the solution should be altered to fit the circumstances. If the speed of the vessel is greater than that of the storm, it is possible for the vessel, if behind the storm, to overtake it. In this case, the only action usually needed is to slow enough to let the storm pull ahead.

(304) In all cases, one should be alert to changes in the direction of movement of the storm center, particularly in the area where the track normally curves toward the pole. If the storm maintains its direction and speed, the ship's course should be maintained as the wind shifts.

(305) If it becomes necessary for a vessel to heave to, the characteristics of the vessel should be considered. A power vessel is concerned primarily with damage by direct action of the sea. A good general rule is to heave to with head to the sea in the dangerous semicircle or stern to the sea in the navigable semicircle. This will result in the greatest amount of headway away from the storm center, and the least amount of leeway toward it. If a vessel handles better with the sea astern or on the quarter, it may be placed in this position in the navigable semicircle or in the rear half of the dangerous semicircle, but never in the forward half of the dangerous semicircle. It has been reported that when the wind reaches hurricane speed and the seas become confused, some ships ride out the storm best if the engines are stopped and the vessel is permitted to seek its own position. In this way, it is said, the ship rides with the storm instead of fighting against it.

(306) In a sailing vessel, while attempting to avoid a storm center, one should steer courses as near as possible to those prescribed above for power vessels. However, if it becomes necessary for such a vessel to heave to, the wind is of greater concern than the sea. A good general rule always is to heave to on whichever tack permits the shifting wind to draw aft. In the northern hemisphere this is the starboard tack in the dangerous semicircle and the port tack in the navigable semicircle.

(307)

Practical rules

(308) When there are indications of a hurricane, vessels should remain in port or seek one if possible. Changes in barometer and wind should be carefully observed and recorded, and every precaution should be taken to avert damage by striking light spars, strengthening moorings and, if a steamer, preparing steam to assist the moorings. In the ports of the southern states hurricanes are generally accompanied by very high tides, and vessels may be endangered by overriding the wharf where moored if the position is at all exposed.

(309) Vessels in the Straits of Florida may not have sea room to maneuver so as to avoid the storm track and should try to make a harbor, or to stand out of the straits to obtain sea room. Vessels unable to reach a port and having sea room to maneuver usually observe the previously discussed general rules for avoiding the storm center, which, for power-driven vessels, are summarized as follows:

(310)

Right or dangerous semicircle

(311) Bring the wind on the starboard bow (045° relative), hold course and make as much way as possible. If obliged to heave to, do so with head to the sea.

(312)

Left or navigable semicircle

(313) Bring the wind on the starboard quarter (135° relative), hold course and make as much way as possible. If obliged to heave to, do so with stern to the sea.

(314)

On storm track, ahead of center

(315) Bring the wind two points on the starboard quarter (157½° relative), hold course and make as much way as possible. When well within the navigable semicircle, maneuver as indicated above.

(316)

On storm track, behind center

(317) Avoid the center by the best practicable course, keeping in mind the tendency of tropical cyclones to curve northward and eastward.

(318)

Coastal effects

(319) The high winds of a hurricane inflict widespread damage when such a storm leaves the ocean and crosses land. Aids to navigation may be blown out of position or destroyed. Craft in harbors, unless they are properly secured, drag anchor or are blown against obstructions. Ashore, trees are blown over, houses are damaged, powerlines are blown down, etc. The greatest damage usually occurs in the dangerous semicircle a short distance from the center, where the strongest winds occur. As the storm continues on across land, its fury subsides faster than it would if it had remained over water.

(320) Along the coast, particularly, greater damage may be inflicted by water than by wind. There are at least four sources of water damage. First, the unusually high seas generated by the storm winds pound against shore installations and craft in their way. Second, the continued blowing of the wind toward land causes the water level to increase perhaps 3 to 10 feet above its normal level. This **storm tide**, which may begin when the storm center is 500 miles or even farther from the shore, gradually increases until the storm passes. The highest storm tides are caused by a slow-moving hurricane of larger diameter, because both of these effects result in greater duration of wind in the same direction. The effect is greatest in a partly enclosed body of water, such as the Gulf of America, where the concave coastline does not readily permit the escape of water. It is least on small islands, which present little obstruction to the flow of water. Third, the furious winds that blow around the wall of the eye often create a ridge of water called a **storm surge**, which strikes the coast and often inflicts heavy damage. The effect is similar to that of a **tsunami (seismic sea wave)** caused by an earthquake in the ocean floor. Both of these waves are popularly called **tidal waves.** Storm surges of 20 feet or more have occurred. About 3 or 4 feet of this is due to the decrease of atmosphere pressure, and the rest to winds. Like the damage caused by wind, damage due to high seas, the storm tide, and the storm surge is greatest in the dangerous semicircle, near the center. The fourth source of water damage is the heavy rain that accompanies a tropical cyclone. This causes floods that add to the damage caused in other ways.

(321) When proceeding along a shore recently visited by a hurricane, a navigator should remember that time is required to restore aids to navigation which have been blown out of position or destroyed. In some instances the aid may remain but its light or sound apparatus may be inoperative. Landmarks may have been damaged or destroyed.

(322)

Ice

(323) The extent to which the harbors of Maine are closed to navigation by ice varies greatly in different years. During some winters most of the harbors are open, while in others the only harbors available for anchorages are Quoddy Narrows, Eastport, Little River, Machias Bay (above Avery Rock Light), Mistake Harbor (not much used), Winter Harbor and Boothbay Harbor. Portland Harbor generally has an open channel in winter, kept so by steamers and tugs. The mouths of the rivers are generally avoided for anchorage in winter and early spring on account of running ice. In the bays and harbors the ice formation is mostly local; beginning at the head, in sheltered places along the shore, it extends outward. During a calm or light winds from northward the local formations rapidly increase, while strong winds break them up and force them as drift ice onto the lee shore. The tidal currents do not prevent the formation of ice or influence its movements in strong winds except in the larger rivers.

(324) In severe winters some of the harbors south of Cape Ann are closed to navigation by ice, and there is more or less drift ice in all the harbors, in Cape Cod Bay and on Monomoy and Nantucket Shoals. In the principal harbors, steamers and tugs usually keep a channel open. See Ice under the different headings in the text.

(325)

Caution

(326) Along the New England coast, ice formation and ice movement caused by wind, tides and currents may result in floating aids to navigation being extinguished, off station, partially submerged or missing. Icing on buoys can result in misleading color characteristics, missing numerals and inoperative sound signals. Fixed aids to navigation may be destroyed.

(327) To protect certain light and sound buoys from damage during icing conditions, they may be removed from station or replaced by unlighted buoys without prior notice. Mariners should exercise discretion in using aids to navigation in areas where icing conditions are known to exist.

(328) The **International Ice Patrol (IIP)** was formed in 1914 to patrol the Grand Banks of Newfoundland, to detect icebergs and to warn mariners of their location. Under the 1974 Safety of Life at Sea (SOLAS) Convention, 17 member nations agree to share the annual cost of operating the patrol. The U.S. Coast Guard conducts the patrol and maintains IIP records.

(329) The IIP is coordinated from its operations center at Groton, Connecticut. The ice season typically runs from February through July but can last longer. Flying out of St. John's, Newfoundland, USCG aircraft cover the ice danger area, a piece of water twice the size of the State of Texas. Its southern boundary is the latitude of New York City, and it reaches halfway across the Atlantic with Newfoundland on the northwest and Greenland and Iceland on its north and northeast. A normal flight lasts seven hours and can cover 35,000 square miles.

(330) Once sighted, a berg's location, size and shape are entered into a computer drift model, used until the berg is re-sighted or melts. The IIP attempts to locate and track all icebergs south of the 50th parallel, and particularly those south of 48°N., which may be hazardous to navigation near the Grand Banks. When sighting data is entered into the drift program, predicted positions of bergs are calculated for 1200 UTC.

(331) All shipping is requested to assist the IIP by reporting all sightings of ice at once to the IIP through any U.S. or Canadian Coast Guard communications station. Ice sightings should include precise position, size and shape of berg; sea surface temperature; and concentration and thickness of sea ice. Reports can be sent to COMINTICEPAT GROTON CT through INMARSAT-C: Code 42. There is no charge for iceberg reports made using Code 42.

(332) The IIP Operations Center can be reached by telephone at 860–441–2626 or via the Coast Guard Atlantic Area Operations Center at 757–398–6231.

(333) A radio facsimile chart of the area depicting 1200Z ice distribution is broadcast three times daily. A list of the radio stations broadcasting IIP Bulletins and frequencies and times of broadcasts is published annually in Local Notices to Mariners of the First Coast Guard District, and in Radio Navigational Aids, a publication issued by the National Geospatial-Intelligence Agency.

(334) The IIP seeks comment on its services to mariners, particularly on the effectiveness of the times and frequencies of radio transmissions. Mariners are requested to mail facsimile charts received at sea to:

(335) International Ice Patrol, 1082 Shennecossett Road, Groton, CT 06340-6095. The frequency used, time of receipt and vessel position at time of receipt should be indicated. Additional customer comments can be directed to IIP Customer Service at 877–423–7287.

(336)

Sizes and Types of Icebergs

Size	Height		Length	
	(feet)	(meters)	(feet)	(meters)
Growler	0-3	0-1	0-19	0-5
Small (S)	4-50	1-15	20-200	6-60
Medium (M)	51-150	16-45	201-400	61-122
Large (L)	151+	46+	401+	123+

Shape	Description
Blocky	Steep sides with fat top. Very solid. Length-height ratio less than 5:1.
Tilted Blocky	Blocky iceberg which has tilted to present a triangular shape from the
Drydock	Eroded such that a large U-shaped slot is formed with twin columns. Slot extends into or near waterline.
Pinnacled	Large central spiral or pyramid.
Dome	Large round smooth top. Solid-type iceberg.
Tabular	Flat-topped iceberg with length-height ratio greater than 5:1.

(337)

Principal ports

(338) The ports within the area of this Coast Pilot that have regular deep-draft commercial traffic are Bucksport, Eastport, Searsport and Portland, ME; Portsmouth, NH; and Gloucester, Salem, and Boston, MA.

(339)

Pilotage

(340) Pilotage is compulsory for foreign vessels and U.S. vessels under register in the foreign trade as follows:

(341) Maine–Eastport, Cobscook Bay, Pennamaquan River and Friar Roads when entered through Head Harbor Passage, Frenchman Bay, Penobscot Bay and River, Kennebec River to Bath and Portland.

(342) New Hampshire–All ports.

(343) Massachusetts–All ports.

(344) Pilotage is optional for coastwise vessels that have on board a pilot properly licensed by the Federal Government for the waters which the vessel travels.

(345) Arrangements for pilots should be made by the ships' agents at least 24 hours in advance at all of the ports. Detailed information on pilotage procedures is given in the text for the ports concerned.

(346)

Towage

(347) Tugs are available at Eastport, Belfast, Boothbay Harbor, Portland, Portsmouth and Boston. At a number of other places power fishing boats and launches can be secured for handling smaller vessels and barges. Arrangements for tugs should be made in advance through ships' agents or the pilots. (See the text for the ports concerned as to the availability of tugs.)

(348)

Vessel Arrival Inspections

(349) Quarantine, customs, immigration and agricultural quarantine officials are stationed in most major U.S. ports. (See Appendix A for addresses.) Vessels subject to such inspections generally make arrangements in advance through ships' agents. Unless otherwise directed, officials usually board vessels at their berths.

(350)

Harbormasters

(351) Harbormasters are appointed for most of the ports. They have charge of the anchorage and berthing of vessels.

(352)

Supplies

(353) Boston, Portland and Portsmouth are the principal ports at which general supplies, provisions and marine supplies can be obtained. Boston, Portland, Bucksport, Salem, Portsmouth and Searsport have stocks of fuel oil. Diesel oil is available at Beverly, Boston, Gloucester, Portsmouth, Searsport, Bucksport, Portland, Rockland and Boothbay Harbor. Yacht and small-boat supplies including gasoline and diesel fuel are available at most of the smaller ports.

(354)

Repairs

(355) Major repairs to large vessels can be made at Boston and to a lesser extent at Bath. Portland is equipped to handle above-water hull and engine repairs of deep-draft vessels. Tugs and large fishing vessels can be hauled out at Boston, Gloucester, Stonington, Rockland and Portland. Smaller vessels, motorboats and yachts can be hauled out, and ordinary repairs to machinery and hull can be made at most of the smaller ports.

(356)

Small-craft facilities

(357) Marine supplies, repair facilities and other services for small craft are available at all of the major ports and most of the coastal towns and villages along the coasts of Massachusetts, New Hampshire and that portion of the Maine coast southwestward of Boothbay Harbor. Northeastward of Boothbay Harbor the coast is less densely populated, the small-craft facilities are usually farther apart and the services rendered are often limited, thereby making careful advance planning prudent. A description of the facilities is given in the geographic text.

(358) **A vessel of less than 65.6 feet (20 meters) in length or a sailing vessel shall not impede the passage of a vessel that can safely navigate only within a narrow channel or fairway. (Navigation Rules, International-Inland Rule 9(b).)**

(359)

Standard Time

(360) The area covered by this Coast Pilot uses eastern standard time (e.s.t.), which is 5 hours slow of Coordinated Universal Time (UTC). For example, when it is 1000 UTC, it is 0500 along this coast.

(361)

Daylight saving time

(362) Throughout the area of this Coast Pilot, clocks are advanced 1 hour on the second Sunday of March and are set back to standard time on the first Sunday of November.

(363)

Legal public holidays

(364) New Year's Day, January 1; Martin Luther King, Jr.'s Birthday, third Monday in January; Washington's Birthday, third Monday in February; Memorial Day, last Monday in May; Independence Day, July 4; Labor Day, first Monday in September; Columbus Day, second Monday in October; Veterans Day, November 11; Thanksgiving Day, fourth Thursday in November; and Christmas Day, December 25. The national holidays are observed by employees of the Federal Government and the District of Columbia and may not be observed by all the states in every case.

(365) In addition, other holidays are observed in the New England states: General Election Day, first Tuesday after first Monday in November, in Maine and New Hampshire; March 17, Evacuation Day, and June 17, Bunker Hill Day, in Boston and Suffolk County, MA; Patriot's Day, third Monday in April, in Maine and Massachusetts; Fast Day, fourth Monday in April, in New Hampshire.

(366)

Canadian Hydrographic Service

(367) Department of Fisheries and Oceans, publishes nautical charts and other related marine publications, including Canadian Sailing Directions. These publications are available from the Canadian Hydrographic Chart Distribution Office and all but the latter are available from the Canadian Communications Group. (See Appendix A for addresses.)

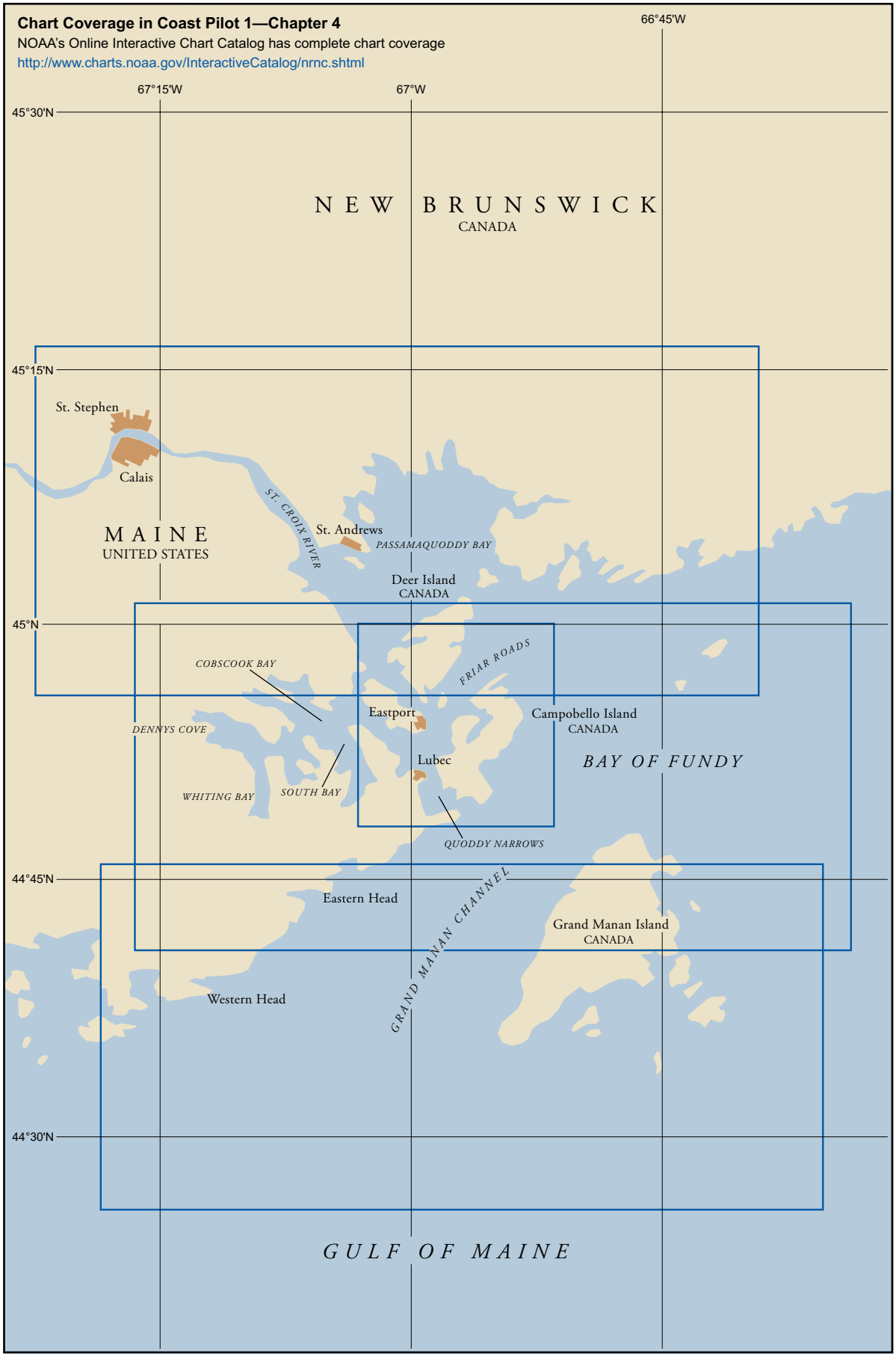


Chart Coverage in Coast Pilot 1—Chapter 4
NOAA's Online Interactive Chart Catalog has complete chart coverage
http://www.charts.noaa.gov/InteractiveCatalog/nrnc.shtml
66°45'W
67°15'W
67°W
45°30'N
45°15'N
45°N
44°45'N
44°30'N
NEW BRUNSWICK
CANADA
MAINE
UNITED STATES
St. Stephen
Calais
ST. CROIX RIVER
St. Andrews
PASSAMAQUODDY BAY
Deer Island
CANADA
COBSCOOK BAY
FRIAR ROADS
DENNYS COVE
Eastport
Campobello Island
CANADA
Lubec
BAY OF FUNDY
WHITING BAY
SOUTH BAY
QUODDY NARROWS
Eastern Head
GRAND MANAN CHANNEL
Grand Manan Island
CANADA
Western Head
GULF OF MAINE

Quoddy Narrows to Calais, Maine

(1) This chapter describes the Maine and New Brunswick coasts from Quoddy Narrows through Lubec Channel, Friar Roads, Western Passage and the St. Croix River to the head of navigation at Calais. Included in the text are discussions of the Maine ports of Lubec, Eastport and Calais; the Canadian ports of St. Stephen and St. Andrews; several small harbors on Campobello Island; and Head Harbour Passage.

(2)

COLREGS Demarcation Lines

(3) The lines established for this part of the coast are described in **33 CFR Part 80.105,** chapter 2.

(4) The approaches to St. Croix River include Quoddy Narrows, Lubec Channel, Friar Roads, Head Harbour Passage, Western Passage and Passamaquoddy Bay. The principal entrance is around the northern end of Campobello Island through Head Harbour Passage. This passage is deep and generally clear of dangers. The channel through Lubec Narrows is also used, especially at high water. The tidal currents are strong in both passages.

(5) **West Quoddy Head**, the easternmost point of the United States, is bold and wooded. **West Quoddy Head Light** (44°48'54"N., 66°57'02"W.), 83 feet above the water, is shown from a 49-foot red and white horizontally banded tower on the eastern edge of the headland. A mariner radio activated sound signal at the light is initiated by keying the microphone five times consecutively on VHF-FM channel 83A. The abandoned Coast Guard lookout tower near the summit of the ridge westward of the light is the most conspicuous landmark in the approach to Quoddy Narrows from seaward.

(6) Between West Quoddy Head and Calais, fluorescent red pyramidal markers define straight line segments and turning points of the United States-Canada boundary.

(7) **Quoddy Narrows (Quoddy Roads)**, between West Quoddy Head and Canada's Campobello Island, is the usual anchorage for vessels seeking shelter or waiting for a favorable tide to pass through Lubec Narrows. The entrance, between West Quoddy Head and The Boring Stone, is about 0.8 mile wide and has a depth of 28 feet near the middle. Winds from east to south generate rough seas in the entrance.

(8) The anchorage affords shelter from northerly and westerly winds in depths of 12 to 25 feet, but is open to winds from the east and south, and protection from northeast gales is reported poor. The northern and western parts of Quoddy Narrows between West Quoddy Head and Lubec are full of shoals which partly uncover.

(9) **Sail Rock** and **Little Sail Rock** are two bare rocks on a ledge about 0.2 mile southeastward of West Quoddy Head Light. The ledge extends more than 100 yards east of the two rocks. As swirls form just southward and eastward of Sail Rock during the strength of the tidal current, the rock should be given a good berth. A lighted whistle buoy is about 0.4 mile southeastward of Sail Rock, about in line with the rock and West Quoddy Head Light. A fairway bell buoy, about 0.5 mile north-northeastward of the light, marks the entrance to Quoddy Narrows and the approach to Lubec Channel.

(10) **Round Rock**, which uncovers, and **The Boring Stone**, 3 feet high and bare, are 500 yards southwest of **Liberty Point**, a bold headland, which is the southern extremity of Campobello Island. Vessels should pass at least 300 yards off the southernmost rock. An islet about 200 yards off Liberty Point is conspicuous, as is **Ragged Point** about 0.4 mile northeastward of it.

(11) **Wormell Ledges**, which partly uncover, are about 400 yards northward of West Quoddy Head and are marked at their northern end by a buoy.

(12) **Middle Ground**, covered 4 feet, is a shoal in the middle of Quoddy Narrows, 0.7 mile north-northwest of West Quoddy Head, and it is marked on its southwestern side by a buoy.

(13) **Lubec Channel** and **Lubec Narrows**, between Quoddy Narrows and Friar Roads, have been improved by dredging. The Federal channel has a project depth of 12 feet. (See Notice to Mariners and the latest edition of the chart for controlling depths.) The channel is marked by a light and buoys. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A. At spring tides the low water may be 3 or 4 feet below the average. Lubec Narrows has strong tidal currents and eddies. Shoals bare on both sides of Lubec Narrows at low water. It is not advisable to use this passage without local knowledge.

(14) The**Franklin D. Roosevelt Memorial Highway Bridge**crosses the narrows from Lubec to Campobello Island at a point about 400 yards southward of the abandoned lighthouse on **Mulholland Point**. The fixed span has a clearance of 47 feet.

(15) **Lubec** is a small town on the west side of Lubec Narrows. Its principal industry is fishing. The most **prominent features** are a tall church spire and a blue water tank on the hill in the town, both of which are visible from Friar Roads and Quoddy Narrows. A prominent school and its gymnasium are about 700 yards westward of **Mowry Point**.

(16)

Current

(17) For current predictions, see the Tidal Current prediction service at *tidesandcurrents.noaa.gov*. Links to a user guide for this service can be found in chapter 1 of this book. Tidal ranges within the area can be affected by atmospheric pressure. Low pressure days can result in tides up to 3 feet higher than predicted.

(18)

North Atlantic Right Whales

(19) The Bay of Fundy is a feeding and nursery area for endangered North Atlantic right whales (peak season: July through October). Mother and calf pairs and groups of right whales may occur in the following areas: north along the New Brunswick coast, along the Campobello-White Horse coast, the Lubec Narrows, the Wolves and along the Grand Manan coast and close to shore from White Head to Swallowtail. (See North Atlantic Right Whales, indexed as such, chapter 3, for more information on right whales and recommended measures to avoid collisions with whales.)

(20) **U.S. Customs and Immigration** officers are stationed at the Roosevelt Memorial Bridge 24 hours daily.

(21) There is a recreational floating pier with water and electricity about 0.2 mile north of the Roosevelt Memorial Bridge.

(22) A 170-foot-long L-shaped commercial fishing wharf with a 62-foot face is on the north waterfront. The depths alongside were reported to be 14 feet, with 6 feet along the outer face. A public small-craft launching ramp with an adjoining float landing is about 250 yards eastward of the wharf.

(23) The **harbormaster** may be contacted through the town hall.

(24) There are no regular repair facilities at Lubec. Due to the large tidal range, boatmen usually ground out their vessels for below-the-waterline repairs. Diesel fuel is available by truck at the commercial fishing wharf on the north side of the town. Ice, provisions and limited marine supplies are available in Lubec. The town has no public transportation.

(25) **Pilotage** is compulsory.

(26) **Johnson Bay**, on the northwest side of Lubec, is a well-sheltered and frequently used anchorage. The approach from southward is through Quoddy Narrows and Lubec Narrows, and the approach from northward is through Friar Roads. An aquaculture site consisting of several floating cages is in Johnson Bay centered at 44°51'48"N., 67°00'12"W.

(27) **Popes Folly** is a thinly wooded islet 0.3 mile north-northwest of Mulholland Point. A bar extends southeastward from the islet to Campobello Island; vessels bound southward to Lubec or through Lubec Narrows cross it. The ledge that extends northeastward from the islet is marked at its outer end by a buoy.

(28) **Dudley Island**, 0.3 mile northwestward of Popes Folly, is high and mostly grass covered. A causeway connects it with Treat Island, 0.2 mile northward.

(29) **Treat Island**, largest of the islands between Lubec Narrows and Eastport, is high and grass covered on the south end and wooded on the north end. **Burial Islet**, small and grass covered, is 300 yards northwestward and bare. **Gull Rock** is 400 yards westward of Treat Island.

(30) Estes Head Cargo Terminal is on the west side of **Estes Head**. The terminal can accommodate ships up to 900 feet. Depths of 64 feet have been reported alongside the pier.

(31) **Broad Cove**, which makes into the south shore of **Moose Island** west of Eastport, is a good anchorage. The head of the cove is shoal for a distance of 0.2 mile. Shackford Ledge, extending 0.3 mile southeast from **Shackford Head**, is marked at its southeastern end by a buoy. An aquaculture farm marked by private buoys is near the center of the entrance to the cove.

(32) **Deep Cove** is the first cove to the northwestward of Broad Cove on Moose Island. A wharf in the cove is in ruins and is no longer usable. A T-shaped pier about 400 feet in length, with floating docks in the summer and a reported depth of 12 feet alongside its outer face, is on a former seaplane launching ramp on the southeastern side of the cove. It is used by an adjacent marine vocational school. A 60-ton travel lift and a repair facility for small vessels are available.

(33) **Snug Cove**, on the west side of Campobello Island eastward of Dudley Island, is of no importance except to small craft. Between Snug Cove and Dudley Island is an unmarked rock covered 14 feet. Vessels entering Friar Roads from the southward pass on either side of the unmarked rock.

(34) **Friars Head**, to the north of Snug Cove, is on the south side of the entrance to **Friars Bay**, on the western side of Campobello Island. Friars Bay is used as an anchorage, and on its northern side is the village of **Welshpool**, where small craft can find protection in all weather at the government wharf, which has a 215-foot face with 14 feet reported alongside.

(35) For a distance of 1.3 miles from Welshpool, the west shore of Campobello Island continues northward to **Bald Head**, a point just south of which is a prominent circular hill 101 feet high. From Bald Head the coast trends northeastward 0.6 mile to **Man of War Head**, which is on the south side of the entrance to Harbour De Lute. The L-shaped breakwater-wharf, in the cove near the northwestern end of Man of War Head, has depths of 5 to 10 feet reported along the inner side of its north face. A light, 23 feet above the water, is shown from a skeleton tower at the outer end of the breakwater. The basin behind the breakwater has been dredged to depths of 5 to 7 feet.

(36) **Harbour De Lute** is used as an anchorage by small vessels, but those without local knowledge should not go beyond the 9-foot spot, known as **Racer Rock**, in the middle of the entrance to the inner harbor. The inner harbor is obstructed by fish weirs. Indenting the north

(49)

shore of Harbour De Lute east of **Windmill Point**, which is on the north side of the entrance, are four coves that are of little importance except to the fishing industry.

(37) In **Curry Cove**, the northernmost, there is an L-shaped wharf with reported depths of 5 feet along the outer and inner sides of its 150-foot outer face. The wharf is unsafe during strong southwest winds.

(38) The harbors on the west side of Campobello Island are used as harbors of refuge by fishing vessels during heavy easterly gales.

(39) **Friar Roads** (Eastport Harbor), which lies between Moose Island and Campobello Island, is approached from northward through Head Harbour Passage and from southward through Quoddy Narrows and Lubec Narrows. Friar Roads is the principal approach to Passamaquoddy Bay and St. Croix River.

(40) **Eastport**, a city situated on the hilly east side of Moose Island, is the easternmost deepwater port in the United States. The docks of the port are along the waterfront on the east shore of the island. The principal industries are forest products, lobstering, scallop harvesting, farming and harvesting salmon and tourism.

(41)

Prominent features

(42) The principal landmarks are a blue standpipe, the customhouse with its square tower and the spire of a church about 300 yards west-southwest of the customhouse. Numerous concrete pylon boundary markers on the tops of the hills are also conspicuous.

(43) A dredged small-craft harbor for commercial and pleasure craft is off the customhouse in Eastport. The harbor is protected on its northerly and easterly sides by a steel piling, solid fill, L-shaped breakwater–wharf onto which fishing vessels can unload their catch into trucks. A town float is on the inner side of the breakwater at the north end of the harbor. Boats usually moor along the inner face of the breakwater. In fair weather, berthing is available along the east and north seaward faces of the breakwater. Electricity is available at all the berths, and diesel fuel can be delivered by truck on short notice. Gasoline and diesel fuel are also available just to the north at the Eastport Chowder House dock. The breakwater is floodlighted at night. The **harbormaster** may be contacted through the town hall. A small-craft launching ramp is in the northwest corner of the harbor. Additional small-craft berths are available 0.2 mile north of the harbor.

(44) Deep-draft vessels may anchor about 0.5 mile off the town with Cherry Island bearing **017°** about 1.1 miles distant. The bottom here is broken and rocky, and the tidal currents are strong. This anchorage is not recommended in easterly weather, when more favorable conditions may be found on the opposite side of the bay off Friars Bay or off Broad Cove on the west side of the island. Anchorage

in Deep Cove is not recommended due to submarine cables.

(45)

Dangers

(46) **Clark Ledge**, marked by a daybeacon, is about 0.5 mile north of the breakwater. Strong rotating currents in the vicinity commonly set vessels onto Clark Ledge.

(47) **Dog Island**, 0.3 mile northwestward of Clark Ledge, has a grassy top and a shelving ledge extending about 100 yards off the high-water line of the island. **Dog Island Light DI** (44°55'07"N., 66°59'21"W.), 35 feet above the water, is shown from a skeleton tower with a square green daymark. A red sector in the light covers Clark Ledge. A mariner radio activated sound signal at the light is initiated by keying the microphone five times consecutively on VHF-FM channel 83A.

(48) Whirlpools and eddies that are dangerous at times for small boats are encountered between Dog Island and **Deer Island Point**, 0.5 mile northeastward. They are reported to be worst about 3 hours after low water. **Old Sow**, the largest whirlpool in the Western Hemisphere, occurs at this location. Transit of large ships through this area is usually planned for periods of relatively slack current. The Tidal Current prediction service at *tidesandcurrents.noaa.gov* should be consulted for specific information about times, directions, and velocities of the current at the numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(50)

Weather, Eastport and vicinity

(51) July is the warmest month in Eastport with an average high of 72 °F and an average minimum of 53 °F. January is the coolest month with an average high of 30 °F and an average minimum of 15 °F. The highest temperature on record for Eastport is 96 °F recorded in

(69) Eastport is a **customs port of entry**. Entry is free between 0800 and 1800; however, there is an afterhours charge.

(70) A former cannery wharf with a 65-foot face and 1 to 5 feet alongside is 100 yards north of the breakwater; fresh water may be available.

(71) The Port of Eastport offers general cargo dockage at the Breakwater Pier. The 420-foot facility can accommodate vessels up to 700 feet in length with a draft up to 36 feet. Water is available at the pier with prior notice; diesel fuel can be delivered by truck. Ice, groceries, and limited marine supplies and full repairs are available. There is a private facility for hauling out craft up to 40 feet in length; contact the harbormaster for additional information.

(72) Eastport has no coastwise steamer service. An automobile ferry connects Eastport with Deer Island in summer only.

(73) There is no railroad service to Eastport, but a good highway parallels the St. Croix River to Calais. There is an airport at Eastport which is used occasionally. Taxi service provides the only connection with the coastal bus service at Perry about 6 miles to the northwestward.

(74) **Western Passage** is between Moose Island and **Deer Island**, the next large Canadian island northwestward of Campobello Island, and connects Friar Roads with Passamaquoddy Bay.

(75) Earth and rock causeways block the shallow passages north and south of **Carlow Island**, which is 0.2 mile northwest of Moose Island and 0.6 mile south of **Pleasant Point**. A prominent red brick Indian mission church with square belfry and numerous houses of the Indian reservation are on Pleasant Point.

(76) **Frost Island** and **Frost Ledge** are at the northern end of Western Passage and between Pleasant Point and **Gleason Cove**, 0.9 mile to the northward. Frost Ledge extends 0.4 mile offshore and is marked by a bell buoy. Between Carlow Island and Frost Island, foul ground extends as much as 400 yards from shore.

(77) The northeastern, or Deer Island, shore of Western Passage is clear; indentations are **Cummings Cove** and **Clam Cove**.

(78) The east coast of **Campobello Island** is mostly clear and can be approached to within a reasonable distance without danger.

(79) **Local magnetic disturbance**.–Differences of as much as 5½° from the normal variation have been observed off the east coast of Campobello Island.

(80) **Herring Cove (Herring Bay)**, near the south end of Campobello Island's eastern shore, is a good temporary anchorage for large vessels. **Schooner Cove**, midway along the eastern shore, and **Mill Cove**, near the northern end, afford temporary anchorage for small craft. A 2-foot spot in the middle of the entrance to Mill Cove is marked by a buoy off its eastern side.

(81) **East Quoddy Head** is the northeasternmost point of Campobello Island. **Head Harbour Light** (44°57.5'N., 66°54.0'W.), 58 feet above the water, is shown from a 47-foot white octagonal tower on the outermost rock. The sound signal is on a small building just north of the light.

(82) **Head Harbour**, between East Quoddy Head and **Head Harbour Island**, 0.2 mile to the southeastward, is one of the best sheltered small-craft harbors in the area. Excellent berthage is available on the westerly side of the harbor about 1.2 miles above East Quoddy Head Light at an L-shaped government breakwater-wharf which is used by commercial fishing vessels. The wharf has a 320-foot outer southeastern face with 17 feet reported alongside. Mooring poles are provided eastward of the pier for pleasure craft. Diesel fuel is available by truck and gasoline is reported to be available at Wilsons Beach. The harbor affords good anchorage and wet winter storage for small vessels. The preferred channel into the harbor is northward of Head Harbour Island. The channel south of the island, shoaler and with numerous fish weirs, should not be used without local knowledge. Shoals, marked by buoys, are on both sides of the northerly channel about 0.6 mile above the entrance light. Fishing craft drawing 9 feet and more frequent the harbor.

(83) **Wilsons Beach** is on the northwest side of Campobello Island about 2 miles southwestward of East Quoddy Head. A small harbor behind a breakwater-wharf affords good protection. Depths of 7 feet are reported along the outer part of the inner side. Crib remains are reported 200 feet west of the outer part. Gasoline and diesel fuel are reported to be available at the wharf.

(84) **Head Harbour Passage** is a deep and clear fairway, about 4 miles long, that follows the northwestern side of Campobello Island from the sea to Friar Roads, opposite Eastport, where it joins Western Passage. The route through Head Harbour Passage and Western Passage is the one usually followed by vessels going to Passamaquoddy Bay and St. Croix River.

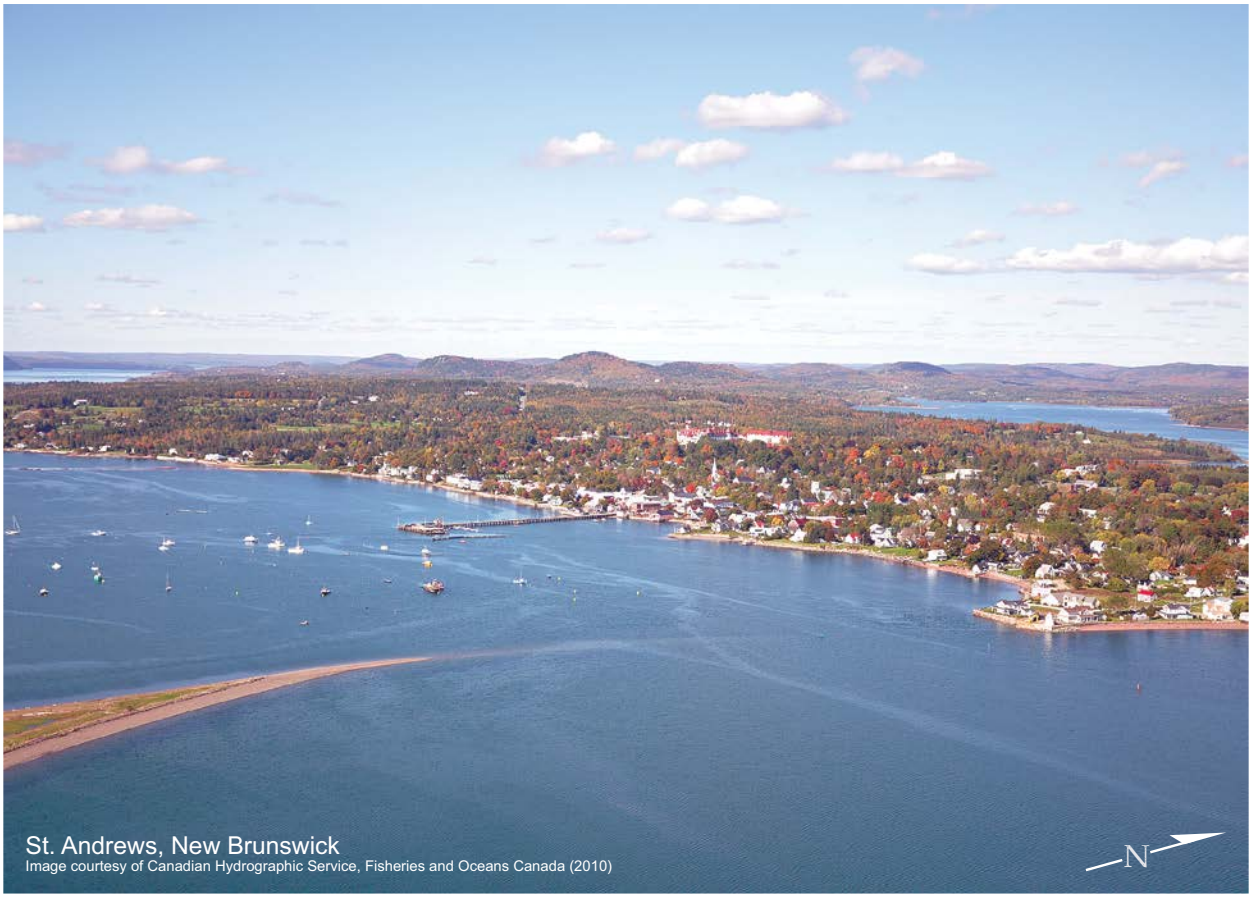
(85) **White Horse Island**, bare, rocky, and 68 feet high, is about 2.3 miles northeastward of East Quoddy Head. The island, whitish in appearance and easily identified, is a good mark for the approach to Head Harbour Passage.

(86) **Little White Horse Ledges**, close northeastward of White Horse Island, are two dangerous unmarked submerged rocks about 250 yards apart. **North Rock**, steep-to and covered 2 feet, is about 0.5 mile northwestward of White Horse Island and is marked by a buoy off its northern side.

(87) **White Island**, 1.6 miles northward of East Quoddy Head, has fringing shoals that extend as much as 400 yards from shore. A group of islets and shoals 0.5 mile northwestward of the island includes **Nubble Island**, **Spectacle Island** and **Hospital Islands**.

(88) **Spruce Island**, 0.8 mile north-northwest of East Quoddy Head, is steep-to on its eastern side. Islets and shoals extend 0.8 mile westward of the island. The westernmost of these dangers are **Tinkers Lower Ledge** and **Tinkers Upper Ledge**, about 300 yards northeastward and 400 yards south-southwestward of **Tinkers Island**, respectively.

(97)



(89) **Black Rock**, small and bare, covered at high-water springs, is 0.8 mile west-northwestward of East Quoddy Head and 0.6 mile southwest of Spruce Island.

(90) **Casco Bay Island**, 0.5 mile southwestward of Black Rock and 0.5 mile from the nearest part of Campobello Island, is 85 feet high. The eastern side of the island is fairly steep-to, but ledges extend 300 yards off its northern end. Several shoals and ledges are within 0.3 mile of the western side of the island, one of which about 0.3 mile to the westward is marked by buoys on its northerly and southerly sides.

(91) **Green Island** is about 0.4 mile southwestward of Casco Bay Island. A 24-foot shoal, near the middle of Head Harbour Passage and marked by a lighted buoy on its southeastern side, is 0.3 mile east-southeastward of Green Island and 0.4 mile from the shore of Campobello Island. **Sandy Ledge** is 500 yards westward of Green Island.

(92) **Popes Island** is 0.5 mile southwestward of Green Island. Shoals extend 300 yards southwestward of Popes Island. **Popes Shoal**, unmarked and covered 9 feet, is 300 yards southeastward of the island. An unmarked 24-foot rocky patch is about 700 yards southeastward of the island. About 0.4 mile westward of Popes Island is Chocolate Shoal, which is covered 8 feet.

(93) **Indian Island**, 109 feet high near its northern end, is 0.4 mile eastward of Deer Island Point, the south end of Deer Island. The channel between Indian and Deer Islands is deep. A shallow bank, on which are three islets, extends about 500 yards off the southeastern part of Indian Island.

(94) **Cherry Island**, at the southeastern end of this bank, is marked at its southeastern end by a light 40 feet above the water and shown from a red and white horizontally banded circular tower; a sound signal is at the light.

(95) **Passamaquoddy Bay** is the large indentation in the shore of New Brunswick east of the mouth of St. Croix River. The principal entrance is by way of Western Passage, which has deep water and is comparatively free from dangers.

(96) **St. Andrews**, a Canadian town in the east side of the entrance to St. Croix River, has some commerce. A large hotel with a red roof and tower is prominent. An abandoned lighthouse is on the seawall at **North Point** on the south side of the town.

(98) Access to St. Andrews is through a channel from the southeastward or from the west, through **Western (Gut) Channel**. Both channels are well marked with buoys and lights. St. Andrews Harbour, between the town and **Navy Island**, can be used by light-draft vessels.

(99) The L-shaped 848-foot government wharf with reported depths of 7 feet alongside its 152-foot outer face is about 0.4 mile northwestward of North Point. A float landing is on the eastern outer end. Gasoline is available by truck at the wharf. Some marine supplies are available

(113)

in the town and water and electricity are available on the wharf.

(100) **St. Croix River** extends north-northwestward for 8 miles from the southern part of Passamaquoddy Bay, then turns westward between **Devils Head** and **Todds Point**. The channel is deep and comparatively clear as far as the turn, then is narrow and winding and has a controlling depth of about 16 feet for some 3 miles to Hills Point (45°09'53"N., 67°13'33"W.).

(101) The channel through St. Croix River above Todds Point is well marked with lighted and unlighted buoys to Calais and St. Stephen on the Canadian side. The two buoys on the north side of the channel opposite **Whitlocks Mill Light 25**, tow under during the strength of the tide. Local knowledge is necessary for the river above Whitlocks Mill.

(102) Small craft up to 40 feet in length can anchor in 14 feet on the west side of the channel just above Whitlocks Mill Light, but larger craft should anchor off Devils Head.

(103) The scattered remains of an old breakwater, which uncover 12 feet in spots, extend southeastward across the mudflats on the south side of St. Croix River for about 300 yards from near channel Buoy 19. The mudflats, which uncover 11 feet, are opposite **The Ledge**, a village on the north side of the river about 9.7 miles above the mouth; caution is advised in this area.

(104)

Ice

(105) St. Croix River above Robbinston is reported to be closed by ice for about one or two weeks in February. The channel to the oil wharf in Calais is usually kept open by the tug and barge bound there. Quoddy Narrows and Eastport Harbor are never closed by ice.

(106)

Freshets

(107) Spring freshets sometimes cause the water to rise above the level of the wharves at Calais and are accompanied by strong current. They are seldom noticeable outside of the river.

(108) **Robbinston** is a village just above **Liberty Point**. A red brick chimney and large green building of an inactive cannery are about 0.7 mile above the point. **Red Beach** is a small village on the west bank about 3 miles north of Robbinston.

(109) On the Canadian side of the river, about 0.3 mile above **Joes Point**, the 3-story brick and concrete building and wharf of the Atlantic Biological Station of Fish and Oceans Canada are conspicuous. The 580-foot wharf has 17 feet alongside.

(110) **St. Croix Island** is in midriver off Red Beach. **St. Croix River Light** (45°07'42"N., 67°08'02"W.), 101 feet above the water, is shown from a 49-foot red and white banded tower on the island. Scattered shoals, covered and awash, fringe the island and extend southeastward

1.1 miles in midriver. The channel eastward of the island is deeper and broader than the channel to the west. The channel east of Little Dochet Island, a wooded islet, is used considerably by local vessels, but is not advisable for strangers as the dangers are not marked.

(111) A wharf, owned by the Canadian government and known as Bayside Marine Terminal, is on the east side of the river about 1.9 miles northward of St. Croix Island. Fish, food stuffs, lumber and wood pulp are handled; water is available. Pilotage for the terminal is not compulsory but is highly recommended. Local pilots board off Head Harbour Light.

(112) **Calais** is a small city on the south bank of the St. Croix River, about 14 miles above the river mouth and 24 miles from Eastport. The city has no waterborne commerce. It is a railroad freight terminus, and the manufacture of woodworking goods and shoes are the primary industries. There is a hospital in town.

(114) Calais is a **customs port of entry**. The customhouse is at the American end of the bridge, as is the immigration office. The city has taxi service and is also served by a bus line from Boston. There is no harbormaster and no known local harbor regulations in force.

(115) The town wharf at Calais has depths of 4 feet reported alongside. At Todd Point, 1.5 miles below the bridge, is a gravel small-craft launching ramp. There are no facilities on the American side of the river for servicing small craft, but gasoline, provisions and some supplies can be obtained in town.

(116) **St. Stephen** is the Canadian town on the opposite side of the river from Calais. Provisions of all kinds and a limited variety of deck and engineroom stores can be obtained. Minor repairs can be made by various machine shops in the town. The town has a hospital and plants which manufacture building materials, steel products and candy. There is a telegraph office, bus service and freight rail connections. A public wharf with float (45°11.5'N., 67°16.6'W.) at St. Stephen is maintained from June to September. In 1984, depths of 3 to 6 feet were reported alongside the float.

(117) Canadian **Customs** and **Immigration** officers are stationed at the International Bridge. St. Stephen is a customs port of entry, and marine documents are issued.

(118) **Cobscook Bay**, extending westward from Moose Island, is large and irregular and has several arms. The approach channel is between Moose Island and **Seward Neck**, about 0.6 mile southwestward. The deepest draft using Cobscook Bay is 14 feet. Local knowledge is needed to navigate the arms of the bay because of the numerous rocks and dangerous currents.

(119) Vessels entering and departing Cobscook Bay should exercise caution in the area of an underwater turbine located at 44°54'36"N, 67°02'45"W. All vessels and persons are advised to avoid anchoring, diving, dredging, dumping, fishing, trawling, laying cable or conducting salvage operations in this area.

(120) Good anchorage can be found in many of the arms or coves in Cobscook Bay, but in most of the channels the currents are too strong and the bottom is too rocky. In the winter, ice obstructs navigation near Whiting at the head of Whiting Bay and Dennysville, and in severe winters other parts of the bay also are affected.

(121) **Bar Harbor**, a shoal arm of the bay northwest of Moose Island, can no longer be used as a shortcut between Cobscook Bay and Western Passage because the eastern passages north and south of Carlow Island have been closed by earth and rock causeways. An overhead power cable crossing the entrance has a clearance of 45 feet.

(122) **Pennamaquan River** empties into Cobscook Bay from northwestward about 4 miles west of Moose Island. The river has ample depth for about 1.7 miles above the entrance, and the principal dangers are marked by buoys. Low-water flats extend 0.8 mile downstream from **Pembroke**, a town 3 miles above the mouth of the river.

(123)

Pilotage, Cobscook Bay

(124) Pilotage is compulsory in Cobscook Bay and Pennamaquan River for foreign vessels and U.S. vessels under register and a draft over 9 feet. (See Pilotage, Eastport, Cobscook Bay and vicinity, indexed as such earlier this chapter.)

(125) **West Pembroke** is about 0.8 mile southwest of Pembroke and on the northwest prong of Pennamaquan River, which uncovers at low water. There is no waterborne commerce and no wharves at West Pembroke.

(126) **Dennysville** is a village about 1 mile up **Dennys River** at the head of **Dennys Bay**, an arm of Cobscook Bay. There is no waterborne commerce, and the river is important only as a salmon fishing stream.

(127) **Whiting** is a village at the head of **Whiting Bay**, which is an arm of Cobscook Bay. Low-water flats filled with boulders extend about a mile below the village. With local knowledge, small craft at high water can go as far as the dam at the mouth of **Orange River** at the village. The channel is unmarked and difficult to follow.

(128)

Current

(129) In Grand Manan Channel, the flood current sets in a general northeast direction and attains a velocity of about 2.3 knots at strength. The ebb sets in a southwesterly direction with a velocity of about 2.4 knots at strength. Daily tidal current predictions can be generated at the Tidal Current prediction service at *tidesandcurrents.noaa.gov*. This service should be consulted for specific information about times, directions, and velocities of the current at numerous locations throughout this area. Links to a user guide for this service can be found in chapter 1 of this book.

(130) Less than 2 miles from the northern shore on the approach to the entrance to Quoddy Narrows, the set of the flood currents is more northward; about 1 mile southeastward of West Quoddy Head the flood sets

directly into the narrows. For a distance of 0.5 mile southeastward of West Quoddy Head the currents are dangerous because of swirls and eddies which, in a light breeze, are apt to draw a vessel onto Sail Rock.

(131) Along the eastern side of Campobello Island the flood current follows the trend of the shore in a northeasterly direction and the ebb sets in the opposite direction.

(132) In Head Harbour Passage the tidal current is said to attain a velocity of 5 knots at times. The flood sets strongly westward toward the islands about 1 mile northward of Campobello Island. The direction of the flood then changes more southward, following the general direction of the passage until nearly to Eastport, where the set is more westerly, toward Western Passage between Deer and Moose Islands and toward the entrance to Cobscook Bay. The ebb generally sets in a reverse direction.

(133) Through Lubec Narrows, the flood current sets northward, following the general trend of the channel; southward of the narrows it has a velocity of about 4 knots at strength, but in the narrows it attains a velocity of about 6 knots during the spring tides. The ebb sets southward, following the general direction of the channel, and in the narrows has a velocity of about 8 knots during spring tides. Below the narrows its velocity is about 4 knots, and the set is in the general direction of the channel. The currents at strength form dangerous eddies on both sides of the channel in the narrows; these are avoided by keeping in midchannel. The duration of slack in the narrows is only 5 to 15 minutes.

(134) Northward of Lubec Narrows, the first of the flood current sets along the west shore of Campobello Island eastward of Popes Folly; it afterwards sets more westward, south of Popes Folly, and across the entrance to Johnson Bay, meeting the flood from Friar Roads westward of Treat Island, and both setting into Cobscook Bay.

(135) The flood current sets northward into Western Passage, and off Deer Island Point, abreast Dog Island, it forms whirlpools and eddies that are dangerous to open boats. The whirlpools and eddies are strongest 2 to 3 hours before high water and during spring tides; the flood then attains a velocity of about 6 to 7 knots. The least disturbance is usually about 300 yards northward of Dog Island, where there is a comparatively narrow direct current which can be readily followed between the whirlpools and eddies on either side. The ebb sets southward but is weaker than the flood.

(136) Above Deer Island Point the flood sets northward with decreasing velocity and follows the general direction of the channel with strong countercurrents and eddies close to the shore, where the configuration of the land is favorable. The ebb sets southward with reduced velocity and disturbance off Deer Island Point, and the inshore reverse currents are less marked than on the flood. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov*. for specific information about times, directions, and velocities of the current at the numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(137) In the St. Croix River, the flood current sets northward with countercurrents inshore on both sides where the conformation of the land is favorable for them. The ebb sets southward with less marked countercurrents. The tidal current normally attains a velocity of about 2 knots between the mouth of the river and Devils Head and 3 to 4 knots between Devils Head and Calais.

(138) In Cobscook Bay and its tributaries the tidal currents follow the general direction of the channels, but in the coves there are strong reverse eddy currents, and heavy overfalls form over the submerged rocks and ledges. The velocity is estimated at 5 to 8 knots, and some of the buoys are towed under when the currents are at strength.

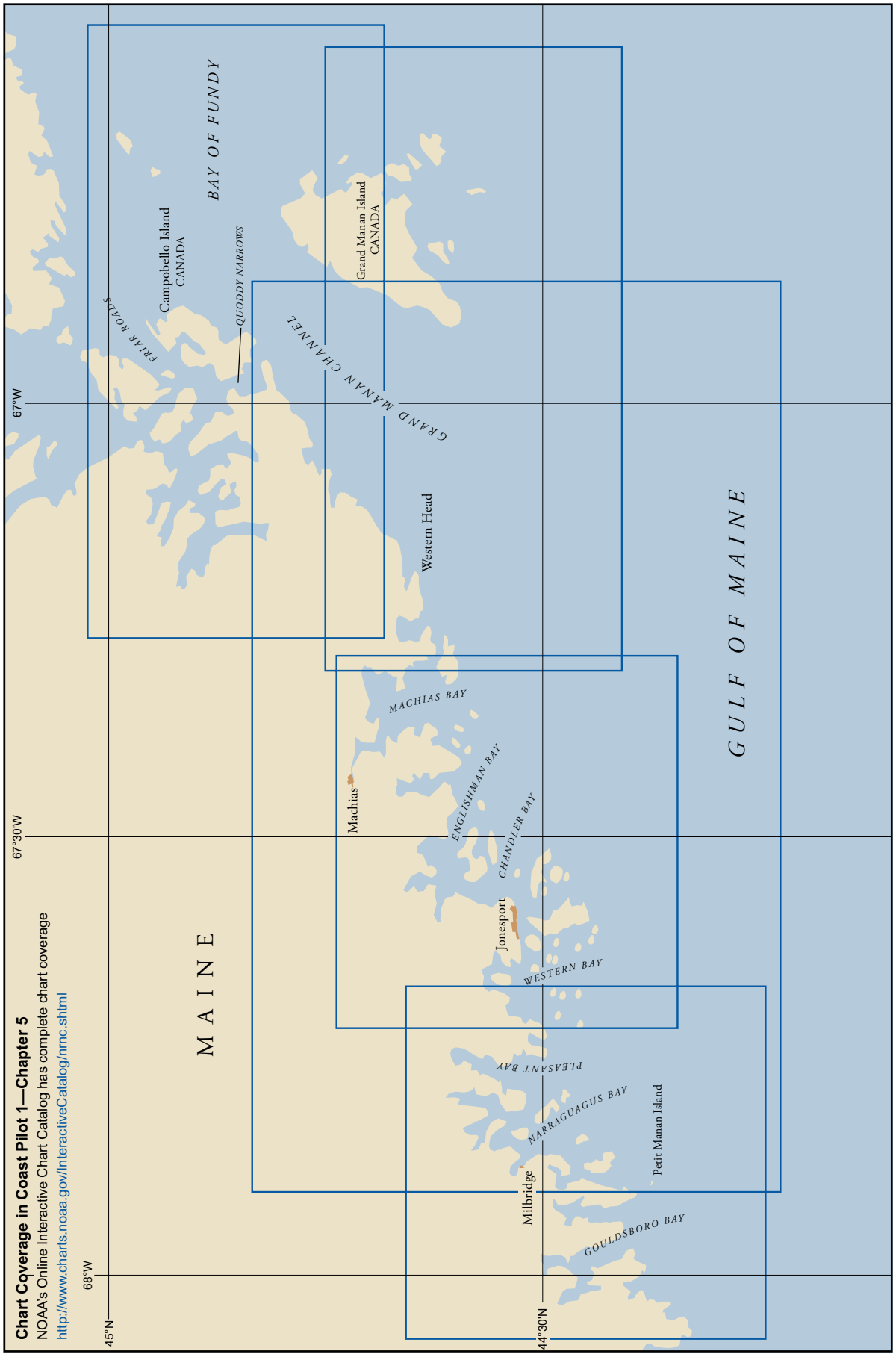


Chart Coverage in Coast Pilot 1—Chapter 5
NOAA's Online Interactive Chart Catalog has complete chart coverage
http://www.charts.noaa.gov/InteractiveCatalog/nrnc.shtml
BAY OF FUNDY
GULF OF MAINE
MAINE
Campobello Island
CANADA
Grand Manan Island
CANADA
FRIAR ROADS
QUODDY NARROWS
GRAND MANAN CHANNEL
Western Head
Machias
MACHIAS BAY
ENGLISHMAN BAY
CHANDLER BAY
Jonesport
WESTERN BAY
PLEASANT BAY
NARRAGUAGUS BAY
Petit Manan Island
Milbridge
GOULDSBORO BAY
67°W
67°30'W
68°W
45°N
44°30'N

Quoddy Narrows to Petit Manan Island, Maine

(1) This chapter describes the rugged Maine coast, with its numerous bays, coves, islands and rivers, from Quoddy Narrows westward to Petit Manan Island. Cutler, Bucks Harbor, Machiasport, Machias, Jonesport, Milbridge and several other coastal towns are discussed.

(2)

COLREGS Demarcation Lines

(3) The lines established for this part of the coast are described in **33 CFR Part 80.105**, chapter 2.

(4) **Grand Manan Channel**, between the coast of Maine and Grand Manan Island, is an approach from westward to Quoddy Narrows and Passamaquoddy Bay. It is the most direct passage for vessels bound up the Bay of Fundy from along the coast of Maine. The channel varies in width from 5.5 miles abreast Campobello Island to 10 miles abreast Southwest Head, the southern point of Grand Manan Island. The western approach is marked by Machias Seal Island Light, which also marks most of the rocks and ledges that lie southwestward of Grand Manan Island. With the exception of the dangers between Machias Seal Island and Grand Manan Island, and the 33-foot unmarked rocky patch known as **Flowers Rock**, 3.9 miles west-northwestward of Machias Seal Island, the channel is free and has a good depth of water. The tidal current velocity is about 2.5 knots and follows the general direction of the channel. Daily predictions are given in the Tidal Current Tables under Bay of Fundy Entrance. Off West Quoddy Head, the currents set in and out of Quoddy Narrows, forming strong rips. Sailing vessels should not approach West Quoddy Head too closely with a light wind.

(5)

North Atlantic Right Whales

(6) The Bay of Fundy is a feeding and nursery area for endangered North Atlantic right whales (peak season: July through October) and includes the Grand Manan Basin, a whale conservation area designated by the Government of Canada. (See North Atlantic Right Whales, indexed as such, chapter 3, for more information on right whales and recommended measures to avoid collisions with whales.)

(7)

Southwest Head to Wallace Ledge

(8) **Southwest Head**, the southern extremity of Grand Manan Island, is a high cliff. It is reported that the fogs often hang close in to the Maine coast between Machias Bay and West Quoddy Head, extending about one-third the way across Grand Manan Channel.

(9) **Machias Seal Island**, 10 miles southwestward of Southwest Head, is about 500 yards long and 28 feet high and marked by a light. The island is steep-to on its western side. Gull Rock, a drying reef, on the end of islet, extends 0.4 mile northeastward. A covered rock is about 300 yards northward of the islet. Depths of 20 feet 0.6 mile eastward and 29 feet 1.2 miles east-northeastward of the island are unmarked as is a 12-foot shoal, sometimes marked by a tide rip, 0.3 mile southeastward of the island.

(10) **Southeast Shoal**, 1.2 miles southeastward of Machias Seal Island, is marked by a V-AIS and covered 9 feet. This shoal breaks in heavy weather and shows a rip during the strength of the tidal current, which reaches a velocity of 3 knots. A depth of 30 feet is about 450 yards southeastward of the shoal.

(11) **North Rock**, marked by a V-AIS is 4 feet high and surrounded by shoal water to a distance of 800 yards, 2.0 miles northward of Machias Seal Island. A 34-foot shoal spot is about 900 yards northeast of North Rock in about 44°32'30"N., 67°04'48"W. Another shoal spot covered 25 feet is 1.4 miles eastward of the rock in about 44°32'18"N., 67°03'16"W.

(12) **North Shoal**, marked by a V-AIS is covered 9 feet, 1.6 miles northward of the light. A depth of 40 feet is 700 yards northward. The shoal breaks in heavy weather, and the whole area is marked by tide rips.

(13) **Middle Shoal**, 5 miles northeastward of Machias Seal Island, is covered 17 feet, with deep water close-to. The shoal shows a tide rip and breaks in heavy weather.

(14) **Bull Rock**, awash at low water and usually breaking, is 6.7 miles eastward of Machias Seal Island and is marked by a lighted whistle buoy. It is surrounded by deep water. **Little Shoal**, a rocky patch covered 28 feet and usually marked by a tide rip, is about midway between Bull Rock and Machias Seal Island. **Guptill Grounds**, covered 29 feet and unmarked, are 1.2 miles south-southwestward of Bull Rock.

(15)

Local magnetic disturbance

(16) Magnetic disturbance has been reported in the vicinity of 44°31.5'N., 66°55.0'W.

(17) **Southeast Ledge**, nearly 6 miles southeastward of Machias Seal Island, covered 24 feet, shows a tide rip and breaks in heavy weather. **Middle Breaker**, a 36-foot patch, marked by tide rips, is 1.4 miles northwestward of this ledge.

(18) **Wallace Ledge**, the northernmost of the Murr Ledges, 3.4 miles northeastward of Bull Rock, uncovers 9 feet. A lighted bell buoy is northwest of the ledge.

(19) Eastward of this area are numerous reefs and ledges. These dangers are described in **Pub. No. 145, Sailing Directions (En route), Nova Scotia, and the St. Lawrence**, published by the National Geospatial-Intelligence Agency, Washington, DC. Some of the dangers are **Murr Ledges**, **Halftide Rock**, **St. Mary Ledge**, **Yellow Ledge**, **Cross Jack Ledge, Long Ledge** and **White (West) Ledge.**

(20)

West Quoddy Head to Moose Cove

(21) The coast southwestward between West Quoddy Head and Moose Cove (44°44.2'N., 67°05.6'W.) is in general rocky, wooded and steep-to and is indented by several coves of slight importance. Along this stretch of coast from West Quoddy Head to Long Point (44°40.1'N., 67°09.3'W.), and particularly off Jims Head (44°45.7'N., 67°03.0'W.), a very rough sea builds up quickly when the wind is contrary to the tidal current, and small craft may find themselves beset and unable to make the shelter of the coves without assistance. In 1986, a similar condition was reported to exist from Long Point as far southwest as Cross Island.

(22) **Carrying Place Cove**, on the west side of West Quoddy Head, has a few buildings at its head. **Wallace Cove**, 1.9 miles southwest of West Quoddy Head Light (44°48.9'N., 66°57.1'W.), and **Hamilton Cove**, 3 miles southwest of the light, have no distinguishing features.

(23) **Morton Ledge**, covered 6 feet and marked by a buoy, is 2.2 miles southwestward of West Quoddy Head Light and 0.3 mile offshore. **Boot Cove**, 4 miles southwestward of the light, has a few small fishermen's houses at the head.

(24) **Baileys Mistake**, 5.5 miles southwest of West Quoddy Head Light, appears from offshore to be a good anchorage, but the holding ground is poor and it is not a good harbor even though a few fishing boats moor here. The village of **South Trescott** is at the head of the harbor. **Bailey Ledge**, which uncovers 5 feet, obstructs the western half of the entrance. A buoy marks the southern side of the ledge. **Jims Head**, on the northeastern side of the entrance, is 160 feet high and prominent. A whistle buoy is 0.2 mile south-southeast of the head.

(25) **Haycock Harbor**, the head of which is **The Pool**, is6.3 miles southwestward of West Quoddy Head Light. The Pool is sometimes entered by small craft at high water. The depth inside is reported to be 7 feet. **Sandy Cove** is an open bight just southwestward of the harbor.

(26) **Moose Cove** is 7.8 miles southwest of West Quoddy Head Light. **Eastern Head**, the eastern extremity of the north entrance point, has a 198-foot hill behind it. **The Porcupine**, a distinctive 280-foot hill, is 1.8 miles northwestward of the head. **Mink Islet**, and **Little Mink Islet**, 6 feet high, are on **Eastern Head Ledges**, which extend over 0.2 miles offshore. **Little Moose Islet**, 10 feet high, is 250 yards northward of the ledges.

(27) **Moose River**, at the head of Moose Cove, bares at low water. There is a small wharf on the south side of the river at its narrowest point. On the north side of the river a rocky spit makes out, forming a natural shelter for small boats.

(28)

Little River to Northeast Harbor

(29) From Moose Cove to Little River, a distance of about 6.5 miles, the coast has no features of importance. The several open, shallow coves include **Bog Brook Cove**, **Holmes Cove, Black Point Cove** and **Long Point Cove**. Just north of Little River are **Almore Cove** and **Money Cove**. An unmarked 13-foot spot is about 0.25 mile southward of **Long Point.**

(30) **Little River** is about 14 miles southwestward of West Quoddy Head Light. In the middle of the entrance is **Little River Island**, which is wooded and has rocky sides. **Little River Light** (44°39'03"N., 67°11'32"W.), 56 feet above the water, is shown from a white conical tower on the northeast corner of the island. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A. A bell buoy, 0.5 mile east-northeastward of the light, marks the entrance to the harbor.

(31) A tree-covered islet on the north side of the entrance, about 350 yards north of Little River Island, and two tree-covered islets off Western Head, on the south side of the entrance, are conspicuous. About 0.5 mile westward of Little River Island, on the prominent point on the south side of Little River, there is a conspicuous white house, and on the north side, about the same distance in, there is a conspicuous white building with a cupola in the village of Cutler.

(32) Little River is small but is easy of access and is an excellent harbor of refuge, sheltered from all winds and with depths of 12 to 30 feet, good holding ground. The channel leads northward of the light and has a depth of about 28 feet. The anchorage just inside of Little River Island is about 0.5 mile long and 0.2 mile wide. The harbor is never obstructed by ice sufficient to prevent vessels from entering.

(33) **Eastern Knubble** is the point on the northern side of the entrance to Little River. Just south of Eastern Knubble and 100 yards offshore is **Little River Ledge**, which uncovers. A buoy westward of the ledge marks the south edge of another small ledge. A ledge extends 100 yards from the south shore, just eastward of a prominent point 0.5 mile west-northwest of Little River Island. With these exceptions, there are no dangers in the harbor if the shores are given a berth of 100 yards. Numerous mooring piles are in the harbor and a lobster car is some distance off the large wharf.

(34) To enter Little River, pass northward of Little River Island, giving it a berth of 150 yards. Anchorage can be selected anywhere in midchannel inside the island. Small local craft anchor off the wharves in depths of 6

to 18 feet. The passage southward of Little River Island has a rocky bar across it with a least found depth of 10 feet in midchannel. This passage should not be used by strangers.

(35) **Cutler** is a village on the north shore of Little River. Many fishing vessels and lobster boats base at the harbor, and it is a popular yacht haven. There are two wharves with float landings; at the largest and westernmost, there is reported to be a depth of 10 feet. Gasoline, diesel fuel, ice and water are available at this wharf and gasoline at the other. Groceries and limited marine supplies are available. Good roads lead to East Machias and to Lubec. A **harbormaster** who supervises the moorings lives at the shore end of the western wharf.

(36) Between Little River and Little Machias Bay there are no features of importance. **House Cove**, 0.6 mile west of **Western Head** (44°38.7'N., 67°11.5'W.), the point on the south side of the entrance to Little River, is a small open bight extending somewhat behind **Great Head. Deer Island**, 1.7 miles west-southwest of Western Head, is a small island close to shore. About 0.3 mile offshore in this vicinity is a series of ledges on which depths as shoal as 13 feet are found.

(37) **Little Machias Bay**, 2.5 miles west of Little River Light (44°39'03"N., 67°11'32"W.), is not used for an anchorage as it is exposed to southerly and southeasterly winds and is close to Little River and Machias Bay, both excellent anchorages. Little Machias Bay is 0.6 mile wide at the entrance, wider inside, and about 2 miles long. **Black Ledges** are bare islets 4 feet and 24 feet high in the middle of the entrance to the bay with deep water close-to on both sides. **Long Ledge**, which uncovers 14 feet, is in the middle of the bay 1 mile inside the entrance. Above Long Ledge the bay is much obstructed by shoals and ledges, two of which are **Upper Ledge** and **Widows Ledge**; both uncover 13 feet. Ruins of fish weirs are reported in the bay. There are some houses on the shores of the bay, but no wharves except for small craft at high water. **North Cutler** is a settlement on the north shore of the bay.

(38) **Old Man** is a small but conspicuous rocky island 71 feet high and grassy on top, 0.5 mile southeastward of Cape Wash Island off the entrance to Little Machias Bay. Old Man is a good mark and may be safely approached as close as 400 yards.

(39) **Cape Wash** and **Cape Wash Island** are on the western side of the entrance to Little Machias Bay. Reefs extend about 0.2 mile south of the island. Just westward of Cape Wash are **Big Holly Cove** and **Little Holly Cove**, which are important only to small craft. **Cutler Peninsula** as far north as Sprague Neck and North Cutler is a naval reservation. Numerous radio towers of various heights on the reservation are conspicuous.

(40) **Cross Island**, 1.6 miles southwestward of Old Man, is the large wooded island on the east side of the main entrance to Machias Bay. A few unpainted shacks on low, flat, **Grassy Point**, the northern extremity of the island, are prominent when approaching Cross Island Narrows from westward. Small thickly wooded **Mink Island** is 0.2 mile off the northeast shore of Cross Island. From seaward, the most conspicuous mark on Cross Island is a skeleton lookout tower on a hill at the eastern end.

(41) **Double Head Shot Islands**, 0.5 mile eastward of Cross Island, are surrounded by ledges that extend out about 150 yards. The larger, southerly island is 84 feet high. The northerly island is less conspicuous. A depth of about 6 feet can be carried between the islands.

(42) **Scotch Island,** 100 yards eastward of Cross Island, is 61 feet high.

(43) **Cross Island Narrows** is a channel leading into Machias Bay northeast of Cross Island. **Thornton Point** and **Quaker Head** are on the northern side of the passage. The channel is much obstructed by rocks, covered or awash at various stages of the tide, and should not be used without local knowledge. Small craft can go through the narrows by closely following the electronic navigation chart (ENC) and not placing too much reliance on the floating aids, which are apt to drag from station during heavy weather. **Dogfish Rocks**, about 350 yards northward of Grassy Point, uncover 8 feet. A buoy is north of the rocks.

(44) Cross Island Narrows is seldom obstructed by ice in the winter and consequently **Northeast Harbor**, the cove southwestward of Mink Island, is much used as a winter anchorage by small fishing boats. In 1982, a ledge awash was reported on the west side of the harbor in about 44°36'57"N., 67°17'20"W.

(45)

Northwest Harbor to Tibbett Narrows

(46) **Northwest Harbor**, a bight in the northwestern shore of Cross Island, has depths of 21 to 54 feet but is little used as an anchorage. A cable area extends across the south half of the harbor.

(47) **Seal Cove Ledge**, extending 400 yards southwestward from Cross Island, has a least depth of 10 feet and is marked on the southwest side by a buoy.

(48) **Machias Bay**, about 22 miles southwestward of West Quoddy Head Light, is the approach to Machias River, and the towns of Machiasport and Machias. The bay is about 6 miles long and 1 to 3 miles wide, is easily entered day or night and affords well-sheltered anchorage for large vessels. The 2-mile-wide main entrance is between Cross Island on the east and Stone Island on the west. Sheep are kept on several of the islands in Machias and Englishman Bays during the summer.

(49) **Libby Islands**, in the middle of the entrance, are two flat grassy islands connected by a bare ledge. Sunken ledges extend about 300 yards off the southern end of the southwestern island and about the same distance off the eastern shores of both of the islands.

(50) **Libby Island Light** (44°34'06"N., 67°22'03"W.), 91 feet above the water, is shown from a 42-foot white tower on the southwestern island. A sound signal at the light is operated by keying the microphone five times

(67) **Holmes Bay**, a large bight in the northeastern part of Machias Bay and northeast of Hog Island, is shallow and has extensive reefs. A seafood-packing plant and wharf which dries at low water are on the north shore of the bay. A white schoolhouse on the point close eastward of the factory and a white church with belfry, about 0.6 mile westward, are conspicuous. Most of the bay dries at low water and is used only by fishermen.

(68) **Machias River**, which empties into the northwestern part of Machias Bay, has a narrow, winding channel leading through flats that are mostly bare at low water. The least depth in the channel to the town of Machiasport is about 17 feet. Above Machiasport, the channel has shoaled to less than 1 foot in the bend below Machias, but with local knowledge 2 to 3 feet can be carried to Machias.

(69) A fixed highway bridge with a clearance of 25 feet crosses the river about 2 miles below Machias. A power plant and milldam cross the river at **Machias Falls** at Machias.

(70) In severe winters, Machias River is closed to navigation by ice, and drift ice will sometimes fill the bay above Avery Rock. In ordinary winters the bay and river are open to Machiasport.

(71) **Machiasport** is a town on the west bank of the Machias River, 2.5 miles above the entrance. Prominent landmarks include a church spire, the cupola of the town hall, northward and below the spire, the tall metal stack of a cannery and a white church with belfry on the slope of the east bank of the river opposite the town. The **harbormaster** can be reached through the town clerk's office.

(72) The cannery wharf has 8 feet reported alongside. There is a boatyard with marine railways capable of hauling out craft up to 50 feet long. Hull and engine repairs are available. A good road connects Machiasport with U.S. Route 1, the main coastal highway, at Machias.

(73) **East Machias River**, which empties into Machias River from northeastward 1 mile above Machiasport, is practically bare at low water at **East Machias**, a village on the railroad 1.5 miles above the entrance. The channel is difficult and is little used except by small craft.

(74) **Machias** is a town of marked historical interest at the head of navigation on Machias River. There is no waterborne commerce. Most of the wharves are in ruins, dry at low water, and is unsafe to lay at anchor due to projecting underpinning. A town concrete ramp is on the west side of the entrance to **Middle River**. The entrance to the river is crossed by an earth-fill causeway with culvert openings. The Machias Boat Club is at Machias. Gasoline, limited marine supplies, banks, groceries, motels, a pharmacy, and hospital are available in the town.

(75) Craft bound for Machias Bay and River from the eastward should not attempt passage through Cross Island Narrows without local knowledge. With the aid of the electronic navigation chart they should have no trouble passing southward of Cross Island, and when clear of Seal Cove Ledge, the buoyed ledge at the southwest end of the island, shape up the bay for Avery Rock. Pass either side of the rock, preferably to the westward, passing eastward of Round Island, then head up for the river entrance, which is marked by a buoy on its southwestern side.

(76) Approaching from the southward and westward, vessels with the aid of the electronic navigation chart (ENC) may pass either side of Libby Islands and head up the bay for Avery Rock, keeping clear of Stone Island Ledge and unmarked Colbeth Rock. Pass westward of Avery Rock and proceed as in the preceding paragraph.

(77) Anchorage may be had anywhere between Avery Rock and Round Island, or eastward or northeastward of the latter at a distance not greater than 0.5 mile, in depths of 30 to 45 feet.

(78) The channel in Machias River is marked by buoys to about 0.8 mile below Machiasport and should be followed with the aid of the electronic navigation chart (ENC). The best time is at low water when the flats are visible and the channel more clearly defined. Small vessels often anchor in the channel off the wharves at Machiasport, or for a distance of about 0.5 mile southward of the wharves.

(79) Above Machiasport, the channel that leads between shoals which uncover is sometimes marked by stakes. Local knowledge is necessary to carry the best water, but strangers in small craft should have no trouble in going to Machias on a rising tide with the aid of the chart.

(80) Englishman and Chandler Bays form a large bight in the coast between Libby Islands and Head Harbor Island. **Roque Island**, 6 miles west of Libby Islands, and numerous smaller islands are in the middle of the bight. The bays join northward of Roque Island and form a good anchorage, with depths of 18 to 32 feet and good holding ground.

(81) **Englishman Bay**, northward of Roque Island, has numerous dangers, most of them unmarked, in the approach to the anchorage northward of Roque Island, but the buoyed channel is broad and is easily followed in daytime and in clear weather with the aid of the chart. The principal entrance to the bay from eastward is between Scabby Islands on the east and The Brothers on the west and affords a straight channel to Shoppee Island above which is the anchorage. The principal dangers are Scabby Island Ledge, Codhead Ledge, Halifax Island Reef and Boundary Ledges.

(82) The bay may be entered from Machias Bay through Foster Channel. Vessels from westward, bound to the anchorage at the head of Englishman Bay or to Chandler River, usually pass through Chandler Bay. Foster Channel and the adjacent islands have been discussed previously.

(83) **Scabby Islands**, on the eastern side of the main entrance to Englishman Bay, are grass covered. A 93-foot mound on the larger Scabby Island is the most prominent mark in approaching Foster Channel from westward. Sheep are kept on Scabby Islands and several other islands in the bay. A covered rock is 400 yards north of Scabby Islands. **Scabby Island Ledge**, awash at low

water and unmarked, is 250 yards southwestward of the islands.

(84) **Codhead Ledge**, awash at low water and marked by a buoy, is 1.5 miles northwestward of the Scabby Islands.

(85) **Shag Ledge**, 0.9 mile eastward of Codhead Ledge, has a low grass-covered islet 13 feet high on its western end. The northeast end of the ledge is covered only at high water, and the south end shelves off to 13 feet. An unmarked shoal covered 5 feet is midway between Codhead and Shag Ledges.

(86) **Pierson Ledge**, 0.4 mile northward of Shag Ledge and 350 yards west of **Point of Main**, uncovers 4 feet.

(87) **Hickey Island**, 0.7 mile northwestward of Shag Ledge and in the entrance to Little Kennebec Bay, is 38 feet high and partly wooded. Sheep are kept on the island. Small craft can find shelter in a small cove in the north side of the island. A rock awash at low water is 250 yards off the east side of the island, and a ledge extends 200 yards south of the island. About 250 yards north of the island is a shoal covered 7 feet.

(88) **Little Kennebec Bay**, which extends northward from the eastern part of the Englishman Bay, is of little commercial importance and is frequented mostly by fishermen. Good well-sheltered anchorage can be found in depths of 12 to 40 feet, soft bottom, northward of **Sea Wall Point**, taking care to avoid the cable area that extends northeast from it. However, this anchorage is seldom used, as nearby Machias Bay and Starboard Cove are much easier to access and are better anchorages. There are fish weirs in the upper part of the bay.

(89) **The Brothers**, grassy islands with rocky shores, are on the southwestern side of the main entrance to Englishman Bay. A bell buoy is off the northeast end of The Brothers. Shoaling to 3 feet extends 300 yards southwest from the islands.

(90) **Green Island**, 0.3 mile north of The Brothers, is grassy. A ledge, partly bare at low water, extends 0.3 mile eastward from the island and is marked by a buoy on its eastern side. A ledge, awash at high water, extends 200 yards westward from Green Island.

(91) **Brothers Passage**, between Green Island and The Brothers, has a depth of 27 feet in midchannel.

(92) **Pulpit Rock**, 1 mile westward of The Brothers, is a bare rocky islet. The southern and eastern sides should be given a berth of at least 300 yards, as a rock covered 7 feet is about 150 yards southeastward of it.

(93) **Jumper Ledge**, about 0.6 mile southward of Pulpit Rock and covered 5 feet, is marked by a buoy. An unmarked 28-foot spot is 0.8 mile east-southeastward of the ledge. **Misery Ledge**, covered 14 feet, about 0.6 mile south-southeastward, is also unmarked.

(94) **Halifax Island**, 0.8 mile northwestward of Green Island, is grass covered with rocky sides, and has a prominent mound at its western end. **Anguilla Island**, **Double Shot Island**, **Great Spruce Island** and **Little Spruce Island**, all westward of Halifax Island, are wooded. A rock, which uncovers 6 feet, is 350 yards southeastward of Halifax Island. A bar with depths of 15 to 26 feet extends from Halifax Island to Green Island. The current is reported to boil over the bar, and this passage should be used with caution. An unmarked 29-foot spot is 0.6 mile east of Halifax Island.

(95) A rock that uncovers 7 feet is 300 yards southward of Double Shot Island. **Shag Rock**, 500 yards eastward of Double Shot Island, is 14 feet high and bare.

(96) **Roque Island Harbor** is formed on the north and west by Roque Island and Lakeman and Bar Islands and on the south by Great Spruce Island and the islands extending eastward to Halifax Island. The harbor affords shelter from all winds and is used by small vessels, but the holding ground is not good except in spots. The best anchorage is in the western or northwestern parts of the harbor where the bottom is soft.

(97) The best entrance to Roque Island Harbor is northward of Halifax Island across a rocky reef with spots of 5 to 10 feet. Kelp is reported to be visible at low water on the reef.

(98) **Lakeman Island**, **Bar Island** and **Marsh Island** are off the east side of Roque Island and on the northern side of the entrance to the harbor.

(99) To enter Roque Island Harbor, follow the electronic navigation chart (ENC) carefully, keeping clear of unmarked dangers in the entrance. The principal dangers in Roque Island Harbor include a spot with 8 feet on it 0.2 mile off the middle of the north side of Great Spruce Island and unmarked **Seal Ledge**, which uncovers 5 feet and is 300 yards westward of the southern point at the eastern end of Roque Island.

(100) **Lakeman Harbor**, on the northeast end of Roque Island Harbor, is a good anchorage for small craft.

(101) The **Thorofare**, connecting the southwest side of Roque Island Harbor with Chandler Bay, has a depth of 8 feet and a reported covered rock in a narrow, crooked channel. The bottom is visible in the shoaler parts of the channel. The Thorofare is used considerably by small vessels with local knowledge. Strangers should avoid it.

(102) **Bunker Cove**, between Great Spruce and Little Spruce Islands and the Thorofare, is a good harbor and is used for winter storage of small craft. Small craft often anchor in its entrance just off the Thorofare.

(103) **Shoppee Island** (44°36.1'N., 67°29.4'W.), 2 miles northwest of Halifax Island, is flat and wooded except at the northwest end. Sheep are kept on the island. **Boundary Ledges** extend northeastward from Roque Island to within 0.4 mile of Shoppee Island. The rocks at the outer end of the ledges, bare at low water, are marked by a buoy.

(104) **Shoppee Point** is on the mainland 0.3 mile north of Shoppee Island. A private wharf with a 45-foot outer face is on the northwest end of the point; a depth of 9 feet is alongside the face.

(105) **Roque Bluffs** is a village 0.6 mile eastward of Shoppee Point. The mouth of **Englishman River**, southeast of Roque Bluffs, is crossed by a fixed highway bridge with a clearance of 3 feet. The river bares at low

water above the bridge. The yellow bluffs at the mouth of the river are prominent from the southward.

(106) **Shorey Cove**, a bight in the north shore of Roque Island, has depths of 7 to 13 feet. The cove is a good anchorage for small vessels but is little used. There are no dangers if the southern and western shores of the cove are given a berth of over 300 yards. There is a private landing in the cove on the east side of **Squire Point**, the northwest extremity of Roque Island.

(107) **Great Cove**, on the northeast side of Englishman Bay above Shoppee Point, has its entrance between **Pond Cove Island**, 1 mile northwestward of Shoppee Island, and **Little Ram Island**, 1 mile north of Roque Island. The entrance is clear of dangers with the exception of **Lapstone Ledge**, which uncovers 3 feet, 300 yards northward of Little Ram Island. Excellent anchorage may be had in depths of 10 to 17 feet, soft bottom, up to 0.6 mile westward or northwestward of Pond Cove Island. The part of the cove northward of Pond Cove Island is shoal. There are numerous fish weirs in the cove.

(108) **Chandler River**, at the head of Englishman Bay, is very narrow and crooked to the head of navigation at **Jonesboro**, a village about 3.5 miles above the mouth. The river is bare at low water at Jonesboro. There are several fish weirs in the river. The channel is unmarked, and strangers should not attempt to enter without local knowledge. Drafts of 14 feet have been taken to **Kilton Point**, about 1.5 miles above the mouth. The only traffic to Jonesboro consists of small boats engaged in fishing and clamming. On the northeast side of the river, 0.5 mile above Kilton Point, are the ruins of a clam factory. Small boats sometimes venture up the river to the highway bridge, above which are the ruins of a former dam; rapids are above the ruins.

(109) Ice closes Chandler River to Kilton Point from about December to April. It is reported that the river seldom freezes below **Deep Hole Point**, 0.3 mile southeastward of Kilton Point, but in extreme winters the bay is said to have been frozen as far as Roque Island.

(110) **Mason Bay**, extending westward from the head of Englishman Bay, is practically bare at low water and has many rocks inside the entrance. An unmarked channel with a depth of 13 feet leads to the entrance from southward. The northern entrance is foul. The small settlement of **Mason Bay** is on the south side of the bay just inside the entrance. Fish weirs are numerous in the vicinity.

(111) **Chandler Bay**, on the west side of Roque Island, extends northward from Mark Island to Squire Point where it joins Englishman Bay. A channel leads eastward of Ballast Island and around Squire Point into Englishman Bay and Chandler River. The principal dangers are buoyed, and the channel can be followed readily during daytime in clear weather with the aid of the chart. The bay is the approach from the westward to Chandler River and the anchorage in Englishman Bay and is the one generally used by strangers. There are no good anchorages in the bay until north of Roque Island. Care should be taken to avoid the unmarked 17-foot rocky patch in the southern entrance 0.7 mile westward of Little Spruce Island.

(112) There are numerous dangers off Chandler Bay. **Big Breaking Ledge**, a pinnacle awash at low water, is on the western side of the approach from sea, 0.2 mile eastward of Head Harbor Island. **Little Breaking Ledge**, covered 9 feet and marked by a lighted gong buoy 200 yards east-northeast of it, is about 0.4 mile north-northeastward of Big Breaking Ledge. **Black Rocks (Eastern Ledges)**, about 600 yards long, are 1.4 miles south of Great Spruce Island. At the easterly end of Black Rocks is a rock that uncovers 4 feet, and at the westerly end is a rock covered 3 feet. **East Black Rock**, 0.7 mile northeastward of Black Rocks uncovers 10 feet and is marked by a daybeacon and almost always by a breaker. **West Black Rock**, covered 7 feet, is 400 yards southwest of Black Rocks. A buoy is about 100 yards southwest of the rock.

(113) In Chandler Bay, **Great Spruce Ledges** are close to the south side of Great Spruce Island; the southernmost rock is 3 feet high. **Wallace Ledge**, 350 yards off the southwestern side of Little Spruce Island, uncovers 4 feet. **Ballast Island**, on the western side of the main channel through Chandler Bay and 0.9 mile west of Roque Island, is grassy; a buoy marks the eastern end of **Ballast Island Ledge**, which extends 200 yards eastward of the island. **Roque Island Ledge**, marked by a buoy at its western end, extends 700 yards off the west side of Squire Point. Just above Squire Point, **Great Bar**extends from the western shore of Chandler Bay for 0.5 mile. The buoy off the end of the bar marks the western side of the channel into Englishman Bay.

(114) Vessels bound into the anchorage northward of Roque Island, either through Englishman Bay or Chandler Bay, should have no difficulty, with the aid of the chart. Due regard should be given to the many unmarked dangers and the necessity for caution if the waters are unfamiliar.

(115) **Moosabec Reach** is the narrow passage west of Chandler Bay leading between the mainland on the north and the group of islands on the south from Chandler Bay to Tibbett Narrows. The passage is an important thorofare and is much used in the daytime by vessels drawing up to 10 feet; a draft of 21 feet can be taken through at high water. **Mark Island**, 123 feet high and heavily wooded, is the prominent guide to the eastern entrance, and Nash Island Light to the western approach. **Kelley Point** is the northeastern entrance point on the mainland, 1.2 miles west of Mark Island.

(116) The channel in Moosabec Reach is well marked. With local knowledge, a depth of about 11 feet can be carried. Vessels can readily follow the channel in daytime with the aid of the chart in clear weather, but strangers should not attempt passage at night. **Emms Rock Light 9** (44°31'44"N., 67°33'57"W.), 28 feet above the water, is shown from a gray skeleton tower with a square green daymark on the south side of the channel. The light is on the north end of the stone breakwater extending southeastward to **Nova Rocks**, about 0.25 mile south of Kelley Point. The breakwater uncovers about 3 feet. A

daybeacon is on **Gilchrist Rock**, 0.5 mile west of Mark Island; and another daybeacon is on **Snows Rock**, about 0.5 mile southwest of Kelley Point.

(117) A fixed highway bridge (under construction 2019) crosses Moosabec Reach from Jonesport to Beals, about 2 miles west of Kelley Point. Lights in the center of the span and on each side of the fender piling mark the navigation channel through the bridge.

(118) Vessels caught by fog in the reach anchor anywhere in the channel where there is swinging room and the bottom is soft.

(119) **Pilots** can usually be obtained from among the local fishermen.

(120) The tidal currents have considerable velocity in the dredged channel, particularly at the light on the stone jetty. The current floods to the eastward and ebbs to the westward. Back eddies form with the beginning of the tidal flow on each side of the bridge. This condition is caused by the solid fill causeway approaches to the bridge. The channel is reported to have been ice free in recent years.

(121) **Jonesport** is a fishing village on the north shore of Moosabec Reach. There is considerable trade in fish and lobsters. Boatbuilding is important, especially sport fishing boats in recent years.

(122) The bridge over the reach is prominent. Conspicuous on the north side of the reach are a green church spire in West Jonesport and a large red brick two-story schoolhouse and the spire of a church eastward of the bridge and just westward of **Sawyer Cove**. On the south side of the reach, the spires of two white churches in Beals are prominent.

(123) A dredged entrance channel leads north from Moosabec Reach into Sawyer Cove to an anchorage basin consisting of a north and south anchorage. In 2022, the controlling depths were 6.6 feet in the entrance channel, 6.7 feet in the south anchorage and 4.5 feet in the north anchorage. The basin is protected by a breakwater on the south side.

(124) **Quarantine, customs, immigration and agricultural quarantine.** (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(125) **Quarantine** is enforced in accordance with regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.)

(126) Jonesport is a **customs port of entry**.

(127) **Jonesport Coast Guard Station** is on the north side of Moosabec Reach, just westward of the bridge. The wharf at the station has a depth of 14 feet alongside. Gasoline and diesel fuel are available at two lobster piers. The pier just east of the bridge has a reported depth of 8 feet alongside its float landing; depths of 3 feet were reported alongside the float landing of the pier on the west side of Sawyer Cove. A boatyard in a cove about 0.7 mile eastward of the bridge builds boats up to 45 feet in length and can haul out on skids boats up to 40 feet in length for hull or engine repairs. Groceries and marine supplies are available in town. A town launching ramp, pier and float are on the west side of Sawyer Cove; depths of 4.5 feet are reported alongside. There are good roads to U.S. Route 1, the primary coastal highway.

(128) **Beals** is a village on the northern end of Beals Island, which is on the south side of Moosabec Reach opposite Jonesport. The main wharf at the northeastern extremity of the island, close eastward of the bridge, has 2 feet alongside. Diesel fuel, gasoline, provisions and limited marine supplies are available at the wharf.

(129) **Beals Harbor** is on the northwest side of Beals Island about 0.2 mile westward of the bridge. The fish wharves in the harbor bare at low water. In 2023, the anchorage basin had depths of 5 to 10 feet.

(130) There are several boatyards on Beals Island where fishing craft up to 65 feet in length are built.

(131) **Indian River** and **West River**, extending northward from near the western end of Moosabec Reach, have crooked unmarked channels fringed by rocks. The rivers are frequented only by local fishermen. There are no landings except for small craft at high water.

(132) **Wohoa Bay**, 3 miles west of Jonesport, is the large bay northeast of **Moose Neck** and south of Bickford Point. Good anchorage is reported between **Carrying Place Island**, 300 yards northeast of Moose Neck, and Fessenden Ledge.

(133) Several islands are adjacent to the usual route through the western part of Moosabec Reach. **Pomp Island**, 1.2 miles west of Beals Island, is wooded. A rock awash and marked by a daybeacon is about 200 yards west of the island. **Hardwood Island**, 0.5 west of Pomp Island, also is wooded and has a house on the north end and a prominent quarry on the south side. A 13-foot shoal is in midchannel between this island and **Fessenden Ledge**, 0.5 mile northwestward. Fessenden Ledge uncovers 1 foot and is marked by a lighted bell buoy. **Shabbit Island Ledge**, 0.5 mile westward of Hardwood Island, uncovers 11 feet and is marked by a buoy. **Shabbit Island**, 1.1 miles southwestward of Hardwood Island, is low and wooded in the center and has several small cottages on it; a buoy is 200 yards west of the island.

(134) **Head Harbor** is between **Head Harbor Island,** the easternmost island of the group, and **Steele Harbor Island**. The harbor affords sheltered anchorage in depths of 14 to 20 feet. It is small and has unmarked rocks bare at low water on both sides of the entrance and anchorage. Strangers are advised to use caution in this area. The rocks of **Man Island** and **Black Head**, the eastern side of the entrance, are dark, while those on the western side are light in color.

(135) **Mistake Harbor**, westward of Steele Harbor Island, is small but affords secure anchorage in depths of 20 to 42 feet. The entrance from southward is through **Main Channel Way**, a deep but narrow channel leading between Steele Harbor Island on the northeast and **Mistake Island** and **Knight Island** on the southwest. With care, the harbor may also be entered through Mud Hole Channel.

(136) **Moose Peak Light** (44°28'28"N., 67°31'55"W.), 72 feet above the water, is shown from a 57-foot white tower on the southeastern extremity of Mistake Island. A mariner radio activated sound signal near the light is initiated by keying the microphone five times on VHF-FM channel 83A.

(137) To enter Mistake Harbor through Main Channel Way, give the south end of Steele Harbor Island a berth of about 0.3 mile when southeastward of it and enter in midchannel. When through the narrowest part of the channel, select anchorage in depths of 20 to 42 feet between the northwest end of Knight Island and the buoy 0.5 mile northwestward of it, taking care to avoid a cable area that extends northwesterly from Knight Island. The buoy, in the middle of Mistake Harbor, marks a rock that uncovers 1 foot.

(138) **Green Island** is the largest of the islands and rocks extending northwestward along the south side of Mistake Harbor. Rocks awash at low water are 300 yards northward and over 400 yards northwestward of the island. Two rocks covered 4 feet are 600 yards northwestward of the island.

(139) **Eastern Bay** is northward of Mistake Harbor between Head Harbor and Great Wass Islands; thorough local knowledge is required to navigate the bay.

(140) **Mud Hole Channel**, 0.5 mile westward of Moose Peak Light, leads northwestward to Mud Hole, which is a narrow cove in Great Wass Island, and to the western end of Mistake Harbor. Good anchorage is available for small vessels at the entrance to Sand Cove and Mud Hole, in depths of 14 to 30 feet, soft bottom.

(141) **Black Ledges**, on the southwest side of Mud Hole Channel and extending about 1 mile in a southeasterly direction from **Little Cape Point** on Great Wass Island, consist of a group of rocks and ledges, some of which uncover 10 feet. **Channel Rock**, the southeasternmost of the Black Ledges, uncovers 14 feet. **Freeman Rock**, 600 yards southwestward of Channel Rock, is bare and about 40 feet high.

(142) The principal dangers on the northeast side of Mud Hole Channel are a rock which uncovers 5 feet, 150 yards off the southwest side of the island on the northeast side of the entrance, and the shoals west of Green Island.

(143) Local knowledge is advisable for passage through Mud Hole Channel from Mistake Harbor to the sea or northward through Eastern Bay to Moosabec Reach as the dangers are numerous and unmarked. Passage can be made in daytime only with the aid of the chart.

(144) In 1968, there were reported at least six abandoned, broken-off weirs on the west side of Eastern Bay from Mink Island north, most of them unbrushed and visible only at near low water.

(145) **Pig Island Gut Channel**, a dredged channel marked by buoys, leads from the head of Eastern Bay through Pig Island Gut and Alley Bay into Moosabec Reach. A dredged anchorage is adjacent to the channel, south of Pig Island. A rock awash is close southwest of the channel in about 44°31'04"N., 67°35'38"W.; several rocks covered about 2 feet are close eastward of the rock awash. In 1984, rocks awash were reported along the northeast channel limit between Buoys 12 and 14.

(146) **Great Wass Island**, 1.5 miles westward of Steele Harbor Island and southward of Jonesport, has a number of coves that are frequented by small craft. A lobster pound is on the west side of the island in **Black Duck Cove**, about 2.0 miles northward of **Pond Point**, the southernmost tip of the island. During the spring and fall, gasoline and diesel fuel are available at the pound's wharf which bares at low water. **Sand Cove**, on the east side of the island about 2.5 miles northward of Pond Point, is used by fishermen; **Mud Hole**, immediately southward of Sand Cove, is occasionally used by fishermen for winter haul-out storage. There are a number of boatyards on the island which build boats up to 45 feet in length. **Red Head**, on the southern side of the island, appears reddish from offshore.

(147) **Western Bay**, westward of Great Wass Island, has numerous groups of islands which lie mostly in a north-and-south direction. Between the groups are passages leading to the western end of Moosabec Reach that are used by vessels with local knowledge.

(148) **Crumple Island**, 0.4 mile southwest of Pond Point, is a high, bare, rocky island with several nubbles. **Fisherman Island**, 0.2 mile northwestward of Crumple Island, and **Browney Island**, 1.2 miles northwestward of Crumple Island, have rocks and ledges between them. The passage between Great Wass Island and Crumple, Fisherman and Browney Islands has numerous unmarked covered rocks. A line of ledges and rocks extends from Browney Island to Great Wass Island. There is a narrow channel with a depth of about 17 feet through these ledges. Strangers, except in small craft drawing less than 5 feet, should not attempt this passage.

(149) **Egg Rock**, a bare rocky islet 15 feet high, is 1.3 miles west-southwestward of Crumple Island. Between the two are numerous rocks and ledges, including **Curlew Rock**, **Green Rock** and **Seal Rock**. **Drown Boys Ledges**, awash at low water, are about 0.7 mile northward of Egg Rock. **Seahorse Rock**, which uncovers 5 feet, is 0.4 mile southwestward of Egg Rock and is marked by a lighted bell buoy about 0.2 mile to the southwest.

(150) **Outer Sand Island**, 44 feet high, is 2.4 miles northwest of Egg Rock. **Inner Sand Island**, 54 feet high, is 0.6 mile north of Outer Sand Island. Both are wooded. **Stanley Ledge**, 0.3 mile southward of Outer Sand Island, is 5 feet high, and **Batson Ledges**, 0.4 mile eastward of Inner Sand Island, are 22 feet high.

(151) **Drisko Island**,**Little Drisko Island** and **Stevens Island**, northward of Sand Islands, are wooded. **Flat Island**, 1.3 miles west of Outer Sand Island, and **Green Island**, 1.5 miles north of Flat Island, are comparatively low and covered with grass. **Plummer Island**, 0.4 mile east of Green Island, is 65 feet high and wooded.

(152) **Black Rock** is a 7-foot-high bare rock, 1.3 miles southward of Flat Island and 2.1 miles southeastward

of Nash Island. An unmarked 35-foot shoal is 2.7 miles southward of Black Rock.

(153) The passage between Seahorse Rock, Drown Boys Ledges, and Ram Island on the east and the Sand and Drisko Islands on the west has a broad channel in its southern part, although there are unmarked dangers on either side. The northern end of the passage on either side of Hardwood Island is foul, and the passages should be used only with local knowledge.

(154) The passage westward of the Sand and Drisko Islands and Shabbit Island and eastward of Black Rock, Flat Island and Plummer Island is comparatively clear, except for 28-foot spots 0.6 mile east and 0.3 mile northeast of Flat Island. Another comparatively clear passage is westward of Black Rock, Flat Island and Green Island and eastward of Big Nash Island. Both passages can be used in the daytime in clear weather with the assistance of the chart.

(155) **Tibbett Narrows** is a narrow buoyed channel southward of Moose Neck on the sheltered inshore route for vessels westbound from Moosabec Reach. It is about 1 mile southwest of the western entrance of the reach. It is about 150 yards wide at its narrowest part and has a depth of 36 feet. Wooded **Tibbett Island** is on its northwestern side, and wooded **Ram Island** is on its southeastern side. An unmarked 25-foot spot is off its eastern entrance, and a 28-foot spot is on the north side of the channel about 250 yards off Tibbett Island.

(156)

Eastern Harbor to Jo Leighton Ground

(157) **Eastern Harbor** (44°30.5'N., 67°43.7'W.), on the west side of **Moose Neck**, is a secure anchorage for small vessels. The buoyed entrance is easily navigated in the daytime. The harbor has extensive flats and ledges, between which is a channel 200 yards or more wide. Fish weirs and fish weir ruins, partly covered at high water, are on both sides of the entrance.

(158) The anchorage with the best swinging room is in depths of 18 to 22 feet in midchannel, about 0.4 mile inside **Eastern Pitch**, the point on the west side of the entrance. Craft of less than 9-foot draft can anchor in depths of 9 to 15 feet in **Otter Cove**, which makes into Moose Neck, 0.6 mile northeastward of Eastern Pitch. Another good anchorage spot, in depths of 8 to 12 feet, is 200 yards northwestward of the wharf on the east side of the harbor, about 0.9 mile above the entrance.

(159) There are several rocky ledges that uncover in the northeastern part of the harbor. The flats are soft mud in places, and small craft sometimes are beached on them. A reef which shows well at low water extends 400 yards southward and southwestward from the point on the east side of the entrance. It is marked on its west side by a buoy.

(160) The wharf of a seafood processing plant is on the east side of Eastern Harbor, about 0.9 mile above the entrance at the village of **South Addison**. Depths of 5 feet are reported alongside the wharf. Gasoline, water and limited marine supplies are available at the wharf or in the village. Engine and hull repairmen are available in the village in an emergency. Boats are usually grounded out for hull repairs.

(161) The approach is clear to Eastern Harbor, between Tibbett Island and Ladle Ledges, if these islands are given a berth of over 400 yards. From westward the approach is clear between the daybeacon south of Norton Island and Pot Rock. The approach from Moosabec Reach is through Tibbett Narrows. Enter the harbor midway between the buoys at the entrance, staying midchannel and keeping a sharp lookout for an old fish weir on the eastern side of the entrance.

(162) Pleasant, Narraguagus, Harrington and Pigeon Hill Bays, which indent the coast between Nash Island on the east and Petit Manan Island on the west, are the approaches to the villages of Addison, Harrington, Milbridge and Cherryfield, all on tributaries of the bays. These waters are frequented mostly by local fishing boats. The bays are separated by islands and rocks, through which are several thorofares.

(163) Good anchorage can be found in Pleasant, Harrington and Narraguagus Bays, the latter being used much as a harbor of refuge.

(164) From December to April, ice usually forms on Pleasant River and Harrington River to their mouths, and very frequently on Harrington Bay. Ice seldom obstructs navigation in Narraguagus River except in January and February, during which time the river usually is frozen to the mouth. In ordinary winters, the ice that forms in these bays goes out with the tides.

(165) **Pleasant Bay**, 1.2 miles westward of Eastern Harbor and 6.5 miles west of Jonesport, is a secure anchorage and is easily entered in the daytime. **Nash Island** and **Big Nash Island**, on the eastern side of the entrance to Pleasant Bay, are grassy. The tower of the former lighthouse on the west side of Nash Island is reported to be prominent. A fairway lighted whistle buoy is about 0.5 mile west of Nash Island. A ledge, the southern end of which uncovers 10 feet, extends about 500 yards southward from Nash Island.

(166) There are numerous islands and ledges in Pleasant Bay, but the important dangers are marked by buoys. A channel with a least width of 0.5 mile, and with depths of 36 feet or more, leads up the bay to the anchorages.

(167) Anchorage is available in depths of 30 to 36 feet westward of **Nightcap Island**, a grassy island with a few bushes on its north side 3.4 miles north of Nash Island, and southward of **Barton Ledge**, a buoyed danger awash at low water 0.4 mile northwest of Nightcap Island. A better anchorage, and the one used most frequently, is in depths of 14 to 18 feet southeastward and eastward of **Birch Islands**, wooded islands 0.7 mile north of Nightcap Island.

(168) No difficulty should be experienced approaching Pleasant Bay anchorage during daytime in clear weather with the aid of the chart. At other times it would not be

prudent for strangers to pass northward of the vicinity of Nash Island, as there are no lighted aids in the bay. If need for shelter demands it, craft can proceed on a **344°** course for 2.2 miles from the lighted whistle buoy 0.5 mile westward of Nash Island to a temporary anchorage in 60 feet in the middle of Pleasant Bay.

(169) **The Ladle** is about 0.9 mile north of Big Nash Island and 0.7 mile southwestward of the entrance to Eastern Harbor. A grass-covered symmetrical 78-foot mound at the northern end is conspicuous. **Pot Rock**, 0.4 mile southwestward of The Ladle, is 6 feet high and bare.

(170) **Flint Island**, on the west side of the entrance to Pleasant Bay and 1.8 miles northwest of Nash Island, is 75 feet high and wooded. The island is a private wildlife sanctuary. Flint Island should be given a berth of 0.4 mile on its eastern and southern sides. **Coles Ledge**, 0.3 mile eastward of the island, is awash at low water and marked on its eastern side by a buoy.

(171) **Flint Island Narrows** is a deep passage leading from Pleasant Bay to Narraguagus Bay between Flint Island and **Dyer Island**, 0.4 mile northward. The passage is used principally by fishing boats. The channel has a width of only about 200 yards at its narrowest part, but in clear weather with the aid of the chart little difficulty should be experienced in its passage, having due regard for the ledges that make out from the south shore of Dyer Island. A buoy, on the northern edge of the ledges making out from the northern end of Flint Island, marks the eastern entrance to the narrows.

(172) **Norton Island**, on the east side of Pleasant Bay and 2 miles north of Nash Island, is grassy. **Norton Island Ledge**, 400 yards westward of Norton Island, is 5 feet high in spots and unmarked. The reef extending 400 yards southward from the island is partly bare at high water and is marked by a buoy at its outer end.

(173) The channel between Norton Island and Cape Split, 0.4 mile eastward, is obstructed inside the entrance by a small unmarked rocky patch, covered 9 feet. Northward of Norton Island the channel is restricted by rocks and ledges.

(174) **Bay Ledge**, 0.7 mile south of **Willard Point** in the northwestern part of Pleasant Bay, uncovers 10 feet. Unmarked 13- and 15-foot rocky patches are 500 yards northeastward and northward of it.

(175) **Nightcap Ledge**, unmarked and extending southward from Nightcap Island, uncovers about 5 feet at the inner end.

(176) **Bunker Ledge**, 0.3 mile westward of the southern Birch Island, uncovers about 5 feet and is marked on its eastern side by a buoy.

(177) **Raspberry Island** and **Mink Island**, northward of Birch Islands, are grassy. **Fort Island** is 0.5 mile north of Raspberry Island.

(178) **Pleasant River** empties into Pleasant Bay from the northward. The channel is marked by buoys and a daybeacon from just below **Look Point** to within 0.7 mile of Addison. The river is seldom used except by fishermen, and the once extensive trade in lumber ceased many years ago. Passage up the river is suitable for small craft only, except with local knowledge, as the river is reported to have shoaled in many places. Above Addison, the river is still navigable to Columbia Falls but is seldom used except by small craft which can clear the bridge just above Addison. Ice obstructs the river from December to April.

(179) The **West Branch** of the river at Addison is closed by highway fill and a bridge-dam that is fitted with clapper valves that prevent the flood tide entering the branch but allow it to drain on the ebb and during freshets. A small clam-processing plant is on the east side of the entrance to West Branch.

(180) **Addison** is a village about 5 miles above the mouth of Pleasant River. There is no waterborne traffic except some fishing vessels. Gasoline and some supplies can be obtained from a general store with a service station. A fixed highway bridge crosses the river just above Addison with a clearance of 11 feet.

(181) **Columbia Falls** is a village with a small shingle mill 5 miles above Addison. The falls and dam at the mill are the head of navigation on the river. The village has railway and bus service, and gasoline and groceries can be obtained here.

(182) **Harrington Bay** is separated from the upper part of Pleasant Bay on the east by **Ripley Neck** and from Narraguagus Bay on the west by **Foster Island**. The bay extends about 2.5 miles in a northerly direction to Harrington River. Good anchorage may be found in Harrington Bay in depths of 30 to 47 feet. The bay and anchorage are seldom used except by local vessels.

(183) **Strout Island**, in the middle of the entrance to Harrington Bay from Pleasant Bay, is wooded. **Strout Island Ledges**, southeastward of Strout Island, uncover 9 feet; the south end of the ledges is marked by a buoy. A rock awash at low water and marked by a buoy is 0.5 mile southeastward of Strout Island. **Shag Islet**, 0.3 mile northward of the island, uncovers 11 feet. The channel between Strout Island and **Otter Island**, 0.2 mile southward, is called **Strout Island Narrows**.

(184) With the aid of the chart, small craft approaching and entering Harrington Bay should have no trouble, but larger vessels and strangers should not attempt it without local knowledge.

(185) **Dyer Island Narrows** is a buoyed passage between Dyer Island and Foster Island that connects Harrington Bay and Narraguagus Bay; the passage has a depth of 8 feet. **Note** that the system of marking is from Harrington Bay to Narraguagus Bay; if entering the channel from Narraguagus Bay, odd numbers are on the right. There are numerous dangers close to the channel, and the buoys must be followed closely. Local boatmen report dangers in this channel and that a boat drawing 3 feet has struck boulders. Strangers should not attempt passage through Dyer Island Narrows without local knowledge.

(186) **Chamberly Island**, 2.5 miles north of Strout Island, is at the head of Harrington Bay.

(187) **Flat Bay** and **Mill River** extending northwestward from the head of Harrington Bay are mostly bare at low water and are seldom used except by boats engaged in clamming.

(188) **Harrington River**, which empties into the head of Harrington Bay from northward, has a narrow, crooked, unmarked channel. The river shoals to a depth of 1 foot near the town of **Harrington**, at the head of navigation, about 4 miles above the mouth. A fixed highway bridge with a clearance of 12 feet crosses the river about 0.4 mile below Harrington. An overhead power cable with a clearance of 48 feet crosses the river on poles, about 0.3 mile below the town. A telephone cable crosses on the same poles about 20 feet below the power cable.

(189) The channel above **Nash Point**, 2.4 miles north of Chamberly Island, is narrow and tortuous. Motorboats and small fishing boats run up to the town at high water, but there are no arrangements for supplying them with gasoline and water. There is an inactive blueberry cannery and wharf on the north bank of the river, about 0.3 mile above the bridge, which is reported to have 2 feet alongside. The town has bus service. Ice forms in the river and bay between December and April as far down as Ripley Neck.

(190) The **Harrington River Seed Mussel Conservation Area**, a Marine Protected Area (MPA), includes the waters of Harrington River and Flat Bay.

(191) **Narraguagus Bay** is about 6.5 miles northward of Petit Manan Island and 3.5 miles westward of Nash Island. The principal dangers in the channel are marked. The lighted whistle buoy west of Nash Island, the light on Petit Manan Island, and the lighted bell buoy eastward of Pond Island are guides for the entrance. The bay is connected with Pleasant Bay by Flint Island Narrows and with Harrington Bay by Dyer Island Narrows, both of which have been described previously.

(192) The lower part of Narraguagus Bay is a well-sheltered anchorage, used as a harbor of refuge in all seasons by vessels up to 18-foot draft. The usual anchorage for vessels seeking shelter in the bay is between Trafton Island and Lower Middle Ground buoy in depths of 21 feet, soft bottom. Vessels of 10-foot draft or less sometimes anchor in depths of 14 to 17 feet north of Trafton Island, about midway between it and the daybeacon on Trafton Halftide Ledge. Vessels bound up to Milbridge anchor in depths of 12 to 16 feet about 0.7 mile east of Mitchell Point, with the cannery pier bearing **274°**. Good anchorage in a depth of 24 feet, soft bottom, may be found about 0.4 mile northeastward of the daybeacon on Trafton Halftide Ledge.

(193) Vessels should experience no trouble in approaching the anchorage in Narraguagus Bay with the aid of the chart in daytime and in clear weather but should not go above Pond Island at night or above the anchorages in daytime without local knowledge.

(194) The **Narraguagus Bay Seed Mussel Conservation Area**, a Marine Protected Area (MPA), includes the waters of Narraguagus Bay.

(195) **Bois Bubert Island**, 5 miles west-southwestward of Nash Island, is on the western side of the approach to Narraguagus Bay. The high wooded island has several cottages along its western shore. **Jerry Ledge**, off the southeast end of Bois Bubert Island, uncovers 2 feet. A 15-foot spot just south of this ledge is marked by a buoy on its southern side.

(196) **Jordans Delight Ledge**, in the middle of the entrance to Narraguagus Bay and 1.6 miles east of Bois Bubert Island, has a least depth of 2 feet. A 5-foot spot, marked by a buoy, is at the south end of the ledge. **Black Ledge**, at the northeast end of Jordans Delight Ledge, uncovers 11 feet. The ground in this vicinity is very broken.

(197) **Mackerel Rock**, unmarked and covered 10 feet, is 0.6 mile north-northeastward from Black Ledge. The rock is slightly west of the range made by Black Ledge and Petit Manan Light. **Jordans Delight**, 3.5 miles west of Nash Island, is a rocky island 91 feet high and sparsely wooded on top. **Pond Island**, 3.7 miles west of Nash Island, shows from southward as a bare conical hill 158 feet high. Narraguagus abandoned lighthouse tower is on the eastern side of the island. This 18-foot tower, white in color and connected with a dwelling, is a conspicuous landmark easily seen to the eastward and southward but cannot be seen to west and northward of it, being hidden by the trees and high land behind it. A cupola of a house near the north end of the island is conspicuous to northward. A lighted bell buoy is 0.3 mile eastward of the tower.

(198) **Douglas Islands**, between Pond Island and Bois Bubert Island, are wooded. **Douglas Island Harbor** is north of the Douglas Islands and west of Pond Island. The harbor has anchorage in depths of 24 to 35 feet but is seldom used, since better anchorage is available above Trafton Island. Considerable sea makes into the harbor in heavy southerly weather. The harbor is clear except at its southwest end where **Douglas Island Ledge**, which uncovers 3 feet in places, extends 350 yards northwestward from the middle Douglas Island.

(199) Entering Douglas Island Harbor northward of Pond Island, pass northward and at least 0.3 mile westward of the buoy 0.3 mile north of the island. The harbor may also be entered from southward between the islands; the best channel is between the wooded island near the southwest end of Pond Island on the east and the easternmost of the three Douglas Islands and the 6-foot high bare rock 250 yards southward from it on the west. Entering by this passage, avoid the aquaculture site located immediately northwest of the wooded island just southwest of Pond Island and two rocks, which uncover, 125 yards southwestward of the wooded islet 250 yards off the middle of the west side of Pond Island.

(200) A narrow unmarked channel leads from Douglas Island Harbor into the head of Pigeon Hill Bay. This channel is bordered on both sides by rocks covered and awash. The best water leads about 100 yards northward of **Currant Island**, which is wooded in the center and 0.3 mile north of Bois Bubert Island. While this channel is

much used by local fishermen who follow the fish weirs as guides during the summer, it is not recommended for strangers.

(201) **Shipstern Island**, 0.5 mile west of Flint Island and on the eastern side at the entrance of Narraguagus Bay, is 95 feet high, round and wooded and has rocky bluffs on its south side.

(202) **Tommy Island**, 0.8 mile northwest of Shipstern Island, is low and sparsely wooded. **Western Reef**, extending 0.4 mile southward from Tommy Island, has a bare rock 2 feet high on it and is marked by a buoy off its south end.

(203) **Trafton Island**, 0.5 mile west of Tommy Island, is 84 feet high and wooded. There is a good channel on either side of Trafton Island. **Trafton Island Ledge**, in the middle of the entrance to the cove on the north side of Trafton Island, is a bare rock. **Trafton Halftide Ledge**, 0.6 mile northward of Trafton Island, is partly bare at low water and marked by a daybeacon. In 1986, submerged piles were reported close east of the ledge.

(204) **Lower Middle Ground**, an extensive shoal on the west side of the bay westward of Trafton Halftide Ledge, is covered 2 feet and marked on its eastern side by a buoy. A private pier with float landing is on the point on the south side of **Stover Cove**, which indents the shore westward of Lower Middle Ground and about 0.8 mile southward of Mitchell Point (44°30.6'N., 67°51.4'W.). A reef extends from the point close southeastward of the pier. There is a lobster pound and wharf, dry at low water, on the south side of **Smith Cove**, immediately northward of Stover Cove. Gasoline and diesel fuel are piped to the wharf. Water is reported to be available.

(205) **Wyman** is a village on the western side of Narraguagus Bay, 1.8 miles northwestward of Trafton Island and westward of **Mitchell Point**. A 300-foot fish cannery pier, with a reported depth of 6 feet at its outer end, is on Mitchell Point. The aluminum painted metal stack at the cannery is conspicuous. Water is piped to the wharf; gasoline and diesel fuel can be delivered to the wharf by truck. A good highway connects with Milbridge.

(206) **Narraguagus River** empties into Narraguagus Bay from the northward. A dredged channel, marked by buoys, leads from the bay to the river, thence to an anchorage off Mitchell Point, thence to two anchorages in the river off Milbridge and thence to the Milbridge town wharf, on the east side of the river about 0.2 mile eastward of the bridge. (See Notice to Mariners and latest editions of charts for controlling depths.) The channel is narrow and crooked and leads between flats that bare at low water. Strangers should navigate the river on a rising tide. Old fish weirs, only part of which show at low water, are on either side of the channel just above the fairway buoy off the entrance. Care should be taken in entering to have this buoy close aboard before heading up for the channel entrance, as the weirs are reported to be not visible at high water.

(207) **Milbridge** is a small town on the west side of the river about 2 miles northward of Wyman. A cannery is just inside the entrance to Wallace Creek, which empties into the west side of Narraguagus River over the mudflats, about 1.6 miles above Mitchell Point. The cannery wharf dries at low water, and the narrow crooked channel leading to it is marked by stakes.

(208) The Milbridge town wharf and float are on the east side of the river, about 0.2 mile east-southeast of the bridge. Depths of 8 to 10 feet are reported alongside the wharf, but less than a foot at the float. The wharf is used by fishermen to unload their catch for the cannery on Wallace Creek. Diesel fuel is available by truck; no other facilities are available at the wharf. A small-craft launching ramp is just west of the wharf.

(209) Care must be taken in the river channel when passing the area between Buoys 13 and 15 to clear the ruins of several piers and a wreck, which are on the west side of the river and extend from the shore to the western edge of the channel. The ruins consist primarily of submerged and partially submerged piling and timbers. Part of the lower ruin, which was formerly known as the upper steamboat wharf, is used occasionally by fishermen to unload their catch. A medical center is at Milbridge, and gasoline, diesel fuel, water, groceries and limited marine supplies are available. U.S. Route 1, the primary coastal highway, passes through the town, and bus service is available.

(210) The Harold West (US Route 1) Bridge crosses the river at Milbridge on a highway fill and two bridges. The bridge over the south channel has a fixed span with a horizontal clearance of 40 feet and a vertical clearance of 7 feet. The bridge over the north channel has a fixed span with a horizontal clearance of 117 feet and a vertical clearance of 5 feet. An overhead telephone cable with a clearance of 25 feet crosses the south channel at the bridge. The south channel is used by craft when taking shelter in the anchorage above the bridge during heavy weather. There is reported to be a depth of 15 feet in the anchorage just above the bridge.

(211) The Narraguagus River above Milbridge as far as the head of navigation at Cherryfield, about 4 miles upstream, is mostly full of boulders that uncover. It is seldom used by any craft. Small boats drawing 3 feet are reported taken to Cherryfield at high water, and then only with local knowledge.

(212) **Cherryfield** is a town at the head of navigation on Narraguagus River. It has railway freight connections and bus service. The town is a center of salmon sport fishing. The primary industries in the town are a lobster trap plant, a lumber mill and two blueberry canneries. An overhead telephone cable crossing the river a short distance below the highway bridge has a clearance of 21 feet.

(213) **Pigeon Hill Bay** (44°26'N., 67°53'W.) is entered eastward of Petit Manan Bar and northward of Petit Manan Island. The bay is not difficult to enter by day with the aid of the chart, but caution should be exercised to avoid the partly submerged fish weirs and fish weir ruins along the shores of the bay. One particularly dangerous fish weir is reported in the bay, about 0.2 mile eastward of **Chitman Point** (44°27.3'N., 67°52.7'W.). The bay

affords good anchorage in depths of 12 to 24 feet but is seldom used except by local fishermen. The channel is unmarked except at the southern entrance. The small settlement of **Pigeon Hill** is on the west side of the bay, about 0.7 mile southward of Chitman Point. Small boats anchor on the west side of the bay off the settlement. There is a lobster pound and float, which bares at low water, about 0.6 mile northwestward of Chitman Point; gasoline and diesel fuel are available at the float.

(214) **Whale Ledge**, 1.4 miles south of Bois Bubert Island and on the east side of the entrance to Pigeon Hill Bay, uncovers about 6 feet. A buoy marks a 13-foot rocky patch 300 yards southward of the ledge.

(215) **Egg Rock**, 1 mile south of Bois Bubert Island, shows at low water as a large ledge of dark boulders, with several high parts that are always out of the water. There is a narrow unmarked channel between Egg Rock and Bois Bubert Island.

(216) **Gull Rocks**, extending 0.2 mile off the west shore of Bois Bubert Island 1.4 miles northward of Egg Rock, consist of a large outer ledge which uncovers 6 feet and smaller ledges inshore. These rocks can be avoided by keeping in midchannel.

(217) **Pigeon Hill**, conspicuous for some distance at sea, is a 317-foot-high, bare-topped hill on the western shore of the bay westward of Chitman Point.

(218) The thorofare connecting the head of Pigeon Hill Bay with Douglas Island Harbor has been described previously.

(219) **Petit Manan Island**, 7.7 miles southwestward of Nash Island, is low and bare and marked by several buildings. **Petit Manan Light** (44°22'03"N., 67°51'52"W.), 123 feet above the water, is shown from a 109-foot gray granite tower on the east side of the island. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A. **Petit Manan Reef**, marked by a buoy at its southern end, extends about 0.4 mile southward from the island.

(220) **Petit Manan Bar** extends from Petit Manan Island to **Petit Manan Point** on the mainland. The bar consists of ledges and large boulders through which is a channel that can be used by small vessels when the sea is smooth. There is 8 feet in the channel which is 1.3 miles northwestward of Petit Manan Light and 0.9 mile southeastward of Petit Manan Point.

(221) **Inner Bar** is another channel across the bar, 0.4 mile southeastward of Petit Manan Point; it is used by small local boats at all stages of the tide, but it is unmarked and difficult and should not be attempted by strangers. The sea breaks along the whole length of the bar with a swell or in heavy weather.

(222) The tidal currents set over the bar with considerable velocity, the flood northeastward and the ebb southwestward.

(223) **Green Island** is 0.4 mile northwest of Petit Manan Island. **Petit Manan Pool**, on the southeast side of Green Island and north of Petit Manan Island, is a small-boat harbor. The pool is bare at low water, but the bottom is soft and boats ground out at low water.

(224) **Simms Rock**, 1.7 miles south-southeastward of Petit Manan Light, is covered 6 feet and marked on its northwest side by a bell buoy. A rock covered 20 feet, about 0.2 mile southeastward, is unmarked.

(225) **Southeast Rock**, 3.2 miles southeastward of Petit Manan Light, is covered 5 feet and is marked by a lighted whistle buoy. An unmarked 41-foot shoal is 1.7 miles southwestward of Southeast Rock.

(226) **Jackson Ledge**, covered 23 feet, is an unmarked danger 2.6 miles east of Petit Manan Light.

(227) **Tibbett Rock**, 3.6 miles east-northeastward of Petit Manan Light, is covered 12 feet, and another rock, about 0.8 mile southeastward of it, is covered 10 feet; both rocks are marked by buoys.

(228) **Jo Leighton Ground**, an unmarked danger covered 15 feet, is 2.1 miles northeastward of Petit Manan Light.

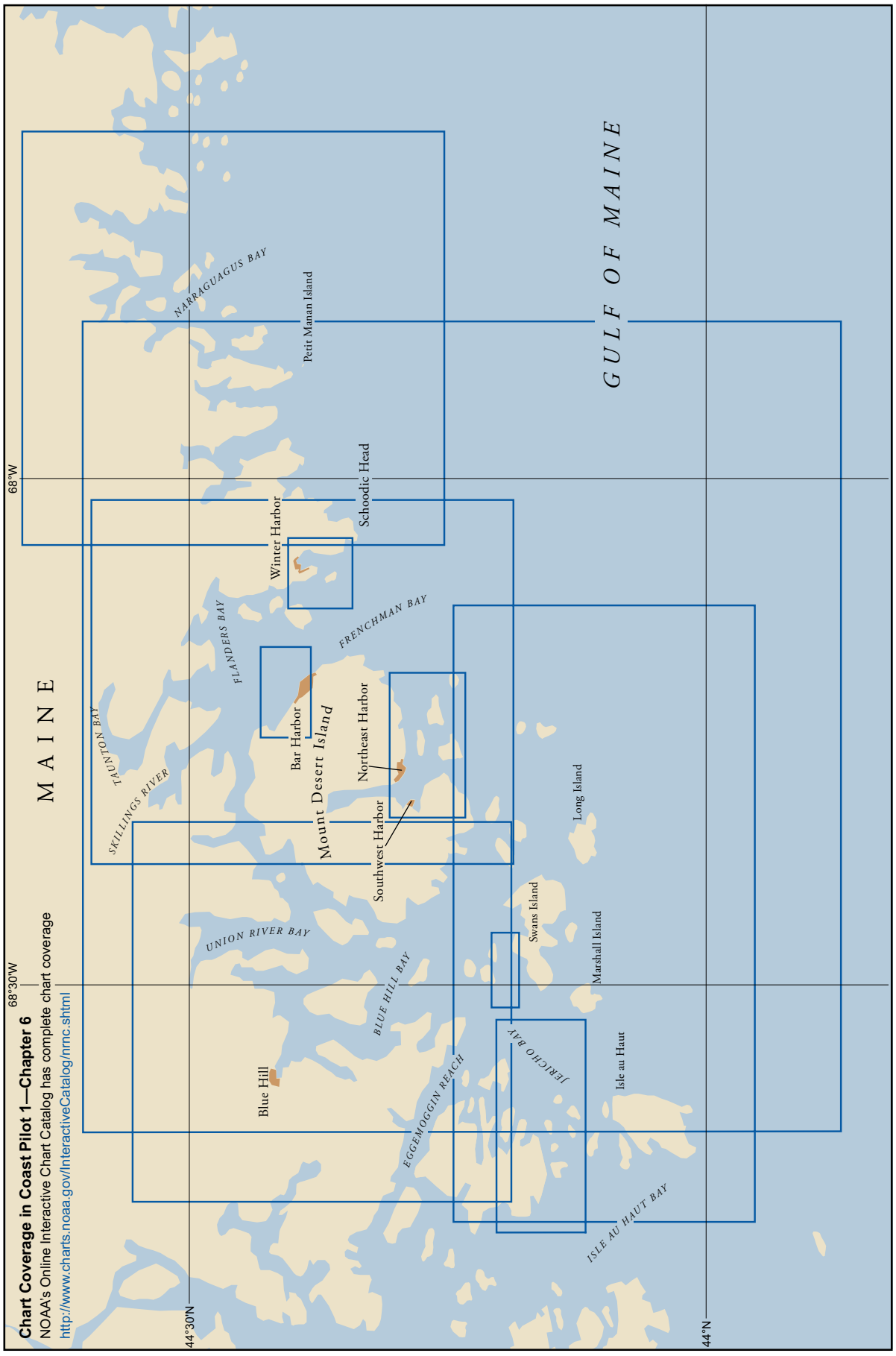


Chart Coverage in Coast Pilot 1—Chapter 6
NOAA's Online Interactive Chart Catalog has complete chart coverage
http://www.charts.noaa.gov/InteractiveCatalog/nrnc.shtml
MAINE
GULF OF MAINE
NARRAGUAGUS BAY
Petit Manan Island
Schoodic Head
Winter Harbor
FLANDERS BAY
FRENCHMAN BAY
TAUNTON BAY
SKILLINGS RIVER
Bar Harbor
Mount Desert Island
Northeast Harbor
Southwest Harbor
Long Island
UNION RIVER BAY
BLUE HILL BAY
Swans Island
Marshall Island
JERICHO BAY
Isle au Haut
Blue Hill
EGGEMOGGIN REACH
ISLE AU HAUT BAY
68°W
68°30'W
44°30'N
44°N

Petit Manan Island to Jericho Bay, Maine

(1) The coast between Petit Manan Point and Jericho Bay is indented by Frenchman Bay, Blue Hill Bay and numerous smaller bays and harbors. **Mount Desert Island**, between Frenchman and Blue Hill Bays, is mountainous and is the highest land feature on the coast of Maine. The summits are rounded, and several of them are nearly the same height, making it difficult to identify individual peaks at a distance.

(2) **Acadia National Park** comprises the greater part of the southern half of Mount Desert Island, particularly the mountainous areas and the lower half of Schoodic Peninsula on the eastern side of Frenchman Bay, including the scenic Schoodic Point, and part of Isle au Haut. **Schoodic Mountain**, about 16 miles northward of Schoodic Point, is visible for a good distance off the coast.

(3) **Mount Desert Rock**, 17.5 miles southward of Mount Desert Island and 11.5 miles outside of the nearest island, is a rocky islet about 20 feet high.

(4) **Mount Desert Light** (43°58'07"N., 68°07'42"W.), 75 feet above the water, is shown from a 65-foot conical gray granite tower on the rock. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A.

(5) **Columbia Ledge**, 0.7 mile southward of the rock, is covered 18 feet and unmarked.

(6)

COLREGS Demarcation Lines

(7) The lines established for this part of the coast are described in **33 CFR Part 80.105**, chapter 2.

(8)

Moulton Ledge to Rolling Island

(9) The bight between Petit Manan Bar and Schoodic Peninsula is the approach to Dyer Bay, Gouldsboro Bay and Prospect Harbor. Local fishermen are the principal users of these waters. Vessels should use caution when crossing broken areas where the charted depth does not considerably exceed the vessel's draft. The most important village is Prospect Harbor. Gouldsboro and Steuben can be reached by small craft at high water.

(10) **Moulton Ledge**, off the entrance to Dyer and Gouldsboro Bays and 3 miles westward of Petit Manan Light, is awash at low water. A lighted bell buoy marks the west side of the ledge. Broken ground, and several unmarked ledges, are in the vicinity of Moulton Ledge; vessels should avoid this area. An 18-foot spot, 0.6 mile to the southward; a 23-foot rocky shoal, 0.3 mile to the southeastward; and **Stone Horse Ledge**, covered 11 feet about 0.8 mile to the northward, are all unmarked.

(11) **Dyer Bay**, westward of Petit Manan Point, has excellent anchorage in depths of 20 to 42 feet. The entrance, 3.3 miles northwestward of Petit Manan Light, and the bay channel are unmarked and seldom used except by small local vessels. A group of ledges and rocks, with narrow and deep passages between them, extends from the westward partway across Dyer Bay entrance.

(12) A good passage nearly 0.5 mile wide is between Petit Manan Point and **The Castle** (44°24.4'N., 67°55.2'W.), the easternmost bare ledge. One mile above The Castle the channel narrows to a width of 250 yards because of rocks and ledges that extend out from both shores and are covered 8 to 11 feet. Above this the channel widens to 0.5 mile and then narrows gradually to 400 yards westward of **Sheep Island**, 3.3 miles north of The Castle. The least depth in midchannel is about 18 feet, but a stranger should not attempt to enter at low water with a vessel drawing more than 8 feet.

(13) Strangers can enter Dyer Bay with the aid of the chart in clear weather in the daytime. Local knowledge should be obtained before attempting it at any other time, as there are many fish weirs covered at or near high water.

(14) Tidal currents are strong in the entrance of Dyer Bay but follow the general direction of the channel except near Dyer Point, on the west side of the entrance, where they set in and out of Gouldsboro Bay.

(15) The Castle; **Bonney Chess Ledge**, 300 yards west of The Castle; and **Little Ledge**, 0.5 mile west of The Castle, are bare and lack distinguishing marks. **Yellow Birch Head**, on the east side of Dyer Bay near the entrance and 0.7 mile northeastward of The Castle, is a high bare bluff.

(16) **Stanley Cove** and **Yeaton Cove** indent the west side of Dyer Bay, 0.7 mile and 1.5 miles northward of Dyer Point, respectively. A commercial lobster pound with service wharf is in each cove. Another lobster pound with service wharf is in the unnamed cove, 0.5 mile eastward of the north end of Sheep Island. Gasoline is available at the wharves, which dry at low water.

(17) **Carrying Place Cove** extends southeastward from Dyer Bay north of Sheep Island. The head of the cove, 300 yards from Pigeon Hill Bay, is soft mud and dries at low water.

(18) **Pinkham Bay**, at the head of Dyer Bay, has numerous rocks and ledges. A narrow crooked channel with a depth of 7 feet leads for some distance toward the head of the bay which dries at low water.

(19) **Dyer Harbor**, a shallow bight in the upper part of Dyer Bay, is dry at low water. It is located about 0.6 mile northwest of Sheep Island and just westward of **Goods**

Point. There is a pier in the cove on the west side of Goods Point, with 2 feet of water at the head.

(20) **Gouldsboro Bay**, separated from Dyer Bay by **Dyer Neck**, is 4 miles northwestward of Petit Manan Light and 6 miles northeastward of Schoodic Island. Excellent anchorage may be had in depths of 12 to 54 feet. The bay is the approach to the villages of Gouldsboro and Steuben, 6.5 and 7 miles, respectively, above the entrance. However, the approaches are unmarked and used only by small craft at high water.

(21) Ice obstructs navigation in the bay from December to March. In severe winters the bay is closed to the entrance. Clusters of piling in the bay, the remains of old fish weirs, are hazardous to small craft. It has been reported that most of the weirs can be avoided by remaining in the middle of the bay.

(22) **Sally Islands**, a chain of small islands and ledges, extend across the entrance to Gouldsboro Bay. Included in the Sally Islands are **Eastern Island**, grassy with tree stumps and 0.5 mile southward of Dyer Neck; **Bald Rock** with sparse grass on it, 700 yards westward of Eastern Island; **Sally Island**, rocky, with grass on top and a small cottage and two lone trees on its northerly side just southwestward of Bald Rock; and **Sheep Island**, thickly wooded in the center and 0.4 mile southwestward of Sally Island. Through the islands are two navigable channels, Eastern Way and Western Passage. When approaching from westward, care must be taken not to mistake the passages as the islands are difficult to recognize. The bay inside the islands is free from outlying dangers, and the water shoals gradually toward the head of the bay.

(23) **Eastern Way** leads into Gouldsboro Bay between Eastern Island and Bald Rock. The passageway is about 300 yards wide between the 18-foot curves and has a spot covered 17 feet about 250 yards west of Eastern Island. A depth of 45 feet is available for a width of 75 yards in the channel between the 30-foot curves. The channel has strong tidal currents; when the current is ebbing, more especially with southerly and easterly winds, small craft or those under sail alone should not attempt the passage. The current sets diagonally across the channel.

(24) **Western Passage**, with a least depth in the channel of 16 feet, leads into the bay between Sally Island and Sheep Island. The passage is about 100 yards wide and is close along the eastern side of Sheep Island and westward of the ledges, bare at half tide, which extend about 500 yards west of Sally Island. It is not advisable for strangers to attempt it. The tidal currents run true with the channel and have a velocity of 2 to 3 knots at strength.

(25) A passage from Dyer Bay to Gouldsboro Bay north of Sally Islands is obstructed by a ledge, which uncovers, that extends 350 yards south of the southern extremity of Dyer Neck, and a shelving ledge covered 8 feet at the end extending 200 yards northeastward from Eastern Island. The channel is about 75 yards wide between the 30-foot curves, and the controlling depth is about 28 feet. The tidal currents have a velocity of 2 to 3 knots at strength through these passages, and in Eastern Way they set diagonally.

(26)

Routes

(27) Approaching Gouldsboro Bay from eastward and entering through Eastern Way from off the fairway bell buoy southward of Petit Manan Light, steer **310°** for 4.5 miles until abeam of the southern tip of Eastern Island, bearing **040°**, distant 550 yards, then steer **000°** through the passage. The tidal currents set across this course with considerable velocity with the flood northeastward and ebb southwestward. Change course as necessary to pass midway between Eastern Island and Bald Rock. When inside the islands, steer **300°** until abeam of the south tip of Youngs Point, then stand up the middle of the bay. The water shoals gradually toward the head, and anchorage can be had anywhere between the entrance and Point Francis by giving the shores a berth of at least 500 yards.

(28) Approaching from westward and entering through Eastern Way, from the lighted bell buoy off Schoodic Island, steer **046°** for 4.4 miles, passing 0.4 mile southeastward of Little Black Ledge to a position where Cranberry Point is in line with Prospect Harbor Point Light. Next steer **028°** for the eastern end of Eastern Island until about 550 yards from the island, and then round up to the northward to pass midway between Eastern Island and Bald Rock. Then continue as directed in the preceding paragraph.

(29) **Point Francis**, on the western shore of Gouldsboro Bay 3.2 miles above Sally Islands, is high and wooded and is prominent from the lower end of the bay.

(30) A lobster pier with float landing is on the east side of the bay on **Dolly Head**, about 1.3 miles north-northeastward of Point Francis. Gasoline is available at the float, and depths of 4 feet are reported alongside. A group of submerged pilings is about 0.2 mile southwestward of Dolly Head. Another lobster pier, dry at low water, is on the west side of the bay, nearly opposite Dolly Head. Marine supplies can be obtained at a store which is within walking distance of this pier.

(31) **Joy Bay**, a shallow body of water 1.5 miles long that extends northward from Gouldsboro Bay, is entered through **The Narrows** between**Rogers Point** and **Garden Point**. In 1985, a visible wreck was reported in the entrance to Joy Bay in about 44°28.3'N., 67°59.0'W. Two coves are at the head of Joy Bay. **Steuben Harbor** extends northeastward, and **Joy Cove** extends westward. The narrow, crooked, and unmarked channels through Joy Cove and Steuben Harbor are nearly bare at low water and are seldom used. **Steuben**, a village at the head of Steuben Harbor, can be reached at high water by vessels of 7 to 8 feet in draft. Groceries and gasoline are available in the village.

(32) **West Bay**, a large shallow arm of Gouldsboro Bay, extends northwestward from the bay for about 2.5 miles. The village of **Gouldsboro**, on the western side of the bay near its head, is of no commercial importance. The

entrance has numerous ledges and rocks. The narrow, crooked, unmarked channel in the bay is nearly dry at low water and seldom used.

(33) The **West Bay Seed Mussel Conservation Area**, a Marine Protected Area (MPA), includes the waters of West Bay and Grand Marsh Bay.

(34) **Corea Harbor** is a small cove between Gouldsboro Bay and Prospect Harbor. A number of islands including grassy **Bar Island**, partially wooded **Outer Bar Island** and wooded **Western Island**, are off the entrance to well-protected Corea Harbor. A rock that uncovers 6 feet about 300 yards southeast of Western Island is marked on the southeast side by a gong buoy. An unmarked rock, covered 23 feet, is 0.7 mile southeast of Western Island. **Corea** is a small village at the head of the harbor. The principal industries are fishing and lobstering. The most prominent objects are a church spire and a group of houses at the head of the harbor and a gray cottage with red roof on Western Island, which are visible for a considerable distance offshore.

(35) The unmarked channel into Corea Harbor leads to the westward of Western Island and then along the northeastern side of the entrance to an anchorage in the middle of the harbor. A ledge extending from the western side of the entrance is cleared by keeping close to the northeastern side. Lobster pots, which are usually placed on the edge of the ledge, are a good indication of the location of the channel. In 1979, a rock awash was reported about 150 yards northwest of Western Island in the approach to Corea Harbor. Kelp growing on the rock was reported visible at low water. Low water is the best time to enter.

(36) In 2022, the controlling depth was 7.6 feet in the channel to the anchorages, thence 4 to 8 feet in the anchorages with lesser depths along the edges. The harbor outside the limits of the anchorage has shoaled considerably. The moorings in the harbor are administered by the **harbormaster**, who may be contacted through either lobster wharf. A heavy surge is sometimes felt in the harbor in southerly winds.

(37) Ice usually obstructs the inner harbor from December to March, but fishing is carried on during the winter from piers on both sides of the entrance to the harbor.

(38) There are numerous piers in the harbor, most of which dry at low water. A lobster wharf on the northeast side of the entrance has a float landing with 8 feet reported alongside. Another lobster wharf, 75 feet long on the western side of the entrance, has a float landing with 6 feet reported alongside. Gasoline and diesel fuel are available at both wharves.

(39) **Prospect Harbor**, 4 miles north-northeastward of Schoodic Island and 6 miles northwestward of Petit Manan Light, is a large deepwater bight, about 1.3 miles wide between **Cranberry Point** and **Prospect Point**. It has ample depth and offers good anchorage for the largest vessels but is exposed to southerly and southeasterly weather. A bell buoy is off Cranberry Point.

(40) The town of **Prospect Harbor** is at the head of the harbor. The upper part of the harbor is divided into two coves by **Prospect Harbor Point**. **Sand Cove**, the eastern branch, has ample depth until near the head with its west side obstructed by rocky ledges.

(41) **Prospect Inner Harbor**, the western branch of the harbor, is used commercially on its western shore; the upper half is obstructed by unmarked rocky ledges. There is no shipping, but fishing and lobstering are important.

(42) The houses around the head of the harbor, the spire of the Community House, and a large green warehouse are conspicuous. The radio antennae of the naval communications station north of Cranberry Point and the dome on Prospect Harbor Point are also prominent.

(43) **Prospect Harbor Point Light** (44°24'12"N., 68°00'47"W.), 42 feet above the water, is shown from a 38-foot white conical tower on the point. The former residence buildings of the light station, now part of a naval communications station, and the radio antennas are conspicuous behind the light.

(44) Anchorage can be found according to draft anywhere in the outer harbor, and in soft bottom in the entrance to the inner harbor about 200 yards northward of a line between **Clark Ledges Daybeacon 5** and the end of the cannery wharf on **Clark Point**.

(45) **Little Black Ledge**, awash and marked by a buoy, **Big Black Ledge**, 5 feet high and **Old Man** and **Old Woman**, which partly uncover 5 feet, are ledges off the entrance to Prospect Harbor. A bell buoy is off the southwest side of Old Woman Ledge. The white sectors of Prospect Harbor Point Light cover the fairways either side of these ledges.

(46) **Clark Ledges**, extending 500 yards eastward of Clark Point, on the southwestern side of the entrance to the inner harbor, have a rock 4 feet high and are marked by a gong buoy and a daybeacon on their easterly side.

(47) The approach to Prospect Harbor and the anchorage can be readily made with the aid of the chart in daytime in clear weather; at night the white sectors of Prospect Harbor Point Light clear all dangers in the approaches. Ice seldom obstructs the harbor.

(48) There is a **harbormaster,** and the moorings are under his control.

(49) There are a number of private piers and commercial wharves on the westerly side of Prospect Inner Harbor. A cannery wharf with a reported depth of 10 feet alongside is on Clark Point. A lobster pier with a reported depth of 2 feet alongside is just northwestward of the cannery wharf. Gasoline is piped to the lobster car moored about 30 yards off. Groceries can be obtained in town.

(50) **Birch Harbor**, on the western side of Prospect Harbor 1.4 miles south of Prospect Harbor Point Light, has a depth of 6 feet for 0.5 mile and then shoals rapidly above this point. The small fishing village of **Birch Harbor** is at the head of the harbor. The landings are bare at low water. The channel is unmarked and difficult. The best water in entering favors the southwest side to avoid

(58)

Navigation Guidelines—Frenchman Bay

The principal guides to the entrance of Frenchman Bay from the sea are Frenchman Bay Lighted Buoy FB (44°19'21"N., 68°07'24"W.) and the lights on Mount Desert Rock, Great Duck Island, Baker Island and Egg Rock.

Recommended Vessel Routes

As the result of a cooperative agreement between the Penobscot Bay and River Pilots Association, fishermen, cruise ship representatives and the U.S. Coast Guard, deep-draft vessels and other commercial vessels transiting through Frenchman Bay are requested to follow designated routes. These routes provide safe, established tracklines for increased commercial vessel traffic and prevent the loss of fishing gear placed in the waters in the approach to and transit through Frenchman Bay. The routes are defined as follows:

Eastern Route

The eastern limit of the route is about 11 miles southeastward of Schoodic Point in about 44°14'54"N., 67°56'18"W. Vessels are requested to begin and end their transit from about this point. Entering and departing vessels should follow tracklines of 300° and 120°, respectively, and intersect the recommended southern approach route 0.4 mile northwest of Frenchman Bay Lighted Buoy FB.

Southern Route

The southern limit of the route is about 18 miles south of Great Duck Island in about 43°52'48"N., 68°17'05"W. Vessels are requested to begin and end their transit from about this point. Entering and departing vessels should follow tracklines of 030° and 210°, respectively. At or about 44°03'12"N., 68°08'36"W., steer course 002° and 182°, respectively and intersect the recommended eastern approach route 0.4 mile northwest of Frenchman Bay Lighted Buoy FB.

The Frenchman Bay recommended route continues northwest of Frenchman Bay Lighted Buoy FB along the following positions:

44°20'00"N., 68°08'54"W.
44°22'36"N., 68°09'36"W.
44°23'42"N., 68°10'24"W.; thence west to
44°23'42"N., 68°11'12"W. at the edge of General Anchorage A and thence northwest to
44°24'24"N., 68°11'42"W. at the edge of General Anchorage B

At no time shall the Navigation Rules, International/Inland, be abridged or amended by these navigation guidelines. These guidelines are intended to enhance safety under conditions wherein navigation is not otherwise constrained.

Roaring Bull, a rock awash at low water in the entrance. A church spire at the head of the cove is conspicuous.

(51) **Bunkers Harbor**, on the west side of Prospect Harbor 0.8 mile south of Birch Harbor, has a small village of fishermen at the head. There are two lobster pounds in the inner harbor; one is at the head and the other is on the southwest side. Gasoline, diesel fuel, water and some marine supplies can be obtained at a pier adjoining the lobster pound at the head of the harbor. The pier's float landing has a reported depth of 6 feet alongside.

(52) The entrance is obstructed by ledges. **Bunkers Ledge**, on the south side of the entrance, has a rock awash at low water at the eastern end and the inner part uncovers 6 feet; a buoy is eastward of the rock. The channel northward of the ledge is said to be the best of the unmarked channels leading into the harbor. An anchorage area has been dredged in the middle of the inner harbor. In 2019, depths of 5 feet were available in the anchorage except for shoaling along the easterly edge.

(53) **Schoodic Harbor**, between Prospect Harbor and Frenchman Bay, has ample depth but is exposed to the sea and rarely used as an anchorage. **Wonsqueak Harbor**, the northern part of Schoodic Harbor, has small fish wharves at its head which dry at low water. An overhead power cable with a clearance of 26 feet crosses near the head of the harbor. Wonsqueak Harbor is difficult to enter. Several islands and ledges are in the entrance to Schoodic Harbor.

(54) **Schoodic Island**, 7.5 miles west-southwestward of Petit Manan Island and on the south side of Schoodic Harbor, is low, wooded on the south end and grassy with a lone tree on the north end. The island is bordered by extensive ledges. **Schoodic Ledge,** 0.4 mile northward of Schoodic Island, uncovers 9 feet, and breakers are always visible on it. The channel between Schoodic Island and Schoodic Ledge is marked by a buoy on either side. It has a depth of 24 feet and is generally used by small local vessels and motorboats bound along the coast.

(55) **Middle Ledge**, 0.8 mile north of Schoodic Island, uncovers 5 feet and is unmarked. **Brown Cow**, a rocky ledge about 300 yards south of **Spruce Point**, on the northeastern side of the entrance to Schoodic Harbor, uncovers 4 feet. A lighted whistle buoy is about 550 yards south of Brown Cow. **Rolling Island**, 0.9 mile north-northwestward of Schoodic Island, is wooded.

(56)

Frenchman Bay

(57) **Frenchman Bay**, westward of Schoodic Peninsula and eastward of Mount Desert Island, is the approach to the towns and important summer resorts of Bar Harbor, Winter Harbor, Southwest Harbor, Seal Harbor, Northeast Harbor and many smaller villages. The bay is frequented by cruise ships, ferry vessels, fishing vessels, yachts and small pleasure craft. The bay proper is about 10 miles long and has an average width of about 4 miles. Near the

center of the bay, a group of islands extends across the bay; between the islands are two deep channels. Vessels of any size and draft can find anchorage. Navigation is not difficult for strangers.

(59) **Cadillac Mountain** (44°21.1'N., 68°13.6'W.), 1,530 feet high, is the highest point on Mount Desert Island and the highest point along the east coastline of the United States. On a clear day the mountain is visible from 35 to 45 miles seaward. An excellent scenic highway leads from Bar Harbor to the summit of Cadillac Mountain.

(60) **Schoodic Head** (44°21.1'N., 68°03.2'W.) on **Schoodic Peninsula**, across the bay from Mount Desert Island, is 440 feet high and is the most prominent land feature at the eastern entrance to the bay.

(61) **Big Moose Island**, the southern extremity of Schoodic Peninsula, is connected to the peninsula by landfill and is part of **Acadia National Park**. A prominent green elevated tank, reported to be a good radar target from offshore, is near the center of the island. **Schoodic Point Observation Spot** and a large parking lot are on the southern extremity of the island.

(62) **Little Moose Island**, rocky and with a few trees, is about 0.3 mile eastward. **Arey Cove**, the bight between the two islands, is unsafe in southerly weather.

(63) The principal entrance to Frenchman Bay is from southward between Schoodic Peninsula and Baker Island, but small vessels can enter from southwestward through Western Way and Eastern Way. Small boats also may enter the head of Frenchman Bay at high water through Mount Desert Narrows.

(64)

Anchorages

(65) Winter Harbor is a good anchorage and is frequently used by vessels entering for shelter; it is usually open throughout the winter. Bar Harbor is partially protected, except against heavy southeasterly winds, but has poor holding ground except near the head of the harbor. Large vessels sometimes anchor northward or northwestward of Bar Island. Stave Island Harbor is a good anchorage. Southwest Harbor is a well-sheltered and frequently used anchorage.

(66) Frenchman Bay is rocky, but the water is deep and in general free from dangers except near the shores. The main part of the bay from a little southward of Egg Rock Light to the entrances of Sullivan Harbor, Skillings River and Eastern Bay, including the channels between Jordan and Long Porcupine Islands, and between Burnt Porcupine and Sheep Porcupine Islands is clear. Vessels navigating the tributaries should proceed with caution when crossing areas where the charted depth does not substantially exceed the draft.

(67) Little difficulty should be experienced in approaching and entering Frenchman Bay at any time, as the approaches are clear, and outlying dangers for the most part are well marked.

(68) Between Bar Harbor and Ironbound Island the flood current velocity is less than 0.3 knot. The ebb velocity at strength is about 0.7 knot and sets southeastward. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov*. for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(69) During mild winters Frenchman Bay is usually clear of ice to Skillings River, but the bays and rivers connected to the northern part of the bay are frozen over. Winter Harbor is reported to be generally clear. It is reported that during foggy weather Frenchman Bay usually clears during the day although the fog remains heavy outside Schooner Head and Ironbound Island.

(70)

Pilotage, Frenchman Bay

(71) Pilotage is compulsory for foreign vessels and U.S. vessels under register in foreign trade with a draft of 9 feet or more. Pilotage is optional for fishing vessels and vessels powered predominately by sail. Pilotage can be arranged through ship's agents by contacting Penobscot Bay and River Pilots Association, telephone 207–548–1077.

(72) Vessels arriving from the east are requested to make their approach at the eastern end of the Frenchman Bay Recommended Route at 44°14'54"N., 67°56'18"W. Vessels should then proceed along the Recommended Route toward Frenchman Bay Lighted Buoy FB. Vessels arriving from the south are requested to approach from the southern end of the Recommended Route, at 44°52'48"N., 68°17'05"W. Vessels then proceed along the Recommended Route to Frenchman Bay Lighted Buoy FB.

(73) The pilots will board arriving vessels 3 miles seaward of Frenchman Bay Lighted Buoy FB, along the recommended route, in about 44°18'06"N., 68°04'00"W. (eastern appproach) and 44°16'06"N., 68°07'48"W. (southern approach). The pilot boat is the FRENCHMAN BAY, a 36-foot white lobster boat displaying a pilot flag and lights for a pilot boat. Arriving vessels should contact the pilot boat one hour prior to arrival. As no pilot boat is kept on station and there are no long-range communications available, arriving vessels are requested to maintain their 24-hour ETA and to not arrive at the pilot station ahead of this schedule.

(74)

Winter Harbor

(75) **Winter Harbor** on the eastern side of Frenchman Bay just inside the entrance is a frequently used harbor of refuge. The principal entrance from southward, 0.7 mile wide between Turtle Island and Schoodic Peninsula, is deep and free of dangers. The entrance from the northward is used only by local vessels drawing 10 feet or less. The aids in the northern approach are colored and numbered for vessels bound north. Good anchorage in depths of 30 to 54 feet, good holding ground, will be found in Winter Harbor. The harbor is comparatively free of danger, and,

although open to the southward, a heavy sea never enters. Ice seldom interferes with navigation.

(76) **Turtle Island**, wooded, is on the western side of the main entrance to Winter Harbor and 0.8 mile west of Schoodic Peninsula. **Turtle Island Ledge** uncovers 5 feet and extends 500 yards off the southwest side of the island.

(77) **Mark Island**, 0.5 mile west of Schoodic Peninsula and on the west side of the entrance to Winter Harbor, is grassy and marked by a conspicuous abandoned lighthouse, a white tower 19 feet high connected to a dwelling. A lighted gong buoy is 0.2 mile south-southeast of the tower. Depths of 12 to 16 feet are up to 300 yards south-southeast of the tower.

(78) **Ned Island**, 0.1 mile north of Mark Island, and **Heron Island**, 0.5 mile northwestward of Turtle Island, are wooded. **Spectacle Island**, just north of Turtle Island, has a conspicuous house on it and is wooded. The outer islands, including **Flat Island** and the **Crow Islands**, are grassy or bare rocks; the largest of the Crow Islands has a house and a few trees on it. All of the islands are surrounded by extensive ledges that uncover at various stages of the tide.

(79) The channel between Turtle Island and Mark and Ned Islands is not recommended for deep-draft vessels because of unmarked 16- and 17-foot spots in midchannel, about 350 yards westward of the north end of Mark Island. **Roaring Bull**, a shoal about 200 yards off the northwestern end of Ned Island, is covered 3 feet and breaks during southerly and easterly weather; it is marked by a buoy off the northwest side.

(80) **Grindstone Neck**, forming the western side of Winter Harbor, is wooded and has many summer homes, several churches and a club hotel. A round gray house, built to resemble a lighthouse and with a glass cupola on top, is on the west side of Grindstone Neck, about 0.9 mile north-northwestward of Grindstone Point. The structure is conspicuous from the southwestward in Frenchman Bay.

(81) **Grindstone Ledge**, which uncovers 5 feet, extends 400 yards southeastward from Grindstone Point and is marked by a daybeacon on the ledge and a buoy south-southeastward of it. A 12-foot shoal is 235 yards southward of the daybeacon and in the middle of the channel between Ned Island and Grindstone Ledge. A narrow unmarked channel, with a depth of 16 feet, is almost midway between the daybeacon and **Grindstone Point**, the southeastern extremity of Grindstone Neck. This channel should not be used without local knowledge. The channel south of the ledge and buoy is the recommended passage.

(82) **Pulpit Ledge**, about 150 yards off the southwestern end of Grindstone Neck, is marked by a daybeacon. The narrow channel between the ledge and neck is used only by small local craft.

(83) The eastern shore of Winter Harbor from **Frazer Point**, opposite Grindstone Point, to Abijah Ledge, 0.5 mile northward, should be given a berth of more than 150 yards. **Abijah Ledge**, near the head of Winter Harbor about 300 yards off the eastern shore, is awash at low water. A buoy is westward of the ledge. Shoal water extends from the ledge to the small cove northeastward.

(84) **Sand Cove**, the northwest arm at the head of Winter Harbor, affords the best anchorage with excellent holding bottom of black mud. Shoal water extends 130 yards off **Harbor Point**, the eastern entrance point of the cove. A buoy marks the southeast side of the shoal. Only partly submerged stones remain of a wharf on the west side of the cove, about 0.2 mile from the head. Winter Harbor Yacht Club, on the west side of Sand Cove, about 0.4 mile from the head, has a pier and float landing with 22 feet alongside. Water is piped to the float. Fish weirs obstruct the upper shoal end of the cove.

(85) **Inner Winter Harbor**, immediately northeastward of Sand Cove, is entered between **Guptill Point** and Harbor Point, 300 yards southward. **Guptill Ledge**, extending southward and southeastward from Guptill Point, is marked by a daybeacon on the ledge and a buoy off its southeastern end. The harbor is secure in all weather and is extensively used by fishing craft, yachts and pleasure craft. An anchorage in the middle of Inner Winter Harbor has depths of about 8 feet. The Winter Harbor town pier and float landing, with approximately 8 feet alongside, are on the west of Guptill Point, just inside the entrance; water and electricity are available. A lobster pier with depths of about 7.8 feet alongside its float landing is on the north side of the harbor about 300 yards westward of the town pier; gasoline, diesel fuel, water and some marine supplies are available.

(86) **Henry Cove**, at the head of Winter Harbor eastward of Guptill Point, is wider but less sheltered than Inner Winter Harbor. It has good holding ground in sticky mud but is open to southerly winds. A sewer outfall extends 1,240 feet from the north end of the cove. Mariners should exercise caution when anchoring in this area. Reefs and shoal water extend about 60 yards from shore on both sides of the entrance. The northern end of the cove is extremely shoal, with the upper 300 yards dry at low water. A large gray private boat shed at the head is conspicuous. A marina with an L-shaped wharf, about 0.2 mile northeast of Guptill Point, has a 10-ton mobile hoist that can handle craft up to 42 feet long for hull and engine repair or open and covered dry winter storage; gasoline, diesel fuel, water, ice, launching ramp, sewage pump-out facilities and marine supplies are also available.

(87) The town of **Winter Harbor** is at the head of Winter Harbor. The principal industries are fishing and lobstering. The homes on Guptill Point, the church spire and a large private boat shed at the head of Henry Cove are conspicuous. Provisions can be obtained in town, and a bank is available. The town **harbormaster** controls the moorings and can be contacted through the town office. Winter Harbor is seldom closed by ice.

(88) Winter Harbor is deep and clear from the entrance to the recommended anchorage in Sand Cove. Little trouble should be experienced when entering at any time with strict attention to the charts and the aids, which mark most

of the important dangers. To enter Inner Winter Harbor and Henry Cove, midchannel courses are recommended, but only small craft should enter these coves for anchorage.

(89)

Egg Rock to Cape Levi

(90) **Egg Rock**, about 2 miles west of the entrance to Winter Harbor, is bare and low. **Egg Rock Light** (44°21'14"N., 68°08'18"W.), 64 feet above the water, is shown from a 40-foot white square tower on a dwelling on the island. It is the most prominent leading mark for this section of the bay. A mariner radio activated sound signal at the light is initiated by keying the microphone consecutively five times on VHF-FM channel 83A. **Handiron Ledge**extends about 0.1 mile northeast of Egg Rock, and another ledge extends 0.4 mile southwestward from the light. Parts of both ledges uncover. A lighted bell buoy is 1 mile southwestward from Egg Rock Light.

(91) **Ironbound Island**,1.5 miles northward of Egg Rock Light, the largest of the islands in Frenchman Bay, is thickly wooded and has high vertical cliffs. **Cod Ledges**, eastward of Ironbound Island, have two critical spots covered 11 feet. Vessels should pass to the eastward of the buoy on the eastern side of the ledges. An unmarked shoal with a depth of 12 feet is about 300 yards off the east shore of Ironbound Island at a point 0.6 mile northeast of **Seal Cove**, a bight in the southern end of the island.

(92) Shoal water extends 100 yards north of **Fish Point**, on the west side of Ironbound Island, and along the cove eastward from the point. A ledge, extending northward from the north end of the island, has depths of 12 feet 250 yards off the shore. A buoy, northward of the ledge, marks the southern side of the northwestern entrance to Halibut Hole.

(93) **Halibut Hole** is the passage between the north end of Ironbound Island and **Jordan Island**, 0.2 mile northeastward. The passage is deep and clear with the exception of a rock covered 19 feet on the northeast side of the passage, 200 yards off the south shore of Jordan Island. The ledge shoals rapidly northward to the beach. The channel is southward of the rock.

(94)

Local magnetic disturbance

(95) Differences of as much as 3° from the normal variation have been observed in the vicinity of Jordan Island.

(96) **Stave Island Harbor** is an excellent harbor of refuge on the eastern shore of Frenchman Bay. The mainland is on the east, Jordan Island is on the south and **Stave Island**, 1.4 miles north of Ironbound Island, is on the north. The anchorage has depths of 21 to 37 feet, soft bottom and is sheltered from all except southwest winds; it is used considerably as an anchorage.

(97) The main entrance to Stave Island Harbor is between Stave and Jordan Islands. An unmarked rock, covered 25 feet, is nearly midway between Jordan and Stave Islands, the deeper channel being southward of it. **Yellow Island**, 200 yards westward from the north end of Jordan Island, is so named from the color of its rocks. The island is wooded. A ledge with a rock awash at low water is 150 yards south of Yellow Island.

(98) Approaching Stave Island Harbor from southward, the mariner will find a broad, clear channel between Ironbound and Long Porcupine Islands; the approach northward of the Porcupines is also clear. The passage from Stave Island Harbor to Flanders Bay east of Stave Island is obstructed by **Stave Island Bar** and is navigable by small craft only at high water. The north end of the harbor eastward of Stave Island is shoal. There is a narrow channel into the harbor from southward over **Jordan Island Bar**, between Jordan Island and the mainland. The channel, which is used only by small local craft, has a depth of 5 feet 100 yards off Jordan Island.

(99) **Summer Harbor** is a small settlement on the east side of **Summer Harbor** in the southeastern part of Stave Island Harbor. The cove is clear with the exception of a rocky ledge, covered 7 feet, making out 250 to 500 yards from shore.

(100) **South Gouldsboro** is a village on the northeastern shore of Stave Island Harbor. A lobster pound is at the head of **Bunker Cove**. A reef extends off the south entrance point to the cove; caution is advised. A lobster pier with 2 feet reported alongside its float landing is on the south side of the entrance. Gasoline is piped to the float. Ice usually obstructs the harbor from December through March.

(101) **Calf Island**, 0.7 mile northwestward of Stave Island, is wooded except on the south where it is low and bare. A house and barn on the southeast side are visible from southward. **Little Calf Island** and **Thrumcap** are partly wooded islands on the extensive shoal extending southward from Calf Island.

(102) **Flanders Bay**, on the northeast side of Frenchman Bay, is protected by Stave and Calf Islands. An excellent anchorage may be found, but the bay is seldom used except by small craft. The villages of West Gouldsboro and East Sullivan are on the eastern shore. The bay can be entered through a narrow marked 8-foot channel across **Calf Island Bar** between Calf and Stave Islands or around the northwest end of Calf Island. The channel northwest of Calf Island has the best water.

(103) An extensive chain of bare and sunken ledges extends through the middle of Flanders Bay from the west end to near the southeast end. **Halftide Ledge**, the southeasterly ledge, uncovers about 5 feet; a buoy is off its southeast end. The channel through the bay is eastward of the buoy.

(104) **Long Ledge**, 0.4 mile northwest of Halftide Ledge, partly uncovers at high water. The ledge southeastward of Long Ledge is covered 5 feet. Between the 5-foot ledge and Long Ledge is an opening with a depth of 19 feet.

(105) **Treasure Island**, on the northwest side of Flanders Bay, is connected to **Waukeag Neck** by a private stone causeway. The area between Treasure Island and Long Ledge is foul. **Junk of Pork**, a small dirt cone of unusual

appearance, **Sheldrake Island** and **Ash Island** are in the area. A buoy is on the eastern side of the foul area.

(106) **Hall Point**, marked by prominent residences, is on the southeast end of **Schieffelin Point**, on the northeast side of the bay.

(107) **West Gouldsboro** is a village at the head of **Jones Cove**, the southeasterly tributary of Flanders Bay. There is a depth of 4 feet to within 500 yards of the village; above this point the channel dries at low water. The channel is unmarked, difficult and seldom used.

(108) **East Sullivan** is a village at the northern end of Flanders Bay. A white church with belfry is conspicuous near the northwestern end of the bay.

(109) **Eastern Point Harbor** is a sheltered anchorage for small craft between Waukeag Neck and the eastern half of **Preble Island**, 0.3 mile west of Calf Island. The head of the harbor is shallow and is separated from Sorrento Harbor by a partly dry reef. The cove on the northwest end of the harbor has been dammed up for a lobster pound. The pier close eastward of the pound has a reported depth of less than 1 foot at the end; gasoline and diesel fuel are available. A boatyard with boatsheds and a marine railway is on the north side of Eastern Point Harbor, about 200 yards eastward of the pier; craft up to 40 feet can be hauled out for hull and engine repairs or winter storage.

(110) **Sorrento Harbor** is a small anchorage used by small pleasure craft in summer on the north side of Frenchman Bay north of Preble Island and **Dram Island**, 0.2 mile west of Preble Island. In bad easterly weather the excursion launches from Bar Harbor sometimes anchor here.

(111) The entrance from southward, which favors Dram Island slightly, is about 125 yards wide between reefs that extend from Dram Island and Preble Island. The entrance has a depth of 23 feet in midchannel. An aquaculture site is about 0.5 mile southeast of the buoy.

(112) The entrance from westward is about 75 yards wide between the 10-foot curves and has a depth of 15 feet in midchannel. A reef that uncovers extends about 150 yards from the north side of the entrance. The best water in this entrance is found about 100 yards from Dram Island on a **091°** course. **Dram Island Ledge**, awash at low water, is 0.2 mile west of Dram Island; a buoy marks the ledge.

(113) **Sorrento**, a summer resort on the north side of Sorrento Harbor, is frequented by small yachts. Some pilings of the former steamer wharf are submerged off the north shore opposite the west end of Preble Island. A town wharf with float having about 7 feet alongside is on the point east of the old steamer wharf ruins. The yacht club uses the town wharf, which has water piped to it. A rock covered 2 feet is reported to be on the north side of the harbor about 30 yards northwest of the town wharf. Numerous summer homes are on the estates on Waukeag Neck northward of Sorrento.

(114) **Sullivan Harbor** is an arm on Frenchman Bay making northward from the north end. It is the approach to the villages of Hancock Point, Mount Desert Ferry, Sullivan and Franklin. The least depth to the falls just above Sullivan is about 25 feet. The channel to Sullivan is marked by a daybeacon and buoys to near Ferry Point.

(115) The main entrance to Sullivan Harbor is between Bean Island and Crabtree Ledge. Vessels also can enter by the buoyed channel eastward of Bean Island, but this channel is seldom used. The channel from the entrance to Sullivan has ledges bare and covered on either side but has ample depth and most of the dangers are marked.

(116) **Bean Island**, in the middle of the entrance to Sullivan Harbor, is partly wooded. The channel used most frequently is westward of the island. **Bean Ledge**, 0.2 mile eastward of Bean Island, has a rock 4 feet high on it.

(117) **Back Cove**, eastward of Bean Island and on the southeast side of Sullivan Harbor, has a boatyard on its southerly side near its head. A rock awash is in the entrance to the cove. A marine railway at the boatyard can handle craft up to 40 feet for hull and engine repairs; dry covered or open winter storage is available.

(118) **Crabtree Ledge**, on the west side at the entrance to Sullivan Harbor and 0.4 mile west of Bean Island, is marked by a bell buoy just east of the ledge. The ruins of an old lighthouse on the ledge are reported to bare at low water.

(119) **Hancock Point** is the southeastern extremity of Crabtree Neck. The village of **Hancock Point** has many summer homes. There is a town wharf with a float landing, which has 7 feet alongside, about 0.4 mile northward of the point. The harbormaster supervises the moorings and may be contacted through the town office. There are no facilities at the landing.

(120) **Mount Desert Ferry** isa settlement on **McNeil Point**, about 1.7 miles northward of Hancock Point. A private pier with a float landing and a conspicuous house are at the point.

(121) **Sullivan** is a small village on the north side of Sullivan Harbor, 3 miles above the entrance. Two private piers, one with a float landing, are at Sullivan.

(122) The channel is unmarked above Ferry Point, has dangerous ledges on both sides and is unsafe without local knowledge.

(123) **Sullivan Falls**, locally known as **The Tidal Falls**, are reversing falls in the constricted reach between **Crabtree Neck** and **Falls Point**, about 0.5 mile northwestward of **Ferry Point**. The channel through the falls is obstructed by ledges and the tidal currents are swift and dangerous. Navigation through the falls is safe only at slack water. Most craft go up on the last of the flood but come out only at high water slack as there is great turbulence whenever the current is running at strength. The cove on the west side just at the bottom of the falls has a lobster pound, and there is a wharf and float landing, with 15 feet alongside, on the north side of the cove. Gasoline, diesel fuel and water are available at the wharf.

(124) The tidal currents through Sullivan Falls are dangerous at strength. High-water slack is 1 hour and 20 minutes and low-water slack 1 hour and 45 minutes later in the falls than below them. Ice obstructs navigation in

Taunton Bay and Sullivan Harbor from January through March.

(125) **West Sullivan**, on the north side of the bay just above Sullivan Falls, has several abandoned quarry wharves at which vessels were formerly loaded. The U.S. Route 1 highway bridge crosses the bay about 0.5 mile above the falls and connects West Sullivan with **Waukeag**. The bridge has a fixed span with a clearance of 17 feet.

(126) **Taunton Bay** joins Sullivan Harbor at Sullivan Falls. An unmarked channel with a depth of about 8 feet leads through the bay to a point about 0.9 mile below the head of the bay. Above this point, the channel is obstructed by rocks and tidal flats. The bay outside of this channel is bare, or nearly so, at low water. The granite quarries along the east side of the bay have been abandoned. **Franklin** is a town at the head of Taunton Bay.

(127) The **Taunton Bay Closed Area**, a Marine Protected Area (MPA), includes the waters of Taunton Bay, Egypt Bay and Hog Bay.

(128) **Skillings River** is an arm of the northern part of Frenchman Bay westward of Sullivan Harbor. The entrance is 1.7 miles wide at the mouth between **Hancock Point** on the east and **Meadow Point** on the west, but it contracts rapidly to a width of 400 yards at **Pecks Point**, about 2 miles above Hancock Point. Above this, the river leads about 4 miles in a northwesterly direction to Franklin Road.

(129) The channel above Pecks Point is narrow and crooked and has numerous rocks and ledges, which makes navigation difficult. Strangers wishing to enter the river should anchor 1.5 miles above Hancock Point in depths of 30 to 42 feet and obtain a pilot from among local boatmen. The river is unmarked and is seldom used except by local fishing craft. The wharves usually are small and bare at low water. Strangers in small craft can enter with the aid of the chart.

(130) **Raccoon Cove** is a large shallow cove on the west side of Skillings River near the entrance. The cove is obstructed by **Shooting Ledge** and other ledges. Boats are often hauled out and stored on the small point of land jutting out from the western end of **Marlboro Beach**, on the north side of the cove.

(131) Large commercial lobster pounds have been formed by damming the bight at and immediately south of **Youngs Point**, 3 miles northwest of Hancock Point at the entrance to **Youngs Bay.**

(132) **Eastern Bay**, together with Mount Desert Narrows and Western Bay, forms a thoroughfare north of Mount Desert Island from the head of Frenchman Bay to Blue Hill Bay. **Googins Ledge**, nearly 0.5 mile long and bare in the center at low water, is near the center of the bay. A buoy is on the southwest side of the ledge. The channel leads southward of the buoy. Except for Googins Ledge the bay is mostly deep and clear in midchannel to the entrance of Mount Desert Narrows, but the eastern half is open and unprotected in easterly and southeasterly weather.

(133) Good anchorage for deep-draft vessels is available westward of Googins Ledge in depths of 36 to 54 feet. There is also good anchorage southward of Googins Ledge, about 0.3 mile from shore off the entrance of Salisbury and Emery Coves in depths of 42 to 48 feet. An unmarked 16-foot spot is off Emery Cove.

(134) The north shore of Eastern Bay 0.7 mile west of Meadow Point is foul; a rock that uncovers 6 feet is 0.2 mile offshore.

(135) **Lamoine Beach** extends about 0.6 mile westward of Meadow Point. At the western end of the beach there are the pilings of a pier that bare at low water.

(136) **Lamoine State Park**, which includes the grounds of a former naval coaling plant, is about 0.8 mile westward of Lamoine Beach. Most of the steel piling of the former coaling pier, which was set in heavy concrete bases, has been removed but caution should be exercised when approaching within 200 feet of the Lamoine State Park pier. The park pier has a float landing and a prominent A-frame structure on its outer end; depths of 15 feet are reported alongside the float. A private small-craft launching ramp is adjacent to the park pier. Two white buildings on the hill behind the pier are conspicuous.

(137) **Sand Point** is on the south shore of Eastern Bay at the entrance. **Salsbury Cove** and **Emery Cove** are 0.8 mile and 1.1 miles westward of Sand Point. On the point between these two coves is a biological experimental station with a float landing.

(138) At **Hadley Point**, 2.2 miles west of Sand Point, Eastern Bay merges with Mount Desert Narrows. **Berry Cove** makes into the northern shore opposite Hadley Point. Good anchorage can be had in 18 to 24 feet off the entrance to this cove, which is shallow at its head.

(139) **Mount Desert Narrows**, northward of Mount Desert Island, connects the head of Frenchman Bay with the head of Blue Hill Bay. The passage is crossed by State Route 3 highway bridge and causeway. The fixed span over the navigation channel has a clearance of 25 feet.

(140) The channel is nearly bare at low water with scattered boulders and ledges of rock and is used at high water by boats up to 9-foot draft. It is narrow, difficult and fringed with reefs. Strangers should not attempt passage with drafts greater than 4 or 5 feet and should go through on a rising tide. The current sets westward on the flood and eastward on the ebb.

(141) Passage through Mount Desert Narrows from Frenchman Bay to Blue Hill Bay should not be attempted without local knowledge. The channel approaches to the bridge from Thomas Island past Trap Rock to deep water in Blue Hill Bay, southward of Haynes Point, are narrow, treacherous, difficult and unmarked. The most favorable time is at high-water slack, as the current at other times is strong and turbulent.

(142) **Jordan River**, making northward from Mount Desert Narrows just west of Berry Cove, has a narrow, crooked and unmarked channel and dries for almost half of its upper length above Lamoine. Local knowledge is necessary for its navigation. The few wharves are in

ruins. Bar Harbor airport is on the west side of the river entrance. The buildings, control tower and aerolight and tower of the airport are conspicuous.

(143) The **Jordan River Seed Mussel Conservation Area**, a Marine Protected Area (MPA), includes the waters of Jordan River.

(144) **Cape Levi** (44°25.8'N., 68°14.8'W.) is on the northeastern shore of Mount Desert Island 1.2 miles east-southeastward of Sand Point. **Sunken Ledge**, covered 5 feet, is 0.5 mile north-northwestward of the cape and is marked on its north side by a buoy. From Cape Levi to Lookout Point, 0.6 mile southward, **Halftide Ledge**, a triangular shoal with scattered boulders which uncover 6 feet makes out from the shore for about 0.6 mile, where it is marked by a buoy. All craft should avoid it.

(145)

Hulls Cove to Rum Key

(146) **Hulls Cove** is a broad open bight on the northeast side of Mount Desert Island. Several dangers are off the cove, but they are marked by buoys. The cove shoals rapidly from the entrance to the head, and the low-water line extends about 200 yards from its head. A boatyard with a 175-foot pier is on the north side of the cove, about 250 yards southwestward of Lookout Point. The yard has a 10-ton marine railway that can handle craft up to 40 feet long for hull and engine repairs or dry open and covered winter storage; gasoline and ice are available and electronic repairs can be made. There is a small private marine railway and boat shed on the south side of the cove near the head.

(147) The Bar Harbor Yacht Club is at **Canoe Point**, the southern point of the entrance to Hulls Cove. A float landing at the club pier is reported to have 20 feet alongside. A shoal with a least depth of 2 feet, just off the entrance to Hulls Cove, is marked on its south side by a buoy. Passage into the cove is on either side of the shoal.

(148) **Bald Rock**, 20 feet high, is about 1.4 miles east-northeastward of Canoe Point. A buoy is northward of a ledge that uncovers 6 feet just north of Bald Rock. **Bald Rock Ledge**, about 0.5 mile in width, is 0.3 mile southwest of Bald Rock. The high part of the ledge uncovers 5 feet. A buoy is on the southwest side of the ledge. The passage between Bald Rock and Bald Rock Ledge is dangerous without complete local knowledge. A dangerous rock awash is near the middle of the passage, about 0.3 mile west-southwestward of Bald Rock.

(149) The shoreline from Hulls Cove to Bar Harbor is backed by many large summer homes. The area between Bald Rock Ledge and Bar Island to the southward is sometimes used as an anchorage by larger yachts.

(150) A **ferry terminal**, about 0.5 mile westward of the west end of Bar Island, is the destination of many cruise ships and ferry vessels in and out of Bar Harbor. Caution should be exercised when selecting anchorage northward and eastward of the terminal due to increased marine traffic in the area. An unmarked ledge that uncovers 5 feet is about 350 yards northwestward of the ferry terminal.

(151) **Recommended Vessel Routes** have been established for deep-draft vessels approaching Frenchman Bay from the south. See **Navigation Guidelines, Frenchman Bay** (indexed as such), under Frenchman Bay earlier in this chapter for details.

(152)

Anchorages

(153) Two general anchorages are on the eastern side of Mount Desert Island, 0.5 and 1.5 mile south of Bald Rock Ledge, respectively. (See **33 CFR 110.1**and **110.130** chapter 2, for limits and regulations.)

(154) **Bar Harbor** is formed by the east shore of Mount Desert Island on the west, **Bar Island** and **Sheep Porcupine Island** on the north, and on the south by a breakwater extending southwesterly from **Bald Porcupine Island** across **Porcupine Dry Ledge** to within 250 yards of the shore. The breakwater, marked by a light at its southwestern end, is covered at high water for most of its length except for a part of Porcupine Dry Ledge.

(155) A deep channel, about 150 yards wide, extends into the harbor between the western end of the breakwater and the shore of Mount Desert Island. This channel is used by small local craft, but extreme caution should be exercised when using it. It has been reported that on extreme high tides with a smooth sea there is no indication of the position of the breakwater by tide rips or otherwise, except for the light marking the southwesterly end of the breakwater.

(156) All the islands surrounding Bar Harbor are high and wooded and have no prominent marks. When approaching from southward, Bald Porcupine Island is distinguishable because of its bare rocky slopes. The bar extending between Bar Island and the town consists of scattered boulders on soft bottom.

(157) The principal entrance is from the eastward, between Bald Porcupine and Sheep Porcupine Islands, and is clear. A rock awash is about 40 yards off the southeastern shore of Bald Porcupine Island. Local vessels sometimes enter from northward between Sheep Porcupine Island and the small islet 2 feet high eastward of Bar Island, where the depth is 13 feet in midchannel. Two private buoys mark the channel.

(158) Some shelter from southerly winds is afforded by the breakwater. A swell makes in during southeast winds, and vessels should not attempt to ride out a gale from that direction in Bar Harbor. **General Anchorage A** is southward and southeastward of the eastern end of Bar Island in depths of 40 to 80 feet, the depths shoaling rapidly toward the bar southward of Bar Island. Large vessels sometimes anchor in **General Anchorage B**, north of Bar Island, in 50 to 80 feet, soft bottom. The western shore is fairly bold. Vessels should keep over 0.4 mile southward of a line joining Bald Rock Ledge and Bald Rock, a bare rocky islet.

(159) **Bar Harbor** is a town on the shore of Bar Harbor with a hospital, pharmacy, several banks and good highway connections. It is an important summer resort and yachting center, and during the summer many sightseeing cruises and fishing trips are scheduled daily for the vicinity of Frenchman Bay and Mount Desert Island.

(160)

Weather, Bar Harbor and vicinity

(161) Bar Harbor averages about two days each year with maximum temperatures in excess of 90 °F. July is the warmest month with an average high of 78 °F and an average minimum of 57 °F. January is the coolest month with an average high of 32 °F and an average minimum of 15 °F. The highest temperature on record for Bar Harbor is 101 °F recorded in August 1975 and the lowest temperature on record is -17 °F recorded in January 1981. About 141 days each year see temperatures below 32 °F. Every month has seen temperatures below 40 °F and every month except June, July and August has recorded temperatures below freezing.

(162) The average annual precipitation for Bar Harbor is 50.3 inches (1278 mm) with an annual maximum during early winter and a minimum during mid-summer. Precipitation falls on about 341 days each year. The wettest month is November with 5.9 inches (150 mm), and the driest, July, averages only 3.0 inches (76 mm). An average of 6 thunderstorm days occur each year with July and August being the most likely months. Snow falls on about 65 days each year and averages about 65 inches (1651 mm) each year. December through March each average greater than a foot (305 mm) per year while February averages 18 inches (457 mm). One-foot (305 mm) snowfalls in a 24-hour period have occurred in each month November through March. About 15 days each year have a snowfall total greater than 1.5 inches (38 mm), and snow has fallen in every month except June through September. Fog is present on average 53 days each year and is evenly distributed throughout the year with a slight maximum during mid-summer.

(163)

Quarantine, customs, immigration and agricultural quarantine

(164) (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(165) **Quarantine** is enforced in accordance with regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.)

(166) Bar Harbor is a **customs port of entry.**

(167) The town **harbormaster** assigns moorings and maintains an office on the municipal pier.

(168) The large municipal pier, on the north shore of the town, has reported depths of 8 to 10 feet at the end. Diesel fuel by truck is available at the pier. Floats for yachts and commercial craft are on each side of the pier; water and electrical shore power are available at the floats. A small-craft launching ramp is at the east and inshore end of the municipal pier. The other wharves at Bar Harbor have depths of less than 1 foot to 5 feet alongside. Gasoline, diesel fuel, water and ice are available at the wharf just west of the municipal pier; marine supplies, provisions and bottled gas can be obtained in town. Charter and excursion boats operate from the municipal pier and the other wharves.

(169) There are no repair facilities in town, but boatyards at Hulls Cove and Sand Point have marine railways and make repairs.

(170) **Cromwell Cove**, westward of the end of the breakwater, is seldom used. A pier in ruins is on the south side of the cove.

(171) **Burnt Porcupine Island**, northeastward of Bar Harbor, is about 0.5 mile eastward of Sheep Porcupine Island. A deep clear channel to the upper part of Frenchman Bay is between the islands. A bell buoy is on the west side of the channel. **Rum Key** is between Burnt Porcupine and Long Porcupine Islands.

(172)

Long Porcupine Island to Otter Cove

(173) Another deep channel to the upper part of Frenchman Bay is between **Long Porcupine Island** (44°24.4'N., 68°09.8'W.), 0.4 mile east of Burnt Porcupine Island, and Ironbound Island. **The Hop** is off the northeast side of Long Porcupine Island.

(174) The southeast shore of Mount Desert Island between Bar Harbor and Seal Harbor (44°17.5'N., 68°14.5'W.) is rocky and precipitous. Several dangers are off the shore, but the most dangerous either show above water or are marked by buoys.

(175) **The Thrumcap**, 1.4 miles southward of Bald Porcupine Island, is a round, rocky island with a clump of trees in its center. It is reported that there are downdraft wind currents around Thrumcap Island, and, accordingly, small sailboats should keep offshore.

(176)

Caution

(177) An 8-foot spot is 0.2 miles north-northwest from The Thrumcap. It is unmarked and has been struck by several yachts navigating along the coast at this point.

(178) **Newport Ledge** is 400 yards from shore, midway between The Thrumcap and Schooner Head. The ledge uncovers at extreme low water; a buoy is on its eastern side. The bottom west of the ledge is broken and should not be crossed by vessels.

(179) **Schooner Head** (44°20.7'N., 68°10.6'W.), 1.2 miles south of The Thrumcap, and **Great Head,** 0.9 mile farther southward, are prominent rocky headlands on the eastern side of Mount Desert Island. On the summit of the eastern hill at Great Head, the ruins of a small round flat-topped stone building are conspicuous. Numerous boulders lie awash between the two headlands.

(180) **Old Whale Ledge**, 350 yards from shore, midway between Schooner Head and Great Head, is awash at low

water. A lighted gong buoy is 300 yards eastward of the ledge.

(181) **Newport Cove**, a small cove westward of Great Head, is exposed southward, has poor holding ground and is never used as an anchorage. **Old Soaker**, a bare rock 6 feet high, is off the entrance. Because of a prominent sand beach at the head of the cove, the area is known locally as **The Sand Beach**.

(182) **Otter Cliff Ledge**, which uncovers 6 feet, is 400 yards eastward of **Otter Point**, 1.5 miles southward of Great Head. A bell buoy is eastward of the ledge.

(183) **Otter Cove** is a long cove making northward between Otter Point and Western Point, 0.5 mile southwestward. The cove has deep water in the entrance and is bare for 0.4 mile from the head. The cove is exposed southward but is used by local fishermen who lay to moorings. A causeway and fixed bridge cross the cove 0.7 mile above the entrance.

(184)

Mount Desert Island to Deadman Point

(185) **Southwest Harbor**, **Somes Sound**, **Northeast Harbor**, **Seal Harbor** and several other coves are in the southeast side of Mount Desert Island, inside a large group of islands and shoals. These waters are the approaches to several important villages and summer resorts and are frequented by many pleasure craft and fishing boats. Southwest Harbor is used extensively as a harbor of refuge. The harbors can be approached through the channels on either side of Sutton Island or through Western Way.

(186)

No-Discharge Zone

(187) The State of Maine, with the approval of the Environmental Protection Agency, has established a No-Discharge Zone (NDZ) in the municipal waters of Mount Desert, Southwest Harbor, portions of Cranberry Isles and Tremont.

(188) Within the NDZ, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZ, discharge of sewage is regulated by **40 CFR 140** (see chapter 2).

(189) **Baker Island**, 3.3 miles south of **Western Point** and the most southeasterly of the group of islands in the vicinity, is mostly wooded but grassy on its northwest end. There are several houses on the island. **Baker Island Light** (44°14'28"N., 68°11'56"W.), 105 feet above the water, is shown from a 43-foot white stone tower in the center of the island. The light is partially obscured by trees. The island is surrounded by ledges, bare and covered, and should be given a berth of at least 0.4 mile.

(190) **The Thumper** is a ledge, which uncovers 5 feet, 300 yards southward of Baker Island. A dangerous rock awash is close southeastward of the ledge. **Southwest Rocks**, which uncover 1 foot, are 500 yards off the southwest shore of the island.

(191) **Harding Ledge**, covered 1½ feet, about 0.2 mile off the east end of Little Cranberry Island; **Gilley Ledge**, covered 11 feet and about 0.55 mile north of Baker Island and the ledges extending northeastward and eastward of Baker Island are all marked by buoys.

(192) **Little Cranberry Island**, about 1 mile northwest of Baker Island, is low and wooded. A large white building with a lookout tower is prominent on the southeast point of the island. **Islesford** is a village on the west side of the island. Three piers with float landings and the ruins of an old stone breakwater-pier, close northwestward, are on the north side of **Hadlock Cove**, a bight off Islesford. Some transient berths are available at the pier landings. The southernmost pier, the village landing, has a reported depth of 9 feet alongside its float. Depths of 7 feet are reported alongside the other two pier landings. Gasoline and diesel fuel are available at the float of the center pier. The mail and passenger ferry uses the village pier. Groceries and limited marine supplies can be obtained in Islesford. A boatyard southward of the piers has several marine railways that can handle craft up to 50 feet for winter storage.

(193) **The Gut**, a passage between Little Cranberry and Great Cranberry Islands, is used at any stage of the tide by small local craft, but it has many unmarked ledges and should not be used by strangers.

(194) Small passenger and mail boats maintain service between the Cranberry Islands and Northeast Harbor the year round and between the islands and Seal Harbor during the summer.

(195) **Cranberry Harbor**, southward of Sutton Island and between Little Cranberry and Great Cranberry Islands, is frequented by small local vessels. Sometimes small coasting vessels anchor in the harbor, but Southwest Harbor offers much better anchorage. The usual anchorage in Cranberry Harbor is in depths of 14 to 20 feet in the middle of the harbor with the wharves at Islesford bearing about 050°. Care must be taken to keep well clear of the buoy on the end of the ledge which extends 350 yards westward from the east side at the entrance. An obstruction, cleared 6 feet, is 0.3 mile northward of Long Point on the west side of the entrance to the harbor.

(196) **The Pool** is a large shallow cove on the east side of Great Cranberry Island southwest of Cranberry Harbor. A rock awash is nearly in midentrance. Several small piers, dry at low water, and a boatyard are on the west side of the pool. A marine railway at the yard can handle craft up to 45 feet for engine repairs; open and covered storage for 60 boats is available.

(197) **Great Cranberry Island**, about 2 miles west of Baker Island, is wooded and has no prominent marks visible from southward. **Cranberry Isles** is a village on the island. **Spurling Cove** makes into the north shore of the island. The 280-foot village pier, the more westerly of two piers on the south side of the cove, has a float landing at which the mail and passenger ferry lands. About 50 yards southeastward of the village pier is a 300-foot commercial pier, also with a float landing. Depths of 8

feet are reported alongside both float landings. Gasoline, diesel fuel and water are available at the commercial pier. **Long Point** is the northeast end of the island. **Crow Island**, northeast of **Deadman Point**, the southeast point of Great Cranberry Island, is 26 feet high and grassy with reefs to the east and southeast.

(198)

South Bunker Ledge and Long Ledge

(199) **South Bunker Ledge** (44°13.6'N., 68°17.0'W.), 0.7 mile southwest of Great Cranberry Island and in the southern approach to Western Way, uncovers about 4 feet. A daybeacon is on the ledge. A 26-foot spot is 1.3 miles southeast of the ledge.

(200) **Long Ledge**, westward of South Bunker Ledge, is a dangerous reef extending 0.5 mile southeastward from Mount Desert Island, on the western side of the southern approach to Western Way. The ledge uncovers with a few rocks which show at high water. A lighted gong buoy is off the southeastern side.

(201)

Western Way to Kings Point

(202) **Western Way**, between the western side of Great Cranberry Island and Mount Desert Island, is a passage frequently used by small vessels bound to Southwest Harbor and vicinity. Also, small vessels bound between points westward and any point in Frenchman Bay use it, except in rough weather. The channel is buoyed, and the least midchannel depth is 13 feet on a bar toward the northern end, but there are unmarked spots of 10 to 12 feet close to the sailing lines. The passage should not be used by strangers in craft drawing more than 10 feet. A lighted fairway bell buoy marks the southern entrance, and a lighted fairway gong buoy marks the northern entrance.

(203) **Cranberry Island Ledge**, covered 9 feet near its southwestern end, about 500 yards from the southwestern end of Great Cranberry Island, and with lesser depths closer to shore, is marked by a buoy. **Flynns Ledge**, covered 2 feet, extends about 0.5 mile southeastward from Seawall Point, where it is marked by a buoy. A bare rock, 6 feet high, is near the middle of the ledge.

(204) **Southwest Harbor**, on the south end of Mount Desert Island, is the approach to the villages of Southwest Harbor and Manset. The harbor is entered from the east and affords an excellent, well-sheltered anchorage. A 360-feet long breakwater sits about 400 yards from the head of Southwest Harbor at (44°16'26"N.,68°19'10"W.); it is marked by a daybeacon.

(205) **Greening Island**, on the north side of the entrance to the harbor, is low and wooded. Several houses are visible on the island, and a large house at the eastern end is prominent. Shoals that border the island on all sides are marked on the southeastern and southwestern ends by buoys. Several private piers with float landings are on Greening Island, and three private boathouses with marine ways are conspicuous.

(206) **Eastern Way**, a well-marked channel approaching Southwest Harbor northward of Sutton Island, is deep and the recommended route for deep-draft and low-powered vessels. This passage is used by all vessels entering Southwest Harbor from the northward in Frenchman Bay and by most vessels entering from the eastward and southeastward.

(207) **Gilley Thorofare**, the channel southward of Sutton Island, has unmarked rocks with cleared depths of 13 to 18 feet. **Spurling Rock**, covered 7 feet, on the south side of the channel about 0.3 mile off the northeastern end of Great Cranberry Island, is marked by a bell buoy to the northward. This passage has been examined by means of a wire drag. With the aid of the chart it is easy to navigate in the daytime, but in hazy weather it should be avoided by all vessels.

(208) Excellent sheltered anchorage, except from southeasterly to southwesterly winds, may be found in Southwest Harbor in depths of 6 to 50 feet. Deep-draft vessels can anchor midway between Greening Island and the southern shore in depths of 34 to 50 feet. Smaller vessels can anchor farther in the harbor; the depths shoal gradually to 12 feet at a point 100 yards eastward of the islet near the head of the harbor.

(209) Vessels approaching Southwest Harbor from any direction in good weather should experience no trouble with strict attention to the chart and following the aids.

(210) Normally, navigation in Southwest Harbor and approaches is not restricted by ice. In very severe winters ice is reported to have formed as far out as the Cranberry Islands but is carried to sea at the first ebb tide by the current out of Somes Sound or is broken up by icebreakers.

(211) A pilot is not required if entering from eastward in daytime with clear weather. Strangers coming from westward and crossing Bass Harbor Bar and Cranberry Island Bar (Western Way) usually can obtain a pilot from among the fishermen at Bass Harbor if desired.

(212) The village of **Southwest Harbor** is on the north side and at the head of Southwest Harbor. The town **harbormaster** supervises moorings in the harbor.

(213) **Southwest Harbor Coast Guard Station** wharf on Clark Point, on the north side of Southwest Harbor, has a reported depth of 15 feet alongside.

(214) The lower town wharf at Clark Point, close westward of the Coast Guard wharf, has three floats, at which there is a reported depth of 12 feet at the outer end.

(215) The upper town landing, about 0.3 mile westward of Clark Point, has float landings with depths of 4 to 6 feet reported alongside. The mail and passenger ferry also calls at this landing.

(216) There are other commercial and private wharves, some with float landings, on the north side of the harbor with depths of 4 to 15 feet alongside. Gasoline, diesel fuel, water, sewage pump-out, limited marine supplies and electronic repairs are available at the wharves.

(217) A shipyard, between the two town facilities, can build craft up to 100 feet in length and can make hull, engine, electrical and electronic repairs. A marine railway at the yard can handle craft up to 100 feet; a 25-ton crane is also available.

(218) A large wharf/marina is on the west side of the harbor. Fuel, water, limited repairs and services are available. Boats up to 150 feet long are accommodated.

(219) **Manset** is a village on the south side of Southwest Harbor. The town pier is about 0.5 mile westward of **Kings Point**, the southern entrance point to the harbor; depths of 5 feet are reported alongside the pier's float landing. A marina, about 0.1 mile eastward of the town pier, has a reported depth of 3 feet alongside its float landing at which gasoline and water are available. A 2½-ton fixed lift, covered winter storage and marine supplies are available at the marina; hull and engine repairs can be made. The large boatyard, about 0.15 mile eastward of the town pier, has moorings, diesel fuel, water, ice and a marine railway that can handle craft up to 70 feet for hull, engine and electronic repairs. A 35-ton mobile lift and open winter storage are also available at the yard. Depths of 10 feet are reported alongside the yard's service float. There are several commercial marine facilities eastward of the town pier and numerous private wharves and float landings westward of the town pier.

(220) The passage between Greening Island and Clark Point has a least depth of 14 feet and is extensively used. A daybeacon is on the west side of the channel northward of Clark Point. The best water from the southward leads 100 to 150 yards westward of the buoy off the southwestern end of Greening Island and 100 yards eastward of the daybeacon.

(221)

Somes Sound to Mount Desert

(222) **Somes Sound** is a fjord, about 4.5 miles long and 0.2 mile wide, making into the south shore of Mount Desert Island. The sound is between steep rocky shores and has a narrow entrance with few dangers. **Middle Rock**, covered 9 feet, on the west side of the entrance to **The Narrows**, is marked by a buoy on its northeast side. Greening Island is in the middle of the approach, with a channel on each side of it. With the aid of the chart, good anchorage can be selected in 54 to 72 feet. Small sailing vessels should be prepared for downdraft winds.

(223) **Norwood Cove**, on the west side of the entrance to Somes Sound, is not navigable and has a causeway with a footbridge across its entrance. **Jesuit Spring**, where the first settlers in 1613 obtained water, is still free flowing; it is near the high-water mark on the north side of **Fernald Cove**, about 200 yards northwest of **Fernald Point.**

(224) Several private float landings are on the east side of the sound above **Manchester Point**, 0.7 mile north of Greening Island.

(225) **Hall Quarry**, the site of an inactive quarry, is a small settlement on the west side of Somes Sound 3 miles above the entrance. A boatyard at Hall Quarry has a 30-ton mobile hoist that can handle craft up to 60 feet long for hull and engine repairs or dry open and covered winter storage. Depths of 20 feet have been reported alongside the float landing. Another boatyard, on the east side of the sound opposite Hall Quarry, has a marine railway that can handle craft up to 40 feet long or 16 tons for minor hull and engine repairs; open or covered winter storage is available.

(226) **Somes Harbor** is a small cove at the head of Somes Sound. The entrance is narrow and is marked by buoys. Several private piers and float landings for pleasure craft are in the harbor.

(227) **Mount Desert** (Somesville) is a village on the west side of Somes Harbor. A church spire in the village is conspicuous.

(228) A boatyard is in the northeast corner of Somes Sound. A marine railway at the yard can handle craft up to 60 feet in length and 7-foot draft for hull repairs or covered storage.

(229)

Gilpatrick Cove to Lewis Rock

(230) **Gilpatrick Cove**, on the east side of the entrance to Somes Sound, is small and shoal, and the upper end dries at low water. A fixed wooden footbridge across the entrance has a clearance of 4 feet. A float landing in about the middle of the bridge has 4 feet alongside. A pier and float landing of the Northeast Fleet Yacht Club is on the east side of the entrance. Many small yachts moor off the entrance to the cove in the summertime. A wharf, with oil storage tanks on it, is about 0.3 mile northwestward of Gilpatrick Cove. Depths of 3 feet were reported alongside, and in 1979, the wharf was in poor condition.

(231) **Gilpatrick Ledge**, just east of the entrance to Gilpatrick Cove, extends about 300 yards southward and is marked by two daybeacons along its southwesterly edge. Vessels should keep south of the daybeacons.

(232) **Northeast Harbor**, 0.6 mile eastward of Gilpatrick Cove, is 300 yards wide at its entrance and extends into the south shore of Mount Desert Island about 0.8 mile. The harbor is an important yachting center, and there is a summer hotel on the north shore overlooking the harbor. Anchorage for small vessels is available in depths of 14 to 28 feet in the lower part of the harbor. This anchorage is about 200 yards wide and favors the western shore. Anchorage is also available in depths of 5.6 to 7 feet in the inner harbor; the uppermost part of the inner harbor is shoal.

(233) In the middle of the entrance to Northeast Harbor is a rock which uncovers 3 feet. A buoy is on the east side of the rock, and a lighted bell buoy is on the west side. The best passage into the harbor is westward of the rock. In average winters the harbor is reported to be clear of ice except at its head, but in severe winters it is reported to freeze as far out as Bear Island.

(234) The summer resort of **Northeast Harbor** is on the western shore of the harbor.

(235) The town pier and ferry landing, on the southwest side of the inner harbor, has float landings and finger floats for transient and seasonal berthing of yachts. All have reported depths of 10 feet alongside. Water is piped to the pier, and electrical shore power is available. The pier is used by the mail and passenger ferry that calls year-round at Cranberry Island. A concrete ramp and a marine railway are close eastward of the town pier. Telephone, washrooms, ample parking and showers are available at the town pier. Gasoline and diesel fuel can be obtained by truck on short notice, and ice, provisions and marine supplies are available. A boatyard, about 150 yards northwestward of the town pier, has a mobile hoist that can handle craft up to 20 tons for hull repairs or open storage. There are many private piers with float landings about the harbor. The **harbormaster** supervises dockage at the town pier; he may be contacted on VHF-FM channels 16 and 68 or at the town pier.

(236) Gasoline, diesel fuel, water, ice and limited marine supplies can also be obtained at a pier on the west side of Northeast Harbor, about 0.4 mile above the entrance. Depths of 25 feet are reported alongside the pier.

(237) **Bear Island**, on the eastern side of the entrance to Northeast Harbor, is high and wooded. Only the stone foundation, covered 2 feet, and a few submerged piles remain of the old Coast Guard wharf on the northwest side of the island; a buoy is about 90 yards westward of the ruins. A pier and float landing are on the north side of the island. The passage north of the island is almost blocked by rocky ledges awash at various stages of the tide, and passage should not be attempted. A lighted bell buoy south of the island marks the western entrance to Eastern Way.

(238) **Sutton Island**, about 1 mile long and wooded, is on the south side of **Eastern Way,** between the south shore of Mount Desert Island and Cranberry Islands. The channel has a depth of 40 feet near the center and depths of 31 and 36 feet near its northern and southern edges, respectively, and is the recommended channel generally used.

(239) On the northern side of this channel opposite Sutton Island and eastward of Bear Island are **Long Pond Shoal**, covered 5 feet, and **Bowden Ledge**, covered 2 feet. Buoys are south of these dangers.

(240) **Sutton**, a summer resort, is on the western part of Sutton Island. The island has many summer cottages along its shores and several piers with float landings, including a town wharf on the western side with a reported depth of 6 feet alongside its float landings. The town wharf is used by the mail and passenger ferry. The approach to the wharf, near **Fernald Point**, leads between two rock ledges; mariners are advised to exercise care in approaching it.

(241) **Bracy Cove**, 0.5 mile northeastward of Bear Island, is exposed to southeast winds, has a rocky and uneven bottom and is unfit for anchorage. There is a private pier with float landing on the east side of the cove.

(242) **Seal Harbor** makes into the south shore of Mount Desert Island about 1 mile east of Bear Island. Anchorage for small vessels may be had in the middle of the harbor in depths of 15 to 18 feet. This anchorage, about 400 yards in diameter, is exposed to southeasterly winds and is reported to experience considerable roll. The approach is between the buoy off Bowden Ledge on the west and a lighted bell buoy on the east. A ledge that uncovers 6 feet at the outer end extends halfway across the entrance from **Crowninshield Point**, the western entrance point. A buoy is off the ledge. A ledge that uncovers 3 feet is 200 yards from the head of the harbor.

(243) The village of **Seal Harbor,** on the shore of the harbor, has numerous summer homes. The town wharf, on the east side of the harbor about 0.25 mile above the entrance, has a reported depth of 9 feet alongside its float landing. Water is available at the wharf. The Seal Harbor Yacht Club, close northward of the town wharf, has a pier and float landing with a reported depth of 8 feet alongside. The town **harbormaster** has an office at the town wharf. A former coal wharf, on the west side of the harbor opposite the town wharf, has private facilities for hauling out and storing yachts.

(244) **East Bunker Ledge**, southeastward of Seal Harbor and 1 mile eastward of Sutton Island, is 0.3 mile long and has two islets 4 feet high on it. A white pyramidal stone structure is on the southwestern islet. A buoy marks **Lewis Rock**, covered 6 feet, which is 200 yards northwestward of the ledge. A lighted gong buoy is about 0.25 mile southeastward of the ledge.

(245)

Bass Harbor Head to Casco Passage

(246) **Bass Harbor Bar** connects Great Gott Island with **Bass Harbor Head**, the southwestern point of Mount Desert Island. **Bass Harbor Head Light** (44°13'19"N., 68°20'14"W.), 56 feet above the water, is shown from a white tower connected to a dwelling on the head.

(247) The channel is on the through route used by vessels drawing 9 feet or less and is sometimes used by vessels drawing slightly more proceeding at high water and with a smooth sea.

(248) In heavy weather breakers occasionally form across the bar. A heavy chop builds up on the bar and off Long Ledge, 1.6 miles eastward, with the wind contrary to the tidal current, which might beset small craft and open boats.

(249) Of the many islands off the entrance to Blue Hill Bay, those southward of Bass Harbor Bar and Casco Passage, and westward from Duck Islands to Swans and Marshall Islands, are discussed in this chapter. The islands in this area are in general wooded and have few conspicuous marks. The only ones having settlements are Swans, Long and Great Gott Islands. The area is very broken and rocky, with numerous bare and submerged ledges,

many of them unmarked. The through route by way of Casco Passage and Bass Harbor Bar is used by many vessels, except that vessels unable to cross the bar pass south of Placentia Island and between Black and Little Gott Islands. The passages through the islands southward are seldom used except by local fishermen and yachtsmen and are described later in this chapter.

(250) **Great Duck Island**, about 5 miles south of Great Cranberry Island, is the most southeasterly of the islands off Blue Hill Bay. The island is partly wooded and from a distance eastward or westward appears as two islands. There are a small white house and a private landing strip on the northern slope of the island.

(251) **Great Duck Island Light** (44°08'31"N., 68°14'45"W.), 67 feet above the water, is shown from a 36-foot white tower on the south end of the island. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A. The buildings of the light station are prominent. The light is partially obscured by trees from about 143° to 206.5°.

(252) **Little Duck Island**, 0.7 mile northward of Great Duck Island, is partly wooded and has no distinguishing marks.

(253) **The Drums**, a dangerous ledge 2.5 miles west of Great Duck Island and 2 miles northeastward of Long Island, is awash at low water. A bell buoy is southeast of the ledge. The range formed by the western ends of Green and Placentia Islands leads well westward of this ledge.

(254) **Horseshoe Ledge**, 1 mile north of The Drums, is awash at low water and marked on its southeast end by a buoy.

(255) **Green Islands**, 4 miles west-northwestward of Great Duck Island Light and 0.7 mile southward of Black Island, are two rocky islets with grass on top.

(256) **Drum Island** is a bare rock 400 yards northeastward of the easterly Green Island.

(257) **Black Island**, 4.5 miles northwestward of Great Duck Island Light, is 157 feet high and wooded. Three ledges are off the east side of the island: **Inner Dawes Ledge**, the northernmost off the northeast side of the island, is bare at high water; **Outer Dawes Ledge**, about 0.4 mile to the southward, is awash at high water; and **Grindstone Ledge**, about 0.4 mile farther southward, uncovers about 5 feet and is marked by a buoy. An unmarked shoal with a clear depth of 11 feet is about 500 yards southeastward of the buoy. **Little Black Island**, off the southwest side of Black Island, is wooded in the center. An aquaculture farm is on the northwest side of Black Island. Mariners are advised to use caution when navigating in the vicinity.

(258) **Placentia Island**, 0.4 mile northwest of Black Island, is 135 feet high and wooded except on its eastern end, which is grassy. Two houses are visible on the slope of the hill on the northeastern extremity. A buoy marks the shoal extending off the northeast end of the island.

(259) **Little Gott Island** and **Great Gott Island**, 0.4 and 1 mile northeastward of Black Island, are mostly wooded. **Gotts Island** is a small town on the west side of Great Gott Island. The approach to the town is via the passage between Little Gott and Great Gott Islands, which can be entered from southward at low water. A bar, bare at low water, obstructs the passage at the northwest end. The current between the islands is reported to be strong, flooding northwestward and ebbing southeastward. The houses are the most prominent marks in this vicinity. There are no wharves.

(260) **Staple Ledge**, between Placentia Island and the northeast end of Swans Island, is awash at low water. A buoy is off the northeast side of the ledge.

(261) **Long Island**, 4.3 miles west-southwestward of Great Duck Island, is the most southerly of the large islands off Blue Hill Bay. The island is 210 feet high and wooded but has no conspicuous marks visible from seaward.

(262) **Lunt Harbor** is a cove in the north side of Long Island.

(263) **Frenchboro** is a village on the shore of the cove. The cove has good holding ground and is used as an anchorage by local boats, but it is somewhat exposed in northeasterly weather. In 2021, the anchorage basins in the north and south parts of the cove had depths of 6 and 4.4 feet, respectively, except for shoaling along the edges. A cable area is in the cove. Ice seldom interferes with navigation. A crib wharf on the northeast side of the harbor has a depth of 6 to 7 feet alongside. The other wharves are bare before low water. Gasoline, diesel fuel and water are available. Mail comes by the Swans Island State auto-passenger ferry that calls at Lunt Harbor.

(264) Northward of Long Island are numerous small islands and ledges. **Harbor Island**, just off Lunt Harbor, is wooded, and a reef which uncovers extends west of the island. A bell buoy is off a 12-foot spot 0.3 mile west of the island. A 28-foot spot is close west of the buoy. **Crow Island**, 0.4 mile north of Long Island, is wooded except at its eastern end which is a bare rock. **Dry Money Ledge**, 400 yards west of Crow Island, has a white rock islet about 10 feet high on it. **Sunken Money Ledge**, 400 yards southwest of Dry Money Ledge, uncovers. **Northeast Ledge**, 0.5 mile northeast of Long Island, is covered 13 feet. **Crow Island Ledge**, 0.3 mile north of Crow Island, is covered 9 feet. A fairway lighted gong buoy is northward of Northeast Ledge and eastward of Crow Island Ledge. **Beaumont Ledge**, 0.5 mile north of Crow Island, is covered 7 feet. **Otter Ledge**, about 0.8 mile north of Crow Island, is awash at low water.

(265) **Sister Islands**, 0.5 mile northwestward of Crow Island, are wooded. **Sister Ledge**, awash at low water, extends 300 yards southward of the westerly of the Sister Islands. **Ram Island**, 0.5 mile north of Sister Islands and 0.2 mile off Swans Island, is marked by a single tree.

(266) A 12-foot spot is 450 yards off **East Point**, the eastern extremity of Swans Island, and 950 yards northeastward of Ram Island.

(267) Westward of Long Island is a deep passage. **Beach Ledge**, about 0.4 mile westward of Long Island, is covered 14 feet. A bell buoy is west of the ledge. **Johns Island**, 1.1 miles west of Long Island, is grassy and has

many dead trees and a few scrub trees. **Johns Island Dry Ledge**, 0.5 mile southwest of Johns Island, is 0.2 mile in diameter and has rocks showing at high water. **Johns Island Sunken Ledge**, 0.6 mile south of Johns Island, is covered 4 feet; a buoy is south of the ledge.

(268) The passage northward between Long and Swans Islands has deep water, but there are many unmarked ledges. The best channel is between the ledge extending from Johns Island and the bell buoy off Beach Ledge, thence between the westerly of the Sister Islands and Ram Island. **Red Point**, on the southeast side of Swans Island 0.4 mile west of Sister Islands, has a low bare reddish bluff. Any of the passages can be used by small craft with the aid of the chart. A 15-foot spot is in midchannel between Red Point and the westernmost of the Sister Islands, and a 24-foot spot is about 0.6 mile southwestward of the same islands; both are unmarked.

(269) **Swans Island**, about 2 miles northwest of Long Island, is the largest of the islands off Blue Hill Bay. The three villages on the island are Atlantic, Swans Island and Minturn. The island has several sheltered coves, but all except Mackerel Cove and Burnt Coat Harbor are foul and little used. There is no piped water supply on the island, but there is a power station and electricity.

(270) **Mackerel Cove** is a good anchorage on the north side of Swans Island south of the eastern entrance to Casco Passage. There are islets and numerous ledges in the cove, but the entrance from northward is easy of access in the daytime. A cable area is in the northeast part of the cove between Crow Island and North Point and extends southeast to **Fir Point**. Vessels are cautioned against anchoring in this area. A ferry vessel operates in and out of Mackerel Cove. Submerged rocks, covered 7 to 11 feet, are near the approach to the ferry pier, southwest of Fir Point. The rocks are marked by a buoy on the west side.

(271) **North Point**, the northernmost point of Swans Island, is on the east side of the northern entrance to Mackerel Cove. A lighted gong buoy, about 0.2 mile northwestward of the point, is off a rock covered 1 foot.

(272) **Crow Island**, about 0.6 mile west of North Point, is on the western side of the northern entrance. A buoy is eastward of a ledge, awash at low water, 400 yards east of the island.

(273) A narrow channel into Mackerel Cove from York Narrows follows close to the shore of Swans Island and passes southward of Orono and Round Islands.

(274) Anchorage can be found in Mackerel Cove between Fir Point Ledge Buoy 3 and the buoy off Crow Island in depths of 24 to 32 feet. Care must be taken to give the eastern shore a berth of 300 yards. Another good berth is between Fir Point Ledge Buoy 3 and a bare ledge 0.3 mile southwestward. A 3-foot spot and a 4-foot spot are 0.5 mile and 0.25 mile northwest of the ledge, respectively. Another 4-foot spot is 500 yards east of the ledge and off the village. All three spots are unmarked.

(275) Most of the dangers in the entrance to Mackerel Cove are buoyed, and, although there are many dangers inside, it should not be difficult even for a stranger to enter and anchor safely by daylight with the aid of a chart. Enter between the lighted gong buoy off North Point and the buoy off Crow Island and steer about **181°** so as to pass westward of Fir Point Ledge Buoy 3.

(276) **Atlantic** is a village on the southeast side of Mackerel Cove. The church spire and several houses are prominent from eastward. The wharves are nearly bare at low water. Gasoline, diesel fuel, provisions and some marine supplies are available at Burnt Coat Harbor. The State auto and passenger ferry operates throughout the year between Atlantic and Bass Harbor. The ferry slip is close southward of Fir Point on the east side of the cove.

(277) **Seal Cove**, on the northwest side of Swans Island just south of **Buckle Island** and York Narrows, and **Toothacher Cove**, on the southwest side of Swans Island, have many unmarked dangers and are important to fishermen. A shoal covered 4 feet is in the middle of the approach to Toothacher Cove. A floating fish pen area is about 400 yards northeast of the 4-foot shoal.

(278) **Burnt Coat Harbor**, a small well-sheltered anchorage on the southwestern side of Swans Island, is much used by fishermen and yachtsmen. **Burnt Coat Harbor Light** (44°08'03"N., 68°26'50"W.), 75 feet above the water, is shown from a square white tower connected to a dwelling on **Hockamock Head**, on the west side of the harbor entrance.

(279) The anchorage, eastward of the light, is about 500 yards wide, with depths of 21 to 34 feet, soft bottom. A good anchorage for small craft is in the channel northward of the light in depths of 13 to 24 feet. A stone wharf and power plant are on the north side of **Long Cove**, on the east side of the harbor near the entrance.

(280) **Swans Island** is a village on the west shore of Burnt Coat Harbor. The largest of several wharves has two floats with 5 to 6 feet alongside. The other wharves have less depth. Transient berths are available. Gasoline, diesel fuel, water, ice, provisions and marine supplies are available. There is a small machine shop that can do minor engine and underwater repairs.

(281) **Minturn** is a small village on the east shore of the harbor. The largest of several fish wharves has 5 feet alongside. Gasoline, diesel fuel, and some provisions and supplies are available at the wharf. Pilots may be obtained from among the fishermen at the harbor.

(282) Off the entrance and approach to Burnt Coat Harbor are several islands and reefs. **Harbor Island**, in the middle of the entrance, is wooded except for its southwest and southeast sides; a house is on the northeastern slope.

(283) **Baker Islands**, 0.3 mile southeast, and **Scrag Island**, 0.2 mile south, respectively, of Harbor Island, are wooded. **Green Island**, 0.3 mile southwest of Scrag Island, and **Gooseberry Island**, 0.6 mile west of Harbor Island, are bare and grassy. **Gooseberry Island Ledge**, 0.2 mile southeastward of Gooseberry Island and on the northwestern side of the approach to the harbor, is awash at low water; a buoy is off the ledge. **High Sheriff**, a bare rock, and **Sheriff Ledge**, awash at low water,

are westward of Gooseberry Island. A buoy, 0.4 mile southwestward of High Sheriff, marks a 24-foot spot. A rocky shoal, covered 18 feet, 0.5 mile southwest of Gooseberry Island, is unmarked.

(284)

Routes

(285) The main entrance to Burnt Coat Harbor is from the southwestward between the daybeacon on the rock off the northwest side of Harbor Island and Burnt Coat Harbor Light. Strangers should have no trouble entering in the daytime with strict attention to the charts and by following the aids.

(286) From the entrance, steer for a position midway between the light and the daybeacon, passing south of the buoy off Gooseberry Island Ledge and favoring the daybeacon slightly. Anchorage may be selected eastward or northeastward from the light or in midchannel north of it.

(287) The passage between Baker Islands and Swans Island is buoyed and is available for small craft entering Burnt Coat Harbor from the eastward. It is used by local craft but is narrow and difficult, and strangers are advised to use it only after obtaining local knowledge and on a rising tide.

(288) **Marshall Island**, 2 miles westward of Harbor Island, is the largest of the group of islands southward of the western end of Swans Island; the island is about 100 feet high and wooded. A rocky shoal, covered 7 feet near its end, extends 0.4 mile north of Marshall Island; a buoy is off the end of the shoal. A large platform is on the island.

(289) **Hat Island**, 0.9 mile north of Marshall Island and just south of the western end of Swans Island, is 111 feet high, bare on the summit and wooded elsewhere. A buoy is 700 yards southwestward of **Hat Island Ledge**, which extends 0.4 mile westward of the island.

(290) **Ringtown Island**, just off the northeast side of Marshall Island, is wooded. **Yellow Ledge**, southeastward of Ringtown Island, has two bare rocks and a considerable area which uncovers. **Brimstone Island**, 1 mile east of Marshall Island, is bare and grassy. **Heron Island**, 0.5 mile south of Brimstone Island, is grassy with trees in the middle.

(291) Many bare and covered rocks and ledges are southward of Brimstone and Heron Islands and southeastward of Marshall Island. Extreme caution must be used in navigating in this area as most of the dangers are not marked. Two small ledges awash at low water are between Brimstone and Heron Islands.

(292) **Heron Island Point Ledge**, 0.2 mile south of Heron Island, is awash at high water. **Mason Ledge**, 0.4 mile west of Heron Island, and **Black Ledge**, 1.2 miles southwest of Heron Island, are bare rocks. About 0.2 mile northeast of Black Ledge is a depth of 6 feet. **Seal Ledge**, 0.9 mile south of Heron Island, is covered 10 feet; a buoy east of the ledge marks a 16-foot spot. **Cod Ledge**, 0.3 mile southwest of Mason Ledge, is covered 3 feet. A depth of 4 feet is 300 yards southwestward of the ledge. **Jobs Ledge**, 0.6 mile south of Marshall Island, and **Sprague Ledge**, 0.3 mile south, are covered 9 and 7 feet, respectively, and are unmarked.

(293) **Spirit Ledge**, 0.7 to 1.2 miles southwestward of Marshall Island, is in two sections. The northern part has a bare rock on it, and the southern part uncovers about 5 feet. **Boxam Ledge**, off the southwest side of Marshall Island, uncovers.

(294) These ledges, together with numerous other ledges and islands, extend across Jericho Bay and southwest across the entrance of that bay to Isle au Haut. The other ledges and islands are discussed in chapter 7.

(295) **Casco Passage** and **York Narrows**, northward of Swans Island and between Swans Island and Black and Johns Islands, form a part of the inland passage between Mount Desert Island and Whitehead Island. The narrow passage separates into two branches in its western part. The eastern end and northern branch form Casco Passage; the southern branch is York Narrows.

(296) **Johns Island**, 1.3 miles northwestward of the northern extremity of Swans Island, **Opechee Island** and **Black Island**, on the north side of the passage, **Orono Island**, **Asa (Phinney) Island**, and **Round Island**, on the south side of the passage, and **Buckle Island**, on the south side of the western end of York Narrows, are, in general, low and wooded. **The Triangles**, a ledge at the eastern end of Casco Passage, has a rock 7 feet high on it, and a reef that uncovers about 5 feet extends 400 yards northward.

(297) **Long Ledge**, bare and awash, and **Hawley Ledge**, covered 6 feet and marked by a buoy, are westward of Orono Island and between the western part of Casco Passage and York Narrows.**Egg Rock**, off the western entrance, is marked by a daybeacon on the ledge. **Sunken Egg Rock**, covered 6 feet, about 0.4 mile south-southwestward from Egg Rock, is marked by a buoy northward of it.

(298) **Hanus Ledge**, covered 9 feet and marked on its south side by a buoy, is 0.8 mile south-southeast of Sunken Egg Rock. In 1981, depths significantly less than those charted and tide rips were reported to extend about 0.3 mile southeast from the ledge; caution is advised.

(299) **The Woodbury**, covered 10 feet and marked by a buoy, is 0.9 mile southwest of Hanus Ledge.

(300) Casco Passage and York Narrows are well marked, the aids being colored and numbered for the passage westward. A fairway bell buoy marks the eastern approach to the passage.

(301) Casco Passage is the straighter and better channel, has a least depth of 12 feet for a width of about 100 yards and is the one recommended. A rock, awash at low water, is 125 yards off the south side of Black Island, and care should be taken to avoid it. There are rocks with little depth over them on each side of the passage.

(302) The current through Casco Passage floods eastward and ebbs westward at a velocity of 0.7 knot. The velocity is influenced greatly by strong winds. See the Tidal

Current prediction service at *tidesandcurrents.noaa.gov*. for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(303) York Narrows is the deeper, with a least depth of 13 feet, but its width is not much over 100 yards, with dangerous unmarked ledges on both sides. It is not recommended. A lighted bell buoy marks the western entrance. Vessels should not attempt passage except with local knowledge, as the currents are reported to be very strong at times.

(304)

Blue Hill Bay to Ellsworth

(305) **Blue Hill Bay**, west of Mount Desert Island, is about 14 miles long. In the bay are several large and some small islands, between which are good channels with deep water. The dangers are comparatively few; the most prominent are marked by buoys. There are numerous coves on both sides of the bay.

(306) The head of the bay is divided into several large arms, the most important of which is Union River Bay. Blue Hill Bay forms the approach to the villages of Bass Harbor, South Blue Hill, Blue Hill Falls, Blue Hill, East Blue Hill and Surry and the city of Ellsworth.

(307) The bay is frequented by many cruise sailing vessels, fishing craft and yachts. Gasoline and provisions are obtainable at most of the villages. Repair yards for small vessels are at Bass Harbor, Bernard and East Blue Hill.

(308) **Routes** for entering Blue Hill Bay are given at the end of this chapter.

(309)

Current

(310) The current in Blue Hill Bay floods northward and ebbs southward. Velocities of 2 knots have been observed near Staple Ledge at the south end of the bay. See the Tidal Current prediction service at tidesandcurrents.noaa.gov. for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(311) **Bass Harbor**, in the southwest end of Mount Desert Island just westward of Bass Harbor Head, is an important fishing port. The harbor is sometimes used as an anchorage by vessels bound through the inside passage. The outer harbor is exposed southward but clear with the exception of **Weaver Ledge**, which is in the middle of the entrance and uncovers 3 feet. Two buoys mark the ledge.

(312) Vessels can enter on either side of Weaver Ledge and anchor between the ledge and the entrance to the inner harbor in depths of 30 to 46 feet, soft bottom in places.

(313) There are four dredged anchorages available in the inner harbor. The anchorages consist of a 10-foot basin in the middle of the harbor with 6-foot basins adjoining to northward and westward and an 8-foot basin adjoining to eastward. (See Notice to Mariners and latest editions of charts for controlling depths.) Buoys mark the inner harbor.

(314) **Bass Harbor** is a village on the east shore of Bass Harbor. The belfry of a church at the head of the harbor is conspicuous. The cannery wharf, on the east side of the inner harbor about 1.1 miles north of Bass Harbor Head Light, has a reported depth of 7 feet alongside. A smaller seafood company wharf, close northward, has a depth of 10 feet reported alongside. Gasoline, diesel fuel, water, ice and some marine supplies are available at this wharf.

(315) A boatyard and machine shop, about 250 yards above the upper seafood wharf, has two marine railways that can handle craft up to 45 feet or 15 tons for hull and engine repairs. Welding and electrical repairs can also be made.

(316) A marina with a float landing is on the east side of the outer harbor, about 400 yards southward of the cannery wharf; depths of 10 feet are reported at the float landing. A 30-ton mobile hoist at the marina can handle craft up to 50 feet for hull and engine repairs. Gasoline, diesel fuel, water, ice and some marine supplies are available. The slip for the state automobile and passenger ferry to Swans Island and Lunt Harbor on Long Island is close northward of the marina.

(317) Groceries, ice, lodgings and some marine supplies can be obtained in town.

(318) **Bernard** is a village on the west side of Bass Harbor. There are two fish and lobster wharves with float landings with 6 feet reported alongside. Gasoline, diesel fuel and some marine supplies can be obtained at the landings.

(319) A boatyard with covered sheds and marine ways, about 400 yards northward of the fish wharves, can build craft up to 55 feet or haul out for hull and engine repairs craft up to 60 feet or 25 tons. It has a pier and float landing with 10 feet reported alongside. Gasoline, water, electronic repairs and covered storage are available.

(320) **Mitchell Cove**, about 0.5 mile northwestward of Bass Harbor, is shoal and foul and has no landings.

(321) **Duck Cove**, about 1.5 miles northwestward, has a boatyard at the head with covered sheds; the yard has a marine way that can handle craft up to 50 feet or 20 tons for hull and engine repairs or open and covered winter storage.

(322) **Goose Cove**, on the eastern side of Blue Hill Bay 2 miles northwestward of Bass Harbor, is frequented by fishing boats. The cove has good holding ground and offers excellent anchorage for small boats except in heavy southwesterly weather. A shoal is in midharbor. **West Tremont** is a village at the head of the cove. A church spire at the village is prominent from seaward. There is a wharf that dries on the east shore about 0.4 mile above the entrance; water can be had from a nearby well.

(323) **Goose Cove Rock**, 0.6 mile northwest of Goose Cove and 0.2 mile offshore, is a rocky islet 5 feet high. **Rumell Island**, 0.6 mile northwestward of Goose Cove Rock, is a rocky islet 4 feet high with grass on top. The ruins of fish weirs are between the island and mainland.

Latty Cove is an indentation between Goose Cove Rock and Rumell Island.

(324) **Seal Cove**, 4 miles northwestward of Bass Harbor, is a sheltered anchorage for small vessels, except in westerly winds. Rocks that uncover 7 feet are about 300 yards offshore just inside **Reed Point**, the northern entrance point, and a ledge partly showing at high water is off the shoal bight just inside **Dodge Point**, on the south side of the entrance. Craft entering in midchannel will find anchorage near the middle of the cove in depths of 11 to 37 feet. There are several private piers and float landings on the south side of the cove near Dodge Point and a paved town launching ramp on the north side of the cove near its head.

(325) **Moose Island**, north of the entrance to Seal Cove, is wooded. A bar which uncovers connects the island to a point 0.3 mile northward of Reed Point. The point has a few buildings, a private wharf and a small flagpole. Small craft anchor northward of the bar. A rocky ledge, awash at low water, is reported to be 125 yards north of Moose Island.

(326) **Hardwood Island**, 0.7 mile northwest of Moose Island, is 113 feet high and wooded. A bar extends 0.3 mile southwestward from the island; a buoy is off a rock covered 11 feet at the end of the bar.

(327) **Sawyer Cove**, on the eastern shore of Blue Hill Bay eastward from the north end of Hardwood Island, is an anchorage for small craft. A ledge awash at high water is in the middle of the entrance. Several float landings are in the cove.

(328) **Pretty Marsh Harbor** makes into the eastern shore of Blue Hill Bay northeastward of Hardwood Island. There is good anchorage in depths of 8 to 37 feet. **Folly Island**, a grassy island with a few trees, is in the entrance. The northern and western sides of the harbor are shoal inside **West Point,** on the western side 0.6 mile northeast of Folly Island. A shoal, covered 9 feet at the end, extends 350 yards southeastward from West Point. There are no dangers away from the shore, except for a ledge, covered 3 feet, about 250 yards east of Folly Island. In 1970, two bare rocks were reported on this ledge; caution is advised. Several float landings are on the east side of the harbor.

(329) **John Island**, an islet 750 yards northward of Folly Island, and a lower islet 400 yards northwestward are grassy. **Birch Island**, 0.4 mile northwestward of John Island, is wooded.

(330) **Bartlett Island**, 0.7 mile northward of Hardwood Island, is 279 feet high and mostly wooded, with a few houses on it. A grass-covered islet is close to the northeast end of the island.

(331) **Bartlett Narrows** is between Mount Desert Island and Bartlett Island. The channel is narrow but has deep water with few dangers and is not difficult. The channel westward of Folly and John Islands is clear in midchannel. If passing eastward of Folly Island, give it a berth of about 400 yards, and give the south end of John Island a berth of 200 yards. The eastern shore of the narrows from West Point to its northern end is bold and should be favored. In the narrowest part keep the eastern shore close aboard, distant 100 yards, to avoid a ledge extending 200 yards southward from a group of bare rocks, southeastward of **Ledges Point**, on Bartlett Island.

(332) The town of Mount Desert has a wharf and float landing on the east side of Bartlett Narrows 0.6 mile north-northeast of John Island; depths of 6 feet are reported alongside. No facilities are available. Moorings off the town float are regulated by the Mount Desert **harbormaster,** whose telephone number is posted at the landing.

(333) A ledge covered 2 to 3 feet is 400 to 600 yards from the eastern shore 0.4 mile northward of Bartlett Narrows. It will be avoided by keeping westward of a range formed by the northwest tangents of Black and Alley Islands.

(334) **Western Bay**, northeastward of Bartlett Island, is a part of the waters that separate Mount Desert Island from the mainland northward. Mount Desert Narrows, described previously, is at the head of Western Bay.

(335) **Black Island**, about 1 mile northeastward of the north end of Bartlett Island, is thickly wooded. **Green Island**, close to the southern shore of the bay, is 0.5 mile east of Black Island. **Alley Island**, the largest island in Western Bay, is 1.2 miles north of Green Island.

(336) Vessels of any size can select anchorage in the bay southwestward of Alley Island in depths of 44 to 64 feet; the rocky broken ground with depths of 34 to 36 feet extending 0.4 mile off the southeast side of **Oak Point**, 1.5 miles westward of Alley Island, should be avoided. With the aid of the chart, good anchorage can be selected also in depths of 21 to 38 feet southeastward and eastward of Alley Island.

(337) The range formed by the summit of Bartlett Island and the middle of Black Island clears the shoal extending 500 yards southeastward from Alley Island.

(338) Foul ground extends about 500 yards from the south shore of Western Bay between Green Island and **Indian Point**, 1 mile northeastward. **Northwest Cove**, eastward of Indian Point, has anchorage in depths of 10 to 20 feet, but a ledge with little water over it extends 500 yards from its southeast shore 600 yards eastward from Indian Point.

(339) **Goose Cove** is a large shallow bight on the north side of Western Bay northward of Alley Island. The villages of **Trenton** and **West Trenton** are on the northern shore. The head of the cove is dry at low water for a distance of 0.5 mile, and thence it deepens gradually to 7 feet 0.5 mile farther out. There are no wharves in the cove.

(340) **Mahoney Island** (44°13.0'N., 68°30.7'W.), on the west side of Blue Hill Bay 7.5 miles west of Bass Harbor Head and just eastward of the entrance to Eggemoggin Reach, is sparsely wooded. **Smuttynose Island**, 0.4 mile northwestward of Mahoney Island, is grassy. **Mahoney Ledge**, southwestward of Mahoney Island, is awash at low water. A buoy southwestward of the ledge marks a shoal covered 5 feet, and another buoy is off the shoal water southeast of Mahoney Island.

(341) **Pond Island**, 1.3 miles eastward of Mahoney Island, is wooded on its eastern side. The western side is grassy.

Lamp Island, 0.2 mile northward of Pond Island, is grassy.

(342) **Pond Island Passage**, the channel north of Pond Island, is used by vessels entering Blue Hill Bay from westward and sometimes by vessels following the inside route eastward or westward. The passage has a least depth of 19 feet in the buoyed channel, but there are dangers close to the sailing lines. The buoys are colored and numbered for vessels bound westward.

(343) Between Pond Island and Casco Passage, 1.5 miles southward, are several islands. Opechee, Johns and Black Islands have been previously discussed. **Sheep Island** is grassy, and **Eagle Island** is wooded. A reef that uncovers 7 feet is 500 yards eastward of Eagle Island.

(344) The passages between these islands are obstructed by reefs.

(345) **Channel Rock**, 1 mile northward of Pond Island, is 3 feet high and has a whitish top. A ledge covered 3 feet extends 0.4 mile eastward of the rock.

(346) **Green Island** is grassy and marked by an abandoned lighthouse tower, white with dwelling, and by **Blue Hill Bay Light 3** (44°14'55"N., 68°29'52"W.), 21 feet above the water and shown from a skeleton tower with a green square daymark. The ledges, of which Green Island is a part, uncover from the island to the shore 1.1 miles northwestward and for a distance of nearly 0.3 mile southward of the island. A buoy is about 0.35 mile southward of the island.

(347) Other islands on the ledges include **Flye Island, Goose Island** and **Gander Island**. A buoy is off the south end of the ledge. **Sand Island**, 0.3 mile northeastward of the light, is bare and nearly covered at high water. A buoy southwestward of the islet and a fairway bell buoy mark **Flye Island Channel** between Green Island and Sand Island.

(348) **Flye Island Ledge**, having rocks covered 7 to 15 feet, extends to a point 1 mile south-southwestward of Blue Hill Bay Light 3.

(349) **Herrick Bay** is a shallow bight on the western side of Blue Hill Bay northwestward of Blue Hill Bay Light 3. **Naskeag Point**, 0.8 mile northwest of Mahoney Island, is on the western side of the approach. The bay dries at low water for nearly 1 mile from its head. There is good anchorage in the approach to the bay 0.5 mile from the western shore northward of Naskeag Point in depths of 24 to 45 feet. In the approach to the anchorage the range formed by the western tangents of Flye and Long Islands leads westward of Flye Island Ledge.

(350) A boatyard on the north shore of Herrick Bay builds fiberglass boats up to 31 feet long and makes hull and engine repairs. In 1979, haul-out and open storage facilities were under construction at the boatyard.

(351) **Ship Island**, **Trumpet Island**, **Bar Island** and **Tinker Island** are a chain of islands 4 miles long in the middle of Blue Hill Bay, eastward of Blue Hill Bay Light 3. The islands are joined by shoals that uncover, except for a channel between Trumpet and Bar Islands that has a depth of 17 feet and is marked by a fairway buoy. Ship and Bar Islands are grassy. A buoy marks the end of the shoal extending 0.3 mile from the northeastern side of Bar Island. Trumpet Island is low and grassy. Tinker Island, partially wooded, has a shack on its southeast end.

(352) **Ship and Barges Ledge**, 0.6 mile southeastward of Ship Island, is 350 yards long and uncovers about 5 feet. The ledge is marked by a daybeacon and by a bell buoy about 0.15 mile northeastward.

(353) **West Barge** is a flat grass-topped rock 0.3 mile westward of Ship Island. **East Barge** is a round grassy islet on the end of the shoal extending 0.1 mile southward from Ship Island.

(354) **Cow and Calf Ledge**, extending 0.4 mile westward and northward from Tinker Island, has several rocks with little water and one rock which uncovers 5 feet; buoys are north and west of the ledge.

(355) **Allen Cove**, on the west side of **Harriman Point** 3.5 miles northwestward of Blue Hill Bay Light 3, is used as an anchorage. The cove is open northward, and the shores are foul. Vessels may anchor in the middle of the cove in depths of 12 to 30 feet. **Harriman Ledge**, covered 10 feet and marked by a buoy, is 0.3 mile eastward of Harriman Point.

(356) **Long Island**, a large uninhabited island in Blue Hill Bay, 1.5 miles west of Bartlett Island, is in general wooded with a few clear sections. **Long Island Hub**, off the south end of Long Island, is conspicuous because of high trees covering it.

(357) **Sand Point** (44°21.4'N., 68°32.8'W.) is on the west shore of Blue Hill Bay 3.3 miles northward of Harriman Point. **South Blue Hill**, a village on the western side of Blue Hill Bay just south of Sand Point, has a wharf with a 50-foot face which is dry at low water. This is all that remains of a former town wharf, the outer 60 yards of which were destroyed by a hurricane. The ruins and foundation of the former wharf, just awash at low water, extend about 100 feet out from the present wharf. Caution is necessary in approaching the wharf. A stone pier is in ruins just north of the wharf.

(358) **Salt Pond** has its entrance about 1.2 miles northward of Sand Point and just south of the entrance to Blue Hill Harbor; tidal falls are at the entrance. The channel is southward of **Mill Island**, on the north side of the entrance. State Route 175 highway fixed bridge crosses the entrance; clearance is 7 feet. The strength and turbulence of the current is such that passage is not recommended except with local knowledge. A private pier with float landing is on the east side of Mill Island.

(359) **Blue Hill Harbor**, northwestward of Long Island, is a large bight in the northwestern part of Blue Hill Bay. **Parker Point**, on the western shore of the harbor 3 miles north of Sand Point, and **Sculpin Point**, on the northern shore of the harbor 0.3 mile northeast of Parker Point, divide the harbor into an inner and outer harbor. The village of Blue Hill is at the head of the inner harbor.

(360) **Blue Hill** (44°26.1'N., 68°35.5'W.), a rounded hill that appears blue in the distance and gives its name to the

Jericho Bay to Penobscot Bay, Maine

(1) This chapter describes the Maine coast from Jericho Bay to but not including Muscongus Bay and the waters and tributaries of East and West Penobscot Bays, Penobscot River and the many passages and thorofares leading into and connecting these waterways. Also discussed are the important ports of Rockland, Searsport, Bucksport and Bangor and many smaller fishing ports and resort towns on these waterways.

(2)

COLREGS Demarcation Lines

(3) The lines established for this part of the coast are described in **33 CFR Part 80.105**, chapter 2.

(4)

Deer Isle

(5) Between Jericho Bay and Penobscot Bay are numerous islands. **Deer Isle**, 10 miles westward of Mount Desert Island, is the largest. Eggemoggin Reach, Deer Island Thorofare and Merchant Row are the three principal passages between the bays. Eggemoggin Reach, between Deer Isle and the mainland, connects Blue Hill Bay and the head of Jericho Bay with Penobscot Bay near its head. The reach is 11 miles long and has a least width of about 0.4 mile at Byard Point. There are several villages along its shores.

(6)

Jericho Bay

(7) **Jericho Bay** is between Swans and Marshall Islands on the east and Isle au Haut and Deer Isle and adjoining islands on the west. The inside routes from Casco Passage and York Narrows to Deer Island Thorofare and Merchant Row, and the passage north of Pond Island to Eggemoggin Reach, lead across the head of Jericho Bay. This section of the bay is used by many craft.

(8) The part of the bay southward of these thorofares has deep water, but there are many ledges, rocks and islets. This area is little used except by local fishermen and yachts.

(9) The dangers in the passages into Jericho Bay from the southward, eastward of Isle au Haut, in the channels between that island and Marshall Island, are for the most part not marked. This is the most direct way from the sea from that direction. There are, however, a number of unmarked shoal spots which must be avoided.

(10) **Halibut Rocks**, in Jericho Bay 0.8 mile northwest of Marshall Island, are two in number. **Halibut Rocks Light** (44°08'03"N., 68°31'32"W.), 25 feet above the water, is shown from a skeleton tower with a red triangular daymark on the northerly rock; a sound signal is at the light.

(11) **West Halibut Rock**, 1 mile westward of Halibut Rocks, is covered 2 feet; a buoy is off the rock. A rock covered 9 feet is 400 yards northeastward of the buoy. **Southern Mark Island Ledge**, 2.3 miles west of Halibut Rocks, has a rock bare at high water.

(12) **Colby Ledge**, 0.8 mile southwest of Southern Mark Island Ledge, uncovers about 5 feet. A daybeacon is on the ledge. A ledge covered 15 feet is 400 yards southward of the daybeacon. **Colby Pup**, covered 3 feet and marked by a buoy, is 0.5 mile south of the daybeacon. Unmarked **Channel Rock**, 0.6 mile southwest of Colby Pup, uncovers 8 feet.

(13) **McGlathery Island** (44°07.5'N., 68°37.0'W.), 2.5 miles southeast of Stonington, is the largest island on the west side of the bay and on the north side of Merchant Row. A rocky ledge with at least two rocks awash and a covered rock extends between the east side of the island and **Gooseberry Island**. The area is foul, and passage through it should be avoided. Vessels rounding the north side of McGlathery Island should take care to avoid the charted rock, reported to cover about 1 foot, about 160 yards north of the island. The remaining islands and dangers in the bay are described in connection with the various channels leading out of the bay.

(14)

Routes

(15) In approaching Jericho Bay from the southeastward, it is advisable to pass between Marshall Island and Swans Island, through Toothacher Bay where most of the dangers are marked, but then only in daytime. In clear weather, strangers should have no trouble navigating any of the passages, or through Merchant Row or Deer Island Thorofare, by giving strict attention to the chart and following the aids, which are colored and numbered for passages to the northward and westward.

(16)

Eggemoggin Reach

(17) **Eggemoggin Reach** is a generally broad and deep thorofare which extends in a general northwesterly-southeasterly direction between the mainland and Deer Isle and joins Jericho Bay with East Penobscot Bay.

(18) The eastern entrance to Eggemoggin Reach is well marked by **Devils Head** (44°13.3'N., 68°32.8'W.), a prominent, high, rock bluff on the south end of **Hog Island**, 2.8 miles west of Pond Island. Off the western

entrance are **Head of the Cape** at the southwest extremity of **Cape Rosier**, high and thickly wooded; a light on Green Ledge, 1.3 miles south of Head of the Cape; and an abandoned lighthouse tower on **Pumpkin Island**, 3.6 miles east of Head of the Cape.

(19) The depth in the main channel through Eggemoggin Reach is sufficient for deep-draft vessels, but the channel is narrow and the bottom is irregular in places. The principal dangers are buoyed and can be easily avoided in the daytime and in clear weather. An unmarked rocky spot, covered 27 feet, lies about 250 yards southeastward of the north tower of the Deer Isle-Sedgwick Bridge.

(20) Vessels can anchor anywhere in the reach where the depth is suitable and the bottom soft, making a lee of either shore, according to the wind. Small craft anchor in the coves off the reach. A submarine cable extends northeasterly from Little Babson Island to the mainland. Caution is advised.

(21)

Devils Head to Billings Cove

(22) A ledge extending 0.3 mile southeastward from **Devils Head** at the eastern end of Eggemoggin Reach, is partly bare at high water; a buoy is off the end of the ledge. **Hay Island Ledge**, 0.5 mile southeastward of Devils Head, is covered 7 feet, and marked by a buoy off its southern side. An unmarked 15-foot spot is 0.7 mile southeastward of the ledge. A fairway bell buoy, 300 yards southward of the ledge, marks the eastern entrance to Eggemoggin Reach.

(23) **Channel Rock**, 900 yards south of Devils Head and covered 2 feet, is marked by a buoy. **The Boulders**, 400yards westward of Channel Rock, uncover 3 feet.

(24) **Greenlaw Cove**, on the southwest side of the eastern entrance to Eggemoggin Reach, has a narrow unmarked channel with shoals on both sides and is suitable only for small craft with local knowledge. **Mountainville** is a village near the head of the cove. The landing is nearly bare at low water.

(25) **White Island**, **Bear Island** and **Conary Island** are off the entrance to Greenlaw Cove and on the southern side of the passage through Eggemoggin Reach. **Conary Ledge**, 0.4 mile north of Conary Island, is covered ½ foot and marked by a buoy north of the ledge.

(26) **Naskeag Harbor**, an anchorage for fishing boats, is north of Hog Island and **Harbor Island**, which is 0.3 mile east of Hog Island. The village of **Naskeag** is on the north side. The harbor can be entered from eastward or westward, but there are many unmarked dangers, and strangers should not attempt to enter except in small craft.

(27) At the eastern approach to the harbor, the bar from the northern shore extends two-thirds of the way across. Between the end of this bar and Harbor Island is a rock reported to uncover 6 feet about 75 yards north of the island. At half tide the bar is marked by ripples.

(28) **The Triangles**, a reef with rocks awash, is in the middle of the western entrance to the harbor. There are several private float landings between Naskeag Harbor and Center Harbor, 2 miles to the northwestward. **Northwest Cove** is a small cove with middle depths of 15 feet, about 2.5 miles northwest of Naskeag Point. **Babson Island** and **Little Babson Island** are wooded islands on the north side of the reach between Naskeag Harbor and Center Harbor. They are occupied only in the summer.

(29) **Torrey Islands** are about 0.9 mile northwest of Little Babson Island. A rocky ledge extends about 350 yards south of the east island. A 12-foot spot, marked by a buoy, is about 0.35 mile west of the south point of the west island. **Torrey Castle**, marked by a daybeacon, is a reef at the end of a ledge that extends 0.2 mile west from the west island. **Torrey Ledge**, covered 2 feet and marked by a buoy, is about 0.35 mile northwest of the west island. A rock awash is between the ledge and the island; mariners should not attempt to pass between the buoy and the island.

(30) **Center Harbor**, an anchorage for small craft only, is a small cove on the north side of the reach northeastward of **Torrey Islands**, 2.2 miles northwest of Hog Island. A buoy 200 yards west-northwest of Chatto Island marks the entrance. The town of **Brooklin** is at the head of the harbor. A white church spire in the town is conspicuous. A rock marked by a daybeacon is in the middle of the entrance northward of **Chatto Island**, which is 0.6 mile north of the eastern Torrey Island. The channel is close southward of the rock. Between the daybeacon and a boatyard near the head, on the north side of the cove, the channel has depths of 8 to 10 feet; above the boatyard it is mostly dry at low water. Good anchorage is available off the entrance in depths of 22 to 24 feet, soft in places.

(31) The main approach to Center Harbor is from westward, but local vessels enter by the channel eastward of Torrey Islands, passing in midchannel on either side of the bare rock 350 yards eastward of the easterly of the islands. This passage should not be attempted by strangers.

(32) A boatyard, about 350 yards east of the daybeacon, has marine railways that can haul out craft up to 30 tons or 50 feet in length for hull or engine repairs or dry open or covered storage. Its pier and float landing has 6 feet alongside. Provisions and marine supplies may be obtained in Brooklin. The yard can build craft up to 50 feet.

(33) The Center Harbor Yacht Club pier and float landing, with 6 feet alongside, is on the north side of the entrance; water is available. The signal mast and clubhouse are conspicuous. There are several float landings in the harbor. Anchorage in soft mud bottom may also be had south of a line between the yacht yard and the daybeacon at the entrance.

(34) **Bridges Point Shoal** extends over 0.5 mile from **Bridges Point**, 4.8 miles northwestward of Naskeag Point, and is covered 5 to 17 feet; a buoy marks the outer end.

(35) **Benjamin River**, the approach to the town of **Sedgwick**, empties into the north side of the reach 5.5

miles northwestward of Naskeag Point. The channel at the entrance northward of **Cape Carter** has a least depth of 19 feet but is restricted on both sides, leaving a passage 100 yards wide at its narrowest part. The channel is marked by seasonal buoys for about 0.5 mile above the mouth. A rock awash is at the outer end of the ledge and sand shoal extending into the river from the east side 0.6 mile northward of Cape Carter. Sedgwick can be reached only at high water as the river dries out some distance below.

(36) On the east side of the river about 1 mile above Cape Carter, there is a boatyard which builds craft up to 40 feet in length. The yard has a 2-ton crane and a marine railway that can haul out craft up to 12 tons or 40 feet in length for hull and engine repairs or dry open or covered storage. Gasoline is available in cans. The boatyard wharf dries at low water. Provisions and some marine supplies are available in Sedgwick. A number of mooring buoys are available off the boatyard.

(37) The village of **West Brooklin** is near the boatyard. A church spire in the village is conspicuous.

(38) A public wharf and float landing is on the west side of the river about 0.5 mile above the entrance. Depths of 8 feet were reported alongside.

(39) **Stump Cove Ledge**, covered 3 feet, is 0.6 mile west of Cape Carter. A buoy is off the southwest side of the ledge.

(40) **North Deer Isle**, on the southern side of Eggemoggin Reach, is a village at the north end of Deer Isle. In 1979, some rock cribbing was the only remnant of the old ferry wharf 0.4 mile westward of Tinker Ledges. There is a rock crib breakwater just east of the old wharf, and the enclosed space between the two is sometimes used for beaching local small craft. The breakwater extends about 200 feet from shore and is covered most of its length at high water. It is not marked and is a danger to all craft approaching close to shore.

(41) **Tinker Ledges**, about 0.7 mile long and covered 13 feet, are on the south side of the reach about 6.7 miles west-northwest of Naskeag Point; a buoy is on the northeast side of the ledges.

(42) A highway causeway extending from the northwest corner of Deer Isle to the eastern side of **Little Deer Isle**, 0.4 mile northwestward, closes the passage between the two islands to all craft. **Stave Island**, just northward of the eastern end of Little Deer Isle, is wooded.

(43) **Billings Cove** is on the northern shore of Eggemoggin Reach 2.3 miles northwestward of Benjamin River and east of **Byard Point**. It dries out 300 yards from the head. Anchorage can be had in the middle of the cove just inside the entrance in depths of about 25 feet. **Sargentville** is a village near the eastern shore of the cove. A private wharf and float are just east of the cove. A good beach for hauling out or launching small craft is just eastward.

(44)

Deer Isle-Sedgwick Ridge to Black Ledges

(45) The **Deer Isle-Sedgwick Bridge** (State Route 175), a suspension-type fixed highway bridge, crosses Eggemoggin Reach between Byard Point and Little Deer Isle. The bridge has a clearance of 85 feet for a midwidth of 200 feet. The village of **Little Deer Isle** is near the south end of the bridge.

(46) **Howard Ledges**, on the south side of Eggemoggin Reach about 1.4 miles northwestward of the bridge, are covered 1 to 9 feet and marked by a buoy on the northwestern end.

(47) **Eggemoggin** is a summer resort with several private float landings at the northwest end of Little Deer Isle, southeastward of Pumpkin Island. Several boatsheds, where small craft are hauled out for winter storage, are at Eggemoggin.

(48) **Bucks Harbor**, on the north side of Eggemoggin Reach opposite Eggemoggin, affords excellent anchorage and is often used by small vessels. **Harbor Island**, in the middle of the harbor, has a good channel around it which forms the anchorage. Shoals extend 250 yards off the northeast side of Harbor Island, and the channel is narrow between them and the shore northeastward. **Harbor Ledge**, covered 5 feet at the north end of the shoals, is marked by a lighted buoy. The channel between the ledge and the northern shore has a depth of 23 feet. Small craft can anchor in the bight on the northeast side of Harbor Island. The best anchorage is west and northwestward of Harbor Island in depths of 28 to 37 feet.

(49) **South Brooksville**, a village at the head of Bucks Harbor, has a marina with 5 to 8 feet reported alongside its float landing. Bucks Harbor Yacht Club, close westward of the marina, has a float landing with 12 feet alongside. Gasoline, diesel fuel, water, ice and some marine supplies are available at the marina.

(50) There are several private float landings in the harbor, and several moorings are available for hire. The village has a general store and guest houses. Engine repairs and electric welding can be made by a garage in the village.

(51) **Orcutt Harbor**, just westward of Bucks Harbor, is about 1.3 miles long and 500 yards wide. Good anchorage is available in depths of 14 to 52 feet in the middle of the harbor northward of a small wooded islet on the western side near the entrance. A reef, awash at low water, extends 300 yards southward from **Condon Point**, on the east side of the entrance. When northward of this reef, favor the eastern side of the entrance to avoid a rock covered 5 feet nearly 200 yards from the western shore and the same distance southward of the wooded islet. In the slight expansion 0.5 mile above the islet, care must be taken to avoid two rocks covered 5 feet, one of which is 200 yards from the western shore and the other 150 yards from the southeast side of the expansion. A boatyard is at the head of the cove making into the east side of the harbor, about 0.6 mile northward of Condon Point. The second

rock described above, covered 5 feet, is on the south side of the entrance. The yard can haul out craft up to 45 feet on skids for hull and engine repairs; open storage is available. Another boatyard with a marine railway is at the head of Orcutt Harbor; craft up to 45 feet in length can be hauled out for engine and minor hull repairs; open storage is available. There are also several private float landings at the head of the harbor.

(52) **Horseshoe Cove** is a long, narrow cove, the entrance to which is 0.6 mile southwestward of Orcutt Harbor. The cove is navigable only for small craft with local knowledge for about 1.4 miles; above that point for another mile it dries out. There are no wharves. Private seasonal aids mark the channel to a boatyard on the west side about 1 mile above the entrance daybeacon. The yard has a marine railway and can build, or haul out for hull and engine repairs, craft up to 50 feet in length and 7-foot draft. Covered and open dry winter storage is available. The yard maintains a number of moorings off the yard. The best anchorage secure in all weather is reported to be in 15 feet, mud bottom, 0.8 mile northward of the entrance, northward of the inner daybeacon.

(53) **Weir Cove**, about 0.7 mile southwestward of Horseshoe Cove, has several private float landings on its east and west sides. **Buck Island** is a wooded islet off the entrance to the cove. A drying ledge, unmarked, extends about 0.2 mile southward from the eastern entrance point. Several rocks awash have been reported on the ledge, and some may exist between the southern extremity of the ledge and Buck Island; mariners are advised to exercise caution in this area. The upper half of Weir Cove is mostly dry at low water.

(54) **Thrumcap Island**, 1 mile northwestward of Pumpkin Island, is grassy and low. **Thrumcap Ledge**, southward of Thrumcap Island, is partly uncovered at high water and marked near its eastern end by a buoy. **Spectacle Island Ledge**, 0.8 mile southwestward of Thrumcap Island, is covered 6 feet. A buoy marks the southerly portion of the ledge; in 1979, it was reported that 6-foot depths extend 75 to 100 yards south of the buoy. **Two Bush Ledge**, 1.2 miles south of Thrumcap Island, is covered 2 feet; a buoy is off its west side. **Merriman Ledge**, awash at low water, is 0.4 mile westward of Pumpkin Island; a buoy is on its north side. In 1984, a 6-foot shoal was reported about 175 yards east of Pumpkin Island in about 44°18'33"N., 68°44'25"W.

(55) About midway between Merriman Ledge and Spectacle Island Ledge is a 13-foot spot marked by a buoy. **Pumpkin Island Ledge**, 0.4 mile northwestward of Pumpkin Island, is covered 12 feet; a buoy is on its west side. **The Triangles**, 0.4 mile northeastward of Pumpkin Island, is a ledge covered 2 feet and marked by buoys on the north and west sides.

(56) A lighted fairway bell buoy, 0.7 mile north of Pumpkin Island, marks the western entrance to Eggemoggin Reach.

(57) Of the islands near the western entrance to Eggemoggin Reach, **Spectacle Islands**, 1.7 miles westward of Pumpkin Island, are grassy. A fairway bell buoy is 0.4 mile southeastward of the islands. **Two Bush Island**, 1.8 miles southwestward of Pumpkin Island, is bare; **Hog Island**, 2.5 miles southwest of Pumpkin Island, has scattered trees, a house and a barn in the center. **Fiddle Head** is a small islet off the northeast end of Hog Island and is connected to it by a bar that uncovers.

(58) **Pond Island**, 0.4 mile northwest of Hog Island, is grassy and has a small clump of trees on the northeast side. **Western Island**, 0.5 mile west of Pond Island, is grassy on its eastern end and has a thick clump of trees on its western end. **Green Ledge**, west of Western Island, is marked by **Green Ledge Light 4** (44°17'25"N., 68°49'42"W.), 31 feet above the water and shown from a white skeleton tower with a red triangular daymark on the ledge; the light marks the western approach to Eggemoggin Reach from East Penobscot Bay. A bell buoy is 0.3 mile southwestward of the light.

(59) **Black Ledges**, awash at low water, are 0.4 mile southwestward of Pond Island.

(60)

Deer Island Thorofare to Sparrow Island

(61) **Deer Island Thorofare** is a narrow passage leading along the south side of Deer Isle, between it and the numerous islands southward. The passage joins Jericho Bay and East Penobscot Bay. It is a link in the chain of inland passages. Stonington is a town on the passage. The thorofare is used occasionally by coastal tankers and extensively by small craft bound through the inland passages. It has a least width of 100 yards in several places and a least depth of 9½ feet in a channel across the bar between Moose and Crotch Islands. Vessels drawing up to 18 feet are reported to use the passage, but there are unmarked rocks covered 9 to 14 feet close to the channel. Local knowledge is advisable. The more important dangers are marked, and the channel is easily followed in the daytime in clear weather.

(62) The standpipe at Stonington and the stiff-leg crane and derricks at the inactive quarries on Crotch Island are prominent from all directions.

(63)

Anchorages

(64) The best anchorage for vessels bound through the thorofare and overtaken by night or bad weather is in Southeast Harbor. When overtaken by fog, they may anchor anywhere near the channel where the bottom is soft and the depth suitable. Small vessels anchor on the north side of the channel off Stonington and between the wharves off Staple Point and the buoy 800 yards eastward. There are a considerable number of moorings off the wharves. A berth at one of these can usually be obtained on application to the harbormaster. There is also a good anchorage north of **Round Island**, 2 miles southeast of Crotch Island. A special anchorage area is off the northeast side of Crotch Island. (See **33 CFR 110.1** and **110.4**, chapter 2, for limits and regulations.)

(65)

Current

(66) The tidal currents at Stonington follow the general direction of the channel and are not strong. The direction of the currents is influenced by the wind; with strong easterly winds the flood and ebb set westward, and with westerly winds they set eastward. When not influenced by the wind, the flood sets eastward and the ebb westward and continues to run about 0.8 hour after high and low waters.

(67) Ice seldom closes Deer Island Thorofare and Southeast Harbor and then is soon broken up by icebreakers. During severe winters, solid ice has existed from Stonington to Isle au Haut.

(68)

Routes

(69) There are two well-marked channels into Deer Island Thorofare from the eastward. The northern channel passes east and south of the buoys marking the ledges off **Green Ledge**, 0.8 mile eastward of Stinson Neck, and enters the thorofare between **Long Ledge**, 0.5 mile south of Green Ledge, and **Potato Ledge**, which extends 0.6 mile northeastward from **Shabby Island**, 20 feet high and wooded. A daybeacon is on Long Ledge, and a bell buoy is south of the ledge. A buoy is north of Potato Ledge. The channel then leads westward, passing south of **Lazygut Ledge**, 0.6 mile west of Long Ledge, and entering the thorofare at **Eastern Mark Island Ledge**, 1.4 miles west of Potato Ledge. The channel then continues between **Sheldrake Ledge** and **Haycock Rock**, marked by a daybeacon, 0.6 mile southwest of Eastern Mark Island Ledge; between **Haskell Ledge**, 0.8 mile west of Haycock Rock, Bold Island Ledges, and several other dangers, most of which are buoyed. The northern channel then joins the southern channel in the thorofare west of Bold Island Ledges 3.5 miles west-southwestward of Potato Ledge.

(70) The southern entrance channel passes south of **Whaleback Ledge**, about 0.8 mile southward of Shabby Island, and runs nearly due west between **Shingle Island**, 1.1 miles southwest of Shabby Island, and **Saddleback Island**, 0.4 mile south of Shingle Island. The channel then swings northwestward and passes between **Bold Island**, 1.3 miles west of Shingle Island, and **Bold Island Ledges**. This channel is well marked by buoys to its junction with the other channels. Its eastern entrance is marked by a fairway bell buoy, about 700 yards east-northeastward of Saddleback Island.

(71) Entering from the westward, the principal leading mark is near **Deer Island Thorofare Light** (44°08'04"N., 68°42'12"W.), 52 feet above the water, shown from a white square tower on the west side of **Mark Island**, a sound signal is at the light. Westward of the light care must be taken to avoid **The Brown Cow**, a ledge 1.3 miles southwestward from the light, and **West Mark Island Ledge**, covered 4 feet, about 0.7 mile northwestward of the light; a buoy is south of the ledge. Passing north of the light and south of **Western Deer Island Ledge**, 5 feet high, and **Bay Ledge**, covered 11 feet, 0.4 mile north of Mark Island, there should be no difficulty in following the aids, which are colored and numbered for passage westward.

(72) **Southeast Harbor** is northwestward of the eastern end of Deer Island Thorofare, between **Stinson Neck** on the east and **Whitmore Neck** on the west. The harbor is an excellent anchorage for vessels using the thorofare. The entrance is easily distinguished and the principal dangers are marked by buoys. **Oceanville** is a village on Whitmore Neck, on the south side of the harbor. A marina is at the village.

(73) **Webb Cove**, about 2 miles southwestward of Southeast Harbor and on the north side of Deer Island Thorofare, has rocks in the entrance but good anchorage inside in depths of 8 to 12 feet. **Grog Island**, **Grog Ledge** and **Humpkins Ledge** are off the entrance, and **Channel Rock** is about in the middle of the entrance. A detached, unmarked and nearly L-shaped 400-foot-long fish weir is in the middle of the cove, about 500 yards above Channel Rock. In 1961, an obstruction, believed to be two pinnacle rocks covered 8 feet, was reported to be about in the middle of the channel between Grog Island and Grog Ledge. In 1987, an obstruction was reported about 500 yards westward of Grog Ledge in about 44°09'45.5"N., 68°38'05.9"W. A 400-foot barge wharf of a stone quarry is on the northeast side at the entrance to the inner half of Webb Cove. In 1979, the wharf was being used by a packing company to offload fish; depths of 7 feet are reported alongside the wharf.

(74) **Stonington**, a town on the north shore of Deer Island Thorofare, has a sizable seafood industry. Many fishing vessels, lobster boats, draggers and some charter and excursion boats operate from the port.

(75) Most of the wharves along the Stonington waterfront are used by commercial vessels. The cannery wharf (44°09'15"N., 68°39'38"W.), on **Staple Point**, has reported depths of 7 feet along its easterly side. A ledge off the wharf has little water on it; a buoy marks the outer end. A lobster wharf, 200 yards west of the cannery wharf, has depths of 7 feet reported alongside. A private pier and float, about 50 yards northwest of the lobster wharf, is used by the Stonington-Isle au Haut mail and passenger ferry; depths of 5 feet are reported alongside. Two lobster wharves, about 250 and 500 yards eastward of the cannery wharf, have reported depths of 10 and 7 feet, respectively, alongside their float landings. There are also several fish wharves at the western end of the harbor, eastward of **Green (Greens) Head.**

(76)

Small-craft facilities

(77) Most of the facilities are along the main waterfront. Provisions and marine supplies can be obtained in town. The nearest vessel repair facility is on Moose Island, just westward of Stonington.

(78) Berthage for transient craft is very limited at Stonington; most vessels anchor off the town or moor to mooring buoys off Staple Point. The town selectmen issue permits for mooring.

(79) Stonington has banks, restaurants, markets, stores, hotels and motels. Good roads on the island connect with the bridge to the mainland.

(80) **Allen Cove**, just west of Stonington and east of **Moose Island**, is protected by a pier and breakwater built out from the southeast end of Moose Island. It is known locally as **Yacht Basin**. Sheds of a shipyard on the southeast end of Moose Island are prominent from westward. A causeway connects Moose Island with Deer Isle. Large lobster pounds occupy the northeast end of the cove and the areas on both sides of the causeway.

(81) The shipyard builds vessels up to 80 feet long and has several marine railways that can handle vessels up to 250 tons or 125 feet long for general hull or engine repairs; electrical and electronic repairs can also be made. A 30-ton mobile hoist and open or covered dry storage are available. Gasoline, diesel fuel, electricity, water, ice and marine supplies can be obtained at the yard's service floats. The shipyard piers have depths of 10 feet reported alongside. Small craft anchor in the cove.

(82) **Crotch Island**, on the south side of Deer Island Thorofare opposite Moose Island, is the site of extensive granite quarries. The large quarry wharf on the north side of the island is reported to have a depth of 12 feet alongside. A 75-ton stiff-leg crane is on the wharf. In 1979, the quarry was inactive; many of the quarry derricks were visible from all around the island.

(83) On the northern side of the western entrance to the thorofare is **Andrews Island**, 60 feet high. Northward of Andrews Island and extending 0.5 mile south of **Fifield Point**, are **The Fort (Fort Island)** and **Second Island**, surrounded by off-lying reefs. A rock, covered 2 feet, is at about 44°09'08"N., 68°42'30"W. A 9-foot shoal 700 yards westward and a 15-foot shoal 700 yards southwestward of Second Island are unmarked and should be avoided. A rock awash at low water is about 150 yards south of Fifield Point, close north of The Fort.

(84) **Burnt Cove**, northeastward of Fifield Point, is secure in all weather except westerlies. Good anchorage is found in mud bottom in midchannel just inside the entrance. The upper half of the cove is shoal and foul. A church spire in the village of **West Stonington (West Deer Isle)**, at the head of the cove, is conspicuous. A lobster company pier and float landing are on the south side of the entrance; depths of 8 feet are reported alongside the float. Gasoline and some marine supplies are available. A boatyard, on the north side of the cove near the head, can haul out boats up to 55 feet in length for dry open and covered winter storage or hull and engine repairs. There are several other private wharves in the cove, but these are mostly dry at low water.

(85) The western shore of Deer Isle is described with East Penobscot Bay.

(86) South of Deer Island Thorofare and north of Merchant Row are many small islands, the more important of which are mentioned below. Navigation among these islands must be considered dangerous, for there are many ledges and the channels are unmarked.

(87) **Barter Island Ledges**, 0.5 mile west of McGlathery Island, are covered at high water; a daybeacon is on the ledges.

(88) **Harbor Island Ledge**, covered 3 feet, is 0.8 mile southward of **George Head Island**, a wooded island 80 feet high, 1.4 miles west of McGlathery Island. A buoy is north of the ledge. A 14-foot spot in midchannel, 0.5 mile south-southwest of George Head Island, is marked by a buoy on its south side.

(89) **Farrel Island**, 40 feet high and 2.6 miles west of McGlathery Island, and **Scraggy Island**, 0.8 mile west of Farrel Island, are wooded. There are several grassy rocks off the south side of Scraggy Island. **Sparrow Island**, 0.5 mile southwest of Farrel Island, is 40 feet high and grassy. **Sparrow Island Ledges** extend 0.5 mile west of the island.

(90) Of the remaining islands in the area, **Bare Island** and **Round Island** are wooded, and **Buckle Island**, **Little Camp Island**, and **Potato Island** are bare. **No Mans Island** is wooded on the western end and grassy elsewhere. **Enchanted Island** has scattered trees. **Camp Island,Rock Island** and **Russ Island** are partly wooded. **Phoebe Island**, **Millet Island, Spruce Island, Coombs Islands, Wreck Island, St. Helena Island, Green Island, Sand Island** and **John Island** are wooded.

(91)

Merchant Row to Isle Au Haut

(92) **Merchant Row** is a passage from Jericho Bay to East Penobscot Bay between the islands and ledges between Deer Isle and Isle au Haut. This passage is used by vessels in winter when Deer Island Thorofare is closed by ice and by deep-draft vessels at all times. It is not quite as direct as Deer Island Thorofare, but the channel is wider and much deeper. There are numerous ledges and rocks on both sides of the passage, but the principal dangers are marked by buoys or daybeacons and the channel can be readily followed in clear weather and daylight.

(93) Deep-draft vessels can enter from the eastward through Toothacher Bay, the passage between Marshall and Swans Islands, through the channels between Marshall Island and Isle au Haut, or from Jericho Bay. Close attention should be given to the chart and the aids, with due regard for unmarked dangers. The description of the dangers, when entering Merchant Row from the southwestward in Isle au Haut Bay, is given later in this chapter.

(94) The islands and reefs on the north side of Merchant Row, including many of those in the channel, have been previously described in this chapter. There are two entrances to Merchant Row from the eastward that are separated by the islands and reefs in Jericho Bay.

(95) In the eastern approach to Merchant Row, south of Halibut Rocks and west of Marshall Island, a series of islands and reefs extend to the eastern entrance to the passage. **Southern Mark Island**, 2.1 miles southwest of Halibut Rocks, is about 30 feet high and grassy. About 1 mile south of Southern Mark Island is **Fog Island**, which is wooded. The numerous ledges east of Fog Island, and between it and Marshall Island, are mostly all bare. The more important of these, since they are closest to the channels, are **North Popplestone Ledge** and **Saddleback** on the north, and **Green Ledge**, **White Ledge** and **Drunkard Ledge** to the south. Saddleback, 1.4 miles east-southeast of Southern Mark Island, in the summer shows some grass on its two conspicuous humps.

(96) **Torrey Ledge**, covered 17 feet, about 0.6 mile southward of Drunkard Ledge, is unmarked. **Blue Hill Rock**, covered 7 feet and about 1.2 miles eastward of Green Ledge, is marked on its southeast side by a buoy.

(97) Of the other islands and ledges on the south side of Merchant Row, **Burnt Island, Pell Island, Bills Island**, **Merchant Island** and **Ewe Island** are wooded; Hardwood Island is round and heavily wooded; and **Ram Island**, 0.3 mile southwestward of **Hardwood Island**, is wooded.

(98) **Channel Rock**, 0.5 mile westward of Ram Island, uncovers 9 feet and is unmarked. **Ram Island Ledge**, awash at low water, about 400 yards southeastward of Channel Rock, is also unmarked. **Scraggy Ledge** is a bare ledge 700 yards westward of Channel Rock. There is foul ground between Scraggy Ledge and **West Halibut Ledges**, bare ledges 0.3 mile northward. **Outer Scrag Ledge**, 1 mile northwestward of Scraggy Ledge, is 4 feet high. **The Brown Cow**, 1mile northwestward of Outer Scrag Ledge, is a ledge with a rock 3 feet high on it and is the westernmost danger at the western end of Merchant Row. A whistle buoy, about 0.6 mile south-southwestward of The Brown Cow, marks the western entrance to Merchant Row.

(99) **Isle Au Haut**, a large wooded island, 543 feet high, 4.2 miles southward of Deer Isle, is one of the principal landmarks of the locality. It has few year-round inhabitants but a considerable number of summer residents. The coast is mostly foul and must be approached with caution. Part of the island is included in Acadia National Park.

(100) **York Island** is about 0.3 mile off the eastern side of Isle au Haut near its northern end. A ridge of shoals and reefs extends about 1 mile northward from York Island, ending in **Airy Ledge**, which has a buoy off its eastern side.

(101) The channel between York Island and Isle au Haut is almost blocked by a group of rocks. Between **Richs Point**, the northeastern end of Isle au Haut, and York Island are numerous reefs and rocks, most of which are marked by kelp. This area should be avoided by all except those having local knowledge.

(102) Foul ground also extends about 1 mile southward of York Island and includes **Turnip Yard**, awash, **Halfway Rock**, which uncovers 6 feet, and **Horseman Ledge**, awash in places at low water. These are unmarked. An unmarked shoal, cleared 15 feet, is about 500 yards eastward of Horseman Ledge; and a cleared depth of 12 feet, in midchannel between Halfway Rock and Little Spoon Island, is also unmarked.

(103) **Little Spoon Island**, **Great Spoon Island**, **White Horse** and **Black Horse** are a group of grass-covered islands about 1.5 miles south-southeast of York Island. **Great Spoon Ledge**, awash at low water, is 0.3 mile north-northeast of Great Spoon Island. **Colt Ledge**, 0.6 mile south of White Horse and covered 8 feet, is marked by a buoy south of it. The chart should be carefully followed in this locality.

(104) **Eastern Ear Ledge**, which has a rock awash at low water on it, is 0.6 mile south-southeast of **Eastern Ear**, a small island close to the southeast corner of Isle au Haut. A buoy is southeastward of the ledge.

(105) **Head Harbor** is a small bight in the south shore of Isle au Haut, just west of 129-foot-high **Eastern Head**, the southeast point of the island. The harbor is used mostly by lobstermen and affords good protection for small boats except in southwest weather. In normal weather, the off-lying ledges break up the swell, causing the water in the northeastern part of the harbor to be fairly calm. The bottom is rocky in general, but some parts are clay. Depths are 9 to 21 feet in the northeastern semiprotected part and 60 feet and more outside. There are a few houses on the shore in the northeastern part.

(106) The inner or northeastern cove of the harbor should not be entered without local knowledge, except in periods of good visibility. The bottom is mostly sand in the cove.

(107) **Roaring Bull Ledge**, 1 mile south-southwestward of Head Harbor, uncovers 4 feet. A lighted bell buoy is about 500 yards southward of the ledge.

(108) **Western Ear** is a wooded island at the southwest end of Isle au Haut. **Western Ear Ledge**, 0.2 mile southward of Western Ear, is awash at low water.

(109) The western side of Isle au Haut is fringed with many rocks and shoals, bare and covered. The westernmost ones visible at high water include one of the three bare rocks of **The Brandies**, whichis 4 feet high about 1 mile westward of the southern part of Isle au Haut, and **Kimball Rock**, which uncovers 10 feet, 0.6 mile westward of **Kimball Island**, off the northwest shore of Isle au Haut.

(110) Several rocky spots with depths of 18 to 30 feet are outside the line joining these rocks. An obstruction, cleared to a depth of 10 feet, is about 0.2 mile northwestward of Kimball Head, and a rock awash is about 200 yards west of the northern extremity of the head.

(111) **Marsh Cove Ledges**, drying ledges which extend about 0.4 mile southwestward of **Marsh Cove**, Kimball Island, are marked by a buoy off the southwest end.

(112) **Duck Harbor**, 1.5 miles north of Western Ear, is a narrow but protected anchorage for craft up to 40 feet on the western side of Isle au Haut. The south shore should be favored on entering. The Stonington ferry lands at the National Park Service float on the south shore with depths of 5 feet reported alongside.

(113) **Moores Harbor** is a cove on the western side of Isle au Haut about 2.5 miles north of Western Ear. This harbor has many outlying ledges off the entrance and in the harbor and is an unsafe anchorage.

(114) **Isle au Haut Thorofare** is on the northwestern side of Isle au Haut, between Isle au Haut and Kimball Island.

(115) A marked 75-foot-wide dredged channel leads across the ledges at the northeast end of Kimball Island. In 2008, the midchannel controlling depth was 4.8 feet.

(116) The thorofare has a width of 100 to 500 yards, being widest at the western end. Secure anchorage for small craft or very small vessels can be had in depths of 33 feet near the village of Isle au Haut.

(117) In 1990, strong unpredictable currents were reported in the thorofare at the change of tide and should be taken into consideration when anchoring.

(118) **Isle au Haut Light** (44°03'53"N., 68°39'05"W.), 48 feet above the water, is shown from a tower with its lower part conical, gray in color, and the upper part cylindrical, white in color, at Robinson Point on the south side of the western entrance. There is a white bridge to the shore. The light has a **034°-060°**white sector that marks the approach to Isle au Haut Thorofare. A buoy and daybeacon are off the two principal dangers on the north side near the western entrance, and buoys mark both ends of the dredged channel.

(119) Between **Moxie Island** and Isle au Haut Thorofare Daybeacon 4, the channel is narrowed by a ledge with a depth of only 2 to 4 feet. The ledge is so close to the charted 16-foot depth that boats either run on it unawares or come to anchor and are set aground by the falling tide. The daybeacon marks the southeast end of the ledge.

(120) Enter Isle au Haut Thorofare between Isle au Haut Light and Sawyer Ledge Buoy 1 and pass southward of Inner Ledge Daybeacon 3, giving it a berth of over 50 yards. Then keep in midchannel except in the choke at the entrance of the anchorage, where the northern side should be favored slightly. Avoid a rock, bare at low water, which is 90 yards from the northwest side of the anchorage.

(121) **Isle au Haut**, the village on the southeastern shore of Isle au Haut Thorofare, has a town wharf and float landing with 6 feet reported alongside. Gasoline, provisions and some marine supplies can be obtained from the village store. Water can be procured from a nearby well. The harbor is reported to be free of ice in winter.

(122) Several other landings in the harbor dry at low water. A white church spire in the village is conspicuous, as is a large building on Point Lookout. A motorboat ferry carries mail and passengers daily between Isle au Haut and Stonington.

(123) **Lookout** is a village and summer resort at the eastern end of Isle au Haut Thorofare. A buoyed channel to the wharf from Merchant Row leads between Merchant and Hardwood Islands, northeastward of Bay Ledges, and westward of grassy **Flake Island** off the village. The wharf has a reported depth of 8 feet alongside. A buoy off the southwestern end of Flake Island marks the turn in the channel to the wharf. In 1965, dangerous rocks, covered at low water, were reported in the channel between Flake Island and Birch Point at the north end of Isle au Haut.

(124)

Penobscot Bay

(125) **Penobscot Bay**, the largest and most important of the many indentations on the coast of Maine, is about 20 miles wide from Isle au Haut on the east to Whitehead Island on the west and 28 miles long from its entrance to the mouth of Penobscot River. A chain of large and small islands divides the bay into two parts, **East Penobscot Bay** and **West Penobscot Bay**. The southern part of East Penobscot Bay is Isle au Haut Bay. **Vinalhaven Island** and **North Haven Island** are large islands dividing the southern part of the bay. Islesboro Island divides the bay near its head. Numerous harbors indent the shores of Penobscot Bay, the most important being Rockland, Rockport, Camden, Belfast and Searsport on the western shore; Castine and Stonington on the eastern shore; and Vinalhaven and North Haven in the center of the bay. The bay is the approach to Penobscot River, on which are several towns and the city of Bangor at the head of navigation. The bay ports collectively are among the leaders for the lobstering industry in Maine.

(126) The sea approaches to the bay are well marked by the lights on Monhegan Island and Matinicus Rock; the entrance is marked by Saddleback Ledge Light on the east and by Whitehead and Two Bush Island Lights on the west side of the bay. The harbors are well lighted, and the more important dangers are marked by buoys or daybeacons. Deep-draft vessels ply the bay throughout the year and recreational vessels are prevalent during the summer. In severe winters many of the harbors are obstructed by ice. The Penobscot River seldom is entirely closed by it as icebreakers usually keep the channel free. The thorofares are only occasionally obstructed by ice and are much used by small vessels bound along the coast.

(127) Penobscot Bay, a region of rocks and ledges, requires extreme caution in navigating. After unusually high tides many logs are present in the bay, particularly from Belfast northward. These logs are dangerous to small craft. Penobscot Bay can be entered from eastward through Eggemoggin Reach, Deer Island Thorofare or Merchant Row and from westward through Muscle Ridge Channel or Two Bush Channel.

(128) Large vessels approaching Penobscot Bay from southward, either from Boston, Cape Cod Canal, or from eastward of Cape Cod, usually make Cape Ann Lighted Whistle Buoy 2 (42°37.9'N., 70°31.2'W.), then shape the course for Manana Island Lighted Whistle Buoy 14M (43°45.3'N., 69°22.5'W.), and then enter through Two Bush or Muscle Ridge Channels. Two Bush Channel is used by most vessels and tows and by all except small local vessels when the visibility is not good. Muscle Ridge Channel has good water, and most of the dangers are marked, but it is narrow in places and has a number

of unmarked 16- to 22-foot spots near the track. It is not recommended for deep-draft vessels.

(129) Vessels entering and departing the eastern channel of Penobscot Bay should exercise caution in the area of a wind turbine at 44°23.1'N., 68°49.4'W. All vessels and persons are advised to avoid transiting, anchoring, diving, dredging, dumping, fishing, trawling, laying cable or conducting salvage operations in this area.

(130) The preceding paragraphs give the simplest directions by pointing out the difficulties and the dangers and especially, when necessary, the need for local knowledge. The channels are well buoyed, most of the dangers well marked, and the approaches clear. No difficulty should be experienced in approaching and entering the bay in clear weather with the aid of the chart and by following the aids.

(131) Two vessel-to-vessel **oil transfer anchorage areas** near the head of Penobscot Bay north of Islesboro Island are discussed later in this chapter, indexed as Oil Transfer Anchorage Area.

(132)

Penobscot Bay and River

(133) The U.S. Coast Guard Captain of the Port, Sector Northern New England, in cooperation with the Maine and New Hampshire Port Safety Forum, has established a Recommended Vessel Route for deep draft vessels entering and departing Penobscot Bay and River. Deep draft vessels are requested to follow the designated routes. These routes were designed to provide safe, established routes for increased deep draft vessels, to prevent the loss of fishing gear placed in the waters in the approaches to Penobscot Bay and River and to reduce the potential for conflicts between less maneuverable deep draft commercial vessels and all other vessels navigating upon these waters. Vessels are responsible for their own safety and are not required to remain inside the route nor are fisherman required to keep fishing gear outside of the 0.4 mile wide route.

(134) **Recommended minimum under-keel clearances for Penobscot Bay and River** have also been established by the aforementioned group, in order to prevent groundings and to promote safety and environmental security of the waterway resources of Penobscot Bay and River. The group recommends that all entities responsible for safe movement of vessels in and through the waters of Penobscot Bay and River operate vessels in such a manner as to maintain a minimum under-keel clearance of 3 feet between the deepest draft of the vessel and the channel bottom when transiting Penobscot Bay and outer Penobscot River, south of Turtle Head on Islesboro Island; 2 feet when transiting Penobscot River north of Turtle Head; and a minimum under-keel clearance of 1 foot at all berthing areas.

(135) The Penobscot Bay & River Pilots distribute educational material to mariners in an effort to reduce right whale ship strikes.

(136) The **Penobscot Bay and River Closed Area**, a Marine Protected Area (MPA), includes all bays, inlets and harbors within Penobscot Bay and River.

(137)

Pilotage, Penobscot Bay and River

(138) Pilotage is compulsory for all foreign vessels, and for U.S. vessels under register in the foreign trade, with a draft of 9 feet or more, entering or departing from any port or harbor within the waters of Penobscot Bay and Penobscot River north of a line drawn from Marshall Point Light at Port Clyde, thence to Matinicus Rock Light and thence to Western Head, Isle au Haut. Pilotage is optional for vessels under enrollment, fishing vessels and vessels powered by sail. Upon departing its loading port enroute to a Penobscot Bay oil terminal, each commercial vessel is requested to furnish via its local shipping agency such information as required by local pilots and receiving facilities. The pre-arrival information includes vessel name and particulars, cargo grades and amounts, estimated time of arrival and any special needs.

(139) Pilot Boarding Area(s): for the eastern entrance to the bay, vessels should make arrival at Penobscot Bay East Approach Lighted Whistle Buoy PBA (43°55'37"N., 68°39'37"W.) and follow the Recommended Route due west. The pilot will board 3 miles east of Penobscot Bay Lighted Buoy WP at 43°55'30"N., 68°48'48"W. For the western entrance, pilots will board at 43°46'36"N., 69°22'36"W. approximately 1 mile north of Manana Island (virtual AIS Aid to Navigation) V-AIS "14M" (43° 45' 19"N., 69° 22' 28"W.) The pilots require 36, 24 and 12 hour ETAs, as no pilot boats are maintained on station.

(140) Pilotage in the waters of Penobscot Bay is available:

(141) **Penobscot Bay and River Pilots Association** offer pilotage for the entire area. The office address is: 18 Mortland Rd, Searsport, ME 04974; telephone number: 207–548–1077 (24-hours); fax: 207–548–1078, email: pilots@penbaypilots.com.

(142) The pilot boats at Monhegan Island (the west entrance to the bay) are local lobster fishing vessels that display a pilot flag by day and lights for a pilot boat by night. The pilot boat for the eastern entrance to the bay is a 48-foot black-hulled pilot boat with a white cabin top and "PILOT" written on the sides and top of the superstructure. The pilot boats can be contacted one hour prior to the last stated ETA on VHF-FM channel 16. The pilot boat will instruct the arriving vessel which side to place the ladder on. The ladder should be ½ meter above the water, well lit and in compliance with IMO specifications. Integrated tug and barge units are required to have a proper pilot ladder rigged from the barge. Pilots will board vessels day and night when weather and sea conditions permit.

(143)

Security Broadcast System, Penobscot Bay

(144) Penobscot Bay and approaches have an established security communication system in which pilots, masters and mates of deep-draft commercial vessels utilize

VHF-FM channels 13 and 16 for security calls when proceeding between the pilot pickup stations and dock or anchorages at the north end of the bay and river.

(145)

Towage

(146) Three tugs up to 1,800 hp are available at Belfast. Arrangements for tugs are usually made through ships' agents; advance notice of 24 hours is required. Large oceangoing vessels require the use of tugs for docking at Searsport and at most of the ports on Penobscot River. A tug usually accompanies large vessels bound upriver to Brewer and other river ports; tugs meet vessels off Fort Point. Vessels bound for Searsport are met by tugs off Sears Island Bell Buoy 2. Tugs monitor VHF-FM channels 16, 13 and 10; work channels 13 and 10.

(147)

Wharves

(148) Approach and mooring criteria for Searsport deepwater facilities are as follows: Large commercial vessels should engage the services of escort tugs for inbound and outbound transits. For inbound transits, the escort tugs should be engaged in the vicinity of Sears Island Bell Buoy 2. Ship-to-shore communications are established at Searsport anchorage. Minimum visibility requirements for Searsport are ¼ mile. Maximum wind speed for entrance into Searsport is at pilot's and master's discretion. Maximum vessel capacity at Searsport is 65,000 DWT and 80,000 DWT at the Maine Port Authority Dry Cargo Pier.

(149) Approach and mooring criteria for Bucksport deepwater facilities are as follows: Large commercial vessels should engage the services of ship-assist tugs for inbound and outbound transits. For inbound transits, the assist tugs should be engaged in the vicinity of Fort Point. Ship-to-tug communication is established below Fort Point. Minimum visibility requirements for the Penobscot River are ½ mile. Maximum wind speed for docking and undocking in Bucksport are at the master and pilot's discretion. Maximum vessel capacity for Bucksport is 65,000 DWT; the maximum draft at MLW is 35 feet.

(150) There is no secure harbor for vessels at any of the islands southward off Penobscot Bay. However, small craft and local fishermen moor at Monhegan Island, in Matinicus Harbor, which is the cove on the eastern side of Matinicus Island northward of Wheaton Island, and in Criehaven Harbor, an indentation in the northwest part of Ragged Island. The waters of this area are well surveyed; deep passages exist between the islands, as shown on the chart. Because of the broken nature of the bottom, vessels, particularly deep-draft ones, should avoid all broken ground having depths less than 60 feet. These waters are frequented mostly by local fishermen. The only settlements are on Monhegan, Matinicus and Ragged Islands. In 1979, passengers bound for Matinicus Island used airplane service or chartered boats from Rockland. From Matinicus Island, a lobster boat can be hired to take passengers to Criehaven Harbor on Ragged Island.

(151)

Dangers

(152) **Seal Island**, the easternmost of the islands off Penobscot Bay, is bare, rocky, about 77 feet high and 1 mile long. **Eastern Ledge**, awash at low water on which the sea usually breaks, extends 350 yards off the east end of the island. **Three Fathom Ledge**, 1.4 miles east-northeast of Seal Island, marked by a lighted buoy, has been cleared to 16 feet. **Gully Ledge**, covered 24 feet, is about 650 yards south of Western Head, the westernmost point of the island.

(153) Seal Island is within the **danger zone** of a naval aircraft bombing target area, centered in 44°53'N., 68°44'W., just eastward of the island. (See 33 CFR 334.10, chapter 2, for limits and regulations.) A wooden target float with a 10-foot tripod equipped with a radar reflector is about 0.4 mile southeast of the island.

(154) **Snippershan Ledge**, about 3.7 miles north-northwestward of Seal Island, has a least depth of 36 feet and is marked with a virtual AIS.

(155) **Malcolm Ledge**, midway between Seal Island and Wooden Ball Island, is 0.4 mile long and marked by a virtual AIS. The north end of the ledge uncovers 9 feet; the south end uncovers 3 feet.

(156) **Wooden Ball Island**, 3 miles southwest of Seal Island, is 62 feet high, 1 mile long and rocky with grass on top. The eastern point of the island is a prominent knob. There are a few small abandoned houses at the low place in the western part of the island and a few summer camps. A lighted bell buoy is about 0.5 mile southwestward of Wooden Ball Island.

(157) **Matinicus Rock**, the southernmost islet in the approach to Penobscot Bay, is 56 feet high and is marked near its south end by **Matinicus Rock Light** (43°47'00"N., 68°51'18"W.), 90 feet above the water, shown from a 41-foot cylindrical gray granite tower. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A.

(158) About 2.3 miles northward of Matinicus Rock, a group of islands and rocks extends about 5 miles northward. Ragged, Tenpound and Matinicus Islands and No Mans Land are the principal islands of the group.

(159) **Ragged Island**, the southernmost, is partly wooded. There are numerous high bare rocks, including **Green Ledge**, **Seal Ledge**, **High Ledge** and **Brig Ledge**, on the east and south sides of the island. Broken ground extends 0.8 mile southward from the island to **Inner Breaker**, which is covered 3 feet and marked by a buoy.

(160) **South Breaker** is a small rock awash 1.7 miles southward of Ragged Island and 1.6 miles northwestward of Matinicus Rock Light. A buoy is south of the rock. A bell buoy is west of **Southwest Ledges**, 0.4 mile southwest of Ragged Island, which uncovers 6 feet in places.

(161) **Criehaven** is a village on **Criehaven Harbor**, on the western side of Ragged Island. There are several year-round residents on the island and some summer visitors. A breakwater extends northward from the southern entrance point; it is marked by **Criehaven Breakwater Light 8** (43°50'05"N., 68°53'33"W.), 32 feet above the water, shown from a skeleton tower with a red triangular daymark. A buoy is 700 yards westward of the harbor entrance off **Harbor Ledges**, which uncovers 4 feet and is on the south side just outside the entrance to the harbor. The best water favors the north side. There are several fish wharves in the harbor that bare alongside at low water. A limited supply of water can be obtained from wells on the island. The harbor affords anchorage for lobster boats moored to communal lines that are rigged across the harbor, but there is no protection from northerly blows. Strangers entering the harbor should exercise caution not to foul these mooring lines. The bottom slopes gently except for one dangerous rock that uncovers 2 feet on the northeast side of the harbor.

(162) **Seal Cove**, on the opposite side of the island from Criehaven Harbor, is reported to be shoal and boulder strewn, with seas that break at the entrance. Fairly good anchorage is available off Seal Cove in depths of 70 feet, flat sand and shell bottom, for vessels up to 100 feet in length; this anchorage is sometimes used by trawlers during northerly blows. However, mariners are cautioned to avoid anchoring in the cable area that extends southeastward from Seal Cove.

(163) **Pudding Island** and **Shag Ledge**, close to the northeast end of Ragged Island, are 35 feet high, bare and rocky. **The Hogshead**, 0.2 mile northward of Shag Ledge, is a small bare rock 9 feet high in the middle of the eastern entrance to Matinicus Roads.

(164) **Tenpound Island**, 0.4 mile north of Ragged Island and 0.3 mile off the southeast side of Matinicus Island, is 41 feet high and grassy. **Matinicus Roads**, between Ragged Island and Tenpound Island, has a controlling depth of about 18 feet. A 7-foot rocky shoal is on the south side of the roads.

(165) **Matinicus Island**, 0.8 mile northward of Ragged Island, is mostly wooded and is the largest of the group. There are about 100 year-round residents, but in the summer many people spend vacations here. A telephone relay tower, about 100 feet high near the center of the island, is prominent from all around the island. **West Black Ledge**, 13 feet high, and **East Black Ledge**, 15 feet high, are bare rocks 0.6 and 0.9 mile eastward of Matinicus Island. **Tuckanuck Ledge**, 200 yards eastward of Eastern Black Ledge, has two rocks which uncover 1 and 8 feet.

(166) **Mackerel Ledge**, 700 yards northward of East Black Ledge, uncovers 7 feet. A buoy is north of the ledge. A rock covered 12 feet is 0.2 mile southward of East Black Ledge. **Greens Ledge**, covered 34 feet, is 0.7 mile northeast of Mackerel Ledge.

(167) **Wheaton Island** is just off the east side of Matinicus Island. The passage between the islands is bare at low water. There are small wharves that dry out in the cove between the two islands, and small craft anchor here. **Old Cove**, westward of the south end of Wheaton Island, is seldom used as an anchorage.

(168) The narrow passage between Wheaton Island and Matinicus Island is used considerably by lobstermen at half tide or better. A small boat attempting this passage should hug the Matinicus side no more than 5 feet from shore because of the rounded ledge that uncovers 3½ feet near midchannel.

(169) **Matinicus Harbor**, on the east side of Matinicus Island, is protected by Wheaton Island and a 450-foot breakwater extending from the north side. A light is close eastward of the breakwater. **Indian Ledge**, 2 feet high, is in about the center of the harbor. Small vessels can anchor in the outer harbor between Wheaton Island and Indian Ledge in depths of 6 to 26 feet. Except during easterly weather, the anchorage is quite calm. Numerous fishing boats moor to communal mooring lines in the inner harbor behind Indian Ledge in depths of 4 to 8 feet.

(170) **Matinicus**, the village at the head of the harbor, has a general store and a restaurant. Gasoline and diesel fuel are available at the main wharf, which has a reported depth of only 1 foot alongside at low water. A limited supply of fresh water may be obtained at a well near the wharf. In 1979, a mailboat worked between Tenants Harbor and Matinicus, but no passengers were carried. Passengers bound for the island generally use airplane service or charter boats from Rockland. A lobster boat can usually be hired to take passengers to Criehaven on Ragged Island.

(171) **Harbor Ledge** is a rock covered 4 feet, 300 yards northeastward of the entrance to Matinicus Harbor. A bell buoy south of the rock marks the entrance to the harbor. **The Barrel**, 300 yards northeastward of Harbor Ledge, is a rock that uncovers 10 feet at the south end of a ledge 300 yards long.

(172) **No Mans Land**, the largest of the rocks and islets northeastward of Matinicus Island, is 51 feet high and grassy. **Two Bush Island**, 22 feet high and grassy on top, is joined to the northeast end of Matinicus Island by a ledge that is covered 2 to 8 feet. **Two Bush Ledge**, 15 feet high and bare, is 0.2 mile southeastward of the island. Rocks covered 3 feet extend 350 yards eastward and northeastward from the ledge.

(173) **Beach Ledges** are two rocks that uncover 1 foot and 3 feet between Two Bush Ledge and Matinicus Island. A buoy marks the southeast end of the ledge. **Whaleback** is a ledge that uncovers 5 feet, 0.3 mile westward of No Mans Land.

(174) **Zephyr Ledges** are two rocks, the easternmost uncovering 3 feet, 0.3 mile north-northeast of No Mans Land. **Zephyr Rock**, the northeast end of the group, is covered 5 feet, 0.6 mile northeastward of No Mans Land. A lighted buoy is north of the rock.

(175) A lighted bell buoy, 0.7 mile north of Matinicus Island, is westward of the dangers northeastward of the island.

(176) Local boats bound to Matinicus Harbor from northward drawing 6 feet pass through the channel between Matinicus Island on the west and Two Bush Island and Beach Ledges on the east, at low water. Strangers should use this passage only in small boats and with a smooth sea, being careful to avoid the ledges.

(177) **Black Rocks**, two rocks 3 feet high, are on a ledge 0.3 mile westward of Matinicus Island.

(178) **Bantam Ledge**, which uncovers 5 feet and is surrounded by deep water, is 2.5 miles westward of Ragged Island. A buoy is southeastward of the ledge.

(179) **Foster Ledges**, 2 miles westward of Matinicus Island, are covered 6 and 13 feet; the southwestern and shoalest rock has a buoy off its west side. A ledge covered 30 feet is about 0.9 mile northward of the buoy.

(180) **Bay Ledge**, covered 3 feet, is about 5 miles northward of No Mans Land. A gong buoy is southwest of the ledge, and a fairway lighted bell buoy is about 1.3 miles southward of it. A ledge covered 24 feet is about 0.5 mile southwestward of Bay Ledge.

(181) **Pigeon Ground**, 3.5 miles west of Matinicus Island and 2 miles southward of Large Green Island, is broken ground about 2 miles long in an east-northeasterly direction. It is covered 15 feet on the western part of the broken ground and 21 feet on the eastern part.

(182) **Large Green Island**, 5 miles northwestward of Matinicus Island, is 43 feet high and grassy, and has some houses on its northern part. **Herring Ledge**, which partly uncovers 7 feet, extends 0.3 mile southward from the island.

(183) **Green Island Seal Ledges** are 0.6 and 0.9 mile south-southeastward of Large Green Island, with broken ground between. The southern part of the ledges uncovers 8 feet; the north end uncovers 5 feet.

(184) **Collins Rock**, about 1 mile north of Large Green Island, is covered 5 feet. **Junken Ledge**, covered 20 feet and marked by a buoy, is about 5 miles north-northeastward of Large Green Island. Two Bush Island Lighted Whistle Buoy TBI is 1.3 miles southwest of Junken Ledge.

(185) **Little Green Island**, about 1 mile northwestward of Large Green Island, is 43 feet high and grassy, and has several cottages on it.

(186) **Northern Triangles**, 1 mile northward of Little Green Island, is a reef about 1 mile long in an east-southeasterly direction. In the western half of the reef are some ledges awash at low water. A buoy is about 0.7 mile northward of the reef. A shoal covered 15 feet is 2 miles northward of Little Green Island and 2 miles southeastward of Two Bush Island Light; the shoal is marked by a buoy.

(187) **Alden Rock**, covered 4 feet and marked by a buoy, is 1.4 miles northwestward of Little Green Island. An unmarked rock covered 14 feet is 0.8 mile west-northwestward of Alden Rock. The 14-foot rock is at the eastern end of broken ground nearly 1 mile long and covered 22 to 30 feet.

(188) **Southern Triangles** are three groups of rocks, awash at low water, located midway between Little Green and Metinic Islands. The southwestern rock, 0.6 mile from the other two, uncovers 3 feet. A buoy is southeastward of the easternmost rock.

(189) **Metinic Island**, 5 miles west-southwestward of Large Green Island, is nearly 2 miles long, 78 feet high near its northern end and partly wooded. The island is occupied during the summer by fishermen. There are no wharves, supplies or mail service available. **Metinic Green Island**, low and grassy, is 0.4 mile southward of Metinic Island, with foul ground and ledges between.

(190) There is a passage for small craft with depths of about 7 feet between Metinic and Metinic Green Islands; local knowledge is advised. A rock covered 2 feet is 300 yards southwest of Metinic Green Island.

(191) A bell buoy, about 0.6 mile northward of Metinic Island, marks **Wheeler Rock**, covered 5 feet, about 0.3 mile northward of the island. **Wheeler Big Rock**, which uncovers 9 feet, is 300 yards northward of the island. **Green Point Shoal**, covered 17 feet, is 0.8 mile eastward of Metinic Island. **Hog Island** and **The Nubble**, both of which are bare, and **Cat Ledge**, which uncovers 3 feet, are close off the east side of Metinic Island.

(192)

Black Rock to Gull Rock Ledge

(193) Broken ground extends 2 to 3 miles westward and southwestward from Metinic Island. **Black Rock**, which uncovers 5 feet, is 0.7 mile westward of Metinic Island. A rock covered 14 feet is 0.3 mile southwestward of Black Rock.

(194) **Metinic Island Ledge**, covered 8 feet and marked by a buoy at its southwest end, is 1.8 miles westward of the northern end of Metinic Island. Kelp is reported on Metinic Island Ledge. A rock covered 26 feet is 0.6 mile northeastward of the ledge. **Hupper Shoal**, covered 17 feet, is 0.6 mile southwestward from the ledge.

(195) **Roaring Bull**, awash at low water and generally marked by breakers, is 2.8 miles westward of Metinic Green Island. A buoy is off the northwest side. A ledge covered 27 feet is 0.6 mile northwestward of Roaring Bull.

(196) **Southeast Breaker** is on a ledge about 0.5 mile long in a northeast direction; the higher part uncovers 4 feet. The ledge is 1.8 miles west-southwestward of Metinic Green Island, on the range of the south end of Metinic Green Island and the north end of Large Green Island. Unmarked shoals covered 10 to 16 feet are within 0.5 mile of the ledge.

(197) **Haddock Ledge**, covered 11 feet, is 1.3 miles southward of Southeast Breaker and 2.5 miles southwestward of Metinic Green Island; it is not marked.

(198) **Monhegan Island**, 9 miles off the mainland and 20 miles westward of Matinicus Rock, is one of the important landmarks for vessels bound along the coast.

The island is 1.4 miles long and 165 feet high and presents a rocky shore with high bluffs in places.

(199) **Monhegan Island Light** (43°45'53"N., 69°18'57"W.), 178 feet above the water, is shown from a white tower connected to a white building, on the middle of the island. Within 3 miles of the island, the light is obscured between west and southwest.

(200) **Monhegan Harbor**, between Monhegan and Manana Islands, is an anchorage for small craft but is exposed southward. The harbor, used principally by local fishermen and yachts, has depths of 15 to 25 feet with poor holding ground and scant room at the anchorage for a small vessel to swing.

(201) The deeper water in the harbor favors Manana Island. Entry into the harbor from the south is clear. A depth of 12 feet can be taken through the northern entrance between the wharf on Monhegan Island and the grass-covered rocky islet on the end of the ledge making out from Manana Island. The channel west of the small islet is shoal and has a depth of only 3 feet.

(202) In entering from the north the best water leads close to the end of the wharf. Even small craft should not attempt to ride out bad weather in this roadstead. During heavy weather the daily mail boat seldom is unable to land at the wharf.

(203) **Monhegan** is a village of fishermen and summer residents on the east side of Monhegan Harbor. The principal wharf has a depth of about 12 feet at the end. The village has telephone communication with the mainland. A diesel-powered motorboat ferry carries mail, freight and passengers from Port Clyde, daily in summer and three times weekly in winter. Gasoline, diesel fuel and provisions are obtainable. There are good hotel accommodations in the summer, and excursion boats from Boothbay Harbor call at Monhegan in the summer.

(204) **Eastern Duck Rock**, 400 yards off the north end of Monhegan Island, is a large, bare rock with some grass on top; the narrow channel between the rock and the island is nearer the rock because of Seal Ledges, which extend from Monhegan Island and show partly at high water. A gong buoy is about 250 yards northwest of the rock. **The Barrel**, arock that uncovers about 5 feet off the northwest shore of Manana Island, is marked by a buoy.

(205) **Duck Rocks**, 0.6 mile off the northwest side of Monhegan Island, are two large, bare rocks. **Sunken Duck Rock**, covered 5 feet, is about 125 yards north-northwestward of the larger Duck Rocks. A bell buoy is northwest of Sunken Duck Rock.

(206) **Allen Shoal**, 1.9 miles northeastward of Monhegan Island Light and cleared to 22 feet, is unmarked.

(207) **Gull Rock Ledge**, covered 20 feet, is 1 mile south-southeastward of Monhegan Island Light. Kelp has been reported on this ledge. Breakers are reported to form on this ledge and on Allen Shoal with heavy swells.

(208)

East Penobscot Bay to Grass Ledge

(209) **East Penobscot Bay** is that part of Penobscot Bay located eastward of Vinalhaven, North Haven and Islesboro Islands. The southern part of it, between Isle au Haut and Vinalhaven Island, is called **Isle au Haut Bay**.

(210) There are many islands and numerous unmarked ledges in Isle au Haut Bay and East Penobscot Bay. The islands have numerous coves and small harbors, but few of these are available as anchorages, except for small craft, because of their shoal depths or obstructed entrances.

(211) The principal traffic through East Penobscot Bay moves in an east-west direction, with access through Eggemoggin Reach, Deer Island Thorofare or Merchant Row from the eastward or through Fox Islands Thorofare or the channels northward of North Haven Island from the westward.

(212) A clear channel good for the deepest-draft vessels, though seldom used, leads through Isle au Haut Bay from Saddleback Ledge Light to the head of East Penobscot Bay. The channel passes eastward of Eagle Island, marked by a light, and a gong buoy northeastward of the light; thence in a northwesterly direction through the islands, northward of Eagle Island; and thence northward passing close westward of Cape Rosier.

(213) The principal dangers in this channel are marked, and the main part of it, with the exception of the areas near the shores, has been swept. The principal thoroughfares east and west have also been swept.

(214) **Saddleback Ledge Light** (44°00'52"N., 68°43'35"W.), 52 feet above the water, is shown from a gray conical tower on a rocky islet in the middle of the southerly entrance to East Penobscot Bay. A sound signal is at the light. There is broken ground between the light and Vinalhaven Island, and deep-draft vessels should enter eastward of the light.

(215) The western side of Isle au Haut Bay is very foul. **Saddleback Ledge Shoal**, covered 2 feet and marked by a buoy, is 0.8 mile northwest of Saddleback Ledge Light. The mile-wide channel west of this shoal and east of 40-foot-high **Diamond Rock** and Diamond Rock Ledge has several shoal spots with depths ranging from 14 to 28 feet. The southern entrance to this channel is nearly 1 mile southwestward of Saddleback Ledge Light. **Diamond Rock Ledge**, covered 2 feet, is marked by a buoy.

(216) Between this channel and the southeast shore of Vinalhaven Island are many islands and reefs dangerous to navigation. A buoyed channel provides a partially protected route through the islands and reefs around the south end of Vinalhaven Island to Carvers Harbor or West Penobscot Bay. In daylight and with good visibility, moderate-sized vessels can run parallel to and about 0.5 to 1 mile offshore on this route.

(217) On the edges of this channel, along the southeast side of Vinalhaven Island, are unmarked **Little Triangle Ledge**; **Triangle Ledge**, **Halibut Ledge**, **Crosby Ledge**, **Sheep Island Ledge**, **Old Duke Ledges**, **Bunker Ledge**, and **Point Ledge**, all marked by buoys, off the entrance to Indian Creek.

(218) Farther inshore, and unmarked, are **Green Island**, **Narrows Island**, **Sheep Island**, **Point Ledge**, **House Ledge**, **Sister Ledge**, **Griffin Ledge**, **Green Ledge**, **Clam Ledges**, **Wreck Ledge** and **Folly Ledge**. Southward of the buoyed channel are **Brimstone Island**, **Hay Islands**, **Roberts Islands**, **Carvers Island** and **Otter Island.**

(219) Surrounding and interspersed between these islands are numerous rocks and ledges, unmarked and dangerous to navigation. **Arey Ledges**, **Colt Ledge**, **Heron Neck Ledge**, **Old Horse Ledge**, **Channel Ledge** and **Knubble Ledge** and **The Breakers**, dangerousledges only partially marked by buoys, lie to the westward, in the southern approach to Carvers Harbor.

(220) **Arey Cove** and **Roberts Harbor**, on the southeast side of Vinalhaven Island, are much obstructed by rocks and ledges and are unsafe for strangers.

(221) The coast northward to Bluff Head and the eastern entrance to Fox Islands Thorofare has many off-lying islands and reefs, extending in some places nearly 1 mile offshore. The coves are small and foul and of no value as harbors.

(222) **Winter Harbor**, locally known as Pleasant River, **Seal Bay** and **Smith Cove** make into the northeastern part of Vinalhaven Island, south of the eastern entrance to Fox Islands Thorofare. They are frequently used by visiting yachtsmen during the summer, but local knowledge is advised.

(223) The islands and dangers on the east sides of Isle au Haut Bay and East Penobscot Bay from the entrance to Barred Island (44°10.0'N., 68°43.2'W.) off the southwestern side of Deer Isle were nearly all described previously in this chapter under the discussion of Deer Island Thorofare, Merchant Row and Isle au Haut. Between Deer Island Thorofare and Eggemoggin Reach, the eastern side of the bay is formed by the western shores of Deer Isle and Little Deer Isle.

(224) The only off-lying dangers from Barred Island off Crockett Cove to Southwest Harbor are **Sellers Rock**, part of which bares at low water and which is marked by a buoy, and the 18-foot spot 700 yards west of it.

(225) **Crockett Cove**, east of Barred Island and northwestward of Burnt Cove, is shoal and foul in its upper half. There is reported to be a good small-craft anchorage, secure in all weather, in 18 feet of water about 0.6 mile inside the entrance. It should be approached only after 3 hours following low water on a rising tide, and favoring the western side of the cove.

(226) **Goose Cove** is a small bight close westward of Crockett Cove. It is foul with rocks awash and submerged in the center. Great care should be taken in entering. A ledge is on the west shore.

(227) Two rocks awash at low water are off the entrance to the cove eastward of Barred Island, and another one farther eastward is off the entrance to Crockett Cove.

(228) **Southwest Harbor** is on the western side of Deer Isle, about 4 miles north of Deer Island Thorofare Light. The harbor is about 0.3 mile wide at the entrance and 1 mile long. The anchorage in depths of 18 to 28 feet is not used much, being open southward. A spire in the village of **Sunset** on the eastern shore of the harbor is prominent. There are no wharves. The western side of the harbor is formed by **Sheephead Island**, from which **Sheephead Island Ledges** extend 0.3 mile southward.

(229) **Mill Pond**, northward of Southwest Harbor, is of little importance. **Sylvester Cove** is northwestward of Mill Pond. The Deer Isle Yacht Club pier and float landing, with 9 feet alongside, and a fish wharf with a town float, dry at low water, are on the north side of the cove. Bare stone cribbing, on the south side of the cove opposite the yacht club float, is all that remains of a former pier. The mail and passenger boat, with year-round service to Eagle Island and summer service to Great Spruce Head Island and Barred Islands, leaves from the yacht club float. Eastward of the yacht club landings, the cove shoals rapidly to a fine shelving beach.

(230) The anchorage in the entrance of the cove is partly sheltered by a long reef on the south side of the entrance; the reef bares at about half tide. A buoy is northwestward of the reef. Caution should be used in rounding this reef, by passing northward of the buoy; it is reported that several small craft have grounded on the reef. There are no services available in Sylvester Cove.

(231) **Dunham Point**, 0.8 mile northwest of Sylvester Cove, is the westernmost point on Deer Isle. **Dunham Point Ledge**, awash at low water, extends 300 yards offshore from the point.

(232) **Pressey Cove**, about 1 mile northeastward of Dunham Point, is shoal and foul. There are islets on the west side and in the middle of the entrance to the cove. The middle one is grassy, and the western one has two trees and is connected with the shore by a sandbar. A number of private homes are around the cove.

(233) **Northwest Harbor**, on the northwestern side of Deer Isle, is about 0.3 mile wide and over 1 mile long. A large part of the upper half of the harbor is shoal and foul and dries out. Good anchorage will be found for small vessels in midharbor in depths of 13 to 17 feet, soft bottom. The harbor is sheltered from all but northwesterly winds. Good anchorage also is off the entrance of the harbor, between Gull Ledge and Heart Island, in depths of 19 to 30 feet. During January and February the harbor is closed by ice.

(234) **Gull Ledge**, partly uncovered at high water, is 0.4 mile northwest of the southern entrance point to Northwest Harbor. Between Gull Ledge and the buoy off the ledge extending from the southern entrance point is a narrow channel. Southwestward of Gull Ledge is a reef with rocks awash at low water, which must be avoided even by small boats if using this channel. Its southwest

end is marked by a buoy, about 0.5 mile from Gull Ledge and 0.4 mile off the main shore.

(235) The village of **Deer Isle** is at the head of the harbor; some marine supplies and provisions are available. A private wharf and float landing are on the northern shore of the harbor about 0.7 mile east of Gull Ledge.

(236) The harbor is seldom used except by pleasure craft as an overnight anchorage in fair weather. With the aid of the chart, little trouble should be experienced in approaching and finding anchorage in midchannel up to 0.4 mile inside the entrance, as the entrance is wide and clear. **Heart Island**, 60 feet high and wooded, is 0.5 mile northward of the entrance.

(237) North of Northwest Harbor, the western shore of Deer Isle extends in a northeasterly direction to Eggemoggin Reach. It was formerly possible for small boats to follow this shore and pass between Little Deer Isle and Deer Isle directly into Eggemoggin Reach. This passage now is closed by a causeway.

(238) The islands off the western end of Eggemoggin Reach were previously described in this chapter.

(239) Between Little Deer Isle and North Haven Island, there is a chain of islands through which are many passes; these passes must be used with care because of the many reefs between the islands.

(240) **Pickering Island**, 1 mile southwest of Little Deer Isle and about 90 feet high, and **Bradbury Island**, 2.5 miles southwest of Little Deer Isle and about 170 feet high, are both wooded and are the principal islands north of the main ship channel through East Penobscot Bay.

(241) **Hardhead Island**, a grassy islet 76 feet high, is 1 mile southward off Bradbury Island. About 0.4 mile northwest of Hardhead Island is **Middle Rock**, a shoal covered 10 feet on the north side of the main ship channel. A buoy is west of the rock.

(242) Southwest of the main ship channel, and between it and North Haven Island, the passes between the islands are nearly obstructed by reefs in many cases. Navigation between these islands, even by small craft, must be done with caution. A few of the reefs are buoyed. Most of the islands are wooded.

(243) **Eagle Island**, 1.5 miles west of Dunham Point, is wooded. **Eagle Island Light** (44°13'04"N., 68°46'04"W.), 106 feet above the water, is shown from a white granite tower on the northeast end of the island. A gong buoy is 320 yards east-northeastward of the light. **Eagle**, a small settlement on the island, has year-round mail and passenger boat service to Sylvester Cove, on the northwest side of Deer Isle. A bell buoy marks a shoal, covered 12 feet, extending 0.6 mile eastward of the island. **The Porcupines** are two high wooded islands off the south end of the island.

(244) **Great Spruce Head Island**, 231 feet high and 2.1 miles northwest of Eagle Island, is the highest island in the group. **Bear Island**, just south of Great Spruce Head Island, has a protected anchorage in a cove at its north end in depths of 12 to 30 feet, rocky bottom. A wharf and float in the cove have a depth of about 9 feet alongside.

(245) **Butter Island**, 186 feet high and 0.5 mile northwest of Eagle Island, is wooded. The passage between Butter Island and the northeast island of the **Barred Islands**, 300 yards westward, uncovers at low water. In 2010, shoaling to 2 feet was reported within the Barred Islands at 44°14'07.6"N., 68°48'23.1"W. **Oak Island**, 1.5 miles west-southwestward of Eagle Island, is grassy and uninhabited. **Burnt Island**, just south of Oak Island, is wooded except for its northeast end, which is grass covered.

(246) There is a passage northward of North Haven Island that is used in winter when Fox Islands Thorofare is closed by ice. To go through this passage, pass about 300 yards southward of Eagle Island and steer for **Spoon Ledge**, 15 feet high with grass on top, about 0.5 mile northwest of Oak Island. On this course pass 400 yards northward of **Grass Ledge**, 15 feet high and grass covered, 0.9 mile east of Oak Island, to a position about 400 yards northward of Oak Island. Then pass midway between Oak Island and Spoon Ledge and steer west-southwesterly, clearing Webster Head by about 600 yards. The least charted depth in this passage is 25 feet.

(247) The preceding paragraphs give the simplest directions for Isle au Haut Bay and East Penobscot Bay by pointing out the difficulties and the dangers and especially, when necessary, the need for local knowledge. By close attention to the chart and following the aids, no difficulty should be experienced in navigating the area in daylight and in clear weather.

(248)

Fox Islands Thorofare to Pulpit Rock

(249) **Fox Islands Thorofare**, leading from East Penobscot Bay to West Penobscot Bay, between North Haven and Vinalhaven Islands, is one of the chain of inshore passages commencing at Bass Harbor and ending at Whitehead. Fox Islands Thorofare is about 7 miles long.

(250)

Prominent features

(251) **Widow Island**, inside the eastern entrance to Fox Islands Thorofare, is marked by a small cottage near its peak, on the southeast side.

(252) **Goose Rocks Light** (44°08'08"N., 68°49'50"W.), 51 feet above the water, is shown from a white conical tower on a black cylindrical foundation. A mariner radio activated sound signal at the light is initiated by keying the microphone five times consecutively on VHF-FM channel 83A. The white sector of the light, from 301° to 304°, marks the fairway for the eastern approach to the thorofare.

(253) **Browns Head Light** (44°06'42"N., 68°54'34"W.), 39 feet above the water, is shown from a white cylindrical tower connected with a dwelling. A mariner radio activated sound signal at the light is initiated by keying the microphone five times consecutively on VHF-FM channel 83A. A white sector in the light, from 050° to 061°, with a red sector on either side of it, marks the

fairway for the western approach to the thoroughfare. A fairway lighted bell buoy, about 2.4 miles southwestward of the light, marks the western entrance.

(254) **Sugar Loaves**, a group of prominent high rocks surrounded by a ledge, are 600 yards northwestward of Browns Head Light. **Fiddler Ledge Daybeacon**, a gray, square stone shaft with a pyramidal top marks **Fiddler Ledge**, which uncovers 5 feet 1.4 miles southwestward of Browns Head Light. It is the most conspicuous mark when approaching from westward.

(255) A large standpipe on the high ground just back of North Haven shows up prominently in approach from either direction.

(256)

Channels

(257) The controlling depth of 18 feet is in midchannel between Iron Point Ledge and Grindstone Ledge. The narrowest part of the channel is about 100 yards wide between **Iron Point Ledge**, marked by a daybeacon, and **Dobbin Rock**, marked by a buoy. Extreme caution should be exercised here as the currents are reported to be strong at times, especially during strong winds from the east or west. At low water, the thoroughfare is seldom used by vessels drawing over 14 feet.

(258)

Anchorages

(259) Good anchorage can be selected in the channel of the thorofare between the entrance of Seal Cove and the western end of the village of North Haven, in depths of 23 to 33 feet, soft bottom.

(260) Good anchorage for vessels of any draft, in depths of 32 to 42 feet, soft bottom, is in the western entrance of Fox Islands Thorofare, westward or northward of Sugar Loaves, and between Amesbury Point and **Crabtree Point Ledge**, 1.7 miles southwestward. A cable area extends across Fox Islands Thorofare in an east-west direction between Calderwood Rock on the north and Sugar Loaves on the south. Care should be taken to avoid anchoring in this area.

(261) Anchorage can be found in **Seal Cove**, a large arm extending 1.5 miles southward from Fox Islands Thorofare southeastward of and on the opposite side of the channel from the village of North Haven. Large areas in the cove have depths of 8 to 12 feet, bottom soft in places, but shoaling has been reported in the middle of the cove. The cable areas in the cove should be avoided when anchoring. Good anchorage in depths of 19 to 23 feet, soft bottom, is in the middle of **Southern Harbor**, which makes northeastward between the **Dumpling Islands** and **Amesbury Point**, near the western end of the thoroughfare. The water shoals gradually toward the head. A 12-foot shoal is in the center of the harbor entrance in about 44°07'52"N., 68°54'11"W.

(262) **Carver Cove**, south of Widow Island at the eastern end of Fox Island Thorofare, is a secure anchorage except during northeast winds, easy of access and convenient for vessels windbound in East Penobscot Bay or passing through the thoroughfare. The anchorage, in depths of 16 to 20 feet, good holding ground, is about 0.5 mile from the head of the cove, and **197°** from the cottage on Widow Island. When entering, the shores should be given a berth of about 300 yards. An unmarked 18-foot rocky patch is about in the middle of the eastern entrance.

(263) **Kent Cove**, in the north shore of the thoroughfare north of Widow Island, is a secure anchorage with depths of 15 to 24 feet, good holding ground. Anchorage in the northeast and northwest arms, in depths of 8 to 11 feet, is preferred. Goose Rocks Light is the prominent guide for entering either day or night, the entrance being westward of the light. **Kent Ledge**, the only outlying danger, covered 3 feet, is 500 yards from the northwest shore of the cove off the entrance.

(264) **Waterman Cove**, in the north shore of the thoroughfare west of Kent Cove, is a good anchorage for small vessels. The water shoals gradually from a depth of 18 feet at the entrance to 4 feet near the head, where a narrow channel leads into the **Cubby Hole**, a shallow cove. The better entrance to Waterman Cove is between the buoys off Fish Point Ledge and Waterman Ledge.

(265)

Dangers

(266) The principal dangers are marked by buoys or daybeacons that can be easily followed in the daytime with clear weather.

(267) On the north side of the eastern entrance to Fox Islands Thorofare are **Babbidge Island**,**Calderwood Island** and **Stimpsons Island**. North of these islands is unmarked Little Thorofare, which can be used by small craft with local knowledge. Ledges extend for over 0.4 mile south and southeast of these islands. A buoy, 0.6 mile southeast of Babbidge Island, is on the north side of the east entrance to Fox Islands Thorofare.

(268) Of the several reefs south of these islands, the most important are **Black Ledge**, **Sunken Black Ledge** and **Channel Rock**. A buoy is just southwestward of Sunken Black Ledge. Channel Rock is marked by a daybeacon.

(269) On the south side of the western approach to Fox Islands Thorofare are **Dogfish Ledges**, **Seal Ledge** and **Inner Bay Ledges**. Seal Leadge is marked by a buoy on the north side and Inner Bay Ledges, forming the westernmost danger in the western approach, is marked by several buoys. The main entrance channel is north of these ledges and is well marked. The channel southeast, between these ledges, is also well buoyed for the guidance of those vessels going to Hurricane Sound and the southern part of Vinalhaven Island.

(270) **Drunkard Ledge**, 0.5 mile westward of Fiddler Ledge Daybeacon, uncovers 7 feet and is marked by a daybeacon on the eastern side. Broken ground, which should be avoided, extends 0.2 mile southward of the line joining the daybeacons. A gong buoy is on the southern extremity of the broken ground.

(271) **Fish Point Ledge**, marked at its southeast end by a buoy, is 400 to 600 yards southeastward of **Fish Point**,

on the eastern side of Waterman Cove. Foul ground is between the point and the ledge. **Waterman Ledge**, covered 4 feet and marked by a buoy, is in the mouth of Waterman Cove 500 yards from the western shore.

(272) **Post Office Ledge**, covered 6 feet, and **Lobster Ledge**, covered 2 feet, are two marked ledges off the town of North Haven.

(273)

Current

(274) The tidal currents in Fox Islands Thorofare are usually not strong. They meet at Iron Point in the middle of the thoroughfare; the flood sets in from both ends and the ebb sets out. However, during periods of strong winds from the eastward or westward, it is reported that strong currents with eddies are apt to be encountered in this vicinity.

(275) The thoroughfare is sometimes closed by ice in winter.

(276)

Pilotage, Fox Islands Thorofare

(277) Pilotage for these waters is discussed in this chapter; see Pilotage, Penobscot Bay and River, indexed as such.

(278) **North Haven** is an important summer resort and yacht center on the north shore of Fox Islands Thorofare. Small craft can anchor on the south side of the channel between Young Point and Hopkins Point where depths allow and on the north side of the channel, taking care to leave a clear channel to the town wharf and ferry slip. The town wharf has a depth of about 12 feet, and the other wharves less.

(279) Water is available at the yacht club float, about 100 yards northwest of the ferry slip, with depths of 4 to 8 feet reported alongside. A boatyard, close eastward of the ferry slip, has marine railways up to 20 tons and a 15-ton mobile hoist that can handle craft up to 45 feet long for hull and engine repairs; gasoline, diesel fuel, ice, marine supplies and winter storage are available. Depths of 6 to 7 feet are reported alongside the boatyard pier. The yard builds craft up to 45 feet long.

(280) State automobile, mail and passenger ferry service to Rockland is maintained the year round. Taxi service, food and lodging are available in the village.

(281) The north shore of Vinalhaven Island, across the thoroughfare from North Haven, has numerous summer residences with private landing floats.

(282) **Perry Creek** is a long narrow arm making westward on the west shore of Seal Cove. The creek is reported to be an excellent anchorage for small craft; avoid the cable area at the entrance. An overhead power cable crosses the northern branch of the creek as shown on the chart.

(283) **Crockett Cove** is just eastward of Crockett Point, the southeastern point at the western entrance to Fox Islands Thorofare. The cove is about 1 mile long and 200 yards wide near the entrance, is obstructed by ledges and is suitable only for small craft with local knowledge.

(284) **Dogfish Island**, 0.4 mile south of Crockett Point and northwestward of Leadbetter Narrows, has a private stone wharf at its eastern end.

(285) **Leadbetter Narrows** is a narrow passage between Vinalhaven Island on the north and **Leadbetter Island**, 0.3 mile southeast of Dogfish Island, on the south. When passing through the narrows, favor the shore of Vinalhaven Island. A buoy marks a rock awash on the south side of the narrows. Continuing south along the eastern side of Leadbetter Island, a passage leads into the northern end of Hurricane Sound.

(286) A small stone wharf is on Leadbetter Island at the narrows. There is also an inactive quarry and wharf on the shore of Vinalhaven Island eastward from the narrows. The cove on Vinalhaven Island just northeast of the east entrance to Leadbetter Narrows is reported to be a good protected anchorage with mud bottom in 5 to 15 feet of water. Leadbetter Narrows should not be attempted by strangers except in launches or small craft.

(287) **Bartlett Harbor**, a small cove with deep water and good anchorage sheltered from all but westerly and northerly winds, is on the western shore of North Haven Island about 2 miles above **Stand-in Point**, the southwestern point of North Haven Island. A rock covered 9 feet is in the middle of the entrance; deep water is close-to around the rock.

(288) **Pulpit Harbor**, on the northwest side of North Haven Island, is 4 miles northeastward of Stand-in Point and 2.5 miles southwestward of **Webster Head**, the high and partly wooded head at the north end of North Haven Island. The entrance has a clear width of over 100 yards, and the harbor is a secure anchorage for small vessels of about 11-foot draft or less.

(289) **Pulpit Rock**, 10 feet high and pointed, is near the end of the reef extending 250 yards northeastward from the western point at the entrance. The rock is a good mark. To enter, give the north side of Pulpit Rock and the eastern shore just northward of the entrance a berth of over 100 yards, and enter in midchannel eastward of Pulpit Rock. Keep in midchannel and anchor in its broad part in depths of 18 to 33 feet.

(290) Another good all-weather anchorage for small craft is reported to be in the southwesterly prong, just inside the entrance, in 18 to 27 feet. More sheltered anchorage in 8 to 10 feet is toward the northeast end of the harbor, where there is a public float landing with 3 feet alongside. There is a telephone at the landing. Gasoline, provisions and most supplies can be obtained by calling North Haven from the landing.

(291)

Lawrys Narrows to Vinalhaven

(292) **Lawrys (Laireys) Narrows**, between Leadbetter Island on the north and Lawrys and Cedar Islands on the south, is a part of the route between Carvers Harbor and Rockland. The principal dangers are buoyed, except for a rock covered 11 feet, reported in 1979 to be 100

yards southwest of the southern end of Leadbetter Island. A dangerous underwater rock ridge is at midchannel in about 44°04'04"N., 68°54'00"W. **Crotch Island**,**Crane Island** and **Spectacle Island** lie southward of Lawrys and Cedar Islands. A safe anchorage is reported to be had in 6 to 8 feet between Crotch Island and the northwestern end of Crane Island.

(293) **The Basin** is a large irregular bight in the west side of Vinalhaven Island, about 2 miles southeast of Crockett Cove. **Barton Island** is in the middle of the entrance, leaving a narrow, crooked, foul and shallow channel north of it. The depth in the basin varies from 10 to 111 feet.

(294) **Hurricane Sound** is bounded on the east by Vinalhaven and Greens Islands and on the west by **Hurricane Island**, 0.7 mile west of Greens Island, and **White Islands**, a group of islands about 1.5 miles northwest of Greens Island and farther north by **Crane Island** and **Cedar Island**. The sound has deep water. Several passages lead into the sound, but there are no good anchorages.

(295) It is reported that there is a good black pebble beach in the cove on the south side of Hurricane Island and that the old stone quarry pier on the northeast side of the island affords a good landing place in good weather. **Outward Bound School**, a summer sailing and survival school, is on the island.

(296) Along this part of West Penobscot Bay, numerous rocks and reefs extend over 2 miles offshore from Vinalhaven Island and the bottom of the bay is irregular with many spots of 10 to 18 feet for about 2 miles farther offshore. The better passes among the islands are buoyed. Great care must be used to avoid the numerous reefs.

(297) **The Reach** is a narrow, much obstructed channel leading northwestward from the entrance of Carvers Harbor, between Greens Island and Vinalhaven Island. The passage is marked and used by vessels bound between Carvers Harbor and Rockland. The channel at its narrowest is only 100 feet wide between the northeastern edge of the ledge marked by Wreck Point Light 3 and a rock covered 10 feet, about 200 feet northeastward of the light. Great care is required in passing this point.

(298) **Old Harbor** is a small cove at the northern end of The Reach and on the opposite side of the channel from the northern end of Greens Island. Caution is necessary in using this harbor because of the many old fish stakes, and a ledge of drying rocks extending southward of the island in the entrance of the harbor.

(299) **Carvers Harbor** is a secure haven in all weather for small vessels on the southwest side of Vinalhaven Island.

(300)

Prominent features

(301) **Heron Neck Light** (44°01'30"N., 68°51'44"W.), 92 feet above the water, is shown from a white conical tower on the southern extremity of **Greens Island**, on the eastern side of the entrance to Hurricane Sound. The light has a white sector from 030° to 063° that marks the fairway of the approach to Carvers Harbor from the southwest. A mariner radio activated sound signal at the light is initiated by keying the microphone five times consecutively on VHF-FM channel 83A.

(302) **Carvers Harbor Entrance Light 2** (44°02'03"N., 68°50'37"W.), 19 feet above the water and shown from an iron spindle with a red triangular daymark on the west end of Green Ledge, marks the entrance to the harbor. Ice seldom closes the harbor. A standpipe on the hill north of the harbor is very prominent.

(303)

Channels

(304) There are four channels in the approaches to Carvers Harbor. The entrance from southwestward is between Heron Neck Ledge and James and Willies Ledge; from the northwestward through The Reach; from the eastward through the channel between Vinalhaven Island and the islands and ledges south of it; and from the southward west of Colt Ledge and between Arey Ledges and The Breakers. The controlling depth in the entrance channel is 19 feet between Potato Island and **Dodge Point** on the north side of the entrance to the harbor.

(305) **Indian Creek**, just eastward of Carvers Harbor, has an entrance from the sea and also a connecting passage to Carvers Harbor. A fixed highway bridge crossing the passage has a clearance of 8 feet. The passage is not safe for strangers. **Lane Island** forms the west side of Indian Creek. The island is grassy with two prominent white houses visible from the southward. **Potato Island** and **Bar Island** are two small islets on the northwest side of Lane Island at the south side of the entrance to the harbor.

(306)

Anchorages

(307) The best anchorage for small craft is reported to be on the east and southeast side of Carvers Harbor; the western side is principally used by commercial craft and fishermen. In 2012, the harbor had depths of about 8 to 15 feet in the center, and the access channel leading to a basin off the town landing had a depth of 6 feet, with 1 to 6 feet available in the basin. An obstruction, covered 3 feet, was reported at 44°02’33”N., 68°50’13”W.; caution is advised.

(308)

Dangers

(309) **James and Willies Ledge**, 5 feet high with rocks awash at the south end, is on the north side of the southwestern approach at its junction with Hurricane Sound. It is part of the extensive ledge area extending southward from Hurricane Island. A buoy is south of the area.

(310) **Heron Neck Ledge**, 7 feet high, is on the southern edge of the southwestern approach and is unmarked.

(311) **Folly Ledge** is a bare unmarked ledge on the west side of the channel at the junction of the eastern, southern and southwestern approaches. **Green Ledge** is on the east side of the channel in the inner approach to Carvers Harbor. It is marked by Carvers Harbor Entrance Light 2.

(312) **Point Ledge**, covered 4 feet and marked by a daybeacon, is 0.7 mile east of Folly Ledge.

(313) The dangers in the eastern approach have been described with Isle au Haut Bay and East Penobscot Bay.

(314) Strangers should bear in mind that many unmarked dangers will, of necessity, have to be passed close-to and should exercise extreme caution by giving strict attention to the chart and following the aids.

(315)

Routes

(316) The preceding paragraphs give the simplest directions by pointing out the difficulties and the dangers and especially, when necessary, the need for local knowledge. Vessels of 12–foot draft or less should experience no difficulty, in daytime and in clear weather, in approaching and entering.

(317) Pilotage for these waters is discussed in this chapter under Pilotage, Penobscot Bay.

(318)

Harbor regulations

(319) There is a **harbormaster** who assigns the moorings in the anchorage. A **speed limit** of 5 miles per hour is enforced within the harbor.

(320) **Sand Cove**, making northward from Carvers Harbor, is foul. There are several wharves and a boatyard at the head at which vessels lie aground at low water.

(321) **Vinalhaven** is a town at the head of Carvers Harbor. There are churches, a library, bank, movies, inns, lodging houses, medical and nursing services, restaurants and picnic areas. The depths at the ferry wharf and float landings vary from 6 to 10 feet. Diesel fuel, gasoline, ice, water, provisions and some marine supplies are available at a landing east of the ferry wharf. There are four boatyards on Vinalhaven Island, one on Indian Creek, one in Sand Cove and two in Carvers Harbor. Craft up to 50 feet in length can be hauled out for hull or engine repairs or dry open or covered winter storage. Emergency radio repairs can be made.

(322) The State maintains mail, automobile and freight service with Rockland the year round. The island has good roads.

(323)

Egg Rock to Resolution Island

(324) North of North Haven Island are numerous islands and reefs extending to Head of the Cape. Most of these have been described previously. The most westerly of the islands and reefs is **Egg Rock**, which is small and grass covered and 2 miles north of Pulpit Harbor. **Egg Rock Ledge**, 0.3 mile south-southwest of Egg Rock, is covered 2 feet. A buoy is northeast of the ledge.

(325) **Compass Island Ledge**, 1.4 miles northeastward of Egg Rock, is covered 8 feet; a buoy is off the ledge. **Compass Island**, 42 feet high, is 0.8 mile northward of Compass Island Ledge. A ledge with a rock at its end, which uncovers 10 feet, extends about 300 yards northeastward from Compass Island; ledges also extend up to 0.3 mile eastward of the island. **Grass Ledge**, a group of rocks 15 feet high, and rocks awash and covered, is between Compass Island, **Scrag Island** and **Little Spruce Head Island**, which is westward of **Great Spruce Head Island**.

(326) **Horsehead Island**, 74 feet high, is about 0.6 mile northward of Little Spruce Head Island. **Colt Head Island**, and still another group of **Barred Islands**, are 0.6 mile and 1.1 miles, respectively, north-northeast of Horsehead Island. **Beach Island**, 81 feet high, is 0.9 mile northward of Great Spruce Head Island. Submerged rocks are reported in the passage between Beach Island and Barred Islands. **Resolution Island**, the northwesterly island of this group of islands between North Haven Island and Cape Rosier, is 93 feet high and wooded.

(327) The passage through these islands, just north of North Haven Island, has been described previously with East Penobscot Bay.

(328)

Islesboro Island

(329) **Islesboro Island** and the adjacent islands and shoals are about 15 miles long and separate East and West Penobscot Bays near their heads. Islesboro Island is nearly divided in the middle. The island is an important summer resort and is frequented by many pleasure boats in summer. Dark Harbor, Islesboro, North Islesboro and Pripet are villages on the island. A state automobile and passenger ferry is operated between Lincolnville, on the mainland, and Grindle Point.

(330)

McIntosh Ledge to Ensign Islands

(331) A chain of islands and rocks, through which are several channels, extends for 5 miles southward from Islesboro Island. **McIntosh Ledge**, the most southerly of the dangers and about 0.7 mile southeastward of Robinson Rock, is awash at low water. A buoy is southeast of the ledge.

(332) **Robinson Rock**, 22 feet high and grassy, is the most southerly visible danger; several smaller bare rocks are around it. Ledges extend for 0.6 mile north-northeast and south-southwest of the rock. There is a bell buoy off the southern end of these ledges.

(333) **Mark Island**, the most southerly wooded island, is high, rounded and prominent. A reef extends southward from the island.

(334) **East Goose Rock**, 0.5 mile northward of Mark Island, is 15 feet high and grassy. **Saddle Island**, 0.7 mile east–northeastward of Mark Island, is high and thickly wooded.

(335) **Lasell Island**, 1.2 miles northeast of Mark Island, is high and wooded except at its north end. **Goose Island** and**Mouse Island**, eastward of Saddle and Lasell Islands, are rocky islets with grass on top. Several bare and covered rocks are between Goose and Mouse Islands. A buoy is 300 yards north of the ledge that uncovers 5 feet

northward of Mouse Island, and a buoy is eastward of the bare rock east of Goose Island.

(336) **Lime Island**, 0.2 mile northeastward of Lasell Island, is low and generally wooded. A rocky spit connects Lime and Lasell Islands at low water. A bare rock is 0.2 mile northward of Lime Island. **Job Island**, 0.7 mile northeastward of Lime Island, is 104 feet high and thickly wooded. A dangerous rock, reported covered at high water, is about 0.2 mile east-southeast of the south point of Job Island. The southerly of the **Ensign Islands**, 0.7 mile west of Job Island, is wooded, and the northerly is wooded in the center with a house on the west side. A landing is on the south end. A lighted bell buoy is about 0.3 mile south of the southerly island.

(337) The channel between Mark, Lasell and Lime Islands on the west and Saddle, Goose and Mouse Islands on the east is used by some vessels bound from Rockland or westward to Eggemoggin Reach or points in the northern part of East Penobscot Bay. The channel is unmarked, and local knowledge is required.

(338)

Dark Harbor to Lobster Rock

(339) **Dark Harbor** is a village, with many summer homes, on the southern part of Islesboro Island. There are grocery and hardware stores, a snack bar and a gas station. **Dark Harbor Cove**, on the eastern side of the island, is crossed by a dam and footbridge just inside the entrance and is seldom used. Small craft visiting the resort tie up at the yacht club or other private floats in Gilkey Harbor.

(340) **Gilkey Harbor**, on the western side of the southern part of Islesboro Island, is between the island and **Seven Hundred Acre Island**,**Warren Island**and **Spruce Island**. The harbor is a secure anchorage with good holding ground and is frequented by many yachts in summer. There are a number of private float landings for small craft but no commercial wharves. The harbor frequently is closed by ice in winter.

(341) Warren Island is a state park. A 200-foot pier with seasonal dockage and 4 feet reported alongside is on the east side of the island.

(342) The Tarratine Yacht Club is on the east side of **Ames Cove**, near Dark Harbor; the clubhouse has a float landing with a depth of 4 feet alongside. Some supplies can be obtained in the village, and water is available at the float. A boatyard, at the east end of Ames Cove, has a machine shop and a 40-foot marine railway. Gasoline, covered and uncovered storage and hull and engine repairs are available. Ames Cove is inaccessible at low water.

(343) **Cradle Cove** is a shallow indentation on the northeast side of Seven Hundred Acre Island. A boatyard, near the eastern entrance point of the cove, has a machine shop and a marine railway capable of hauling out craft to 60 feet long or 25 tons for hull and engine repairs and dry covered or open winter storage. Electric and electronic repairs can generally be made, and the yard has a small crane and pile driver. Gasoline and diesel fuel are available at the 300-foot pier and float landing, which has 6 feet reported alongside. Water, ice, marine supplies and electricity are available, and the yard maintains guest moorings.

(344)

Channels

(345) The main entrance to Gilkey Harbor is from southwestward between Job Island and Ensign Islands; the controlling depth is about 27 feet in midchannel between Minot Island and Seven Hundred Acre Island. Unmarked rocks of less depth are near the sides. The channel is partially buoyed and easily entered. The entrance from the northward is marked by **Grindel Point Light** (44°16'53"N., 68°56'35"W.), 39 feet above the water, shown from a white square tower with a green square daymark, close to an abandoned lighthouse on the north side of the entrance. The state ferry slip, and a municipal float landing with 12 feet reported alongside, are close southeastward of the light; a municipal small-craft launching ramp is adjacent eastward of the shore end of the ferry auto ramp. A lighted bell buoy is west of the entrance, and the channel into Gilkey Harbor is partially marked by buoys.

(346) Provisions and some marine supplies can also be obtained from Islesboro.

(347) Small craft can also enter Gilkey Harbor through narrow, crooked **Bracketts Channel**, westward of the south end of Islesboro Island and eastward of Job Island and **Minot Island**. The unmarked channel is said to have a controlling depth of about 6 feet. The best water favors the east side.

(348) No difficulty should be experienced in entering Gilkey Harbor from the southward or northwestward with close attention to the chart and bearing in mind a number of unmarked 14- to 18-foot spots in the northern half of the harbor. Wooded **Thrumcap (Thrumcap Island)**, near the middle of the harbor, has a reef extending westward from it that is marked by a buoy. **Lobster Rock**, awash 0.6 mile east-southeast of Grindle Point, is unmarked.

(349)

Gooseberry Point to Parker Cove

(350) **Gooseberry Point**, 0.6 mile northward of Grindel Point, is low and flat, with a clump of trees at its outer end. **Crow Cove**, 2.4 miles northeast of Grindel Point, is an anchorage for small craft only.

(351) **Seal Harbor**, on the western side of Islesboro Island about 3 miles northward of Grindel Point, offers good anchorage sheltered from all but southwest winds. This harbor, easy of access, is used by vessels bound up or down the bay as an anchorage for the night. Vessels of any size can anchor with ample swinging room about 0.5 mile eastward of Flat Island, in depths of 54 to 60 feet. Anchorage can also be had in depths of 48 to 57 feet in the middle of the harbor, keeping the southern and eastern shores distant about 500 yards. The northern side of the

harbor is foul. The wreck of a schooner, covered 12 feet, is about 0.2 mile from the head of the harbor.

(352) The entrance to Seal Harbor from the southward is deep and clear. The entrance from westward is 400 yards wide, with depths of 22 to 29 feet between Seal Island and a shelving ledge that extends 500 yards northward from Flat Island.

(353) The approach from the northward east of Seal and Ram Islands has a controlling midchannel depth of about 15 feet but is unmarked and should not be used except with local knowledge because of the many unmarked shoal spots close to the channel edges.

(354) **Flat Island** is a private bird sanctuary on the western side of the southern entrance to Seal Harbor. The island is grassy, with a few trees and scattered brush. **Seal Island**, 0.6 mile north of Flat Island, is wooded and has a brown house with a black roof on its western side. A private pier and float landing are on the east side of the island. **Ram Island**, 0.3 mile northward of Seal Island, is wooded. The ledge extending 0.3 mile northward from the island has three rocks awash.

(355) **Islesboro Harbor** is an open bight in the east side of Islesboro Island, 2.7 miles westward and on the opposite side of East Penobscot Bay from the head of Cape Rosier. The harbor affords good shelter in westerly winds and has depths of from 31 to 42 feet, rocky bottom. **Hewes Ledge**, off the southern point at the entrance and awash at low water, is marked by two buoys. Vessels can pass on either side of the ledge, being guided by the buoys. Foul ground extending over 0.2 mile from the western shore will be avoided by keeping the knoll northward of the harbor open from the north point of the harbor.

(356) The village of **Islesboro** is on the south side of the harbor. The village has a general store where marine supplies are available.

(357) **Sabbathday Harbor** is a small cove in the eastern side of Islesboro Island, about 2 miles northward of **Hewes Point**, the high point on the south side at the entrance to Islesboro Harbor. **Ryder Cove**, the northern part of the harbor, dries at low water. Sabbathday Harbor is open southward and provides anchorage for small vessels in depths of 6 to 20 feet. A dangerous sunken rock is about 150 yards southward of the western entrance point, and stonecribs are reported on the east side of the harbor, about 300 yards above the entrance. The village of **North Islesboro**, on the west side of the harbor, has a general store, filling station and restaurant.

(358) **Sprague Ledge**, 0.5 mile northward of Ram Island and about 0.5 mile off the west shore of Islesboro Island, is covered 2 feet. **Barley Ledge**, 0.3 mile northeastward of Sprague Ledge, is awash at low water.

(359) **Marshall Point**, near the north end of Islesboro Island, is marked by prominent yellowish bluffs. A fairway lighted bell buoy is 0.8 mile west of the point.

(360) **Turtle Head Cove**, a broad bight in the north end of Islesboro Island, is sheltered from southerly and easterly winds and has good anchorage in depths of 18 to 37 feet, soft bottom. The anchorage has a clear width of about 700 yards and is in the eastern part of the cove. The eastern shore must be given a berth of 250 yards and the south end of the cove 500 yards.

(361) In the western half of the cove, a shoal awash in one spot at low water and covered 9 feet near its north edge extends 600 yards from shore. The north end of Turtle Head bearing anything eastward of **070°** clears the shoal.

(362) **Turtle Head**, the north end of Islesboro Island, is a prominent wooded head joined to the island by a long, narrow, wooded neck. The village of **Pripet** is southward of Turtle Head. A crib wharf at Pripet is reported in ruins.

(363) **Parker Cove**, on the east side of Islesboro Island 2.2 miles south of Turtle Head, is a shallow cove used only as an anchorage by small local craft. **Islesboro Ledge**, covered 8 feet, is southeastward of the entrance; a buoy is off the east side of the ledge.

(364)

Two Bush Channel to Sheep Island Shoals

(365) Two Bush Channel and Muscle Ridge Channel are entrances to West Penobscot Bay from westward, the former leading southward and the latter northward of an extensive group of islands and shoals.

(366) **Two Bush Channel** is broad and deep, and the principal dangers are buoyed. This channel is used in preference to Muscle Ridge Channel by large vessels and tows and is generally used at night by all except small local vessels.

(367) **Two Bush Island**, the southeastern island of the group between the two channels, is marked by **Two Bush Island Light** (43°57’51”N., 69°04’26”W.), 65 feet above the water, shown from a 42-foot white tower on the north side of Two Bush Channel. A mariner radio activated sound signal at the light is initiated by keying the microphone five times consecutively on VHF-FM channel 83A. The light is the principal guide to the channel.

(368) **Halibut Rock**, awash at low water and marked by a buoy, is 1.6 miles northeastward of Two Bush Island Light. **False Halibut Ledge**, covered 6 feet and unmarked, is 0.3 mile northeastward. **Northeast Pond Ledge**, 0.6 mile northeastward of **Andrews Island**, is awash at low water, and **Sunken Pond Ledge**, covered 6 feet, is 500 yards southeastward; neither is marked.

(369) The larger islands between Two Bush Channel and Muscle Ridge Channel are mostly wooded, and of little importance. The small islands are bare and grassy, and there are many bare and covered rocks. Privately owned **Dix Island**, 2.7 miles north of Two Bush Island, is wooded. **High Island**, 0.2 mile northeastward of Dix Island, has an abandoned quarry on it. **Birch Island**, just east of Dix Island and south of High Island, is about 20 feet high. **Fisherman Island**, about 5 miles north-northeast of Two Bush Island, is 43 feet high and grassy. **Marblehead Island**, 0.3 mile south of Fisherman Island, is 46 feet high and bare. **Grindstone Ledge**, covered 2 feet, is 0.3 mile northwestward of Fisherman Island.

(370) **Muscle Ridge Channel** is used in daylight and clear weather because it is sheltered and affords anchorage in case of bad weather. The channel is deep but narrow in places, especially between Sheep Island and Hendrickson Point, where the channel is only 85 yards wide but has a depth of 38 feet in midchannel. From the entrance at Whitehead Island, the channel extends in a northeasterly direction about 6 miles to Sheep Island, passing between numerous rocks and ledges. Shoal depths of 13 to 22 feet are close to the channel, but these dangers are well marked, and in daylight and clear weather no difficulty should be experienced.

(371) **Whitehead Island** is on the west side of the southern entrance to Muscle Ridge Channel. **Whitehead Light** (43°58'43"N., 69°07'27"W.), 75 feet above the water, is shown from a gray granite tower on the east end of Whitehead Island. A mariner radio activated sound signal at the light is initiated by keying the microphone five times consecutively on VHF-FM channel 83A.

(372) There is a small wharf in the cove on the northeast side of the island 300 yards northward from the light. The narrow channel between Whitehead Island and **Norton Island**, 500 yards westward, is blocked by a reef which uncovers about 6 feet.

(373) The following information is given to identify the dangers close to the sailing line through Muscle Ridge Channel. **South Breaker**, 0.4 mile southward and on the opposite side of the channel from Whitehead Light, is awash at low water and marked on the southwest end by a bell buoy. **Yellow Ledge**, on the opposite side of the channel from Whitehead Light, is awash at high water and marked by a daybeacon. **Yellow Ridge Islet**, close eastward, is 15 feet high and bare.

(374) **Lower Gangway Ledge**, 0.4 mile north of Yellow Ledge, is covered 6 feet; a buoy is west of the ledge. **Hurricane Ledge**, 1 mile northeastward of Yellow Ledge, is awash at low water and marked on its northwest side by a buoy. **Garden Island**, about 2 miles northeastward of Whitehead Island, is 15 feet high and bare except for a little grass on top. **Garden Island Ledge**, 0.3 mile east-northeastward of Garden Island, uncovers about 5 feet; a daybeacon is on the ledge. **Sunken Ledge**, covered 4 feet, about 0.4 mile southward, is marked by a buoy off its south end.

(375) **Wiggins Rock**, covered 9 feet, is about 750 yards north-northeastward of Garden Island Ledge, and there are two rock patches covered 10 and 12 feet, respectively, about 500 yards northwestward of Wiggins Rock. **High Clam Ledge**, 0.8 mile northeast of Hurricane Ledge, is bare and grassy at its south end and awash at low water at its north end. **Channel Rock**, 0.4 mile north of High Clam Ledge, uncovers 10 feet and is unmarked.

(376) **Otter Island**, 0.5 mile north of Dix Island, is 31 feet high and wooded; a daybeacon is on the northwest end of the island. **Otter Island Ledge**, 0.3 mile northwestward on the opposite side of the channel from Otter Island, uncovers about 5 feet; a daybeacon is on the ledge. **Upper Gangway Ledge**, 0.6 mile north-northeast of Otter Island, is covered 5 feet and marked by a buoy. **Inner Grindstone Ledge**, awash at low water, is 0.3 mile east of Upper Gangway Ledge; a buoy is north of the ledge.

(377) **Seal Harbor**, an anchorage formerly much used by coasters, is on the western side of Muscle Ridge Channel between Whitehead Island and **Sprucehead Island**, 0.8 mile northward. The outer part of the harbor has depths of 15 to 39 feet with soft bottom, and thence depths decrease in the inner part of the harbor. The harbor is easily entered in the daytime; the principal dangers in the entrance are buoyed. The chart is the guide. A dangerous ledge extends about 100 yards east and 200 yards northeast of **Slins Island** (43°59'54"N., 69°08'00"W.) on the west side of the entrance to the north part of the harbor. A rock with a prominent black bullseye painted in the center is on the east end of Sprucehead Island.

(378) A causeway and highway bridge connecting **Elwell Point** with Sprucehead Island has a fixed span with a clearance of 7 feet. A town ramp is at the north end on the west side of the bridge. A lobster wharf on the northeast side of Elwell Point is dry at low water.

(379) **Spruce Head** is a village on the north side of Seal Harbor. There are several private wharves in the harbor. A service wharf and float landing are on the east side of the cove in the south side of Sprucehead Island; depths of 7 feet are reported alongside the float. Gasoline, diesel fuel, water, ice, provisions and some marine supplies are available.

(380) **Seal Island** is about 500 yards northward of the light on Whitehead Island. **Seal Island Ledge**, which uncovers for almost 500 yards north of Seal Island, is the principal danger on the south side of the entrance. A buoy marks the east side of the ledge. **Long Ledge**, 0.2 to 0.5 mile north of Whitehead Island, shows in two places at high water.

(381) **Burnt Island**, connected to Sprucehead Island by a private bridge, has a summer home. Small craft use the passage between these two islands. The highway bridge has a fixed span with a clearance of 12 feet. **Burnt Island Ledge**, 150 yards south of Burnt Island and marked by a buoy, is covered 2 feet.

(382) **Dix Island Harbor** is an anchorage off the southeast side of Muscle Ridge Channel between Andrews, Birch and Dix Islands. The harbor is entered from southwestward through a narrow and crooked channel leading between the ledges north of **Hewett Island**, 1.5 miles north of Two Bush Island. The channel and harbor are unsafe for strangers.

(383) On the west side of **The Neck**, an island just westward of Andrews Island, is a wharf with 2 feet alongside. There is a stone wharf with good water reported alongside for small craft on the west side of **High Island**. Care must be exercised in approaching it to clear a rock covered 3 feet about 200 yards westward of the wharf. A good all-weather anchorage for small craft is reported between High Island, **Dix Island** and **Little Green Island**.

(384) **Weskeag River** empties into the western side of Muscle Ridge Channel at the head of the bight westward

of **Ash Island**, a 54-foot-high wooded island, about 3 miles northeast of Sprucehead Island. The channel between Ash Island and Ash Point is shoal and foul. **Lark Ledges**, dangerous unmarked ledges with several rocks, sunken and awash; **Grace Rock**, covered 2 feet and also unmarked; and a number of other unmarked patches and ledges obstruct the approach to Weskeag River.

(385) **Spaulding Island**, wooded, is about in the middle of the river entrance between **Otter Point** and **Thorndike Point**. There is a stone town wharf with 2 feet at the head on the west side just inside the entrance abreast of Spaulding Island. There are no facilities. The river has a narrow, crooked unmarked channel south of Spaulding Island that is not safe for strangers.

(386) The village of **South Thomaston** is at the head of navigation, 2 miles above its mouth; the greatest draft taken to the village is 9 feet at high water. Vessels seldom enter. Provisions and some marine supplies can be obtained at the village general store. There is a town wharf, which bares at low water, a small-craft launching ramp and parking at the head of the harbor near the store. A private marine railway is 100 yards west of the town wharf. There are good roads to the interior. Local knowledge is advisable for entering and anchoring.

(387) **Fisherman Island Passage** leads from Muscle Ridge Channel to Penobscot Bay, between Fisherman Island and Sheep Island. Several dangers are in this passage, but the principal ones are marked by buoys and can be easily avoided in the daytime in clear weather. A strong southwesterly flow is reported on the ebb; make allowance for this current when approaching from the east.

(388) **Sheep Island Shoals**, which extend over 0.3 mile south of Sheep Island and uncover in places, are on the north side of the channel; the shoals are marked on the south end by a buoy. **Emery Ledge**, covered 5 feet, and **Northwest Ledge**, covered 4 feet, are on the south side; both are marked by buoys.

(389)

Owls Head Bay to Camden Harbor

(390) **Owls Head Bay** is between Sheep and Monroe Islands, about 6.5 miles north-northeastward of Two Bush Island on the east and the mainland on the west. The bay is a continuation of Muscle Ridge Channel northward of Fisherman Island Passage. The channel through Owls Head Bay is very narrow on the western side of Sheep Island between Sheep Island Bar and **Hendrickson Point**, where the width is only 85 yards between the 5-fathom curves, and the depth 38 feet. It is marked by two buoys. Vessels caught by fog can anchor in the middle of the bay abreast Monroe Island in depths of 41 to 68 feet.

(391) Small vessels can anchor in the entrance to **Owls Head Harbor**, on the west side of the bay, between **Dodge Point** and the bare ledge 0.2 mile southwestward, in depths of 9 to 25 feet. Anchorage in depths of about 6 feet is available inside, in about the middle of the harbor. A lobster pound and wharf and a fish and lobster wharf with 7 feet reported alongside are on the western shore. Gasoline, diesel fuel and fishing supplies are available at the southerly wharf; the town float landing is at the end of this wharf. Ice, provisions and some supplies can be obtained at a general store in the village of **Owls Head**. There is a good firm beach where small boats may be launched from trailers at any stage of tide. There are good roads to the interior.

(392) **Owls Head** is a prominent headland at the northeast entrance to Owls Head Bay and on the south side of the entrance to Rockland Harbor. **Owls Head Light** (44°05'32"N., 69°02'38"W.), 100 feet above the water, is shown from a white tower on the headland; a sound signal is at the light and is activated by keying the microphone five times consecutively on VHF-FM channel 83A. The light is obscured from 324° to 354° by Monroe Island.

(393) **Emery Island** is a small islet 0.8 mile west of the southerly end of, and on the opposite side of the channel from, Sheep Island. A rock 350 yards eastward of Emery Island is awash at low water; a daybeacon marks the rock. **Dodge Point Ledge,** eastward of Dodge Point, uncovers about 5 feet and is marked by a daybeacon. **Owls Head Ledge**, southeastward of Owls Head and awash at low water, is marked by a buoy.

(394) In West Penobscot Bay, eastward of Monroe Island, the tidal current has velocities up to 0.6 knot at strength. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov*. for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(395) **Rockland Harbor**, one of the most important harbors in Penobscot Bay, is on the west shore of West Penobscot Bay between Owls Head on the south and **Jameson Point**, 2.1 miles northwestward, on the north. The harbor offers anchorage for large vessels but is somewhat exposed to easterly winds. Northeasterly winds raise a heavy sea in the southwestern part of the harbor, but shelter may be found behind the breakwater.

(396) The breakwater extends 0.7 mile southward from Jameson Point. **Rockland Harbor Breakwater Light** (44°06'15"N., 69°04'39"W.), 39 feet above the water, is shown from a red brick tower attached to a building on a granite pier at the outer end of the breakwater. A mariner radio activated sound signal at the light is initiated by keying the microphone five times consecutively on VHF-FM channel 83A.

(397) **Rockland**, a city on the western shore of the harbor, has some trade in fish and petroleum products. Mail, freight, automobile and passenger ferries leave the Rockland Port Terminal in **Lermond Cove** several times daily for North Haven and Vinalhaven.

(398) There are banks, hotels, motels, restaurants, a general hospital, library, shops, churches and schools in Rockland. The city has many small metal, textile and woodworking industries and seafood processing and

fruit packing plants. Several seasonal coastal cruising schooners operate out of Rockland, as well as from Rockport and Camden.

(399)

Prominent features

(400) The most prominent objects in approaching Rockland Harbor are the radio tower of station WVOM, located on Benner Hill about 2 miles westward of the harbor, the radio tower (44°06.3'N., 69°06.4'W.) and the signal mast at **Rockland Coast Guard Station** on **Crockett Point** (44°06.3'N., 69°06.3'W.). The light on Owls Head and the light at the end of the breakwater are also conspicuous.

(401)

Channels

(402) A federal project provides for an approach channel and three branch channels, each with a turning basin. In 2008, the controlling depth in the entrance channel was 17.6 feet, thence 12.1 feet in the southwestern channel and 14 feet in the basin; 10 feet in the channel leading north to Crockett Point; 13 feet in the northern channel with depths of 14 feet available in the northern basin, thence 11 feet in the western channel and turning basin. All channels are buoyed.

(403)

Anchorages

(404) Two general anchorages, one in the northern part of the harbor and the other in the southern part, and a small-craft anchorage in the western part are available in Rockland Harbor. (See **33 CFR 110.1**,**110.4**, and **110.132**, chapter 2, for limits and regulations.)

(405)

No-Discharge Zone

(406) The State of Maine, with the approval of the Environmental Protection Agency, has established a No-Discharge Zone (NDZ) in the coastal waters of Camden, Rockport, Rockland and portions of Owls Head.

(407) Within the NDZ, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZ, discharge of sewage is regulated by **40 CFR 140** (see chapter 2).

(408)

Dangers

(409) Standing westward in the harbor the water shoals gradually toward the wharves.

(410) Several rocks and ledges are in the harbor. The visible ones are **Shag Rock**, on a cluster of bare rocks, 0.3 mile northwestward of Owls Head and marked by a daybeacon; **Lowell Ledge**, a cluster of rocks awash at low water on the south shore of the harbor near **Battery Point**; and **Seal Ledge**, which uncovers about 5 feet, in the southwest end of the bay and marked by a daybeacon. A buoy marks **Spears Rock**, covered 5 feet, about 300 yards northeastward of Lowell Ledge.

(411)

Weather, Rockland and vicinity

(412) July is the warmest month in Rockland with an average high of 77 °F and an average minimum of 56 °F. January is the coolest month with an average high of 33 °F and an average minimum of 13 °F. The highest temperature on record for Rockland is 99 °F recorded in August 1944 and again in August 1948 and the lowest temperature on record is -18 °F recorded in January 1971. Every month has seen temperatures at or below 40 °F and every month except July and August has recorded temperatures below freezing.

(413) The average annual precipitation for Rockland is 47.4 inches (1204 mm) with an annual maximum during early winter and a minimum during mid-summer. Precipitation falls on about 340 days each year. The wettest month is November with 5.6 inches (142 mm) and the driest, August, averages only 2.7 inches (69 mm). Snow falls on about 89 days each year and averages about 60 inches (1524 mm) each year. January through March each average greater than a foot (305 mm). One-foot (305 mm) snowfalls in a 24-hour period have occurred in each month December through March. Snow has fallen in every month except June through September. Fog is present on average 63 days each year with a minimum occurrence during mid-winter and a maximum during July and August.

(414)

Routes

(415) Approaching Rockland Harbor, Rockland Breakwater Light may be steered for on any safe course, using the chart as a guide. Enter the harbor southward of the breakwater light, giving it a berth of 100 yards or more.

(416) **Pilotage** for Rockland Harbor is discussed in this chapter under Pilotage, Penobscot Bay.

(417)

Towage

(418) Two small motor launches, used as tugs, are available at Rockland. Tugs up to 1,800 hp are available at all times at Belfast; see Towage, Penobscot Bay, this chapter for details.

(419)

Quarantine, customs, immigration, and agricultural quarantine

(420) (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(421) Rockland is a **customs port of entry.**

(422) **Quarantine** is enforced in accordance with the regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.)

(423)

Coast Guard

(424) **Rockland Coast Guard Station** is on the east side of Crockett Point.

(425) **Harbor regulations** can be obtained from the **harbormaster,** whose office is at the municipal landing,

0.5 mile northwest of **Atlantic Point**. The harbormaster monitors VHF-FM channel 16.

(426)

Wharves

(427) The Rockland Port District Terminal Wharf, on the west side of Lermond Cove, is the ferry terminal. The wharf is 280 feet long and has a ferry slip, a lift bridge, and ramp; depths of 11 feet are reported alongside. A private light on the wharf operates only upon the approach of a ferry. In addition, there are several private and public wharves and piers on the west side of the harbor that are used by vessels and barges engaged in coastwise shipping. Depths at these facilities are reported to range from about 6 to 14 feet.

(428)

Supplies

(429) Gasoline, diesel fuel, water, ice and marine supplies are available at several of the wharves. Provisions and most supplies are available in town.

(430)

Repairs

(431) A commercial fishing corporation has a repair yard with marine railways and cranes up to 50 tons on Atlantic Point. The largest railway can handle vessels up to 225 feet in length, 40-foot beam, 1,200-ton displacement and 16-foot draft. Hull, engine and electronic repairs can be made. The nearest port where vessels can be drydocked is Boston. Several boatyards, catering to small craft, are also available in the harbor. Hull, engine and electronic repairs can be made. Marine railways up to 75 feet and lifts up to 35 tons are available.

(432)

Small-craft facilities

(433) There are three marinas in the harbor. A municipal dock with 6 feet reported alongside its float landing is on the west side of the harbor, about 0.3 mile northwestward of Atlantic Point. Water and electricity are available at the float, and guest moorings are maintained. Sewage pump-out is available.

(434)

Communications

(435) The port has railway freight, bus, truck and taxi service and microwave telephone communication with North Haven, Vinalhaven and the inhabited islands off the entrance to Penobscot Bay.

(436) Charter planes are available at two nearby airports for trips to several towns in the bay. The town is a terminus of a Class II railroad that connects with the main line at Brunswick.

(437) **Clam Cove**, on the west side of West Penobscot Bay, about 2 miles northward of Rockland Harbor, is shoal at the head and is not a good anchorage. The ruins of a stone wharf are on the north side of **Brewster Point**, 1.2 miles north of Jameson Point. **Glen Cove** is a village near the head of the cove.

(438) **Brewster Point Ledge**, extending over 700 yards southeastward from Brewster Point on the south side of the entrance, is awash at high water; a buoy is southeastward of the ledge.

(439) **Ram Island** is a grass-covered rock 400 yards northeastward of Brewster Point. The shoal that extends northeastward from the island is marked by a buoy.

(440) **Rockport Harbor**, on the west side of West Penobscot Bay about 4 miles northward of Rockland Harbor, is a good anchorage for vessels of any size, sheltered from all but southerly winds, and is easy of access. The harbor is about 0.7 mile wide at the entrance between Indian Island and the western shore and gradually narrows to the head.

(441) **Rockport** is a town at the head of the harbor at the entrance to Goose River.

(442)

Prominent features

(443) **Lowell Rock Light 2** (44°09'46"N., 69°03'37"W.), 25 feet above the water, is shown from a spindle with a triangular red daymark on the south end of **Lowell Rock**. A fairway bell buoy 0.4 mile southeast of the light marks the entrance to the harbor. A clock tower at the head of the harbor is conspicuous. **Beauchamp Point**. the eastern point of Rockport Harbor north of Indian Island, is prominent.

(444)

Channels

(445) The entrance is deep and clear with the exception of Porterfield Ledge in the middle of the entrance. The depths in the channel range from over 50 feet in the entrance to 8.3 feet at the head. Passage is sometimes made by local small craft at high water across the ledge between Indian Island and Beauchamp Point.

(446)

Anchorages

(447) Vessels can anchor anywhere between the entrance and a point 1 mile southward of the head, in depths of 42 to 63 feet, soft bottom. Small vessels and motorboats can find anchorage nearer the head.

(448)

No-Discharge Zone

(449) The State of Maine, with the approval of the Environmental Protection Agency, has established a No-Discharge Zone (NDZ) in the coastal waters of Camden, Rockport, Rockland and portions of Owls Head.

(450) Within the NDZ, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZ, discharge of sewage is regulated by **40 CFR 140** (see chapter 2).

(451)

Dangers

(452) **Porterfield Ledge**, in the middle of the entrance to Rockport Harbor, uncovers several feet at low water; a daybeacon is on the ledge. **Indian Island**, on the eastern side of the entrance, is grassy and marked at its south end by an abandoned lighthouse. An unmarked fish weir is on

the west side of the harbor, about 0.3 mile southward of the entrance to Goose River.

(453) **Seal Ledge**, uncovered at low water, on the east side of the harbor about 0.7 mile northward of the light, is marked by a daybeacon.

(454)

Routes

(455) Vessels can enter Rockport Harbor on either side of Porterfield Ledge Daybeacon, giving the daybeacon a berth of at least 150 yards. When in the harbor stand northward in midharbor until 0.3 mile from the head, then slightly favor the eastern side.

(456)

Supplies

(457) Gasoline, diesel fuel, ice, provisions and some marine supplies can be obtained in Rockport.

(458)

Small-craft facilities

(459) A public float landing, maintained by the town of Rockport, is at the east side of the entrance to Goose River, at the head of the harbor. Depths of 3 feet are reported alongside the float; water is available. The Rockport Yacht Club, close westward, has a float landing with 3 feet reported alongside. There are a number of private wharves in the harbor.

(460) A boatyard, close eastward of the public landing, has a 12-ton mobile hoist and facilities for open or covered winter storage. Depths of 8 to 10 feet are reported alongside the floats. Hull and engine repairs can be made, and gasoline, diesel fuel, water, ice, electricity, marine supplies and transient moorings are available. The yard builds boats up to 45 feet long.

(461) A municipal marina park is on the west side of the harbor, about 120 yards west of the boatyard. Transient berths, with depths of 3 to 8 feet reported alongside the floats, are available.

(462) The town **harbormaster** can usually be found at the park. A **speed limit** of 5 mph is enforced in the harbor.

(463)

Communications

(464) Local taxi service is available as is through bus service.

(465) **Deadman Point** is about 0.4 mile northeast of Indian Island. **Hog Cove**, on the north side of Deadman Point, has two private piers with float landings. **Hog Cove Ledge** extends about 0.4 mile above Deadman Point and forms the eastern side of the cove. **Goose Rock**is a bare ledge about 0.2 mile north of Hog Cove Ledge. A 12-foot spot 0.4 mile northeast of Deadman Point is marked on the east side by a buoy.

(466) **The Graves**, about 1 mile offshore, midway between the entrance to Rockport and Camden Harbors, is a ledge showing bare rocky heads at high water and a large area that uncovers at low water. A lighted gong buoy is just eastward of The Graves.

(467) **Camden Harbor**, on the west side of West Penobscot Bay about 6 miles north of Rockland Harbor, is the approach to the town of Camden. The harbor is frequented by many yachts and small craft.

(468) **Camden**, the town on the inner harbor, is important as a yachting center and as the homeport of several seasonal cruising schooners. The nearest railway freight point is Rockland. There is a public park and picnic area. Swimming, boat rental, parking, country clubs, banks, churches, hospital, restaurant, and markets and shops of all kinds are available in the town.

(469)

Prominent features

(470) The most conspicuous feature seen in entering Camden Harbor is **Mount Battie** (44°13'22"N., 69°04'10"W.), 800 feet high. A small stone memorial tower on the summit shows as a long ridge from off the harbor.

(471) **Curtis Island**, on the southern side of the entrance, is prominent. **Curtis Island Light** (44°12'05"N., 69°02'56"W.), 52 feet above the water, is shown from a white tower on the southeast end of the island.

(472) **Northeast Point**, on the northeast side of the entrance, is marked off its south side by **Northeast Point Light 2** (44°12'31"N., 69°02'47"W.), 20 feet above the water, shown from a white skeleton tower with a red triangular daymark.

(473)

Channels

(474) The main channel into the harbor is from southward and is deep and clear; it is marked by a bell buoy at the entrance and by buoys and Curtis Island Light. The inner harbor, westward of Eaton Point, has depths of about 7.7 to 10 feet in the middle and 4.4 to 6.1 feet along the east and west shores with lesser depths along the north shore. **Northeast Passage**, with a depth of about 19 feet, is a narrow channel leading into Camden Harbor between Northeast Point and Inner Ledges. The deeper water favors the light off Northeast Point. A fairway bell buoy is 0.3 mile northeastward of the entrance to Northeast Passage. This channel is used by local vessels but should be used with great caution by strangers. The passage between Curtis Island and **Dillingham Point** is shoal and foul. Rocks awash are about 110 yards southwest of the light and about 150 yards northwestward of the island.

(475)

Anchorages

(476) The outer harbor is easy of access and affords good anchorage in depths of 13 to 33 feet, soft bottom. The depths in the outer harbor shoal gradually northward to a depth of 12 feet about 500 yards from the head of **Sherman Cove**, in the northern part of Camden Harbor. Above the 12-foot curve the cove is shoal. Two special anchorage areas are in the harbor. (See **33 CFR 110.1** and **110.4**, chapter 2, for limits and regulations.)

(477) The greater part of the inner harbor west of **Eaton Point** is occupied by small pleasure and fishing craft. There are numerous private and some public moorings.

(478)

No-Discharge Zone

(479) The State of Maine, with the approval of the Environmental Protection Agency, has established a No-Discharge Zone (NDZ) in the coastal waters of Camden, Rockport, Rockland and portions of Owls Head.

(480) Within the NDZ, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZ, discharge of sewage is regulated by **40 CFR 140** (see chapter 2).

(481)

Dangers

(482) **Northeast Ledge**, consisting of **Inner Ledges** and **Outer Ledges**, is southward of Northeast Point and constricts the main entrance to Camden Harbor to a width of about 400 yards. The higher parts of Inner and Outer Ledges uncover about 5 feet. The ledges are marked by buoys and a daybeacon.

(483) **Dillingham Ledge**, having a buoy off its east side, is 0.5 mile offshore and 1.3 miles northeast of Camden Harbor.

(484) A shoal extends 80 yards from the north shore off Eaton Point and the shipyard at the entrance of the inner harbor. Ruins of a pier extend about 150 yards southeast from Eaton Point.

(485) **Ice** sometimes forms in the harbor from January to March but is not dangerous for vessels in the outer harbor. Westerly winds clear the harbor of ice if it is broken up.

(486)

Routes

(487) Entering Camden Harbor by the main channel, vessels can steer for Curtis Island Light on any safe course, taking care to avoid The Graves, until close to the entrance bell buoy, thence select anchorage in the outer harbor, as shown on the chart. If going to the inner harbor, pass 100 yards northeastward of Camden Harbor Buoy 7 and steer for the entrance of the inner harbor, clearing the shoal on the north side off Eaton Point at the entrance to the inner harbor, and haul northward in midharbor.

(488) To enter by Northeast Passage, from the fairway bell buoy, steer for the north end of Curtis Island until close to the buoy at the northerly end of Outer Ledges. Pass northward of this buoy and steer westward between Northeast Point Light 2 and Inner Ledges Daybeacon 3, favoring the light.

(489)

Pilotage, Camden Harbor

(490) Pilotage for Camden Harbor is discussed in this chapter. See Pilotage, Penobscot Bay and River, indexed as such.

(491) The town **harbormaster** supervises the moorings and enforces the local regulations; he can be contacted at the town wharf.

(492)

Wharves

(493) The town wharf, on the west side of the inner harbor near the head, has two float landings with depths of 8 feet reported alongside. The wharf is used by several excursion schooners which operate along the Maine coast from Camden during the summer. The Camden Yacht Club, about 150 yards southward of the town wharf, has several float landings with depths of 7 feet reported alongside. Water is available at the floats, and guest moorings are maintained by the club in the outer harbor.

(494) A boatyard, close eastward of Eaton Point, has a 125-ton marine railway that can handle craft up to 100 feet and an 18-ton mobile hoist for hull and engine repairs or dry open and covered winter storage. Gasoline, diesel fuel, transient berths with water and electricity and marine supplies are available.

(495)

Small-craft facilities

(496) Most of the facilities are in the inner harbor at Camden.

(497)

Communications

(498) Bus, both local and coastal, and taxi service are available, and a number of coastal cruising schooners operate from the harbor on weekly schedules in the summer. The nearest air transportation is at Rockland.

(499) **Mount Megunticook** (44°14.5'N., 69°04.1'W.), 1,385 feet high, is 2 miles northward from Camden. The mountain shows as a flat-topped peak with a steep shoulder on its southern side.

(500)

Ducktrap Harbor to Kenduskeag Stream

(501) **Ducktrap Harbor** is a broad open bight in the west shore of West Penobscot Bay, 5 miles northeastward of Camden Harbor. Good anchorage, sheltered from northerly and westerly winds, is 600 yards from the north shore of the harbor, bottom soft in places. **Haddock Ledge**, the only outlying danger, is a rock covered 3 feet about 0.6 mile from the western shore and the same distance southwestward of Spruce Head, the northeast point of the harbor. A buoy is on the southwest side of the ledge. With this exception, danger will be avoided by giving the shore of the harbor a berth of about 500 yards. A bell buoy marks the southern approach to the harbor.

(502) **Lincolnville** is a village at the southwest end of Ducktrap Harbor. A state automobile and passenger ferry operates between Lincolnville and Grindel Point, Islesboro Island. The ferry terminals are marked by private lights. A public float landing, with 3 feet reported alongside, is on the north side of the ferry pier, and a small-craft launching ramp is close northward. The village **harbormaster** can be contacted through the town office. The ferry pier and shed are prominent from offshore. A church with a white spire, 0.5 mile northward of Lincolnville, is conspicuous from the bay. The viaduct of the main coastal highway where it crosses the valley at the head of the harbor is conspicuous.

(503) **Great Spruce Head**, 2 miles northward of Spruce Head, is bold.

(504) **Saturday Cove** is a small cove on the west side of West Penobscot Bay, 9 miles northeastward of Camden harbor. The village of **Northport** is on the south side of the cove. Private float landings are usually maintained near the entrance.

(505) **Temple Heights** is a small summer settlement on the western shore of the northern end of West Penobscot Bay, about 0.5 mile north of Saturday Cove.

(506) **Bayside** is a summer settlement on the southwest side of Penobscot Bay, 2.1 miles north-northwestward of Temple Heights. A red elevated water tank on the hill back of the village is prominent. The wharf has a float landing maintained by the village of Northport. There is reported to be a depth of 5 feet alongside the floats. Water is available at the wharf, and gasoline, oil, provisions and some marine supplies are obtainable in the village. A small-craft launching ramp is just northward of the wharf.

(507) **Belfast Bay** and **Passagassawakeag River** empty into the head of Penobscot Bay from northwestward and form the approach to the town of **Belfast** and village of City Point, about 2 miles above Belfast.

(508) The Passagassawakeag River is a Marine Protected Area (MPA).

(509) Two **vessel-to-vesseloil transfer anchorage areas** in Penobscot Bay have been designated by the State of Maine, Department of Environmental Protection. One area is 1.8 miles northwest of Islesboro Island within a circle having a diameter of 2 miles, centered in 44°24'15"N., 68°55'25"W. The other area is 2.1 miles northeast of **Turtle Head**, the northernmost point on Islesboro Island, within a circle having a diameter of 1 mile centered in 44°25'00"N., 68°50'43"W.

(510)

Channels

(511) The main channel in Belfast Bay is wide and clear between Steels Ledge and the western shore with depths of 50 feet at the entrance gradually decreasing to 14 feet off the mouth of Goose River. The channel is partially marked by buoys to a point about 0.3 mile below the first bridge, a fixed footbridge. The channel above this point requires local knowledge and is little used except by small craft. It is narrow, crooked and unmarked and bares in places at low water.

(512)

Anchorages

(513) The bay affords good anchorage, exposed to southeasterly winds, and is easy of access. Good anchorage can be had off the entrance to the river westward of Steels Ledge, in depths of 19 to 28 feet; also in the river south of Goose River in midchannel, or by favoring the western shore, in depths of 11 to 16 feet, soft bottom. Above this point, shoals extend halfway across the harbor from the northeast side and for a short distance below the bridge extend two-thirds of the distance across. Small vessels can anchor about 75 yards off the upper wharves of the city in depths of 10 to 22 feet.

(514)

Dangers

(515) **Steels Ledge**, on the north side of Belfast Bay, is an extensive ledge with a least depth of 1 foot, marked by a lighted bell buoy. An abandoned lighthouse also marks the ledge. The passage between the ledge and the north shore should not be used because of shoals to the eastward.

(516)

Bridges

(517) The former highway swing bridge crossing the Passagassawakeag River has been converted to a footbridge that has a bascule span with a horizontal clearance of 25 feet and a vertical clearance of 9 feet. U.S. Route 1 highway bridge about 250 yards northwestward has a fixed span with a clearance of 68 feet. About 0.7 mile upstream, an overhead power cable crossing the river has a clearance of 30 feet.

(518)

Ice

(519) Ice obstructs navigation throughout the river and bay in severe winters. The bay has been frozen over to Islesboro Island.

(520)

Routes

(521) Vessels entering Belfast Bay can shape the course to pass anywhere between the bell buoy southward of Steels Ledge and the western shore, then head north-northwestward in midchannel.

(522)

Pilotage, Belfast

(523) Pilotage for Belfast is discussed in this chapter under Pilotage, Penobscot Bay (indexed as such).

(524)

Towage

(525) Three modern tugs up to 1,800 hp are available at Belfast. They berth at the Marshall Wharf. See Towage, Penobscot Bay, this chapter for details.

(526)

Quarantine, customs, immigration, and agricultural quarantine

(527) (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(528) Belfast is a **customs port of entry.**

(529) **Quarantine** is enforced in accordance with the regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.)

(530)

Wharves

(531) Wharves at Belfast include the Stinson Canning Wharf, Maineport Towboats (Marshall Wharf) and the wharf at Belfast Boatyard, adjacent to the towing company wharf. The canning company wharf, used by fishing vessels, is on the south side of the river just below the footbridge; depths of 8 to 10 feet are reported along its 50-foot northeasterly face. The towage company

wharf, about 0.2 mile southeastward, is used primarily for mooring tugs; depths of 15 feet are reported alongside.

(532) The town wharf and float landing, on the southeast side of the towage company wharf, has a depth of 8 feet reported alongside. A breakwater east of the city landing offers protection. The remainder of the piers and wharves at Belfast are in ruins.

(533)

Supplies

(534) Gasoline can be obtained at the town wharf. Provisions and some marine supplies are available in town.

(535)

Repairs

(536) The boatyard has a 40-foot marine railway and can make complete hull repairs. There are machine shops in town.

(537)

Communications

(538) The main coastal highway, U.S. Route 1, passes through the town. Taxi and coastal bus services are available.

(539) **Searsport Harbor**, at the head of Penobscot Bay about 4 miles east of Belfast, is a broad bight open to the southward. The town of **Searsport** is at the head of the harbor. The commercial development of the harbor is at **Mack Point**, 1 mile east of Searsport. There is considerable traffic in oil, potatoes, general cargo and dry bulk commodities.

(540)

Prominent features

(541) Three steam cranes, oil tanks on Mack Point and a spire in the village are conspicuous.

(542)

Channels

(543) Natural depths in the main channel of Penobscot Bay provide depths of over 40 feet to within 1 mile of the facilities at Mack Point, thence depths of about 35 feet to a dredged access channel which leads to a turning basin off the facilities, both with a project depth of 35 feet. (See Local Notice to Mariners and the latest editions of charts for controlling depths.)

(544)

Anchorages

(545) Good anchorage, used by all classes of vessels, may be had in depths of 18 to 32 feet, soft bottom, sheltered from northerly winds, within a mile southward of Mack Point.

(546)

Dangers

(547) **Long Cove Ledge**, covered 1 foot near its southern end, is 400 to 800 yards south of the west end of Mack Point. A lighted bell buoy is off the southeast side of the ledge, and buoys are on the east and west sides of it. A rocky ledge, covered 32 feet and marked with a lighted buoy, is about 2.5 miles south-southwestward of the railroad pier at Mack Point, in the northern section of the Oil Transfer Area.

(548) Ledges make off 0.3 mile from the western shore of the southern half of Sears Island; one of these ledges, **Sears Island Ledge**, is bare at low water. A bell buoy is about 0.5 mile southwest of Sears Island and at the south end of the ledge. Two other buoys mark the limit of the ledge westward of the island.

(549)

Routes

(550) The approach to Mack Point piers is between Sears Island and the lighted bell buoy off the southeast side of Long Cove Ledge.

(551)

Pilotage, Searsport

(552) Pilotage for Searsport is discussed in this chapter under Pilotage, Penobscot Bay (indexed as such).

(553)

Towage

(554) Large vessels require tug assistance in docking at Searsport. Three modern tugs up to 1,800 hp are available at Belfast. See Towage, Penobscot Bay, this chapter for details.

(555)

Quarantine, customs, immigration and agricultural quarantine

(556) (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(557) Belfast is the **customs port of entry** for Searsport.

(558) **Quarantine** is enforced in accordance with the regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.)

(559)

Wharves

(560) There are two deep-water piers on Mack Point; a bulk liquid dock and a dry cargo facility capable of handling bulk, break-bulk and containerized cargoes. Both docks are operated by Sprague Energy, which can be contacted at 207-548-2531 for further details. These facilities have highway connections and are served by a Class II railroad.

(561) The Maine Port Authority Dry Cargo Pier, the larger of the two, is on the southeast end of Mack Point. The pier is a pile-supported concrete structure measuring 801 in length by 100 in width. There are portable hoppers on the dock for the discharge of bulk cargo. The pier has 544 feet of berthing space on the east and west sides. The east side is dredged to 42 feet at MLW and can accommodate vessels of up to 750 feet in length and 106 feet beam, up to 80,000 DWT. The west side of the pier is dredged to 32 feet. Warehouses in excess of 30,000 square feet are located on land adjacent to the dock.

(562) The Sprague Energy Liquid Bulk Pier, about 250 feet westward of the dry cargo pier, is 660 feet long and provides 625 feet of berthing space along its easterly side; depths of 36 feet are alongside. Liquid bulk cargoes of

chemicals and petroleum products are handled. Vessels can receive bunker C fuel oil at the pier.

(563) At Searsport, west of the entrance to **Mill Brook**, there is a town landing with 4 feet reported alongside the float and pier. A small-craft launching ramp is adjacent to the landing.

(564)

Supplies

(565) Gasoline, diesel fuel and diesel oil are available by tank truck; bunker C fuel oil is available at the C. H. Sprague & Son Co., Searsport pier. Provisions, water, ice and some marine supplies can be obtained in Searsport.

(566)

Repairs

(567) Aboard ship repairs can be handled by a firm in Searsport. There are no marine railways, and the nearest drydocks for large vessels are at Boston.

(568)

Communications

(569) Searsport is the ocean terminus for a Class II railroad. Local taxi service is available and bus service at the main coastal highway, U.S. Route 1, about 0.7 mile from the terminals.

(570) **Long Cove** is eastward of Searsport Harbor between the northwestern shore of Sears Island □ □ a C II railroad.

(613)

Structures across Penobscot River

Name•Description•Type	Location	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
Penobscot Narrows Bridge (fxed)	44°33'36"N., 68°48'07"W.	428	143	
US Route 1 Bridge (fxed)	44°34'09"N., 68°47'18"W.	160	19	Bridge crosses Eastern Channel
Overhead power cable	44°34'08"N., 68°47'19"W.		42	Cable crosses Eastern Channel
Overhead power cable	44°34'18"N., 68°48'31"W.		142	
Overhead power cable	44°40'41"N., 68°48'52"W.		131	
I-395 Bridge (fxed)	44°47'03"N., 68°46'30"W.	270	74	
US Route 1A Bridge (fxed)	44°47'45"N., 68°46'08"W.	152	22	
State Street Bridge (fxed)	44°47'57"N., 68°45'49"W.	195	23	

depths of 19 to 58 feet, soft bottom, and shelter can be found there in all winds.

(590) Another small-craft anchorage is in what is locally known as **Hospital Cove** between **Nautilus Island, Holbrook Island** and **Ram Island** of the northern extremity of Cape Rosier. This cove can be entered from the westward through the channel, marked by a buoy, between **Nautilus Rock** and Holbrook Island. In 2021, off the northeast end of Holbrook Island, an obstruction covered 4 feet was reported in about 44°22'15"N., 68°48'25"W.; caution is advised. The southwesterly channel, between Ram and Holbrook Islands and Cape Rosier, is unmarked and difficult to navigate. Anchorage, secure in all weather, can be had in good holding ground in 13 to 37 feet in the westerly half of the cove. The holding ground in the channel southeastward of Castine is poor.

(591)

Dangers

(592) **Henry Point** is on the east side of the approach to Bartlett Cove. Dangers to be avoided in the cove are the middle-ground ledge, awash and marked on the east side by a buoy, about 0.5 mile south of Henry Point, and a rock covered 3 feet 300 yards west of **Sheep Island**, near the southern part of the cove. In addition, there are numerous unmarked bare and submerged rocks along the edges of the cove and caution should be exercised.

(593) **Otter Rock Shoal**, awash at low water and marked by a buoy at its south end, extends 0.2 mile off the north shore at the entrance near Dice Head.

(594) **Hosmer Ledge**, a drying ledge on the south side of the channel, extends 0.2 mile off the north end of Cape Rosier about 0.9 mile east of Dice Head Light. A daybeacon is on the ledge.

(595) **Middle Ground**, which uncovers 2 feet and is marked on its west end by a buoy, is off the east side of the channel 1.4 miles above the entrance.

(596) **Trott Ledge**, which uncovers 5 feet and is marked by a buoy, is on the west side of the channel about 1.8 miles above the entrance.

(597) A rocky ledge, covered 4 feet and marked by a buoy, is on the west side of the channel 2.6 miles above the entrance, 0.6 mile south of Esther and Emanuel Islands.

(598) Numerous other rocks and ledges, mostly unmarked, are on both sides of the channel above the **Narrows**, a constricted part of the channel about 4.5 miles above the entrance.

(599)

Currents

(600) Currents of almost 5 knots have been observed at Jones Point, about 4 miles above the entrance.

(601)

Ice

(602) The river is usually free from ice at Castine and for some distance above, but in very severe winters the river is entirely closed.

(603)

Routes

(604) Craft entering Castine Harbor will find the eastern shore northward and southward of the entrance is bold and can be followed at a distance of 0.3 mile. Pass close to the fairway bell buoy on either side and, keeping a reasonable distance offshore and south of the buoy marking Otter Rock Shoal, steer into the harbor on a midchannel course. By close attention to the chart, anchorage can be found in Bartlett Cove about 200 to 500 yards south of Henry Point, or for small craft south of Sheep Island near the head of the cove, taking care to avoid the charted dangerous rock previously mentioned.

(605) There are no commercial facilities in Castine Harbor. The Maine Maritime Academy, at the western end of the Castine waterfront, maintains an excellent wharf with 26 feet alongside at which the large training vessel moors.

(606)

Small-craft facilities

(607) The town wharf and float landing, just eastward of the Academy wharf, has 12 feet reported alongside. A boatyard is 150 yards northeast of the town wharf. A 20-ton marine railway at the yard can handle craft up to 45 feet long for hull or engine repairs or dry open and covered winter storage; gasoline, diesel fuel, water, ice and some marine supplies are available.

(608) Castine Yacht Club, about 100 yards eastward of the boatyard, has a depth of 8 feet reported alongside its float landing. The stone foundation of an old stone pier is

between the shore and the northeast end of the yacht club float; care should be taken to avoid it in coming alongside.

(609) **West Brooksville** is a village on the south side of the river 1.5 miles above Castine Harbor, and **North Castine** is a village on the west side 2 miles above Castine.

(610) **North Brooksville** is a village on the southern branch of Bagaduce River, about 6 miles above Castine. At high water, small boats sometimes go to the bridge crossing the river at the village, but the channel is unmarked and unsafe for strangers.

(611) **Penobscot** is a village on **Northern Bay** at the head of navigation on the north branch of the Bagaduce River, 6.5 miles above Castine. The approach to the village is bare at low water.

(612) **Penobscot River**, emptying into the head of Penobscot Bay, forms the approach to the towns of Bucksport and Winterport and the cities of Bangor and Brewer; the last two are at the head of navigation about 24 miles above Fort Point Light at the entrance. The deepest draft ordinarily trading to Bangor is about 16 feet. The channel is marked by buoys, daybeacons and a lighted buoy to a point about 1.5 miles below South Brewer.

(614) The most difficult sections for vessels are off Lawrence and Luce Coves where it is difficult to mark the best water and off Frankfort Flats where large vessels experience difficulty with the sharp turns.

(615) The channel in Penobscot River is crooked and narrow in places, and frequent changes occur. Strangers should not attempt to carry drafts greater than 10 feet to Bangor at low water. With a deeper draft a pilot or towboat should be used; 14 to 18 feet is carried to Bangor and Brewer at high water, and deeper drafts occasionally to the oil berth at South Brewer. The safest time is on a rising tide. Navigation of the river at night is extremely dangerous due to lack of lighted navigational aids. After unusually high tides many logs, dangerous to small craft, are in the river. At times of maximum ebb currents, buoys are occasionally pulled under. The paragraphs describing the river give the simplest directions by pointing out the difficulties and the dangers and especially the need for local knowledge. The chart and the aids must be carefully followed.

(616)

Dangers

(617) **Fort Point Ledge**, 0.3 to 0.6 mile southward of Fort Point Light, uncovers about 5 feet. A light marks the ledge.

(618) **Odom Ledge**, a drying ledge in the middle of the main channel 2.7 miles above Fort Point, is marked by a daybeacon on a square stone base on the highest point of the ledge and by a lighted buoy off the southwest side.

(619) Vessels drawing 30 feet or more should exercise caution when proceeding between Fort Point and Odom Ledge.

(620)

Anchorages

(621) The usual anchorage for coastal vessels waiting at the river entrance for passage upstream is northward of Fort Point on either side of the channel. Anchorage in the river is not advised because vessels tend to drag anchor on strong ebb tides. Larger vessels usually anchor south of the point or off Searsport Harbor and start up the river about 1 to 3 hours after low water.

(622) **Morse Cove**, on the east side at the entrance to the river, is sometimes used by pleasure boats for temporary anchorage in depths of 7 to 18 feet, soft bottom.

(623) **Fort Point Cove**, on the west side of the river northward of Fort Point, is used frequently as an anchorage. The depths are from 3 to 21 feet, shoaling gradually westward. A state park is on the south side of the cove. Seasonal dockage with 12 feet reported alongside is available at a 200-foot pier at the park. It is reported that this mooring becomes very rough in a north wind.

(624)

Currents

(625) Currents of 3 knots are not unusual from Odom Ledge to Orrington, and during spring runoff, currents reported to exceed 5 knots may be encountered. Because of these currents, larger vessels should use caution navigating the river. Passage up the river is more advisable during flood current.

(626) **Ice** impedes but usually does not prevent navigation above Winterport for nearly 5 months each year, beginning about the end of November. During extreme winters the river is closed to the mouth. The most difficult place below Winterport is abreast Fort Knox, where ice jams occur. If vessels can pass this point they usually can go to Winterport. The river is kept open by an ice breaker, which prevents much of the damage that might otherwise be caused by ice and freshets.

(627) Freshets occur in the river during March and April; at times they are dangerous to vessels.

(628)

Pilotage, Penobscot River

(629) Pilotage for Penobscot River is discussed in this chapter under Pilotage, Penobscot Bay and River, indexed as such.

(630)

Towage

(631) Large vessels bound upriver usually take a tug to assist in making the turns and in docking. Three tugs up to 1,800 hp are available at Belfast. (See Towage, Penobscot Bay, this chapter for details.)

(632) **Fort Point**, on the west side at the entrance to Penobscot River, is partly wooded. **Fort Point Light** (44°28'02"N., 68°48'42"W.), 88 feet above the water, is shown from a white house. A mariner radio activated sound signal at the light is initiated by keying the microphone five times on VHF-FM channel 83A. There are several houses farther back on the north side of the

point. A buoy marks the extremity of the shoal extending eastward from the point.

(633) About 2.5 miles above Fort Point Light, Penobscot River is divided by **Verona Island** into two channels. The principal channel is on the west side of the island, and the **Eastern Channel (Eastern River)** is on the east side. The channels unite north of Verona Island at the town of Bucksport.

(634) **Orland River**, flowing into Eastern Channel from a northeasterly direction, is a shallow stream navigable for small boats and fishermen at high water to the dam at the village of **Orland**, about 2.2 miles above the mouth. The channel is crooked, unmarked and bare at low water a little below Orland.

(635) **Sandy Point** is a village on the west bank of Penobscot River about 1 mile above **Sandy Point**, the northern entrance to Fort Point Cove. A large two-story mansion, with three small houses behind, and locally known as **Herseys Retreat**, is on the bluff at Sandy Point and reported to be conspicuous when coming up the river from seaward.

(636) The ruins of a large pier extend east from the sand beach just north of Sandy Point. In 1984, submerged pilings extended northward about 40 yards from the easternmost visible pilings.

(637) **Verona Park** is a small summer settlement on the west side of Verona Island about 1 mile below Bucksport.

(638) **Bucksport** is a town on the east bank of the river 6.5 miles above Fort Point; paper manufacture and oil distribution are the principal industries. There are markets, banks, restaurants, lodging and other conveniences in town.

(639)

Quarantine, customs, immigration and agricultural quarantine

(640) (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(641) Bucksport is a **customs station.**

(642) **Quarantine** is enforced in accordance with the regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.)

(643)

Wharves

(644) There is one deep-draft facility at Bucksport in general use. Most of the other wharves are in ruins with only broken pilings and stone foundations remaining.

(645) The paper mill wharf (44°34.4'N., 68°48.1'W.), on the southeast side of the point just northwest of the town, has about 400 feet of berthing space with depths of 5 to 24 feet reported alongside. It is used principally to load small vessels and barges with paper. An occasional pulp shipment is discharged from oceangoing vessels.

(646) A petroleum handling berth, consisting of nine concrete pile clusters supporting a handling platform, extends from a former railway wharf and provides a 700-foot berth with depths of 35 feet alongside and can accommodate vessels up to 65,000 DWT. A one-foot under-keel clearance is required when alongside Penobscot Bay and River oil facilities.

(647)

Small-craft facilities

(648) A public float landing, with depths of 10 feet reported alongside, is about 200 feet east of the petroleum berth. Gasoline, diesel fuel, provisions, marine supplies and services are available nearby.

(649) **Fort Knox**, a restored fort of imposing appearances across the river from Bucksport, is a state park. **Prospect Ferry** is just above Fort Knox. Nothing remains of the old ferry landing except the stone foundation that uncovers at low water.

(650) **Harriman Cove** is on the east side of Penobscot River, 1.3 miles above Bucksport. It has a prominent tall concrete silo.

(651) **Frankfort Flats**, marked by buoys, are 3 miles above Bucksport. The channel crosses from the east side of the river to the west side at this point, and it is difficult to carry the best water. Frequent changes occur here, and large steamers experience difficulty making the sharp turns without the aid of a tug.

(652) **Marsh River** is a shallow stream flowing into the west side of Penobscot River from a southerly direction just westward of Frankfort Flats. **Frankfort**is a small village on North Branch of the river. The channel in North Branch is bare at low water and full of boulders a little below the village.

(653) **Winterport** is a town on the west bank of Penobscot River about 12 miles above Fort Point. A 270-foot barge, moored 300 feet offshore about 0.3 mile south of the town, with 25 feet alongside and a floating deck height of 8 feet, is used to ship potatoes and frozen foods in foreign trade. A series of conveyors are used to load the vessels; water and electricity are available. A marina and boatyard at the former upper potato wharf has a depth of 15 feet reported alongside. Transient moorings, limited transient berths with electricity and water, and gasoline, diesel fuel, marine supplies and open or covered storage facilities are available. A 15-ton mobile hoist is available; hull and minor engine repairs can be made.

(654) **Bucks Ledge**, about 3.2 miles above Winterport, is marked by a daybeacon at the center and bares at low water. The ledge extends north and south of the daybeacon.

(655) **South Orrington**, 0.6 mile north of Bucks Ledge daybeacon, has a public launching ramp. The ramp may be inaccessible at low water.

(656) **Hampden**, a small town on the west bank of the river, is 20 miles above Fort Point. The village of **Orrington** is on the east bank opposite Hampden. **East Hampden**, on the west bank 1.5 miles below Bangor, has facilities for small tankers discharging oil. A public launching ramp and float dock are available.

(657) **Brewer** is a city on the east bank of the river opposite Bangor. The city has banks, markets, stores of all kinds, motels, restaurants and other conveniences.

(658) **Bangor** is an important city on the west bank of Penobscot River at the head of navigation. The principal water traffic to Bangor is in petroleum and asphalt. Most of the river in front of the city has been dredged where necessary to obtain a depth of 14 feet. Considerable shoaling has been reported in some places on the Brewer side. The bottom is rocky with poor holding ground, and there are submerged obstructions and a few rocks with less than 14 feet over them. The city has banks, a library, a general hospital, markets, shops of all kinds, hotels, motels, restaurants, churches, schools and public parks.

(659)

Quarantine, customs, immigration and agricultural quarantine

(660) (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(661) Bangor is a **customs port of entry.**

(662) **Quarantine** is enforced in accordance with the regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.)

(663)

Wharves

(664) Bangor has three oil wharves and an asphalt wharf with depths of 7 to 14 feet reported alongside.

(665) A public float landing, just below the mouth of Kenduskeag Stream, has a depth of 3 feet reported alongside, but no facilities are available. Gasoline, diesel fuel, provisions and some marine supplies can be obtained in Bangor.

(666)

Communications

(667) The port is served by multiple Class II railroads and numerous trunklines. Bus service, both local and coastal, and taxi service are available. The Bangor International Airport is 2 miles west of the city.

(668) **Kenduskeag Stream** empties into the Penobscot River from the westward at the north end of Bangor. A railroad swing bridge with a clearance of 6 feet crosses the stream at the entrance; the swing span is inoperative. Just above the railroad bridge is a fixed highway bridge. There is no navigation on the stream.

(669)

Norton Island Ledges to Mill River

(670) The following is a description of the coast from Muscle Ridge Channel to Georges Islands. Muscle Ridge Channel and Seal Harbor have been described previously in this chapter.

(671) **Norton Island Ledges** are 0.6 to 1.2 miles westward of Whitehead Island (43°58.8'N., 69°07.8'W.). A bare rock is near the southwest end of the ledge, and rocks awash at low water are 600 yards southeastward and southwestward of the bare rock.

(672) **Seavey Ledges**, westward of Norton Island Ledges, have four rocks awash at high water. There is a depth of 5 feet north of the ledges marked by a buoy and another 5-foot depth, unmarked, at the southern end of the ledges, about 300 yards southwestward of the southerly group of three bare rocks.

(673) **Wheeler Bay** and **Clark Cove**, northward of Seavey Ledges, are foul. A wharf at the head of Wheeler Bay has a float landing with 4 feet alongside. **Clark Island** is a village on the northwest side of Clark Cove, northward of **Clark Island.** There is an inactive granite quarry with stone wharf on Clark Island. The derricks are conspicuous. The island is joined to the mainland by a causeway.

(674) **Makertown Cove** is on the east side of Wheeler Bay northeastward of **Calf Island**. It has seven fish wharves and a float landing at the entrance with 5 feet alongside. Gasoline and some provisions are available. The cove is difficult to enter without local knowledge.

(675) **Tenants Harbor**, 3 miles westward of Whitehead Light, is an excellent anchorage frequently used as a harbor of refuge by small vessels and is easy of access. **Southern Island**, on the southern side of the entrance, is marked on its east side by an abandoned lighthouse, a white tower connected to a dwelling. A lighted bell buoy is east of the island. **Northern Island** is on the north side of the entrance. There are depths of 8 to 25 feet in the harbor.

(676) The **anchorage** with most swinging room in Tenants Harbor is halfway from the western ends of Northern and Southern Islands to the stone pier on the north side. Small craft anchor toward the head of the harbor. The bottom is mostly soft mud and good holding ground and shoals gradually westward. The north side of the harbor eastward of the stone pier is clear, while westward of it are spots with depths of 4 to 9 feet. The south side of the harbor abreast the western entrance point of Long Cove should be given a berth of 200 yards because of a ledge covered 2 feet making out into the harbor from the south shore. The harbor is open eastward, and an easterly gale raises a choppy sea in the harbor, but vessels with good ground tackle can ride in safety. It is reported that a strong chop can also develop with a southwest wind and that craft of 6-foot draft or less can find calm anchorage in Long Cove on the north side of the harbor. Ice often obstructs the harbor during February; during extremely cold weather it is sometimes frozen to Southern Island.

(677) Vessels entering Tenants Harbor can pass midway between Southern and Northern Islands and steer **268°** into the harbor, slightly favoring the northern side.

(678) The channel between Southern Island and Hart Neck is shoal and foul with rocks awash at its northwestern end.

(679) The village of **Tenants Harbor** is on the northern shore near the head of the harbor. There are two service facilities and a boatyard along the northern side of the harbor with depths of 4 to 8 feet reported alongside their float landings. The facility adjacent to the town wharf has water, ice and marine supplies and maintains guest moorings. The boatyard has a marine railway that can handle craft up to 50 feet in length for hull and engine repairs or dry covered or open winter storage. Mariners are advised to avoid taking a direct route from one facility

to another as partially bare ledges extend from the shore between the facilities. Lodging is available in the village, and a good road leads to Thomaston.

(680) **Long Cove**, making northward from the entrance to Tenants Harbor, has several stone quarries that are not in operation. A lobster pound and fish pier are on the west side of the cove. Several private float landings are in the cove. The cove is reported to afford excellent anchorage for craft up to 6-foot draft.

(681) The entrance to the cove is about 150 yards west of the southwestern end of Northern Island between reefs partly bare at low water. A buoy marks the east side of the entrance to the cove. A bare rock is on the eastern end of the reef on the western side of the entrance. In 1982, shoaling was reported in about the center of the cove.

(682) **Hart Ledge** extends nearly 500 yards from shore 0.7 mile southward of Southern Island. A rock awash is near the northeast end of the ledge, and another rock awash at low water is near its southwest end. A buoy is off the northeast side of the ledge.

(683) **Mosquito Harbor**, 2 miles southwest of Tenants Harbor, is shoal and used by a few fishing boats. The landings bare at low water. **Martinsville** is a settlement at the head of the harbor. **Mosquito Head**, on the eastern side of the entrance, is 92 feet high and wooded and looks like an island from a distance.

(684) **Mosquito Island**, off the entrance to Mosquito Harbor, is 60 feet high and wooded. The islets southwestward of Mosquito Island, including **Hay Ledge**, 15 feet high, **The Brothers**, 18 to 20 feet high, and **Gunning Rocks** are rocky with grass on top. Three-foot-high **Black Rock**, 0.6 mile southwest of Gunning Rocks, is bare. **Hart Bar**, extending 0.5 mile northwestward of **Hart Island**, 1.8 miles west of Mosquito Island, is partly awash at low water. There are many unmarked submerged ledges in this vicinity.

(685) The passage south of Mosquito Island and north of these rocks and islands is part of the inside route used by many vessels drawing 12 feet or less. The principal dangers are buoyed, but there are two unmarked rocks including **Barter Shoal** that are covered 14 and 19 feet.

(686) **Old Cilley Ledge**, 1 mile southward of Hart Island, is about 0.5 mile long. The eastern end of the ledge is covered 2 feet, and its western end uncovers 9 feet. A bell buoy is 0.3 mile eastward of the eastern end.

(687) **Marshall Point** is 7 miles southwestward of Whitehead Light and on the east side of the southern entrance to Port Clyde. **Marshall Point Light** (43°55'03"N., 69°15'41"W.), 30 feet above the water, is shown from a white tower. A mariner radio activated sound signal at the light is initiated by keying the microphone five times on VHF-FM channel 83A.

(688) **Port Clyde** is a small but excellent harbor and anchorage between Marshall Point and Hupper Island, about 9 miles north-northeastward of Monhegan Island. Fishermen and coasters use it as a harbor of refuge. A bar, with boulders and covered 2 to 6 feet, obstructs the northern entrance. Vessels of 15-foot draft have been taken over this bar at high water by local pilots, but strangers should not attempt it.

(689) The anchorage is anywhere in the channel inside of Marshall Point in depths of 23 to 35 feet, good holding ground; there is a clear width of 200 to 250 yards. Good anchorage is also found, in southerly weather, northward of Hupper Island eastward of a line between Blubber Island and Hupper Point in depths of 21 to 24 feet.

(690) **Ice** usually does not interfere with navigation. In very severe winters the harbor may be frozen over for a short time.

(691) There are entrances from northward on either side of **Raspberry Island**, a small islet about 20 feet high about in the middle of the northern entrance. The passages on both sides of the island have depths of about 5 feet, but they are narrow and difficult and should not be attempted without local knowledge, except in small craft. The easterly channel is best for strangers in small craft. The best water follows the eastern shore at a distance of about 70 yards and passes eastward of a reef which makes eastward from a small islet.

(692) The village of **Port Clyde**, the base of many fishing boats, is on the eastern side of the harbor. The village has no rail connections, but a highway runs to Thomaston. Fields Wharf, 0.5 mile northward of Marshall Point Light, is used by the ferry that maintains mail, passenger and freight service with Monhegan Island; gasoline and diesel fuel are available at the wharf. The town float, with depths of 5 to 10 feet reported alongside, and a small-craft launching ramp are just northward of the wharf. There are a number of other wharves in the harbor, some with float landings, with depths of 6 to 18 feet alongside. Numerous other wharves are bare. Gasoline, diesel fuel, water, ice and some marine supplies can be obtained at some of these facilities.

(693) The town **harbormaster,** usually found at the town float, controls moorings in the harbor and enforces local **harbor regulations.** A **speed limit** of 5 mph is enforced.

(694) A general store, hotel and restaurant are in the village. A small private boatyard, about 0.4 mile northwestward of Fields Wharf, has a machine shop and a marine railway that can handle craft up to 30 feet in length in an emergency only.

(695) **St. George River** entrance is about 9 miles southwestward of Whitehead Island and north-northeastward of Monhegan Island. Marshall Point Light marks the eastern approach and Franklin Island Light the western. The Georges Islands extend 6 miles south-southwestward from the middle of the entrance, which also is obstructed by numerous ledges and rocks, the most prominent of which are marked. St. George River extends 10 miles in a northeasterly direction to the town of Thomaston, above which it is shallow and of no commercial importance.

(696) The channel depths in the river up to Broad Cove range from about 22 feet to over 80 feet; above this, the depths gradually decrease and the channel narrows to a small stream through extensive flats that bare at low

water. The channel in the upper river is marked by buoys, some of which are uncharted because they are frequently shifted to mark the best water. The sharp bend in the dredged channel near Thomaston is marked by a light.

(697) Good anchorage for large vessels is found eastward of Caldwell Island in depths of 33 to 53 feet, soft bottom; above this, vessels anchor anywhere in the channel where the depth is not too great or in Turkey, Maple Juice, Otis or Broad Coves.

(698) Ice closes the river to navigation from December to March in severe winters. In ordinary winters, it is not usually closed entirely for more than 1 month, although ice sufficient to interfere with navigation may be encountered at any time for a period of 3 months.

(699) The approach to the entrance of St. George River has very broken and irregular bottom, with numerous ledges bare and submerged. Strangers should proceed with caution and avoid crossing broken areas where the charted depth does not greatly exceed the draft.

(700) In approaching and entering St. George River no difficulty should be experienced by the navigator closely following the chart and the aids, having due regard for the unmarked dangers, some of which have been described in the preceding paragraphs. The anchorages in the lower river have been described previously. Above the Narrows at Bird Point, there is excellent anchorage near the middle of the river off Otis Cove. Passage in the river above the Narrows should be guided by the chart and the buoys. The safest time is at low water and on a rising tide when the flats are bare.

(701) **Georges Islands** are a group of islands and rocks extending about 6.5 miles south-southwestward from the middle of the entrance to St. George River. The larger islands are in general wooded, and the smaller ones grassy or rocky; there are few prominent landmarks. Several channels lead between the islands; the most important are Davis Strait, the channel between McGee and Seavey Islands and the channel northwestward of Caldwell Island.

(702) **Old Man Ledge** is the most southerly of the dangers. A lighted whistle buoy is about 0.3 mile south of the ledge. **Old Woman Ledge**, 0.6 mile northward of Old Man Ledge, uncovers 3 feet.

(703) **Burnt Island**, the eastern large island at the south end of Georges Islands, is privately owned, about 160 feet high, wooded and marked on its summit by a conspicuous lookout tower. At low water there is no passage between Burnt Island and **Little Burnt Island**, just north of the peninsula.

(704) **Georges Harbor** is between **Allen Island**, 0.5 mile west of Burnt Island, and **Benner Island**, off the northwest side of Allen Island. There is a small settlement of fishermen, and small craft sometimes anchor there. The best water is midchannel in entering the thoroughfare from northeastward. Entering from southwestward favor the south side. In 1966, a submerged mooring cable, suspended above the bottom, was reported extending across Georges Harbor between Allen Island and Benner Island.

(705) **Davis Island**, 0.9 mile northwestward of Burnt Island, is grassy and has two knolls with a saddle between. **Davis Strait** is the passage between Davis Island on the south and **Thompson Island** and other small islands on the north. The two southernmost islets on the north side of Davis Strait are grassy, and the others are wooded. The passage is part of the through route used by many vessels drawing 12 feet or less. It is reported that barges drawing 16 feet use this strait. It has ample depth, but **Griffin Ledge**, in midchannel, has a depth of 10 feet over it. On the south side of the ledge is a buoy, and the channel, which is southeastward of this buoy, is only 75 yards wide.

(706) Between Thompson and Hupper Islands the bottom is very broken, and there are numerous dangers, most of which are marked or visible at some stage of the tide. **The Sisters**, 1.3 miles east of Thompson Island, are two small ledges awash at low water; a buoy is on the northwest side of the ledges. **Old Horse Ledge**, 0.4 mile northwestward of The Sisters, uncovers at low water and is marked by a daybeacon.

(707) **Outer Shag Ledge**, 0.3 mile westward of Old Horse Ledge, uncovers about 5 feet, and **Inner Shag Ledge**, 0.3 mile west of Outer Shag Ledge, is awash at high water. **Kelp Ledges**, 300 yards west of Hupper Island, are awash at low water. **Gig Rock**, 0.6 mile southwest of The Sisters, is covered 7 feet; a bell buoy is off the northwest side of the rock.

(708) **Bar Island**, 0.7 mile west of Hupper Island, is low and grassy. There is ample depth in the channel between Bar Island and **Seavey Island** on the northeast and larger McGee Island on the southwest. Some of the dangers are buoyed, but there are unmarked dangers close to the channel. The passage is used by small boats. **Jenks Ledge**, the most westerly danger off the passage, is awash at low water and is marked by a buoy. A submerged obstruction of unknown depth is about 0.3 mile southwestward of the ledge.

(709) **Deep Cove**, on the eastern shore just north of the northern entrance to Port Clyde, has good anchorage in depths of 21 to 43 feet, soft bottom. A dangerous rock awash is in the north part of the cove, and a 12-foot spot is about 200 yards south. **Caldwell Island** is at the northern end of Georges Islands and the middle of the entrance to St. Georges River.

(710) **Gay Cove** is a shallow and unimportant cove in the eastern shore of Gay Island, the western point at the entrance of the river. It is reportedly sometimes used by yachts.

(711) **Pleasant Point Gut** separates Gay Island from the mainland. Its western part is bare at low water. **Pleasant Point**, a village of fishermen, is along the shore of the mainland. There are several lobster wharves, one of which has a depth of 4 feet alongside; the others are bare or have depths of less than 3 feet alongside. Gasoline, oil and some supplies can be obtained at the float landing

with 6 feet alongside. A private wharf is on **Gay Island** on the south side of the harbor. Local small craft are reported to make passage through the western entrance to the gut about half tide. Strangers should not attempt it.

(712) **Turkey Cove**, on the eastern shore of the river about 1.5 miles above Caldwell Island, has good anchorage in depths of 15 to 27 feet, soft bottom, about midway between the points at the entrance.

(713) **Maple Juice Cove** is a long, shallow cove on the west shore about 2 miles above Caldwell Island. Good anchorage is found at the entrance in depths of 13 to 24 feet.

(714) **Otis Cove**, broad but shallow at its head, is on the eastern shore about 1.7 miles above Turkey Cove. There is good anchorage off the entrance in depths of 20 to 27 feet. There are no wharves.

(715) **Broad Cove**, on the western shore about 4.5 miles above Caldwell Island, is shallow. The village of **Cushing** is near the northern shore. There is a wharf that bares at low water. An aquaculture farm marked by private buoys is in the center of the cove. **Bailey Ledge**, off the southern entrance, is bare at low water and marked on the southeast side by a buoy.

(716) **Watts Cove** is a shallow cove on the eastern shore opposite Broad Cove. The village of **St. George** is at the head of the north arm of the cove.

(717) **Thomaston** is a town near the head of navigation on the St. George River. There is no waterborne commerce. Two towers of a cement plant, a large red boatyard building and a railroad bridge across the mouth of Mill River on the east end of the town are conspicuous.

(718) There are two boatyards at Thomaston. Craft up to 150 feet can be built, and hull and engine repairs can be made. Open and covered dry winter storage facilities are also available. The public landing has a float landing with 15 feet reported alongside and a small-craft launching ramp. Gasoline, diesel fuel, water, dry open and covered winter storage, limited marine supplies and hull and engine repairs are available at marinas on either side of the river just below the bridge. The **harbormaster** can be contacted through the Thomaston Police Department.

(719) St. George River is crossed at Thomaston by a fixed highway bridge that is under construction (2016). The nearby overhead power and telephone cables have a clearance of 40 feet.

(720) Two fixed bridges, a railroad and a highway bridge, cross the river about 2 miles above the bridge at Thomaston; least clearance is 5 feet. The piles of a former wooden bridge just below the railroad bridge project about 5 feet above the river bottom and form obstructions in the channel. There is little traffic, except for small boats, in this part of the river. The fixed railroad bridge across the mouth of **Mill River**, east of Thomaston, has a clearance of 25 feet for a width of 28 feet.

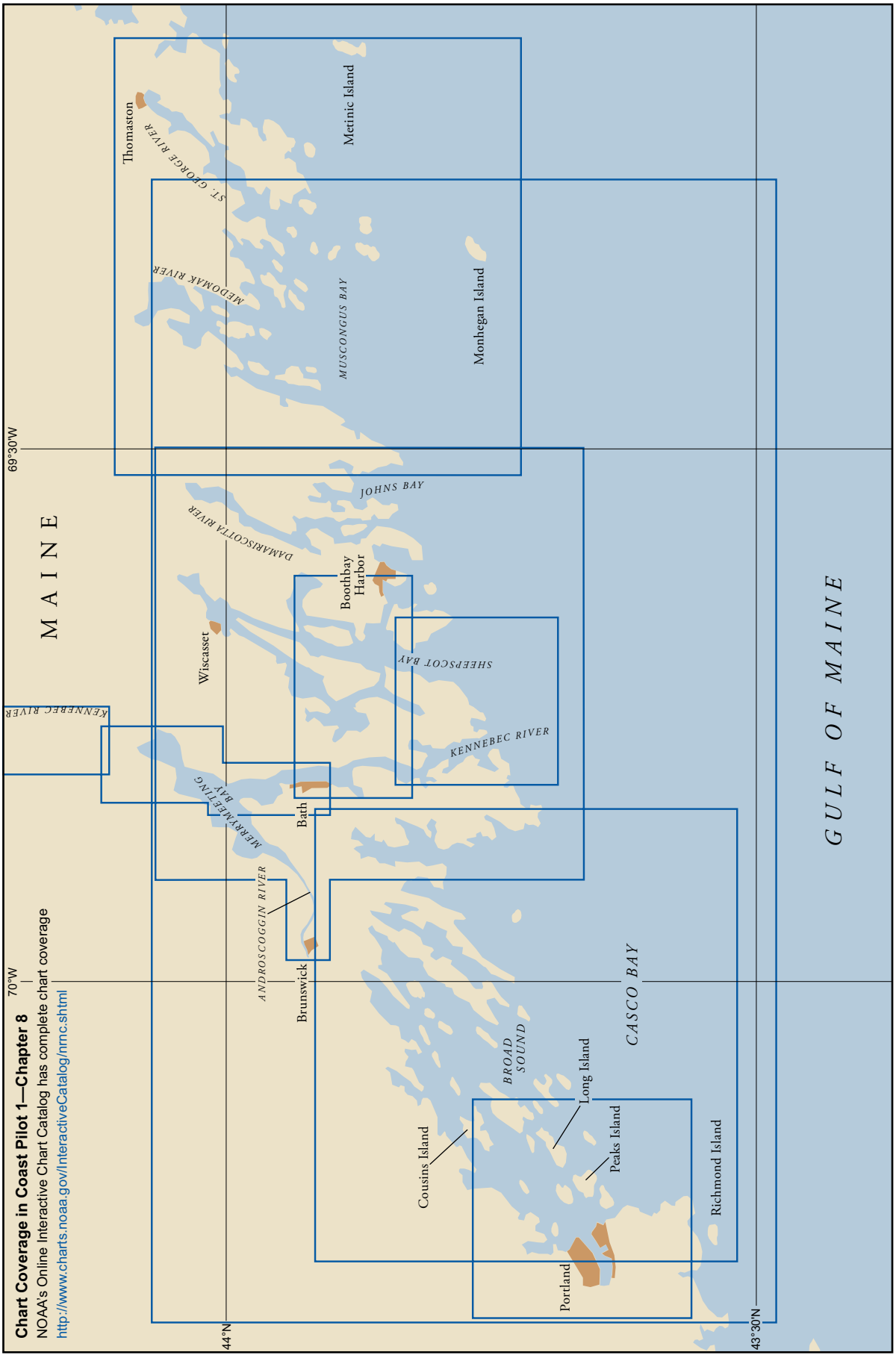


Chart Coverage in Coast Pilot 1—Chapter 8
NOAA's Online Interactive Chart Catalog has complete chart coverage
http://www.charts.noaa.gov/InteractiveCatalog/nrnc.shtml
MAINE
GULF OF MAINE
Thomaston
ST. GEORGE RIVER
Metinic Island
MEDOMAK RIVER
MUSCONGUS BAY
Monhegan Island
69°30'W
JOHNS BAY
DAMARISCOTTA RIVER
Boothbay Harbor
Wiscasset
SHEEPSCOT BAY
KENNEBEC RIVER
KENNEBEC RIVER
MERRYMEETING BAY
Bath
ANDROSCOGGIN RIVER
Brunswick
70°W
CASCO BAY
BROAD SOUND
Long Island
Cousins Island
Peaks Island
Portland
Richmond Island
44°N
43°30'N

Muscongus Bay to Cape Elizabeth, Maine

(1) This chapter describes Muscongus, Booth, Sheepscot and Casco Bays; Medomak, Damariscotta, Sheepscot, Kennebec and New Meadows Rivers; and the ports of Portland, Bath, Boothbay Harbor and Wiscasset. This area has many islands, rocks and long peninsulas. Many of the islands have been joined by fixed highway bridges; hence, so far as masted vessels are concerned, whole groups become additional peninsulas. In general, the outer islands and rocks rise from deep water and the lower parts of the rivers are deep.

(2)

COLREGS Demarcation Lines

(3) The lines established for this part of the coast are described in **33 CFR Parts 80.105**, and **80.110**, chapter 2.

(4)

Muscongus Bay to Lower Narrows

(5) **Muscongus Bay**, between the Georges Islands on the east and Pemaquid Neck on the west, forms the approach to Meduncook and Medomak Rivers and Muscongus Sound; the villages of Friendship, Round Pond and Medomak; and the town of Waldoboro. The bay is frequented by small pleasure and fishing craft. It is obstructed by numerous islands and ledges and much foul ground. Many of the dangers are marked by buoys.

(6) **Moser Ledge**, the outermost of the dangers, covered 15 feet and marked by a buoy, lies about in the middle of the entrance to the bay, about on line between the north end of Monhegan Island and Pemaquid Point Light (43°50'12"N., 69°30'21"W.).

(7) **Franklin Island Light** (43°53'31"N., 69°22'29"W.), 57 feet above the water shown from a white tower on the northwestern side of **Franklin Island**, is the principal aid to the approach and passage through the bay. The light is obscured by trees from **253°** to **352°.**

(8) Access to the eastern side of the bay, between Allen Island and Franklin Island, is obstructed by an area of islands and mostly unmarked shoals and ledges. The area, about 3 miles long north and south and 2 miles east and west, is bounded on the west by **South Ledge**, an unmarked ledge covered 13 feet; **Egg Rock South Ledge**, covered 7 feet; **Eastern Egg Rock**, 23 feet high and bare and marked on its north side by a daybeacon; **Egg Rock North Ledge**, marked on its southeast side by a buoy; **Hough Ledge; Little Franklin Ledge**; and Franklin Island. Its eastern side is bounded by **Shark Island**; unmarked **Little Egg Rock Shoals; Little Egg Rock**, 28 feet high; **Seal Ledges**, marked on their north end by a buoy; and **The Kegs**, marked by a daybeacon. On the north end is **Gangway Ledge**, an unmarked bare rock and ledge area.

(9) Three deep, natural, mostly unmarked channels, narrow in places, lead in a northerly and northeasterly direction past or through the area and into the St. George River. The eastern channel leads west of Georges Islands. The western channel leads westward of the area of islands, shoals and ledges near the center of Muscongus Bay and westward of Eastern Egg Rock and Franklin Island. **Old Hump Channel** leads through the center of the area.

(10) A buoyed channel marked for a westerly crossing, known as **Davis Strait Passage**, os used mostly by small craft proceeding between Pemaquid Point and Port Clyde or Penobscot Bay, via Muscle Ridge Channel. From a fairway bell buoy off its western entrance between Eastern Egg Rock and Egg Rock North Ledge, this passage crosses Old Hump Channel, then passes between **Old Hump Ledge** and Seal Ledges; thence through Davis Strait; thence northeastward past Gig Rock; thence between Old Horse Ledge and The Sisters; thence southward of Hupper Island and northward of Allen Ledge to the entrance to Port Clyde. Craft proceeding farther eastward continue on, passing southward of Marshall Ledge; thence between Gunning Rocks and Mosquito Ledge; and thence southeastward around Mosquito Island and Barter Shoal before rounding up to the northeastward for Muscle Ridge Channel.

(11) A group of islands in the middle of the bay, extending 3 miles southwestward from Friendship Long Island, separates the approaches of the St. George and Meduncook Rivers from the Medomak River. This group includes **Crane Island, Harbor Island, Hall Island, Black Island, Otter Island, Cranberry Island** and **Morse Island**. Surrounding and interspersed between these islands are numerous rocks and ledges. **Harbor Island Rock**, **Black Island Ledge**, **Otter Island Ledge** and **Beyer Ship Ledge** are all unmarked. **Morse Ledge** is marked by a daybeacon. The passages between these islands and ledges are mostly shoal, foul and unmarked and of interest only to local craft.

(12) In the western part of the bay, islands and ledges extend 3 miles southward from Louds Island. **Bar Island**, close south of Louds Island, is grassy, **Haddock Island** is wooded and **Ross Island** is grassy. **Haddock Island Kelp Ledge**, covered 8 feet, is marked on its south side by a buoy. **Webber Dry Ledge** uncovers at low water; **Webber Sunken Ledge**, with a rock awash at low water and marked by a buoy, extends 0.3 mile south of it. **Browns Head Ledge**, covered 13 feet, is marked by a

buoy. **Bar Island Ledge**, 0.2 mile long and awash at low water, is marked on its south end by a buoy.

(13) The most southerly of these ledges are **New Harbor Sunken Ledges**, awash at low water at the north end and marked at their south end by a buoy. The reef extending 0.3 mile eastward of grass-covered **Western Egg Rock**, the southeasternmost of this group of ledges, is covered 4 feet and marked by a buoy.

(14) **Devils Elbow**, which uncovers 1 foot; **Devils Back**, which uncovers 8 feet; **Devils Limb**, awash; bare **Wreck Island Ledges** and **Garden Island South Ledge**, awash, unmarked and dangerous; 48-foot-high **Wreck Island**; and 23-foot-high **Jones Garden Island** are all on a line about 1.5 miles eastward of Louds Island; Jones Garden Island is the northeasternmost of the group.

(15) **Haddock Island**, **Ross Island**, **Marsh Island**, **Killick Stone Island**, **Thief Island** and **Indian Island** are all part of the western group and lie southward and eastward of Louds Island.

(16) **Meduncook River** is an estuary making in a general northeasterly direction, just westward of St. George River; the entrance forms one of the approaches to Friendship Harbor and is a good anchorage with depths of 10 to 30 feet. The approaches to the entrance are the same as for St. George River. In 1983, the entrance, between Gay Island and Morse Island, was reported to be extremely hazardous because it was obstructed by lobstering equipment. Local knowledge is advised.

(17) The river above the anchorage has a narrow, crooked channel and is obstructed by numerous unmarked rocks and ledges, so that local knowledge is necessary for its navigation.

(18) **Back River** enters Meduncook River about 0.6 mile above **Crotch Island**; a tidal boatyard is on Back River.

(19) **Friendship Harbor** (43°58.0'N., 69°20.5'W.) is west of Meduncook River, from which it is separated by **Friendship Long Island** and **Garrison Island**; between these two islands a buoyed channel leads from the anchorage in Meduncook River into Friendship Harbor.

(20) A passage, foul and dry at half tide, but used by some local fishermen, leads into the harbor between Garrison Island and the mainland. Overhead power and telephone cables over the passage have a clearance of 20 feet.

(21) Friendship Harbor, about 1 mile long with good anchorage in 21 to 28 feet, is used extensively by fishermen and yachtsmen. It is reported that the harbor may be closed by ice during January and February.

(22) A ledge extends 300 yards southwestward from **Jameson Point** to a rock, uncovered at low water, which is marked by a daybeacon. On the southern side of the entrance, opposite the daybeacon, an unmarked shoal with a cleared depth of 7 feet at its outer end extends about 300 yards into the channel from Friendship Long Island.

(23) Above the wharves the northern and eastern side of the harbor should be given a berth of over 200 yards. The southeast side of the harbor should be given a berth of over 200 yards. **Murphy Ledge** is a rock that uncovers about 4 feet and is marked by a daybeacon, 200 yards from the southeast side of the harbor abreast Jameson Point. A shoal with a cleared depth of 14 feet, about 200 yards northward of the daybeacon on Murphy Ledge, is unmarked. In the eastern part of the harbor a shoal extends 350 yards northeastward from the northeast end of Friendship Long Island; a buoy marks the outer end.

(24) **Friendship** is a town on the north shore of Friendship Harbor. There are numerous wharves and piers with float landings on the north side of the harbor on Jameson Point; depths of 2 to 12 feet are reported alongside. Gasoline, diesel fuel and water are available at several of the landings and marine supplies at some. Engine repairs can be made. The town wharf, one of the more northerly facilities, has a float landing on its northerly side with a reported depth of 2 feet alongside. Rocks, some submerged, extend northeasterly from the outer end of the town wharf; mariners are advised to use caution when approaching the town float landing. Provisions and lodging can be obtained in town.

(25) **Hatchet Cove** is a shallow cove making northward at the western end of Friendship Harbor. A narrow unmarked channel with a least depth of 11 feet leads northeastward into the cove near the western point at its entrance. It is unimportant as an anchorage, and the landings bare at low water. A boatyard is on the east side at the head of the cove. Some hull repairs and dry open winter storage are available. The town-owned small-craft launching ramp, usable at half tide or better, adjoins the boatyard.

(26) **Gull Rock**, in the western entrance to Friendship Harbor, includes two rocks bare at high water. A ledge cleared to 20 feet at its southwestern end is about 0.4 mile eastward of Gull Rock.

(27) **Medomak River** enters the head of Muscongus Bay westward of **Martin Point**, the western point at the entrance to Friendship Harbor. Strangers should take a local pilot because of the many unmarked dangers, narrow and crooked channels and strong tidal currents that require local knowledge.

(28) The lower part of the river is about 2 miles wide but is separated by several islands into two approaches; these have three narrow and crooked channels by which entrance is made to the upper river. The approaches to these channels are through Muscongus Bay or Muscongus Sound.

(29) The eastern approach is 0.5 mile wide and comparatively clear of dangers. At its upper end are two passages leading into the river, one through **Back River Cove** and the other through **Flying Passage.** Both of these passages are narrow and unmarked and have shoal rocky areas near the middle and on their edges which, together with the strong tidal currents, make them difficult to navigate.

(30) **Hockomock Channel**, the western approach, has much better water and is the preferred channel despite the fact that it is narrow in places and has strong tidal currents.

(31) There are two fish wharves on **Keene (Hockomock) Neck**, on the west side of the channel, at which gasoline, diesel fuel and some supplies can be obtained. One of these, behind **Oar Island**, has a lobster pound adjoining it and a float landing with 5 to 6 feet reported alongside. Some protection from east and southeast winds is afforded this landing by the hulk of the five-masted schooner CORA CRESSY, which has been hauled up on the reef between Oar Island and the neck. Provisions and some supplies can be obtained in the village of Medomak.

(32) The channel in Medomak River has ample depth for 5 miles above the entrance. Some of the dangers are marked, but there are unmarked ones close to the channel. For the next 2.5 miles to within 1.6 miles of Waldoboro, the channel leads between flats nearly bare at low water. Up to 5 feet is available with local knowledge.

(33) There is no commercial shipping and only limited fishing and small-boat activity on the river to Waldoboro. The channel can best be followed at low water when the flats are visible, or on a rising tide.

(34)

Current

(35) Tidal currents in **The Narrows**, between **Locust Island** and **Havener Ledge**, are reported to be very strong.

(36) **Medomak** is a village on the western side of Hockomock Channel. There are a town wharf and float landing with 2 feet alongside and a fish wharf with a depth of 4 feet, about 0.5 mile southward of the village. Gasoline and diesel fuel are piped to the fish wharf.

(37) **Broad Cove**, on the west side of Medomak River, is used by a few fishermen. The channel into the cove is unmarked.

(38) The **Medomak River Closed Area**, a Marine Protected Area (MPA), includes the waters of the Medomak River north of Hardy Island, including Broad Cove, Eastern Branch and Western Branch.

(39) **Waldoboro**, at the head of navigation on Medomak River, is a town on a branch of a Class II railroad with markets, restaurants, motels and a library. There is no commercial waterborne commerce from the town. An old steamer wharf, in disrepair, is on the east side of the river, and a town landing is on the west side. There is little water alongside these wharves. Provisions, ice, and some marine supplies are available in town. Gasoline can be obtained from a filling station and diesel fuel by tank truck.

(40) A lobster wharf with 5 feet reported alongside its float landing is on the east side of Medomak River, about 2.1 miles northward of Martin Point. Gasoline is available at the float landing.

(41) **Pemaquid Neck**, a wooded peninsula, is on the west side of Muscongus Bay. **Pemaquid Point** is the south point of the neck. A radio tower on the point is prominent. Pemaquid Point Light (43°50'12"N., 69°30'21"W.), 79 feet above the water, is shown from a white conical tower on the southeast point of the neck about 0.5 mile northeast of Pemaquid Point. The town of **Pemaquid Point** is on the southern end of the neck. A gong buoy is 500 yards south of the point.

(42) **Pemaquid Ledge**, 1 mile south of the point, has a cleared depth of 10 feet and is marked by a buoy on its southeast side. An unmarked 23-foot patch is about 550 yards southward of the ledge.

(43) A 1-mile radius naval test area is centered 7.9 miles **169°** from Pemaquid Point Light. (See **33 CFR 334.30**, chapter 2, for limits and regulations.) Mariners are cautioned against proceeding through the area while operations are in progress.

(44) An abandoned 1-mile radius naval test area centered 3 miles **162°** from Pemaquid Point Light is known to be foul with unexploded ordnance. Caution should be exercised against dragging operations in the area; any material inadvertently picked up should be discarded immediately with a minimum of handling.

(45) **Pumpkin Cove Ledge**, 1 mile east-northeastward of Pemaquid Point Light, is covered 19 feet and is unmarked. The sea breaks on it in heavy weather.

(46) **New Harbor Dry Ledges**, 2 miles northeastward of Pemaquid Point Light, extend 0.3 mile from the shore. The ledges are 0.3 mile long with a bare rock near each end and no safe passage for strangers between them and the shore. An unmarked rock, covered 3 feet, is 200 yards offshore about 0.3 mile southwestward of the ledge. **Little Island**, showing a clump of trees, is 200 yards from the shore 0.2 mile southward of the entrance to New Harbor. It is the highest part of a ledge about 0.3 mile long.

(47) **New Harbor** is on the western shore of Muscongus Bay, about 2.5 miles northeastward of Pemaquid Point Light. A lighted bell buoy is off the entrance to the harbor. A church spire in the village of **New Harbor** at the head is prominent. The cove offers anchorage to small craft only and is open eastward. The channel is narrow between a shelving ledge extending northeastward from the south point at the entrance and a ledge just inside it that extends halfway across from the north side and is marked at its end by a buoy. A 100-foot-wide channel then leads northward of a daybeacon between ledges to dredged anchorage basins. Enter about 100 feet north of the daybeacon. The channel and basins are subject to shoaling, particularly along the edges. It is reported that ice does not prevent navigation in the winter.

(48) There are two service wharves with float landings on the north side of the harbor at which gasoline, diesel fuel, water, ice and marine supplies can be obtained. Depths of 10 feet are reported alongside the service landings. Several fish and lobster wharves are throughout the harbor. A fleet of seiners operates from the harbor, and ground fish are shipped from the port by truck. Markets, provisions, restaurants and lodging are available in town. There is no marine railway, but local fishermen ground out their boats for repairs.

(49) **Back Cove**, a southwesterly arm of New Harbor, is used by local pleasure and fishing craft. A dredged channel leads to an anchorage basin that extends to near

the head of the cove.The channel and basin are subject to shoaling, particularly along the edges. There are a number of private and fish piers, but no facilities.

(50) **Long Cove**, about 0.6 mile northward of New Harbor, is about 0.5 mile long and 250 yards wide at the entrance. It affords good anchorage in from 14 to 53 feet to within 400 yards of its head in all but southerly weather. It is used by local pleasure craft. The approach to the cove from the southward is clear from northward of **Salt Pond Ledge**, an unmarked ledge covered 8 feet about 0.4 mile south of the entrance. There are no facilities in the cove.

(51) **Louds Island** is inhabited mostly by fishermen; there are also some farms on the island. **Loudville** is a village on the cove on the east side of the island northwestward of the northern end of Marsh Island. There is reported to be a wharf in the cove, which dries out at low water.

(52) **Marsh Harbor**, on the southeast side of Louds Island between it and Marsh Island, is seldom used as an anchorage.

(53) **Muscongus Sound** is on the western side of Muscongus Bay, between Louds and Hog Islands on the east and the mainland on the west. It is about 0.5 mile wide and 5 miles long and has several rocks and ledges near its southern entrance, the most important of which are marked by buoys. Above the Poland Ledges to abreast Muscongus Harbor, the depths in the sound decrease gradually from 48 to 24 feet, and anchorage can be selected by the chart.

(54) Webber Sunken Ledge, Webber Dry Ledge, Browns Head Ledge and Bar Island Ledge, all previously described, and **Webber North Ledge**, covered 15 feet and unmarked, are dangers in the southern entrance to Muscongus Sound.

(55) **Poland South Ledge** is covered 9 feet, but **Poland North Ledge** is awash at low water. Both are marked by buoys. The better channel leads eastward of them. An unmarked ledge cleared to 18 feet is about 350 yards southeastward of Poland North Ledge.

(56) **Round Pond** is a small harbor with 10 to 17 feet in its middle on the west shore of Muscongus Sound, westward of the north end of Louds Island. It affords good anchorage for small vessels. The village of Round Pond is at the head of the harbor. The northeast and southwest ends of the harbor should be given a berth of 350 yards and the west side 200 yards. The best water in entering favors the north side, northward of the buoy marking a 7-foot rock near the end of a reef making northward from the southern point of the entrance.

(57) There is a town landing with 6 feet reported alongside its float in the northwest part of Round Pond. Two lobster piers with float landings are near the town landing; depths of 3 to 6 feet are reported alongside. Gasoline, diesel fuel and some marine supplies can be obtained at these landings. A boatyard, close southwestward of the town landing, has a marine railway that can handle craft up to 45 feet or 30 tons for hull and engine repairs. Guest moorings and open dry winter storage are available. There is a general store and restaurant in the village, and ice can sometimes be had. There is a ramp for launching small craft from trailers, and lodging and parking are available.

(58)

Current

(59) Off the entrance to Round Pond there is practically no flood current; the ebb has a velocity of 0.5 knot at strength.

(60) **Muscongus Harbor** is a small cove on the west shore of the sound about 1.5 miles above Round Pond. The village of **Muscongus** is on the north shore of the harbor. A marina on the north shore has 4 feet reported alongside; gasoline, diesel fuel and limited marine supplies are available. During the summer, fishing and pleasure boats anchor just inside the entrance in 5 to 8 feet of water.

(61) **Greenland Cove** is the extreme northern end of the sound. It is shallow and of no importance. It is reportedly often used by yachts. **Bremen** is a small village at the head of the cove.

(62) **Lower Narrows**, leading into the head of Muscongus Sound north of **Hog Island**, has a depth of about 13 feet. A rock awash on the north side of the narrows is marked by a buoy on its east side. There is a group of boulders, reported to be about 4 feet high, at the northwest end of Hog Island on the south side of Lower Narrows. Mariners should not attempt passage between the rocks and Hog Island. Local knowledge is necessary to carry the best water.

(63) The Audubon Society of America maintains a camp on the northeast point of Hog Island, at which there are several buildings and a float landing.

(64)

Johns Bay to Pig Cove

(65) **Johns Bay** (43°50.0'N., 69°32.0'W.) is westward of Pemaquid Neck, between it and **Rutherford Island**, its entrance is about 1.4 miles wide, and the length of the bay is 2 miles to Johns Island, above which Pemaquid River empties into the northeastern end. Johns River flows into the northwestern part. Depths in the bay are very irregular, and there are several ledges and rocks. A high square observatory tower on Rutherford Island is prominent.

(66) Though not commercially important, the bay has summer resorts on its shores and is used as an anchorage by fishermen and yachtsmen. The holding ground is poor except in a few spots near the head of the bay and in the coves. Port Clyde, eastward, and Boothbay Harbor, westward, are preferable at all times.

(67) **Pemaquid Harbor** (43°52.5'N., 69°32.0'W.) is at the entrance to Pemaquid River, northeastward of Johns Island. The bottom is rocky and irregular, but there is a fair anchorage for small vessels in 36 feet in the eastern part of the harbor between **Fish Point** and the entrance of Pemaquid River. The preferred anchorage for small craft, although crowded, is said to be north of the fort where the

bottom is soft in places. The village of **Pemaquid Harbor** is on the north side of the entrance to the harbor. There are a number of private float landings and boatsheds.

(68) **Pemaquid River** extends northeastward about 2 miles to the village of **Pemaquid**. The river is dry at low water near its head and has a narrow, crooked channel marked by private buoys. On the point marking the southern entrance to Pemaquid River there is a prominent stone tower marking the position of the former **Fort William Henry.**

(69) The pier and float landing of a lobster wharf are on the north side of Pemaquid River about 0.5 mile northeastward of the old fort. Depths of 3 feet are reported alongside the float; gasoline, diesel fuel and some marine supplies are available.

(70) **Pemaquid Beach** is a village on the south side of Pemaquid River at the entrance. There is a private wharf with a float at the old fort. A pier and float landing are at a state park, close northeastward of the private wharf. Depths of 10 feet are reported alongside the float. Parking, a restaurant and a small-craft launching ramp are available at the state park. Groceries and lodging can be obtained in the village nearby.

(71) A reef almost bare in places at low water extends offshore between the private wharf and the state park pier. Several small fish wharves are to the eastward on the south side of the river.

(72) A ledge, partly bare at half tide, extends 225 yards north-northeastward from the north end of Johns Island, where it is marked by a spindle, and another shoal cleared to 13 feet is about 0.3 mile south of the island.

(73) **Thurston Ledges** are mostly bare rocks extending 300 yards southward from **Thurston Point** on the north side at the entrance of Pemaquid Harbor, their south edge being 300 yards northward of Beaver Island.

(74)

Routes

(75) Pemaquid Harbor can be entered from westward by passing midchannel between **Beaver Island**, the high rounded islet with some trees, 300 yards northward of Johns Island, and Thurston Ledges. From the southward, when 0.5 mile or more southward of Johns Island, steer so as to pass 150 yards eastward of Johns Island, being careful to avoid the 13-foot shoal southward of the island, and then westward of the western bare rocks of **Knowles Rocks.**

(76) **McFarlands Cove** is on the western side of Johns Bay, northward and westward of **Witch Island.** A steep 150-foot hill is on the west shore of the cove. There is good anchorage in 24 to 36 feet in the cove for a small vessel about 300 yards northward of Witch Island.

(77) **McFarlands Ledges**, about 450 to 800 yards north-northeastward of Witch Island, have a rock that uncovers 6 feet near the north end and one that uncovers at low water near the south end. A buoy marks the south end of the ledges. **Corvette Ledge**, about 200 yards northeastward of Witch Island, is covered 3 feet; a buoy marks its north end. When entering the cove from eastward between the buoys marking these two ledges, take care to avoid the rock awash off the northwestern point of Witch Island.

(78) The Gut, a thorofare connecting McFarlands Cove with Damariscotta River, is described under the description of that river.

(79) **Johns River** extends northward about 2 miles above McFarlands Cove and separates into two branches. **Eastern Branch** is the eastern, and **North Branch** is the western. **Poorhouse Cove** makes into the western shore of Johns River above High Island. Good anchorage is available in depths of 18 to 24 feet southeastward and eastward of **Sproul Point**. The river is little used. Two boatyards that haul out and store yachts up to 35 feet in length are on Johns River, one on Sproul Point and the other at the head of **Bradstreet Cove**, a western arm of Poorhouse Cove.

(80)

Routes

(81) Stand up the middle of the bay, heading for the eastern shoulder of High Island, pass 400 yards westward of Johns Island, avoiding unmarked **Pollock Rock** and an unmarked 11-foot spot 350 yards southwest of Thurston Point, and pass about 300 yards off the eastern shore northward of Pemaquid Harbor. Then keep in midchannel until abreast of High Island, and then pass about 50 yards westward of the buoy marking a rock covered 10 feet, about 350 yards northeastward of High Island; anchor near midriver, about 400 yards northward of the buoy, in 18 to 24 feet.

(82) **Thread of Life** is a narrow deep channel, lying between Thread of Life Ledges and Crow Island on the east and the southern part of Rutherford Island and Turnip Island on the west. It is used by small local vessels entering Johns Bay from westward or from Damariscotta River. **Thrumcap Island** is partly wooded in its northern part and has a prominent house on it. **Thread of Life Ledges** are bare or grassy islets; **Turnip Island**, partly wooded, has a house on it. **Crow Island** is wooded. A shelving ledge awash at low water and marked by a buoy extends 300 yards southward from **Hay Island**, which is wooded. The channel westward of **Birch Island**, northward of Hay Island, has been dammed off to form lobster pounds. The hulk of an old tug, aground, rests against the southernmost dam; a fish pier with float landing is at the northernmost dam.

(83) To pass through Thread of Life from westward, after clearing Fisherman Island Passage, steer for the north end of Thrumcap Island with Ram Island Light astern. Pass 400 yards southward of The Bulldog, that uncovers 3 feet, and then 500 yards southward of the two rocks which uncover 6 feet about 350 yards eastward of Inner Heron Island. When about 400 yards from the north end of Thrumcap Island, round up to the northward keeping 200 yards off Thread of Life Ledges, and pass midway between them and Turnip Island. Continuing in midchannel to the north end of the passage, pass into

Johns Bay between the buoy south of Hay Island and the buoy marking the ledge extending 200 yards northward of Crow Island.

(84) **Damariscotta River** extends about 14 miles northward to the twin towns of Damariscotta and Newcastle, thence another 2 miles to **Damariscotta Mills** at the mouth of **Damariscotta Lake.**

(85) The entrance to Damariscotta River is about 3.2 miles west-southwestward of Pemaquid Point Light (43°50'12"N., 69°30'21"W.) and 1.3 miles northeastward of Ram Island Light (43°48'14"N., 69°35'57"W.) The tidal current is strong. Although some of the dangers are marked by buoys, strangers in anything but small craft should not pass through or above The Narrows at Fort Island without a pilot.

(86) The channel of the river is crooked. In many places it is very narrow because of the constricting islands and ledges. For a distance of 11 miles above the mouth of the river a least depth of 20 feet may be carried in the channel, although there are many unmarked 16- to 18-foot spots on each side of the channel. Above this point the water shoals to 10 feet just below the town of Damariscotta.

(87) The channel had a controlling depth of 9 feet in 1958 and for 2 miles south of the Damariscotta-Newcastle Bridge is bordered with mudflats on both sides; care should be exercised in piloting. Above the bridge, navigation is impossible except at high-water slack and with local knowledge due to the rapids and falls at Damariscotta Mills.

(88) The **White Islands**, about 1.5 miles south of the entrance to Damariscotta River, are prominent. The northern island is grassy with conspicuous standing trunks of dead trees. The southern island is partly wooded on the northern two-thirds and is bare rock on the southern third. There is a house on the island. Give the south and west side of the island a berth of at least 300 yards to avoid a rock covered 2 or 3 feet reported about 150 yards off the southern tip and the ledges and rocks making out from the west side. A ledge extends about 100 yards north at the northern point of the island.

(89) Southward and southwestward of the White Islands, **Outer Heron Island**, wooded, and **Pumpkin Island**, together with their off-lying ledges, extend about 2.5 miles. **Outer Heron Island Ledge**, covered 6 feet, about 0.9 mile east-southeastward of Outer Heron Island, is marked on its east side by a buoy. **Southeast Breaker**, covered 19 feet, about 0.7 mile southeastward of Pumpkin Island, and **Pinkham Shoal**, covered 8 feet, about 0.5 mile southwestward of Pumpkin Island, are unmarked. An unmarked rocky area cleared to 10 feet is about 0.5 mile eastward of the southern tip of the island.

(90)

Anchorages

(91) Vessels bound into the river usually go as far as Meadow Cove, just above East Boothbay, where good anchorage is available in 30 to 48 feet, keeping 150 yards offshore. This is as far as a stranger should attempt to go without local knowledge. Above The Narrows vessels can anchor anywhere in the channel where the bottom and depth are suitable.

(92)

Routes

(93) Extreme caution is necessary in this region where there are many rocks and ledges and very broken bottom.

(94) With the aid of the chart, enter the river midway between the gong buoy off Little River and the buoy marking Inner Heron Ledge, keeping in midchannel for about 1.5 miles above Inner Heron Island.

(95) There are unmarked 16-, 18-, and 23-foot spots in the channel between **Farnum Point** and Rutherford Island, and an 8-foot shoal marked by a buoy about 0.3 mile southeastward of the point. Favor **Jones Point** when passing the shoals eastward of **Montgomery Point** and, when clear, round up to the northwestward for the anchorage off **Meadow Cove.**

(96) Small craft should have no trouble in going to the head of navigation with the aid of the chart. The best time is on a rising tide. It is reported that the buoy at the entrance to The Narrows tows under during strength of the current.

(97)

Current

(98) The tidal current in the constricted sections attains an estimated velocity of 5 knots. The ebb lasts about 2 hours after low water in the upper part of the river and is usually stronger than the flood. The currents follow the general direction of the channel. Off Cavis Point the velocity at strength of current is about 1 knot. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov*. for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(99) **Ice** closes the river for a distance of 4 miles below Damariscotta during January, February and March.

(100)

Pilotage

(101) Pilotage is compulsory for all foreign vessels and for U.S. vessels under registry with a draft of 9 feet or more. Pilotage is optional for fishing vessels and vessels powered predominantly by sail.

(102) **Little River**, a long narrow inlet in Linekin Neck on the west side of the entrance of Damariscotta River, has a number of private float landings and fish wharves. A junction gong buoy about 350 yards south of **Reeds Island** marks the entrance. The channel is narrow and constricted at the entrance, but secure anchorage can be found in 5 to 12 feet in the outer section and 13 to 18 feet in the inner section of the inlet above the fish wharves on the east side. Small craft anchor near the head of the inlet above the narrows.

(103) A ledge, locally known as **The Bull**, is in the middle of the entrance; local knowledge is required to carry the best water.

(104) **Treasure Island**, with a house on it and connected to the shore by a fixed trestle bridge, is on the northeast side of the entrance to the inner harbor.

(105) There is a good holding ground in 13 to 18 feet, mud bottom, in midchannel from abreast the first fish wharf on the east side to the private pier with float landing just above the fish wharf on the west side, about 0.6 mile above the daybeacon. Above that point the harbor shoals rapidly. Local knowledge is advised. Gasoline may be obtained at the first fish wharf on the east side.

(106) **Inner Heron Island** (43°49.8'N., 69°34.0'W.), on the eastern side of the entrance to Damariscotta River, is thickly wooded. Two private float landings are on the northeast side; depths of about 12 feet are at their ends. Boats going to the landing must avoid the reef that uncovers about 5 feet extending northward from the island; it is marked by a buoy.

(107) **Inner Heron Island Ledge**, 0.2 mile southwestward of the south end of Inner Heron Island, is covered 2 feet and marked on the southwest side by a buoy. **The Bulldog**, 300 yards southward of the island, uncovers 3 feet. The rock 350 yards eastward of the south end of the island uncovers 6 feet.

(108) Other unmarked dangers exist between Inner Heron Island and the shore of Rutherford Island; this passage should not be used by strangers.

(109) **Christmas Cove** (43°50.8'N., 69°33.3'W.), 0.7 mile north-northeastward of Inner Heron Island, offers good protection for small craft. The narrow entrance to the cove proper is midway between two bare rocks, the one on the southeast side being marked by Middle Ledge Daybeacon 2. Daybeacon 3 marks the north side of the channel, and Steamboat Wharf Daybeacon 4 marks the point of a ledge near the south side close westward of the town landing.

(110) A high square observatory tower, about a mile northeastward of the cove, is conspicuous.

(111) The summer resort of **Christmas Cove** is on the eastern side. The village residents maintain a private sport, social and yachting club. The town landings with reported depths of 4 to 12 feet alongside are on the southeast side of the cove. There are several private float landings and moorings in the cove. **Anchoring** is reported to be difficult due to the densely packed moorings and numerous lobster pot buoys and is not recommended.

(112) The **harbormaster** for the town of Christmas Cove can be reached by telephone at 207–644–8120.

(113) A marina-restaurant with 12 feet reported alongside its float landing is on the west side of the cove. Electricity, water, pump-out and a small-craft launch are available. The marina can be contacted at 207–644–8282.

(114) **The Gut** (43°51'44″N., 69°33'24″W.) is a thoroughfare connecting Damariscotta River at South Bristol with McFarlands Cove and Johns Bay. A submerged rock ledge is reported on the south side of The Gut, about 300 yards eastward of the bridge. Route 129 highway bridge over The Gut is a bascule bridge with a horizontal span of 28 feet, a vertical clearance of 5 feet when closed, and 55 feet when opened. Overhead power and telephone cables at the bridge have a least clearance of 55 feet. Daybeacons mark ledges on the south side of the western entrance and on the north side just west of the bridge. The Gut east of the bridge is thickly congested with moorings and lobster pot buoys, but the harbormaster keeps a 100-foot channel clear.

(115) A small-craft launching ramp is 250 yards eastward of the bridge on the north side. A marina is on the north side of The Gut, west of the bridge. The marina has two finger piers with limited transient dockage. Gasoline, diesel fuel, water and a lift to 25 tons are available. The marina can be contacted at 207–644–8181 during business hours and weekends.

(116) **South Bristol** is a village on The Gut. There are a number of wharves with float landings. Four on the north shore east of the bridge are lobster wharves with depths of 4 to 12 feet reported alongside their floats; gasoline and diesel fuel are available. A general store is on the wharf by the bridge. Some marine supplies, ice and provisions may be obtained. The town wharf on the south shore close west of the bridge has a reported depth of 3 feet alongside. The town **harbormaster** may be contacted at 207–644–8120.

(117) **East Boothbay** is a village on the west bank of Damariscotta River, about 3 miles above the mouth. A church spire, lighted at night, and the large buildings of three boatyards are prominent. Three wharves are in general use and have float landings and berthing space with water and electricity and 10 feet reported alongside. The yards maintain guest moorings in the anchorage off the wharves; the controlling depth is about 7 feet in the anchorage.

(118) The yards can build craft up to 200 feet in length and 1,000 tons and are equipped with complete facilities for hull and engine repairs. Machine, carpenter and pipe shops a sail loft and two marine railways are available. The larger of the railways can handle craft up to 100 feet. Gasoline, diesel fuel, water, ice, provisions, marine supplies, pump-out facilities, a 35-ton travel lift and open, covered, wet and dry winter storage are available at the yards. The marina can be contacted at 207–633–0773.

(119) **Kelp Ledge**, just south of the approach to the boatyards, and 150 feet from the shore, is awash at low water and is marked by a buoy north of the ledge.

(120) At **The Narrows**, 1.3 miles above East Boothbay, the channel is contracted to a width of 100 yards, and the tidal currents are strong with swirls. **Western Ledge**, with a rock awash at low water 550 yards south of Fort Point, in midchannel, is marked by a buoy to the southeast; the buoy tows under at full current strength.

(121) **Eastern Ledge**, extending 100 yards from the eastern shore, is a rock covered 2 feet. A buoy marks its southwest side. This buoy almost tows under during full strength of the current. On the west side of The Narrows is a ledge,

mostly covered and with rocks awash on it, extending 250 yards southwestward and 75 yards eastward from Fort Point. There are other ledges, one covered 4 feet, in this vicinity.

(122) At the **Back Narrows** leading westward of Fort Island the channel is foul with rocks. Fish wharves and private float landings are in the two coves westward of Fort Island.

(123) **Seal Cove** and **Long Cove**, on the east side just above The Narrows, have many unmarked dangers and are seldom entered.

(124) **Carlisle Island** is a low island close off the east side of Carlisle Point about 2 miles above The Narrows. The channel between the island and the point is not recommended because of an unmarked 2-foot spot at its southern end.

(125) **Miller Island**, a low wooded island in midchannel east of Carlisle Point, divides the river into two channels. The western channel is the more direct but has an 18-foot spot at its northern end. The eastern channel is deep and passes close to Clark Cove.

(126) **Clark Cove**, on the east side, 2.5 miles above The Narrows, is a broad bight, shoal near the shores.

(127) **Pleasant Cove** is on the western shore of the river opposite Clark Cove, and makes in nearly 1.5 miles southwestward. Good anchorage can be had in the mouth of this cove just northwestward of **Carlisle Point**, in 15 to 30 feet, soft bottom. **Pleasant Cove Ledges**, extending northward of the cove, uncover 8 feet and are marked by a buoy at the north end. There is a private float landing in the cove.

(128) **Lowes Cove** indents the east shore for about 800 yards between **McGuire Point** and **Wentworth Point** but dries out for most of its length. It is only about 100 yards wide. Anchorage in 15 feet can be had in the entrance.

(129) Anchorage can also be had behind Pleasant Cove Ledges on the west side in Wadsworth Cove.

(130) **Kelsey Point**, about 1 mile north of Wentworth Point, is low, but the land behind it rises abruptly to about 160 feet. A rock off Kelsey Point is covered 2 feet and is marked by a buoy.

(131) **Salt Marsh Cove**, on the west side southwest of Kelsey Point, dries out. **Merry Island**, off the western shore northwestward of Kelsey Point, is wooded. A daybeacon is on a bare rock off the island.

(132) **Mears Cove**, eastward of Merry Island and between Kelsey Point and **Lower Fitch Point**, affords excellent anchorage in 20 to 25 feet.

(133) **Fitch Point** is a low point making out from the east shore about 5.5 miles above The Narrows. Small **Baker Islet** is on **Glidden Ledge**, which extends about 350 yards from Fitch Point. A daybeacon is on the outer end of the ledge. The river channel is only about 100 yards wide at the point, and strong tidal currents are reported to sweep across the ledge and through the channel on the ebb.

(134) **Dodge Point** is a high bluff headland 1.2 miles above Fitch Point. **Perkins Point**, 100 feet high and cleared, is on the west shore about 1 mile above Dodge Point. A buoy marks the channel off the point, and a daybeacon marks the shoal water 0.4 mile northward of the point.

(135) About 0.8 mile above Perkins Point, the river is again narrowed to about 100 yards by **Goose Ledge**, which extends 0.3 mile southward of Hall Point on the east shore. **Hog Island**, a small wooded island, is in the entrance to **Huston Cove**, eastward of **Hall Point**. The cove dries out.

(136) Between Hall Point and **Little Point** on the west bank, the river is only about 250 yards wide and the channel less than 100 yards wide. A strong ebb tidal current is reported to run between the two points.

(137) A midchannel drying bank is northeastward of Little Point. The channel leads eastward of the shoal and is marked on the western edge. The channel then trends northward to **Jacks Point** and to the anchorage off the towns of Damariscotta and Newcastle.

(138) **Damariscotta** on the east bank and **Newcastle** on the west bank, about 14 miles above the mouth of the river, are connected by the Damariscotta River Bridge (U.S. Route 1 bypass). The bridge has a fixed span with a clearance of 5 feet; a center pier in the bridge obstructs the channel. Old Indian shell mounds are on the west bank on **Glidden Point** 1 mile above the bridge. The Glidden Point Bridge (U.S. Route 1) crosses the river at Glidden Point. The fixed span has a clearance of 31 feet. The river between the bridges is obstructed by reversing rapids; passage is possible at high water slack. Newcastle is on a Class II railroad. The towns have banks, a hospital, motels, hotels, inns, restaurants, markets, laundromats and shops of all kinds. Taxi and through coastal bus services are available. The **harbormaster** can be reached at 207–563–3398 or on weekends at 207–563–3504.

(139) There is little traffic by water except for yachts and small fishing boats.

(140) A small-craft launching ramp is on the east bank just below the bridge. The town landing and municipal parking lot are adjacent to the launching ramp.

(141) Small craft can pass under U.S. Route 1 highway bridge at high water slack.

(142) A boatyard, on the west bank in the cove below Jacks Point, builds craft up to 35 feet long. The yard has a marine railway that can haul out craft up to 35 feet in length at high water for hull and engine repairs or dry open or covered winter storage. Marine supplies are available; gasoline and diesel fuel can be obtained by truck. The float and the marine railway dry at low water.

(143)

Anchorages

(144) Anchorage in 11 feet, soft bottom, is available off the landings.

(145) **Booth Bay** and **Linekin Bay** are between Linekin Neck and Fisherman Island on the east and **Southport Island** on the west. They form the approach to the town

of Boothbay Harbor and many summer resorts. They are frequented by many vessels and by a large number of fishing and pleasure craft in summer.

(146) Islands and rocks extend about 5 miles southward from the south end of Linekin Neck. The ground is very broken, rocks rising abruptly from deep water.

(147) **Bantam Rock**, awash at low water, the most southerly visible danger, is 1.3 miles southward of Damariscove Island. The wreck of the SS HARTWELSON, broken in two parts on Bantam Rock, is no longer visible. It is marked by a lighted buoy.

(148) **Damariscove Island**, on the southeast side of the entrance to Booth Bay, is 1.7 miles long, bare, and nearly divided in the middle. **Damariscove Harbor**, at the south end, is used as a small-boat harbor by local fishermen. Conspicuous objects are two lookout towers and the buildings of a former Coast Guard station on the highest parts of the southerly section of the island.

(149) A fairway gong buoy is 0.5 mile south of the entrance to the harbor. **The Motions**, a ledge extending 0.3 mile south-southwestward of the southwest end of Damariscove Island, is awash at low water. An unmarked shoal cleared 32 feet is 0.8 mile southward of the southeast end of the island.

(150) **Poor Shoal**, covered 33 feet and unmarked, is 1.7 miles south of the island.

(151) **Fisherman Island**, northeastward of Damariscove Island, is bare. A large stone house on the highest part of the north section of the island is prominent.

(152) **Ram Island**, on the south side of Fisherman Island Passage, is a grassy island marked on the northwest side by **Ram Island Light** (43°48'14"N., 69°35'57"W.), 36 feet above the water, shown from a white tower. A mariner radio activated sound signal at the light is initiated by keying the microphone five times on VHF-FM channel 83A. The light has two white sectors that cover two approaches to Fisherman Island Passage; the eastern from 258° to 261°, and the southwestern from 030° to 046°.

(153) **The Hypocrites** is a long ledge with two low bare rocks eastward of Fisherman Island. A buoy marks the north end, and a daybeacon is at the south end. There is an unmarked channel between The Hypocrites and the ledges, which extend 500 yards eastward of Fisherman Island. The southerly part of The Hypocrites was formerly known as **Smedrick Ledge.**

(154) **The Cuckolds** are two bare islets off **Cape Newagen**, the southern extremity of Southport Island, on the west side of the entrance to Booth Bay. The westerly islet is 12 feet high and the easterly 10 feet high. The easterly islet is marked by **The Cuckolds Light** (43°46'46"N., 69°39'00"W.), 59 feet above the water, shown from a white octagonal tower on a dwelling. A mariner radio activated sound signal is at the light, initiated by keying the microphone five times on VHF-FM channel 83A. When approaching The Cuckolds, the easterly islet is more prominent and appears to be the larger and higher of the two.

(155) **Cape Harbor**, between Cape Island and Cape Newagen, accommodates small craft; yachts and fishermen use it mostly. **Cape Island** is wooded in the center. **Newagen** is a village on the harbor. There are two entrances to the harbor. The easterly one, leading between **The Ark** and Cape Newagen, reported to have a depth of 3 feet, is used by fishermen in good weather but should not be attempted by strangers without local knowledge.

(156) The main entrance, from the westward between Hunting Island Daybeacon 4 and the shore, has a depth of about 10 feet. Pass north of Hunting Island Daybeacon 4 because the passage between **Hunting Island** and Cape Island is foul. A buoy marks the west side of the ledge that extends southwest from Hunting Island.

(157) Depths in the harbor are from 6 to 16 feet. There are a town wharf and float landing with 2 feet alongside and a service pier with gasoline available that has 3 to 6 feet alongside. There is a large summer inn in the village, and there are also many summer homes. The inn also maintains a float landing to which water is piped in summer on the southwest side of the harbor.

(158) **Squirrel Island**, in the middle of Booth Bay, is an important summer resort. It is wooded and has many large homes. Water pipelines, submarine power cables, and telephone cables extend to the north end of the island from the southern tip of Spruce Point. The ferry from Boothbay Harbor lands passengers, mail and freight at a float in the northerly of the two coves on the west side of the island. A ledge extending northwestward from the island is marked by a lighted buoy.

(159) **Squirrel Cove**, the southerly of the two coves on the west side, is sometimes used as an anchorage by small craft. A float landing in the cove has 8 to 10 feet alongside. A daybeacon marks the ledge at the south side of the entrance.

(160) **Linekin Bay**, the northeasterly arm of Booth Bay, is northeastward of Squirrel Island and between Linekin Neck and Spruce Point. The principal dangers are buoyed. Good anchorage can be found, the depths being 40 to 75 feet in the lower part of the bay and 30 to 36 feet in the upper portion. There are several private float landings.

(161) **Spruce Point Ledges**, awash at low water, are in the middle of the entrance; they are marked by two buoys at the south and north ends. The preferred channel is outside the southern buoy.

(162) In the narrow channel between the northern buoy and Spruce Point, give the point a berth of over 150 yards. A **028°** course with the southeast point of Squirrel Island astern will lead through the southern channel, thence **024°** to the head of the bay.

(163) **Ocean Point**, the point and village at the southern entrance to Linekin Bay, is marked by many summer homes and hotels. A depth of 3 feet is reported 275 yards westward of the point. A public wharf and float landing with 10 feet reported alongside is maintained in **Card Cove**, 700 yards north of the point. A ledge, which partially uncovers at low water, extends about 150 yards

from shore just southward of the wharf; mariners are advised to use caution when approaching the wharf.

(164) South and southwest of Ocean Point, **Card Ledge, Dictator Ledge**, and **Gangway Ledge**, the main dangers in Fisherman Island Passage, are buoyed. Passage through the area between the buoys and Ocean Point should not be attempted because of the numerous dangers with little water over them. Broken bottom extends southwestward of Dictator Ledge to **Wylie Rock.**

(165) The principal dangers in Linekin Bay above Spruce Point Ledges, from south to north, include **Tibbits Ledge**, covered 8 feet and marked on its southwestern side by a buoy; **Cabbage Island**, wooded and with a house in the center, and the buoyed ledge that extends south from it; **Holbrook Ledge**, which uncovers 3 feet and is marked on its northwest side by a buoy; a rock covered 12 feet 200 yards westward of the south end of Holbrook Ledge; **Seal Rock**, awash at low water and marked off the southeast side by a buoy; a depth of 19 feet about 150 yards east of the buoy; a rock awash at low water reported 120 yards northward of Seal Rock, which several boats have reported striking; and a ledge on the east side surrounding **Perch Island** marked by a buoy at the southwest end.

(166) **Fish Hawk Island**, about 0.4 mile northward of Seal Rock, has several trees, and a ledge, which uncovers about 4 feet, extends southward of it. The narrow unmarked channel westward of Seal Rock should be used with caution. There are numerous unmarked rocks at the head of the bay. **Spruce Point**, the north entrance point to Linekin Bay, is wooded.

(167) East of Tibbits Ledge is a yacht yard which builds craft up to 65 feet in length and manufactures marine hardware. The yard has a marine railway and a machine shop but does not solicit repair work. There is a depth of 7 feet at its float landing; the yard maintains guest moorings.

(168) **Capitol Island** (43°49.4'N., 69°39.0'W.), on the west side of Booth Bay, is connected at its northern end by a footbridge to Southport Island. There is a private float landing at the bridge. Capitol Island, a summer colony, is on the island. Daybeacons mark the ledges off the south and east sides of the island.

(169) **Pig Cove**, between the island and Southport Island, has anchorage in 11 to 63 feet for three-fourths of its length but is shoal and foul at its northern end above the narrows. Fish wharves, a lobster pound, and a number of private float landings are in the cove. There are no facilities.

(170)

Boothbay Harbor to Westport Island

(171) **Boothbay Harbor**, the western arm of Booth Bay, is one of the best anchorages on the Maine coast. The harbor is spacious and well sheltered and has good holding ground. The town of **Boothbay Harbor**, at the head of the harbor, is an important summer resort and yachting center, with a hospital, hotels and motels. Fishing, boatbuilding and summer tourists are its main industries. A number of excursion, sightseeing, charter and party fishing boats operate from the harbor to the outlying islands and surrounding waters in the summer.

(172)

Prominent features

(173) **Burnt Island**, partly wooded, is marked on the southeast side by **Burnt Island Light** (43°49'31"N., 69°38'25"W.), 61 feet above the water, shown from a white conical tower with covered way to a dwelling. A mariner radio activated sound signal at the light is initiated by keying the microphone five times consecutively on VHF-FM channel 83A. White sectors in the light from 307° to 316° cover the fairway in the approach eastward of Squirrel Island from Fisherman Island Passage and from 355° to 008°, the approach westward of the island from the south and westward.

(174) **Mouse Island**, northward of Burnt Island, is wooded; it has a private float landing on the north side with a depth of about 12 feet and a pier and float landing on the east side. A flagstaff on the east side of the island and several homes are prominent.

(175) The tower and buildings of the Maine Department of Marine Resources fish hatchery and laboratory on **McKown Point** and the footbridge across the head of the harbor are conspicuous. A tower with a flashing red light above the harbor was also reported to be very prominent.

(176)

Channels

(177) Two deep natural channels lead into the harbor. The easterly and widest leads between Spruce Point on the east and Squirrel, Burnt and Mouse Islands on the west. The westerly one leads between those islands and Southport Island on the west but is narrow in places. Most of the dangers are marked and have been described. The chart and the aids if carefully followed should be sufficient guidance for strangers to enter at any time.

(178)

Anchorages

(179) Anchorage can be found in 24 to 42 feet for large vessels in the outer harbor northward of Tumbler Island and off McKown Point. The inner harbor has depths of 6 to 24 feet. The anchorage most used by small craft is on the northwest side of the inner harbor, northeastward of McFarland Island, where there are general depths of 10 to 12 feet, when clear of the ledge around the island.

(180) Most craft anchor off the wharves, but there are numerous private moorings, guest moorings maintained by the yacht clubs, and those for hire by the various service facilities. However, it is sometimes difficult to secure adequate swinging room.

(181)

No-Discharge Zone

(182) The State of Maine, with the approval of the Environmental Protection Agency, has established a No-Discharge Zone (NDZ) for the waters of Boothbay Harbor.

(183) Within the NDZ, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZ, discharge of sewage is regulated by **40 CFR 140** (see chapter 2).

(184)

Dangers

(185) The approaches to the harbor are generally deep and clear with most of the dangers marked. **Tumbler Island Ledge**, off the west side of Spruce Point, covered 9 feet, is marked on its west side by a buoy. A lighted buoy, about 225 yards west-northwestward of **Tumbler Island**, marks the ledges extending westward and northwestward of the island. The wooded island has a house and a prominent flagpole on it, and a pier with float landing extends from its northeastern end.

(186) The passage between Tumbler Island and Spruce Point should not be attempted by strangers as it is shoal and foul; strangers should not anchor there.

(187) **Clam Rock**, about 700 yards northeastward of Tumbler Island, close to shore, is unmarked, as are 10- and 12-foot rocky ledges, 250 yards southwestward and 150 yards westward, respectively, of the rock. A 14-foot rocky ledge, about 300 yards southwestward of **McFarland Island**, is unmarked, but the ledges surrounding the island are marked on the south side by a lighted buoy.

(188)

Caution

(189) In summer the inner harbor is nearly filled with all types of fishing and pleasure craft. At night, many of these are often unlighted, and great care should be exercised in approaching the anchorage to avoid fouling them or any of the numerous unoccupied moorings, which also are often unlighted. The footbridge across the head of the harbor has a small drawspan with a clearance of 4 feet.

(190)

Current

(191) Tidal currents have little velocity in the harbor.

(192)

Ice

(193) In severe winters, ice occasionally obstructs navigation above Tumbler Island during February and March. In normal winters the harbor is free of ice to the footbridge.

(194)

Pilotage, Boothbay Harbor

(195) Pilotage is compulsory for all foreign vessels and U.S. vessels under registry with a draft of 9 feet or more. Pilotage is optional for fishing vessels and vessels powered predominantly by sail. The pilots' address is Penobscot Bay and River Pilots Association, 18 Mortland Rd, Searsport, ME 04974; telephone, 207-548-1077; email: pilots@penbaypilots.com. The pilot station monitors radiotelephone VHF-FM channels 13 and 16 when expecting traffic.

(196) The pilot boarding stations will be coordinated by the pilot association in advance. The pilot boarding location varies according to the sea condition. The pilot boat will display the standard running lights. When needed, the ship's pilot ladder should be rigged one meter above the water. A 48-hour and a 24-hour advance notice of arrival is requested from the vessel's agent.

(197)

Towage

(198) Tugs are stationed at Bath and Southport. Contact the pilots for tug service.

(199) A **hospital** with pier and float landing is about 0.4 mile north-northeast of McKown Point.

(200) A **Boothbay Harbor Coast Guard Station** is on McKown Point.

(201)

Harbor regulations

(202) Harbor regulations and moorings in the harbor are under the supervision of the **harbormaster,** who can be reached through the town office, through any of the service facilities along the waterfront or on VHF-FM channel 6 or 16. A **speed limit** of 5 knots in the harbor is enforced.

(203)

Wharves

(204) There are service wharves and marinas, almost all with float landings, that have reported depths of 4 to 15 feet alongside. A town float landing with a reported depth of 6 feet alongside is at the draw of the swing footbridge at the northeastern end of the harbor; another town landing is on the west side of the harbor. Piers and buildings of several seafood processing plants are along the easterly shore of the harbor.

(205) The Boothbay Harbor Yacht Club operates from float landings on the south shore of the village of West Boothbay Harbor, northward of McKown Point; depths of 12 feet are reported alongside the landings. The club maintains several guest moorings.

(206)

Small-craft facilities

(207) There are excellent shipbuilding, boatbuilding and small-craft repair facilities along the entire town waterfront in the eastern part of the harbor. Pump-out facilities are available.

(208)

Communications

(209) Taxi service, both local and to coastal bus service at Wiscasset, is available. Ferry service to the islands is maintained throughout the year.

(210) The Inside Passage from Boothbay Harbor to Bath is about 11 miles long and leads between the islands located between Boothbay Harbor and Kennebec River. The protected route is used by excursion boats, yachts and fishing boats.

(211) The aids are colored and numbered for passage westward. In the vicinity of Cameron Point Light 7, on the north end of Southport Island, is one of the most difficult places to make in the thoroughfare; craft entering

from the westward at this point should be careful to pass southward of the buoy marking the ledge extending southward from Indiantown Island.

(212) The channel is very narrow in places, has strong tidal currents, and is much obstructed by rocks and shoals. Though most dangers are marked, strangers drawing 7 feet or more should not attempt it at low water. The passage leads through Townsend Gut, across Sheepscot River and through Goose Rock Passage and Knubble Bay into Sasanoa River.

(213) Goose Rock Passage is marked by a directional light, buoys and a daybeacon. About midway through Sasanoa River the channel crosses the southern part of Hockomock Bay and then continues through Sasanoa River, coming out in the Kennebec River opposite the city of Bath. In 1958, the least depth in Sasanoa River was 7 feet at the southern end of **Hanson Bay** and near the northern entrance to the river. In the spring logs and driftwood may be present.

(214) Two highway bridges cross the thoroughfare. State Route 27 highway bridge at Townsend Gut has a swing span with a clearance of 10 feet. (See **33 CFR 117.1** through **117.59** and **117.537**,chapter 2, for drawbridge regulations.) State Route 127 highway bridge over Sasanoa River near its junction with Kennebec River has a fixed span with a clearance of 51 feet. An overhead power cable at the bridge has a clearance of 75 feet.

(215)

Routes

(216) This passage is narrow and crooked, has strong tidal currents and requires local knowledge to carry the best water. Strangers on larger vessels or yachts should pick up a pilot at Boothbay Harbor or Bath. With the aid of the chart, strangers in small craft drawing 7 feet or less should be able to go through. The best time is on a rising tide. The channel is well marked but careful navigation is required.

(217)

Caution

(218) At strength of current in the narrow places, the buoys are often run under for short periods.

(219) The thorofare is usually closed by ice for about 2 months, but in mild winters it has been known to remain open all winter. Several summer resorts and other landings are along the route.

(220) **Townsend Gut** is a narrow, crooked thoroughfare connecting Boothbay Harbor with Sheepscot River. The shores of Townsend Gut are lined with private docks and floats. Mariners are advised to use prudent speed to avoid wake damage. A **speed limit** of 5 knots is enforced through the State Route 27 highway swing bridge at Townsend Gut. There are unmarked rocks with little depth close to the channel. A rock, covered 5 feet and marked by a buoy, is about 100 yards southeastward of the swing bridge. This rock can be cleared by keeping lined up with the center island, but avoid being set to the eastward while waiting for the bridge to open.

(221) **Deckers Cove**, on the west side of Townsend Gut about 0.4 mile above the southern entrance, is crossed by State Route 27 highway bridge which has a 15-foot fixed span with a clearance of 6 feet. East of the north end of the bridge is a former fish wharf with a depth of 17 feet alongside, at which large yachts are moored for winter storage. There are several boatsheds and float landings in the cove.

(222) **Southport** is a village and summer resort on the west side of the gut near the western end. There are numerous float landings on both sides.

(223) **Hodgdon Cove**, on the northeast side of the Gut opposite Southport, is shoal and foul at the head and around the edges with numerous sunken rocks but affords good sheltered anchorage in from 12 to 27 feet, mud bottom, in the middle of the outer part of the cove.

(224) **Moffat Cove** is just northwestward of Hodgdon Cove on the east side of the Gut. Good anchorage in 10 to 22 feet with a mud bottom is available.

(225) **Cameron Point Light 7** (43°51'05"N., 69°40'06"W.), 24 feet above the water, shown from a white skeleton tower with a green square daymark, marks the ledge extending northward from **Cameron Point**, the northern extremity of Southport Island. A ledge that uncovers 1 foot extends southward from **Indiantown Island** and is marked at the south end by a buoy.

(226) **Isle of Springs**, 0.6 mile northwest of Cameron Point, is a summer resort at the north end of Townsend Gut. The island is wooded and has an elevated tank at its summit. The ledge extending off the north end of the island is marked by a daybeacon. There is a wharf with float landing, with 10 feet alongside, on the northeastern side of the island from which, in summer, a private motorboat ferry runs north to Clam Cove, at the southwest end of Sawyer Island.

(227) **Sawyer Island**, northward of Isle of Springs, is connected to the mainland by a highway bridge at its southeast corner that has a fixed span with a clearance of about 2 feet. The current is strong in this locality. It is also connected at the northeast end to the south end of Hodgdon Island by a fixed highway bridge (under construction) that has a 35-foot fixed span with a clearance of 6 feet.

(228) **Goose Rock Passage** leads from Sheepscot River into Sasanoa River northward of MacMahan Island, and forms a part of the inside route. It has ample depth but is narrow in places; principal dangers are marked.

(229) At the western end of the passage, **Goose Rock Passage Light 5** (43°50'56"N., 69°43'15"W.), 16 feet above the water and shown from a white triangular tower with a green square daymark on a caisson, marks the best water through the passage and into Knubble Bay. **MacMahan Island Ledge**, a drying reef off **Northeast Point** on **MacMahan Island**, is marked by a daybeacon. **Sixfoot Rock**, off the northwest corner of the island, is marked by a buoy on its north side.

(230) **Boiler Rock**, covered 3 feet and marked by a buoy on its southeast side, is at the western end of the passage. This buoy is reported to tow under during the strength

of the current. **Goose Rock**, a bare rock on a ledge that uncovers and gives the passage its name, is about 150 yards northwestward of Boiler Rock. Intensified beams of Goose Rock Passage Light 5 mark the best water through the passage and northwestward into Knubble Bay and Sasanoa River.

(231) **Little Sheepscot River** is a narrow passage westward of MacMahan Island leading from Sheepscot River into Sasanoa River. The channel is narrow, being less than 50 yards wide at its narrowest part. The best entrance from the southward is west of **Turnip Island**. Craft of more than 4-foot draft should avoid passing through the channel between Turnip Island and the southern end of MacMahan Island at low water.

(232) Little Sheepscot River is marked by a buoy about midway through the passage, west of **MacMahan Ledge**. Another buoy at the northern end, marking Sixfoot Rock, should be passed well to westward when entering Goose Rock Passage from Little Sheepscot River. There is an unmarked drying ledge, with two rocks that uncover 4 feet, on the west side of the channel, about 200 yards south-southwest of the buoy marking MacMahan Ledge. Two float landings are on MacMahan Island eastward of it.

(233) **MacMahan** is a summer resort on the west side of MacMahan Island.

(234) **Sasanoa River**, part of the Inside Passage from Booth Bay to Bath, is an estuary leading from Sheepscot River to Kennebec River, north of Georgetown and Arrowsic Islands. It has numerous coves and bays, none of which are of commercial importance, making off northward and southward. The general trend of this river is northwest and southeast.

(235) The principal coves and bays making southward are **Robinhood Cove**, **Riggs Cove**and **Hall Bay**. Northward are **Heal Cove** and **Hockomock Bay**, **Montsweag Bay** and **Brookings Bay** lead northward from Hockomock Bay. Montsweag Bay separates Westport Island from the mainland and joins the Sheepscot River at Wiscasset through Back River.

(236) **Knubble Bay** is the broadest part of the river after passing Robinhood Cove and the **Knubble**, before entering Hockomock Bay when coming from eastward.

(237) **Lower Hell Gate** is the crooked passage from Knubble Bay into Hockomock Bay. **Upper Hell Gate**, about 2 miles from the western entrance to the river, is the narrowest part, about 60 yards wide. A daybeacon marks Lime Rock on the southwest side of Upper Hell Gate.

(238) **Halftide Ledge**, about 400 yards southeastward of Upper Hell Gate, is marked by a daybeacon. In 1958, shoaling to 4 feet was reported about 50 yards south of the daybeacon. A rock awash at low water was also reported on the southern side of the channel about 150 yards 190° from the daybeacon. **Extreme caution** should be exercised in this area.

(239) The Sasanoa River is a Marine Protected Area (MPA).

(240)

Currents in Sasanoa River

(241) The velocity of the tidal current at strength is 1.8 knots off Lowe Point; 3.0 knots on the flood and 3.5 knots on the ebb at Lower Hell Gate; and about 1.0 knot at Upper Hell Gate. Velocities up to 9.0 knots have been observed in the vicinity of **The Boilers** at Lower Hell Gate causing dangerous eddies and whirlpools; navigation through this area should be attempted only at or near slack water. The current floods to the northwestward and ebbs southeastward generally. It has been reported that the ebb current sometimes runs for 8 or 9 hours at Upper Hell Gate. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov*. for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(242) **Robinhood** is a village on the western side of the entrance to Robinhood Cove. There is a marina and yacht yard with a wharf and floats on the south side of Riggs Cove at the village. The yard has a 40-ton mobile lift and a 5-ton hoist and can make hull, engine and electrical repairs. Gasoline, diesel fuel, water, ice, berthing, marine supplies and storage facilities are available. Depths of about 10 feet are reported alongside the wharf and floats.

(243) There is good anchorage in 20 to 70 feet, blue clay bottom, northeastward of the wharf. The harbor is reported to be free of ice.

(244) **Blacksmithshop Ledge**, eastward of Robinhood on the west side of the entrance to Robinhood Cove, uncovers 2 feet and is marked by a daybeacon. **Back River**, which also connects Sheepscot River to Kennebec River, crosses Sasanoa River at Hockomock Bay. This river has a general north-south direction. South of Hockomock Bay the river separates **Arrowsic Island** and **Georgetown Island**, is unmarked and is crossed by a fixed highway bridge with a clearance of 6 feet. The river shoals between the bridge and Hockomock Bay.

(245) North of Hockomock Bay a natural channel, marked by private buoys, leads through Montsweag Bay, the upper part of the river, and Cowseagan Narrows, separating Westport Island from the mainland, and joins Sheepscot River just below Wiscasset.

(246) It is reported that the incoming tide up Montsweag Bay meets the tide of Back River in the vicinity of **Young Point** (43°56.3'N., 69°42.6'W.).

(247) Currents are strong and erratic through Back River and in the vicinity of the fixed Cowseagan Narrows Bridge, clearance 48 feet, that crosses **Cowseagan Narrows** about 2 miles south of Wiscasset. The ledges and shoals in the narrows make the channel quite narrow at this point. Mariners are advised that passage through the narrows should not be attempted without local knowledge, and then only by small boats at slack water.

(248) **Westport Island** is 9 miles long, about 1.7 miles wide and wooded. It has little commercial importance. There are a number of summer homes and camps on the

island. A general store is on the main road about 1.5 miles south of the bridge across Cowseagan Narrows. Gasoline, provisions and some supplies can be obtained there.

(249) Abandoned ferry landing ramps remain on both sides of Back River about 1 mile south of the bridge.

(250)

Sheepscot River

(251) **Sheepscot River** is the approach to several small villages in the lower end and to the city of Wiscasset, 14 miles above the entrance. The entrance is about 5 miles northeastward of Seguin Island, between The Cuckolds and Griffith Head (43°47.0'N., 69°43.4'W.).

(252)

Recommended minimum under-keel clearances for Sheepscot River

(253) The U.S. Coast Guard, in cooperation with the Maine and New Hampshire Port Safety Forum, has established recommended minimum under-keel clearances for Sheepscot River, in order to prevent groundings and to promote safety and environmental security of the waterway resources of Sheepscot River. The group recommends that all entities responsible for safe movement of vessels in and through the waters of Sheepscot River operate vessels in such a manner as to maintain a minimum under-keel clearance of 2 feet between the deepest draft of their vessel and the channel bottom when transiting the river inside Entrance Lighted Buoy 2SR and a minimum under-keel clearance of 1 foot at all berthing areas.

(254)

Recommended Vessel Route

(255) The U.S. Coast Guard Captain of the Port, Sector Northern New England, in cooperation with the Maine and New Hampshire Port Safety Forum, has established a Recommended Vessel Route for vessels entering and departing the Sheepscot River. Vessels are requested to follow the designated route. Local fisherman in this area normally standby on VHF-FM channels 7 (156.350 MHz) and 10 (156.500 MHz). This route was designed to provide a safe, established route for increased vessel traffic, to prevent the loss of fishing gear placed in the waters of, and approaches to, the Sheepscot River and to reduce the potential for conflicts between larger, less maneuverable vessels and all other vessels navigating upon these waters. Vessels are responsible for their own safety and are not required to remain inside the route, nor are fisherman required to keep fishing gear outside of the recommended vessel route.

(256)

Channels

(257) The channel in Sheepscot River is deep, and the principal dangers are marked. It is a region of rocks and ledges, many of them rising abruptly from deep water. The channel has a depth of over 30 feet to Wiscasset and is navigable for small craft at high water for about 4 miles above Wiscasset to the village of Sheepscot. There is a 25-foot shoal in midchannel in the bend below Wiscasset, about 270 yards south-southwestward of the tower of old **Fort Edgecomb** on the southwestern end of **Davis Island.**

(258)

Anchorages

(259) Ebenecook Harbor is the first anchorage available for vessels drawing up to 20 feet entering the river. Above **Stover Ledge**, about 2 miles north-northeast of Northeast Point on MacMahan Island, anchorage can be had in the channel, the depths being usually 72 feet or less. **Colby Cove**, in the west bank about 2.2 miles above Cross River, affords anchorage in 47 to 60 feet; Merril Ledge is northeastward of the anchorage. To reduce risk of vessel collisions and increase safety of life and property in Edgecomb, the Coast Guard establishes a **general anchorage area** just off the southwestern end of Davis Island. (See **33 CFR 110.1** and **110.131**, chapter 2, for limits and regulations.) The anchorage at Wiscasset is below the bridge near the town wharves in 28 to 50 feet.

(260) The **Sheepscot River Closed Area**, a Marine Protected Area (MPA), includes the waters of the Sheepscot River north of Wiscasset.

(261)

Dangers

(262) There are several unmarked rocky areas with depths of 20 to 30 feet near the middle of the river southward of Bull Ledge. The river should be navigated with extreme caution. With the aid of the chart and by following the aids, little trouble should be experienced in reaching Wiscasset. Detailed information on the dangers in Sheepscot River is given with the description of the river.

(263)

Pilotage, Sheepscot River

(264) Pilotage is compulsory for all foreign vessels and U.S. vessels under registry with a draft of 9 feet or greater. Pilotage is optional for fishing vessels and vessels powered predominately by sail. Pilot services are available to all vessels. The pilot boarding station is located in the vicinity of Buoy 2SR. (See Pilotage, Boothbay Harbor, for additional information concerning arrangements for pilot and tug services, and communications.)

(265)

Current

(266) The tidal currents in the river generally set in the direction of the channel and have considerable velocity in the narrow parts. At the entrance of Cross River the flood sets onto **Quarry Point**. The ebb sets onto **Clough Point**. On the falling tide a strong set to westward is felt near Bull Ledge and a strong set to the eastward near Middle Ledge. These sets are not noticeable on a rising tide. There is a strong ebb current near the entrance to Cross River. Off Barter Island the tidal current has an average velocity at strength of about 1 knot. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov*. for specific information about times, directions, and

velocities of the current. Links to a user guide for this service can be found in chapter 1 of this book.

(267) **Ice** usually does not interfere with navigation below Wiscasset. The river above Wiscasset is usually closed in winter.

(268)

Wharves

(269) The only deepwater wharf on the river is at the powerplant at Wiscasset. Wharves for small craft are at the small ports along the river, and information on them is given in the description of the river.

(270) The Cuckolds Light and Cape Harbor, on the east side at the entrance, were described previously. Rocks, bare and covered, extend 0.5 mile westward of the point in this vicinity.

(271)

Tom Rock to Hendricks Harbor

(272) **Tom Rock**, 2.4 miles northeastward of Seguin Light (43°42.5'N., 69°45.5'W.), awash at low water and marked by a buoy on the southwest side, is the outermost danger in the entrance to Sheepscot River.

(273) **The Sisters**, 0.5 mile northward of Tom Rock and 1.5 miles from the northwestern shore at the entrance to Sheepscot River, are a number of small, bare rocks on an extensive ledge area. A buoy is 0.2 mile northwestward of the ledges.

(274) **The Black Rocks**, 1 mile from the northwestern side of the entrance to Sheepscot Bay, are three groups of bare and covered rocks and ledges that extend over a distance of about 0.7 mile. The highest bare rock in the middle of the group is 15 feet high. The southern part of the ledge is reported to uncover just after the start of the ebb and should be given a wide berth. The islet on the northern group is 10 feet high.

(275) The channel between The Black Rocks and the buoy marking **Sloop Ledge**, 0.4 mile northwestward, which is covered 5 feet, should be used with caution. The area between the buoy and the northern shore is very broken and should not be crossed because of **Little River Ledges**, which are awash in places.

(276) **Griffith Head**, white and rocky, is on the west side of the entrance to Sheepscot River, about 5 miles northward of Seguin Island Light. **Outer Head**, a bare rocky islet, is 200 yards eastward. A buoy, 0.4 mile east of the islet, marks **Griffith Head Ledge**, which is covered 4 feet. Unmarked shoals, cleared to 35 and 25 feet, are on the western side of the main channel 0.8 mile and 1.3 miles northward of Griffith Head Ledge, respectively.

(277) Griffith Head and a considerable amount of the surrounding area are included in **Reid State Park**, a public picnic area, open in the summer. There are swimming beaches, bath houses, showers, restrooms and a snack bar. There are no landings. A dam and highway bridge cross the mouth of the creek at the head of the cove on the north side of the head. The cove is foul.

(278) **Lower Mark Island**, on the eastern side just inside the entrance to Sheepscot Bay, is 12 feet high, wooded and a good landmark. A ledge that uncovers 4 feet extends 400 yards eastward of the island. Broken ground with 19- and 23-foot spots extends about 0.6 mile northwestward of the island. The 23-foot spot is marked by a gong buoy. Unmarked **Cranberry Ledge**, covered 10 feet, is 0.4 mile southeastward of Lower Mark Island.

(279) **Cat Ledges** and **Dry Ledge** are a group of islets and ledges extending 0.5 to 1 mile northward of Lower Mark Island. Dry Ledge, the northwesterly end, is 4 feet high, and the southeasterly end of Cat Ledges uncovers 3 feet. The coves in Southport Island eastward of these ledges are foul and of no importance.

(280) **Harmon Harbor** is a long, narrow cove making northward on the western side of the river about 1.5 miles above Griffith Head. It has good anchorage, except during southerly gales, in 24 to 36 feet, but has a very narrow entrance between a bare ledge near the west shore and a dangerous reef, awash at low water, extending 275 yards southwestward from **Wood Island**, on the eastern side of the entrance, south of **Dry Point**. A buoy marks the southwest end of the reef. There are no public landings in the harbor. There is a prominent hotel on the west side near the middle of the harbor, and a small settlement at the head.

(281) **Five Islands Harbor**, a narrow passage between Five Islands and the western shore north of Dry Point, forms a secure harbor for small craft, with depths of 18 to 30 feet. The main entrance is northward of **Malden Island**, the largest wooded island, which is 30 feet high. A colony of summer homes is on the island, and a private float landing is on its northwestern side. Malden Island is connected to the island close westward of it by a bridge. A rock awash, north of Malden Island in the harbor entrance, is marked by a buoy.

(282) Boats also can enter the harbor from the northwestward, following the western shore and passing inside of all islands and shoals. **Crow Island Ledge**, extending west from Crow Island at the northern entrance, is marked by a daybeacon. Northwestward of the daybeacon, an unmarked ledge makes out from the Georgetown Island shore. Care should be taken to avoid it by favoring the Crow Island side of the channel slightly and passing close westward of the daybeacon. The southern entrance, nearly blocked by rocks and ledges that uncover about 4 feet, should not be used without local knowledge. There is also a clear channel from the eastward south of Malden Island.

(283) **Five Islands** is a village on Georgetown Island on the western side of the harbor. There are several float landings. A marina has depths of 6 to 10 feet reported alongside its float landings. Transient berths, gasoline and some marine supplies are available. A 10-ton fixed lift can handle craft up to 40 feet for hull and engine repairs or dry open or covered winter storage. The village landing, adjacent southward, has 12 feet alongside. Provisions can

be obtained at a store at the landings, and there is a snack bar.

(284) **Gotts Cove**, close northwestward of Five Islands Harbor, has a private wharf and two float landings, one north and one south, along the north shore. The north float has reported depths alongside to 15 feet and the south float has 6 feet. Diesel fuel, moorings and marine supplies are available at the wharf.

(285) **Cozy Harbor** is a cove on the eastern side of Sheepscot River. The entrance is 0.4 mile southeastward of **Hendricks Head Light** (43°49'21"N., 69°41'23"W.), 43 feet above the water, shown from a 39-foot white square tower on the head.

(286) The harbor is frequented by local pleasure and fishing craft and in summer by many cruising yachts. The narrow entrance channel, marked by two daybeacons and a buoy, has depths of 15 to 8 feet. The harbor, though small, is secure with depths of 2 to 8 feet in the anchorage. In 2022, the area just northward of Daybeacon 5 was shoal to bare.

(287) The Southport Yacht Club in the harbor has 4 feet alongside its float landing. A service wharf adjacent to the club landing, with 2 feet alongside its float landing, has gasoline and water.

(288) A general store, restaurant, bowling alley and telephone are on the wharf. Provisions, bottled gas, lobsters and some marine supplies can be obtained. There is a ramp; parking and picnic areas are in the vicinity.

(289) The village of **West Southport** is at the harbor. There are fish wharves and private landings in the harbor. A causeway and fixed bridge with a clearance of about 3 feet connects Southport Island with **Pratts Island** at the south end of the harbor.

(290) **Hendricks Harbor**, shoal and foul, is on the east side of Hendricks Head. There are no landings in the harbor.

(291)

Ebenecook Harbor to Greenleaf Ledge

(292) About 6 miles of Sheepscot River is shown on the chart of the inside passage from Boothbay Harbor to Bath. The chart should used as a guide going into Ebenecook Harbor or any of the channels except the main river.

(293) **Ebenecook Harbor**, making into the northwest end of Southport Island, is an excellent anchorage for vessels up to 20-foot draft. Its entrance, about 1 mile above Hendricks Head on the eastern side of Sheepscot River, leads between Dogfish Head on the south and the Green Islands on the north. It is the first large anchorage available for craft entering the river. The entrance is narrow.

(294) The southern part of the harbor divides into three arms, the outer sections of which afford good, sheltered anchorage; the inner sections to the heads are shoal and foul and should be avoided.

(295) **Maddock Cove**, the westerly arm, has a large marina and yacht yard on its eastern shore. The yard has a wharf with float landings that have 8 feet alongside. Gasoline, diesel fuel and water are available at the floats; ice, provisions, bottled gas and marine supplies are available. Overnight berthing is permitted, and the yard maintains guest moorings. The yard has a 30-ton mobile hoist that can haul out craft to 60 feet for open or covered dry winter storage. General hull, engine, electric and electronic repairs can be made, and the yard has machine, paint and carpentry shops. There is a telephone on the wharf. Anchorage can be had in midchannel off and to the northwestward of the marina in 8 to 16 feet.

(296) When entering Maddock Cove care should be taken to avoid the ledges on the east side of the cove. A buoy marks the north end of a 7-foot spot on the east side of the entrance.

(297) **Pierce Cove**, the middle arm, has several private float landings.

(298) **Love Cove**, the eastern arm, has excellent anchorages in 8 to 9 feet in midchannel southward of the submarine power and telephone cables crossing the entrance to the cove to Little Island, the smaller of the two islands on the western side of the entrance to the cove. Three private float landings are on the cove, and a guest mooring is maintained by the pilot for the area, who resides on the east shore of the cove. The head of the cove is shoal and foul.

(299)

Routes

(300) Entering Ebenecook Harbor, vessels should give the eastern shore of Sheepscot River a berth of 300 yards from Hendricks Head Light north for 1 mile to **Dogfish Head**, rocky and grass covered with a prominent domed house on the northwest end. Pass in midchannel between Dogfish Head and the southern extremity of Green Islands, avoiding a 7-foot rock patch marked by a buoy inside the entrance. Small craft may choose anchorage in any of the coves in the southern part of the harbor or, if preferred, anchor in the northern part where desirable. A rocky unmarked ledge, covered 14 feet, is about in the middle of the harbor.

(301) Extending northward from Ebenecook Harbor to Sawyer Island is a channel, affording good anchorage in places, which is used by small pleasure craft in summer. The channel is a part of the Inside Passage used by local vessels between Boothbay Harbor and Bath. Navigation of its northern part, as well as the passages between the islands and ledges on its western side, requires some local knowledge.

(302) The principal islands and rocks are wooded **Green Islands**; a rock that uncovers at low water 200 yards northeastward of them and marked by a buoy southeastward of it; a ledge, with an islet 5 feet high in its middle, between Green Islands and Boston Island; and a rock, covered 6 feet, 250 yards westward of the ledge.

(303) **Boston Island** is high and partly wooded and has two houses and a boat landing. A ledge extends about 250 yards southeast from the island. **Spectacle Islands** are

partly wooded. A ledge awash at low water is 150 yards westward of their southwest end.

(304) Townsend Gut, Isle of Springs and Sawyer Island, on the eastern side of Sheepscot River, and Little Sheepscot River and Goose Rock Passage on the western side were described previously under the Inside Passage.

(305) **Bull Ledge**, west of Dogfish Head and 1 mile northward of Hendricks Head, uncovers at the north end at low water and is marked at the south end by a buoy.

(306) **Middle Mark Island**, a small, round, bare islet 12 feet high, is in the middle of a ledge 0.3 mile long located 0.3 mile east of MacMahan Island and 1.5 miles above Hendricks Head. **Mark Island Ledge**, covered 7 feet, is 250 yards southwestward of the island. The main channel leads eastward of the island.

(307) **Middle Ledge**, 600 yards eastward of the southern side of the entrance to Goose Rock Passage, is covered 8 feet, but less depth has been reported on this ledge. A buoy marks its northern side.

(308) **Clous Ledge**, 0.2 mile eastward from wooded **Whittum Island**, at the entrance to Goose Rock Passage, uncovers about 4 feet and is marked by a daybeacon on the middle of the ledge and a bell buoy off its northern end.

(309) **Powderhorn Island**, 25 feet high and grassy, is on the eastern side of the river 2 miles above Hendricks Head. **Powderhorn South Ledge**, which uncovers 6 feet, extends 0.3 mile southward from the island and is marked at its south end by a buoy. A narrow channel is between the buoy and the north end of **Harding Ledge**, covered 5 feet and marked at its south end by a buoy.

(310) **Powderhorn Ledge**, covered 3 feet, is 200 to 350 yards northward of Powderhorn Island and is marked on its western side by a lighted buoy.

(311) **Fourfoot Rock** is on the west side of the channel about 0.2 mile northward of Clous Ledge.

(312) **Long Ledge**, 0.2 mile long and awash at low water, is about 0.2 mile westward of Fourfoot Rock.

(313) **Ram Island Ledge**, which uncovers 5 feet in spots, is on the east side of the channel and extends 0.3 mile in a north-northeasterly direction from **Ram Islands** to the entrance to Back River. A ledge that uncovers 6 feet, marked by a daybeacon and a buoy at its north end, is eastward of Ram Island Ledge. These aids also are guides to the narrow channel leading northward from Ebenecook Harbor.

(314) **Upper Mark Island**, about 0.5 mile northwestward of Ram Islands, is a low grassy islet 8 feet high from which a shoal extends 600 yards northward.

(315) **Jewett Cove** and **Long Cove** are unimportant coves on the west side of Sheepscot River westward of the entrance to Back River.

(316) **Back River** is a shallow, narrow, and unmarked stream between **Barters Island** and the mainland. Its southern entrance is on the eastern side of the Sheepscot about 3 miles northward of Hendricks Head; its northern entrance is from Cross River. Only small craft use it; local knowledge is required for its navigation. The entrance to Back River is marked by a buoy 300 yards westward of the southern end of Barters Island and a buoy 300 yards northeastward of Ram Island Ledge. Buoys and a daybeacon mark the critical points in the channel between Barters Island on the northerly side and Sawyer, Hodgdon and Merrow Islands on the southerly side.

(317) There are several private float landings on the south end of Barters Island, just inside the entrance. A bridge across the river between Hodgdon Island and the south end of Barters Island was under construction (2018). Overhead cables at the bridge have a minimum clearance of 38 feet reported.

(318) **Trevett** is a small village at the Hodgdon Island end of the drawbridge. It has a general store. A highway bridge with a clearance of 8 feet connects Hodgdon Island with the mainland. Two small-craft launching ramps are at the north and south sides of the west end of the bridge.

(319) **Merrow Island**, **Miles Island**, **Tibbet Island** and **Gooseberry Island**, all wooded, are on the eastern side of the channel in Back River. Merrow and Tibbet Islands are connected with the mainland by fixed bridges having small clearances. There is no traffic through them as the water is shoal and foul.

(320) **Tarbox Landing** is a small settlement just north of **Tarbox Cove** on the west side of the Sheepscot River. **Hodgdon Ledge**, 250 yards eastward of Tarbox Cove, uncovers 5 feet and is marked on the southeast end by a buoy.

(321) **Stover Ledge**, on the east side of the river about 0.5 mile northward of the southern end of Barters Island, uncovers 5 feet and is marked by a buoy off its southwestern edge.

(322) A **195°45'–015°45'** measured nautical mile is off the west side of Barters Island. Shore ranges about 1 and 2 miles from the north end of Barters Island mark the ends of the course.

(323) **Greenleaf Ledge**, on the west side of Sheepscot River just south of the entrance to Cross River, uncovers 5 feet and is marked by a buoy. Unmarked shoals are in the bight in the western shore westward of the ledge.

(324)

Cross River to Marsh River

(325) **Cross River** empties into the east side of Sheepscot River about 6 miles above Hendricks Head. Its entrance is marked by a lighted buoy. It has a deep channel for over 1 mile to Oven Mouth where the river is confined to a narrow channel between high cliffs.

(326) Cross River southeast of **Oven Mouth** requires local knowledge to navigate. In 1979, a rock covered 2 feet was reported on the east side of the river about 0.9 mile above Oven Mouth. Burleigh Hill Yacht Club, a boys' camp on the east side of Cross River about 1 mile above Oven Mouth, has a float landing with 10 feet alongside. There are no facilities.

(327) **Merrill Ledge**, on the east side of Sheepscot River 2.4 miles above the entrance to Cross River, uncovers

about 4 feet in the middle. The south end is marked by a daybeacon, and a lighted buoy is on the west side. The channel leads westward of it.

(328) An unmarked rock, covered 13 feet, is about 500 yards southward of **Clough Point**, the north end of Westport Island. The rock is on the west side of the channel, just eastward of a line connecting the buoy off Clough Point and the buoy just above **Hilton Point.**

(329) There is an unmarked 25-foot shoal in midchannel in the bend at Clough Point, about 270 yards south-southwestward of the tower, or blockhouse, of old Fort Edgecomb, on the southwestern end of Davis Island. **Seal Rock**, 550 yards westward of Clough Point, uncovers 6 feet and is marked on the north by a buoy. During times of strong currents the buoy is reported to tow under.

(330) Montsweag Bay and Back River form a thoroughfare from Sasanoa River and Hockomock Bay to Sheepscot River near Wiscasset. They have been previously described under the Inside Passage. The thoroughfare is hazardous because of currents which are reported to reach 5 knots on the ebb and flood. A fixed bridge at Cowseagan Narrows has a vertical clearance of 48 feet. Passage should not be attempted without local knowledge.

(331) **Wiscasset** is a town on the west side of Sheepscot River 14 miles above the entrance. It is on U.S. Highway No. 1 and on a Class II railroad. Seasonal passenger rail service is available on the rail line running between Brunswick and Rockland. The wharves are in ruins, and there is virtually no commerce by water.

(332) The town landing and Wiscasset Yacht Club, both with float landings reported to have 15 feet alongside, are at the south end of town below the wharf ruins. Water is available at the yacht club float. Overnight berthing is permitted at both landings, and the yacht club maintains a guest mooring. A small-craft launching ramp is between the two landings.

(333) Gasoline and diesel fuel can be obtained by tank truck at the landing, and ice, provisions and marine supplies are available in town.

(334) Berthage with electricity and gasoline are available at the float landing of a marina and lodge on the east side of Sheepscot River, about 0.8 mile southward of Davis Island. A small-craft launching ramp is also available here.

(335) Wiscasset has hotels, motels and restaurants and bus and taxi service.

(336) The Whites Island Swimming Club with a float landing is about 200 yards southwest of the yacht club.

(337) Anchorage in 25 to 30 feet in muddy bottom can be had south and southwestward of the landings. There are ample parking facilities and picnic areas in the vicinity.

(338) The U.S. Route 1 fixed highway bridge over Sheepscot River at Wiscasset has a clearance of 25 feet. The railroad bridge 1 mile above Wiscasset has a 40-foot bascule span with a clearance of 8 feet. In 1981, the railroad bridge was being maintained in the closed position. (See **33 CFR 117.533**, chapter 2, for drawbridge regulations.) An overhead power cable 0.2 mile north of the bascule bridge has a clearance of 75 feet.

(339) The depth is reported to be about 10 feet for 4 miles above Wiscasset to rapids in the river. Boats of about 4-foot draft can go through the rapids at high-water slack and for about 3 miles above. **Sheepscot** is a village just above the rapids. A highway bridge crossing the river at Sheepscot has a 48-foot fixed span with a clearance of 10 feet. The channel is unmarked above Wiscasset, and local knowledge is required for its navigation.

(340) **Marsh River**, a tributary, enters the Sheepscot River about 2 miles above Wiscasset. Small craft are reported to go up the river for 3 or 4 miles for salmon fishing. The railroad bridge about 2 miles above the mouth has a 33-foot fixed span with a clearance of 22 feet.

(341)

Kennebec River

(342) The mouth of the **Kennebec River** is northward of Seguin Island and 20 miles eastward of the entrance of Portland Harbor. It is the approach to the cities of Bath, Augusta, Richmond and Gardiner and smaller river towns. Waterborne commerce in the area consists mainly of traffic to and from the shipyard in Bath.

(343) With the aid of the charts, small craft should have no trouble reaching Augusta, the head of navigation on the Kennebec River. Vessels with a draft approaching the depth of the channel should employ a pilot. The channel above Bath is reported to be subject to considerable changes annually caused by freshets.

(344)

Prominent features

(345) **Seguin Light** (43°42'27"N., 69°45'29"W.), 180 feet above the water, shown from a 53-foot white brick tower connected to a dwelling, is on the summit of 145-foot, grassy **Seguin Island**. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A.This light is the most prominent mark in the vicinity.

(346) **Cape Small** is the wooded point about 4 miles westward of the mouth of the river. The distinguishing marks are an elevated tank 1.4 miles northward from the end and visible from eastward or westward; **Bald Head**, a bare round knob on the west side of the point; and **Bald Head Ledge**, bare at half tide and marked by a bell buoy.

(347) A **danger zone** of a naval aircraft practice mining range is close southeastward of Cape Small and westward of Seguin Island. (See **33 CFR 334.20**, chapter 2, for limits and regulations.)

(348) **Fuller Rock Light** (43°41'45"N., 69°50'01"W.), 39 feet above the water, is shown from a white skeleton tower with a red and white diamond-shaped daymark on a low bare islet of the same name, about 0.3 mile southward of Cape Small.

(349) **Pond Island**, about 30 feet high, is a grassy island on the west side of the entrance to Kennebec River. **Pond Island Light** (43°44'24"N., 69°46'13"W.), 52 feet above

the water, is shown from a white tower on the summit of the island; a sound signal is at the light and is activated by keying the microphone five times consecutively on VHF-FM channel 83A. The light shows a higher intensity beam up and down the river.

(350) **Fort Popham Memorial** is an unfinished and abandoned fort, now a state historical landmark, on Hunnewell Point. **Fort Popham Light**(43°45'18"N., 69°47'00"W.), 27 feet above the water, is shown from a cylindrical iron stand on the parapet of the old fort. The light shows higher intensity beams up and down the river.

(351)

Channels

(352) There are two approaches to the entrance. The eastern, east of Seguin Island, which leads between Whaleback Rock and Pond Island, is the main channel. The western, west of Seguin Island, leads between Pond Island Shoal gong buoy and the shoals eastward. Both are used, but vessels drawing more than 18 feet usually enter by the eastern channel. The entrance has strong tidal currents, and if the wind is opposed to the current an ugly chop sea is encountered that is at times dangerous for small craft.

(353) A Federal project in Kennebec River provides for a channel 27 feet deep, from the mouth to a point about 0.6 mile above the bridge at Bath, thence 16 to 18 feet to Gardiner, thence 11 feet to the head of navigation at Augusta.

(354)

Anchorages

(355) Large vessels awaiting the pilot may anchor safely in the vicinity of White Ledge Lighted Bell Buoy 1 (43°43'49"N., 69°44'54"W.), in 50 to 65 feet. Small craft may find suitable anchorage northwest of Hunnewell Point (43°45'17"N., 69°47'04"W.).

(356) Farther upstream, anchorage is also available on the eastern side of the channel southward of Kennebec River Buoy 12, in 36 to 48 feet. On the eastern edge of the channel at the anchorage, the depths shoal abruptly from 30 feet to a few feet. Drift ice coming down the river generally follows the western shore.

(357) Anchorage for small vessels can be had on the western side of the channel off Parker Flats, about 4 miles above the entrance, in 20 to 36 feet. Above Parker Flats, vessels anchor wherever they find good holding ground and suitable depth, keeping out of the strength of the current.

(358) **General anchorages** are at Bath. (See **33 CFR 110.1** and **110.133**, chapter 2, for limits and regulations.)

(359)

Dangers

(360) This is a region of rock and very broken ground; therefore, strangers should proceed with extreme caution and avoid crossing broken ground where the charted depths do not substantially exceed the draft.

(361) The principal dangers in the river are marked, but the channel is narrow in places. The narrowest place below Bath is between North Sugarloaf and Popham Beach, where the deep channel is only about 100 yards wide. Some sections of the dredged channel between the south end of Swan Island and Augusta are not marked well enough to help strangers keep in them.

(362) The entrance to Kennebec River is somewhat obstructed by an area of islands and rocks and very broken ground, extending for a distance of 4.5 miles. The most southerly known danger is **Seguin SSW Ledge**, covered 33 feet; it is 2.6 miles southwest of Seguin Island Light.

(363) During freshets, pulp logs are sometimes washed over the dam above Augusta and present a serious navigational hazard, especially to small craft. Log booms are maintained at Brown Island and on the east side of the river below Shepard Point to facilitate recovery of the drifting logs. The booms are not lighted but are outside the navigation channel.

(364) The presence of deadheads, known locally as **tide walkers,** are a constant hazard in the river, especially to small craft. These water-logged boom logs, weighted at one end by parts of mooring chains, with one end down and the other end at the surface or just under, shift position with the tidal or river currents and are hard to detect, especially at night. A sharp lookout for them should be kept.

(365) The dangers outside of Seguin Island are **Mile Ledge**, covered 20 feet and marked by Mile Ledge Lighted Bell Buoy 20ML, and **Camel Ground**, 1 mile west-southwestward of Seguin Island Light, which has been cleared to 23 feet. Camel Ground is unmarked, and the sea breaks on it in heavy weather.

(366) Westward of Seguin Island, **Buttonmold Ledges**, covered 11 feet, and **Bill Wallace Ground**, covered 19 feet, lie between Fuller Rock and Bald Head Ledge and are unmarked. **Halibut Rocks**, an extensive ledge covered 24 feet about 0.6 mile eastward of Fuller Rock, are unmarked. There are rocks and very broken ground in the vicinity of Cape Small.

(367) **Ellingwood Rock**, 400 yards northward of the north end of Seguin Island, is a bare islet about 6 feet high.

(368)

Local magnetic disturbance

(369) Differences of as much as 8° from the normal variation have been observed in an area around Ellingwood Rock for approximately 1 mile in all directions.

(370) **Seguin Ledges**, 0.4 mile northeastward of Ellingwood Rock, have a bare islet about 5 feet high and have covered ledges extending 300 yards northeastward and 400 yards southward from the bare islet, all unmarked.

(371) **White Ledge** is an unmarked 11-foot spot 0.4 mile northward of Seguin Ledges.

(372) **Jackknife Ledge**, covered 8 feet, is about 1.3 miles northwestward of Seguin Light and is marked on the east by a buoy.

(373) **Pond Island Shoal** is the rocky shoal southward and southeastward of Pond Island. It has depths of from 5 to 21 feet over it and in heavy gales is covered with breakers.

A gong buoy 0.7 mile south-southeastward of Pond Island Light marks the southeastern end of the shoal. Vessels should not pass between this buoy and Pond Island. Small craft entering the river from the westward often cut across this shoal, but it is not advisable to do so in southerly weather when a heavy chop is built up by the ebb tidal current from the river; this often causes heavy breakers to form on it.

(374) The dangers eastward of the entrance, including Tom Rock and The Sisters, were included in the description of Sheepscot River. The dangers in Kennebec River are included in the description of the river.

(375)

Current

(376) Tidal currents between the entrance and Bath have average velocities at strength of 2 to 3 knots. Ebb velocities up to 6 knots have been observed, and considerably larger velocities may be expected during freshets. Above Bath similar velocities are believed to occur, but no definite information is available. The direction of the current at the entrance is influenced by strong winds, especially easterly gales. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov.* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(377) **Freshets** occur in March and April, and also after heavy rains in the fall, but are not dangerous to shipping unless accompanied by ice. A height of 9 feet above high water usually occurs several times a year at Augusta, but the height diminishes rapidly southward.

(378) **Ice** usually closes the river above Bath from December to April. Steamers are rarely delayed by ice below Bath, as the channel is kept clear by ice breakers.

(379)

Pilotage, Kennebec River

(380) Pilotage is compulsory for foreign vessels and U.S. vessels under registry. The pilot boarding station is located at the White Ledge Lighted Bell Buoy I. Shipping Services, Inc. offers pilotage for the Kennebec River. See Pilotage, Boothbay Harbor, earlier in this chapter, for contact information for Shipping Services, Inc., and Bath Iron Works Pilot.

(381) Ships bound for the shipyard at Bath usually obtain the services of the yard's pilot. The pilot uses either the yard tug or a lobster boat as a pilot boat. The tug has a black hull and red superstructure and monitors VHF-FM channels 13 and 16 when working ships. Arrangements for pilot, tug and boarding place should be made in advance through the shipyard, telephone 207–443–3311, or by calling 207–374–2217.

(382)

Towage

(383) Tugs are available at Bath. Bath shipyard tug handles primarily shipyard traffic. If desired, commercial tugs can be obtained from Bath, Southport, Boothbay Harbor, Belfast or Portland; arrangements for this service should be made in advance through ships' agents.

(384)

Quarantine, customs, immigration and agricultural quarantine

(385) (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(386) **Quarantine** is enforced in accordance with regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.)

(387) Bath is a **customs port of entry.**

(388)

Harbor regulations

(389) Regulations for the ports along the river are under control of the various harbormasters.

(390)

Wharves

(391) There are no usable deep-water commercial wharves at Bath, except those at the shipyard. Wharves along the river are included in the description of the river ports.

(392)

Supplies

(393) Supplies are available at Bath, including marine supplies, fuel and provisions. Detailed information is given later in the text.

(394)

Repairs

(395) The large shipyard at Bath has drydocking facilities. Repair facilities are available at the boatyard at Bath and the few marinas on the river. Detailed information on the facilities is given later in the text.

(396)

Communications

(397) Taxi and coastal bus services are available at all the river ports, and the passenger rail service is provided between Brunswick and Rockland.

(398) The following description of the river from the entrance to Augusta affords a means of navigating the river by acquainting the stranger with its various features, anchorages, dangers, important aids to navigation and the facilities at the river ports.

(399)

Sprague River to Lee Island

(400) **Sprague River** and **Morse River** between Cape Small and the entrance of Kennebec River, are nearly bare at low water at their entrances and seldom entered even by local boats. **Heron Islands** and **Fox Islands** are groups of wooded islands off the mouth of Morse River.

(401) **Wood Island**, 0.3 mile westward of Pond Island, is high and wooded. The channel between Wood and Pond Islands should not be used by strangers.

(402) **Whaleback Rock**, 8 feet high and bare, is on the eastern side of the entrance to the river and 0.6 mile eastward of Pond Island. A shoal extends about 100 yards southward from it. **Salter Island**, northward of

Whaleback Rock, is wooded. **Stage Island**, 0.5 mile northwestward, is also wooded.

(403) **Stage Island Bay**, **Sagadahoc Bay** and **Heal Eddy**, on the east side of Kennebec River at the entrance, are shoal inside, have no wharves and are of little importance.

(404) **North Sugarloaf** and **South Sugarloaf** are high, rounded, bare and rocky islets in the middle of Kennebec River just inside the entrance. A ledge extends 100 yards southward from South Sugarloaf. **Jack Rock**, near the end of a ledge extending 200 yards northeastward from South Sugarloaf, is awash at low water and is marked by a daybeacon. A rock awash is about 125 yards southeastward of the daybeacon. A ledge extends 250 yards southeastward of North Sugarloaf. Another ledge, covered 17 feet and marked by a buoy, extends 200 yards northwestward from North Sugarloaf; the narrowest part of the channel between the entrance and Bath is westward of this ledge.

(405) **Popham Beach** is a summer resort on the west side of Kennebec River just inside the entrance. An abandoned Coast Guard station is on the beach; its L-shaped wharf is located close westward of Fort Popham and has 9 feet alongside. Only ruins of some cribbing remains of an old wharf in the bight southwestward of the fort and the long Government pier extending northward from **Sabino Head** is also in ruins.

(406) Old Fort Popham is now a state park, and Popham Beach is believed to be the site of the first settlement in New England. The ship VIRGINIA was built here in 1608. There is a park service float landing with 2 feet alongside, a ramp west of the Coast Guard wharf, a store and a restaurant.

(407) **Atkins Bay**, a large bay west of **Hunnewell Point**, dries out for most of its length.

(408) **Bay Point** is a village on the east side of Kennebec River entrance, opposite Fort Popham. A lobster wharf has 4 feet alongside. Another private wharf close southward is in ruins. Water is available from nearby wells. Craft approaching the wharf should avoid getting too far northward, as a bar which bares at half tide extends nearly all the way across the entrance to Long Island Narrows from Gilbert Head.

(409) **Gilbert Head**, the southern extremity of Long Island, is high and wooded except near the south end, where there is a large gray house. The house is very conspicuous and a good mark in hazy weather when surface aids are obscured or not readily discernible.

(410) **Shag Rock**, on the eastern side of the channel, southeastward of Cox Head, is 3 feet high. It is marked by a lighted buoy about 75 yards west of it. The wreck of a schooner stranded on Long Island, eastward of Shag Rock, is reported visible at low water.

(411) **Cox Head** is about 140 feet high and wooded. **Todd Bay**, on the east side of Kennebec River northeastward of Cox Head, is almost bare at low water.

(412) **Dix Island**, 0.2 mile northward of Cox Head, has a ledge that uncovers, extending northward of it. A buoy is northward of the ledge.

(413) **Perkins Island**, on the east side of the main channel 3 miles above the entrance, is wooded on the north end and bare on the south end. **Perkins Island Light** (43°47'12"N., 69°47°07"W.), 41 feet above the water, is shown from a white octagonal tower on the west side of the island. A 5-foot shoal, about 350 yards westward of the light, is marked on its southeast side by a buoy. **Perkins Island Ledge**, covered 7 feet, is about 0.3 mile south-southwestward of the island; a buoy is about 200 yards southwestward of the ledge.

(414) **Parker Head** is a village on the west side of the river westward of **Parker Head**, a prominent headland. The approach to the village is by a narrow channel, shoaling gradually from 3 feet to 1 foot. The channel is sometimes marked by bush stakes, and there are several old pilings along its sides southeast of the former milldam. A buoy marks the easterly edge of shoal water extending about 0.6 mile north-northeastward of Parker Head.

(415) **Back River** is a narrow, crooked and unmarked thoroughfare connecting Kennebec River with Sasanoa River, Hockomock Bay and Sheepscot River. It is described with the Inside Passage. **West Georgetown** is a village on the east side of Back River, just inside its southern entrance, which is marked by a buoy. A ledge extends about 350 yards southwestward of **Crow Islands**, which are in the middle of the entrance. A buoy is southwest of the ledge.

(416) **Seal Rocks**, on the west side of the channel at the upper end of **Parker Flats**, is a ledge that uncovers 5 feet. A buoy is northeastward of a rock awash at the outer end of the ledge.

(417) **Phippsburg** is a village on the west side of Kennebec River 5.5 miles above the entrance. A conspicuous white church spire in Phippsburg is a good leading mark for the reach from Bald Head to Squirrel Point.

(418) **Squirrel Point**, the southwestern extremity of Arrowsic Island, is marked by **Squirrel Point Light** (43°48'59"N., 69°48'09"W.), 25 feet above the water and shown from a white octagonal tower.

(419) **Goat Island**, 700 yards northwestward of Squirrel Point, is wooded, and the smaller islands near it are bare and grassy. The ledge extending southward of the island, which uncovers 4 feet, is marked by a buoy on its southeastern side. A ledge that uncovers 4 feet extends 300 yards northward of the island.

(420) **Pettis Rocks**, in the middle of the river 6.5 miles above the entrance, are bare at the highest part and marked at the south end by a light. This is a dangerous part of the river, and vessels inbound, after passing the southern end of Lee Island, should cross over to and favor the east side of the river to avoid the shoals extending from Pettis Rocks and Ram Island.

(421) **Ram Island**, just northward of Pettis Rocks, is low and bushy. Ledges that uncover 5 feet extend nearly 200 yards northward and 75 yards eastward of the island. A light marks the eastern ledge.

(422) **Lee Island**, 128 feet high and wooded, is on the west side of the river westward of Pettis Rocks and Ram

Island. A rock awash off the southeastern shore of the island is marked by a buoy.

(423)

Indian Point to Woolwich

(424) **Indian Point** (43°50.6'N., 69°47.9'W.), on the west bank of Kennebec River, about 0.4 mile above Ram Island, is low. A ledge covered 7 feet, about 500 yards northward of Indian Point, is marked on its southeastern side by a buoy. At **Bluff Head**, 1 mile above Ram Island, the river narrows. The upper part of this section is marked by Doubling Point Lighted Range. The lights are shown from white octagonal towers on the bearing **359°**.

(425) This range passes very close to and eastward of **Lithgow Rock** and **Fiddler Ledge**, both of which are covered 27 feet and unmarked. It will be better to steer a little eastward of the range rather than take any chance of the vessel getting to the westward of it. An unmarked 29-foot rock ledge is close eastward of the range about midway between Lithgow Rock and Fiddler Ledge. Care should be taken in deep-draft vessels not to get too far eastward and foul this rock.

(426) Just northward of Fiddler Ledge the channel takes a sharp turn to the west through **Fiddler Reach. Doubling Point** at the right angle turn from Fiddler Reach into **Long Reach** is marked by **Doubling Point Light** (43°52'57"N., 69°48'24"W.), 23 feet above the water, shown from a white octagonal tower on a square gray pier with a footbridge to the shore.

(427) There are reported to be strong back eddies on both ends of this turn, and great care should be taken to keep the vessel well under control. Caution should be exercised by vessels bound downriver on a strong ebb when rounding Doubling Point from Long Reach into Fiddler Reach.

(428) **Winnegance** is a village on **Winnegance Creek**, 0.5 mile from the main channel of Kennebec River. The channel is shoal and navigable only by small craft. Old pilings extending across the creek northeast of the highway causeway and dam are partly covered at high water.

(429) **Bath** is a city on the west side of Kennebec River 12 miles above the entrance. There is little waterborne traffic to Bath, except for barge traffic to the shipyard and vessels bound for repairs. In 1979, the maximum draft carried to the shipyard was 26 feet.

(430) Bath was the most important U.S. shipbuilding center in the 19th century; the HENRY B. HYDE, a three-masted full-rigged wooden ship, and the six-masted schooner WYOMING, the largest of their kind ever built in America, were constructed here. The Maine Maritime Museum is close to the waterfront. There are many historical points of interest.

(431) The city has churches, hospitals, a library, banks, hotels, motels, laundry, markets and stores of all kinds.

(432) The railroad lift bridge crosses the Kennebec River at Bath to Woolwich. The vertical lift span has a clearance of 10 feet down and 135 feet up. (See **33 CFR 117.1**through**117.59**and**117.525**, chapter 2, for drawbridge regulations.) The U.S. Route 1 highway bridge, just north of the railroad bridge, has a fixed span with a clearance of 70 feet.

(433)

Wharves

(434) Bath has deep-draft facilities on the west side of the Kennebec River. The alongside depths given for the facilities described are reported depths. (For the latest depths, contact the operator.) The facilities described have highway and railway connections.

(435) **Bath Iron Works, Outfitting Pier:** 733 yards below U.S. Highway 1 Bridge; has 600 feet of berthing space; 26 to 50 feet alongside; deck height, 10 feet; one 94-ton traveling gantry crane, one 30-ton wingwall crane and three additional cranes with capacities of 30 to 50 tons; one 8,400-ton floating drydock; mooring vessels for outfitting and repair and is owned and operated by Bath Iron Works.

(436) **Bath Iron Works, South Wharf:** 460 yards below U.S. Highway 1 Bridge; 26 feet alongside; deck height, 9 feet, one 25-ton traveling gantry crane, one fixed 97 to 220-ton crane; three shipbuilding ways; mooring vessels for outfitting; owned and operated by Bath Iron Works.

(437) **Bath Iron Works, North Wharf:** below U.S. Highway 1 Bridge; 32 feet alongside; deck height, 8 feet; one 25-ton traveling gantry crane, one 5-ton crane; mooring vessels for repair; owned and operated by Bath Iron Works.

(438) A marina with a float landing is on the west side of the river, about 0.1 mile above the bridge at Bath; gasoline, water, a small-craft launching ramp, some marine supplies and berthage with electricity are available. The town float landing is just northward of the marina. Another marina about 150 yards northward of the town landing has gasoline, diesel fuel, water, and some marine supplies and can make engine repairs. Depths of 15 feet are reported alongside these float landings.

(439)

Supplies

(440) Provisions, gasoline, diesel fuel, ice, bottled gas and some marine supplies are available in town.

(441) Bath has bus and taxi service.

(442)

Repairs

(443) A shipyard on the east side of the river at Woolwich, about 500 yards north of the bridge, builds steel vessels up to 120 feet long. A boatyard, on the west side of the river about 1.3 miles below the bridge, has a marine railway that can handle craft up to 50 feet in length. Hull, engine and electrical repairs can be made, and dry open winter storage and some marine supplies are available. Gasoline and water can be obtained at the yard's float landing; depths of 12 feet are reported alongside the float. The yard maintains guest moorings and permits overnight berthing at the float.

(444) The Sasanoa River entering Kennebec River between **Preble Point**, the northern extremity of Arrowsic Island, and **Sasanoa Point**, the southern extremity of **Towesic Neck**, is described under the Inside Passage. See caution note contained in tidal current data for the Kennebec River in this chapter.

(445) **Woolwich** is a village on Towesic Neck, opposite Bath. The asphalt pier there is reported to have 22 feet alongside. Only piling remains of the old coal wharf and ferry slips just below this pier.

(446)

Winslow Rocks to Trotts Rock

(447) About 1 mile above the bridge at Bath, Kennebec River is divided into two channels by an extensive area of rocks awash and covered ledges in midriver; the principal hazards on it are **Winslow Rocks** and **Stetson Rocks**, parts of which are awash at low water. Obstruction buoys mark the northern and southern ends of the area, and the eastern side is marked by channel buoys.

(448) The main or eastern channel is deep and favors the eastern bank of the river. The western channel is not marked and is used only by small craft. Ledges south of **Days Ferry**, on the east bank of the river, north of Stetson Rocks, are marked by a buoy. The channel past **Thorne Head** is deep and clear.

(449) Two miles above Bath, Kennebec River divides into three channels. The eastern, or **Burnt Jacket Channel**, is the most direct and has a depth of 14 feet. It is unmarked and extremely foul, difficult at its northern end and is used mostly by small craft. Local knowledge is necessary to navigate it safely.

(450) The main channel or **West Branch**, the widest, has a depth of about 22 feet and is partly buoyed, clear, and easily followed by aid of the chart. **Thorne Island Ledge**, covered 4 feet and marked by a buoy near its southeast edge, **Thorne Island**, and **Lines Island** are all on the northeast side of the channel, and **Woods Island**, **Crawford Island** and **Ram Island** are on the southwest. A ledge making out from the northeast side of Woods Island is buoyed. A rock bare at low water is 50 yards off the west side of Lines Island, and a rock awash is off the southwestern end of the island. Near the northern end of the channel, **Grace Rock**, covered 2 feet, is marked on its west side by a buoy.

(451) The third channel trends to the southwestward between Woods, Crawford and Ram Islands and the mainland. It is unmarked, foul and little used.

(452) **Chops** is the narrow passage between two headlands, **Chops Point** and **West Chops Point**, about 4.5 miles above Bath. Two high steel transmission towers on the points are very prominent. The overhead power cables have a clearance of 145 feet.

(453) **Trotts Rock**, with a least depth of 3 feet and marked on its west side by a buoy, is about 0.4 mile northward of Chops Point.

(454)

Brunswick and vicinity

(455) **Merrymeeting Bay** is a shoal bay making westward from Kennebec River 17 miles above the entrance. The bay is the approach to the towns of **Brunswick** and **Topsham** on the **Androscoggin River** and **Bowdoinham** on the **Cathance River**, 8 and 4 miles, respectively, above Kennebec River. The channels are narrow and most are unmarked; local knowledge is necessary.

(456) A fixed highway bridge, with a clearance of 40 feet, crosses the Androscoggin River about 7.7 miles above the entrance to the bay. The railroad bridge crosses the river just above the highway bridge and has a fixed span with a clearance of 20 feet. The U.S. Route 201 highway bridge at Brunswick is the head of navigation, above which are a dam and falls.

(457) Several overhead power cables about 1.5 miles below the railroad bridge have clearances of 44 feet in the west channel and 45 feet in the east channel. The power cables over Cathance River 0.3 and 0.6 mile above the mouth have clearances of 55 and 65 feet, respectively. The overhead power cable about 2 miles above Bowdoinham has a clearance of 40 feet.

(458) It is reported that heavy storms and winter ice change the shoals and depths in Androscoggin River.

(459) **Brunswick** is the site of Bowdoin College and a community with a number of restaurants and hotels with various services available.

(460) The **Kennebec River Closed Area**, a Marine Protected Area (MPA), includes the waters of Merrymeeting Bay and Androscoggin River.

(461)

Weather, Brunswick and vicinity

(462) Brunswick, on the Androscoggin River near the junction with the Kennebec River in "downeast" Maine, averages about three days each year with maximum temperatures in excess of 90 °F. July is the warmest month with an average high of 78 °F and an average minimum of 59 °F. January is the coolest month with an average high of 30 °F and an average minimum of 11 °F. The highest temperature on record for Brunswick is 104 °F recorded in August 1975 and the lowest temperature on record is -30 °F recorded in January 1981. About 147 days each year see temperatures below 32 °F and an average 24 days each year record temperatures below 5 °F. Every month except July has seen temperatures below 40 °F and every month except June, July and August has recorded temperatures below freezing.

(463) The average annual precipitation for Brunswick is 45.2 inches (1148 mm), which is fairly evenly distributed throughout the year. Precipitation falls on about 186 days each year. The wettest month is November with 5.0 inches (127 mm) and the driest, July, averages only 3.0 inches (76 mm). An average of 29 thunderstorm days occur each year with July and August being the most likely months. Snow falls on about 59 days each year

and averages about 72 inches (1829 mm) each year. December through March each average greater than one foot (305 mm) per year while January averages 19 inches (483 mm). One – foot (305 mm) snowfalls in a 24-hour period have occurred in each month November through March and 21 inches (533 mm) fell in one 24-hour period during December 1970. About 15 days each year have a snowfall total greater than 1.5 inches (38 mm) and snow has fallen in every month, October through April. Fog is present on average 207 days each year and is evenly distributed throughout the year with a slight maximum during late summer.

(464) The prevailing wind direction in Brunswick is the north during the winter and south during the warmer months. March is the windiest month.

(465)

Abagadassett Point to Richmond

(466) **Abagadasset Point** (44°00.3'N., 69°49.4'W.), on the west bank of the river about 1.6 miles above the Chops, should be given a wide berth to avoid the shoals extending from it to the northward. A buoy marks the northeastern extremity of the shoals.

(467) Overhead power cables over Kennebec River at Abagadasset Point have a clearance of 105 feet. An 5-foot spot on the west side of the channel opposite Twing Point is marked by a buoy. **Ames Ledge**, on the east side of the river north of **Twing Point**, is marked by a buoy off its northwest side.

(468) **Swan Island**, about 1.8 miles above Abagadasset Point, divides Kennebec River into two channels. The main channel, east of the island, is marked by buoys and by a daybeacon on **Beef Rock**. The channel leading westward of the island is not maintained and is only partially marked by private buoys. A rock covered 4 feet is reported at the entrance in about 44°01.7'N., 69°49.1'W. East of the main channel, a riprap training wall extends from off **Carney Point** to **Green Point.**

(469) **Eastern River** enters Kennebec River between Carney Point (44°02.0'N., 69°48.0'W.) and the flats and training wall extending 1.4 miles southwestward of **Green Point.** The river follows the eastern shore to **South Dresden**. It is unmarked and crossed by three highway bridges. State Route 128 bridge, about 2 miles above the mouth, has a fixed span with a clearance of 16 feet. Telephone and power cables on the south side of the bridge have a clearance of 22 feet.

(470) In 1970, the river was reported navigable to Dresden Mills with a draft of 4 feet and above that by small outboard craft for several miles through beautiful woodland. There are several private landings on the river but no facilities. Remains of old wharves can be seen at Dresden Mills and other points.

(471) **Richmond**, westward of Swan Island, is a town on the west bank of Kennebec River 23 miles above the entrance. There are several landings at the town. The town float landing, at the mill with a conspicuous red brick stack, has 16 feet alongside. There are no facilities at the landing, but gasoline, diesel fuel, water, provisions and some marine supplies can be obtained in town.

(472) The submerged ruins of a jetty extend northeasterly from the northeastern tip of Swan Island to near channel Buoy 33.

(473) State Route 197 highway bridge crosses the river at a point just north of Swan Island between Richmond and Dresden; the bridge is under construction (2013).

(474)

Cedar Grove to Augusta

(475) **Cedar Grove** is a small settlement on the east bank of Kennebec River, 2 miles above the north end of Swan Island and 0.6 mile above **Courthouse Point** (44°06.4'N., 69°46.0'W.).

(476) **Hathorn Rock**, covered 8 feet about 1.7 miles north of Courthouse Point, is marked on the east side by a buoy. A rocky area is reported on the west side of the river, about 0.5 mile northward of Hathorn Rock.

(477) **South Gardiner**, about 4.5 miles above Courthouse Point, is a village on the west side of Kennebec River 30 miles above the entrance. There are several private landings at the village.

(478) **Gardiner**, about 3.5 miles above South Gardiner, is a town on the west side of the river 33.5 miles above the entrance. The town wharf and float landing has 12 feet alongside but no facilities. A public parking lot is on the wharf. The old coal wharf just upstream has 15 feet reported alongside but is seldom used.

(479) **Randolph**, a village on the east side of the river opposite Gardiner, has a wharf with 12 feet alongside and oil connections but is seldom used. Kennebec Boating Association has a float landing and ramp at the wharf. Ice and provisions are available. A hardware store adjoins the landing, and restaurants are in the vicinity.

(480) A fixed highway bridge with a clearance of 35 feet crosses the river between Gardiner and Randolph just north of the waterfront facilities.

(481) Ruins of an old pier and power plant are on the west side of Kennebec River just above **Farmingdale**. Rock-filled cribs extend over 100 yards offshore and are marked by a buoy at the eastern end. No attempt should be made by small craft to pass between them and the west bank as the area is extremely foul. The east bank should be favored.

(482) A foul area, reported to be deadheads, is on the west side of the river off Farmingdale, about 0.4 mile northward of the bridge at Gardiner.

(483) At **Browns Island**, about 1.5 miles above Gardiner, the river is crossed by two sets of power cables that have clearances of 59 feet. Log booms extend southwestward and northwestward from the island. They are unmarked and are used to catch drifting pulp logs that are washed over the dams above Augusta by spring floods and freshets. A shoal with a least depth of 3 feet makes out to the north and northwestward of the island.

(484) **Hallowell**, about 3.5 miles above Gardiner, is a town on the west side of the river 37 miles above the entrance; it has a small town landing.

(485) A pinnacle rock, covered 5 feet, is on the east side of the channel about 500 yards southwestward of the wharf on Oil Cloth Point. It is marked by a buoy on its northwest side. A submerged obstruction, reported in 1965, is in the channel about 300 yards southwestward of the pinnacle rock and about 50 yards offshore.

(486) **Augusta**, the capital of Maine, is at the head of navigation on the Kennebec River 39 miles above the mouth. The city has no waterborne commerce. There is a public float landing on the east bank just above the southernmost bridge with 4 feet reported alongside. The landing is also used by the Augusta Yacht Club; a parking lot is available, but there are no other services.

(487)

Bridges

(488) The four bridges at Augusta have fixed spans. The first, U.S. Routes 201–202 highway bridge, has a clearance of 70 feet for a width of 67 feet; the second, a city highway bridge at the upper end of the turning basin, has a clearance of 27 feet. The head of navigation is at this bridge as the river is very shallow above it, and not even small craft venture there. The third bridge is now in ruins. The railroad bridge adjacent to and above the third bridge has a clearance of 23 feet.

(489) Gasoline, diesel fuel, lubricants, provisions, ice and marine supplies can be obtained in Augusta. Bus, taxi and railroad freight services are available.

(490)

Casco Bay to Cousins River

(491) **Casco Bay** is a very extensive area between Cape Small and Cape Elizabeth, a distance of 17.8 miles. Between these two capes the bay extends up into the land an average distance of about 12 miles. The number of islands in Casco Bay is 136, and many are fertile and under cultivation; nearly all are inhabited. Nearly every large island extends northeast and southwest, which is the general course of the bay and of all rivers and coves contained within its limits.

(492) A **vessel-to-vessel oil transfer anchorage area** in Casco Bay, about 3.5 miles northeastward of Portland, has been designated by the State of Maine Department of Environmental Protection. The area is 1 square mile beginning at Hussey Sound Buoy 12.

(493) 43°42'10"N., 70°09'46"W.; thence north to

(494) 43°43'10"N., 70°09'46"W.; thence west to

(495) 43°43'10"N., 70°11'09"W.; thence south to

(496) 43°42'10"N., 70°11'09"W.; thence east to origin.

(497)

Anchorages

(498) In the eastern part of Casco Bay, the best anchorage for strangers is in New Meadows River. Local fishermen and yachtsmen frequently use Sebasco and Cundy Harbors. Potts Harbor, Harpswell Harbor and Mackerel Cove are good anchorages in the middle of the bay for small vessels and yachts.

(499) Merriconeag Sound and Harpswell Sound and the whole Casco Bay westward of Harpswell Neck afford good anchorage for large vessels, except in heavy northeast gales.

(500) Vessels can enter through Broad Sound, Luckse Sound or Hussey Sound and select an anchorage under the lee of some of the many islands, a suitable depth and good holding ground being found in most places. Portland Harbor is a secure anchorage on the western side of the bay and is the one used mostly by larger vessels.

(501) Most of the dangers are marked, and the waters are well charted, so that, with the aid of the chart, no difficulty should be experienced in navigating Casco Bay in clear weather.

(502)

COLREGS Demarcation Lines

(503) The lines established for Casco Bay are described in **80.110**, chapter 2.

(504)

No-Discharge Zone

(505) The State of Maine, with the approval of the Environmental Protection Agency, has established a No-Discharge Zone (NDZ) in Casco Bay. The NDZ includes all waters north and east of 43°33'56"N., 70°11'48"W. at Cape Elizabeth Light to 43°42'17"N., 69°51'17"W. at Bald Point in Phippsburg; see the chart for limits.

(506) Within the NDZ, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZ, discharge of sewage is regulated by **40 CFR 140** (see chapter 2).

(507)

Current

(508) The velocity of the tidal current at strength is about 1 knot in the entrance to Portland Harbor and in Hussey and Broad Sounds. In the open waters of the bay it is generally 0.5 knot or less. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov*. for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(509)

Ice

(510) Considerable ice forms at the heads of the numerous arms extending northward in Casco Bay, but the principal anchorages are available at any season of the year.

(511) The part of Casco Bay between Cape Small on the east and Halfway Rock Light and Harpswell Neck on the west is full of small islands, ledges and rocks. Between them, narrow but deep channels lead to the bays and sounds at the head. These arms afford good anchorage for small vessels but are used only by local fishing and pleasure craft. There are several small villages in this part of the bay, but no towns.

(512) **Temple Ledge**, about 1.8 miles southwestward of Cape Small and covered 25 feet, is unmarked. **Lumbo Ledge**, 2 miles west of Temple Ledge and 2.6 miles south of**Ragged Island**, is covered 17 feet and marked by a buoy on its south side.

(513) **Spoonbowl Ledge**, about 0.3 mile westward of Cape Small and about 0.4 mile southwest of **Gooseberry Island**, is covered 5 feet and unmarked. Craft bound from Cape Small to Small Point Harbor should be careful to avoid it.

(514) **East Brown Cow**, 1.6 miles west-northwestward of Cape Small, is 12 feet high and bare. **Mark Island**, 0.8 mile northward of East Brown Cow, is high and thickly wooded. **Mark Island Ledge**, 0.3 mile southwestward of Mark Island, uncovers 3 feet and is marked on its west side by a buoy. **Wyman Ledge**, 0.5 mile east of Mark Island, covered 4 feet, is marked on its eastern side by a buoy.

(515) **White Bull**, 1 mile westward of Mark Island, is a high, round and bare islet. **White Bull Lighted Gong Buoy WB,** about 0.4 mile southeastward of the island, marks the southwestern approach to New Meadows River. **Bold Dick**, an unmarked rock about 0.7 mile west-southwestward of White Bull, uncovers 7 feet.

(516) **Small Point Harbor**, between Wood and Little Wood Islands on the west and **Hermit Island** on the east, is on the east side of Casco Bay 1.5 miles northward of Bald Head, the southwestern extremity of Cape Small. The harbor is an anchorage for local fishermen and yachts but is open to southerly winds.

(517) The principal dangers are **Gooseberry Island Ledge**, extending about 0.3 mile southwestward of Gooseberry Island, awash at low water and marked by a buoy; **Wood Island South Ledge**, covered 5 feet at the end and extending about 0.3 mile south of Wood Island, marked by a lighted bell buoy about 350 yards westward of the south end of the ledge; **Middle Ledge**, awash and marked by a buoy on its southwestern side; **Pitchpine Ledges**, covered 6 feet and marked on its western side by a buoy; and a 3-foot shoal, marked off its southwest side by a buoy, about 0.2 mile southwestward of Carrying Place Head.

(518)

Anchorages

(519) There is good anchorage in the harbor for small craft and in the tributary harbors of Fish House Cove, West Point Harbor and Cape Small Harbor, but the bottom shoals too rapidly in **Tottman Cove**, north of **Flat Point**, for good anchorage.

(520) Small Point Harbor can be entered either southward of Wood Island or northward of Little Wood Island. **Wood Island** is rocky and partly wooded, and **Little Wood Island** is thickly wooded. **Small Point**, a village on the eastern side of the harbor, has an improved highway to Bath, the nearest city.

(521) **Cape Small Harbor**, between Hermit Island and Cape Small, affords good anchorage for small craft, but its entrance, with 4 feet at low water, is narrow and difficult, and should be entered only with local knowledge or at high water. The harbor entrance is marked by private unlighted buoys off Mill Point. A private camping ground is on Hermit Island. A large white hotel northeastward of Goose Rock may be used as a mark to clear the rock when entering northward of it.

(522) The passage between Goose Rock and **Mill Point** is sometimes used by local fishing craft but is not recommended for strangers. The best water is reported to be obtained by entering northward of Goose Rock and then favoring the eastern shore until abeam of the northern extremity of Mill Point, then favoring the west side of the two islets eastward of Mill Point until southward of the fish pier at the lobster pound. The channel eastward of the islets should be used only at high water. There is reported to be 18 feet at the fish pier; gasoline is available. A marine railway at the pound can haul out craft up to 60 feet in length.

(523) Anchorage is in midchannel southward of the pier where swinging room can be found in 8 to 10 feet.

(524) A restaurant, open in the summer, is at **Head Beach** at the south end of the harbor, which joins Hermit Island to Small Point. A woodland road leads from the beach to the various camping sites on the island and to the lobster pound. Restrooms and picnic area are available at the restaurant when open.

(525) There are a number of private float landings and many moorings in the harbor, which is secure in all weather.

(526) **Carrying Place Cove** is a narrow, partially bare thoroughfare on the north side of Small Point Harbor. The thoroughfare is entered just westward of **West Point** and leads northward between **Carrying Place Head** on the west and the village of West Point on the east. It is reported that 5 feet can be carried through the thoroughfare at high water; local knowledge is advised. There is a small islet with a house on it in the middle of the passage. Two overhead power cables crossing the thoroughfare have a minimum clearance of 30 feet. The southern part of Carrying Place Cove is also known locally as **West Point Harbor.**

(527) There are numerous fish wharves and several service wharves along the east side of the thoroughfare at the fishing village of **West Point.** Two of the service

are no landings. A rock awash is almost in the middle of the anchorage, and The Basin shoals in its eastern half. In 1980, depths considerably greater than those charted were reported in The Basin; care should be taken to allow sufficient scope of chain when anchoring.

(550) **Winnegance Bay**, on the east side of New Meadows River 3 miles north of the entrance, is a large bight with secure anchorage in 18 to 24 feet. There are a few private landings. The southeast side of the bay is foul. **Bushy Islet** and **Hen Islet** are near the edge of the foul ground; **Hen Island Ledge**, awash at its southwest end at low water, extends 500 yards west-southwestward from the south islet, where it is marked by a daybeacon. The north side of the bay is clear. There is considerable yachting activity in this bay, and good anchorage is available in **Brighams Cove** at the head of the bay.

(551) A light on **Birch Point**, on the northwestern side of the entrance to the bay, shows a higher intensity beam downstream and marks the reach in the river from Sheep Island Ledge to the entrances to Winnegance Bay and the upper river. The light is the only lighted aid in the river northward of Bear Island.

(552)

Anchorages

(553) Good anchorage can be found in the long coves on either side of **Rich Hill**, about 2 miles northward of Birch Point.

(554) **New Meadows** and **Harding** are small villages on the highway at the bridge crossing New Meadows River at the head of navigation. The former dam has been converted to a causeway with a narrow, low-level bridge; caution is advised in the vicinity because of swift currents reported at the bridge. The remains of the piers of an old highway bridge, 0.3 mile below the former dam, are covered at high water. Caution should be used in passing between them. An overhead cable with a clearance of 50 feet is just above the bridge ruins. A town float and a launching ramp are on the west bank in the vicinity of the old bridge ruins.

(555) An inn is on the east bank at the causeway; lodging and a restaurant are available. Gasoline and non-potable water are available at the float of a marina on the west bank at the causeway; depths of 5½ feet are reported alongside the float. Guest berths and some marine supplies are available, and provisions can be obtained nearby. A 5-ton fixed lift at the marina can handle craft up to 24 feet for hull and engine repairs or winter storage. Good anchorage in 10 feet is off the landings.

(556) **Ridley Cove** is eastward of **Yarmouth Island** and just westward of the entrance to New Meadows River. The cove has good anchorage in 23 to 37 feet, but is exposed to southerly and southwesterly winds. It should be avoided by strangers because of the numerous unmarked ledges and rocks off the entrance. On the end of **West Cundy Point** is a large one-story house, which is very conspicuous from Ragged Island to Small Point.

(557) From the northern end, a narrow deep channel leads close westward of **George Island** into Hen Cove. Another narrow channel with a reported depth of 3 feet, obstructed and suitable only for small craft in the absence of local knowledge, leads into Quahog Bay. **Hen Cove** has extensive shoals but is a good anchorage for small craft.

(558) **Little Yarmouth Island**, close westward of Yarmouth Island, has ruins of a wharf on its north end.

(559) Dangers off the entrance to Ridley Cove include **Jenny Island**, 10 feet high and grassy; **North Jenny Ledge**, covered 2 feet and marked by a buoy at the south end; **Jenny Ledge**, which uncovers 5 feet; **Ballaststone Ledge**, with grassy **Duck Rock** 5 feet high on it; and numerous bare spots on **Yarmouth Ledges. Flash Island** is a small islet on the extensive ledge area southward of Yarmouth Island.

(560) **Quahog Bay** is a narrow arm extending about 4 miles in a northeasterly direction. It offers good anchorage for small vessels. Numerous unmarked ledges and many small islands are off its entrance, which is between Yarmouth Island and Ledges on the east and **Long Point Island** on the west.

(561) The buoyed channel from New Meadows River to Orrs and Bailey Islands leads across the entrance.

(562) There is also a good channel between Saddleback Ledge, Ragged Island, Blacksnake Ledge, Yellow Rock and Two Bush Island on the east and Round Rock, **Middle Ground Rock** and Cedar Ledges on the west.

(563) **Saddleback Ledge** uncovers 5 feet; **Ragged Island**, about 50 feet high and scantily wooded on top, has a house on it; **Blacksnake Ledge** uncovers; **Yellow Rock**, 4 feet high, and **Two Bush Island**, are grassy; **Round Rock**, marked off its south side by a lighted buoy, uncovers 7 feet; and **Cedar Ledges**, 2 feet high, are bare.

(564) Several unmarked ledges and sunken rocks are in Quahog Bay. **South Ledges**, covered at high water and marked by a buoy on the west side, and **North Ledge**, awash, extend 0.4 mile southwestward and northeastward, respectively, from **Pole Island.**

(565) **Card Cove**, on the west side of the bay and west of Pole Island, is used by small fishing boats, but the entrance is only 50 yards wide between ledges off **Pinkham Point** and the point on the south side. There are a few private wharves. Good anchorage in 29 feet is in the cove.

(566) On the east side of Pinkham Point in the channel between it and Pole Island are two lobster wharves with float landings. Depths of 15 to 20 feet are reported alongside. Gasoline and diesel fuel are piped to the floats. Excellent anchorage is off the wharves.

(567) The upper end of Quahog Bay, about 1.7 miles above Pinkham Point off **Dyer Cove** (43°49'15"N., 69°55'11"W.), affords one of the best anchorages on the coast for cruising craft, and the swimming in the warm water of **Mill Cove** northward of **Snow Island** is reported to be excellent.

(568) In **Orrs Cove**, 0.4 mile above Dyer Cove, is a boatyard and marina on the west bank about 0.4 mile

below the head. Limited transient berths and moorings, gasoline, diesel fuel, water, electricity, ice, marine supplies and a pump-out station are available. Craft up to 50 feet long or 15 tons are handled on skids for limited hull, engine and electronic repairs or open winter storage. In 2004, depths of 6 feet were reported alongside the fuel pumps.

(569) **Gun Point Cove**, westward of Quahog Bay, is a narrow arm of no importance making northward on the east side of Orrs Island. There are no wharves. **Gun Point** on the east side is wooded and has a house on the end. **Hen Island** and **Oak Island** are islets on the ledge area southward of Gun Point and **Long Point Island**. A channel across this ledge area, marked by a buoy, is part of an inside passage for small craft from New Meadows River westward to Orrs and Bailey Islands.

(570) A passage with a depth of 4 feet extends from the north end of Gun Point Cove into Harpswell Sound. This passage is crossed by State Route 24 highway bridge, which has a fixed span with a clearance of 10 feet. The passage is difficult because of strong currents and unmarked ledges and should not be attempted by strangers. There are several lobster wharves with float landings in the vicinity of the bridge.

(571) **Lowell Cove**, in the south end of Orrs Island, is used as an anchorage by local fishermen. There are a number of fish and lobster wharves in the cove, most of which dry at low water. Ice, provisions and some supplies can be had at a general store in the village of **Orrs Island**, at the head of the cove. The supply of water is very limited. There is a good road to the mainland.

(572) **Water Cove**, southward of Lowell Cove, makes into the north end of **Bailey Island**. The cove is foul near its shores and is little used.

(573) **Ram Island** and **Pond Island**,southeastward of Lowell Cove, are round and grassy. **Pond Island Ledges**, awash at high water, extend 0.6 mile southwestward of Pond Island, have many spots bare at low water and are unmarked.

(574) **Halfway Rock**, about in the middle of the southern part of Casco Bay, is a low, rocky islet marked by **Halfway Rock Light** (43°39'21"N., 70°02'12"W.), 76 feet above the water, shown from a white tower attached to a dwelling. A mariner radio activated sound signal at the light is initiated by keying the microphone five times consecutively on VHF-FM channel 83A. Ledges extend 0.2 mile southwestward and northward from it. **Webster Rock**, covered 8 feet at the end of the ledge extending northward, is marked by a buoy.

(575) **Drunkers Ledges**, 2 miles north-northeastward of Halfway Rock, consist of two ledges 0.3 mile apart. The southeast one, **Eastern Drunkers Ledge**, is covered 4 feet and is marked on its southwest end by a buoy. The northwest one uncovers about 3 feet and is also marked by a buoy.

(576) Between Drunkers Ledges and Jaquish Island is **Mark Island Ledge** covered 4 feet and marked at its north end by a buoy. An area of broken ground with depths of 4 to 22 feet extends south-southwestward from Jaquish Island to Eastern Drunkers Ledge. In heavy weather the sea breaks on the shoalest places on it.

(577) **Little Mark Island**, on the west side of Merriconeag Sound at the entrance, is 37 feet high and grassy. It is marked by **Little Mark Island Monument Light**, 74 feet above the water, shown from a black and white stone pyramidal monument. **Great Mark Island** is 24 feet high, bare, and grassy.

(578) **Whale Rock**, 5 feet high, is 0.4 mile southwestward of Little Mark Island.

(579) **Merriconeag Sound** and **Harpswell Sound** are of little commercial importance, but they form the approach to a good and convenient anchorage. Vessels of the deepest draft can enter and find anchorage in 21 to 60 feet, good holding ground.

(580) The entrance is 3.5 miles north-northeastward of Halfway Rock Light and is marked on its western side by the light on Little Mark Island. The two sounds extend in a northeasterly direction for 10 miles to **Harpswell Cove**, and for the first 4 miles the important dangers are marked. Above this, strangers should not go without a pilot, as the channel is narrow, and flats make out some distance from the shore in several places.

(581)

Anchorages

(582) **Special anchorages** are in Harpswell Sound, at Harpswell Harbor, Mackerel Cove, Beals Cove and the yacht club anchorage off the southwestern end of Orrs Island. (See **33 CFR 110.1** and**110.5**, chapter 2, for limits and regulations.)

(583) **Jaquish Island**, 29 feet high and grassy, on the east side of the entrance to Merriconeag Sound, and **Turnip Island,** 17 feet high and grassy, are conspicuous. **Turnip Island Ledge**, about 0.2 mile southwestward of Turnip Island, is awash at its southern end and is marked by a lighted gong buoy about 0.1 mile to the westward. **Jaquish Gut**, between Jaquish Island and Bailey Island to the northward, is reported to have a controlling depth of 7 feet; local knowledge is advised. A bell buoy is about 0.5 mile southeastward of Jaquish Island.

(584) **Charity Ledge**, eastward of Jaquish Island and covered 11 feet, is marked by a buoy.

(585) **Mackerel Cove**, in the southwestern shore of Bailey Island on the eastern side of the entrance to Merriconeag Sound, is a good anchorage in 30 to 48 feet for small craft, which use it frequently; it is open southwestward, but a heavy sea seldom enters.

(586)

Anchorage

(587) A **special anchorage** is in Mackerel Cove. (See **33 CFR 110.1** and **110.5**, chapter 2, for limits and regulations.)

(588) The village of **Bailey Island** is on Mackerel Cove. A ledge, marked by a lighted buoy, extends southward from **Abner Point**, on the west side of the entrance. There are no known dangers in the channel except for unmarked

ledges that extend from the shores and an unmarked rock, reported to uncover about 2 feet, near the head of the cove. The water shoals gradually toward the head. It is reported that the west side of the channel should be favored in making the wharf and marina on the west side of the cove near the head. An uncharted large white house, on the west shore, is reported to be a good guide. Several fish piers are in the cove.

(589) A wharf and marina on the west side near the head has gasoline, diesel fuel, water and electricity at the floats, which have 6 to 10 feet reported alongside; ice, some provisions and some marine supplies are available. A restaurant and motel are at the wharf. Overnight berthing is permitted, and guest moorings are maintained. Lodging and a store are available in the village.

(590) There is a good road to the interior. Ice seldom obstructs the cove in winter.

(591) The southern point of Bailey Island is marked by two high observation towers and a house.

(592) A boatyard is on the north side of the unnamed cove on the west side of the island, just northward of Mackerel Cove. The yard has a 1½-ton crane, a marine railway that can handle craft up to 50 feet in length for hull and limited engine repairs and dry, covered or open winter storage. Electric and electronic repairs can also be made, and some marine supplies can be obtained.

(593) A marina in the cove has gasoline and water available. Depths of 4 feet are reported alongside the floats.

(594) Small boats can be launched from the hard beach at the head of the cove making into the north end of Bailey Island, west of the bridge over Wills Gut. The cove is protected on its westerly side by a spit. A lobster pound with wharf and float landing is on the end of the spit. Depths of 6 feet are reported alongside the float; gasoline is available. Parking and picnic areas adjoin the restaurant on the wharf.

(595) A small-craft launching ramp, usable only at high water, is available at a small marina on the north end of Bailey Island just east of the bridge.

(596) **Wills Gut** is a thoroughfare between the south end of Orrs Island and the north end of Bailey Island. It is used by local fishing boats, but the channel is very narrow and difficult. Strangers using the channel should await low water, when the ledges bare enough on each side to indicate the channel course. Bailey Island Bridge crosses the channel and has a fixed span with a clearance of 13 feet. An overhead power cable at the bridge has a clearance of 41 feet.

(597) A summer resort is on the southwest end of Orrs Island. Only a few piles remain of the old steamer wharf there. The Orrs-Bailey Yacht Club has a float with 15 feet alongside close northward of the ruins of the old steamer wharf. Gasoline and water are available at the float. A **special anchorage** is off the club. (See **33 CFR 110.1** and **110.5**, chapter 2, for limits and regulations.)

(598) There are several wharves southward of the club landing toward the Wills Gut bridge, on one of which is a general store where ice, provisions and some marine supplies may be obtained. There is a boat ramp with parking and picnic areas at the store.

(599) The approach to the wharves from Merriconeag and Harpswell Sounds is northward of a buoy and a daybeacon marking the end of **Cox Ledge**, which uncovers at low water and extends from the northwestern point of Bailey Island.

(600) **Pinkham Island**, on the west side of Merriconeag Sound northward of the thoroughfare leading westward into Potts Harbor, has one house and is 31 feet high, bare and grassy. Ledges, bare and covered, extend 500 yards southward of the island to a buoy. A channel northwestward of the island has a depth of at least 8 feet. It leads between shoals and should not be used by strangers.

(601) **Harpswell Harbor**, on the west side of Harpswell Sound 3.5 miles above Little Mark Island, is a good anchorage in 18 to 36 feet, shoaling gradually to the head. There are private float landings for small craft on the west side, and the small settlement of **West Harpswell** is on the main road back of the landing. The waters of Harpswell Harbor are a **special anchorage.** (See **33 CFR 110.1** and **110.5**, chapter 2, for limits and regulations.)

(602) **Beals Cove**, a shoal, foul cove on the west side of Orrs Island, is a **special anchorage**. (See **33 CFR 110.1** and **110.5,** Chapter 2, for limits and regulations.)

(603) **Reed Cove**, on the west side of Orrs Island, is reported to have good anchorage in 12 to 18 feet with protection from south and southwesterly winds; sand over mud bottom.

(604) There is a thoroughfare from the north end of Harpswell Sound through **Ewin Narrows**, **Prince Gurnet**, **Long Reach** and **Gurnet Strait** to New Meadows River. It is occasionally used by local boats. The channel is narrow, has a depth of about 6 feet and has many dangers; the tidal currents are strong, and the thoroughfare should not be used by strangers. It is sometimes marked by bush stakes.

(605) A fixed highway bridge with a clearance of 30 feet crosses the southern part of Ewin Narrows.

(606) Gurnet Strait is crossed by State Route 24 highway bridge, which has a fixed span with a vertical clearance of 10 feet and a horizontal clearance of 42 feet. The depth at the bridge is reported to be about 6 feet. In 1979, shoaling to bare was reported in the northerly part of the channel.

(607)

Currents

(608) The **tidal current** through Gurnet Strait is very strong at strength-estimated at 7 to 8 knots at times–and boats go through only at slack water. The ebb current runs eastward. Low-water slack occurs a little before low water at Portland.

(609) At the strength of the current, there is a difference of elevation of probably 3 feet in the level of the water on either side of the bridge. The flood currents meet in the reach between Prince Gurnet and Gurnet Strait.

(610) **Gurnet** is a village on the shores of Gurnet Strait. There are several wharves with float landings. Restaurants

are available on the wharves or nearby. Provisions can be obtained.

(611) The part of Casco Bay westward of **Harpswell Neck** has numerous sounds, bays and rivers, separated by islands mostly lying in a northeasterly and southwesterly direction. Portland Harbor, at the western end of the bay, is the principal port of Maine. Many summer resorts and landings are on the islands and shores of the bay, and small vessels from Portland run as far east as Bailey Island and call at the state piers on the islands.

(612) There are broad channels into the bay through Broad, Luckse and Hussey Sounds, and secure anchorage for vessels of any draft can be found. The bay is frequented by many yachts and small pleasure craft and some fishing boats. The ferries running to the state piers are of 4 to 6 feet in draft.

(613) Through **Hussey Sound**, 42 feet can be taken on either side of Soldier Ledge, which has been cleared to a depth of 40 feet. The inshore channel extends from Peaks Island along the north shore of Long Island. A submerged obstruction is reported close westward of the lighted gong buoy on the south side of the eastern entrance to Hussey Sound.

(614) The Coast Guard Captain of the Port, Sector Northern New England, advises the minimum visibility requirements for deep draft vessels for Hussey Sound and its approach channels are ¼ mile. Pilots and tugs are available at Portland; see Pilotage, Portland, indexed as such. Tug/barge units should not change mode of towing in the vicinity of the Hussey Sound entrance due to the restricted navigation area available.

(615) From the fairway bell buoy in the entrance to **Broad Sound**, 42 feet can be carried to good anchorages in upper Broad Sound to Middle Bay and to the vicinity of French Island and the north end of Great Chebeague Island.

(616) Also, 42 feet can be taken through **Luckse Sound** to the vicinity of **Bangs Island**. There is a minimum effective cleared depth of 20 feet westward of the island to off the north point of Great Chebeague Island.

(617) An inshore channel used by interisland ferries, yachts and fishing craft extends from the south point of Great Chebeague Island around either side of Bangs Island, across Broad Sound, and through Potts Harbor to Merriconeag Sound.

(618) **Potts Harbor** is a large irregular bight in the southern end of Harpswell Neck, between Potts Point on the east and Basin Point on the west, and **Haskell Island** and **Upper Flag Island** and the ledges between them on its southern side. The harbor affords good anchorage in depths of 24 to 33 feet.

(619) **South Harpswell** is a village on the east side of Potts Harbor. A town wharf with float landing is on the east side of the harbor, about 0.4 mile above **Potts Point**; depths of 6 feet are reported alongside the float. A fish wharf with 7 feet reported alongside in 1979 is on the west side of **Ash Point**, at the entrance to Basin Cove; gasoline and diesel fuel are available. A marina and restaurant are on the west side of the harbor about 0.1 mile above **Basin Point**. Transient berthage, gasoline, diesel fuel, water, electricity and ice are available; depths of 5 feet are reported alongside the float landing. A trailer at the marina can handle craft up to 35 feet for some engine repairs and open winter storage.

(620)

Anchorage

(621) A **special anchorage** is on the northeast side of Basin Point. (See **33 CFR 110.1** and **110.5**, chapter 2, for limits and regulations.)

(622) There are two entrances to the harbor. The eastern one, from Merriconeag Sound, is marked by buoys and a daybeacon. It has a depth of about 14 feet but is narrow and crooked with strong tidal currents. It is suitable only for small craft or small vessels with local knowledge.

(623) The western entrance, between Upper Flag Island and Horse Island, is straight and about 225 yards wide at its narrowest part, between Horse Island and the edge of the shoal between Upper Flag Island and **Thrumcap**, a grass-covered rock. **Upper Flag Island**, 59 feet high, **Little Birch Island**, 14 feet high, and **Horse Island**, 23 feet high, are grass covered.

(624) A ledge extending southwestward from Little Birch Island is marked by a bell buoy, and a shoal covered 3 feet, about 0.2 mile west of Upper Flag Island, is marked on its southwestern end by a buoy.

(625)

Anchorage

(626) **Basin Cove** is a **special anchorage**. (See **33 CFR 110.1** and **110.5**, chapter 2, for limits and regulations.)

(627) The entrance to the cove is obstructed by the remains of an old dam that is covered at high water. Entry into this cove is dangerous at all times and should not be attempted without local knowledge.

(628) **Outer Green Island**, 4 miles westward of Halfway Rock Light (43°39.4'N., 70°02.3'W.), is grassy. **Junk of Pork**, a high rock with surrounding bare ledges, is 250 yards southward of it. **Johnson Rock**, 0.1 mile northeastward of Outer Green Island, with foul ground between, is covered 8 feet and is marked on its north side by a buoy.

(629) **Green Island Reef**, about 0.2 mile long and bare at low water, is 0.7 mile northeastward of Outer Green Island. It is marked on its southwest end by a buoy. **Green Island Passage**, leading between the buoys marking Green Island Reef and Johnson Rock, has a width of 400 yards and a depth of 44 feet and is used by small vessels. **Inner Green Island**, 0.4 mile northeastward of Green Island Reef with foul ground between, is 15 feet high and grassy.

(630) **Jewell Island** and **Cliff Island**, northward of Inner Green Island, are partly wooded. Numerous homes and several private landings are on the northwest shore of Cliff Island; the state pier and public float landing are on the west shore of the island about 0.7 mile from the south end. There is 22 feet at the head of the pier. Gasoline is available at a pier and float landing with 6 feet alongside

on the east shore of the island. Provisions can be obtained at a store near the pier. The old steamer wharf on the west shore of Jewell Island is reported to be in disrepair.

(631) **Johns Ledge**, covered 3 feet in places, extends 0.4 mile southwestward from the southern end of Cliff Island. Its end, covered 16 feet, is marked by a bell buoy. There is no safe passage for vessels between the bell buoy and the island. A buoy marks the broken ground and shoals southeastward of the island.

(632) A cove on the northwest side of Jewell Island has excellent anchorage with good protection in 10 feet, mud bottom. A prominent stone tower is on the south end of Jewell Island, and the ruins of two old wharves and a house are on the west side. There are no facilities.

(633) **Broken Cove**, about 1 mile northeastward of Jewell Island, is formed by a group of bare rocks and small islets connected by ledges extending 0.6 mile northeastward from **West Brown Cow**, a 36-foot-high grass-covered islet. The daybeacon on Stockman Island in range with or open eastward of the northeast point of Ministerial Island leads eastward of the ledges, which are marked on the northeast side by a gong buoy.

(634) **Eagle Island** is 64 feet high, wooded and prominent, and has a house and flagstaff on the northeast side. A ledge, which uncovers about 6 feet, extends 300 yards westward of the island; a buoy is on the southwest side of the ledge. The State of Maine maintains a pier, float, ramp and at least four guest moorings for summer visitors on Eagle Island.

(635) **Eagle Island Ledge**, awash at high water, is 300 yards southeastward of the southern end of Eagle Island. Ledges covered 4 to 5 feet extend 300 yards southeastward and 500 yards northeastward from Eagle Island Ledge. Partly bare ledges extend about 250 yards northward from Eagle Island.

(636) **Bates Island**, 29 feet high, and **Ministerial Island**, 24 feet high, both westward of Eagle Island, are grassy. They are surrounded by extensive ledges. **Stave Island** is sparsely wooded. **Stave Island Ledge** uncovers 2 feet and is marked by a buoy at its northeast end.

(637) **Hope Island**, in Luckse Sound opposite Cliff Island, is 90 feet high and wooded except on the southwest end, which is marked by a large house and flagpole. **Rogues Island**, 16 feet high, and **Sand Island**, northeastward of Hope Island, are grassy. The channel between them is marked by buoys. **Crow Island**, 0.8 mile north of Sand Island, is 15 feet high and grassy and has one house in the center.

(638) **Bangs Island**, 66 feet high, and **Stockman Island**, 36 feet high, are bare and grassy. Stockman Island has a daybeacon at the southwest end. **Goose Nest** is a grassy islet about 4 feet high, and **Goose Nest Ledge**, part of which uncovers about 7 feet, is northward of Goose Nest. A ledge extending 400 yards south of Goose Nest is marked by a buoy.

(639) **Whaleboat Island** is 74 feet high and wooded on the north end, the highest part, and 56 feet high and grassy at the south end. A light shown from a white skeleton tower is near the southerly point. A 23-foot shoal, about 0.2 mile south of the light, is marked by a buoy. The buoy also marks the junction of two deep channels leading to a naval fuel depot and wharf on the west shore of Harpswell Neck in Middle Bay. The tanks and other features of the naval fuel depot are conspicuous. The T-head pier is reported to have 35 feet alongside.

(640) The eastern channel leads between **Birch Island Ledge**, which uncovers 6 feet and is marked on its southwestern end by a buoy, and a buoyed 27-foot spot on the east, and Whaleboat Island on the west. The western channel leads between Whaleboat and Little Whaleboat Islands. It is buoyed.

(641) A **038°–218°** measured course, 5,946 feet long and marked by shore ranges, is on the west side of Whaleboat Island.

(642) **Little Whaleboat Island** is 35 feet high and wooded. Extensive ledges extend about 0.7 mile northward, westward and southwestward of the island. **Little Whaleboat Ledge**, covered 3 feet, and **Whaleboat Ledge**, covered 6 feet, are about 0.8 and 1 mile, respectively, southwestward of Little Whaleboat Island. Both are marked by buoys.

(643) **Middle Bay** makes northeastward on the west side of Harpswell Neck. **Harpswell Center** is a village on the main road of Harpswell Neck. The bay has good anchorage but is seldom used. **Lower Goose Island**, 73 feet high, and **Upper Goose Island**, 85 feet high, on the west side of the bay, are wooded.

(644) Gasoline and diesel fuel can be obtained at the float landing of a lobster pier on **Lookout Point**, on the east side of Middle Bay opposite Upper Goose Island. Depths of 3 feet are reported alongside the float.

(645) **Merepoint Bay**, shallow and obstructed by flats at its northern end, is between **Birch Island**, about 50 feet high, and **White Island** on the east and **Merepoint Neck** on the west. It is the center of considerable yachting activity in the summer season. The Merepoint Yacht Club on the neck is an organization of summer residents without formal clubhouse or landing. Several private float landings of members are used. **Merepoint** is a village on the neck.

(646) A marina with 3 feet reported alongside its float is on the east side of the neck, about 1 mile above **Mere Point**. Limited moorings, gasoline, diesel fuel, water, ice, some marine supplies and a pump-out facility are available. A marine lift to 22 tons and dry winter storage are also available.

(647) **Maquoit Bay** makes northeastward on the westward side of Merepoint Neck; the entrance is north of the Goose Islands. Most of the bay is shoal and is obstructed by flats covered 1 to 4 feet. Through the flats a channel with 19 to 24 feet leads for a distance of 2 miles northwestward from its entrance.

(648) A boatyard is on the west side of Merepoint Neck, about 2 miles above Mere Point. The marine railway at the yard can handle craft up to 35 feet in length for

hull and engine repairs, and dry, covered or open winter storage. A small-craft launching ramp is at the yard.

(649) Southward of Maquoit Bay, the chain of islands between **Sister Island** and **Bustins Island** are wooded, and there are flats between and northward of them. **Sister Island Ledge**, northward of 41-foot-high Sister Island, is partly bare at high water. Bustins Island, 83 feet high, has numerous cottages. A public landing is on the southeast side of the island with a post office and store nearby. Gasoline is available.

(650) Rocks, awash at low water, are 50 yards southeastward and 75 yards southwestward from the landing. Eastward of the landing is a house on a ledge about 100 yards offshore. **Little Bustins Island**, 15 feet high, is marked by a house and a clump of trees in the center.

(651) **Bustins Ledge**, southeastward of Bustins Island, is about 4 feet high in one spot. **French Island**, 62 feet high, is wooded. **Little French Island**, also wooded, is on ledges which extend northward of the island.

(652) **Harraseeket River** is west of Maquoit Bay. The approach is between Bustins Island on the east and **Moshier Island**, 91 feet high and wooded, on the west. The entrance to the river, between **Moore Point** and **Stockbridge Point,** is narrow and marked by buoys. Except for a dangerous midchannel rock, covered 2 feet, reported to lie in the entrance about 100 yards southwest of Pound of Tea Island, it has a depth of 23 feet.

(653) From the entrance the channel leads between flats, mostly bare at low water, in a northerly direction to **Weston Point**. Thence a shoal unmarked channel leads to **Porter Landing**, to which small craft drawing up to 6 feet or less are reported to go at high water with local knowledge.

(654)

Anchorage

(655) A **special anchorage** is between Stockbridge Point and Weston Point. (See **33 CFR 110.1** and **110.5**, chapter 2, for limits and regulations.)

(656) **South Freeport**, on the west side of Harraseeket River, about 0.7 mile above the entrance, has a town wharf with a depth of 15 feet reported alongside its float landing. Guest berths and moorings, electricity, gasoline, diesel fuel, water, ice, most marine supplies and a pump-out facility can be obtained at the float landings of marinas on either side of the town wharf; depths of 12 to 16 feet are reported alongside the floats. Lifts to 36 tons are available, and complete hull, engine and electronic repairs can be made. Dry open winter storage is also available.

(657) The Harraseeket Yacht Club with 19 feet reported alongside its float landing is about 300 yards southward of the town wharf; guest moorings are maintained. A motorboat passenger ferry operates from the town wharf to Bustins Island during the summer.

(658) At Porter Landing, 1.4 miles north-northeast of the South Freeport town wharf, is a boatyard that builds boats up to 40 feet and makes hull and engine repairs. The landing is reported accessible for a draft of 6 feet within 2½ hours of high tide.

(659) Prominent landmarks include a large stone turreted tower at South Freeport, a tank and standpipe at Yarmouth and the two stacks and green painted powerplant and oil tanks on Birch Point, the southwestern end of Cousins Island, which are visible from every section of Casco Bay. The stacks are marked by flashing lights.

(660) **Littlejohn Island** and **Cousins Island**, northward of Great Chebeague Island, are connected by a wooden bridge. The passage between the islands dries at low water. An overhead telephone cable crosses the passage just north of the bridge. There is a wharf on the southeast side of Littlejohn Island that has a reported depth of 3 feet alongside and is seldom used.

(661) There is a large powerplant on the north side of **Birch Point**, the southwestern end of Cousins Island; the two stacks and green painted powerplant and oil tanks are conspicuous throughout the bay. The two stacks are marked by flashing lights. The plant's T-head pier with dolphins can accommodate vessels 715 feet in length and 32 feet in draft. In 1979, depths of 33 feet were reported alongside; bottom is mud and rock. The pier is used by tankers that re-supply the powerplant with fuel oil. Vessels normally moor starboardside-to and require tugs and a line boat to handle bow and stern lines. Pilots and tugs are available at Portland; see Pilotage, Portland, indexed as such. Only fresh water is available; bunker fuel oil and diesel oil can be obtained in Portland. Vessels should engage the services of one or more escort tugs for transit to the Wyman Station Oil Terminal on Cousins Island. For inbound transits the escort tugs should be engaged in the vicinity of Cow Island. The use of escort tugs for vessels with an operational bow thruster will be at the discretion of the pilot.

(662) There is a wharf and float landing on **Doyle Point** on the east side of Cousins Island. No facilities are available. A motorboat taxi service operates from the landing to a stone wharf on the northwest side of Great Chebeague Island throughout the year.

(663) Cousins Island is connected to the mainland by a highway bridge. The fixed span over the main navigation channel has a clearance of 25 feet. Overhead power cables with a clearance of 68 feet over the main channel cross the waterway northeastward of the bridge.

(664) **Royal River** is a narrow crooked stream southwestward of Harraseeket River. The river is entered northward of Cousins Island through a dredged channel that leads from the northwestern part of Casco Bay to the river entrance between **Parker Point** on the south and **Fogg Point** on the north, and thence to the head of river navigation at the anchorage basin, about 0.7 mile below the town of **Yarmouth**. The channel is marked by buoys. The approach section of the channel is state maintained. Falls in the river are about 0.3 mile above the turning basin.

(665)

Small-craft facilities

(666) There are several small-craft facilities on the river below the bridge, with electricity, gasoline, diesel fuel, water, ice, provisions, marine supplies, pump-out stations, small-craft launching ramps and dry, covered or open winter storage available. Lifts to 60 tons and a marine railway that can handle craft up to 50 feet long or 20 tons are available for hull, engine and electronic repairs.

(667) **Cousins River**, a narrow shallow stream marked by private seasonal uncharted buoys, empties into the mouth of Royal River from northward. U.S. Route 1 and Interstate 295 highway bridges crossing the river about a mile above its mouth have 46-foot fixed spans with clearances of 3 feet. A boatyard on the west side of the river about 0.9 mile above the mouth builds boats up to 70 feet in length. The yard has a 3½-ton fixed crane and a marine railway that can handle craft up to 40 feet for hull and engine repairs; guest moorings are maintained.

(668)

Great Chebeague Island Prescumpscot River

(669) **Great Chebeague Island** is one of the largest islands in Casco Bay. **Indian Point**, a sandspit at the southwestern end of the island, has a house and a lone tree on it.

(670) **Chandler Cove** is formed by a bight in the southwestern end of Great Chebeague Island and by Little Chebeague Island; it is a good anchorage with 30 to 60 feet but is little used. Mariners are cautioned to avoid anchoring in the cable area that extends across the southeast part of Chandler Cove. There is passenger and freight service from Portland to the state pier and public float landing in Chandler Cove, at the south end of the island. The pier has a depth of 15 feet at the head. A water taxi service carries passengers from the wharf at Doyle Point on Cousins Island to the stone wharf on the northwest shore of Great Chebeague Island; there is 6 feet alongside the wharf.

(671) A boatyard is on the east side of Great Chebeague Island, in the bight west of Crow Island. Gasoline, limited water and sewage pumpout facilities are available.

(672) **Chebeague Island** is a village located in the north central part of the island.

(673) **Little Chebeague Island** has a patch of woods in its center and a few houses. The old landing, on the east side, is in disrepair.

(674) **Long Island**, southwestward of Great Chebeague Island, has several landings on its northwest side. **Mariner** and **Long Island** are villages near the northern and western ends, respectively. A passenger and freight ferry from Portland calls at **Ponce Landing** on the northwest shore. The ruins of three piers are northeastward of Ponce Landing.

(675) The passage between **Crow Island**, 6 feet high, and the north point of Long Island, is closed by scuttled vessels with hulls showing above high water.

(676) The islands southward of Long Island are described with Portland Harbor.

(677) **Broad Cove** (43°46.0'N., 70°11.0'W.), in the northwestern part of Casco Bay, is shallow. Good anchorage is available in the middle of the cove, southwest of **Prince Point**, in 15 to 17 feet. It is open southward and eastward.

(678) **Falmouth Foreside**, a yachting center 4.3 miles north of Portland, has a boatyard that builds boats to 55 feet and a large marina with mobile hoists to 15 tons, where craft up to 50 feet can be hauled out for hull and engine repairs and covered or open dry winter storage. Electric and electronic repairs can be made. The marina has float landings with 6 to 10 feet reported alongside. Gasoline, diesel fuel and water are piped to the floats, and electricity and most marine supplies are available. There is a restaurant at the yard and lodging in the vicinity.

(679) There are numerous private moorings, and the yard maintains guest moorings. Ice, provisions, marine supplies and bus and taxi services are available.

(680) The Portland Yacht Club, close northeastward of the yard, has a float landing with 10 feet reported alongside; water and electricity are available. The club has a restaurant and club facilities for members and guests.

(681) The waters off Falmouth Foreside shore from Prince Point (43°42.7'N., 70°13.0'W.), northeastward for 1.8 miles are a **special anchorage**. (See **33 CFR 110.1**and**110.5,** chapter 2, for limits and regulations.) The **harbormaster** supervises the moorings; he can be reached by telephone (207-781-4673).

(682) In approaching the landings from the southward, care should be taken to pass eastward and northward of the buoy close eastward of **York Ledge** before rounding up to the northwestward. A number of small craft cutting too close to the buoy have hung up on the ledge. A daybeacon is on the ledge.

(683) **Sturdivant Island**, 51 feet high and covered with grass and bushes, is partly wooded and has a house on it.**Sturdivant Island Ledges**, about 0.4 mile south-southwest of the island, uncover 7 feet in places and are marked on the southern, eastern and western edges by buoys. **Underwood Ledge**, to the westward, is covered 4 feet and is marked on its eastern side by a buoy. **Basket Island** is wooded.

(684) **Upper Basket Ledge** and **Lower Basket Ledge** both uncover 5 feet and are marked by daybeacons. A 12-foot spot, about 700 yards westward of the daybeacon on Lower Basket Ledge, is marked by a buoy on its southwestern side. **Clapboard Island** is 50 feet high and wooded and has a private landing on its west side. The island is surrounded by ledges, bare and covered.

(685) **Cow Island Ledge Light** (43°42'11"N., 70°11'19"W.), 23 feet above the water, is shown from a spindle tower with a red and white diamond-shaped daymark on a red caisson; the light marks the ledge

between Clapboard Island and Cow Island. A buoy marks the northeast extremity of the ledge.

(686) **Waites Landing** is 1.7 miles southward of Falmouth Foreside. **The Brothers** are two low, flat islets, 6 and 11 feet high.

(687) **Mackworth Island** is connected to **Mackworth Point**, the eastern entrance point of Presumpscot River, by a stone causeway and highway bridge on piles with a 17-foot fixed span and a clearance of 5 feet at the navigation channel through the bridge.

(688) **Presumpscot River**, the entrance of which is between Mackworth Island and **Martin Point**, has a narrow crooked channel with a depth of 13 feet to the U.S. Route 1 highway fixed bridge at the entrance. The bridge has a clearance of 12 feet. For about 1 mile above the bridge, the channel has a depth of 6 feet.

(689) Three fixed spans, Interstate 295 Highway bridge, a railroad bridge, and State Route 9 Highway bridge, cross the river about 2 miles above the U.S. Route 1 fixed bridge; the minimum clearance is 9 feet. An overhead power cable between the railroad bridge and State Route 9 Highway bridge has a clearance of 42 feet. There is no waterborne commerce on the river and the channel is unmarked. On Martin Point, the buildings of the former marine hospital are conspicuous.

(690)

Portland Harbor to Back Cove

(691) **Portland Harbor**, at the western end of Casco Bay, is the most important port on the coast of Maine. The ice-free harbor offers secure anchorage to deep-draft vessels in all weather. There is considerable domestic and foreign commerce in petroleum products, wood pulp, paper, seafood products and general cargo. It is also the Atlantic terminus of pipeline shipments of petroleum products to Canada.

(692) The **outer harbor** comprises the area westward of Cushing, Peaks, House, Great and Little Diamond Islands from the entrance at Portland Head to the entrance of Fore River at **Fish Point**—including three general anchorages and the oil discharging berth westward of **Spring Point**. The Coast Guard Captain of the Port, Sector Northern New England, advises the minimum visibility requirements for deep draft vessels for Portland Outer Harbor are 0.5 mile.

(693) The inner harbor is considered to be in two sections; the outer part or **Main Harbor**, extending from the entrance of Fore River to the Casco Bay Bridge, and the inner part, or **Fore River**, from Casco Bay Bridge to the head of deepwater navigation at the combined fixed railroad and highway bridge. The Coast Guard Captain of the Port, Sector Northern New England, advises the minimum visibility requirements for deep draft vessels for Portland Inner Harbor are 0.25 mile.

(694) **Portland**, an important manufacturing, fishing and industrial center, is on the north side of the inner harbor with all the railroad, bulk and general cargo terminals and piers. **South Portland** is on the south side of the harbor with all of the petroleum handling terminals and pipeline facilities along its waterfront.

(695) The main approaches to the harbor are from the southward from Portland Lighted Whistle Buoy P, or from the eastward from Halfway Rock Light to the entrance of the harbor between Portland Head and Ram and Cushing Islands.

(696) **Traffic Separation Scheme (Portland)** has been established in the approaches to Portland Harbor.

(697) The Scheme is composed basically of **directed traffic lanes** each with one-way inbound and outbound traffic lanes separated by **defined separation zones** and a **precautionary area.** The Scheme is recommended for use by vessels approaching or departing from Portland Harbor but is not necessarily intended for tugs, tows or other small vessels which traditionally operate outside of the usual steamer lanes or close inshore.

(698) The Traffic Separation Scheme has been designed to aid in the prevention of collisions at the approaches to major harbors but is not intended in any way to supersede or alter the applicable Navigation Rules. Separation zones are intended to separate inbound and outbound traffic lanes and to be free of ship traffic and should not be used except for crossing purposes. Mariners should use extreme caution when crossing traffic lanes and separation zones. (See **33 CFR 167.1** through **167.15** and **167.50** through **167.53**, chapter 2, for limits and regulations and Traffic Separation Schemes, chapter 1, for additional information.)

(699) **Recommended Two-Way Routes** have been established in the approaches to Portland Harbor and Casco Bay, through Hussey Sound to Cousins Island and through Broad Sound to Harpswell, Maine. Deep draft vessels including tugs and barges are requested to follow the designated routes. The routes were designed to provide safe, established routes for vessels; to reduce the potential for conflict with recreational boaters, fishing gear, and other small craft; and to reduce the potential for grounding or collision. Vessels are responsible for their own safety and are not required to remain inside the routes nor are fishermen required to keep fishing gear outside the routes. Other vessels, while not excluded from these routes, should exercise caution in and around these areas and monitor VHF-FM Channels 16 or 13 for information concerning deep draft commercial vessels (including tugs and barges) transiting these routes.

(700)

Prominent features

(701) **Portland Lighted Whistle Buoy P** (43°31'36"N., 70°05'28"W.) is located about 5.3 miles east-southeastward of Cape Elizabeth Light and is equipped with a sound signal and racon.

(702) **Cape Elizabeth**, the southern entrance point of Casco Bay, is marked by **Cape Elizabeth Light** (43°33'58"N., 70°12'00"W.), 129 feet above the water, shown from a 67-foot white conical tower. A mariner radio activated sound

(714)

Security Broadcast System
Portland Harbor

The Coast Guard Captain of the Port, Sector Northern New England, has established a voluntary system of radiotelephone broadcast/reporting procedures designed to give masters and pilots real-time information on marine traffic in Portland Harbor. The system supplements the Vessel Bridge-to-Bridge Radiotelephone Regulations contained in **33 CFR 26** (see chapter 2), and all vessels subject to these regulations are urged to participate in the system. Nothing in these procedures shall supersede the **Navigation Rules** or relieve the master of the vessel of his responsibility for the safe navigation of the vessel. These recommended procedures are designed to give notice of unseen vessels, give notice of intended movement, clear VHF-FM channel 13 of traffic unrelated to navigation and give vessels information on other vessels within the immediate vicinity.

All participating vessels are requested to use VHF-FM channel 13 for all bridge-to-bridge communications, including listening watches and security calls, except when calling a small vessel not responding on channel 13, in which case channel 16 is appropriate.

Participating vessels shall maintain a listening watch commencing 30 minutes prior to getting underway or 30 minutes prior to reaching the vicinity of Portland Lighted Whistle Buoy P inbound.

Security calls shall be made as follows:
- 15 minutes prior to getting underway
- Getting underway (including route)
- Passing Portland Lighted Whistle Buoy P (inbound)
- 15 minutes prior to passing Willard Rock (43°36.1'N., 70°13.4'W.), Witch Rock (43°37.4'N., 70°10.6'W.), or entering Hussey Sound (43°39.9'N., 70°10.0'W.)—For inbound vessels not passing Portland Lighted Whistle Buoy P
- Passing Willard Rock Lighted Gong Buoy 7, Witch Rock Lighted Bell Buoy 2, or Hussey Sound Lighted Gong Buoy 3 (include destination)
- Passing Spring Point Ledge Light (43°39.1'N., 70°13.4'W.)
- Passing Casco Bay Bridge
- Mooring or anchoring.

During periods of low visibility, security calls should be made at more frequent intervals.

If a call is made to a ship or station to pass any of the above information on channel 13, an additional security call is unnecessary. Example: tug and barge in Fore River calling Casco Bay Bridge 15 minutes prior to getting underway to arrange for an opening.

Vessels carrying passengers or cargo and not required by law to comply with the Vessel Bridge-to-Bridge Radiotelephone Regulations are encouraged to monitor and respond on channel 13. During periods of low visibility these vessels should follow the security call procedures discussed above, except that security calls 15 minutes prior to getting underway, when passing Portland Lighted Whistle Buoy P, and 15 minutes prior to passing Witch Rock, Willard Rock, or entering Hussey Sound should not be made.

Coast Guard Sector Northern New England monitors channel 16 and will receive and transmit information when necessary.

signal near the light is initiated by keying the microphone five times consecutively on VHF-FM channel 83A—the sound signal is about 266 yards southeast of the light. An abandoned lighthouse tower is about 300 yards to the southwest. Numerous houses are near the light.

(703) A stone tower about 0.5 mile southward of Portland Head Light is conspicuous as is **Chimney Rock** about 300 yards southeastward of the tower.

(704) **Portland Head Light** (43°37'23"N., 70°12'28"W.), 101 feet above the water, shown from an 80-foot white tower, marks the south side of the entrance. A mariner radio activated sound signal at the light is initiated by keying the microphone five times consecutively on VHF-FM channel 83A. A directional light, 23 feet above the water, is shown from the same structure.

(705) **Ram Island Ledge Light** (43°37'53"N., 70°11'15"W.), 77 feet above the water, shown from a light gray, conical tower about 400 yards south of **Ram Island**, and marks the north side of the entrance. A mariner radio activated sound signal at the light is intitiated by keying the microphone five times on VHF-FM channel 83A. The light is at the end of a ledge which is awash at low water.

(706) **Cushing Island**, on the northeast side of the entrance, is mostly grass covered. **White Head** is a bluff at its northeastern end. A pier is in Spring Cove on the north side.

(707) Two old observation towers on the island are conspicuous. One is on White Head at the northeast end of the island; another is 500 yards southwestward of it.

(708) **House Island**, also on the east side of the main channel, northwestward of Cushing Island, is the site of

the abandoned quarantine station. Old **Fort Scammel** on the southwest end is conspicuous, and the summit of the northeastern part of the island is marked by a house and flagpole. Lights mark the north and south ends of the island.

(709) **Spring Point** is on the west side of the channel about 1.8 miles northwest of Portland Head Light. The buildings at **Fort Preble** on and southward of the point are conspicuous. **Spring Point Ledge Light** (43°39'08"N., 70°13'26"W.), 54 feet above the water, is shown from a white tower on the end of a breakwater extending 300 yards northeast of Spring Point. A mariner radio activated sound signal at the light is initiated by keying the microphone five times on VHF-FM channel 83A. The light shows a white sector over the entrance fairway.

(710) **Fort Gorges**, a conspicuous gray stone structure, is on **Diamond Island Ledge**, 0.8 mile northwestward of House Island. The ledge has a large area which uncovers, and a few spots bare at high water. **Diamond Island Ledge Light** marks the west end of the ledge. The south and east side of the ledge are marked by buoys. The wreck of a six-masted schooner about 700 yards **018°** from Fort Gorges is visible at low water.

(711) On the bluff above and westward of Fish Point on the north side of the entrance is the city of Portland. There are numerous conspicuous landmarks on the bluff and in the city, most of which are charted. One of the most conspicuous and historical is the old observatory tower which resembles a lighthouse. The microwave towers on the telephone building are very conspicuous.

(712)

COLREGS Demarcation Lines

(713) The lines established for Casco Bay are described in **80.110**, chapter 2.

(715)

Recommended minimum under-keel clearances for the Port of Portland

(716) The U.S. Coast Guard, in cooperation with the Maine and New Hampshire Port Safety Forum, has established recommended minimum under-keel clearances for the Port of Portland, in order to prevent groundings and to promote safety and environmental security of the waterway resources for the Port of Portland. The group recommends that all entities responsible for safe movement of vessels in and through the waters of the Port of Portland operate vessels in such a manner as to maintain a minimum under-keel clearance as follows:

(717) (a) 3 feet, when transiting inside a line drawn between Ram Island Ledge Light and Portland Head Light to Dredged Channel Lighted Buoy 5 at the entrance to Fore River,

(718) (b) 2 feet, when transiting Dredged Channel Lighted Buoy 5, including Fore River,

(719) (c) 2 feet, when transiting via Hussey Sound inside a line drawn between Lighted Gong Buoy 3 and Lighted Buoy 4. The minimum under-keel clearance should be between the deepest draft of the vessel and the channel bottom; a minimum under-keel clearance of 1 foot is recommended for all berthing areas.

(720)

Channels

(721) The main entrance is from the southward, between Ram and Cushing Islands on the north and Portland Head on the south.

(722) In addition to the main entrance from the southward, there are several entrances and channels from eastward and northward between and westward of the islands, some of which have been described previously. These are seldom used except by local vessels familiar with them or by small craft.

(723) A Federal project provides for a 45-foot channel from the sea to Fort Gorges, thence 35 feet in the Inner Harbor and Fore River to a turning basin at the head of the project near the combination railroad and highway bridge, a 45-foot anchorage in Diamond Island Roads and a 30-foot anchorage off Fish Point. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A.

(724) **Whitehead Passage**, between Cushing and Peaks Islands, is an important and direct way out of Portland Harbor for vessels headed to the east. It is also used by the commercial day passenger sailing fleet. The prevailing winds in the channel are southwest. The channel is narrow but the principal dangers in the passage are marked. In 2007, it was reported that an obstruction covered 5 feet was located in the channel in about 43°38.9'N., 70°11.6'W. Mariners transiting the channel are advised to use caution between Daybeacon 3 and Buoy 6.

(725) **Diamond Island Pass**, between Peaks Island and Little and Great Diamond Islands, is marked by lighted and unlighted buoys, and is used mainly by small craft. To carry the best water, pass 50 yards off the old and former Coast Guard buoy pier on Little Diamond Island and the wharf on the south end of Great Diamond Island.

(726) A channel, marked by buoys, is westward of Great and Little Diamond Islands and connects Hussey Sound with Portland Harbor.

(727) A channel dredged to 15 feet and marked with daybeacons and buoys leads from the main channel in Fore River to the two mole-type piers of **South Portland Coast Guard Station** in South Portland, about 0.4 mile northeast of Casco Bay Bridge.

(728) **Fore River** constitutes the Inner Harbor of Portland. The Casco Bay Bridge, about 1.5 miles above the entrance to the river, has a bascule span with a clearance of 55 feet. (See **33 CFR 117.1** through **117.59**, chapter 2, for drawbridge regulations.) The Casco Bay Bridge is often considered to pose the greatest risk to large vessels that transit into or out of the Fore River. Great care and prudent seamanship must always be exercised by mariners who transit through this bridge span. Mariners are cautioned

that strong crosscurrents on both the ebb and flood tides frequently tend to set vessels to the South Portland side of the bridge. The dual railroad and highway bridge at the head of deep water navigation on the river has a fixed span with a clearance of 10 feet. All vessels passing through the bridge span should observe the voluntary speed limit of a maximum of 4 to 6 knots over the ground, except when additional speed is necessary to maintain proper steerage way. All vessels should also note the special procedures in Pilotage, Portland, this chapter, indexed as such.

(729) **Note:** The city councils of Portland and South Portland request that mariners voluntarily refrain from requesting draw openings of the Casco Bay Bridge during the peak hours of highway commuter traffic on Mondays through Fridays from 0700 to 0900 and from 1600 to 1800. It is also requested that mariners desiring draw openings of the Casco Bay Bridge on Saturdays, Sundays and holidays during June, July and August notify the bridgetender at the bridge by radiotelephone and also their agents and tug companies of the expected time of arrival at the bridge; a minimum of 1 hour's notice is desired. The draw of the bridge will be opened for transit of vessels upon arrival at the bridge. The bridgetender monitors VHF-FM channel 16 and works on channel 13; call sign KQU-653. The bridgetender may also be contacted by telephone at 207–774–3534.

(730)

Anchorages

(731) Secure anchorage for any vessel is available at all times in Portland Harbor. General and special anchorages are in the harbor. (See **33 CFR 110.1**, **110.6**, **110.6a** and **110.134**, chapter 2, for limits and regulations.) An abandoned cable area extends southeast from House Island through part of the available anchorage area.

(732) **Diamond Island Roads** with depths of 40 to 45 feet, is the principal deepwater anchorage in the outer harbor. The anchorage eastward of Fish Point, called the **30-foot anchorage**, has depths of 25 to 30 feet with lesser depths to 20 feet in the northeast part.

(733) A **special anchorage** is between Little Diamond Island and Great Diamond Island. (See **33 CFR 110.1** and **110.6**, chapter 2, for limits and regulations.)

(734) **Naval anchorage berths** extend north of Great Diamond Island to the south end of Cousins Island.

(735) A vessel-to-vessel **oil transfer anchorage area** in Casco Bay, about 3.5 miles northeastward of Portland, has been designated by the Maine State Department of Environmental Protection. The area is 1 mile square beginning at Hussey Sound Buoy 12, (43°42'10"N., 70°09'46"W.); thence north to

(736) 43°43'10"N., 70°09'46"W.; thence west to

(737) 43°43'10"N., 70°11'09"W.; thence south to

(738) 43°42'10"N., 70°11'09"W.; thence east to origin.

(739)

Dangers

(740) There are numerous isolated dangers in the approaches to the harbor and the most important ones are marked. **West Cod Ledge**, a 6.5-mile-long area of broken ground and isolated shoals, sets across the entrance from northeastward and southwestward. These include **Bulwark Shoal**, **Bache Rock**, **West Cod Ledge Rock**, **Corwin Rock**, **Alden Rock**, **Old Anthony Rock**, **East Hue and Cry** and **West Hue and Cry**. The most important and largest of the shoal areas are buoyed, and the deep natural channels between them afford a clear approach to the harbor in clear weather from several directions.

(741) A second area of shoals extending from Ram Island Ledge to Cape Elizabeth includes **Witch Rock**, **Jordan Reef**, **Pine Tree Ledge**, **Willard Rock**, **Trundy Reef**, **Broad Cove Rock** and **Mitchell Rock**, all of which are marked by buoys except Trundy Reef and Mitchell Rock. Several deep clear channels between them afford approach and entry well into the harbor by deep-draft vessels.

(742) In 1982, unexploded depth charges were reported in the western end of Portland Eastern Approach Traffic Lane and in the eastern part of the precautionary area within a circle having a radius of 3 miles centered in 43°31'03"N., 70°00'08"W. Mariners are cautioned not to conduct dragging operations in this area.

(743) Platts Bank (43°09'N., 69°37'W.) between the Portland Eastern and Southern Approaches, approximately 40 miles east of Portland, is a feeding area for endangered North Atlantic right whales (peak season: April through June and October through December).

(744) The City of Portland, Department of Transportation and Waterfront, distributes educational material to mariners in an effort to reduce right whale ship strikes.

(745)

Current

(746) The velocity at strength of the tidal current in the channel is about 1 knot southwest of Cushing Island and southwest of Diamond Island Ledge; within the harbor it is about 0.5 knot. See the Tidal Current prediction service at tidesandcurrents.noaa.gov. for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(747)

Weather, Portland and vicinity

(748) As a rule, Portland has very pleasant summers and falls, cold winters with frequent thaws, and disagreeable springs. Very few summer nights are too warm and humid for comfortable sleeping. Autumn has the greatest number of sunny days and the least cloudiness. Winters are quite severe, but begin late and then extend deeply into the normal springtime.

(749) Heavy seasonal snowfalls, over 100 inches (2540 mm), normally occur about each 10 years and extreme events as well as true blizzards are very rare. The White Mountains, to the northwest, keep considerable snow from reaching the Portland area and also moderate the temperature. The 24-hour record snowfall for Portland is 22 inches (559 mm), recorded in December 1970. Snow falls on an average of 59 days each year and has fallen during each month except June, July and August. Average annual snowfall is 71 inches (1803 mm) and about 15 days each year have snowfall greater than 1.5 inches (38 mm).

(750) Normal monthly precipitation is remarkably uniform throughout the year, averaging 43.6 inches (1107 mm) each year. The wettest month is November with about five inches (127 mm) and the driest months are July and August, each averaging 2.9 inches (74 mm). About 29 days each year have precipitation greater than 0.5 inch (13 mm) and precipitation falls an average of 185 days each year. Thunderstorms occur about 16 days each year and most frequently during June, July and August.

(751) Winds are generally quite light with the highest velocities being confined mostly to April and May. The prevailing winds are southerly during the summer and northerly during the winter. During all seasons the heaviest gales are usually from the northeastward or eastward. The occasional northeasterly gales have usually lost much of their severity before reaching the coast of Maine.

(752) Fogs occur most frequently during June through September but happen during each month of the year. At the head of the bays and within rivers it is often comparatively clear when it is thick outside. Winds from the east to the southwest by way of south bring fog; westerly and northerly winds clear it away. An average of 167 days each year have foggy conditions. During August and September it is occasionally foggy or smoky in the harbor in early morning when it is clear outside.

(753) The warmest month is July with an average high of 79 °F and an average low of 58 °F. The coolest month is January with an average high of 31 °F and an average low of 12 °F. The warmest temperature on record is 103 °F in August 1975. Temperatures well below zero are recorded frequently each winter. Cold waves sometimes come in on strong winds, but extremely low temperatures are generally accompanied by light winds. The average freeze-free season at the airport station is 139 days. May 12 is the average date of the last freeze in spring, but this has been as early as April 22 and as late as May 31. The average date of the first freeze in fall is September 27, with the earliest and latest occurrences on September 17 and October 10. Every month has seen extreme minimum temperatures of 40 °F or below. The coldest temperature on record is -26 °F recorded in January 1971. The average year has 155 days with minimums below 32 °F and 23 days with minimums of 5 °F or below.

(754) Ice seldom obstructs navigation; when it does it is only for a limited time. Tugs keep a clear channel to the wharves.

(755)

North Atlantic Right Whales

(756) Endangered North Atlantic right whales may occur in the vicinity of or within the Portland approaches. The northernmost portion of Jeffreys Ledge, a well known feeding area for right whales, abuts the Traffic Separation Scheme (Portland) Southern Approach Outbound Lane. (See North Atlantic Right Whales, indexed as such, chapter 3, for more information on right whales and recommended measures to avoid collisions with whales.)

(757)

Pilotage, Portland

(758) Pilotage is compulsory for all foreign vessels and U.S. vessels under register in the foreign trade drawing over 9 feet. Pilotage is optional for coastwise or fishing vessels under enrollment or license who have on board a pilot licensed by the federal government. Pilotage is provided by Portland Pilots, Inc., 48 Union Wharf, Portland, ME 04101-4607, telephone 207–774–5623, FAX 207–774–5683. The pilot office monitors VHF-FM channels 16 and 11, Monday through Friday from 0900 to 1700 hours. A pilot can be reached during non-office hours by calling 207–774–5623 or through the vessel's port agent. The pilot boat monitors VHF-FM channels 16, 13, 11, 10 and 7A when underway; works on channel 11. A pilot boarding area with a two-mile radius is centered at 43°31'36.6"N., 70°05'30.0"W. The pilot boat has a black hull with white superstructure and the word PILOT on either side of the superstructure; one boat is 48-foot and the other is 65-foot in length. The pilot boat maintains station only in anticipation of prescheduled vessel movement. Arrangements for pilots should be made in advance through the ship's agent. Vessels are requested to give a 48-hour and a 24-hour notice of their ETA at the above buoy and an update of any appreciable change of ETA. The Coast Guard Captain of the Port, Sector Northern New England, recommends the following for deep draft vessels:

(759) Vessels awaiting the boarding of a pilot should anchor about 1.5 miles north and east of Portland Lighted Whistle Buoy P to allow the unrestricted passage of other vessels through the precautionary area.

(760) All self-propelled vessels over 375 feet length-overall (LOA) and all tank vessels should engage the services of one or more escort tugs for inbound and outbound transits through the Casco Bay Bridge. The use of escort tugs for vessels with an operational bow thruster will be at the discretion of the pilot. For inbound transits the escort tugs should be engaged in the vicinity of Spring Point and Portland Pipeline Corp. Pier 2.

(761) All tug/barge units should only pass through the Casco Bay Bridge in the pushing ahead or towing alongside modes. All light tug/barge should use flood tide when making a transit outbound through the Casco Bay Bridge. All light tug/barge units with a capacity of 70,000 barrels or more should also engage the services of an assist tug.

(774)

Harbor Regulations
Portland Harbor

The Board of Harbor Commissioners in Portland has jurisdiction over the piers to the pierhead line, checks on harbor pollution, establishes pilot rates, appoints harbor pilots and establishes harbor regulations. The harbormaster, appointed by the City of Portland, enforces the regulations and maintains an office at the state pier. The following speed regulations for Portland Harbor have been promulgated by the Board of Harbor Commissioners:

Restricted Speed Area—The term Restricted Speed Area shall mean the following areas of Portland Harbor:
Main Ship Channel
• Between a line drawn from Portland Head Light to Portland Harbor Main Approach Lighted Bell Buoy 12, and a line drawn from Spring Point, where the breakwater meets the shore, to Fort Scammel Point Light 2 on House Island, the speed of vessels shall not exceed 20 knots.
• Between a line drawn from Spring Point, where the breakwater meets the shore, to Fort Scammel Point Light 2 on House Island, and a line drawn from South Portland Breakwater Light to Diamond Island Ledge Light, the speed of vessels shall not exceed 15 knots.
• Between a line drawn from South Portland Breakwater Light to Diamond Island Ledge Light, and a line drawn from the Maine State Pier to Portland Pipe Line Pier #1, the speed of vessels shall not exceed 10 knots.
• Between a line drawn from the Maine State Pier to Portland Pipe Line Pier #1, and a line created by the Veterans Memorial Bridge, the speed of vessels shall be headway speed only, not to exceed 10 knots, with NO WAKE.
Portland
(1) All areas north of the following line – from a point located at the intersection of (I) a line from Fish Point Lighted Buoy 1 to the north end of the draw in the bridge and (II) a line formed by extending in a southerly direction the westerly side of the Maine State Pier to the northerly end of Portland Pipe Line Pier #1.
(2) All areas within a 150 yard radius of the gas pumps on Ted Rand's marina on the easterly side of Little Diamond Island.
South Portland
All areas south of the following line – from the northerly end of Portland Pipe Line Pier #1, to Lighthouse Channel Lighted Buoy 1, to Lighthouse Channel Lighted Buoy 2, to Mill Cove Daybeacon 1, to south end of draw in the bridge

Restricted Wake Area—The term Restricted Wake Area shall mean the following areas of Portland Harbor:
• **Inner harbor –** all areas between (a) a line formed by extending in a southerly direction the westerly side of the Maine State Pier to the northerly end of Portland Pipe Line Pier #1, and (b) the bridge.
• **Diamond Island Pass –** All areas in Diamond Island Pass extending northeasterly from Diamond Island Pass Buoy 6, off the southwesterly point of Little Diamond Island, to Diamond Island Pass Lighted Buoy 1, off the northeast point of Peaks Island.
• **Whitehead Passage –** (a) All areas south of a line between the point on the westerly side of Spicers Cove and the northernmost point of White Head on Cushings Island; and (b) All areas north of the following line: from Brackett Point on Peaks Island southeasterly to Catnip Island, and from Catnip Island to Daybeacon 4 on Trotts Rock.
• **Spring Point –** All areas west of a line between the easterly end of Portland Pipe Line Pier #2 and the easterly end of the pier of Southern Maine Community College.
• **Willard Beach –** Simonton Cove, also known as Willard Beach.
• **South Portland Public Landing –** All areas within a 200 foot radius of the end of the dock at the South Portland Public Landing.
• **Peaks Island Public Landing –** All areas within a 200 foot radius of the end of the dock at the Peaks Island Public Landing.
• **East End Beach and Mooring Area –** All areas west of a line from Fish Point to Pomeroy Rock, and from Pomeroy Rock continuing at a distance of 300 feet offshore to railroad swing bridge at the entrance of Back Cove.

Bridge—The term *bridge* shall mean the bridge between Portland and South Portland known as the *Casco Bay Bridge*.

Vessel—The term *vessel* shall mean any watercraft used or capable of being used for transportation.

Speed of Vessels—It shall be unlawful to operate a vessel within the **restricted speed area** at a speed (a) in excess of 5 mph or (b) that endangers any person or property

Wake—It shall be unlawful to operate a vessel within the **restricted wake area** in such a manner as to cause a wash, wake or waves that damage, endanger or unreasonably disturb any person, wharf, float, anchored or moored vessel, or vessel tied up at any pier, float, dock, wharf or marina.

Exception—Nothing in these speed or wake regulations shall make unlawful any action necessary to (a) navigational safety, (b) observance of rules of the road, and (c) emergency missions by emergency or public safety craft. The burden shall be upon any person asserting the exception provided by this paragraph as a defense to a prosecution for violation of any speed or wake regulation.

Penalty—A violation of any speed or wake regulation shall be penalized by a fine of $100, to be collected by the harbormaster in District Court.

Buoys—The harbormaster shall establish speed and wake signs on the Maine State Pier and at such other locations and on such buoys at points on the perimeter of the **restricted speed area** and **restricted wake area** as he finds necessary.

(762) All tug/barge units in the towing astern mode should refrain from transiting the area of Portland Head to Spring Point while this area is being transited by large vessels. Enhanced bridge-to-bridge communications should be used to avoid meeting situations in restricted navigation areas. Tug/barge units towing astern should change mode of towing before the Portland Pipeline Corp. Pier 2, weather conditions permitting, in preparation for entrance into Portland Harbor. Inbound laden tug/barge units with a capacity of 70,000 barrels or more towing astern past Portland Head Light to Spring Point should engage the services of an assist tug when transitioning the mode of towing.

(763)

Towage

(764) A fleet of modern tugs up to 3,500 hp is available at Portland. Tugs meet vessels off Spring Point and use VHF-FM channel 7. Arrangements for tugs are made through ships' agents or direct by telephone or cable; telephone 207–772–8319; cable address MORTOW. Most large vessels use tugs when docking.

(765)

Tug Escort System, Portland Harbor

(766) The Port of Portland has established a voluntary system of tug escorts for crude oil tankers of any size or tonnage entering, departing or moving within Portland Harbor. The system is designed to assist vessel operators in the safe navigation of western Casco Bay and Portland Harbor. Nothing in these procedures shall supersede the Navigation Rules or relieve the master of the vessel of his responsibility for the safe navigation of the vessel.

(767) All participating vessels are requested to arrange for an escort by tugs of sufficient horsepower at Portland Harbor Eastern Approach Lighted Gong Buoy 1 (43°37.1'N., 70°09.8'W.). Any movement of these vessels within the confines of the harbor or the channel must be under the escort of tugs with sufficient horsepower for docking, undocking or assistance in maneuvering.

(768)

Quarantine, customs, immigration, and agricultural quarantine

(769) (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(770) **Quarantine** is enforced in accordance with the regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.) Two private hospitals are available in Portland.

(771) Portland is a **customs port of entry.**

(772)

Coast Guard

(773) **U.S. Coast Guard Sector Northern New England** is located in South Portland. (See Appendix A for address.)

(775)

Wharves

(776) Deepwater facilities at Portland include seven petroleum terminals, one general cargo terminal and one international ferry terminal. All have highway connections and most have railroad connections. The alongside depths are reported; for information on the latest depths contact the operator. The other active facilities in the port are used as repair berths and by fishing vessels, small craft, barges, tugs, ferries and other miscellaneous craft.

(777) **Facilities on the north side of Fore River at Portland**:

(778) **State Pier:** about 0.8 mile northeastward of Casco Bay Bridge; 115-foot face, 35 feet alongside; southwest side in two sections, 587 feet and 430 feet long; 14 to 35 feet alongside; deck height, 16 feet; moorage for ferry, police boats, city fireboat and transient vessels; owned by the City of Portland and operated by the City of Portland and Casco Bay Lines.

(779) **International Marine Terminal:** immediately northeastward of Casco Bay Bridge; 707-foot wharf, 27 feet alongside; deck height, 19 feet; trailer-truck marshalling area adjacent; passenger and vehicle ferry operates between this terminal and Yarmouth, Nova Scotia, Canada; roll-on/roll-off facility for trailer trucks; owned by the City of Portland and operated by Prince of Fundy Cruises, Ltd.

(780) **Facilities on the south side of Fore River at South Portland:**

(781) **Bancroft and Martin L-Dock:** (43°38'27"N., 70°17'06"W.); 114-foot face; 440 feet of berthing space with dolphins; 35 feet alongside; deck height, 15 feet; water connections; pipelines extend to storage tanks, 2 1/5 -million-barrel capacity; receipt and shipment of petroleum products and asphalt; owned by Bancroft and Martin, Inc., and operated by various oil companies.

(782) **Bancroft and Martin T-Dock:** immediately southeastward of L-Dock: 115-foot face; 250 feet of berthing space with dolphins; 25 to 20 feet alongside; deck height, 15 feet; pipelines extend to storage tanks, 220,000-barrel capacity; receipt and shipment of petroleum products; owned by Bancroft and Martin, Inc., and operated by Gibbs Oil Division of BP. Oil, Inc.

(783) **Mobil Oil Corp. South Portland Terminal Dock:** about 1.1 miles westward of Casco Bay Bridge; 600 feet with dolphins; 35 feet alongside; deck height, 14½ feet; water connections; receipt and shipment of petroleum products; owned and operated by Mobil Oil Corp.

(784) **Texaco Wharf:** about 0.3 mile southwestward of Casco Bay Bridge; 900 feet with dolphins; 39 feet alongside; deck height; 19 feet; water connections; receipt and shipment of petroleum products and bunkering vessels; owned and operated by Texaco, Inc.

(785) **Portland Pipe Line Corp. Pier No. 1:** about 0.85 mile northeastward of Casco Bay Bridge; southwest and northeast sides 850 feet of berthing space; 34 feet alongside; deck height, 16 feet; water connections; pipelines extend to storage tanks, 816-barrel capacity; receipt of crude oil; owned and operated by Portland Pipe Line Corp. A wreck, covered 27 feet, is near the pier at 43°39'11"N., 70°14'33"W.

(786) **Chevron USA, South Portland Terminal Dock:** about 0.95 mile northeastward of Casco Bay Bridge; 100-foot face; 750 feet total berthing space; 28 to 32 feet alongside; deck height, 15 feet; water connections; receipt and shipment of petroleum products; owned and operated by Chevron, U.S.A., Inc.

(787) **Portland Pipe Line Corp. Pier 2:** (43°39.3'N., 70°13.8'W.); northwest and southwest sides; 910 feet with dolphins; 48 feet alongside; deck height, 20 feet; water connections; receipt of crude oil; owned and operated by Portland Pipe Line Corp.

(788)

Supplies

(789) All grades of fuel oil are available. Bunkers can be obtained at the oil terminals or at the piers from barges. Water is available at most of the piers. Marine supplies and provisions are available in any quantity.

(790)

Repairs

(791) A boatyard at South Portland about 0.7 mile northeastward of the Casco Bay Bridge has three marine railways, the largest of which can handle craft up to 210 feet long, 1,200 tons displacement and 16-foot draft for practically any type of repair work. A machine shop is at the yard; rental mobile cranes can be obtained.

(792) There are several ship repair firms in the port that have fully equipped machine, pipe, joiner and welding shops and can handle above-the-water hull and engine repairs. A 100-ton fixed derrick, floating cranes up to 20 tons and a 65-ton mobile crane are available in the port.

(793)

Small-craft facilities

(794) There are ample facilities in the port where all services can be obtained either at the piers on the Portland side of the river or at the facilities on the South Portland side. The Centerboard Yacht Club in South Portland is between the Coast Guard station and the boatyard. The yacht club has a float landing with depths of 3 to 5 feet reported alongside. Water is available at the float. The public landing and boat ramp are 150 yards west of the abandoned lighthouse on the south entrance point to Fore River. The approach channel to the landing is marked by private seasonal buoys.

(795) A marina in South Portland is in the cove entered northeastward of the Casco Bay Bridge; depths of 2 to 6 feet are reported alongside the berths. A 20-ton mobile hoist at the marina can handle craft up to 40 feet for minor engine repairs or dry open winter storage. A privately dredged and marked channel leads to the marina's service float at which gasoline and diesel fuel can be obtained. A second marina is at the old shipyard, between Spring Point Ledge Light and Portland Pipe Line Corp. Pier 2. A depth of 8 feet is reported at the marina; a 35-ton mobile lift, gasoline, diesel fuel, storage facilities, a pump-out station and marine supplies are available. Hull, engine and electronic repairs can be made.

(796) There are marinas with service piers at the old buoy depot on Little Diamond Island and Peaks Island and several on the Portland waterfront from the state pier to the combination railroad and highway bridge.

(797) There are also public landings at the state pier in Portland and at most of the state piers on the islands in Casco Bay.

(798) Gasoline, diesel fuel and water can be obtained at the service piers of the marinas or from fuel barges which serve vessels in the stream. Marine supplies, food and ice can be obtained in any quantity in the port.

(799)

Communications

(800) Portland is served by both Class I and Class II railroads. Passenger and freight ferries serve the nearby islands. Interstate bus lines offer transportation to all sections of the United States and Canada. Portland International Jetport is on the southwest side of the city. Scheduled airlines operate from the airport, and charter and air taxi service is available. A scheduled passenger and vehicular ferry operates between Portland and Yarmouth, Nova Scotia, Canada. Numerous truck lines serve the greater Portland area with interstate and intrastate service.

(801) **Ship Cove**, **Maiden Cove**, **Danford Cove**, **Broad Cove** and **Simonton Cove**, small coves on the west side of the main channel south of Spring Point, are important only as summer anchorages for local pleasure craft.

(802) **Peaks Island** is the large island northeastward of Cushing Island. It has communications with Portland by automobile and passenger ferries. Several wharves are on the west side. The ferries land at **Forest City Landing** at the village of **Peaks Island** on the west side of the island. The Casco Bay boats dock at the state pier just south of Forest City Landing where there is a public float landing. **Trefethen** is a village at the north end of the island. **Pumpkin Nob**, 51 feet high, is about 150 yards north of the northern extremity of Peaks Island; a lighted bell buoy is off its eastern side.

(803) **Great Diamond Island** and **Little Diamond Island**, northwestward of Peaks Island, are connected by a sandbar covered at high water. Little Diamond has many houses visible on it and a former Coast Guard buoy pier on its east side. Casco Bay passenger ferries from Portland call at the landings at wharves on the south end of both islands.

(804) Gasoline, water, some provisions and supplies are available at the marina at the former buoy pier.

(805) The red-roofed community building on the state pier on Little Diamond Island is very conspicuous.

(806)

Anchorage

(807) It is reported that excellent anchorage in 18 feet with good protection from all but easterly winds can be had in **Diamond Cove**, at the northeast end of Great Diamond Island. Anchor beyond the 16-foot contour to ensure adequate swinging room.

(808) **Back Cove** is on the north side of Portland. The cove is now of little commercial importance and mostly dries out. There is an oil-handling berth that has 3 feet reported alongside on the north side of the entrance to the cove, outside the railroad bridge.

(809) An approach channel to Back Cove, north of Fish Point, has a project depth of 30 feet to the Canadian National Railway bridge. Above this bridge, the project depths are 14 feet to the highway bridge, thence 12 feet in the channel along the east side of Back Cove.

(810) The Canadian National Railway bridge crossing the entrance to Back Cove has a swing span with a clearance of 5 feet. The bridge is maintained in the closed position. (See **33 CFR 117.521**, chapter 2, for drawbridge regulations.) U.S. Interstate 295 bridge, about 500 yards above the railway bridge, has a fixed span with a clearance of 29 feet.

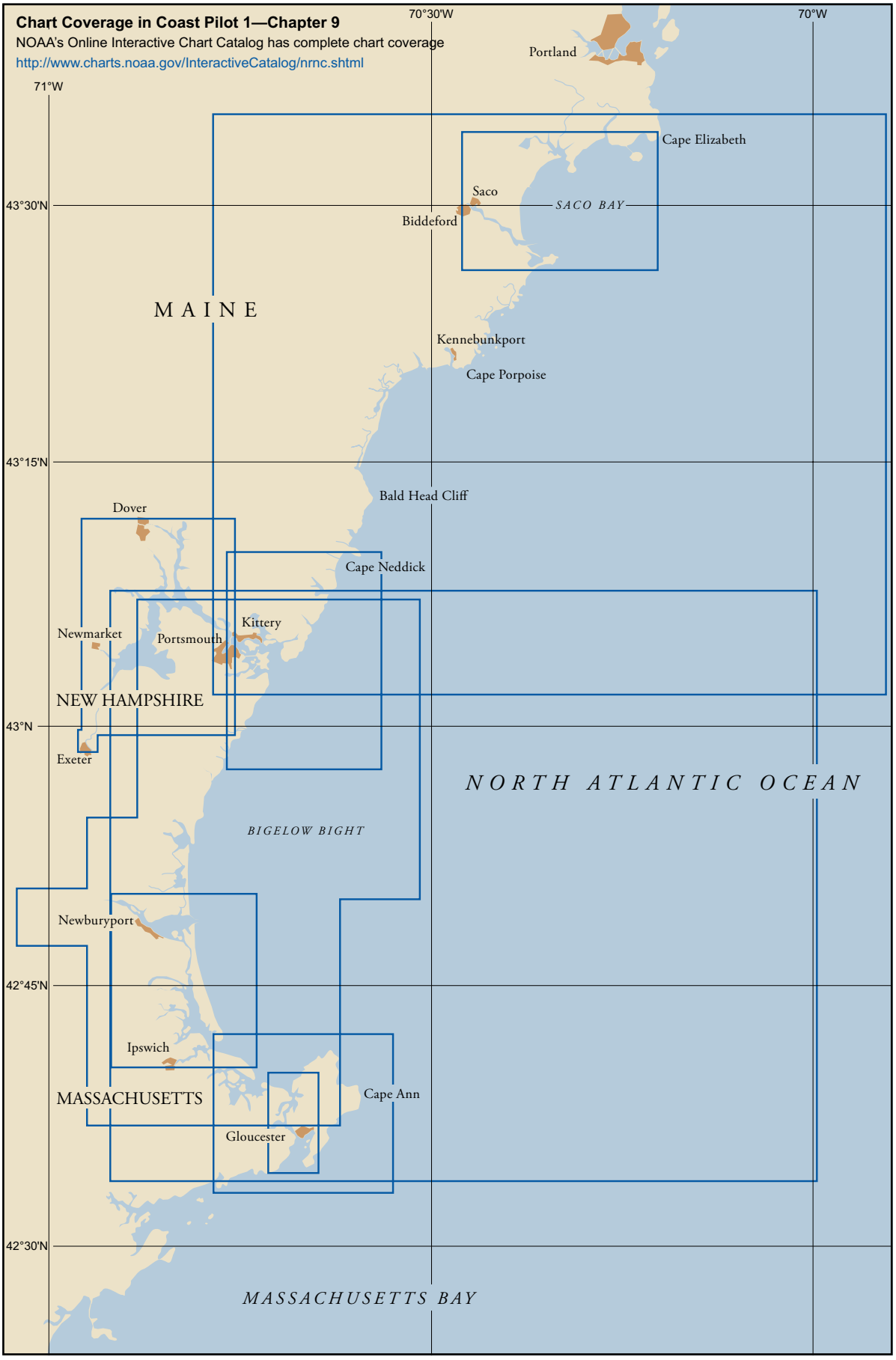


Chart Coverage in Coast Pilot 1—Chapter 9
NOAA's Online Interactive Chart Catalog has complete chart coverage
http://www.charts.noaa.gov/InteractiveCatalog/nrnc.shtml
70°30'W
70°W
71°W
Portland
Cape Elizabeth
Saco
SACO BAY
Biddeford
43°30'N
MAINE
Kennebunkport
Cape Porpoise
43°15'N
Bald Head Cliff
Dover
Cape Neddick
Kittery
Newmarket
Portsmouth
NEW HAMPSHIRE
43°N
Exeter
NORTH ATLANTIC OCEAN
BIGELOW BIGHT
Newburyport
42°45'N
Ipswich
MASSACHUSETTS
Cape Ann
Gloucester
42°30'N
MASSACHUSETTS BAY

Cape Elizabeth, Maine to Cape Ann, Massachusetts

(1) From Cape Elizabeth the coast of Maine continues southwestward for about 37 miles to the Piscataqua River and the deepwater port of Portsmouth, NH. The few harbors along this part of Maine are suited mostly to fishing vessels, yachts and small pleasure craft. This is a summer-resort area, and many of the buildings are large and prominent. Two tall water tanks, one westward of Wood Island Light and one at Cape Porpoise Harbor, are the most prominent objects between Portland and Portsmouth.

(2) Extending south-southwestward from Portsmouth Harbor is the 13-mile coast of New Hampshire; the Isles of Shoals are 6 miles southeast of the harbor. Southward and eastward from the New Hampshire line the extreme northern part of the Massachusetts coast extends about 23 miles to Cape Ann Light. The Merrimack River approach to Newburyport, MA, is about 3 miles south of the New Hampshire boundary.

(3)

COLREGS Demarcation Lines

(4) The lines established for this part of the coast are described in **33 CFR Part 80.115**, chapter 2.

(5)

No-Discharge Zones

(6) The State of New Hampshire, with the approval of the Environmental Protection Agency, has established a No-Discharge Zone (NDZ) covering all coastal waters of New Hampshire, extending about 3 miles offshore.

(7) The State of Massachusetts, with the approval of the Environmental Protection Agency, has established a No-Discharge Zone (NDZ) in all coastal waters of Massachusetts described in this volume, extending about 3 miles offshore.

(8) Within the NDZs, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZs, discharge of sewage is regulated by **40 CFR 140** (see chapter 2).

(9)

Seal Cove to Dansbury Reef

(10) Cape Elizabeth Light and Portland Lighted Whistle Buoy P were described in chapter 8.

(11) **Seal Cove**, on the southeast side of Cape Elizabeth and northeastward of Richmond Island, has numerous rocks and ledges. **The Sisters**, awash, and **Seal Rock**, which uncovers about 4 feet, are dangers near the center of the cove. The eastern extremity of the ledge extending eastward of Seal Rock is marked by a buoy that facilitates entrance to the anchorage north of the ledge. The holding ground in the cove is sand and poor, but some shelter is afforded in easterly weather north of a line between **McKenney Point** and Seal Rock. Care should be taken to stay clear of unmarked **Crowell Rock. Stevens Rock**, covered 6 feet, about 650 yards southward of Seal Rock is also unmarked. A small-craft launching ramp is in **Ship Cove**, 0.4 mile northeastward of Seal Rock, but no services are available. A bell buoy, about 0.5 mile southeastward of **Watts Ledge** off the eastern end of Richmond Island, marks the entrance to Seal Cove.

(12) **Richmond Island**, about 0.5 mile south of Cape Elizabeth and connected to it by a breakwater, is partly wooded with a conspicuous barn on it. Parts of the breakwater are covered at high water, and caution should be exercised in the vicinity.

(13) Small craft seeking refuge from westerly and southerly winds anchor in **Broad Cove** in the lee of **East Point**, the northeast point of Richmond Island, directly off the shore opposite a long, low barn. The bottom is sand and mud.

(14) **Richmond Island Harbor**, westward of Richmond Island and the breakwater, is sheltered from northerly and westerly winds but is exposed to southwesterly and southerly winds. Foul ground extends 0.4 mile from the northern side of the harbor. The depths shoal gradually from 45 feet at the entrance to 15 feet 350 yards from the breakwater at the head. The holding ground is good, sand and mud. The anchorage is used by yachts and small craft.

(15) **Chimney Rock**, 0.3 mile from the north shore of Richmond Island Harbor, awash at low water, is marked by a buoy. Vessels must pass southward of the buoy. A rock covered 16 feet is 0.2 mile east-southeastward of Chimney Rock; an 18-foot spot 0.3 mile east-northeastward and a 12-foot spot about 0.5 mile east-southeastward are all unmarked.

(16) An unmarked rocky ledge covered 16 feet near its southwest end is about 0.4 mile westward of **Ram Island**, low and grassy, which is 0.2 mile northwestward of Chimney Rock.**The Brothers**, a ledge that uncovers, is 300 yards north-northeastward of Chimney Rock.

(17) **Spurwink River**, 1.6 miles northwestward of Richmond Island, can be entered only by small craft at half tide or higher with a smooth sea. **Higgins Beach**, on the west side at the entrance, has many visible cottages. The river is narrow and crooked, and there are no facilities. A bridge crossing the river about 1.7 miles above the mouth has a clearance of 5 feet. An obstruction, covered 8 feet, is about 500 yards off the entrance to the river.

(18) **Old Proprietor**, a ledge that uncovers at low water, 0.9 mile from shore and 1.8 miles westward of Richmond Island, is marked on its south side by a buoy. A ledge covered 11 feet about 0.5 mile and a 17-foot spot about 0.7 mile north-northeastward of Old Proprietor are both unmarked.

(19) Between Richmond Island and Wood Island Light, a distance of about 6 miles, the shore forms a large open bight, the southern part of which is **Saco Bay.**

(20) **Prouts Neck**, a conspicuous point 3 miles westward of Richmond Island, is the northern point of Saco Bay. The neck is partly wooded and has many houses. A standpipe on **Blue Point Hill** 2.3 miles northwestward of Prouts Neck is conspicuous.

(21) The Prouts Neck Yacht Club, float landing and moorings are on the west side and close northward of a short stone breakwater. There is reported to be a depth of 5½ feet at the float; water is available at the float.

(22) **Scarborough River** enters the sea about 0.6 mile northwestward of Prouts Neck. The river and its tributaries, the Libby and Nonesuch Rivers, are used by local fishing and pleasure craft in considerable number at half tide or higher. There are many fishing piers and private float landings on these rivers, most of which are dry at low water.

(23) A channel leads across the bar from Saco Bay, thence into Scarborough River to the town dock about 0.3 mile above **Pine Point**. The channel is marked by buoys; the buoys in Scarborough River are not charted due to frequent relocations to mark the best water. Following protracted spells of bad weather the positions of the buoys should not be relied upon as they often do not indicate the best water. A jetty extends in a southerly direction from Pine Point on the west side of the entrance.

(24) The town dock has a depths of 7 to 8 feet at the float landing. Gasoline, electricity, water, ice and some marine supplies are available here; guest moorings are maintained. A small-craft launching ramp, usable at or near high tide, is close eastward of the dock.

(25) Provisions and lodging are obtainable in the village of **Pine Point** a short distance from the town pier.

(26) Along the shore of Saco Bay from northward to southward are **Grand Beach, Old Orchard Beach** and **Ferry Beach**. The large hotels and the standpipe at Old Orchard Beach are prominent.

(27) **Bar Ledge**, covered 11 feet, is 0.9 mile from shore off Grand Beach and is marked on its southern side by a buoy. About 0.6 mile westward of the buoy and 0.7 mile northeastward of the pier at Old Orchard Beach, **Little River Rock**, covered 2 feet and extending 0.5 mile from shore, is unmarked.

(28) **Goosefare Brook** enters the sea at the south end of Old Orchard Beach. The brook is foul, and the piles of an old highway bridge block the river near the entrance. About 150 yards farther upstream is State Highway No. 9 bridge with little or no vertical clearance.

(29) **Stratton Island** and **Bluff Island**, 20 feet high and grass covered, are off the northern part of Saco Bay, 1 mile southward of Prouts Neck. Deep water is between the islands and Prouts Neck, but between the islands are numerous ledges. Ledges, awash at low water, are 0.3 mile off the eastern side of Stratton Island and 0.2 mile off the southwestern side.

(30) Islands and ledges in the southern end of Saco Bay extend up to 1.5 miles from the shore. Inside of the islands are Wood Island Harbor and the entrance to Saco River.

(31) **Eagle Island**, 2.5 miles southwest of Stratton Island, and **Ram Island**, 0.7 mile south of Eagle Island, are rocky and grass-covered; vessels should pass eastward of these islands, giving them a berth of at least 0.5 mile. There is a house on Ram Island.

(32) **Saco River**, with its entrance in the south end of Saco Bay west-northwestward of Wood Island, is the approach to the cities of **Biddeford**, on the south bank, and **Saco** on the north bank. The cities are at the head of navigation 5 miles above the mouth of the river. Private piers and a public boat ramp are located along the river. There has been limited commercial traffic on the Saco River in recent years, except for fishing vessels moored inside the mouth of the river. A party fishing boat operates from the pier at Camp Ellis, a settlement on the north bank of the river at its mouth. The **harbormaster** for the river resides there: telephone 207–284–6288.

(33)

Prominent features

(34) **Wood Island Light** (43°27'25"N., 70°19'45"W.), 71 feet above the water, is shown from a white conical tower connected to a dwelling, on the east end of the island; a mariner-activated sound signal is at the light, initiated by keying the microphone five times on VHF-FM channel 83A.

(35) A small island, (43°27'26.1"N., 70°20'29.6"W.) is just westward of Wood Island. **Stage Island**, 0.6 mile west of Wood Island, is 20 feet high and marked by a prominent stone monument.

(36) **Basket Island**, 0.3 mile west of Stage Island, is 20 feet high and grassy and has several cottages.

(37)

Channels

(38) Saco River is entered through a marked channel that leads over the bar between two jetties, thence to **Factory Island**, the head of river navigation at Biddeford and Saco. A fairway bell buoy, 0.3 mile eastward of Ram Island Ledge, marks the inner approach entrance from Saco Bay. The outer 0.6 mile of the south jetty and the outer 0.4 mile of the north jetty are covered at high water. Daybeacons mark the south jetty about mid-length and the outer end of the north jetty. A basin northwest of Cow Island surrounds the bare mudflats in the middle of the river. The area in the vicinity of the submerged pilings at the southeast end of the flats should be avoided.

(39) Small craft can enter the river with a smooth sea and on a rising tide by passing between Ram Island Ledge and Stage Island, then following the buoyed channel over the bar.

(40) The river channel, marked by buoys and daybeacons, is narrow, crooked and bordered closely by shoals. In 1985, an obstruction was reported northward of Brimstone Point in about 43°27'54"N., 70°23'38"W, and in 2019, there was shoaling to 2 feet in about 43°29'03"N., 70°25'41"W. The bar is subject to change; local knowledge is advised. No attempt should be made by small craft to cross the bar in either direction on the ebb with an easterly wind. Several small craft have grounded in attempting to do so.

(41)

Dangers

(42) **Ram Island Ledge**, extending 0.5 mile east of Ram Island and covered 6 feet, is marked by a buoy on its eastern side. **Stage Island Shoal**, partly bare at low water, extends 300 yards east-northeastward from the island and is marked at its end by a buoy. Wood Island Harbor, southeastward of the island, is described following the discussion of Saco River.

(43) Ledge (43°27'41"N., 70°20'15"W.) 0.2 mile north of Wood Island is covered 8 feet and marked on its north side by a buoy. Ledges also extend southwestward from Wood Island; a buoy marks the end of the southwest ledge.

(44)

Current

(45) From March to May heavy freshets are liable to change the channel depths by as much as 8 feet above high water at Saco; this condition also causes dangerous currents.

(46)

Ice

(47) Ice closes the river from January to April.

(48) At Saco, the float landings and moorings of the Saco Yacht Club are on the north shore of the river just northeastward of the eastern end of Factory Island. Depths of 7 feet are reported alongside the float; a small-craft launching ramp is at the club.

(49) A boatyard is on the south side of the river at Biddeford, about 0.2 mile below the bridge to Factory Island. Depths of about 10 feet are reported alongside the floats. The yard can build craft up to 55 feet in length and has a 15-ton mobile hoist that can handle craft up to 40 feet in length for hull and engine repairs and open or covered winter storage. Gasoline, diesel fuel by truck, water, ice and marine supplies are available. Moorings are maintained north and west of the channel.

(50) A marina with depths of 10 feet reported alongside its floats is on the north side of the river, about 3.5 miles upriver from the entrance, or 2 miles below Saco. Gasoline, water and open winter storage facilities are available. Provisions and marine supplies can be obtained at Saco and Biddeford. Provisions can also be obtained near the wharf at Camp Ellis.

(51) On the south bank of the river about 2.5 miles below Saco is a state park; a large parking area for cars and trailers and a small-craft launching ramp are available.

(52) At Biddeford an overhead power cable crossing the river from Factory Island has a clearance of 123 feet.

(53) **Wood Island Harbor**, south-southwest of Wood and Stage Islands, is an anchorage for small and moderate-sized vessels. It is reported that larger yachts also anchor in this general area.

(54) Small craft can proceed to the southwestern part of Wood Island Harbor and anchor in depths of 6 to 18 feet. In entering this part of the harbor it is well to give the eastern side a good berth. The bottom in this inner anchorage is reported to be soft mud.

(55) **The Pool** is a shallow bay making southwestward from Wood Island Harbor inside **Fletcher Neck**, the south shore of Wood Island Harbor. The entrance is about 50 yards wide.

(56) A dredged channel, just southeast of Stage Island, leads through Wood Island Harbor to the entrance of The Pool. In 2021, the controlling depth was 5.4 feet. A dredged anchorage basin is just inside the entrance to The Pool. In 2021, the basin shoaled to 1.9 feet along the edges with greater depths in the entrance and the southwest corner. Biddeford Pool Channel Buoy 10 marks the entrance to the basin. Three stone icebreakers are along the northeastern side of the basin. Care should be taken by strangers not to anchor too close to them. They are difficult to see at night at or near high water. Neither should they attempt to go between the northeasternmost icebreaker and the fish wharf because of a partially submerged breakwater between the breaker and the wharf.

(57) Small craft anchor just inside the inner end of the entrance, which is locally known as **The Gut**, if there is room. No attempt should be made to anchor in The Gut as the tidal currents have considerable velocity and holding ground is poor. Local fishing and pleasure craft usually occupy most of the moorings, but permission can usually be obtained to occupy one of the unoccupied ones.

(58) **Biddeford Pool** is a village on the south side of Wood Island Harbor, extending from The Pool nearly to the eastern point of Fletcher Neck. There are small wharves on each side of the Gut. There is a **harbormaster** at Biddeford Pool: telephone 207–282–0803.

(59) The Biddeford Pool Yacht Club wharf with 20 feet reported alongside the floats is at the inner end of The Gut on the east side, with a private wharf just to the northeast. A fish wharf close eastward of the yacht club wharf has 2 feet reported alongside.

(60) Gasoline, diesel fuel in small quantities and water are seasonally available at the yacht club float; the yacht club maintains two marked guest moorings in the outer harbor. Provisions can be obtained at a store near the landings. Meals, lodging and most services are obtainable in the village.

(61)

Routes

(62) To enter Wood Island Harbor from the northeast, keep about 0.5 mile north of Wood Island until near the fairway bell buoy eastward of Ram Island Ledge. Pass about 100 yards southeastward of this buoy, heading past the monument on Stage Island.

(63) If proceeding to the southwestern or lower end of the harbor, pass about 100 yards eastward of the buoy 0.2 mile northeastward of Stage Island, from the channel between Wood and Stage Islands head in a southwesterly direction for the gut, being careful to give the east side a good berth. Select anchorage northwestward of Halftide Rock Daybeacon 9.

(64) If continuing on to the anchorage basin in The Pool, favor the northwesterly side until in The Gut, then in midchannel to the buoy at the inner end.

(65) If anchorage is desired southward of Wood Island, the best approach from northward is to the eastward of Wood Island. From a position 300 yards due east of Wood Island Light, head for the end of the bluff on the eastern extremity of Fletcher Neck until the monument on Stage Island opens up south of Wood Island, then bear around to the westward and head for the daybeacon on Philip Rock. Select anchorage from 150 to 250 yards off the middle of the island eastward of the cable area.

(66) If coming from the southeastward, head for the middle of Wood Island to pass midchannel between the buoy marking Washman Rock and the buoy southward of Dansbury Reef. When about 200 to 250 yards off Wood Island on this leg bear sharp around to the westward and select anchorage from 150 to 250 yards off the middle of the island.

(67) If coming from the southeastward and bound for Wood Island Harbor, continue as in the preceding paragraph to pass 50 to 100 yards south of the buoy, southwestward of Wood Island. Hold this course until The Gut opens up westward of the buoy and daybeacon marking Halftide Rock. Then bear around to the southwestward and select anchorage northwestward of Halftide Rock Daybeacon 9; or, if prudent, continue inward through The Gut into The Pool.

(68) The chart must be the guide at all times. Proceed no farther until each aid to navigation is properly identified and passed correctly.

(69) **Washman Rock**, which uncovers 9 feet, is near the end of a reef that extends 600 yards southeastward from the eastern point of Fletcher Neck and is marked close southeastward by a buoy.

(70) **Dansbury Reef**, 0.5 mile southward of Wood Island Light, is a small ledge covered 2 feet and is marked on its southeast side by a buoy. There are several shoal spots between the reef and Wood Island, and strangers should not pass between them.

(71) Numerous rocks and ledges extend 0.6 mile southeastward of Fletcher Neck. The cupola and signal towers of a former Coast Guard station, on the east side of Fletcher Neck, are conspicuous, as are the many large homes on the neck.

(72)

Hussey Rock to Mount Agamenticus

(73) **Hussey Rock** (43°25.8'N., 70°20.5'W.), covered 5 feet, is about 0.5 mile south of Fletcher Neck and is marked on its south side by a buoy.

(74) **Goosefare Bay**, 5.4 miles southwestward of Wood Island Light, is a shallow cove with numerous rocks and ledges. The coast between Fletcher Neck and Goosefare Bay is lined with summer homes, some very large and prominent.

(75) **Little River** and **Batson River** empty into Goosefare Bay. Both are used by small pleasure craft. There are no facilities in Little River. Overhead power and telephone cables with clearances of 25 feet cross Little River about 0.5 mile above the mouth.

(76) Only small craft use Batson River. There are no facilities. Navigation is terminated by a dam at the highway bridge about 1 mile above the mouth.

(77) **Stage Island Harbor**, 6.7 miles southwestward of Wood Island Light, is a small slough used by small local craft. The entrance is about 75 yards wide between the reefs making northward from **Cape Island** and southward from **Little Stage Island**; it is not safe for strangers. The ruins of a house are on Little Stage Island, the southern half of Stage Island.

(78) **Cape Porpoise Harbor**, about 7.5 miles southwestward of Wood Island Light, is a safe and protected harbor. It is ideal for the many fishing and pleasure craft that base there. It is midway between Portsmouth and Portland and is often a welcome haven for cruising craft caught in a blow on this stretch of coast.

(79) Seiners sometimes enter for shelter, though the anchorage is somewhat restricted by size and depth for the larger vessels.

(80) The village of **Cape Porpoise**, around **Porpoise Cove**, is at the head of the harbor. Lobstering, fishing, and summer tourism are the principal industries.

(81)

Prominent features

(82) The principal mark for approaching Cape Porpoise Harbor is **Goat Island Light** (43°21'28"N., 70°25'30"W.), 38 feet above the water, shown from a brick concial tower on the south end of Goat Island on the east side of the entrance. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A. A bell buoy about 0.4 mile southeastward of the light marks the approach.

(83) A water tank and a church spire are at the head of the harbor.

(84)

Channels

(85) Cape Porpoise Harbor is entered by a dredged channel that leads from the entrance to a combined channel and anchorage to the town wharf, and thence through Porpoise Cove to the head of the harbor. The channel is marked by buoys and daybeacons.

(86)

Anchorages

(87) The anchorage basin is usually occupied by local fishing and pleasure craft. The holding ground is good, and a hole can usually be found to drop anchor in.

(88)

Dangers

(89) The **Old Prince**, a ledge with a rock awash, extends from 400 to 500 yards southeastward of Goat Island Light. Local craft sometimes cut between Old Prince and Goat Island in entering; this passage is not advisable for strangers.

(90) Ledges extending up to 0.3 mile south of **Folly Island** are unmarked. A daybeacon marks the ledges extending northeastward from the island. This daybeacon is 180 feet from the westerly edge of the entrance to the dredged bar channel and should be given a berth of at least 250 feet in entering.

(91) Another daybeacon is on the ledge, bare at low water about 370 feet southwestward of Goat Island Light. The daybeacon is about 30 feet from the easterly edge of the bar channel and should be given a berth of about 150 feet when entering.

(92) The principal hazards in approaching and entering are the numerous lobster pot buoys, which are in the channel and outlying waters in the summer. Care should be taken to avoid these, especially at night or during periods of low visibility.

(93)

Wharves

(94) A private wharf, formerly the town wharf, is on the east side of Cape Porpoise Harbor about 0.6 mile above the entrance. The wharf, 200 feet long with 6½ to 8 feet alongside, is used by commercial fishermen to offload their catches and by transients for temporary berthage. Gasoline, diesel fuel, water and limited marine supplies are available. Small cranes are on the wharf; restaurants and lodging are close by.

(95)

Supplies

(96) Ice, provisions and marine supplies can be obtained in or on order from the village. A telephone is on the dock. There are no marine railways or repair yards; the nearest is at Kennebunkport.

(97) Good roads connect the landing with the village and nearby towns and cities. Taxi service is available.

(98) Most of **Paddy Creek**, just west of Cape Porpoise Harbor, dries at low water.

(99) **Turbats Creek**, westward of Paddy Creek, has several private landings and considerable small-craft activity, but no service facilities.

(100) Southwestward of Goat Island Light is an area of broken ground, with depths of 16 to 34 feet, extending as much as 2 miles offshore in places.

(101) On the point locally known as **Walkers Point**, 1.8 miles west-southwestward of Goat Island Light, a large mansion with four large stone chimneys is one of the most conspicuous landmarks in the area.

(102) Near the head of the cove, west of the point, is a stone breakwater behind which is a town float landing. Local pleasure craft moor in the cove, and the reported depth at the landing is 8 feet. There are no facilities.

(103) A **security zone** at Walkers Point, including the coves on both sides, extends about 0.5 mile southward to its southernmost boundary, which extends about 0.5 mile south-southeastward from **Cape Arundel** on the east side of the approaches to Kennebunk River. (See **33 CFR 165.102**, chapter 2, for exact limits and regulations.)

(104) **Kennebunk River**, about 2.5 miles southwestward of Goat Island Light, is the approach to the popular summer resort and yachting center of **Kennebunkport.**

(105)

Prominent features

(106) The beach for 0.8 mile eastward and 1.7 miles westward of the entrance is lined with hotels and summer homes, the largest and most conspicuous of which is a large white hotel with cupola on the east side of the entrance to the river.

(107) The entrance to the river is between two stone jetties, the outer end of the easterly one being marked by **Kennebunkport Breakwater Light 6** (43°20'46"N., 70°28'34"W.), 25 feet above the water, shown from a white skeleton tower with a red triangular daymark.

(108)

Channels

(109) A federal channel leads from the sea to a point about 60 yards below the highway bridge at Kennebunkport, about 1 mile above the jetties; the channel has a project depth of 8 feet to nearly Buoy 10, thence 6 feet to the limit of the project. Anchorages with a project depth of 6 feet lie both east and west of the channel about midway from the jetties to the highway bridge. Greater depths can be had using care and local knowledge. Buoys and a daybeacon mark the channel. It is reported that the entrance channel between the jetties is subject to frequent change.

(110)

No-Discharge Zone

(111) The State of Maine, with the approval of the Environmental Protection Agency, has established a No-Discharge Zone (NDZ) for the waters of Kennebunk, Kennebunkport and Wells, see the chart for limits.

(112) Within the NDZ, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZ, discharge of sewage is regulated by **40 CFR 140** (see chapter 2).

(113)

Dangers

(114) **Fishing Rock**, about 0.6 mile southward of Kennebunkport Breakwater Light 6, is marked by a daybeacon and a buoy on the east side. A reef with a least depth of 7 feet extends 0.8 mile southward of Fishing Rock where it is marked by a lighted bell buoy. **Oaks Reef**, an extensive foul ledge area with a number of drying rocks

and rocks awash, extends about 0.5 mile southward of Kennebunk Beach and is marked by a daybeacon.

(115) The State Route 9 highway bridge crossing the river at Kennebunkport was under construction in 2016.

(116)

Routes

(117) The chart should be the guide, keeping well clear of dangers and following the aids. In southerly weather with heavy seas running it is hazardous to enter through the jetties on the ebb.

(118) The approach to the port is marked by two buoys and two spindle daybeacons, which also mark the principal dangers. The best approach is to the eastward of the buoys.

(119) Some local craft prefer to approach the entrance through the passage between these two daybeacons, but strangers are advised against it.

(120) The best time to make the passage upriver is just after low water on a rising tide when the mudflats are still visible.

(121)

Wharves

(122) There are numerous private piers and float landings on the river, most of which are along the east bank. There are also a number of fish wharves and shipping plants on the upper river near the bridge.

(123) The town landing on the east bank about 500 yards inside the entrance is about 200 feet long with 6 feet reported alongside. A restaurant is nearby.

(124) The Kennebunk River Yacht Club is on the east bank about 150 yards above the town landing. Its basin, protected by a stone jetty covered at high water, has floats with 2 to 6 feet reported alongside. The upper and lower ends of the jetty are marked by stone pylons. Water is available at the floats.

(125) The Arundel Yacht Club has a pier and float landing on the east bank about 400 yards below the bridge.

(126) Small pleasure and fishing craft secure to moorings placed wherever there are sufficient depth and swinging room in the river. The Kennebunkport **harbormaster** can be contacted through the town manager's office or the local police department.

(127)

Small-craft facilities

(128) There are several marinas and boatyards on both sides of the Kennebunk River. Most of these facilities can provide gasoline, diesel fuel, water, ice and marine supplies, and some can make hull, engine and electrical repairs. The facilities with the greatest capacity include a 40-foot marine railway and a 15-ton mobile hoist. Storage facilities are also available.

(129) Marine supplies and provisions can be obtained in Kennebunkport. The town has taxi service to Kennebunk with connections for bus service to other coastal and inland points.

(130) **Kennebunk Beach** is a village extending 1 mile westward of Kennebunk River entrance. Ledges extend 0.8 mile from shore southward of the village. **Great Hill**, a prominent yellow bluff at the western end of Kennebunk Beach, marks the mouth of the Mousam River. Several of the houses on the bluff are conspicuous.

(131) **Mousam River** is used by small craft with local knowledge. A fixed highway bridge, with a clearance of about 3 feet each side of the center pier, crosses the river about 0.3 mile above the mouth. There are private landings on the river, but no services.

(132) From Mousam River, a beach extends southwestward about 1.3 miles to another inlet into which **Little River** and its tributaries, **Branch Brook** and **Merriland River**, flow. A large house with a brick chimney, on a jutting point about the middle of the beach, is discernible among the other summer homes that line the beach. The inlet is not passable except for very small craft with local knowledge.

(133) **Drakes Island Beach**, extending from this inlet to the jettied entrance at Wells Harbor about 1 mile southwestward, is a resort of numerous summer homes. A foul area with many rocks awash is about 0.7 mile off Drakes Island Beach and is unmarked.

(134) **Wells Harbor**, about 6 miles west-southwestward of Goat Island Light, is used by local fishing and pleasure craft. **Webhannet River,** which flows into Wells Harbor from the southward, has no services. The harbor is protected at the entrance by two jetties marked by lights.

(135)

Prominent features

(136) The principal landmarks along this stretch of beach from Kennebunkport to Ogunquit are the large resort hotels at Bald Head Cliff; Ogunquit, Wells and Kennebunk beaches; a church spire about 1.3 miles southward of Wells; and the standpipes at Ogunquit and Kennebunk. The numerous summer homes, some large mansions, also stand out.

(137) **Wells Beach** extends about 2 miles southward from the entrance to Wells Harbor to a bluff on which are a number of prominent homes, one of which has a conspicuous pointed cupola.

(138)

Channels

(139) The entrance channel to the harbor leads between two jetties, marked on the outer ends by lights, to an anchorage basin about 0.5 mile inside; the south jetty should be favored. The approach to the entrance is marked by a lighted bell buoy and the channel is marked by a buoy and daybeacons to the anchorage basin. It is reported that even during a moderate sea, swells break across the entrance making entry hazardous.

(140)

Anchorages

(141) Three anchorages are in Wells Harbor. (See **33 CFR 110.1** and **110.9**, chapter 2, for limits and regulations.)

(166)

(142)

No-Discharge Zone

(143) The State of Maine, with the approval of the Environmental Protection Agency, has established a No-Discharge Zone (NDZ) for the waters of Kennebunk, Kennebunkport and Wells; see the chart for limits.

(144) Within the NDZ, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZ, discharge of sewage is regulated by **40 CFR 140** (see chapter 2).

(145)

Dangers

(146) The principal outlying dangers off these beaches are an unmarked shoal and foul area that extends about 0.5 mile off Wells Beach and has a rock that uncovers 3 feet and rocks awash on it. **Bibb Rock**, which uncovers 2 feet, about 0.8 mile off the point at the north end of Moody Beach, is marked on its east side by a buoy.

(147)

Small-craft facilities

(148) There are town piers and small-craft launching ramps on both the east and west sides of the anchorage basin at Wells Harbor. The pier on the east side has a depth of about 6 feet reported alongside its float landing but no services. The pier on the west side has a depth of about 10 feet reported alongside its float landing; gasoline, diesel fuel and water are available. A marina adjacent to southward is reported to have a marine railway that can handle craft up to 40 feet for engine repairs and dry open storage. A restaurant is nearby. The **harbormaster** maintains an office on the westerly pier, telephone 207–646–3236.

(149) Groceries and other services are available in the villiage of **Wells**, just westward of the harbor.

(150) **Moody Beach** extends southward 1.2 miles where it joins **Ogunquit Beach**, which extends 1.2 miles farther to the entrance of **Ogunquit River**. The river runs southward, draining the marshes behind these beaches, and enters the ocean at Ogunquit, 4.7 miles southward of Wells Harbor. Some small craft use the river above the highway bridge about 0.3 mile above the entrance, which has a 26-foot fixed span with a clearance of 6 feet.

(151) The entrance to the river is not marked, and the swells break across it making it difficult and dangerous to enter even in calm weather. There are no services, but there are restaurants, a parking lot and picnic areas on the beach.

(152) **Ogunquit** is a summer resort of historical importance. **Israels Head**, a prominent headland, overlooks the entrance to the river on the south.

(153) **Perkins Cove**, at the mouth of **Josias River**, 1 mile southeastward of Ogunquit, is a small landlocked harbor, very popular with yachtsmen, at which a number of fishing, pleasure and party fishing boats base.

(178)

(154) The facilities of the harbor are controlled by the village corporation, and the moorings are under supervision of the **harbormaster**, who usually can be found at the town float landing on the north side of the harbor by the footbridge.

(155) Perkins Cove is entered by a narrow entrance channel that leads to an anchorage basin at the head of the harbor, known as **Flat Pond.** In 2006, the controlling depths were 4.7 feet in the entrance channel, thence 5 feet in the anchorage basin, except for shoaling to 4.6 feet along the southern edge and 2.1 feet along the western edge. The channel to the anchorage is marked by two buoys and a daybeacon; a lighted bell buoy is about 0.8 mile northeastward of the entrance.

(156) The harbor is a safe haven for small craft in this stretch of coast in a sudden blow, but no attempt should be made to enter once the sea has made up, as heavy swells break clear across the entrance during easterly weather and for as long as 2 days after a heavy blow. Small craft may broach to in attempting to enter under such conditions.

(157) The harbor is crossed, just above the town float, by a wooden double bascule footbridge, which is operated by the harbormaster on request. The bridge has a channel width of 20 feet and a clearance of 16 feet.

(158) Diesel fuel by truck and water are available at the town float, which has 5 feet reported alongside. Seasonal stores, lodging and restaurants are at the harbor. Ice, provisions and marine supplies are also available at the harbor or at Ogunquit.

(159) Taxi and other services are available, and the main coastal highway passes a short distance from the harbor.

(160) A marine railway that can handle craft up to 40 feet is on the east bank at the town wharf. Open winter storage and use of the railway for repairs are on a do-it-yourself basis.

(161) **Bald Head Cliff**, 11 miles southwestward of Cape Porpoise, is a prominent high point on which are two conspicuous white buildings.

(162) **Mount Agamenticus**, 691 feet high, is the highest and southernmost of three hills on a ridge 5 miles westward of Bald Head Cliff. The hill is a prominent landmark for vessels cruising along this section of the coast.

(163)

Weare Point to Bragdon Island

(164) **Weare Point** (43°11.2'N., 70°35.9'W.), 2.3 miles southward of Bald Head Cliff, is a headland with several large houses on it.

(165) **Cape Neddick Harbor** is a small open bight between Weare Point and **Barn Point** about 1 mile northwestward of Cape Neddick. The entrance is marked, but the dangers inside the entrance are not marked. There is good anchorage in 9 to 30 feet in the middle of the bight, which is protected by the reefs on each side of the entrance

said to be plainly visible if the water is clear, is south of Black Rocks and is marked by a buoy.

(193) The ledge extending northeastward from Western Point is marked by a buoy about 200 yards northeastward of the point. These two buoys are the first pair in entering the harbor and should be passed in midchannel, with York Harbor Entrance Leading Light 8 dead ahead on a course of **270°**.

(194) A rock covered 3 feet, part of a ledge extending 100 yards southeastward of **Fort Point**, the eastern end of Stage Neck, is marked on its south side by a buoy.

(195) **Rocks Nose**, a bare ledge extending 150 yards northeastward from the shore on the south side of the entrance channel, is marked by a buoy.

(196) A buoy marks the ledge off the southwestern extremity of Stage Neck and the sharp turn from the entrance channel up into the inner harbor. In making this turn, sharp seamanship is needed, especially on the strength of ebb, to avoid setting over to the westward and bringing up on the rock ledge covered 1½ feet which is eastward of **Harris Island**; give the daybeacon marking the east side of the ledge a good berth.

(197) The ledge off the eastern end of **Bragdon Island** is covered 3 feet and should be given a good berth when proceeding into the inner harbor. The northeast end of the ledge is marked by a buoy that also marks the turn of the river to the northwestward off the wharves.

(198) The **currents** are strong in the constricted sections of the channel, where the buoys are reported to tow under at times.

(199) The **harbormaster** will, on request, meet visiting craft outside the harbor and pilot them in. He can usually be contacted through the marinas or be found about the harbor.

(200)

Bridges

(201) State Route 103 highway bridge about 1.15 miles above the entrance has a fixed span with a clearance of 15 feet. The second fixed highway bridge, **Sewall Bridge**, about 1.7 miles above the entrance, was rebuilt in 1940 as a replica of the first pile drawbridge built on the site in the colonial days of 1761. The present bridge has an imitation bascule drawspan that is not operable and has a clearance of 3 feet.

(202)

Routes

(203) Craft entering York Harbor in daylight with the aid of the chart and following the aids should have no problems. The most difficult problem is making the sharp turn at the buoy at the southwestern end of Stage Neck.

(204) After making the bell buoy off the entrance, it is well to bring the leading light ahead on the bearing **270°** and, if at night, to run in on the intensified beam.

(205) It would be prudent, however, at night, if the sea and swell are not too heavy, to anchor in the hole eastward of Fort Point, just out of the channel in line with the two nun buoys, and wait for daylight before attempting the run into the harbor and negotiating the turn around Stage Neck.

(206)

Small-craft facilities

(207) The facilities for yachts and small craft in the harbor are full and complete. All services can be had, and ice, provisions and supplies of all kinds are available or can be obtained on short notice. There are three service facilities along the waterfront with wharves and float landings with 8 to 12 feet reported alongside. Gasoline, diesel fuel and water are available. Overnight berthing at the landings is permitted.

(208) A well-equipped marina and boatyard is on Harris Island in the cove westward of Stage Neck. There is a reported depth of 8 feet at the floats, and gasoline, diesel fuel, water and electricity are available. Its marine railways can haul out sail or motor craft up to 50 feet long or 100 tons for hull and engine repairs or dry winter storage. Marine supplies, lodging and parking are available. Taxi and car rental service are available.

(209) Two town piers and floats are available. One is at the north end of Bragdon Island about 75 yards east of State Route 103 highway bridge. The second is on the east side of the causeway connecting Bragdon and Harris Islands, midway between them. The wharves have no services; docking is limited to 30 minutes.

(210)

Boon Island to Sanders Ledge

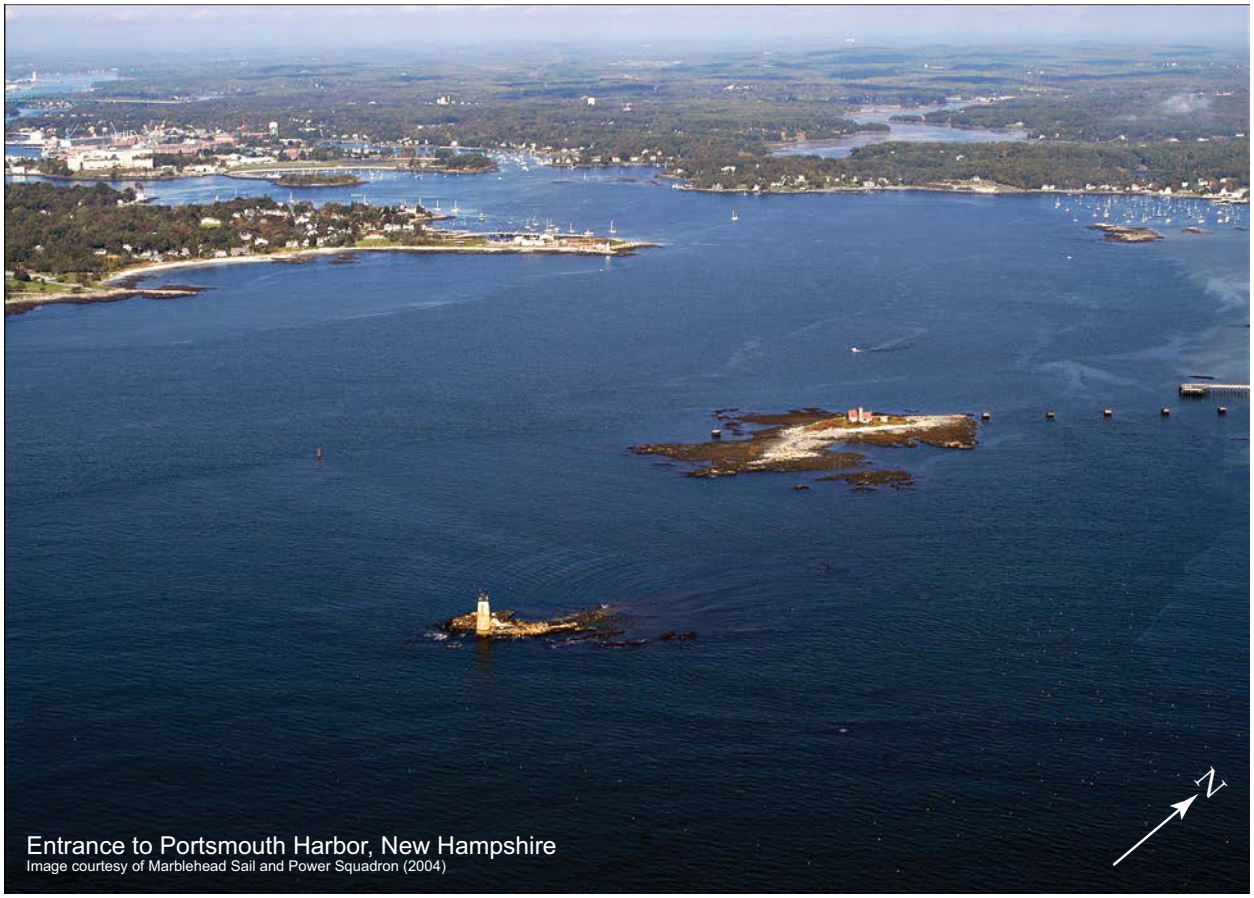
(211) Vessels must observe caution to avoid the offshore dangers in the northern approach to Portsmouth. **Boon Island**, 5.7 miles southeastward of Cape Neddick, is a small, low, rocky islet, marked by **Boon Island Light** (43°07'17"N., 70°28'35"W.), 137 feet above the water, and shown from a 133-foot gray granite conical tower. A sound signal is at the light.

(212) Boon Island is surrounded by deep water, but there are numerous detached ledges in the vicinity. The easternmost is **Boon Island Ledge**, 2.8 miles eastward of the light, which is awash at low water and has a lighted whistle buoy off its southeast end.

(213) Vessels should not pass between this buoy and Boon Island Light as there is a shoal area covered 16 feet between them. If passing westward of the light, give it a berth of 2 miles or more to assure staying in a depth of more than 30 feet as there is an unmarked rocky area covered 25 feet, about 1.6 miles west-southwestward of it. Depths of 26 feet are up to 1.3 miles southward of the light.

(214) **Pollock Rock**, covered 17 feet, and **Southeast Shoal**, covered 21 feet, are 0.7 mile southwest and southeastward, respectively, from Boon Island Light. **Sanders Ledge**, covered 28 feet, is about 1.2 miles south of Boon Island.

(234)



(215)

Caution

(216) U.S. Naval vessels may be operating with submarines in the area south and eastward of Boon Island. Escorting naval surface vessels usually display a red flag or the international code flag signal **NE 2,** meaning: **You should proceed with great caution; submarines are exercising in this area**.

(217) All vessels should keep well clear of vessels displaying this signal and should obey promptly any orders that may be given by commanding officers of navy vessels.

(218) Between Cape Neddick and the entrance to Portsmouth Harbor, a distance of 8 miles, the shore is indented by York Harbor, already described; Godfreys Cove, a shallow bight seldom entered; and Brave Boat Harbor.

(219)

Brave Boat Harbor to Portsmouth

(220) **Brave Boat Harbor** (43°06.0'N., 70°39.6'W.), 2 miles southwestward of York Harbor, has a few private landings but no facilities. Some local small craft were observed there, but the surf is reported to break clear across the entrance with the least sign of weather. Two old railway trestles cross the streams entering into it about 0.2 mile above the entrance. A large mansion on **Raynes Neck**, the point about 0.35 mile northeastward of the entrance, is conspicuous.

(221) **Cutts Island**, on the south side of the entrance, is connected with Gerrish Island to the south of it by a natural seawall of stones and rock thrown up by winter gales. It is conspicuous. A public beach is at the north end of the seawall.

(222) **Moores Rock**, covered 5 feet and unmarked, is about 0.5 mile eastward of the entrance to Brave Boat Harbor. A long reef that uncovers 4 feet is about 0.3 mile southeastward of the entrance.

(223) Two dangerous ledges are 2.5 miles offshore. **York Ledge**, the northernmost, covered 3 feet and 2.9 miles southeastward of York River, is marked on the east side by a buoy. **Murray Rock**, 1.5 miles south-southwestward of York Ledge, is covered 6 feet and is marked by a buoy off its southwest side. Between these ledges and the shore, the bottom is very broken and vessels are advised to pass 1 mile east of the ledges. In 1997, a dangerous rock covered by 24 feet of water, protruding from a rocky ledge, was reported in about 43°03'45"N., 70°35'59"W., about 0.7 mile southeast of Murray Rock. Broken ground covered 24 to 39 feet, extends 2 miles south-southeastward of the buoy marking Murray Rock.

(224) **Portsmouth Harbor**, 37 miles southwestward of Cape Elizabeth and about 25 miles northward of Cape Ann Light, is the only harbor of refuge for deep-draft vessels between Portland and Gloucester. No large vessel

(235)

should proceed northward of Kitts Rocks Lighted Whistle Buoy 2KR (43°03.0'N., 70°41.5'W.) without a pilot, as the anchorage area is limited.

(225) Portsmouth Harbor is at the mouth of Piscataqua River and is the approach to the cities of Portsmouth and Dover and the towns of New Castle, Kittery, Newmarket, Durham, Newington and Exeter.

(226) Several U.S. Navy activities, including the **Portsmouth Naval Shipyard** and a regional medical clinic, are on **Seavey Island** at Kittery, on the north side of the harbor opposite Portsmouth.

(227) A **regulated navigation area** has been established in the vicinity of the Portsmouth Naval Shipyard on Seavey Island. (See **33 CFR 165.1** through **165.13** and **165.101**, chapter 2, for limits and regulations.)

(228) A moving safety zone is established surrounding tank vessels carrying liquified petroleum gas (LPG) while transiting Bigelow Bight, Portsmouth Harbor and the Piscataqua River. (See **33 CFR 165.20**, **165.23** and **165.103**, chapter 2, for limits and regulations)

(229) **Restricted areas** are at the east end of Seavey Island in the cove between Clarks, Seavey and Jamaica Islands and at the west end of Seavey Island from Henderson Point along the shore to the combined highway and railroad bridge across Back Channel. (See **33 CFR 334.50**, chapter 2, for limits and regulations.)

(230) A security barrier has been established inside the regulated navigation area and the western restricted area.

(231)

COLREGS Demarcation Lines

(232) The lines established for Portsmouth Harbor are described in **180.15**, chapter 2.

(233) **Portsmouth** is a city on the south bank of Piscataqua River about 4 miles above the entrance to the harbor.

(236) Foreign trade is in petroleum products gypsum, frozen fish, fish products, and salt. Oil shipments in tankers, drawing as much as 35 feet, arrive frequently, except during the summer.

(237) Coastwise trade is in arrivals of oil tankers drawing up to 35 feet. The shipment of cable from Newington is of major importance.

(238) The harbor, of sufficient depth to accommodate large deep-draft ships, is open throughout the year, though vessels may be hampered somewhat in passing through the two lift bridges to deepwater berths above the city.

(239) **New Castle**, a village on the south side of the harbor and the northern part of **New Castle Island**, is reached from Portsmouth by a highway connecting the islands on the south side of the harbor. The island is of considerable importance as a summer resort.

(240) **Kittery** is a town on the north bank of Piscataqua River opposite Portsmouth.

(241)

Prominent features

(242) **Gerrish Island**, forming the east side of the harbor entrance, has many summer homes. A park and government reservation, with conspicuous buildings, are on the southwestern end. The old observation tower on the south end of the island is most conspicuous. A long pier is at the southwestern end of the island. The area just northwest of the pier is used as a bathing beach; boaters either beach their craft or anchor offshore. The park has picnic tables and other facilities.

(243) For craft approaching Portsmouth, the large hotel with a charted cupola at the southwest end of New Castle Island is prominent. Other landmarks are: the stone building and square tower of the former naval prison and the water tank on Seavey Island; Whaleback Light; the weathered buildings with conspicuous cupola of the abandoned Coast Guard station on Wood Island; and numerous standpipes, elevated tanks, church spires and stacks in the area, most of which are charted. The old blockhouse and parapets of **Fort McClary**, on Kittery Point, just westward of the entrance channel range lights, are also conspicuous.

(244) **Whaleback Light** (43°03'32"N., 70°41'47"W.), 59 feet above the water, is shown from a gray granite conical tower on Whaleback Reef at the northeast side of the outer entrance. A mariner-activated sound signal that operates by keying the microphone five times on VHF-FM channel 83A is at the light.

(245) **Portsmouth Harbor (New Castle) Light** (43°04’16”N., 70°42’31”W.), 52 feet above the water, is shown from a white conical tower attached to a house on **Fort Point**, the northeast end of New Castle Island. A mariner radio-activated sound signal at the light is initiated by keying the microphone five times on VHF-FM channel 83A.

(246) **Portsmouth Harbor Coast Guard Station** and lookout tower are on Fort Point.

(247)

Security Broadcast System, Portsmouth Harbor

(248) The Coast Guard Captain of the Port, Sector Northern New England, has established a voluntary system of radiotelephone broadcast/reporting procedures designed to give masters and pilots real-time information on marine traffic in Portsmouth Harbor. The system supplements the Vessel Bridge-to-Bridge Radiotelephone Regulations contained in 33 CFR 26 (see chapter 2), and all vessels subject to these regulations are urged to participate in the system. Nothing in these procedures shall supersede the Navigation Rules or relieve the master of the vessel of his responsibility for the safe navigation of the vessel. These recommended procedures are designed to give notice of unseen vessels, give notice of intended movement, clear VHF-FM channel 13 of traffic unrelated to navigation and give vessels information on other vessels within the immediate vicinity.

(249) All participating vessels are requested to use VHF-FM channel 13 for listening watches and security calls, except when calling a small vessel not responding on channel 13, in which case channel 16 is appropriate.

(250) Participating vessels shall maintain a listening watch commencing 30 minutes prior to getting underway or 30 minutes prior to reaching the vicinity of Gunboat Shoal Lighted Bell Buoy 1 (43°01.4'N., 70°41.9'W.).

(251) Security calls shall be made as follows: 15 minutes prior to getting underway; when getting underway, including route; when passing Gunboat Shoal Lighted Bell Buoy 1, or from north when approaching Wood Island Lighted Buoy 2 (43°03'40"N., 70°42'04"W.), including destination if inbound; and when mooring or anchoring.

(252) Arrangements for bridge openings are made on channel 13.

(253) If a call is made to a ship or station to pass any of the above information on channel 13, an additional call is unnecessary. Example: a ship calling a bridge 15 minutes prior to getting underway to arrange for an opening.

(254) Vessels carrying passengers or cargo and not required by law to comply with Vessel Bridge-to-Bridge Radiotelephone Regulations are encouraged to monitor and respond on channel 13. During periods of low visibility, it is appropriate to follow security call procedures discussed above, except that security calls 15 minutes prior to getting underway should not be made.

(255) Portsmouth Harbor Coast Guard Station monitors VHF-FM channel 13.

(256)

Recommended minimum under-keel clearances for the Port of Portsmouth

(257) The U.S. Coast Guard, in cooperation with the Navigation Subcommittee of the Maine and New Hampshire Port Safety Forum, has established recommended minimum under-keel clearances for the Port of Portsmouth, in order to prevent groundings and to promote safety and environmental security of the waterway resources of the Port of Portsmouth. The group recommends that all entities responsible for safe movement of vessels in and through the waters of the Port of Portsmouth operate vessels in such a manner as to maintain a minimum under-keel clearance of 3 feet between the deepest draft of their vessel and the channel bottom when transiting Portsmouth Harbor and the Piscataqua River inside Kitts Rock Lighted Whistle Buoy 2KR; a minimum under-keel clearance of 1 foot is recommended at berthing areas.

(258) The Maine and New Hampshire Port Safety Forum, in cooperation with U.S. Coast Guard Sector Northern New England, requests vessels to follow the mooring recommendations for the Piscataqua River listed below.

(259) **Recommendation:** Due to the very strong ebb and flood tidal currents on the Piscataqua River and its tributaries, a mooring plan will be provided by the

Portsmouth Pilots upon boarding, for the intended terminal.

(260) Vessels shifting at the dock must only do so during periods of slack water. It is extremely dangerous to attempt to shift a vessel at moorings on the Piscataqua River at any other time and should not be attempted. Masters should be particularly vigilant in minding and tending to their vessel's moorings.

(261) No vessel shall rely solely upon automatic tensioning winches while moored at any facility on the Piscataqua River.

(262) Vessels meeting **all** of the following criteria are recommended to obtain the services of a mooring master while moored on the Piscataqua River. Intentions for obtaining the services of a mooring master shall be included in the vessel's 24-hour advance notice of arrival.

(263) **Parameters for mooring master:**

(264) **Vessels meeting the maximum Length Over All (LOA) for the following terminals:**

(265) Portsmouth-Schiller: 621 feet (189.28 meters);

(266) Sprague Avery Lane: 648 feet (197.51 meters);

(267) Sprague River Road: 661 feet (201.47 meters)

(268) **Range of Tide:** 12 feet (3.66 meters) or greater, as per Boston HW and LW

(269) **Vessel draft:** Greater than 32 feet (9.75 meters).

(270) **NOTE:** Vessels meeting the above criteria that do not obtain the services of a mooring master must obtain permission from the Coast Guard Captain of the Port, Sector Northern New England, via the vessel agent or the U.S. Coast Guard Marine Safety Detachment, Portsmouth, NH.

(271) All vessels must maintain minimum under-keel clearance of 1 foot while moored at any terminal and 3 feet during transits.

(272) IMO Ship Safety Bulletin 13/95, "Safety of Ships Carrying Solid Bulk Cargoes," provides a checklist for vessels and terminals. The checklist is recommended for use by terminals and vessels conducting bulk cargo transfers on the Piscataqua River. A copy of this checklist can be obtained from U.S. Coast Guard Marine Safety Detachment, Portsmouth, NH, the Portsmouth Pilots or vessel agents.

(273)

Channels

(274) Depths of about 34 feet can be carried in the marked channel through Portsmouth Harbor to the Memorial (U.S. Route 1) Highway Bridge. From this point, a Federal project provides for a channel 35 feet deep for about 3.5 miles to a turning basin about 0.4 mile above Frankfurt Island in Piscataqua River. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A. The entrance and harbor channels are marked by lights, lighted ranges, lighted and unlighted buoys and daybeacons.

(275) **Portsmouth Harbor Channel Lighted Range** on Kittery Point leads into the harbor on the bearing **352°45'**. The range structures are in a narrow clearing of trees on Kittery Point. Outbound vessels are cautioned that the range lights will come into line soon after the rear light becomes visible. Vessels should commence their turn onto the range line early enough to avoid overrunning it. The rear light may be visible earlier during the winter months.

(276) **Pierce Island Lighted Range** marks the main channel to Portsmouth on bearing **266.6°**.

(277) A small-boat channel, privately marked by seasonal buoys, leads northerly from the main ship channel about 100 yards below the combined U.S. Route 1 Bypass highway and the railroad bridge and passes under a retractable span of the railroad bridge. In 1968, the reported controlling depth in the channel was 6 feet. Clearances for the retractable span are given under bridges for Portsmouth Harbor.

(278) **Back Channel**, between Seavey Island and Kittery, is limited principally to small craft and is covered in geographical sequence in the description of the harbor features.

(279) The channel in Piscataqua River above the bridges is covered in the description of the river.

(280)

Anchorages

(281) The anchorage for medium-sized vessels is anywhere on the east and north sides of the channel between Wood Island, north of Whaleback Light, and Clarks Island, the small island on the north side about 0.8 mile above Fort Point, in 18 to 71 feet. Space is limited, however, to one medium-sized vessel northward of Fort Point.

(282) Strangers should not go above Kitts Rocks in deep-draft vessels without a pilot. Because of the strong currents and eddies in the bend around Fort Point, it is difficult for any large vessel to make the swing without the assistance of a tug. It is not advisable to proceed above Wood Island without a tug and pilot. Most large vessels awaiting tug and pilot or favorable mooring or docking conditions anchor temporarily between Gunboat Shoal and the lighted whistle buoy south of Kitts Rocks.

(283) With southerly wind, the best anchorage is above Fort Point on the south side of the channel in 49 to 58 feet, bottom generally clay. There is swinging room there for only one medium-sized vessel without encroaching on the channel ranges. There is no room to anchor in the channel above Clarks Island.

(284) Yachts and smaller vessels usually anchor in Pepperrell Cove or northward of New Castle Island, southward of the range line.

(285) A **special anchorage** is off the north side of New Castle Island. (See **33 CFR 110.1** and **110.10**, chapter 2, for limits and regulations.)

(290)

Structures across Piscataqua River and its Tributaries

Name•Description•Type	Location	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
Back Channel				
Highway Bridge (fxed)	43°05'06"N., 70°44'09"W.	90	8	Note 1 Vertical clearance is at center of span
US Navy Bridge 1 (highway/railroad–fxed)	43°05'05"N., 70°44'30"W.	56	7	Note 2
Overhead cable	43°05'04"N., 70°44'32"W.		65	
Piscataqua River				
Memorial (US Route 1) Bridge (vertical lift)	43°04'45"N., 70°45'09"W.	275	21 (down), 150 (up)	Notes 3 and 4
Memorial (US Route 1) Bridge (fxed)	43°05'00"N., 70°45'04"W.	98	11	Between Badgers Island and Kittery Vertical clearance is at center of span.
Sarah Mildred Long Bridge (highway/railroad–vertical lift)	43°05'10"N., 70°45'40"W.	200	10 (down), 135 (up)	Notes 3, 5, 6 and 7
Interstate 95 Highway Bridge (fxed)	43°05'34"N., 70°46'00"W.	440	134	
Overhead power cables	43°05'58"N., 70°47'00"W.		165	
Overhead power cables	43°10'18"N., 70°49'36"W.		65	
Salmon Falls River				
Toll Bridge (highway–fxed)	43°11'25"N., 70°49'30"W.	36	5	
Cocheco River				
Overhead power cable	43°11'44"N., 70°51'37"W.		52	
Overhead power cable	43°11'49"N., 70°51'45"W.		34	
Overhead power cable	43°11'49"N., 70°51'48"W.		44	
Overhead power cable	43°11'50"N., 70°51'53"W.		34	
Overhead power cable	43°11'50"N., 70°52'12"W.		35	
Little Bay				
General Sullivan Bridge (highway/pedestrian–fxed)	43°07'02"N., 70°49'31"W.	200	33	Bridge is under construction
Bellamy River				
Scammel Bridge (highway–fxed)	43°07'45"N., 70°05'56"W.	87	11	
Overhead power cable	43°09'48"N., 70°51'28"W.		32	
Squamscott River				
Boston & Maine Railroad Bridge (fxed)	43°03'10"N., 70°54'45"W.	30	5	
Overhead power cable	43°02'35"N., 70°55'15"W.		50	
Route 108 Bridge (fxed)	43°02'22"N., 70°55'42"W.	108	11	
Overhead power cable	43°00'44"N., 70°56'23"W.		55	
Route 101 Bridge (fxed)	43°00'00"N., 70°56'23"W.	146	26	

Note 1 – Navigation channel is reported to be northward of the center pier.
Note 2 – Navigation channel is reported to be under the second span from the south end of the bridge.
Note 3 – See 33 CFR 117.1 through 117.59 and 117.531, chapter 2, for drawbridge regulations.
Note 4 – Bridgetender monitors VHF-FM channel 16 and works on channel 13; call sign KBK-472.
Note 5 – Bridgetender monitors VHF-FM channel 16 and works on channel 13; call sign KAW-766.
Note 6 – Regulated Navigation Area surrounds bridge. See, 33 CFR 165.1 through 165.13 and 165.T01-0554, chapter 2, for limits and regulations.
Note 7 – The railroad bridge has a retractable span near the north end (5 feet closed and 36 feet open.) The span is kept open except for the passage of one train per day, Tuesday through Saturday, April to November.

(286)

Dangers

(287) The principal outlying dangers are marked so that no difficulty should be experienced when entering in clear weather, day or night.

(288) **Gunboat Shoal**, rocky and covered 20 feet, on the west side of the entrance about 2.2 miles southward of Whaleback Light, is marked on its northeast end by a lighted bell buoy. An area of rocks and ledges, some of which uncover up to 5 feet, extends about 1.5 miles eastward of Whaleback Light and up to 0.6 mile offshore. They include **West Sister**, which uncovers 3 feet and is marked by a buoy off its southeast end; **East Sister**, an unmarked ledge which uncovers 2 feet about 0.5 mile northeastward of West Sister; **Phillips Rock**, unmarked and covered 4 feet, about 0.2 mile southwestward of West Sister; **Horn Island**, surrounded by a drying reef; and 4-foot-high **White Island** and **White Island Reef**, southeastward of which are a number of unmarked rocks.

(289) **Kitts Rocks**, covered 11 feet, are on the east side of the channel, about 0.4 mile southward of Whaleback Light, and are marked by a lighted whistle buoy to the

southward. **Wood Island Ledge**, extending 0.2 mile off **Wood Island**, is marked off its southwest end by a lighted buoy. **Stielman Rocks**, covered 2 feet, are on the west side of the entrance about 500 yards southward of Fort Point Light; they are marked by a daybeacon on the rocks and a buoy on the northeast end. **Cod Rock**, covered 17 feet, is 225 yards northwestward of Fort Point. The rock is marked by a distinct, violent eddy just before low water slack. The remaining dangers in the harbor are described in geographic sequence.

(291)

Weather, Portsmouth and vicinity

(292) Portsmouth, located on the extreme north coast of New Hampshire, has an average annual temperature of 47.9 °F. July is the warmest month with an average high of 79 °F and an average minimum of 61 °F. January is the coolest month with an average high of 31 °F and an average minimum of 15 °F. The highest temperature on record for Portsmouth is 101 °F recorded in July 1964, and the lowest temperature on record is -16 °F recorded in January 1957. An average of six days each year record temperatures in excess of 90 °F, 135 days have temperatures below freezing and 14 days drop below 5 °F. Every month has seen temperatures below 50 °F and every month except June, July and August has recorded temperatures below freezing.

(293) The average annual precipitation for Portsmouth is 42.8 inches (1087 mm), which is fairly evenly distributed throughout the year. Precipitation falls on about 180 days each year. The wettest month is November with 5.1 inches (130 mm) and the driest, August, averages only 2.3 inches (58 mm). An average of 18 thunderstorm days occur each year with June, July and August being the most likely months. Snow falls on about 59 days each year and averages about 68 inches (1727 mm) each year. December, January and February each average about 17 inches (432 mm) of snowfall each year. Seventeen inch (432 mm) snowfalls in a 24-hour period occurred in January 1961 and again in December 1961. About 12 days each year have a snowfall total greater than 1.5 inches (38 mm), and snow has fallen in every month except June through September. Fog is present on average 168 days each year and is evenly distributed throughout the year with a slight maximum in the summer.

(294) The prevailing wind direction in Portsmouth is the west. February is the windiest month.

(295)

Current

(296) The tidal currents are strong, and special care is required especially in the restricted sections of the channel above and below the bridges. Daily predictions can be generated at the Tidal Current prediction service at *tidesandcurrents.noaa.gov*. Links to a user guide for this service can be found in chapter 1 of this book.

(297) In the cove on the northwest side of Fort Point, the current is reported to frequently flow counter to the current in the harbor for a period after slack water.

(298)

Pilotage, Portsmouth

(299) Pilotage is compulsory for all foreign vessels and United States vessels under register in the foreign trade. Pilotage is optional for coastwise vessels under enrollment or license who have on board a pilot licensed by the federal government.

(300) Pilotage is provided by Portsmouth Pilots, Inc., 34 Ceres Street Wharf, Portsmouth NH 03801, or Portsmouth Pilots, Inc., P.O. Box 72, Portsmouth, NH 03801; telephone 603–436–1209, FAX 603–436–0417. The pilot office usually monitors VHF-FM channels 16 and 13, between 0800 and 1600, daily. When tugs are required, the tugs are used as pilot boats. The tugs have green hulls, dark red superstructure and a white letter "M" on black stacks. When a tug is not required, the pilot boat is a white 23-foot outboard launch with a cuddy cabin. The tugs monitor VHF-FM channel 16 and 13 and usually work on channel 7A or 77. The launch when underway monitors channel 13. Pilots board about 1 mile south-southeast of Kitts Rocks Lighted Whistle Buoy 2KR (43°02'58"N., 70°41'28"W.). Vessels with freeboard greater than 10 feet should provide a boarding ladder 3 feet above the water. Vessel movements are coordinated with minimum current and may be canceled during periods of fog. Pilots are generally arranged for through ship's agents. A 24-hour advance notice of ETA is requested.

(301) Maximum wind for pilot boarding and transit is normally 40 knots but may be extended to 50 knots on a case-by-case basis as determined by the vessel's master and the pilot. A minimum of ½ mile visibility is required for transit.

(302) As all commercial wharves now in use, except fish piers, are above the first bridge, Memorial Highway Bridge, all large vessels, including coastal tankers, take a pilot and tug from the outer anchorage.

(303) The strong currents in the narrow channel make the approach to and passage through the bridges very difficult. The largest vessels usually require two or more tugs and are taken through at or near the nearest slack water, depending on draft.

(304) A pilot to the outer anchorage is not necessary in clear weather when the aids are seen, but strangers should not go beyond Kitts Rocks at any time. In fog or low visibility no vessel of any size should proceed northward of Wood Island.

(305) The larger vessels awaiting a pilot or tide usually anchor between Kitts Rocks Lighted Whistle Buoy 2KR and Gunboat Shoal.

(306) Due to extremely strong currents on the Piscataqua River and its tributaries, vessels are recommended to follow the Coast Guard Captain of the Port mooring plans. The mooring plans are accepted as the minimum guidelines, and an even more conservative assessment of the local conditions should be made when determining whether the vessel is sufficiently moored. **All Liquefied Petroleum Gas vessels are required to comply with the**

mooring plans. The plans are available from the local pilots and shipping agents.

(307)

Towage

(308) Tugs up to 3,000 hp are available at Portsmouth. They are also used as pilot boats; see Pilotage, Portsmouth Harbor, this chapter for a description of the tugs and radio frequencies used. Naval and other vessels docking at Seavey Island usually require tug assistance. Inbound laden tug/barge units carrying 70,00 barrels or more of oil and towing astern inside of Kitts Rock Lighted Whistle Buoy 2KR should engage the services of an assist tug when transitioning the mode of towing.

(309)

Quarantine, customs, immigration and agricultural quarantine

(310) (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(311) Portsmouth is a **customs port of entry.**

(312) **Quarantine** is enforced in accordance with the regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.)

(313) Portsmouth has several public and private hospitals.

(314)

Harbor Regulations

(315) Regulations for Portsmouth Harbor are established by the New Hampshire State Port Authority and are enforced by the **harbormaster.** The Authority maintains offices at the New Hampshire State Port Authority Marine Terminal; the harbormaster can be contacted through the Authority.

(316)

Wharves

(317) All of the commercial deep-draft facilities in use are on the south bank of the Piscataqua River between the first bridge, Memorial Highway Bridge and Dover Point. All of the facilities have highway connections, and all except the Defense Fuel Support Point, Newington Dock, have rail connections. The alongside depths given for each facility described are reported; for information on the latest depths, contact the operator. Cargo discharge is curtailed at the discretion of the facility during severe electrical storms and at wind speeds above 50 miles per hour dependent on wind direction. Only the major facilities are described.

(318) **Granite State Minerals Dock:** about 0.3 mile above the Memorial Highway Bridge; 300-foot marginal wharf; 32 feet alongside; deck height, 18 feet; 2 acres of open storage; two crawler cranes with 2½-cubic yard clamshell buckets for combined lifting capacity of 20 tons; 2½-cubic yard front-end loader; 130-ton mobile crane; water and electrical shore power connections; receipt of salt, receipt and shipment of dry bulk cargoes and heavy lift items; owned and operated by Granite State Minerals, Inc.

(319) **New Hampshire State Port Authority, Marine Terminal Wharf:** about 0.45 mile above the Memorial Highway Bridge and immediately southeastward of the second bridge; 578-foot face; 35 feet alongside; deck height, 14 feet; 43,000 square feet covered storage and 10 acres open storage; mobile cranes up to 165 tons and fork lift trucks; receipt and shipment of containerized and conventional general cargo and shipment of scrap metals; owned by New Hampshire State Port Authority and operated by New Hampshire State Port Authority and John T. Clark and Son of New Hampshire, Inc.

(320) **National Gypsum Co., Portsmouth Plant Wharf:** about 0.9 mile above the Memorial Highway Bridge; 300-foot marginal wharf; 35 to 34 feet alongside; deck height, 14 feet; hopper conveyor-belt system for handling gypsum rock; receipt of gypsum rock by self-unloading vessels and receipt of petroleum products; owned by Gold Bond Building Products, division of National Gypsum Co. and operated by National Gypsum Co., and Northeast Petroleum Corp. of New Hampshire.

(321) **Mobil Oil Corp., Portsmouth Terminal Wharf:** about 1.75 miles above the Memorial Highway Bridge; offshore wharf; 250 feet with dolphins; 37 feet alongside; deck height, 10 feet; receipt of petroleum products; owned by Public Service Co. of New Hampshire and operated by Mobil Oil Corp.

(322) **C. H. Sprague and Son Co. Wharf:** immediately northward of Mobil Oil Corp. Wharf; 405-foot offshore wharf, 700 feet with dolphins; 37 feet alongside; deck height, 11 feet; water connections; receipt of coal and fuel oil; owned by Public Service Co. of New Hampshire and operated by C. H. Sprague and Son Co.

(323) **Simplex Wire and Cable Co. Wharf**: about 2.3 miles above the Memorial Highway Bridge; 130-foot offshore wharf, 690 feet with dolphins; 30 feet alongside; deck height, 15 feet; special equipment for loading cable; water connections; receipt and shipment of wire and submarine cable; owned and operated by Simplex Wire and Cable Co.

(324) **Defense Fuel Support Point, Newington Dock:** about 2.8 miles above the Memorial Highway Bridge; 344-foot offshore wharf; 32 feet alongside; deck height, 15 feet; occasional receipt and shipment of petroleum products; owned by U.S. Government, Department of Defense Logistics Agency and operated by New England Tank Industries of New Hampshire, Inc.

(325) **Storage Tank Development Corp. Dock:** about 2.9 miles above the Memorial Highway Bridge; 250-foot offshore wharf, 700 feet with dolphins; 38 feet alongside; deck height, 14 feet; pipelines extend to storage tanks, 900,000-barrel capacity; receipt and shipment of petroleum products and receipt of asphalt and LPG; owned and operated by Storage Tank Development Corp.

(326) **Sprague Energy Newington Terminal Wharf:** about 3.5 miles above Memorial Highway Bridge; 225-foot offshore wharf; 780 feet with dolphins; 35 feet alongside; deck height, 14 feet; receipt and shipment of petroleum products, asphalt, tallow and caustic soda; owned and operated by C. H. Sprague & Son Co.

made. The marina monitors VHF-FM channels 16, 68 and 71. A charter fishing boat operates from the marina in summer.

(340) A narrow thoroughfare, partially dredged and marked by buoys, connects the northwestern end of Little Harbor with Portsmouth Harbor. The dredged section of the thoroughfare extends from just below the highway bridge across Little Harbor to a point about 0.8 mile above the bridge. Above this point, the thoroughfare leads between Shapleigh Island and Goat Island into Portsmouth Harbor. (See Notice to Mariners and the most recent chart editions for controlling depths.) The thoroughfare has a number of private float landings. A highway bridge with a 48-foot fixed span and a clearance of 14 feet crosses the thoroughfare between Shapleigh Island and Goat Island.

(341) Portsmouth Harbor can also be reached through another part of the thoroughfare that leads westward of **Shapleigh Island** and **Peirce Island** from above the dredged section. Two fixed highway bridges cross it. State Route 1B highway bridge from Shapleigh Island to **Frame Point** has a clearance of 10 feet. The other bridge from Pierce Island to the Portsmouth mainland has a clearance of 16 feet. Depths through this part of the thoroughfare are about 1 foot. A bare spot and a dangerous rock, which uncovers, are in midchannel about 0.3 mile and 0.2 mile southward of the first bridge, respectively; the chart is the guide. The entrance to the thoroughfare from Portsmouth Harbor is marked by buoys.

(342) **Sagamore Creek** empties into Little Harbor from the westward, about 0.2 mile above the highway bridge across the harbor. The creek is entered by a marked dredged channel that leads to a highway bridge 0.8 mile above the entrance; an anchorage basin is about 0.3 mile above the entrance. (See Notice to Mariners and the latest edition of the chart for controlling depths.) The creek has considerable small-craft activity.

(343) A marina is on the south side of Sagamore Creek, about 0.5 mile above the mouth. Depths of 3 to 6 feet are alongside the floats. Berths with electricity, gasoline, guest moorings and a small-craft launching ramp are available. A 10-ton and a 25-ton mobile hoist at the marina can handle craft up to 55 feet in length for hull and engine repairs and open and covered winter storage. Ice, provisions and marine supplies can be obtained. Party fishing boats operate from the marina daily in the summer. A restaurant is on a pier close eastward.

(344) The fixed highway bridge crossing the creek 0.8 mile above the entrance has a clearance of 11 feet. An overhead power cable with a reported clearance of 16 feet crosses the creek about 750 yards above the bridge. There are several private landings on the creek.

(345) **Pepperrell Cove** is on the eastern side of the harbor, northeastward of Portsmouth Harbor Light, and on the north side of Fishing Island, which is grassy. The cove is subject to shoaling and has depths of about 7 to 11 feet. It is mainly used by fishing vessels, yachts and small craft. An anchorage area is in the cove and a buoy northwestward of Fishing Island marks the entrance to the cove.

(346) **Kittery Point**, a village on the north side of the cove, has a public wharf and float landings with 12 feet reported alongside. Gasoline and water are available at the float, and ice, provisions and marine supplies are available at the wharf. A small-craft launching ramp is alongside the wharf. The Pepperrell Cove Yacht Club, also at the wharf, has a float landing on the east side of the wharf and maintains guest moorings.

(347) Moorings in the cove are under the supervision of the **harbormaster**, who can be found at the landing or contacted through the yacht club, market or local police.

(348) **Chauncey Creek**, which empties into the east side of Pepperrell Cove, has its entrance between Gooseberry Island and **Phillips Island** and extends about 1.2 miles eastward between Gerrish Island and the mainland. The creek is crossed by an overhead power cable with a reported clearance of 40 feet and a fixed bridge. There is considerable small-craft activity in the creek, which dries in its upper half.

(349) **Clarks Island**, close southeastward of Seavey Island, is joined with Seavey Island by a rock-fill causeway. The island is marked on its south side by a light. The cove is a restricted area. (See **33 CFR 334.50**, chapter 2, for limits and regulations.)

(350) **Hick Rocks**, a drying ledge with sections that uncover 11 and 7 feet, extends 350 yards from the southwest end of Kittery Point and is marked by a daybeacon on the ledge and by a buoy at its southern end.

(351) **Back Channel**, with its eastern entrance between Clarks Island and Hick Rocks, extends westward between Seavey Island and the Kittery mainland. It rejoins Piscataqua River westward of Badgers Island. The easterly half of the channel is marked by buoys. There are landings for small craft and several wharves with depths of 8 to 9 feet that are no longer used commercially with the exception of some fishing. A town wharf and float landing are about 125 yards westward of the westernmost bridge to Seavey Island. Back Channel has several dangers and is used principally by small craft and fishermen.

(352) **Spruce Creek** empties into the north side of Portsmouth Harbor at the eastern end of Back Channel. The creek has a narrow unmarked channel with a least depth of 12 feet for about 1.2 miles above the entrance and lesser depths shoaling gradually to 1 foot or less to a point about 0.8 mile farther upstream. The creek dries out about 0.2 mile below the dam about 2 miles above the entrance at the fixed highway bridge of the main coastal highway, U.S. Route 1. Extensive mudflats border the channel for most of its length.

(353) Just above the entrance, State Route 103 highway bridge, a fixed span with a clearance of 6.8 feet, crosses the creek and joins Kittery Point with Kittery. About 0.2 mile above this bridge, the remains of an old railway trestle cross the creek; some of the trestle and its piling have been removed from the channel; horizontal clearance

at the bridge is 24 feet. The creek has private landings but no services.

(354)

Small-craft facilities in Portsmouth Harbor

(355) Portsmouth Yacht Club is on the north side of New Castle Island close westward of Salamander Point. Reported depths of 9 feet are at its float landings at which gasoline, diesel fuel, water, ice and electricity are available. Guest moorings are maintained by the club, and other moorings in the special small-vessel anchorage are available for hire.

(356) A boatyard in the cove westward of the club has a marine railway that can haul out craft up to 30 feet in length for repairs or winter storage. The **harbormaster** for Portsmouth and New Castle can be reached through the yacht club or local police.

(357) Prescott Park Wharf is a public facility on the south bank of Piscataqua River, about 100 yards eastward of the Memorial Highway Bridge. Depths of 5 to 15 feet are reported alongside the float landings. Berthing for periods not to exceed 24 hours is available to small craft.

(358) There is a boat repair and storage yard in Kittery at the eastern end of Back Channel, northeastward of **Jamaica Island**. Its marine railway can haul out craft up to 60 feet long or 80 tons for hull and engine repairs or dry open or covered storage. Water, ice, provisions and most marine supplies can be obtained. Another yard with a machine shop is on the south side of Badgers Island west of the bridge. Water is available at its 100-foot pier, which has a depth of 11 feet reported alongside. Two marine railways can handle craft up to 65 feet in length for repairs or storage. The yard maintains guest moorings and permits overnight berthing. Provisions, electricity, diesel fuel by truck and most marine supplies can be provided.

(359) The Pepperrell Cove Yacht Club and the other facilities in Pepperrell Cove, Chauncey Creek and Sagamore Creek were covered in the description of those places. The small-craft facilities on Piscataqua River above Portsmouth are covered in geographic sequence with the description of the river that follows.

(360)

Piscataqua River to Exeter

(361) The **Piscataqua River**, above Portsmouth, forms the approach to Salmon Falls, Cocheco, Bellamy, Oyster, Lamprey and Squamscott Rivers. It is also the approach to the towns of Newington, Durham, Newmarket and Exeter and the city of Dover; all have rail freight service.

(362) The river has ample depth for large vessels for about 3.5 miles above the second lift bridge at Portsmouth to its confluence with its western branch at the fork at Dover Point. Most of the dangers in this section of the river are marked.

(363) The main river continues northward for 3.5 miles to the confluence of the Salmon Falls and Cocheco Rivers, both of which are described later.

(364) The Piscataqua River is buoyed to a point about 2.5 miles above Dover Point, and its western branch in Little Bay is marked for about 4.8 miles above Dover Point to a point in Great Bay, about 1 mile above Adams Point in Furber Strait. The western branch, Little and Great Bays and their tributaries are also described later in the text.

(365) The channels in all the tributary rivers are narrow, crooked, shoal at the heads and unmarked; local knowledge is necessary to navigate them.

(366) Some of the buoys in the river are reported to tow under sometimes in the strong currents, and, in particular, Buoys 13 and 16, which mark extensive shoals extending from the west and east banks, respectively, in the vicinity of Dover Point. A number of wooden pile dolphins marking the southern and western edges of the shoal extending from the east bank are covered at high water and reported to be dangerous to small craft.

(367)

Current

(368) General navigation throughout the entire length of the Piscataqua River system is severely hampered by rapid tidal currents. The velocities of these currents differ at various locations because of the irregularities in the width and depth of the river and its tributaries.

(369) The maximum average velocity in the river occurs off Nobles Island and off Dover Point at the entrance to Little Bay, and amounts to over 4 knots on the ebb. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov*. for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(370) The irregularities of width and depth plus the abrupt directional changes of course result in changes in the direction of the currents which at some locations do not coincide with the direction of the channel and cause hazardous crosscurrents.

(371) As a result of the combination of rapid tidal currents and hazardous crosscurrents, navigation of deep-draft vessels is limited to the 3-hour period from 1.5 hours before to 1.5 hours after slack water.

(372) The harbor pilots indicate that deep-draft vessels proceeding to the wharves above the lift bridges usually require more than one tug.

(373) Pilots and tugs can be obtained at Portsmouth. Traffic above Dover Point is confined to yachts, fishing boats and other small craft.

(374) **Spinney Creek**, about 0.1 mile above the I-95 bridge, is crossed by a causeway dam, with culvert, about 300 yards above its entrance. The cove thus formed, marked on the south side of the entrance by a lighted buoy, is a snug haven for small craft out of the strong currents of the river.

(375) The east bank has several private float landings. A boatyard and marina on the northwest bank of the cove has a marine railway that can haul out craft up to 60 feet in length for hull and engine repairs, or dry open or

wet winter storage. Gasoline, electricity and water are available at the floats which have 12 to 25 feet reported alongside. Diesel fuel can be obtained by truck. The pier has a snack bar, and ice, provisions and some marine supplies can be obtained. There is good anchorage in the cove in up to 25 feet, soft mud bottom. The yard has a small-craft launching ramp.

(376) On the west bank of the river, about 0.7 mile westward of the entrance to Spinney Creek, are two wharves. The lower one is the Mobil Oil Co. Wharf, and the upper one is the C. H. Sprague and Son Co. Wharf. These wharves were described earlier in this chapter under Wharves, Portsmouth Harbor.

(377)

Caution

(378) Mariners are advised to exercise caution when approaching these wharves as strong currents tend to sweep toward them. Also, the channel at this point may be reduced in width when large tankers drawing up to 35 feet are alongside these wharves.

(379) All vessels except the smaller tankers usually have the assistance of more than one tug when maneuvering the area.

(380) Vessels should exercise caution and pass this area with very little headway to avoid interference with or damage to the moored vessels or installations when unloading operations are in progress.

(381) An overhead power cable with a clearance of 165 feet crosses the river about 0.8 mile west-northwestward of the entrance to Spinney Creek.

(382) The Simplex Wire and Cable Co. Wharf, about 0.5 mile upstream of the C. H. Sprague and Son Co. Wharf, and the other deepwater wharves farther upstream were described earlier in this chapter under Wharves, Portsmouth Harbor.

(383) Prominent on this section of the river are the elevated tanks at the cable and gypsum plants, the coal transporter on the C. H. Sprague and Son Co. Wharf, the powerplant and its lighted stack, 0.4 mile west-northwest of the Sprague Wharf, and the General Sullivan Bridge at Dover Point.

(384) From **Dover Point** the river extends 3.5 miles to the confluence of Salmon Falls and Cocheco Rivers.

(385) On the east side of Dover Point, **Hilton State Park** has a pier, float landing, gravel-surfaced ramp for launching small craft from trailers, special parking facilities for cars and boat trailers and picnic areas. Water is available at the float, and restaurants, lodging and telephones are nearby.

(386) About 1.9 miles northward of Dover Point, on the west bank, is a boatyard and marina with space for transients. A marine travel lift can handle craft to 35 tons and 65 feet in length. Both open and indoor winter storage is available. Marine supplies, professional marine services, a fuel dock and restaurant are also available.

(387) **Sturgeon Creek**, on the east bank about 2 miles north of Dover Point, dries out at low water and is foul. Small craft have been known to moor in the narrow crooked channel. There are some private landings on the creek but no service facilities. A fixed bridge crosses the creek about 0.5 mile from the entrance.

(388) Piscataqua River is buoyed to about 2.5 miles north of Dover Point and has a fairly deep and clear channel for 1.8 miles in midriver. Above that point the river is unmarked and shoals gradually. About 3.2 miles north of Dover Point, overhead power cables crossing the river have a clearance of 65 feet.

(389) About 4 miles above Dover Point, Piscataqua River divides at a confluence known locally as **Three Rivers**, the north fork continuing northward as Salmon Falls River and the northwest fork as Cocheco River.

(390) **Salmon Falls River** is said to be navigable for small craft for about 3 miles to just below **South Berwick**, Maine. The channel is narrow, crooked and unmarked. In 1970, no small-craft activity was observed on the river.

(391) **Cocheco River** has a crooked channel from Piscataqua River to the head of navigation at a dam at the city of **Dover**, about 10 miles above Portsmouth. In 2023, the controlling depth was 3.1 feet to the head of the project at Dover; mariners are advised to consult local knowledge for channel conditions. The channel is privately marked with stakes.

(392) There is no commercial traffic on the river, but there is small-craft activity. A marina is on the north bank of the river, about 0.5 mile below the dam; hull and outboard engine repairs can be made; and gasoline, water, ice and some marine supplies are available. Depths of 6 feet are reported alongside the marina's float. Meals and lodgings are available nearby.

(393) **Little Bay** extends about 1.7 miles westward from its confluence with the main river, as far as **Fox Point**. It then trends southward to a junction off Adams Point in **Furber Strait** with **Great Bay**, the upper section of the western branch, about 3.8 miles above the General Sullivan Bridge.

(394) Most of the important dangers in Little and Great Bays are marked, and a buoyed channel can be followed from the mouth to a point in Great Bay about 0.35 mile above Furber Strait.

(395) Little Bay is deep and generally clear in the middle as far as **Goat Island**, but there are several unmarked shoal spots up to that point, and the edges are shoal with drying flats extending 200 to 300 yards offshore in places.

(396) Just inside the entrance to Little Bay on the west side of Dover Point, there is a marina where gasoline, water, storage facilities, marine supplies, a small-craft launching ramp and a 1½-ton forklift are available. Engine repairs can be made.

(397) A large marina, protected on its westerly side by a stone breakwater, is on the south bank of Little Bay, about 0.4 mile westward of the General Sullivan Bridge. Depths of 9 feet are reported alongside the floats. Berths with electricity, gasoline, diesel fuel, ice, water, marine supplies, a small-craft launching ramp, storage facilities,

restaurant and professional marine services are available. A 35-ton travel lift handles vessels to 65 feet.

(398) **Bellamy River** flows into Little Bay from the north. The river has a reported depth of less than 4 feet in a narrow, crooked and unmarked channel for about 1.4 miles above the Scammel Bridge across the mouth. Local knowledge is necessary to keep in the narrow unmarked channel, which is seldom used except by small craft.

(399) **Oyster River**, which flows into Little Bay westward of Fox Point, has a narrow, crooked and unmarked channel, bare in places at low water, to the village of Durham, 8.2 miles above Portsmouth.

(400) **Durham**, site of the University of New Hampshire, has many historical colonial connections. There are several private landings, including the University of New Hampshire Sailing Club, but no service facilities. Local knowledge of the river is essential to its passage.

(401) **Great Bay**, a large expanse mostly of mudflats about 2 miles long and 3 miles wide, is the upper section of the western branch of the Piscataqua River. Into it flow the Lamprey and Squamscott Rivers. Deep water extends up the middle of the bay for about 1 mile above **Adams Point** in Furber Strait.

(402) From that point a crooked, unmarked and somewhat foul channel leads to the mouths of the two rivers. Some small-craft activity was noted about the shores of the bay in 1970, but there were no service facilities.

(403) The University of New Hampshire's Jackson Estuarine Laboratory is on Adams Point. The two-story red brick laboratory building is prominent. The float landing at the facility has a depth of 6 feet reported alongside but no services. A rock, covered 3 feet, about 70 yards east of the landing, should be avoided.

(404) A public small-craft launching ramp is about 0.3 mile northward of Adams Point.

(405) The **Great Bay National Estuarine Research Reserve**, a Marine Protected Area (MPA), includes the waters of Great Bay and a portion of Little Bay.

(406) **Lamprey River** has a depth of about 2 feet in a narrow, crooked and privately marked channel to the village of **Newmarket**, 12 miles above Portsmouth. Small craft navigate the river, and local knowledge is necessary to its passage. Much of the river is reported to dry at low water, but there is always a narrow channel in which small craft can, and do, get through.

(407) There is a marina and boatyard on the west bank just below the dam and mill that straddle the river at the village. Depths of 8 feet are reported alongside the floats; gasoline and water are available. A 3½-ton mobile hoist can handle craft up to 30 feet for hull and engine repairs. Boats up to 30 feet can be built. Provisions and other essentials can be obtained in the village.

(408) There is room and depth for small craft to anchor off the marina.

(409) An overhead power cable crossing the river at the Lower Narrows has a clearance of 54 feet.

(410) **Squamscott River**, which flows into the western end of the head of Great Bay, had a depth of 2.5 feet to **Oxbow Cut** in 2006. From there to the town of Exeter, about 16.5 miles above Portsmouth, the channel is reported to dry in places. Local knowledge is advised to navigate the river to the head of navigation at the dam at Exeter.

(411) **Exeter** is the site of Phillips Exeter Academy and a town of antiquity and colonial historical importance. The buildings of the academy and public buildings of the town are impressive. There is a stone launching ramp for small boats at the town. During the spring, summer and fall, the river from the launching ramp to the Route 101 Bridge is used extensively by the academy rowing team. Caution should be exercised while navigating in this area. A ramp for launching small craft from trailers is at the northeast side of the Route 108 Bridge.

(412)

Odiornes Point to Lunging Island

(413) From Portsmouth Harbor entrance for 5 miles to Rye Ledge, the coast has a general southwesterly trend with no marked indentations. It presents the appearance of a succession of sand beaches separated by ledges extending out about 0.5 mile with occasional hotels and many summer homes back of the high-water line.

(414) **Odiornes Point** (43°02.5'N., 70°42.8'W.), is about 0.8 mile south of Jaffrey Point on New Castle Island. The point is part of Odiornes Point New Hampshire State Park. A launching ramp is on the Little Harbor side of the park. About 0.7 mile southward of Odiornes Point is a conspicuous round concrete observation tower. This is an outstanding landmark for vessels approaching Portsmouth or Little Harbors from the southward.

(415) **High Rock**, covered 2 feet, and **Pulpit Rock** and **Seal Rocks**, which uncover 6 and 3 feet, respectively, are part of a foul area extending about 0.4 mile offshore southward of Odiornes Point. They are unmarked.

(416) Cruising small craft approaching Little Harbor or Portsmouth from the southward, when passing inside Gunboat Shoal, should keep at least 0.7 mile offshore in order to avoid this area, before coming up to Portsmouth Harbor Channel Range.

(417) **Concord Point** is about 3 miles southwestward of Whaleback Light. **Foss Ledges**, which uncover 3 feet, extend 0.5 mile offshore from the point and are marked by a buoy at the outer end.

(418) **Rye Harbor**, 4.2 miles southwestward of Whaleback Light, is a small cove used by pleasure and fishing boats. A stone breakwater extending southward from **Ragged Neck Point** is marked at the end by a light. Another breakwater extends northeastward from the point at the south side of the entrance to Rye Harbor. These breakwaters are about 6 feet above high water. A rocky ledge, covered 3½ feet, extends to within 10 feet of the entrance channel on the south side and is marked by buoys. A lighted whistle buoy marks the approach about 0.75 mile southeastward of the harbor entrance. A dredged channel leads through the breakwaters to anchorage basins on the north and south

(429)

sides of the channel and state anchorage at the western limit.

(419) About 500 yards westward of the north breakwater, a stone jetty extends about 150 yards in a southwesterly direction from the north side of the harbor. Rye State Park includes Ragged Neck, the north side of the harbor, and the head that has been diked and backfilled to form a public landing. Two state piers, the southerly one for commercial vessels and the northerly for pleasure craft, are at the landing. There are reported depths of 7 to 8 feet at the piers. The northerly pier has float landings with over 200 feet of berthing space. Both piers and floats are floodlighted at night, and water and electricity are available. The landing has a parking area.

(420) Water is available at the floats of a service wharf on the south side of the harbor; depths of 6 feet are reported alongside the floats. Party fishing boats and a charter fishing boat are available for hire at the wharf.

(421) The **harbormaster,** who can be contacted by calling 603–431–1779 or 603–436–8500, controls and assigns the moorings in the harbor. Occasionally some guest moorings become available. The harbor is small and congested but safe for strangers attempting to enter during heavy easterly weather.

(422) **Lockes Neck (Straw Point)**, 0.5 mile south of Rye Harbor, is marked by a prominent white flagpole. **Rye Ledge** is 1.2 miles southward of Lockes Neck. The ledge, partly bare at high water, extends 0.4 mile from shore and is unmarked. The buildings and control tower of an Air Force installation on shore northwestward of the ledge are very conspicuous.

(423) **Isles of Shoals**, about 5 to 6 miles offshore and about the same distance southeastward of Portsmouth Harbor entrance, consist of a group of eight main islands and a number of islets, rocks and ledges. They extend about 3 miles in a northeast-southwest direction and on a clear day can be seen for 10 miles. The islands first drew attention in 1614 when Captain John Smith on one of his voyages of exploration northward from the Jamestown Colony drew a chart of the New England Coast and named the islands the Smith Isles. However, the group had been known as the Isles of Shoals sometime before his arrival.

(424) Earlier, fishermen, mostly from England, had found it profitable to sail from home in early spring and return in the fall with rich cargoes of fish caught and cured at the isles. The isles are now frequented by fishermen and summer visitors, but only a few winter residents inhabit the isles in winter. Three of the islands, Star, Lunging and White, are within the political jurisdiction of the town of Rye, New Hampshire; the others, Cedar, Smuttynose, Malaga, Appledore and Duck are in the town of Kittery, Maine. The state boundary line passes through the center of Gosport Harbor and between Star and Cedar Islands.

(425) **Gosport Harbor**, formed by breakwaters joining Star, Cedar, Smuttynose and Malaga Islands of the group, is used as an anchorage by local fishermen and

yachts and sometimes by small coasting vessels seeking shelter. It offers protection from all but westerly winds; however, the bottom is reported to be rocky and foul and caution should be exercised in strong winds. A diesel-powered ferry carries passengers, mail and supplies from Portsmouth to the 200-foot stone wharf on the north side of Star Island.

(426)

Prominent features

(427) **Isles of Shoals Light** (42°58'02"N., 70°37'24"W.), 82 feet above the water, is shown from a 58-foot white conical tower with covered way to a dwelling on the south end of **White Island**, the southernmost island of the group. A mariner radio-activated sound signal at the light is initiated by keying the microphone five times on VHF-FM channel 83A. The light covers the entire horizon but is obscured by the houses on the island to the northward of it.

(428) The more prominent landmarks are the large white hotel and other buildings around it and a flagpole on Star Island; a former Coast Guard station with cupola, an old tall concrete observation tower, and five old abandoned stone houses on Appledore Island; and a house on Lunging Island.

(430)

Channels

(431) Several channels between the islands lead into Gosport Harbor and are mostly deep and clear. The narrow channel between Appledore and Smuttynose Islands has a depth of 20 feet, though there is an unmarked rock, covered 6 feet, in its eastern approach. A fairway bell buoy marks the western approach to Gosport Harbor. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A.

(432)

Dangers

(433) Ledges surround most of the islands, but most of the detached shoals are marked. **Cedar Island Ledge**, 0.4 mile southeastward of Cedar Island, uncovers 4 feet and is marked by a buoy. It should be given a berth of at least 0.5 mile.

(434) **Anderson Ledge**, which uncovers 4 feet and is marked by a buoy off its south side, is about 1 mile east-southeastward of Isles of Shoals Light. The ledge, the outermost danger, is about 200 yards in diameter and has deep water around it.

(435) **Halfway Rocks**, a ledge that uncovers 2 feet, marked on its west side by a buoy, is in midchannel between Star and Lunging Islands.

(436) Bare **Square Rock** and a ledge that uncovers 3 feet, both unmarked, are off the west shore of Lunging Island.

(437) **Appledore Ledge**, covered 7 feet and marked on its west side by a buoy, is off the northwest end of Appledore Island. An unmarked 22-foot spot is about 500 yards off the north end of the island, and a rock covered 6 feet is off the southeast shore.

(438) **Southwest Ledge** and **Jimmies Ledge**, both drying ledges, and bare **Mingo Rock** and **Eastern Rocks** are off the 18-foot-high bare **Duck Island.** A **danger zone** of a naval target area is centered on **Shag Rock** off the east side of the island. (See **33 CFR 334.40**, chapter 2, for limits and regulations.)

(439) All dangers surrounding Isles of Shoals can be avoided by passing 0.5 mile to westward and 1.5 miles to eastward.

(440) Trawlers and other vessels conducting bottom operations within a 6.7-mile radius seaward of Isles of Shoals Light should exercise caution because of Jet Assist Take-Off racks and associated debris on the ocean floor.

(441) **Star Island**, the most important of the group, is the site of many religious conventions and seminars held in the hotel. There are many points of historical interest on the island. An old stone church, a graveyard, a 40-foot memorial obelisk, and a monument to Captain John Smith are near the south central part of the island. In clear weather Boon Island, Mount Agamenticus on the mainland and even Cape Ann, 20 miles to the southward, can be seen from the island.

(442) **Appledore Island** is the largest of the group; Cornell University's Shoals Marine Laboratory maintains a small wharf on the west side of the island. A landing can also be made in Babbs Cove on the west side at the old Coast Guard boathouse. The laboratory maintains a picnic ground; fires are prohibited.

(443) **Cedar Island** with four houses on it and **Smuttynose Island** with three are northward of Star Island. **Haley Cove**, formed by a stone breakwater joining Smuttynose Island to **Malaga Island**, is occasionally used by recreational boaters in summer. Boats with over 1-foot draft should not enter Haley Cove because of reported uncharted rocks in the entrance channel. The boats lie aground at low water. There are no piers or moorings.

(444) **Lunging Island**, a bare low rocky islet about 0.5 mile west of Star Island, has a refuge hut on it.

(445)

Fox Hill Point to Breaking Rocks

(446) From **Fox Hill Point** (42°57.9'N., 70°46.2'W.) to Merrimack River entrance, there are about 9 miles of sandy beaches, several rocky headlands and offlying reefs and ledges up to 1 mile from shore. A large house with three chimneys on Fox Hill Point is very prominent. Summer resorts line the beaches, and hotels and prominent summer homes are on the headlands. Salt marshes between the beaches and the coastal ridge about 2 to 2.5 miles westward are drained by small rivers, most of which flow into the inlet at Hampton Harbor.

(447) **Little Boars Head** is a yellow bluff 7 miles southwestward of Whaleback Light. A summer resort of the same name extends over 0.5 mile northeastward from the bluff; a large mansion on the head is conspicuous. A

use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A.

(456)

Anchorages

(457) Anchorages are available in the basins or in the narrow channels of the Hampton and Blackwater Rivers and other rivers and creeks northward and southward of the inlet.

(459) In 1984, the navigable entrance to Blackwater River was reported to have shifted about 220 yards north from its currently charted position.

(460)

Dangers

(461) Extensive rocky ledges obstruct the approaches to the entrance to the inlet. **Hampton Shoal Ledge**, covered 19 feet, about 2.8 miles eastward of the entrance, is unmarked.

(462) About 0.5 mile off the entrance is an extensive area of drying and covered rocky ledges consisting of **Old Cellar Rock, Inner Sunk Rocks, Outer Sunk Rocks** and other rocks between Inner and Outer Sunk Rocks; a buoy is northeastward of the area.

(463) State Route 1A Highway bridge crosses the inner end of the inlet. It has a 40-foot bascule span with a clearance of 18 feet. (See **33 CFR 117.1** through **117.59** and **117.697**, chapter 2, for drawbridge regulations.) It is reported that the flood velocity under the bridge is 1.5 to 2.2 knots and the ebb velocity 2 to 3.2 knots.

(464)

Routes

(465) For craft entering or leaving, the chart should be the guide; follow the aids with due attention to existing conditions. In heavy weather, the harbor may be closed because of heavy breakers across the entrance.

(466)

Small-craft facilities

(467) Several party fishing boats operate from the float landing of the state park inside the harbor, close northward of the bridge, and from a sport fishing pier and a service landing in the cove close to the northwestward of the park float. Water is available at the float, and a restaurant is on the pier.

(468) A marina is in a privately dredged basin protected by wooden jetties, about 0.4 mile northward of the bridge. There are slips with floats for 135 boats up to 60 feet in length with reported depths of 5 to 7 feet along side. Water and electricity are available at all of the berths. The marina has a 25-ton mobile hoist to haul out craft for engine or hull repairs and dry or open winter storage. The marina may be contacted on VHF-FM channels 16 or 10. Ice, a pump-out station, provisions and marine supplies are available. Motels, hotels, restaurants, markets and many other conveniences are nearby. There is a small-craft launching ramp north of the basin.

(469) A state park is across the road. Motels, restaurants, lodging, markets and other conveniences are available at the village at Hampton Beach.

(470) Taxi and bus services are available.

(471) There are a town wharf and two service wharves with 3 feet reported alongside at Seabrook at the southern end of the harbor from which a number of party and charter fishing boats operate. Water is available at the floats of the service wharves. A snack bar and refreshments are on the wharves, and a restaurant is nearby. A narrow dredged channel leads southward to it from the inlet. Numerous small craft are usually found moored in the channel as well as barges and workboats used in the construction of the Seabrook Nuclear Power Station, Public Service Company of New Hampshire.

(472) From Hampton Harbor, **Seabrook Beach** and **Salisbury Beach** extend 4.3 miles in a southerly direction to the entrance of Merrimack River. Unmarked ledges and foul and broken ground extend up to 0.8 mile offshore and among them a number of rocks awash, including **Thomas Rock** and **Round Rock**. **Breaking Rocks** a ledge covered 3 feet, is 0.7 mile offshore and nearly 2 miles south of Hampton River. It is marked at its northeast end by a buoy.

(473)

Merrimack River

(474) **Merrimack River** is the largest river in the eastern part of Massachusetts. It is the approach to the cities of Newburyport and Haverhill and to the towns of Amesbury, Merrimacport, Groveland and Bradford. The river is used by vessels of 6-foot draft at high water up to Haverhill and about 12-foot draft at high water to Newburyport. The head of navigation is at the dam just above Broadway Bridge in Lawrence, 25.7 miles above the mouth. The river is seldom entered for refuge and has virtually no commercial traffic.

(475) The shifting bar at the entrance is usually dangerous to cross in heavy weather. The whole entrance breaks in easterly gales. A lighted fairway whistle buoy, about 1 mile off the jetties, marks the approach.

(476) **Newburyport** is a city on the south bank of the river, 3 miles above the entrance. **Merrimack River Coast Guard Station** is on the south side of the river west of the American Yacht Club.

(477)

Prominent features

(478) In the approach to the entrance of Merrimack River, the most important objects are the elevated water tank 1.5 miles north of the entrance and the large bathing pavilion and bath houses of the state park near the southern end of Salisbury Beach, just north of the entrance. A large water tank, three-bladed wind turbine, standpipe, the bridges, church spires, several stacks and a cupola, all in Newburyport, are conspicuous.

(479) **Newburyport Harbor Light** (42°48'55"N., 70°49'08"W.), 50 feet above the water, is shown from a

(489)

Structures across Merrimack River

Name•Description•Type	Location	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
U.S. Route 1 Bridge (bascule)	42°48'57"N., 70°52'22"W.	35	50 (open) 100 (closed)	Notes 1, 2 and 3
Massachusetts Bay Transportation Authority Bridge (swing)	42°48'57"N., 70°52'24"W.	69 (north draw) 64 (south draw)	13	Note 1 Channel is through the north draw
Essex Merrimac Bridge (fxed)	42°50'01"N., 70°54'24"W.	220	28	Belleville to Deer Island
1st LT. Derek S. Hines Memorial Bridge (swing)	42°50'07"N., 70°54'26"W.	54	14	Note 1. Deer Island to Salisbury Point
John Greenleaf Whittier/I-95 Bridge (fxed)	42°50'06"N., 70°54'43"W.	465 (main) 271 (aux)	55 (main) 48 (aux)	
Overhead power cable	42°49'27"N., 70°58'29"W.		50	
Overhead power cable	42°48'44"N., 71°00'03"W.		50	
Main Street/Rocks Village Bridge (swing)	42°48'39"N., 71°00'01"W.	57 (east draw) 53 (west draw)	16	Note 1
Bates/Groveland-Haverhill Bridge (bascule)	42°45'50"N., 71°02'05"W.	70	16	Note 1
Brasilere Bridge (fxed)	42°46'26"N., 71°04'34"W.	75	25	
Overhead power cable	42°46'16"N., 71°05'11"W.		50	
MBTA Commuter Railroad Bridge (fxed)	42°46'16"N., 71°05'11"W.	121	31	Bridge is under construction
Comeau Bridge (fxed)	42°46'14"N., 71°05'15"W.	139	30	

Note 1 – See 33 CFR 117.1 through 117.59 and 117.605, chapter 2, for drawbridge regulations.
Note 2 – Bridgetender monitors VHF-FM channel 16 and works on channel 13; call sign WQA-806.
Note 3 – In the open position, the draws overhang the channel above a height of 55 feet.

white conical tower near the western end of **Plum Island Point**, the southern point of the entrance. The light is obscured in several sectors by shore structures.

(480)

Channels

(481) A dredged channel leads into Merrimack River between two jetties and upriver to the U.S. Route 1 Highway Bridge at Newburyport, about 3 miles above the mouth. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A. The river channel above Newburyport to Haverhill, about 18 miles above the mouth of the river, has not been maintained for several years. Depths can range from 1 to 6 feet in the channel at low water. An area with numerous obstructions and shoaling is in the channel between the bridge at Groveland and Haverhill. If navigation is necessary in this section of the river, mariners are urged to do so at high water. The river channel to Newburyport is marked by lighted and unlighted buoys. Buoys mark the channel above Newburyport to Haverhill.

(482) The jetties extend from both points at the entrance out to the bar and are difficult to see at high water, particularly at night and in periods of low visibility. About 240 yards of the outer end of the north jetty is submerged at high water.

(483)

Anchorages

(484) At Newburyport the usual and best anchorage is in the channel about 400 yards below the highway bridge, favoring the north side of the channel and keeping clear of the two charted cable areas. The current is reported to run strongest along the south shore here. The holding ground is good.

(485) The yacht club maintains guest moorings as do many of the service facilities and marinas. Numerous private moorings are maintained off Newburyport and in the upper river as far as Haverhill. They are under control of the **harbormasters** at Newburyport, Amesbury and Haverhill.

(486) Public floats are along the south side of the river at Newburyport, about 0.2 mile west of **Merrimack River Coast Guard Station**. In 1979, 8 feet was reported alongside the floats. Berthing is under the control of the Newburyport harbormaster.

(487)

Dangers

(488) Endangered North Atlantic right whales have been reported swimming in shallow waters off of Plum Island and Ipswich, MA.

(490)

Routes

(491) A lighted whistle buoy is about 1 mile outside the bar at the entrance to Merrimack River. The channel across the bar is marked by lights on the outer ends of the jetties, lighted and unlighted buoys; the chart should be the guide following the aids. Considerable chop is experienced on the bar with the wind against the tide.

(492) Small craft may enter when the sea is smooth and on a rising tide, following the buoys. The river cannot be entered during a heavy sea. The outer ends of the jetties are awash at high water.

(493) After the bar is crossed, the channel is well marked and easily followed to Newburyport.

(494) The channel between Newburyport and Haverhill is marked by buoys at the most difficult points but is narrow, crooked and leads close to rocks in places. Local knowledge is required to keep in it.

(495) The Coast Guard provided the following information to assist the mariner in crossing the bar when outbound from the Merrimack River.

(496) The bar area between the beach and Merrimack River Entrance Lighted Buoy 2, north and south of the jetties, is subject to breaking seas—particularly on an ebb tide with easterly winds. The ebb tide runs out of Merrimack River from 3 to 6 knots. Vessels should proceed slowly through the channel, evaluating the bar well inside of the two jetties. If the decision is made to cross, proceed all the way out beyond the breakers and do not attempt to turn around if the bar is breaking.

(497) The area southward of the outer 240 yards of the north jetty and the channel is a shoaling sand bar subject to constant change in depth. This area and a portion of the channel just south are extremely hazardous. Avoid crossing the sunken jetty or sandbar, and use caution in the channel to the south of it.

(498) Ocean swells meeting an outgoing tide in the river mouth result in breaking seas. The most dangerous period is from about 1 hour before low water and 1 hour after low water. Even on the calmest days the tidal conditions may be such that small boats will be endangered at this period. Mariners should learn the stages of the tide when local conditions are the most favorable for bar crossing.

(499) Due to the sandy nature of the river bottom, one can expect unannounced changes in the bar shoals depending upon prevailing winds and currents. These changing bars and shallow areas may not be marked on the charts.

(500)

Current

(501) Currents are strong in the river, and yachts sometimes drag when anchored off the American Yacht Club. Strangers should use a mooring, if available. Current predictions for the entrance and at Newburyport can be generated at the Tidal Current prediction service at *tidesandcurrents.noaa.gov*. Links to a user guide for this service can be found in chapter 1 of this book.

(502) **Freshets** occur in the spring but do not interfere with navigation, as a rule.

(503) **Ice** occasionally obstructs navigation below the bridge at Newburyport. Westerly winds carry the drift ice out to sea and, during their continuance, the flood current has no effect upon the local formation of drift ice. With the wind from any other direction, the flood current will prevent the drift ice from leaving the river.

(504) Above the Newburyport bridges the river is liable to be closed by ice from January to March.

(505)

Pilotage, Merrimack River

(506) Two pilots for the river reside in Haverhill telephone 617–372–3420 and 617–372–3745. Information on the river can be obtained from the local boatmen at Plum Island Point or any of the service facilities or marinas at Newburyport.

(507)

Towage

(508) There are no tugs at Newburyport, but there are three at Portsmouth.

(509)

Harbor regulations

(510) A **no-wake, headway-only speed limit** is enforced in the vicinity of boat docks along the Merrimack River.

(511) A hospital is at Newburyport.

(512)

Supplies

(513) Gasoline, diesel fuel, water, ice, provisions, bottled gas and marine supplies can be obtained.

(514)

Small-craft facilities

(515) The port has a number of small-craft facilities along the waterfront.

(516) Sewage pump-out is available at Cashman Park on the south bank west of the highway bridge and by contacting the Newburyport harbormaster.

(517) A town wharf and float landing are on the north bank east of the highway bridge. A municipal marina and launching ramp are on the south bank about 0.1 mile east of the highway bridge.

(518) The American Yacht Club at the east end of town has 14 feet alongside its float landing. Gasoline and water are available at the float. Guest moorings and club facilities are available to visiting yachtsmen. The North End Yacht Club, open to members only, is at the west end of town above the bridge.

(519)

Communications

(520) A Class I railroad and bus and truck lines serve the port; there is taxi service.

(521) **Amesbury** is a city on the **Powwow River**, 1 mile above its confluence with the Merrimack. Four highway bridges cross the river between the mouth and Amesbury. A 36-foot fixed span at the mouth has a clearance of 8 feet, twin 40-foot fixed spans 0.5 mile above the mouth have clearances of 12 feet, and a fixed span 0.6 mile above the mouth has a clearance of 8 feet. A railroad bridge at Amesbury has an 11-foot bascule span with a clearance of 4 feet. (See **33 CFR 117.1** through **117.59**, chapter 2, for drawbridge regulations.) An overhead power cable crossing the river 0.5 mile below the bascule bridge has a clearance of 30 feet.

(522) On the west side of the mouth of the Powwow River is a large marina and boatyard that has two marine railways. Craft up to 42 feet long or 25 tons can be handled for hull repairs or dry open or covered winter storage. Gasoline, diesel fuel, water and electricity are available at the float landings, which have a reported 12 feet alongside. Ice, provisions, bottled gas and marine supplies can be furnished. There is a launching ramp. Overnight berthing is permitted, and several guest moorings are maintained. Good restaurants, hotels, markets and stores are in Amesbury. Taxi service is available.

(523) The **harbormaster** can be contacted through the Amesbury Police Department.

(524) Sewage pump-out is available at a marina approximately 0.3 mile westward of the mouth of the Powwow River.

(525) About 0.7 mile westward of the Powwow, on the north bank, is another marina. Gasoline, water and electricity are available at the floats, which have a reported 10 feet alongside. A marine railway at the marina can haul out craft up to 50 feet in length for hull and engine repairs or dry covered or open winter storage. There is a gravel small-boat launching ramp and parking. Marine supplies and ice are available.

(526) **Merrimacport** is a village on the north bank of Merrimack River about 10 miles above the entrance. Two natural ramps for launching small craft from trailers and a float landing with 2 to 3 feet alongside are on the north bank at the town.

(527) **Groveland** is a town on the south bank of the river, 15 miles above the entrance.

(528) **Haverhill** is a city on the north bank, at the usual head of navigation of the Merrimack River, 18 miles above the entrance. The wharves are in disrepair. There has been no commerce by water for many years.

(529) There is a marina and boatyard at Riverside on the north bank 0.3 mile eastward of the Groveland highway bridge. The yard has two float landings with 9 feet alongside, a 20-ton crane, and a marine railway that can handle craft up to 200 tons or 140 feet long for hull or engine repairs or dry open winter storage.

(530) Diesel fuel and water are available at the floats. Ice, provisions, marine supplies and bottled gas can be obtained. Haverhill Riverside Airport with an 1,800-foot landing strip is adjacent to the marina; a seaplane, landplane and helicopter are available. The owner and manager of the marina is also the **harbormaster,** pilot for the river and chief of the Merrimack River Rescue Service. The service, which operates the police boats, an amphibious craft and a helicopter can be contacted directly, 617–372–3420, or through the Haverhill Police Department, 617–373–1212. There are two ramps at the facility, one of which is hard surfaced.

(531) Another marina and boatyard, about 0.7 mile below the bridge on the north bank, has two float landings with a reported 4 feet alongside. Gasoline, water and electricity are available at the floats. There is a hard-surfaced ramp and a 3½-ton crane. Hull and engine repairs can be made, and dry open or covered storage is available. Guest moorings are maintained.

(532) **Bradford**, a town on the south bank of the river, is connected by two highway bridges and a railroad bridge with Haverhill. The Haverhill (Crescent) Yacht Club, on the south bank east of the lower bridge, has 6 feet at its float landing. Guest moorings are maintained. Small craft anchor or secure to moorings off the club. Fuel, provisions and supplies can be obtained.

(533) At **Mitchells Falls**, about 2 miles above the upper highway (County) bridge at Haverhill, the river becomes foul and full of rocks, virtually impassable at low water, but at high water small craft are reported to navigate the river to the dam at Lawrence.

(534) **Plum Island River** forms a thoroughfare for small craft between Merrimack River, just inside its entrance, and Plum Island Sound. It is bare in places at low water and is said to have a depth of 7 feet at high water, but the deepest draft that is taken through at high water with local knowledge is reported to be about 6 feet. The unmarked channel is narrow and does not always lead in midchannel. Local knowledge is necessary for its navigation. It is crossed by a highway bridge that has a 40-foot bascule span with a clearance of 13 feet. (See **33 CFR 117.1** through **117.59** and **117.615**, chapter 2, for drawbridge regulations.) An overhead power cable with a reported clearance of 60 feet is just northward of the bridge.

(535) The approach to the north end of the thoroughfare is between the east side of **Woodbridge Island** and the west end of the breakwater, which uncovers about 3 feet.

(536) From Merrimack River entrance the seacoast, formed by **Plum Island**, is sand dunes and trends southward for about 7.5 miles to the entrance of Plum Island Sound and Ipswich River. There are many cottages in the town of Plum Island on the north end of the island at Merrimack River entrance and scattered cottages southward along the beach for about 0.5 mile. The remainder of the island southward to Ipswich Bay is a federal wildlife sanctuary for the most part.

(537) **Ipswich Bay** is the bight between the northern point of Cape Ann and the south end of Plum Island. Between these points it is about 6 miles wide and makes in about 3 miles. The bay is the approach to Plum Island Sound and to the Essex and Annisquam Rivers. It has depths of 20 to 70 feet, except in its southern and southwestern sides where the shore should be given a berth of a little over 1 mile to avoid the shoals off the river entrances. Several rocks covered 2 to 5 feet and one that uncovers 4 feet are in the southern part of the bay about 0.9 mile westward of Annisquam Harbor Light and about 0.3 to 0.5 mile offshore.

(538) **Ipswich Light** (42°41'07"N., 70°45'58"W.), 30 feet above the water, shown from a white skeleton tower with a red and white diamond-shaped daymark, is on Castle Neck at the south side of the entrance to Plum Island Sound. A seasonal lighted bell buoy 1.6 miles eastward of

(541)

the light marks the approach to Ipswich River and Plum Island Sound.

(539) The Crane mansion known as **The Castle**, on **Castle Hill**, is the most prominent landmark on this stretch of coast and can be seen for a great distance. The north side of **Steep Hill**, about 0.5 mile northwest of Ipswich Light, is a conspicuous bare rocky face.

(540) **Plum Island Sound**, the approach to several small rivers, is frequented by many small craft. The bar channel at the entrance to the sound is subject to continual changes. The entrance is marked by a seasonal lighted buoy. The buoys on the bar are not charted because they are frequently shifted in position. The buoys marking the channels across the bar and through the sound and rivers inside are seasonal.

(542) In 1979, local boatmen reported that with local knowledge 6 feet could be taken over the bar and through the entrance into Plum Island Sound, except in heavy easterly weather.

(543) **Bass Rock**, a stone ledge southward of Plum Island, is marked by a daybeacon. Shoaling extends from Plum Island to a point 200 yards southward of the daybeacon on Bass Rock, constricting the entrance channel at this point to a width of less than 100 yards. Rocks covered 4 feet are reported to extend 250 yards southwest of the daybeacon; caution is advised.

(544) A number of the buoys in Plum Island Sound are reported to tow under during the strength of tide, and too great reliance should not be placed on them as marking the best water. Local knowledge is recommended for strangers attempting passage through the sound for the first time.

(545) **Ipswich River**, emptying into the south end of Plum Island Sound from the westward, leads to the town of **Ipswich** about 2.5 miles above the entrance to the river at Little Neck. In 2024, the river was very shallow and bare in several places between Little Neck and the town landing at Ipswich.

(546) The launching ramp of the Ipswich Boat Club and two floats with 2 feet alongside are on the north bank at the town.

(547) The town of Ipswich is of great colonial antiquity and importance historically. It has railroad, bus and taxi services and markets.

(548) **Little Neck**, a summer settlement on a prominent hill on Plum Island Sound on the north side of the entrance to Ipswich River, has a landing on the west end of the neck, with 2 feet reported alongside its float. There are no services at the float.

(549) **Great Neck** is a distinctive headland on the west side of the south end of Plum Island Sound. It has two high hills, **North Ridge** and **Plover Hill**, that are very conspicuous. A tank on Plover Hill is very prominent.

(550) The Ipswich Bay Yacht Club is on the east side of North Ridge on the neck. Gasoline and water are available at the float landing, which has 4 to 8 feet reported

alongside. The club has a snack bar, ice and limited accommodations for visiting yachtsmen. Sewage pump-out, ice, provisions and marine supplies can be obtained from Ipswich.

(551) During the summer many yachts moor off the landing in 10 to 15 feet, sand and mud bottom. The club maintains moorings.

(552) **Rowley River**, which empties into Plum Island Sound at **Hog Island Point**, about 1 mile north of Great Neck, dries in many places and is marked, during the summer, by stakes that are topped with red or black cans. Several landings are on the river. A town landing and a yacht club are about 250 yards above the Boston and Maine trestle bridge; clearance at the bridge is 11 feet. Little water is reported alongside the town landing and yacht club, and no services are available. The railroad station is only a short distance from the town landing. The town of **Rowley** is about 0.5 mile from the station.

(553) **Parker River**, emptying into the north end of Plum Island Sound from westward, has a depth of about 4 feet in a very narrow channel to State Route 1A highway bridge at **Newbury Old Town**, 1.6 miles above the entrance. The bridge has a fixed span with a clearance of 12 feet. The town is principally a summer settlement.

(554) The channel in Parker River is difficult to follow. In 1979, local boatmen reported that 4 feet could be taken to Newbury Old Town with local knowledge.

(555) Numerous pleasure craft of all sizes frequent the river.

(556) There are two marinas on the south bank at the bridge. In 2002, the marina on the east side had a reported approach and alongside depth of 3 feet, and provides a one-ton lift and dry winter storage. It maintains guest moorings and has a snack bar nearby.

(557) The large marina on the west side of the bridge has guest moorings and berthage for 50 boats; overnight berthing is permitted. Electricity, gasoline, water, marine supplies, sewage pump-out and a small-craft launching ramp are available. A 14-ton mobile hoist is available and craft up to 45 feet in length can be hauled out for dry open or covered winter storage or hull or engine repair. The yard also builds craft up to 24 feet in length.

(558) A town wharf and a float landing with 2 feet reported alongside are on the north bank just eastward of the bridge. The Old Town Yacht and Country Club is on the south bank about 0.3 mile below the bridge. The depth alongside the club float is 5 feet.

(559) Above Newbury Old Town, the river is reported to be navigable for several miles but is seldom used. This section of the river is crossed by three bridges. A railroad bridge 2.6 miles above the entrance has a 41-foot fixed span with a clearance of 7 feet. The U.S. Route 1 bridge 4.3 miles above the entrance has two fixed openings; the southern opening has a 37-foot span with a clearance of 7 feet. The Middle Street Bridge about 5 miles above the entrance has a 41-foot fixed span with a clearance of 3 feet.

(560) The **Parker River National Wildlife Refuge**, a Marine Protected Area (MPA), includes the waters of Plum Island Sound, Plum Island River and Parker River.

(561)

Essex Bay

(562) **Essex Bay** and **Essex River** are about midway between Ipswich and Annisquam Harbor Lights. The entrance is through a shifting bar over which, with local knowledge, 5 feet can usually be carried. With onshore winds on an ebb tide, a heavy chop builds up and during heavy weather the bar is often impassable. Caution is always indicated, especially for smaller boats.

(563) The river is navigable for small craft to the town of **Essex**, about 5 miles above the entrance. Local fishermen and numerous pleasure craft use the river.

(564) The entrance is marked by a seasonal lighted bell buoy, and the bay channel is marked from the bar to about 2 miles above the entrance by a daybeacon and seasonal buoys. The bay channel is subject to continual change, and the buoys marking it are not charted because they are frequently shifted. Above **Conomo Point**, the town of Essex maintains seasonal midchannel spar buoys. The channel is narrow and difficult to follow. Mariners should obtain local knowledge before navigating the river.

(565) There are several small-craft facilities just below the bridge at Essex.

(566) Restaurants, lodging and motels are on or near the waterfront; the town has markets, a bank and taxi services.

(567) A private residential yacht club is at Conomo Point.

(568)

Annisquam River

(569) The **Annisquam River** and **Blynman Canal** form a thoroughfare leading from the eastern part of Ipswich Bay, northwest of Cape Ann, to Gloucester Harbor, on the south side of the cape.

(570) **Annisquam** is a village and summer resort on the east side of Annisquam River just inside its north end. **Lobster Cove**, on the southeast side of the town, is the scene of much small pleasure-boat activity during the summer.

(571)

COLREGS Demarcation Lines

(572) The lines established for the Annisquam River and Blynman Canal are described in **33 CFR 80.115**, chapter 2.

(573)

Prominent features

(574) **Annisquam Harbor Light** (42°39'43"N., 70°40'53"W.), 45 feet above the water, is shown from a white cylindrical tower with elevated walk to a dwelling on **Wigwam Point** at the east side at the northern entrance to Annisquam River. A red sector in the light from 180° to 217° covers the shoals on the eastern side of the

approach to the bar channel from the north. A lighted bell buoy marks the approach, and a sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A.

(575)

Local magnetic disturbance

(576) Differences of as much as 3° from the normal variation have been observed in the vicinity of Annisquam.

(577)

Channels

(578) A marked channel with dredged sections across the bar at the northern entrance to Annisquam River and in the river and Blynman Canal leads from Ipswich Bay to Western Harbor at the north end of Gloucester Harbor; the project depth is 8 feet. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A.

(579) This thoroughfare is narrow but is adequately marked by lights, daybeacons and buoys and is extensively used by small craft. Strangers should have no trouble getting through with a smooth sea and by the use of the chart. The bar at the northern entrance is difficult to cross in a heavy sea—the best time is on a rising tide. During strong tides and high vessel traffic, extreme caution is required while passing under Blynman Bridge. Reported in 2024, channel depths of less than 8 feet close North of the Blynman Bridge.

(580)

Anchorages

(581) Craft anchor in the coves, creeks or estuaries of the waterway or moor at the marinas. The entrance of **Lobster Cove**, near the north end of the waterway east of Annisquam, has been dredged as far as the bridge. In 1997, the entrance had depths of less than 1 foot in the south part, gradually deepening to over 5 feet at the north edge; thence general depths of 5 to 8 feet were available in the middle of the anchorage.

(582)

Dangers

(583) No special directions are necessary—the chart is the best guide. In passing from north to south in the Annisquam River and Blynman Canal, take care to avoid the unmarked rocky area covered 4 feet on the east side of the channel about 0.35 mile north of the Annisquam Harbor Light, just southeast of Buoy 3; several rocks, submerged and awash, on the east side of the channel, marked by Daybeacon 7; a rock, reported covered 2½ feet and marked by a buoy, on the east channel edge about 125 yards northward of Annisquam Channel Light 25; and the rocks on the south side of the southern entrance to Blynman Canal, marked by a daybeacon.

(584)

Bridges

(585) About 2.5 miles south of Annisquam Harbor Light, State Route 128 crosses the waterway on a fixed span that has a clearance of 65 feet for a center width of 100 feet. A railroad bridge, under construction (2018), is about 0.7 mile southward of the fixed span. At the southern end of the waterway, State Route 127 highway bridge has a 38-foot bascule span with a clearance of 8 feet. The bridgetender monitors VHF-FM channel 13; call sign, WQA-834. (See **33 CFR 117.1** through **117.59** and **117.586**, chapter 2, for drawbridge regulations.)

(586)

Current

(587) Currents at Annisquam Harbor Light average 1.3 knots at strength. Tidal currents at the southern entrance to Blynman Canal average over 3 knots at strength, but greater velocities to 10 knots were reported in 1992 in the vicinity of Blynman Bridge (State Route 127). Mariners are advised to use caution when approaching the bridge, especially during maximum flood and ebb.

(588)

Harbor regulations

(589) The Gloucester Chief of Police is also **harbormaster** for Annisquam River and Blynman Canal. The deputy harbormaster supervises the moorings and anchorages. A **speed limit** of 4 knots is enforced on the river and in Lobster Cove.

(590)

Small-craft facilities

(591) There is a marina on the west bank of Lobster Cove and several private float landings around the cove. Gasoline, diesel fuel and water are available at the floats of the marina, which have 12 feet reported alongside. Ice, provisions and marine supplies are available. Overnight berthing is permitted, and guest moorings are maintained.

(592) A footbridge with a horizontal clearance of 30 feet and a vertical clearance of 6 feet crosses the Lobster Cove about 0.3 mile above the entrance. A town float landing is on the south side of the bridge.

(593) A private marine railway that can haul out craft up to 40 feet in length in an emergency is on the west side of the cove near the entrance.

(594) The Annisquam Yacht Club is on the point on the west side of the entrance. The usual courtesies are extended by the club to visiting members of accredited yacht clubs. Showers, restrooms and limited guest accommodations are available to visiting yachtsmen. Water is available at the float, ice is obtainable and guest moorings are maintained by the club. A daybeacon and a buoy mark dangerous ledges south of the yacht club.

(595) **Mill River** is a tributary of Annisquam River, on the east side, 0.4 mile southward of Annisquam. Two rocks covered 2 feet are near the middle of the entrance to Mill River. There are numerous summer homes and float landings on the river, which is used by many small craft in the summer.

(596) On the east side of Annisquam River, just north of the fixed highway bridge at **Ferry Hill**, is a boatyard that builds wooden craft up to 35 feet long or handles craft up to 30 feet long for repairs or dry open or covered winter storage.

(597) A marina on **Rust Island** just west of **Biskie Head** on the north side of **Little River** has float landings with 5 feet reported alongside. Gasoline, ice, a small-craft launching ramp, marine supplies and a restaurant are available.

(598) On the west bank of the waterway at the north end of Blynman Canal there is a marina with 12 feet reported at the floats. Gasoline, water, ice, berths with electricity, a pump-out station and some marine supplies are available; hull, engine and electronic repairs can be made. On the east bank opposite the marina is a 142-foot concrete ramp with float landings. No services are available.

(599) Blynman Canal and Gloucester Harbor are described in chapter 10, Cape Ann to Boston Harbor.

(600)

Cape Ann to Cape Hedge

(601) **Cape Ann** is very rocky and broken, 235 feet high at **Pool Hill**, its highest point, with numerous summer homes and has several abandoned granite quarries. Communication is by railroad to Gloucester and Rockport and by highway entirely around the cape.

(602) **Bay View** is a village on **Hodgkins Cove** on the west shore of Cape Ann, 0.8 mile northeastward of Annisquam Harbor Light. The University of Massachusetts Marine Station has a wharf on the outer southwest side of the long stone pier on the east side of the cove. In 1966, there was a depth of 12 feet on the outer half of the southwest side, in a channel about 70 feet wide. The cove at the inner end of the pier on the northeast side has a depth of about 2 feet at the entrance and mostly dry inside. Unmarked rocks are at the entrance.

(603) **Lanes Cove**, 1.4 miles northeastward of Annisquam Harbor Light, is a small cove protected by stone breakwaters at the entrance, forming a harbor for small craft. It has a depth of 12 feet at the entrance and 10 feet in the middle inside. **Lanesville** is a village on the cove. Many fishing and pleasure craft moor in the harbor. Provisions, ice and some supplies are available from a market in the village.

(604) **Folly Cove** is on the north side of Cape Ann, 2.4 miles northeast of Annisquam Harbor Light. A 3-foot spot is about 100 yards north of **Folly Point**, the west entrance point, in about 42°41'25.5"N., 70°38'41.0"W. The cove has a stone wharf on the east side with about 16 feet alongside. A 3-foot spot is about 100 yards westward of the wharf. A restaurant is on the wharf and a motel at the head of the cove, the latter open only in summer. **Halibut Point** forms the northern extremity of Cape Ann.

(605) **Ocean View** is a summer resort on **Andrews Point** at the north end of Sandy Bay.

(606) **Sandy Bay** is a large bight in the northeastern shore of Cape Ann between Straitsmouth Island on the east and Andrews Point on the west. The bay is 2 miles wide between these points and about 1.5 miles long to its head.

(607) A breakwater has been partially completed to form a harbor of refuge. It extends 1,200 yards northward from Avery Ledge, then 830 yards northwestward toward Andrews Point. In 1979, it was awash at low water except for a distance of about 300 yards near the middle where it was above high water. About 400 yards of each end of the breakwater are covered at low water. A lighted gong buoy is off the northwest end, and a lighted buoy is off the south end. It is reported that several boats have grounded on the breakwater. This can be avoided by keeping on the correct sides of the buoys marking the ends.

(608) Depths inside the breakwater are 30 to 88 feet, with several rocky spots of 14 to 30 feet in the southern part. **Ninefoot Rock** on the south side of the bay is marked on its northern side by a buoy. The bay is sometimes used as an anchorage but is exposed to north and northeasterly weather and at such times Gloucester or Salem Harbors are generally used.

(609) The entrance to Sandy Bay between Straitsmouth Island and the lighted buoy marking **Avery Ledge** has broken bottom and a rocky spot covered 24 feet in the middle. Strangers may be unable to avoid this and should not use this channel when drawing more than 18 feet.

(610) On the south side of this channel, a ledge which uncovers in places and is covered 17 feet near the end extends about 330 yards northeastward from the northeast end of Straitsmouth Island. The northern entrance to the bay westward of the lighted gong buoy at the northwest end of the breakwater is deep and clear.

(611) **Pigeon Cove**, 0.8 mile south of Andrews Point, is a small cove protected by a breakwater; the entrance is marked by a buoy. A channel leads northwest into the cove to an anchorage basin. In 2014, the controlling depth was 10 feet in the entrance channel and 8 feet in the anchorage basin. The most prominent features of Pigeon Cove are the high concrete stack of the foundry and the tank on **Pigeon Hill**.There are bulkhead wharves around the harbor, a public float landing with 6 feet reported alongside and a small-craft launching ramp. A number of fishing and pleasure craft lay at moorings in the cove.

(612) The best water is on the northeast side. **Pigeon Rock**, 50 yards south of the east point outside the jetty, is nearly uncovered at extreme low water. A 5-foot spot is near the entrance about 80 yards southward of Pigeon Rock.

(613) Two old stone quarry breakwaters are built out from the shore 0.3 and 0.5 mile southward of Pigeon Cove. The southerly one forms a harbor that is used by fishing and pleasure craft. Mooring is not allowed alongside the stone wharves. A small-craft launching ramp is at the head.

(614) A small basin at **Rowe Point**, about 0.7 mile southward of Pigeon Cove, is now a lobster pound.

(615) **Dodge Rock**, Bartlett Rock and **Mitchell Rock** are in a cluster of rocks about 300 yards from the western shore of Sandy Bay. Dodge Rock, awash at low water, is

(619)

is currently unmarked. The western end of the rock is 100 yards offshore, and the southern rock, covered 9 feet, is about 150 yards southeastward of the daybeacon.

(616) Mitchell Rock, covered 3 feet, and another rock, covered 10 feet, are 280 and 400 yards, respectively, northward of the daybeacon. **Bartlett Rock**, awash at low water, is about 125 yards north of the daybeacon. These rocks are unmarked.

(617) **Sandy Bay Ledge** is partly bare at high water and extends 200 yards from the western shore of Sandy Bay at Rowe Point. In fair weather, vessels up to 150 feet long are reported to anchor in the cove south of Sandy Bay Ledge.

(618) **Rockport Harbor** at the southwest end of Sandy Bay is reported to be open to strong northeasterly to easterly winds but can be entered at any time. The harbor is protected by two breakwaters, one of which extends eastward from **Bearskin Neck** on the northwest side of the harbor.

(620) The other breakwater extending in a northerly direction from **The Headlands** is a short one.

(621) The harbor consists of an outer basin and two inner basins that are separated by the town wharf. In 2004, the outer basin had a least depth of 8 feet, the northern inner basin had depths of 4.5 to 8.0 feet and the southern inner basin had depths of 6.1 to 8.0 feet.

(622) **Rockport**, the town, has communication by railroad, bus and taxi service. Banks, churches, restaurants, hotels and guest houses, hospitals and markets are available.

(623)

Prominent features

(624) **Straitsmouth Island**, low and grassy, is marked on its eastern end by **Straitsmouth Light** (42°39'44"N., 70°35'17"W.), 46 feet above the water and shown from a white cylindrical tower, near the northeast end of the island. A mariner radio-activated sound signal at the light is initiated by keying the microphone five times on VHF-FM channel 83A. The radio tower and buildings of a former Coast Guard station are conspicuous on **Gap Head**, the peninsula westward of Straitsmouth Island. A standpipe on the summit of a hill south of the harbor is also prominent. Passage should not be attempted between Straitsmouth Island and Gap Head at low water without local knowledge.

(625) **Rockport Breakwater Light 6** (42°39'39"N., 70°36'43"W.), 32 feet above the water, is shown from a spindle with a red triangular daymark on the end of the north breakwater.

(626)

Channels

(627) The entrance channel between the breakwaters is about 26 yards wide with depths from 8 to 10 feet. It is not advisable, however, to enter with drafts greater than

7 feet without local knowledge. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A.

(628)

Harbor regulations

(629) Moorings and berths in the harbor are under control of the harbormaster, who can be contacted through the local police department. A **speed limit** of 4 miles per hour is enforced within the harbor limits. There are no guest moorings, but one can usually be arranged for through the harbormaster.

(630) The basin on the southeast side of the town wharf is used to moor small sailing craft, and the northwesterly basin, or commercial basin, is used by fishing and lobster boats. In 1993, a section of the town wharf had broken off and had reportedly created an obstruction in the northwesterly basin in about 42°39'32.7"N., 70°36'55.5"W. A town ramp, dry at low water and with 3 feet at high water, is at the head of the basin.

(631)

Dangers

(632) **Flat Ground**, a dangerous ledge 0.5 mile long covered 1 to 15 feet, is 1 to 1.5 miles north-northeastward of Straitsmouth Light. The ledge is marked by a buoy at its south end and a bell buoy at the north end.

(633) The engine block of the liberty ship CHARLES S. HAIGHT was reported, in 1979, to be still visible on the reef at low water.

(634) **Dry Salvages** is a bare ledge about 15 feet high near the middle of a reef about 500 yards long in a northerly direction. A lighted bell buoy is 0.5 mile northeastward of the ledge.

(635) **Little Salvages** is a ledge showing well bare at low water and with parts awash at high water. It is about 500 yards westward of Dry Salvages. Shoal water extends out a little more than 200 yards from the western side of the bare part of the ledge, and a rock bare at lowest tides and a sunken wreck are between it and Dry Salvages.

(636) **Harbor Rock**, covered 2 feet, is about 130 yards northeastward of the end of the north breakwater at the entrance of Rockport Harbor; a buoy is about 0.1 mile northeastward of the rock. Inshore of the rock, a shelving unmarked ledge extends 75 yards northeastward from the end of the north breakwater.

(637) The edges of the harbor are shoal and foul, with ledges near the shores, particularly on the north side northward of a line between the end of the north breakwater and the end of the first wharf on the north side. All except light-draft craft should stay out of that area.

(638)

Wharves

(639) The first wharf, in the northwest part of the harbor, is a private wharf locally known as **Tuna Wharf.** The second wharf, locally known as **Bradley Wharf**, has overnight berthage that can be arranged through the harbormaster. In 1979, 6 feet was reported alongside Bradley Wharf, with no services available.

(640) The town float landing, with 6 feet reported alongside, is at the head of the town wharf; sewage pump-out is available. Party fishing boats operate from the landing in the summer. Parking is available on the town wharf.

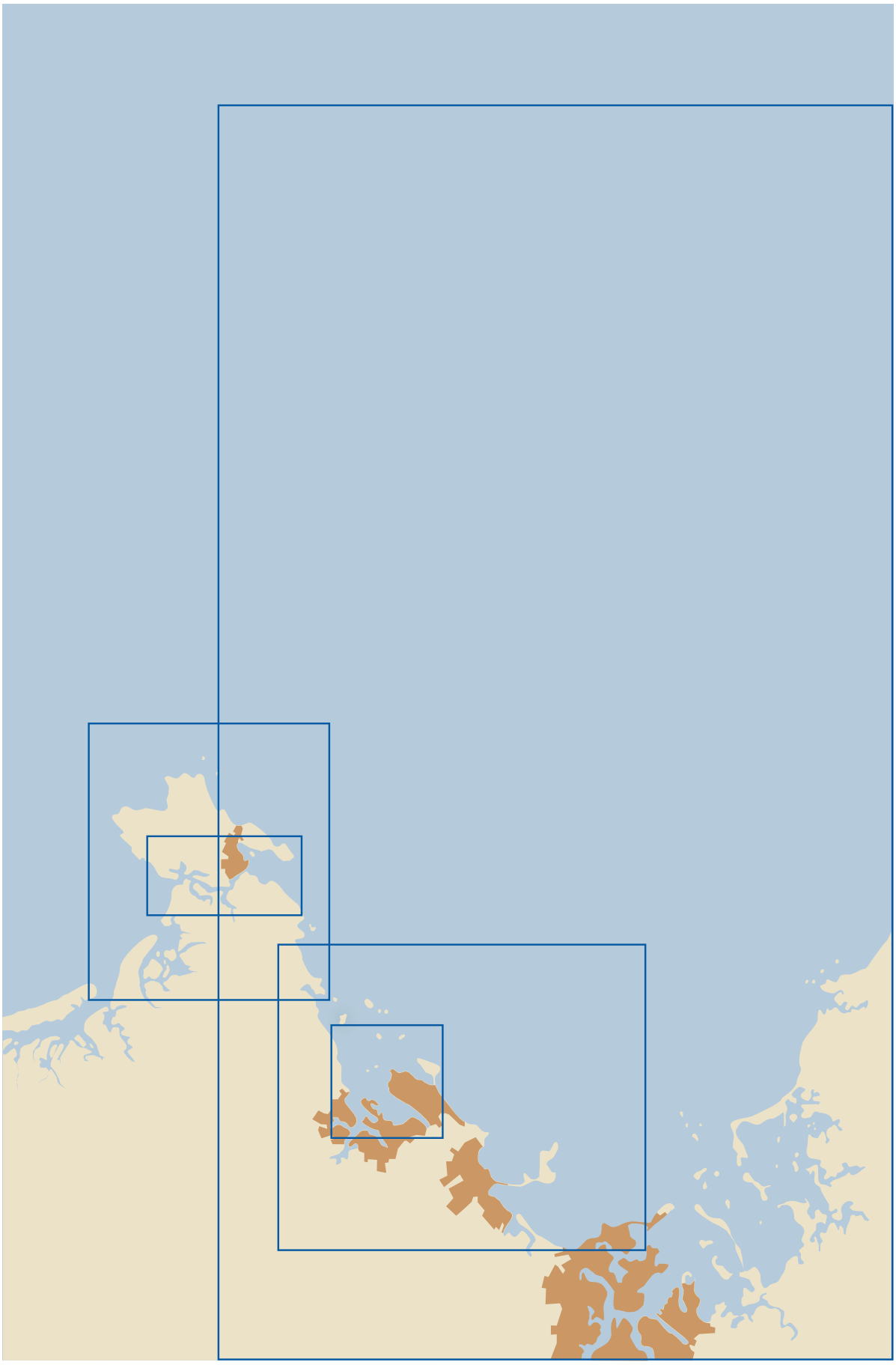
(641) On the southeast side of the head of the town landing is the Sandy Bay Yacht Club, which has float landings with 6 feet alongside. The club has restrooms available to visiting yachtsmen. Water, electricity and ice are available at the floats.

(642) **Cape Ann Light** (42°38'12”N., 70°34'30”W.), 166 feet above the water, is shown from the southerly of two identical 124-foot gray stone towers on the east side of **Thacher Island**, 1.3 miles south-southeast of Straightsmouth Island. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A. The northerly tower is marked by a private light. **Oak Rock**, covered 6 feet and marked on its east side by a buoy, lies between Thacher Island and **Emerson Point**.

(643) **Londoner**, a ledge about 0.4 mile long in a northeasterly direction, covered 1 to 11 feet, is 0.5 mile east-southeastward of Cape Ann Light. Near the center of the ledge, on a cluster of rocks that uncover at low water, is a pole. Between Londoner and Thacher Island is a passage with depths of 16 to 38 feet. This passage should not be attempted by a stranger.

(644) **Milk Island**, about 0.4 mile southward of Emerson Point, is connected with that point and Thacher Island by two bars covered 3 feet. A rock awash is about 0.2 mile north-northeast of the north point of Milk Island. **Salt Island Ledge**, 1.3 miles southwestward of Milk Island, is awash at extreme low water. A buoy marks the southeast end of the ledge.

(645) There are numerous reddish brown bare bluffs along the coast between **Cape Hedge** and Eastern Point. The most prominent of these are on Cape Hedge, 50-foot **Salt Island**, the points to the north and west of Salt Island, the points on both sides of the entrance to **Brace Cove**, and on the southern part of Eastern Point.



Cape Ann to Boston Harbor, Massachusetts

(1) This chapter describes the Massachusetts coast along the northwestern shore of Massachusetts Bay from Cape Ann southwestward to but not including Boston Harbor. The harbors of Gloucester, Manchester, Beverly, Salem, Marblehead, Swampscott and Lynn are discussed as are most of the islands and dangers off the entrances to these harbors.

(2)

COLREGS Demarcation Lines

(3) The lines established for this part of the coast are described in **33 CFR 80.120** and **80.125**, chapter 2.

(4)

No-Discharge Zone

(5) The State of Massachusetts, with the approval of the Environmental Protection Agency, has established a No-Discharge Zone (NDZ) in all coastal waters of Massachusetts described in this volume, extending about 3 miles offshore.

(6) Within the NDZ, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZ, discharge of sewage is regulated by **40 CFR 140** (see chapter 2).

(7)

Massachusetts Bay

(8) **Massachusetts Bay** is the body of water lying westward of a line connecting Cape Ann Light on Teacher Island with Race Point Light on the northwestern extremity of Cape Cod, about 38 miles south-southeastward. It includes Boston Harbor, described in chapter 11, and Cape Cod Bay, described in Chapter 12. Between Cape Ann Light and Boston Harbor, 24 miles to the southwestward, the principal harbors are Gloucester, Beverly, Salem, Marblehead and Lynn, all available to vessels of moderate draft. The coast is rocky and generally bold with numerous detached islands, rocks and sunken dangers.

(9)

Gloucester Harbor

(10) **Gloucester Harbor** is one of the most important fishing ports in the United States and an important harbor of refuge. It is 5 miles southwestward of Emerson Point, the easternmost point of Cape Ann, 26 miles from Boston and 234 miles from New York. The entrance is marked on its eastern side by Eastern Point Light. There is an outer and inner harbor, the former having depths generally of 18 to 52 feet and the latter, depths of 13 to 25 feet.

(11) **Gloucester Inner Harbor** limits begin at a line between Black Rock Danger Daybeacon and **Fort Point.**

(12) **Gloucester** is a city of great historical interest, the first permanent settlement having been established in 1623. The city limits cover the greater part of Cape Ann and part of the mainland as far west as Magnolia Harbor. Its principal industries are directly or indirectly connected with the fishing or related industries in the processing, freezing, canning or shipment of fish and lobsters.

(13) The principal import is seafood. Limited amounts of general cargo are the principal export.

(14)

Prominent features

(15) **Eastern Point Light** (42°34'49"N., 70°39'52"W.), 57 feet above the water, is shown from a 36-foot white conical tower with a covered way to a dwelling.

(16) A breakwater extends 750 yards west-northwest from the shore near Eastern Point Light and is marked at its outer end by **Gloucester Breakwater Light** (42°34'57"N., 70°40'20"W.), 45 feet above the water and shown from a 37–foot white house and tower on a brown square skeleton framework structure. A sound signal at the light is activated by keying the microphone five times on VHF-FM channel 83A.

(17) **Normans Woe**, on the west side at the entrance to Gloucester Harbor, is a rocky headland split by a deep cleft, known as **Rafes Chasm**, into which the sea enters during heavy weather. **Normans Woe Rock**, 0.3 mile northeastward of Normans Woe and over 0.1 mile offshore, is a rounded rocky islet 45 feet high, surrounded by extensive ledges. A bell buoy is about 0.2 mile east-southeast of Normans Woe Rock. The stone building and double tower of the John Hays Hammond Museum, about 0.3 mile northward of the rock, are conspicuous from seaward.

(18) **Tenpound Island Light** (42°36'07"N., 70°39'56"W.), 57 feet above the water, is shown from a white conical tower on the west side of Tenpound Island off the entrance to Inner Harbor; a sound signal is at the light. The ruins of several piers are on the north side of the island. The island is owned by the city of Gloucester.

(19)

COLREGS Demarcation Lines

(20) The lines established for Gloucester Harbor are described in **33 CFR 80.120**, chapter 2.

(21)

Channels

(22) The entrance westward of the breakwater between Dog Bar and **Mussel Point** is about 0.6 mile wide. About

500 yards westward of Round Rock Shoal is an unmarked rocky ledge covered 23 feet. This leaves only a channel about 400 yards wide with depths of 38 to 55 feet into the outer harbor.

(23) **Dog Bar Channel** between the end of Dog Bar and the eastern edge of Round Rock Shoal is only about 150 yards wide with depths of 16 to 22 feet.

(24) During heavy southeasterly gales, the sea at times breaks nearly the whole distance across the entrance. Strangers should enter by the deepest channel westward of Round Rock Shoal, where there is reported to be a space known not to break.

(25) A Federal project provides for a 20-foot channel from the northeasterly part of Gloucester Harbor into Inner Harbor, connecting with 20-foot north and south access channels which lead on either side of the Gloucester State Fish Pier to the head of the harbor. An 18-foot channel and a 16-foot channel lead from the Inner Harbor entrance channel into Harbor Cove and Smith Cove, on the northwestern and southeastern sides of Inner Harbor, respectively. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A. The channels are marked by buoys.

(26) The southern entrance to **Blynman Canal** and Annisquam River is through Blynman Bridge at the head of Western Harbor. This is the inside route to Ipswich Bay on the north side of Cape Ann, described in chapter 9. The school tower 500 yards north-northwest of the bridge is prominent. Rocks, marked by a daybeacon, are close southward of the canal entrance.

(27)

Anchorages

(28) The best anchorage in the outer harbor for vessels coming in for shelter or bound to Gloucester is **Southeast Harbor**, the cove in the eastern part of Gloucester Harbor northward of **Black Bess Point** and southward of Tenpound Island, known locally as **Pancake Ground**. This is the one most frequently used. It has good anchorage, soft mud and clay bottom in about 23 to 30 feet, and is also used by vessels taking shelter.

(29) In **Western Harbor**, the semicircular cove northwestward of Tenpound Island in the northern part of Gloucester Harbor, there is also good anchorage, soft mud and clay bottom in 20 to 27 feet. Give the shore a berth of 300 yards. In 2003, a wreck covered 16 feet was reported in 42°36'26"N., 70°40'12"W. Debris was still present at that location in 2025. There are no wharves. The city of Gloucester maintains a parkway along the shore of Western Harbor to **Stage Head**. The Gloucester Fisherman statue faces the harbor from this parkway about 200 yards eastward of the entrance to Blynman Canal.

(30) A dredged anchorage, about 300 yards southwestward of the Gloucester State Fish Pier, is available in Inner Harbor. The anchorage is partially marked by buoys. Mooring permits for the Inner Harbor are issued by the deputy harbormaster, who patrols the harbor in a police/fire boat; the patrol boat monitors VHF-FM channel 16 when underway.

(31) **Smith Cove**, in the southeast part of Inner Harbor, provides good anchorage for small craft in 6 to 15 feet but is somewhat congested with moorings. **Harbor Cove**, on the northwest side of the entrance to Inner Harbor, has depths of 15 feet in the dredged area with lesser depths along the edges. **Gloucester Coast Guard Station** is on its northeast side. A dredged anchorage is on the east side of the entrance to Harbor Cove.

(32)

Dangers

(33) Gloucester Harbor and approaches have very broken ground and many rocks and ledges, some of them unmarked; careful navigation is necessary, especially in thick weather.

(34) The principal dangers are marked for vessels of 24-foot draft or less to an anchorage in Southeast Harbor and for 18-foot draft or less into the inner harbor. Strangers are advised not to bring in greater drafts without a pilot.

(35) **Dog Bar**, on which the breakwater is built, extends 100 yards westward of the end of the breakwater where it is marked by a buoy.

(36) **Round Rock Shoal**, a rocky ledge about 400 yards in extent northeast to southwest and covered 13 feet, extends from 0.15 to 0.3 mile westward of the breakwater light. It is marked on its northeastern edge by a buoy and on its southwestern edge by a lighted buoy.

(37) **Green Rock**, 175 yards eastward of Tenpound Island, is marked by a daybeacon. The passage between Tenpound Island and Rocky Neck is shoal and foul and should not be attempted, especially by strangers.

(38) **Tenpound Island Ledge** and **Mayflower Ledge**, on the eastern side of the approaches to the Inner Harbor, are covered 18 and 17 feet, respectively; a buoy is off the northwest side of Mayflower Ledge. Two shoal spots, covered 17 and 20 feet and marked by a buoy, are about 220 yards southwestward of Tenpound Ledge. There are also unmarked depths of 19 feet about 230 yards north-northwestward of Tenpound Island Light. **Prairie Ledge**, on the western side of the approach, is covered 4 feet and marked on its eastern end by a lighted buoy. **Babson Ledge**, 500 yards north of Tenpound Island, is covered 13 feet and marked on its south side by a buoy.

(39) **Rocky Neck,** a high and partly wooded island on the east side at the entrance to Inner Harbor, is connected with the easterly shore by a causeway. **Black Rock**, about 100 yards off the western end of Rocky Neck, is marked by a daybeacon.

(40)

Current

(41) The tidal currents in Gloucester Harbor do not greatly interfere with the movements of vessels, as they set directly in and out of the harbor and their velocity is comparatively small. However, the tidal currents in the entrance to Blynman Canal average over 3 knots at strength, and greater velocities to 10 knots were reported in 1992 in the vicinity of Blynman Bridge (State Route 127). Mariners are advised to use caution when approaching the bridge, especially during maximum flood and ebb.

(42)

Ice

(43) Ice seldom extends outside Tenpound Island at the entrance to Inner Harbor. The movement of boats generally keeps Inner Harbor open.

(44)

Pilotage, Gloucester

(45) Pilotage is compulsory for all foreign originating vessels over 350 gross tons and/or 7-foot draft and for U.S. vessels over 10,000 gross tons.

(46) Pilotage service for Gloucester is provided by Eastern Point Pilots, P.O. Box 705, Rowley, MA 01969, telephone 978–948–3900. Office is manned 24 hours; pilots are available 24 hours. Pilot operations are based out of Gloucester. Eastern Point Pilots are state and federal pilots for the ports of Massachusetts north of Nahant to the New Hampshire border. This includes the ports of Marblehead, Salem, Beverly, Manchester, Rockport and Newburyport. Pilotage is compulsory for vessels defined above entering these ports.

(47) The 27-foot pilot boat, VIGILANT, has a 300-horsepower jet drive and a gray hull and white superstructure.

(48) Pilot boarding station is in 42°33'35"N., 70°39'50"W., about 1.3 miles S of Eastern Point Light. About 1 hour before vessel arrival the pilot boat monitors VHF-FM channel 16. Working channels are 12 or 06. Arrival notice is requested 24 hours ahead of ETA with an update at minus 8 hours. For departure sailings, a 4-hour minimum notice is requested. Vessels are requested to provide a leeward side for boarding with secure pilot ladder with a 1 meter (about 3 feet) above the water clearance. Daylight restrictions apply for vessels in the port of Salem when the combined LOA and maximum beam exceeds 750 feet. Tide restrictions may apply to loaded vessels due to ranges in tide fluctuation and weather.

(49)

Towage

(50) Tugs are available from Boston.

(51)

Quarantine, customs, immigration and agricultural quarantine

(52) (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(53) **Quarantine** is enforced in accordance with the regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.)

(54) Gloucester is a **customs port of entry.**

(55)

Coast Guard

(56) The Gloucester Coast Guard Station is on the east side of Harbor Cove at Gloucester.

(57)

Harbor regulations

(58) The **harbormaster** can be reached at 978–325–5750. A **speed limit** of 5 miles per hour is enforced in the Inner Harbor.

(59)

Wharves

(60) There are many wharves along the Inner Harbor at Gloucester. Most of these facilities are used by the fishing industry. A description of several of the principal wharves follows.

(61) **Gloucester State Fish Pier**, at the head of Inner Harbor, has 1,000 feet of berthing space along its north side with reported depths of 22 to 24 feet alongside. Several piers with shoal depths alongside extend from the south side of the pier. The State Fish Pier is used to tranship and process seafood. Water is available, and diesel fuel can be obtained by lighter.

(62) The Quincy Market Cold Storage and Warehouse Company operates three wharves at Gloucester. The wharves are used to unload imported frozen seafood products. Cold storage facilities with a combined capacity of 4 million cubic feet are available. Mobile cranes and forklifts are available, and diesel fuel can be obtained by lighter. A description of the wharves follows.

(63) **Rogers Street Wharf** (42°36'45"N., 70°39'28"W.), on the north side of Inner Harbor, has a 300-foot face with depths of 22 to 24 feet reported alongside.

(64) **Rowe Square Wharf**, about 100 yards northeastward of the Rogers Street Wharf, has a 450-foot face with depths of 22 to 24 feet reported alongside.

(65) **East Main Street Wharf**, on the south side of Inner Harbor and on the north side of the entrance to Smith Cove, has a 360-foot face with depths of 21 feet reported alongside.

(66)

Supplies

(67) Fuel oil is not available in bunker quantities, but diesel fuel can be had as desired from tank trucks and lighters. Marine and most other supplies are obtainable in town. Water is available at most of the wharves.

(68)

Repairs

(69) Gloucester has ship repair plants on Rocky Neck and on the northwest side of the harbor. The two plants have machine and other shops and can carry out all repairs to wood and steel vessels. The shipyard pier on Rocky Neck is 270 feet long with 15 to 16 feet reported alongside. The

yard has a 10-ton crane. The largest marine railway can haul out craft up to 145 feet in length and up to 600 tons in weight. Radio and electronic repairs can be made.

(70) Boston is the nearest port where large vessels can be drydocked for extensive repairs to hull and machinery.

(71)

Small-craft facilities

(72) A town float landing, with a reported 2 feet alongside, is at the head of Harbor Cove. Gloucester has many small-craft facilities along the waterfront. Sewage pump-out, marine railways up to 50 feet and several 20-ton mobile hoists are available.

(73) Party fishing boats operate from several points in the harbor during the summer, and charter boats, rowboats and outboards can be hired.

(74) Eastern Point Yacht Club is on the north side of the entrance to the cove at the inner end of Gloucester Breakwater. There is a depth of 8 feet at its float landing, to which water is piped. The club maintains guest moorings and restrooms, showers and dining room and provides certain services for visiting members of yacht clubs.

(75)

Communications

(76) Gloucester has rail and highway connections, and taxi and bus service. Schools, churches, banks, restaurants, lodging, a library and a hospital are in the city.

(77)

Magnolia Harbor

(78) **Magnolia Harbor** is a cove about 1.5 miles westward of the entrance to Gloucester Harbor and just north of **Kettle Island.** The summer resort of **Magnolia** is conspicuous on the eastern side. It has a public pier and float with 3 feet reported alongside. A rock that uncovers is just southward of the outer end of this pier, and the partial submerged ruins of two piers are about 150 yards farther southward. The harbor, used by many small craft, does not have very good holding ground and is exposed to southerly weather.

(79) There are no facilities at the public pier, but gasoline can be obtained in cans from garages at the head of the harbor. Magnolia has hotels, restaurants, markets and stores and provisions and marine supplies can be obtained.

(80)

Prominent features

(81) Prominent features are the large summer homes on **Magnolia Point**, an old wartime white concrete observation tower on **Coolidge Point** and a large pavilion and several bathhouses on the beach at the head of the harbor. The edges and head of the harbor are shoal and foul, and none of the dangers are marked. The harbor at best is temporary anchorage for small craft in foul weather or offshore winds.

(82) **Kettle Cove**, west of Magnolia Harbor, is shoal and foul at the head. There is one private landing on the east side of the entrance, but no facilities. Large private homes on Coolidge Point and Crow Island are conspicuous. There are several rocks awash in the entrance to the cove.

(83)

Manchester Harbor

(84) Off the shore eastward of Manchester Harbor entrance, between Gloucester entrance and House Island, are many islands, rocks and ledges extending about 0.8 mile offshore. The farthest outlying ones, named in order from eastward are **Great Egg Rock**, 34 feet high and bare; **Paddock Rock**, covered 13 feet; **Boohoo Ledge**, covered 1 foot; **Salt Rock**, showing at high water; **Pickett Ledge**, part of which uncovers 3 feet; **Gales Ledge**, covered 5 feet; a ledge, covered 17 feet and marked by a buoy, about 350 yards south of Gales Ledge; and **Pilgrim Ledge**, covered 18 feet. Of the several coves, only Manchester Harbor is of interest to navigation.

(85) The shoreline of this section of the coast is lined with summer homes, many of which are large and conspicuous. The beach at **Eagle Head** is conspicuous.

(86) **Manchester Harbor**, about 5 miles west-southwest of Gloucester Harbor, is an arm of **Manchester Bay** extending in a northeasterly direction for 1 mile west of Gales Point to the town of **Manchester**. The entrance to Manchester Bay is northward of Bakers Island Light, between **House Island**, partly wooded, on the east, and **Great Misery Island** on the west. The ruins of two stone houses, one in the center and another on the west end, are on Great Misery Island.

(87) Manchester Harbor is principally a yachting center, with only a small amount of local commercial fishing. The harbor above **Proctor Point** is practically landlocked and secure in all weather.

(88)

Prominent features

(89) Conspicuous objects include a white square observation tower on Gales Point, a large standpipe on **Moses Hill** north of the town, the bascule span of the railroad bridge, and the fishing pavilion at the end of the town wharf next to the yacht club at **Tucks Point.**

(90)

Channels

(91) The approach channel to Manchester Bay from the eastward, between Whaleback and House Island, is clear and about 250 yards wide; the approach from southward, westward of Whaleback, is 500 yards wide.

(92) **Manchester Channel**, privately dredged and marked by buoys, leads from Manchester Bay to an anchorage basin at the head of Manchester Harbor; dredged anchorage basins are on either side of the channel about 300 yards northeast of Proctor Point. Shoaling to an unknown extent was reported in about 42°33'35"N., 70°47'12"W. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are

available through a USACE hydrographic survey website listed in Appendix A. The railroad bridge, about 1 mile above the entrance and just above the anchorage basin, has a 48-foot bascule span with a clearance of 6 feet. (See **33 CFR 117.1** through **117.59** and **117.603**, chapter 2, for drawbridge regulations.) A mooring basin with depths of about 7 feet is above the bridge.

(93)

Anchorages

(94) By local regulations, vessels over 45 feet in length must anchor in Manchester Bay. The anchorage is northward of a line between Great Misery and House Islands as far as Manchester Channel Buoy 5. Those desiring to anchor only overnight, or from head winds, may find fair holding ground and good shelter except in southerly gales.

(95) The anchorage basins in Manchester Harbor are restricted to craft not over 45 feet in length. This regulation is strictly enforced.

(96)

Dangers

(97) There is a bad ledge locally known as **Bow Bell**, with a rock awash on it, on the east side of the channel opposite the yacht club and public landing on Tucks Point, just above Proctor Point. A buoy marks the northwestern edge of the ledge. It is usually covered, and the only indication of it is a hole, or clear spot, amidst the craft moored or anchored in the vicinity. Care should be taken to avoid anchoring on the ledge.

(98) **Whaleback**, a dangerous ledge in the entrance to Manchester Bay, is about 400 yards long east and west, and 200 yards wide. Near the middle of its northern side is a rock awash at low water, marked by a daybeacon.

(99) **Sauli Rock**, which uncovers 9 feet, is 300 to 400 yards eastward of the northeast end of Great Misery Island and is marked by a daybeacon.

(100) **White Ledge**, awash at low water, is 300 yards northwestward of House Island and is marked by a buoy on its west side. **Halftide Rocks**, which uncover, are 250 yards northward of White Ledge, and are marked by a buoy off the west side.

(101) **Chubb Islet**, bare and rocky, is 300 yards from the north shore of Manchester Bay and should be given a berth of more than 200 yards.

(102)

Harbor regulations

(103) In addition to the local regulations restricting the size of craft using the anchorage basins in Manchester Harbor, a **speed limit** of 5 miles per hour is enforced within the harbor.

(104) The **harbormaster** and deputies supervise the moorings and on application will usually find a vacant one for a visitor or advise where best to anchor. The yacht yards maintain guest moorings.

(105)

Wharves

(106) A town pier and float landing, used by commercial lobstermen, are on the east side of the harbor, about 0.5 mile above Proctor Point; depths of 6 feet are reported at the face of the pier and 3 feet at the float; water and electricity are available.

(107)

Small-craft facilities

(108) There are commercial and private float landings in the harbor. Four public landings and two small-craft launching ramps are available; depths of 5 to 10 feet are reported alongside the landings.

(109) Two yacht yards with marine railways with capacities up to 70 feet or 80 tons are on the west side of the harbor. Hull, engine, electrical and electronic repairs can be made; a machine shop is available. The yard can provide gasoline, diesel fuel, water, ice, provisions, marine supplies, a pump-out facility and dry covered or open winter storage; lifts up to 35 tons are also available. An outboard marina is on the west side of the harbor just above the bridge. The Manchester Yacht Club, at Tucks Point, has depths of 10 feet alongside its floats; water is available.

(110) Manchester has rail, bus and taxi services.

(111)

Salem Harbor, Beverly Harbor and Marblehead Harbor

(112) **Salem Harbor**, **Beverly Harbor** and **Marblehead Harbor**, each of which in turn will be described in detail, form a large irregular indentation in the shore of Massachusetts Bay, 11 miles southwestward of Cape Ann and 12 miles northeastward of Boston Harbor entrance. Gales Point is the northern end and Marblehead Neck the southern point at the entrance to this large indentation, which includes within its limits the harbors of Manchester, Beverly, Salem and Marblehead, the distance between the two points being 4 miles. This wide space is studded with islands, bare rocks and sunken ledges, through which are several channels leading into the harbors.

(113) Salem has some commercial shipping with the importation of coal and petroleum products. Marblehead is principally a yachting center.

(114)

Prominent features

(115) The most conspicuous landmarks approaching the harbor are Bakers Island, Great Misery Island, Childrens Island and Marblehead Neck; a white square observation tower 90 feet high on Gales Point; a large standpipe on Moses Hill back of Manchester Harbor; the church spires in Beverly; the large powerplant with three stacks on Salem Neck; the red brick tower of Abbott Hall and two standpipes, one with conical top in Marblehead; and the light, large mansions and homes on Marblehead Neck. Two large mansions on the estate along the north shore

westward of Manchester Harbor also stand out. **Bowditch Ledge Daybeacon**, about 1.9 miles southwest of Gales Point, is 30 feet high with red and white diamond-shaped daymark on a conical granite monument and is readily discernible to anyone approaching the harbor.

(116) **Bakers Island Light** (42°32'11"N., 70°47'09"W.), 111 feet above the water, is shown from a white conical tower on the north end of Bakers Island. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A. Many summer homes are on the island, and there is a ferry landing on the west side.

(117) **Marblehead Light** (42°30'19"N., 70°50'01"W.), 130 feet above the water, is shown from a 105-foot brown square skeleton tower with black top on the northern extremity of **Marblehead Neck**, a high, rocky promontory connected with the mainland by a sandbar and causeway.

(118) **Marblehead Rock**, 500 yards east-southeast of the northern end of Marblehead Neck, is a high bare rock.

(119) **Halfway Rock**, 1.7 miles south-southeast of Bakers Island, about 60 feet high and resembling a sugarloaf, has deep water around it. It is one of the most distinctive marks in the approach to the harbor.

(120) **North Gooseberry Island** and **South Gooseberry Island** are rocky islets on the extensive ledges southward of Bakers Island. **Dry Breakers**, the southerly part of the ledges, show at high water as a low bare ledge. An unmarked 13-foot spot is about 0.3 mile southwestward of South Gooseberry Island.

(121) **Childrens Island**, about 0.8 mile northeastward of Marblehead Neck, is privately owned, bare and more than 0.3 mile long. The island has several houses toward its center and is used by the Marblehead YMCA as a summer day camp. There is a pier and float landing on the southwest side. Extensive ledges, bare and submerged, surround the island.

(122) **Eagle Island**, 0.7 mile north-northeastward of the north end of Childrens Island, is small, grassy and rocky.

(123) **Coney Island**, northward of Marblehead, is a low grassy islet.

(124)

COLREGS Demarcation Lines

(125) The lines established for Salem, Beverly, and Marblehead Harbors are described in **33 CFR 80.120,** chapter 2.

(126)

Channels

(127) Three main channels leading through the islands and rocks at the entrance are known as Salem Channel, Childrens Island Channel and Marblehead Channel. Several other channels of less importance are used only by local boats. Most of the dangers adjacent to, or on the channel edges, are marked. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A.

(128) **Salem Channel**, the deep-draft and most northerly channel, leads westward between Bakers and Great Misery Islands and through **Salem Sound** for about 3 miles, thence southwestward through a dredged section to a turning basin on the northwest side of Salem Harbor. The entrance is marked by **Hospital Point Lighted Range** on bearing **276.3°**. Several buoyed dangers, described later under dangers, are close to the sailing line. Salem Channel is well marked.

(129) **Childrens Island Channel**, the middle one, has its entrance near Halfway Rock. It leads in a northwesterly direction between Childrens Island on the west and Satan Rock, Brimbles and Eagle Island on the east. The least depth in Childrens Island Channel, about 26 feet, is between Eagle Island and Childrens Island.

(130) **Eagle Island Channel**, deep, clear, sheltered and buoyed, leads from Salem Channel in a southwesterly direction between Hardy Rocks and Eagle Island on the northwest and Bakers Island, Pope Head Shoal and Brimbles on the southeast. It is used by most craft bound to Marblehead Harbor from the northeastward.

(131) **Marblehead Channel**, the southwesterly one, leads in a northwesterly direction between Childrens Island and Marblehead Rock, thence northeasterly between Chappel Ledge and Childrens Island, thence northerly between Coney Ledge and Eagle Bar to Salem Sound. Dangers of less than 18 feet in Marblehead Channel are marked except for a 17-foot spot in about 42°29'40"N., 70°49'10"W. **Fifteen-foot Rock**, about midway between Marblehead Rock and Cormorant Rock, is marked by a lighted bell buoy.

(132) **South Channel** leads westerly along the northern shore of the peninsula between Marblehead and Salem Harbors. The channel, fringed by rocks and ledges, is less than 100 yards wide in its narrowest part and is not recommended for strangers drawing more than 10 feet.

(133)

Anchorages

(134) Special anchorages are in Salem, Beverly and Marblehead Harbors and in Bass River. (See **33 CFR 110.1**, **110.25**, and **110.26**, chapter 2, for limits and regulations.) A submerged pipelines extends from Cat Cove to just south of Folger Point. Mariners are advised to exercise extreme caution when transiting or anchoring in this area.

(135)

Dangers

(136) The approaches to Salem, Beverly and Marblehead Harbors have very broken ground, and all of the channels lead between islands and rocks, bare and submerged. Caution is necessary at all times. Strangers should not attempt to enter or leave in thick weather.

(137) An extensive area of rocky patches and reefs, marked by buoys on its northeasterly, easterly and southerly sides, extends about 2 miles southeastward of

Bakers Island, ending with **Newcomb Ledge**, which is covered 20 feet. A lighted whistle buoy is about 0.7 mile east-southeastward of the ledge. In this area are **Searle Rock, Middle Breakers, Southeast Breakers, Inner Breakers** and **Davis Ledge**, all unmarked. Middle and Inner Breakers are partly bare at low water, and all break in heavy weather.

(138) **Hardy Shoal**; of which **Hardy Rocks** uncovers 7 feet, and **Rising States Ledge**, covered 3 feet, are a part; extends from 0.6 to 0.8 mile westward of Bakers Island Light. The shoal is marked on its eastern side by a daybeacon.

(139) **Bowditch Ledge**, 1.2 miles west-northwestward of Bakers Island Light and 300 yards southward of Salem Channel, is marked by a daybeacon 30 feet high with red and white diamond-shaped daymark on a conical granite monument; the daybeacon is very conspicuous when approaching the harbor. The extension of the ledge northeastward is marked by a lighted buoy. **House Ledge**, covered 10 feet, and two ledges, covered 14 and 16 feet, all unmarked, are 0.4 mile eastward, 0.1 mile eastward, and 0.2 mile southeastward of Bowditch Ledge, respectively. **Powers Rock**, covered 14 feet and marked by a lighted bell buoy, is on the northern end of Bakers Island Shoals. These dangers are all south of the Salem Channel range line.

(140) North of Salem Channel, a ledge covered 12 feet and marked by a buoy extends southwestward from Little Misery Island. **Misery Ledge**, covered 18 feet and unmarked, is about 0.5 mile westward of Little Misery Island. **John Ledge**, covered 13 feet and about 0.1 mile westward of Misery Ledge, is marked close to the southward by a lighted buoy.

(141) **Great Haste** is a bare rock surrounded by ledges on the south side of Salem Channel, 2.4 miles westward of Bakers Island. **Little Haste**, close northwestward of Great Haste, is awash at low water and marked by a daybeacon. A 17–foot spot on the northwest end of **Haste Shoal**, about 650 yards north-northwest of the daybeacon, is marked on the north side by a lighted buoy.

(142) **Pope Head** (42°31.7'N., 70°47.8'W.), 300 yards northwestward of North Gooseberry, is a rugged, bare rock surrounded by ledges to a distance of 150 yards. A buoy marks **Pope Head Shoal**, the western extremity of the ledges around the islet.

(143) **Satan Rock**, 0.5 mile east-southeastward of Childrens Island, is a small bare rock marked by a daybeacon. The rock should be given a berth of over 200 yards.

(144) **Brimbles** is a rock awash at low water 0.3 mile south-southeastward of Eagle Island. It is marked by a red and white diamond-shaped daymark on an iron spindle. The daybeacon should be given a berth of over 200 yards.

(145) **Eagle Bar**, an extensive foul ledge and shoal area extending from Eagle Island to the northward, eastward and westward and terminating with **Cutthroat Shoal**, covered 6 feet, on the northeast and **Midchannel Rock** covered 5 feet, on the southwest, is marked on its eastern, southern, and western extremities by buoys.

(146) **Grays Rock**, 0.7 mile northwestward of Childrens Island, is 10 feet high. **Chappel Ledge**, covered 14 feet, about 350 yards eastward of Grays Rock, is marked to the eastward by a lighted buoy. **Coney Ledge**, an extensive ledge extending southeastward from Coney Island, is marked at its easterly end by a buoy.

(147) Islands and rocks, sunken and bare, extend 2 miles southward and southwestward from the south end of Marblehead Neck. Some of these dangers are marked by buoys, and the channels between them are used by local boats, but the area should be avoided by strangers. A lighted bell buoy marks **Outer Breakers**, covered 8 feet, the southeastern end of the broken ground.

(148) Dangers showing above water are **Great Pig Rocks**, bare at high water; **Southwest Breakers**, awash at low water; **Sammy Rock**, awash at low water and marked by a buoy; **Ram Islet**, high, rocky and grassy; **Little Pig Rocks**, awash at high water; **Roaring Bull**, bare at low water and marked by a daybeacon; and **Tinkers Island**, marked by several houses.

(149) A bar with little depth connects Tinkers Island with **Flying Point**, the southern end of Marblehead Neck.

(150) **Tom Moore Rock**, at the eastern end of a reef extending 500 yards eastward from the middle of Marblehead Neck, uncovers 6 feet and is marked by a buoy.

(151) No attempt has been made to describe all of the dangers, shoals, rocks and ledges, as all those known are charted. Most of those in or near the fairways, or near the channel edges, are marked. The chart should be the guide and due caution exercised. Important dangers within the limits of the three harbors will be described where necessary under the description of each individual harbor and its facilities.

(152) A buried high-pressure natural gas pipeline extends along the sea floor from south of Woodbury Point in Beverly eastward along the bottom into Salem Sound and then southeast into Broad Sound. Mariners are advised to use extreme caution when anchoring in these areas.

(153)

Current

(154) The tidal current in Salem and Marblehead Harbors has little velocity. In Beverly Harbor it has considerable velocity and sets across the channel in places. During the first half of the ebb the current sets across the shoal extending northeastward from Monument Bar.

(155)

Ice

(156) The head of Salem Harbor on the flats usually is closed by ice every winter during January and February, but the formations rarely extend beyond the coal pier except in unusually severe winters. Northerly and northwesterly winds are most favorable to local formation in Salem Harbor.

(157) Winds from southward and westward, during light formation, have a tendency to carry the ice off to sea, while those from eastward usually break up the formation both in the harbor and its approaches.

(158) Ice rarely obstructs Marblehead Harbor to such an extent as to hinder navigation. Fishermen have made it a refuge when it was impossible to get into Gloucester, Salem or Lynn Harbors. The formation of ice in Marblehead is entirely local, and it remains only a short time.

(159)

Pilotage, Salem Harbor, Beverly Harbor and Marblehead Harbor

(160) Pilotage is compulsory for all foreign vessels and for U.S. vessels under register in the foreign trade that draw over 7 feet. Pilotage is optional for coastwise vessels who have on board a pilot licensed by the federal government, but most deep-draft vessels take a pilot. The pilot usually boards in the vicinity of Eastern Point Lighted Whistle Buoy 2 (42°34'14"N., 70°39'50"W.). See Pilotage, Gloucester, earlier this chapter, indexed as such, for additional information.

(161)

Towage

(162) Tug assistance is usually needed for deep-draft vessels. The nearest tugs are based at Boston.

(163)

Quarantine, customs, immigration and agricultural quarantine

(164) (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(165) **Quarantine** is enforced in accordance with the regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.)

(166) Salem is the **customs port of entry** for Manchester, Salem, Beverly, Danversport, Marblehead, Peabody and Lynn.

(167)

Harbor regulations

(168) Moorings in the three harbors are under supervision of their respective harbormasters. A **speed limit** of 5 miles per hour is enforced within the limits of the harbors by the harbormasters and by police patrol boats.

(169)

Wharves

(170) An oil storage terminal is in operation at Salem Terminal. Other wharves are in ruins or in various stages of disrepair and disuse. Several fish wharves in Salem are in operation.

(171)

Supplies

(172) Fuel, water, provisions, ice and marine supplies can be obtained at the ports.

(173)

Repairs

(174) There are machine shops at Beverly and Salem that can carry out repairs. There are no drydocks; the nearest is at Boston. There are numerous yacht and boatyards which do repairs, haul out and store. The largest of the marine railways in the harbors, located at Salem, can haul out craft up to 100 feet in length. Dry covered and open winter storage is available.

(175) **Beverly Harbor** is north of Salem Neck at the western end of Salem Sound and is formed by the confluence of Danvers River, Bass River and North River. It forms the approach to the city of Beverly, a manufacturing and resort city on the north side of the harbor just inside the entrance.

(176)

Channels

(177) **Beverly Channel** leads from Salem Sound to **Tuck Point** and the marinas on the north side just below the State Route 1A Highway bridge. The channel is buoyed, and most of the dangers are marked. Monument Bar and Lobster Rocks, both on the south side of the channel, are marked by daybeacons.

(178) An unmarked channel, known as **Rams Horn Channel**, is entered just westward of Monument Bar Daybeacon. The channel leads southward from Beverly Channel to an anchorage area and **Collins Cove**. Shoaling of unknown extent has been reported just inside the channel entrance. Collins Cove is mostly bare and seldom used. **Rams Horn Rock**, marked by a daybeacon, is on the southeast side of the channel. An overhead power cable across Collins Cove, about 0.2 mile above the entrance, has a clearance of 50 feet.

(179)

Anchorages

(180) Special anchorages are in Beverly Harbor. These anchorages are on **Monument Bar**, north of Salem Neck, in Collins Cove, in the southwestern part of the harbor, and in **Beverly Cove** and **Mackerel Cove**, in the northern part of the harbor. (See **33 CFR 110.1**, and **110.25**, chapter 2, for limits and regulations.) A **pipeline area** extends from Tuck Point to Salem Neck; care should be taken to avoid anchoring in this area.

(181)

Harbor regulations

(182) The harbor is patrolled, and a **speed limit** of 5 miles per hour is enforced.

(183) The **harbormaster** has an office at the town pier just east of the State Route 1A bridge.

(184)

Pilotage, Beverly

(185) For Pilotage at Beverly and vicinity see Pilotage, Gloucester, indexed as such, this chapter.

(189)

Structures across Danvers River

Name•Description•Type	Location	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
Route 1A Bridge (fxed)	42°32'21"N., 70°53'15"W.	104	49	
Overhead power cable	42°32'21"N., 70°53'17"W.		85	
MBTA/Amtrack Bridge (swing)	42°32'21"N., 70°53'15"W.	40	10	Notes 1 and 2 Call sign WRD-626 Bridge is under construction.
Kernwood Avenue Bridge (swing)	42°32'35"N., 70°53'54"W.	50	8	Notes 1 and 2 Call sign WRD-625

Note 1 – See 33 CFR 117.1 through 117.59 and 117.595, chapter 2, for drawbridge regulations.
Note 2 – Bridgetender monitors VHF-FM channel 16 and works on channel 6.

(186)

Small-craft facilities

(187) Jubilee Yacht Club at Tuck Point had a reported alongside depth of 14 feet in 2006. Gasoline, diesel fuel, water, ice and a few moorings are available. A small-craft launching ramp is on Tuck Point.

(188) Several marinas with depths of 10 to 20 feet alongside their floats are on the north bank east of the State Route 1A Highway bridge. These facilities can provide gasoline, diesel fuel, water, ice, marine supplies, pumpout facilities, provisions and guest moorings. Mobile hoists up to 35 tons can handle craft up to 55 feet in length for hull or engine repairs or dry open or covered winter storage; electronic repairs can be made.

(190)

Danvers River

(191) **Danvers River**, the continuation of Beverly Harbor northwestward, has a depth of about 7 feet for 2 miles above Beverly to **Danversport**. The channel, buoyed for about 1.5 miles to the mouth of Waters River, is narrow and leads between flats that uncover at low water. A marina with a pier and float landing is on the south bank of the river just east of the Kernwood Avenue Highway bridge; depths of 6 to 7 feet are alongside the floats. A small-craft launching ramp and water are available. State Route 35 highway bridge at Danversport has a fixed span and is the head of navigation on Waters River, which is shoal above the bridge.

(192) **Crane River** empties into the west side of Danvers River, about 0.3 mile above the entrance to Waters River. A privately dredged channel leads from the entrance to a mooring basin and public marina at the head.

(193) **Porter River**, a northwesterly tributary of the Danvers River, has its entrance just eastward of the entrance to Crane River. A privately dredged channel leads from the entrance for about 0.4 mile to a mooring basin just below the State Route 35 fixed highway bridge, the head of navigation. In 2008, the controlling depth was 4 feet in the river.

(194)

Small-craft facilities

(195) There are several small-craft facilities on Porter River. Marine railways to 38 tons, lifts to 50 tons, marine supplies, storage facilities and other services are available; hull and engine repairs can be made. The Danversport Yacht Club at the entrance and west side of the river has a reported depth of 5 feet alongside its floats. Transient berths, electricity, gasoline, water, a pump-out station and other services are available.

(196)

Harbor regulations

(197) The **harbormaster** for Danversport maintains an office on the west side of Porter River at the town landing immediately below the highway bridge; all moorings are under his control. The harbormaster can be reached by VHF radio on channels 13 and 16.

(198)

North River to Marblehead

(199) **North River**, a tributary of Danvers River entering from southward just above the bridges, is nearly bare at low water in a narrow, unmarked channel that is seldom used. A boatyard with a marine railway that can haul out craft up to 40 feet in length for minor repairs or winter storage is on the west side of the river just below State Route 114 highway bridge at the head of navigation. Overhead power cables crossing the mouth of North River have clearances of 75 feet, and where they cross Collins Cove they have a clearance of 50 feet. The towers are conspicuous from seaward.

(200) **Bass River** empties into the north side of Danvers River opposite the entrance to North River. A depth of about 6 feet can be carried to the Hall-Whitaker highway swing bridge, about 0.7 mile above the entrance, thence about 4 feet to the yacht club 0.3 mile above the bridge. The channel leads between flats bare at low water and is most easily followed at that stage. In 1978, an unknown obstruction was reported on the east edge of the channel about 0.2 mile above the entrance. Buoys mark the channel to a point about 300 yards below the bridge. The bridge has a swing span with a width of 40 feet and a clearance of 5 feet. (See **33 CFR 117.1** through **117.59**

and **117.588**, chapter 2, for drawbridge regulations.) A 24-hour advance notice is required to open the hand-cranked swing span. Requests should be made to the Massachusetts Department of Public Works in Danvers; telephone, 617–774–3190. An overhead power cable on the south side of the bridge has a clearance of 48 feet.

(201)

Anchorages

(202) A **special anchorage** is about 300 yards northward of the bridge. (See **33 CFR 110.1** and **110.25**, chapter 2, for limits and regulations.)

(203) A private yacht club is about 0.3 mile above the bridge, and a ramp is just above the bridge; both are on the east bank of the river. There are no services, except a service station near the ramp. The boatyard builds or hauls out for repairs or storage craft up to 35 feet in length.

(204) **Salem Harbor** is about 1.5 miles long in a southwesterly direction. Dominion Energy's power plant dominates the harbor. The city of **Salem** is on the western side of the harbor. The principal industries are tourism and electrical production. The city has many points of historical interest, including museums devoted to maritime subjects. Waterborne commerce is principally in petroleum, coal, seafood products and tourism. Salem Harbor is approached from the northward through a dredged section of Salem Channel.

(205)

Anchorages

(206) A special anchorage is in the south and east parts of Salem Harbor and extends eastward along the south side of South Channel to Peachs Point. (See **33 CFR 110.1** and **110.25(c)**, chapter 2, for limits and regulations.) The harbor is sometimes used as a harbor of refuge, especially during the autumn. Good anchorage is available in what is known as the outer anchorage southward of the main channel and northward and eastward of Little Haste.

(207)

Pilotage, Salem

(208) For Pilotage at Salem and vicinity see Pilotage, Gloucester, indexed as such, this chapter.

(209) On the western side of the dredged approach is Salem Neck. **Salem Willows Park**, the Salem Willows Yacht Club and a 400-foot public pier are on the northeastern extremity of Salem Neck. Excursion and party fishing boats operate from the three landings in the summer. Rowboats can also be hired. Depths of 4 feet are reported alongside the yacht club floats; gasoline and water are available. Reported depths of 10 feet are at the head of the public pier and 3 to 5 feet at the floats on each side of it.

(210) **Juniper Point**, the eastern extremity of the neck, has many summer homes.

(211) **Juniper Cove**, a shallow foul cove which is mostly dry at low water, lies between Salem Neck and **Winter Island**. A boat and yacht yard at the head of the cove can handle hull and engine repairs. In summer, small craft moor in the entrance to the cove between **Abbot Rock**, marked by a daybeacon, and Juniper Point. The cove is open to easterly weather, but the holding ground in the entrance is reported to be good.

(212) A private light is shown from a white conical tower on the southeastern point of Winter Island. A ramp, pier and seasonal floats are located on the southern half of the island at Winter Island Maritime Park.

(213) **Great Aquavitae**, a shoal area on the east side of the dredged section of Salem Channel, is marked by a daybeacon.

(214) Conspicuous on Salem Neck are the oil tanks of the Salem Terminal and the three stacks of the power plant adjacent to the north. The stacks are lighted.

(215) A marina is about 300 yards southwestward of Salem Terminal Wharf. Depths of 5 to 10 feet are reported alongside the floats; water, electricity, ice, storage facilities and hull and engine repairs are available.

(216)

Wharves

(217) **Dominion Energy, Salem Harbor, L.C.C. Power Plant Wharf** (42°31'27"N., 70°52'40"W.): 700-foot face; 900 feet of berthing space with dolphins; 35 feet alongside; deck height, 14 feet; pipelines extend to storage tanks with 650,000-barrel capacity; receipt and shipment of fuel oil and coal; owned and operated by Dominion Energy, L.C.C.

(218) **Derby Wharf**, on the west side of Salem Harbor about 0.4 mile southwestward of Salem Terminal Wharf, is a stone jetty about 0.3 mile long, marked at its outer end by a private light. The wharf and the old customhouse on its shore end are now part of the Salem Maritime National Historic Site under the U.S. Department of the Interior.

(219) A lighted buoy marks the entrance to two dredged channels leading along the east and west sides of Derby Wharf, respectively. The northerly channel leads to an anchorage basin on the east side of the wharf. In 1993, the channel had a controlling depth of 7 feet with 1 to 8 feet available in the basin except for shoaling to bare along the northern and northeastern sides. A public float is midway along the wharf; moorings are available in the basin. The southerly channel, on the west side of the wharf, leads to South River.

(220) **South River**, a short estuary extending into the city of Salem, has its entrance through a dredged channel leading along the west side of Derby Wharf.

(221) On the west side of South River, large industrial buildings and warehouses are conspicuous. The fixed bridge across South River has a 43-foot span with a vertical clearance of 7 feet. A pumpout facility is at the Congress Street Landing. Gasoline, water and most services are available at the floats of the marina on the north bank. It has been reported that navigation through the Congress Street Bridge is hazardous during periods of maximum flood and ebb.

(222) A channel south of Long Point and marked by private buoys leads westward from the South River entrance channel to a yacht club in **Palmer Cove.**

(223) The southwestern part of Salem Harbor is shoal and at the head particularly foul, southward of Pickering Point.

(224) South of **Palmer Point**, a large yacht yard with marina has a marine railway, the largest in the area, that can haul out craft up to 100 feet in length. A privately dredged 8-foot channel, marked by piles, leads to an anchorage basin off the yard. Water, open and covered winter storage, overnight berthage with electricity and most services are available at the yard. The yard maintains guest moorings.

(225) **Marblehead Harbor**, 1 mile long and 700 yards wide, is formed on the east and south by Marblehead Neck and **Back Beach**, a narrow strip beach on the south side of the causeway and seawall connecting the south end of Marblehead Neck with the mainland. Marblehead Light marks the easterly point of the entrance. Marblehead Harbor is an excellent anchorage used mostly by yachts during the summer. The anchorage is reported uncomfortable for yachts when the wind is northeast.

(226) The depths in the harbor up to **Skinner Head** are from 21 to 30 feet, with the exception on the east side, northerly of **Boden Point**, where **Boden Rocks** are covered 9 feet.

(227) Southward of Skinner Head, the depths shoal gradually with 8 feet available to within about 300 yards of the head and east side. It is reported that this part of the harbor is extremely uncomfortable for small craft during a northeaster.

(228)

Anchorages

(229) A **special anchorage** occupies most of Marblehead Harbor. (See **33 CFR 110.1** and **110.26**, chapter 2, for limits and regulations.)

(230)

Harbor regulations

(231) The harbor, often referred to as the "yachting capital of the world," is somewhat congested during the summer. The harbormaster reports that there are about 2,300 moorings in the harbor.

(232) The moorings are under the supervision and control of the **harbormaster**, who issues permits for them. The harbor is patrolled by a police boat, and a **speed limit** of 5 miles per hour is enforced. The harbormaster may be contacted through the local police department and may, on request, direct a stranger to a mooring or a quiet spot for anchoring.

(233) No directions are deemed necessary for entering. The chart should be the guide. All known dangers are charted, and most of them are marked. Some difficulty may be experienced, once in the harbor, in finding a mooring or good swinging room if the harbor is congested. In that case, the harbormaster may be of help.

(234) **Marblehead**, a combined business and residential community on the west side, and **Marblehead Neck**, all residential, on the east side of the harbor, are important summer resorts.

(235) There are numerous float landings on both sides of the harbor at which there are depths of 6 to 25 feet. Overnight berthing, as a rule, is not permitted at any of the service or yacht club floats.

(236) There are six yacht clubs on the harbor: Eastern, Corinthian and Pleon on the east side; and Boston, Dolphin and Marblehead along the west side. All have facilities to a varying degree, either private or available to visiting yachtsmen.

(237)

Small-craft facilities

(238) There are many small-craft facilities in the harbor. Wood, steel or fiberglass, sail or motor craft up to 65 feet in length can be built at Marblehead. There are two public float landings and two small-craft launching ramps usable at half tide or better; pumpout facilities are available.

(239) Lodging, restaurants, markets, a hospital and churches are in Marblehead; bus and taxi services are available.

(240)

Phillips Point to Revere

(241) **Phillips Point**, 3.5 miles southwestward of Marblehead Light, is 50 feet high and rocky with woods and large homes along its shore. A rock ledge covered 12 feet is 600 yards eastward of Littles Point, the eastern part of Phillips Point. A fishing net extends eastward from Littles Point during the summer. A reef with bare heads extends 350 yards southward from Phillips Point. **Dread Ledge**, 500 yards southward of the point, uncovers 5 feet and is marked by a daybeacon.

(242) **Nahant Bay** is 2 miles wide between Phillips Point and Nahant. Temporary anchorage, exposed to easterly and southerly winds, can be had in the bay in 18 to 36 feet but is seldom used. The usual anchorage is off Swampscott, northwestward of Lincoln House Point. Many small craft moor here in the summer. Several mooring buoys, used by local craft, are in the cove westward of the point, off **Fishermans Beach**. A 440-foot pier with 50 feet of floats at the end extends off the beach; there are depths of 5 feet at the floats, but no services. The Swampscott Yacht Club is at the shore end of the pier. A **harbormaster**, who maintains an office at the yacht club, supervises the moorings.

(243) There is a public small-boat ramp with parking space adjoining the pier. Water and guest moorings are available. Gasoline can be obtained in cans from a nearby service station. Ice, provisions and marine supplies are available from the nearby markets and stores.

(244) Nahant Bay is mostly clear. **Lincoln House (Fishing) Point, Blaney Rock** and **Red Rock** are rocky points on the northern side of Nahant Bay. A dangerous submerged rock, marked on its south side by a buoy, is about 125 yards south of Lincoln House Point. Two other rocks, covered 16 feet and 18 feet, are about 400 yards

southward and about 700 yards south-southwestward of Lincoln House Point, respectively.

(245) The town of **Swampscott** is on the northern shore of Nahant Bay. A large blue standpipe and the school cupola are conspicuous. A church spire, lighted at night, on the shore drive at Red Rock, and an observatory, about 0.9 mile northwestward of Red Rock, are also prominent.

(246) **Long Beach** is a narrow strip of sand about 1 mile long in a southerly direction separating Nahant Bay from Lynn Harbor. **Little Nahant**, a high grassy head with many houses, is joined to Nahant by **Short Beach**, a strip of beach 0.4 mile long. The white buildings and signal tower of the former Coast Guard station on Short Beach are just southward of Little Nahant.

(247) **Egg Rock**, 60 feet high and bare, is on the southern side of the entrance to Nahant Bay. It is a bird sanctuary.

(248) **Nahant** is a high peninsula about 1.5 miles long with bluff seaward faces. The town of Nahant is connected to Lynn by a highway. Among the most prominent objects are four nearly identical concrete observation towers; two are on the outer end of the peninsula just north of **Pea Island**, one is about 0.4 mile to the northward, and the fourth is on the west side of the harbor. The top section of the southerly of the two towers just north of Pea Island is topped with a wind break painted red and reportedly is especially prominent. A former military installation, now used by Northeastern University as a marine research facility, is on **East Point**, the easternmost point of the peninsula. A 60-foot windmill on East Point is reported to be prominent.

(249) **Nahant Harbor** is the bight on the south side of Nahant. On entering between **Joe Beach Ledge**, covered 3 feet and marked by a buoy, and **The Spindle (Bass Rock)**, awash and marked by a buoy, select temporary anchorage off the wharf in 18 to 24 feet, hard bottom. The town wharf on the east shore at the head has about 6 feet alongside the float landing; a hard-surface launching ramp for small boats is on the north side of the shore end of the wharf. Water is available at the float. There is a sailing club on the wharf, and a **harbormaster** controls the moorings. The Boston pilot boats land and pick up pilots at this wharf and maintain a mooring off the wharf.

(250) **Shag Rocks** are bare rocks extending 300 yards southward from the southeast end of Nahant. A ledge, awash at lowest tides, extends 100 yards southward from the southernmost Shag Rock. A lighted buoy is south of the ledge.

(251) **Broad Sound**, about 4 miles wide between Nahant on the northeast and Deer Island on the southwest, forms the approach to Nahant and the city of Lynn at its north end, and the northern approach to Boston Harbor at its south end. It has depths of 18 to 48 feet in the entrance but is shoal near the shores. A foul area with submerged rocks, some covered ½ foot, is offshore on the west side of the sound, about 0.9 mile east-northeast of the tower at Revere Beach. Pilings, awash and unmarked, about 300 yards northeastward of the foul area, are in 42°25'09"N., 70°57'48"W.

(252) **Lynn Harbor**, the northerly end of Broad Sound, is mostly sand and mudflats that largely bare at low water and through which a channel has been dredged to the city of Lynn, an industrial community of major importance. In 1979, it had little waterborne commerce. Of the many diversified industries, a large General Electric plant, which manufactures airplane engines and electronic products, is the largest employer. In 1979, several of the wharves were being used as marinas, and the remainder were not in general use or were in ruins or disrepair.

(253)

Prominent features

(254) Landmarks in approaching Lynn Harbor are the standpipe on Winthrop Head, two apartment houses on Winthrop Highlands, the observation towers at Nahant, a white tank at the head of the harbor, and the radio towers on the Saugus River and the Pines River.

(255)

Channels

(256) A dredged channel leads from Broad Sound, at a point about 0.8 mile westward of Bass Point, to a turning basin at the head of Lynn Harbor. A privately maintained subsidiary channel leads from the basin in a southwesterly direction to the wharf of a powerplant. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A.

(257) **Black Rock Channel**, a branch channel leading along the western side of Nahant, is unmarked and suitable only for small craft. Sand flats, bare or nearly bare at low water, are on each side. In 1979, it was reported that the channel had shoaled to 5 feet but that it was still being used by lobstermen and small craft.

(258) **Western Channel**, westward of the main channel to Lynn Harbor, leads from Broad Sound to the General Edwards Bridge and the **Saugus River**. **Pines River** is entered from Saugus River just westward of the bridge.

(259) Western Channel is marked by buoys to just below the General Edwards Bridge. Dangerous rocks awash, on the northern side of the channel, extend about 200 yards south-southeastward from a point on the north shore about 200 yards below the General Edwards Bridge; the southern extremity of the rocks is marked by a buoy. Above the bridge, the channel is unmarked and local knowledge is advised.

(260)

Anchorages

(261) Small craft moor off the marina and boatyard north of Bass Point and eastward of the turning basin at the head of Lynn Harbor. (See **33 CFR 110.1** and **110.27**, chapter 2, for limits and regulations.) In Saugus River, some moor off the yacht club close eastward of General Edwards Bridge. Above the bridge they moor on the west side of the channel and above Fox Hill Bridge in the channel wherever space permits. In Pines River

(266)

Structures across Saugus River

Name•Description•Type	Location	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
General Edwards Bridge (bascule)	42°26'37"N., 70°57'58"W.	100	27	Notes 1 and 2
Overhead power cable	42°26'42"N., 70°58'14"W.		85	
MBTA Railroad Bridge (bascule)	42°26'50"N., 70°58'19"W.	50	7	Note 2
Fox Hill Bridge (bascule)	42°27'03"N., 70°58'44"W.	40	6	Note 1 Bridge is under construction

Note 1 – See 33 CFR 117.1 through 117.59 and 117.618, chapter 2, for drawbridge regulations.
Note 2 – Bridgetender monitors VHF-FM channel 16 and works on channel 13.

they moor in the narrow channel. Moorings are under control and supervision of the harbormaster, who can be contacted through the local police department. Depths in the anchorages are off Bass Point from 5 to 30 feet, at the head of Lynn Harbor from 7 to 17 feet, and in Saugus and Pines Rivers from 3 to 9 feet.

(262) Some local knowledge or assistance will be needed in finding swinging room or a vacant mooring off the yacht clubs at the head of Lynn Harbor, as this area is usually heavily congested with small craft.

(263) Many small craft moor on the east side of the harbor near Bass Point, and the boatyard there maintains about 150 moorings, with usually a few vacant.

(264)

Dangers

(265) The principal dangers in the approach to Lynn Harbor are **Flip Rock**, covered 14 feet and marked by a gong buoy, 0.6 mile south of The Spindle; **Nahant Rock**, covered 19 feet and marked by a buoy, 0.8 mile southwest of Bass Point; and two unmarked shoals, covered 15 and 19 feet, about 0.7 mile northward of Nahant Rock.

(267)

Current

(268) The average velocity at strength of the tidal current at Lynn Harbor entrance is 0.5 knot.

(269)

Harbormasters

(270) There is a harbormaster at Lynn and one at Saugus. The former can be contacted through the Volunteer Yacht Club at Lynn; the latter through the Saugus Police Department. They supervise and control the moorings.

(271)

Wharves

(272) In 1979, only the wharf of the General Electric plant on Saugus River was in periodic use. Depths of 9 feet were reported alongside. A development plan for the greater use of the Lynn waterfront was being implemented in 1979.

(273)

Supplies and repairs

(274) There are no bunkering facilities, drydocks, or major repair facilities for large vessels at Lynn. The nearest drydocks or repair facilities for large vessels are at Boston. Marine supplies, provisions and machine shops are available in the city.

(275)

Small-craft facilities

(276) There are several marinas, boatyards, and private yacht clubs at Lynn and on the Saugus and Pines Rivers. Most of these facilities have gasoline, marine supplies, storage and berthing facilities, water, ice and launching ramps. The facilities on Pines River can make hull and engine repairs; an 18-ton mobile hoist and a 50-foot marine railway are available. Pump-out facilities are available at Seaport Landing Marina at the head of Lynn Harbor and at Revere through the harbormaster.

(277) The yacht clubs include the Bass Point Boat Club, on Bass Point; the Lynn and Volunteer Yacht Clubs at the head of the harbor; the Point of Pines Yacht Club, on the south side of Saugus River, just eastward of the General Edwards Bridge; the Fox Hill Yacht Club, near the Fox Hill Bridge; and the Saugus River Yacht Club, about 0.7 mile above the Fox Hill Bridge. The clubs all have berthing facilities, guest moorings and other services available to members and guests.

(278) Lynn has railroad and bus services, a hospital, hotels, banks, churches and many other conveniences.

(279) **Revere** is a city and summer beach resort on the west side of Broad Sound. At the southern end of **Revere Beach**, a breakwater extends out from the shore on **Cherry Island Bar**, forming an anchorage for small craft in 3 to 4 feet of water, but it is exposed. Parts of the breakwater are covered at high water. A pumpout facility is in Revere.

(280) Westward of the breakwater are the ruins of a 600-foot-long pier. A shorter pier to the eastward of the ruins dries at low water.

(281) **Winthrop Highlands**, about 0.8 mile southward of Cherry Island Bar, has two conspicuous apartment houses just southward. The structures are about 100 feet high.

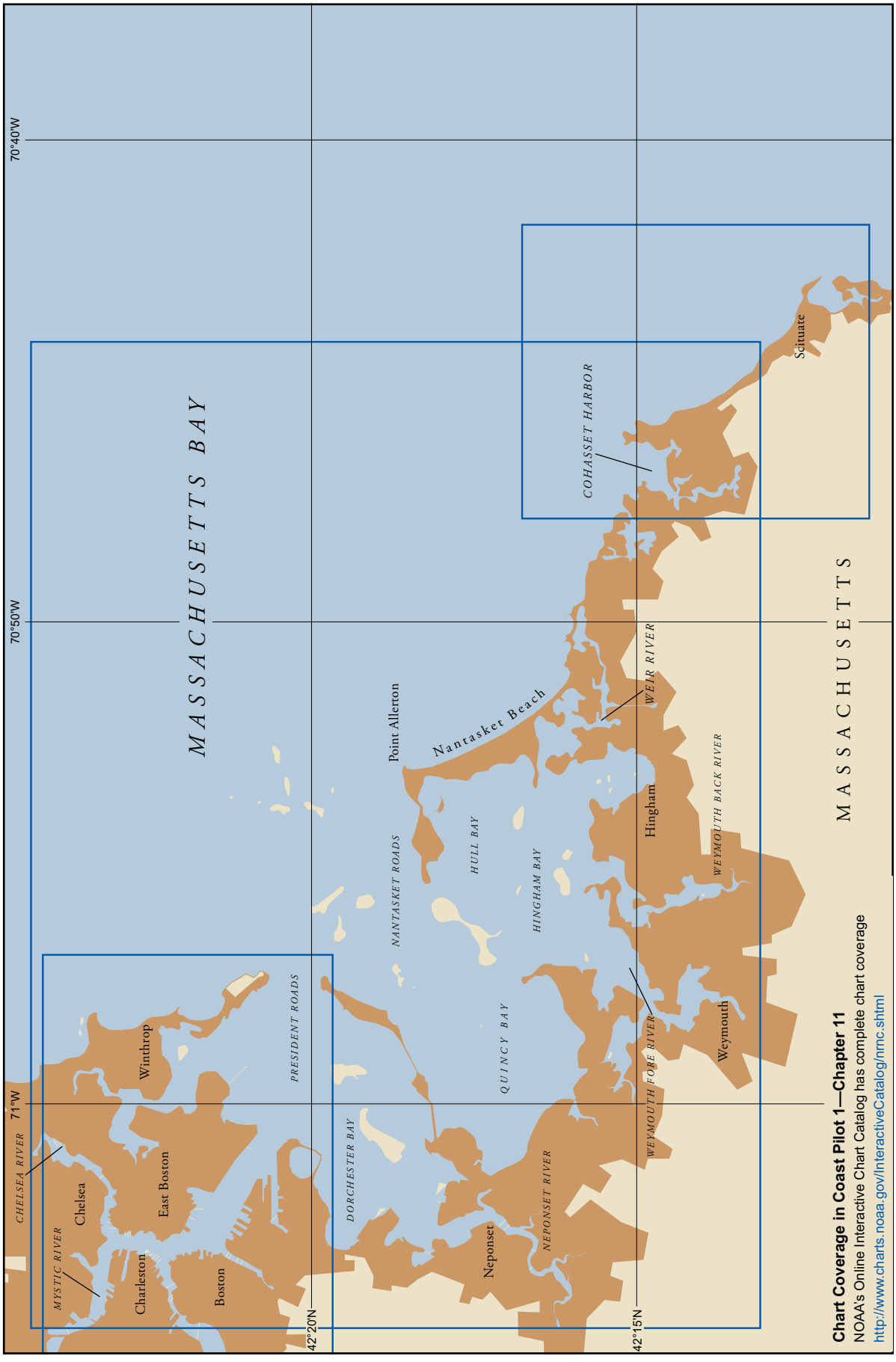


Chart Coverage in Coast Pilot 1—Chapter 11
NOAA's Online Interactive Chart Catalog has complete chart coverage
http://www.charts.noaa.gov/InteractiveCatalog/nrnc.shtml

Boston Harbor and Approaches

(1) This chapter describes Boston Harbor, its approaches and tributaries, and the major commercial facilities in the port of Boston. The more important tributaries include Charles, Chelsea, Mystic and Weymouth Fore Rivers and Dorchester and Hingham Bays.

(2)

COLREGS Demarcation Lines

(3) The lines established for Boston Harbor are described in **33 CFR 80.130**, chapter 2.

(4)

No-Discharge Zone

(5) The State of Massachusetts, with the approval of the Environmental Protection Agency, has established a No-Discharge Zone (NDZ) in all coastal waters of Massachusetts described in this volume, extending about 3 miles offshore.

(6) Within the NDZ, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZ, discharge of sewage is regulated by **40 CFR 140** (see chapter 2).

(7)

Boston Harbor

(8) **Boston Harbor**, the largest seaport in New England, includes all the tidewater lying within a line from the southern extremity of Deer Island to Point Allerton, about 4 miles to the southeastward. Numerous dangers lie in the approaches to the harbor. The northeastern approach is obstructed by islands and shoals that extend 4 miles from the entrance; between them are the dredged channels which lead into the harbor. In the southeastern approach, broken ground extends as much as 3 miles from shore. The approaches are marked by a number of powerful lights, and the principal dangers are buoyed.

(9) **Traffic Separation Scheme (Boston)** has been established in the approach to Boston Harbor; see charts for limits.

(10) The Scheme is composed basically of **directed traffic lanes** each with one-way inbound and outbound traffic lanes separated by a **defined separation zone** and two **precautionary areas.** The Scheme is recommended for use by vessels approaching or departing from Boston Harbor but is not necessarily intended for tugs, tows or other small vessels which traditionally operate outside of the usual steamer lanes or close inshore.

(11) The Traffic Separation Scheme has been designed to aid in the prevention of collisions at the approaches to major harbors but is not intended in any way to supersede or alter the applicable Navigation Rules. Separation zones are intended to separate inbound and outbound traffic lanes and to be free of ship traffic and should not be used except for crossing purposes. Mariners should use extreme caution when crossing traffic lanes and separation zones. (See **33 CFR 167.1** through **167.15** and **167.75** through **167.77**, chapter 2, for limits and regulations and Traffic Separation Schemes, chapter 1, for additional information.)

(12) A **precautionary area** is at the junction of Traffic Separation Scheme (Boston) and the Eastern Approach Off Nantucket to Traffic Separation Scheme Off New York. (See U.S. Coast Pilot 2, Atlantic Coast, Cape Cod to Sandy Hook, for a description of Traffic Separation Scheme Off New York. Consult the charts for the Off New York Scheme. The precautionary area is bounded on the east by a circle with a radius of 15.5 miles centered in

(13) 40°35'01"N., 69°59'58"W. and intersected by the Traffic Separation Schemes at points in

(14) 40°23'45"N., 69°13'57"W. and

(15) 40°50'28"N., 68°58'40"W., and is bounded on the west by a line connecting the schemes at points in

(16) 40°36'46"N., 69°15'08"W. and

(17) 40°48'02"N., 69°02'57"W.

(18) The **precautionary area** in the approach to Boston Harbor has a radius of 6.17 miles centered on Boston Lighted Whistle Buoy B (42°22'42"N., 70°46'58"W.), excluding that area of the circle bounded by an imaginary line extending between the outer limits of the inbound and outbound traffic lanes.

(19) The **separation zone** is a 1-mile zone centered in the following positions:

(20) (i) 42°20'44"N., 70°39'04"W.

(21) (ii) 42°18'17"N., 70°01'08"W., and

(22) (iii) 40°49'15"N., 69°00'49"W.

(23) **Inbound traffic lane** is a 1.5-mile-wide lane with a length of about 129 miles. Entering the traffic lane at a point in about 40°50'00"N., 68°59'30"W., a course of **333°** for about 100 miles, thence a course of **275°** for about 29 miles follows the centerline of the traffic lane to the junction with the precautionary area.

(24) **Outbound traffic lane** is a 1.5-mile-wide lane with a length of about 127 miles. Entering the traffic lane at a point in about 42°19'30"N., 70°39'50"W., a course of **95°** for about 28 miles, thence a course of **153°** for about 99 miles follows the centerline of the traffic lane to the junction with the precautionary area; thence steer usual courses to destination.

(25) The Traffic Separation Scheme is buoyed.

(26) The area around the Traffic Separation Scheme (Boston) is an important feeding ground for North Atlantic right whales. All vessels 65 feet or greater in length overall (LOA) and subject to the jurisdiction of the United States are restricted to speeds of 10 knots or less in three seasonal management areas. The first, near Great South Channel, is defined as the waters bounded by:

(27) 42°30'N., 69°45'W.;

(28) 42°30'N., 67°27'W.;

(29) 42°09'N., 67°08'24"W.;

(30) 41°00'N., 69°05'W.;

(31) 41°40'N., 69°45'W.; thence back to starting point and the restrictions apply between April 1 and July 31. The second area, off Race Point, is defined as the waters bounded by:

(32) 42°04'56.5"N., 70°12'W.;

(33) 42°12'N., 70°12'W.;

(34) 42°12'N., 70°30'W.;

(35) 42°30'N., 70°30'W.;

(36) 42°30'N., 69°45'W.;

(37) 41°41'N., 69°45'W.; thence due west to the shore and the restrictions apply between March 1 and April 30. The final area is defined as all waters of Cape Cod Bay with a northern boundary of 42°41'56.5"N., 70°12'W., to 42°12'N., 70°12'W., thence due west back to shore and the restrictions apply between January 1 and May 15. (See **50 CFR 224.105** in chapter 2 for regulations, limitations and exceptions.)

(38) Approaches to Boston Harbor lie within the WHALESNORTH Mandatory Ship Reporting Area. Each self-propelled ship of 300 gross tons or greater entering WHALESNORTH must participate in the Mandatory Ship Reporting System (See **33 CFR 169**, chapter 2, for limits and regulations, and chapter 3 for sample reports). Sovereign immune vessels are exempt from the requirement to report but are encouraged to participate.

(39)

Regulated Navigation Areas

(40) Safety and security zones, no-anchoring areas and areas to be avoided are in the vicinity of the Neptune Deepwater Port and the Northeast Gateway Deepwater Port located northeast and east of the Traffic Separation Scheme (Boston) precautionary area in the approach to Boston Harbor. (See **33 CFR 150.940** and **165.117**, chapter 2, for limits and regulations.)

(41)

Area to be avoided

(42) In order to significantly reduce the risk of ship strikes to the North Atlantic right whale, an area to be avoided was established in the Great South Channel, east of the Boston Harbor traffic lanes. Ships of 300 gross tons and above should avoid the area bounded by lines connecting the following geographical positions:

(43) 41°44'08"N., 69°34'50"W.;

(44) 42°10'00"N., 68°31'00"W.;

(45) 41°24'53"N., 68°31'00"W.; and

(46) 40°50'28"N., 68°58'40"W. between the period of April 1 through July 31.

(47)

Prominent features

(48) **Boston Lighted Whistle Buoy B** (42°22'42"N., 70°46'58"W.) is about 7.8 miles east-northeastward of Deer Island. The buoy is equipped with a sound signal and racon. (See Light List for details of operation.)

(49) Conspicuous to a vessel approaching Boston Harbor from northeastward is the tall red, white and blue standpipe on Winthrop Head. From eastward, the most prominent island in the entrance is Great Brewster. On the south side of the entrance, a turreted tower is conspicuous on Point Allerton; also prominent are the tank and standpipe on Strawberry Hill. Two miles south of Point Allerton are two radio towers which are illuminated at night.

(50) The outstanding landmarks in the city of **Boston** are the John Hancock Building, the Prudential Building, the bridge over Mystic River, the control tower at Logan International Airport, the pointed tower of the customhouse and a large gas tank in Everett. Also prominent is the John F. Kennedy Federal Building in Boston.

(51) **Security Broadcast System, Boston Harbor**. In conjunctionwith various maritime interests, the Coast Guard has developed a system of recommended radiotelephone procedures for Boston Harbor that is designed to supplement the Vessel Bridge-to-Bridge Radiotelephone Regulations (33 CFR 26) (see chapter 2). These voluntary procedures consist of security calls to be made by vessel masters, pilots or operators on VHF-FM channel 13 at designated points. The procedures are designed to give notice of unseen vessels, give notice of intended movement, clear channel 13 of traffic unrelated to navigation and give each vessel information on all others in the immediate vicinity and to do so at little cost and with as little radiotelephone traffic as possible. These recommendations do not relieve a master, pilot or operator of any requirements of law or regulation. There is no guarantee that every vessel will follow them. These security calls are also repeated on VHF FM channel 16.

(52) Vessels inbound for Boston should make Security calls when boarding a pilot (1.5 miles east of Boston Approach Lighted Buoy BG at 42°23'27"N., 70°51'29"W.), when entering Boston North Channel at Entrance Lighted Gong Buoy NC, in President Roads abreast Deer Island, in the vicinity of Commonwealth Pier in South Boston (state whether bound for Mystic River or Chelsea River) and near the confluence of Mystic and Chelsea Rivers. Outbound vessels should make Security calls when singled up, when underway off their dock, near the confluence of Mystic and Chelsea Rivers, in the vicinity of Commonwealth Pier in South Boston, when approaching Deer Island in President Roads and when disembarking the pilot (give destination information).

(53) Vessels inbound for Quincy or Braintree should make security calls when approaching the pilot pickup point in the vicinity of Thieves Ledge, when approaching Boston Light and when picking up tugs in Nantasket Roads off Georges Island. Outbound vessels should make Security calls when singled up, when underway off the dock, abreast Germantown Point starting out the channel and when approaching Hull.

(54)

Northeastern approach

(55) **Deer Island**, on the northwest side of the entrance to Boston Harbor, is about 1 mile long and is joined to the mainland by a fill. A sewage treatment facility with numerous egg-shaped holding tanks is a conspicuous landmark on the south part of the island.

(56) **Deer Island Light** (42°20'22"N., 70°57'16"W.), 53 feet above the water, is on the outer end of a ledge that extends 0.3 mile southward from the island. A mariner radio activated sound signal at the light is initiated by keying the microphone five times consecutively on VHF-FM channel 83A.

(57) **Winthrop Head**, about 1 mile northward of the northwestern end of Deer Island, is a 100-foot hill covered with buildings and a tall red, white and blue standpipe on top that is the most prominent mark in the vicinity. Sewage pump-out is available. **Winthrop Beach** lies along the shore just northward of Winthrop Head. About 0.2 mile off and parallel to Winthrop Beach is a breakwater about 0.4 mile long which is bare several feet at the highest tides and is fairly prominent. Small craft moor behind the breakwater; there are no landings or facilities.

(58) **Great Faun**, the inner part of the shoal ground extending from the northeastern side of Deer Island, is a partly drying flat, marked on its outer part by a buoy that is about 1 mile northeastward of Deer Island Light and 0.3 mile northwestward of Boston North Channel. **Little Faun**, which uncovers on its inner part, extends 0.5 mile eastward from the southern end of Deer Island.

(59) **Finns Ledge**, covered 27 feet, lies on the western side of the entrance to Boston North Channel, the principal approach to the harbor. The ledge, marked by a lighted bell buoy, is at the outer end of shoal ground covered less than 36 feet. The shoal ground extends about 2 miles northeastward from Deer Island. Careful navigation is required in the channel entrance, especially when incoming and outgoing vessels meet.

(60) **The Graves**, a group of bare rocks and ledges about 4 miles east-northeastward of the southern point of Deer Island, extend 0.5 mile north-northeastward and 0.1 mile south-southwestward from The Graves Light. **Northeast Grave**, the northernmost of these rocks, uncovers 3 feet, and from it shoal ground extends about 0.3 mile northeastward; a lighted whistle buoy is moored 0.5 mile northeast of Northeast Grave. **The Graves Light** (42°21'54"N., 70°52'09"W.), 98 feet above the water, is shown from a light gray conical granite tower. A mariner-activated sound signal at the light is initiated by keying the microphone five times on VHF-FM channel 83A.

(61) **Roaring Bulls**, which partly uncover, are a group of rocks which lie from 0.5 to 0.9 mile southwestward of The Graves Light; the highest rocks uncover 8 feet.

(62) **Green Island**, 44 feet high, is 1.2 miles southwestward of The Graves Light and 0.4 mile west-southwestward of the Roaring Bulls. The island is on a drying reef, with several other islets on it, which extends 0.3 mile southwestward from the island.

(63) **Commissioners Ledge**, **Devils Back** and **Aldridge Ledge** lie 0.5 to 1 mile westward of Green Island and close southeastward of Boston South Channel; Devils Back is covered less than 1 foot over its northeast end. Between these ledges and Green Island are **Maffitt Ledge**, covered 17 feet, and **Halftide Rocks**, which uncover 3 feet. A dangerous wreck and a reported submerged obstruction are 250 to 300 yards south-southwestward of Maffitt Ledge.

(64) Between Boston South Channel and Boston North Channel is a large area of shoal ground. The shoalest spot in this area is covered 11 feet and lies 1.6 miles northeastward of Deer Island Light.

(65)

Southeastern approach

(66) **Point Allerton**, on the southeast side of the entrance to Boston Harbor, is 3.7 miles from the southern end of Deer Island. On the point is a 100-foot hill covered with buildings; a seawall protects the seaward base of the hill. A turreted tower on the hill is conspicuous.

(67) From Point Allerton the shore extends westward for 2 miles to **Windmill Point**, which is marked by a light. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A. The Coast Guard has a boathouse on the southern side of Windmill Point and another about 0.3 mile eastward of the point at the **Point Allerton Coast Guard Station. Telegraph Hill**, 1 mile west of Point Allerton, is about 100 feet high and is marked by a stone tower with a conical top; the town of **Hull** is on the western slopes of the hill. Excursion vessels from Boston call at the town wharf in summer and stop at Georges Island on the way.

(68) **Nantasket Beach**, extending about 3 miles south-southeastward from Point Allerton, is covered with buildings. Grassy **Strawberry Hill**, 1.2 miles southward of Point Allerton, is about 100 feet high. Two other grassy hills, **White Head** and **Sagamore Hill**, are on the southern part of the beach 2 and 2.4 miles, respectively, from Point Allerton. From Nantasket Beach to Cohasset Harbor, about 3 miles to the east-southeastward, the coast presents a general hilly appearance, and rocks and sunken ledges extend 0.5 mile offshore in places.

(69) **Atlantic Hill**, **Center Hill**, and **Green Hill** are prominent on the stretch of coast between Nantasket Beach and Cohasset Harbor. **Gun Rock** is off the west

point of the entrance of a cove off **Crescent Beach**, between Center Hill and Green Hill. The cove is protected by a breakwater extending 135 yards northward from the foreshore at Green Hill; thence 330 yards westward to the vicinity of **Seal Rock**, which is about 500 yards eastward of Gun Rock.

(70) **Black Rocks** are a group of rocky islets off Green Hill. The large 20-foot-high islet has a house on it.

(71) Shoals extend eastward and northward from Point Allerton. Two drying rocks, about 0.1 mile apart, lie about 0.2 mile northward of the point; the eastern rock uncovers 5 feet. A lighted bell buoy is moored about 0.5 mile northward of the point. **Ultonia Ledge**, the eastern end of the broken ground, has unmarked spots covered 13 to 24 feet extending 1.3 miles east-northeastward from the point.

(72) **Harding Ledge**, 1.5 miles eastward of Point Allerton, uncovers before low water. A detached rock, which uncovers 1 foot, is 300 yards southwest of the ledge. Between Harding Ledge and Point Allerton, the bottom is very uneven, and vessels should pass outside the lighted bell buoy that is moored 0.3 mile northeast of the ledge.

(73) **Thieves Ledge**, 2.3 miles east-northeastward of Point Allerton, has a least depth of 27 feet and is marked on its northeast side by a lighted whistle buoy. Patches covered 33 feet and 34 feet are 0.5 mile east by northeastward and 1 mile east-southeastward, respectively, of the lighted whistle buoy. In heavy easterly gales the sea sometimes breaks on the ledge and the patches.

(74) **Three and One-half Fathom Ledge**, cleared to a depth of 18 feet, about 3 miles northeastward of Point Allerton, is marked by a lighted bell buoy about 0.2 mile southeastward of the ledge. **Martin Ledge**, covered 16 feet, is 0.8 mile southwest of Three and One-half Fathom Ledge and is marked on its eastern side by a buoy. **Boston Ledge**, covered 17 feet, is 1.4 miles southwest of Three and One-half Fathom Ledge and is marked by a buoy.

(75) **Shag Rocks**, 1.2 miles northward of Point Allerton and 0.3 to 0.6 mile east-northeastward of Boston Light, are 20 feet high and surrounded by extensive covered ledges and foul ground. Reefs and foul ground extend 0.5 mile east-northeastward to within 0.2 mile of Boston Ledge and west-southwestward to within 0.2 mile of Boston Light.

(76) **Nash Rock Shoal**, covered 19 feet, lies about 0.4 mile southwest of Boston Light.

(77) **Boston Light** (42°19'41"N., 70°53'24"W.), 102 feet above the water, is shown from an 89-foot white conical tower, on the southeast side of **Little Brewster Island**, about 1 mile northward of Point Allerton. A mariner radio-activated sound signal at the light is initiated by keying the microphone five times on VHF-FM channel 83A.

(78) **Great Brewster Island**, 0.4 mile northwest of Little Brewster Island, is 103 feet high and has a bluff at the north end. Little Brewster and Great Brewster Islands lie on the northern side of the southeastern approach on a drying bank, of which **Great Brewster Spit**, the western part, extends about 1 mile west-southwestward from Great Brewster Island. The western end of the spit is marked by **The Narrows Light 4**. Shoal ground extends about 0.3 mile southward from Great Brewster Spit, and on this extension are **Kelp Ledges**, awash, about 0.8 mile westward of Boston Light.

(79) From the northern end of the bank on which Great Brewster Island lies, reefs extend about 1 mile east-northeastward and 0.7 mile northward; on the eastern extension are **Middle Brewster Island** and **Outer Brewster Island**, and on the northern extension are **Calf Island** and **Little Calf Island**. On these reefs are several islets, and off-lying them are numerous shoals, the area between the Brewsters and Shag Rocks being particularly foul. Among these dangers is **Tewksbury Rock**, covered 9 feet, which is about midway between Outer Brewster Island and Martin Ledge.

(80) **Georges Island** is about 1.6 miles west-southwestward of Boston Light and 0.8 mile north-northwestward across Nantasket Roads from Windmill Point. The island is the site of historic **Fort Warren** and has several other buildings on it. A state recreation park is on the island, and a state landing is in a protected basin at the wharf on the west shore of the island. Daytime berthing and a limited amount of water are available. A seasonal ferry runs from the Boston waterfront to this wharf and from here to several nearby islands and to Boston.

(81) **Lovell Island** is 0.3 mile northward across The Narrows from Georges Island and on the south side of Boston South Channel. A pier is on the southwest side of the island. The ruins of Fort Standish are on the island, and a cement lookout tower on the southern end of the island is prominent. **Ram Head Flats** and **Ram Head** extend up to 0.8 mile northeastward from the island; Ram Head partly uncovers.

(82) **Gallops Island**, 0.3 mile northwestward of Georges Island and 0.2 mile westward across The Narrows from Lovell Island, is high with trees and brush on its northern side. The island is marked by a light off its eastern end. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A. A pier, protected by a breakwater, is on the southwest side. A reef named **Nixes Mate** lies on the outer part of the shoal ground that extends 0.4 mile northwestward from Gallops Island; near the center of the reef is a low islet marked by a daybeacon.

(83) **Long Island**, 0.6 mile westward of Gallops Island and 0.8 mile southward across President Roads from Deer Island, is 1.5 miles long in a northeast-southwest direction and has a greatest width of about 0.25 mile. A large standpipe and the buildings of Long Island Hospital are prominent. There is a wharf on the northwest side of the island. **Long Island Head Light** (42°19'49"N., 70°57'28"W.), 120 feet above the water, is shown from a white brick tower on the north side of the island. The piers of a former fixed bridge are between **West Head**,

the south end of Long Island, and Moon Head. Private lights mark the main channel through the piers and the remaining piers are marked by private white lights.

(84) **Spectacle Island** is on the south side of President Roads and 0.7 mile westward of Long Island. A marina and ferry dock are on the southwest side of the island; sewage pump-out is available.The island is open to the public via ferry for recreational activities.

(85)

Channels

(86) Boston North Channel, Boston South Channel, and The Narrows are the main entrances from the sea to President Roads. Several other channels of less importance are used by local vessels.

(87) **Boston North Channel** leads from Broad Sound to President Roads from the northeastward. It is the principal entrance to Boston Harbor. A Federal project provides for a channel 1,500 feet wide dredged to 40 feet in the eastern 900 feet and 35 feet in the western 600 feet. The channel is well marked by lighted buoys. (See Notice to Mariners and the latest edition of the chart for controlling depths.)

(88) **Boston South Channel** leads from Broad Sound in a southwesterly and westerly direction to President Roads. A Federal project provides for a channel 1,200 feet wide dredged to 30 feet deep. The channel is marked by lighted and unlighted buoys. (See Notice to Mariners and the latest edition of the chart for controlling depths.)

(89) Pilots of deep-draft vessels use the North Channel most of the time. The South Channel is rarely, if ever, used, because deep-draft vessels have a tendency to feel the bottom, making steering difficult.

(90) **President Roads** is the area between Deer Island and Governors Island Flats, north of Long Island, and north-northeastward of Spectacle Island. A Federal project provides for a channel 40 feet deep. (See Notice to Mariners and the latest edition of the chart for controlling depths.) North Channel, South Channel, The Narrows, Nubble Channel and Sculpin Ledge Channel are entrances to Boston Main Channel and converge at President Roads. This is a dense traffic area for fast ferries, merchant ships, tug and barge units and recreational boaters. Tug and barge units frequent the sewage treatment plant on Deer Island at the northeast end of President Roads. Its northern part is used as a general and quarantine anchorage for ships awaiting berth in Boston or Nantasket Roads. President Roads has a sand and mud bottom. The south part of President Roads is Boston Main Channel and the entrance into Boston Inner Harbor.

(91) **Nantasket Roads**, westward of the southern entrance to The Narrows, is a good anchorage with depths up to 50 feet. There are numerous shoals in it that must be avoided by deep-draft vessels; the chart is the guide.

(92) **The Narrows** is a channel that extends from Nantasket Roads northwest to President Roads. It is bounded on the northeast side by Great Brewster Spit and Lovell Island and on the southwest side by Georges Island and Gallops Island. Depths of about 26 feet can be carried in the well-marked channel; however, shoals with considerably lesser depths are along the edges of the channel.

(93) Because of the strong currents and sharp turns, it is necessary to conn a ship by eye through the approaches and in The Narrows channel. The navigator must take precautions to prevent being set off course by crosscurrents sweeping in or out of Black Rock Channel and the channel between Gallops Island and Georges Island.

(94) **Hypocrite Channel** is a natural channel leading between Green Island on the north and Little Calf Island on the south. The channel has several unmarked dangers and is not recommended for strangers or for large vessels.

(95) **Black Rock Channel** leads into The Narrows from northeastward between Great Brewster Spit on the southeast and Lovell Island and Ram Head Flats on the northwest. The channel is marked by buoys. There is an unmarked rock covered 8 feet nearly in midchannel. The channel is used only by small local craft and is not recommended for strangers.

(96) A channel 250 yards wide leads into The Narrows from westward between Georges Island and Gallops Island. A light is on the north side of the channel near the end of the shoal off the southeast end of Gallops Island. The channel is suitable only for quick-working vessels on account of the sharp turn into The Narrows.

(97) **Nubble Channel** leads from Nantasket Roads to President Roads between Nixes Mate and Long Island; the channel is marked by buoys.

(98) **Sculpin Ledge Channel** leads between Long Island and Spectacle Island. It will accommodate vessels of about 8-foot draft to Hingham Bay by the passage southward of Peddocks Island. The deeper water favors Long Island, and in coming from President Roads the island should be followed at a distance of about 400 yards until up with the buoy southward of Sculpin Ledge. Pass about 300 yards southeastward of the buoy and round the southwesterly end of Long Island at a distance of 300 yards. The channel then leads between the piers of a former fixed bridge; the piers are marked by lights.

(99) A fish haven, covered 14 feet, is in Sculpin Ledge Channel. The reef is west of the standpipe on Long Island along a northeast-southwest axis in about 42°19'26"N., 70°58'15"W.

(100) A rock, covered 2 feet, is about 200 yards northeast of Sculpin Ledge.

(101) The channel leading from Nantasket Roads to Boston, southward of Long Island and Spectacle Island, is partially marked by buoys and can be used by boats of 8-foot draft with the aid of the chart.

(102) **Boston Main Channel** extends from President Roads to the mouths of the Chelsea and Mystic Rivers and to Charlestown Bridge on the Charles River. The federal project depth in this channel is 35 to 40 feet.

(103) The waters adjacent to the piers and wharves extending northward from Northern Avenue Bridge to

and including Pier 4 along the Boston waterfront, west of the Boston Main Channel, are declared **nonnavigable** through Public Law 90-312. Uncharted hazards or dangers may exist in these waters. Strangers are advised to seek local knowledge before entering, and all mariners are advised to exercise caution in the area.

(104)

Anchorages

(105) General, explosives and special anchorages are in Boston Harbor. (See **33 CFR 110.1**, **110.29**, **110.30** and **110.138**, chapter 2, for limits and regulations.)

(106) The anchorage on the north side of President Roads is the most commonly used anchorage for ships and barges in Boston Harbor. The anchorage in Nantasket Roads north of Peddocks Island has scattered shoal areas and rocks. The anchorage on the westerly side of Georges Island has a better bottom and is frequently used by vessels seeking shelter from easterly winds.

(107) A buried high-pressure natural gas pipeline extends along the sea floor from the northwest quadrant of the precautionary area, southwest toward the south side of Georges Island, and through Anchorage 5 to West Gut. The pipeline then travels to the south, and southwest along the north side of Weymouth Fore River. Mariners are advised to use extreme caution when anchoring in these areas.

(108)

Currents

(109) See the Tidal Current prediction service at *tidesandcurrents.noaa.gov*. for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(110) For some distance northwestward of Cape Cod the tidal currents have a slight set into Cape Cod Bay on the flood and out of the bay on the ebb. Along the north shore of Massachusetts Bay the flood sets in a general southwesterly direction and the ebb in a northeasterly direction. The velocity of the currents is influenced greatly by the force and direction of the wind. Off the entrance to Boston Harbor, the flood sets westward and the ebb eastward, increasing slightly in velocity as the entrance is approached.

(111) The currents at Boston Lighted Whistle Buoy B are described in chapter 3.

(112) In Broad Sound the velocity of the current at strength in most places is less than 0.8 knot. This increases to about 1 knot or more on approaching the entrances of the channels leading into Boston Harbor.

(113) In Boston South Channel, north of Ram Head, the velocity at strength is about 1.2 knots. In the channel between Deer Island Light and Long Island Head the velocity at strength is about 1.4 knots. In Hypocrite Channel the velocity at strength is about 1 knot. In Black Rock Channel the velocity at strength is between 0.5 and 1 knot. The flood sets southwestward through the channel and the ebb northeastward. This should be kept in mind when passing through The Narrows.

(114) Near the middle of the channel between Boston Light and Point Allerton the velocity at strength is about 1.5 knots. On the northern side of the channel southward of Great Brewster Spit the velocity is about half as great. In the middle of the channel in Nantasket Roads the velocity at strength is about 1.5 knots. In Nantasket Gut the velocity at strength is about 2.5 knots.

(115) Between Georges Island and Gallops Island the velocity at strength is about 1 knot. The flood sets westward and the ebb northeastward.

(116) Between Gallops Island and Long Island Head the velocity at strength is about 1 knot. The flood current sets southward to southwestward and the ebb in the opposite directions.

(117) Between Moon Head and Long Island, the current is rotary, turning counterclockwise. The average velocity at strength is about 0.2 knot. Usually, strength of flood sets southwestward and strength of ebb eastward. Between Thompson Island and Spectacle Island the velocity at strength is about 0.5 knot. The flood sets northwestward and the ebb southeastward.

(118) In Boston Main Channel from Spectacle Island to the mouth of the Charles River the velocity at strength varies between 0.5 and 1 knot.

(119)

Weather, Boston and vicinity

(120) Three important influences are responsible for the main features of Boston's climate. First, the latitude (42°N.) places the city in the zone of prevailing west to east atmospheric flow in which are encompassed the northward and southward movements of large bodies of air from tropical and polar regions. This results in variety and changeability of the weather elements. Secondly, Boston is situated on or near several tracks frequently followed by systems of low air pressure. The consequent fluctuations from fair to cloudy or stormy conditions reinforce the influence of the first factor, while also assuring a rather dependable precipitation supply. The third factor, Boston's east-coast location, is a moderating factor affecting temperature extremes of winter and summer.

(121) Hot summer afternoons are frequently relieved by the locally celebrated "sea-breeze," as air flows inland from the cool water surface to displace the warm westerly wind component. This refreshing east wind is more commonly experienced along the shore than in the interior of the city or the western suburbs. In winter, under appropriate conditions, the severity of cold waves is reduced by the nearness of the then relatively warm water. The average date of the last occurrence of freezing temperature in spring is April 8; the latest is May 3, 1874 and 1882. The average date of the first occurrence of freezing temperature in autumn is November 7; the earliest on record is October 5, 1881. In suburban areas, especially away from the coast, these dates are later in spring

and earlier in autumn by up to one month in the more susceptible localities. The average annual temperature in Boston is 51.8 °F. July is the warmest month with an average maximum of 82 °F and an average minimum of 65 °F. January is the coolest with an average high of 37 °F and an average low of 22 °F. Extremes include 102 °F in August 1975 and again in July 1977 and an extreme minimum of -12 °F in January 1957.

(122) Boston has no dry season. The average annual rainfall is 42.9 inches (1090 mm). For most years the longest run of days with no measurable precipitation does not extend much more than 2 weeks. This "dry spell" may occur at any time of year. Precipitation falls an average of 187 days each year. November is the wettest month averaging 4.4 inches (112 mm) and July the driest averaging 2.8 inches (71 mm). On 28 days each year precipitation can be expected to surpass one-half inch (13 mm) in a 24-hour period.

(123) Much of the rainfall from June to September comes from showers and thunderstorms. During the rest of the year, low-pressure systems pass more or less regularly and produce precipitation on an average of roughly one day in three. Coastal storms, or "nor'easters," are prolific producers of rain and snow. The main snow season extends from December through March. The average annual snowfall total is 43 inches (1092 mm). Snow falls an average 50 days in any given year and has fallen during each month October through May. The number of days with 1.5 inches (38 mm) or more of snowfall is eight per season. Periods when the ground is bare or nearly bare of snow may occur at any time in the winter.

(124) Relative humidity has been known to fall as low as 5 percent (May 10, 1962), but such desert dryness is very rare. Heavy fog occurs on an average of about 2 days per month with its prevalence increasing eastward from the interior of Boston Bay to the open waters beyond. Fog, in whatever thickness, occurs an average of 135 days each year. Winds from the east to southwest bring fog; westerly and northerly winds clear it away.

(125) At all seasons the heaviest gales are usually from the northeastward or eastward. Although winds of 27 knots (about 32 miles per hour) or higher may be expected on at least one day in every month of the year, gales are both more common and more severe in winter. The predominant wind direction is west through northwest.

(126) The National Weather Service office is in the Customhouse. Barometers may be compared at the Logan International Airport in East Boston.

(127) **Fogs** are prevalent throughout the year. Winds from the east to southwest bring fog; westerly and northerly winds clear it away.

(128)

Ice

(129) The channels of Boston Harbor are navigable throughout the year. Ice rarely forms in the main channel. Occasionally during severe winters the greater part of the harbor is frozen, but towboats and steamers keep the main channels open. The Charles, Mystic and Chelsea Rivers and the minor passages in the harbor sometimes are frozen during severe winters. They are almost invariably kept open, however, by tugboat traffic. When ice is prevalent, the buoys may be displaced or even carried away. Local towboats can be employed for breaking ice.

(130)

Routes

(131) Boston Harbor and approaches have very broken rocky bottom, and caution is required.

(132)

Approaching Boston from Cape Ann

(133) The soundings in the vicinity of Cape Ann are very irregular and cannot be depended on to locate even approximately the vessel's position. A **228°** course from 0.2 mile off the lighted whistle buoy, 2.5 miles eastward of Cape Ann Light, clears the offshore dangers between Cape Ann and Nahant, and leads close to the lighted gong buoy marking the entrance to Boston North Channel.

(134) At night the lighted aids are sufficiently numerous to locate the position by cross bearings. In clear weather the course should be shaped to pass well northward of The Graves Light and enter through Boston North Channel.

(135)

Approaching Boston from Cape Cod

(136) Approaching the easterly side of the cape, soundings of 20 fathoms indicate a distance of 3 to 3.5 miles from shore, but off the north side of the cape, the 20-fathom curve draws closer inshore and the soundings are not so regular. Vessels standing to clear Boston Lighted Whistle Buoy B on a course of **297°** from the locality of Peaked Hill Bar Lighted Whistle Buoy 2 PH will cross the southwesterly end of Stellwagen Bank in depths of 12 to 15 fathoms. Soundings on Stellwagen Bank cannot be depended on to locate a position, except near the extreme southwest end of the bank where the shoalest depth of 10 fathoms is found. The recommended route, however, for deep-draft vessels is via the **Boston Traffic Separation Scheme,** which is described at the beginning of this chapter.

(137) As the entrance to Boston Harbor is approached, after crossing Stellwagen Bank, soundings of 20 fathoms or more ensure a distance of at least 5 miles from the shore and well outside of outlying rocks. Inside the depths of 20 fathoms, the soundings are very irregular and cannot be depended upon as a rule to keep a vessel out of danger. Northeast of Nahant the 20-fathom curve runs closer inshore and some of the dangers extend offshore nearly to the curve.

(138) In the approach to Boston Lighted Whistle Buoy B from the southward, the coast from Scituate to Minots Ledge Light should be given a berth of 4 miles to avoid the broken ground of Stellwagen Ledges.

(139)

North Atlantic Right Whales

(140) Endangered North Atlantic right whales may occur in the Stellwagen Bank and Jefferys Ledge area

year-round (peak season for Jefferys Ledge: October through December; peak season in the Stellwagen Bank: early spring). This area has been designated as the **Gerry E. Studds-Stellwagen Bank National Marine Sanctuary** and includes the North Atlantic Right Whale Critical Habitat Northeastern Foraging Area (See **50 CFR 226.203**, chapter 2, for limits and regulations.) The Pilots distribute educational material to mariners in an effort to reduce right whale ship strikes. (See North Atlantic Right Whales, indexed as such, chapter 3, for more information on right whales and recommended measures to avoid collisions with whales.)

(141) All vessels 65 feet or greater in length overall (LOA) and subject to the jurisdiction of the United States are restricted to speeds of 10 knots or less in two Seasonal Management Areas. The first area is defined as all waters of Cape Cod Bay with a northern boundary of 42°04'56.5"N., 70°12'W., to 42°12'N., 70°12'W., thence due west back to shore, and the restrictions apply between January 1 and May 15. The second area, off Race Point, is defined as the waters bounded by:

(142) 42°04'56.5"N., 70°12'W.

(143) 42°12'N., 70°12'W.

(144) 42°12'N., 70°30'W.

(145) 42°30'N., 70°30'W.

(146) 42°30'N., 69°45'W.

(147) 41°41'N., 69°45'W.; thence due west to the shore, and the restrictions apply between March 1 and April 30. (See 50 CFR 224.105 in chapter 2 for regulations, limitations and exceptions.)

(148) The waters of Cape Cod Bay lie within the WHALESNORTH Mandatory Ship Reporting Area. Each self-propelled ship of 300 gross tons or greater entering WHALESNORTH must participate in the Mandatory Ship Reporting System (See **33 CFR 169**, chapter 2, for limits and regulations, and chapter 3 for sample reports). Sovereign immune vessels are exempt from the requirement to report but are encouraged to participate.

(149) The **Outer Boston Harbor Restricted Area**, a Marine Protected Area (MPA), extends 5.5 miles offshore from Hull to Nantasket.

(150)

Entering Boston Harbor in fog

(151) In thick weather a course should be laid to clear Boston Lighted Whistle Buoy B by a safe distance when approaching from either Cape Ann or Cape Cod, and the water should not be shoaled to less than 20 fathoms until the buoy is located by radar or other means. Occasionally, vessels anchored in Broad Sound have been mistaken on radar for navigational aids; caution is advised. From the buoy, steer a course to pass 0.4 mile northward of Boston Approach Lighted Buoy BG and enter the harbor via Boston North Channel. Unless Boston Lighted Whistle Buoy B is located, no attempt should be made to enter the harbor.

(152) If a vessel in the vicinity of Cape Cod is overtaken by fog or thick weather, she may find it convenient to anchor in Provincetown Harbor or on the west side of the cape south of Provincetown, where there is a good lee and the holding ground is in 7 to 12 fathoms.

(153)

Pilotage, Boston and vicinity

(154) Pilotage is compulsory for all foreign vessels and for U.S. vessels under register in the foreign trade. Pilotage is available for coastwise vessels not carrying an officer licensed as a pilot by the federal government for these waters. Pilots for Boston, Mystic, Chelesa River, Quincy, Weymouth, Weymouth Fore River, Back River, Town River, Lynn, Saugus and Saugus River are available from Boston Pilot Association, Pier No. 1, S. Bremen Street, East Boston, MA 02128. Telephone: office, 617–569–4500; FAX 617–569–4502; Pilot boats, 617–962–4670 (night); Internet address, *bostonpilots.com*. The pilot office and boats monitor and work on VHF-FM channels 13, 16 and 20 INT. Pilot services are generally arranged for in advance through ship's agents.

(155) The pilot boats meet vessels 1.5 miles east of Boston Approach Lighted Whistle Buoy BG at 42°23'27"N., 70°51'29"W. During winter, the pilot boats may seek shelter from northwest winds under Nahant Head.

(156) The pilot boats, CHELSEA and MYSTIC, both 53 feet, have black hulls with orange superstructure and the word PILOT in black letters on the sides; both are equipped with DSC and AIS. The pilot boats will provide boarding instructions by radiotelephone on VHF-FM channel 20 INT.

(157) When about 2 hours away vessels are requested to provide an updated ETA.

(158)

Towage

(159) Tugs to 4,400 hp are available at Boston. The tugs maintain radio communications on VHF-FM channels 18A and 5A. Tugs are available from Boston Towing and Transportation Co., telephone 617–567–9100 and Constellation Towing, telephone 617–561–0223. Inbound vessels are usually met in the vicinity of Anchorage areas 1 or 2. Arrangements for tugs are usually made in advance through ships' agents. Fireboats are also available; the call for the fireboat is five prolonged blasts of the ship's whistle.

(160)

Quarantine, customs, immigration and agricultural quarantine

(161) (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(162) **Quarantine** is enforced in accordance with regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.) Quarantine anchorages for Boston Harbor are on the north side of President Roads and on Bird Island Flats.

(163) Boston is a **customs port of entry.**

(164)

Coast Guard

(165) **Coast Guard Sector Boston** is located in the north end of Boston (see Appendix A for address). **Coast Guard Station Boston** is located on the south side of the mouth of the Charles River. **Coast Guard Station Port Allerton** is located at the end of Windmill Point in Hull. **Coast Guard Station Scituate** is located at the entrance to Scituate Harbor (seasonal station).

(166)

Harbor regulations

(167) The rules and regulations of the Commonwealth of Massachusetts, the city of Boston and metropolitan area communities are enforced by the Harbor Patrol Unit of the Boston Police Department, the Massachusetts State Police Marine Unit and the Massachusetts Environmental Police. The harbor patrol boathouse is on the south side of Charles River, about 0.5 mile above the entrance. Information on boating laws can be obtained from the Massachusetts Environmental Police at 617–727–8760 or 508–790–9360 or from the Boat and Recreation Vehicle Safety Bureau, 1019 Iyanough Road (Rt 132), 2nd floor, Hyannis, MA 02601 or the Department of Fish and Game, 251 Causeway Street, Suite 400, Boston, MA 02114.

(168) The Distrigas liquified natural gas (LNG) facility on Mystic River is within a **safety zone**. Additionally, the waters surrounding loaded LNG vessels transiting Boston Harbor are a **safety zone**. (See **33 CFR 165.110**, chapter 2, for limits and regulations.)

(169)

Wharves

(170) The Port of Boston has many piers and wharves, most of which are located on the main channel at South Boston, East Boston and Charlestown and on the Chelsea River and Mystic River.

(171) The piers and wharves generally are of open-pile concrete deck construction, extending from stone or timber bulkheads with solid fill. Only the deep-draft facilities are described; the other active facilities in the port are used as repair berths and by government vessels, fishing vessels, small craft and barges. The alongside depths for the facilities described are reported; for information on the latest depths contact the Massachusetts Port Authority or the private operator. All of the facilities have direct highway connections, and some have railroad connections. Water connections are available at most piers and wharves.

(172) General cargo at the port is usually handled by ship's tackle, except container vessels use shore side equipment; special handling equipment, if available, is mentioned in the description of the particular facility.

(173) Numerous warehouses and cold storage facilities adjacent to the waterfront are available.

(174) All of the large general cargo terminals are owned or leased by the Massachusetts Port Authority. Containerized cargo is handled at the Paul W. Conley Marine Terminal. Most of the deepwater oil and bulk terminals are on the Chelsea River and Mystic River.

(175) The office of the Port Director is at One Harborside Drive, Suite 200S, East Boston, MA 02128-2909; telephone 617–946–4413.

(176) **Facilities at South Boston**:

(177) **Massachusetts Port Authority, Paul W. Conley Marine Terminal, Berths 11-15:**

(178) Berth 15 (42°20'25"N., 71°00'41"W.): southwest side of Boston Main Channel; 570-foot face, 35 feet alongside; deck height, 15 to 16 feet; owned and operated by Massachusetts Port Authority.

(179) Berth 14 (42°20'29"N., 71°00'47"W.): southwest side of Boston Main Channel; 630-foot face, 35 feet alongside; deck height, 15 to 16 feet; owned and operated by Massachusetts Port Authority.

(180) Berth 13: adjacent westward of Berth 14 on south side of Reserved Channel; 1,000 feet of berthing space; 40 feet alongside; deck height, 15 to 16 feet.

(181) Berths 11-12: adjacent westward of Berth 13 on south side of Reserved Channel; 2,040 feet of berthing space; 45 feet alongside; deck height, 15 to 16 feet; 101 acres open storage; two 40-ton container cranes and two 50-ton container cranes; owned and operated by Massachusetts Port Authority.

(182) **Coastal Oil New England, South Boston Ship Dock** (42°20'30"N., 71°01'37"W.): south side of Reserved Channel; 85-foot, 700 feet of berthing space with dolphins; 38 feet alongside; deck height, 14 feet. Owned by El Paso Gas Company. This facility is presently closed.

(183) **Boston Marine Industrial Park, Berths 1-9:**

(184) Berth 6 (42°20'37"N., 71°01'31"W.): north side of Reserved Channel, 540-foot face; 35 feet alongside; deck height, 17½ feet; owned by Massachusetts Port Authority.

(185) Berths 1-3 (42°20'38"N., 71°01'17"W.): 964-foot face (upper side), 352-foot face (lower side); 35 feet alongside; deck height, 17½ feet; pipelines extend to storage silos with capacity of 18,700 tons of cement at a rate of 420 tons per hour; receipt of bulk cement; owned by Massachusetts Port Authority and operated by Boston Marine Industrial Park, Massachusetts Port Authority, and Coastal Cement Corp.

(186) **Black Falcon Cruise Terminal Berths 7, 8 and 9** (42°20'36"N., 71°01'56"W.): north side of Reserved Channel, approximately 2,000 feet from the entrance, 2,100-foot face; 35 feet alongside; deck height 17½ feet; passenger terminal for cruise vessels; owned and operated by Massachusetts Port Authority. A security zone is in effect around cruise ships moored at this facility (See **33 CFR 165.116**, chapter 2).

(187) **Massachusetts Port Authority, Marine Terminal Wharf** North Jetty (42°20'57"N., 71°01'44"W.): 800-foot face; 35 feet alongside; deck height, 16 feet; 47 acres of open storage; receipt of miscellaneous dry bulk commodities; owned by Massachusetts Port Authority. This wharf is closed to all vessels except government vessels.

(188) **Massachusetts Port Authority, Commonwealth Pier** (42°21'11"N., 71°02'21"W.): 400-foot face, southeast and northwest sides, 1,200 feet long; 33 to 38 feet alongside; deck height, 18 feet; mooring excursion vessels and passenger ferry terminal for cruise vessels; owned by Massachusetts Port Authority and operated by World Trade Center.

(189) **Facilities at Charlestown:**

(190) **Mystic Piers 48-50** (42°22'58"N., 71°02'51"W.): 423-foot face (northeast side), 360-foot face (east side), 560-foot face (south side); 25 feet alongside; deck height, 18 feet; open storage; receipt of dry bulk cargo; owned by Massachusetts Port Authority. This facility is closed.

(191) **Facilities on Mystic River, south bank:**

(192) **United States Gypsum Co. Wharf** (42°23'02"N., 71°02'58"W.): immediately westward of Mystic River-Tobin Memorial Bridge; 492 feet of berthing space; 27 feet alongside; deck height, 14½ feet; open storage for 38,000 tons; storage silos for 25,000 tons; receipt of gypsum rock from self-unloading vessels; owned and operated by United States Gypsum Co.

(193) **John F. Moran Terminal Wharf** (42°23'04"N., 71°03'12"W.): immediately westward of United States Gypsum Co. Wharf; 1,100-foot face, 40 feet alongside; deck height, 17½ feet; 50 acres open storage; receipt and shipment of automobiles and general cargo; owned by Massachusetts Port Authority and operated by Boston Autoport, LLC.

(194) **Lafarge Cement Boston Terminal** (42°23'05"N., 71°03'35"W.): approximately 0.5 mile above Mystic River - Tobin Memorial Bridge; 610-foot face, 31 feet alongside; deck height, 16.4 feet.

(195) **Medford Street Terminal** (42°23'06"N., 71°03'41"W.): approximately 0.6 mile above Mystic River-Tobin Memorial Bridge; this facility is closed.

(196) **Facilities on Mystic River, north bank:**

(197) **Global Companies, Chelsea Terminal Wharf** (42°23'06"N., 71°02'40"W.): 560 feet of berthing space with dolphins; 35 feet alongside; deck height, 14 feet; storage tanks with 630,500-barrel capacity; receipt and shipment of petroleum products; bunkering vessels; owned and operated by Global Companies.

(198) **Exxon Co., U.S.A., Everett Terminal Wharf, Berth 1** (42°23'17"N., 71°03'21"W.): on west side of Island End River at junction with Mystic River; 300 feet of berthing space; 21 feet alongside; deck height, 15 feet; receipt and shipment of petroleum products; owned and operated by Exxon Co., U.S.A.

(199) **Exxon Co., U.S.A., Everett Terminal Wharf, Berths 3 and 4** (42°23'17"N., 71°03'29"W.): Berth 3, 155-foot face; Berth 4, 90-foot face; 950 feet total berthing space with dolphins; 35 to 40 feet alongside; deck height, 15 feet; pipelines extend to storage tanks with 2½-million-barrel capacity; receipt and shipment of petroleum products,; receipt of asphalt; owned and operated by Exxon Co., U.S.A.

(200) **Distrigas of Massachusetts Corp. Wharf** (42°23'19"N., 71°03'43"W.): 0.1 mile west of Exxon Co., U.S.A. Wharf; 67-foot face, 1,000 feet of berthing space with dolphins; 36 feet alongside; deck height, 17 feet; pipelines to storage tanks with 974,000-barrel capacity; receipt of liquefied natural gas; owned and operated by Distrigas of Massachusetts Corp.

(201) **Prolerized New England Co. Scrap Metal Wharf:** 0.2 mile west of Exxon Co., U.S.A. Wharf; 320-foot face, 820 feet of berthing space with dolphins; 40 feet alongside; deck height, 14 feet; 35-ton crawler crane; conveyor system with loading rate of 1,000 tons per hour; shipment of scrap metal; owned by Hugo Neu Steel Products, Inc., and Prolerized Transport Systems, Inc., and operated by Prolerized New England Co.

(202) **Facilities on Chelsea River, north bank:**

(203) **Gulf Oil Limited Partnership, Chelsea Terminal Tanker Wharf** (42°23'35"N., 71°01'03"W.): 60-foot face, 710 feet of berthing space; 36 feet alongside; deck height, 14 feet; pipelines extend to storage tanks with 1.1-million-barrel capacity; receipt and shipment of petroleum products; bunkering vessels; owned and operated by Gulf Oil Limited Partnership.

(204) **Facilities on Chelsea River, south bank:**

(205) **CONOCO/Phillips East Boston Terminal Wharf** (42°23'06"N., 71°01'28"W.): 600 feet of berthing space; 38 feet alongside; deck height, 16 feet; pipelines extend to storage tanks with 1-million-barrel capacity; receipt and shipment of petroleum products; owned and operated by Mobil Oil Corp.

(206) **Irving Oil Co., Revere Marine Terminal, Ship Pier** (42°23'48"N., 71°00'44"W.): north side has 600 feet of berthing space; 36 feet alongside; south side has 21 feet alongside; deck height, 15 feet; pipelines extend to storage tanks with 760,000-barrel capacity; receipt and shipment of petroleum products; bunkering vessels.

(207) **Global Revco Terminal, Ship Pier** (42°23'51"N., 71°00'48"W.): south side has 600 feet of berthing space; 37 feet alongside; deck height, 15 feet; pipelines extend to storage tanks with 1¼-million-barrel capacity; receipt and shipment of petroleum products; bunkering vessels.

(208) **BP Oil Co. Pier** (42°23'48"N., 71°00'45"W.): north and south sides 550 feet of berthing space; north side 10 to 21 feet alongside, south side 10 to 35 feet alongside; deck height, 15 feet; pipelines extend to storage tanks with 1¼-million-barrel capacity; receipt and shipment of petroleum products; owned and operated by BP Oil Co. and Global Petroleum Co.

(209) **Coastal Oil New England, Ship Pier** (42°23'51"N., 71°00'48"W.): south side has 600 feet of berthing space; 37 feet alongside; deck height, 15 feet; pipelines extend to storage tanks with 1¼-million-barrel capacity; receipt and shipment of petroleum products; bunkering vessels; owned and operated by Coastal Oil New England, Inc.

(210)

Supplies

(211) Provisions and marine supplies of all kinds are available in the port of Boston. All grades of heavy marine bunker fuel, lubricants and diesel fuel can be obtained.

Vessels may bunker directly at several of the marine oil terminals or may be serviced by barges at anchor or at loading berths. Gasoline can be obtained at the marinas or from barges anchored in the stream in the summer.

(212)

Repairs

(213) The port of Boston has excellent facilities for making all types of hull and engine repairs to vessels of all sizes. Several of these firms operate waterfront facilities for the construction, repair and conversion of oceangoing vessels, tugs and towboats, barges and various types of small vessels. In addition, there are a number of firms without waterfront facilities that are engaged in marine repair work. These companies maintain shops and portable equipment for making above-waterline repairs and for installing equipment, gear and machinery on all types of craft at their berths. There are several drydocks and marine railways available in the port. The largest repair facility is located in South Boston. The South Boston yard has a graving dock with a length of 1,175 feet and a width of 127 feet at the entrance and a depth of 36 feet over the sill. Cranes to 50 tons are at the yard.

(214) A marine railway in the port can handle vessels to 180 feet long and 1,000 tons.

(215) Several smaller repair facilities in the port cater to yachtsmen and small-craft operators.

(216)

Communications

(217) Boston is the terminus of two trunk railroads. About 35 commercial shipping lines serve the port in foreign trade to or from over 175 world ports. There is little or no coastwise traffic except in bulk gypsum, liquid sulfur, cement and petroleum.

(218) Several major airlines provide frequent scheduled services between Logan International Airport in East Boston and domestic and overseas points.

(219) Boston has through bus and rail service to all points. There are numerous trucking firms engaged in long- and short-haul freight service from the port.

(220)

Small-craft facilities

(221) Public float landings for small craft are at Summer Street, Northern Avenue, on Charles River, and several other places along the waterfront. Small-craft facilities at Boston and Charlestown can provide berths with electricity, water, ice, marine supplies, wet storage and sewage pump-out; complete hull, engine and electronic repairs are available.

(222)

Boston Habor

(223) **East Boston**, on the northeastern side of Boston Harbor, is separated from the city of Chelsea by Chelsea River. The waterfront has modern piers and a large ship repair yard. These facilities were described earlier in this chapter under Wharves, Boston Harbor.

(224) The Jeffries Yacht Club is in the cove adjacent westward of Logan International Airport. A boatyard, close southwestward of the yacht club, has pump-out facilities and a marine railway that can handle vessels up to 100 tons for hull and engine repairs; a 10-ton crane is also available. A **fireboat** moors on the east side of the cove.

(225) **Chelsea** is separated from Charlestown, on the western side of the harbor, by the Mystic River. **Charlestown** is separated from Boston proper by the Charles River. **Charleston Navy Yard** is located on the north side of the mouth of the Charles River and is home to the **U.S.S. Constitution** and **U.S.S. Cassin Young**.

(226) A **safety zone** has been established around the Constitution when moored and underway and the Cassin Young while underway. (See **33 CFR 165.20**, **165.23**, **165.111**, and **165.112**, chapter 2, for limits and regulations.)

(227) **South Boston** is on the peninsula southeast of the city proper, from which it is separated by Fort Point Channel.

(228) A "no-wake"**speed limit** is enforced in the inner harbor from Pier #7 in South Boston to the northwest corner of Logan International Airport.

(229) **Logan International Airport** is between Governors Island Flats and East Boston. The airport area, almost entirely filled land, is low, flat and quite extensive. The airport control tower is conspicuous. **Governors Island**, on the northeast side of Boston Main Channel and at the southerly end of the airport, is a low grass-covered peninsula.

(230) **Vessel Reporting Advisory: Operations in Vicinity of Logan International Airport.** Vessels operating near Boston's Logan International Airport have the potential to affect procedures for aircraft landing and departing at the airport. Vessels with an air draft of 85 feet or greater require that advisories be issued to aircraft by air traffic controllers and in some instances that runway use be restricted. Notification by these vessels of maximum air draft when operating in the vicinity of the airport is intended to provide aircraft important notice of potential obstructions.

(231) Vessels with an air draft height of 85 feet (26 meters) or greater are advised to report their maximum air draft heights (including masts, cranes, antenna or other projections) when anchored or in transit in Boston Harbor between Deer Island and the World Trade Center. Notification is to be made to the Massachusetts Port Authority's Communication Center, preferably via telephone, at 617–561–1919. Vessels without telephone capability are requested to provide notification to Coast Guard Sector Boston on VHF channel 81, to be relayed via telephone to the Massachusetts Port Authority's Communication Center.

(232) Please supply the vessel's name, maximum air draft (including masts, cranes, antenna and other projections), destination, and estimated time that the vessel will be transiting/anchored/moored in Boston Harbor between

(235)

Structures across the Tributaries of Boston Harbor

Name•Description•Type	Location	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
Reserved Channel				
Summer Street Bridge (fxed)	42°20'32"N., 71°02'10"W.	40	6	
Fort Point Channel				
Northern Avenue Bridge (fxed)	42°21'16"N., 71°02'58"W.	76	7	
Northern Avenue Bridge (fxed)	42°21'14"N., 71°03'00"W.	75	16	
Congress Street Bridge (fxed)	42°21'07"N., 71°03'04"W.	75	6	
Summer Street Bridge (fxed)	42°21'03"N., 71°03'05"W.	51	8	
Charles River				
Charlestown Bridge (fxed)	42°22'06"N., 71°03'34"W.	50	23	Bridge is under construction (2018)
I-93 Mainline Bridge (fxed)	42°22'09"N., 71°03'49"W.	581	38*	
Storrow Drive Bridge (fxed)	42°22'08"N., 71°03'52"W.	344	44**	
MBTA/Amtrak Bridges (bascule)	42°22'08"N., 71°03'55"W.	65	3***	Note 1
Charles River Dam Bridges (bascule)	42°22'02"N., 71°04'08"W.	50	5***	Note 1
Longfellow Bridge (fxed)	42°21'41"N., 71°04'31"W.	187	29*	Bridge is under construction (2017)
Little Mystic Channel				
Chelsea Street Bridge (fxed)	42°22'48"N., 71°03'07"W.	75	9	
Route 1 Bridge (fxed)	42°22'48"N., 71°03'07"W.	75	100	
Mystic River				
Mystic River-Tobin Memorial Bridge (fxed)	42°23'05"N., 71°02'51"W.	600	135	
Malden/Alford Street Bridges (highway/bascule)	42°23'22"N., 71°04'16"W.	74	12	Note 2
MBTA Railroad Bridge (fxed)	42°23'35"N., 71°04'28"W.	74	30	
railroad bridge (fxed)		74	30****	
Wellington/Middlesex Avenue Bridge (fxed)	42°23'58"N., 71°05'01"W.	50	16****	Note 2
Harvard Street-General Lawrence Bridge (fxed)	42°24'22"N., 71°05'51"W.	50	13****	
Route 93 Bridge (fxed)	42°24'53"N., 71°06'14"W.	81	16****	
Malden River				
Woods Memorial/Revere Beach Pkwy (fxed)	42°24'14"N., 71°04'21"W.	80	21*	Bridge is under construction
Chelsea River				
Andrew P. McArdle Bridge (bascule)	42°23'08"N., 71°02'21"W.	175	20	Notes 4 and 5
Chelsea Street Bridge (vertical lift)	42°23'10"N., 71°01'22"W.	175	9 (down), 175 (up)	Notes 4 and 5

* Vertical clearances are referenced to Mean Basin Level, which is 7.2 feet above mean low water.
** Vertical clearances are referenced to Mean Sea Level.
*** Vertical clearances are referenced to High Water.
**** Vertical clearances are referenced to Normal Pool Elevation, which is 6.2 feet above mean low water.
Note 1 – See 33 CFR 117.1 through 117.59 and 117.591, chapter 2, for drawbridge regulations.
Note 2 – See 33 CFR 117.1 through 117.59 and 117.609, chapter 2, for drawbridge regulations.
Note 3 – See 33 CFR 117.1 through 117.59 and 117.593, chapter 2, for drawbridge regulations.
Note 4 – The bridgetender monitors VHF-FM channel 16 and works on channel 13.

Deer Harbor and the World Trade Center. Also supply a point of contact and call-back method of communication for the vessel while operating in the specified area.

(233) **Castle Island**, on the southwest side of Boston Main Channel 1 mile northwestward of Spectacle Island, is marked by **Fort Independence**. It is connected to the shore westward by filled land. Several boulders bare at low water are a short distance southeastward of Castle Island. This area should be avoided. On the northeast corner of the island is the 52-foot granite **Donald McKay Monument**, erected in 1933 to commemorate the famous East Boston builder of clipper ships.

(234) **Pleasure Bay**, just westward of Castle Island, is closed by an earth-filled dam extending from the southern end of the island to the jetty light southeastward of **City Point.**

(236) **Reserved Channel**, 0.5 mile northwestward of Castle Island, is a dredged unmarked channel that leads westward from the Boston Main Channel for about 1 mile.

(237) There are modern and extensive freight terminals on the north and south sides of Reserved Channel; these facilities were described earlier in this chapter under Wharves, Boston Harbor.

(238) **Fort Point Channel** separates Boston proper from South Boston. A dredged channel leads from the entrance to the Summer Street Bridge. Vessels bound for Fort Point Channel may require the assistance of a tug.

(239)

Anchorage

(240) A **special anchorage** is just north of the entrance to Fort Point Channel. (See **33 CFR 110.1** and **110.29**, chapter 2, for limits and regulations.)

(241) **Charles River**, on the western side of the harbor between Boston proper and Charlestown, is the approach by water to **Cambridge** and Watertown. There are two dams over the Charles River. The first is about 0.5 mile above the entrance to the river. The dam has three locks; the large north lock has a usable length of 300 feet and width of 40 feet with 14 feet over the sill; the other two locks have usable lengths of 200 feet with widths of 25 feet and 6 feet over the sills. The second dam is about 1 mile above the entrance. This dam has a single lock with usable dimensions of 350 feet by 45 feet with 17 feet over the sill; the lock is no longer in use and is maintained in the open position. (See **33 CFR 207.10**, chapter 2, for regulations governing the use, administration and navigation of the locks.) A *no wake* speed limit is enforced between the entrance to Charles River and the first Charles River Dam.

(242) Charles River above the dams is maintained at a height of 7.2 feet above mean low water. The river above the dams is used by many yachts and small craft. No toll is charged for passage through the locks. There are four yacht clubs on the river, some college sailing and rowing clubs, a large marina below the dams and two public float landings above the dams. Pump-out facilities are available at Cambridge and Watertown.

(243) **Little Mystic Channel** is a slip about 0.5 mile long 0.2 mile south-southeast of the mouth of the Mystic River at Charlestown. A wreck, covered 18 feet, is just eastward of the bridge in the middle of the channel and a wreck, covered 12 feet, is about 0.2 mile westward of the bridge.

(244) **Chelsea River**, locally known as Chelsea Creek, emptying into Boston Harbor from eastward between East Boston and Chelsea, is the approach to important wharves and facilities and to the city of **Revere** at the head, 2.6 miles above the entrance. The dredged channel and turning basin in the river have a project depth of 38 feet.

(245) Chelsea River has a heavy traffic of deep-draft oil tankers. The tankers berth at the oil company terminals and storage areas on both banks of the river. These facilities were described earlier in this chapter under Wharves, Boston Harbor.

(246) **Mystic River**, which empties into Boston Harbor opposite Chelsea River, is the approach by water to the towns of **Medford** and **Malden.**

(247)

Anchorages

(248) **Special anchorages** are on either side of the north end of the Mystic River-Tobin Memorial Bridge. (See **33 CFR 110.1** and **110.29**, chapter 2, for limits and regulations.)

(249) Amelia Earhart Dam, an earth-filled dam with 3 locks, crosses the Mystic River about 1.6 miles above the mouth. The largest lock, a commercial-vessel type, has a length of 325 feet, a width of 45 feet, and depths of 15.5 feet over the lower sill and 11.5 feet over the upper sill. Two smaller parallel locks just westward have lengths of 120 feet, widths of 22 feet, and depths of 6.5 feet over the lower sills and 0.5 foot over the upper sills. (See **33 CFR 207.9**, chapter 2, for regulations governing the use, administration and navigation of the locks.) There are no overhead vertical restrictions on any of the locks.

(250) A large marina is on the north bank of the river, just westward of the railroad bridge. Gasoline, water, ice, marine supplies, storage facilities, a small-craft launching ramp and a 15-ton mobile hoist are available; hull, engine and electronic repairs can be made.

(251) There are two yacht clubs on the river above the mouth of the Malden River: the Winter Hill at Somerville and the Riverside at Medford. The Chelsea Yacht Club is on the north bank on the east side of the Mystic River-Tobin Memorial Bridge. Gasoline, diesel fuel, water and electricity are available at the floats, which have 30 feet alongside.

(252) **Island End River** is a tributary of the Mystic River entering from northward, 0.5 mile above the entrance. Three companies maintain wharf facilities on the west side of the river. A rocky area on the east side of the entrance, and the current of Mystic River running across the entrance, make navigation difficult for large vessels. A tug usually is employed to assist such vessels. A dredged channel leads to a marina basin near the head of the river; the channel is marked by daybeacons. A pump-out facility is available at the marina.

(253) **Malden River**, a tributary of Mystic River from northward, has a privately dredged channel for a distance of 1.6 miles upstream.

(254)

Winthrop to Worlds End

(255) North and west of President Roads is an area of flats, much of which bares at low water. Between **Deer Island Flats** and **Governors Island Flats**, a buoyed channel leads to Cottage Park Yacht Club at **Winthrop**. Another buoyed channel, branching to the east, leads to Winthrop Head and is marked by a light just west of the entrance. **Snake Island**, on the westerly side of the channel, is 10 feet high with low trees.

(256) The Winthrop Yacht Club, a wharf with depths of about 5 feet alongside its floats, and a marina are on the east bank of Winthrop Head. Gasoline, diesel fuel, water, ice, a 10-ton lift, a small-craft launching ramp, some marine supplies, and limited overnight berthage are available at the marina; hull and engine repairs can be made. Pumpout service is available from the Town of Winthrop.

(257) The westerly channel leading to **Belle Isle Inlet** has a controlling depth of about 15 feet. A **special anchorage**

(264)

Structures across Neponset River

Name•Description•Type	Location	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
Dorchester Bay Basin				
William T. Morrisey Boulevard Bridge (bascule)	42°18'14"N., 71°02'53"W.	65	12	Note 1
Neponset River				
Old Colony Railroad Bridge (fxed)	42°17'08"N., 71°02'18"W.	109	30	
Route 3A Bridge (fxed)	42°17'04"N., 71°02'24"W.	80	30	
Neponset River Bridge (highway/fxed)	42°16'39"N., 71°02'56"W.	130	33	
Granite Avenue Bridge (bascule)	42°16'39"N., 71°03'12"W.	50	6	Note 2

Note 1 – See 33 CFR 117.1 through 117.59 and 117.597, chapter 2, for drawbridge regulations.
Note 2 – See 33 CFR 117.1 through 117.59 and 117.611, chapter 2, for drawbridge regulations.

is off the Pleasant Park Yacht Club, just south of the inlet. (See **33 CFR 110.1** and **110.29**, chapter 2, for limits and regulations.) The highway bridge over the mouth of the inlet has a 35-foot fixed span with a clearance of 7 feet. Marinas near the yacht club can provide berths, some marine supplies, covered storage, a 50-foot marine railway; hull and engine repairs. Farther west of the mouth of the inlet are the Orient Heights and East Boston Yacht Clubs. Fuel, water and various services are available at the yacht clubs.

(258) **Dorchester Bay** extends southwestward from President Roads between **Spectacle Island** and **Thompson Island** on the east and South Boston on the west. The bay is filled with extensive flats, large areas of which are nearly bare at low water and rise abruptly from the edge of the channel. There is no deep-draft vessel traffic in Dorchester Bay. The John F. Kennedy Memorial Library on **Columbia Point** (42°18'54"N., 71°02'22"W.) is prominent. A state-maintained small boat channel leads from the main channel in Dorchester Bay to a turning basin near the JFK Memorial Library.

(259)

Anchorages

(260) **Special** and **general anchorages** are in Dorchester Bay. (See **33 CFR 110.1**, **110.30** and **110.138**, chapter 2, for limits and regulations.) The yacht anchorage most commonly used is south and east of City Point, clear of the cable area.

(261) **Old Harbor**, on the west side of Dorchester Bay, just south of South Boston, is filled with flats having little water over them. A channel with a least depth of 5 feet leads to the yacht clubs and the public float in the northeastern part of the harbor, westward of City Point. Gasoline and diesel fuel are available.

(262) **Squantum Channel** leads from the main channel in Dorchester Bay to a marina east of **Squantum Point**. The channel to the basin, which is enclosed and protected by stone breakwaters, is marked by buoys. Gasoline, diesel fuel, water and electricity can be obtained at the floats, and a 30-ton mobile hoist, storage facilities, marine supplies and a pump-out facility are available; hull, engine and electronic repairs can be made.

(263) **Dorchester Bay Basin**, on the southwest side of Dorchester Bay, is entered about 0.2 mile westward of **Commercial Point**, the western entrance point to Neponset River. A channel, privately marked by seasonal buoys, leads to a yacht club on the northwest side of the basin. Gasoline, water, ice and a pump-out station are available. A rock awash is reported immediately eastward of the north draw of the highway bridge.

(265) **Neponset River** enters Dorchester Bay from the south between Commercial Point and Squantum Point. A dam is at **Milton**, 3 miles above the mouth. Small craft with local knowledge navigate to Milton during times of high water.

(266) Several yacht clubs are on the river. Gasoline, diesel fuel, water, ice, marine supplies, a pump-out facility, storage facilities and lifts up to 40 tons are available at a large marina on the west side about 1 mile above the mouth; complete hull, engine and electronic repairs can be made.

(267) **Quincy Bay** indents the southerly shore of Boston Harbor between the peninsulas of Squantum and Houghs Neck. Depths in the bay are in general 8 to 10 feet, but shoals partly bare at low water extend 0.5 to 0.7 mile from its southerly side.

(268)

Anchorages

(269) **Special anchorages** are in Quincy Bay. (See **33 CFR 110.1** and **110.30**, chapter 2, for limits and regulations.)

(270) The wharf extending from the south side of **Rainsford Island**, at the northeastern entrance to Quincy Bay, is in ruins. **Quarantine Rocks** extend 0.5 mile southward of the island.

(271) **Sunken Ledge**, bare at low water, is about 1 mile southward of Rainsford Island. A daybeacon is on the ledge, and a lighted buoy is 0.2 mile southeast of it. A buoy marks a channel west of the ledge.

(272) **Hangman Island**, small and rocky, is near the middle of the entrance to Quincy Bay, 0.6 mile southwestward of Sunken Ledge. The end of a reef extending 0.2 mile southwestward from the islet is marked by a daybeacon. Two charted rocks with depths of 6 feet and 4 feet are 0.4 mile northward of the daybeacon.

(273) **Wreck Rock**, 0.6 mile southeast of Hangman Island, is covered 3 feet and marked on the north end by a buoy.

(274) **Squantum** is on the west side of Quincy Bay. **Moon Head**, which can be recognized by the grassy hill and bluff on its easterly end, is connected to Squantum by a causeway. A Boston Fire Department training center is prominent on the northeast side.

(275) A channel leads northward from Quincy Bay between Moon Head and Long Island, and the piers of a former fixed bridge. The piers on either side of the channel are marked by private lights. The channel continues northwesterly and connects with Presidents Roads via **Western Way**, between Thompson and Spectacle Islands, or by **Sculpin Ledge Channel**, westward of Long Island.

(276) **Wollaston Channel**, privately maintained, leads southwestward from the westerly end of Quincy Bay to the small basin of the Squantum and Wollaston Yacht Clubs. With local knowledge, 6 feet was reported to be carried in the entrance channel and basin. The channel is marked by buoys.

(277) **Houghs Neck**, on the southeast side of Quincy Bay, is marked at its northeasterly end by **Quincy Great Hill**, 100 feet high and mostly settled. **Nut Island**, marked by a power station and stack, is connected by a causeway 300 yards northward of Quincy Great Hill.

(278) A private daybeacon marks the outer end of an overflow pipeline extension with riprap cover, off the northeast end of Nut Island. The pipeline extends 150 yards into the bay and is submerged at high water. **Pig Rock**, about 0.6 mile eastward of Nut Island, is visible at all stages of the tide. Rocks awash are close east of Pig Rock.

(279) The Quincy Yacht Club is on the eastern side of Houghs Neck. A channel, marked by buoys, leads to the club wharf.

(280)

Anchorages

(281) A **special anchorage** is off the east side of Houghs Neck. (See **33 CFR 110.1** and **110.30**, chapter 2, for limits and regulations.)

(282) **Spiers Stand**, which uncovers, is about 575 yards off the east side of Houghs Neck and is marked by buoys off its north and south ends. Give Spiers Stand a wide berth, as several groundings have been reported in its vicinity.

(283) **Peddocks Island**, 0.5 mile northeast of Houghs Neck, is long and narrow with hills in the middle and at both ends, with low land between. The remains of the brick buildings of old Fort Andrews are visible. Caution is advised against approaching the island too closely because of rocks awash. West Head is reserved as a bird sanctuary.

(284) **West Gut** is a buoyed channel leading into Hingham Bay between Nut Island and Peddocks Island. The channel through West Gut has a controlling depth of about 19 feet. A rock covered 17 feet is in the channel about 0.3 mile east-northeastward of Nut Island.

(285) **Hingham Bay** is that part of Boston Harbor southeastward of Peddocks Island. It is the approach to Weymouth Fore River, Weymouth Back River, Hingham Harbor and Weir River. Extensive shoals make out from the southerly shore and surround the islands in the bay. **Hull Bay**, the eastern part of the bay, also has many shoal areas.

(286)

Anchorages

(287) **Special anchorages** are in Hull Bay. (See **33 CFR 110.1** and **110.31**, chapter 2, for limits and regulations.)

(288) The easterly entrance to Hingham Bay is through Hull Gut, but the entrance through West Gut, southward of Peddocks Island, is frequently used by vessels bound into Weymouth Fore or Weymouth Back Rivers.

(289) **Hull Gut**, a dredged channel between Peddocks Island and Windmill Point, leads into Hingham Bay from Nantasket Roads and is a section of the Weymouth Fore River Channel improvement. Unmarked submerged rocks cleared to a depth of 16 feet are about 500 yards southward of Windmill Point Light WP.

(290)

Current

(291) The tidal currents have an average velocity of about 2 knots at strength and generally follow the direction of the channel; the flood sets southward and the ebb northward.

(292) A channel about 13 feet deep, commencing just southward of the buoy marking **Inner Seal Rock**, about 0.6 mile southeastward of Windmill Point Light, leads to a wharf on the northwest side of **Spinnaker (Hog) Island**. The island is connected to the mainland by a causeway and fixed bridge.

(293)

Anchorages

(294) A channel, privately marked by seasonal buoys, eastward of Spinnaker Island, leads northward to a **special anchorage** in **Allerton Harbor**. (See **33 CFR 110.1** and **110.31**, chapter 2, for limits and regulations.) The Hull Yacht Club is on the north side of the harbor.

(295) A marina with depths of about 6 feet alongside its floats is at **Waveland**, about 0.7 mile southeastward of Spinnaker Island. Gasoline, diesel fuel, water, ice, electricity, marine supplies, a small-craft launching ramp and lifts up to 30 tons are available; hull, engine and electronic repairs can be made.

(296) **Weymouth Fore River** has its entrance on the southwest side of Hingham Bay between Houghs Neck and **Grape Island** and is the approach by water to **Quincy Point, Weymouth,East Braintree** and several landings. An electric power plant and several other industries are on the river. Waterborne commerce is principally in petroleum products.

(297) **Grape Island**, on the south side of Hingham Bay, has a recreational pier on the southerly side of the island. Ruins of several buildings, nature trails and picnic areas are on the island. Primitive camping is permitted.

(304)

Structures across Weymouth Fore and Weymouth Back Rivers

Name•Description•Type	Location	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
Weymouth Fore River				
State Route 3A Bridge (vertical lift)	42°14'39"N., 70°58'03"W.	175	175 (open) 60 (closed)	Notes 1 and 2
Overhead power cable	42°14'10"N., 70°57'51"W.		150	
Overhead power cable	42°13'53"N., 70°57'42"W.		100	
Overhead power cable	42°13'40"N., 70°57'40"W.		60	
State Route 53 Bridge (fxed)	42°13'25"N., 70°58'08"W.	51	11	
Weymouth Back River				
Lincoln Street Bridge (fxed)	42°14'49"N., 70°55'54"W.	70	36	
Overhead power cables	42°14'49"N., 70°55'54"W.		55	

Note 1 – See 33 CFR 117.1 through 117.59 and 117.621, chapter 2, for drawbridge regulations.
Note 2 – The Bridgetender monitors VHF-FM channel 16 and works on channel 13; call sign WRD-634

(298) The following are prominent upon entering Weymouth Fore River; a tank on Weymouth Great Hill, the bridge at Quincy Point and the large stacks of the power plant on the east side of the river just above the bridge.

(299) Sewage pump-out is available at the Wessagusset Yacht above Weymouth Great Hill.

(300)

Channels

(301) A dredged channel leads from the sea through Nantasket Roads, Hull Gut and Hingham Bay, thence into Weymouth Fore River; a turning basin is on the south side of the channel at Kings Cove. The channel through West Gut is described earlier in the chapter. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A.

(302)

Anchorages

(303) **Special anchorages** are in Weymouth Fore River. (See **33 CFR 110.1** and **110.30**, chapter 2, for limits and regulations.)

(305)

Weather, South Weymouth and vicinity

(306) **South Weymouth** is about 3 miles south-southwest of Weymouth. South Weymouth averages about 11 days each year with maximum temperatures in excess of 90 °F. July is the warmest month with an average high of 82 °F and an average minimum of 63 °F. January is the coolest month with an average high of 36 °F and an average minimum of 19 °F. The highest temperature on record for South Weymouth is 102 °F recorded in August 1975 and the lowest temperature on record is -15 °F recorded in January 1981. About 121 days each year see temperatures below 32 °F and an average seven days each year record temperatures below 5 °F. Every month except July has seen temperatures below 40 °F and every month except June, July and August has recorded temperatures below freezing.

(307) The average annual precipitation for South Weymouth is 44.3 inches (1125 mm), which is fairly evenly distributed throughout the year. Precipitation falls on about 188 days each year. The wettest month is November with 4.6 inches (117 mm) and the driest, July, averages only 2.8 inches (71 mm). An average of 20 thunderstorm days occur each year with June, July and August being the most likely months. Snow falls on about 51 days each year and averages about 43 inches (1092 mm) each year. Due to the strong maritime influence, only January and February average greater than one-foot (305 mm) of snow. One-foot (305 mm) snowfalls in a 24-hour period have occurred in each month December through April and 19 inches (483 mm) fell in one 24-hour period during February 1994. About eight days each year have a snowfall total greater than 1.5 inches (38 mm) and snow has fallen in every month, October through May. Fog is present on average 204 days each year and is evenly distributed throughout the year with a slight maximum during mid-summer.

(308) The prevailing wind direction in South Weymouth is west-northwest during the winter and south-southwest during the warmer months. March is the windiest month—see the South Weymouth climatological table in this chapter.

(309) **Town River Bay** is a branch of Weymouth Fore River north of Quincy Point. A Federal project provides for a channel 35 feet deep from the junction with Weymouth Fore River to a point 1 mile upstream, with a turning basin 35 feet deep at the inner end; thence 15 feet deep to a point just below the Quincy Electric Light & Power Company plant, 1.2 miles above the mouth. (See Notice to Mariners and latest edition of the chart for controlling depths.) About 175 yards above the head of the project on Quincy Reach, an overhead power cable has a clearance of 35 feet.

(310)

Wharves

(311) There are two deep-draft facilities on Weymouth Fore River and two on Town River Bay.

(312) **Twin Rivers Technologies (TRT)**, consists of East and North Docks. **East Dock** (42°14'49"N., 70°58'00"W.): 150-foot face with 435 feet berthing space with dolphins; 30 feet alongside; deck height, 15½ feet; receipt of vegetable oil and caustic soda. **North Dock** (42°14'54"N., 70°58'04"W.): 90-foot face with 600 feet berthing space with dolphins; 29 feet alongside; deck height, 16½ feet; pipelines extend to storage tanks in rear with 22½-million-gallon capacity; receipt of coconut oil, vegetable oil and caustic soda; owned and operated by Twin Rivers Technologies.

(313) **Citgo Petroleum Corp., Braintree Terminal Wharf** (42°14'12"N., 70°58'05"W.): 248-foot face, 700 feet berthing space; 38 feet alongside; deck height, 14 to 16 feet; pipelines extend to storage tanks with a 1¼-million barrel capacity; receipt and shipment of petroleum products; owned and operated by Citgo Petroleum Corp.

(314) **Sprague Energy, Quincy Terminal Wharf** (42°15'12"N., 70°59'10"W.): 77-foot face with 700 feet berthing space with dolphins; 35 feet alongside; deck height, 14 feet; receipt and shipment of petroleum products; bunkering vessels; owned and operated by Sprague Energy, Inc.

(315) There are several private piers that are used occasionally for mooring barges and small vessels on Town River Bay.

(316) A marina and a yacht club are on the south bank about 0.5 mile and 0.7 mile, respectively, above the entrance. Gasoline, diesel fuel, ice, supplies, pump-out facilities and a 15-ton hoist are available; hull and engine repairs can be made.

(317) **Weymouth Back River** is just eastward of Weymouth Fore River and southward of Grape Island. A wharf, in ruins, of a former fertilizer works is on the north side of the river on Eastern Neck. The channel is buoyed.

(318)

Anchorages

(319) A **special anchorage** is in Weymouth Back River. (See **33 CFR 110.1** and **110.30 (l)**, chapter 2, for limits and regulations.)

(320)

Small-craft facilities

(321) Small-craft facilities on Weymouth Back River can provide berths with electricity, gasoline, diesel fuel, water, ice, marine supplies, a launching ramp and lifts to 37 tons; hull, engine and electronic repairs are available. Sewage pump-out is available on the south side of the river near the end of the channel. A public launching ramp is on the north side of the river about 1.5 miles above the entrance.

(322) Hingham Harbor and Weir River in the southeasterly end of Hingham Bay are shallow. Their common entrance is close westward of Bumkin Island. The channel leads in a southeasterly direction for about 0.5 mile from the westerly end of **Bumkin Island** and then divides. The branch leading eastward is Weir River. Bumkin Island has a recreational pier on the southwest side. Ruins of several buildings, nature trails and picnic areas are on the island. Primitive camping is permitted.

(323) The narrow channel leading to Hingham Harbor trends southward to **Crow Point** and is marked by buoys. The Hingham Yacht Club has a clubhouse, pier and floats at Crow Point. It is reported that considerable shoaling has occurred along the face of the pier and the northern half cannot be approached by large vessels, except at half tide or higher; water is available at the pier.

(324) **Hingham Harbor** is a cove 1 mile in length, with an average width of about 0.6 mile. At low water it is a dry flat through which a narrow and tortuous buoyed channel winds to the town of Hingham.

(325)

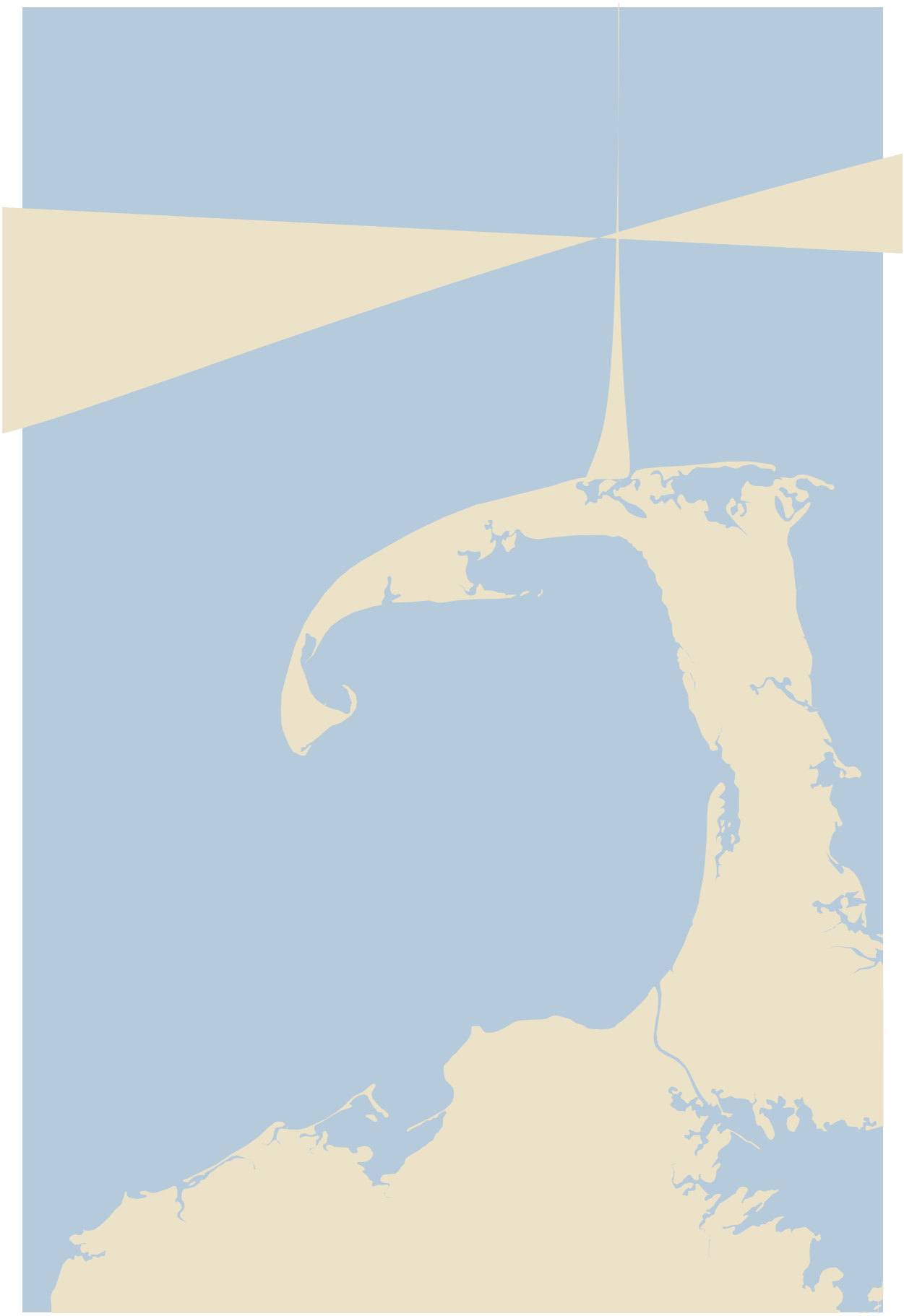
Anchorages

(326) **Special anchorages** are eastward of Crow Point at the entrance to Hingham Harbor and at the southern end of the harbor. (See **33 CFR 110.1** and **110.32**, chapter 2, for limits and regulations.)

(327) The small-boat basin at the south end of the harbor has depths of 2 to 7 feet. A town float landing and ramp are on the south side of the basin, and there are two service wharves where gasoline, diesel fuel by truck, water and most other services are obtainable. Boat rental and a pump-out facility are available; outboard repairs are made.

(328) **Weir River** leads to the wharf at **Nantasket Beach**. Extensive flats, mostly bare at low water, are on both sides of the river. The channel is marked by buoys and is used by excursion boats running from Boston to Nantasket Beach during the summer. In 2000, the large wharf at Nantasket Beach had depths of 10.9 and 9.2 feet alongside the 175-foot southwest and northwest faces, respectively. Two small marinas, one north and one south of the wharf can provide berths with electricity, gasoline, ice, a launching ramp, pump-out, a 5-ton lift for emergencies, open and covered storage.

(329) A safe anchorage for small craft with good holding ground, mud bottom, is reported close eastward of **Worlds End**, the south entrance point to Weir River. The anchorage is unmarked; local knowledge is advised.



Minots Ledge to Provincetown, Massachusetts

(1) This chapter describes the Massachusetts coast southward from Minots Ledge, off Cohasset Harbor, to and including Cape Cod Bay. Also discussed are the principal harbors of Cohasset, Scituate, Green, Duxbury, Plymouth, Barnstable, Sesuit, Rock, Wellfleet, Pamet and Provincetown and New Inlet and its tributaries. Provincetown and Plymouth with their seafood handling and processing facilities and fleets of fishing vessels are the principal commercial harbors. Pleasure boating is prevalent with some commercial fishing at the other harbors.

(2)

No-Discharge Zone

(3) The State of Massachusetts, with the approval of the Environmental Protection Agency, has established a No-Discharge Zone (NDZ) in all coastal waters of Massachusetts described in this volume, extending about 3 miles offshore.

(4) Within the NDZ, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZ, discharge of sewage is regulated by **40 CFR 140** (see chapter 2).

(5)

Minots Ledge Light to Scituate Harbor

(6) **Minots Ledge Light** (42°16'11"N., 70°45'33"W.), 85 feet above the water, is shown from a gray conical tower on **Outer Minot**. A mariner-radio-activated sound signal at the light is initiated by keying the microphone five times on VHF-FM channel 83A. This ledge, which uncovers 3 feet, is about 6 miles southeastward of Point Allerton and 1 mile north-northeastward of **Strawberry Point**, the northeastern extremity of **Scituate Neck**. Outer Minot is the outermost of the visible dangers off the entrance to Cohasset Harbor.

(7) Submerged rocks and very broken ground, on which the sea breaks in heavy weather, extend more than 1 mile northeastward and 2.5 miles eastward of the light. This area should be avoided.

(8) Numerous rocks and ledges extend westward and southward from the light across the entrances to Cohasset Harbor. **East Shag Rock**, 7 feet high and marked by a buoy, and **West Shag Rock**, 6 feet high, are the most prominent southwestward of the light. Shifting boulders are reported on the shoal just eastward of **Barrel Rock** (42°15.5'N., 70°47.1'W.), marked by a daybeacon.

(9) Three natural channels lead into **Cohasset Harbor** through the area of rocks and ledges: **Western Channel**, which enters between **Brush Ledge** and **Chittenden Rock**; **The Gangway**, a passage that leads between **The Grampuses** and **West Hogshead Rock**; and **Eastern Channel**, which leads between **Enos Ledge** and **West Willies**. Although all three channels are marked by buoys, there are numerous unmarked dangers. One notable outlying danger is **Rumrunner Rock**, just south of **Brush Island**, in the vicinity of 42°15'23.5"N., 70°47'34.5"W.

(10) The Gangway passage is the widest, but there are unmarked covered 9- and 10-foot rocks in the middle of it, and it should be used only in clear weather and with a smooth sea, even in small craft. Eastern Channel is the clearest and deepest of the three. The best time to enter is on a rising tide.

(11)

COLREGS Demarcation Lines

(12) The lines established for this part of the coast are described in **33 CFR 80.135**, chapter 2.

(13) **Cohasset Harbor** is a large shallow bight southwestward of Minots Ledge Light and about 6 miles southeastward of Point Allerton. The harbor is frequented by numerous yachts and fishing craft. A prominent lookout tower is near the summit of a hill eastward of **The Glades** on the east side of the harbor.

(14)

Anchorages

(15) Anchorage is available in depths of 6 to 10 feet in the outer harbor.

(16) **Cohasset Cove**, the inner harbor, is protected by a breakwater that extends about 0.1 mile northward from near the westerly end of **Bassing Beach**. The breakwater is partially covered at high water.

(17) A dredged channel leads southward from the outer harbor to an anchorage basin southward of Bryant Point in Cohasset Cove. There are three additional dredged anchorage areas: one is immediately southward of the Cohasset Cove anchorage; one in **Bailey Creek**, in the southeastern part of the inner harbor; and one immediately westward of the southern end of the Cohasset Cove anchorage. The channel into Cohasset Cove is marked by lights and buoys; a light is off **Bryant Point.**

(18) A rock, which uncovers 6½ feet, is in 42°14'21"N., 70°47'15"W., close to the southerly edge of the channel leading to the anchorage in Bailey Creek. Another rock, covered about 1 foot, is reported in the westerly anchorage, about 65 yards northeastward of the town landing on the southerly side of the anchorage; caution is necessary when maneuvering around the service wharves eastward of this landing.

(20)

(19) **Cohasset** is a town on the west side of the inner harbor. There is some fishing, but the town is mostly residential. The Cohasset Yacht Club, close westward of Bryant Point, has depths of 5 to 8 feet reported alongside its float landing; water is available. The town maintains four float landings in various parts of the inner harbor; depths of 3 to 5 feet are reported alongside these landings. Sewage pump-out is available.

(21)

Harbormaster

(22) The harbormaster maintains an office in a cottage that overlooks the town wharf southwestward of the entrance to Bailey Creek.

(23) The Cohasset Sailing Club, about 100 yards eastward of this town landing, has a depth of 3 feet reported alongside its float landing. A small-craft launching ramp is about 150 yards eastward of the sailing club.

(24) A boatyard is just westward of the dam at the head of the inner harbor. Depths of 9 feet are reported alongside the yard's float landing. The marine railway at the yard can handle craft up to 55 feet in length or 80 tons for hull and engine repairs or open or covered storage; gasoline and water are available.

(25)

Ice

(26) Cohasset Harbor is usually closed by ice for about 2 months during normal winters.

(27) **Stellwagen Ledges**, consisting of rocks awash and covered, extend 3.8 miles south-southeastward from Davis Ledge to Tar Pouch. Some of these ledges lie over 1 mile from shore and are covered 5 to 16 feet in surrounding depths of 4 to 9 fathoms. Most of them are unmarked. Strangers should keep over 3 miles from shore.

(28) **Davis Ledge**, covered 15 feet and marked by a lighted gong buoy, is about 0.4 mile eastward of Minots Ledge Light. **Tobias Ledge**, about 0.25 mile eastward of Strawberry Point, is marked by a daybeacon. **Tar Pouch**, covered 14 feet and marked on the northeast side by a buoy, is about 1 mile northeastward of the entrance to Scituate Harbor.

(29) **Scituate Harbor**, about 4 miles southeastward of Cohasset Harbor, is used mostly by yachts and fishermen and occasionally as a harbor of refuge by draggers.

(31) **Cedar Point**, on the north side of the harbor, is marked by a concrete lighthouse tower maintained by the Coast Guard. The harbor is partially protected by breakwaters.

(32) The north breakwater extends about 300 yards southeastward from the southeast extremity of Cedar Point. **Scituate North Jetty Light 2A** (42°12'11"N., 70°42'47"W.), 23 feet above the water, is shown from a white skeleton tower with tankhouse and a red triangular daymark on the seaward end of the north breakwater.

(30)

The south jetty extends about 100 yards northward from the southern point of the entrance. A lighted gong buoy, 0.7 mile eastward of the north breakwater, marks the approach to the harbor.

(33) On the high land about 2 miles westward of the entrance to Scituate Harbor, there is a conspicuous high tower with pointed top that is visible many miles from seaward.

(34) It is reported that the bar at the entrance breaks entirely across the channel at low tide and in heavy weather. The most unfavorable weather is from the northeast. The outer harbor is free of ice most of the winter.

(35)

Coast Guard

(36) Coast Guard Station Scituate is located at the entrance to Scituate Harbor. The station is a seasonal station open during the summer boating season.

(37)

Channels

(38) A dredged channel leads westward between two breakwaters into Scituate Harbor, thence turns southwest and south terminating in an anchorage basin adjacent to the Scituate waterfront; the channel is marked by buoys.

(39) The Scituate Harbor Yacht Club is on the west shore of the harbor, about 0.5 mile westward of the jetty light where water and berths are available. Varying depths of 3 to 8 feet are reported throughout the harbor and anchorage areas; refer to the chart and Local Notice to Mariner's for depth and channel conditions. Southward are the Satuit Boat Club and the Satuit Waterfront Club. Between them, there are two small-craft launching ramps and a marina where a service float, gasoline, diesel fuel, water and ice, are available. The Scituate town pier is on the west side of the harbor, about 0.6 mile above the jetty light; sewage pump-out is available. Another marina with several floats is close southward of the town pier; electricity, gasoline, diesel fuel, water, ice and sewage pump-out are available. A large parking lot is at the marina. A public dock and a small-craft launching ramp are just southward of the marina.

(40) A boat storage yard, at the head of the cove at the southeast end of the harbor, has a 10-ton marine railway that can handle craft up to 30 feet in length for hull and minor engine repairs. Electricity, water, ice and some marine supplies are available at the service float, which has a reported depth of 6 feet alongside.

(41) Marine supplies, tackle and most services are available at the marinas and the boat storage yard. Groceries and lodging are available within walking distance.

(42) On the south side of the entrance to Scituate Harbor is a bluff known as **First Cliff**. A similar formation, known as **Second Cliff**, is about 0.6 mile south of the entrance.

(43)

New Inlet to South River

(44) **New Inlet**, on the north side of **Fourth Cliff** and 2 miles southward of Scituate Harbor, is the approach to North River and South River. The inlet is marked by a lighted bell buoy off the entrance and the channel is marked by buoys that are frequently shifted with changing conditions. The channel is subject to change and is never entered except by small craft with local knowledge. Strangers should not attempt to cross the bar on the ebb with an easterly wind or in heavy seas as waves break across the bar. The bar consists of boulders that are reported to be particularly numerous on the south side of the inlet. A strong current flows out of the inlet during the falling tide.

(45) Sand and gravel were formerly shipped from a wharf on the east bank about 1 mile above the mouth of **Herring River**, a tributary of North River from the north. Uncharted private buoys that are frequently shifted with changing conditions mark the river. In 1979, it was reported that with local knowledge about 4 feet could be carried to the wharf and to a marina in a basin about 0.4 mile above the wharf. The marina boatyard has a 25-ton mobile hoist that can haul out craft up to 60 feet in length for hull and engine repairs or dry covered or open winter storage. Gasoline, diesel fuel, electricity, water and a pump-out facility are available at the floats, which have a reported 3 to 6 feet alongside. Ice, provisions and marine supplies can be obtained at the marina, and restaurants are available nearby.

(46) **North River** formerly emptied into the sea near **Rexhame**, but its present outlet dates from the great storm of 1898. The river has been partly cleared of boulders to **Hanover**, 10 miles above the entrance. The depth to this point is about 2 feet. Local knowledge is advisable to navigate the river. Navigation at spring tides in excess of 9 feet above mean low water is difficult because of flooding of large areas of marshland on either side of the river. The channel to the State Route 3A bridge is partially marked by privately maintained stakes in the summer.

(47) About 1.4 miles above the mouth, the south abutment of an abandoned railway bridge, which has been removed, is used as a town landing. There are no services at the float, which has 6 feet reported alongside. The channel is very narrow here, and the currents flow strongly, especially on the ebb.

(48) State Route 3A highway bridge crossing the river about 1.65 miles above the mouth has a fixed span with a clearance of 17 feet. In 2002, an obstruction (submerged piling) was reported in North River directly under State Highway Route 3A bridge; the piling is reported visible at low tides. The second highway bridge about 4 miles above the mouth has a 27-foot bascule span with a clearance of 6 feet. (See **33 CFR 117.1** through **117.59** and **117.613**, chapter 2, for drawbridge regulations.)

(49) There are two marinas at the first highway bridge. The one on the north bank just east of the bridge is principally for outboards; a pump-out facility, a small-craft launching ramp and a 20-ton crane are at the facility. The marina on the south bank just west of the bridge has gasoline, a pump-out facility and water available at a float that had 3 feet of water reported alongside and a paved small-craft launching ramp. Outboard boat rental and bait are available.

(50) About 1.5 miles above the first highway bridge, at Kings Landing, is a boatyard. Boats up to 40 feet in length are hauled out on skids for hull and engine repairs or open winter storage. The river has a posted **speed limit** of 5 miles per hour.

(51) **South River**, emptying through New Inlet from southward, is used by fishermen and yachtsmen. **Humarock** is a small village on the beach between South River and the ocean, 1.5 miles southward of New Inlet. Local knowledge of the river channel is advisable to navigate to the town. In 1979, the reported controlling depth was 3 feet from the entrance to the first bridge and thence shoaling to bare about 350 yards above this bridge. In 1985, a sunken wreck was reported in the channel in about 42°08'50"N., 70°42'10"W. The channel is marked by privately maintained and uncharted buoys that are frequently shifted with changing conditions.

(52) A highway bridge 2.2 miles above the river's mouth has a 45-foot fixed span with a clearance of 8 feet. There is a marina on the east bank just north of the bridge and another on the west bank just south of the bridge. Both marinas have small-craft launching ramps and service floats with 2 to 5 feet reported alongside; water, gasoline and electricity are available. The marina on the east bank has a 14-ton mobile hoist that can handle craft up to 43 feet, and the marina on the west bank has a marine railway that can haul out boats up to 40 feet in length for hull and engine repairs or dry open or covered winter storage. Ice, provisions and marine supplies are obtainable, and restaurants and lodging are nearby. Guest moorings are maintained by the marinas and pump-out facilities are available.

(53) A **speed limit** of 5 miles per hour is posted on the river.

(54) The Marshfield Yacht Club is on the west bank about 0.3 mile above the first highway bridge; a depth of 4 feet is at the float landings. Water and electricity are available at the floats. The **harbormaster** can usually be found here. There is a boatyard on **Littles Creek** about 0.5 mile northwestward of the first bridge. Boats up to 40 feet in length are hauled out at high water for dry winter storage and minor repairs.

(55) The second highway bridge about 2.5 miles above the mouth has a 24-foot fixed span with a clearance of 5 feet; there is little or no navigation above the second bridge.

(56)

Cape Cod Bay

(57) **Cape Cod Bay** is contained between the peninsula of Cape Cod, on the east and south, and the mainland of Massachusetts on the west. Between these limits the bay is about 20 miles in diameter with depths ranging from 10 to 32 fathoms, except close to the shore and in its southeasterly part. Race Point, the northwesterly extremity of Cape Cod, is the eastern point; and Gurnet Point, on the north side of the entrance to Plymouth Bay, is the western point of the entrance to Cape Cod Bay.

(58) Within the limits of Cape Cod Bay are several harbors, including those of Plymouth on the western shore, Sandwich and Barnstable on the southern shore and Wellfleet and Provincetown on the eastern shore. It is also the approach to Cape Cod Canal, which connects Cape Cod Bay with Buzzards Bay.

(59) The shallow harbors of Cape Cod Bay, such as Plymouth, Barnstable and Wellfleet, usually are closed to navigation by ice a part of each winter. This ice, together with the ice that forms in the shallower parts of Cape Cod Bay in severe winters, is driven by the winds out into the bay. There it masses into heavy fields or windrows, sometimes as much as 10 feet or more thick, making navigation in parts of the bay unsafe or impractical. The prevailing northerly winds drive the ice down to the southern end of the bay, but on a few occasions it has been known to obstruct Provincetown Harbor for several days. The movements of the ice depend largely on the winds; the tidal currents apparently have little or no effect.

(60) Deep-draft vessels entering Cape Cod Bay from the northward should pass eastward of the lighted whistle buoy that is about 7 miles northeastward of Brant Rock and well east of the extremity of the broken bottom extending over 4 miles offshore in this direction.

(61)

North Atlantic Right Whales

(62) Cape Cod Bay lies within the federally designated critical habitat for North Atlantic right whales. The Northeastern Foraging Area is a primary winter/spring feeding area and may be inhabited by right whales year-round (peak season: December through May). (See **50 CFR 226.101** and **226.203**, Chapter 2 for habitat boundary and regulations.) It is illegal to approach closer than 500 yards of any right whale (see **50 CFR 224.103(c)**, chapter 2, for limits and regulation).

(63) **Recommended two-way Whale Avoidance Routes** have been established in Cape Cod Bay to reduce the likelihood of ship strikes of endangered North Atlantic right whales. All vessels are encouraged to use recommended routes when traveling in Cape Cod Bay. A **Recommended two-way Whale Avoidance Track** has also been established for vessels traveling between the Cape Cod Canal and Provincetown. **CAUTION:** Full bottom coverage surveys have not been conducted within the entire route, so uncharted dangers may exist.

(64) It is recommended that all large vessels (over 100 gross tons) avoid operating in the critical habitat during the period of peak right whale occurrence. When the area cannot be avoided, precautionary measures should be taken to reduce the risk of ship strike. (See North Atlantic Right Whales, indexed as such, in chapter 3 for more information on right whales and recommended measures to avoid collisions with whales.)

(65) All vessels 65 feet or greater in length overall (LOA) and subject to the jurisdiction of the United States are restricted to speeds of 10 knots or less in the Cape Cod Bay Seasonal Management Area between January 1 and May 15. The area is defined as all waters of Cape Cod Bay with a northern boundary of 42°04'56.5"N., 70°12'W., to 42°12'N., 70°12'W., thence due west back to shore. (See **50 CFR 224.105** in chapter 2 for regulations, limitations and exceptions.)

(66) The waters of Cape Cod Bay lie within the WHALESNORTH Mandatory Ship Reporting Area. Each self-propelled ship of 300 gross tons or greater entering WHALESNORTH must participate in the Mandatory Ship Reporting System (See **33 CFR 169**, chapter 2, for limits and regulations, and chapter 3 for sample reports). Sovereign immune vessels are exempt from the requirement to report but are encouraged to participate.

(67) **Cape Cod Bay Restricted Areas**, Marine Protected Areas (MPAs), encompass all of Cape Cod Bay.

(68)

Brant Rock to Plymouth

(69) **Brant Rock** is a village about 5 miles southward of Fourth Cliff. The village derives its name from **Brant Rock**, a distinctive bare rocky islet about 300 yards offshore that is joined to the shore by a stone jetty. A square concrete tower in the village is very prominent. For about 1 mile south of Brant Rock to the entrance of Green Harbor River, foul ground extends offshore for nearly 1.5 miles to **Farnham Rock**, which is covered 14 feet. A lighted bell buoy is just eastward of the rock.

(70) **Green Harbor River** has its entrance west of **Blackmans Point** at the southern end of **Green Harbor Point**. **Bartlett Rock**, which uncovers 2 feet, and **Howland Ledge**, covered 7 feet and marked by a buoy, are 0.6 and 1.2 miles eastward of the entrance, respectively. An obstruction reported to be covered 6 feet is about 275 yards south-southeast of Bartlett Rock. In 1987, an obstruction was reported 200 yards south of Bartlett Rock in about 42°04'34.2"N., 70°37'49.0"W. Jetties are on each side of the entrance; the east jetty is marked by a light. A channel leads to a turning basin about 0.6 mile above the seaward ends of the jetties. An anchorage basin is on the east side of the channel off the town wharf. Lighted and unlighted buoys mark the entrance and channel inside.

(71) Local fishermen adjust their arrival and departure times so that they are not in the entrance channel 90 minutes on either side of low water. A current flowing out

(74)

of the entrance channel during the falling tide reportedly sets up a rip just inside the entrance jetties.

(72) The town wharf and a marina are on the east bank about 0.4 mile above the jetties. The facilities have approach and alongside depths of 6 feet. Berths, electricity, gasoline, diesel fuel, water, ice, sewage pump-out, launching ramps, a 12.5-ton lift, storage and hull and engine repairs are available.

(73) **Green Harbor** is a small village on the west side of the river. Four prominent radio towers are just southwest of the village and 5 miles northward of the entrance to Plymouth Bay. A marina and the Green Harbor Yacht Club are on the west bank near the head of the harbor close southward of the causeway. Berthage, electricity, gasoline, water, marine supplies and a small-craft launching ramp are available at the marina. The service float has 6 feet reported alongside. A 15-ton mobile hoist can haul out vessels for hull, engine, electrical and electronic repairs and for open winter storage.

(75) **High Pine Ledge**, awash at low water and marked on its easterly side by a buoy, is about 0.8 mile off **Duxbury Beach** and 2 miles northward of Plymouth Light. The ledge extends from the buoy nearly to the shore; vessels should not attempt to pass westward of the buoy.

(76) **Plymouth Bay** is about 20 miles southeastward of Minots Ledge Light. From its entrance, between Gurnet Point and Rocky Point, it extends about 2.5 miles westward to **Plymouth Beach. Warren Cove**, the southern part of Plymouth Bay, is sometimes used as a temporary anchorage.

(77) **Plymouth Harbor** is about 1 mile wide at its northern end, gradually narrowing to its southern end. Most of the harbor is dry at low water. The channels in Plymouth Harbor and tributaries usually have soft bottoms. The channel through the entrance is well marked and easily followed in clear weather.

(79) **Plymouth** is a town on the southwestern side of Plymouth Harbor. There is some waterborne commerce at Plymouth, most of it being fishing vessels and excursion boats. At the town wharf, fishing craft unload fish, scallops and lobsters for shipment to New York and southern markets.

(80) **Duxbury Bay** is between Duxbury Beach on the east, Saquish Neck on the southeast, and the mainland on the west. It is about 3 miles long, with an average width of 2 miles. The bay is full of flats, mostly bare at low water, through which are several narrow and crooked channels. Shoals covered in spots by little water rise abruptly on both sides of these channels, and at low water the shoal edges are usually revealed by discolored water.

(81) **Duxbury**, a town on the west shore of the bay, is a summer yachting and residential resort.

(82) **Kingston Bay**, between the mainland and the western point of Duxbury Bay, is about 1.5 miles wide and has numerous flats. Caution and local knowledge are

(78)

advised. The village of **Kingston** is nearly 1 mile back from its western shore on **Jones River**. This bay is of little importance either as a harbor or port.

(83)

Prominent features

(84) **Gurnet Point**, on the north side of the entrance to the bay, is marked by **Plymouth Light** (42°00'13"N., 70°36'02"W.), 102 feet above the water and shown from a white octagonal pyramidal tower with a white dwelling. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A.

(85) **Rocky Point**, on the south side of the entrance, is about 3 miles south of Gurnet Point. The rectangular reactor building of the Pilgrim Nuclear Power Station southeast of Rocky Point is conspicuous.

(86) **Duxbury Pier Light** (41°59'15"N., 70°38'55"W.), 35 feet above the water, is shown from a brown conical tower with white upper half; a sound signal is at the light and is activated by keying the microphone five times consecutively on VHF-FM channel 83A. The light, locally known as "Bug Light," marks the north side of the channel and the south end of the shoal between the main channel and **Cowyard.**

(87) **Captains Hill**, on the peninsula between Duxbury and Kingston Bays, is about 200 feet high. On its summit is **Standish Monument**, 291 feet high, which can be seen from all directions when approaching the harbor. **Manomet Hill**, about 5 miles southward of Gurnet Point, is 390 feet high, heavily wooded, and conspicuous in approaching the entrance.

(88) The monument at Plymouth and a standpipe are conspicuous. A lookout tower on **Monks Hill**, about 2.5 miles westward of Plymouth, and the buildings and stack of the former cordage company at **Cordage** are prominent.

(89) **Plymouth Harbor Channel** is a dredged channel which leads southward from Plymouth Bay from a point 0.3 mile southwestward of Duxbury Pier Light to the State Pier at Plymouth, about 2 miles above the entrance, thence to a turning basin off the Town Wharf, about 0.2 mile above the State Pier. An anchorage basin, protected by a breakwater on the north and northeast sides, is in the harbor. A Federal project provides for a depth of 15 feet in the dredged channel to and in the turning basin; the anchorage basin in the harbor has a project depth of 8 feet. (See Notice to Mariners and latest editions of charts for controlling depths.) The channel is marked by a light with a white sector marking the entrance, buoys, another light and private lighted ranges. The range structures are difficult to identify in the daytime.

(90) A channel leads from the southerly end of Cowyard, westward of Duxbury Pier Light to the wharf of the former Plymouth Cordage Company, about 1.5 miles northwest of the Plymouth Town Wharf. The channel is marked by buoys. In 1964, shoaling to 9 feet was reported in the

channel for about 0.6 mile westward of The Nummet. The section of the channel approaching the wharf is privately marked.

(91)

Duxbury Bay Channels

(92) Where the several bay channels come together in the locality westward of Duxbury Pier Light, a channel extends northward up Duxbury Bay until west of **Clarks Island**. This channel, **Cowyard**, about 200 yards wide and with depths of 20 to 35 feet, offers good anchorage for small craft. The channel splits at a point westward of Clarks Island. The eastern branch, **Beach Channel**, is reported to be marked by private seasonal buoys and continues up the easterly side of Duxbury Bay. A highway bridge at **Powder Point**, at the junction of **Back River** with Duxbury Bay, has a 25-foot fixed span with a clearance of 5 feet.

(93) The western branch has a deep natural channel for about 1.5 miles from the area of Clarks Island northward of the fork in the channel. The channel to this point is buoyed and easily followed and at this point connects with a dredged channel that leads northwesterly to an anchorage basin at the village of Duxbury. The dredged channel, marked by buoys, has a project depth of 8 feet. (See Notice to Mariners and the latest edition of the chart for controlling depths.)

(94)

Anchorages

(95) Vessels waiting to enter may anchor on the north side of the entrance channel southeast of Saquish Head and eastward of the buoy marking the extremity of the shoal that makes southward from that head, or they may proceed to the intersection of the bay channels and anchor where the swinging room is greatest, between 500 and 600 yards westward of Duxbury Pier Light.

(96) The best anchorage is in Cowyard, but small light-draft vessels often find good anchorage under the lee of Plymouth Beach. Yachts and small craft anchor in the anchorage basin off the wharves at Plymouth. In 1983, the basin had depths of 8 feet except for shoaling in the northwest corner and along the north edge.

(97)

Dangers

(98) **Outer Tautog Rock**, with 2 feet over it, is part of an unmarked shoal extending about 0.5 mile northward of Rocky Point.

(99) **Browns Bank** is in the central part of Plymouth Bay. Northward of Browns Bank, and between it and **Saquish Neck** and **Saquish Head**, is the entrance channel to Plymouth Harbor, Kingston Bay and Duxbury Bay.

(100) The unmarked channels in Kingston Bay and Duxbury Bay are narrow and crooked and lead between flats bare or nearly so at low water; local knowledge is required to carry the best water. The best time for strangers to navigate these channels inside the harbor is at low water when the flats are visible.

(101)

Current

(102) In the channel between Gurnet Point and Duxbury Pier the tidal current at strength has a velocity of about 1.4 knots. The set is generally in the direction of the channel; but the ebb sets southward and eastward across Browns Bank, while the flood sets northward and westward above Saquish Head, and sweeps strongly around Duxbury Pier Light northward into Cowyard.

(103) **Ice** often closes the harbor from about the first of January through February. When there is ice in the harbor, Cowyard is not a safe anchorage. In winter the safest anchorage from ice is in the channel southward or eastward of Saquish Head, and vessels sometimes go to sea on account of drift ice at this anchorage. Westerly winds tend to carry the ice out in fields. Normally the channel to the pier at Cordage is open to traffic all winter. Northwesterly winds sometimes bring ice in, but southerly winds clear it out.

(104)

Quarantine, customs, immigration and agricultural quarantine

(105) (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(106) **Quarantine** is enforced in accordance with regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.)

(107) Plymouth is a **customs port of entry.**

(108)

Harbor regulations

(109) The several towns have harbor regulations that are enforced by the various **harbormasters**, who control the moorings. The Plymouth and Duxbury harbormasters maintain offices on their respective town wharves. The **speed limit** is 6 miles per hour in Plymouth Harbor.

(110)

Wharves

(111) Fishing vessels discharge their catches at the town wharf at Plymouth and Duxbury. The town wharf at Plymouth has a reported 10 feet alongside its service float; a small-craft launching ramp is close by. The State Pier at Plymouth has 12 feet alongside. The replica of the MAYFLOWER is berthed on the south side of the State Pier.

(112)

Supplies

(113) Gasoline, diesel fuel and water are available at the Plymouth and Duxbury town wharves and at most of the marinas and boatyards. Ice, provisions, bottled gas and marine supplies are available at Plymouth and Duxbury. Motels, hotels, restaurants, laundromats, shops and markets are in the area.

(114)

Repairs

(115) There is a boatyard at Plymouth, one at **North Plymouth** and several at Duxbury. These facilities can

make hull, engine and electronic repairs and have storage facilities and marine supplies. The boatyard at Plymouth, about 0.2 mile southeastward of the State Pier, has a 10-ton crane, 60-ton boat lift, berths with electricity, gasoline and diesel fuel. The largest marine railway at Duxbury can handle craft up to 50 feet in length; lifts up to 40 tons are also available. The boatyard at North Plymouth, close southward of the pier at Cordage, has a 75-foot marine railway and a 40-ton lift.

(116)

Small-craft facilities

(117) There are well-equipped marinas at Duxbury and Plymouth at which gasoline and diesel fuel, water, berthing with electricity, pump-out facilities and most yacht services are available. Small-boat launching ramps, both public and private, are available. The Duxbury Yacht Club, at the northwest corner of the turning basin at Duxbury, and the Plymouth Yacht Club, about 0.3 mile southward of the State Pier at Plymouth, offer various services to visiting yachtsmen. Gasoline is available at the Plymouth Yacht Club float.

(118)

Communications

(119) Plymouth has local taxi service and bus service to Boston and other inland points. Numerous truck lines serve the area.

(120)

Rocky Point to Sagamore Beach

(121) Between Rocky Point and Manomet Point, there are several outlying rocks which will be avoided by giving the shore a berth of 1 mile. The shore is backed by high wooded hills, the most conspicuous of which is Manomet Hill, 390 feet high. **Manomet Point** is a bluff about 2.4 miles southeast of Rocky Point. Also prominent is the large rectangular reactor housing of the Pilgrim Nuclear Power Station, about 0.4 mile southeastward of Rocky Point. Seaward of the power station, stone breakwaters enclose a basin from which cooling water is obtained. This basin is hazardous to approach in heavy weather since seas break over the breakwaters. A private buoy is about 0.3 mile eastward of Rocky Point.

(122) **White Horse Beach** is a summer resort northwest of Manomet Point. **White Horse Rocks**, 10 feet high, and a group of rocks awash are part of an unmarked shoal that extends about 0.7 mile northward from White Horse Beach.

(123) **Mary Ann Rocks**, two rocks that uncover about 5 feet, are 0.7 and 0.9 mile southeastward of the northerly end of Manomet Point and are marked by a lighted whistle buoy, 0.8 mile eastward from the outer rock. **Stone Horse Rocks,** awash at low water, are southwestward of Mary Ann Rocks and form a part of a reef extending about 1 mile southeastward from Manomet Point.

(124) **Stellwagen Rock**, covered 7 feet and unmarked, is 1.7 miles southward of Manomet Point and 0.8 mile from shore.

(125) From Manomet Point to **Peaked Cliff**, a distance of 7 miles, the shore is a line of high bluffs backed by woods. Shoals with little water in places extend 0.6 mile from shore just southward of **Center Hill Point**. A standpipe on 140-foot-high **Indian Hill**, about 1.5 miles north of Center Hill Point, is prominent.

(126) **Ellisville Harbor** is a small-boat harbor about 0.4 mile northward from **Lookout Point**. The entrance, which is almost bare, is protected by a small jetty on the northern side. The basin is shoal and available for small craft only at half tide or better.

(127) From Peaked Cliff the shore is low and trends southeastward. At the resort town of **Sagamore Beach**, 2 miles northwestward of Cape Cod Canal, a standpipe is prominent.

(128)

Cape Cod Canal

(129) **Cape Cod Canal** is a deep-draft sea-level waterway that extends westward from Cape Cod Bay to the head of Buzzards Bay. The waterway has a project depth of 32 feet and a least overhead clearance of 135 feet. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A. The eastern entrance to the canal is marked by a lighted **244°54'** range, lighted and unlighted buoys and a light and sound signal. A tall strobe-lighted stack and buildings of the power plant on the south bank of the canal about 0.75 mile above the eastern entrance, is prominent.

(130) Endangered North Atlantic right whales have been sighted within the Cape Cod Canal and in the vicinity of both the east and west entrances.

(131) Cape Cod Canal Marine Traffic Controllers provide information regarding North Atlantic right whale sightings and locations.

(132) The Northeast Marine Pilots distribute educational material to mariners in an effort to reduce right whale ship strikes.

(133) All vessels 65 feet or greater in length overall (LOA) and subject to the jurisdiction of the United States are restricted to speeds of 10 knots or less in the Cape Cod Bay Seasonal Management Area between January 1 and May 15.The area is defined as all waters of Cape Cod Bay with a northern boundary of 42°41'56.5"N., 70°12'W. to 42°12'N., 70°12'W. Thence due west back to shore. (See **50 CFR 224.105**, chapter 2, for regulations, limitations, and exceptions.)

(134) The waters of Cape Cod Bay lie within the WHALESNORTH Mandatory Ship Reporting Area. Each self-propelled ship of 300 gross tons or greater entering WHALESNORTH must participate in the Mandatory Ship Reporting System (See **33 CFR 169**, chapter 2, for limits and regulations, and chapter 3 for sample reports). Sovereign immune vessels are exempt

from the requirement to report, but are encouraged to participate.

(135)

COLREGS Demarcation Lines

(136) The line established for the eastern entrance to the Cape Cod Canal is described in **33 CFR 80.135**, chapter 2.

(137) A detailed description of the Cape Cod Canal and its facilities is given in **United States Coast Pilot 2, Atlantic Coast, Cape Cod to Sandy Hook.**

(138)

Sandwich Harbor to Scorton Harbor

(139) **Cape Cod** is a long peninsula forming the easterly extremity of Massachusetts. It makes out from the mainland in an easterly direction for 31 miles, thence extends northward and westward for over 25 miles. The portion of Cape Cod between Cape Cod Canal and Chatham is known as the **Upper Cape**. This region is wooded and has numerous towns and villages. The northern trend of Cape Cod, forming what is sometimes termed the **Hook of the Cape**, is known as the **Lower Cape**. This section is well settled and composed almost entirely of sandy lands with high, bare sand dunes and low, nearly level plains. Much of the outer shore of the lower cape is part of the Cape Cod National Seashore under the U.S. Department of the Interior.

(140) **Sandwich Harbor**, 1 mile southeastward of the eastern entrance of Cape Cod Canal, is the approach to the town of **Sandwich**. The shore in front of the town is low marsh, faced by a sand beach. The channel to the town can shoal to bare at low water. The harbor is suitable only for small craft; currents are weak and variable. A pump-out facility is available at Sandwich Harbor; fuel, water and other services are obtainable at the East Boat Basin just inside the entrance to Cape Cod Canal.

(141) **Springhill Beach** extends 2.3 miles southeastward from Sandwich Harbor to Scorton Harbor. The latter harbor is surrounded by sand dunes backed by cultivated lowlands.

(142) **Scorton Harbor**, 3.5 miles southeastward of the entrance of Cape Cod Canal, has a narrow jettied entrance bare at low water. Small local pleasure craft sometimes enter the harbor at half tide or higher. The harbor has no wharves. **Scorton Ledge**, an unmarked 12-foot ledge, is 0.7 mile north-northeastward of the entrance.

(143)

Barnstable Harbor to Nobscusset Point

(144) **Barnstable Harbor**, 10 miles eastward of Cape Cod Canal entrance, is the approach to the town of **Barnstable** and the village of **Yarmouth Port**. It is used mostly by local fishing and charter fishing boats and pleasure boats. A lighted bell buoy, about 2 miles northward of Beach Point, marks the approach. Mariners are urged to use caution when approaching the entrance and inside the harbor as there are a fair amount of lobster traps in the area, mostly outside the entrance.

(145)

Prominent features

(146) Prominent landmarks include the privately owned tower of a former lighthouse on the south side of Beach Point, a standpipe and a lighted radio tower at Barnstable and a spire in Yarmouth.

(147)

Channels

(148) The channel into Barnstable Harbor is marked by lighted and unlighted buoys. The bar channel and the channel through the harbor leading to **Maraspin Creek** are subject to continual change; strangers should obtain local information before entering. The harbor is nearly filled by flats and shoals; the chart is the best guide for navigating the channels. With northerly winds a heavy sea makes on the bar, and vessels bound to Barnstable should take shelter in the eastern entrance to Cape Cod Canal or anchor in Plymouth or Provincetown Harbors until the weather moderates.

(149) Maraspin Creek Lighted Buoy 1, about 0.2 mile northward of Blish Point, marks the entrance to the channel; this private aid is maintained from May to November. From the lighted buoy to **Blish Point**, the channel is marked by private daybeacons.

(150)

Anchorages

(151) Sheltered anchorage is available north of the channel between Beach Point and Maraspin Creek entrance. Care should be taken to avoid the cable area extending southward from Beach Point.

(152)

Dangers

(153) The south side of the harbor is very foul with covered rocks and ledges most of which are unmarked; extreme caution should be exercised if heading for the yacht club without local knowledge. Several rocks near the channel leading to the yacht club are marked by private seasonal buoys; these aids should not be taken as marking the entrance to Maraspin Creek.

(154)

Current

(155) Velocities of the tidal current in the entrance at strength average 1.3 knots, flooding southward and ebbing northward.

(156) **Ice** generally obstructs the harbor during a part of the winter.

(157)

Harbor regulations

(158) Berthage at the float landings along the western and southern sides of Maraspin Creek are under the control of the **harbormaster,** who may be contacted at 508–790–6273 or VHF-FM channel 16.

(167)

(159)

Small-craft facilities

(160) A marina on the west side of the creek, about 200 yards southward of Blish Point, has 5 to 6 feet reported alongside its floats. Gasoline and diesel fuel are available at the service float, and ice, pump-out facility, provisions and marine supplies are obtainable nearby. There is a 20-ton mobile hoist that can haul out boats up to 55 feet in length for hull and engine repairs and dry covered or open winter storage.

(161) A marina, on the east side of the creek, is used primarily by outboard boats. Gasoline, water and electricity are available at the float landings, which have a reported 3 feet alongside. Outboard motor repair and open winter storage are available for small craft that are hauled out using trailers. A paved town small-craft launching ramp is on the north side of this marina, and the state-run Blish Point boat ramp is just north of the small-craft ramp.

(162) Limited transient berths with 3 to 6 feet alongside with water and electricity available are maintained by the town. A snack bar and other conveniences are at the harbor, and lodging, a grocery store and restaurants are within walking distance.

(163) The Barnstable Yacht Club is on the southern side of Barnstable Harbor about 0.6 mile westward of the entrance to Maraspin Creek. Another paved town small-craft launching ramp is about 0.3 mile westward of the yacht club.

(164) In 1979, the channel to **Yarmouth** had shoaled over; it bares before low water, and the town landing is no longer used.

(165) **North Dennis** is a village 3.2 miles east-northeastward of Beach Point. **Scargo Hill**, 170 feet high and the highest hill in the vicinity, is southeastward of North Dennis.

(166) **Nobscusset Point**, 4.2 miles east-northeast of Beach Point, has a small breakwater that formerly provided a limited anchorage for small craft, but the area inside the breakwater is bare at low water.

(168)

Sesuit Harbor to Great Island

(169) Between Barnstable and Wellfleet are several creeks which are used by local boats and launches at high water. All are dry at low water, except Sesuit Harbor. The 18-foot curve is 0.2 to 0.3 mile from shore between North Dennis and Sesuit Harbor, but eastward of the latter it is 0.5 to 1.5 miles from shore.

(170) **Sesuit Harbor**, 5 miles eastward of Barnstable Harbor, has two jetties. The west jetty is marked by a light, and the east jetty by a daybeacon. A lighted bell buoy, about 1 mile north-northwestward of the entrance, marks the approach. The channel between the jetties and

(171)

the harbor are subject to frequent shoaling, and local knowledge should be obtained before entering.

(172)

Anchorages

(173) **East Dennis** is a village 0.5 mile inland. The waters of the harbor are a **special anchorage.** (See **33 CFR 110.1** and **110.37**, Chapter 2, for limits and regulations.)

(174)

Harbor regulations

(175) The moorings and berths at the town marina are under the control of the **harbormaster**, whose office is on the west side at the town landing. A **speed limit** of 4 miles per hour is enforced in the harbor.

(176) Members of the Dennis Yacht Club moor their boats in the small bight on the west side of the channel just inside the west jetty. A marina, on the west side of the harbor about 0.35 miles southward of the jetty light, has depths of 6 feet reported alongside its service floats. The marina has a 20-ton mobile hoist for dry covered or open winter storage. Gasoline, diesel fuel, water, ice, pump-out facility, provisions, marine supplies, guest berths and charter fishing boats are available.

(177) About 250 yards southward of the marina is the town landing with ramps, two piers and float landings at which berthing with electricity and water are available.

(178) A public small-craft launching ramp and an adjoining float landing are on the east side of the harbor, about 0.4 mile southward of the jetty light. Ample parking is available, and lodging can be obtained in town.

(179) **Rock Harbor**, on the south side of **Rock Harbor Creek**, is about 7 miles eastward of Sesuit Harbor. The centerline of the channel forms part of the boundary between the towns of **Orleans** and **Eastham**. A seasonal lighted bell buoy is about 1.7 miles west of the entrance, and a private **100°** lighted range marks the entrance. The channel is marked by private seasonal bush stakes.

(180) In 1979, the approach from about 0.7 mile offshore to the channel entrance was bare at low water. The harbor is usually entered 2 hours on either side of high water; local knowledge is advised.

(181) The Orleans town wharf and marina extends along the south and east sides of the harbor from the south jetty to the head. Party boats, draggers, yachts and other small craft moor at the berths at which water and electricity are available; depths of 5 to 6 feet are reported alongside the berths. Gasoline and diesel fuel are available at a service wharf on the east side of the lower bend in the creek; depths of 5 feet are reported alongside the wharf. A pump-out facility is available at the town pier. The Eastham town marina, on the west bank of the creek just above the lower bend, has a small-craft launching ramp. Another launching ramp is on the southern side of the harbor near the jetty.

(186)

(182)

Harbormaster

(183) There is a harbormaster; the harbor is under the jurisdiction of the Selectmen of the towns of Orleans and Eastham.

(184) A naval aircraft bombing target **danger area** is centered in 41°49'46"N., 70°02'54"W. on the hulk of the former liberty ship JAMES LONGSTREET. (See **33 CFR 334.60**, chapter 2, for limits and regulations.) The ship has been scuttled in 14 feet of water, with her hull showing above high water, about 2.5 miles northwestward of Rock Harbor. A seasonal lighted buoy is 250 yards westward of the hulk.

(185) **Wellfleet Harbor** is on the western side of the hook of Cape Cod, near its southern end. **Wellfleet** is a town at the head of the harbor. **Mayo Beach** is also at the head of the harbor. The sandspit extending eastward from **Shirttail Point** is protected by stone revetment and is paved for a parking area for the town wharf and marina. The basin north of the spit has been developed into a large marina with floats and berths for small craft and yachts.

(187)

Prominent features

(188) **Wellfleet Harbor Breakwater Light 14** (41°55'31"N., 70°02'07"W.), 16 feet above the water, is shown from a spindle with a red triangular daymark on the end of the breakwater that protects the inner harbor and anchorages. The easterly of two church spires in the town of Wellfleet and a fire lookout tower in South Wellfleet are also prominent.

(189)

Channels

(190) A dredged channel, marked by lighted and unlighted buoys and a light, leads from deep water in Wellfleet Harbor to a dredged anchorage basin southward of the town wharf at Wellfleet.The channel is subject to frequent changes, and the buoys are moved to mark the best water.

(191) Private seasonal floats mark the channel leading from the anchorage into **Duck Creek** to the basin of the town marina. In 1979, the controlling depth in this channel was reported to be about 1 foot.

(192)

Anchorages

(193) The inner harbor offers the best anchorages off the Wellfleet town wharf. In the outer harbor, northeast of **Smalley Bar**, the anchorage in depths of 12 to 21 feet is somewhat exposed in westerly winds. In northerly gales vessels sometimes anchor on the lee side of **Billingsgate Shoal** in 12 to 42 feet; the shoal breaks the sea so that vessels with good ground tackle can ride out a heavy gale from northward.

(194)

Dangers

(195) Extensive shoals are in the entrance. Billingsgate Shoal extends about 5.5 miles westward of **Billingsgate Island**, on the western side of the entrance to the harbor. The island is covered at high water. A lighted bell buoy marks the southwest end of the shoal. Numerous sunken wrecks are south and west of the shoal; mariners are advised to exercise caution.

(196) The approach channel into Wellfleet outer harbor leads between the shoals and is narrow in places, but it is marked by unlighted and seasonal lighted buoys and is easily followed in daytime in clear weather. In 2021, a rock, covered 10 feet, was reported at 41°51'22"N., 70°02'55"W.; caution is advised. The breakwater that protects the inner harbor is reported to cover at extreme high tides. Bush stakes mark the clam and oyster flats in the inner harbor.

(197)

Current

(198) The tidal currents at strength in the harbor entrance, north of Smalley Bar, average 0.7 knot on the flood and 0.5 knot on the ebb.

(199) The harbor is usually closed by **ice** during a part of each winter.

(200)

Harbor regulations

(201) The town wharf, landings, and moorings in the harbor are under the control of the **harbormaster**, whose office is on the town wharf.

(202)

Small-craft facilities

(203) The town pier and the town wharf extend southward and eastward, respectively, from Shirttail Point. The town pier has depths of 3½ to 7 feet reported along its south side, and 3 to 4 feet are reported alongside the floats on its north side; gasoline, diesel fuel by truck, water, electricity and a pump-out facility are available. Floats with electricity and water are available on the north side of the town wharf; a small-craft launching ramp is on the south side. Hydraulic trailers at the town ramp can handle craft up to 45 feet for hull and engine repairs by local firms. Guest moorings in the anchorage area are under the control of the harbormaster. A snack bar, restaurant and marine supply store are at the shoreward end of the town wharf. Groceries and lodging are available within walking distance. The Wellfleet Yacht Club is at the west end of Mayo Beach.

(204) **Great Island**, on the western side of Wellfleet Harbor, is now part of the Cape Cod National Seashore under the U.S. Department of the Interior. Its beaches are open to pleasure boatmen who can either beach their boats or anchor a short distance offshore. Great Island has no facilities.

(205)

Pamet Harbor to Provincetown

(206) **Pamet Harbor**, at the mouth of **Pamet River**, about 5.5 miles southeast of Provincetown, is a small harbor frequented by yachts and a few fishermen. Pamet River leads eastward to the town of **Truro**. The ruins of a railroad trestle are near the mouth of the river at the head of the harbor. The harbor is entered by a privately dredged channel that leads eastward between two jetties thence southeastward to an anchorage basin, about 0.3 mile above the jetties. In 2011, it was reported that the harbor should only be entered within 3 hours before or after high tide. The shoals which extend 1 mile off the entrance are changeable.

(207) A town small-craft launching ramp, beach and parking lot are on the east side of the anchorage basin. The Pamet Harbor Yacht Club is just southward of the ramp. Water is available at the club. The harbor is reported to be a good small-craft refuge during hurricanes.

(208) **Provincetown Harbor**, formed by a turn in the northern end of the hook of Cape Cod, has a diameter of about 2 miles. It is one of the best harbors on the Atlantic Coast, having a sizable anchorage area in depths of 12 to 57 feet with excellent holding ground. Coasters and fishermen find protection here in gales from any direction.

(209) The historical town of **Provincetown**, on the northwestern side of the harbor, is at the site of the first landing of the MAYFLOWER in the New World. It is the home port of numerous fishing, lobster, charter, pleasure and sightseeing boats.

(210)

Prominent features

(211) **Pilgrim Monument**, a slim stone structure 348 feet above the water, which rises 252½ feet above **High Pole Hill** in Provincetown, is the most prominent landmark on the cape. **Race Point Light** (42°03'44"N., 70°14'35"W.), 41 feet above the water, is shown from a white tower on the northwest point of Cape Cod.

(212) **Wood End Light** (42°01'17"N., 70°11'37"W.), 45 feet above the water, is shown from a white square tower, near the water on the southern end of the hook of the cape. A mariner radio activated sound signal at the light is initiated by keying the microphone five times consecutively on VHF-FM channel 83A. **Long Point Light** (42°01'59"N., 70°10'07"W.), 36 feet above the water, is shown from a white square tower at the eastern end of Long Point on the western side of the harbor entrance. A mariner radio activated sound signal at the light is initiated by keying the microphone five times consecutively on VHF-FM channel 83A.

(213) Two standpipes about 0.2 mile westward of the monument; a tank, about 1.5 miles northeastward of the monument; and several church spires in Provincetown are prominent from the bay. Several radar domes in North Truro are also prominent. A large white bathhouse, part of

(215)

the Cape Cod National Seashore, is prominent on Herring Cove about 1.7 miles northwestward of Wood End Light.

(214) Prominent from the north are the observation tower and buildings of the former Race Point Coast Guard Station, about 1.4 miles northeastward of Race Point Light, and the aerolight at Provincetown Municipal Airport just southward of the station. The cupola of the Cape Cod National Seashore's Visitors Center, on **Ocean View Hill** about 0.8 mile southeastward of the station, is prominent from the north and east. At night Highland Light will show over the land westward of it when the entrance is approached on certain bearings.

(216)

Anchorages

(217) Excellent anchorage may be had in Provincetown Harbor. Numerous fishing vessels work out of Provincetown during the year. During the summer months, private floats are set out that are capable of mooring vessels up to 40 feet in length. Larger vessels anchor from south to southwest of the westerly end of the breakwater, depending on draft. In addition, small craft sometimes anchor in **Herring Cove**, 0.8 mile southward of Race Point Light. A temporary lee from easterly winds is found well inshore in depths of 10 to 24 feet. Anchorage inside the breakwater is reported to be poor to fair due to soft bottom with much debris. The marina close southwest of MacMillan Wharf maintains 100 mooring buoys on the west side of the harbor.

(218) The Coast Guard Captain of the Port, Providence, has established a fairway 100 yards wide extending from 42°02'00.4"N., 70°09'33.1"W. to 42°02'43.4"N., 70°10'59.1"W., and in the area extending 100 yards around the piers in Provincetown. Anchorage is prohibited in this fairway.

(219)

Dangers

(220) **Shank Painter Bar**, which extends to a maximum distance of 0.6 mile offshore between Race Point and Wood End Lights, rises abruptly from deep water. **Wood End Bar** is the continuation of the shoal that makes sharply into Wood End. A lighted bell buoy is about 0.6 mile southwestward of Wood End Light. A 2,500-foot stone breakwater is about 300 yards southeastward of the end of the town pier at Provincetown. The breakwater extends northeastward from a point in 42°02'45"N., 70°10'55"W., approximately parallel to the shoreline. The east and west ends of the breakwater are each marked by a light. Strangers should exercise caution when operating in the area.

(221)

Caution

(222) Shipping should keep a sharp lookout when navigating in the vicinity of Race Point, especially during

periods of darkness and low visibility, because of the numerous fishing craft which operate in the area. There are large fish weirs in the harbor.

(223)

Current

(224) The tidal current velocities between Race Point and Highland Light are very strong but diminish to less than 1 knot between Highland Light and Chatham Light. The flood sets southwestward, and the ebb northeastward. Tide rips occur during heavy weather when the wind is against the current. Westward of the stretch of coast between Wood End and Race Point, the velocity at strength is about 1 knot. In this locality the ebb current sets northwesterly and the flood sets southeasterly. At the entrance and in the harbor the tidal currents have little velocity. See the Tidal Current prediction service at t*idesandcurrents.noaa.gov*. for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(225)

Ice

(226) Ice forms only in severe winters in the harbor, and then only for short periods. There are recorded cases of fields of ice being driven northward from the shallow harbors of Cape Cod Bay into the harbor so as to close it briefly, but such cases are rare.

(227)

North Atlantic Right Whales

(228) Endangered North Atlantic right whales may occur year round in the vicinity of Race Point and Wood End. (See North Atlantic Right Whales, indexed as such, chapter 3, for more information on right whales and recommended measures to avoid collisions with whales.)

(229) All vessels 65 feet or greater in length overall (LOA) and subject to the jurisdiction of the United States are restricted to speeds of 10 knots or less in two Seasonal Management Areas. The first area is defined as all waters of Cape Cod Bay with a northern boundary of 42°04'56.5"N., 70°12'W., to 42°12'N., 70°12'W., thence due west back to shore and the restrictions apply between January 1 and May 15. The second area, off Race Point, is defined as the waters bounded by:

(230) 42°04'56.5"N., 70°12'W.

(231) 42°12'N., 70°12'W.

(232) 42°12'N., 70°30'W.

(233) 42°30'N., 70°30'W.

(234) 42°30'N., 69°45'W.

(235) 41°41'N., 69°45'W.; thence due west to the shore, and the restrictions apply between March 1 and April 30. (See **50 CFR 224.105** in chapter 2 for regulations, limitations, and exceptions.)

(236) Approaches to Pamet Harbor and Provincetown Harbor lie within the WHALESNORTH Mandatory Ship Reporting Area. Each self-propelled ship of 300 gross tons or greater entering WHALESNORTH must participate in the Mandatory Ship Reporting System (See **33 CFR 169**, chapter 2, for limits and regulations, and chapter 3 for sample reports). Sovereign immune vessels are exempt from the requirement to report but are encouraged to participate.

(237)

Harbor regulations

(238) Moorings and berths at the town pier and all moorings in the harbor are under the control of the **harbormaster,** whose office is at the end of the town pier. The harbormaster monitors VHF-FM channel 16.

(239) **Provincetown Coast Guard Station** is about 0.4 mile southwest of the town pier. The finger pier close northeast of the Coast Guard pier is in ruins. Mariners should exercise caution while navigating in this area.

(240) Provincetown is a **customs station.**

(241)

Wharves

(242) The town pier, known as MacMillan Wharf, is a long finger pier extending 1,300 feet into the bay from a large municipal parking lot. Two seafood-packing plants are on the outer end of the pier, which has a reported 13 feet alongside. There are several float landings along the southwestern side of the pier inshore of the packing plants, which are used by charter and sightseeing craft. A pump-out facility is available.

(243) The finger pier of a marina is about 120 yards southwestward of the town pier. Float landings are along the northeastern side of the pier, and gasoline and diesel fuel are available at the service float, which has 14 feet reported alongside. Guest berths with water and electricity are available. In 1979, depths of 10 feet were reported in the basin between the town pier and the marina's pier except for shoaling along the edges.

(244) In 1981, an 8-foot shoal spot was reported in about 42°02'50"N., 70°10'56"W., in the approach to MacMillan Wharf and the finger pier about 120 yards southwestward of it. Mariners are advised to exercise caution when operating in this area.

(245)

Supplies

(246) Marine supplies, restaurants, laundromats, lodging, groceries and shops of all kinds are available within walking distance in town.

(247)

Repairs

(248) There is a boatyard with a marine railway about 0.5 mile southwestward of the town pier. Boats up to 80 tons or 70 feet long can be hauled out. Hull repairs can be made, and machine shop repairs can be made on short notice. Engine repairs are available.

(249)

Communications

(250) Bus and taxi services are available throughout the year. During the summer, a commuter ferry runs between Provincetown and Boston. Throughout the year, regularly

scheduled flights to Boston depart from the Provincetown Municipal Airport, which is about 2 miles northwestward of the town pier.

(251)

Highlands, Peaked Hill Bar and Highland Light

(252) From Race Point the Cape Cod shore curves northeastward, eastward and then southeastward to the **Highlands**, a total distance of about 9 miles, and is composed of bare sand dunes of various heights. On the approach to the Highlands, the sand dunes are covered with brownish-looking growth of grass and the land is higher. The pitch of the cape at this point shows a high bluff on which stands Highland Light. At the Highlands, the shore may be safely approached as close as 0.5 mile, but the water shoals somewhat abruptly, and care must be taken not to go inside the 5-fathom curve. Much of the shoreline area of this portion of the lower cape is part of the Cape Cod National Seashore.

(253) **Peaked Hill Bar** includes shoals with a least depth of 10 feet about 3.5 miles northeast of Race Point Light. The bar is about 0.6 mile offshore and extends for about 4 miles paralleling the coastline. Vessels have grounded here, mainly because of failure to take soundings. This area should be given a berth of at least 2 miles. A lighted whistle buoy is about 2.5 miles off Peaked Hill Bar and about 5 miles northwestward of Highland Light. For northbound vessels, keeping in a depth of 20 fathoms will ensure passing 2.5 to 3 miles off the eastern side of Cape Cod and will lead to the lighted whistle buoy off Peaked Hill Bar.

(254)

Current

(255) Between Race Point and Chatham Light, tidal current velocities are generally less than 1 knot. Strengths of flood and ebb set northward and southward, respectively, along the coast. The time of current changes rapidly, strength of flood or ebb occurring about 2 hours later off Nauset Beach Light than off Chatham Light.

(256) **Highland Light** (42°02'22"N., 70°03'39"W.), 170 feet above the water, is shown from a 66-foot white tower, with covered way to the dwelling, situated on the brow of a hill at the north end of the Highlands. A crenelated tower, a red brick stack and three spherical radar domes on the summit of a hill, 0.5 mile south of the light, are prominent.

(257) The eastern side of Cape Cod is described in **United States Coast Pilot 2, Atlantic Coast, Cape Cod to Sandy Hook**.

Navigation Rules

(1) Following is an amalgamation of the **International (72 COLREGS) and Inland Navigation Rules**, their Annexes, and associated Federal rules and regulations.

(2) Text unique to Inland Rules is *italicized* and set apart in a text box or within *‹‹ double angle brackets ››*. International Rules are set apart in a text box or denoted with ‹ single angle brackets ›.

(3) Text within {curly brackets} denotes additions made by the U.S. Coast Guard Office of Navigation Systems.

(4) Disparate paragraph or section numbering are shown side by side separated by a dagger, i.e. (a)╪(b).

(5) Instances of "…§§83.xx / in / with / of … this section / subpart / part of this Rule, etc." are redacted, and herein are shown as the enumerated rule(s) they referred to, i.e. 72 COLREGS Rule 18(e) states: "...with the Rules of this Part" and the same Inland Rule states: "...with the Rules of this Subpart (Rules 4-19) (§§83.04 through 83.19)", but, herein it is stated as "...with Rules 4-19.

(6) Instances of paragraph / section (x) are redacted, and herein are shown as §(x).

(7) Rules denoted with an asterisk also have an associated implementing or interpretative rule (i.e. 33 CFR 81-90), which can be found in chapter 2.

(8)

Part A—General

(9)

Rule 1—Application (International)
(a) These Rules shall apply to all vessels upon the high seas and in all waters connected therewith navigable by seagoing vessels.
(b) Nothing in these Rules shall interfere with the operation of special rules made by an appropriate authority for roadsteads, harbors, rivers, lakes, or inland waterways connected with the high seas and navigable by seagoing vessels. Such special rules shall conform as closely as possible to these Rules.
(c) Nothing in these Rules shall interfere with the operation of any special rules made by the Government of any State with respect to additional station or signal lights, shapes or whistle signals for ships of war and vessels proceeding under convoy, or with respect to additional station or signal lights or shapes for fishing vessels engaged in fishing as a fleet. These additional stations or signal lights, shapes or whistle signals shall, so far as possible, be such that they cannot be mistaken for any light, shape, or signal authorized elsewhere under these Rules.

Rule 1—Application (International)
(d) Traffic separation schemes may be adopted by the Organization for the purpose of these Rules.
(e) Whenever the Government concerned shall have determined that a vessel of special construction or purpose cannot comply fully with the provisions of any of these Rules with respect to number, position, range or arc of visibility of lights or shapes, as well as to the disposition and characteristics of sound-signaling appliances, such vessel shall comply with such other provisions in regard to number, position, range or arc of visibility of lights or shapes, as well as to the disposition and characteristics of sound-signaling appliances, as the Government shall have determined to be the closest possible compliance with these Rules in respect to that vessel.

(10)

Rule 1—Application (Inland)
(a) These rules apply to all vessels upon the inland waters of the United States, and to vessels of the United States on the Canadian waters of the Great Lakes to the extent that there is no conflict with Canadian law. These Rules have preemptive effect over State or local regulation within the same field.
(b)(i)These rules constitute special rules made by an appropriate authority within the meaning of Rule 1(b) of the International Regulations for Preventing Collisions at Sea, 1972, including annexes currently in force for the United States ("International Regulations"). *(ii) All vessels complying with the construction and equipment requirements of the International Regulations are considered to be in compliance with these Rules.*
(c) Nothing in these Rules shall interfere with the operation of any special rules made by the Secretary of the Navy with respect to additional station or signal lights and shapes or whistle signals for ships of war and vessels proceeding under convoy, or by the Secretary with respect to additional station or signal lights and shapes for fishing vessels engaged in fishing as a fleet. These additional station or signal lights and shapes or whistle signals shall, so far as possible, be such that they cannot be mistaken for any light, shape or signal authorized elsewhere under these Rules. Notice of such special rules shall be published in the Federal Register and, after the effective date specified in such notice, they shall have effect as if they were a part of these Rules.
(d) Traffic separation schemes may be established for the purposes of these Rules. Vessel traffic service regulations may be in effect in certain areas.

Rule 1—Application (Inland)

(e) Whenever the Secretary determines that a vessel or class of vessels of special construction or purpose cannot comply fully with the provisions of any of these Rules with respect to the number, position, range, or arc of visibility of lights or shapes, as well as to the disposition and characteristics of sound-signaling appliances, the vessel shall comply with such other provisions in regard to the number, position, range, or arc of visibility of lights or shapes, as well as to the disposition and characteristics of sound-signaling appliances, as the Secretary shall have determined to be the closest possible compliance with these Rules. The Secretary may issue a certificate of alternative compliance for a vessel or class of vessels specifying the closest possible compliance with these Rules. The Secretary of the Navy shall make these determinations and issue certificates of alternative compliance for vessels of the Navy.

(f) The Secretary may accept a certificate of alternative compliance issued by a contracting party to the International Regulations if it determines that the alternative compliance standards of the contracting party are substantially the same as those of the United States.

(g) The operator of each self-propelled vessel 12 meters or more in length shall carry, on board and maintain for ready reference, a copy of these Rules.

(11)

Rule 2—Responsibility

(12) (a) Nothing in these Rules shall exonerate any vessel, or the owner, master, or crew thereof, from the consequences of any neglect to comply with these Rules or of the neglect of any precaution which may be required by the ordinary practice of seamen, or by the special circumstances of the case.

(13) (b) In construing and complying with these Rules due regard shall be had to all dangers of navigation and collision and to any special circumstances, including the limitations of the vessels involved, which may make a departure from these Rules necessary to avoid immediate danger.

(14)

Rule 3—General Definitions

(15) For the purpose of these Rules, except where the context otherwise requires:

(16) (a) The word “vessel” includes every description of watercraft, including non-displacement craft, WIG craft, and seaplanes, used or capable of being used as a means of transportation on water.

(17) (b) The term “power-driven vessel” means any vessel propelled by machinery.

(18) (c) The term “sailing vessel” means any vessel under sail provided that propelling machinery, if fitted, is not being used.

(19) (d) The term “vessel engaged in fishing” means any vessel fishing with nets, lines, trawls, or other fishing apparatus which restrict maneuverability, but does not include a vessel fishing with trolling lines or other fishing apparatus which do not restrict maneuverability.

(20) (e) The term “seaplane” includes any aircraft designed to maneuver on the water.

(21) (f) The term “vessel not under command” means a vessel which through some exceptional circumstance is unable to maneuver as required by these Rules and is therefore unable to keep out of the way of another vessel.

(22) (g) The term “vessel restricted in her ability to maneuver” means a vessel which from the nature of her work is restricted in her ability to maneuver as required by these Rules and is therefore unable to keep out of the way of another vessel. The term “vessels restricted in their ability to maneuver” shall include but not be limited to: (i) A vessel engaged in laying, servicing, or picking up a navigational mark, submarine cable or pipeline; (ii) A vessel engaged in dredging, surveying or underwater operations; (iii) A vessel engaged in replenishment or transferring persons, provisions or cargo while underway; (iv) A vessel engaged in the launching or recovery of aircraft; (v) A vessel engaged in mine clearance operations; (vi) A vessel engaged in a towing operation such as severely restricts the towing vessel and her tow in their ability to deviate from their course.

(23)

Rule 3h (International)

(h) The term “vessel constrained by her draft” means a power-driven vessel which because of her draft in relation to the available depth and width of navigable water is severely restricted in her ability to deviate from the course she is following.

(24) (i) The word “underway” means that a vessel is not at anchor, or made fast to the shore, or aground.

(25) (j) The words “length” and “breadth” of a vessel mean her length overall and greatest breadth.

(26) 26)4The words “length32 -13.824 ref/GS2 Tm[maneuve46 0 0 6 3129 (f

Rules 3n–3s (Inland)
(o) "Great Lakes" means the Great Lakes and their connecting tributary waters including the Calumet River as far as the Thomas J. O'Brien Lock and Controlling Waters (between mile 326 and 327), the Chicago River as far as the east side of the Ashland Avenue Bridge (between mile 321 and 322), and the Saint Lawrence River as far east as the lower exit of Saint Lambert Lock.
(p) "Secretary" means the Secretary of the Department in which the Coast Guard is operating.
(q) "Inland Waters" means the navigable waters of the United States shoreward of the navigational demarcation lines {30 CFR 80} dividing the high seas from harbors, rivers and other inland waters of the United States and the waters of the Great Lakes on the United States side of the International Boundary.
(r) "Inland Rules" or "Rules" means these Inland Navigational Rules and the annexes thereto, which govern the conduct of vessels and specify the lights, shapes, and sound signals that apply on inland waters.
(s) "International Regulations" means the International Regulations for Preventing Collisions at Sea, 1972, including annexes currently in force for the United States.

(30) **Implementing Rule**—See **33 CFR 89.25**, chapter 2, for regulations.

(31)

Part B—Steering and Sailing Rules

(32)

I—Conduct of Vessels in Any Condition of Visibility

(33)

Rule 4—Application

(34) Rules 4 through 10 apply in any condition of visibility.

(35)

Rule 5—Lookout

(36) Every vessel shall at all times maintain a proper look-out by sight and hearing as well as by all available means appropriate in the prevailing circumstances and conditions so as to make a full appraisal of the situation and of the risk of collision.

(37)

Rule 6—Safe Speed

(38) Every vessel shall at all times proceed at a safe speed so that she can take proper and effective action to avoid collision and be stopped within a distance appropriate to the prevailing circumstances and conditions. In determining a safe speed the following factors shall be among those taken into account:

(39) (a) By all vessels:

(40) (i) The state of visibility; (ii) The traffic density including concentrations of fishing vessels or any other vessels; (iii) The maneuverability of the vessel with special reference to stopping distance and turning ability in the prevailing conditions; (iv) At night, the presence of background light such as from shore lights or from back scatter from her own lights; (v) The state of wind, sea and current, and the proximity of navigational hazards; (vi) The draft in relation to the available depth of water.

(41) (b) Additionally, by vessels with operational radar:

(42) (i) The characteristics, efficiency and limitations of the radar equipment; (ii) Any constraints imposed by the radar range scale in use; (iii) The effect on radar detection of the sea state, weather and other sources of interference; (iv) The possibility that small vessels, ice and other floating objects may not be detected by radar at an adequate range; (v) The number, location and movement of vessels detected by radar; (vi) The more exact assessment of the visibility that may be possible when radar is used to determine the range of vessels or other objects in the vicinity.

(43)

Rule 7—Risk of Collision

(44) (a) Every vessel shall use all available means appropriate to the prevailing circumstances and conditions to determine if risk of collision exists. If there is any doubt such risk shall be deemed to exist.

(45) (b) Proper use shall be made of radar equipment if fitted and operational, including long-range scanning to obtain early warning of risk of collision and radar plotting or equivalent systematic observation of detected objects.

(46) (c) Assumptions shall not be made on the basis of scanty information, especially scanty radar information.

(47) (d) In determining if risk of collision exists the following considerations shall be among those taken into account:

(48) (i) Such risk shall be deemed to exist if the compass bearing of an approaching vessel does not appreciably change.

(49) (ii) Such risk may sometimes exist even when an appreciable bearing change is evident, particularly when approaching a very large vessel or a tow or when approaching a vessel at close range.

(50)

Rule 8—Action to Avoid Collision

(51) (a) Any action taken to avoid collision shall be taken in accordance with Rules 4 through 19 and shall if the circumstances of the case admit, be positive, made in ample time and with due regard to the observance of good seamanship.

(52) (b) Any alteration of course and/or speed to avoid collision shall, if the circumstances of the case admit, be large enough to be readily apparent to another vessel observing visually or by radar; a succession of small alterations of course and/or speed should be avoided.

(53) (c) If there is sufficient sea room, alteration of course alone may be the most effective action to avoid a close-quarters situation provided that it is made in good time, is substantial and does not result in another close-quarters situation.

(54) (d) Action taken to avoid collision with another vessel shall be such as to result in passing at a safe distance. The

effectiveness of the action shall be carefully checked until the other vessel is finally past and clear.

(55) (e) If necessary to avoid collision or allow more time to assess the situation, a vessel shall slacken her speed or take all way off by stopping or reversing her means of propulsion.

(56) (f)(i) A vessel which, by any of these Rules, is required not to impede the passage or safe passage of another vessel shall, when required by the circumstances of the case, take early action to allow sufficient sea room for the safe passage of the other vessel.

(57) (ii) A vessel required not to impede the passage or safe passage of another vessel is not relieved of this obligation if approaching the other vessel so as to involve risk of collision and shall, when taking action, have full regard to the action which may be required by Rules 4 through 19.

(58) (iii) A vessel, the passage of which is not to be impeded remains fully obliged to comply with Rules 4 through 19 when the two vessels are approaching one another so as to involve risk of collision.

(59)

Rule 9—Narrow Channels

(60) (a) ‹‹*(i)*›› A vessel proceeding along the course of a narrow channel or fairway shall keep as near to the outer limit of the channel or fairway which lies on her starboard side as is safe and practicable.

(61)

Rule 9a (Inland)

(ii) Notwithstanding Rule 9(a)(i) and Rule 14(a), a power-driven vessel operating in narrow channel or fairway on the Great Lakes, Western Rivers, or waters specified by the Secretary, and proceeding downbound with a following current shall have the right-of-way over an upbound vessel, shall propose the manner and place of passage, and shall initiate the maneuvering signals prescribed by Rule 34(a)(i), as appropriate. The vessel proceeding upbound against the current shall hold as necessary to permit safe passing.

(62) (b) A vessel of less than 20 meters in length or a sailing vessel shall not impede the passage of a vessel ‹ which ›‹‹ *that* ›› can safely navigate only within a narrow channel or fairway.

(63) (c) A vessel engaged in fishing shall not impede the passage of any other vessel navigating within a narrow channel or fairway.

(64) (d) A vessel ‹ shall ›‹‹ *must* ›› not cross a narrow channel or fairway if such crossing impedes the passage of a vessel which can safely navigate only within that channel or fairway. The latter vessel ‹ may ›‹‹ *must* ›› use the signal prescribed in Rule 34(d) if in doubt as to the intention of the crossing vessel.

(65)

Rule 9e (International)

(e)(i) In a narrow channel or fairway when overtaking can take place only if the vessel to be overtaken has to take action to permit safe passing, the vessel intending to overtake shall indicate her intention by sounding the appropriate signal prescribed in Rule 34(c)(ii). The vessel to be overtaken shall, if in agreement, sound the appropriate signal prescribed in Rule 34(c)(i) and take steps to permit safe passing. If in doubt she may sound the signals prescribed in Rule 34(d).

Rule 9e (Inland)

(e)(i) In a narrow channel or fairway when overtaking, the power-driven vessel intending to overtake another power-driven vessel shall indicate her intention by sounding the appropriate signal prescribed in Rule 34(c) and take steps to permit safe passing. The power-driven vessel being overtaken, if in agreement, shall sound the same signal and may, if specifically agreed to, take steps to permit safe passing. If in doubt she shall sound the signal prescribed in Rule 34(d).

(66) (e)(ii) This rule does not relieve the overtaking vessel of her obligation under Rule 13.

(67) (f) A vessel nearing a bend or an area of a narrow channel or fairway where other vessels may be obscured by an intervening obstruction shall navigate with particular alertness and caution and shall sound the appropriate signal prescribed in Rule 34(e).

(68) (g) Any vessel shall, if the circumstances of the case admit, avoid anchoring in a narrow channel.

(69)

Rule 10—Traffic Separation Schemes

(70) (a) This Rule applies to traffic separation schemes ‹ adopted by the Organization › and does not relieve any vessel of her obligation under any other rule.

(71) (b) A vessel using a traffic separation scheme shall:

(72) (i) Proceed in the appropriate traffic lane in the general direction of traffic flow for that lane.

(73) (ii) So far as is practicable keep clear of a traffic separation line or separation zone.

(74) (iii) Normally join or leave a traffic lane at the termination of the lane, but when joining or leaving from either side shall do so at as small an angle to the general direction of traffic flow as practicable.

(75) (c) A vessel, shall so far as practicable, avoid crossing traffic lanes but if obliged to do so shall cross on a heading as nearly as practicable at right angles to the general direction of traffic flow.

(76) (d)(i) A vessel shall not use an inshore traffic zone when she can safely use the appropriate traffic lane within the adjacent traffic separation scheme. However, vessels of less than 20 meters in length, sailing vessels and vessels engaged in fishing may use the inshore traffic zone.

(77) (ii) Notwithstanding Rule 10(d)(i), a vessel may use an inshore traffic zone when en route to or from a port, offshore installation or structure, pilot station or any other place situated within the inshore traffic zone, or to avoid immediate danger.

(78) (e) A vessel, other than a crossing vessel or a vessel joining or leaving a lane shall not normally enter a separation zone or cross a separation line except:

(79) (i) in cases of emergency to avoid immediate danger;

(80) (ii) to engage in fishing within a separation zone.

(81) (f) A vessel navigating in areas near the terminations of traffic separation schemes shall do so with particular caution.

(82) (g) A vessel shall so far as practicable avoid anchoring in a traffic separation scheme or in areas near its terminations.

(83) (h) A vessel not using a traffic separating scheme shall avoid it by as wide a margin as is practicable.

(84) (i) A vessel engaged in fishing shall not impede the passage of any vessel following a traffic lane.

(85) (j) A vessel of less than 20 meters in length or a sailing vessel shall not impede the safe passage of a power-driven vessel following a traffic lane.

(86) (k) A vessel restricted in her ability to maneuver when engaged in an operation for the maintenance of safety of navigation in a traffic separation scheme is exempted from complying with this Rule to the extent necessary to carry out the operation.

(87) (l) A vessel restricted in her ability to maneuver when engaged in an operation for the laying, servicing or picking up of a submarine cable, within a traffic separation scheme, is exempted from complying with this Rule to the extent necessary to carry out the operation.

(88)

II—Conduct of Vessels in Sight of One Another

(89)

Rule 11—Application

(90) Rules 11 through 18 apply to vessels in sight of one another.

(91)

Rule 12—Sailing Vessels

(92) (a) When two sailing vessels are approaching one another, so as to involve risk of collision, one of them shall keep out of the way of the other as follows:

(93) (i) when each has the wind on a different side, the vessel which has the wind on the port side shall keep out of the way of the other;

(94) (ii) when both have the wind on the same side, the vessel which is to windward shall keep out of the way of the vessel which is to leeward;

(95) (iii) if a vessel with the wind on the port side sees a vessel to windward and cannot determine with certainty whether the other vessel has the wind on the port or on the starboard side, she shall keep out of the way of the other.

(96) (b) For the purposes of this Rule, the windward side shall be deemed to be the side opposite that on which the mainsail is carried or, in the case of a square-rigged vessel, the side opposite to that on which the largest fore-and-aft sail is carried.

(97)

Rule 13—Overtaking

(98) (a) Notwithstanding anything contained in the Rules 4 through 18, any vessel overtaking any other shall keep out of the way of the vessel being overtaken.

(99) (b) A vessel shall be deemed to be overtaking when coming up with a another vessel from a direction more than 22.5 degrees abaft her beam, that is, in such a position with reference to the vessel she is overtaking, that at night she would be able to see only the sternlight of that vessel but neither of her sidelights.

(100) (c) When a vessel is in any doubt as to whether she is overtaking another, she shall assume that this is the case and act accordingly.

(101) (d) Any subsequent alteration of the bearing between the two vessels shall not make the overtaking vessel a crossing vessel within the meaning of these Rules or relieve her of the duty of keeping clear of the overtaken vessel until she is finally past and clear.

(102)

Rule 14—Head-on Situation

(103) (a) *‹‹ Unless otherwise agreed ››* when two power-driven vessels are meeting on reciprocal or nearly reciprocal courses so as to involve risk of collision each shall alter her course to starboard so that each shall pass on the port side of the other.

(104) (b) Such a situation shall be deemed to exist when a vessel sees the other ahead or nearly ahead and by night she could see the masthead lights of the other in a line or nearly in a line and/or both sidelights and by day she observes the corresponding aspect of the other vessel.

(105) (c) When a vessel is in any doubt as to whether such a situation exists she shall assume that it does exist and act accordingly.

(106)

Rule 14d (Inland)

(d) Notwithstanding Rule 14(a), a power-driven vessel operating on the Great Lakes, Western Rivers, or waters specified by the Secretary, and proceeding downbound with a following current shall have the right-of-way over an upbound vessel, shall propose the manner of passage, and shall initiate the maneuvering signals prescribed by Rule 34(a)(i), as appropriate.

(107)

Rule 15—Crossing Situation

(108) (a) When two power-driven vessels are crossing so as to involve risk of collision, the vessel which has the other on her own starboard side shall keep out of the way and shall, if the circumstances of the case admit, avoid crossing ahead of the other vessel.

(109)

Rule 15b (Inland)

(b) Notwithstanding Rule 15(a), on the Great Lakes, Western Rivers, or water specified by the Secretary, a power-driven vessel crossing a river shall keep out of the way of a power-driven vessel ascending or descending the river.

(110)

Rule 16—Action by Give-way Vessel

(111) Every vessel which is directed to keep out of the way of another vessel shall, so far as possible, take early and substantial action to keep well clear.

(112)

Rule 17—Action by Stand-on Vessel

(113) (a)(i) Where one of two vessels is to keep out of the way, the other shall keep her course and speed.

(114) (ii) The latter vessel may, however, take action to avoid collision by her maneuver alone, as soon as it becomes apparent to her that the vessel required to keep out of the way is not taking appropriate action in compliance with these Rules.

(115) (b) When, from any cause, the vessel required to keep her course and speed finds herself so close that collision cannot be avoided by the action of the give-way vessel alone, she shall take such action as will best aid to avoid collision.

(116) (c) A power-driven vessel which takes action in a crossing situation in accordance with Rule 17(a)(ii) to avoid collision with another power-driven vessel shall, if the circumstances of the case admit, not alter course to port for a vessel on her own port side.

(117) (d) This Rule does not relieve the give-way vessel of her obligation to keep out of the way.

(118)

Rule 18—Responsibilities Between Vessels

(119) Except where Rules 9, 10, and 13 otherwise require:

(120) (a) A power-driven vessel underway shall keep out of the way of: (i) a vessel not under command; (ii) a vessel restricted in her ability to maneuver; (iii) a vessel engaged in fishing; (iv) a sailing vessel.

(121) (b) A sailing vessel underway shall keep out of the way of: (i) a vessel not under command; (ii) a vessel restricted in her ability to maneuver; (iii) a vessel engaged in fishing.

(122) (c) A vessel engaged in fishing when underway shall, so far as possible, keep out of the way of: (i) a vessel not under command; (ii) a vessel restricted in her ability to maneuver.

(123)

Rule 18d (International)

(d)(i) Any vessel other than a vessel not under command or a vessel restricted in her ability to maneuver shall, if the circumstances of the case admit, avoid impeding the safe passage of a vessel constrained by her draft, exhibiting the signals in Rule 28.

(ii) A vessel constrained by her draft shall navigate with particular caution having full regard to her special condition.

(124) (e) A seaplane on the water shall, in general, keep well clear of all vessels and avoid impeding their navigation. In circumstances, however, where risk of collision exists, she shall comply with Rules 4 through 19.

(125) (f)(i) A WIG craft shall, when taking off, landing and in flight near the surface, keep well clear of all other vessels and avoid impeding their navigation;

(126) (ii) a WIG craft operating on the water surface shall comply with Rules 4 through 19 as a power-driven vessel.

(127)

III—Conduct of Vessels in Restricted Visibility

(128)

Rule 19—Conduct of Vessels in Restricted Visibility

(129) (a) This Rule applies to vessels not in sight of one another when navigating in or near an area of restricted visibility.

(130) (b) Every vessel shall proceed at a safe speed adapted to the prevailing circumstances and conditions of restricted visibility. A power-driven vessel shall have her engines ready for immediate maneuver.

(131) (c) Every vessel shall have due regard to the prevailing circumstances and conditions of restricted visibility when complying with Rules 4 through 10.

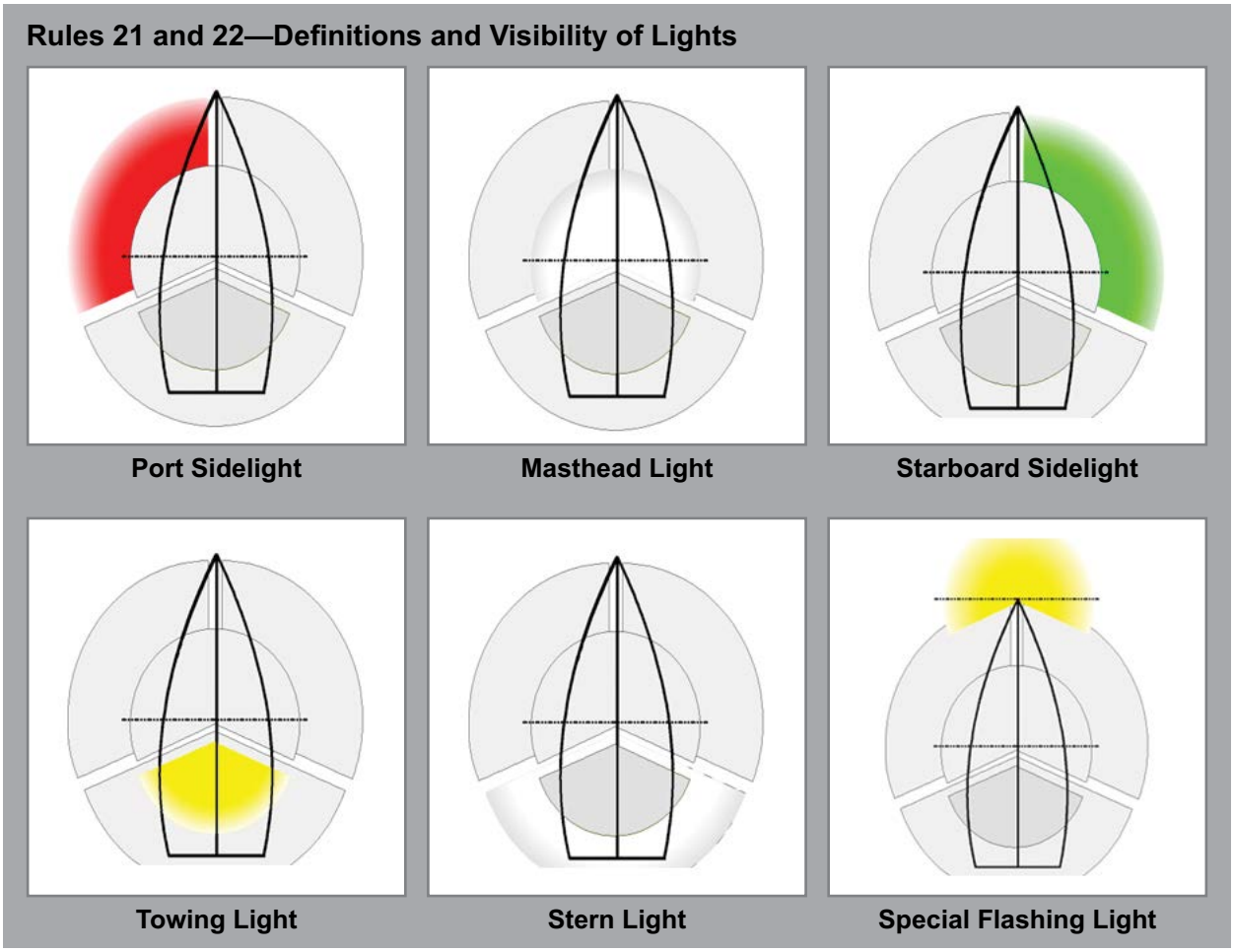
(132) (d) A vessel which detects by radar alone the presence of another vessel shall determine if a close-quarters situation is developing and/or risk of collision exists. If so, she shall take avoiding action in ample time, provided that when such action consists of an alteration in course, so far as possible the following shall be avoided:

(133) (i) An alteration of course to port for a vessel forward of the beam, other than for a vessel being overtaken;

(134) (ii) An alteration of course toward a vessel abeam or abaft the beam.

(135) (e) Except where it has been determined that a risk of collision does not exist, every vessel which hears apparently forward of her beam the fog signal of another vessel, or which cannot avoid a close-quarters situation with another vessel forward of her beam, shall reduce her speed to be the minimum at which she can be kept on her course. She shall if necessary take all her way off and in any event navigate with extreme caution until danger of collision is over.

(158)



(136)

Part C—Lights and Shapes

(137)

Rule 20—Application

(138) (a) Rules 20 through 31 shall be complied with in all weathers.

(139) (b) The Rules concerning lights shall be complied with from sunset to sunrise, and during such times no other lights shall be exhibited, except such lights which cannot be mistaken for the lights specified in these Rules or do not impair their visibility or distinctive character, or interfere with the keeping of a proper look-out.

(140) (c) The lights prescribed by these Rules shall, if carried, also be exhibited from sunrise to sunset in restricted visibility and may be exhibited in all other circumstances when it is deemed necessary.

(141) (d) The Rules concerning shapes shall be complied with by day.

(142) (e) The lights and shapes specified in these Rules shall comply with the provisions of Annex I of these Rules.

(143)

Rule 20f (Inland)

(f) A vessel's navigation lights and shapes may be lowered if necessary to pass under a bridge.

(144)

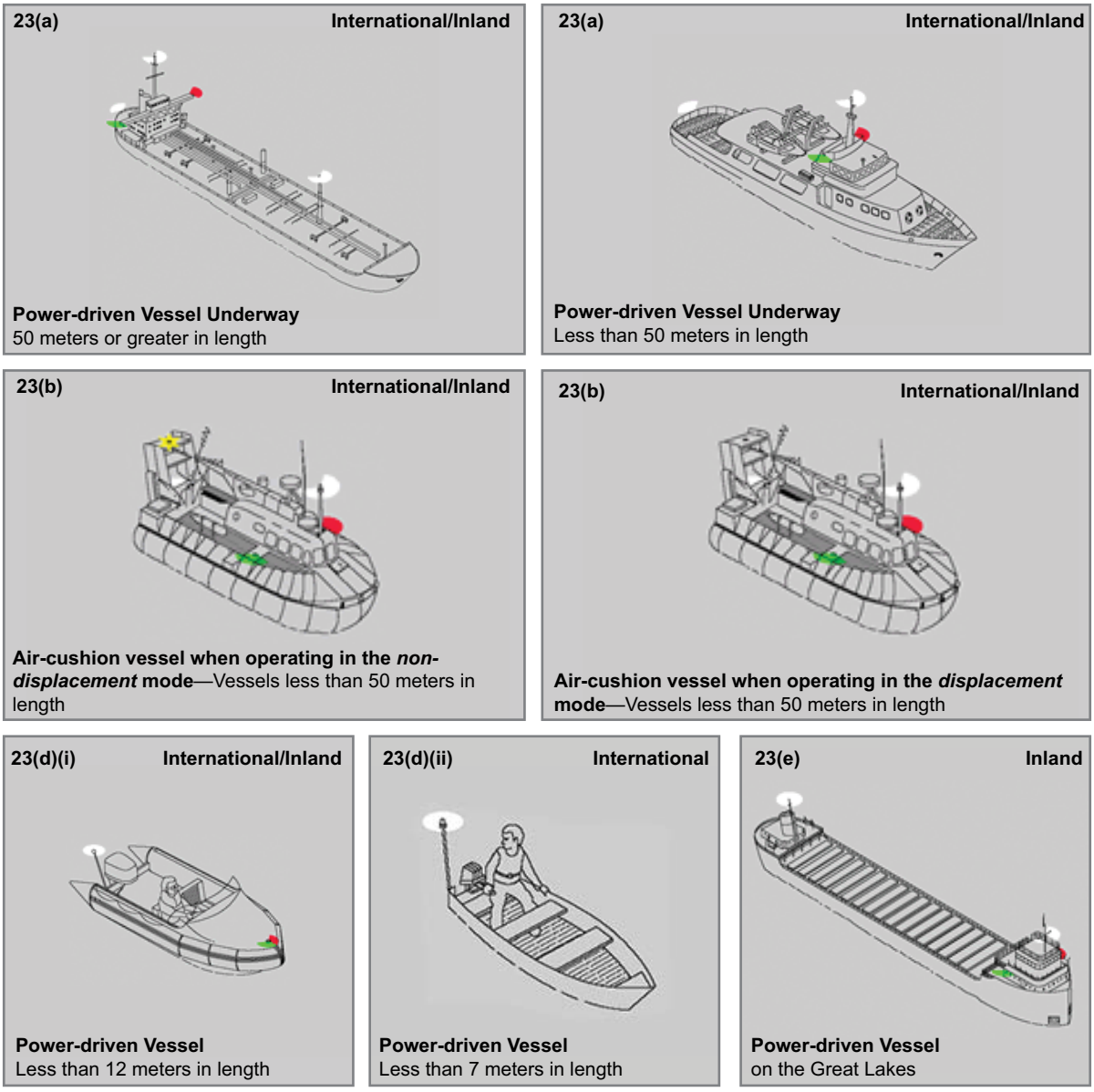
Rule 21—Definitions

(145) (a) "Masthead light" means a white light placed over the fore and aft centerline of the vessel showing an unbroken light over an arc of the horizon of 225° and so fixed as to show the light from right ahead to 22.5° abaft the beam on either side of the vessel *‹‹ except that on a vessel of less than 12 meters in length the masthead light shall be placed as nearly as practicable to the fore and aft centerline of the vessel ››*.

(146) (b) "Sidelights" means a green light on the starboard side and a red light on the port side each showing an unbroken light over an arc of the horizon of 112.5° and so fixed as to show the light from right ahead to 22.5° abaft the beam on its respective side. In a vessel of less than 20 meters in length the sidelights may be combined in one lantern carried on the fore and aft centerline of the vessel *‹‹ , except that on a vessel of less than 12 meters in length the sidelights when combined in one lantern*

(166)

Rule 23—Power-driven Vessels Underway



shall be placed as nearly as practicable to the fore and aft centerline of the vessel ››.

(147) (c) "Sternlight" means a white light placed as nearly as practicable at the stern showing an unbroken light over an arc of the horizon of 135° and so fixed as to show the light 67.5° from right aft on each side of the vessel.

(148) (d) "Towing light" means a yellow light having the same characteristics as the "sternlight" defined in Rule 21(c).

(149) (e) "All-round light" means a light showing an unbroken light over an arc of the horizon of 360°.

(150) (f) "Flashing light" means a light flashing at regular intervals at a frequency of 120 flashes or more per minute.

(151)

Rule 21g (Inland)

(g) "Special flashing light" means a yellow light flashing at regular intervals at a frequency of 50 to 70 flashes per minute, placed as far forward and as nearly as practicable on the fore and aft centerline of the tow and showing an unbroken light over an arc of the horizon of not less than 180 degrees nor more than 225 degrees and so fixed as to show the light from right ahead to abeam and no more than 22.5 degrees abaft the beam on either side of the vessel.

(152)

Rule 22—Visibility of Lights

(153) The lights prescribed in these Rules (Subpart C) shall have an intensity as specified in Annex I to these

Rules (33 CFR part 84), so as to be visible at the following minimum ranges:

(154) (a) In a vessel of 50 meters or more in length: (i) a masthead light, 6 miles; (ii) a sidelight, 3 miles; (iii) a sternlight, 3 miles; (iv) a towing light, 3 miles; (v) a white, red, green or yellow all-round light, 3 miles; «*and (vi) a special flashing light, 2 miles*. »

(155) (b) In a vessel of 12 meters or more in length but less than 50 meters in length: (i) a masthead light, 5 miles; except that where the length of the vessel is less than 20 meters, 3 miles; (ii) a sidelight, 2 miles; (iii) a sternlight, 2 miles; (iv) a towing light, 2 miles; (v) a white, red, green or yellow all-round light, 2 miles; «*and (vi) a special flashing light, 2 miles*. »

(156) (c) In a vessel of less than 12 meters in length: (i) a masthead light, 2 miles; (ii) a sidelight, 1 mile; (iii) a sternlight, 2 miles; (iv) A towing light, 2 miles; (v) a white, red, green or yellow all-round light, 2 miles; «*and (vi) a special flashing light, 2 miles*. »

(157) (d) In an inconspicuous, partly submerged vessel or objects being towed: (i) A white all-round light, 3 miles. (ii) [Reserved]

(159)

Rule 23—Power-driven Vessels Underway

(160) (a) A power-driven vessel underway shall exhibit: (i) a masthead light forward; (ii) a second masthead light abaft of and higher than the forward one; except that a vessel of less than 50 meters in length shall not be obliged to exhibit such a light but may do so; (iii) sidelights; and (iv) a sternlight.

(161) (b) An air-cushion vessel when operating in non-displacement mode shall, in addition to the lights prescribed in Rule 23(a) Air Cushion Vessel in Displacement Mode, exhibit an all-round flashing yellow light *‹‹ , where it can best be seen ››*.

(162) (c) A WIG craft only when taking off, landing and in flight near the surface shall, in addition to the lights prescribed in Rule 23(a), exhibit a high intensity all-round flashing red light.

(163) (d)(i) A power-driven vessel of less than 12 meters in length may in lieu of the lights prescribed in Rule 23(a) exhibit an all-round white light and sidelights.

(164)

Rule 23d (International)

(ii) a power-driven vessel of less than 7 meters in length whose maximum speed does not exceed 7 knots may in lieu of the lights prescribed in Rule 23(a) exhibit an all-round white light and shall, if practicable, also exhibit sidelights.

(iii) the masthead light or all-round white light on a power-driven vessel of less than 12 metres in length may be displaced from the fore and aft centre line of the vessel if centreline fitting is not practicable, provided that the sidelights are combined in one lantern which shall be carried on the fore and aft centre line of the vessel or located as nearly as practicable in the same fore and aft line as the masthead light or the all-round white light.

Rule 23e (Inland)

(e) A power-driven vessel when operating on the Great Lakes may carry an all-round white light in lieu of the second masthead light and sternlight prescribed in Rule 23(a). The light shall be carried in the position of the second masthead light and be visible at the same minimum range.

(165) Regulations containing specifics on **Law Enforcement** and **Public Safety Vessel** lighting are in **Annex V–Pilot Rules, 33 CFR 88.05** and **33 CFR 88.07**, chapter 2.

(167)

Rule 24—Towing and Pushing

(168) (a) A power-driven vessel when towing astern shall exhibit: (i) instead of the light prescribed in Rule 23(a)(i) or 23(a)(ii), two masthead lights in a vertical line. When the length of the tow, measuring from the stern of the towing vessel to the after end of the tow, exceeds 200 meters, three such lights in a vertical line; (ii) sidelights; (iii) a sternlight; (iv) a towing light in a vertical line above the sternlight; and (v) when the length of the tow exceeds 200 meters, a diamond shape where it can best be seen.

(169) (b) When a pushing vessel and a vessel being pushed ahead are rigidly connected in a composite unit they shall be regarded as a power-driven vessel and exhibit the lights prescribed in Rule 23.

(170) **Interpretive Rule**—See **33 CFR 90.3** and **33 CFR 82.3**, chapter 2, for regulations.

(171) (c) A power-driven vessel when pushing ahead or towing alongside, except ‹ in the case of a composite unit *›‹‹ as required by Rules 24(b) and (i) ››*, shall exhibit: (i) instead of the light prescribed in Rule 23(a)(i) or 23(a)(ii), two masthead lights in a vertical line; (ii) sidelights; and (iii) ‹ a sternlight *›‹‹ two towing lights in a vertical line ››*.

(172) (d) A power-driven vessel to which paragraphs (a) or (c) of this Rule applies shall also comply with Rule 23 *‹‹ (a)(i) and ››* (a)(ii).

(173) (e) A vessel or object being towed, other than those ‹ mentioned *›‹‹ referred ››* in Rule 24(g), shall exhibit: (i) sidelights; (ii) a sternlight; (iii) when the length of the tow exceeds 200 meters, a diamond shape where it can best be seen.

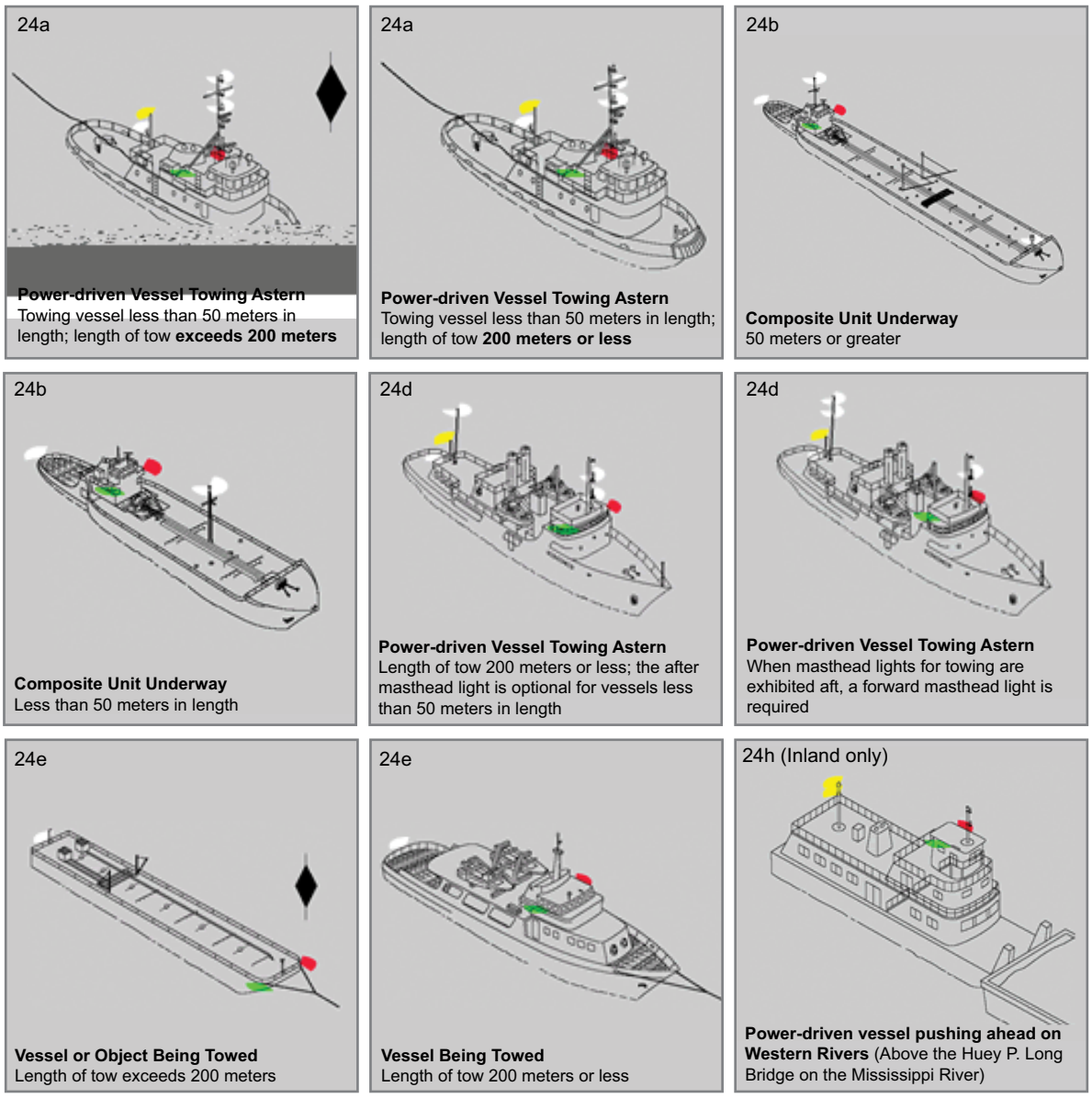
(174) (f) Provided that any number of vessels being towed alongside or pushed in a group shall be lighted as one vessel *‹‹ except as provided in Rule 24(f)(iii) ››*.

(175) (i) a vessel being pushed ahead, not being part of a composite unit, shall exhibit at the forward end, sidelights, and *‹‹ a special flashing light ››*;

(176) (ii) a vessel being towed alongside shall exhibit a sternlight and at the forward end, sidelights, and *‹‹ a special flashing light ››*;

(188)

Rule 24—Towing and Pushing (International/Inland)



(177)

Rule 24f (Inland)

(iii) when vessels are towed alongside on both sides of the towing vessel a sternlight shall be exhibited on the stern of the outboard vessel on each side of the towing vessel, and a single set of sidelights as far forward and as far outboard as is practicable, and a single special flashing light;

(178) (g) An inconspicuous, partly submerged vessel or object, or combination of such vessels or objects being towed, shall exhibit:

(179) (i) if it is less than 25 meters in breadth, one all-round white light at or near the forward end and one at or near the after end except that dracones need not exhibit a light at or near ‹ the forward ›‹‹ *each* ›› end.

(180)

Rule 24g (International)

(ii) if it is 25 meters or more in breadth, two additional all-round white lights at or near the extremities of its breadth;

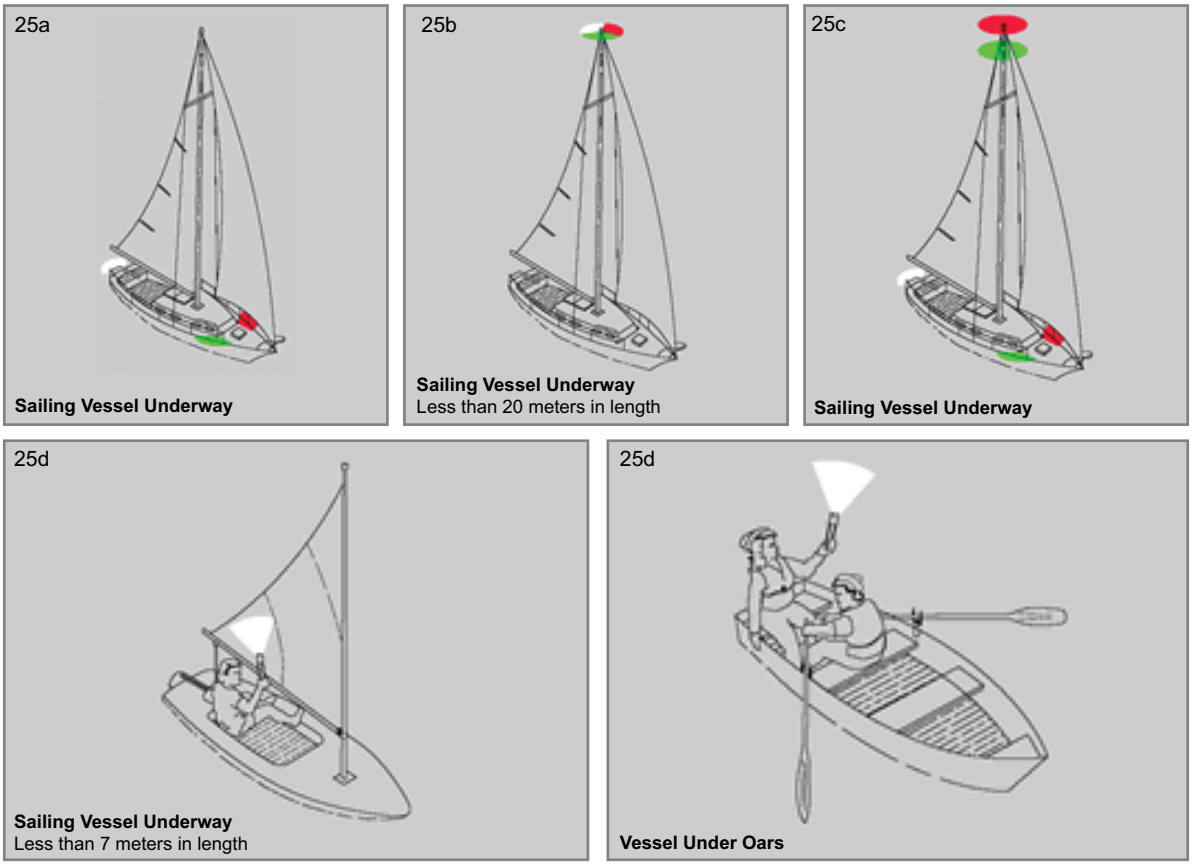
Rule 24g (Inland)

(ii) if it is 25 meters or more in breadth, four all-round white lights to mark its length and breadth;

(181) (iii) if it exceeds 100 meters in length, additional all-round white lights between the lights prescribed in Rule 24(g)(i) ‹‹ *and (ii)* ›› and so that the distance between the lights shall not exceed 100 meters. ‹‹ *Provided that any*

(196)

Rule 25—Sailing Vessels Underway and Vessels Under Oars (International/Inland)



vessels or objects being towed alongside each other shall be lighted as one vessel or object ››.

(182) (iv) a diamond shape at or near the aftermost extremity of the last vessel or object being towed; and ‹ if the length of the tow exceeds 200 meters an additional diamond shape where it can best be seen and located as far forward as is practicable. ›

(183)

Rule 24g (Inland)

(v) the towing vessel may direct a searchlight in the direction of the tow to indicate its presence to an approaching vessel.

(184) (h) Where from any sufficient cause it is impracticable for a vessel or object being towed to exhibit the lights or shapes prescribed in Rule 24(e) or (g), all possible measures shall be taken to light the vessel or object towed or at least to indicate the presence of ‹ such *›‹ the unlighted ››* vessel or object.

(185) **Interpretive Rule**—See **33 CFR 90.7** and **33 CFR 82.7**, chapter 2, for regulations.

(186) (i) Where from any sufficient cause it is impracticable for a vessel not normally engaged in towing operations to display the lights prescribed by paragraph (a), (c), «*or (j)*» of this Rule, such vessel shall not be required to exhibit those lights when engaged in towing another vessel in distress or otherwise in need of assistance. All possible measures shall be taken to indicate the nature of the relationship between the towing vessel and the vessel being towed ‹ as authorized by Rule 36, in particular by illuminating the towline ›«*and the vessel being assisted. The searchlight authorized by Rule 36 may be used to illuminate the tow*».

(187)

Rule 24j (Inland)

(i) Notwithstanding paragraph (c) of this Rule, on the Western Rivers (except below the Huey P. Long Bridge at mile 106.1 Above Head of Passes on the Mississippi River) and on waters specified by the Secretary, a power-driven vessel when pushing ahead or towing alongside, except as paragraph (b) of this Rule applies, shall exhibit: (i) sidelights; and (ii) two towing lights in a vertical line.

(189)

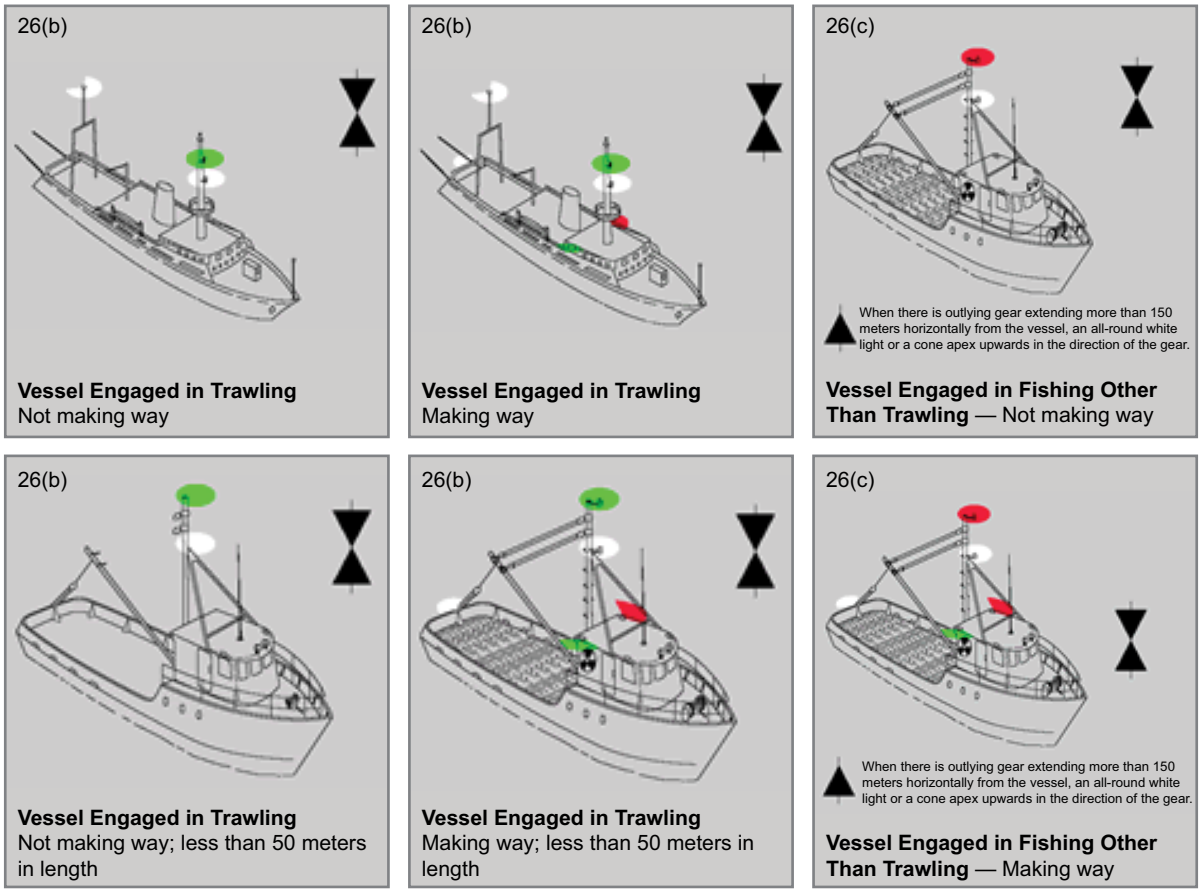
Rule 25—Sailing Vessels Underway and Vessels Under Oars

(190) (a) A sailing vessel underway shall exhibit: (i) sidelights; (ii) a sternlight.

(191) (b) In a sailing vessel of less than 20 meters in length the lights prescribed in Rule 25(a) may be combined in one lantern carried at or near the top of the mast where it can best be seen.

(210)

Rule 26—Fishing Vessels (International/Inland)



(192) (c) A sailing vessel underway may, in addition to the lights prescribed in Rule 25(a), exhibit at or near the top of the mast, where they can best be seen, two all-round lights in a vertical line, the upper being red and the lower green, but these lights shall not be exhibited in conjunction with the combined lantern permitted by Rule 25(b).

(193) (d)(i) A sailing vessel of less than 7 meter in length shall, if practicable, exhibit the lights prescribed in Rule 25(a) or (b), but if she does not, she shall *« exhibit an all around white light or »* have ready at hand an electric torch or lighted lantern showing a white light which shall be exhibited in sufficient time to prevent collision.

(194) (ii) A vessel under oars may exhibit the lights prescribed in this rule for sailing vessels, but if she does not, she shall *« exhibit an all around white light or »* have ready at hand an electric torch or lighted lantern showing a white light which shall be exhibited in sufficient time to prevent collision.

(195) (e) A vessel proceeding under sail when also being propelled by machinery shall exhibit forward where it can best be seen a conical shape, apex downwards. *« A vessel of less than 12 meters in length is not required to exhibit this shape, but may do so.»*

(197)

Rule 26—Fishing Vessels

(198) (a) A vessel engaged in fishing, whether underway or at anchor, shall exhibit only the lights and shapes prescribed in this Rule.

(199) (b) A vessel when engaged in trawling, by which is meant the dragging through the water of a dredge net or other apparatus used as a fishing appliance, shall exhibit: (i) two all-round lights in a vertical line, the upper being green and the lower white, or a shape consisting of two cones with their apexes together in a vertical line one above the other; (ii) a masthead light abaft of and higher than the all-round green light; a vessel of less than 50 meters in length shall not be obliged to exhibit such a light but may do so; (iii) when making way through the water, in addition to the lights prescribed in this paragraph, sidelights and a sternlight.

(200) (c) A vessel engaged in fishing, other than trawling, shall exhibit: (i) two all-round lights in a vertical line, the upper being red and the lower white, or a shape consisting of two cones with their apexes together in a vertical line one above the other; (ii) when there is outlying gear extending more than 150 meters horizontally from the vessel, an all-round white light or a cone apex upwards in

the direction of the gear; (iii) when making way through the water, in addition to the lights prescribed in this paragraph, sidelights and a sternlight.

(201)

Rule 26d (International)

(d) The additional signals described in Annex II to these Regulations apply to a vessel engaged in fishing in close proximity to other vessels engaged in fishing.

(202) (e) A vessel ‹when› not engaged in fishing shall not exhibit the lights or shapes prescribed in this Rule, but only those prescribed for a vessel of her length.

(203) «*(f) Additional signals for fishing vessels in close proximity.*» {Same as International Rules Annex II}

(204) 1ǂ(i) The lights mentioned herein shall ‹,if exhibited in pursuance of Rule 26(d),› be placed where they can best be seen. They shall be at least 0.9 meters apart but at a lower level than lights prescribed in Rule 26. ‹(b)(i) and (c)(i)› The lights shall be visible all round the horizon at a distance of at least 1 mile but at a lesser distance from the lights prescribed by ‹these Rules›«*Rule 26(a)-(c)*» for fishing vessels.

(205) 2ǂ(ii) Signals for trawlers.

(206) (a)ǂ(1) Vessels ‹of 20 meters or more in length› when engaged in trawling, whether using demersal or pelagic gear, ‹shall›«*may*» exhibit: (i)ǂ(A) when shooting their nets—two white lights in a vertical line; (ii)ǂ(B) when hauling their nets—one white light over one red light in a vertical line; (iii)ǂ(C) when the net has come fast upon an obstruction—two red lights in a vertical line.

(207) (b)ǂ(2) ‹A›«*Each*» vessel ‹of 20 meters or more in length› engaged in pair trawling ‹shall›«*may*» exhibit: (i)ǂ(A) by night, a searchlight directed forward and in the direction of the other vessel of the pair; (ii)ǂ(B) when shooting or hauling their nets or when their nets have come fast upon an obstruction, the lights prescribed in Rule 26(f)(2)(a)ǂ(f)(ii)(1).

(208) 3ǂ(iii) Signals for purse seiners.

(209) (a)ǂ(1) Vessels engaged in fishing with purse seine gear may exhibit two yellow lights in a vertical line. These lights shall flash alternately every second and with equal light and occultation duration. These lights may be exhibited only when the vessel is hampered by its fishing gear.

(211)

Rule 27—Vessels Not Under Command or Restricted in Their Ability to Maneuver

(212) (a) A vessel not under command shall exhibit: (i) two all-round red lights in a vertical line where they can best be seen; (ii) two balls or similar shapes in a vertical line where they can best be seen; (iii) when making way through the water, in addition to the lights prescribed in this paragraph, sidelights and a sternlight.

(213) (b) A vessel restricted in her ability to maneuver, except a vessel engaged in mineclearance operations, shall exhibit: (i) three all-round lights in a vertical line where they can best be seen. The highest and lowest of these lights shall be red and the middle light shall be white; (ii) three shapes in a vertical line where they can best be seen. The highest and lowest of these shapes shall be balls and the middle one a diamond; (iii) when making way through the water, a masthead light(s), sidelights and a sternlight in addition to the lights prescribed in Rule 27(b)(i); (iv) when at anchor, in addition to the lights or shapes prescribed in Rule 27(b)(i) and (ii), the light, lights, or shapes prescribed in Rule 30.

(214) (c) A power-driven vessel engaged in a towing operation such as severely restricts the towing vessel and her tow in their ability to deviate from their course shall, in addition to the lights or shape prescribed in Rule 27(b)(i) and (ii), exhibit the lights or shape prescribed in Rule 24.

(215) (d) A vessel engaged in dredging or underwater operations, when restricted in her ability to maneuver, shall exhibit the lights and shapes prescribed in Rules 27(b)(i), (ii) and (iii) and shall in addition when an obstruction exists, exhibit: (i) two all-round red lights or two balls in a vertical line to indicate the side on which the obstruction exists; (ii) two all-round green lights or two diamonds in a vertical line to indicate the side on which another vessel may pass; and (iii) when at anchor, the lights or shapes prescribed in this paragraph instead of the lights or shapes prescribed in Rule 30.

(216)

Rule 27d (Inland)

(iv) Dredge pipelines that are floating or supported on trestles shall display the following lights at night and in periods of restricted visibility.

(1) One row of yellow lights. The lights must be: (A) flashing 50 to 70 times per minute, (B) visible all round the horizon, (C) visible for at least 2 miles, (D) not less than 1 and not more than 3.5 meters above the water, (E) approximately equally spaced, and (F) not more than 10 meters apart where the pipeline crosses a navigable channel. Where the pipeline does not cross a navigable channel the lights must be sufficient in number to clearly show the pipeline's length and course.

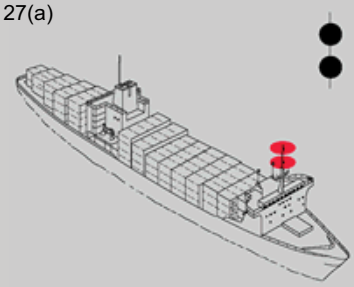
(2) Two red lights at each end of the pipeline, including the ends in a channel where the pipeline is separated to allow vessels to pass (whether open or closed). The lights must be: (A) visible all round the horizon, and (B) visible for at least 2 miles, and (C) one meter apart in a vertical line with the lower light at the same height above the water as the flashing yellow light.

(217) (e) Whenever the size of a vessel engaged in diving operations makes it impracticable to exhibit all lights and shapes prescribed in Rule 27(d), the following shall be exhibited: (i) Three all-round lights in a vertical line where they can best be seen. The highest and lowest of these lights shall be red and the middle light shall be white; (ii) a rigid replica of the International Code flag "A" not less than 1 meter in height. Measures shall be taken to ensure its all-round visibility.

(221)

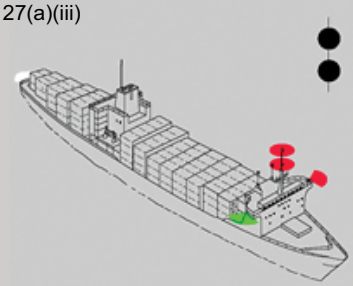
Rule 27—Vessels Not Under Command or Restricted in Their Ability to Maneuver (International/Inland)

27(a)



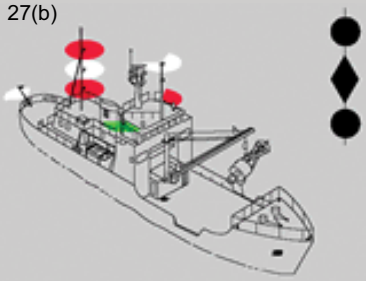
Vessel Not Under Command
Not making way

27(a)(iii)



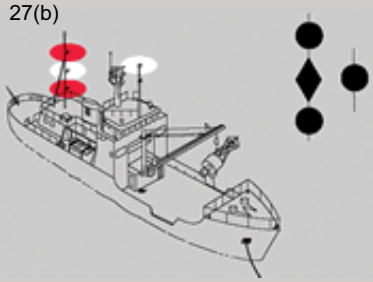
Vessel Not Under Command
Making way

27(b)



Vessel Restricted in Her Ability to Maneuver—Making way; less than 50 meters in length

27(b)

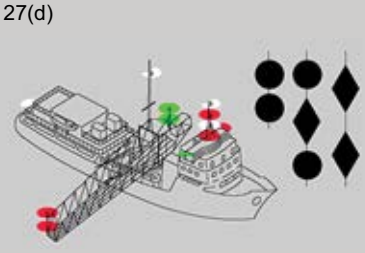


Vessel Restricted in Her Ability to Maneuver—At anchor; less than 50 meters in length

27(c)

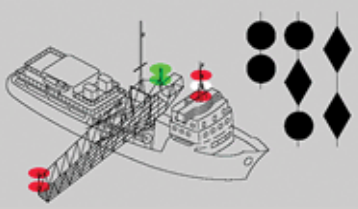
Vessel engaged in towing operation which severely restricts towing vessel and her tow in their ability to deviate from their course—Length of tow does not exceed 200 meters; towing vessel less than 50 meters in length.

27(d)



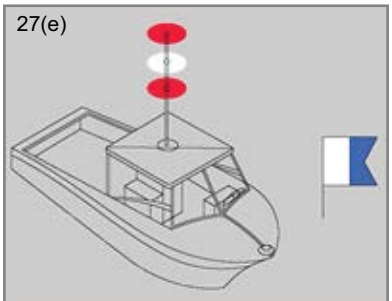
Vessel engaged in dredging or underwater operations when restricted in ability to maneuver—Making way with an obstruction on the starboard side.

27(d)



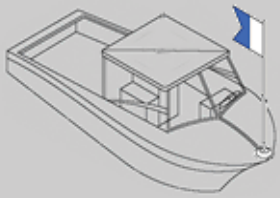
Vessel engaged in dredging or underwater operations when restricted in ability to maneuver—Not making way with an obstruction on the starboard side.

27(e)



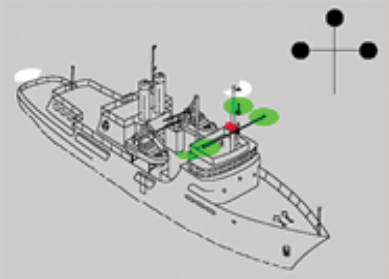
Small vessel engaged in diving operations

27(e)



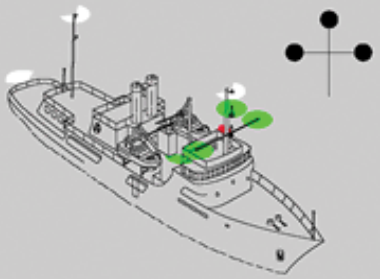
Small vessel engaged in diving operations

27(f)



Vessel engaged in mineclearance operations
Vessel less than 50 meters in length.

27(f)

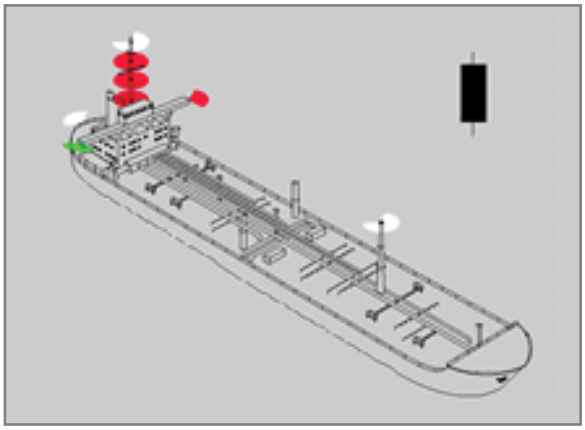


Vessel engaged in mineclearance operations
Vessel 50 meters or greater in length.

(224)

Rule 28—Vessel Constrained by Their Draft (International)

A vessel constrained by her draft may, in addition to the lights prescribed for power-driven vessels in Rule 23, exhibit where they can best be seen three all-round red lights in a vertical line, or a cylinder.



(218) (f) A vessel engaged in mine clearance operations shall, in addition to the lights prescribed for a power-driven vessel in Rule 23 or to the lights or shape prescribed for a vessel at anchor in Rule 30 as appropriate, exhibit three all-round green lights or three balls. One of these lights or shapes shall be exhibited near the foremast head and one at each end of the fore yard. These lights or shapes indicate that it is dangerous for another vessel to approach within 1000 meters of the mineclearance vessel.

(219) (g) Vessels of less than 12 meters in length, except ‹ those ›«*when* » engaged in diving operations, ‹ shall not be ›«*is not* » required to exhibit the lights ‹ and ›«*or* » shapes prescribed in this Rule.

(220) (h) The signals prescribed in this Rule are not signals of vessels in distress and requiring assistance. Such signals are contained in Annex IV to these Rules.

(222)

Rule 28—Vessels Constrained by Their Draft

(223) See graphic, **Rule 28—Vessels Constrained by Their Draft**

(225)

Rule 29—Pilot Vessels

(226) (a) A vessel engaged on pilotage duty shall exhibit: (i) at or near the masthead, two all-round lights in a vertical line, the upper being white and the lower red; (ii) when underway, in addition, sidelights and a sternlight; (iii) when at anchor, in addition to the lights prescribed in Rule 29(a)(i), the light, lights, or shape prescribed in Rule 30 for vessels at anchor.

(227) (b) A pilot vessel when not engaged on pilotage duty shall exhibit the lights or shapes prescribed for a similar vessel of her length.

(229)

Rule 30—Anchored Vessels and Vessels Aground

(230) (a) A vessel at anchor shall exhibit where it can best be seen: (i) in the fore part, an all-round white light or one ball; (ii) at or near the stern and at a lower level than the light prescribed in Rule 30(a)(i), an all-round white light.

(231) **Interpretive Rule**—See **33 CFR 90.5** and **33 CFR 82.5**, chapter 2, for regulations on vessels at anchor.

(232) (b) A vessel of less than 50 meters in length may exhibit an all-round white light where it can best be seen instead of the lights prescribed in Rule 30(a).

(233) (c) A vessel at anchor may, and a vessel of 100 meters and more in length shall, also use the available working or equivalent lights to illuminate her decks.

(234) (d) A vessel aground shall exhibit the lights prescribed in Rule 30(a) or (b) and in addition, if practicable, where they can best be seen: (i) two all-round red lights in a vertical line; (ii) three balls in a vertical line.

(235) (e) A vessel of less than 7 meters in length, when at anchor not in or near a narrow channel, fairway or where other vessels normally navigate, shall not be required to exhibit the lights or shape prescribed in Rule 30(a) and (b).

(236) (f) A vessel of less than 12 meters in length, when aground, shall not be required to exhibit the lights or shapes prescribed in Rule 30(d)(i) and (ii).

(237)

Rule 30 (Inland)

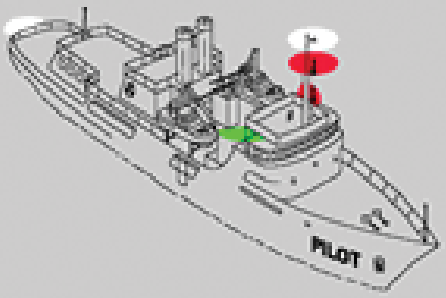
(g) A vessel of less than 20 meters in length, when at anchor in a special anchorage area designated by the Coast Guard, shall not be required to exhibit the anchor lights and shapes required by this Rule.

(h) The following barges shall display at night and if practicable in periods of restricted visibility the lights described in Rule 30(i):
(i) Every barge projecting into a buoyed or restricted channel.
(ii) Every barge so moored that it reduces the available navigable width of any channel to less than 80 meters.
(iii) Barges moored in groups more than two barges wide or to a maximum width of over 25 meters.
(iv) Every barge not moored parallel to the bank or dock.

(i) Barges described in Rule 30(h) shall carry two unobstructed all-round white lights of an intensity to be visible for at least 1 nautical mile and meeting the technical requirements as prescribed in Annex I.

(228)

Rule 29—Pilot Vessels (International/Inland)



Vessel Engaged on Pilotage Duty
Underway

Vessel Engaged on Pilotage Duty
At anchor; vessel less than 50 meters in length

Rule 30 (Inland)
(j) A barge or a group of barges at anchor or made fast to one or more mooring buoys or other similar device, in lieu of the provisions of Rule 30, may carry unobstructed all-round white lights of an intensity to be visible for at least 1 nautical mile that meet the requirements of Annex I and shall be arranged as follows: *(i) Any barge that projects from a group formation, shall be lighted on its outboard corners.* *(ii) On a single barge moored in water where other vessels normally navigate on both sides of the barge, lights shall be placed to mark the corner extremities of the barge.* *(iii) On barges moored in group formation, moored in water where other vessels normally navigate on both sides of the group, lights shall be placed to mark the corner extremities of the group.*
(k) The following are exempt from the requirements of Rule 30: *(i) A barge or group of barges moored in a slip or slough used primarily for mooring purposes.* *(ii) A barge or group of barges moored behind a pierhead.* *(iii) A barge less than 20 meters in length when moored in a special anchorage area designated in accordance with 33 CFR 109.10.*
(l) Barges moored in well-illuminated areas are exempt from the lighting requirements of Rule 30. These areas are as follows:
CHICAGO SANITARY SHIP CANAL
(1) Mile 293.2 to 293.9 — *(15) Mile 314.6*
(2) Mile 295.2 to 296.1 — *(16) Mile 314.8 to 315.3*
(3) Mile 297.5 to 297.8 — *(17) Mile 315.7 to 316*
(4) Mile 298 to 298.2 — *(18) Mile 316.8*
(5) Mile 298.6 to 298.8 — *(19) Mile 316.85 to 317.05*
(6) Mile 299.3 to 299.4 — *(20) Mile 317.5*
(7) Mile 299.8 to 300.5 — *(21) Mile 318.4 to 318.9*
(8) Mile 303 to 303.2 — *(22) Mile 318.7 to 318.8*
(9) Mile 303.7 to 303.9 — *(23) Mile 320 to 320.3*
(10) Mile 305.7 to 305.8 — *(24) Mile 320.6*
(11) Mile 310.7 to 310.9 — *(25) Mile 322.3 to 322.4*
(12) Mile 311 to 311.2 — *(26) Mile 322.8*
(13) Mile 312.5 to 312.6 — *(27) Mile 322.9 to 327.2*
(14) Mile 313.8 to 314.2
CALUMET SAG CHANNEL

Rule 30 (Inland)	
(28) Mile 316.5	
LITTLE CALUMET RIVER	
(29) Mile 321.2	*(30) Mile 322.3*
CALUMET RIVER	
(31) Mile 328.5 to 328.7	*(34) Mile 331.4 to 331.6*
(32) Mile 329.2 to 329.4	*(35) Mile 332.2 to 332.4*
(33) Mile 330 west bank to 330.2	*(36) Mile 332.6 to 332.8*
CUMBERLAND RIVER	
(37) Mile 126.8	*(38) Mile 191*

(238)

Rule 31—Seaplanes

(239) Where it is impracticable for a seaplane or a WIG craft to exhibit lights or shapes of the characteristics or in the positions prescribed in Rules 20 through 31 she shall exhibit lights and shapes as closely similar in characteristics and position as is possible.

(240)

Part D—Sound and Light Signals

(241)

Rule 32—Definitions

(242) (a) The word “whistle” means any sound signaling appliance capable of producing the prescribed blasts and which complies with the specifications in Annex III to these Rules.

(243) (b) The term “short blast” means a blast of about one seconds duration.

(244) (c) The term “prolonged blast” means a blast of from four to six seconds duration.

(245)

Rule 33—Equipment for Sound Signals

(246) (a) A vessel of 12 meters or more in length shall be provided with a whistle, a vessel of 20 meters or more in length shall be provided with a bell in addition to a

whistle, and a vessel of 100 meters or more in length shall, in addition be provided with a gong, the tone and sound of which cannot be confused with that of the bell. The whistle, bell and gong shall comply with the specifications in Annex III to these Regulations. The bell or gong or both may be replaced by other equipment having the same respective sound characteristics, provided that manual sounding of the prescribed signals shall always be possible.

(247) (b) A vessel of less than 12 meters in length shall not be obliged to carry the sound signaling appliances prescribed in Rule 33(a) but if she does not, she shall be provided with some other means of making an efficient signal.

(248)

Rule 34—Maneuvering and Warning Signs (International)

(a) When vessels are in sight of one, a power-driven vessel underway, when maneuvering as authorized or required by these Rules, shall indicate that manoeuvre by the following signals on her whistle:
–One short blasts to mean "I am altering my course to starboard"
–Two short blasts to mean "I am altering my course to port"
–Three short blasts to mean "I am operating astern propulsion"

(b) Any vessel may supplement the whistle signals prescribed in Rule 34(a) by light signals, repeated as appropriate, while the maneuver is being carried out:
(i) these signals shall have the following significance:
(ii) the duration of each flash shall be about one second, the interval between flashes shall be about one second, and the interval between successive signals shall not be less than ten seconds.
(iii) the light used for this signal shall, if fitted, be an all-round white, visible at a minimum range of 5 miles, and shall comply with the provisions of Annex I to these Regulations.
–One flash to mean "I am altering my course to starboard"
–Two flashes to mean I am altering my course to port"
–Three flashes to mean "I am operating astern propulsion".

(c) When in sight of one another in a narrow channel or fairway:
(i) a vessel intending to overtake another shall in compliance with Rule 9(e)(i) indicate her intention by the following signals on her whistle:
–Two prolonged blasts followed by one short blast to mean "I intend to overtake you on your starboard side"
–Two prolonged blasts followed by two short blasts to mean "I intend to overtake you on your port side".
(ii) the vessel about to be overtaken when acting in accordance with Rule 9(e)(i) shall indicate her agreement by the following signal on her whistle:
–one prolonged, one short, one prolonged and one short blast, in that order.

(d) When vessels in sight of one another are approaching each other and from any cause either vessel fails to understand the intentions or actions of the other, or is in doubt whether sufficient action is being taken by the other to avoid collision, the vessel in doubt shall immediately indicate such doubt by giving at least five short and rapid blasts on the whistle. Such signal may be supplemented by at least five short and rapid flashes.

Rule 34—Maneuvering and Warning Signs (International)

(e) A vessel nearing a bend or an area of a channel or fairway where other vessels may be obscured by an intervening obstruction shall sound one prolonged blast. This signal shall be answered with a prolonged blast by any approaching vessel that may be within hearing around the bend or behind the intervening obstruction.

(f) If whistles are fitted on a vessel at a distance apart of more than 100 meters, one whistle only shall be used for giving maneuvering and warning signals.

(249)

Rule 34—Maneuvering and Warning Signs (Inland)

(a) When power-driven vessels are in sight of one another and meeting or crossing at a distance within half a mile of each other, each vessel underway, when maneuvering as authorized or required by these Rules,
(i) shall indicate that maneuver by the following signals on her whistle:
–One short blasts to mean "I intend to leave you on my port side"
–Two short blasts to mean "I intend to leave you on my starboard side"
–Three short blasts to mean "I am operating astern propulsion"
(ii) upon hearing the one or two blast signal of the other shall, if in agreement, sound the same whistle signal and take the steps necessary to effect a safe passing. If, however, from any cause, the vessel doubts the safety of the proposed maneuver, she shall sound the signal specified in Rule 34(d) and each vessel shall take appropriate precautionary action until a safe passing agreement is made

(b) Any vessel may supplement the whistle signals prescribed in Rule 34(a) by light signals:
(i) these signals shall have the following significance:
(ii) the duration of each flash shall be about one second.
(iii) the light used for this signal shall, if fitted, be an all-round white or yellow, visible at a minimum range of 2 miles, synchronized with the whistle and shall comply with the provisions of Annex I to these Regulations.
–One flash to mean "I intend to leave you on my port side"
–Two flashes to mean "I intend to leave you on my starboard side"
–Three flashes to mean "I am operating astern propulsion"

(c) When in sight of one another:
(i) a power-driven vessel intending to overtake another power-driven vessel shall indicate her intention by the following signals on her whistle:
–One short blast to mean "I intend to overtake you on your starboard side"
–Two short blasts to mean "I intend to overtake you on your port side"
(ii) the power-driven vessel about to be overtaken shall, if in agreement, sound a similar sound signal. If in doubt she shall sound the signal prescribed in Rule 34(d).

(d) When vessels in sight of one another are approaching each other and from any cause either vessel fails to understand the intentions or actions of the other, or is in doubt whether sufficient action is being taken by the other to avoid collision, the vessel in doubt shall immediately indicate such doubt by giving at least five short and rapid blasts on the whistle. Such signal may be supplemented by at least five short and rapid flashes.

Rule 34—Maneuvering and Warning Signs (Inland)

(e) A vessel nearing a bend or an area of a channel or fairway where other vessels may be obscured by an intervening obstruction shall sound one prolonged blast. This signal shall be answered with a prolonged blast by any approaching vessel that may be within hearing around the bend or behind the intervening obstruction.

(f) If whistles are fitted on a vessel at a distance apart of more than 100 meters, one whistle only shall be used for giving maneuvering and warning signals.

(g) When a power-driven vessel is leaving a dock or berth, she shall sound one prolonged blast.

(h) A vessel that reaches agreement with another vessel in a head-on, crossing, or overtaking situation, as for example, by using the radiotelephone as prescribed by the Vessel Bridge-to-Bridge Radiotelephone Act (85 Stat. 164; 33 U.S.C. 1201 et seq.), is not obliged to sound the whistle signals prescribed by this Rule, but may do so. If agreement is not reached, then whistle signals shall be exchanged in a timely manner and shall prevail.

(250)

Rule 35—Sound Signals in Restricted Visibility

(251) In or near an area of restricted visibility, whether by day or night the signals prescribed in this Rule shall be used as follows:

(252) (a) A power-driven vessel making way through the water shall sound at intervals of not more than 2 minutes one prolonged blast.

(253) (b) A power-driven vessel underway but stopped and making no way through the water shall sound at intervals of no more than 2 minutes two prolonged blasts in succession with an interval of about 2 seconds between them.

(254) (c) A vessel not under command, a vessel restricted in her ability to maneuver «*whether underway or at anchor*», ‹ a vessel constrained by her draft › , a sailing vessel, a vessel engaged in fishing and a vessel engaged in towing or pushing another vessel shall, instead of the signals prescribed in Rule 35(a) or (b), sound at intervals of not more than 2 minutes three blasts in succession, namely one prolonged followed by two short blasts.

(255)

Rule 35d (International)

(d) A vessel engaged in fishing, when at anchor, and a vessel restricted in her ability to maneuver when carrying out her work at anchor, shall instead of the signals prescribed in Rule 35(g) sound the signal prescribed in Rule 35(c).

(256) (e) A vessel towed or if more than one vessel is towed the last vessel of the tow, if manned, shall at intervals of not more than 2 minutes sound four blasts in succession, namely one prolonged followed by three short blasts. When practicable, this signal shall be made immediately after the signal made by the towing vessel.

(257) (f) When a pushing vessel and a vessel being pushed ahead are rigidly connected in a composite unit they shall be regarded as a power-driven vessel and shall give the signals prescribed in Rule 35(a) or (b).

(258) (g) A vessel at anchor shall at intervals of not more than 1 minute ring the bell rapidly for about 5 seconds. In a vessel 100 meters or more in length the bell shall be sounded in the forepart of the vessel and immediately after the ringing of the bell the gong shall be sounded rapidly for about 5 seconds in the after part of the vessel. A vessel at anchor may in addition sound three blasts in succession, namely one short, one long and one short blast, to give warning of her position and of the possibility of collision to an approaching vessel.

(259) (h) A vessel aground shall give the bell signal and if required the gong signal prescribed in Rule 35(g) and shall, in addition, give three separate and distinct strokes on the bell immediately before and after the rapid ringing of the bell. A vessel aground may in addition sound an appropriate whistle signal.

(260) (i) A vessel of 12 meters or more but less than 20 meters in length shall not be obliged to give the bell signals prescribed in Rule 35(g) and (h). However, if she does not, she shall make some other efficient sound signal at intervals of not more than 2 minutes.

(261) (j) A vessel of less than 12 meters in length shall not be obliged to give the above mentioned signals but, if she does not, shall make some other efficient sound signal at intervals of not more than 2 minutes.

(262) (k) A pilot vessel when engaged on pilotage duty may, in addition to the signals prescribed in Rule 35(a), (b) or (g), sound an identity signal consisting of four short blasts.

(263)

Rule 35 (Inland)

(l) The following vessels shall not be required to sound signals as prescribed in Rule 35(g) when anchored in a special anchorage area designated by the Coast Guard:

(i) a vessel of less than 20 meters in length; and

(ii) a barge, canal boat, scow, or other nondescript craft.

(264)

Rule 36—Signals to Attract Attention

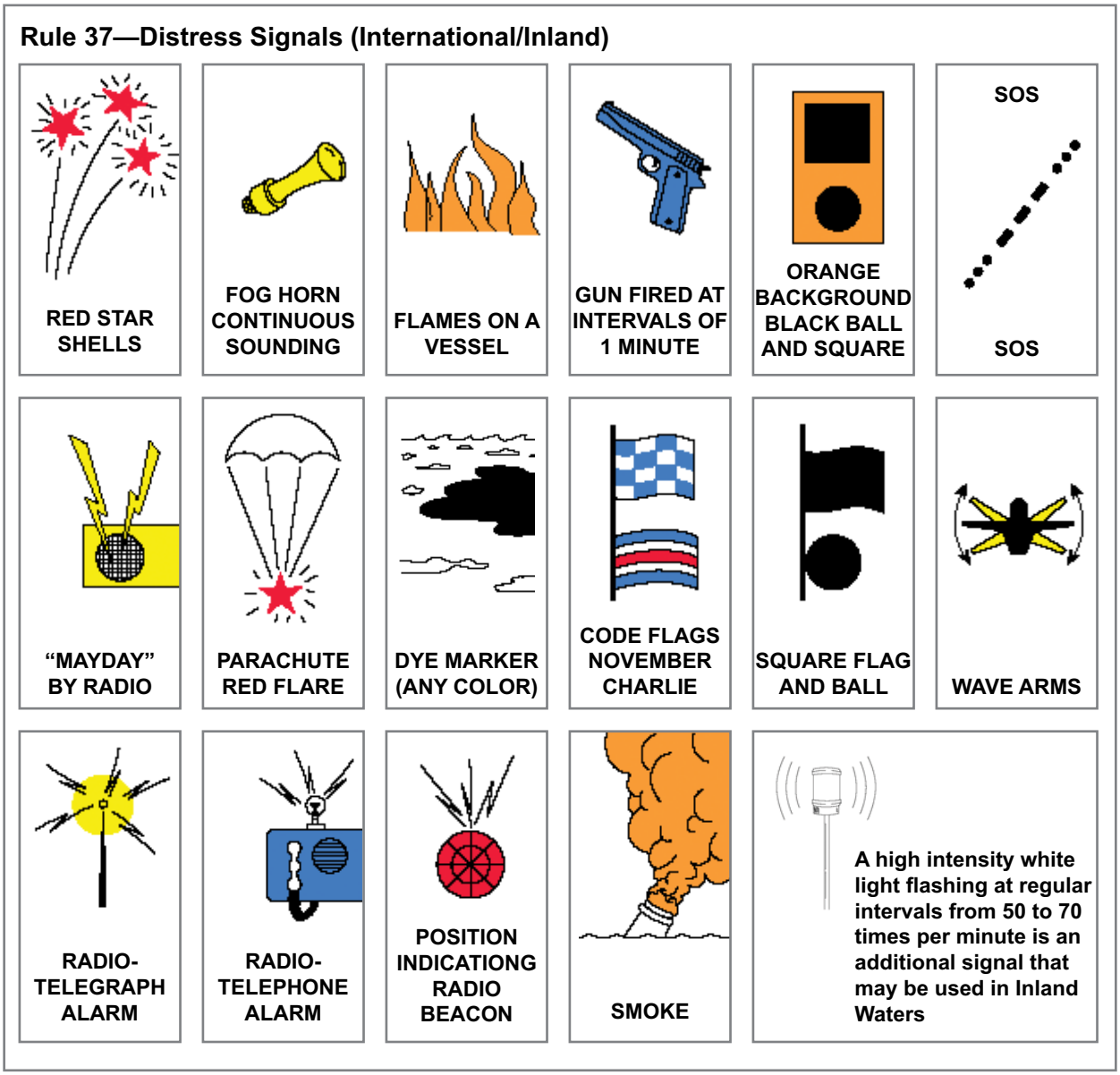
(265) If necessary to attract the attention of another vessel, any vessel may make light or sound signals that cannot be mistaken for any signal authorized elsewhere in these Rules, or may direct the beam of her searchlight in the direction of the danger, in such a way as not to embarrass any vessel.

(266)

Rule 36 (International)

Any light to attract the attention of another vessel shall be such that it cannot be mistaken for any aid to navigation. For the purpose of this Rule the use of high intensity intermittent or revolving lights, such as strobe lights, shall be avoided.

(269)



(267)

Rule 37—Distress Signals

(268) When a vessel is in distress and requires assistance she shall use or exhibit the signals described in Annex IV to these Rules. (See graphic, **Rule 37—Distress Signals**).

(270)

Part E—Exemptions

(271)

Rule 38—Exemptions (International)

Any vessel (or class of vessel) provided that she complies with the requirements of — the International Regulations for the Preventing of Collisions at Sea, 1960, the keel of which is laid or is at a corresponding stage of construction before the entry into force of these Regulations may be exempted from compliance therewith as follows:

Rule 38—Exemptions (International)

(a) The installation of lights with ranges prescribed in Rule 22, until 4 years after the date of entry into force of these Regulations.

(b) The installation of lights with color specifications as prescribed in §7 of Annex I to these Regulations, until 4 years after the entry into force of these Regulations.

(c) The repositioning of lights as a result of conversion from Imperial to metric units and rounding off measurement figures, permanent exemption.

(d)(i) The repositioning of masthead lights on vessels of less than 150 meters in length, resulting from the prescriptions of §3 (a) of Annex I to these Regulations, permanent exemption.
(ii) The repositioning of masthead lights on vessels of 150 meters or more in length, resulting from the prescriptions of §3 (a) of Annex I to these Regulations, until 9 years after the date of entry into force of these Regulations.

Rule 38—Exemptions (International)
(e) The repositioning of masthead lights resulting from the prescriptions of §2(b) of Annex I to these Regulations, until 9 years after the date of entry into force of these Regulations.
(f) The repositioning of sidelights resulting from the prescriptions of §2(g) and 3(b) of Annex I to these Regulations, until 9 years after the date of entry into force of these Regulations.
(g) The requirements for sound signal appliances prescribed in Annex II to these Regulations, until 9 years after the date of entry into force of these Regulations.
(h) The repositioning of all-round lights resulting from the prescription of §9(b) of Annex I to these Regulations, permanent exemption.

(272)

Rule 38—Exemptions (Inland)
Any vessel or class of vessels, the keel of which was laid or which is at a corresponding stage of construction before December 24, 1980, provided that she complies with the requirements of —
(a) The Act of June 7, 1897 (30 Stat. 96), as amended (33 U.S.C. 154-232) for vessels navigating the waters subject to that statute;
(b) §4233 of the Revised Statutes (33 U.S.C. 301-356) for vessels navigating the waters subject to that statute;
(c) The Act of February 8, 1895 (28 Stat. 645), as amended (33 U.S.C. 241-295) for vessels navigating the waters subject to that statute; or
(d) §§3, 4, and 5 of the Act of April 25, 1940 (54 Stat. 163), as amended (46 U.S.C. 526 b, c, and d) for motorboats navigating the waters subject to that statute; shall be exempted from compliance with the technical Annexes to these Rules as follows: *(i) The installation of lights with ranges prescribed in Rule 22, vessels of less than 20 meters in length are permanently exempt;* *(ii) The installation of lights with color specifications as prescribed in §7 of Annex I to these Rules, until 4 years after the effective date of the Inland Navigational Rules Act of 1980 (Pub. L. 96-591), except that vessels of less than 20 meters in length are permanently exempt;* *(iii) The repositioning of lights as a result of a conversion to metric units and rounding off of measurement figures, are permanently exempt.* *(iv) The horizontal repositioning of masthead lights prescribed by Annex I to these Rules, vessels of less than 150 meters in length are permanently exempted.* *(v) Power-driven vessels of 12 meters or more but less than 20 meters in length are permanently exempt from the provisions of Rule 23(a)(i) and 23(a)(iv) provided that, in place of these lights, the vessel exhibits a white light aft visible all-around the horizon.*

(273) **Implementing Rule**—See **33 CFR 81.20**, chapter 2, for regulations.

(274)

Part F—Verification of Compliance with the Provisions of the Convention

(275)

Rule 39—Definitions

(276)

Rule 39 (International)
(a) "Audit" means a systematic, independent and documented process for obtaining audit evidence and evaluating it objectively to determine the extent to which audit criteria are fulfilled. (b) "Audit Scheme" means the IMO Member State Audit Scheme established by the Organization and taking into account the guidelines developed by the Organization*. (c) "Code for Implementation" means the IMO Instruments Implementation Code (III Code) adopted by the Organization by resolution A.1070(28). (d) "Audit Standard" means the Code for Implementation.

(277)

Rule 40—Application

(278)

Rule 40 (International)
Contracting Parties shall use the provisions of the Code for Implementation in the execution of their obligations and responsibilities contained in the present Convention.

(279)

Rule 41—Verification of Compliance

(280)

Rule 41 (International)
(a) Every Contracting Party shall be subject to periodic audits by the Organization in accordance with the audit standard to verify compliance with and implementation of the present Convention.
(b) The Secretary-General of the Organization shall have responsibility for administering the Audit Scheme, based on the guidelines developed by the Organization*.
(c) Every Contracting Party shall have responsibility for facilitating the conduct of the audit and implementation of a programme of actions to address the findings, based on the guidelines developed by the Organization*.
(d) Audit of all Contracting Parties shall be:
(i) based on an overall schedule developed by the Secretary-General of the Organization, taking into account the guidelines developed by the Organization*; and
(ii) conducted at periodic intervals, taking into account the guidelines developed by the Organization*.
* Refer to the Framework and Procedures for the IMO Member State Audit Scheme, adopted by the Organization by resolution A.1067(28).

(281)

Annex I—Positioning and Technical Details of Lights and Shapes

(282)

Definitions

(283) (a) The term "height above the hull" means height above the uppermost continuous deck. This height shall be measured from the position vertically beneath the location of the light.

(284)

Annex I (Inland)

(b) High-speed craft means a craft capable of maximum speed in meters per second (m/s) equal to or exceeding: $3.7\nabla^{0.1667}$; where ∇ = displacement corresponding to the design waterline (cubic meters).

Note: The same formula expressed in pounds and knots is maximum speed in knots (kts) equal to exceeding 1.98(lbs) $3.7\nabla^{0.1667}$; where ∇=displacement corresponding to design waterline in pounds.

(c) The term "practical cut-off" means, for vessels 20 meters or more in length, 12.5 percent of the minimum luminous intensity (Table 14(b)) corresponding to the greatest range of visibility for which the requirements of Annex I are met.

(d) The term "Rule" or "Rules" has the same meaning as in Rule 3(r).

(285)

Vertical Positioning and Spacing of Lights

(286) (a) On a power-driven vessel of 20 meters or more in length the masthead light shall be placed as follows: (i) The forward masthead light, or if only one masthead light is carried, then that light, at a height above the hull of not less than ‹ 6 ›«*5*» meters, and, if the breadth of the vessel exceeds ‹ 6 ›«*5*» meters, then at a height above the hull not less than such breadth, so however that the light need not be placed at a greater height above the hull than ‹ 12 ›«*8*» meters; (ii) when two masthead lights are carried the after one shall be at least ‹ 4.5 ›«*2*» meters vertically higher than the forward one.

(287) (b) The vertical separation of the masthead lights of power-driven vessels shall be such that in all normal conditions of trim the after light will be seen over and separate from the forward light at a distance of 1000 meters from the stem when viewed from ‹ sea ›«*water*» level.

(288) (c) The masthead light of a power-driven vessel of 12 meters but less than 20 meters in length shall be placed at a height above the gunwale of not less than 2.5 meters.

(289)

Annex I (International)

(d) A power-driven vessel of less than 12 meters in length may carry the uppermost light at a height of less than 2.5 meters above the gunwale. When, however, a masthead light is carried in addition to sidelights and a sternlight or the all-round light prescribed in Rule 23(d)(i) is carried in addition to sidelights, then such masthead light or all-round light shall be carried at least 1 meter higher than the sidelights.

Annex I (Inland)

(d) The masthead light, or the all-round light described in Rule 23(d), of a power-driven vessel of less than 12 meters in length shall be carried at least 1 meter higher than the sidelights.

(290) (e) One of the two or three masthead lights prescribed for a power-driven vessel when engaged in towing or pushing another vessel shall be placed in the same position as either the forward masthead light or the after masthead light, provided that ‹ , if carried on the after mast, › the lowest after masthead light shall be at least ‹ 4.5 ›«*2*» meters vertically higher than the «*highest*» forward masthead light.

(291) (f)(i) The masthead lights or lights prescribed in Rule 23(a) shall be so placed as to be above and clear of all other lights and obstructions except as described in §(f)(ii).

(292) (ii) When it is impracticable to carry the all-round lights prescribed by Rule 27(b)(i) ‹ or Rule 28 › below the masthead lights, they may be carried above the after masthead light(s) or vertically in between the forward masthead light(s) and after masthead light(s), provided that in the latter case the requirement of §3(c) shall be complied with.

(293) (g) The sidelights of a power-driven vessel shall be placed at ‹ a height above the hull not greater than three quarters of that ›«*least 1 meter lower*» of ‹ the ›«*than*» forward masthead light. They shall not be so low as to be interfered with by deck lights.

(294)

Annex I (International)

(h) The sidelights, if in a combined lantern and carried on a power-driven vessel of less than 20 meters in length, shall be placed not less than 1 meter below the masthead light.

(295) (i) When the Rules prescribe two or three lights to be carried in a vertical line, they shall be spaced as follows: (i) On a vessel of 20 meters in length or more such lights shall be spaced not less than ‹ 2 ›«*1*» meter apart, and the lowest of these lights shall, except where a towing light is required, be placed at a height of not less than 4 meters above the hull. (ii) On a vessel of less than 20 meters in length such lights shall be spaced not less than 1 meter apart and the lowest of these lights shall, except where a towing light is required, be placed at a height of not less than 2 meters above the gunwale. (iii) When three lights are carried they shall be equally spaced.

(296) (j) The lower of the two all-round lights prescribed for a vessel when engaged in fishing shall be at a height above the sidelights not less than twice the distance between the two vertical lights.

(297) (k) The forward anchor light prescribed in Rule 30(a)(i), when two are carried, shall not be less than 4.5 meters above the after one. On a vessel of 50 meters or more in length this forward anchor light shall be placed at a height or not less than 6 meters above the hull.

(298)

Horizontal Positioning and Spacing of Lights

(299) (a) «*Except as specified in §1(e)*, » when two masthead lights are prescribed for a power-driven vessel, the horizontal distance between them must not be less than one- ‹ quarter ›«*half*» of the length of the vessel but need not be more than ‹ 100 ›«*50*» meters. The forward light must be placed not more than one- ‹ quarter ›«*half*» of the length of the vessel from the stem.

(300) (b) On a power-driven vessel of 20 meters or more in length the sidelights shall not be placed in front of the forward masthead lights. They shall be placed at or near the side of the vessel.

(301) (c) When the lights prescribed in Rule 27(b)(i) ‹ or Rule 28 › are placed vertically between the forward masthead light(s) and the after masthead light(s), these all-round lights shall be placed at a horizontal distance of not less than 2 meters from the fore and aft centerline of the vessel in the athwartship direction.

(302) (d) When only one masthead light is prescribed for a power-driven vessel, this light must be exhibited forward of amidships. For a vessel of less than 20 meters in length, the vessel shall exhibit one masthead light as far forward as is practicable.

(303)

Annex I (Inland)

(e) On power-driven vessels 50 meters but less than 60 meters in length operated on the Western Rivers, and those { waters specified by the Secretary }, the horizontal distance between masthead lights shall not be less than 10 meters.

(304)

Details of Location of Direction-Indicating Lights for Fishing Vessels, Dredgers and Vessels Engaged in Underwater Operations

(305) (a) The light indicating the direction of the outlying gear from a vessel engaged in fishing as prescribed in Rule 26(c)(ii) shall be placed at a horizontal distance of not less than 2 meters and not more than 6 meters away from the two all-round red and white lights. This light shall be placed not higher than the all-round white light prescribed in Rule 26(c)(i) and not lower than the sidelights.

(306) (b) The lights and shapes on a vessel engaged in dredging or underwater operations to indicate the obstructed side and/or the side on which it is safe to pass, as prescribed in Rule 27(d)(i) and (ii), shall be placed at the maximum practical horizontal distance, but in no case less than 2 meters, from the lights or shapes prescribed in Rule 27(b)(i) and (ii). In no case shall the upper of these lights or shapes be at a greater height than the lower of the three lights or shapes prescribed in Rule 27(b)(i) and (ii).

(307)

Screens ‹ For Sidelights ›

(308) (a) The sidelights of vessels of 20 meters or more in length shall be fitted with ‹ inboard screens painted › matt black, «*inboard screens*» and meet ‹ ing › the requirements of §‹ 9 ›«*15*». On vessels of less than 20 meters in length, the sidelights, if necessary to meet the requirements of §‹ 9 ›«*15*», shall be fitted with ‹ inboard › matt black «*inboard*» screens. With a combined lantern, using a single vertical filament and a very narrow division between the green and red sections, external screens need not be fitted.

(309)

Annex I (Inland)

(b) On power-driven vessels less than 12 meters in length constructed after July 31, 1983, the masthead light, or the all-round light described in Rule 23(d) shall be screened to prevent direct illumination of the vessel forward of the operator's position.

(310)

Shapes

(311) (a) Shapes shall be black and of the following sizes: (i) A ball shall have a diameter of not less than 0.6 meter; (ii) a cone shall have a base diameter of not less than 0.6 meter‹s› and a height equal to its diameter; ‹ (iii) a cylinder shall have a diameter of at least 0.6 meter and a height of twice its diameter; › (iv)╪(iii) a diamond shape shall consist of two cones as defined in §(a)(ii) having a common base.

(312) (b) The vertical distance between shapes shall be at least 1.5 meter ‹ s ›.

(313) (c) In a vessel of less than 20 meters in length shapes of lesser dimensions but commensurate with the size of the vessel may be used and the distance apart may be correspondingly reduced.

(314)

Color Specification of Lights

(315) (a) The chromaticity of all navigation lights shall conform to the following standards, which lie within the boundaries of the area of the diagram specified for each color by the International Commission on Illumination (CIE). ‹ , in the "Colors of Light Signals", which is incorporated by reference. It is Publication CIE No. 2.2. (TC-1.6), 1975, and is available from the Illumination Engineering Society, 345 East 47th Street, New York, NY 10017 and is available for inspection at the Coast Guard, Shore Infrastructure Logistics Center, Aids to Navigation and Marine Environmental Response Product Line (CG-SILC-ATON/MER), 2703 Martin Luther King, Jr. Ave SE, Mailstop 7714, Washington, DC 20593-7714. It is

also available for inspection at the National Archives and Records Administration (NARA). For information on the availability of this material at NARA, call 202–741–6030, or go to: *http://www.archives.gov/federal_register/code_of_federal_regulations/ibr_locations.html*. This incorporation by reference was approved by the Director of the Federal Register. ›

(316) (b) The boundaries of the area for each color are given by indicating the corner coordinates, which are as follows:

(317)

(i) White						
x	0.525	0.525	0.452	0.310	0.310	0.443
y	0.382	0.440	0.440	0.348	0.283	0.382
(ii) Green						
x	0.028	0.009	0.300	0.203		
y	0.385	0.723	0.511	0.356		
(iii) Red						
x	0.680	0.660	0.735	0.721		
y	0.320	0.320	0.265	0.259		
(iv) Yellow						
x	0.612	0.618	0.575	0.575		
y	0.382	0.382	0.425	0.406		

(318)

Intensity of Lights

(319) (a) The minimum luminous intensity of lights shall be calculated by using the formula:

(320)

$I = 3.43 \times 10^6 \times T \times D^2 \times K^{-D}$
I is luminous intensity in candelas under service conditions.
T is threshold factor 2×10^{-7} lux.
D is range of visibility (luminous range) of the light in nautical miles.
K is atmospheric transmissivity. For prescribed lights the value of K shall be 0.8, corresponding to a meteorological visibility of approximately 13 miles.

(321) (b) A selection of figures derived from the formula is given in the following table:

(322)

Range of visibility (luminous range) of light in nautical miles D	Minimum luminous intensity of light in candelas for K = 0.8 I
1	0.9
2	4.3
3	12
4	27
5	52
6	94

(323) ‹ Note: The maximum luminous intensity of navigation lights should be limited to avoid undue glare. This shall not be achieved by a variable control of the luminous intensity. ›

(324)

Horizontal Sectors

(325) (a)(i) In the forward direction, sidelights as fitted on the vessel shall show the minimum required intensities. The intensities shall decrease to reach practical cut-off between 1 and one degrees outside the prescribed sectors.

(326) (ii) For sternlights and masthead lights and at 22.5 degrees abaft the beam for sidelights, the minimum required intensities shall be maintained over the arc of the horizon up to 5 degrees within the limits of the sectors prescribed in Rule 21. From 5 degrees within the prescribed sectors the intensity may decrease by 50 percent up to the prescribed limits; it shall decrease steadily to reach practical cut-off at not more than 5 degrees outside the prescribed sectors.

(327) (b)(i) All-round lights shall be so located as not to be obscured by masts, topmasts or structures within angular sectors of more than 6 degrees, except anchor lights prescribed in Rule 30, which need not be placed at an impracticable height above the hull « , *and the all-round white light described in Rule 23(e), which may not be obscured at all*».

(328) (ii) If it is impracticable to comply with §(b)(i) by exhibiting only one all-round light, two all-round lights shall be used suitably positioned or screened so that they «*to*» appear, as far as practicable, as one light at a «*minimum*» distance of 1 «*nautical*» mile.

(329) «*Note: Two unscreened all-round lights that are 1.28 meters apart or less will appear as one light to the naked eye at a distance of 1 nautical mile.*»

(330)

Vertical Sectors

(331) (a) The vertical sectors of electric lights as fitted, with the exception of lights on sailing vessels underway «*and on unmanned barges*», shall ensure that: (i) At least the required minimum intensity is maintained at all angles from 5 degrees above to 5 degrees below the horizontal; (ii) at least 60 percent of the required minimum intensity is maintained from 7.5 degrees above to 7.5 degrees below the horizontal.

(332) (b) In the case of sailing vessels underway the vertical sectors of electric lights as fitted shall ensure that: (i) At least the required minimum intensity is maintained at all angles from 5 degrees above to 5 degrees below the horizontal; (ii) at least 50 percent of the required minimum intensity is maintained from 25 degrees above to 25 degrees below the horizontal.

(333)

Annex I (Inland)
(c) In the case of unmanned barges the minimum required intensity of electric lights as fitted shall be maintained on the horizontal.

(334) (c)╪(d) In the case of lights other than electric lights these specifications shall be met as closely as possible.

(335)

Intensity of Non-electric Lights

(336) Non-electric lights shall so far as practicable comply with the minimum intensities, as specified in the «*Intensity of Lights*» Table.

(337)

Maneuvering Light

(338) «*Notwithstanding the provisions of §2(f)*», the maneuvering light described in Rule 34(b) shall be placed «*approximately*» in the same fore and aft vertical plane as the masthead light or lights and, where practicable, at a minimum height of ‹ 2 ›«*1.5*» meter vertically above the forward masthead light, provided that it shall be carried not less than ‹ 2 ›«*1.5*» meter vertically above or below the after masthead light. On a vessel where only one masthead light is carried, the maneuvering light, if fitted, shall be carried where it can best be seen, not less than ‹ 2 ›«*1.5*» meters vertically apart from the masthead light.

(339)

High-speed Craft

(340) (a) The masthead light of high-speed craft may be placed at a height related to the breadth «*of the craft*» lower than that prescribed in §2(a)(i), provided that the base angle of the isosceles triangle formed by the sidelights and masthead light when seen in end elevation is not less than 27 degrees.

(341) (b) On high-speed craft of 50 meters or more in length, the vertical separation between foremast and mainmast light of 4.5 meters required by §‹ 2(a)(ii) ›«*2(k)*» may be modified provided that such distance shall not be less than the value determined by the following formula:

(342)

$$y = \frac{y = (a+17\Psi)\ C}{1000} + 2$$

y the height of the mainmast light above the foremast light in meters.

a is the height of the foremast light above the water surface in service condition in meters

Y is the trim in service condition in degrees.

C is the horizontal separation of masthead lights in meters.

Note: Refer to the International Code of Safety for High-Speed Craft, 1994 and the International Code of Safety for High-Speed Craft, 2000.

(343)

Approval

(344) The construction of lights and shapes and the installation of lights on board the vessel ‹ shall be to the satisfaction of the appropriate authority of the State whose flag the vessel is entitled to fly ›«*must satisfy the Commandant, U. S. Coast Guard*».

(345)

Annex II—Additional Signals for Fishing Vessels Fishing in Close Proximity

(346) See Rule 26(f).

(347)

Annex III—Technical Details of Sound Signal Appliances

(348) (a) Frequencies and range of audibility. The fundamental frequency of the signal shall lie within the range 70-700 Hz. The range of audibility of the signal from a whistle shall be determined by those frequencies, which may include the fundamental and/or one or more higher frequencies, which lie within the range 180-700 Hz (+/- 1 percent) for a vessel of 20 meters or more in length, or 180-2100 Hz (+/- 1 percent) for a vessel of less than 20 meters in length and which provide the sound pressure levels specified in §1(c).

(349) (b) Limits of fundamental frequencies. To ensure a wide variety of whistle characteristics, the fundamental frequency of a whistle shall be between the following limits: (i) 70-200 Hz, for a vessel 200 meters or more in length; (ii) 130-350 Hz, for a vessel 75 meters but less than 200 meters in length; (iii) 250-700 Hz, for a vessel less than 75 meters in length.

(350) (c) Sound signal intensity and range of audibility. A whistle fitted in a vessel shall provide, in the direction of maximum intensity of the whistle and at a distance of 1 meter from it, a sound pressure level in at least one one-third octave band within the range of frequencies 180-700 Hz (+/- 1 percent) for a vessel of 20 meters ‹ or more in length, or 180-2100 Hz (+/- 1 percent) for a vessel of less than 20 meters in length ›, of not less than the appropriate figure given in the table below.

(351)

Length of vessel in meters	One-third octave band level at 1 meter in dB referred to 2 x 10^{-5}N/m^2	Audible range in nautical miles
200 or more	143	2
75 but less than 200	138	1.5
20 but less than 75	130	1
Less than 20	120* 115** 111***	0.5

* When the measured frequencies lie within the range 180-450 Hz
** When the measured frequencies lie within the range 450-800 Hz
*** When the measured frequencies lie within the range 800-2100 Hz

(352) The range of audibility in the table is for information and is approximately the range at which a whistle may be heard on its forward axis with 90 percent probability in conditions of still air on board a vessel having average background noise level at the listening posts (taken to be 68 dB in the octave band centered on 250 Hz and 63 dB in the octave band centered on 500 Hz). «*It is shown for informational purposes only.*» In practice, the range

at which a whistle may be heard is extremely variable and depends critically on weather conditions; the values given can be regarded as typical but under conditions of strong wind or high ambient noise level at the listening post the range may be reduced.

(353) (d) Directional properties. The sound pressure level of a directional whistle shall be not more than 4 dB below the ‹ prescribed › sound pressure level ‹ on the axis at ›, «*specified in §(c)*» any direction in the horizontal plane within +/- 45 degrees of the axis. The sound pressure level at «*of the whistle in*» any other direction in the horizontal plane shall be not more than 10 dB ‹ below the prescribed ›«*less than the*» sound pressure level ‹ on the ›«*specified for the forward*» axis, so that the range «*audibility*» in any direction will be at least half the range «*required*» on the forward axis. The sound pressure level shall be measured in that one-third octave band which determines the audibility range.

(354) (e) Positioning of whistles.

(355) (i) When a directional whistle is to be used as the only whistle on ‹ a vessel, it shall be installed with its maximum intensity directed straight ahead ›«*the vessel and is permanently installed, it shall be installed with its forward axis directed forward*».

(356) (ii) A whistle shall be placed as high as practicable on a vessel, in order to reduce interception of the emitted sound by obstructions and also to minimize hearing damage risk to personnel. The sound pressure level of the vessel's own signal at listening posts shall not exceed 110 dB(A) and so far as practicable should not exceed 100 dB(A).

(357) (f) Fitting of more than one whistle. If whistles are fitted at a distance apart of more than 100 meters, ‹ it shall be so arranged that they are ›«*they shall*» not «*be*» sounded simultaneously.

(358)

Annex IIIg (International)

(g) Combined whistle systems.

If due to the presence of obstructions the sound field of a single whistle or of one of the whistles referred to in §(f) is likely to have a zone of greatly reduced signal level, it is recommended that a combined whistle system be fitted so as to overcome this reduction. The whistles of a combined system shall be located at a distance apart of not more than 100 meters and arranged to be sounded simultaneously. The frequency of any one whistle shall differ from those of the others by at least 10 Hz.

Annex IIIg (Inland)

(g) Combined whistle systems.

(i) A combined whistle system is a number of whistles (sound emitting sources) operated together. For the purposes of the Rules a combined whistle system is to be regarded as a single whistle.

(ii) The whistles of a combined system shall:

(1) Be located at a distance apart of not more than 100 meters;

(2) Be sounded simultaneously;

(3) Each have a fundamental frequency different from those of the others by at least 10 Hz; and

(4) Have a tonal characteristic appropriate for the length of vessel which shall be evidenced by at least 2-thirds of the whistles in the combined system having fundamental frequencies falling within the limits prescribed in §(b) of this section, or if there are only two whistles in the combined system, by the higher fundamental frequency falling within the limits prescribed in paragraph (b) of this section.

Note: If, due to the presence of obstructions, the sound field of a single whistle or of one of the whistles referred to in §(f) of this section is likely to have a zone of greatly reduced signal level, a combined whistle system should be fitted so as to overcome this reduction.

(359) For the purposes of the Rules a combined whistle system is to be regarded as a single whistle. ‹ (ii) › The whistles of a combined system shall:

(360) (1) Be located at a distance apart of not more than 100 meters;

(361)

Annex III(h) (Inland)

(h) Towing vessel whistles

A power-driven vessel normally engaged in pushing ahead or towing alongside may, at all times, use a whistle whose characteristic falls within the limits prescribed by §1(b) for the longest customary composite length of the vessel and its tow.

(362)

Bell or Gong

(363) (a) Intensity of signal. A bell or gong, or other device having similar sound characteristics shall produce a sound pressure level of not less than 110 dB at ‹ a distance of › 1 meter ‹ from it ›.

(364) (b) Construction. Bells and gongs shall be made of corrosion-resistant material and designed to give clear tone. The diameter of the mouth of the bell shall be not less than 300 mm for vessels of 20 meters or more in length. Where practicable, a power-driven bell striker is recommended to ensure constant force but manual operation shall be possible. The mass of the striker shall be not less than 3 percent of the mass of the bell.

(365)

Approval

(366)

Annex III (International)

The construction of sound signal appliances, their performance and their installation on board the vessel shall be to the satisfaction of the appropriate authority of the State whose flag the vessel is entitled to fly.

(367)

Annex IV—Distress Signals

(368)

«*Need of Assistance*»

(369) The following signals, used or exhibited either together or separately, indicate distress and need of assistance:

(370) (a) a gun or other explosive signal fired at intervals of about a minute;

(371) (b) a continuous sounding with any fog-signaling apparatus;

(372) (c) rockets or shells, throwing red stars fired one at a time at short intervals;

(373) (d) a signal made by any signaling method consisting of the group . . . - - - . . . (SOS) in the Morse Code;

(374) (e) a signal sent by radiotelephony consisting of the spoken word "Mayday";

(375) (f) the International Code Signal of distress indicated by N.C.;

(376) (g) a signal consisting of a square flag having above or below it a ball or anything resembling a ball;

(377) (h) flames on the vessel (as from a burning tar barrel, oil barrel, etc.);

(378) (i) a rocket parachute flare or a hand flare showing a red light;

(379) (j) a smoke signal giving off orange-colored smoke;

(380) (k) slowly and repeatedly raising and lowering arms outstretched to each side;

(381) (l) a distress alert by means of digital selective calling (DSC) transmitted on: (i) VHF channel 70, or (ii) MF/HF on the frequencies 2187.5 kHz, 8414.5 kHz, 4207.5 kHz, 6312 kHz, 12577 kHz or 16804.5 kHz;

(382) (m) a ship-to-shore distress alert transmitted by the ship's Inmarsat or other mobile satellite service provider ship earth station;

(383) (n) signals transmitted by emergency position-indicating radio beacons;

(384) (o) approved signals transmitted by radiocommunication systems, including survival craft radar transponders «*meeting the requirements of 47 CFR 80.109*».

(385) «*(p) A high intensity white light flashing at regular intervals from 50 to 70 times per minute.*»

(386)

«*Exclusive Use*»

(387) The use or exhibition of any of the foregoing signals except for the purpose of indicating distress and need of assistance and the use of other signals which may be confused with any of the above signals is prohibited.

(388)

«*Supplemental Signals*»

(389) Attention is drawn to the relevant sections of the International Code of Signals, the International Aeronautical and Maritime Search and Rescue Manual, Volume III, ‹ the International Telecommunication Union Radio Regulations, › and the following signals:

(390) (a) A piece of orange-colored canvas with either a black square and circle or other appropriate symbol (for identification from the air);

(391) (b) A dye marker.

(392)

Annex V—Pilot Rules

(393)

§88.01 Purpose and applicability.

(394) This part applies to all vessels operating on United States inland waters and to United States vessels operating on the Canadian waters of the Great Lakes to the extent there is no conflict with Canadian law.

(395)

§88.03 Definitions.

(396) The terms used in this part have the same meaning as the terms defined in part 83 of this subchapter.

(397)

§88.05 Law enforcement vessels.

(398) (a) Law enforcement vessels may display a flashing blue light when engaged in direct law enforcement or public safety activities. This light must be located so that it does not interfere with the visibility of the vessel's navigation lights.

(399) (b) The blue light described in this section may be displayed by law enforcement vessels of the United States and the States and their political subdivisions.

(400)

§88.07 Public safety activities.

(401) (a) Vessels engaged in government sanctioned public safety activities, and commercial vessels performing similar functions, may display an alternately flashing red and yellow light signal. This identification light signal must be located so that it does not interfere with the visibility of the vessel's navigation lights. The identification light signal may be used only as an identification signal and conveys no special privilege. Vessels using the identification light signal during public safety activities must abide by the Inland Navigation Rules, and must not presume that the light or the exigency gives them precedence or right of way.

(402) (b) Public safety activities include but are not limited to patrolling marine parades, regattas, or special water celebrations; traffic control; salvage; firefighting; medical assistance; assisting disabled vessels; and search and rescue.

(403)

Implementing Rules

(404) **Alternative Compliance**—see **33 CFR 81** and **33 CFR 89**, chapter 2, for regulations.

(405) **Vessel Bridge-to-Bridge Radiotelephone Regulations**—see **33 CFR 26**, chapter 2, for regulations.

Appendix A

(1)

Sales Information

(2) NOAA publications, nautical charts and unclassified National Geospatial-Intelligence Agency (NGA) nautical charts are sold by authorized sales agents in many U.S. ports and in some foreign ports. Information on obtaining charting products and a listing of authorized agents can be found at *www.nauticalcharts.noaa.gov*.

(3)

Products and Services–NOAA

(4)

Reporting corrections to Nautical Charts and Coast Pilots

(5) Users are requested to report all significant discrepancies or additions to NOAA navigational products, including depth information in privately maintained channels and basins; obstructions, wrecks, and other dangers; new, relocated, or demolished landmarks; uncharted fixed private aids to navigation; deletions or additions of small-craft facilities and any other information pertinent to safe navigation. This information may be submitted using the NOAA Office of Coast Survey website: https://www.nauticalcharts.noaa.gov/customer-service/assist/

(6)

Department of Commerce, NOAA
Nautical Data Branch
N/CS26, Station 7505
1315 East-West Highway
Silver Spring, Maryland 20910
ocs.ndb@noaa.gov

(7)

Nautical Charts

(8) NOAA maintains the nautical charts and publications for the coast of the United States and the Great Lakes. Over a thousand charts cover 95,000 miles of shoreline and 3.4 million square nautical miles of water. Access to charts, publications and chart catalogs is available through *www.nauticalcharts.noaa.gov*.

(9)

Dates of Latest Editions

(10) Information concerning the dates of latest editions for the full suite of NOAA's nautical charts and U.S. Coast Pilot volumes can be found at *www.nauticalcharts.noaa.gov/mcd/dole.htm*.

(11)

U.S. Coast Pilots

(12)

U.S. Coast Pilot 1—Atlantic Coast: Eastport to Cape Cod
U.S. Coast Pilot 2—Atlantic Coast: Cape Cod to Sandy Hook
U.S. Coast Pilot 3—Atlantic Coast: Sandy Hook to Cape Henry
U.S. Coast Pilot 4—Atlantic Coast: Cape Henry to Key West
U.S. Coast Pilot 5—Gulf Coast, Puerto Rico and Virgin Islands
U.S. Coast Pilot 6—Great Lakes: Huron, Ontario, Michigan, Erie, Superior, and St. Lawrence River
U.S. Coast Pilot 7—Pacifc Coast: California
U.S. Coast Pilot 8—Alaska: Dixon Entrance to Cape Spencer
U.S. Coast Pilot 9—Alaska: Cape Spencer to Beaufort Sea
U.S. Coast Pilot 10—Pacifc Coast: Oregon, Washington, Hawaii, and Pacifc Islands

(13)

Distance Tables

(14) Distances Between United States Ports is available at *https://nauticalcharts.noaa.gov/publications/docs/distances.pdf*

(15)

Center for Operational Oceanographic Products and Services

(16)

1305 East-West Highway
Silver Spring, Maryland 20910
301–713–2815 (phone)
301–713–4500 (fax)
www.tidesandcurrents.noaa.gov

(17)

National Weather Service Offices

(18) The following offices provide forecasts, current conditions, local information and climatological data. This data can be accessed through the websites listed after each office below.

(19)

Maine

NWS Forecast O f ce Gray/Portland – *www.weather.gov/gyx*
P.O. Box 1208, 1 Weather Lane, Gray, ME 04039

Massachusetts

NWS Forecast O f ce Boston/Taunton – *www.weather.gov/box*
445 Myles Standish Boulevard, Taunton, MA 02780

(20)

NOAA Weather Radio

(21) National Weather Service VHF-FM radio stations provide mariners with continuous FM broadcasts of

weather warnings, forecasts, radar reports, and selected weather observations. Reception range is typically 20 to 40 nautical miles from the antenna site, but can be as much as 100 nautical miles depending on elevation, terrain, type of receiver, and antenna used. The following VHF-FM radio stations are located in or near the area covered by this Coast Pilot.

(22)

Call Sign	Station	Location	Frequency
WNG-543	Jonesboro, ME	44°40′N., 67°35′W.	162.450
KEC-93	Ellsworth, ME	44°32′N., 68°30′W.	162.40
WSM-60	Dresden, ME	44°07′N., 69°40′W.	162.475
KDO-95	Portland, ME	43°45′N., 70°19′W.	162.55
WNG-574	Gloucester, MA	42°37′N., 70°40′W.	162.425
KHB-35	Boston, MA	42°12′N., 71°06′W.	162.475
KEC-73	Hyannis, MA	41°41′N., 70°20′W.	162.55

(23) The National Weather Service provides **Radio Facsimile Weather Information** for east coast waters through the Coast Guard Communications Station Boston (NMF). Broadcasts are made on the following frequencies: 4235 (02z, 08z), 6340.5, 9110, 12750 (14z) kHz. For carrier frequency, subtract 1.9 kHz. Fax schedules are transmitted at 0243 and 1405 GMT and provide area coverage and descriptions of services. For further information on Marine Radiofax Charts, visit: *https://www.weather.gov/marine/radiofax_charts*

(24)

Marine Weather Forecasts

(25) Coastal Marine Forecasts are issued four times daily by National Weather Service Offices. For further information on coastal marine forecasts as well as additional types of forecasts, visit: *https://weather.gov/marine/forecast* -and- *https://nowcoast.noaa.gov/*

(26)

Space Weather Prediction Center (SWPC)

(27) The Space Weather Prediction Center provides real-time monitoring and forecasting of solar and geophysical events that impact satellites, power grids, communications, navigation and many other technological systems.

(28)

NOAA, National Weather Service
National Centers for Environmental Predictions
Space Weather Prediction Center, W/NP9
325 Broadway
Boulder, Colorado 80305
www.swpc.noaa.gov

(29)

National Weather Service Port Meteorological Officers (PMOs)

(30) **Port Meteorological Officers** provide assistance on matters of weather chart interpretation, instruments, marine weather communications, and requirements affecting ship operations. (See **National Weather Service**, chapter 1, for further details.) PMO offices in the area covered by this Coast Pilot are as follows:

(31) **New York City**–NOAA/NWS P.O. Box 366, Flemington, NJ 08822 (908–216–3477).

(32)

Products and Services–Other U.S. Government Agencies

(33) A partial list of publications and charts considered of navigational value is included for the ready reference of the mariner. In addition to the agents located in the principal seaports handling publication sales, certain libraries have been designated by the Congress of the United States to receive the publications as issued for public review.

(34)

Government Publishing Office

(35)

U.S. Government Publishing O f ce
710 North Capitol Street, NW
Washington, DC 20401-0001
202-512-1800
866-512-1800
www.gpo.gov/
ContactCenter@gpo.gov

(36)

Hydrographic Surveys

(37) U.S. Army Corps of Engineers hydrographic survey activity is available at: *https://www.mvr.usace.army.mil/Missions/Navigation/Hydrographic-Surveys/HydrographicSurveysMap/*

(38)

Nautical Charts

(39) **Apalachicola, Chattahoochee and Flint Rivers Navigation Charts**, **Alabama River Charts** and **Black Warrior-Tombigbee Rivers River Charts**—available from the U.S. Army Corps of Engineers Mobile District for purchase in bound hard copy or as a free download in PDF at *www.sam.usace.army.mil*.

(40) **Flood Control and Navigation Maps of the Mississippi River, Cairo, Illinois to the Gulf of America**—available from the U.S. Army Corps of Engineers Memphis District as a free download in PDF at *www.mvm.usace.army.mil*.

(41) **Upper Mississippi River Navigation Charts (Mississippi River, Cairo, Illinois to Minneapolis, Minnesota) and Charts of the Illinois Waterway, from Mississippi River at Grafton, Illinois to Lake Michigan at Chicago and Calumet Harbors**—available from the U.S. Army Corps of Engineers Rock Island District for purchase in hard copy format or as a free download in PDF at *www.mvr.usace.army.mil*.

(42)

Publications and Services

(43) **Local Notice to Mariners** are posted weekly by the U.S. Coast Guard Navigation Center at *www.navcen.uscg.gov*. The National Geospatial-Intelligence Agency, U.S. Notice to Mariners are available at *msi.nga.mil/NGAPortal/MSI.portal.*

(44) **Special Notice to Mariners** are issued annually in National Geospatial-Intelligence Agency Notice to Mariners 1. These notices contain important information of considerable interest to all mariners. Interested parties are advised to read these notices.

(45) **Light List**—maintained by the United States Coast Guard and available online at *www.navcen.uscg.gov*. Also see Light List, Chapter 1, for additional information.

(46) **List of Lights**, **Sailing Directions**, **Radio Navigational Aids (Pub. 117)**, **American Practical Navigator (Pub. 9)** and **International Code of Signals (Pub. 102)**—issued by the National Geospatial-Intelligence Agency and available at *msi.nga.mil/NGAPortal/MSI.portal*.

(47) **The Nautical Almanac**, the **Air Almanac** and **Astronomical Almanac**—available through the United States Naval Observatory—*https://www.public.navy.mil/fltfor/cnmoc/Pages/usno_test_page.aspx* -and- *https://bookstore.gpo.gov/agency/united-states-naval-observatory-usno*

(48) **Dissemination of Marine Weather Information**, maintained by National Weather Service on the internet at *https://www.weather.gov/marine/nws_dissemination* -and- **NWS Marine Weather Services** at *https://www.weather.gov/marine/*

(49) **Navigation Rules and Regulations Handbook**—publication produced by the United States Coast Guard Navigation Standards Branch, which contains International and Inland Rules of the Road and Navigation Regulations. Available for download or viewing at *www.navcen.uscg.gov.* Navigation Rules are also found near the end of each individual Coast Pilot volume.

(50)

(51)

Offices and Services–Other U.S. Government Agencies

(52)

U.S. Army Corps of Engineers (USACE) Offices

(53)

District/Division Office	Contact Information
New England District O f ce 696 Virginia Road Concord, MA 01742	*www.nae.usace.army.mil* 978–318–8238

(54)

Environmental Protection Agency (EPA) Offices

(55)

Regional Areas, States and Information
Region 1 New Hampshire, Vermont, Maine, Massachusetts, Connecticut, Rhode Island *www.epa.gov/aboutepa/epa-region-1-new-england*
Region 2 New Jersey, New York, Puerto Rico, Virgin Islands *www.epa.gov/aboutepa/epa-region-2*
Region 3 Delaware, Maryland, Virginia, District of Columbia, Pennsylvania *www.epa.gov/aboutepa/epa-region-3-mid-atlantic*
Region 4 Alabama, Florida, Georgia, Mississippi, South Carolina, North Carolina *https://www.epa.gov/aboutepa/about-epa-region-4-southeast*
Region 5 Illinois, Indiana, Michigan, Minnesota, Ohio, Wisconsin *www.epa.gov/aboutepa/epa-region-5*
Region 6 Louisiana, Texas *www.epa.gov/aboutepa/epa-region-6-south-central*
Region 9 California, Hawaii, Guam *www.epa.gov/aboutepa/epa-region-9-pacific-southwest*
Region 10 Alaska, Oregon, Washington *www.epa.gov/aboutepa/epa-region-10-pacific-northwest*

(56)

(57)

U.S. Coast Guard Navigation Center (NAVCEN)

(58) The Coast Guard Navigation Center provides cutting-edge services for safe, secure and efficient maritime transportation. The center operates the Navigation Information Service (NIS), the Maritime Differential GPS (DGPS) and the developing Nationwide Differential Global Positioning System (NDGPS). In addition, NAVCEN serves as the civilian interface for the Global Positioning System and manages other navigation-related projects.

(59) For further information and/or operational questions regarding GPS and DGPS, visit *navcen,uscg.gov* or contact:

(60)

Commanding O f cer
U.S. Coast Guard Navigation Center
NAVCEN MS 7310
7323 Telegraph Road
Alexandria, VA 20598-7310

(61)

Coast Guard District Offices

(62)

Districts, Boundary Description and Contact Information
First Coast Guard District Maine, New Hampshire, Vermont, Massachusetts, Rhode Island, Connecticut, New York except that part north of latitude 42°N and west of longitude 74°39'W; that part of New Jersey, north of latitude 40°18'N, east of longitude 74°30.5'W, and northeast of a line from 40°18'N., 74°30.5'W., north-northwesterly to the New York, New Jersey and Pennsylvania boundaries at Tristate; all U.S. Naval reservations on shore at Newfoundland; the ocean area encompassed by the Search and Rescue boundary between Canada and the United States easterly to longitude 63°W.; thence due south to latitude 41°N; thence southwesterly along a line bearing 219°T to the point of intersection at 37°N., 67°13'W., with a line bearing 122°T from the New Jersey shoreline at latitude 40°18'N, (just south of the Shrewsbury River); thence northwesterly along this line to the coast. 408 Atlantic Avenue Boston, MA 02110-3350 617–223–8356

(63)

Coast Guard Sector Offices

(64) Note: A Sector Office combines the functions of the Captain of the Port and Marine Inspection Office.

(65)

Sectors	Contact Information
Sector Northern New England	259 High Street South Portland, ME 04106 207–767–0320
Sector Boston	427 Commercial Street Boston, MA 02109 617–223–3123
Sector Southeastern New England	1 Little Harbor Road Woods Hole, MA 02543 866–819–9128

(66)

Coast Guard Stations

(67) The stations listed are in the area covered by this Coast Pilot. They have search and rescue capabilities and may provide lookout, communication and/or patrol functions to assist vessels in distress. The National VHF-FM Distress System provides continuous coastal radio coverage outwards to 20 miles on channel 16. After contact on channel 16, communications with the Coast Guard should be on channel 1022 (previously channel 22A). If channel 1022 is not available to the mariner, communications may be made on channel 12. Selected stations guard the International Radiotelephone Distress, Safety and Calling Frequencies.

(68)

Maine	
Station Eastport	On west side of Friar Roads, north of Eastport Breakwater.
Station Jonesport	Near north end of bridge over Moosabec Reach.
Station Southwest Harbor	At the southerly end of Clark Point.
Station Rockland	On west side of Rockland Harbor.
Station Boothbay Harbor	About 100 yards southwest of the northeast tip of McKown Point.
Station South Portland	In South Portland on the south bank of the Fore River.
New Hampshire	
Station Portsmouth Harbor	On New Castle Island, at Portsmouth Harbor Light.
Massachusetts	
Station Merrimack River	On the south bank of the Merrimack River west of the American Yacht Club.
Station Gloucester	East side of Harbor Cove at Gloucester.
Station Boston	In Boston Harbor, on the south bank of Charles River at the mouth.
Station Point Allerton	About 0.4 mile east of Windmill Point on Hull Bay.
Station Scituate	On the southwestern shore of Scituate Harbor.
Station Cape Cod Canal	East entrance to the canal, near Sandwich, MA
Station Provincetown	On southwest side of harbor, about 0.4 mile southwest of town pier.

(69)

Coast Guard Radio Broadcasts

(70) Urgent, safety and scheduled marine information broadcasts are made by Coast Guard stations. In general, these broadcasts provide information vital to vessels operating in the approaches and coastal waters of the United States including the Great Lakes, Puerto Rico and U.S. Virgin Islands. Types of broadcasts are as follows:

(71) **Scheduled broadcasts**—U.S. Coast Guard stations make scheduled broadcasts on a pre-published schedule in intervals of 12 hours. After the preliminary announcements on VHF-FM channel 16, the station advises shifting to working frequency VHF-FM channel 1022 (previously channel 22A). Scheduled broadcasts are made on the following stations and times (UTC):

(72)

Coast Guard Radio Station	Scheduled Broadcast Times (UTC)
Sector Northern New England Portland, ME (NMF-31)	1105 and 2305
Sector Boston (NMF, NIK, NMF-7)	1035 and 2235
Sector Southeastern New England Woods Hole, MA (NMF-2, NMF-3)	1300 and 2300

(73) **Safety broadcasts**—U.S. Coast Guard stations that make scheduled broadcasts issue safety broadcasts upon receipt and on the next scheduled broadcast. Safety broadcasts are preceded by the safety signal SECURITY. After the preliminary signal on VHF-FM channel 16, the station may announce shifting to working frequency: VHF-FM channel 1022 (previously channel 22A).

(74) **Urgent broadcasts**—U.S. Coast Guard stations that make scheduled broadcasts issue urgent broadcasts upon receipt and on schedule until canceled. Urgent broadcasts are preceded by the urgent signal, PAN-PAN. Both the

urgent signal and message may be transmitted on VHF-FM channel 16.

(75)

U.S. NAVTEX Transmitting Stations

(76) NAVTEX is an international automated medium frequency direct-printing service informing mariners of navigational and meteorological warnings and forecasts, as well as urgent marine safety information. Coverage is reasonably continuous to 200 NM off the U.S. East, Gulf and West Coasts; Puerto Rico; Southwest Alaska; Hawaii; and 100 NM off Guam. U.S. Coast Guard NAVTEX broadcast stations and message content for the areas covered by this Coast Pilot are shown below.

(77)

Station	ID	WX Broadcast Schedule (UTC)
Boston (NMF)	F	0050, 0450, 0850, 1250, 1650, 2050
Chesapeake (NMN)	N	0210, 0610, 1010, 1410, 1810, 2210
Charleston (NMN)	E	0040, 0440, 0840, 1240, 1640, 2040
Miami (NMA)	A	0000, 0400, 0800, 1200, 1600, 2000
San Juan (NMR)	R	0250, 0650, 1050, 1450, 1850, 2250
New Orleans (NMG)	G	0100, 0500, 0900, 1300, 1700, 2100

(78)

Customs Ports of Entry and Stations

(79) Vessels arriving in the United States from a foreign port or place are required to report their arrival to Customs and Border Protection immediately. Field Operations Offices and contact information is listed below.

(80)

Field Operations Office	Contact Information
Boston	10 Causeway Street Room 801 Boston, MA 02222 617–565–6208

(81)

Public Health Service Quarantine Stations

(82)

Quarantine Stations and Addresses
c/o CDC New York Quarantine Station JFK International Airport Terminal 4, Room 219.016 2nd Floor, East Concourse Jamaica, NY 11430-1081 617–561–5701 (24 hours)

(83) At other ports, quarantine and/or medical examinations are usually performed by Public Health Service contract personnel or by quarantine inspectors from the nearest quarantine station. Inquiries concerning quarantine matters should be directed to the nearest quarantine station.

(84)

Food and Drug Administration (FDA) Regional Offices

(85)

Northeast Region	158-15 Liberty Avenue Jamaica, New York 11433 718–340–7000
Central Region	20 North Michigan Avenue Suite 510 Chicago, Illinois 60602 215–597–4390
Pacifc Region	1301 Clay Street Room 1180N Oakland, California 94612 510–287–2700
Southeast Region	60 Eighth Street NE Atlanta, Georgia 30309 404–253–1171
Southwest Region	4040 North Central Expressway Suite 900 Dallas, Texas 75204 214–253–4901

(86)

Department of Agriculture, Animal and Plant Health Inspection Service (APHIS)

(87) Information on the importation of plants, animals, and plant and animal products is available from APHIS, Department of Agriculture, 4700 River Road, Riverdale, MD 20737. Visit *aphis.usda.gov* for more information.

(88)

USDA Animal and Plant Inspection Service Animal Import Centers:
Los Angeles Animal Import Center (LAAIC) 222 Kansas Street El Segundo, CA 90245 310-955-3311
Miami Animal Import Center (MAIC) 6300 NW 36th Street Miami, FL 33122 305-876-2200
New York Animal Import Center (NYAIC) 474 Animal Import Center Newburg, NY 12550 845-838-5500
John F. Kennedy Airport Office 230-59 Rockaway Blvd. Suite 100, Room 101 Jamaica, NY 11413 718-553-3570
Agriculture Select Service Agents 4700 River Road, Unit 2 Riverdale, MD 20737 AgSAS@aphis.usda.gov 301-851-3300 (select option 3)

(89)

U.S. Citizenship and Immigration Service Offices

(90)

Maine	
Portland Field Office	176 Gannett Drive South Portland, ME 04106
Massachusetts	

Boston Field O f ce	John F. Kennedy Federal Building Government Center Boston, MA 02203

(91)

Federal Communications Commission Office

(92) **District Field Office**:

(93) Boston MA: 1 Batterymarch Park, Quincy, MA 02169-7495.

(94) Telephone toll-free: 888–225–5322; (888–CALL–FCC) to report radio communications interference issues.

(95)

Radio shore stations providing medical advice

(96) To obtain radio medical advice by reliable voice radio communications urgent calls for assistance may be broadcast using the normal Urgency prowords PAN PAN as follows:

(97)

Broadcast Language	Details
PAN-PAN	(3 times)
All Stations	(3 times or specifc station if known)
This is ship name	(3 times)
Call sign	(call sign)
In Position	(give position)
I require medical advice	
Over	

(98)

Measured Courses

(99) The positions of measured courses are shown on the charts and their descriptions are included in the Coast Pilots when information is reported to NOAA. Courses are located in the following places covered by this Coast Pilot:

(100) Sheepscot River, west side of Barters Island.

(101) Casco Bay, west side of Whaleboat Island.

(102) The pages in the text describing the courses can be obtained by referring to the index for the geographic places; chart numbers follow the names.

(103)

Canadian Government Agencies

(104)

Canadian Hydrographic Service

(105)

Fisheries and Oceans Canada
Communications Branch
200 Kent Street, 13th Floor, Station 13E228
Ottawa, Ontario K1A 0E6 Canada
1–800–465–7735
1–613–993–0999

(106)

Canadian Border Services Agency

(107)

Canadian Border Services Agency
Ottawa, Ontario Canada
K1A 0LB
1–800–461–9999 (within Canada)
1–204–983–3500 (outside Canada)
1–506–636–5064

(108)

St. Lawrence Seaway Management Corporation (Canada)

(109)

St. Lawrence Seaway Management Corporation
202 Pitt Street
Cornwall, Ontario
CANADA K6J 3P7
613–932–5170
marketing@seaway.ca

Weekly Record of Updates

Week of	Action	Chapter	Paragraph(s)	User notes
05 MAR 2025				U.S. Coast Pilot 1, 55th Edition has been issued.
09 MAR 2025	**No Correction**			
16 MAR 2025	**No Correction**			
23 MAR 2025	**No Correction**			
30 MAR 2025	**No Correction**			
06 APR 2025	**No Correction**			
13 APR 2025	**No Correction**			
20 APR 2025	**No Correction**			
27 APR 2025	**No Correction**			
04 MAY 2025	**No Correction**			
11 MAY 2025	**No Correction**			
18 MAY 2025	**No Correction**			
25 MAY 2025	**No Correction**			
01 JUN 2025	**No Correction**			
08 JUN 2025	**No Correction**			
15 JUN 2025	**No Correction**			
22 JUN 2025	**No Correction**			
29 JUN 2025	**No Correction**			
06 JUL 2025	**No Correction**			
13 JUL 2025	**No Correction**			
20 JUL 2025	**No Correction**			
27 JUL 2025	**No Correction**			
03 AUG 2025	**No Correction**			
10 AUG 2025	**No Correction**			
17 AUG 2025	**No Correction**			
24 AUG 2025	**No Correction**			
31 AUG 2025	**No Correction**			
07 SEP 2025	**No Correction**			
14 SEP 2025	**No Correction**			
21 SEP 2025	**No Correction**			
28 SEP 2025	**No Correction**			
05 OCT 2025	**No Correction**			
12 OCT 2025	**No Correction**			
19 OCT 2025	**No Correction**			
26 OCT 2025	**No Correction**			

This record is intended as a log for critical updates applied to this volume. For online versions or Print on Demand (POD) copies, all weekly critical updates issued and applied to this edition at time of download or purchase are listed.
Affected paragraphs within the chapters are indicated by a gray highlight for ease of identification; e.g. (215)

Updates are available from NOAA at http://nauticalcharts.noaa.gov/nsd/cpdownload.htm.

Week of	Action	Chapter	Paragraph(s)	User notes
02 NOV 2025	**No Correction**			
09 NOV 2025	**No Correction**			
16 NOV 2025	**No Correction**			
23 NOV 2025	**No Correction**			
30 NOV 2025	**No Correction**			
07 DEC 2025	**No Correction**			
14 DEC 2025	**No Correction**			
21 DEC 2025	**No Correction**			
28 DEC 2025	**No Correction**			
04 JAN 2026	**No Correction**			
11 JAN 2026	**No Correction**			
18 JAN 2026	**No Correction**			
25 JAN 2026	**Change**	1	433.0020	
	Insert	1	433.0030	
	Insert	1	433.0040	
01 FEB 2026	**No Correction**			
08 FEB 2026	**No Correction**			

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Updates are available from NOAA at http://nauticalcharts.noaa.gov/nsd/cpdownload.htm.

Week of	Action	Chapter	Paragraph(s)	User notes

Week of	Action	Chapter	Paragraph(s)	User notes

Index

A

Abagadasset Point 274
Abbot Rock 340
Abijah Ledge 192
Abner Point 279
Acadia National Park 187
Adams Point. 316
Addison 181
Agriculture, Department of. 26
Aids to navigation 10, 134
Airy Ledge 219
Alden Rock 224, 288
Aldridge Ledge. 347
Allen Cove 207, 218
Allen Island 248
Allen Shoal 225
Allerton Harbor 359
Alley Island 206
Almore Cove. 172
Amendments 1
American Practical Navigator (Bowditch) 409
Amesbury. 322
Amesbury Point 228
Ames Cove 232
Ames Ledge 274
Ammen Rock 139
Anchorage Regulations 50
Anderson Ledge 318
Andrews Island. 218, 233
Andrews Point 327
Androscoggin River. 273
Anguilla Island. 176
Animal and Plant Health Inspection Service. 26
Annisquam 325
Annisquam Harbor Light 325
Annisquam River. 325
Appledore Island 318
Appledore Ledge 318
Area to be avoided 141
Arey Cove. 191, 226
Arey Ledges 226
Army Corps of Engineers 28
Arrowsic Island 263
Articulated Daybeacons 11
Articulated Lights 11
Asa (Phinney) Island 204
Ash Island. 194, 235
Ash Point 281
Asia Rip 141
Atkins Bay 271
Atlantic. 203
Atlantic Hill 347
Atlantic Point 237
Atlantis Canyon 142
Augusta 275
aurora borealis. 151
Automated Mutual Assistance Vessel Rescue System (AMVER). 14
Automatic Identification System (AIS) Aids to Navigation 12
Avery Ledge 327
Avery Rock 174

B

Babbidge Island 13308 228
Babson Island 214
Babson Ledge 332
Bache Rock 288
Back Beach 341
Back Channel 308, 313
Back Cove. 194, 253, 293
Back Narrows 258
Back River . . . 252, 263, 267, 271, 370
Back River Cove 252
Bagaduce River 242
Bailey Creek. 363
Bailey Island. 279
Bailey Ledge. 172, 249
Baileys Mistake 172
Baker Island. 198
Baker Islands 203
Baker Islet 258
Bakers Island Light 336
Bald Head. 162, 268
Bald Head Cliff. 302
Bald Head Ledge 268
Bald Porcupine Island. 196
Bald Rock. 188, 196
Bald Rock Ledge 196
Ballast Island 177
Ballast Island Ledge. 177
Ballaststone Ledge 278
Bangor 246
Bangs Island. 281, 282
Bantam Ledge 224
Bantam Rock 259
Bare Island 174, 218
Bar Harbor 168, 196, 197
Bar Island. . 174, 176, 189, 196, 207, 230, 248, 251
Bar Island Ledge 252
Bar Ledge. 296
Barley Ledge 233
Barn Point 302
Barnstable 372
Barnstable Harbor 372
Barred Islands 227, 231
Barrel Rock 363
Barrel, The 13301 225
Barter Island Ledges 218
Barter Shoal. 247
Barters Island 267
Bartlett Harbor 229
Bartlett Island 206
Bartlett Narrows 206
Bartlett Rock 328, 367
Bartlett (Smith) Cove 242
Barton Island 230
Barton Ledge 180
Basin Cove 281
Basin Point 281
Basket Island 284, 296
Bass Harbor 205
Bass Harbor Bar 201
Bass Harbor Head 201
Bassing Beach 363
Bass River. 339
Bass Rock. 324
Bates Island 282
Bath 272
Batson Ledges 179
Batson River. 298
Battery Point 236
Bay Ledge. 181, 217, 224
Bay Point 271
Bayside. 240
Bay View 327
Beach Channel. 370
Beach Island. 231
Beach Ledge. 202
Beach Ledges 223
Beals. 178
Beals Cove 280
Beals Harbor 178
Bean Island 194
Bean Ledge 194
Bearings 1
Bear Island 201, 214, 227, 277
Bearskin Neck 328
Beauchamp Point 237
Beaumont Ledge 202
Beaver Island 255
Beef Rock. 274
Belfast 240
Belfast Bay 240
Bellamy River 316
Belle Isle Inlet 357
Benjamin River 214
Benner Island 248
Bernard 205
Berry Cove 195
Beverly Channel 338
Beverly Cove 338
Beverly Harbor. 335, 338
Beyer Ship Ledge. 251
Bibb Rock. 301
Biddeford 296
Biddeford Pool. 297
Big Black Ledge 189
Big Breaking Ledge. 177
Big Holly Cove. 173
Big Moose Island 191
Big Nash Island 180
Billings Cove 215
Billingsgate Island 376
Billingsgate Shoal 375

Bills Island 219
Bill Wallace Ground. 269
Birch Harbor 189
Birch Island 206, 233, 255, 282
Birch Island Ledge 282
Birch Islands 180
Birch Point 278, 283
Bird Rock 209
Biskie Head 327
Black Bess Point 332
Black Duck Cove 179
Black Head 178
Black Horse 219
Black Island 202, 204, 206, 251
Black Island Ledge 251
Black Ledge 182, 204, 228
Black Ledges 173, 179, 216
Blackmans Point 367
Black Point 209
Black Point Cove 172
Black Rock . 166, 179, 209, 224, 247, 332
Black Rock Channel 342, 349
Black Rocks 224, 303, 348
Black Rocks (Eastern Ledges) 177
Blacksmithshop Ledge 263
Blacksnake Ledge 278
Blackwater River 319
Blaney Rock 341
Blish Point 372
Blue Hill 207, 208
Blue Hill Bay 205, 209
Blue Hill Bay Light 3 207
Blue Hill Harbor 207
Blue Hill Rock 219
Blue Point Hill. 296
Bluff Head 272
Bluff Island 296
Blynman Canal 325, 332
Boden Point 341
Boden Rocks 341
Bog Brook Cove 172
Boiler Rock 262
Bois Bubert Island 182
Bold Dick 276
Bold Island 217
Bold Island Ledges 217
Bonney Chess Ledge 187
Boohoo Ledge 334
BookletCharts 3
Boon Island 304
Boon Island Ledge 304
Boot Cove. 172
Booth Bay. 258
Boothbay Harbor 260
Boston 346
Boston Harbor 345
Boston Island 266
Boston Ledge 348
Boston Light 348
Boston Main Channel 349
Boston North Channel 349
Boston South Channel 349
Boundary Ledges 176
Bow Bell 335
Bowden Ledge 201
Bowditch Ledge 337
Bowditch Ledge Daybeacon 336
Bowdoinham 273
Boxam Ledge 204
Brace Cove 329
Bracketts Channel 232
Bracy Cove 201
Bradbury Island 227
Bradford 323
Bradstreet Cove 255
Bragdon Island. 304
Branch Brook 300
Brant Rock 367
Brave Boat Harbor 305
Breakers, The 13305 226
Breaking Rocks 320
Bremen 254
Brewer 245
Brewster Point. 237
Brewster Point Ledge 237
Bridge and Cable Clearances 6
Bridge Lights and Clearance Gages . . 12
Bridges and Cables 1
Bridges Point 214
Bridges Point Shoal. 214
Brighams Cove. 278
Brig Ledge 222
Brimbles 337
Brimstone Island 204, 226
Broadcast Notices to Mariners 9
Broadcast Notice to Mariners 17
Broad Cove . 162, 249, 253, 284, 292, 295
Broad Cove Rock 288
Broad Sound 281, 342
Broken Cove. 282
Brookings Bay 263
Brooklin 214
Brothers Passage 176
Brown Cow 190
Brown Cow, The 13305 219
Browney Island 179
Browns Bank 139, 370
Browns Head Ledge. 251
Browns Head Light 227
Browns Island 274
Brunswick 273
Brush Ledge. 363
Bryant Point. 363
Buck Island 216
Buckle Island 203, 204, 218
Bucks Harbor 174, 215
Bucks Head 174
Bucks Ledge. 245
Bucks Neck 174
Bucksport. 245
Bull Ledge 267
Bull Rock 171
Bulwark Shoal 288
Bumkin Island 361
Bunker Cove 176, 193
Bunker Ledge 181, 226
Bunkers Harbor 190
Bunkers Ledge. 190
Buoys 11
Burial Islet 162
Burnt Coat Harbor 203
Burnt Coat Island 277
Burnt Cove 218
Burnt Island. . . . 219, 227, 234, 248, 260
Burnt Island Ledge 234
Burnt Jacket Channel 273
Burnt Porcupine Island 197
Bushy Islet 278
Bustins Island 283
Bustins Ledge 283
Butter Island 227
Buttonmold Ledges 269
Byard Point 215

C

Cabbage Island. 260
Cable ferries 2
Cadillac Mountain 191
Calais 168
Calderwood Island 228
Caldwell Island. 248
Calf Island 193, 246, 348
Calf Island Bar 193
Cambridge 357
Camden 238
Camden Harbor 238
Camel Ground 269
Cameron Point. 262
Camp Island. 218
Campobello Island 165
Canoe Point 196
Cape Ann 327
Cape Arundel 299
Cape Carter 215
Cape Cod 372
Cape Cod Bay 367
Cape Cod Canal 371
Cape Elizabeth. 285
Cape Harbor. 259
Cape Hedge 329
Cape Island 259, 298
Cape Jellison 242
Cape Levi 196
Cape Neddick 303
Cape Neddick Harbor 302
Cape Neddick Light 303
Cape Neddick Nubble 303
Cape Newagen 259
Cape Porpoise 298
Cape Porpoise Harbor 298
Cape Rosier 214
Cape Sable 139
Cape Small 268
Cape Small Harbor 276

Cape Wash 173
Cape Wash Island. 173
Capitol Island 260
Captains Hill 369
Card Cove. 259, 278
Card Ledge 260
Carlisle Island 258
Carlisle Point 258
Carlow Island 165
Carney Point 274
Carrying Place Cove . . . 172, 187, 276
Carrying Place Head 276
Carrying Place Island 178
Carver Cove 228
Carvers Harbor 230
Carvers Harbor Entrance Light 2 . . 230
Carvers Island 226
Casco Bay. 275
Casco Bay Island 166
Casco Passage 204
Cashes Ledge 139
Castine 242
Castine Harbor. 242
Castle Hill 324
Castle Island. 356
Cathance River. 273
Cat Ledge 224
Cat Ledges 265
Cedar Grove 274
Cedar Island. 230, 318
Cedar Island Ledge 318
Cedar Ledges 278
Cedar Point 364
Center for Operational Oceanographic Products and Services (CO-OPS) 26
Center Harbor 214
Center Hill 347
Center Hill Point 371
Chamberly Island. 181
Chance Island 174
Chandler Bay 177
Chandler Cove 284
Chandler River. 177
Channel Ledge. 226
Channel Markers 12
Channel Rock 179, 207, 213, 214, 217, 219, 228, 234
Chappel Ledge 337
Charity Ledge 279
Charles River 357
Charleston Navy Yard 355
Charlestown. 355
Chart Accuracy 5
Chart Datum, Tidal Waters 5
Chart No. 1 6
Chart Projections. 4
Chart Scale 4
Chart Symbols, Abbreviations and Terms 6
Chatto Island 214
Chauncey Creek 313
Chebeague Island. 284
Chelsea. 355
Chelsea River 357
Cherryfield 183
Cherry Island 166
Cherry Island Bar 343
Childrens Island 336
Childrens Island Channel 336
Chimney Rock 286, 295
Chitman Point 183
Chittenden Rock 363
Chops 273
Chops Point 273
Christmas Cove 257
Chubb Islet 335
Citizenship and Immigration Services . 31
City Point. 356
Clam Cove 165, 237
Clam Ledges. 226
Clam Rock 261
Clapboard Island 284
Clark Cove 246, 258
Clark Island 246
Clark Ledge 164
Clark Ledges 189
Clark Ledges Daybeacon 5 189
Clark Point 189
Clarks Island 313, 370
Cliff Island 281
Closson Point 208
Clough Point 264, 268
Clous Ledge 267
Coastal Warning Display 28
Coast Pilot 1
Cobscook Bay 168
Cocheco River 315
Codhead Ledge. 176
Cod Ledge 204
Cod Ledges 193
Cod Rock 310
Cohasset 364
Cohasset Cove 363
Cohasset Harbor 363
Colbeth Rock 174
Colby Cove 264
Colby Ledge 213
Colby Pup. 213
Coles Ledge 181
Collins Cove. 338
Collins Rock. 224
COLREGS Demarcation Lines . . 45, 134
Colt Head Island 231
Colt Ledge 219, 226
Columbia Falls. 181
Columbia Ledge 187
Columbia Point 358
Commerce, Department of 26
Commercial Maritime Coast Stations and Weather Nets 17
Commercial Point 358
Commissioners Ledge. 347
Compass Island 231
Compass Island Ledge. 231
Compass Roses. 9
Conary Island 214
Conary Ledge 214
Conary Nub 209
Conary Point 209
Concord Point 316
Condon Point 215
Coney Island 336
Coney Ledge. 337
Conomo Point 325
Continental Shelf. 141
Coolidge Point. 334
Coombs Islands 218
Cordage 369
Corea 189
Corea Harbor 189
Corsair Canyon 141
Corvette Ledge. 255
Corwin Rock 288
COSPAS-SARSAT. 14
Courses 2
Courthouse Point 274
Cousins Island. 283
Cousins River 284
Cow and Calf Ledge. 207
Cow Island Ledge Light 284
Cowseagan Narrows. 263
Cowyard 369, 370
Cox Head 271
Cox Ledge. 280
Cozy Harbor. 266
Crabtree Ledge. 194
Crabtree Neck 194
Crabtree Point Ledge 228
Cradle Cove 232
Cranberry Harbor 198
Cranberry Island 251
Cranberry Island Ledge 199
Cranberry Isles. 198
Cranberry Ledge 265
Cranberry Point 189
Crane Island. 230, 251
Crane River 339
Crawford Island 273
Crescent Beach 348
Criehaven. 223
Criehaven Breakwater Light 8 . . . 223
Criehaven Harbor 223
Crockett Cove 226, 229
Crockett Point 236
Cromwell Cove. 197
Crosby Ledge 226
Cross Island 173
Cross Island Narrows 173
Cross Jack Ledge 172
Cross River 267
Crotch Island 218, 252
Crotch Island 13305, 13315 230
Crow Cove 232
Crowell Rock 295
Crow Island . 199, 202, 203, 255, 282, 284
Crow Island Ledge 202, 265

Crow Islands 192, 271
Crowninshield Point 201
Crow Point 361
Crumple Island 179
Cubby Hole 228
Cultivator Shoal 140
Cummings Cove 165
Cundy Harbor 277
Cundys Harbor. 277
Curlew Rock. 179
Currant Island 182
Currents 2
Curry Cove 163
Curtis Island 238
Cushing 249
Cushing Island. 286
Customs Ports of Entry and Stations . 411
Cutler 173
Cutler Peninsula 173
Cutthroat Shoal 337
Cutts Island 305

D

Damariscotta 258
Damariscotta Lake 256
Damariscotta Mills 256
Damariscotta River 256
Damariscove Harbor 259
Damariscove Island 259
Danford Cove 292
Dansbury Reef 298
Danversport 339
Danvers River 339
Dark Harbor. 232
Dark Harbor Cove 232
Darling Island 208
Darling Ledge 208
Davis Island 248, 264
Davis Ledge 337, 364
Davis Strait 248
Davis Strait Passage. 251
Daybeacons 11
Days Ferry 273
Deadman Point 199, 238
Deckers Cove 262
Deep Cove 162, 248
Deep Hole Point 177
Deer Island 165, 173, 347
Deer Island Flats 357
Deer Island Point. 164
Deer Island Thorofare 216
Deer Island Thorofare Light 217
Deer Isle 213, 227
Deer Isle-Sedgwick Bridge 215
Defence Point 242
Defense, Department of 28
Dennys Bay 168
Dennys River 168
Dennysville 168
Department of Agriculture 26
Department of Agriculture, Animal and Plant Health Inspection Service (APHIS) Offices 411
Department of Commerce 26
Department of Defense 28
Department of Health and Human Services 30
Department of Homeland Security. . . 31
Depths 2
Derby Wharf. 340
Designated Critical Habitat. 127
Destructive Waves 19
Devils Back 252, 347
Devils Elbow 252
Devils Head 167, 213, 214
Devils Limb 252
Diamond Cove 292
Diamond Island Ledge. 287
Diamond Island Pass 287
Diamond Island Roads. 288
Diamond Rock 225
Diamond Rock Ledge 225
Dice Head Light 242
Dictator Ledge 260
Digital Selective Calling (DSC) 14
Dillingham Ledge 239
Dillingham Point. 238
Dingley Island 277
Disposal areas 8
Disposal Sites 8
Distances 2
Distress: Communication Procedures . 13
Dix Island. 233, 234, 271
Dix Island Harbor 234
Dobbin Rock 228
Dodge Point 206, 230, 235, 258
Dodge Point Ledge 235
Dodge Rock 327
Dog Bar 332
Dog Bar Channel 332
Dogfish Head 266
Dogfish Island 229
Dogfish Ledges. 228
Dogfish Rocks 173
Dog Island 164
Dolly Head 188
Donald McKay Monument 356
Dorchester Bay 358
Dorchester Bay Basin 358
Double Head Shot Islands 173
Double Shot Island 176
Doubling Point 272
Doubling Point Light 272
Douglas Island Harbor 182
Douglas Island Ledge 182
Douglas Islands 182
Dover 315
Dover Point 315
Doyle Point 283
Drakes Island Beach 300
Drawbridge Operation Regulations . . 57
Dread Ledge. 341
Drisko Island 179
Drown Boys Ledges 179
Drum Island. 202
Drunkard Ledge 219, 228
Drunkers Ledges 279
Dry Breakers 336
Dry Ledge. 265
Dry Ledges 277
Dry Money Ledge. 202
Dry Point 265
Dry Salvages. 329
Duck Cove 205
Duck Creek 375
Duck Harbor 219
Duck Island.. 318
Duck Rock 278
Duck Rocks 225
Ducktrap Harbor 239
Dudley Island 162
Dumping Grounds 8
Dumping of dredged material. 32
Dumpling Islands 228
Dunham Point. 226
Dunham Point Ledge 226
Durham 316
Duxbury 368
Duxbury Bay 368
Duxbury Bay Channels 370
Duxbury Beach 368
Dyer Bay 187
Dyer Cove. 278
Dyer Harbor. 187
Dyer Island 181
Dyer Island Narrows 181
Dyer Neck. 188

E

Eagle 227
Eagle Bar 337
Eagle Head 334
Eagle Island . . . 207, 227, 282, 296, 336
Eagle Island Channel 336
Eagle Island Ledge 282
East Barge 207
East Black Rock 177
East Blue Hill 208
East Boothbay 257
East Boston 355
East Braintree 359
East Brown Cow 276
East Bunker Ledge 201
East Dennis 374
Eastern Bay 179, 195
Eastern Black Ledge 223
Eastern Branch 255
Eastern Channel 363
Eastern Channel (Eastern River . . . 245
Eastern Drunkers Ledge 279
Eastern Duck Rock 225
Eastern Ear 219

Eastern Ear Ledge 219
Eastern Egg Rock 251
Eastern Harbor 180
Eastern Head 172, 219
Eastern Head Ledges 172
Eastern Island 188
Eastern Knubble 172
Eastern Ledge 222, 257
Eastern Mark Island Ledge 217
Eastern Pitch 180
Eastern Point Harbor 194
Eastern Point Light 331
Eastern River 274
Eastern Rocks 318
Eastern Way 188, 199, 201
East Goose Rock 231
Eastham 374
East Hampden 245
East Hue and Cry 288
East Machias 175
East Machias River 175
East Penobscot Bay 220, 225
East Point 202, 295, 303, 342
Eastport 163
East Quoddy Head 165
East Shag Rock 363
East Sister 309
East Sullivan 194
Eaton Point 238
Ebenecook Harbor 266
Echo Soundings 9
Eggemoggin 215
Eggemoggin Reach 213
Egg Rock 179, 184, 193, 231, 342
Egg Rock Ledge 231
Egg Rock North Ledge 251
Egg Rock South Ledge 251
Electronic Navigational Charts (NOAA ENC®) 4
Electronic Positioning Systems 13
Ellingwood Rock 269
Ellisville Harbor 371
Ellsworth 209
Elwell Point 234
Emergency Position Indicating Radiobeacons (EPIRB) 14
Emerson Point 329
Emery Cove 195
Emery Island 235
Emery Ledge 235
Emms Rock Light 9 177
Enchanted Island 218
Endangered Marine and Anadromous Species 125
Englishman Bay 175
Englishman River 176
Enos Ledge 363
Ensign Islands 232
Environmental Protection Agency (EPA) 32
Environmental Protection Agency (EPA) Offices 409
Essex 325
Essex Bay 325
Essex River 325
Estes Head 162
Ewe Island 219
Ewin Narrows 280
Exeter 316

F

Factory Island 296
Falls Point 194
Falmouth Foreside 284
False Halibut Ledge 233
Farmingdale 274
Farnham Rock 367
Farnum Point 256
Farrel Island 218
Federal Communications Commission . 32
Federal Communications Commission Office 412
Federal Water Pollution Control Act (FWPCA) 21
Fernald Cove 200
Fernald Point 200, 201
Ferry Beach 296
Ferry Hill 327
Ferry, Mount Desert 13318 194
Ferry Point 194
Fessenden Ledge 178
Fiddle Head 216
Fiddler Ledge 228, 272
Fiddler Ledge Daybeacon 228
Fiddler Reach 272
Fifield Point 218
Fifteen-foot Rock 336
Finns Ledge 347
Fippennies Ledge 139
Fir Point 203
First Cliff 365
Fisherman Island 179, 233, 259
Fisherman Island Passage 235
Fishermans Beach 341
Fish havens 8, 30
Fish Hawk Island 260
Fish House Cove 277
Fishing Rip 141
Fishing Rock 299
Fish Point 193, 228, 254, 285
Fish Point Ledge 228
Fishtrap areas 8
fishtraps 30
Fitch Point 258
Five Islands 265
Five Islands Harbor 265
Flag Island 277
Flag Island Ledge 277
Flake Island 220
Flanders Bay 193
Flash Island 278
Flat Bay 182
Flat Ground 329
Flat Island 179, 192, 233
Flat Point 276
Flat Pond 302
Fletcher Neck 297
Flint Island 181
Flint Island Narrows 181
Flip Rock 343
Float Plan 16
Flowers Rock 171
Flye Island 207
Flye Island Channel 207
Flye Island Ledge 207
Flying Passage 252
Flying Point 337
Flynns Ledge 199
Fogg Point 283
Fog Island 219
Folly Cove 327
Folly Island 206, 299
Folly Ledge 226, 230
Folly Point 327
Food and Drug Administration (FDA) . 30
Fore River 285, 287
Forest City Landing 292
Fort Edgecomb 264
Fort Gorges 287
Fort Independence 356
Fort Island 181
Fort Knox 245
Fort McClary 307
Fort Point 244, 277, 304, 307, 331
Fort Point Channel 356
Fort Point Cove 244
Fort Popham Light 269
Fort Popham Memorial 269
Fort Preble 287
Fort Scammel 287
Fort Warren 348
Fort William Henry 255
Foss Ledges 316
Foster Channel 174
Foster Island 174, 181
Foster Ledges 224
Foster Point 277
Fourfoot Rock 267
Fourth Cliff 366
Fox Hill Point 318
Fox Islands 270
Fox Islands Thorofare 227
Fox Point 315
Frame Point 313
Frankfort 245
Frankfort Flats 245
Franklin 195
Franklin D. Roosevelt Memorial Highway Bridge 161
Franklin Island 251
Frazer Point 192
Freeman Rock 179
Frenchboro 202
French Island 283
Frenchman Bay 190

Friar Roads 163
Friars Bay. 162
Friars Head 162
Friendship 252
Friendship Harbor 252
Friendship Long Island 252
Frost Island 165
Frost Ledge 165
Frost Point 312
Fuller Rock Light 268
Furber Strait 315

G

Gales Ledge 334
Gallops Island 348
Gander Island 207
Gangway Ledge 251, 260
Gap Head 328
Garden Island 234
Garden Island Ledge 234
Garden Island South Ledge. 252
Garden Point 188
Gardiner 274
Garrison Island 252
Gay Cove 248
Gay Island 249
General Endangered and Threatened Marine Species 124
Geographic Coordinates 2
George Head Island 218
George Island 278
Georges Bank 140
Georges Harbor 248
Georges Island 348
Georges Islands 248
Georges Shoal 140
Georgetown Island 263
Gerrish Island 307
Gerry E. Studds–Stellwagen Bank National Marine Sanctuary . 140
Gerry E. Studds-Stellwagen Bank National Marine Sanctuary 352
Gig Rock 248
Gilbert Canyon. 142
Gilbert Head. 271
Gilchrist Rock 178
Gilkey Harbor 232
Gilley Ledge 198
Gilley Thorofare 199
Gilpatrick Cove 200
Gilpatrick Ledge 200
Gleason Cove 165
Glen Cove. 237
Glidden Ledge 258
Glidden Point 258
Global Maritime Distress and Safety System (GMDSS). 13
Global Positioning System (GPS) . . . 13
Gloucester 331
Gloucester Breakwater Light 331
Gloucester Harbor 331
Gloucester Inner Harbor. 331
Goat Island 271, 315
Goat Island Light. 298
Godfreys Cove 303
Goods Point 187
Googins Ledge 195
Gooseberry Island . . 203, 213, 267, 276
Gooseberry Island Ledge 203, 276
Gooseberry Point. 232
Goose Cove 205, 206, 226
Goose Cove Rock 205
Goosefare Bay 298
Goosefare Brook 296
Goose Island. 207, 231
Goose Ledge. 258
Goose Nest 282
Goose Nest Ledge 282
Goose Rock 238, 263
Goose Rock Passage. 262
Goose Rocks Light 227
Gosport Harbor 317
Gotts Cove 266
Gotts Island 202
Goudy Ledge 277
Gouldsboro 188
Gouldsboro Bay 188
Governors Island 355
Governors Island Flats 357
Grace Rock 235, 273
Grand Beach 296
Grand Manan Banks 139
Grand Manan Channel 171
Grape Island. 359
Grass Ledge 227, 231
Grassy Point. 173
Grays Rock 337
Great Aquavitae 340
Great Bar 177
Great Bay 315, 316
Great Boars Head. 319
Great Brewster Island 348
Great Brewster Spit. 348
Great Chebeague Island 284
Great Cove 177
Great Cranberry Island 198
Great Diamond Island 292
Great Duck Island 202
Great Egg Rock 334
Great Faun 347
Great Gott Island. 202
Great Haste 337
Great Head 173, 197
Great Hill 300
Great Island 376
Great Mark Island 279
Great Misery Island 334
Great Neck 324
Great Pig Rocks 337
Great Pond Cove 174
Great South Channel 141
Great Spoon Island 219
Great Spoon Ledge 219
Great Spruce Head 239
Great Spruce Head Island . . . 227, 231
Great Spruce Island. 176
Great Wass Island 179
Green (Greens) Head 217
Green Harbor 368
Green Harbor Point. 367
Green Harbor River. 367
Green Hill 347
Greening Island 199
Green Island 166, 176, 179, 184, 203, 206, 207, 218, 226, 347
Green Island Passage 281
Green Island Reef 281
Green Islands 202, 266
Green Island Seal Ledges 224
Greenland Cove 254
Greenlaw Cove. 214
Greenleaf Ledge 267
Green Ledge. 216, 217, 219, 222, 226, 230
Green Point 274
Green Point Shoal 224
Green Rock 179, 332
Greens Island 230
Greens Ledge 223
Griffin Ledge 226, 248
Griffith Head 265
Griffith Head Ledge. 265
Grindel Point Light 232
Grindstone Ledge 192, 202, 233
Grindstone Neck 192
Grindstone Point. 192
Grog Island 217
Grog Ledge 217
Groveland. 323
Gulf of Maine 133
Gull Ledge 226
Gull Rock. 162, 252
Gull Rock Ledge 225
Gull Rocks 184
Gully Ledge 222
Gunboat Shoal. 309
Gunning Rocks 247
Gun Point. 279
Gun Point Cove 279
Gun Rock. 347
Guptill Grounds 171
Guptill Ledge 192
Guptill Point 192
Gurnet 280
Gurnet Point 369
Gurnet Strait 280

H

Haddock Island 251, 252
Haddock Island Kelp Ledge. 251
Haddock Ledge 224, 239
Hadley Point 195
Hadlock Cove 198
Haley Cove 318

Halftide Ledge 193, 196, 263
Halftide Rock 172
Halftide Rocks 335, 347
Halfway Rock 219, 279, 336
Halfway Rocks 318
Halibut Hole 193
Halibut Ledge 226
Halibut Point 327
Halibut Rock 233
Halibut Rocks 213, 269
Halibut Rocks Light 213
Halifax Island 176
Hall Bay 263
Hall Island 251
Hallowell 275
Hall Point. 194, 258
Hall Quarry 200
Hamilton Cove. 172
Hampden 245
Hampton Beach 319
Hampton Harbor 319
Hampton River 319
Hampton Shoal Ledge. 320
Hancock Point 194
Handiron Ledge 193
Hangman Island 358
Hanover 366
Hanson Bay 262
Hanus Ledge 204
Harbor Cove. 332
Harbor Island 202, 203, 214, 215, 251, 277
Harbor Island Ledge 218
Harbor Island Rock 251
Harbor Ledge 215, 223
Harbor Ledges 223
Harbor Point 192
Harbor Rock. 329
Harbour De Lute 162
Hardhead Island 227
Harding 278
Harding Ledge 198, 267, 348
Hardwood Island 178, 206, 219
Hardy Rocks. 337
Hardy Shoal 337
Harmon Harbor 265
Harpswell Center. 282
Harpswell Cove 279
Harpswell Harbor 280
Harpswell Neck 281
Harpswell Sound 279
Harraseeket River 283
Harriman Cove 245
Harriman Ledge 207
Harriman Point 207
Harrington 182
Harrington Bay 181
Harrington River 182
Harris Island 304
Hart Bar 247
Hart Island 247
Hart Ledge 247
Haskell Island 281
Haskell Ledge 217
Haste Shoal 337
Hatchet Cove 252
Hathorn Rock 274
Hat Island. 204
Hat Island Ledge 204
Havener Ledge. 253
Haverhill 323
Hawley Ledge 204
Haycock Harbor 172
Haycock Rock 217
Hay Island 255
Hay Island Ledge 214
Hay Islands 226
Hay Ledge 247
Head Beach 276
Head Harbor 178, 219
Head Harbor Island 178
Head Harbour 165
Head Harbour Island 165
Head Harbour Passage 165
Head of the Cape 214
Heal Cove. 263
Heal Eddy. 271
Health and Human Services, Department of 30
Heart Island 227
Heights. 3
Hen Cove 278
Hendricks Harbor 266
Hendricks Head Light 266
Hendrickson Point 235
Hen Island 279
Hen Island Ledge. 278
Hen Islet 278
Henry Cove 192
Henry Point 243
Hermit Island 276
Heron Island 192, 204
Heron Island Point Ledge 204
Heron Islands 270
Heron Neck Ledge 226, 230
Heron Neck Light 230
Herrick Bay 207
Herring Cove 377
Herring Cove (Herring Bay) 165
Herring Ledge 224
Herring River 366
Herseys Retreat 245
Hewes Ledge 233
Hewes Point. 233
Hewett Island 234
Hickey Island 176
Hick Rocks 313
Higgins Beach 295
High Clam Ledge. 234
High Island 233, 234
Highland Light 379
Highlands. 379
High Ledge 222
High Pine Ledge 368
High Pole Hill 376
High Rock 316
High Sheriff. 203
Hilton Point. 268
Hingham Bay 359
Hingham Harbor. 361
Hockamock Head. 203
Hockomock Bay 263
Hockomock Channel 252
Hodgdon Cove 262
Hodgdon Ledge 267
Hodgkins Cove. 327
Hog Cove 238
Hog Cove Ledge 238
Hog Island . 174, 213, 216, 224, 254, 258
Hog Island Point 325
Holbrook Island 243
Holbrook Ledge 260
Holmes Bay 175
Holmes Cove 172
Homeland Security, Department of . . 31
Hook of the Cape 372
Hope Island 282
Horizontal Datum 5
Horn Island 309
Horsehead Island. 231
Horse Island. 281
Horseman Ledge 219
Horseshoe Cove 216
Horseshoe Ledge 202
Hosmer Ledge 243
Hospital Cove 243
Hospital Islands 165
Hospital Point Lighted Range. 336
Hough Ledge 251
Houghs Neck 359
House Cove 173
House Island 286, 334
House Ledge 226, 337
Howard Cove 174
Howard Ledges 215
Howard Point 174, 277
Howland Ledge 367
Hull 347
Hull Bay 359
Hull Gut 359
Hulls Cove 196
Humarock 366
Humpkins Ledge 217
Hunnewell Point 271
Hunting Island. 259
Hupper Shoal 224
Hurricane Island 230
Hurricane Ledge 234
Hurricanes 152
Hurricanes and Tropical Storms. . . . 19
Hurricane Sound. 230
Hussey Rock. 298
Hussey Sound 281
Huston Cove 258
Hydrographer Canyon. 142
Hypocrite Channel 349

I

Immersion Hypothermia. 20
Indian Cove 174
Indian Creek 230
Indian Hill 371
Indian Island 166, 237, 252
Indian Ledge 223
Indian Point. 206, 272, 284
Indian River. 178
Indiantown Island 262
Inner Bar 184
Inner Bay Ledges. 228
Inner Breaker 222
Inner Breakers. 337
Inner Dawes Ledge 202
Inner Green Island 281
Inner Grindstone Ledge 234
Inner Heron Island 257
Inner Heron Island Ledge 257
Inner Ledges 239
Inner Sand Island 179
Inner Seal Rock 359
Inner Shag Ledge 248
Inner Sunk Rocks 320
Inner Winter Harbor 192
International Ice Patrol (IIP) 157
Ipswich. 324
Ipswich Bay 323
Ipswich Light 323
Ipswich River 324
Ironbound Island. 193
Iron Point Ledge 228
Island End River 357
Isle au Haut 219, 220
Isle au Haut Bay 225
Isle au Haut Light 220
Isle au Haut Thorofare 220
Isle of Springs 262
Islesboro 233
Islesboro Harbor 233
Islesboro Island(.)TjSpanActualTextFEFFFFFDFFFDFFFD BDC ( )TjEMC (.)TjSpanActualTextFEFFFFFDFFFDFFFD BDC ( )TjEMC (.)TjSpanActualualTextFE 233

List of Lights (Foreign Countries) . . 409
Lithgow Rock 272
Little Babson Island. 214
Little Bay 315
Little Birch Island 281
Little Black Island 202
Little Black Ledge 189
Little Boars Head. 318
Little Breaking Ledge 177
Little Brewster Island 348
Little Burnt Island 248
Little Bustins Island 283
Little Calf Island 193, 348
Little Camp Island 218
Little Cape Point 179
Little Chebeague Island 284
Little Cranberry Island 198
Little Deer Isle. 215
Little Diamond Island 292
Little Drisko Island 179
Little Duck Island 202
Little Egg Rock 251
Little Egg Rock Shoals 251
Little Faun 347
Little Franklin Ledge 251
Little French Island. 283
Little Gott Island 202
Little Green Island 224, 234
Little Harbor 312
Little Haste 337
Little Holly Cove 173
Little Island 253
Littlejohn Island 283
Little Kennebec Bay 176
Little Ledge 187
Little Machias Bay 173
Little Mark Island 279
Little Mark Island Monument Light . 279
Little Mink Islet 172
Little Moose Island 191
Little Moose Islet. 172
Little Mystic Channel 357
Little Nahant 342
Little Neck 324
Little Pig Rocks 337
Little Point 258
Little Ram Island. 177
Little River . . . 172, 256, 298, 300, 327
Little River Island 172
Little River Ledge 172
Little River Ledges 265
Little River Rock 296
Little Sail Rock 161
Little Salvages 329
Littles Creek. 366
Little Sheepscot River 263
Little Shoal 171
Little Spoon Island 219
Little Spruce Head Island 231
Little Spruce Island. 176
Little Stage Island 298
Little Triangle Ledge 226
Little Whaleboat Island 282
Little Whaleboat Ledge 282
Little White Horse Ledges 165
Little Wood Island 276
Little Yarmouth Island 278
Lobster Cove 325, 326
Lobster Ledge 229
Lobster Rock 232
Local Magnetic Disturbances 9
Local Notices to Mariners 9
Lockes Neck (Straw Point) 317
Logan International Airport 355
Londoner 329
Long Beach 342
Long Cove . 203, 242, 247, 254, 258, 267
Long Cove Ledge 241
Long Island 202, 207, 284, 348
Long Island Head Light 348
Long Island Hub 207
Long Ledge . 172, 173, 193, 199, 204, 217, 234, 267, 277
Long Point 172, 199
Long Point Cove 172
Long Point Island 278, 279
Long Point Light 376
Long Pond Shoal 201
Long Porcupine Island 197
Long Reach 272, 280
Lookout 220
Lookout Point 282, 371
Look Point 181
LORAN-C 13
Lord Rock. 209
Louds Island. 254
Loudville 254
Love Cove. 266
Lovell Island. 348
Lowell Cove 279
Lowell Ledge 236
Lowell Rock 237
Lowell Rock Light 2. 237
Lower Basket Ledge. 284
Lower Cape 372
Lower Fitch Point 258
Lower Gangway Ledge. 234
Lower Goose Island 282
Lower Hell Gate 263
Lower Mark Island 265
Lower Middle Ground 183
Lower Narrows. 254
Lowes Cove 258
Lubec 161
Lubec Channel. 161
Lubec Narrows. 161
Luckse Sound 281
Lumbo Ledge 276
Lunging Island. 318
Lunt Harbor. 202
Lurcher Shoal 139
Lydonia Canyon 142
Lynn Harbor 342

M

Machias 175
Machias Bay 173
Machias Falls 175
Machiasport 175
Machias River 175
Machias Seal Island 171
Mackerel Cove 203, 279, 338
Mackerel Ledge 223
Mackerel Rock 182
Mack Point 241
Mackworth Island 285
Mackworth Point 285
MacMahan 263
MacMahan Island. 262
MacMahan Island Ledge 262
MacMahan Ledge. 263
Maddock Cove 266
Maffitt Ledge 347
Magnolia 334
Magnolia Harbor 334
Magnolia Point. 334
Mahoney Island 206
Mahoney Ledge 206
Maiden Cove. 292
Main Channel Way 178
Main Harbor. 285
Makertown Cove 246
Malaga Island 277, 318
Malcolm Ledge. 222
Malden 357
Malden Island 265
Malden River 357
Manchester 334
Manchester Bay 334
Manchester Channel 334
Manchester Harbor 334
Manchester Point. 200
Man Island 178
Manning Requirements (in part) . . 124
Man of War Head 162
Manomet Hill 369
Manomet Point 371
Manset 200
Maple Juice Cove 249
Maquoit Bay. 282
Maraspin Creek 372
Marblehead 341
Marblehead Channel 336
Marblehead Harbor 335, 341
Marblehead Island 233
Marblehead Light. 336
Marblehead Neck 336, 341
Marblehead Rock 336
Marine Pollution 21
Marine Product Dissemination Information. 28
Marine Protected Area (MPA) 25
Mariner 284
Marine Sanitation Device Standard . 121
Marine Weather Forecasts 27

Mark Island . . . 177, 192, 217, 231, 276
Mark Island Ledge 267, 276, 279
Marlboro Beach 195
Marshall Island 204
Marshall Point 233, 247
Marsh Cove 219
Marsh Cove Ledges 219
Marsh Harbor 254
Marsh Island 176, 252
Marsh River 245, 268
Martin Ledge 348
Martin Point. 252, 285
Martinsville 247
Mary Ann Rocks 371
Mason Bay 177
Mason Ledge 204
Massachusetts Bay 331
Matinicus 223
Matinicus Harbor. 223
Matinicus Island 223
Matinicus Roads 223
Matinicus Rock 222
Matinicus Rock Light 222
Mayflower Ledge 332
Mayo Beach 375
McFarland Island. 261
McFarlands Cove 255
McFarlands Ledges 255
McGlathery Island 213
McGuire Point 258
McHeard Cove 208
McIntosh Ledge 231
McKenney Point 295
McKown Point 260
McNeil Point 194
Meadow Cove 256
Meadow Point 195
Mears Cove 258
Medford 357
Medical Advice 14
Medomak 253
Medomak River 252
Meduncook River. 252
Merchant Island 219
Merchant Row 218
Merepoint. 282
Mere Point 282
Merepoint Bay 282
Merepoint Neck 282
Merriconeag Sound 279
Merriland River 300
Merrill Ledge 267
Merrimack River 320
Merrimacport 323
Merriman Ledge 216
Merrow Island 267
Merry Island. 258
Merrymeeting Bay 273
Metinic Green Island 224
Metinic Island 224
Metinic Island Ledge 224
Midchannel Rock. 337
Middle Bay 282
Middle Breaker. 171
Middle Breakers 337
Middle Brewster Island 348
Middle Ground. 161, 208, 243
Middle Ground Rock 278
Middle Ledge 190, 267, 276
Middle Mark Island 267
Middle River. 175
Middle Rock 200, 227
Middle Shoal 171
Milbridge 183
Mile Ledge 269
Miles Island 267
Milk Island 329
Mill Brook 242
Millbury Ledge. 303
Mill Cove 165, 209, 278
Miller Island. 258
Millet Island. 218
Mill Island 207
Mill Point 276
Mill Pond 226
Mill River 182, 249, 326
Milton 358
Mingo Rock 318
Ministerial Island. 282
Mink Island 173, 181, 208
Mink Islet. 172
Minot Island. 232
Minots Ledge Light 363
Minturn 203
Misery Ledge 176, 337
Mistake Harbor 178
Mistake Island 178
Mitchell Cove 205
Mitchell Point 183
Mitchell Rock 288, 327
Mitchells Falls 323
Moffat Cove 262
Money Cove 172
Monhegan 225
Monhegan Harbor 225
Monhegan Island 224
Monhegan Island Light 225
Monks Hill 369
Montgomery Point 256
Montsweag Bay 263
Monument Bar. 338
Moody Beach 301
Moon Head 359
Moore Point 283
Moores Harbor. 220
Moores Rock 305
Moosabec Reach 177
Moose Cove 172
Moose Island 162, 206, 218
Moose Neck 178, 180
Moose Peak Light 179
Moose River 172
Morgan Bay 208
Morse Cove 244
Morse Island. 251
Morse Ledge. 251
Morse River 270
Morton Ledge 172
Moser Ledge. 251
Moses Hill 334
Moshier Island. 283
Mosquito Harbor 247
Mosquito Head. 247
Mosquito Island 247
Moulton Ledge. 187
Mount Agamenticus 302
Mountainville 214
Mount Battie 238
Mount Desert 200
Mount Desert Island 187
Mount Desert Light. 187
Mount Desert Narrows 195
Mount Desert Rock 187
Mount Megunticook 239
Mousam River 300
Mouse Island 231, 260
Moxie Island. 220
Mud Hole 179
Mud Hole Channel 179
Mulholland Point. 161
Murphy Ledge 252
Murray Rock 305
Murr Ledges. 172
Muscle Ridge Channel. 234
Muscongus 254
Muscongus Bay 251
Muscongus Harbor 254
Muscongus Sound 254
Mussel Point 331
Mystic River 357

N

Nahant 342
Nahant Bay 341
Nahant Harbor. 342
Nahant Rock 343
Nantasket Beach 347, 361
Nantasket Roads 349
Nantucket Shoals. 140
Narraguagus Bay 182
Narraguagus River 183
Narrows 243
Narrows Island. 226
Narrows, The 13324. 188
Nash Island 180
Nash Point 182
Nash Rock Shoal 348
Naskeag 214
Naskeag Harbor 214
Naskeag Point 207
National Data Buoy Center Meteorological Buoys 27
National Environmental Satellite, Data, and Information Service (NESDIS) 28

National Geospatial-Intelligence Agency (NGA) 28
National Institute of Standards and Technology (NIST) 19
National Marine Sanctuary Program Regulations 35
National Ocean Service (NOS) 26
National Weather Service Offices . . . 27
National Weather Service Port Meteorological Officers (PMOs) 28
Nautical Chart–New Editions and Corrections 4
Nautical Chart Numbering System. . . 4
Nautilus Island. 243
Nautilus Rock 243
Naval Observatory 30
Naval Shipyard, Portsmouth 306
Navigational Warnings, Information and Weather 16
Navigation Guidelines, Boston Harbor 346
Navigation Regulations 113
NAVTEX 17
Navy Island 166
Ned Island 192
Neponset River. 358
Newagen 259
Newbury Neck 208
Newbury Old Town 325
Newburyport 320
Newburyport Harbor Light 320
Newcastle. 258
New Castle 306
New Castle Island 306
Newcomb Ledge 337
New Harbor 253
New Harbor Dry Ledges 253
New Harbor Sunken Ledges 252
New Inlet 366
Newmarket 316
New Meadows 278
New Meadows River. 277
Newport Cove 198
Newport Ledge. 197
Nightcap Island 180
Nightcap Ledge 181
Ninefoot Rock 327
Nixes Mate 348
NOAA Weather Radio 27
NOAA Weather Radio Broadcasts . . . 17
Nobscusset Point. 373
No-Discharge Zones 21
No Mans Island 218
No Mans Land 223
Normans Woe 331
Normans Woe Rock. 331
North Branch 255
North Brooksville 244
North Castine 244
North Cutler 173
North Deer Isle 215
North Dennis 373
Northeast Bank 139
Northeast Grave 347
Northeast Harbor. 173, 198, 200
Northeast Ledge 202, 239
Northeast Passage 238
Northeast Point 238, 262
Northeast Pond Ledge. 233
Northern Bay 244
Northern Island 246
Northern Triangles 224
North Gooseberry Island. 336
North Haven 229
North Haven Island. 220
North Islesboro 233
North Jenny Ledge 278
North Ledge. 278
North Plymouth 370
North Point 166, 203
North Popplestone Ledge 219
Northport. 240
North Ridge 324
North River 339, 366
North Rock 165, 171
North Shoal 171
North Sugarloaf 271
Northwest Cove 206, 214
Northwest Harbor 173, 226
Northwest Ledge 235
Norton Island 181, 234
Norton Island Ledge 181
Norton Island Ledges 246
Norwood Cove 200
Notices to Mariners 9
Notification of Arrival (NOA) 25
Nova Rocks 177
Nubble Channel 349
Nubble Island 165
Nut Island 359

O

Oak Island 227, 279
Oak Point. 206
Oak Rock 329
Oaks Reef 299
Oar Island. 253
Obstructions 3
Ocean Dumping 21
Oceanographer Canyon 142
Ocean Point 259
Ocean View 327
Ocean View Hill 377
Oceanville 217
Odiornes Point. 316
Odom Ledge. 244
Ogunquit 301
Ogunquit Beach 301
Ogunquit River 301
Oil Spill Reporting 21
Oil transfer anchorage area. . . 240, 275
Old Anthony Rock 288
Old Cellar Rock 320
Old Cilley Ledge 247
Old Cove 223
Old Duke Ledges 226
Old Harbor 230, 358
Old Horse Ledge 226, 248
Old Hump Channel 251
Old Hump Ledge 251
Old Man 173, 189
Old Man Ledge. 248
Old Orchard Beach 296
Old Prince 299
Old Proprietor 296
Old Soaker 198
Old Sow 164
Old Whale Ledge 197
Old Woman 189
Old Woman Ledge 248
Opechee Island. 204
Orange River 168
Orcutt Harbor 215
Orland 245
Orland River. 245
Orleans. 374
Orono Island 204
Orrington. 245
Orrs Cove 278
Orrs Island 279
Otis Cove 249
Otter Cliff Ledge 198
Otter Cove 180, 198
Otter Island 181, 226, 234, 251
Otter Island Ledge 234, 251
Otter Ledge 202
Otter Point 198, 235
Otter Rock Shoal 243
Outer Bar Island 189
Outer Breakers. 337
Outer Brewster Island 348
Outer Dawes Ledge 202
Outer Green Island 281
Outer Head 265
Outer Heron Island 256
Outer Heron Island Ledge 256
Outer Ledges 239
Outer Minot. 363
Outer Sand Island 179
Outer Scrag Ledge 219
Outer Shag Ledge 248
Outer Sunk Rocks 320
Outer Tautog Rock 370
Oven Mouth. 267
Owls Head 235
Owls Head Bay. 235
Owls Head Harbor 235
Owls Head Ledge 235
Owls Head Light 235
Oxbow Cut 316
Oyster River. 316

P

Paddock Rock 334

Paddy Creek 299
Palmer Cove 340
Palmer Point 341
Pamet Harbor 376
Pamet River 376
Pancake Ground 332
Paper Print on Demand Nautical Charts 3
Parker Cove 233
Parker Flats 271
Parker Head 271
Parker Point 207, 283
Parker River 325
Passagassawakeag River 240
Passamaquoddy Bay 166
Patten Bay 209
Pea Island 342
Peaked Cliff 371
Peaked Hill Bar 379
Peaks Island 292
Pecks Point 195
Peddocks Island 359
Peirce Island 313
Pell Island 219
Pemaquid 255
Pemaquid Beach 255
Pemaquid Harbor 254
Pemaquid Ledge 253
Pemaquid Neck 253
Pemaquid Point 253
Pemaquid River 255
Pembroke 168
Pennamaquan River 168
Penobscot 244
Penobscot Bay 220
Penobscot River 244
Pepperrell Cove 313
Perch Island 260
Perkins Cove 301
Perkins Island 271
Perkins Island Ledge 271
Perkins Point 258
Perry Creek 229
Peters Point 208
Petit Manan Bar 184
Petit Manan Island 184
Petit Manan Light 184
Petit Manan Point 184
Petit Manan Pool 184
Petit Manan Reef 184
Pettis Rocks 271
Phelps Bank 141
Phillips Island 313
Phillips Point 341
Phillips Rock 309
Phippsburg 271
Phoebe Island 218
Pickering Island 227
Pickett Ledge 334
Pierce Cove 266
Pierson Ledge 176
Pig Cove 260
Pigeon Cove 327
Pigeon Ground 224
Pigeon Hill 184, 327
Pigeon Hill Bay 183
Pigeon Rock 327
Pig Island Gut Channel 179
Pig Rock 359
Pilgrim Ledge 334
Pilgrim Monument 376
Pilotage, Belfast 240
Pilotage, Beverly 338
Pilotage, Boothbay Harbor 261
Pilotage, Boston and vicinity 352
Pilotage, Camden Harbor 239
Pilotage, Cobscook Bay 164, 168
Pilotage, Eastport, Cobscook Bay and vicinity 164
Pilotage, Fox Islands Thorofare . . . 229
Pilotage, Frenchman Bay 191
Pilotage, Gloucester 333
Pilotage, Kennebec River 270
Pilotage, Merrimack River 322
Pilotage, Penobscot Bay and River . . 221
Pilotage, Penobscot River 244
Pilotage, Portland 289
Pilotage, Portsmouth 310
Pilotage, Salem 340
Pilotage, Salem Harbor, Beverly Harbor and Marblehead Harbor. . . 338
Pilotage, Searsport 241
Pilotage, Sheepscot River 264
Pine Point 296
Pines River 342
Pine Tree Ledge 288
Pinkham Bay 187
Pinkham Island 280
Pinkham Point 278
Pinkham Shoal 256
Piscataqua River 314
Pitchpine Ledges 276
Placentia Island 202
Platts Bank 139
Pleasant Bay 180
Pleasant Cove 258
Pleasant Cove Ledges 258
Pleasant Point 165, 248
Pleasant Point Gut 248
Pleasant River 181
Pleasure Bay 356
Plover Hill 324
Plum Island 323
Plum Island Point 321
Plum Island River 323
Plum Island Sound 324
Plummer Island 179
Plymouth 368
Plymouth Bay 368
Plymouth Beach 368
Plymouth Harbor 368
Plymouth Harbor Channel 369
Plymouth Light 369
Point Allerton 347
Point Francis 188
Point Ledge 226, 231
Point of Main 176
Poland North Ledge 254
Poland South Ledge 254
Pole Island 278
Pollock Rock 255, 304
Pomp Island 178
Ponce Landing 284
Pond Cove Island 177
Pond Island . . . 182, 206, 216, 268, 279
Pond Island Ledges 279
Pond Island Passage 207
Pond Island Shoal 269
Pond Point 179
Pool Hill 327
Pool, The 13947 172
Poorhouse Cove 255
Poor Shoal 259
Pope Head 337
Pope Head Shoal 337
Popes Folly 162
Popes Island 166
Popes Shoal 166
Popham Beach 271
Porcupine Dry Ledge 196
Porpoise Cove 298
Portable Document Format (PDF) Nautical Charts 3
Port Clyde 247
Porterfield Ledge 237
Porter Landing 283
Porter River 339
Portland 285
Portland Harbor 285
Portland Head Light 286
Portland Lighted Whistle Buoy P . . 285
PORTS® (Physical Oceanographic Real-Time System) 26
Ports and Waterways Safety-General . . 76
Portsmouth 306
Portsmouth Harbor 305
Portsmouth Harbor (New Castle) Light 307
Post Office Ledge 229
Potato Island 218, 230
Potato Ledge 217
Pot Rock 181
Potts Harbor 281
Potts Point 281
Powderhorn Island 267
Powderhorn Ledge 267
Powderhorn South Ledge 267
Powder Point 370
Powers Rock 337
Powwow River 322
Prairie Ledge 332
Pratts Island 266
Preble Island 194
Preble Point 273
President Roads 349
Pressey Cove 226
Presumpscot River 285
Pretty Marsh Harbor 206

Prince Gurnet 280
Prince Point 284
Pripet 233
Proctor Point 334
Prospect Ferry 245
Prospect Harbor 189
Prospect Harbor Point 189
Prospect Inner Harbor 189
Prospect Point 189
Prouts Neck 296
Provincetown 376
Provincetown Harbor 376
Public Health Service 30
Public Health Service Quarantine Stations 411
Pudding Island 223
Pulpit Harbor 229
Pulpit Ledge 192
Pulpit Rock 176, 229, 316
Pumpkin Cove Ledge 253
Pumpkin Island 214, 256
Pumpkin Island Ledge 216
Pumpkin Nob 292

Q

Quahog Bay 278
Quaker Head 173
Quarantine, animal and plant 26
Quarantine Rocks 358
Quarry Point 264
Quincy Bay 358
Quincy Great Hill 359
Quincy Point 359
Quoddy Narrows (Quoddy Roads) . . 161
Qurantine stations, human 411

R

Raccoon Cove 195
Race Point Light 376
Racer Rock 162
Radio Facsimile Weather Information 408
Radio Navigational Aids 3
Radiotelephone Distress Message . . . 13
Rafes Chasm 331
Ragged Island 222, 276, 278
Ragged Neck Point 316
Ragged Point 161
Rainsford Island 358
Ram Head 348
Ram Head Flats 348
Ram Island . 174, 180, 202, 219, 233, 237, 243, 259, 271, 273, 279, 286, 295, 296
Ram Island Ledge 219, 267, 297
Ram Island Ledge Light 286
Ram Islands 267
Ram Islet 337
Rams Horn Channel 338
Rams Horn Rock 338
Randolph 274
Ranges 3
Raspberry Island 181, 247
Raster Navigational Charts (NOAA RNC®)4
Raynes Neck 305
Red Beach 167
Red Head 179
Red Point 203
Red Rock 341
Reed Cove 280
Reed Point 206
Reeds Island 256
Regulated Waters 22
Reid State Park 265
Reported information 3
Reserved Channel 356
Resolution Island 231
Revere 343, 357
Revere Beach 343
Rexhame 366
Rich Hill 278
Richmond 274
Richmond Island 295
Richmond Island Harbor 295
Richs Point 219
Ridley Cove 278
Riggs Cove 263
Right whales, North Atlantic 135
Ringtown Island 204
Ripley Neck 181
Rising States Ledge 337
Roaring Bull 189, 192, 224, 337
Roaring Bull Ledge 219
Roaring Bulls 347
Roaring Rock Point 303
Robbinston 167
Roberts Harbor 226
Roberts Islands 226
Robinhood 263
Robinhood Cove 263
Robinson Rock 231
Rock Harbor 374
Rock Harbor Creek 374
Rock Island 218
Rockland 235
Rockland Harbor 235
Rockland Harbor Breakwater Light . 235
Rockport 237, 328
Rockport Harbor 237, 328
Rocks Nose 304
Rocky Neck 332
Rocky Point 369
Rogers Point 188
Rogue Island 277
Rogues Island 282
Rolling Island 190
Roque Bluffs 176
Roque Island 175
Roque Island Harbor 176
Roque Island Ledge 177
Ross Island 251, 252
Round Island 174, 204, 216, 218
Round Pond 254
Round Rock 161, 278, 320
Round Rock Shoal 332
Rowe Point 327
Rowley 325
Rowley River 325
Royal River 283
Rumell Island 205
Rum Key 197
Russ Island 218
Rust Island 327
Rutherford Island 254
Ryder Cove 233
Rye Harbor 316
Rye Ledge 317

S

Sabbathday Harbor 233
Sabino Head 271
Saco 296
Saco Bay 296
Saco River 296
Saddleback 219
Saddleback Island 217
Saddleback Ledge 278
Saddleback Ledge Light 225
Saddleback Ledge Shoal 225
Saddle Island 231
Sagadahoc Bay 271
Sagamore Beach 371
Sagamore Creek 313
Sagamore Hill 347
Sail Rock 161
Salem 340
Salem Channel 336
Salem Harbor 335, 340
Salem Sound 336
Salem Willows Park 340
Salisbury Beach 320
Sally Island 188
Sally Islands 188
Salmon Falls River 315
Salsbury Cove 195
Salter Island 270
Salt Island 174, 329
Salt Island Ledge 329
Salt Marsh Cove 258
Salt Pond 207
Salt Pond Ledge 254
Salt Rock 334
Sammy Rock 337
Sand Cove 179, 189, 192, 231
Sanders Ledge 304
Sand Island 207, 282
Sand Point 195, 207
Sandwich 372
Sandwich Harbor 372
Sandy Bay 327
Sandy Bay Ledge 328
Sandy Cove 172

Sandy Ledge 166
Sandy Point 245
Saquish Head 370
Saquish Neck 370
Sargentville 215
Sasanoa Point 273
Sasanoa River 263
Satan Rock 337
Saturday Cove 240
Saugus River 342
Sauli Rock 335
Sawyer Cove 178, 206
Sawyer Island 262
Scabby Island Ledge 175
Scabby Islands 175
Scarborough River 296
Scargo Hill 373
Schieffelin Point 194
Schoodic Harbor 190
Schoodic Head 191
Schoodic Island 190
Schoodic Ledge 190
Schoodic Mountain 187
Schoodic Peninsula 191
Schoodic Point Observation Spot . . 191
Schooner Cove 165
Schooner Head 197
Scituate Harbor 364
Scituate Neck 363
Scorton Harbor 372
Scorton Ledge 372
Scotch Island 173
Scraggy Island 218
Scraggy Ledge 219
Scrag Island 203, 231
Sculpin Ledge 208
Sculpin Ledge Channel 349
Sculpin Point 207
Seabrook Beach 320
Seahorse Rock 179
Seal Bay 226
Seal Cove 193, 203, 206, 223, 228, 258, 295
Seal Cove Ledge 173
Seal Harbor 198, 201, 232, 234
Seal Island 222, 233, 234
Seal Island Ledge 234
Seal Ledge . 176, 204, 209, 222, 228, 236, 238
Seal Ledges 251
Seal Rock 179, 260, 268, 295, 348
Seal Rocks 271, 316
Search and Rescue 13
SEARCH AND RESCUE 13
Searle Rock 337
Sears Island 242
Sears Island Ledge 241
Searsport 241
Searsport Harbor 241
Seashore Ledge 174
Seavey Island 248, 306
Seavey Ledges 246
Sea Wall Point 176
Sebasco 277
Sebasco Estates 277
Sebasco Harbor 277
Second Cliff 365
Second Island 218
Sedgwick 214
Seguin Island 268
Seguin Ledges 269
Seguin Light 268
Seguin SSW Ledge 269
Seiche 20
Sellers Rock 226
Sesuit Harbor 373
Seven Hundred Acre Island 13305 . . 232
Sewall Bridge 304
Seward Neck 168
Shabbit Island 178
Shabbit Island Ledge 178
Shabby Island 217
Shackford Head 162
Shag Islet 181
Shag Ledge 176, 223
Shag Rock 176, 236, 271, 318
Shag Rocks 342, 348
Shank Painter Bar 377
Shapleigh Island 313
Shark Island 251
Sheephead Island 226
Sheephead Island Ledges 226
Sheep Island . . . 187, 188, 207, 226, 243
Sheep Island Ledge 226
Sheep Island Shoals 235
Sheep Porcupine Island 196
Sheepscot 268
Sheepscot River 264
Sheldrake Island 194
Sheldrake Ledge 217
Sheriff Ledge 203
Sherman Cove 238
Shingle Island 217
Ship and Barges Ledge 207
Ship Cove 292, 295
Ship Island 207
Shipping 124
Ship Reporting Systems 109
Shipstern Island 183
Shirttail Point 375
Shooting Ledge 195
Shoppee Island 176
Shoppee Point 176
Shorey Cove 177
Short Beach 342
Simms Rock 184
Simonton Cove 292
Sister Island 283
Sister Island Ledge 283
Sister Islands 202
Sister Ledge 202, 226
Sixfoot Rock 262
Skillings River 195
Skinner Head 341
Slins Island 234
Sloop Ledge 265
Smalley Bar 375
Small Point 276
Small Point Harbor 276
Smedrick Ledge 259
Smith Cove 183, 226, 332
Smuttynose Island 206, 318
Snake Island 357
Snippershan Ledge 222
Snow Island 278
Snows Rock 178
Snug Cove 162
Somes Harbor 200
Somes Sound 198, 200
Sorrento 194
Sorrento Harbor 194
Sound Signals 12
South Addison 180
South Berwick 315
South Blue Hill 207
South Boston 355
South Breaker 222, 234
South Bristol 257
South Brooksville 215
South Bunker Ledge 199
South Channel 336
South Dresden 274
Southeast Breaker 224, 256
Southeast Breakers 337
Southeast Harbor 217, 332
Southeast Ledge 171
Southeast Rock 184
Southeast Shoal 171, 304
Southern Harbor 228
Southern Island 246
Southern Mark Island 219
Southern Mark Island Ledge 213
Southern Triangles 224
South Freeport 283
South Gardiner 274
South Gooseberry Island 336
South Gouldsboro 193
South Harpswell 281
South Ledge 209, 251
South Ledges 278
South Orrington 245
Southport 262
Southport Island 258
South Portland 285
South River 340, 366
South Sugarloaf 271
South Thomaston 235
South Trescott 172
Southwest Bank 139
Southwest Breakers 337
Southwest Harbor 198, 199, 226
Southwest Head 171
Southwest Ledge 318
Southwest Ledges 222
Southwest Rocks 198
South Weymouth 360
Space Weather Prediction Center (SWPC)

. 28
Sparrow Island 218
Sparrow Island Ledges 218
Spaulding Island 235
Spears Rock 236
Spectacle Island . 165, 192, 230, 349, 358
Spectacle Island Ledge 216
Spectacle Islands 216, 266
Spiers Stand 359
Spinnaker (Hog) Island 359
Spinney Creek 314
Spirit Ledge 204
Spoil areas 8
Spoonbowl Ledge 276
Spoon Ledge 227
Sprague Ledge 204, 233
Sprague River 270
Springhill Beach 372
Spring Point 285, 287
Sproul Point 255
Spruce Creek 313
Spruce Head 234
Sprucehead Island 234
Spruce Island 165, 218, 232
Spruce Point 190, 260
Spruce Point Ledges 259
Spurling Cove 198
Spurling Rock 199
Spurwink River 295
Squamscott River 316
Squantum 359
Squantum Channel 358
Squantum Point 358
Square Rock 318
Squire Point 177
Squirrel Cove 259
Squirrel Island 259
Squirrel Point 271
Stage Head 332
Stage Island 271, 296
Stage Island Bay 271
Stage Island Harbor 298
Stage Island Shoal 297
Stage Neck 303
Standard Abbreviations for Broadcasts . 17
Stand-in Point 229
Standish Monument 369
St. Andrews 166
Stanley Cove 187
Stanley Ledge 179
Staple Ledge 202
Staple Point 217
Starboard 174
Starboard Cove 174
Starboard Island 174
Starboard Island Bar 174
Starboard Island Ledge 174
Star Island 318
Stave Island 193, 215, 282
Stave Island Bar 193
Stave Island Harbor 193
Stave Island Ledge 282
St. Croix Island 167
St. Croix River 167
Steele Harbor Island 178
Steep Hill 324
Steets Ledge 240
Stellwagen Bank 139
Stellwagen Ledges 364
Stellwagen Rock 371
St. Elmo's fire 151
Stetson Rocks 273
Steuben 188
Steuben Harbor 188
Stevens Island 179
Stevens Rock 295
St. George 249
St. George River 247
St. Helena Island 218
Stielman Rocks 310
Stimpsons Island 228
Stinson Neck 217
St. Mary Ledge 172
Stockbridge Point 283
Stockman Island 282
Stockton Harbor 242
Stockton Springs 242
Stone Horse Ledge 187
Stone Horse Rocks 371
Stone Island 174
Stone Island Ledge 174
Stones Rock 303
Stonington 217
Storm Surge 20
Stover Cove 183
Stover Ledge 264, 267
Straitsmouth Island 328
Stratton Island 296
Strawberry Hill 347
Strawberry Point 363
Strout Island 181
Strout Island Ledges 181
Strout Island Narrows 181
St. Stephen 168
Stump Cove Ledge 215
Sturdivant Island 284
Sturdivant Island Ledges 284
Sturgeon Creek 315
Submarine Cables and Submerged Pipelines 8
Sugar Loaves 228
Sullivan 194
Sullivan Falls 194
Sullivan Harbor 194
Summer Harbor 193
Sunken Black Ledge 228
Sunken Duck Rock 225
Sunken Egg Rock 204
Sunken Ledge 196, 234, 358
Sunken Money Ledge 202
Sunken Pond Ledge 233
Sunset 226
Surry 209
Sutton 201
Sutton Island 201
Swampscott 342
Swan Island 274
Swans Island 203
Sylvester Cove 226

T

Tarbox Cove 267
Tarbox Landing 267
Tar Pouch 364
Taunton Bay 195
Telegraph Hill 347
Temple Heights 240
Temple Ledge 276
Tenants Harbor 246
Tenpound Island 223
Tenpound Island Ledge 332
Territorial Sea 23
Tewksbury Rock 348
Thacher Island 329
The Ark 259
The Boilers 263
The Boring Stone 161
The Boulders 13316 214
The Brandies 219
The Brandies 13313 219
The Brothers 176
The Brown Cow 217
The Bulldog 257
The Castle 187, 324
The Cuckolds 259
The Drums 202
The Glades 363
The Graves 347
The Gut 198, 257
The Hop 197
The Hypocrites 259
The Ledge 167
The Narrows 349
The Pool 172, 198, 297
The Porcupine 172
The Sisters 295
The Thrumcap 197
The Thumper 198
The Triangles 204
The Woodbury 204
Thief Island 252
Thieves Ledge 348
Thomas Rock 320
Thomaston 249
Thompson Island 248, 358
Thorndike Point 235
Thorne Head 273
Thorne Island 273
Thorne Island Ledge 273
Thornton Point 173
Thorofare 176
Thread of Life 255
Thread of Life Ledges 255
Three and One-half Fathom Ledge . . 348

Three Dory Ridge. 139
Three Fathom Ledge 222
Three Rivers. 315
Thrumcap 193, 281
Thrumcap Island 216, 255
Thrumcap Ledge 216
Thrumcap (Thrumcap Island). . . . 232
Thurston Ledges 255
Thurston Point 255
Tibbet Island 267
Tibbett Island 180
Tibbett Narrows 180
Tibbett Rock. 184
Tibbits Ledge 260
Tides 3
Time 3
Tinker Island 207
Tinker Ledges 215
Tinkers Island 165, 337
Tinkers Lower Ledge 165
Tinkers Upper Ledge 165
Tobias Ledge 364
Todd Bay 271
Todds Point 167
Tom Moore Rock 337
Tommy Island 183
Tom Rock. 265
Toothacher Cove 203
Topsham 273
Torrey Castle 214
Torrey Islands 214
Torrey Ledge 214, 219
Tottman Cove 276
Towesic Neck 273
Town River Bay 360
Townsend Gut 262
Traffic Separation Scheme (Boston) . 345
Traffic Separation Scheme (Portland) 285
Trafton Halftide Ledge. 183
Trafton Island 183
Trafton Island Ledge 183
Treasure Island 193, 257
Treat Island 162
Trefethen 292
Trenton. 206
Trevett 267
Triangle Ledge 226
Triangles 208
Trott Ledge 243
Trotts Rock 273
Trumpet Island 207
Trundy Reef 288
Truro 376
Tsunamis 19
Tuckanuck Ledge. 223
Tuck Point 338
Tucks Point 334
Tumbler Island. 261
Tumbler Island Ledge 261
Tupper Ledge 209
Turbats Creek 299
Turkey Cove 249
Turnip Island 255, 263, 279
Turnip Island Ledge. 279
Turnip Yard 219
Turtle Head 233, 240
Turtle Head Cove 233
Turtle Island. 192
Turtle Island Ledge 192
Twing Point 274
Two Bush Channel 233
Two Bush Island . . . 216, 223, 233, 278
Two Bush Ledge 216, 223

U

Ultonia Ledge 348
Under-keel clearances 2
Underwood Ledge 284
Union River 209
Union River Bay 209
Upper Basket Ledge. 284
Upper Cape 372
Upper Flag Island. 281
Upper Gangway Ledge. 234
Upper Goose Island 282
Upper Hell Gate 263
Upper Ledge. 173
Upper Mark Island 267
U.S. Army Corps of Engineers (USACE) Offices 409
U.S. Citizenship and Immigration Service Offices 411
U.S. Coast Guard 31
U.S. Customs and Border Protection . . 31
U.S.S. Cassin Young. 355
U.S.S. Constitution 355

V

Veatch Canyon 142
Verona Island 245
Verona Park 245
Vessel Bridge-to-Bridge Radiotelephone Regulations 43
Vessel Identification 14
Vessel Response Plans 25
Vinalhaven 231
Vinalhaven Island 220
Voluntary Observing Ship Program (VOS) 19

W

Waites Landing 285
Waldoboro 253
Walkers Point 299
Wallace Cove 172
Wallace Ledge 171, 177
Warren Cove 368
Warren Island 232
Washman Rock 298
Water Cove 279
Waterman Cove 228
Waterman Ledge 229
Watts Cove 249
Watts Ledge 295
Waukeag 195
Waukeag Neck 193
Waveland 359
Weare Point 302
Weather, Atlantic Coast, Eastport to Cape Cod 145
Weather, Bar Harbor and vicinity . . 197
Weather, Boston and vicinity 350
Weather, Brunswick and vicinity . . 273
Weather, Eastport and vicinity . . . 164
Weather, Portland and vicinity . . . 288
Weather, Portsmouth and vicinity . . 310
Weather, Rockland and vicinity . . . 236
Weather, South Weymouth and vicinity 360
Weaver Ledge 205
Webb Cove 217
Webber Cove 209
Webber Dry Ledge 251
Webber North Ledge 254
Webber Sunken Ledge. 251
Webhannet River 300
Webster Head 229
Webster Rock 279
Weir Cove. 216
Weir River 361
Welker Canyon. 142
Wellfleet 375
Wellfleet Harbor 375
Wells. 301
Wells Beach 300
Wells Harbor 300
Welshpool. 162
Wentworth Point 258
Weskeag River 234
West Barge 207
West Bay 188
West Black Rock 177
West Branch. 181, 273
West Brooklin 215
West Brooksville 244
West Brown Cow 282
West Chops Point. 273
West Cod Ledge 288
West Cod Ledge Rock 288
West Cundy Point 278
Western Bay. 179, 206
Western Black Ledge 223
Western Channel 342, 363
Western Deer Island Ledge 217
Western Ear 219
Western Ear Ledge 219
Western Egg Rock 252
Western (Gut) Channel 166
Western Harbor 332
Western Head 173
Western Island. 189, 216
Western Ledge 257

Western Passage 165, 188
Western Point 198, 303
Western Reef 183
Western Way 199, 359
West Georgetown. 271
West Gouldsboro 194
West Gut 359
West Halibut Ledges 219
West Halibut Rock 213
West Harpswell 280
West Head 348
West Hogshead Rock 363
West Hue and Cry 288
West Mark Island Ledge 217
Weston Point 283
West Pembroke 168
West Penobscot Bay. 220
West Point 206, 276
Westport Island 263
West Quoddy Head 161
West River 178
West Shag Rock 363
West Sister 309
West Southport 266
West Stonington (West Deer Isle) . . 218
West Sullivan 195
West Tremont 205
West Trenton 206
West Willies 363
Weymouth 359
Weymouth Back River. 361
Weymouth Fore River 359
Weymouth Point 209
Whaleback 223, 335
Whaleback Ledge. 217
Whaleback Light 307
Whaleback Rock 270
Whaleboat Island. 282
Whaleboat Ledge 282
Whale Ledge. 184
Whale Rock 279
Wheaton Island 223
Wheeler Bay. 246
Wheeler Big Rock 224
Wheeler Rock 224
White Bull 276
White Head 286, 347
Whitehead Island. 234
Whitehead Light 234
Whitehead Passage 287
White Horse. 219
White Horse Beach 371
White Horse Island 165
White Horse Rocks 371
White Island. . . 165, 214, 282, 309, 318
White Island Reef. 309
White Islands 230, 256
White Ledge. 219, 269, 335
White (West) Ledge 172
Whiting 168
Whiting Bay. 168
Whitlocks Mill Light 25 167
Whitmore Neck 217
Whittum Island 267
Widow Island 227
Widows Ledge 173
Wiggins Rock 234
Wigwam Point 325
Wildlife and Fisheries 124
Willard Point 181
Willard Rock 288
Wills Gut 280
Wilsons Beach 165
Wind Chill and Frostbite. 20
Windmill Point 163, 347
Winnegance 272
Winnegance Bay 278
Winnegance Creek 272
Winslow Rocks. 273
Winter Harbor 191, 192, 226
Winter Island 340
Winterport 245
Winthrop 357
Winthrop Beach 347
Winthrop Head 347
Winthrop Highlands 343
Wiscasset 268
Witch Island. 255
Witch Rock 288
Wohoa Bay 178
Wollaston Channel 359
Wonsqueak Harbor 190
Woodbridge Island 323
Wooden Ball Island 222
Wood End Bar 377
Wood End Light 376
Wood Island 265, 270, 276, 310
Wood Island Harbor. 297
Wood Island Ledge 310
Wood Island Light 296
Wood Island South Ledge 276
Woods Island 273
Woods Point. 208
Woolwich 273
Worlds End 361
Wormell Ledges 161
Wreck Island 218, 252
Wreck Island Ledges 252
Wreck Ledge 226
Wreck Rock 359
Wylie Rock 260
Wyman. 183
Wyman Ledge 276

Y

Yacht Basin 218
Yarmouth. 283, 373
Yarmouth Island 278
Yarmouth Ledges. 278
Yarmouth Port. 372
Yeaton Cove 187
Yellow Birch Head 187
Yellow Head 174
Yellow Island 193
Yellow Ledge 172, 204, 234
Yellow Ridge Islet 234
Yellow Rock 278
York Beach 303
York Harbor 303
York Island 219
York Ledge 284, 305
York Narrows 204
York River 303
York Village 303
Young Point 263
Youngs Bay 195
Youngs Point 195

Z

Zephyr Ledges 223
Zephyr Rock. 223